Gauge-Covariant Variational Free Energy and Exact Coarse-Graining

Agents as Local Sections at One Base Point: Collective Evidence, Attention, Inference Histories, and Formation

Robert C. Dennis

Independent Researcher

Date: September 2026

Contents

I Single-Point Foundations **1**

1 Program, Scope, and Standing Conventions **2**
1.1 Program aim . . . 2
1.2 Objects and the local-first architecture . . . 3
1.3 Results and reading map . . . 4
1.4 Standing conventions and scope . . . 5
1.5 Status of statements . . . 6

2 Geometry at One Base Point **8**
2.1 The base point, the principal torsor, and associated law fibers . . . 8
2.2 Frames, passive re-expression, and the active gauge group at c_* . . . 9
2.2.1 Pointwise frame comparisons and the active gauge group . . . 9
2.2.2 The common frame is a gauge choice . . . 10
2.3 Agents and cross-associated-bundle morphisms . . . 11
2.4 Independent links at c_*: vertex law, holonomy, edge twists . . . 12
2.5 Running example: three agents on one loop . . . 16

3 Probability Semantics at c_* **18**
3.1 Latent and record spaces at c_* . . . 18
3.2 Structural data and kernel signatures . . . 19
3.3 Reference measures . . . 20
3.4 Kernels first, densities second . . . 20
3.5 Absolute continuity . . . 21
3.6 The relative-entropy chain rule and data processing . . . 22
3.7 Recognition marginals, couplings, and selection . . . 23
3.8 Levels of description . . . 25

4 The Fixed Normalized Generative Law **28**
4.1 Parameters, the directed latent law, and the typing prohibition . . . 28
4.2 Normalized record extension and the undirected alternative . . . 30
4.3 Covariance under passive frame re-expression . . . 32

5 One Evidence Bound and Local-Collective Free Energy **35**
5.1 Recognition coupling and total correlation . . . 35
5.2 Evidence domain . . . 36

5.3 The extended evidence bound and its record presentation . . . 37
5.4 Restriction principle . . . 40
5.5 The factorwise decomposition . . . 42
5.6 Observations as interaction records . . . 43
5.7 Local conditional free energy and the potential identity . . . 44
5.8 Additive accounting . . . 48
5.9 Model coordinates and accepted updates . . . 49
5.10 What remains open . . . 51

6 Attention and Selective Communication 53
6.1 Attention as a latent source label . . . 53
6.2 Communication as a normalized channel . . . 59
6.3 Factored links and loops . . . 60
6.4 Comparison, content, and the transformer correspondence . . . 62
6.4.1 Exact receiver update and transformer attention . . . 63
6.5 Compositions as records . . . 66
6.6 Information loss and descent . . . 67

7 Fiber Information Geometry and Inference Histories 70
7.1 Fiber statistical tensors . . . 71
7.2 Divergence jets and the orbit Fisher form . . . 72
7.3 Markov coarse morphisms and the vertical Fisher defect . . . 73
7.4 Configuration manifolds and the exact free-energy lift . . . 75
7.4.1 Canonicity boundary of the lift . . . 78
7.5 A finite correlated lift: the parity witness . . . 79
7.6 Natural-gradient orbits . . . 81
7.7 Fisher duration . . . 83
7.8 Orthogonal clocks . . . 85
7.9 Record clocks . . . 86
7.10 The coarse configuration map and oriented semiconjugacy . . . 87
7.11 Semiconjugacy on symmetry quotients . . . 90
7.12 Coarse-graining depth, orbit parameter, and duration . . . 91

8 Two-Timescale Dynamics 93
8.1 Block descent at fixed parameter . . . 93
8.2 Fast recognition and slow model coordinates . . . 94
8.3 Parent-sensitive inference and the prepared parent . . . 100

9 Lie-Algebra Links as Latent Fields 103
9.1 The exponential chart of the links . . . 105
9.2 Curvature on a noncompact structure group . . . 107
9.3 The generator prior . . . 110
9.4 Latent generators in the collective evidence bound . . . 113

9.5 The link field equation . . . 117
9.6 Held links as a limit, and conditional statements . . . 122
9.7 Directed and undirected matter . . . 124
9.8 Inference dynamics of the links . . . 125
9.9 Link histories across learning episodes . . . 128
9.10 The linearized link equation . . . 132

II Exact Coarse-Graining at One Base Point 137

10 Coarse Channels 138
10.1 Markov coarse maps . . . 138
10.2 KL contraction, equality, and recovery . . . 139
10.3 The common-channel VFE identity and the pointwise parent datum . . . 141
10.4 Pushforward, precomposition, and Galerkin restriction . . . 143
10.4.1 Galerkin operators and message passing . . . 144
10.5 Gauge and permutation equivariance . . . 145
10.6 Scale diagrams . . . 147
10.7 Labeled families and retained label information . . . 147
10.8 Retained-label Fisher asymptotics . . . 150

11 Gauge-Retained Network State and Fixed Sectors 153
11.1 Levelwise fiber state and scale intertwiners at c_* . . . 153
11.2 Rooted trees, based holonomy, and dressed edges . . . 155
11.3 Linear feature tier and tree dependence . . . 157
11.4 Holonomy-conditioned marginal-law modes . . . 160
11.5 Structural fixed sectors . . . 165
11.6 Cut links and rectangular sheaf closure . . . 167
11.7 The frame sector . . . 172
11.8 An operational record at c_* sensitive to loop holonomy . . . 173

12 Exact Agent-Network Coarse-Graining 175
12.1 Exact effective action . . . 175
12.2 Bounded action calculus and contraction . . . 177
12.3 Finite-network interaction coordinates . . . 180
12.4 The exact interaction step, its residual, and finite-step multipliers . . . 182
12.5 Exact closure generically requires hyperedges . . . 187
12.6 Fine–meta bridge kernels . . . 188
12.7 Exact attention between meta-agents . . . 189
12.8 Path laws and memory . . . 192
12.8.1 The hidden history and its filter . . . 194
12.8.2 Finite-memory projection . . . 195
12.9 The exact retained state . . . 197

13 The Multiscale Graph VFE and Closing the Tower 200
13.1 The full answer before decomposition . . . 201
13.2 A normalized hierarchical graph model . . . 201
13.3 The scale-local two-channel graph energy . . . 204
13.4 Partition variables and event pushforwards . . . 205
13.4.1 A random partition catalog . . . 206
13.4.2 Membership kernels and event pushforwards . . . 207
13.5 Coarse links and what the parent retains . . . 208
13.6 How a parent influences its children . . . 214
13.7 Closing the tower . . . 214
13.7.1 A declared root . . . 215
13.7.2 Three loci of participation . . . 215
13.7.3 The remaining hierarchy question . . . 219

III Formation and Interpretation 222

14 Formation Witnesses 223
14.1 The pointwise carrier and its restoring flag . . . 225
14.2 One normalized experiment with a nonlinear source record . . . 226
14.3 Physical posterior, restricted optimizer, and recognition . . . 228
14.4 Exact marked parents and their inherited inference . . . 229
14.5 Complete model scores and dissipation . . . 230
14.6 Preparation and successive activation . . . 231
14.7 The finite-depth formation theorem . . . 234
14.8 A two-triangle instance and the remaining scope . . . 237
14.9 An endogenous four-cycle witness . . . 239

15 Interpretation and Its Limits 245
15.1 Scaffolding, idle posits, and structural readings . . . 245
15.2 Participation, formation, and testability . . . 247

16 Related Work 252
16.1 Gauge theory, free energy, and multi-agent active inference . . . 252
16.2 Connection Laplacians, sheaves, and equivariant networks . . . 253
16.3 Lattice gauge theory, induced actions, and learning rules . . . 253
16.4 Averaging and path-space coarse-graining . . . 254
16.5 Information-geometric and Bayesian renormalization . . . 254
16.6 Network renormalization, graph coarsening, and multigrid . . . 256

A Measure-Theoretic Lemmas and Integrability Witnesses 258
A.1 Kernel measurability and proofs for the probability chapter . . . 258
A.2 The relative-log representation and integrability witnesses . . . 261

B Deferred Proofs **263**
B.1 Proofs for Attention and Selective Communication 263
B.2 Proofs for Fiber Information Geometry and Inference Histories 266
B.3 Proofs for Lie-Algebra Links 270
B.4 Proofs for Coarse Channels 283
B.5 Proofs for Exact Agent-Network Coarse-Graining 288

C Proofs for Finite-Depth Formation **295**
C.1 Exact marginals, rates, and parameter derivatives 295
C.2 Prepared laws and coupled continuation 298
C.3 Entropy estimates 301
C.4 Exact fast marginals and complete source histories 306
C.5 Coherence, source movement, and continuous paths 310
C.6 The Gaussian core across an early interface 315
C.7 The endogenous four-cycle 317
C.8 Depth one under exclusive source gating 323
C.9 Proof of Pairing Selection 325
C.10 Proof of Joint Selection Along a Schedule 336
C.11 Coherence and Persistence of the Selected Pairing 341
C.11.1 The first contrast on the original episode window 344
C.12 Activation from the Actual Conditioned Recognition Law 346
C.13 Early Activation on the Seeded Contrast 353

D Notation **367**

E Open Problems **371**

F Numerical Index **378**

List of Tables

1.1 Status of statements. 6

6.1 Correspondence between transformer attention and the pointwise attention network at c_*. 65

9.1 Symbols of the latent-link construction. 104

12.1 The retained state of exact agent-network coarse-graining at c_*. 198

14.1 Chapter-local symbols of the formation families, shared with the proof appendix. . 224

C.1 Lemma dependencies of the finite-depth formation theorem. 295

D.1 Notation of the main manuscript. 367

E.1 Open problems of the main manuscript. 371

F.1 Numerical index of the main manuscript. 378

Part I

Single-Point Foundations

Chapter 1

Program, Scope, and Standing Conventions

This manuscript asks what collective inference is available to agents that are local sections of a belief and a model associated bundle of one principal bundle, placed at one base point and interacting through independent links under one normalized joint law, and which parts of that inference survive exact coarse-graining of the agents into meta-agents. The answer has five main parts. The whole population has one evidence bound, an identity in the extended reals whose gap vanishes exactly at the posterior as a measure (Corollary 5.7 and theorem 5.8). Each agent's local objective is a conditional free energy whose differential is that of the collective free energy (Theorem 5.23). The links can themselves be inferred as Lie-algebra generators under a prior that is normalized for every structure group, and they then enter the free energy through one relative entropy (Propositions 9.12 and 9.18). Coarse channels of the agent network preserve the evidence and decompose the free energy exactly (Theorem 10.7), with an effective action for the retained coordinates (Theorem 12.3). A finite-depth family forms successive meta-agents under free-energy descent with learning, in a stated scope (Theorem 14.17). Section 1.3 maps these results to the chapters, and Section 1.4 fixes, once for the whole manuscript, the defaults and the claims it declines.

1.1 Program aim

The program is a participatory multi-agent theory. Agents are local sections of associated bundles, collectives form through the agents' coupled inference, and parents act back on their children through normalized laws rather than through quantities that read the children's beliefs. Three requirements follow from that aim, and the manuscript is organized around them. Every collective object must be a complete probability object, constructed from the agents' mechanisms and interaction records or selected explicitly, never an additional agent and never a sum of local objectives that happens to look like one. Every statement about a parent must hold for the normalized parent law that coarse-graining actually produces. And formation of a parent must be a theorem about a declared family, with its scope stated, rather than a reading of a formula.

The links between agents are inferred as well. Written in the exponential chart, their Lie-algebra generators are latent variables under a normalized prior, and they move with the agents under descent of the one collective free energy (Chapter 9, Proposition 9.28). A result stated at held link

values is a statement about the conditional laws given the generators (Proposition 9.26), and held links are the zero-width limit of the latent model (Theorem 9.25).

The present phase fixes one base point c_* and settles the theory there before any transport across the base is considered. At c_* the exact normalized parent law exists for every coarse channel of the agent network (Corollary 10.11), and Theorem 14.17 gives a formation witness under a supplied hierarchy of interaction strengths. On a homogeneous four-cycle a random initial source assignment already makes a collective form from every Gaussian initial law, with source learning idle (Proposition 15.4). At each sufficiently small fixed temperature, slow source learning selects one matching in the assignment law for almost every Gaussian initial mean, while the frozen-prior pairing difference vanishes at long times (Theorem 15.5). How a parent's own dynamics are determined by the network it summarizes is Open Problem 12.40, the defining open problem of the phase.

1.2 Objects and the local-first architecture

Agent i is the object $\mathcal{A}^i = (\mathcal{C}_i; q_i^b, q_i^m, u_i)$ of Definition 2.8: a belief section and a model section of the associated bundles $\mathcal{E}_b$ and $\mathcal{E}_m$ of one principal G-bundle, expressed in one common frame u_i, a gauge choice by Proposition 2.7. The two channels may carry inequivalent representations, and cross morphisms between them are additional structure (Definition 2.9). Interactions are normalized record kernels K_a on the scopes of an interaction multigraph, and the links Θ_e^x on its edge copies are independent data with loop holonomy and per-edge invariant classes (Proposition 2.12), held as structural data, moved as parameters, or inferred as latent generators under a declared prior (Definition 9.15). The primitive descriptive data are the agents, their local recognition laws, their evaluated mechanisms, and the record kernels. Population laws enter only by construction (Proposition 4.4) or by an explicit selection (Definition 3.15). The architecture is

$$\{\mathcal{A}^i, G_{\theta,i}, \mathbb{Q}_{i,o,X}, K_a\}_{i\in V, a\in\mathfrak{A}} \longrightarrow \begin{cases} \mathbb{P}_{\theta,V} & \text{normalized construction,} \\ \mathbb{Q}_{V,o,X} \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}) & \text{measurable selection,} \end{cases} \longrightarrow \mathcal{L}_V, \tag{1.1}$$

so the constructed joint and the selected coupling precede every local and collective objective. On the evidence domain of Hypothesis 5.3 there is one identity,

$$\log p_{\theta,V}(o \mid X) = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + D_{\mathrm{KL}}\big(\mathbb{Q}_{V,o,X} \big\| \mathbf{\Pi}_{\theta,V,o,X}\big), \tag{1.2}$$

with equality of the bound and the log evidence exactly when the recognition coupling is the posterior as a measure, which matching coordinate marginals does not achieve (Theorem 5.8). The identity is invariant under passive re-expression of the frames when the kernels are covariant (Proposition 4.9 and corollary 4.10). Excluding the posterior from a recognition family makes every pointwise gap positive, but it does not make the optimized gap positive or attained (Section 5.4).

1.3 Results and reading map

Part I builds the single-point foundations. Chapter 2 fixes the geometry at c_*: the active gauge group is one copy of G, links carry per-edge twist classes, and three notions of agreement are separated. Chapter 3 fixes kernels before densities, proves the chain rule used everywhere, and distinguishes sample, law-point, and configuration levels. Chapter 4 builds the normalized joint on arbitrary record cycles and states the typing prohibition. Chapter 5 derives the evidence bound and the local-collective potential identity, with observations entering as interaction records (Theorem 5.17). Chapter 6 places attention inside the fixed joint as a latent source label (Proposition 6.2), builds selective communication, and records its correspondence with transformer attention (Table 6.1). Chapter 7 develops fiber information geometry, Fisher contraction under Markov morphisms (Theorem 7.8), and inference histories measured by a Fisher duration that is invariant under reparameterization (Theorem 7.36). Chapter 8 types fast beliefs and slow models (Definition 8.2) and proves the reduced slow flow (Proposition 8.4). Chapter 9 infers the links as Lie-algebra generators under a Gaussian prior reweighted by a loop action, normalized for every structure group (Proposition 9.12); their stationarity condition is a link field equation (Proposition 9.21), and held links are the zero-width limit of the latent model (Theorem 9.25).

Part II coarse-grains at one base point. Chapter 10 separates Markov coarse channels, energy restriction, and parameter aggregation (Section 10.4) and proves the exact coarse free energy and the pointwise parent datum (Theorem 10.7 and corollary 10.11). Chapter 11 gives the gauge-retained network state, the kernel–holonomy fixed sectors (Proposition 11.19), and the holonomy-conditioned marginal law (Theorem 11.10). Chapter 12 is exact coarse-graining of the agent network: effective action, residual, finite-step multipliers, and the native growth bound (Theorems 12.3 and 12.4 and proposition 12.21), with the retained state collected in Table 12.1 and a parent-sensitive mechanism with path laws and an exact finite-memory cost (Section 8.3). Chapter 13 assembles the multiscale free energy and closes a tower of priors (Section 13.7).

Part III turns to formation and interpretation. Chapter 14 constructs, for every finite depth, a family in which successive coherence episodes come from linear relaxation of a prepared mean through a supplied hierarchy of interaction strengths; source jumps and learning are perturbations the proof controls, not the cause of formation, and a separate witness produces a four-cycle endogenously (Proposition 14.22). Chapter 15 states what the mathematics does and does not support, and Chapter 16 positions the construction. The appendices hold measure-theoretic lemmas (Appendix A), the formation proofs (Appendix C), the notation (Appendix D), the open problems (Appendix E), and the numerical index (Appendix F).

Two supplements accompany the manuscript, and no result here depends on them. Supplement G develops the multivariate-Gaussian realization. Supplement B collects base transport, covers and connection descent, multi-point designs, infinite networks with their asymptotic apparatus, and the lattice cluster.

A first reading can take Chapters 2 to 4 for the objects and the joint, then Chapter 5 for the bound and Chapter 6 for attention, then Chapters 10 and 12 for exact coarse-graining and Chapter 14 for

formation; proofs, witnesses, and the ledger can be recovered through their cross-references on a second reading.

1.4 Standing conventions and scope

Everything takes place at one base point. The design is $\{c_*\}$, design and incidence indices are dropped, and every agent's frame, beliefs, and models are points of the fibers over c_*. Base maps and base curves, neighborhoods, connection descent, transport along the base, section spaces over $\mathcal{C}$, and continuum law theory are deferred to the base supplement named in Section 1.3.

The following defaults hold throughout, and a statement mentions only departures from them. Measurable spaces are nonempty and standard Borel, and $\mathcal{P}(\mathsf{S})$ denotes the probability laws on a measurable space S. Laws and kernels are normalized and measurable. A coarse channel is a Markov kernel on latent coordinates that does not depend on the recognition law, leaves observations unchanged, and is vertical over the identity of c_*. An admitted observation has finite positive evidence. Structural data are parameters unless a configuration prior or a structural channel is declared, and the declaration names its law; the latent-link model of Definition 9.15 is such a declaration for the links. The interaction multigraph at c_* is a structural parameter whose edge copies come in reversal pairs; it contains every copy in the support of a receiving row's source prior and, unless a model adds further copies, is the reversal closure of the union of these supports. Each support is complete unless masked, and supports need not be symmetric: a receiver may attend to a source that does not attend to it. Gibbs attention weights are strictly positive on positive-prior edges when their costs are finite; the extended-cost rows of Proposition 13.9 can assign zero weight to infinite-cost edges. Dynamic structure is carried by attention weights, link values, and, in the latent-link model, the inferred generators. Discrete creation and deletion of edges is not part of the model; it is Open Problem 6.9. Coarse multigraphs are induced structure. Absolute continuity of a recognition law with respect to the posterior is not a default: it is clause (H3) of Hypothesis 5.3, and its failure is the singular branch of the extended bound.

Three transports must remain distinct. A pointwise comparison T_{ij} relates two frames in the same principal fiber and involves no curve (Proposition 2.5). A link Θ_e^x is an independent variable on an edge copy of the interaction multigraph (Definition 2.11). Parallel transport along a base curve is not used. Graph links are not base transport, and graph-link holonomy is neither the holonomy of a base connection nor a property of the topology of the principal bundle.

The word coarse has two uses that must not be merged. A Markov channel discards sample information while preserving normalization; a hard restriction substitutes a lower-dimensional configuration into an energy. They can agree in a special realization, but a shared word transfers no theorem, and parameter aggregation inside a closed family is a third operation (Section 10.4). Likewise, an inference history is an oriented unparameterized orbit in a configuration space, an ordered interaction-record path retained by exact coarse-graining is a sample coordinate, coarse-graining depth is a scale index, and a Fisher duration is a statistical length.

Scope. The manuscript declines the following claims. None of the inference index, flow parameter, transition index, ordered path coordinate, coarse-graining depth, or Fisher duration is identified with physical time; such an identification would need a named target clock, a calibration map with an uncertainty model, causal and compositional consistency, and discriminating evidence against alternative parameterizations. The contextual base is not spacetime and carries no Lorentzian structure. Sections, laws, and meta-agents are mathematical objects: calling a boundary node an agent is a naming decision, no claim of autonomous agency or of what exists is made, and population laws, population samples, and blocks are not agents. No canonical coupling selector, partition, or membership selector is asserted. No renormalization semigroup with rescaling, fixed points, or beta functions is constructed; Supplement B holds that asymptotic apparatus. The identification of any limiting object with a physical law is Open Problem 15.7. Declining a claim is not refuting it, and a refuted statement appears instead as an established negative result. [NOT-CLAIMED]

1.5 Status of statements

Every definition, standing hypothesis, lemma, proposition, theorem, corollary, conjecture, open problem, and claim-bearing remark is a numbered statement. Definitions, standing hypotheses, conjectures, and open problems carry their status in their heading type. Every lemma, proposition, theorem, corollary, and claim-bearing remark carries exactly one of three tags, one for all of its conclusions. Text that motivates, summarizes, or points carries no tag and makes no new claim. The fourth tag of Table 1.1, for declined claims, appears once in each scope paragraph and nowhere else. Open problems are collected in Table E.1, and numerical statements with their checks in Table F.1.

Table 1.1: Status of statements.

Tag or heading	Meaning
[ESTABLISHED]	Proved in this manuscript. A refutation is an established negative statement.
[STANDARD]	A known result, cited to a checked primary source with a theorem or section locator.
[NUMERICAL]	Supported only by a named computation, indexed with its control and seed; computation is not proof.
Definition	A declared type, construction, or convention; nothing is proved.
Hypothesis	A standing restriction adopted for a chapter or part; it states what it excludes.
Conjecture	A precise unproved statement with a named observation that would refute it.
Open problem	An unsettled question with what holds, what is unproved, and what would settle it.
[NOT-CLAIMED]	A claim declined in a scope paragraph; declining is not refuting.

Physicist's summary

The manuscript studies agents that are local sections of belief and model bundles, all at one base point, coupled through independent links and interaction records under a single normalized joint law. The links can be held fixed or inferred, as Lie-algebra generators under a prior that is normalized for any structure group. There is one evidence bound for the whole population, the local objectives are exact conditional pieces of it, and coarse-graining the agent network into meta-agents is done with Markov channels that preserve the evidence and split the free energy exactly. Formation of meta-agents is proved for a declared finite-depth family in a stated regime and conjectured more generally. Physical time, spacetime, agency, and transport across the base are not claimed; they are either deferred to the base supplement or declined, and the defaults stated above are not repeated in later chapters.

Chapter 2

Geometry at One Base Point

This chapter answers what geometric data an agent carries at one base point c_* and which of those data are invariant. The active gauge group at c_* is one copy of G inside the passive vertex group $\prod_{i\in V} G$ (Proposition 2.6); every independent link carries a gauge-invariant conjugacy class, and loop holonomy is the telescoped product of these per-edge twists (Proposition 2.12); frame agreement, agreement of an equivariant statistic, and law coherence up to holonomy stabilizers are distinct (Proposition 2.18). Chapter 4 uses passive re-expression for covariance of the generative law, Chapter 6 builds attention on link-transported comparisons, and Chapter 11 turns holonomy and the trivializing criterion (Proposition 2.13) into fixed sectors of coarse channels.

Scope. Normalizability of a law on links built from the loop penalties of Remark 2.16 against the Haar reference is not asserted for a noncompact or general linear G (the compact unitary case is in the remark, and a Gaussian reference on the Lie algebra normalizes the generator prior of Proposition 9.12 for every G), and whether law coherence bears on the selection of a partition into meta-agents is part of Open Problem 10.20. [NOT-CLAIMED]

2.1 The base point, the principal torsor, and associated law fibers

Definition 2.1 (Contextual base and base point). Let $\mathcal{C}$ be a finite-dimensional, second countable, Hausdorff smooth manifold whose points c are called contexts, and fix one base point $c_* \in \mathcal{C}$.

Definition 2.2 (Common principal gauge bundle and agent frames). Let π be a smooth principal right G-bundle for a Lie group G and, for each agent $i \in V$, let u_i be one local frame on an open support $\mathcal{C}_i \ni c_*$:

$$\pi : \mathscr{P}_G \longrightarrow \mathcal{C}, \tag{2.1}$$

$$u_i : \mathcal{C}_i \longrightarrow \mathscr{P}_G, \qquad \pi \circ u_i = \mathrm{id}_{\mathcal{C}_i}\,. \tag{2.2}$$

The fiber $(\mathscr{P}_G)_{c_*}$ is a right G-torsor and the values $u_i := u_i(c_*)$ are points of it. For any reference point $\sigma \in (\mathscr{P}_G)_{c_*}$ the frames are $u_i = \sigma \cdot U_i^{-1}$ for unique $U_i \in G$.

The same frame u_i serves the belief and the model channel; Proposition 2.7 shows that this is a gauge choice. On the belief and model sample fibers $(\mathsf{K}, \mathscr{K})$ and $(\mathsf{M}, \mathscr{M})$ declare law fibers, measurable representations, which need not be equivalent, faithful, or of equal dimension, and their

pushforward actions on laws,

$$\mathcal{B}_b \subseteq \mathcal{P}(\mathrm{K}), \qquad \mathcal{B}_m \subseteq \mathcal{P}(\mathrm{M}), \tag{2.3}$$

$$\rho_b : G \longrightarrow \mathrm{Aut}(\mathrm{K}), \qquad \rho_m : G \longrightarrow \mathrm{Aut}(\mathrm{M}), \tag{2.4}$$

$$\widehat{\rho}_b(g)q = (\rho_b(g))_{\#}q, \qquad \widehat{\rho}_m(g)q = (\rho_m(g))_{\#}q, \tag{2.5}$$

so that $(\widehat{\rho}_x(g)q)(A) = q(\rho_x(g)^{-1}A)$.

Hypothesis 2.3 (Selected smooth tier). Whenever a differential statement is made, $\mathcal{B}_b$ and $\mathcal{B}_m$ are finite-dimensional smooth parametrized-measure models, each closed under its G-action, and the actions $(g, q) \mapsto (\rho_x(g))_{\#}q$ are smooth. The hypothesis excludes stratified and infinite-dimensional law fibers; a statistical reading of the tier further requires differentiability in quadratic mean and the integrability each tensor uses, and a Fisher tensor is a metric only where it is nondegenerate.

Definition 2.4 (Belief and model associated bundles). On invariant law fibers define the bundles below, with projections $\varpi_x : \mathcal{E}_x \to \mathcal{C}$ and the quotient convention of the second line,

$$\mathcal{E}_b = \mathscr{P}_G \times_{\widehat{\rho}_b} \mathcal{B}_b, \qquad \mathcal{E}_m = \mathscr{P}_G \times_{\widehat{\rho}_m} \mathcal{B}_m, \tag{2.6}$$

$$(u \cdot g, \beta) \sim (u, \widehat{\rho}_x(g)\beta). \tag{2.7}$$

At c_* the frame u_i identifies $\mathcal{B}_x$ with the fiber $(\mathcal{E}_x)_{c_*}$ by $\beta \mapsto [u_i, \beta]$. This is the standard associated-bundle construction [67, Section I.5], smooth under Hypothesis 2.3.

2.2 Frames, passive re-expression, and the active gauge group at c_*

A passive re-expression of agent i is a choice $g_i \in G$ that replaces the frame as in the first line below. Since $[u_i, \beta] = [u_i' \cdot g_i, \beta] = [u_i', \widehat{\rho}_x(g_i)\beta]$, associated coordinates change by $\beta' = \widehat{\rho}_x(g_i)\beta$, and the represented sample-coordinate maps are those of the second line:

$$u_i' = u_i \cdot g_i^{-1}, \tag{2.8}$$

$$R_i^b = \rho_b(g_i), \qquad R_i^m = \rho_m(g_i), \qquad k_i' = R_i^b k_i, \quad m_i' = R_i^m m_i. \tag{2.9}$$

The group element, not a represented operator, multiplies the frame, and the tuple $(g_i)_{i \in V}$ ranges over the passive vertex group $\prod_{i \in V} G$.

2.2.1 Pointwise frame comparisons and the active gauge group

Proposition 2.5 (Frame comparisons at one base point). For $i, j \in V$ there is a unique $T_{ij} \in G$ with the first relation below, for every reference point σ. The comparisons satisfy $T_{ii} = e_G$, $T_{ji} = T_{ij}^{-1}$,

and $T_{ij}T_{jl} = T_{il}$, and under passive re-expressions they transform by the second line:

$$u_j = u_i \cdot T_{ij}, \qquad T_{ij} = U_i U_j^{-1}, \tag{2.10}$$

$$T'_{ij} = g_i T_{ij} g_j^{-1}, \qquad U'_i = g_i U_i. \tag{2.11}$$

[ESTABLISHED]

Proof. Freeness and transitivity of the torsor give uniqueness of T_{ij}, and $u_j = \sigma \cdot U_j^{-1} = u_i \cdot U_i U_j^{-1}$ gives the formula; the identities follow from uniqueness. From $u'_j = u_j \cdot g_j^{-1} = u'_i \cdot g_i T_{ij} g_j^{-1}$ and $u'_i = \sigma \cdot U_i^{-1} g_i^{-1}$ follow both laws in (2.11). □

Proposition 2.6 (Active gauge image at one base point). Let $F \in \mathrm{Aut}_G((\mathscr{P}_G)_{c_*})$, the group of right-$G$-equivariant bijections of the principal fiber over c_*, have local functions $F(u_i) = u_i \cdot \varphi_i$. Then

$$\varphi_j = T_{ij}^{-1} \varphi_i T_{ij}, \qquad i, j \in V, \tag{2.12}$$

and $F \mapsto (\varphi_i)_{i \in V}$ is a group isomorphism onto the subgroup $\mathcal{A}_{c_*} \subseteq \prod_{i \in V} G$ cut out by (2.12). For each $i_0 \in V$ the projection $(\varphi_i)_{i \in V} \mapsto \varphi_{i_0}$ is an isomorphism $\mathcal{A}_{c_*} \cong G$. In the frame u_i, F changes the coordinates of both associated fibers by $\beta \mapsto \widehat{\rho}_x(\varphi_i)\beta$, which is the coordinate change of the passive re-expression $g_i = \varphi_i$. Hence the active transformations at c_* form one copy $\mathcal{A}_{c_*} \cong G$ of the passive vertex group, a proper subgroup of $\prod_{i \in V} G$ when G is nontrivial and $|V| \geq 2$, of dimension $\dim G$ against $|V| \dim G$ for a Lie group. The local functions at c_* of every gauge automorphism of $\mathscr{P}_G$ form one element of $\mathcal{A}_{c_*}$. [ESTABLISHED]

Proof. Equivariance gives $F(u_j) = F(u_i \cdot T_{ij}) = u_i \cdot \varphi_i T_{ij} = u_j \cdot T_{ij}^{-1} \varphi_i T_{ij}$, and uniqueness in the torsor gives (2.12). Since $F(F'(u)) = F(u \cdot \varphi') = u \cdot \varphi\varphi'$, the map is a homomorphism. The value φ_{i_0} determines F on the whole torsor, so the map is injective, and for $\varphi_{i_0} \in G$ the map $u_{i_0} \cdot g \mapsto u_{i_0} \cdot \varphi_{i_0} g$ is an equivariant bijection, which proves $\mathcal{A}_{c_*} \cong G$. The coordinate statement is $F[u_i, \beta] = [u_i \cdot \varphi_i, \beta] = [u_i, \widehat{\rho}_x(\varphi_i)\beta]$, the passive coordinate change with $g_i = \varphi_i$. If $G \neq \{e_G\}$ and $i \neq j$, the tuple with $\varphi_i = a \neq e_G$ and all other entries e_G violates (2.12), so the subgroup is proper. A gauge automorphism restricts to an equivariant bijection of each fiber. □

2.2.2 The common frame is a gauge choice

Proposition 2.7 (Two atlases reduce to one by a passive re-expression). Suppose belief and model coordinates were given in two frames u_i^b, u_i^m of the same fiber. Then there is a unique $h_i \in G$ with the first relation below, and under independent re-expressions $u_i^{b\prime} = u_i^b \cdot (g_i^b)^{-1}$ and $u_i^{m\prime} = u_i^m \cdot (g_i^m)^{-1}$ it transforms by the second:

$$u_i^m = u_i^b \cdot h_i, \tag{2.13}$$

$$h'_i = g_i^b h_i (g_i^m)^{-1}. \tag{2.14}$$

In particular the re-expression $g_i^b = e_G$, $g_i^m = h_i$ of the model atlas alone makes the two frames equal. In two atlases the belief and model comparisons, the local functions $F(u_i^x) = u_i^x \cdot \varphi_i^x$ of a

fiber automorphism F, and the representatives of a cross morphism are related by

$$T_{ij}^{m} = h_i^{-1} T_{ij}^{b} h_j, \tag{2.15}$$

$$\varphi_i^{m} = h_i^{-1} \varphi_i^{b} h_i, \tag{2.16}$$

$$\Phi_i^{\mathrm{conv}} = \widehat{\rho}_m(h_i) \circ \Phi_i^{bm}, \tag{2.17}$$

where Φ_i^{bm} is the representative of Φ with source frame u_i^b and target frame u_i^m, and Φ_i^{conv} is its representative with both frames equal to u_i^b. All three relations become identities when $h_i = e_G$. [ESTABLISHED]

Proof. Uniqueness of h_i is the torsor property. Substituting the re-expressions gives $u_i^{m\prime} = u_i^b \cdot h_i (g_i^m)^{-1} = u_i^{b\prime} \cdot g_i^b h_i (g_i^m)^{-1}$, which is (2.14). The two expressions $u_j^m = u_j^b \cdot h_j = u_i^b \cdot T_{ij}^b h_j$ and $u_j^m = u_i^m \cdot T_{ij}^m = u_i^b \cdot h_i T_{ij}^m$ give (2.15). Equivariance gives $F(u_i^m) = F(u_i^b) \cdot h_i = u_i^b \cdot \varphi_i^b h_i = u_i^m \cdot h_i^{-1} \varphi_i^b h_i$. Finally, $\Phi[u_i^b, \beta] = [u_i^m, \Phi_i^{bm} \beta] = [u_i^b, \widehat{\rho}_m(h_i) \Phi_i^{bm} \beta]$ by (2.7). □

No energy, kernel, or evaluator reads h_i, so the common frame

$$u_i^b = u_i^m = u_i, \qquad h_i = e_G, \qquad T_{ij}^b = T_{ij}^m = T_{ij} \tag{2.18}$$

is adopted throughout.

2.3 Agents and cross-associated-bundle morphisms

Definition 2.8 (Agent). Agent i is the section-bearing object $\mathcal{A}^i$ of the first line below, with the sections of the second line, and a population sample at c_* is a point y of a product of sample fibers:

$$\mathcal{A}^i = \big(\mathcal{C}_i; q_i^b, q_i^m, u_i\big), \tag{2.19}$$

$$q_i^b \in \Gamma(\mathcal{C}_i, \mathcal{E}_b|_{\mathcal{C}_i}), \qquad q_i^m \in \Gamma(\mathcal{C}_i, \mathcal{E}_m|_{\mathcal{C}_i}), \tag{2.20}$$

$$y = \big((k_1, m_1), \ldots, (k_N, m_N)\big) \in \prod_{i \in V} (\mathsf{K} \times \mathsf{M}). \tag{2.21}$$

The values of q_i^b, q_i^m at c_* are points of $(\mathcal{E}_b)_{c_*}$ and $(\mathcal{E}_m)_{c_*}$, whose coordinates in the frame u_i are laws in $\mathcal{B}_b$ and $\mathcal{B}_m$; the same letters denote these coordinate laws, which Definition 3.13 identifies with the local recognition marginals. Agents need not have equal supports.

Definition 2.9 (Cross morphisms). Cross morphisms are measurable fiber-preserving maps $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ and $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$, $\varpi_m \circ \Phi = \varpi_b$ and $\varpi_b \circ \widetilde{\Phi} = \varpi_m$. At c_* they are maps between the fibers $(\mathcal{E}_b)_{c_*}$ and $(\mathcal{E}_m)_{c_*}$. They are not principal-bundle morphisms, gauge automorphisms, same-channel transports, or assumed inverses, and their existence is additional structure. Under Hypothesis 2.3 a differential statement also assumes them smooth. In the frame u_i their representatives $\Phi_i : \mathcal{B}_b \to \mathcal{B}_m$ and $\widetilde{\Phi}_i : \mathcal{B}_m \to \mathcal{B}_b$ are defined by $\Phi[u_i, \beta] = [u_i, \Phi_i \beta]$ and $\widetilde{\Phi}[u_i, \beta] = [u_i, \widetilde{\Phi}_i \beta]$.

Proposition 2.10 (Representatives and induced cross morphisms). The representatives transform under passive re-expression by the first two lines below, and those of different agents are related by the third:

$$\Phi_i' = \widehat{\rho}_m(g_i) \circ \Phi_i \circ \widehat{\rho}_b(g_i)^{-1}, \tag{2.22}$$

$$\widetilde{\Phi}_i' = \widehat{\rho}_b(g_i) \circ \widetilde{\Phi}_i \circ \widehat{\rho}_m(g_i)^{-1}, \tag{2.23}$$

$$\Phi_i \circ \widehat{\rho}_b(T_{ij}) = \widehat{\rho}_m(T_{ij}) \circ \Phi_j, \qquad \widetilde{\Phi}_i \circ \widehat{\rho}_m(T_{ij}) = \widehat{\rho}_b(T_{ij}) \circ \widetilde{\Phi}_j. \tag{2.24}$$

If maps $\Phi_0 : \mathcal{B}_b \to \mathcal{B}_m$ and $\widetilde{\Phi}_0 : \mathcal{B}_m \to \mathcal{B}_b$ satisfy the first line below for every $g \in G$, then the maps of the second line

$$\Phi_0 \circ \widehat{\rho}_b(g) = \widehat{\rho}_m(g) \circ \Phi_0, \qquad \widetilde{\Phi}_0 \circ \widehat{\rho}_m(g) = \widehat{\rho}_b(g) \circ \widetilde{\Phi}_0, \tag{2.25}$$

$$[p, \beta] \longmapsto [p, \Phi_0 \beta], \qquad [p, \beta] \longmapsto [p, \widetilde{\Phi}_0 \beta] \tag{2.26}$$

are well-defined cross morphisms, whose representatives in every frame are Φ_0 and $\widetilde{\Phi}_0$; conversely, $[p, \beta] \mapsto [p, \Phi_0 \beta]$ is well defined only if Φ_0 satisfies (2.25). Not every cross morphism is induced in this way, and for a linear specialization with irreducible, inequivalent representations every equivariant linear map between the sample fibers vanishes, while cross morphisms still exist. [ESTABLISHED]

Proof. From $[u_i, \beta] = [u_i', \widehat{\rho}_b(g_i)\beta]$ we get $\Phi[u_i', \beta'] = [u_i, \Phi_i \widehat{\rho}_b(g_i)^{-1}\beta'] = [u_i', \widehat{\rho}_m(g_i)\Phi_i \widehat{\rho}_b(g_i)^{-1}\beta']$, which is (2.22); interchanging the channels gives (2.23). Evaluating $\Phi[u_j, \beta] = \Phi[u_i, \widehat{\rho}_b(T_{ij})\beta]$ in both frames gives (2.24). The representatives $(p \cdot g, \beta)$ and $(p, \widehat{\rho}_b(g)\beta)$ of one class have the same image under (2.26) exactly when $\widehat{\rho}_m(g)\Phi_0\beta = \Phi_0 \widehat{\rho}_b(g)\beta$, which proves both directions of the induced statement. For a cross morphism that is not induced, let $G = SO(2)$ act on the laws on $\mathbb{R}^2$ in both channels by the defining representation, fix $\sigma \in (\mathscr{P}_G)_{c_*}$, and put $\Phi[\sigma, \beta] = [\sigma, D_\# \beta]$ with $D = \mathrm{diag}(2, 1)$, which is well defined because every class has a unique representative with principal component σ. By (2.22) its representative in the frame $\sigma \cdot g^{-1}$ is $\widehat{\rho}_m(g) D_\# \widehat{\rho}_b(g)^{-1}$, and for the rotation $R_{\pi/2}$ one has $D_\#(R_{\pi/2})_\# \delta_{(1,0)} = \delta_{(0,1)} \neq \delta_{(0,2)} = (R_{\pi/2})_\# D_\# \delta_{(1,0)}$, so the representative depends on the frame. In the linear specialization, the kernel and image of an equivariant linear map between irreducible representation spaces are invariant subspaces, so the map is zero or an isomorphism, and inequivalence excludes an isomorphism. □

Law-level equivariance is weaker than linear equivariance: an equivariant Markov kernel induces an equivariant map on laws without supplying a linear map between the sample fibers.

2.4 Independent links at c_*: vertex law, holonomy, edge twists

Definition 2.11 (Typed graph links). Let $\mathfrak{I} = (V, E, {}^-)$ be a finite interaction multigraph at c_*: a finite set E of oriented edge copies $e : j \to i$ with a fixed-point-free reversal $e \mapsto \bar{e}$, parallel copies allowed. For each channel $x \in \{b, m\}$ and each $e : j \to i$ declare a link Θ_e^x as in the first line below, with link map $\mathfrak{t}_e^x(u_j \cdot g) = u_i \cdot \Theta_e^x g$ of the fiber; endpoint notation Θ_{ij}^x is used only when the copy $j \to i$ is unique. Under passive re-expression the links obey the vertex law of the second and third lines,

with $R_i^x = \rho_x(g_i)$, because $\mathrm{t}_e^x(u'_j \cdot g) = u_i \cdot \Theta_e^x g_j^{-1} g = u'_i \cdot g_i \Theta_e^x g_j^{-1} g$. For a walk $p = (e_0, \dots, e_{r-1})$ from j to i, that is, with $e_a : i_{a+1} \to i_a$, $i_0 = i$, and $i_r = j$, the path product and, when $i = j$, the holonomy are those of the fourth line, and the common-link specialization of the fifth line declares that the two link maps coincide in the common frame:

$$\Theta_e^x \in G, \qquad \Theta_{\bar{e}}^x = (\Theta_e^x)^{-1}, \tag{2.27}$$

$$\Theta_e^{x\prime} = g_i \Theta_e^x g_j^{-1}, \tag{2.28}$$

$$\rho_x(\Theta_e^{x\prime}) = R_i^x \rho_x(\Theta_e^x)(R_j^x)^{-1}, \tag{2.29}$$

$$\Theta_p^x = \Theta_{e_0}^x \Theta_{e_1}^x \cdots \Theta_{e_{r-1}}^x, \qquad \mathrm{Hol}^x(p) = \Theta_p^x, \tag{2.30}$$

$$\Theta_e^b = \Theta_e^m = \Theta_e. \tag{2.31}$$

Proposition 2.12 (Per-edge twist invariance at one base point). Fix a reference point σ with $u_i = \sigma \cdot U_i^{-1}$, and define the per-edge twist $\mathring{\Theta}_e^x$ of $e : j \to i$ by the first line below; a closed walk p based at i_0 then has the holonomy of the second line:

$$\mathrm{t}_e^x(\sigma) = \sigma \cdot \mathring{\Theta}_e^x, \qquad \mathring{\Theta}_e^x = U_i^{-1} \Theta_e^x U_j, \tag{2.32}$$

$$\mathrm{Hol}^x(p) = U_{i_0} \mathring{\Theta}_{e_0}^x \mathring{\Theta}_{e_1}^x \cdots \mathring{\Theta}_{e_{r-1}}^x U_{i_0}^{-1}, \qquad \mathrm{Hol}^x(p)' = g_{i_0} \mathrm{Hol}^x(p) g_{i_0}^{-1}. \tag{2.33}$$

Then t_e^x is a right-G-equivariant bijection that does not depend on the frames, and $\mathrm{t}_{\bar{e}}^x = (\mathrm{t}_e^x)^{-1}$. Every twist is unchanged by arbitrary passive re-expressions; a change of reference point $\sigma' = \sigma \cdot g_0$ conjugates every twist by the same g_0; and an active F with $F(\sigma) = \sigma \cdot \kappa$ sends t_e^x to $F \circ \mathrm{t}_e^x \circ F^{-1}$ and every twist to $\kappa \mathring{\Theta}_e^x \kappa^{-1}$. Each edge copy therefore carries the conjugacy class $[\mathring{\Theta}_e^x]$, invariant under all passive and active changes, and two link data sets on the same multigraph are related by one active automorphism exactly when their twist tuples, computed from one reference point, are related by one simultaneous conjugation. The conjugacy class of $\mathrm{Hol}^x(p)$ and the condition $\mathrm{Hol}^x(p) = e_G$ are invariant. [ESTABLISHED]

Proof. Every point of the fiber is $u_j \cdot g$ for a unique g, so t_e^x is a well-defined equivariant bijection whose inverse $u_i \cdot g \mapsto u_j \cdot (\Theta_e^x)^{-1} g$ is $\mathrm{t}_{\bar{e}}^x$, and by the computation in Definition 2.11 re-expressed frames and links define the same map. Since $\sigma = u_j \cdot U_j$, $\mathrm{t}_e^x(\sigma) = u_i \cdot \Theta_e^x U_j = \sigma \cdot U_i^{-1} \Theta_e^x U_j$; the left side depends only on t_e^x and σ, which proves passive invariance, and $\mathrm{t}_e^x(\sigma \cdot g_0) = (\sigma \cdot g_0) \cdot g_0^{-1} \mathring{\Theta}_e^x g_0$. For the active statement, $F \mathrm{t}_e^x F^{-1}(\sigma) = F(\sigma \cdot \mathring{\Theta}_e^x \kappa^{-1}) = \sigma \cdot \kappa \mathring{\Theta}_e^x \kappa^{-1}$; an equivariant self-map of the torsor is determined by its value at σ, and $F \mapsto \kappa$ is a bijection onto G, which gives the criterion for two link data sets. Finally $\Theta_{e_a}^x = U_{i_a} \mathring{\Theta}_{e_a}^x U_{i_{a+1}}^{-1}$ telescopes, and $U'_{i_0} = g_{i_0} U_{i_0}$ by (2.11). □

Proposition 2.13 (Trivializing criterion). Fix a subfamily $I \subseteq V$ and elements $\Xi_e \in G$ on the internal oriented edge copies of I. The following are equivalent:

$$\begin{aligned} &\exists V_i \in G \ (i \in I) \quad \Xi_e = V_i V_j^{-1} \quad \text{for every internal } e : j \to i, \\ &\Xi_{e_0} \Xi_{e_1} \cdots \Xi_{e_{r-1}} = e_G \quad \text{for every closed internal walk } (e_0, \dots, e_{r-1}). \end{aligned} \tag{2.34}$$

When they hold, the potentials V_i are unique up to right multiplication by one element of G on each connected component of the internal multigraph. The statement and proof use only the group structure, so they hold for any group in place of G, in particular for a subgroup of $\mathrm{GL}(K)$ containing represented links. [ESTABLISHED]

Proof. The first condition telescopes around every closed walk. Conversely, backtracks $(e, \bar{e})$ are closed walks, so $\Xi_{\bar{e}} = \Xi_e^{-1}$ and reversed walks have inverse products. Choose a base vertex in each connected component and set $V_i = \Xi_p$ for an internal walk p from that vertex to i, with products ordered as in (2.30); two choices differ by a closed walk and agree, and extending a walk to j by $e : j \to i$ gives $V_i = \Xi_e V_j$. If also $\Xi_e = W_i W_j^{-1}$, then $W_i^{-1} V_i = W_j^{-1} V_j$ along every internal edge, so $W_i^{-1} V_i$ is constant on each component. □

Definition 2.14 (Graph-flat, jointly trivializing, and frame-aligned links). The links of channel x are graph flat on $I \subseteq V$ when Proposition 2.13 holds for $\Xi_e = \Theta_e^x$, and I is jointly trivializing when this holds in both channels. The links are frame aligned when $\Theta_e^x = T_{ij} = U_i U_j^{-1}$ for every $e : j \to i$, equivalently when every twist $\mathring{\Theta}_e^x$ equals e_G and every link map $\mathfrak{t}_e^x$ is the identity of the fiber.

Remark 2.15 (Frame alignment, graph flatness, and per-edge classes). Frame alignment implies graph flatness, because $T_{ij} = U_i U_j^{-1}$ is a coboundary, but not conversely: on two vertices joined by one edge copy $e : 2 \to 1$ and its reversal, every closed walk is a backtrack, so any $\Theta_e^x \neq T_{12}$ is graph flat with potentials $V_1 = \Theta_e^x$, $V_2 = e_G$, while $[\mathring{\Theta}_e^x] \neq \{e_G\}$. Graph flatness of the links is graph flatness of the twists, since $\Theta_e^x = V_i V_j^{-1}$ exactly when $\mathring{\Theta}_e^x = W_i W_j^{-1}$ with $W_i = U_i^{-1} V_i$. A single link coordinate between distinct vertices carries no invariant: the passive re-expression $g_i = e_G$, $g_j = \Theta_e^x$ sends Θ_e^x to e_G. On a tree every loop product is trivial, yet the per-edge classes can all differ from $\{e_G\}$. Graph flatness is a property of the links alone: the one fiber over c_* is always a trivial torsor, links on a triangle with nontrivial loop product are not graph flat, and all-identity links are graph flat whatever the topology of $\mathscr{P}_G$. [ESTABLISHED]

Remark 2.16 (Invariant loop penalties). Let p be a closed walk based at i_0 and $\mathsf{U} = \rho_x(\mathrm{Hol}^x(p))$. If ρ_x is unitary of dimension n, then $\frac{1}{2n} \operatorname{Re} \operatorname{Tr}(2I_n - \mathsf{U} - \mathsf{U}^*) = \frac{1}{2n} \operatorname{Tr}((I_n - \mathsf{U})^*(I_n - \mathsf{U})) \geq 0$, with equality exactly when $\mathsf{U} = I_n$, and the value is invariant under passive re-expression, which conjugates U by the unitary $R_{i_0}^x = \rho_x(g_{i_0})$. For $n \geq 2$ no norm on $\mathfrak{gl}(n, \mathbb{R})$ is invariant under conjugation by all of $\mathrm{GL}(n, \mathbb{R})$: conjugation by $\mathrm{diag}(2, 1, \ldots, 1)$ doubles the matrix unit E_{12}. A positive invariant penalty for general linear links therefore needs further data: if a symmetric positive definite fiber metric $\mathsf{G}^{\mathrm{fib}}$ at i_0 is declared to transform by $\mathsf{G}^{\mathrm{fib}} \mapsto (R_{i_0}^x)^{-\top} \mathsf{G}^{\mathrm{fib}} (R_{i_0}^x)^{-1}$, then $\operatorname{Tr}\big((\mathsf{G}^{\mathrm{fib}})^{-1}(I_n - \mathsf{U})^\top \mathsf{G}^{\mathrm{fib}}(I_n - \mathsf{U})\big)$ is invariant, nonnegative, and zero exactly when $\mathsf{U} = I_n$. For compact G with ρ_x unitary, the penalty $\frac{1}{2n} \operatorname{Tr}((I_n - \mathsf{U})^*(I_n - \mathsf{U}))$ of the first sentence lies in $[0, 2]$, so for finitely many closed walks p with holonomies $\mathsf{U}_p = \rho_x(\mathrm{Hol}^x(p))$ and coefficients $\mathrm{c}_p \geq 0$ the law on links with density $\exp\big(-\sum_p \mathrm{c}_p \frac{1}{2n} \operatorname{Tr}((I_n - \mathsf{U}_p)^*(I_n - \mathsf{U}_p))\big)$ against the product, over one orientation of each edge copy, of the Haar probability measure of G has normalizer in $[e^{-2\sum_p \mathrm{c}_p}, 1]$. [ESTABLISHED]

Proof. Unitarity gives $(I_n - \mathsf{U})^*(I_n - \mathsf{U}) = 2I_n - \mathsf{U} - \mathsf{U}^*$, whose trace is real and equals the squared Frobenius norm of $I_n - \mathsf{U}$. With $\Delta = I_n - \mathsf{U}$ and $\mathsf{G} = \mathsf{G}^{\mathrm{fib}}$, passive re-expression sends Δ to $R^x_{i_0}\Delta(R^x_{i_0})^{-1}$, cyclicity of the trace gives invariance, and $\mathrm{Tr}(\mathsf{G}^{-1}\Delta^\top\mathsf{G}\Delta) = \|\mathsf{G}^{1/2}\Delta\mathsf{G}^{-1/2}\|_{\mathrm{F}}^2$. A norm $\|\cdot\|_\star$ with $\|E_{12}\|_\star > 0$ cannot satisfy $\|2E_{12}\|_\star = \|E_{12}\|_\star$. For the compact case, $\mathrm{Tr}(2I_n - \mathsf{U} - \mathsf{U}^*) = 2n - 2\,\mathrm{Re}\,\mathrm{Tr}\,\mathsf{U}$ and $|\mathrm{Tr}\,\mathsf{U}| \le n$ for unitary U, so that penalty lies in $[0, 2]$ and the exponent in $[-2\sum_p \mathrm{c}_p, 0]$; each U_p is a continuous function of the links, and integrating against a probability measure gives the two bounds on the normalizer. □

In the exponential chart of Chapter 9, with the fiber metric at each vertex induced by one matrix at the reference point, one half of the general linear penalty above is the loop defect of Definition 9.8 (Proposition 9.10); and for $G = \mathrm{GL}(K, \mathbb{R})$ or $\mathrm{GL}^+(K)$, on a closed walk that traverses some reversal pair exactly once, every continuous penalty of the holonomy that is unchanged by passive re-expression takes at every unipotent holonomy its value at trivial holonomy (Corollary 9.7), so a penalty that vanishes only at trivial holonomy needs data beyond the holonomy, such as the fiber metric.

Definition 2.17 (Law coherence). Let $I \subseteq V$ induce a connected internal multigraph. Laws $q_i \in \mathcal{B}_x$, $i \in I$, are law coherent on I when

$$q_i = \widehat{\rho}_x(\Theta^x_e)q_j \qquad \text{on each internal edge copy } e : j \to i. \tag{2.35}$$

For a G-set S and a map $s : \mathcal{B}_x \to S$ with $s(\widehat{\rho}_x(g)q) = g \cdot s(q)$, the laws are s-agreeing when $s(q_i) = \Theta^x_e \cdot s(q_j)$ on each internal edge copy; the mean of laws on a linear representation space is the basic instance. The links have frame agreement on I when they are graph flat on I.

Proposition 2.18 (Three agreements). Let I be as in Definition 2.17 and fix $i_0 \in I$. (i) The laws are law coherent exactly when q_{i_0} is fixed by $\widehat{\rho}_x(\mathrm{Hol}^x(p))$ for every closed internal walk p based at i_0 and $q_i = \widehat{\rho}_x(\Theta^x_{p_i})q_{i_0}$ for some, hence every, internal walk p_i from i_0 to i; for an equivariant statistic the same holds with $s(q_{i_0})$ fixed by the holonomy, so that frame agreement and mean agreement on edges, the two conditions of the first line below, are conditions on the links and on the means, and mean agreement holds exactly when the base mean satisfies the second line and the other means are its transports:

$$\mathrm{Hol}^x(p) = e_G \text{ for every closed internal walk } p, \quad \mu_i = \rho_x(\Theta^x_e)\mu_j \text{ on each internal } e : j \to i, \tag{2.36}$$

$$\mu_{i_0} \in \bigcap_{p \text{ closed at } i_0} \ker\big(\rho_x(\mathrm{Hol}^x(p)) - \mathrm{id}\big). \tag{2.37}$$

(ii) Law coherence implies s-agreement for every equivariant statistic s, and for the mean the converse fails. (iii) Frame agreement neither implies nor is implied by law coherence or by mean agreement. (iv) With several independent cycles, trivial holonomy or a fixed base mean for one loop is necessary but not sufficient. In particular, for $G = \mathrm{GL}(2, \mathbb{R})$ acting on laws on $\mathbb{R}^2$, three agents with the common law $\mathcal{N}(0, \Sigma)$, $\Sigma = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and links $\Theta_{12} = \Sigma^{1/2}R(0.7)\Sigma^{-1/2}$ and $\Theta_{23} = \Theta_{31} = I_2$,

with $R(\phi)$ the rotation by ϕ, are law coherent while the triangle holonomy Θ_{12} is not the identity, is not orthogonal, and fixes no nonzero vector. [ESTABLISHED]

Proof. (i) Composing (2.35) along an internal walk p from j to i gives $q_i = \widehat{\rho}_x(\Theta^x_p)q_j$, which yields both conditions. Conversely, let the conditions hold; for $e : j \to i$ and walks p_i, p_j from i_0, the walk that follows p_j, then e, then p_i backward is closed at i_0 with holonomy $(\Theta^x_{p_i})^{-1}\Theta^x_e\Theta^x_{p_j}$, so $\widehat{\rho}_x(\Theta^x_e)q_j = \widehat{\rho}_x(\Theta^x_{p_i})\widehat{\rho}_x\big((\Theta^x_{p_i})^{-1}\Theta^x_e\Theta^x_{p_j}\big)q_{i_0} = q_i$. The statistic version is the same argument in S, and the vectors left unchanged by the linear map $\rho_x(\mathrm{Hol}^x(p))$ form $\ker(\rho_x(\mathrm{Hol}^x(p)) - \mathrm{id})$. (ii) Apply s to (2.35). For the converse take two agents joined by one edge copy with identity link and the laws $\mathcal{N}(0, I_2)$ and $\mathcal{N}(0, 2I_2)$: the means agree and the laws do not. (iii) The same two agents with the laws $\mathcal{N}(0, I_2)$ and $\mathcal{N}(v, I_2)$, $v \neq 0$, have frame agreement without mean agreement or law coherence. In the displayed witness, $\Theta_{12}\Sigma\Theta_{12}^\top = \Sigma^{1/2}R(0.7)R(0.7)^\top\Sigma^{1/2} = \Sigma$, so every transported law equals $\mathcal{N}(0, \Sigma)$ and all means vanish, while Θ_{12} is similar to a rotation by 0.7, hence has no eigenvalue 1, and $\Theta_{12}\Theta_{12}^\top \neq I_2$ because Σ is not a multiple of I_2. (iv) Two vertices joined by three parallel copies with $\Theta_e = \Theta_f \neq \Theta_g$ have trivial holonomy on the walk through e and $\bar{f}$ and nontrivial holonomy on the walk through e and $\bar{g}$. □

2.5 Running example: three agents on one loop

Take $G = SO(3)$, ρ_b the defining representation on $\mathsf{K} = \mathbb{R}^3$, and ρ_m the trivial representation on $\mathsf{M} = \mathbb{R}$, so the channels carry inequivalent representations. Let $V = \{1, 2, 3\}$, let $\mathfrak{I}$ be the triangle with edge copies $e_1 : 2 \to 1$, $e_2 : 3 \to 2$, $e_3 : 1 \to 3$ and their reversals, and, with $R_z(\phi)$ and $R_x(\phi)$ the rotations by ϕ about the third and first coordinate axes, declare the belief links of the first line below; model links act trivially. Choose $\sigma = u_1$ and, for display, $u_2 = u_3 = u_1$, so $U_i = e_G$ and each twist equals its link. The per-edge classes are the angles $\pi/2$, $\pi/2$, and 0, and the loop $p = (e_1, e_2, e_3)$ based at agent 1 has the holonomy of the second line,

$$\Theta^b_{e_1} = R_z(\tfrac{\pi}{2}), \qquad \Theta^b_{e_2} = R_x(\tfrac{\pi}{2}), \qquad \Theta^b_{e_3} = I_3, \tag{2.38}$$

$$\mathrm{Hol}^b(p) = R_z(\tfrac{\pi}{2})R_x(\tfrac{\pi}{2}), \qquad \mathrm{Tr}\,\mathrm{Hol}^b(p) = 0, \tag{2.39}$$

a rotation by $2\pi/3$; two quarter turns about one axis would give a rotation by π with the same classes, so the class tuple does not determine the holonomy, while the tuple modulo one simultaneous conjugation does (Proposition 2.12). A passive re-expression g_2 changes $\Theta^b_{e_1}$ to $\Theta^b_{e_1}g_2^{-1}$ and $\Theta^b_{e_2}$ to $g_2\Theta^b_{e_2}$ but changes no twist, recomputed with $U_2' = g_2$, and not the holonomy. The loop is not graph flat, and the isotropic laws $q_i^b = \mathcal{N}(0, s^2 I_3)$ are law coherent on it, because every rotation fixes them (Proposition 2.18). One record reads the transported discrepancy of agents 1 and 2: the Markov kernel from $\mathsf{K} \times \mathsf{K}$ to $\{0, 1\}$ with

$$K_{a_1}\big(k_1, k_2, \{1\}\big) = \exp\Big(-\tfrac{1}{2}\big|k_1 - \rho_b(\Theta^b_{e_1})k_2\big|^2\Big), \tag{2.40}$$

which is gauge inert, since passive re-expressions rotate $k_1 - \rho_b(\Theta^b_{e_1})k_2$ by $\rho_b(g_1)$.

Physicist's summary

At one base point all frames lie in one principal fiber, so they are compared directly by $T_{ij} = U_i U_j^{-1}$, the only active symmetry is one global rotation of that fiber, and per-agent frame changes are bookkeeping. Each independent link carries an invariant, its conjugacy class after dressing by the frames, whereas in lattice gauge theory only closed-loop products are invariant [106]. Agreement comes in three kinds, flat links, agreement of a statistic such as the mean, and law coherence, which survives nontrivial holonomy when the base law has a nontrivial stabilizer.

Chapter 3

Probability Semantics at c_*

On which measurable objects, and in which order, does every probabilistic statement at c_* rest? Laws and kernels come first, densities follow as Radon–Nikodym derivatives against references fixed for a whole family, and logarithms of density ratios enter only under Definition 3.10; this order makes the evidence bound of Chapter 5 an identity in the extended reals. The main results are common domination of a mixed family exactly when its atoms are countable (Theorem 3.5), the relative-entropy chain rule in $[0, +\infty]$ (Lemma 3.12) that every later decomposition cites, and three levels of description, at which a law-point energy is not a sample energy and point-mass recognition of law points has infinite cost (Propositions 3.20 and 3.21). Chapter 4 builds the joint on these spaces, Chapter 5 applies the chain rule, and Chapters 6 and 14 use the levels.

Scope. Injectivity of a model evaluator is not assumed, and no regularity of the quotient of presentations with the same evaluated kernel is supplied; presentations are retained. The coupling selector is applied once, and its equivariance under frame re-expression (rerunning it on transformed marginals returns the transformed coupling) is neither assumed nor claimed. Open are a normalized law-point prior (Open Problem 3.23) and likelihoods for record families without a common dominating measure (Open Problem 3.9). [NOT-CLAIMED]

3.1 Latent and record spaces at c_*

Definition 3.1 (Sample spaces and product sigma-algebras). For each agent $i \in V$ let $(\mathsf{K}_i, \mathscr{K}_i)$ and $(\mathsf{M}_i, \mathscr{M}_i)$ be its belief and model sample fibers at c_*, identified with K and M by the frame u_i, and put $\mathsf{Y}_i = \mathsf{K}_i \times \mathsf{M}_i$. Let $\mathfrak{A}$ be a finite set of records; each $a \in \mathfrak{A}$ has a scope $\partial a \subseteq V$ and a record space $(\mathsf{O}_a, \mathscr{O}_a)$. The latent and observation spaces are

$$\mathsf{Y} = \prod_{i \in V} \mathsf{Y}_i, \qquad \mathscr{Y} = \bigotimes_{i \in V} (\mathscr{K}_i \otimes \mathscr{M}_i), \tag{3.1}$$

$$\mathsf{O} = \prod_{a \in \mathfrak{A}} \mathsf{O}_a, \qquad \mathscr{O} = \bigotimes_{a \in \mathfrak{A}} \mathscr{O}_a, \tag{3.2}$$

with points $y = ((k_i, m_i))_{i \in V}$ and $o = (o_a)_{a \in \mathfrak{A}}$, and $\mathsf{Y}_{\partial a} = \prod_{i \in \partial a} \mathsf{Y}_i$. An observation of a single agent is a record with singleton scope.

A population sample is a point of Y, and evaluating a sample field at c_* returns

$$\mathrm{Ev}(y) = \big((k_i(c_*), m_i(c_*))\big)_{i\in V} \in \mathsf{Y}, \tag{3.3}$$

whereas the section values q_i^b, q_i^m at c_* are laws on K_i and M_i, which become latent coordinates only at the law-point level of Section 3.8. Missing or aggregated observations are represented by an observation kernel into a coarser record space, which changes O and leaves Y unchanged.

3.2 Structural data and kernel signatures

Definition 3.2 (Structural configuration and kernel signatures). Fix a measurable configuration space $(\mathsf{X}, \mathscr{X})$. Every $X \in \mathsf{X}$ uses the same agent set V, the same records and scopes, and the same fiber identifications, so $(\mathsf{Y}, \mathscr{Y})$ and $(\mathsf{O}, \mathscr{O})$ do not vary with X. A configuration carries the interaction multigraph $\mathfrak{I}$ of Definition 2.11, its links Θ_e^x, the frames u_i, and the parameters of the mechanisms. With $(\mathsf{I}_i, \mathscr{I}_i)$ an access space for each $i \in V$, the access maps, local recognition kernels, record kernels, and population laws have the signatures

$$\mathrm{Acc}_i : \mathsf{O}\times\mathsf{X} \longrightarrow \mathsf{I}_i, \quad \mathbb{Q}_i : \mathsf{I}_i \rightsquigarrow \mathsf{Y}_i, \qquad \mathbb{Q}_{i,o,X} := \mathbb{Q}_i(\mathrm{Acc}_i(o,X),\cdot), \tag{3.4}$$

$$K_a : (\mathsf{X}\times\mathsf{Y}_{\partial a}) \rightsquigarrow \mathsf{O}_a, \qquad a \in \mathfrak{A}, \tag{3.5}$$

$$\mathbb{P}_{\theta,V} : (\mathsf{X},\mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{Y}\times\mathsf{O}, \mathscr{Y}\otimes\mathscr{O}\big), \qquad X \longmapsto \mathbb{P}_{\theta,V}(\cdot \mid X), \tag{3.6}$$

$$\mathbb{Q}_V : (\mathsf{O}\times\mathsf{X}, \mathscr{O}\otimes\mathscr{X}) \longrightarrow \mathcal{P}\big(\mathsf{Y},\mathscr{Y}\big), \qquad (o,X) \longmapsto \mathbb{Q}_{V,o,X}, \tag{3.7}$$

where each record coordinate occurs once, $\mathbb{P}_{\theta,V}$ is constructed in Chapter 4, and $\mathbb{Q}_{V,o,X}$ is the coupling selected in Section 3.7; neither is primitive data. The identity access map is the unrestricted interface; a stricter one restricts what the local law may depend on, not the population law (Proposition 3.18). An agent's generative-model structure is the pair (q_i^m, ev_i) of its model law and the evaluator of Definition 3.3. The generative kernel reads only θ, X, and latent coordinates (Hypothesis 4.3), and a configuration prior, when declared, is a law on $(\mathsf{X}, \mathscr{X})$.

Definition 3.3 (Model presentations and evaluated kernels). For each $i \in V$ and structural value X, let $(\mathsf{S}_i^X, \mathscr{S}_i^X)$ and $(\mathsf{T}_i^X, \mathscr{T}_i^X)$ contain every source and output coordinate of agent i's generative factor, and let $(\mathsf{M}_i, \mathscr{M}_i)$ be its presentation space, so that a model sample is $m_i \in \mathsf{M}_i$ and the model law is $q_i^m \in \mathcal{P}(\mathsf{M}_i)$. The evaluator is a map

$$\mathrm{ev}_i : \mathsf{M}_i \longrightarrow \mathrm{Kern}(\mathsf{S}_i^X, \mathsf{T}_i^X), \qquad m_i \longmapsto K_{i,m_i}^X, \tag{3.8}$$

with $(m_i, z) \mapsto K_{i,m_i}^X(z, B)$ measurable for every $B \in \mathscr{T}_i^X$ and $K_{i,m_i}^X(z, \mathsf{T}_i^X) = 1$. Each source space retains every observation, structural, or parent coordinate its factor requires. In the definite case $q_i^m = \delta_{m_i}$. Neither a model law, a recognition parameter, nor a posterior is an argument of an evaluated kernel.

3.3 Reference measures

Definition 3.4 (Reference measures). Each Euclidean coordinate carries its Borel sigma-algebra and Lebesgue measure, and each countable coordinate its power set and counting measure. A mixed real coordinate carries one parameter-independent reference for the whole family of laws placed on it, generative and recognition alike: when the union A_* of the atom sets of that family is countable, it carries ν_{mix} of (3.10); when A_* is uncountable no density model is declared on that coordinate (Theorem 3.5). Any other fiber carries an explicitly declared sigma-finite measure. With ν_i^k, ν_i^m, ν_a the component measures, the product references are

$$\nu^{\mathsf{Y}} = \bigotimes_{i\in V} (\nu_i^k \otimes \nu_i^m), \qquad \nu^{\mathsf{O}} = \bigotimes_{a\in\mathfrak{A}} \nu_a. \tag{3.9}$$

Theorem 3.5 (Common domination for a mixed family). Let $(\mu_t)_{t\in\mathbb{T}}$ be any family of Borel probability measures on $\mathbb{R}$ with no singular-continuous part, written $\mu_t = f_t \,\mathrm{Leb} + \sum_{a\in A_t} p_t(a)\delta_a$ with $A_t := \{a \in \mathbb{R} : \mu_t(\{a\}) > 0\}$. A single sigma-finite Borel measure dominates every μ_t if and only if $A_* := \bigcup_{t\in\mathbb{T}} A_t$ is countable. When $A_* = \{a_1, a_2, \ldots\}$, one such parameter-independent measure is

$$\nu_{\mathrm{mix}} := \mathrm{Leb} + \sum_{n\geq 1} 2^{-n}\delta_{a_n}. \tag{3.10}$$

In particular, the moving-atom family $\mu_t = \frac{1}{2}\mathcal{N}(0,1) + \frac{1}{2}\delta_t$, $t \in \mathbb{R}$, has no common sigma-finite dominating measure. [ESTABLISHED]

The proof is in Appendix A.

Remark 3.6 (Dirac families are not dominated). On an uncountable standard Borel space, an uncountable family of Dirac laws admits no common sigma-finite dominating measure, so density charts cannot be inferred from a general law fiber. [ESTABLISHED]

Proof. A dominating measure must give positive mass to every atom of the family, and a sigma-finite measure has at most countably many points of positive mass: on each finite-measure member of a countable cover, the points of mass at least $1/n$ are finitely many. □

3.4 Kernels first, densities second

Definition 3.7 (Normalized kernels; densities as derivatives). The objects (3.6) and (3.7) are probability kernels. When $\mathbb{P}_{\theta,V}(\cdot \mid X) \ll \nu^{\mathsf{Y}} \otimes \nu^{\mathsf{O}}$, its density, the Radon–Nikodym derivative $p_{\theta,V}(o, y \mid X)$, and the evidence, its marginal density, are determined by

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = p_{\theta,V}(o, y \mid X)\nu^{\mathsf{Y}}(dy)\nu^{\mathsf{O}}(do), \tag{3.11}$$

$$p_{\theta,V}(o \mid X) = \int_{\mathsf{Y}} p_{\theta,V}(o, y \mid X)\nu^{\mathsf{Y}}(dy). \tag{3.12}$$

When $\mathbb{Q}_{V,o,X} \ll \nu^{\mathsf{Y}}$ its density is written $q_{V,o,X}(y)$. Densities are determined up to null sets of the reference, and versions that agree outside such a set denote the same density. Measurability in (o, X) of the evidence and of recognition expectations is Proposition A.3, and jointly measurable densities are Corollary A.2; an unnormalized energy in a later chapter comes with a finite positive normalizer.

Hypothesis 3.8 (Regular-conditional convention). For fixed (θ, X) the observation marginal and the posterior, a regular conditional probability, are

$$\mathbb{P}^{\mathsf{O}}_{\theta,V,X}(A) := \mathbb{P}_{\theta,V}(\mathsf{Y} \times A \mid X), \qquad A \in \mathscr{O}, \tag{3.13}$$

$$\mathbf{\Pi}_{\theta,V,o,X}(B) = \mathbb{P}_{\theta,V}(B \mid o, X), \qquad B \in \mathscr{Y}, \tag{3.14}$$

the posterior being unique for $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$-almost every o. When the domination of (3.11) holds, (3.12) is a density of $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$, and after a change on a ν^{O}-null set there is a measurable set $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ with

$$\mathbb{P}^{\mathsf{O}}_{\theta,V,X}\big(\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}\big) = 1, \qquad 0 < p_{\theta,V}(o \mid X) < \infty \quad \text{on } \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}, \tag{3.15}$$

$$\mathbf{\Pi}_{\theta,V,o,X}(B) = \frac{\int_B p_{\theta,V}(o, y \mid X)\nu^{\mathsf{Y}}(dy)}{p_{\theta,V}(o \mid X)}, \qquad o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}. \tag{3.16}$$

Every fixed-observation posterior, evidence-bound, or free-energy statement is asserted for $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$-almost every o, represented by $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$. The hypothesis excludes everywhere-pointwise statements unless a jointly measurable density, an evidence representative, and a conditional version are declared as model data; the joint measure alone does not determine a posterior at an exceptional observation (Remark A.5).

Open Problem 3.9 (Nondominated likelihood families). The density formulas here use, for every record kernel K_a, one sigma-finite ν_a dominating $\{K_a(X, y_{\partial a}, \cdot)\}$ and a jointly measurable density. For a family without one, such as $K_a(X, y_{\partial a}, \cdot) = \delta_{h(y_{\partial a})}$ with h taking uncountably many values (Remark 3.6), the joint law and its regular conditional posterior exist, but the likelihood $dK_a(X, y_{\partial a}, \cdot)/d\nu_a(o_a)$ is undefined and the evidence density can fail to exist. Open is a treatment of the evidence bound, its local disintegrations, and the coarse channels at the level of joint measures and regular conditional laws, or hypotheses restoring a jointly measurable likelihood, with a rule for propagating record-null choices of versions. It is settled by such a treatment with proofs that the exact identities of Chapters 5 and 10 hold in it, or by a record family for which they provably have no measure-level counterpart.

3.5 Absolute continuity

Definition 3.10 (Logarithmic density ratios and relative entropy). Wherever a logarithmic density ratio or its expectation is written, the numerator measure is absolutely continuous with

respect to the denominator measure. Relative entropy is

$$D_{\mathrm{KL}}(Q\|P) = \begin{cases} \displaystyle\int \log\Big(\frac{dQ}{dP}(y)\Big) Q(dy), & Q \ll P, \\ +\infty, & \text{otherwise}, \end{cases} \tag{3.17}$$

the first branch being the divergence of Kullback and Leibler [69] and the value $+\infty$ off absolute continuity the convention of Csiszár [28]. The negative part of the integrand is Q-integrable, since $\int_{\{dQ/dP<1\}} -\log(dQ/dP)dQ \le 1/e$, so the first branch lies in $[0,+\infty]$.

Proposition 3.11 (Density criterion for absolute continuity). Let Q and P be probability measures with $Q \ll \nu$ and $P \ll \nu$ for a sigma-finite ν, with densities q and p, and put $Z = \{p = 0\}$. Then

$$Q \ll P \iff Q(Z) = 0 \iff q = 0 \quad \nu\text{-almost everywhere on } Z, \tag{3.18}$$

$$Q(S) = 1, \qquad P(S) = 0, \tag{3.19}$$

where the first line holds, and whether or not Q is dominated by ν, a measurable S satisfying the second gives $Q \not\ll P$ and $D_{\mathrm{KL}}(Q\|P) = +\infty$. [ESTABLISHED]

Proof. First $P(Z) = \int_Z p(y)\nu(dy) = 0$, so $Q \ll P$ gives $Q(Z) = 0$. Conversely, if $Q(Z) = 0$ and $P(A) = 0$, then $\nu(A \cap \{p > 0\}) = 0$, so $Q(A \cap \{p > 0\}) = 0$, while $Q(A \cap Z) \le Q(Z) = 0$. The second equivalence follows from $Q(Z) = \int_Z q(y)\nu(dy)$. The last claim is the definition of absolute continuity together with (3.17). □

A law carried by a Lebesgue-null set, such as a proper affine subspace, has no Lebesgue density, and (3.19) shows its singularity (Proposition 5.4).

3.6 The relative-entropy chain rule and data processing

Lemma 3.12 (Relative-entropy chain rule, data processing, and bijection invariance). Let P, Q be laws on $\mathsf{Y}_1 \times \mathsf{Y}_2$ with first marginals P_1, Q_1 and conditional kernels $P_{2|1}, Q_{2|1}$. Then:

(a) the map $y_1 \mapsto D_{\mathrm{KL}}(Q_{2|1}(y_1,\cdot)\|P_{2|1}(y_1,\cdot))$ is measurable and

$$D_{\mathrm{KL}}(Q\|P) = D_{\mathrm{KL}}(Q_1\|P_1) + \int D_{\mathrm{KL}}\big(Q_{2|1}(y_1,\cdot)\big\|P_{2|1}(y_1,\cdot)\big) Q_1(dy_1) \tag{3.20}$$

in $[0,+\infty]$, with a right side that does not depend on the chosen conditional versions;

(b) on a finite product $\mathsf{Y} = \prod_{b=1}^{B} \mathsf{Y}_b$, for any law Q with marginals Q_b and any laws ρ_b on Y_b,

$$D_{\mathrm{KL}}\Big(Q\Big\|\bigotimes_b \rho_b\Big) = D_{\mathrm{KL}}\Big(Q\Big\|\bigotimes_b Q_b\Big) + \sum_b D_{\mathrm{KL}}(Q_b\|\rho_b) \tag{3.21}$$

in $[0,+\infty]$;

(c) for a law μ on Y_1, the minimum of $D_{\mathrm{KL}}(Q\|P)$ over all laws Q with $Q_1 = \mu$ is $D_{\mathrm{KL}}(\mu\|P_1)$, attained by $\mu(dy_1)P_{2|1}(y_1, dy_2)$, uniquely when $D_{\mathrm{KL}}(\mu\|P_1) < \infty$;

(d) for laws P, Q on Y and every Markov kernel $K : \mathsf{Y} \rightsquigarrow \mathsf{Z}$,

$$D_{\mathrm{KL}}(QK\|PK) \le D_{\mathrm{KL}}(Q\|P) \qquad \text{in } [0, +\infty]; \tag{3.22}$$

(e) for a bimeasurable bijection $A : \mathsf{Y} \to \mathsf{Y}'$,

$$\frac{d(A_\# Q)}{d(A_\# P)} = \frac{dQ}{dP} \circ A^{-1} \quad A_\# P\text{-almost surely, if } Q \ll P, \tag{3.23}$$

$$D_{\mathrm{KL}}(A_\# Q\|A_\# P) = D_{\mathrm{KL}}(Q\|P) \quad \text{in } [0, +\infty]. \tag{3.24}$$

[ESTABLISHED]

The proof is in Appendix A. Clauses (a), (b), and (d) need no absolute-continuity or finiteness hypothesis.

3.7 Recognition marginals, couplings, and selection

Definition 3.13 (Local recognition laws and their coupling class). With $\mathrm{pr}_i : \mathsf{Y} \to \mathsf{Y}_i$ the agent-block projection and $\mathrm{pr}_i^k, \mathrm{pr}_i^m$ the coordinate projections of Y_i, for each admitted (o, X) put

$$\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}) := \Big\{\mathbb{J} \in \mathcal{P}(\mathsf{Y}) : (\mathrm{pr}_i)_\# \mathbb{J} = \mathbb{Q}_{i,o,X} \text{ for every } i \in V\Big\}, \tag{3.25}$$

$$q_i^{b;o,X} := (\mathrm{pr}_i^k)_\# \mathbb{Q}_{i,o,X}, \qquad q_i^{m;o,X} := (\mathrm{pr}_i^m)_\# \mathbb{Q}_{i,o,X}. \tag{3.26}$$

The belief and model recognition marginals $q_i^{b;o,X}, q_i^{m;o,X}$ are the values at c_*, in the frame u_i, of agent i's sections q_i^b, q_i^m at the admitted (o, X). They are local data; a member of (3.25) is population-level dependence data.

Proposition 3.14 (Existence and convexity of the coupling class). For every admitted (o, X) the class $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$ is nonempty and convex, and

$$(o, X) \longmapsto \bigotimes_{i \in V} \mathbb{Q}_{i,o,X} \tag{3.27}$$

is a measurable product-coupling kernel. [ESTABLISHED]

Proof. On measurable rectangles the product is a product of measurable functions, and a monotone-class argument extends measurability; its i-th marginal is $\mathbb{Q}_{i,o,X}$ because the other factors have unit mass. Marginalization is affine, so convex combinations of couplings are couplings. □

Definition 3.15 (Selected population recognition kernel). A selection rule S is declared, with law spaces carrying their evaluation sigma-algebras, such that the population recognition kernel

$$\mathbb{Q}_V : (\mathsf{O} \times \mathsf{X}, \mathscr{O} \otimes \mathscr{X}) \longrightarrow \mathcal{P}(\mathsf{Y}, \mathscr{Y}), \qquad (o, X) \longmapsto \mathbb{Q}_{V,o,X}, \tag{3.28}$$

$$\mathbb{Q}_{V,o,X} = \mathsf{S}_{o,X}\big((\mathbb{Q}_{i,o,X})_{i\in V}\big) \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X}), \tag{3.29}$$

has $(o, X) \mapsto \mathbb{Q}_{V,o,X}(B)$ measurable for every $B \in \mathscr{Y}$. The law may correlate agents and channels; the product coupling is one admissible selection, not the default, and copulas, common latent variables, ordered conditionals, or graphical families supply others.

Hypothesis 3.16 (Constrained coupling selection). When the selection must lie in an imposed subfamily $\mathrm{Cpl}^{\mathrm{res}}(\mathbb{Q}_{\bullet,o,X}) \subseteq \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$, every fiber $\mathrm{Cpl}^{\mathrm{res}}(\mathbb{Q}_{\bullet,o,X})$ is assumed nonempty, and a selector S from these fibers is assumed with $(o, X) \mapsto \mathsf{S}_{o,X}((\mathbb{Q}_{i,o,X})_{i\in V})(B)$ measurable for every $B \in \mathscr{Y}$. The hypothesis excludes restricted families whose fibers are empty or admit no measurable selection; the two conditions are independent, and the product coupling proves neither.

Proposition 3.17 (Coordinate marginals do not determine a sufficiently rich joint). For a finite family of standard Borel blocks, the class $\mathrm{Cpl}((\mu_i)_{i\in V})$ is a singleton if and only if at most one marginal μ_i is non-Dirac. Hence, when two local recognition laws are non-Dirac, the local marginals do not determine $\mathbb{Q}_{V,o,X}$. [ESTABLISHED]

Proof. Let μ_1, μ_2 be non-Dirac. A non-Dirac law charges some measurable set with probability in $(0, 1)$, since otherwise the members or complements of probability one of a countable separating generator would meet in a set of probability one with at most one point. Choose such sets A, B and, for $0 < |\varepsilon| < 1$, put

$$d\mathbb{J}_\varepsilon(x, x') = \big[1 + \varepsilon(\mathbf{1}_A(x) - \mu_1(A))(\mathbf{1}_B(x') - \mu_2(B))\big] d\mu_1(x) d\mu_2(x'). \tag{3.30}$$

The bracket is at least $1 - |\varepsilon| > 0$ and integrating in either variable removes the perturbation, so $\mathbb{J}_\varepsilon$ couples μ_1, μ_2, while $\mathbb{J}_\varepsilon(A \times B) = \mu_1(A)\mu_2(B) + \varepsilon\mu_1(A)(1 - \mu_1(A))\mu_2(B)(1 - \mu_2(B))$ varies with ε; tensoring with the other marginals gives distinct couplings of the family. Conversely, if at most one marginal is non-Dirac, every coupling is carried by the Dirac atoms in the other coordinates, and its law there is the remaining marginal. □

Proposition 3.18 (Limits of access restriction). A restriction of the access maps constrains the local laws and not the population law, and it has an evidence price that no selector removes.

(i) Let $V = \{1, 2\}$ with binary blocks, let both local laws be $\mathrm{Ber}(\frac{1}{2})$ at every observation, so that every access map is respected, and fix $0 < r < 1$. The selector that returns the coupling with $\mathbb{Q}_{V,o,X}(y_1 = y_2) = (1 + r)/2$ when a record o_2 of agent 2 equals 1 and $(1 - r)/2$ otherwise satisfies (3.29), and the population law depends on o_2 while no local law does. A decentralized information claim therefore also requires a selector indexed by $((\mathrm{Acc}_i(o, X))_{i\in V}, X)$ alone.

(ii) Let $V = \{1, 2\}$ with binary latents, the prior with mass 0.45 on each of $(0, 0)$ and $(1, 1)$ and 0.05 on each of $(0, 1)$ and $(1, 0)$, and singleton records o_1, o_2 that report y_1, y_2 correctly with probability 0.8, independently given y. Let $\mathrm{Acc}_1(o, X) = o_1$, so that $\mathbb{Q}_{1,o,X} = \mathrm{Ber}(t)$ is the same at $o = (1, 0)$ and $o = (1, 1)$. The posterior marginals of y_1 at these observations, written as the probabilities of $y_1 = 0$ and $y_1 = 1$, are $\Pi_1^{(0)} = (37/89, 52/89)$ and $\Pi_1^{(1)} = (13/161, 148/161)$, and every recognition law with this access restriction satisfies

$$\begin{aligned}\max_{o_2\in\{0,1\}} D_{\mathrm{KL}}\big(\mathbb{Q}_{V,(1,o_2),X}\big\|\mathbf{\Pi}_{\theta,V,(1,o_2),X}\big) &\geq \max_{o_2\in\{0,1\}} D_{\mathrm{KL}}\big(\mathrm{Ber}(t)\big\|\Pi_1^{(o_2)}\big)\\ &\geq \sup_{\xi\in[0,1]}\big(-\log Z_\xi\big) = 0.08849\ldots,\end{aligned} \tag{3.31}$$

where $Z_\xi = \sum_{y_1} \Pi_1^{(0)}(y_1)^\xi \Pi_1^{(1)}(y_1)^{1-\xi}$. The last bound is sharp: the supremum is attained at $\xi \approx 0.548$, and at $t \approx 0.783$ the middle term equals it. Hence the equality condition of Theorem 5.8 lies outside the access-restricted class, and the exact E-coordinate of Corollary 5.22, which optimizes over every law with a prescribed complement marginal, need not be access restricted.

[ESTABLISHED]

Proof. (i) Both couplings have $\mathrm{Ber}(\frac{1}{2})$ marginals. (ii) Bayes' rule gives the posterior marginals, and projection onto y_1 with Lemma 3.12(d) gives the first inequality. For $\xi \in [0, 1]$ let $\Pi^{[\xi]} = Z_\xi^{-1}(\Pi_1^{(0)})^\xi(\Pi_1^{(1)})^{1-\xi}$. Expanding logarithms gives $\xi D_{\mathrm{KL}}(\mathrm{Ber}(t)\|\Pi_1^{(0)}) + (1-\xi)D_{\mathrm{KL}}(\mathrm{Ber}(t)\|\Pi_1^{(1)}) = D_{\mathrm{KL}}(\mathrm{Ber}(t)\|\Pi^{[\xi]}) - \log Z_\xi \geq -\log Z_\xi$, and a maximum dominates every convex combination. The concave function $\xi \mapsto -\log Z_\xi$ has derivative $\mathbb{E}_{\Pi^{[\xi]}} \log(\Pi_1^{(1)}/\Pi_1^{(0)})$, which vanishes at its interior maximizer $\xi_\star$, where the two divergences from $\Pi^{[\xi_\star]}$ are therefore equal; the identity with $\mathrm{Ber}(t) = \Pi^{[\xi_\star]}$ shows that both equal $-\log Z_{\xi_\star}$, the Chernoff information [24, Section 11.9]. The value and maximizers follow from the closed forms, and every access-restricted law has a positive gap at one of the two observations. □

3.8 Levels of description

Definition 3.19 (Sample, law-point, and configuration levels). At the sample level the agent block is Y_i of Definition 3.1, the population recognition law $\mathbb{Q}_{V,o,X}$ is a law of samples, and the belief of agent i is the recognition marginal $q_i^{b;o,X}$ of (3.26). Generative kernels and the record kernels (3.5) read sample coordinates. At the law-point level, law points are latent coordinates, and the law-point blocks are

$$\mathsf{Y}_i^{\mathrm{law}} = (\mathcal{E}_b)_{c_*} \times (\mathcal{E}_m)_{c_*} \ni \big(\mathfrak{q}_i^b, \mathfrak{q}_i^m\big), \qquad \mathsf{Y}^{\mathrm{law}} = \prod_{i\in V} \mathsf{Y}_i^{\mathrm{law}}, \tag{3.32}$$

possibly augmented by further latent coordinates; through the frame u_i, $\mathfrak{q}_i^b$ and $\mathfrak{q}_i^m$ are laws on K_i and M_i. A generative law at this level is a law on $\mathsf{Y}^{\mathrm{law}} \times \mathsf{O}$ whose kernels may read law points, and a recognition law is a law over law points. At the configuration level the latent state is $(w, y) \in \mathsf{Y}^{\mathrm{cfg}} \times \mathsf{Y}$, where the configuration space $\mathsf{Y}^{\mathrm{cfg}}$ carries a measurable presentation map $\mathrm{pres} : \mathsf{Y}^{\mathrm{cfg}} \to \mathsf{Y}^{\mathrm{law}}$

and may carry further latent coordinates such as random links. A lifted generative law is a law $\mathbb{P}^{\mathrm{cfg}}_{\theta,V}(dw, dy, do \mid X)$, and a lifted recognition law has the form $\mathbb{J}(dw, dy) = \mathbb{J}^{\mathrm{cfg}}(dw)\kappa^{\mathrm{pres}}(w, dy)$ with a presentation kernel $\kappa^{\mathrm{pres}} : \mathsf{Y}^{\mathrm{cfg}} \rightsquigarrow \mathsf{Y}$ whose marginal at each agent is the law presented by the corresponding component of $\mathrm{pres}(w)$.

The levels define different optimization problems, so an energy enters a free energy only after its level is named. The source energies of (6.2) and every record kernel are sample-level. The transported divergences (13.12) and the role divergences (6.35) are measurable $[0, +\infty]$-valued functions of law points (Corollary A.4), admissible as generative energies when their arguments are latent law points; read at the sample level they would make a generative factor read recognition marginals, which Hypothesis 4.3 excludes. A block admitted by Hypothesis 5.5 with law points is a law-point or configuration block, and the formation model (14.1) is configuration-level, with records that read the configuration.

Proposition 3.20 (A law-point energy is not a sample energy). Let q_i be a receiver law and p_1, p_2 source laws in $\mathcal{B}_x$ with $D_{\mathrm{KL}}(q_i \| \widehat{\rho}_x(\Theta^x_e)p_k) < \infty$ for $k = 1, 2$, whose transported laws have densities against a common dominating measure that differ on a set of positive q_i-measure. Then for $0 < s < 1$

$$D_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta^x_e)(sp_1 + (1-s)p_2)\big) < sD_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta^x_e)p_1\big) + (1-s)D_{\mathrm{KL}}\big(q_i \| \widehat{\rho}_x(\Theta^x_e)p_2\big),$$

and no recognition-independent sample energy D satisfies $\mathbb{E}_{q_i \otimes q} D = D_{\mathrm{KL}}(q_i \| \widehat{\rho}_x(\Theta^x_e)q)$ at $q = p_1$, at $q = p_2$, and at one strict mixture of them. [ESTABLISHED]

Proof. Pushforward preserves mixtures. Let α, β be the q_i-densities of the absolutely continuous parts of the two transported laws. Finite divergences give $q_i \ll$ each transported law, so $\alpha, \beta > 0$ q_i-almost everywhere and the divergences are $\int -\log \alpha(y) q_i(dy)$ and $\int -\log \beta(y) q_i(dy)$, well defined because $\int (\log \alpha(y))^+ q_i(dy) \le \int \alpha(y) q_i(dy) \le 1$. The mixture has relative density $s\alpha + (1-s)\beta$, and $-\log(s\alpha + (1-s)\beta) \le -s \log \alpha - (1-s) \log \beta$, strictly on the set $\{\alpha \neq \beta\}$ of positive q_i-measure; integrating gives the strict inequality. An energy D contributes $\mathbb{E}_{q_i \otimes q} D$, affine in q along mixtures, so matching the divergence at p_1 and p_2 forces the affine value at the mixture, which the strict inequality excludes. □

Proposition 3.21 (Point-mass recognition of law points has infinite cost). At the law-point level let $\mathbb{P}^{\mathrm{law}}_{\theta,V}(\cdot \mid X)$ be a generative law with latent marginal $\mathbb{P}^{\mathrm{law},\mathsf{Y}}_{\theta,V}$ and posterior $\mathbf{\Pi}^{\mathrm{law}}_{\theta,V,o,X} \ll \mathbb{P}^{\mathrm{law},\mathsf{Y}}_{\theta,V}$ at an observation with finite positive evidence; the domination holds when every record kernel has a density against a fixed reference. If $\bar{\mathsf{q}}$ is not an atom of the prior, then

$$D_{\mathrm{KL}}\big(\delta_{\bar{\mathsf{q}}} \big\| \mathbf{\Pi}^{\mathrm{law}}_{\theta,V,o,X}\big) = +\infty, \tag{3.33}$$

and the extended evidence bound of $\delta_{\bar{\mathsf{q}}}$ is $-\infty$. Consequently, against a non-atomic law-point prior, a row or an update computed at fixed law points is not the recognition optimum of a finite evidence bound at the law-point level: with the law points latent it is a generative posterior quantity conditional

on them, and with the law points declared as parameters in (θ, X) it is a function of parameters whose change is an M-coordinate subject to Definition 5.28. [ESTABLISHED]

Proof. Singletons are measurable, and domination gives $\mathbf{\Pi}^{\mathrm{law}}_{\theta,V,o,X}(\{\bar{\mathfrak{q}}\}) = 0$, so (3.19) holds for $\delta_{\bar{\mathfrak{q}}}$ with $S = \{\bar{\mathfrak{q}}\}$. The extended bound (5.15) subtracts this divergence from a finite log evidence, and a recognition optimum with finite bound has finite divergence. □

Proposition 3.22 (Configuration-lift identity). Let $\kappa^{\mathrm{cfg}} : \mathsf{Y} \rightsquigarrow \mathsf{Y}^{\mathrm{cfg}}$ be a Markov kernel, and let the lifted generative law be $\mathbb{P}_{\theta,V}(dy, do \mid X)\kappa^{\mathrm{cfg}}(y, dw)$, so that its (y, o) marginal is the sample-level law and its records read only sample coordinates. Then at every $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ the evidence is unchanged, the lifted posterior may be selected as $\mathbf{\Pi}_{\theta,V,o,X}(dy)\kappa^{\mathrm{cfg}}(y, dw)$, and every lifted recognition law $\mathbb{J}$ with sample marginal $\bar{\mathbb{Q}}$ and conditional kernel $\mathbb{J}(dw \mid y)$ satisfies

$$\mathcal{L}^{\mathrm{ext,cfg}}_V(\mathbb{J}; X, o) = \mathcal{L}^{\mathrm{ext}}_V(\bar{\mathbb{Q}}; X, o) - \mathbb{E}_{\bar{\mathbb{Q}}}\Big[D_{\mathrm{KL}}\big(\mathbb{J}(\cdot \mid Y)\big\|\kappa^{\mathrm{cfg}}(Y, \cdot)\big)\Big] \tag{3.34}$$

in $[-\infty, \log p_{\theta,V}(o \mid X)]$, where $\mathcal{L}^{\mathrm{ext,cfg}}_V$ is (5.15) for the lifted law. The lift never raises the bound of the induced sample law, and when that bound is finite it leaves it unchanged exactly when $\mathbb{J}(\cdot \mid y) = \kappa^{\mathrm{cfg}}(y, \cdot)$ for $\bar{\mathbb{Q}}$-almost every y. [ESTABLISHED]

Proof. The (y, o) marginal of the lifted law is $\mathbb{P}_{\theta,V}$, so the evidence is unchanged, and disintegrating along o gives the stated posterior. Lemma 3.12(a), with y as the first coordinate, gives the lifted divergence as $D_{\mathrm{KL}}(\bar{\mathbb{Q}}\|\mathbf{\Pi}_{\theta,V,o,X})$ plus the conditional term in $[0, +\infty]$, and the conditional term vanishes exactly under the stated almost-everywhere equality. Substitution in (5.15) subtracts two terms in $[0, +\infty]$ from a finite log evidence. □

Open Problem 3.23 (A normalized law-point prior). No chapter constructs a normalized generative law on $\mathsf{Y}^{\mathrm{law}}$ for a general class of law points, which the law-point energies (13.12) and (6.35) need to be evidence-bound terms rather than a separately declared potential. The law fibers supply no canonical sigma-finite reference ($\mathcal{P}(\mathsf{K})$ is in general infinite dimensional), and the Haar measure of a noncompact G is infinite, so a Haar prior on a free orbit is not normalizable; the formation model supplies one normalized configuration-level prior, (14.7), for finite-dimensional Gaussian presentations with a proper, non-invariant reference. It is settled for a class of law points by a sigma-finite reference on the law-point or presentation space, a finite normalizer for the exponentiated energies, the transformation law of the prior under the active gauge action, record kernels that read law points, and a compatibility relation between the law-point posterior and the sample-level recognition marginals.

Physicist's summary

Everything probabilistic at c_* is a law or a kernel, with densities against references fixed per family, and one chain rule in $[0, +\infty]$ does all later bookkeeping. Local recognition laws do not determine their coupling, restricted access has an exact evidence price, and attention's transported divergence is a law-point energy.

Chapter 4

The Fixed Normalized Generative Law

This chapter answers how one normalized joint law of latent samples and records is built at c_* and what passive re-expression of the frames does to it. The joint is normalized for every finite record hypergraph, cyclic ones included (Proposition 4.4), local normalization of reciprocal latent conditionals supplies no joint (Proposition 4.6), and under kernel covariance the evidence and the bound of a transported coupling are invariant (Proposition 4.9 and corollary 4.10). Its evidence is bounded in Chapter 5, and attention and coarse channels act on it in Chapters 6 and 10.

Scope. The cross morphisms supply no generative factor, and a model with factors in both directions needs its own normalized joint. No quotient of the parameter space by passive re-expression is formed, and no parameter count subtracts the dimension of the vertex group, whose action can have nontrivial stabilizers. [NOT-CLAIMED]

4.1 Parameters, the directed latent law, and the typing prohibition

Definition 4.1 (Parameters, structural data, and generative factors). Fix the agent set $V = \{1, \dots, N\}$ and the spaces and references of Chapter 3. Structural data $X \in \mathsf{X}$ contain a finite directed acyclic graph Γ on V with parent sets $\mathrm{pa}(i)$, the interaction multigraph $\mathfrak{I}$ with its links, and the frames; the numerical parameters that a learning coordinate may move are denoted θ. The split between X and θ is declared, not derived, and every typing statement below applies to both. For each agent i declare the Markov kernels of the first line below, its model and state factors, which are its model and belief priors and are generative factors, not components of the agent of Definition 2.8; at a root, where $\mathrm{pa}(i) = \varnothing$, the first is a proper model law and the second a bridge from the agent's own model coordinate to its belief coordinate. The state factor is the evaluated kernel of Definition 3.3 with source $\mathsf{S}_i^X = \prod_{j \in \mathrm{pa}(i)} \mathsf{K}_j$, the empty product being a singleton $\{*\}$, and target $\mathsf{T}_i^X = \mathsf{K}_i$ (second line):

$$P^m_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X), \qquad P^k_{\theta,i}(dk_i \mid k_{\mathrm{pa}(i)}, m_i, X), \tag{4.1}$$

$$\mathrm{ev}_i(m_i) = K^X_{i,m_i}, \qquad K^X_{i,m_i}(k_{\mathrm{pa}(i)}, B) := P^k_{\theta,i}(B \mid k_{\mathrm{pa}(i)}, m_i, X). \tag{4.2}$$

With $\mathsf{Y}_{\mathrm{pa}(i)} = \prod_{j\in\mathrm{pa}(i)} \mathsf{Y}_j$, the agent mechanism is the ordered composite below, which reads the whole parent block $y_{\mathrm{pa}(i)}$ and has unit mass:

$$G_{\theta,i} : (\mathsf{X} \times \mathsf{Y}_{\mathrm{pa}(i)}) \rightsquigarrow \mathsf{Y}_i, \tag{4.3}$$

$$G_{\theta,i}\big(d(k_i, m_i) \mid y_{\mathrm{pa}(i)}, X\big) := P^m_{\theta,i}(dm_i \mid m_{\mathrm{pa}(i)}, X) K^X_{i,m_i}(k_{\mathrm{pa}(i)}, dk_i). \tag{4.4}$$

Definition 8.2 divides these coordinates into fast recognition and slow model coordinates. At a root the state factor induces, for a model law s on M_i, the belief law

$$\widetilde{\Phi}^{\mathrm{gen}}_i(s)(B) = \int_{\mathsf{M}_i} P^k_{\theta,i}(B \mid m, X) s(dm), \qquad B \in \mathscr{K}_i, \tag{4.5}$$

which can serve as the representative at c_* of the cross morphism $\widetilde{\Phi}$ of Definition 2.9 when that identification is declared; at a nonroot the evaluated kernel is indexed by the parent belief tuple and is a kernel-valued bridge, not a map of law fibers.

Definition 4.2 (Finite directed latent law). Choose a topological ordering $v_1, \dots, v_N$ of Γ. The latent law is the ordered composition of the first line below, where the arrow denotes ordered application of kernels as in the second line, for bounded measurable f; every conditioning coordinate is introduced before its kernel is applied:

$$\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X) := \overrightarrow{\prod_{l=1}^{N}} G_{\theta,v_l}\big(dy_{v_l} \mid y_{\mathrm{pa}(v_l)}, X\big), \tag{4.6}$$

$$\int f(y) \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X) = \int G_{\theta,v_1}(dy_{v_1} \mid X) \cdots \int G_{\theta,v_N}\big(dy_{v_N} \mid y_{\mathrm{pa}(v_N)}, X\big) f(y). \tag{4.7}$$

When the factors have densities against the component references, the densities are

$$g_{\theta,i}(k_i, m_i \mid y_{\mathrm{pa}(i)}, X) = p^m_{\theta,i}(m_i \mid m_{\mathrm{pa}(i)}, X) p^k_{\theta,i}(k_i \mid k_{\mathrm{pa}(i)}, m_i, X), \tag{4.8}$$

$$p^{\mathsf{Y}}_{\theta,V}(y \mid X) = \prod_{i\in V} g_{\theta,i}(k_i, m_i \mid y_{\mathrm{pa}(i)}, X), \qquad \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X) = p^{\mathsf{Y}}_{\theta,V}(y \mid X) \nu^{\mathsf{Y}}(dy). \tag{4.9}$$

Hypothesis 4.3 (Typing prohibition). Each factor in (4.1) and each record kernel is a function only of θ, X, and its displayed latent coordinates; no generative kernel takes as an argument the recognition law $\mathbb{Q}_{V,o,X}$, one of its marginals or parameters, or the posterior $\mathbf{\Pi}_{\theta,V,o,X}$. In the two-argument form $\mathcal{F}_V(\mathbb{Q};\theta)$ of the free energy [81], every evaluation holds the generative argument (θ, X) fixed while the recognition argument varies, which excludes the composite $\mathbb{Q} \mapsto \mathcal{F}_V(\mathbb{Q};\theta(\mathbb{Q}))$ inside one evaluation: a generative factor that reads the recognition law being scored, such as a transition built from a live peer recognition marginal, is a map from recognition laws to kernels rather than a factor of one joint. A parameter may depend on the observation or on an earlier recognition law if it is held fixed while the bound is evaluated, so slow evolution of an agent's model or prior is admissible as recognition dynamics of the model channel under one fixed joint or as accepted

M-coordinates on θ (Propositions 5.27 and 5.30 and definition 5.28), each accepted value defining a new fixed joint to which the evidence identity applies afresh.

Coupling through latents and loops of inference updates under a fixed joint remain admissible (Section 13.7), and a family of joints indexed by recognition laws designates no target (Proposition 5.11).

The links in X can also be latent: the joint of Definition 9.15 draws the generators of the links from a prior that reads only θ and X, lets every factor read them as a latent coordinate, and has unit mass on every record hypergraph, with the generators as a root block (Proposition 9.16).

4.2 Normalized record extension and the undirected alternative

Proposition 4.4 (Normalization by latent composition and record attachment). The iterated integral (4.7) defines a probability kernel $X \mapsto \mathbb{P}^{\mathsf{Y}}_{\theta,V}(\cdot \mid X)$. If the record coordinates are conditionally independent given (X, y), the population law is the kernel of the first line below, and the second line holds for every finite record hypergraph, cyclic or not, and every (θ, X):

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X) \prod_{a \in \mathfrak{A}} K_a(X, y_{\partial a}, do_a), \tag{4.10}$$

$$\mathbb{P}_{\theta,V}(\mathsf{Y} \times \mathsf{O} \mid X) = 1. \tag{4.11}$$

If $K_a(X, y_{\partial a}, do_a) = \exp\big(-E_{a,o}(y_{\partial a})\big)\nu_a(do_a)$ with a jointly measurable negative log density $E_{a,o}$ and $\nu^{\mathsf{O}} = \bigotimes_a \nu_a$, then

$$p_{\theta,V}(o, y \mid X) = p^{\mathsf{Y}}_{\theta,V}(y \mid X) \exp\Big(-\sum_{a \in \mathfrak{A}} E_{a,o}(y_{\partial a})\Big) \tag{4.12}$$

is its density. [ESTABLISHED]

Proof. Integrating a bounded measurable function against a probability kernel preserves boundedness and measurability in the remaining variables, so induction along the topological order defines a measurable kernel in X, and $f \equiv 1$ gives unit mass; finite kernel composition gives measurability of (4.10). By Tonelli's theorem, integrate the record coordinates first, in any order: each K_a has unit mass in o_a, so the finite product disappears and leaves $\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X)$ whatever the overlaps of the scopes, since the records are conditionally independent children of one latent configuration. The same computation with the Radon–Nikodym derivatives proves (4.12). □

Each record kernel is owned once by the joint, while the coordinate-local free energies of Chapter 5 include every factor incident to a coordinate.

Running example. The agents of Section 2.5 carry the directed joint with $\mathrm{pa}(1) = \varnothing$, $\mathrm{pa}(2) = \mathrm{pa}(3) = \{1\}$, independent $\mathcal{N}(0,1)$ model coordinates, $k_1 \mid m_1 \sim \mathcal{N}(0, e^{-m_1} I_3)$, and children transported from agent 1, $k_2 \mid k_1, m_2 \sim \mathcal{N}\big(\rho_b(\Theta^b_{\bar{e}_1})k_1, e^{-m_2} I_3\big)$ and $k_3 \mid k_1, m_3 \sim \mathcal{N}\big(\rho_b(\Theta^b_{e_3})k_1, e^{-m_3} I_3\big)$. Two binary records are attached: a_1 with scope $\{1, 2\}$ as in (2.40), and a_2 with scope $\{2, 3\}$ and success probability $\exp(-\frac{1}{2}|k_2 - \rho_b(\Theta^b_{e_2})k_3|^2)$. The records close the latent tree into the triangle, and Proposition 4.4 normalizes the joint. At the conditional means of the children the residual read

by a_2 is $\rho_b(\Theta^b_{\bar{e}_1})k_1 - \rho_b(\Theta^b_{e_2}\Theta^b_{e_3})k_1 = \rho_b(\Theta^b_{e_1})^{-1}(I_3 - \rho_b(\mathrm{Hol}^b(p)))k_1$, whose length is $\sqrt{3}$ times that of the component of k_1 orthogonal to the rotation axis of the holonomy (2.39), so the records see the loop holonomy through an invariant.

Definition 4.5 (Normalized undirected model). For nonnegative potentials on the agents, on the edges of an undirected graph with edge set E^{u}, and on the records, put $\widetilde{p}_X$ and Z_X as in the first two lines below. When $0 < Z_X < \infty$, a condition that can depend on the parameters, the undirected model is the law with the density of the third line:

$$\widetilde{p}_X(o,y) = \prod_{i\in V}\psi_i(k_i,m_i;X)\prod_{\{i,j\}\in E^{\mathrm{u}}}\psi_{ij}(y_i,y_j;X)\prod_{a\in\mathfrak{A}}\psi_a(o_a,y_{\partial a};X), \tag{4.13}$$

$$Z_X = \int \widetilde{p}_X(o,y)\nu^{\mathsf{O}}(do)\nu^{\mathsf{Y}}(dy), \tag{4.14}$$

$$p^{\mathrm{Gibbs}}_{\theta,V}(o,y \mid X) = Z_X^{-1}\widetilde{p}_X(o,y). \tag{4.15}$$

Proposition 4.6 (Local normalization is insufficient). Normalized agent potentials and an everywhere positive, finite, continuous edge potential need not give a finite Z_X. [ESTABLISHED]

Proof. Take two agents with scalar belief coordinates $k_1, k_2 \in \mathbb{R}$ under Lebesgue measure, singleton model and record spaces with unit point mass, and potentials with product exponent $-\frac{1}{2}k^{\top}\mathsf{W}_{\mathrm{c}}k$,

$$\psi_i(k_i,m_i;X) = (2\pi)^{-1/2}e^{-k_i^2/2}, \qquad \psi_{12}(y_1,y_2;X) = e^{\mathrm{c}k_1k_2}, \qquad \mathrm{c}\geq 1, \tag{4.16}$$

$$\mathsf{W}_{\mathrm{c}} = \left(\begin{smallmatrix} 1 & -\mathrm{c} \\ -\mathrm{c} & 1 \end{smallmatrix}\right), \qquad \operatorname{spec}\mathsf{W}_{\mathrm{c}} = \{1-\mathrm{c}, 1+\mathrm{c}\}; \tag{4.17}$$

in orthonormal sum and difference coordinates the integral factorizes, and the factor in the direction of the eigenvalue $1-\mathrm{c}\leq 0$ diverges. □

Remark 4.7 (Three uses of a reciprocal energy). Let U be a measurable energy on Y, bounded below, that couples coordinates reciprocally. If U enters a potential (4.13) with $Z_X = +\infty$, no law is defined and there is no evidence to bound. If it enters with $0 < Z_X < \infty$, the undirected model is a legitimate joint to which Chapter 5 applies. If U is appended to the free energy of a joint $\mathbb{P}_{\theta,V}$ that does not contain it, the evidence of that joint is unaffected, and for every recognition law Q in the classical domain of Hypothesis 5.3 with $\mathbb{E}_Q|U| < \infty$,

$$\mathcal{F}_V[Q;X,o] + \mathbb{E}_Q U = \widetilde{\mathcal{F}}_V[Q;X,o] - \log Z_U, \qquad Z_U = \int e^{-U(y)}\mathbb{P}_{\theta,V}(dy,do \mid X) \in (0,\infty), \tag{4.18}$$

where $\widetilde{\mathcal{F}}_V$ is the free energy of the tilted joint $Z_U^{-1}e^{-U(y)}\mathbb{P}_{\theta,V}(dy,do \mid X)$. The appended objective is therefore an evidence bound of that tilted joint only after $\log Z_U$, which depends on (θ, X), is restored. [ESTABLISHED]

Proof. Since U is bounded below, $0 < Z_U \leq \sup e^{-U} < \infty$, and the tilted joint has density $p_{\theta,V}(o,y \mid X)e^{-U(y)}/Z_U$. With $\mathcal{F}_V[Q;X,o] = \mathbb{E}_Q[\log q_{V,o,X}(Y) - \log p_{\theta,V}(o,Y \mid X)]$, adding $\mathbb{E}_Q U$

and splitting $\log Z_U$ off the tilted log density gives (4.18); all expectations are finite under the stated integrability. □

The directed construction is used because its normalization does not depend on the parameters.

4.3 Covariance under passive frame re-expression

A passive re-expression $(g_i)_{i\in V}$ acts on samples through $R_i^x = \rho_x(g_i)$ of (2.9), and on the latent and joint spaces by the first two lines below, so records are gauge inert. The structural data transform to X' with the frames $u_i \cdot g_i^{-1}$ and, by the vertex law (2.28), the links of the third line; a structural constraint tying parallel copies $e, e' : j \to i$ to one link is preserved, because both copies transform by the same pair (fourth line). Active invariance is invariance under the one copy $\mathcal{A}_{c_*} \cong G$ of Proposition 2.6.

$$R = \bigoplus_{i\in V} (R_i^b \oplus R_i^m) : \mathsf{Y} \longrightarrow \mathsf{Y}, \tag{4.19}$$

$$T(y, o) = (Ry, o), \tag{4.20}$$

$$\Theta_e^{x\prime} = g_i \Theta_e^x g_j^{-1}, \qquad e : j \to i,\ x \in \{b, m\}, \tag{4.21}$$

$$\Theta_e^x = \Theta_{e'}^x \quad \Longrightarrow \quad g_i \Theta_e^x g_j^{-1} = g_i \Theta_{e'}^x g_j^{-1}. \tag{4.22}$$

Hypothesis 4.8 (Kernel covariance). The transformed parameters (θ', X') are such that every factor obeys the first two lines below, with the empty parent tuple at roots, and every record kernel obeys the third, where $R_{\partial a}$ is the restriction of R to the incident blocks; for a record with scope $\{i\}$ the third line reads as the fourth:

$$P^m_{\theta',i}\big(\cdot \mid R^m_{\mathrm{pa}(i)} m_{\mathrm{pa}(i)}, X'\big) = (R_i^m)_{\#} P^m_{\theta,i}\big(\cdot \mid m_{\mathrm{pa}(i)}, X\big), \tag{4.23}$$

$$P^k_{\theta',i}\big(\cdot \mid R^b_{\mathrm{pa}(i)} k_{\mathrm{pa}(i)}, R_i^m m_i, X'\big) = (R_i^b)_{\#} P^k_{\theta,i}\big(\cdot \mid k_{\mathrm{pa}(i)}, m_i, X\big), \tag{4.24}$$

$$K_a\big(X', R_{\partial a} y_{\partial a}, \cdot\big) = K_a\big(X, y_{\partial a}, \cdot\big), \qquad a \in \mathfrak{A}, \tag{4.25}$$

$$K_a\big(X', (R_i^b k_i, R_i^m m_i), \cdot\big) = K_a\big(X, (k_i, m_i), \cdot\big). \tag{4.26}$$

The hypothesis excludes mechanisms whose coordinate form is compatible with no re-expression of the parameters.

Proposition 4.9 (Joint measure covariance and evidence invariance). Under Hypothesis 4.8, with R and T as in (4.19)–(4.20), the first line below holds as probability measures, without a density or reference hypothesis, and the observation marginals agree, so the second holds:

$$\mathbb{P}_{\theta',V}(\cdot \mid X') = T_{\#} \mathbb{P}_{\theta,V}(\cdot \mid X), \tag{4.27}$$

$$p_{\theta',V}(o \mid X') = p_{\theta,V}(o \mid X) \quad \text{for } \nu^{\mathsf{O}}\text{-almost every } o. \tag{4.28}$$

If latent densities exist and $R_{\#}\nu^{\mathsf{Y}} \ll \nu^{\mathsf{Y}}$, then with j_R of the first line below the second holds $(\nu^{\mathsf{Y}} \otimes \nu^{\mathsf{O}})$-almost everywhere, so composing a density with R^{-1} represents the pushforward for every density exactly when $j_R = 1$ almost everywhere. If every represented block is an invertible linear map between Euclidean spaces of equal dimension, with Lebesgue reference (equal within each channel, with $\dim \mathsf{K} \neq \dim \mathsf{M}$ allowed), then $j_R = |\det R|^{-1}$ and the third line holds:

$$j_R(y') := \big(d(R_{\#}\nu^{\mathsf{Y}})/d\nu^{\mathsf{Y}}\big)(y'), \tag{4.29}$$

$$p_{\theta',V}(o, y' \mid X') = p_{\theta,V}(o, R^{-1}y' \mid X) j_R(y'), \tag{4.30}$$

$$p_{\theta',V}(o, y' \mid X') = |\det R|^{-1} p_{\theta,V}(o, R^{-1}y' \mid X). \tag{4.31}$$

Finally, the root law bridges (4.5) of θ and θ' satisfy $\widetilde{\Phi}_i^{\mathrm{gen}\prime} = \widehat{\rho}_b(g_i) \circ \widetilde{\Phi}_i^{\mathrm{gen}} \circ \widehat{\rho}_m(g_i)^{-1}$, the law (2.23) of a cross-morphism representative. [ESTABLISHED]

Proof. Apply (4.23) and (4.24) along the topological ordering: finite kernel induction gives $\mathbb{P}^{\mathsf{Y}}_{\theta',V}(\cdot \mid X') = R_{\#}\mathbb{P}^{\mathsf{Y}}_{\theta,V}(\cdot \mid X)$. For bounded measurable F on $\mathsf{Y} \times \mathsf{O}$, write $\mathbb{P}_{\theta',V}(\cdot \mid X')$ by (4.10), substitute $y' = Ry$ in the latent integral, and replace the record kernels at $(X', R_{\partial a} y_{\partial a})$ by those at $(X, y_{\partial a})$ by (4.25); this gives $\int F(y,o)\mathbb{P}_{\theta',V}(dy, do \mid X') = \int F(T(y,o))\mathbb{P}_{\theta,V}(dy, do \mid X)$, which is (4.27), and projecting to o gives (4.28). For $Q = q\nu^{\mathsf{Y}}$ and measurable B, $R_{\#}Q(B) = \int_{R^{-1}B} q(y)\nu^{\mathsf{Y}}(dy) = \int_B q(R^{-1}y')(R_{\#}\nu^{\mathsf{Y}})(dy')$, so $R_{\#}Q = (q \circ R^{-1}) j_R \nu^{\mathsf{Y}}$; applied to the joint density this is (4.30), and linear change of variables gives $j_R = |\det R|^{-1}$. For the bridge, (4.24) at a root gives $\widetilde{\Phi}_i^{\mathrm{gen}\prime}(\widehat{\rho}_m(g_i)s) = \int P^k_{\theta',i}(\cdot \mid R_i^m m, X')s(dm) = \widehat{\rho}_b(g_i)\widetilde{\Phi}_i^{\mathrm{gen}}(s)$. □

Corollary 4.10 (Independent channel-frame invariance). Under Hypothesis 4.8, let $\mathbb{Q}_{V,o,X}$ be selected before the re-expression and transported as a whole, $\mathbb{Q}'_{V,o,X'} = R_{\#}\mathbb{Q}_{V,o,X}$. Then for $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$-almost every o the posterior transforms as $\mathbf{\Pi}_{\theta',V,o,X'} = R_{\#}\mathbf{\Pi}_{\theta,V,o,X}$, and

$$D_{\mathrm{KL}}(\mathbb{Q}'_{V,o,X'} \| \mathbf{\Pi}_{\theta',V,o,X'}) = D_{\mathrm{KL}}(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}), \quad \mathcal{L}'^{\mathrm{ext}}_V(\mathbb{Q}'_{V,o,X'}; X', o) = \mathcal{L}^{\mathrm{ext}}_V(\mathbb{Q}_{V,o,X}; X, o) \tag{4.32}$$

in the extended reals, the singular branch included, with $\mathcal{L}^{\mathrm{ext}}_V$ as in (5.15); on the classical domain of both sides the same holds for $\mathcal{L}_V$. [ESTABLISHED]

Proof. By (4.27), for measurable $B \subseteq \mathsf{Y}$ and $A \subseteq \mathsf{O}$, $\mathbb{P}_{\theta',V}(B \times A \mid X') = \mathbb{P}_{\theta,V}(R^{-1}B \times A \mid X) = \int_A (R_{\#}\mathbf{\Pi}_{\theta,V,o,X})(B)\mathbb{P}^{\mathsf{O}}_{\theta,V,X}(do)$, and the observation marginals agree, so $R_{\#}\mathbf{\Pi}_{\theta,V,o,X}$ is a version of the new posterior. Lemma 3.12(e) gives equality of the divergences in $[0, +\infty]$. The evidence agrees by (4.28), and substitution in (5.15) subtracts equal quantities from equal finite numbers; the classical functional equals the extended one on its domain. □

The running example satisfies Hypothesis 4.8 with $\theta' = \theta$, since ρ_m is trivial and its isotropic state covariances, link-transported means, and record lengths are compatible with (4.21).

Physicist's summary

The generative model is one normalized joint of models, beliefs given parents, and records, which may close arbitrary cycles; reciprocal latent conditionals define no joint. No factor may read the recognition law it is scored against, and passive frame changes preserve evidence and bound.

Chapter 5

One Evidence Bound and Local-Collective Free Energy

Chapter 4 fixed one normalized joint law and chapter 3 one correlated recognition coupling of the whole population. This chapter asks what exact relation binds the two, and how the objective of one agent, or of a block of agents, is related to that of the population. The answer is one extended evidence bound, defined for every probability law on the latent space (definition 5.6 and corollary 5.7), and the local–global potential identity (theorem 5.23): with the outside marginal held fixed, the change of the conditional local free energy of a block equals the change of the collective free energy, so the agents' local objectives form an exact potential game (corollary 5.24). Every local object is a disintegration of the one bound through lemma 3.12. The bound is the objective of chapters 6 to 8, and chapters 10 and 13 transport it along evidence-preserving channels.

Scope. The chapter does not claim that a maximizer of the bound over a restricted recognition family exists, that the bound is tight for any particular restricted family, that the alternating scheme of exact coordinates converges (open problem 5.32), or that a block of agents together with its local free energy is a meta-agent. No criterion internal to one bound that selects a recognition structure is known (open problem 5.14). [NOT-CLAIMED]

5.1 Recognition coupling and total correlation

The population recognition law is the coupling $\mathbb{Q}_{V,o,X} \in \mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$ selected by definition 3.15, and every restriction used later is a subfamily $\mathcal{Q}'$ of proposition 5.9. Its marginals do not determine it once two of them are non-Dirac (proposition 3.17), so they cannot replace it in an exact entropy or evidence calculation; the next two statements give the error its exact value and sign.

Corollary 5.1 (Extended total-correlation chain identity). Let $Y = (Y_j)_{j\in\mathcal{J}}$ take values in a finite product of standard Borel spaces, let Q be a probability law with marginals Q_j, and let $(\varrho_j)_{j\in\mathcal{J}}$ be arbitrary probability laws. With the total correlation

$$\mathrm{TC}(Q) := D_{\mathrm{KL}}\Big(Q \,\Big\|\, \bigotimes_j Q_j\Big) \in [0,+\infty], \tag{5.1}$$

the identity

$$D_{\mathrm{KL}}\Big(Q \,\Big\|\, \bigotimes_j \varrho_j\Big) = \mathrm{TC}(Q) + \sum_j D_{\mathrm{KL}}(Q_j \| \varrho_j) \tag{5.2}$$

holds in $[0, +\infty]$, including every singular branch. The infinite branch occurs for a continuous diagonal: if $U \sim \mathrm{Unif}[0,1]$ and $Q = \mathrm{law}(U,U)$, both marginals are uniform and $\mathrm{TC}(Q) = +\infty$. [ESTABLISHED]

Proof. The identity is (3.21) of lemma 3.12 with its first term named. The diagonal is null for the uniform law on the square and carries all the mass of Q, so Q is singular with respect to $Q_1 \otimes Q_2$. □

Proposition 5.2 (Finite-density total correlation and the two signs). Partition the latent coordinates into blocks j, let $\mathbb{Q}_{V,o,X}$ have density $q_{V,o,X}$ with respect to ν^{Y} and block marginal densities q_j, and suppose the joint and block differential entropies are finite. Then the total correlation $\mathrm{TC}(\mathbb{Q}_{V,o,X}) \geq 0$ of the blocks, (5.1), satisfies

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[\log q_{V,o,X}(Y)\big] = \sum_j \mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[\log q_j(Y_j)\big] + \mathrm{TC}(\mathbb{Q}_{V,o,X}), \tag{5.3}$$

with equality of the two log-density expectations exactly when the blocks are independent. If in addition $\mathbb{E}_{\mathbb{Q}_{V,o,X}}|\log p_{\theta,V}(o, Y \mid X)| < \infty$, the substituted free energy and its pseudo-ELBO,

$$\widetilde{\mathcal{F}} := \mathbb{E}_{\mathbb{Q}_{V,o,X}}\Big[-\log p_{\theta,V}(o, Y \mid X) + \sum_j \log q_j(Y_j)\Big], \qquad \widetilde{\mathcal{L}} := -\widetilde{\mathcal{F}}, \tag{5.4}$$

satisfy

$$\widetilde{\mathcal{F}} = \mathcal{F}_V - \mathrm{TC}(\mathbb{Q}_{V,o,X}), \qquad \widetilde{\mathcal{L}} = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + \mathrm{TC}(\mathbb{Q}_{V,o,X}), \tag{5.5}$$

where $\mathcal{L}_V$ and $\mathcal{F}_V$ are the classical functionals of theorem 5.8. The pseudo-ELBO exceeds the ELBO by the total correlation and need not bound the log evidence. [ESTABLISHED]

Proof. Take $\varrho_j = \mathbb{Q}_{V,o,X,j}$ in corollary 5.1; the entropy hypotheses make every term finite and allow the logarithm of $q_{V,o,X} / \prod_j q_j$ to be split, which is (5.3). Subtracting (5.4) from $\mathcal{F}_V = \mathbb{E}_{\mathbb{Q}_{V,o,X}}[-\log p_{\theta,V} + \log q_{V,o,X}]$ leaves TC, and negation gives the second identity. □

5.2 Evidence domain

Hypothesis 5.3 (Evidence-domain hypotheses). The extended bound of section 5.3 uses (H1)–(H2); its classical split into two expected log densities also uses (H3)–(H4).

(H1) Types and version. The joint $\mathbb{P}_{\theta,V}(\cdot \mid X)$ of chapter 4 has the density $p_{\theta,V}(o, y \mid X)$ relative to its reference, and $\mathbf{\Pi}_{\theta,V,o,X}$ is the version (3.16) on the regular set $\mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$ of hypothesis 3.8, so every posterior and bound below holds for $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$-almost every o. A comparison of finitely many parameter values uses an observation in the intersection of their regular sets. No density of the recognition law is required.

(H2) Positive finite evidence. $0 < p_{\theta,V}(o \mid X) < \infty$; this excludes the observations at which the posterior formula would divide by zero or by infinity.

(H3) Absolute continuity. $\mathbb{Q}_{V,o,X} \ll \mathbf{\Pi}_{\theta,V,o,X}$. Since the posterior is dominated by ν^{Y}, the recognition law then has a density $q_{V,o,X} = d\mathbb{Q}_{V,o,X}/d\nu^{\mathsf{Y}}$, with a jointly measurable version for parameterized kernels under theorem A.1 and corollary A.2. Clause (H3) excludes every recognition law that charges a posterior-null set; such a law lies on the singular branch of the extended bound, where the bound equals $-\infty$ (proposition 5.4).

(H4) Log-integrability. The generative and recognition halves

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left|\log p_{\theta,V}(o, Y \mid X)\right| < \infty, \tag{5.6}$$

$$\mathbb{E}_{\mathbb{Q}_{V,o,X}}\left|\log q_{V,o,X}(Y)\right| < \infty, \tag{5.7}$$

both hold. Clause (H4) excludes recognition laws whose tails make the two classical expectations individually infinite. It is not implied by (H1)–(H3), and the extended bound is well defined without it; proposition A.7 gives a Cauchy witness whose formal difference is $-\infty$ and a countable witness whose two expectations are both $-\infty$.

Proposition 5.4 (Posterior-null support has infinite cost). Let Q be a probability law on the latent space with $Q(S) = 1$ and $\mathbf{\Pi}_{\theta,V,o,X}(S) = 0$ for some measurable S. Then (H3) fails for Q, $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) = +\infty$, and $\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = -\infty$. In particular this holds when the posterior is dominated by Lebesgue measure on $\mathbb{R}^d$ and Q is supported on a proper affine subspace. A proper closed support alone does not suffice: the uniform law on $[0, \frac{1}{2}]$ has divergence $\log 2$ from the uniform law on $[0, 1]$. [ESTABLISHED]

Proof. The first claims are the null-set clause of proposition 3.11 and (5.15). A proper affine subspace is Lebesgue null, and the uniform law on $[0, \frac{1}{2}]$ has density 2 on its support relative to the uniform law on $[0, 1]$. □

5.3 The extended evidence bound and its record presentation

Hypothesis 5.5 (Agent state and interaction records). Agent i has a block space $(\mathsf{Y}_i, \mathscr{Y}_i)$, and $(\mathsf{Y}, \mathscr{Y}) = \prod_{i \in V}(\mathsf{Y}_i, \mathscr{Y}_i)$. The block may contain auxiliary latent variables and law points of $(\mathcal{E}_b)_{c_*}$ and $(\mathcal{E}_m)_{c_*}$, which are then latent coordinates at the law-point or configuration level of definition 3.19, not the recognition marginal of agent i. The baseline is the latent population kernel $\mathbb{P}^{\mathsf{Y}}_{\theta,V} : (\mathsf{X}, \mathscr{X}) \rightsquigarrow (\mathsf{Y}, \mathscr{Y})$ of definition 4.2. Each record $a \in \mathfrak{A}$ occurs once and has a finite scope $\partial a \subseteq V$, a record space $(\mathsf{O}_a, \mathscr{O}_a)$, and a record kernel K_a of the type (3.5), all of whose values $K_a(X, y_{\partial a}, \cdot)$ are dominated by one σ-finite measure ν_a with a jointly measurable negative log density $E_{a,o}$ in $(-\infty, +\infty]$,

$$K_a(X, y_{\partial a}, do_a) = e^{-E_{a,o}(X, y_{\partial a})}\nu_a(do_a), \tag{5.8}$$

where $E_{a,o}$ depends on o only through o_a. A record shared by agents i and j is one element of $\mathfrak{A}$, and writing both E_{ij} and E_{ji} asserts two directed records. Binary success records $K_a(y_{\partial a}, \{1\}) = e^{-E(y_{\partial a})}$ with measurable $E : \mathsf{Y}_{\partial a} \to [0, +\infty]$ are admitted. The kernels and $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$ read neither a recognition law, a variational parameter, nor a posterior (hypothesis 4.3), and under a common vertex gauge the coordinate presentation of each K_a is equivariant under the simultaneous change of the incident frames, as in (4.25), so the kernel itself is frame independent. The hypothesis excludes record families without a common dominating measure (open problem 3.9) and every generative factor that reads recognition.

Correlated baseline states are permitted; a graph-local Markov blanket needs in addition the corresponding locality of the regular conditionals of $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$. Frame independence can be realized by giving record a its own frame and a link $\Theta^x_{a\leftarrow i}$ on each leg, so that the density reads $\rho_x(\Theta^x_{a\leftarrow i})y^x_i$; comparing the belief and model channels still requires the cross morphisms of section 2.3. From here on one $X \in \mathsf{X}$ is fixed and suppressed from K_a, $E_{a,o}$ and $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$.

Definition 5.6 (Record likelihood, slice measure, and extended bound). Assume hypothesis 5.5 and (H1)–(H2), and fix $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,V,X}$. The record likelihood and the interaction energy are

$$L_o(y) := \prod_{a\in\mathfrak{A}} e^{-E_{a,o}(y_{\partial a})} \in [0, +\infty), \tag{5.9}$$

$$H_o(y) := -\log L_o(y) = \sum_{a\in\mathfrak{A}} E_{a,o}(y_{\partial a}) \in (-\infty, +\infty]; \tag{5.10}$$

H_o has one term for each record, never equals $-\infty$ because each factor density is finite, and can be negative. With $\nu^{\mathsf{O}} = \bigotimes_a \nu_a$ the joint, normalized on every finite record hypergraph, cyclic or not, by proposition 4.4, is

$$\mathbb{P}_{\theta,V}(dy, do \mid X) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy) \prod_{a\in\mathfrak{A}} K_a(y_{\partial a}, do_a) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy) L_o(y) \nu^{\mathsf{O}}(do). \tag{5.11}$$

The evidence and the posterior slice are

$$p_{\theta,V}(o \mid X) = \int_{\mathsf{Y}} L_o(y) \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy) \in (0, +\infty), \tag{5.12}$$

$$\mathbf{\Pi}_{\theta,V,o,X}(dy) = p_{\theta,V}(o \mid X)^{-1} L_o(y) \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy), \tag{5.13}$$

and the slice measure is

$$M_{\theta,V,o,X}(S) := \int_S p_{\theta,V}(o, y \mid X) \nu^{\mathsf{Y}}(dy) = \int_S L_o(y) \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy) = p_{\theta,V}(o \mid X) \mathbf{\Pi}_{\theta,V,o,X}(S). \tag{5.14}$$

For every probability law Q on $(\mathsf{Y}, \mathscr{Y})$ the extended ELBO and free energy, which are well-defined extended reals because $\log p_{\theta,V}(o \mid X)$ is finite and relative entropy lies in $[0, +\infty]$, are

$$\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) := \log p_{\theta,V}(o \mid X) - D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big) \in \big[-\infty, \log p_{\theta,V}(o \mid X)\big], \tag{5.15}$$

$$\mathcal{F}_V^{\mathrm{ext}}[Q; X, o] := -\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = -\log p_{\theta,V}(o \mid X) + D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big), \tag{5.16}$$

and with θ, V and X fixed the collective free energy is written

$$\mathcal{F}_o^{\mathrm{ext}}(Q) := -\mathcal{L}_o^{\mathrm{ext}}(Q) := \mathcal{F}_V^{\mathrm{ext}}[Q; X, o]. \tag{5.17}$$

No domination, density, or mean-field condition on Q enters.

Corollary 5.7 (Extended gap identity and energy form). For every probability law Q on the latent space,

$$\log p_{\theta,V}(o \mid X) - \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \mathcal{F}_V^{\mathrm{ext}}[Q; X, o] + \log p_{\theta,V}(o \mid X) = D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big), \tag{5.18}$$

so $\mathcal{L}_V^{\mathrm{ext}} \le \log p_{\theta,V}(o \mid X)$ with equality exactly when $Q = \mathbf{\Pi}_{\theta,V,o,X}$ as measures. If moreover $Q \ll \mathbb{P}^{\mathsf{Y}}_{\theta,V}$, $D_{\mathrm{KL}}(Q \| \mathbb{P}^{\mathsf{Y}}_{\theta,V}) < \infty$, and the negative part of H_o is Q-integrable, then

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = D_{\mathrm{KL}}\big(Q \| \mathbb{P}^{\mathsf{Y}}_{\theta,V}\big) + \mathbb{E}_Q H_o \quad \text{in } (-\infty, +\infty]. \tag{5.19}$$

No formula $\log p_{\theta,V} = \mathcal{L}_V^{\mathrm{ext}} + D_{\mathrm{KL}}$ is asserted on the singular branch, where it would read $-\infty + \infty$. [ESTABLISHED]

Proof. The gap identity adds or subtracts the finite number $\log p_{\theta,V}(o \mid X)$ in (5.15)–(5.16), and the equality case is the zero case of relative entropy. For the energy form let $r' = dQ/d\mathbb{P}^{\mathsf{Y}}_{\theta,V}$. If $Q \ll M_{\theta,V,o,X}$, then $dQ/dM_{\theta,V,o,X} = r' e^{H_o}$ Q-almost surely, and proposition A.6 gives $\mathcal{F}_o^{\mathrm{ext}}(Q) = \int (\log r' + H_o) dQ$; the first term integrates to the finite $D_{\mathrm{KL}}(Q \| \mathbb{P}^{\mathsf{Y}}_{\theta,V})$ and the second is quasi-integrable, so the integral splits. If $Q \not\ll M_{\theta,V,o,X}$, some set null for $M_{\theta,V,o,X}$ has positive Q-mass; since $M_{\theta,V,o,X}$ and $\mathbb{P}^{\mathsf{Y}}_{\theta,V}$ agree in null sets on $\{L_o > 0\}$ and $Q \ll \mathbb{P}^{\mathsf{Y}}_{\theta,V}$, that mass lies in $\{H_o = +\infty\}$, and both sides equal $+\infty$. □

Theorem 5.8 (Classical finite evidence identity). Under hypothesis 5.3, the extended bound has the separately integrable representation

$$\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) := \mathbb{E}_{\mathbb{Q}_{V,o,X}}\big[\log p_{\theta,V}(o, Y \mid X) - \log q_{V,o,X}(Y)\big] = \mathcal{L}_V^{\mathrm{ext}}(\mathbb{Q}_{V,o,X}; X, o), \tag{5.20}$$

$$\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] := -\mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o), \tag{5.21}$$

and for $\mathbb{P}^{\mathsf{O}}_{\theta,V,X}$-almost every observation

$$\log p_{\theta,V}(o \mid X) = \mathcal{L}_V(\mathbb{Q}_{V,o,X}; X, o) + D_{\mathrm{KL}}\big(\mathbb{Q}_{V,o,X} \| \mathbf{\Pi}_{\theta,V,o,X}\big). \tag{5.22}$$

Consequently $\mathcal{L}_V \le \log p_{\theta,V}(o \mid X)$, with equality if and only if $\mathbb{Q}_{V,o,X} = \mathbf{\Pi}_{\theta,V,o,X}$ as probability measures on $(\mathsf{Y}, \mathscr{Y})$. Equality of all coordinate marginals is not sufficient. [ESTABLISHED]

Proof. The posterior and the slice measure have the same null sets, so (H3) gives $\mathbb{Q}_{V,o,X} \ll M_{\theta,V,o,X}$, and since the two measures have the densities $q_{V,o,X}$ and $p_{\theta,V}(o,\cdot \mid X)$ relative to ν^{Y}, $d\mathbb{Q}_{V,o,X}/dM_{\theta,V,o,X} = q_{V,o,X}/p_{\theta,V}(o,\cdot \mid X)$ $\mathbb{Q}_{V,o,X}$-almost surely. Proposition A.6 gives $\mathcal{L}_V^{\mathrm{ext}} = -\int \log(q_{V,o,X}/p_{\theta,V}(o,\cdot \mid X)) d\mathbb{Q}_{V,o,X}$, and (H4) splits the integral, which is (5.20). The bound is then finite, so (5.15) rearranges to (5.22), and the equality case is that of corollary 5.7. Distinct measures with equal coordinate marginals are supplied by proposition 3.17. □

5.4 Restriction principle

Proposition 5.9 (Restriction principle). Fix (θ, X) and o as in definition 5.6, and write $P^\star = \mathbf{\Pi}_{\theta,V,o,X}$, so that for every law Q

$$\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \log p_{\theta,V}(o \mid X) - D_{\mathrm{KL}}(Q \| P^\star). \tag{5.23}$$

For arbitrary sets of probability laws $\mathcal{Q}' \subseteq \mathcal{Q}$,

$$\sup_{Q \in \mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = \log p_{\theta,V}(o \mid X) - \inf_{Q \in \mathcal{Q}'} D_{\mathrm{KL}}(Q \| P^\star), \tag{5.24}$$

the supremum over $\mathcal{Q}'$ is at most the supremum over $\mathcal{Q}$, and when both infima are finite

$$\sup_{\mathcal{Q}} \mathcal{L}_V^{\mathrm{ext}} - \sup_{\mathcal{Q}'} \mathcal{L}_V^{\mathrm{ext}} = \inf_{\mathcal{Q}'} D_{\mathrm{KL}}(Q \| P^\star) - \inf_{\mathcal{Q}} D_{\mathrm{KL}}(Q \| P^\star) \ge 0. \tag{5.25}$$

No attainment of the suprema or infima is required. [ESTABLISHED]

Proof. Identity (5.23) is (5.15). Taking suprema turns a finite constant minus a nonnegative extended divergence into that constant minus its infimum, including the value $-\infty$. Inclusion gives monotonicity, and subtraction is legitimate only when both infima are finite. □

The evidence in (5.23) is constant because the generative kernel does not read Q (hypothesis 4.3); the next two propositions show what fails when a model moves with Q.

Remark 5.10 (Orientation, separation, and attainment). The infimum in (5.24) is a reverse information projection, since Q is the first argument of the divergence. Excluding $P^\star$ from a family does not make the optimum positive or attained: for $P^\star = \mathcal{N}(0,1)$ the family $\{\mathcal{N}(\mu,1) : \mu > 0\}$ has gaps $\mu^2/2$ with unattained infimum zero. A positive optimized gap is the condition $\inf_{Q \in \mathcal{Q}} D_{\mathrm{KL}}(Q \| P^\star) > 0$. A family constrained to product laws need not be convex, because mixtures of product laws need not be product laws. Strict convexity of relative entropy on the ambient law space therefore does not establish uniqueness on that family; on a convex feasible family it does give at most one finite minimizer. [ESTABLISHED]

Proposition 5.11 (No distinguished target). Let $\mathcal{Q}$ be a nonempty family of recognition laws, and replace one factor of (4.6) by a nonconstant Q-indexed family, giving a normalized joint $P_{\theta,Q}$ for each $Q \in \mathcal{Q}$. Assume at the observation that $0 < p_{\theta,Q}(o \mid X) < \infty$, that $Q \ll P_{\theta,Q}(\cdot \mid o, X)$, and that the log densities defining the ELBO are integrable. Then for each fixed Q

$$\log p_{\theta,Q}(o \mid X) = \mathcal{L}(Q; X) + D_{\mathrm{KL}}\big(Q \| P_{\theta,Q}(\cdot \mid o, X)\big), \tag{5.26}$$

and with $\mathsf{e}(Q) = \log p_{\theta,Q}(o \mid X)$ two members satisfy

$$\mathcal{L}(Q'; X) - \mathcal{L}(Q; X) = \mathsf{e}(Q') - \mathsf{e}(Q) - \big[D_{\mathrm{KL}}\big(Q' \| P_{\theta,Q'}(\cdot \mid o, X)\big) - D_{\mathrm{KL}}\big(Q \| P_{\theta,Q}(\cdot \mid o, X)\big)\big]. \tag{5.27}$$

On a level set of e the change of the bound is the negative change of the divergence; off it the two changes differ by $\mathsf{e}(Q') - \mathsf{e}(Q)$. The family selects no joint as the model. [ESTABLISHED]

Proof. Apply theorem 5.8 to each fixed pair $(Q, P_{\theta,Q})$ and subtract. No member is distinguished because no selection rule was declared. □

Proposition 5.12 (The stronger consequences fail). Nonconstant dependence of $P_{\theta,Q}$ on Q does not imply that e varies, that no ascent improves a fixed quantity, or that no single member's evidence bounds the family of bounds. Explicitly, for one binary latent y, one binary observation, and $Q_{\mathrm{p}} = \mathrm{Ber}(\mathrm{p})$ with $\mathrm{p} \in (0, 1)$, the joint

$$P_{\theta,Q_{\mathrm{p}}}(o, y \mid X) = \tfrac{1}{2}\,\mathrm{Ber}\big(\tfrac{1}{2}\mathrm{p} + \tfrac{1}{4}\big)(y) \tag{5.28}$$

is normalized and injective in p, has $\mathsf{e}(Q_{\mathrm{p}}) = \log \frac{1}{2}$ for every p, and has gaps

$$\tfrac{9}{10}\log\tfrac{9}{7} + \tfrac{1}{10}\log\tfrac{1}{3} = 0.1163218\ldots, \qquad \tfrac{7}{10}\log\tfrac{7}{6} + \tfrac{3}{10}\log\tfrac{3}{4} = 0.0216009\ldots, \qquad 0 \tag{5.29}$$

at $\mathrm{p} = \frac{9}{10}, \frac{7}{10}, \frac{1}{2}$. [ESTABLISHED]

Proof. The observation marginal is $\frac{1}{2}$ for every p, and the posterior is $\mathrm{Ber}(\frac{1}{2}\mathrm{p} + \frac{1}{4})$, so the gaps are the Bernoulli relative entropies displayed. The bound increases to the fixed evidence along the three values, and the member at $\mathrm{p} = \frac{1}{2}$ has evidence equal to the supremum of the bounds. □

Proposition 5.13 (ELBO values do not order unrelated joints). Let P_θ and $P_{\theta'}$ be normalized joints over the same observation, possibly with different latent inventories, with recognition families $\mathcal{Q}$ and $\mathcal{Q}'$ on their latent spaces, and let o lie in the regular set of each. Suppose both infima of the posterior relative entropy over the respective recognition families are finite. If no evidence-preserving channel relates the two joints, their optimized bounds satisfy

$$\mathcal{L}^\star - \mathcal{L}'^\star = \big[\log p_\theta(o \mid X) - \log p_{\theta'}(o \mid X)\big] - \Big[\inf_{\mathcal{Q}} D_{\mathrm{KL}} - \inf_{\mathcal{Q}'} D_{\mathrm{KL}}\Big], \tag{5.30}$$

which combines an evidence difference, a model-comparison quantity, with a tightness difference, a recognition-quality quantity. The sign of the left side orders neither the models nor the recognition families: with binary O and Y, constants $\mathrm{c}_1 \le 0 \le \mathrm{c}_2$, and the normalized joints $P(o, 1) = e^{\mathrm{c}_1 - \mathrm{c}_2}$,

$P(o,0) = e^{c_1}(1-e^{-c_2})$, $P(\bar{o},0) = 1-e^{c_1}$, $P(\bar{o},1) = 0$ under the singleton family $\{\delta_1\}$, the pairs $(c_1,c_2) = (-1,0),(0,2)$ give bounds $-1 > -2$ with the reverse evidence order, and the pairs $(0,1),(-2,0)$ give bounds $-1 > -2$ with the reverse tightness order. [ESTABLISHED]

Proof. Apply (5.24) to each model and subtract. In the witness, $\log P(O=o) = c_1$ and $P(Y = 1 \mid o) = e^{-c_2}$, so the minimized gap of $\{\delta_1\}$ is c_2 and the optimized bound is $c_1 - c_2$. An unused deterministic coordinate appended to either model, with its point mass appended to the recognition law, makes the latent inventories literally different and changes no evidence, divergence, or bound. □

Open Problem 5.14 (An intrinsic criterion). No criterion internal to one evidence bound at a fixed model is known that selects a nondegenerate recognition structure. Along nested recognition families the optimized bound is monotone by inclusion (proposition 5.9), so minimizing the gap selects the largest family; an arbitrary replacement joint supplies no common evidence (proposition 5.13); and evidence-preserving channels, which do order bounds across latent spaces, do so without selecting a nondegenerate partition (section 10.3). The problem is settled positively by a functional on admissible channels or partitions, expressible relative to one fixed joint and not monotone toward an endpoint, whose optimizer is nondegenerate, and negatively by a proof of endpoint degeneracy for a precisely stated class of such functionals.

5.5 The factorwise decomposition

Proposition 5.15 (Exact factorwise decomposition). Assume (H1)–(H3) of hypothesis 5.3 and factorwise log-integrability: the absolute logarithm of every component density of (4.8) and every record energy $E_{a,o}$ is $\mathbb{Q}_{V,o,X}$-integrable. Then (i) the generative half (5.6) of (H4) holds, and (H4) holds exactly when the recognition half (5.7) does; (ii) under (H4) the free energy has exactly one decomposition into factor terms,

$$\begin{aligned}\mathcal{F}_V[\mathbb{Q}_{V,o,X}; X, o] = &\sum_{i\in V} \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^m_{\theta,i}(m_i \mid m_{\mathrm{pa}(i)}, X)\right] + \sum_{i\in V} \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[-\log p^k_{\theta,i}(k_i \mid k_{\mathrm{pa}(i)}, m_i, X)\right] \\ &+ \sum_{a\in\mathfrak{A}} \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[E_{a,o}(Y_{\partial a})\right] + \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log q_{V,o,X}(Y)\right], \end{aligned} \tag{5.31}$$

with $\mathrm{pa}(i) = \varnothing$ at the roots, $2N + |\mathfrak{A}|$ generative factors, and a final term that is one full-joint recognition log density, not a sum of marginal entropies; (iii) factorwise integrability does not imply the recognition half, and the generative half does not imply factorwise integrability. [ESTABLISHED]

Proof. By (4.8) and hypothesis 5.5,

$$\log p_{\theta,V}(o, y \mid X) = \sum_{i\in V}\left[\log p^m_{\theta,i}(m_i \mid m_{\mathrm{pa}(i)}, X) + \log p^k_{\theta,i}(k_i \mid k_{\mathrm{pa}(i)}, m_i, X)\right] - \sum_{a\in\mathfrak{A}} E_{a,o}(y_{\partial a}), \tag{5.32}$$

a finite sum, so the triangle inequality gives (i). Under (H4), taking $\mathbb{Q}_{V,o,X}$-expectations of the negative of (5.32) term by term and adding $\mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_{V,o,X}]$ gives (5.31) from (5.20). For (iii),

appendix A.2 gives a model with bounded factor logarithms whose recognition density has an infinite mean logarithm, and a root law $\mathrm{c}/(m(\log m)^2)$ whose state bridge cancels its logarithm, so that the complete log joint is integrable while one factor is not. □

5.6 Observations as interaction records

Definition 5.16 (Operational environment node). An operational environment node is a standard Borel state space with a normalized state law and a measurable message kernel.

Theorem 5.17 (Observation–interaction kernel equivalence). Every normalized observation kernel $K(do \mid y)$ from a measurable space to a standard Borel space is the message of an operational environment node: there are $U \sim \mathrm{Unif}[0,1]$ and a measurable map F such that

$$O = F(Y,U), \qquad \Pr\big(F(y,U) \in A\big) = K(A \mid y). \tag{5.33}$$

Conversely, marginalizing any environment state and message policy produces a normalized observation kernel. [ESTABLISHED]

Proof. The forward direction is the randomization lemma for kernels [61]. By Kuratowski's theorem a standard Borel space admits a Borel isomorphism ϕ onto a Borel subset of $\mathbb{R}$. The function $G(y,x) = K(\phi^{-1}(-\infty,x] \mid y)$ is measurable in y and a distribution function in x, so the quantile $F_0(y,u) = \inf\{x : G(y,x) \geq u\}$ on $(0,1)$ satisfies $\{F_0 \leq x\} = \{(y,u) : u \leq G(y,x)\}$; hence F_0 is jointly measurable and $\Pr(F_0(y,U) \leq x) = G(y,x)$. The value $F_0(y,U)$ lies in the image of ϕ almost surely; setting $F = \phi^{-1} \circ F_0$ there and a fixed point elsewhere gives (5.33). Conversely, a state law $\varrho_{\mathrm{env}}(du)$ and a message policy $\kappa_{\mathrm{env}}(y,u,A)$ give $K(A \mid y) = \int \kappa_{\mathrm{env}}(y,u,A)\varrho_{\mathrm{env}}(du)$, which is measurable in y, countably additive in A, and normalized. □

Proposition 5.18 (Conditioning information and retained statistics). Let (Y,O) have joint law P on standard Borel spaces with marginals P^Y and P^O, let $I(Y;O) = D_{\mathrm{KL}}(P \| P^Y \otimes P^O)$, and for a measurable statistic $S = T(O)$ let $I(Y;O \mid S) = \int D_{\mathrm{KL}}(P_{Y,O|s} \| P_{Y|s} \otimes P_{O|s}) P^S(ds)$. The information carried by the record is

$$I(Y;O) = \int D_{\mathrm{KL}}\big(P(dY \mid o) \| P^Y\big) P^O(do) \in [0,+\infty]. \tag{5.34}$$

Moreover

$$I(Y;O) = I(Y;S) + I(Y;O \mid S) \quad \text{in } [0,+\infty], \tag{5.35}$$

so replacing O by S leaves the posterior unchanged exactly when $I(Y;O \mid S) = 0$, up to the choice of conditional versions. Deleting the record makes S constant and returns the prior; for $Y \sim \mathrm{Ber}(1/2)$ and $O = Y$, $I(Y;O) = \log 2$ and $P(dY \mid O = o) = \delta_o$. A kernel that does not depend on y carries zero information. [ESTABLISHED]

Proof. Disintegrating $D_{\mathrm{KL}}(P\|P^Y \otimes P^O)$ over O with lemma 3.12, whose O-marginals agree, gives (5.34). Since $o \mapsto (T(o), o)$ is a bimeasurable bijection onto its graph, relative entropy is unchanged by passing to (Y, S, O) (3.24). Disintegrating $D_{\mathrm{KL}}(P_{Y,S,O}\|P^Y \otimes P_{S,O})$ over (Y, S) gives $I(Y;S) + \mathbb{E}_{P_{Y,S}} D_{\mathrm{KL}}(P_{O|Y,S}\|P_{O|S})$, because under the reference O is independent of Y given S with law $P_{O|S}$; for each s, disintegrating $D_{\mathrm{KL}}(P_{Y,O|s}\|P_{Y|s} \otimes P_{O|s})$ over Y and integrating in s identifies the second term with $I(Y;O \mid S)$, which proves (5.35); the discrete case is Cover and Thomas [24, Sec. 2.5]. Posterior sufficiency is the vanishing of the conditional term, the Bernoulli values are direct, and a y-independent kernel makes $P = P^Y \otimes P^O$. □

Remark 5.19 (Environment nodes in the variational problem). Theorem 5.17 is exact for the joint law of (Y, O), and it keeps the record variable O and hence the conditioning σ-algebra $\sigma(O)$, but it is not exact for a variational problem over the enlarged state (Y, U). For P^O-almost every o the posterior of (Y, U) is concentrated on the level set $\{F(y, u) = o\}$. For $Y \sim \mathcal{N}(0, 1)$ and $F(y, u) = y + F_{\mathcal{N}}^{-1}(u)$, with $F_{\mathcal{N}}$ the standard normal distribution function, which realizes $K(do \mid y) = \mathcal{N}(y, 1)$, that level set is the Lebesgue-null curve $\{u = F_{\mathcal{N}}(o - y)\}$, so every recognition law on (Y, U) with a Lebesgue density has relative entropy $+\infty$ by proposition 5.4, whereas the fiber-restricted law $q(dy)\delta_{F_{\mathcal{N}}(o-y)}(du)$, the image of q under the injective map $y \mapsto (y, F_{\mathcal{N}}(o - y))$ that also carries the posterior of Y to that of (Y, U), has relative entropy $D_{\mathrm{KL}}(q\|P(dY \mid o))$. A finite variational problem therefore integrates U out or restricts recognition to that fiber. [ESTABLISHED]

5.7 Local conditional free energy and the potential identity

Fix a nonempty block $B \subseteq V$, one agent or any aggregate, and write $y = (y_B, y_{-B})$; the probability model does not change with B.

Lemma 5.20 (Block disintegration of the posterior). Let $\mathfrak{A}_B = \{a \in \mathfrak{A} : \partial a \cap B \neq \varnothing\}$ and define the incident and outside likelihoods and energies

$$L_{B,o}(y_B; y_{-B}) = \prod_{a \in \mathfrak{A}_B} e^{-E_{a,o}(y_{\partial a})}, \qquad L_{\bar{B},o}(y_{-B}) = \prod_{a \notin \mathfrak{A}_B} e^{-E_{a,o}(y_{\partial a})}, \tag{5.36}$$

$$H_{B,o}(y_B; y_{-B}) = -\log L_{B,o} = \sum_{a \in \mathfrak{A}_B} E_{a,o}(y_{\partial a}), \qquad H_{\bar{B},o}(y_{-B}) = -\log L_{\bar{B},o}(y_{-B}), \tag{5.37}$$

so that $L_o = L_{B,o} L_{\bar{B},o}$ and $H_o = H_{B,o} + H_{\bar{B},o}$ in $(-\infty, +\infty]$. Fix a measurable version of the baseline disintegration

$$\mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy_B, dy_{-B}) = \mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}(dy_{-B}) \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(dy_B \mid y_{-B}), \tag{5.38}$$

and define the local evidence and the likelihood of the outside state,

$$Z_B(y_{-B}) = \int L_{B,o}(y_B; y_{-B}) \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(dy_B \mid y_{-B}), \tag{5.39}$$

$$\widetilde{L}_{B,o}(y_{-B}) = \int L_o(y_B, y_{-B}) \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(dy_B \mid y_{-B}). \tag{5.40}$$

Then (i) the evidence and the outside posterior marginal are

$$p_{\theta,V}(o \mid X) = \int \widetilde{L}_{B,o} d\mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}, \qquad \mathbf{\Pi}_{\theta,V,o,X,-B}(dy_{-B}) = p_{\theta,V}(o \mid X)^{-1} \widetilde{L}_{B,o}(y_{-B}) \mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}(dy_{-B}); \tag{5.41}$$

(ii) the set $\Omega_{B,o}$ of outside states at which $0 < \widetilde{L}_{B,o} < \infty$, $0 < L_{\bar{B},o} < \infty$, $0 < Z_B < \infty$, and $\widetilde{L}_{B,o} = L_{\bar{B},o} Z_B$ has $\mathbf{\Pi}_{\theta,V,o,X,-B}$-measure one, though it need not be $\mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}$-full; (iii) the kernel

$$\mathbf{\Pi}_{\theta,V,o,X,B}(dy_B \mid y_{-B}) = Z_B(y_{-B})^{-1} L_{B,o}(y_B; y_{-B}) \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(dy_B \mid y_{-B}) \qquad (y_{-B} \in \Omega_{B,o}), \tag{5.42}$$

extended by any measurable probability kernel off $\Omega_{B,o}$, satisfies

$$\mathbf{\Pi}_{\theta,V,o,X}(dy_B, dy_{-B}) = \mathbf{\Pi}_{\theta,V,o,X,-B}(dy_{-B}) \mathbf{\Pi}_{\theta,V,o,X,B}(dy_B \mid y_{-B}). \tag{5.43}$$

The posterior determines this kernel, and every pointwise local quantity built from it, only $\mathbf{\Pi}_{\theta,V,o,X,-B}$-almost everywhere; two baseline versions give local posteriors that agree there. [ESTABLISHED]

Proof. Tonelli's theorem applied to (5.12)–(5.13) with (5.38) gives (i); defining $\widetilde{L}_{B,o}$ directly avoids the product $L_{\bar{B},o} Z_B$, which can be $0 \cdot \infty$ off $\Omega_{B,o}$. For (ii), finite evidence makes $\widetilde{L}_{B,o} < \infty$ off a $\mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}$-null set, and by (i) the outside posterior charges only $\{\widetilde{L}_{B,o} > 0\}$. There $L_{\bar{B},o}$ is finite because each factor density is, positive because $L_{\bar{B},o} = 0$ would force $\widetilde{L}_{B,o} = 0$, and independent of y_B, so it leaves the conditional integral and $Z_B = \widetilde{L}_{B,o} / L_{\bar{B},o} \in (0, \infty)$. For (iii), a bounded measurable f has $\int f d\mathbf{\Pi}_{\theta,V,o,X} = p_{\theta,V}(o \mid X)^{-1} \int L_{\bar{B},o} \int f L_{B,o} d\mathbb{P}^{\mathsf{Y}}_{\theta,V,B} d\mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}$, and on $\Omega_{B,o}$ the inner integral is Z_B times the integral of f against (5.42); (i) and (ii) turn the outer weight into $\mathbf{\Pi}_{\theta,V,o,X,-B}$. Almost-everywhere uniqueness is uniqueness of disintegrations; two baseline versions agree $\mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}$-almost everywhere, hence $\mathbf{\Pi}_{\theta,V,o,X,-B}$-almost everywhere by (i). □

Definition 5.21 (Local conditional free energy). For $y_{-B} \in \Omega_{B,o}$ and every probability law $r_B(\cdot \mid y_{-B})$ on Y_B, the local conditional ELBO and free energy of the block are

$$\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; y_{-B}) := -\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; y_{-B}) := -\log Z_B(y_{-B}) + D_{\mathrm{KL}}\big(r_B(\cdot \mid y_{-B}) \| \mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid y_{-B})\big). \tag{5.44}$$

For $B = \{i\}$ this is the local free energy of agent i. It contains every record incident to the block and no other.

Corollary 5.22 (Local Gibbs bound and the exact block E-coordinate). (i) $\mathcal{L}^{\mathrm{ext}}_{B,o}(r_B; y_{-B}) \le \log Z_B(y_{-B})$, with equality exactly at $r_B = \mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid y_{-B})$. If $r_B \ll \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(\cdot \mid y_{-B})$ with finite

relative entropy and the negative part of $H_{B,o}(\cdot; y_{-B})$ is r_B-integrable,

$$\mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; y_{-B}) = D_{\mathrm{KL}}\big(r_B(\cdot \mid y_{-B}) \| \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(\cdot \mid y_{-B})\big) + \mathbb{E}_{r_B(\cdot|y_{-B})} H_{B,o}(Y_B; y_{-B}). \tag{5.45}$$

(ii) If $Q(dy_B, dy_{-B}) = Q_{-B}(dy_{-B}) r_B(dy_B \mid y_{-B})$ and $Q_{-B} \ll \mathbf{\Pi}_{\theta,V,o,X,-B}$, then in $[0, +\infty]$, with the baseline conditional of (5.38) in the second identity,

$$D_{\mathrm{KL}}\big(Q \| \mathbf{\Pi}_{\theta,V,o,X}\big) = D_{\mathrm{KL}}\big(Q_{-B} \| \mathbf{\Pi}_{\theta,V,o,X,-B}\big) + \mathbb{E}_{Q_{-B}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{-B}) \| \mathbf{\Pi}_{\theta,V,o,X,B}(\cdot \mid Y_{-B})\big), \tag{5.46}$$

$$D_{\mathrm{KL}}\big(Q \| \mathbb{P}^{\mathsf{Y}}_{\theta,V}\big) = D_{\mathrm{KL}}\big(Q_{-B} \| \mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}\big) + \mathbb{E}_{Q_{-B}} D_{\mathrm{KL}}\big(r_B(\cdot \mid Y_{-B}) \| \mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(\cdot \mid Y_{-B})\big). \tag{5.47}$$

(iii) Let $\mathcal{B}$ be any block of latent coordinates with complement $-\mathcal{B}$, let P_- be the $-\mathcal{B}$ marginal of $P = \mathbf{\Pi}_{\theta,V,o,X}$, let R be any law on the $-\mathcal{B}$ coordinates, and let $\kappa_{\mathcal{B}}(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}})$ be a Markov kernel that agrees P_--almost everywhere with a regular conditional law of P (for an agent block, (5.42)). Among all laws with $-\mathcal{B}$ marginal R, the divergence from P has minimum $D_{\mathrm{KL}}(R \| P_-) \in [0, +\infty]$, attained by

$$\mathbb{Q}^{\mathrm{new}}_{V,o,X}(dY_{\mathcal{B}}, dY_{-\mathcal{B}}) = \kappa_{\mathcal{B}}(dY_{\mathcal{B}} \mid Y_{-\mathcal{B}}) R(dY_{-\mathcal{B}}), \tag{5.48}$$

which therefore maximizes $\mathcal{L}^{\mathrm{ext}}_V$ on that class. The minimizer is unique when $D_{\mathrm{KL}}(R \| P_-) < \infty$; otherwise every law in the class has extended bound $-\infty$. If the minimum is finite and

$$\mathbb{E}_{\mathbb{Q}^{\mathrm{new}}_{V,o,X}}\big[|\log p_{\theta,V}(o, Y \mid X)| + |\log q^{\mathrm{new}}_{V,o,X}(Y)|\big] < \infty, \tag{5.49}$$

the new law satisfies (H3)–(H4) and maximizes the classical $\mathcal{L}_V$ among the laws with marginal R that satisfy hypothesis 5.3. Finite complement relative entropy does not imply (5.49).

(iv) If $\mathcal{F}^{\mathrm{ext}}_o(Q) < \infty$ for $Q = Q_{-B} r_B$, then for $\mathrm{w} \in [0, 1]$ the damped kernel $r^{\mathrm{w}}_B = (1 - \mathrm{w}) r_B + \mathrm{w} \mathbf{\Pi}_{\theta,V,o,X,B}$ satisfies

$$\mathcal{F}^{\mathrm{ext}}_o(Q_{-B} r^{\mathrm{w}}_B) \le (1 - \mathrm{w}) \mathcal{F}^{\mathrm{ext}}_o(Q) + \mathrm{w} \mathcal{F}^{\mathrm{ext}}_o(Q_{-B} \mathbf{\Pi}_{\theta,V,o,X,B}) \le \mathcal{F}^{\mathrm{ext}}_o(Q), \tag{5.50}$$

so replacing r_B by the local posterior with Q_{-B} fixed does not increase $\mathcal{F}^{\mathrm{ext}}_o$. [ESTABLISHED]

Proof. For (i), the conditional model with reference $\mathbb{P}^{\mathsf{Y}}_{\theta,V,B}(\cdot \mid y_{-B})$ and likelihood $L_{B,o}(\cdot; y_{-B})$ has evidence $Z_B(y_{-B}) \in (0, \infty)$ and posterior (5.42), and corollary 5.7 applies to it. Part (ii) is lemma 3.12 with the disintegrations (5.43) and (5.38). For (iii), the fixed-marginal clause of lemma 3.12, with the $-\mathcal{B}$ coordinates first, gives the minimum $D_{\mathrm{KL}}(R \| P_-)$, the minimizer (5.48), and uniqueness when the minimum is finite; then $R \ll P_-$, the completion of $\kappa_{\mathcal{B}}$ off a P_--null set is irrelevant, and $d\mathbb{Q}^{\mathrm{new}}_{V,o,X}/dP = (dR/dP_-)(Y_{-\mathcal{B}})$. When the minimum is infinite, the data-processing clause (3.22) applied to the projection makes every divergence in the class infinite. Definition 5.6 converts divergence into the extended bound, and theorem 5.8 gives the classical statement. The countable witness of appendix A.2 has $R = P_-$ and fails (5.49). For (iv), finiteness makes $Q_{-B} \ll \mathbf{\Pi}_{\theta,V,o,X,-B}$ by data processing, and (5.46) leaves the outside term unchanged; the exact

replacement sets the conditional term to zero, and convexity of relative entropy in its first argument, applied for each y_{-B}, gives (5.50). □

The exact full E-step is case (iii) with $\mathcal{B}$ equal to all coordinates; a coordinate that optimizes inside a product family does not have the form (5.48) and leaves the gap of proposition 5.9. By (iv) a sequential sweep of exact or damped block updates does not increase the collective free energy; replacing the conditionals of several correlated blocks at once is not covered, because each replacement changes the outside marginal of the others.

Theorem 5.23 (Exact local–global potential identity). Let $Q = Q_{-B} r_B$ and $Q' = Q_{-B} r'_B$ have the same outside marginal with $Q_{-B} \ll \mathbf{\Pi}_{\theta,V,o,X,-B}$, and suppose both have finite divergence from $\mathbf{\Pi}_{\theta,V,o,X}$. Then

$$\mathcal{F}^{\mathrm{ext}}_o(Q') - \mathcal{F}^{\mathrm{ext}}_o(Q) = \mathbb{E}_{Q_{-B}}\big[\mathcal{F}^{\mathrm{ext}}_{B,o}(r'_B; Y_{-B}) - \mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; Y_{-B})\big]. \tag{5.51}$$

When in addition every baseline relative entropy and the positive and negative parts of the energies are integrable,

$$\mathcal{F}^{\mathrm{ext}}_o(Q) = D_{\mathrm{KL}}\big(Q_{-B} \| \mathbb{P}^{\mathsf{Y}}_{\theta,V,-B}\big) + \mathbb{E}_{Q_{-B}}\big[H_{\bar{B},o}(Y_{-B}) + \mathcal{F}^{\mathrm{ext}}_{B,o}(r_B; Y_{-B})\big]. \tag{5.52}$$

Every exact local update is thus a coordinate update of the collective free energy. The fixed outside marginal is load bearing. [ESTABLISHED]

Proof. Apply (5.46) to both laws. The finite divergences make both conditional terms integrable, the outside terms agree, the constant $-\log p_{\theta,V}(o \mid X)$ cancels, and so does $-\log Z_B(Y_{-B})$ inside the two local functionals. The decomposition follows from (5.47), (5.45), and (5.19). □

Corollary 5.24 (Exact potential game). Let the blocks be singletons and choose a nonempty strategy set $\mathcal{R}_i$ for each agent. Suppose every profile $(r_i) \in \prod_{i \in V} \mathcal{R}_i$ defines a product recognition law $Q = \bigotimes_{i \in V} r_i$ with $D_{\mathrm{KL}}(Q \| \mathbf{\Pi}_{\theta,V,o,X}) < \infty$ and $\mathbb{E}_{Q_{-i}} |\log Z_{\{i\}}(Y_{-i})| < \infty$ for every i, where $Q_{-i} = \bigotimes_{j \neq i} r_j$. The outside-averaged local free energy

$$\overline{\mathcal{F}}_i(r_i; r_{-i}) := \mathbb{E}_{Q_{-i}} \mathcal{F}^{\mathrm{ext}}_{\{i\},o}(r_i; Y_{-i}) \tag{5.53}$$

is finite, and between two such profiles that differ only in r_i it changes by exactly the change of $\mathcal{F}^{\mathrm{ext}}_o$. The players V, strategy sets $\mathcal{R}_i$, and costs $\overline{\mathcal{F}}_i$ therefore form an exact potential game with potential $\mathcal{F}^{\mathrm{ext}}_o$ in the sense of Monderer and Shapley [79, Sec. 2], written for costs instead of payoffs. [ESTABLISHED]

Proof. Finite divergence gives $Q_{-i} \ll \mathbf{\Pi}_{\theta,V,o,X,-i}$ by data processing, and (5.46) makes the averaged conditional divergence finite; with the hypothesis on $\log Z_{\{i\}}$ this makes (5.53) finite. A unilateral change leaves Q_{-i} fixed, and theorem 5.23 with $B = \{i\}$ gives the equality of changes. □

Figure 5.1 shows which records enter a block potential; block-gradient dynamics are in section 8.1.

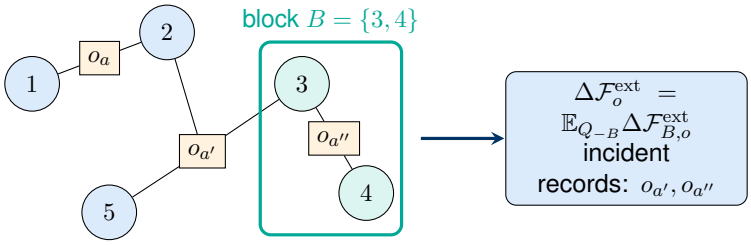


Figure 5.1: A block-local free energy contains every record incident to the block: $o_{a'}$ crosses the boundary and enters the block conditional, while o_a lies outside and cancels.

5.8 Additive accounting

The local free energies of definition 5.21 are coordinate potentials of the collective free energy, not shares of it; an additive single-count ledger is a different construction.

Corollary 5.25 (Correlated ledger and incident overcounting). Suppose $\mathbb{P}_{\theta,V}^{\mathsf{Y}} = \bigotimes_{i\in V} \varrho_i$ and let Q be any law on the latent space. Then (i) in $[0, +\infty]$,

$$D_{\mathrm{KL}}\big(Q\|\mathbb{P}_{\theta,V}^{\mathsf{Y}}\big) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\varrho_i); \tag{5.54}$$

(ii) whenever (5.19) holds,

$$\mathcal{F}_o^{\mathrm{ext}}(Q) = \mathrm{TC}(Q) + \sum_i D_{\mathrm{KL}}(Q_i\|\varrho_i) + \mathbb{E}_Q\Big[\sum_{a\in\mathfrak{A}} E_{a,o}(Y_{\partial a})\Big], \tag{5.55}$$

in which each record appears once and $\mathrm{TC}(Q) = 0$ exactly when Q is a product law, so mean field is a special case of the ledger and not a premise of it; (iii) if every record has a nonempty scope, the singleton incident energies satisfy, pointwise in $(-\infty, +\infty]$,

$$H_{\{i\},o}(y_i; y_{-i}) = \sum_{a: i\in\partial a} E_{a,o}(y_{\partial a}), \qquad \sum_{i\in V} H_{\{i\},o}(y_i; y_{-i}) = \sum_{a\in\mathfrak{A}} |\partial a| E_{a,o}(y_{\partial a}), \tag{5.56}$$

so the singleton local free energies count each record $|\partial a|$ times and are not the single-count ledger; (iv) this is a statement about these local free energies, not a bar to additive decompositions: iterating lemma 3.12 along any ordering of V writes $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X})$ as one term per agent, each conditioned on the agents before it. [ESTABLISHED]

Proof. Part (i) is corollary 5.1 with $\varrho_j = \varrho_i$, and (ii) substitutes it into (5.19). For (iii), every $E_{a,o}$ lies in $(-\infty, +\infty]$ and both index sets are finite, so the rearrangement is valid pointwise. Part (iv) is repeated application of the chain rule, peeling one agent at a time. □

The same chain rule accounts for link values once they are latent; the interaction multigraph itself stays structural (section 1.4).

Proposition 5.26 (Collective free energy with random link values). Let the link family $\Theta = (\Theta_e^x)$ on the edge copies of $\mathfrak{I}$ be a latent variable with values in a standard Borel space $\mathscr{T}$. Let $\nu^{\mathscr{T}}$ be a probability law on $\mathscr{T}$, let $\mathscr{S} : \mathscr{T} \to (-\infty, +\infty]$ be a measurable action with $0 < Z_{\mathscr{S}} = \int e^{-\mathscr{S}} d\nu^{\mathscr{T}} < \infty$, and let the link prior be $Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}} \nu^{\mathscr{T}}$. For each Θ let the record model of hypothesis 5.5 with links Θ have evidence $p(o \mid \Theta) \in (0, \infty)$, measurable in Θ, posterior $\mathbf{\Pi}_\Theta$, and free energy $\mathcal{F}_{\mathrm{matter}}(Q_\Theta; o, \Theta) = -\log p(o \mid \Theta) + D_{\mathrm{KL}}(Q_\Theta \| \mathbf{\Pi}_\Theta)$, and assume the joint evidence $p(o) = \int p(o \mid \Theta) Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}(\Theta)} \nu^{\mathscr{T}}(d\Theta)$ is finite. For a recognition law $Q(d\Theta, dy) = R(d\Theta) Q_\Theta(dy)$ with $D_{\mathrm{KL}}(R \| \nu^{\mathscr{T}}) < \infty$ and $\mathbb{E}_R |\mathscr{S}| < \infty$, the extended free energy of the joint model is

$$\mathcal{F} = \mathbb{E}_R \mathcal{F}_{\mathrm{matter}}(Q_\Theta; o, \Theta) + \mathbb{E}_R \mathscr{S} + D_{\mathrm{KL}}(R \| \nu^{\mathscr{T}}) + \log Z_{\mathscr{S}} \quad \text{in } (-\infty, +\infty]. \tag{5.57}$$

Omitting the link entropy term or the normalizer changes the free energy; when the links are structural parameters instead of latent variables, only $\mathcal{F}_{\mathrm{matter}}$ appears. [ESTABLISHED]

Proof. The posterior of the joint model has conditional law $\mathbf{\Pi}_\Theta$ given Θ and link marginal $\mathbf{\Pi}^{\mathscr{T}}(d\Theta) = p(o)^{-1} p(o \mid \Theta) Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}(\Theta)} \nu^{\mathscr{T}}(d\Theta)$, so the chain rule (3.20) gives $D_{\mathrm{KL}}(Q \| \mathbf{\Pi}) = D_{\mathrm{KL}}(R \| \mathbf{\Pi}^{\mathscr{T}}) + \mathbb{E}_R D_{\mathrm{KL}}(Q_\Theta \| \mathbf{\Pi}_\Theta)$ in $[0, +\infty]$, with a measurable integrand. Since $\mathbb{E}_R |\mathscr{S}| < \infty$, R charges no set where $\mathscr{S} = +\infty$, so $R \ll \mathbf{\Pi}^{\mathscr{T}}$ and $\log(dR/d\mathbf{\Pi}^{\mathscr{T}}) = \log(dR/d\nu^{\mathscr{T}}) + \mathscr{S} + \log Z_{\mathscr{S}} - \log p(o \mid \Theta) + \log p(o)$. The first two terms are R-integrable, and so is the positive part f of $\log p(o \mid \cdot)$: for $f_n = \min(f, n)$, the nonnegative divergence from the law proportional to e^{f_n} times the prior gives $\mathbb{E}_R f_n \le D_{\mathrm{KL}}(R \| Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}} \nu^{\mathscr{T}}) + \log(1 + p(o))$, and that divergence equals $D_{\mathrm{KL}}(R \| \nu^{\mathscr{T}}) + \mathbb{E}_R \mathscr{S} + \log Z_{\mathscr{S}}$. Integrating term by term and adding the conditional term, whose integrand is nonnegative, turns $\mathcal{F} = -\log p(o) + D_{\mathrm{KL}}(Q \| \mathbf{\Pi})$ into (5.57); every summand has an integrable negative part, so no difference of infinities occurs. □

Chapter 9 supplies, for every Lie group G, data that satisfy the hypotheses of this proposition: the latent variable is the family of Lie-algebra generators of the links, the base law is Gaussian and the action a nonnegative combination of loop defects, the joint has unit mass on every record hypergraph (Proposition 9.16), and for a Gaussian recognition law of the generators the link sector of (5.57) is the relative entropy from the generator prior (Proposition 9.18).

5.9 Model coordinates and accepted updates

In the two-argument form $\mathcal{F}_V(\mathbb{Q}; \theta)$ of hypothesis 4.3, block coordinates move $\mathbb{Q}$ at fixed θ and a model coordinate moves θ at fixed $\mathbb{Q}$; each accepted value defines a new fixed joint.

Proposition 5.27 (Exact M-coordinate). Hold $\mathbb{Q}_{V,o,X}$ fixed, and suppose the parameterized family supplies a common regular-observation set or jointly measurable pointwise versions for the candidates compared, with $\mathbb{E}_{\mathbb{Q}_{V,o,X}} |\log q_{V,o,X}(Y)| < \infty$ and $\mathbb{E}_{\mathbb{Q}_{V,o,X}} |\log p_{\theta',V}(o, Y \mid X)| < \infty$ for every candidate θ'. If a maximizer exists in the declared parameter space, then every

$$\theta^{\mathrm{new}} \in \operatorname*{argmax}_{\theta'} \mathbb{E}_{\mathbb{Q}_{V,o,X}} \left[\log p_{\theta',V}(o, Y \mid X) \right] \tag{5.58}$$

maximizes $\mathcal{L}_{V,\theta'}(\mathbb{Q}_{V,o,X}; X, o)$ over the same space. The finiteness hypotheses cannot be dropped. [ESTABLISHED]

Proof. The recognition term $\mathbb{E}_{\mathbb{Q}_{V,o,X}}[\log q_{V,o,X}]$ is finite and does not depend on θ'. For the last sentence, on $\{2, 3, \ldots\}$ let $q_n = \mathrm{c}/(n(\log n)^2)$ with c normalizing, $h_n = \mathbf{1}_{\{n=2\}} - q_2$, and $r_n = 1 + \mathrm{c}_2 h_n$ with $0 < \mathrm{c}_2 < 1/q_2$, so that r is bounded, strictly positive, nonconstant, and $\sum_n q_n r_n = 1$. With a trivial record, let the two candidates have latent laws $p_0 = q$ and $p_1 = qr$, and let the recognition law be q. The entropy of q is infinite (appendix A.2), so $\mathbb{E}_q \log p_0 = \mathbb{E}_q \log p_1 = -\infty$ and both candidates maximize the expected complete log-likelihood, whereas their extended bounds are $\mathbb{E}_q \log(p_0/q) = 0$ and $\mathbb{E}_q \log r = -D_{\mathrm{KL}}(q \| p_1) < 0$. □

Definition 5.28 (Generalized M-coordinate acceptance). A generalized M-coordinate need not attain a maximum, but it must pass the acceptance condition

$$\mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}_{V,o,X}; X, o) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}_{V,o,X}; X, o), \tag{5.59}$$

evaluated at an observation in the regular sets of both candidates, or with jointly measurable versions declared as part of the parameterized model. No recognition density and no (H4) is required; on the classical domain the condition is the same inequality for (5.20), and (5.58) is a sufficient condition.

Corollary 5.29 (Evidence-free acceptance test). Suppose $\mathbb{Q}_{V,o,X} \ll M_{\theta^{\mathrm{old}},V,o,X}$, the two slice measures (5.14) are mutually absolutely continuous, and both extended values are finite. Then

$$\mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}_{V,o,X}; X, o) - \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}_{V,o,X}; X, o) = \mathbb{E}_{\mathbb{Q}_{V,o,X}}\left[\log \frac{p_{\theta^{\mathrm{new}},V}(o, Y \mid X)}{p_{\theta^{\mathrm{old}},V}(o, Y \mid X)}\right], \tag{5.60}$$

so the test (5.59) needs no evidence value. [ESTABLISHED]

Proof. With $r_x = d\mathbb{Q}_{V,o,X}/dM_{\theta^x,V,o,X}$ for $x \in \{\mathrm{old}, \mathrm{new}\}$, the Radon–Nikodym chain rule gives $\log r_{\mathrm{old}} - \log r_{\mathrm{new}} = \log(dM_{\theta^{\mathrm{new}},V,o,X}/dM_{\theta^{\mathrm{old}},V,o,X})$ almost surely, and this derivative is the ratio of the two joint densities at o. By proposition A.6 the negative part of each $\log r_x$ is integrable, so finiteness makes each $\log r_x$ integrable, and subtracting the two representations (A.4) proves the identity. □

Proposition 5.30 (Evidence monotonicity). Suppose (H1)–(H2) hold for the fixed $\mathbb{Q}_{V,o,X}$ at both θ^{old} and θ^{new}. If the E-coordinate is exact at the old parameter, $\mathbb{Q}_{V,o,X} = \mathbf{\Pi}_{\theta^{\mathrm{old}},V,o,X}$, and the M-coordinate satisfies (5.59), then

$$\log p_{\theta^{\mathrm{new}},V}(o \mid X) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{new}}}(\mathbb{Q}_{V,o,X}; X, o) \geq \mathcal{L}^{\mathrm{ext}}_{V,\theta^{\mathrm{old}}}(\mathbb{Q}_{V,o,X}; X, o) = \log p_{\theta^{\mathrm{old}},V}(o \mid X). \tag{5.61}$$

Consequently a schedule that places a full exact E phase before each accepted M phase has nondecreasing evidence values, which converge in $(-\infty, +\infty]$ and to a finite limit when bounded above. Block E-coordinates and accepted M-coordinates make the compared bound values

nondecreasing on a common functional; the transfer to the evidence in (5.61) uses the full E phase. [ESTABLISHED]

Proof. The first inequality is corollary 5.7 at θ^{new}, the second is (5.59), and the equality is the zero-gap case at θ^{old}; the singular-branch sum $\mathcal{L}^{\mathrm{ext}} + D_{\mathrm{KL}}$ is never written. A nondecreasing sequence of extended reals converges. Each block coordinate does not decrease the bound by corollary 5.22, and each accepted step does not decrease it by (5.59); only the final equality of (5.61) uses the exact E phase. Under (H3)–(H4) at both candidates this is the classical conclusion of Dempster et al. [32] in the reading of Neal and Hinton [81]. □

Convergence of the parameter sequence needs regularity conditions of the kind identified for the classical algorithm by Wu [108]. A finite step along a natural-gradient or other optimizer direction is a proposal, not an accepted coordinate, and it can decrease the bound.

Proposition 5.31 (Finite natural-gradient steps need not be monotone). Let $\mathsf{G} \succ 0$, write $\nabla\mathcal{L}_V$ for the gradient of $\theta' \mapsto \mathcal{L}_{V,\theta'}(\mathbb{Q}_{V,o,X}; X, o)$, and consider the step $\theta^+ = \theta' + \mathrm{h}\mathsf{G}^{-1}\nabla\mathcal{L}_V$ of length $\mathrm{h} > 0$. At a nonstationary point the direction is an ascent direction,

$$\frac{d}{d\mathrm{h}}\mathcal{L}_{V,\theta^+}\Big|_{\mathrm{h}=0} = \nabla\mathcal{L}_V^{\top}\mathsf{G}^{-1}\nabla\mathcal{L}_V > 0. \tag{5.62}$$

If $\mathcal{L}_{V,\theta''} = \mathcal{L}^\star - \frac{1}{2}(\theta'' - \theta^\star)^{\top}\mathsf{H}(\theta'' - \theta^\star)$ with $\mathsf{H} \succ 0$ for θ'' in a neighborhood $\mathcal{U}$, $d_{\max}$ is the largest generalized eigenvalue of (H, G), and $\mathbf{z}$ is a corresponding eigenvector with $\mathbf{z}^{\top}\mathsf{G}\mathbf{z} = 1$, then at every $\theta' = \theta^\star + \mathrm{c}_1\mathbf{z} \in \mathcal{U}$ with $\mathrm{c}_1 \neq 0$ and every step whose endpoint lies in $\mathcal{U}$,

$$\mathcal{L}_{V,\theta^+} - \mathcal{L}_{V,\theta'} = \tfrac{1}{2}d_{\max}\mathrm{c}_1^2\big[1 - (1 - \mathrm{h}d_{\max})^2\big] < 0 \qquad \text{whenever} \qquad \mathrm{h} > \frac{2}{d_{\max}}. \tag{5.63}$$

[ESTABLISHED]

Proof. The first display is the chain rule. With $\Delta\theta = \theta' - \theta^\star = \mathrm{c}_1\mathbf{z}$, $\nabla\mathcal{L}_V = -\mathsf{H}\Delta\theta$ and $\mathsf{H}\mathbf{z} = d_{\max}\mathsf{G}\mathbf{z}$ give $\theta^+ - \theta^\star = (1 - \mathrm{h}d_{\max})\mathrm{c}_1\mathbf{z}$; substituting in the quadratic gives the second display, whose bracket is negative exactly when $\mathrm{h}d_{\max} > 2$. □

Monotonicity at a finite step therefore needs a line search, a trust region, or another rule that verifies (5.59) on the same functional; a better preconditioner does not remove the requirement, because (5.63) is stated in the preconditioned spectrum.

5.10 What remains open

Open Problem 5.32 (Convergence of the alternating scheme). Proposition 5.30 gives monotone convergence of the evidence values under a full exact E phase and nothing about the parameters, and since (5.59) is a weak inequality the null step is an accepted M-coordinate, so the general scheme of block E-coordinates (5.48) and accepted M-coordinates need not move its parameter at all. The problem concerns models of hypothesis 5.5 with finitely many latent states, parameters in an open cone of a Euclidean space, and joint densities twice continuously

differentiable in the parameters, under the scheme that alternates a full exact E phase with the exact M-coordinate (5.58) whose maximizer is unique. The regularity conditions of Wu [108] concern the limit points of the parameter sequence and its convergence when the stationary points at its limit value are isolated; isolation is not among the hypotheses of the class, in which uniqueness of the maximizer leaves open that the evidence is constant along a curve of parameters on which the complete-data likelihood is not. It is settled by conditions checkable on the model, including compactness of the upper level sets of the evidence in the cone, that are met by some model of the class with parameter dimension at least two whose stationary set at the limit value of an orbit is not discrete, under which every orbit of the scheme converges to one stationary point of the extended bound, or by a model of that class with compact upper level sets whose orbit has more than one limit point.

Physicist's summary

The population carries one generative law and one recognition law, and the only evidence bound is $\log p(o) - D_{\mathrm{KL}}(Q\|\mathbf{\Pi})$, defined for every Q and equal to $-\infty$ when Q charges what the posterior cannot see; everything else is bookkeeping of that relative entropy through the chain rule. Substituting marginal entropies overstates the bound by exactly the total correlation. Observations are records attached to latent states, and an environment can be rewritten as boundary nodes emitting them. Each block of agents has a local free energy built from the records that touch it; with the rest held fixed, lowering it lowers the collective free energy by the same amount, so the population is an exact potential game. Summed, the local free energies count every shared record once per participant. Parameters move only between evaluations, by accepted steps, and the evidence rises monotonically when each step follows an exact posterior refresh; convergence is open.

Chapter 6

Attention and Selective Communication

Which agent does a receiver listen to, and what does it hear? Attention enters the joint of chapter 4 as a latent source label: the softmax row is its generative posterior at a latent state, and the optimal recognition row held constant in the latent state is a different softmax, over averaged energies (proposition 6.2), at the exact cost of proposition 6.3. Communication enters as normalized encoder, transmission, and decoder kernels, and the exact receiver update of a comparison record fixes the value map by the query and key maps (proposition 6.19). The row calculus and the reduced curvature of proposition 6.8 feed chapter 8, the event law of definition 6.4 is the attention state that chapter 12 coarse-grains, and the comparison records supply the endpoint energies of chapter 11. Rows, events, links, and divergences are indexed by oriented edge copies $e : j \to i$; the pair ij names a unique copy, a pair quantity is the sum over parallel copies, and the construction serves both channels, with a model-channel row written β^m where the channel must be named. Longer proofs are deferred to appendix B.

This chapter does not claim that transformer attention as a whole is derived (table 6.1; feed-forward blocks and layer normalization have no counterpart), that declared per-head rows follow from a label model, or that an interface map, its rank, or its coding establishes a channel capacity, a reconstruction of the sender's belief, or shared meaning. Capacity needs an admissible input family and, for a nonzero continuous Gaussian channel, an input resource constraint; a coding theorem also needs repeated uses, errors, and rates [92]; a task-dependent distortion or sufficient statistic is further model data. Open are the label-level multi-head model (open problem 6.22), discrete creation and deletion of interaction edges (open problem 6.9), and formation with evolving selective interfaces (open problem 6.27). [NOT-CLAIMED]

6.1 Attention as a latent source label

Definition 6.1 (Per-receiver source-label model). Let $\mathcal{I}_{\mathrm{att}}$ be a finite set of receiving rows, each an agent $i \in V$ or an agent–head pair (i, h) that inherits its agent's latent state (written i when no head is needed). Row i carries a source label J_i valued in the incoming edge copies of i, with a structural prior row π_i independent of recognition whose support is the source set $\mathcal{N}_i$; source sums and indices range over $\mathcal{N}_i$. These supports determine the interaction multigraph at c_* as in section 1.4, and whether $\mathcal{N}_i$ contains the self edge is part of the model. The labels augment the

latent baseline independently,

$$\mathbb{P}^{Y,\mathrm{aug}}_{\theta,V}(dy, dj) = \mathbb{P}^{Y}_{\theta,V}(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \pi_i(dj_i). \tag{6.1}$$

Each row has a temperature $\tau_i > 0$ and finite measurable energies $D_{ij} : \mathsf{Y} \to \mathbb{R}$ that contain no recognition law and are gauge-invariant functions of sample-level latent states (definition 3.19) and of the links, and at the selected record row i has the record density

$$L_i(o_i \mid y, J_i = j) = \mathrm{c}_i(o_i, y) \exp[-D_{ij}(y)/\tau_i], \tag{6.2}$$

with c_i independent of j; the kernel off the selected slice is part of the model. When $\mathrm{c}_i(o_i, \cdot)$ depends only on y_i and each D_{ij} only on (y_i, y_j), the record is one hyperedge factor of hypothesis 5.5 with scope $\{i\} \cup \mathcal{N}_i$, and through (5.36) it enters the local free energy of the receiver and of every candidate source. Each row carries exactly one label, so receiver occupancy is fixed; inferring it needs a null label in the source alphabet or a different joint label model with its own likelihood, such as definition 6.5.

The slice form (6.2) is inhabited: a binary success record with success probability $\exp[-D_{ij}/\tau_i]$ and every $D_{ij} \geq 0$ realizes it with $\mathrm{c}_i = 1$, as in (5.10), and so does the comparison record of proposition 6.19 at every record value.

Proposition 6.2 (Exact posterior and recognition attention). Assume definition 6.1 and label exclusivity at the complete selected record o: the augmented likelihood factors as

$$L^{\mathrm{aug}}_o(y, j) = L^{Y}_o(y) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \mathrm{c}_i(o_i, y) \exp[-D_{ij_i}(y)/\tau_i], \tag{6.3}$$

with $L^{Y}_o : \mathsf{Y} \to [0, +\infty)$ finite, measurable, and free of every label, and no other record or factor of the generative law reads any J_i. (a) On $\{L^{Y}_o \prod_k \mathrm{c}_k(o_k, \cdot) > 0\}$ the generative posterior row given y and the record is

$$\beta^{P}_{ij}(y) = \pi_{ij} e^{-D_{ij}(y)/\tau_i} \Big/ \sum_k \pi_{ik} e^{-D_{ik}(y)/\tau_i}, \tag{6.4}$$

a version that may also be chosen off that set; it is strictly positive exactly on $\mathcal{N}_i$, and if row i has nonnegative energies, an included self edge with $D_{ii} = 0$, and equal prior weights, no source has a larger posterior weight than the self edge. (b) Restrict the augmented recognition law to the full factorization

$$\mathbb{Q}(dy, dj) = \mathbb{Q}_Y(dy) \prod_{i \in \mathcal{I}_{\mathrm{att}}} \beta^{Q}_i(dj_i), \qquad \beta^{Q}_i(dj_i) \text{ independent of } y, \tag{6.5}$$

which makes the labels conditionally independent and every row independent of the latent state, and assume $\mathbb{E}_{\mathbb{Q}_Y} |D_{ij}| < +\infty$ for every source and $\mathbb{E}_{\mathbb{Q}_Y} |\log \mathrm{c}_i(o_i, Y)| < +\infty$. Then row i contributes

(6.6) to the collective VFE, whose first term does not depend on β_i^Q, with (6.7):

$$-\mathbb{E}_{\mathbb{Q}_Y}\log \mathrm{c}_i(o_i,Y)+\mathcal{F}_i^{\mathrm{att}}(\beta_i^Q), \tag{6.6}$$

$$\mathcal{F}_i^{\mathrm{att}}(\beta_i^Q)=D_{\mathrm{KL}}(\beta_i^Q\|\pi_i)+\frac{1}{\tau_i}\sum_j \beta_{ij}^Q\mathbb{E}_{\mathbb{Q}_Y}D_{ij}. \tag{6.7}$$

Its unique minimizer on the open simplex over $\mathcal{N}_i$ is

$$\beta_{ij}^{Q\star}=\pi_{ij}e^{-\mathbb{E}_{\mathbb{Q}_Y}D_{ij}/\tau_i}\Big/\sum_k \pi_{ik}e^{-\mathbb{E}_{\mathbb{Q}_Y}D_{ik}/\tau_i}, \tag{6.8}$$

and the row functional $\sum_j \beta_{ij}\mathbb{E}_{\mathbb{Q}_Y}D_{ij}+\tau_i D_{\mathrm{KL}}(\beta_i\|\pi_i)$ equals $\tau_i\mathcal{F}_i^{\mathrm{att}}$ with the same minimizer, while the collective VFE contains $\mathcal{F}_i^{\mathrm{att}}$ itself. (c) For an arbitrary recognition law $\mathbb{Q}_Y(dy)\mathbb{Q}_{J|Y}(dj\mid y)$ with row marginals $\mathbb{Q}_{J_i|Y}$, and with $-\mathbb{E}_{\mathbb{Q}_Y}\log L_o^Y(Y)$ counted in the state sector, the attention-label contribution is

$$\begin{aligned}&-\sum_{i\in\mathcal{I}_{\mathrm{att}}}\mathbb{E}_{\mathbb{Q}_Y}\log \mathrm{c}_i(o_i,Y)+\mathbb{E}_{\mathbb{Q}_Y}\,\mathrm{TC}(\mathbb{Q}_{J|Y})\\&+\sum_{i\in\mathcal{I}_{\mathrm{att}}}\mathbb{E}_{\mathbb{Q}_Y}\left[D_{\mathrm{KL}}(\mathbb{Q}_{J_i|Y}\|\pi_i)+\frac{1}{\tau_i}\mathbb{E}_{\mathbb{Q}_{J_i|Y}}D_{iJ_i}(Y)\right],\end{aligned} \tag{6.9}$$

where $\mathrm{TC}(\mathbb{Q}_{J|Y=y})=D_{\mathrm{KL}}\big(\mathbb{Q}_{J|Y=y}\|\prod_i\mathbb{Q}_{J_i|Y=y}\big)$. Its minimum over unrestricted conditional laws is attained by the product of the posterior rows (6.4), and row marginals independent of y do not by themselves give (6.7), since a correlated conditional law keeps the nonnegative total-correlation term. [ESTABLISHED]

Proposition 6.3 (Cost of the constant-row restriction). Assume the label exclusivity of proposition 6.2, fix $\mathbb{Q}_Y$ with $\mathbb{E}_{\mathbb{Q}_Y}|D_{ij}|<+\infty$ for every row i and $j\in\mathcal{N}_i$, write $d_{ij}=\mathbb{E}_{\mathbb{Q}_Y}D_{ij}$, and define

$$R_i^Q:=-\log\sum_j \pi_{ij}e^{-d_{ij}/\tau_i},\qquad R_i^P(y):=-\log\sum_j \pi_{ij}e^{-D_{ij}(y)/\tau_i}. \tag{6.10}$$

Over constant rows (6.5) the minimum of $\mathcal{F}_i^{\mathrm{att}}$ is R_i^Q, attained at (6.8); over all conditional label laws $\mathbb{Q}_{J|Y}$ the minimum of the label terms of (6.9) other than $-\sum_i\mathbb{E}_{\mathbb{Q}_Y}\log \mathrm{c}_i(o_i,Y)$ is $\sum_i\mathbb{E}_{\mathbb{Q}_Y}R_i^P(Y)$, attained by the product of the posterior rows (6.4). For each row the difference is exactly

$$R_i^Q-\mathbb{E}_{\mathbb{Q}_Y}R_i^P(Y)=\mathbb{E}_{\mathbb{Q}_Y}D_{\mathrm{KL}}\big(\beta_i^{Q\star}\|\beta_i^P(Y)\big)\geq 0, \tag{6.11}$$

which vanishes exactly when every difference $D_{ij}(Y)-D_{ik}(Y)$ is $\mathbb{Q}_Y$-almost surely constant. If further $D_{ij}=\widetilde{D}_{ij}\circ\psi_i$ for one measurable map $\psi_i:\mathsf{Y}\to\mathsf{Z}_i$ into a finite-dimensional real vector space with finite $\mathbb{Q}_Y$-mean $\bar{\psi}_i$, and every $\widetilde{D}_{ij}-\widetilde{D}_{ik}$ is affine, the recognition row is the posterior row at the recognition mean,

$$\beta_{ij}^{Q\star}=\pi_{ij}e^{-\widetilde{D}_{ij}(\bar{\psi}_i)/\tau_i}\Big/\sum_k \pi_{ik}e^{-\widetilde{D}_{ik}(\bar{\psi}_i)/\tau_i}. \tag{6.12}$$

In a linear sample fiber, the energies $\frac{1}{2}(y_i-\mathring{y}_{ij})^\top W_i(y_i-\mathring{y}_{ij})$ with $W_i \succ 0$ and structural templates $\mathring{y}_{ij}$ have affine differences. If instead $D_{ij}(y) = \frac{1}{2}r_{ij}^\top W_i r_{ij}$ with $r_{ij} = y_i - \rho_b(\Theta_{ij}^b)y_j$ for a unique edge copy, and Y_i and Y_j are uncorrelated under $\mathbb{Q}_Y$ with finite second moments, then $\beta_i^{Q\star}$ is the posterior row at the recognition mean with the logit of source j lowered by $\operatorname{Tr}(W_i\rho_b(\Theta_{ij}^b)\operatorname{Cov}(Y_j)\rho_b(\Theta_{ij}^b)^\top)/(2\tau_i)$. [ESTABLISHED]

The proofs of propositions 6.2 and 6.3 are in appendix B.1. Coarse-graining a row, a conditional law of a source given a receiver, needs a receiver law, which the next two definitions supply.

Definition 6.4 (Receiver occupancy and typed event law). For a role mark $x \in \{P, \Pi, Q\}$, a receiver occupancy α_i^x, a probability on the rows, and a row β_i^x on $\mathcal{N}_i$ give the marked edge-event law

$$\eta_{ij}^x = \alpha_i^x \beta_{ij}^x, \qquad \sum_{i,j} \eta_{ij}^x = 1. \tag{6.13}$$

The mark names the law: P is the generative joint conditioned on the latent state and the record, so η^P, α^P, and β^P are functions of y; Π is the posterior slice, with $\eta_{ij}^\Pi = \mathbb{E}_\Pi \eta_{ij}^P(Y)$; and Q is the selected recognition coupling. The event law carries no head index. The occupancy is not supplied by definition 6.1, in which every row carries its own label, and it enters in one of two typed ways, each use naming which: a structural receiver-selection law, fixed with the coarse channel before observation and independent of recognition, or the receiver marginal of a single joint edge label as in definition 6.5, with the label prior as its prior and its posterior from Bayes' formula. On $\{\alpha_i^x > 0\}$ the row is recovered by disintegration, $\beta_{ij}^x = \eta_{ij}^x/\alpha_i^x$; a conditional row alone is not recursive state.

Definition 6.5 (Single-edge joint-label attention model). This generative model differs from definition 6.1: the baseline is augmented by one joint edge label (i, j), a single receiver–source pair for the selected record, with recognition-independent prior

$$\mathbb{P}_{\theta,V}^{\mathrm{att}}(dy, di, dj) = a_i \pi_{ij} \mathbb{P}_{\theta,V}^Y(dy), \qquad \sum_i a_i = 1, \quad \sum_j \pi_{ij} = 1, \tag{6.14}$$

and the augmented likelihood at the complete selected record o factors as

$$L_o^{\mathrm{edge}}(y, i, j) = L_o^Y(y)\widetilde{\mathrm{c}}_i(o, y)\exp[-D_{ij}(y)/\tau_i], \qquad \tau_i > 0, \tag{6.15}$$

with L_o^Y and $\widetilde{\mathrm{c}}_i$ finite, nonnegative, and measurable, L_o^Y free of (i, j), $\widetilde{\mathrm{c}}_i$ free of j, each D_{ij} finite and measurable, and no other record or factor of the generative law reading the label; this exclusivity excludes every model in which several receivers attend at once. In general $\widetilde{\mathrm{c}}_i$ is not the c_i of (6.2): if the selected receiver's record has slice density $\mathrm{c}_i(o_i, y)\exp[-D_{ij}(y)/\tau_i]$ while every other receiver's record keeps a label-free kernel $g_k(o_k \mid y)$, then where every $g_k(o_k \mid y) > 0$ the factorization holds with $L_o^Y = \prod_k g_k(o_k \mid y)$ and $\widetilde{\mathrm{c}}_i = \mathrm{c}_i/g_i(o_i \mid y)$, a likelihood ratio against the non-event kernel. The evidence-weighted occupancy of this model is proposition 12.32; the per-receiver model collapses with its declared selection law instead (proposition 12.33).

Corollary 6.6 (Self edges and blocking). Let $N \geq 2$ agents carry one row each with uniform receiver occupancy $\alpha_i = 1/N$, partition them into blocks of equal size n dividing N, and push an event law with rows independent of the latent state along the partition by block sums, $\eta^{\mathrm{c}}_{IJ} = \sum_{i\in I, j\in J} \eta_{ij}$, with $\alpha^{\mathrm{c}}_I = \sum_J \eta^{\mathrm{c}}_{IJ}$ and $\beta^{\mathrm{c}}_{IJ} = \eta^{\mathrm{c}}_{IJ}/\alpha^{\mathrm{c}}_I$. If every fine row is uniform over $j \neq i$, the block event masses are $n^2/(N(N-1))$ from I to $J \neq I$ and $n(n-1)/(N(N-1))$ from I to itself, the block occupancy is n/N, and

$$\beta^{\mathrm{c}}_{II} = \frac{n-1}{N-1}, \qquad \beta^{\mathrm{c}}_{IJ} = \frac{n}{N-1} \quad (J \neq I). \tag{6.16}$$

If every fine row is uniform over all N sources, including the self edge, every coarse row is uniform with weight n/N. A fine no-self convention is therefore not preserved: the coarse source set of I contains I. [ESTABLISHED]

Proof. Without self edges each fine event has mass $1/(N(N-1))$, a block pair $I \neq J$ contains n^2 of them, $I \times I$ contains $n(n-1)$, and the block occupancy is n/N; with self edges each event has mass $1/N^2$ and every block pair contains n^2 events. Dividing gives both cases. □

Corollary 6.7 (Row flows, dissipation, and parameter paths). Hold $\mathbb{Q}_Y$, and hence every $\mathbb{E}_{\mathbb{Q}_Y} D_{ij}$, fixed. With row relaxation rate $\mathsf{r}_i \geq 0$, the Fisher natural gradient of (6.7) on the open simplex over $\mathcal{N}_i$ is the replicator equation (6.17), with partial row-free-energy rate (6.18):

$$\dot\beta^Q_{ij} = -\mathsf{r}_i \beta^Q_{ij} \left(\mathsf{c}_{ij} - \sum_k \beta^Q_{ik} \mathsf{c}_{ik} \right), \qquad \mathsf{c}_{ij} = \log \frac{\beta^Q_{ij}}{\pi_{ij}} + \frac{\mathbb{E}_{\mathbb{Q}_Y} D_{ij}}{\tau_i}, \tag{6.17}$$

$$\frac{d}{dt} \mathcal{F}^{\mathrm{att}}_i \Big|_{\beta_i} = -\mathsf{r}_i \operatorname{Var}_{\beta^Q_i}(\mathsf{c}_{ij}) \leq 0. \tag{6.18}$$

If instead $\mathbb{Q}_{Y,\lambda(t)}$ moves along a C^1 curve λ in an open coordinate domain, with every $d_{ij}(\lambda) = \mathbb{E}_{\mathbb{Q}_{Y,\lambda}} D_{ij}$ of class C^1, and the row follows (6.17) with $d_{ij}(\lambda(t))$ in place of $\mathbb{E}_{\mathbb{Q}_Y} D_{ij}$, then $\mathcal{F}^{\mathrm{att}}_i(\beta_i; \lambda) = D_{\mathrm{KL}}(\beta_i \| \pi_i) + \tau_i^{-1} \sum_j \beta_{ij} d_{ij}(\lambda)$ satisfies

$$\frac{d}{dt} \mathcal{F}^{\mathrm{att}}_i(\beta^Q_i(t); \lambda(t)) = -\mathsf{r}_i \operatorname{Var}_{\beta^Q_i}(\mathsf{c}_{ij}) + \frac{1}{\tau_i} \sum_j \beta^Q_{ij} \nabla_\lambda d_{ij}(\lambda) \cdot \dot\lambda, \tag{6.19}$$

whose second term has no sign in general. Along any differentiable path on $\{\alpha_i > 0\}$ an event law and its row satisfy (6.20), and a softmax row $\beta_j = \pi_j e^{-E_j/\tau} / \sum_k \pi_k e^{-E_k/\tau}$ on a strictly positive prior support, with (π, E, τ) differentiable and $\tau > 0$, satisfies (6.21):

$$\dot\beta_{ij} = \frac{\dot\eta_{ij} - \beta_{ij} \dot\alpha_i}{\alpha_i}, \tag{6.20}$$

$$\dot\beta_j = \beta_j \left(\chi_j - \sum_k \beta_k \chi_k \right), \qquad \chi_j = \partial_t \log \pi_j - \frac{\dot E_j}{\tau} + \frac{E_j \dot\tau}{\tau^2}. \tag{6.21}$$

[ESTABLISHED]

The proof is in appendix B.1. No scale map determines the path in (6.21), and on $\{\alpha_i = 0\}$ the flow variable is η, not the quotient row.

Proposition 6.8 (Attention-reduced potential and its curvature). Fix a row i, an open set U of a finite-dimensional real vector space, and $E_{ij} \in C^2(U)$ for $j \in \mathcal{N}_i$, and define

$$R_i(x) = -\log \sum_j \pi_{ij} e^{-E_{ij}(x)/\tau_i}, \qquad \beta_{ij}(x) = \pi_{ij} e^{-E_{ij}(x)/\tau_i} \Big/ \sum_k \pi_{ik} e^{-E_{ik}(x)/\tau_i}. \tag{6.22}$$

Then $R_i \in C^2(U)$, and

$$\nabla R_i = \frac{1}{\tau_i} \sum_j \beta_{ij} \nabla E_{ij}, \qquad \nabla^2 R_i = \frac{1}{\tau_i} \sum_j \beta_{ij} \nabla^2 E_{ij} - \frac{1}{\tau_i^2} \operatorname{Cov}_{\beta_i}(\nabla E_i), \tag{6.23}$$

where $\operatorname{Cov}_{\beta_i}(\nabla E_i) = \sum_j \beta_{ij} (\nabla E_{ij} - \bar{g})(\nabla E_{ij} - \bar{g})^\top$ and $\bar{g} = \sum_j \beta_{ij} \nabla E_{ij}$. With $\mathscr{J}_i(x, \beta) = D_{\mathrm{KL}}(\beta \| \pi_i) + \tau_i^{-1} \sum_j \beta_j E_{ij}(x)$, the gradient is the frozen-weight gradient $\partial_x \mathscr{J}_i(x, \beta_i(x))$, and the Hessian is the Schur complement, over the tangent space $\{w : \sum_j w_j = 0\}$ of the source simplex, of the joint Hessian of $\mathscr{J}_i$ at $(x, \beta_i(x))$. The potential arises under both label treatments: under the constant-row restriction, with $x = \lambda$ the coordinate of a recognition family $\mathbb{Q}_{Y,\lambda}$ whose averaged energies $E_{ij}(\lambda) = \mathbb{E}_{\mathbb{Q}_{Y,\lambda}} D_{ij}$ are C^2, $R_i(\lambda)$ is the row-optimized value R_i^Q of proposition 6.3; under exact label integration, with $x = y$ and $E_{ij} = D_{ij}$ on an open set of latent coordinates where every D_{ij} is C^2, summing over J_i in (6.3) replaces the label factor by $\mathrm{c}_i(o_i, y) e^{-R_i(y)}$, so $R_i = R_i^P$ enters the negative log posterior density of the latent state relative to its baseline. In both cases the correction along a direction v is $-\tau_i^{-2} \operatorname{Var}_{\beta_i}(\nabla E_{ij} \cdot v)$, so positivity of the frozen-weight curvature $\sum_j \beta_{ij} \nabla^2 E_{ij}$ does not imply positivity of $\nabla^2 R_i$, and in the following normalized witness the correction reverses the sign. Let the latent $X \in \mathbb{R}$ have baseline $\mathcal{N}(0, \tau/\omega)$ with $\omega, \tau > 0$, let one row have source alphabet $\{-1, 1\}$ with uniform prior, and let the record $O \in \mathbb{R}$ have kernel $\mathcal{N}(x - j\delta, \tau)$ given $J = j$, with $\delta > 0$. This kernel is $\mathrm{c} \exp[-D_j^o(x)/\tau]$ with $\mathrm{c} = (2\pi\tau)^{-1/2}$ and $D_j^o(x) = (o - x + j\delta)^2/2$, so (6.2) holds at every record value, and the evidence at $o = 0$, $\exp[-\delta^2/(2\tau(1 + \omega^{-1}))]/\sqrt{2\pi\tau(1 + \omega^{-1})}$, is finite and positive. Exact label integration gives the negative log posterior density

$$\frac{(\omega + 1)x^2}{2\tau} - \log\cosh\frac{\delta x}{\tau} + \text{const}, \qquad \text{with second derivative } \frac{1}{\tau}\left(\omega + 1 - \frac{\delta^2}{\tau}\right) \text{ at } x = 0, \tag{6.24}$$

whereas the prior curvature plus the frozen-weight curvature is $(\omega + 1)/\tau$. Under the constant-row restriction with Gaussian recognition $\mathcal{N}(\mathsf{m}, \mathsf{v})$, $\mathsf{v} > 0$, the averaged energies $((\mathsf{m} - j\delta)^2 + \mathsf{v})/2$ have differences affine in m, and the m-curvature of the row-optimized collective VFE at $\mathsf{m} = 0$ is again $(\omega + 1 - \delta^2/\tau)/\tau$. For $(\omega, \delta, \tau) = (1, 2, 1)$ the frozen-weight value is 2 and the reduced value -2, although every source energy and the prior are convex, so the symmetric point is a strict local maximum of the reduced functional along the mean. [ESTABLISHED]

The proof is in appendix B.1.

Open Problem 6.9 (Discrete creation and deletion of interaction edges). At c_* the source sets, and with them the interaction multigraph, belong to the structural data. Every row of this chapter is strictly positive on its source set (proposition 6.2), and the row flows change no support at finite time: along (6.17), $\beta_{ij}^Q(t) = \beta_{ij}^Q(0)\exp[-\int_0^t \mathsf{r}_i(\mathsf{c}_{ij} - \bar{\mathsf{c}}_i)ds]$ with $\bar{\mathsf{c}}_i = \sum_k \beta_{ik}^Q \mathsf{c}_{ik}$ whenever the costs stay finite. Kinetics that change the support lie outside the model of this manuscript, and the problem fixes one rule so that the question is which normalized free energy, if any, such kinetics descend. Let the source sets be sets of copies of a multigraph on V with edge multiplicity at most a fixed bound, let a comparison energy D_{ij} be declared for every candidate copy $j \to i$, present in the source set of i or absent, and let the rows on the current support follow (6.17). The threshold rule deletes a present copy at a rate that is a continuous nonincreasing function of its recognition row weight β_{ij}^Q, positive below a threshold in $(0, 1)$ and zero above it, and creates an absent copy at a rate that is a continuous nonincreasing function of its averaged energy $\mathbb{E}_{\mathbb{Q}_Y} D_{ij}$, positive below a finite threshold and zero above it; at a deletion the receiver's row is restricted to the remaining copies and renormalized, and at a creation the new copy enters the row at the deletion threshold and the row is renormalized. The rule reads the recognition law and the averaged energies and no law over multigraphs. The problem is settled, for some pair of such rate functions, by a normalized joint at c_* in which the source sets are a structural coordinate, declared as a configuration prior or structural channel with a named law and coupled to the link, interface, and label variables, such that the collective VFE decomposition of definition 5.6 holds with the source sets as a coordinate, the event law of definition 6.4 and its pushforward in section 12.7 extend to the random support with the zero-occupancy convention used there, and the recognition law over supports, evolving by the master equation of the rule with the rows on each support following (6.17), has nonincreasing collective VFE from every initial support and recognition law; negatively, by a proof that for every pair of such rate functions no joint of that class has this property.

6.2 Communication as a normalized channel

Definition 6.10 (Directed communication interface). For an oriented edge copy $e : j \to i$, whose independent link is that of definition 2.11, and a channel $x \in \{b, m\}$, let S_j^x be the source sample space. Choose transmitted and received message spaces H_e^- and H_e^+, a record space O_e, receiver side information v in a measurable space, and encoding, transmission, and decoding kernels, in the sense of the communication model of Shannon [92], jointly measurable in v, whose composition is the composed value kernel K_e^v:

$$\mathsf{E}_e^v : \mathsf{S}_j^x \rightsquigarrow \mathsf{H}_e^-, \qquad \mathsf{N}_e^v : \mathsf{H}_e^- \rightsquigarrow \mathsf{H}_e^+, \qquad \mathsf{D}_e^v : \mathsf{H}_e^+ \rightsquigarrow \mathsf{O}_e, \tag{6.25}$$

$$K_e^v(s, do) = \int \mathsf{E}_e^v(s, d\xi^-)\mathsf{N}_e^v(\xi^-, d\xi^+)\mathsf{D}_e^v(\xi^+, do). \tag{6.26}$$

Corollary 6.11 (Normalization and the kernel-level gauge law). (a) K_e^v is a Markov kernel jointly measurable in v, and a source law q_j^x gives the law $q_j^x K_e^v$ on O_e, a candidate law when O_e is a receiver sample space, whose membership in a restricted receiver family requires a separate

closure condition. (b) Let S_j^x be the channel sample fiber on which G acts measurably by ρ_x as in (2.4), O_e the receiver's channel sample fiber, the side information gauge inert, and $\phi_e^{\pm}$ separately chosen bimeasurable recoordinatizations of the message spaces, and write g_i for the old-to-new bijection of the channel-x sample fiber induced at agent i by the vertex re-expression of (2.28). The re-expressed kernels

$$\begin{gathered}\mathsf{E}_e^{v\prime}(s',\cdot)=(\phi_e^-)_{\#}\mathsf{E}_e^v(\mathsf{g}_j^{-1}s',\cdot),\qquad \mathsf{N}_e^{v\prime}(\xi',\cdot)=(\phi_e^+)_{\#}\mathsf{N}_e^v((\phi_e^-)^{-1}\xi',\cdot),\\ \mathsf{D}_e^{v\prime}(\xi',\cdot)=(\mathsf{g}_i)_{\#}\mathsf{D}_e^v((\phi_e^+)^{-1}\xi',\cdot)\end{gathered}\tag{6.27}$$

compose to $K_e^{v\prime}(s',\cdot)=(\mathsf{g}_i)_{\#}K_e^v(\mathsf{g}_j^{-1}s',\cdot)$, so $q_j'K_e^{v\prime}=(\mathsf{g}_i)_{\#}(q_jK_e^v)$ for $q_j'=(\mathsf{g}_j)_{\#}q_j$; for a gauge-inert record space the pushforward by g_i is omitted. (c) For the represented link $\rho_x(\Theta_e^x)$, a bimeasurable bijection, $F_e^v(s,\cdot):=K_e^v(\rho_x(\Theta_e^x)^{-1}s,\cdot)$ satisfies $K_e^v(s,\cdot)=F_e^v(\rho_x(\Theta_e^x)s,\cdot)$, and the link law gives $F_e^{v\prime}(s',\cdot)=(\mathsf{g}_i)_{\#}F_e^v(\mathsf{g}_i^{-1}s',\cdot)$; this separation uses the invertibility of the link and does not determine the link from K_e^v. [ESTABLISHED]

The proof is in appendix B.1. A kernel may encode a belief-law or model-law point only when that point is a latent coordinate of hypothesis 5.5 at the law-point level of definition 3.19; by hypothesis 4.3 it never reads the live recognition law over that point, and on the model channel a compatible evaluator is additional data.

6.3 Factored links and loops

Definition 6.12 (Factored matrix communication). On the finite-dimensional vector tier, write E_i for the sample fiber of agent i in channel x, distinct from the law fiber $(\mathcal{E}_x)_{c_*}$, and suppress x. For each edge copy $e:j\to i$ choose linear maps, possibly depending on the copy and with message spaces of possibly different dimensions,

$$\mathsf{L}_e^{\mathrm{enc}}:E_j\to H_e^-,\qquad \mathsf{L}_e^{\mathrm{tr}}:H_e^-\to H_e^+,\qquad \mathsf{L}_e^{\mathrm{dec}}:H_e^+\to E_i,\qquad \mathsf{L}_e=\mathsf{L}_e^{\mathrm{dec}}\mathsf{L}_e^{\mathrm{tr}}\mathsf{L}_e^{\mathrm{enc}},\tag{6.28}$$

separately in the belief and model representations, crossing channels only through the morphisms of section 2.3. The Markov kernel of a linear realization is the composed value kernel K_e^v, not L_e. For represented frames $U_j:\mathbb{R}^{\dim E}\to E_j$, a value extraction $\mathsf{M}^{\mathrm{val}}:\mathbb{R}^{\dim E}\to H$ and an output placement $\mathsf{M}^{\mathrm{out}}:H\to\mathbb{R}^{\dim E}$ give the shared-head specialization

$$\mathsf{L}_e^{\mathrm{enc}}=\mathsf{M}^{\mathrm{val}}U_j^{-1},\qquad \mathsf{L}_e^{\mathrm{tr}}=I_H,\qquad \mathsf{L}_e^{\mathrm{dec}}=U_i\mathsf{M}^{\mathrm{out}},\qquad \mathsf{L}_e=U_i\mathsf{M}^{\mathrm{out}}\mathsf{M}^{\mathrm{val}}U_j^{-1}.\tag{6.29}$$

With $\mathsf{G}_j=U_j^{-\top}U_j^{-1}$, sending uses $U_j\,\mathrm{im}(\mathsf{M}^{\mathrm{val}})^{\top}$ and placement uses $U_i\,\mathrm{im}\,\mathsf{M}^{\mathrm{out}}$; these and the comparison directions can overlap and need not partition a frame, and $\mathsf{M}^{\mathrm{out}}\mathsf{M}^{\mathrm{val}}$ need be neither orthogonal nor idempotent.

Proposition 6.13 (Endpoint covariance and rank). (a) Under the old-to-new maps g_i of corollary 6.11, here linear, and independent message coordinate changes $\phi_e^-\in\mathrm{GL}(H_e^-)$ and

$\phi_e^+ \in \mathrm{GL}(H_e^+)$,

$$\mathsf{L}_e^{\mathrm{enc}\prime} = \phi_e^- \mathsf{L}_e^{\mathrm{enc}} \mathsf{g}_j^{-1}, \qquad \mathsf{L}_e^{\mathrm{tr}\prime} = \phi_e^+ \mathsf{L}_e^{\mathrm{tr}} (\phi_e^-)^{-1}, \qquad \mathsf{L}_e^{\mathrm{dec}\prime} = \mathsf{g}_i \mathsf{L}_e^{\mathrm{dec}} (\phi_e^+)^{-1},$$
$$\mathsf{L}_e' = \mathsf{g}_i \mathsf{L}_e \mathsf{g}_j^{-1}, \tag{6.30}$$

and the message-basis freedom does not enlarge the structure group. (b) Direct composition gives

$$\ker \mathsf{L}_e^{\mathrm{enc}} \subseteq \ker \mathsf{L}_e, \qquad \mathrm{im}\, \mathsf{L}_e \subseteq \mathrm{im}\, \mathsf{L}_e^{\mathrm{dec}}, \qquad \mathrm{rank}\, \mathsf{L}_e \le \min(\mathrm{rank}\, \mathsf{L}_e^{\mathrm{enc}}, \mathrm{rank}\, \mathsf{L}_e^{\mathrm{tr}}, \mathrm{rank}\, \mathsf{L}_e^{\mathrm{dec}}), \tag{6.31}$$

so the encoder accesses the quotient $E_j/\ker \mathsf{L}_e^{\mathrm{enc}}$, the potential placement space is $\mathrm{im}\, \mathsf{L}_e^{\mathrm{dec}}$, and the reachable linear signal is the possibly smaller $\mathrm{im}\, \mathsf{L}_e$. An SPD source metric G_j identifies the quotient with $(\ker \mathsf{L}_e^{\mathrm{enc}})^{\perp_{\mathsf{G}_j}} = \mathsf{G}_j^{-1} \mathrm{im}(\mathsf{L}_e^{\mathrm{enc}})^\top$, and $\mathsf{G}_j' = \mathsf{g}_j^{-\top} \mathsf{G}_j \mathsf{g}_j^{-1}$, so a distinguished sending subspace requires a metric while the kernel and quotient do not. Every finite-dimensional L_e factors through its rank. (c) A weighted sum $\sum_h w_h \mathsf{L}_h$ of maps $E_j \to E_i$, each factoring through a space H_h, has rank at most $\min(\dim E_i, \sum_h \dim H_h)$, attained with rank-one factors: for $\dim E_i = \dim E_j = 2$, identity frames, $\mathsf{L}_h = \mathbf{e}_h \mathbf{e}_h^\top$, and positive weights, the sum is $\mathrm{diag}(w_1, w_2)$. [ESTABLISHED]

Proposition 6.14 (Connection data and communication loops). Keep the represented invertible link Θ_e of $e : j \to i$ on the vector tier, with $\Theta_{\bar{e}} = \Theta_e^{-1}$. (a) For any L_e and a chosen link, $F_e := \mathsf{L}_e \Theta_e^{-1}$ gives

$$\mathsf{L}_e = F_e \Theta_e, \qquad F_e' = \mathsf{g}_i F_e \mathsf{g}_i^{-1}, \tag{6.32}$$

and if $\mathsf{L}_e^{\mathrm{enc}} = \widetilde{\mathsf{L}}_e^{\mathrm{enc}} \Theta_e$ for some $\widetilde{\mathsf{L}}_e^{\mathrm{enc}} : E_i \to H_e^-$, then $F_e = \mathsf{L}_e^{\mathrm{dec}} \mathsf{L}_e^{\mathrm{tr}} \widetilde{\mathsf{L}}_e^{\mathrm{enc}}$. The separation does not recover Θ_e from communication, and a receiver-only processing map $F_e = F_i$ exists exactly when $\mathsf{L}_e \Theta_e^{-1}$ agrees on every incoming edge copy at i. (b) Communication on the reverse copy is separately specified and need not be an inverse, an adjoint, or of the same rank. For a closed walk $p = (e_0, \dots, e_{r-1})$ with $e_a : i_{a+1} \to i_a$ and $i_r = i_0$, the communication round trip

$$\mathsf{L}_p = \mathsf{L}_{e_0} \cdots \mathsf{L}_{e_{r-1}} \qquad \text{satisfies} \qquad \mathsf{L}_p' = \mathsf{g}_{i_0} \mathsf{L}_p \mathsf{g}_{i_0}^{-1}, \tag{6.33}$$

even when the maps are singular, so its rank, characteristic polynomial, and trace are invariant; a singular round trip records processing and possible loss of distinguishability and is not the link holonomy $\mathrm{Hol}^x(p)$ of (2.30). (c) For two sites with identity frames and identity links, $e : 1 \to 2$ reading coordinate one and writing coordinate two, and $f : 2 \to 1$ the reverse,

$$\mathsf{L}_e = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right), \qquad \mathsf{L}_f = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right), \qquad \mathsf{L}_f \mathsf{L}_e = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right), \qquad \mathsf{L}_e \mathsf{L}_f = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right) \tag{6.34}$$

give round trips that preserve one direction and discard the other despite trivial link holonomy, and $2\mathsf{L}_f$ in place of L_f adds gain at the same rank. The shared-head family (6.29) has length-r product $U_{i_0}(\mathsf{M}^{\mathrm{out}} \mathsf{M}^{\mathrm{val}})^r U_{i_r}^{-1}$, which depends on a path only through its length and endpoints; independent edge maps remove that restriction. (d) Link holonomy and communication rank are independent: for $G = \mathrm{GL}(2)$ in its defining representation and two parallel copies $a, b : 1 \to 2$ with $\Theta_a = I_2$ and

$\Theta_b = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the loop through b and $\bar{a}$ has nonidentity holonomy, while arbitrary singular $\mathsf{L}_a, \mathsf{L}_b$ are admissible through $F_a = \mathsf{L}_a \Theta_a^{-1}$ and $F_b = \mathsf{L}_b \Theta_b^{-1}$. [ESTABLISHED]

The proofs of propositions 6.13 and 6.14 are in appendix B.1. General matrix link fields also occur in linear lattice gauge theory [104], whose compact-group action and continuum conclusions do not follow from this endpoint algebra.

6.4 Comparison, content, and the transformer correspondence

Definition 6.15 (Role comparison, quadratic disagreement, and mean operator). For an edge copy $e : j \to i$, comparison uses data separate from content transmission: a comparison space H_e^{cmp} and query and key maps $S_e^Q : E_i \to H_e^{\mathrm{cmp}}$ and $S_e^K : E_j \to H_e^{\mathrm{cmp}}$, or, in the measurable setting, measurement kernels with a common output space. For law points q_i, q_j the role divergence is

$$d_e = D_{\mathrm{KL}}\big((S_e^Q)_{\#} q_i \| (S_e^K)_{\#} q_j\big), \tag{6.35}$$

a law-point-level energy (definition 3.19) admitted where it is finite; this cospan compares observables and chooses no map back into E_i. Reference maps with $S_e^Q = S_{\mathrm{ref}}^Q U_i^{-1}$ and $S_e^K = S_{\mathrm{ref}}^K U_j^{-1}$ reproduce projected query and key comparison; with independent links the sender can instead be measured by $S_e^K = S_{\mathrm{ref}}^K U_i^{-1} \Theta_e$, which recovers the reference formula when $\Theta_e = U_i U_j^{-1}$ without imposing it. For an SPD state metric G_i and an SPD comparison precision W_e, the quadratic disagreement is $V_e(k_i, k_j) = \frac{1}{2}(S_e^Q k_i - S_e^K k_j)^\top W_e (S_e^Q k_i - S_e^K k_j)$. Declared per-head rows $\beta_{ij}^{(h)}$, value extractions $\mathsf{M}_h^{\mathrm{val}} : \mathbb{R}^{\dim E} \to H_h$, and output placements $\mathsf{M}_h^{\mathrm{out}} : H_h \to \mathbb{R}^{\dim E}$ define the mean operator

$$\mu_i^{\mathrm{new}} = \mu_i + \sum_{j,h} \beta_{ij}^{(h)} U_i \mathsf{M}_h^{\mathrm{out}} \mathsf{M}_h^{\mathrm{val}} U_j^{-1} \mu_j, \tag{6.36}$$

which in reference coordinates is value extraction, weighted aggregation, output placement, and a residual, the operator form of multi-head attention [101]. The operator is declared: definition 6.1 constructs one label per row, and nothing here selects the content maps.

Proposition 6.16 (Natural-gradient response to a quadratic disagreement). Fix the maps, an SPD state metric G_i, and an SPD precision W_e. The negative natural gradient of V_e in k_i is

$$-\operatorname{grad}_{\mathsf{G}_i} V_e = -\mathsf{G}_i^{-1} S_e^{Q\top} W_e S_e^Q k_i + \mathsf{G}_i^{-1} S_e^{Q\top} W_e S_e^K k_j. \tag{6.37}$$

Descent of this energy thus determines one receiving map $\mathsf{G}_i^{-1} S_e^{Q\top} W_e S_e^K$ and one self term, so a declared insertion such as (6.36) is a natural-gradient step of a quadratic disagreement only when its maps have this form; identifying G_i with a Fisher metric requires a regular statistical model that supplies it (chapter 7). [ESTABLISHED]

Proof. Differentiate V_e in k_i and multiply by G_i^{-1}. □

Corollary 6.17 (Softmax over role divergences). (a) When the latent coordinates include the law points compared in (6.35), at the law-point level of definition 3.19, a latent configuration y gives the source energies $D_{ij}(y) = d_{ij}$. On a finite positive-prior source set with finite costs and $\tau_i > 0$, the categorical row functional satisfies

$$\sum_j \beta_{ij} d_{ij} + \tau_i D_{\mathrm{KL}}(\beta_i \| \pi_i) = \tau_i D_{\mathrm{KL}}\big(\beta_i \| \beta_i^P(y)\big) - \tau_i \log \sum_j \pi_{ij} e^{-d_{ij}/\tau_i}, \tag{6.38}$$

so the posterior row (6.4) is its unique minimizer: softmax attention over divergences between current law points is the posterior row at a latent configuration. The recognition optimum (6.8) reduces to it for $\mathbb{Q}_Y = \delta_y$, and when the latent baseline gives $\{y\}$ zero mass that point mass is not absolutely continuous with respect to the posterior slice, so its relative entropy and VFE are infinite (proposition 5.4); the law-point prior that such a latent requires is open problem 3.23. (b) Let the projected key covariance be the same SPD matrix Σ^K for every source, and put $\mu_i^Q = S_{\mathrm{ref}}^Q U_i^{-1} \mu_i$ and $\mu_j^K = S_{\mathrm{ref}}^K U_j^{-1} \mu_j$. For Gaussian law points

$$d_{ij} = d_i^\circ - (\mu_i^Q)^\top (\Sigma^K)^{-1} \mu_j^K + \tfrac{1}{2} (\mu_j^K)^\top (\Sigma^K)^{-1} \mu_j^K, \tag{6.39}$$

where d_i° collects every term independent of j, including the query covariance terms, so whitening gives a dot-product logit together with a key-norm bias and $\log \pi_{ij}$. The bias disappears when the whitened key norms are constant, or when a source prior cancels it at these comparison laws, and such compensation in a population model requires its own source law. Unequal projected key covariances generally retain additional source-dependent terms. [ESTABLISHED]

The proof is in appendix B.1.

Corollary 6.18 (Content changes routing). Let $p(j, y, v) = \pi_j p_j(y) r_j(v \mid y)$ and $q(j, y, v) = \beta_j \widetilde{q}(y) s_j(v \mid y)$ be normalized laws on a source label, a comparison state, and a content variable, with $q \ll p$ and finite terms. Then

$$D_{\mathrm{KL}}(q \| p) = D_{\mathrm{KL}}(\beta \| \pi) + \sum_j \beta_j \left[D_{\mathrm{KL}}(\widetilde{q} \| p_j) + \mathbb{E}_{\widetilde{q}} D_{\mathrm{KL}}(s_j(\cdot \mid Y) \| r_j(\cdot \mid Y)) \right], \tag{6.40}$$

so the conditional content term generally changes the optimal routing, and matching $s_j = r_j$ removes it. [ESTABLISHED]

Proof. Apply lemma 3.12 to the disintegration over j, then over y given j. □

6.4.1 Exact receiver update and transformer attention

Proposition 6.19 (Exact receiver update in the comparison-record model). Fix a receiver i with a finite source set on which $\pi_{ij} > 0$, each source j reaching i through one edge copy $e : j \to i$, and suppose that these copies share the comparison space H_i^{cmp}, the query map $S_i^Q := S_e^Q$, and an SPD precision W_i, while the key maps S_e^K may depend on the copy. Let the attention record of

definition 6.1 be the comparison record

$$o_i \mid k_i, J_i = j, y_{-i} \sim \mathcal{N}\big(S_i^Q k_i - S_e^K k_j, \tau_i W_i^{-1}\big), \qquad \tau_i > 0, \tag{6.41}$$

observed at $o_i = 0$, where y_{-i} collects every latent coordinate other than k_i and J_i and so fixes the source states k_j. At $o_i = 0$ its density is $\mathrm{c}_i \exp[-V_e(k_i, k_j)/\tau_i]$, with V_e of definition 6.15 and $\mathrm{c}_i = (2\pi\tau_i)^{-\dim H_i^{\mathrm{cmp}}/2}(\det W_i)^{1/2}$, so it realizes (6.2). Suppose that it is the only factor reading J_i, as in (6.3), and that, given y_{-i} and every record other than o_i, the law of k_i is $\mathcal{N}(\mu_i, \Sigma_i)$ with $\Sigma_i \succ 0$. Write $\widetilde{W}_i = W_i/\tau_i$ and

$$\Sigma_i^+ = \big(\Sigma_i^{-1} + S_i^{Q\top}\widetilde{W}_i S_i^Q\big)^{-1}, \qquad \Sigma_i^{\mathrm{pred}} = \widetilde{W}_i^{-1} + S_i^Q \Sigma_i S_i^{Q\top}, \qquad r_{ij} = S_i^Q \mu_i - S_e^K k_j. \tag{6.42}$$

(a) Given y_{-i} and all records, J_i has the posterior weights w_{ij}, normalized over the source set, k_i has the conditional components, and the conditional mean of k_i is

$$w_{ij} \propto \pi_{ij} \exp\big[-\tfrac{1}{2} r_{ij}^\top (\Sigma_i^{\mathrm{pred}})^{-1} r_{ij}\big], \qquad k_i \mid J_i = j \sim \mathcal{N}\big(\Sigma_i^+(\Sigma_i^{-1}\mu_i + S_i^{Q\top}\widetilde{W}_i S_e^K k_j), \Sigma_i^+\big), \tag{6.43}$$

$$\mathbb{E}[k_i \mid y_{-i}, o] = \Sigma_i^+\Sigma_i^{-1}\mu_i + \sum_j w_{ij} \mathsf{L}_e^{\mathrm{val}} k_j, \quad \mathsf{L}_e^{\mathrm{val}} = \Sigma_i^+ S_i^{Q\top}\widetilde{W}_i S_e^K, \quad \Sigma_i^+\Sigma_i^{-1} = I - \Sigma_i^+ S_i^{Q\top}\widetilde{W}_i S_i^Q. \tag{6.44}$$

The value map is thus fixed by the query map, the key map, the record precision, and the receiver covariance, and the skip connection is replaced by the precision-weighted term $\Sigma_i^+\Sigma_i^{-1}\mu_i$; equivalently the update is $\mu_i + \sum_j w_{ij}\Sigma_i^+ S_i^{Q\top}\widetilde{W}_i(S_e^K k_j - S_i^Q \mu_i)$. (b) The weights use Σ_i^{pred} in place of $\widetilde{W}_i^{-1}$ and need not equal the posterior row (6.4) at $k_i = \mu_i$: for a scalar receiver with $S_i^Q = S_e^K = \widetilde{W}_i = \Sigma_i = 1$, $\mu_i = 0$, a uniform prior, and sources $k_1 = 0$ and $k_2 = 2$, one has $w_{i2} = e^{-1}/(1 + e^{-1})$ and $\beta_{i2}^P = e^{-2}/(1 + e^{-2})$ at $k_i = 0$, while $\mathsf{L}_e^{\mathrm{val}} = \Sigma_i^+\Sigma_i^{-1} = 1/2$. (c) The Woodbury identity $\Sigma_i^+ S_i^{Q\top}\widetilde{W}_i = \Sigma_i S_i^{Q\top}(\Sigma_i^{\mathrm{pred}})^{-1}$ holds, and for full-row-rank S_i^Q and $\widetilde{W}_i^{-1} \to 0$ the value map tends to $R_e S_e^K$ with $R_e = \Sigma_i S_i^{Q\top}(S_i^Q \Sigma_i S_i^{Q\top})^{-1}$, the Σ_i^{-1}-minimal right inverse of S_i^Q. (d) With copy-dependent query maps, precisions, or comparison spaces, the same computation gives $w_{ij} \propto \pi_{ij}\mathcal{N}(0; r_{ij}, \Sigma_{ij}^{\mathrm{pred}})$ and component covariances Σ_{ij}^+, so the normalizer of the predictive law enters the weights. [ESTABLISHED]

The proof is in appendix B.1. Table 6.1 compares the pointwise attention network with the attention layer of Vaswani et al. [101] row by row; the metric lift $R_e S_e^K$ of (c) is developed for Gaussian beliefs in Supplement G.

Proposition 6.20 (A shared receiver state couples head labels). Give two heads $h = 1, 2$ of one receiver their own exclusive labels $J_{i,h}$ and comparison records of the form (6.41), conditionally independent given the latent state. At fixed k_i the label posterior factors by head. For a scalar receiver with $k_i \sim \mathcal{N}(0, 1)$ given the rest, $S^Q = S^K = \widetilde{W} = 1$ for both heads, a uniform prior, and sources with states $k_1 = 0$ and $k_2 = a \neq 0$, the posterior mass of $(J_{i,1}, J_{i,2}) = (j, l)$ after k_i is integrated is proportional to

$$\exp\big[-\tfrac{1}{3}(k_j^2 + k_l^2) + \tfrac{1}{3} k_j k_l\big], \tag{6.45}$$

Table 6.1: Correspondence between transformer attention and the pointwise attention network at c_*.

Transformer	Pointwise attention network	Standing
Token position	Agent $i \in V$ at c_*, its embedding read as a coordinate of its latent state.	declared
Query and key projections	$S_e^Q = S_{\mathrm{ref}}^Q U_i^{-1}$, $S_e^K = S_{\mathrm{ref}}^K U_j^{-1}$ and the role divergence (6.35); with a common projected key covariance, a dot-product logit plus a key-norm bias, (6.39).	derived, corollary 6.17
One key map shared by all edges	Independent links enter $S_e^K = S_{\mathrm{ref}}^K U_i^{-1}\Theta_e$ only through the represented edge twist $U_i^{-1}\Theta_e U_j$; the shared map is the identity-twist case $\Theta_e = U_i U_j^{-1}$, and a nonidentity twist, including nontrivial link holonomy, has no counterpart in that architecture.	derived, proposition 2.12
Softmax row	Generative posterior of a categorical source label, (6.4); over law-point divergences, the posterior row at a latent configuration, (6.38), not the recognition optimum.	derived, proposition 6.2 and corollary 6.17
Attention mask	Zero prior weight: the source lies outside the support of π_i, the source set $\mathcal{N}_i$, where the posterior row vanishes.	derived, proposition 6.2
Temperature, inverse square root of key width	The temperature $\tau_i > 0$ of (6.2); for $\Sigma^K = \varsigma^2 I$ the dot-product part of the logit $-d_{ij}/\tau_i$ is $(\mu_i^Q)^\top \mu_j^K/(\varsigma^2\tau_i)$, and $\varsigma^2\tau_i = \sqrt{\dim H_i^{\mathrm{cmp}}}$ reproduces the scaled dot product, a choice not derived here.	declared
Value and output maps, residual connection	Declared in (6.36); in the comparison-record model the exact update fixes the value map $\mathsf{L}_e^{\mathrm{val}}$ and weights the residual by $\Sigma_i^+\Sigma_i^{-1}$.	derived, proposition 6.19
Multiple heads	Declared per-head rows $\beta_{ij}^{(h)}$; no label-level construction, and no head index in the event law (6.13).	absent, open problem 6.22
Feed-forward block, layer normalization	No counterpart is modeled.	absent

and its odds ratio between agreeing and disagreeing label pairs is $e^{a^2/3} \neq 1$, which excludes a product law. [ESTABLISHED]

The proof is in appendix B.1.

Remark 6.21 (The Gaussian criterion for head-label coupling). For two heads with independent positive label priors, let their stacked predictive record covariance Σ^{rec} be positive definite and independent of the labels. Write $\Lambda_{12}^{\mathrm{rec}}$ for the off-diagonal block of its inverse, and $\mathsf{a}_j, \mathsf{b}_l$ for the head-specific key means. The positive integrated posterior table p_{jl} satisfies

$$\log \frac{p_{jl}p_{j'l'}}{p_{jl'}p_{j'l}} = -(\mathsf{a}_j - \mathsf{a}_{j'})^\top \Lambda_{12}^{\mathrm{rec}}(\mathsf{b}_l - \mathsf{b}_{l'}). \tag{6.46}$$

Thus the labels are independent exactly when every displayed pairing vanishes. Nonzero covariance between projected queries alone does not imply label dependence. [ESTABLISHED]

Proof. Expand the Gaussian log likelihood with precision $(\Sigma^{\mathrm{rec}})^{-1}$. Its mixed term is $-\mathsf{a}_j^\top \Lambda_{12}^{\mathrm{rec}} \mathsf{b}_l$; the prior and one-head terms cancel from the cross ratio. A positive table factors exactly when all its

cross ratios equal one. For example, a receiver $k_i \sim \mathcal{N}(0, I_3)$ with query maps $S_1^Q k_i = (k_{i,1}, k_{i,2})$ and $S_2^Q k_i = (k_{i,1}, k_{i,3})$, independent record noises of covariance I_2, and key means $(0,0), (0,a)$ and $(0,0), (0,b)$ has query cross-covariance $\mathrm{diag}(1,0)$ but every mixed key pairing zero. For $a, b \neq 0$ both labels remain informative. □

Open Problem 6.22 (Label-level multi-head attention). The per-head rows of (6.36) are declared, and the event law (6.13) carries no head index. Give a receiver i at least two heads h, each an exclusive label $J_{i,h}$ with a comparison record of the form (6.41) and its own query map $S_{i,h}^Q$, key maps, precision, and temperature, the records conditionally independent given the latent state, in the setting of proposition 6.19, where k_i has the conditional law $\mathcal{N}(\mu_i, \Sigma_i)$ given the rest. The heads read a common receiver coordinate when the projected queries $S_{i,h}^Q k_i$ and $S_{i,h'}^Q k_i$ are correlated under that law for some pair $h \neq h'$; require also that the integrated positive label table does not factor by head. These conditions define the coupled class of the problem. For two heads with label-independent predictive covariance, the second condition is the nonvanishing of some pairing in eq. (6.46); proposition 6.20 supplies one instance. The factorizing subclass is excluded, including heads that read disjoint channels through untied links under a receiver law that is a product over channels. What holds in the class is the per-row label posterior at a fixed receiver state, proposition 6.2 with rows (i, h); the event law of definition 6.4 carries no head index, so the pushforward of proposition 12.31 gives a parent no row per head. A parent whose receiver state is one coarse coordinate of its block reads every row of its children through that coordinate, so the exact recursion multiplies the number of rows read through one coordinate at each depth unless a head-merging channel, which the recursion does not declare, is imposed; once declared, its conditional relative entropy enters (10.4). The problem is settled by an exact recursion with bounded head count in the class: a family of comparison-record models, closed under the pushforward of proposition 12.31 composed with a declared head-merging channel without increase of the number of heads per receiving row, in which the parent's receiver state is a coarse coordinate of its block with the pushed-forward conditional law, its per-head rows are the disintegrations of the merged event law, and its posterior mean given its records and the rest is the coarse coordinate of its children's posterior means; negatively, by a member of the class and a partition into blocks for which the posterior mean of a block's coarse coordinate, as a function of the source states, is reproduced by no comparison-record model at the block with fewer heads than the block has rows.

6.5 Compositions as records

Choosing $s = s(y_j)$ and $v = v(X, y_{\partial e})$ measurably in (6.26) makes the output one owned record of hypothesis 5.5, and replacing record kernels by these compositions leaves the joint (5.11) normalized on cycles by proposition 4.4, with distinct sending and receiving interfaces and under the typing of hypothesis 4.3. Records that share a realized random message retain it once as a latent or declare their joint kernel, since conditional independence cannot be assumed after it is marginalized; independent draws from one latent law point are independent given that point. Fixed-record density formulas keep the family domination and regular-slice hypotheses of hypothesis 5.5, under which

the collective VFE of definition 5.6 and the local–global potential identity of theorem 5.23 apply. Composed records realize rectangular endpoint maps on cycles (a linear Gaussian realization is in Supplement G) but settle neither a canonical compression of such maps nor coarse-graining through moving interfaces (proposition 6.26).

Definition 6.23 (Learned and random interfaces). If interfaces or links vary as generative parameters, their score includes every varying normalizer, baseline, and record term. If they are random relation coordinates that evolve during inference, they are latent coordinates with a normalized prior and declared dynamics in the augmented state, and natural-gradient evolution needs a regular configuration metric or a justified quotient; endpoint covariance specifies neither the metric nor the learning rule. In both cases the dynamics move link values and interface maps on the edge copies of the interaction multigraph; creating or deleting edge copies is open problem 6.9.

For links the latent option is realized in chapter 9: the generator prior is a probability law for every Lie group, compact or not, and the joint with latent generators has unit mass (propositions 9.12 and 9.16); with undirected matter normalized jointly with the generators, the construction requires a finite joint normalizer, which finiteness of the prior's normalizer does not imply (proposition 9.27); and the dynamics and the configuration metric are those of the natural-gradient recognition flow of proposition 9.28.

6.6 Information loss and descent

Proposition 6.24 (Conditional data processing through an interface). For a source law and a side-information law, let S_j be the source sample in S_j^x, Υ the random receiver side information with values v, ξ^- and ξ^+ the random messages, and O the decoded record. Construction (6.26) makes $S_j \to \xi^- \to \xi^+ \to O$ a Markov chain conditional on Υ, and

$$I(S_j; O \mid \Upsilon) \le I(S_j; \xi^+ \mid \Upsilon) \le I(S_j; \xi^- \mid \Upsilon) \qquad \text{in } [0, +\infty]. \tag{6.47}$$

[ESTABLISHED]

The proof is in appendix B.1.

Proposition 6.25 (Communicated agreement leaves conditional information). Let q, p be laws on one standard Borel space with $q \ll p$ and finite relative entropy, and let M be one measurable measurement into a standard Borel space with output laws q_M, p_M. Then

$$D_{\mathrm{KL}}(q\|p) = D_{\mathrm{KL}}(q_M\|p_M) + \int D_{\mathrm{KL}}\big(q(\cdot \mid M = z)\|p(\cdot \mid M = z)\big) q_M(dz). \tag{6.48}$$

For $M = (1, 0)$, the laws $\mathcal{N}((0, a), I_2)$ and $\mathcal{N}((0, b), I_2)$ have identical communicated laws and full relative entropy $(a - b)^2/2$. [ESTABLISHED]

Proof. Append the deterministic measurement to both laws, which leaves the relative entropy unchanged, and apply the disintegration form of lemma 3.12 over M. The example has equal first coordinates and Gaussian relative entropy $(a-b)^2/2$. □

The identity needs one measurement: distinct roles $S_e^Q \neq S_e^K$ need not ask the same question, and separate head marginals need not fix their joint law (proposition 3.17).

Proposition 6.26 (Descent through a selective interface). Let E_j, E_i, E and $\overline{E}_j, \overline{E}_i, \overline{E}$ be finite-dimensional real vector spaces, and let $M_j : E_j \to \overline{E}_j$, $M_i : E_i \to \overline{E}_i$, and $M : E \to \overline{E}$ be surjective linear measurements. (a) Static intertwiner: for a linear $\mathsf{L}_e : E_j \to E_i$, a linear $\overline{\mathsf{L}}_e : \overline{E}_j \to \overline{E}_i$ with $M_i\mathsf{L}_e = \overline{\mathsf{L}}_e M_j$ exists exactly when

$$\mathsf{L}_e \ker M_j \subseteq \ker M_i, \tag{6.49}$$

and it is then unique. (b) Autonomous linear flow: for $\mathcal{A} \in \mathrm{End}(E)$, a linear $\overline{\mathcal{A}} \in \mathrm{End}(\overline{E})$ with $Me^{t\mathcal{A}} = e^{t\overline{\mathcal{A}}}M$ for every $t \in \mathbb{R}$, equivalently with $M\mathsf{u}$ solving $\dot{\bar{\mathsf{u}}} = \overline{\mathcal{A}}\bar{\mathsf{u}}$ for every solution of $\dot{\mathsf{u}} = \mathcal{A}\mathsf{u}$, exists exactly when

$$M\mathcal{A} = \overline{\mathcal{A}}M \tag{6.50}$$

is solvable, that is, exactly when $\mathcal{A}\ker M \subseteq \ker M$, and $\overline{\mathcal{A}}$ is then unique. (c) Moving measurement: on an open interval (t_0, t_1), let $t \mapsto M_t : E \to \overline{E}$ be a C^1 family of surjections and $t \mapsto \mathcal{A}_t$ continuous. A family $\overline{\mathcal{A}}_t \in \mathrm{End}(\overline{E})$ with $\frac{d}{dt}(M_t\mathsf{u}(t)) = \overline{\mathcal{A}}_t M_t \mathsf{u}(t)$ for every solution of $\dot{\mathsf{u}} = \mathcal{A}_t\mathsf{u}$ on (t_0, t_1) exists exactly when

$$\dot{M}_t + M_t\mathcal{A}_t = \overline{\mathcal{A}}_t M_t \qquad (t_0 < t < t_1) \tag{6.51}$$

is solvable, that is, exactly when $\dot{M}_t + M_t\mathcal{A}_t$ vanishes on $\ker M_t$ for every t; the family is then unique and continuous, with $\overline{\mathcal{A}}_t = (\dot{M}_t + M_t\mathcal{A}_t)M_t^+$ for any continuous right inverse M_t^+ of M_t. (d) If $\mathsf{L}_e \ker M_j \not\subseteq \ker M_i$, no linear map of the retained coordinates reproduces $M_i\mathsf{L}_e$; a coordinate swap with $M_i = M_j = (1, 0)$ is an instance, in which a hidden coordinate becomes visible. If $\mathcal{A}\ker M \not\subseteq \ker M$, two solutions with the same retained initial value have different retained velocities, so the retained coordinates do not evolve autonomously, and the same holds at each time for a moving measurement that violates (6.51). (e) A low-rank interface does not by itself give an instantaneous Markov blanket: for a joint law of a private state z^{in}, an interface z^{bd}, and an exterior z^{ex}, the blanket condition is $P(d\mathsf{z}^{\mathrm{in}}, d\mathsf{z}^{\mathrm{ex}} \mid \mathsf{z}^{\mathrm{bd}}) = P(d\mathsf{z}^{\mathrm{in}} \mid \mathsf{z}^{\mathrm{bd}})P(d\mathsf{z}^{\mathrm{ex}} \mid \mathsf{z}^{\mathrm{bd}})$, and if z^{in} is nondegenerate and independent of z^{bd} while $\mathsf{z}^{\mathrm{ex}} = \mathsf{z}^{\mathrm{in}}$, the conditional law of $(\mathsf{z}^{\mathrm{in}}, \mathsf{z}^{\mathrm{ex}})$ given z^{bd} is concentrated on the diagonal and is not a product. [ESTABLISHED]

The proof is in appendix B.1.

Open Problem 6.27 (Formation with evolving selective interfaces). Take a connected block of agents on the vector tier with Gaussian belief laws whose means are the fast coordinates, linear factored communication (definition 6.12) through surjective linear interfaces M_i of one fixed rank below the fiber dimension, and comparison records (6.41) whose query and key maps read the receiver and the transported source through the receiver's interface, $S_i^Q = M_i$ and $S_e^K = M_i\Theta_e$, with the labels of definition 6.1. The means follow the gradient flow of the sample-level energy of

the current label assignment with identity mobility and diffusion covariance $2\mathrm{T}I\,dt$ at a temperature $\mathrm{T} > 0$, and the assignment is redrawn at a rate-one clock from the receiver posterior rows, as in definition 14.21; the interfaces are generative parameters in the sense of definition 6.23 that move by the natural-gradient flow of the collective VFE under a declared metric on interfaces of that rank, at a mobility $\epsilon_{\mathrm{M}} > 0$. This is the class of the problem. Along such a path the $\dot{M}_t$ term of (6.51) is not controlled by a fixed-interface construction, because the path reads the state, and neither is the label switching it induces. A parent for the block is a retained mechanism in the sense of section 8.3: a Markov process on a state space of fixed dimension, below the dimension of the block's means and assignment and independent of T, ϵ_{M}, and the horizon, whose state determines the communicated coordinates $M_{i,t}\mu_i$ of every agent of the block. The admitted state space is a smooth manifold, the stochastic coefficients between jumps are locally Lipschitz, initialization and jump maps are continuous in their native inputs, and the communicated-coordinate readout is C^1 on each assignment stratum. Any additional conditional-law filter obeys definition 12.39. These are explicit regularity restrictions on this problem; dimension alone would allow arbitrary Borel encodings of the full state or history. What holds is the exact alternative that retains the private coordinates with their conditional laws at the dimension of the block (theorem 12.44 in finite state), and the pointwise descent criterion of proposition 6.26(c) for a supplied C^1 interface path. The problem is settled by a parent of that kind for which the relative entropy of the fine path law of the communicated coordinates with respect to the parent's path law, on horizons of order $1/\epsilon_{\mathrm{M}}$, on which the interfaces move by an amount of order one, tends to zero as $\epsilon_{\mathrm{M}} \to 0$ uniformly over T in an interval $(0, \mathrm{T}_0]$; negatively, by a member of the class in which the learned interface path has a rest point at which no parent of that kind is projectable, or in which every parent of that kind has that relative entropy bounded below on those horizons as $\epsilon_{\mathrm{M}} \to 0$ and $\mathrm{T} \to 0$ along a stated relation.

Physicist's summary. Attention is inference over a hidden pointer: a categorical label names whom a receiver listens to, its prior's support is the receiver's source set, and its record's likelihood falls with a comparison energy. Bayes' formula gives the softmax row, masks are zero prior weights, and the temperature is a free parameter of the record, of which the inverse-square-root key-width scaling is one choice. A recognition row held constant over the latent state is a different softmax, over averaged energies, at an exact expected relative-entropy cost that penalizes uncertain sources. Integrating the label out gives a log-sum-exp potential whose curvature is the frozen-weight curvature minus an attention covariance, which can turn a convex landscape concave, the seed of symmetry breaking in the two-timescale and formation chapters. Communication is a normalized encoder–channel–decoder kernel, covariant under separate recoordinatizations of each message space; it may be singular, and a singular round trip is processing, not holonomy. For a Gaussian comparison record the exact receiver update has the transformer shape, but the value map is fixed by the query and key maps, the residual is precision weighted, and the weights use the predictive covariance. Heads sharing one receiver state can have coupled labels; an exact recursion with bounded head count for the coupled class remains open. Coarse-graining carries the receiver–source event law, not the conditional row, and a coarse source set contains the block itself.

Chapter 7

Fiber Information Geometry and Inference Histories

This chapter asks how the collective free energy of chapter 5 selects an inference history of the agents at c_* and what duration the Fisher geometry of their law fibers assigns to it. A chosen metric makes the positive ray of the natural gradient select an oriented orbit, whose Fisher length is an information clock invariant under reparameterization (theorem 7.36). A parameter-independent Markov channel projects scores and loses Fisher information by an exact conditional-covariance defect (theorem 7.8), so a same-path Markov image of a history, such as the law of a meta-agent, accumulates no more duration (corollary 7.42), while a coarse orbit recomputed from its own free energy compares with the fine one only through an oriented semiconjugacy, whose rate cancels from every duration (theorems 7.47 and 7.49). A three-agent parity witness shows that a collective free energy and its Fisher tensor are not functions of the agents' marginal laws. Chapter 8 uses the mobility and clock results, chapter 10 the Fisher-loss theorem, and chapters 12 and 13 the coarse configuration map and the record clocks. Longer proofs are deferred to appendix B.

Scope. Law fibers and kernel fibers are distinct, and no manifold, dominating measure, affine chart, cone, or Fisher metric is inferred from the words belief or model. The Fisher metric is chosen: no invariance axiom that characterizes it (proposition 7.4) is imposed, and no uniqueness is asserted. The parity hyperedge (definition 7.27) is supplied generative data, not derived from pairwise locality or a canonical agent inventory, and the promoted parity coordinate (corollary 7.26) is not thereby an agent or an intervention variable. No fine-to-coarse semiconjugacy is asserted for coarse maps that discard directions on which the fine free energy depends. Open are correlated lifts beyond the parity witness (open problem 7.29), global Fisher quotients and canonical block data (open problem 7.15), the quantitative meta-agent clock ratio (open problem 7.43), and the slow-manifold limit of the information-discarding case (open problem 8.7). [NOT-CLAIMED]

Notation. Tangent vectors are Z, Z_1, Z_2, Z_3, and s_Z is the score in the direction Z; $\xi_{\mathcal{M}}$ is the fundamental vector field of ξ in the Lie algebra $\mathfrak{g}$ of G on a G-manifold $\mathcal{M}$. A parameter-independent Markov kernel is K, with law map $K_\star p = pK$ and joint law $p \otimes K(dy, d\bar{y}) = p(dy)K(y, d\bar{y})$. A history is Q with curve parameter t, a is a positive rate factor, $\mathcal{I}_6$ indexes the bits of the parity witness and par is its parity character, $\mathcal{N}(\mu, \varsigma^2)$ is the normal family, and upright letters $\mathrm{c}, \mathrm{c}_1, \mathrm{c}_2, \mathrm{q}$ are constants local to one statement.

7.1 Fiber statistical tensors

At c_* a frame u in the G-torsor $(\mathscr{P}_G)_{c_*}$ identifies each law fiber $\mathcal{B}_x$ with $(\mathcal{E}_x)_{c_*}$ through $\iota_u : \beta \mapsto [u, \beta]$, and $\iota_{u\cdot g} = \iota_u \circ \widehat{\rho}_x(g)$ by (2.7), so a tensor on $\mathcal{B}_x$ defines one on $(\mathcal{E}_x)_{c_*}$ exactly when it is invariant under the represented action.

Hypothesis 7.1 (Regular channel models). Each law fiber $\mathcal{B}_x$, $x \in \{b, m\}$, and each coarse fiber $\bar{\mathcal{B}}_x$ used below is a finite-dimensional smooth parametrized-measure model with smooth represented action, as in hypothesis 2.3, differentiable in quadratic mean with a positive-definite Fisher form; every score direction is integrable to the third power, the second- and third-order score moments depend smoothly on the point, and the model is dominated enough to differentiate the divergences of section 7.2 under the integral through third order. The hypothesis excludes singular statistical strata, families whose support moves without this differentiability, and law fibers given only by a chart without a sample-space action.

Definition 7.2 (Fiber statistical tensors and the model-parameter Fisher tensor). With s_Z the score of $\mathcal{B}_x$ at p in the direction $Z \in T_p\mathcal{B}_x$, the Fisher and Amari–Chentsov tensors of the channel and, for the generative parameter θ, structural data X, tangent vectors Z_1, Z_2 of the parameter space of θ, and a probability law $\mathsf{r}(dX)$ on structural data, the model-parameter Fisher tensor of the evidence $p_{\theta,V}(o \mid X)$ and its average are

$$g^F_{x,p}(Z_1, Z_2) = \mathbb{E}_p[\mathrm{s}_{Z_1}\mathrm{s}_{Z_2}], \tag{7.1}$$

$$T^F_{x,p}(Z_1, Z_2, Z_3) = \mathbb{E}_p[\mathrm{s}_{Z_1}\mathrm{s}_{Z_2}\mathrm{s}_{Z_3}], \tag{7.2}$$

$$\mathsf{I}_X(\theta)(Z_1, Z_2) = \mathbb{E}_\theta\left[\partial_{Z_1} \log p_{\theta,V}(O \mid X)\partial_{Z_2} \log p_{\theta,V}(O \mid X) \,\middle|\, X\right], \tag{7.3}$$

$$\mathsf{I}_\mathsf{r}(\theta) = \int \mathsf{I}_X(\theta)\mathsf{r}(dX). \tag{7.4}$$

Each of I_X and I_r is called a metric only where it is nondegenerate.

On the latent sample space Y an interaction energy is a function, not a metric; the law fibers and the configuration manifolds of section 7.4 carry g^F; the parameter space of θ carries I_X or I_r, of rank depending on X or r, one available metric for θ in the fast and slow coordinates of definition 8.2, whose model-channel recognition coordinates carry g^F_m. Equal components of a quadratic energy on Y and a Fisher metric, as in the Gaussian mean sector of Supplement G, are a coincidence of coordinates, not an identification.

Proposition 7.3 (Descent of the statistical tensors). Under hypothesis 7.1, every $\widehat{\rho}_x(g)$ preserves g^F_x and T^F_x, so $(\iota_u^{-1})^* g^F_x$ and $(\iota_u^{-1})^* T^F_x$ are tensors on $(\mathcal{E}_x)_{c_*}$ that do not depend on the frame u; they are denoted by the same symbols. The sample-space origin of the action cannot be replaced by invariance of the metric alone: for the normal family, whose Fisher metric is $(d\mu^2 + 2d\varsigma^2)/\varsigma^2$, the map induced by $z \mapsto (1+z)/(1-z)$ on $z = \mu/\sqrt{2} + \mathrm{i}\varsigma$ preserves g^F, fixes $(\mu, \varsigma) = (0, 1)$, and does not preserve T^F. [ESTABLISHED]

The proof is in appendix B.2.

Proposition 7.4 (Invariance characterizations of the Fisher metric). On the open simplex of positive laws on a finite set, the Fisher metric is, up to a positive factor, the unique Riemannian metric invariant under congruent Markov embeddings [102, Lemma 11.3, p. 161]. On the cone of positive finite measures on a finite set, with coordinates x_i and total mass $|\mathsf{x}| = \sum_i \mathsf{x}_i$, the positive-definite metrics invariant under the same maps are exactly those with components $g_{\mathsf{x}}(\partial_i, \partial_j) = \mathrm{f}_2(|\mathsf{x}|) + \delta_{ij}|\mathsf{x}|\mathrm{f}_1(|\mathsf{x}|)/\mathsf{x}_i$ for smooth functions with $\mathrm{f}_1 > 0$ and $\mathrm{f}_1 + \mathrm{f}_2 > 0$ [18]; on tangent vectors of the simplex, where $\sum_i Z_i = 0$, the f_2 term vanishes and the metric is $\mathrm{f}_1(1)$ times the Fisher metric. For parametrized measure models on general sample spaces, every local continuous quadratic form field invariant under sufficient statistics is a combination of the Fisher form and the square of the differential of total mass with coefficients that are functions of the total mass, and on statistical models, where the total mass is one, the Fisher form is the unique weakly continuous such field up to a constant [3, Main Theorem 2.10]; the finite case of this characterization is Campbell's [72, Section 1]. Uniqueness up to a constant also holds for strongly continuous forms that are monotone under statistics [72, Theorem 1.4], for diffeomorphism-invariant smooth weak Riemannian metrics on smooth positive densities on a closed manifold of dimension at least two [8, Theorem in the introduction], and, on exponential families and their derived families, for Riemannian metrics invariant under independent and identically distributed extensions and canonical sufficient statistics [34, Theorem 1]. [STANDARD]

7.2 Divergence jets and the orbit Fisher form

Definition 7.5 (Regular invariant contrast and its jets). A regular invariant contrast on a law fiber $\mathcal{B}$ is a function $\mathscr{D} : \mathcal{B} \times \mathcal{B} \to \mathbb{R}$, C^3 near the diagonal, with $\mathscr{D} \geq 0$ and, near the diagonal, $\mathscr{D}(p, q) = 0$ exactly when $p = q$, invariant under the simultaneous represented action, $\mathscr{D}(\widehat{\rho}(g)p, \widehat{\rho}(g)q) = \mathscr{D}(p, q)$, and with positive-definite mixed second jet. With $(Z, 0)$ and $(0, Z)$ the lifts of $Z \in T_p\mathcal{B}$ to the two factors, extended as constant coordinate fields in one chart, the jets of $\mathscr{D}$ are

$$g_{\mathscr{D}}(Z_1, Z_2) = -[(Z_1, 0)(0, Z_2)\mathscr{D}]_{(p,p)}, \tag{7.5}$$

$$\Gamma_{\mathscr{D}}(Z_1, Z_2, Z_3) = -[(Z_1, 0)(Z_2, 0)(0, Z_3)\mathscr{D}]_{(p,p)}, \tag{7.6}$$

$$\Gamma^*_{\mathscr{D}}(Z_1, Z_2, Z_3) = -[(0, Z_1)(0, Z_2)(Z_3, 0)\mathscr{D}]_{(p,p)}, \tag{7.7}$$

$$T^F_{\mathscr{D}} = \Gamma_{\mathscr{D}} - \Gamma^*_{\mathscr{D}}. \tag{7.8}$$

The two third-order arrays are connection coefficients in the chart, their difference is a symmetric cubic tensor. Invariance makes the metric and cubic jets define tensors, and the connection jets define affine connections, on the associated fiber independently of the frame. The contrast $(p, q) \mapsto (p - q)^4$ on a one-dimensional fiber is positive off the diagonal with zero second jet, so it is excluded.

Proposition 7.6 (KL divergence jets). On a fiber satisfying hypothesis 7.1, the relative entropy $D_{\mathrm{KL}}(p\|q)$ is a regular invariant contrast, its metric jet is g^F, and its cubic jet $T^F_{D_{\mathrm{KL}}}$ is the Amari–Chentsov tensor T^F. [ESTABLISHED]

Corollary 7.7 (Orbit Fisher form). Assume hypothesis 7.1 in one channel. For $q \in \mathcal{B}$ define

$$\mathfrak{k}_q(\xi,\eta) = g^F_q\big(\xi_{\mathcal{B}}(q), \eta_{\mathcal{B}}(q)\big), \qquad \mathfrak{g}_q = \{\xi \in \mathfrak{g} : \xi_{\mathcal{B}}(q) = 0\}. \tag{7.9}$$

Then $\mathfrak{k}_q$ is positive semidefinite, its radical is exactly $\mathfrak{g}_q$, and $\mathfrak{g}_q$ is the Lie algebra of the stabilizer G_q of q. For every $\xi \in \mathfrak{g}$, as $t \to 0$,

$$D_{\mathrm{KL}}\left(q\|\widehat{\rho}(\exp t\xi)q\right) = \frac{t^2}{2}\mathfrak{k}_q(\xi,\xi) + O(t^3), \tag{7.10}$$

and for $\xi \in \mathfrak{g}_q$ the left side vanishes for every t. The same expansion holds for every regular invariant contrast, with g^F replaced by its metric jet $g_{\mathscr{D}}$. [ESTABLISHED]

The proofs of proposition 7.6 and corollary 7.7 are in appendix B.2.

In the running example of section 2.5, a loop holonomy $\exp t\xi$ acting on an agent's law gives the loop a transported divergence $\frac{t^2}{2}\mathfrak{k}_q(\xi,\xi) + O(t^3)$ that vanishes to all orders exactly when $\xi \in \mathfrak{g}_q$, the second-order form of law coherence up to holonomy stabilizers (definition 2.17).

With the link as a variable, the orbit Fisher form is the Hessian of the transported divergence at an exact link, and the minimizers of that divergence are unions of double cosets of the stabilizers, hence not isolated when a stabilizer has positive dimension (proposition 9.23).

7.3 Markov coarse morphisms and the vertical Fisher defect

Theorem 7.8 (Score projection and Fisher loss). Let $\{P_\lambda\}$ be a family of laws on Y, indexed by an open subset of $\mathbb{R}^d$, that is differentiable in quadratic mean at λ_0 with centered square-integrable score $\mathbf{s}_{\lambda_0} : \mathsf{Y} \to \mathbb{R}^d$, and let $K : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be a Markov kernel that does not depend on λ and satisfies $K(y,\bar{\mathsf{Y}}) = 1$ for every y. Under the joint law $P_{\lambda_0} \otimes K$, the coarse family $\{P_\lambda K\}$ is differentiable in quadratic mean at λ_0 with score

$$\bar{\mathbf{s}}_{\lambda_0}(\bar{Y}) = \mathbb{E}\left[\mathbf{s}_{\lambda_0}(Y) \mid \bar{Y}\right] \quad P_{\lambda_0}K\text{-almost surely}, \tag{7.11}$$

and its Fisher information obeys

$$\mathsf{I}_{\bar{\mathsf{Y}}}(\lambda_0) \preceq \mathsf{I}_{\mathsf{Y}}(\lambda_0), \qquad \mathsf{I}_{\mathsf{Y}}(\lambda_0) - \mathsf{I}_{\bar{\mathsf{Y}}}(\lambda_0) = \mathbb{E}\,\mathrm{Cov}\left(\mathbf{s}_{\lambda_0}(Y) \mid \bar{Y}\right) \succeq 0. \tag{7.12}$$

Equality in a direction $z \in \mathbb{R}^d$ holds exactly when $z^\top \mathbf{s}_{\lambda_0}(Y)$ is $\bar{Y}$-measurable. No common dominating measure is assumed. [ESTABLISHED]

The proof is in appendix B.2.

Remark 7.9 (Fisher equality is not recovery). Equality in (7.12) at one parameter value is local score sufficiency, not recovery of the experiment. Let $Y = (Y_1, Y_2)$ have independent Bernoulli coordinates with success probabilities $\frac{1}{2} + \frac{\lambda}{4}$ and $\frac{1}{2} + \frac{\lambda^2}{4}$, and let K discard Y_2. At $\lambda = 0$ the score of Y_2 vanishes, so the fine and coarse Fisher informations both equal $\frac{1}{4}$, yet no parameter-independent kernel K' satisfies $(P_\lambda K)K' = P_\lambda$ for every λ in a neighborhood of zero: applying any fixed K' to the Bernoulli law $P_\lambda K$ makes every output event probability affine in λ, whereas the required output event $Y_2 = 1$ has probability $\frac{1}{2} + \frac{\lambda^2}{4}$. Kagan and Shepp [60] give an insufficient statistic that preserves Fisher information at every parameter value, and Pollard [86, Theorem 7] explains such examples by the criterion of theorem 7.8: preservation at λ holds exactly when the score is almost surely a function of the statistic. [ESTABLISHED]

At c_* the theorem applies to maps between fibers, for instance from the joint fiber of a block of agents to the fiber of their meta-agent.

Definition 7.10 (Markov coarse morphism and its vertical defects). Let $\bar{\mathcal{B}}_x$ be a second law fiber in channel x, with represented action $\widehat{\bar{\rho}}_x$ of G, associated bundle $\bar{\mathcal{E}}_x$, and frame maps $\bar{\iota}_u$. A Markov coarse morphism in channel x at c_* is a map $\mathscr{K} : (\mathcal{E}_x)_{c_*} \to (\bar{\mathcal{E}}_x)_{c_*}$ with $\bar{\iota}_u^{-1} \circ \mathscr{K} \circ \iota_u = K_\star|_{\mathcal{B}_x}$ for one, and hence every, frame u, where $K : \mathsf{Y}_x \rightsquigarrow \bar{\mathsf{Y}}_x$ is a Markov kernel that does not depend on the law parameter, satisfies $K(y, \bar{\mathsf{Y}}_x) = 1$ for every y, maps $\mathcal{B}_x$ smoothly into $\bar{\mathcal{B}}_x$, and is equivariant, $K_\star \circ \widehat{\rho}_x(g) = \widehat{\bar{\rho}}_x(g) \circ K_\star$ for every g; no common dominating measure is assumed. The vertical Fisher defect and the Amari defect of $\mathscr{K}$ are

$$\Delta_F^{\mathscr{K}} = g_x^F - (T\mathscr{K})^* \bar{g}_x^F, \tag{7.13}$$

$$\Delta_T^{\mathscr{K}} = T_x^F - (T\mathscr{K})^* \bar{T}_x^F. \tag{7.14}$$

Equivariance makes the frame representative independent of u, since $\bar{\iota}_{u\cdot g}^{-1} \mathscr{K} \iota_{u\cdot g} = \widehat{\bar{\rho}}_x(g)^{-1} K_\star \widehat{\rho}_x(g)$, and by proposition 7.3 both defects are tensors on $(\mathcal{E}_x)_{c_*}$.

Restriction to a submodel, precomposition of an energy with a map, a kernel that depends on the law parameter, a fitted coarse approximation, and a comparison across a mismatch of generative and recognition parameters are Markov coarse morphisms, and fall under the results below, only when one kernel that realizes them is exhibited, along the whole history when histories are compared. One kernel serves every law on one fixed-precision fiber of a Gaussian family; Supplement G shows that Gaussian Galerkin aggregation has this form on each fixed-precision fiber and no such form across precisions.

Corollary 7.11 (Vertical Fisher defect). For a Markov coarse morphism $\mathscr{K}$ and every tangent vector Z at p, under the joint law $p \otimes K$,

$$\Delta_F^{\mathscr{K}} \succeq 0, \tag{7.15}$$

$$\Delta_F^{\mathscr{K}}(Z, Z) = \mathbb{E}\,\mathrm{Var}\left(\mathrm{s}_Z(Y) \mid \bar{Y}\right), \tag{7.16}$$

and $\Delta_F^{\mathscr{K}}(Z, Z) = 0$ exactly when $\mathrm{s}_Z(Y)$ is $\bar{Y}$-measurable. [ESTABLISHED]

Proof. In a frame $\mathscr{K}$ is $K_\star$. For a smooth curve p_t in $\mathcal{B}_x$ with velocity Z at p, theorem 7.8 gives the image curve $K_\star p_t$ the score $\mathbb{E}[\mathrm{s}_Z(Y) \mid \bar{Y}]$, which by uniqueness of the quadratic-mean score is the score of $\bar{\mathcal{B}}_x$ in the direction $TK_\star Z$, with second moment $\bar{g}^F_x(TK_\star Z, TK_\star Z)$; the defect identity (7.12) for this one-parameter family is (7.16), with the stated equality clause. □

Theorem 7.12 (Vertical Fisher defect cocycle). Let $\mathscr{K}_{01} : (\mathcal{E}_0)_{c_*} \to (\mathcal{E}_1)_{c_*}$ and $\mathscr{K}_{12} : (\mathcal{E}_1)_{c_*} \to (\mathcal{E}_2)_{c_*}$ be smooth maps between fibers carrying Fisher tensors g^F_0, g^F_1, g^F_2. Then, unconditionally,

$$\Delta_F^{\mathscr{K}_{12}\circ\mathscr{K}_{01}} = \Delta_F^{\mathscr{K}_{01}} + (T\mathscr{K}_{01})^* \Delta_F^{\mathscr{K}_{12}}, \tag{7.17}$$

and the Amari defects obey the same identity. If both maps are Markov coarse morphisms, both terms of (7.17) are positive semidefinite. No nonzero Amari defect is sign definite, since $\Delta_T^{\mathscr{K}}(-Z,-Z,-Z) = -\Delta_T^{\mathscr{K}}(Z,Z,Z)$; Markov coarse-graining therefore gives no monotonicity for T^F, and $\Delta_T^{\mathscr{K}} = 0$ is an additional property of $\mathscr{K}$. [ESTABLISHED]

Proof. Contravariance of pullback gives $\Delta_F^{\mathscr{K}_{12}\circ\mathscr{K}_{01}} = g^F_0 - (T\mathscr{K}_{01})^* g^F_1 + (T\mathscr{K}_{01})^*[g^F_1 - (T\mathscr{K}_{12})^* g^F_2]$, which is (7.17), and the same computation applies to T^F. Positivity is corollary 7.11 for each Markov arrow, and oddness is homogeneity of degree three. □

7.4 Configuration manifolds and the exact free-energy lift

Definition 7.13 (Configuration manifold at c_*). The configuration manifold $\mathcal{Q}_i$ of agent i at c_* is the product $(\mathcal{E}_b)_{c_*} \times (\mathcal{E}_m)_{c_*}$ of its belief and model law fibers, identified with $\mathcal{B}_b \times \mathcal{B}_m$ through its frame u_i, or an embedded submanifold of that product (a regular submodel), with points $Q_i = (q^b_i, q^m_i)$. The collective configuration manifold is $\mathcal{Q}_V = \prod_{i\in V} \mathcal{Q}_i$, extended by further coordinate manifolds, such as open simplices of attention rows, when those coordinates move. A configuration manifold carries a metric G that must be specified. The channel-weighted metric of agent i is $\mathsf{G}_i = w_b g^F_b \oplus w_m g^F_m$ with positive weights, under which a C^1 curve $Q_i(t)$ has squared speed

$$\|\dot{Q}_i\|^2_{\mathsf{G}_i} = w_b g^F_b(\dot{q}^b_i, \dot{q}^b_i) + w_m g^F_m(\dot{q}^m_i, \dot{q}^m_i); \tag{7.18}$$

on $\mathcal{Q}_V$ the metric is either block diagonal, $\bigoplus_i \mathsf{G}_i$, or the pullback of a joint Fisher metric along an exact lift (definition 7.17).

These metrics are frame independent by proposition 7.3; their base-direction version is in Supplement B.

Theorem 7.14 (Fisher radical and quotient of a pulled-back tensor). Let $\mathrm{ret} : \mathcal{Q} \to \mathcal{Q}'$ be a smooth map of finite-dimensional manifolds, g a positive-semidefinite symmetric tensor on $\mathcal{Q}'$, and $h = \mathrm{ret}^*\, g$. Then

$$\operatorname{rad} h_Q = (d\,\mathrm{ret}_Q)^{-1} \left(\operatorname{rad} g_{\mathrm{ret}(Q)}\right), \tag{7.19}$$

so $\operatorname{rad} h_Q = \ker d\,\mathrm{ret}_Q$ exactly when $\operatorname{im} d\,\mathrm{ret}_Q \cap \operatorname{rad} g_{\mathrm{ret}(Q)} = \{0\}$. Where $\operatorname{rad} h$ has constant rank and h is C^1, $\operatorname{rad} h$ is a C^1 subbundle and h induces a positive-definite form on the quotient bundle

$T\mathcal{Q}/\operatorname{rad} h$. A quotient manifold additionally requires the null leaves to form a Hausdorff manifold leaf space; when $\operatorname{rad} h = \ker d\,\mathrm{ret}$ and the connected leaves form such a space $\mathcal{Q}/\!\sim$ with quotient projection a smooth submersion, h is the pullback of a positive-definite tensor on $\mathcal{Q}/\!\sim$. For a positive C^3 family on a finite set, with ret the law map into the open simplex and g its Fisher metric, h is the Fisher tensor of the family, of class C^2, positive definite exactly where ret is an immersion, and there ret is locally injective and h has a C^1 Levi-Civita connection. For softmax logits $z \in \mathbb{R}^k$ with $p = \mathrm{softmax}(z)$, $h(z) = \mathrm{diag}(p) - pp^\top$ has radical $\mathbb{R}\mathbf{1}$ at every z, and for an objective that depends on z only through p the solutions Z of $hZ = \nabla\mathcal{F}$ form one class $Z_0 + \mathbb{R}\mathbf{1}$, whose Moore–Penrose representative $h^+\nabla\mathcal{F}$ is Euclidean-orthogonal to $\mathbf{1}$. [ESTABLISHED]

The proof is in appendix B.2.

Open Problem 7.15 (Global Fisher quotients and canonical block data). Theorem 7.14 gives a positive-definite quotient bundle on every constant-rank stratum and a quotient manifold once a Hausdorff leaf space is supplied, and proposition 12.49 shows that the law and its Fisher tensor alone select no splitting into agent blocks. Open are a criterion, checkable on the configuration manifolds of this chapter, under which the null foliation of a pulled-back Fisher tensor has a Hausdorff manifold leaf space, and block data on that quotient determined by a stated typing of agents and channels. Such a criterion with a proof, for instance exhibiting the leaves as the orbits of a free and proper group action, together with a construction of natural block data from the typing or a counterexample showing that the typing does not determine it, settles the problem.

Hypothesis 7.16 (Standing configuration tier). Wherever this chapter asserts a natural-gradient vector field or a Fisher duration on a configuration manifold $\mathcal{Q}$, either $\mathcal{Q}$ is a finite-dimensional configuration manifold as in definition 7.13, a finite product of regular statistical manifolds or an embedded submodel of one, with a C^1 positive-definite metric G, or $\mathcal{Q}$ is the quotient of such a manifold by the null leaves of a C^1 positive-semidefinite Fisher tensor of constant rank with a Hausdorff manifold leaf space, as in theorem 7.14, and the objective is constant on those leaves. The objective $\mathcal{F}$ is C^2, and every minimal regular exponential family with its Fisher metric is an instance. The hypothesis excludes spaces of sections over a base, which belong to Supplement B, semidefinite metrics without a constructed quotient, and configuration metrics called exact joint-law Fisher metrics without the lift of definition 7.17.

Definition 7.17 (Exact multi-agent free-energy lift). Fix a block B of latent coordinates, an outside recognition marginal $\mathbb{Q}_{B^c}$, and a manifold $\mathscr{R}_B$ of admissible conditional recognition kernels $r_B(dy_B \mid y_{B^c})$. An exact lift of the configuration manifold $\mathcal{Q}_i$ consists of a smooth extraction map $\mathrm{ext}_i : \mathscr{R}_B \to \mathcal{Q}_i$ and a smooth right inverse $\iota_i : \mathcal{Q}_i \to \mathscr{R}_B$, $Q_i \mapsto r_B^{Q_i}$, $\mathrm{ext}_i \circ \iota_i = \mathrm{id}_{\mathcal{Q}_i}$, such that, for every Q_i, the support, finiteness, and integrability hypotheses of definition 5.21 and theorem 5.23 hold for $\mathbb{Q}_{B^c} r_B^{Q_i}$, the lifted family is C^2, and for every chart ch of $\mathcal{Q}_i$ with coordinates λ and every compact set C in its image there is $\mathrm{e}_{\mathsf{C}} \in L^1(\mathbb{Q}_{B^c})$ with, for jointly measurable versions of the

derivatives,

$$\sup_{\lambda\in\mathsf{C}}\left|\partial_\lambda^\alpha \mathcal{F}^{\mathrm{ext}}_{B,o}(r_B^{\mathrm{ch}^{-1}(\lambda)}; y_{B^c})\right| \le \mathrm{e}_{\mathsf{C}}(y_{B^c}) \quad \text{for } \mathbb{Q}_{B^c}\text{-almost every } y_{B^c} \text{ and } |\alpha|\le 2. \tag{7.20}$$

The lifted objective is the outside-averaged local free energy

$$\overline{\mathcal{F}}_{B,o}(Q_i) = \mathbb{E}_{\mathbb{Q}_{B^c}}\left[\mathcal{F}^{\mathrm{ext}}_{B,o}\left(r_B^{Q_i}; Y_{B^c}\right)\right], \tag{7.21}$$

a history of it is an exact free-energy history, and the exact recognition Fisher metric of $\mathcal{Q}_i$ is $\mathsf{G}_i = \iota_i^* \mathsf{G}_{\mathscr{R}_B}$ on a nondegenerate stratum, or its quotient as in theorem 7.14.

The lift carries every correlation that the configuration does not determine, and $\mathrm{ext}_i \circ \iota_i = \mathrm{id}$ makes it a lift rather than a family of laws indexed by configurations. With the outside marginal held fixed, (7.21) differs from the collective free energy along $Q_i \mapsto \mathbb{Q}_{B^c} r_B^{Q_i}$ by a term independent of Q_i (theorem 5.23), so the two have the same differential and, for one metric, the same natural-gradient ray.

Proposition 7.18 (Joint-versus-weighted-marginal score residual and lift dependence). Let P_λ be a regular dominated family of joint laws of blocks $(Y_1,\dots,Y_n)$ with square-integrable joint score $\mathbf{s} = \nabla_\lambda \log p_\lambda(Y)$, embedded marginal scores $\mathbf{s}_i = \mathbb{E}[\mathbf{s} \mid Y_i]$, and scalar weights w_i, and define

$$\mathsf{G}_{\mathrm{J}} = \mathbb{E}[\mathbf{s}\mathbf{s}^\top], \qquad \mathsf{G}_i = \mathbb{E}[\mathbf{s}_i\mathbf{s}_i^\top], \qquad \mathsf{G}_w = \sum_i w_i \mathsf{G}_i, \tag{7.22}$$

$$\mathbf{r} = \mathbf{s} - \sum_i \mathbf{s}_i, \qquad \mathfrak{K} = \mathbb{E}[\mathbf{r}\mathbf{r}^\top], \qquad \mathfrak{C} = \sum_{i\neq j} \mathbb{E}[\mathbf{s}_i\mathbf{s}_j^\top]. \tag{7.23}$$

The cross tensor $\mathfrak{C}$ is symmetric and can be indefinite,

$$\mathsf{G}_{\mathrm{J}} - \mathsf{G}_w = \mathfrak{K} - \mathfrak{C} + \sum_i (1-w_i)\mathsf{G}_i, \tag{7.24}$$

and on a tangent subspace $\mathcal{T}$, $\mathsf{G}_{\mathrm{J}} = \mathsf{G}_w$ holds exactly when the right side of (7.24) vanishes on $\mathcal{T}\times\mathcal{T}$. For two blocks and unit weights, in a direction Z with scores $\mathbf{s}_Z = Z^\top\mathbf{s}$ and $\mathbf{s}_{i,Z} = Z^\top\mathbf{s}_i$,

$$\|\mathbf{s}_Z\|^2 - \|\mathbf{s}_{1,Z}\|^2 - \|\mathbf{s}_{2,Z}\|^2 = \|\mathbf{s}_Z - \mathbf{s}_{1,Z} - \mathbf{s}_{2,Z}\|^2 - 2\langle \mathbf{s}_{1,Z}, \mathbf{s}_{2,Z}\rangle, \tag{7.25}$$

whose first term is nonnegative and whose second is signed. If the two blocks are independent under every P_λ, then $\mathsf{G}_{\mathrm{J}} = \mathsf{G}_1 + \mathsf{G}_2$, and $\mathsf{G}_{\mathrm{J}} = w_1\mathsf{G}_1 + w_2\mathsf{G}_2$ forces $w_1 = 1$ as soon as some direction has $\mathsf{G}_1(Z,Z) > 0 = \mathsf{G}_2(Z,Z)$, and symmetrically for w_2. No Loewner order between G_{J} and $\mathsf{G}_1+\mathsf{G}_2$ holds in general: for the bivariate normal location family with precision $\left(\begin{smallmatrix}1 & \mathrm{c}\\ \mathrm{c} & 1\end{smallmatrix}\right)$ and $0<\mathrm{c}<1$, the difference is $\left(\begin{smallmatrix}\mathrm{c}^2 & \mathrm{c}\\ \mathrm{c} & \mathrm{c}^2\end{smallmatrix}\right)$, with value $2\mathrm{c}(1+\mathrm{c})>0$ along $(1,1)$ and $2\mathrm{c}(\mathrm{c}-1)<0$ along $(1,-1)$. Distinct right inverses of one extraction map can induce distinct pullback metrics: for $\mathcal{N}((\lambda_1,\lambda_2),\mathrm{I}_2)$, with Fisher metric $d\lambda_1^2 + d\lambda_2^2$, and the extraction $(\lambda_1,\lambda_2)\mapsto\lambda_1$, the lifts $\lambda_1\mapsto(\lambda_1,0)$ and $\lambda_1\mapsto(\lambda_1,1)$ induce $d\lambda_1^2$, whereas $\lambda_1\mapsto(\lambda_1,\lambda_1)$ induces $2d\lambda_1^2$. [ESTABLISHED]

The proof is in appendix B.2.

7.4.1 Canonicity boundary of the lift

A lift supplies the dependence that the marginal configurations do not determine. The following finite results show that naturality under local kernels forces the product law and that admitting correlated refinements leaves no natural selector.

Definition 7.19 (Finite marginal selection problem). For a finite nonempty typed list $\mathcal{X} = (\mathcal{X}_i)_{i=1}^n$ of finite sets, let $\mathcal{J}_\mathcal{X} = \Delta(\prod_i \mathcal{X}_i)$, $\mathcal{J}^1_\mathcal{X} = \prod_i \Delta(\mathcal{X}_i)$, and $\mathrm{mar}_\mathcal{X} : \mathcal{J}_\mathcal{X} \to \mathcal{J}^1_\mathcal{X}$ the map that extracts the one-coordinate marginals. A selector is a map $\mathrm{sel}_\mathcal{X} : \mathcal{J}^1_\mathcal{X} \to \mathcal{J}_\mathcal{X}$ with $\mathrm{mar}_\mathcal{X} \circ \mathrm{sel}_\mathcal{X} = \mathrm{id}$. A local morphism is a tensor product $K = \bigotimes_i K_i$ of finite Markov kernels of the same arity, and a kernel $K : \prod_i \mathcal{X}_i \rightsquigarrow \prod_j \mathcal{Y}_j$ that changes arity is marginal compatible when $\mathrm{mar}_\mathcal{Y} K_\star = \bar{K} \mathrm{mar}_\mathcal{X}$ for some map $\bar{K} : \mathcal{J}^1_\mathcal{X} \to \mathcal{J}^1_\mathcal{Y}$. A family of selectors is natural on a class of morphisms when $K_\star \mathrm{sel}_\mathcal{X} = \mathrm{sel}_\mathcal{Y} \bar{K}$ for every morphism in the class; the class is part of the data, and not every finite kernel is admitted.

Theorem 7.20 (Tensor-local naturality selects the product law). If a family of selectors is natural under every local morphism, then $\mathrm{sel}_\mathcal{X}(\mu_1, \dots, \mu_n) = \bigotimes_{i=1}^n \mu_i$. Among classes of marginal-compatible kernels that contain every local morphism, a natural selector exists exactly when every admitted kernel sends product laws to the product of their image marginals, and then it is the product selector; the product-preserving marginal-compatible kernels therefore form the largest such class. [ESTABLISHED]

Proof. For each i let the kernel from a one-point set send its point to μ_i. Their tensor product sends the unique law on the one-point product to $\bigotimes_i \mu_i$, while the induced marginal map sends the unique marginal tuple to $(\mu_i)_i$, so naturality forces the product selector. With it forced, naturality for an added kernel says exactly that it sends each product law to the product of its image marginals. □

Theorem 7.21 (Admitted correlated refinements refute an absolute selector). For a source bit y and target bits (y', y''), admit the split kernels $K_\mathrm{c}^{\mathrm{split}}(y', y'' \mid y) = \mathbf{1}_{\{y'=y\}}(1+\mathrm{c})/2$ if $y'' = y$ and $\mathbf{1}_{\{y'=y\}}(1-\mathrm{c})/2$ if $y'' \neq y$, for $\mathrm{c} \in \{1/3, 1/2\}$. Both are marginal compatible, and no single-valued selector is natural under both; the argument uses no local naturality. In particular marginalization is not injective on positive laws, so no selector returns every joint law from its marginals. [ESTABLISHED]

Proof. The two kernels push the fair source law to laws with fair marginals and atom multisets $\{1/3, 1/3, 1/6, 1/6\}$ and $\{3/8, 3/8, 1/8, 1/8\}$. Naturality would require the value of the target selector at the fair marginal pair to equal both laws, which differ even after relabeling, and the two positive laws share their marginals. □

Selection relative to a positive reference law is a different problem. A finite-relative-entropy minimizer over a convex family of moment-constrained laws, when it exists, is unique by strict convexity of relative entropy. Both that minimizer and, for a deterministic coarse map, the reference conditional

completion of a prescribed coarse marginal (lemma 3.12) depend on the reference law as additional data.

7.5 A finite correlated lift: the parity witness

Definition 7.22 (Three-agent parity lift). Let $\mathcal{I}_6 = \{1,2,3\} \times \{b,m\}$, $y = (y_{i,x}) \in \{0,1\}^{\mathcal{I}_6}$, and $\lambda = (\lambda_{i,x}) \in (0,1)^{\mathcal{I}_6}$, where typed agent permutations preserve the channel label, and set

$$p_0(y;\lambda) = \prod_{(i,x)} \lambda_{i,x}^{y_{i,x}} (1-\lambda_{i,x})^{1-y_{i,x}}, \qquad \mathrm{D}(\lambda) = \prod_{(i,x)} \lambda_{i,x}(1-\lambda_{i,x}), \tag{7.26}$$

$$p_\varkappa(y;\lambda) = p_0(y;\lambda) + \varkappa\,\mathrm{par}(y)\mathrm{D}(\lambda), \qquad \mathrm{par}(y) = (-1)^{\sum_{(i,x)} y_{i,x}}, \qquad |\varkappa| < 1. \tag{7.27}$$

The map $\iota_\varkappa : (0,1)^6 \to \Delta_{63}^\circ$ sends λ to $p_\varkappa(\cdot;\lambda)$. The parity amplitude $\varkappa$ is fixed data of the lift, not an inferred coordinate, and its hyperedge correlation is $\bar{\varkappa} = \varkappa/64$.

Theorem 7.23 (Positive smooth lift with paired relabeling symmetry). For every $|\varkappa| < 1$, the law (7.27) is strictly positive and normalized, and every proper marginal is the corresponding marginal of p_0. Consequently $\iota_\varkappa$ is a smooth right inverse of the map that extracts the six one-bit marginals, and it is an immersion. For each marginal tuple λ, the positive laws with those marginals form a 57-dimensional set, of which varying $\varkappa$ traces one line. Let $H_{\mathrm{pair}} = (\mathbb{Z}_2)^3 \rtimes S_3$ act by complementing both bits of a selected agent and by typed agent permutations, and act on λ by the same complements and permutations. Then

$$\iota_\varkappa(g\lambda) = g_\# \iota_\varkappa(\lambda) \qquad \text{for every } g \in H_{\mathrm{pair}}, \tag{7.28}$$

whereas the complement $F_{i,x}$ of one bit satisfies

$$(F_{i,x})_\# p_\varkappa(\cdot;\lambda) = p_{-\varkappa}(\cdot; F_{i,x}\lambda). \tag{7.29}$$

Odd complements are therefore equivariant only if $\varkappa$ changes sign with them, or if $\varkappa = 0$. [ESTABLISHED]

The proof is in appendix B.2.

Corollary 7.24 (Agent-block local–global free-energy identity). Fix an agent block $B_i = \{(i,b),(i,m)\}$ and the four outside parameters $\lambda_{B_i^c}$, and let $r_{i,\lambda_{B_i}} = p_\varkappa(y;\lambda)/p_{0,B_i^c}(y_{B_i^c};\lambda_{B_i^c})$ be the conditional law of y_{B_i} given $y_{B_i^c}$. For any interior block parameters λ_{B_i} and λ'_{B_i} and any strictly positive posterior $\mathbf{\Pi}_o$ on the Boolean cube, all relative entropies are finite, and for every block

tangent Z

$$\mathcal{F}_o^{\mathrm{ext}}\big(p_\varkappa(\cdot;\lambda'_{B_i},\lambda_{B_i^c})\big)-\mathcal{F}_o^{\mathrm{ext}}\big(p_\varkappa(\cdot;\lambda_{B_i},\lambda_{B_i^c})\big)$$
$$=\mathbb{E}_{p_{0,B_i^c}}\big[\mathcal{F}_{B_i,o}^{\mathrm{ext}}(r_{i,\lambda'_{B_i}};Y_{B_i^c})-\mathcal{F}_{B_i,o}^{\mathrm{ext}}(r_{i,\lambda_{B_i}};Y_{B_i^c})\big], \tag{7.30}$$
$$d\mathcal{F}_o^{\mathrm{ext}}[Z]=\mathbb{E}_{p_{0,B_i^c}}\big[d\mathcal{F}_{B_i,o}^{\mathrm{ext}}[Z]\big], \tag{7.31}$$

with both differentials evaluated at the same lifted conditional law. The outside-averaged block free energy is thus the restriction of one collective free energy up to a constant. [ESTABLISHED]

Proof. By theorem 7.23 the outside marginal is exactly p_{0,B_i^c}, independent of λ_{B_i}, and strict positivity makes $r_{i,\lambda_{B_i}}$ a positive conditional law. The finite state space and the positive posterior make every relative entropy finite, so (7.30) is theorem 5.23 with this outside marginal, and differentiating the finite smooth identity gives (7.31). □

Corollary 7.25 (Parity-family Fisher residual). For the family of definition 7.22, every pair of bits is independent, so $\mathfrak{C}=0$, and with unit weights $\mathsf{G}_{\mathrm{J}}-\sum_{(i,x)\in\mathcal{I}_6}\mathsf{G}_{i,x}=\mathfrak{K}\succ 0$ when $\varkappa\neq 0$. At $\lambda=\frac{1}{2}\mathbf{1}$, and there with $\varkappa=\frac{1}{2}$,

$$\mathsf{G}_{\mathrm{J}}=\frac{4}{1-\bar{\varkappa}^2}\mathrm{I}_6,\qquad \mathsf{G}_{\mathrm{J}}-\sum_{(i,x)\in\mathcal{I}_6}\mathsf{G}_{i,x}=\frac{4\bar{\varkappa}^2}{1-\bar{\varkappa}^2}\mathrm{I}_6, \tag{7.32}$$
$$\mathsf{G}_{\mathrm{J}}=\frac{65536}{16383}\mathrm{I}_6,\qquad \mathsf{G}_{\mathrm{J}}-\sum_{(i,x)\in\mathcal{I}_6}\mathsf{G}_{i,x}=\frac{4}{16383}\mathrm{I}_6, \tag{7.33}$$

and at the center a weighted marginal metric equals the joint one on the full tangent space exactly when

$$w_{i,x}=\frac{1}{1-\bar{\varkappa}^2}\qquad\text{for every } (i,x)\in\mathcal{I}_6, \tag{7.34}$$

so unit weights are exact only at $\varkappa=0$. Since the one-bit marginal families do not depend on $\varkappa$, no rule that reads only them returns every joint Fisher tensor. [ESTABLISHED]

The proof is in appendix B.2.

Corollary 7.26 (Promoted parity has one hidden direction). For the seven-parameter family $(\lambda,\varkappa)\mapsto p_\varkappa(\cdot;\lambda)$ on $(0,1)^6\times(-1,1)$, the derivative of the joint law has rank seven everywhere, while the one-bit marginal map has rank six and

$$\ker d\,(\mathrm{mar}\circ p)=\mathrm{span}\{\partial_\varkappa\}. \tag{7.35}$$

Thus $\varkappa$ is identifiable from the joint law and invisible to the one-bit marginals. [ESTABLISHED]

Proof. The one-bit success probabilities are exactly λ, so a vanishing joint derivative has zero λ-component, and the remaining derivative is a multiple of $\partial_\varkappa p_\varkappa=\mathrm{par}\,\mathrm{D}$, which is nowhere zero on the open domain; hence the joint derivative is injective. The same identity gives rank six for the marginal map and (7.35). □

At the center $\lambda = \frac{1}{2}\mathbf{1}$, where $\mathrm{D} = 4^{-6}$, the lift (7.27) is

$$p_{\varkappa}\big(y; \tfrac{1}{2}\mathbf{1}\big) = \tfrac{1}{64}\,(1 + \bar{\varkappa}\,\mathrm{par}(y))\,. \tag{7.36}$$

Definition 7.27 (Three-agent parity hyperedge). At the center (7.36), take the uniform latent prior $\mathrm{Unif}(y) = 1/64$ and the binary record kernel

$$K_{\mathrm{hyp}}(1 \mid y) = \tfrac{1}{2}\,(1 + \bar{\varkappa}\,\mathrm{par}(y))\,, \qquad K_{\mathrm{hyp}}(0 \mid y) = \tfrac{1}{2}\,(1 - \bar{\varkappa}\,\mathrm{par}(y))\,. \tag{7.37}$$

Proposition 7.28 (Exact posterior and tight free energy for the hyperedge). The record $o = 1$ has evidence $\frac{1}{2}$, and its posterior is exactly $p_{\varkappa}(\cdot; \frac{1}{2}\mathbf{1})$. For the free energy $\mathcal{F}_1(Q) = D_{\mathrm{KL}}(Q \| \,\mathrm{Unif}) - \mathbb{E}_Q \log K_{\mathrm{hyp}}(1 \mid Y)$ the posterior attains the tight value $\mathcal{F}_1 = \log 2$, the product law of its one-bit marginals is Unif, and

$$\mathcal{F}_1(\mathrm{Unif}) - \log 2 = -\tfrac{1}{2}\log(1 - \bar{\varkappa}^2). \tag{7.38}$$

Consequently, when $\varkappa \neq 0$, no function of the one-bit marginals of the recognition law, the one-bit marginals of the posterior, and the evidence equals the free energy on every recognition law. [ESTABLISHED]

The proof is in appendix B.2.

Open Problem 7.29 (Scope beyond the finite parity witness). Theorem 7.23 gives one smooth correlated right inverse of the one-bit marginal map, equivariant under paired complements and typed permutations, and theorem 7.21 refutes an absolute selector on the admitted split kernels, so that question is closed. Open are the existence, for a stated regular family of agent laws and a stated class of morphisms that contains non-product-preserving kernels, of a smooth right inverse natural under that class, and the existence of correlated lifts equivariant under a connected structure group acting on continuous law fibers. A construction with proof for a stated family and class, or a no-go theorem for it, settles each part.

7.6 Natural-gradient orbits

Definition 7.30 (Oriented unparameterized history and natural-gradient history). A representative of a history in a configuration manifold $\mathcal{Q}$ is a C^1 curve $Q : \mathcal{T} \to \mathcal{Q}$ on an open interval with nowhere-vanishing velocity, and two representatives define the same oriented history when

$$\widetilde{Q} = Q \circ \phi, \qquad \phi : \widetilde{\mathcal{T}} \longrightarrow \mathcal{T} \text{ an orientation-preserving } C^1 \text{ diffeomorphism.} \tag{7.39}$$

The resulting abstract oriented curve $\mathscr{H}$ comes with its traversal map $\jmath : \mathscr{H} \to \mathcal{Q}$, which retains order and multiplicity; for an embedded representative $\mathscr{H}$ is its oriented image. Under hypothesis 7.16 the natural gradient of $\mathcal{F}$ is defined by the first line below, the descent ray at a noncritical Q

is $\mathcal{R}^-_{\mathcal{F}}(Q) = \{-\mathsf{a}\,\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}(Q) : \mathsf{a} > 0\}$, and a natural-gradient history is an oriented history whose representatives satisfy the second:

$$\mathsf{G}\big(\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}, Z\big) = d\mathcal{F}[Z] \quad \text{for every } Z \in T\mathcal{Q}, \tag{7.40}$$

$$\dot{Q}(t) = -\mathsf{a}(t)\,\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}(Q(t)), \qquad \mathsf{a}(t) > 0. \tag{7.41}$$

When $\mathcal{F}$ is the lifted objective of definition 7.17, such a history is an exact free-energy history.

Proposition 7.31 (Scalar mobility preserves the oriented orbit). On a noncritical domain, multiplying the natural-gradient field by a smooth positive function changes only the parameterization of its orbits. A positive-definite anisotropic mobility can change the orbit: on the Euclidean plane, for $\mathcal{F} = \frac{1}{2}(z_1^2 + z_2^2)$ and $\mathsf{M} = \mathrm{diag}(1, 2)$, gradient descent from (z_1^0, z_2^0) with $z_1^0 z_2^0 \neq 0$ follows the line $z_2 = (z_2^0/z_1^0)z_1$, whereas $\dot{z} = -\mathsf{M}\nabla\mathcal{F}$ follows

$$z_2 = z_2^0 \left(z_1/z_1^0\right)^2. \tag{7.42}$$

A natural-gradient history is not in general a geodesic: on the same Euclidean plane the natural-gradient flow of $\mathcal{F} = \frac{1}{2}(z_1^2 + 2z_2^2)$ is $\dot{z}_1 = -z_1$, $\dot{z}_2 = -2z_2$, whose orbits are the curves (7.42), while the geodesics are straight lines. [ESTABLISHED]

Proof. If $\mathscr{V} = -\,\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}$ and $\mathscr{V}' = \mathsf{a}\mathscr{V}$ with $\mathsf{a} > 0$, the integral curves have the same tangent lines and orientation, and the change of parameter defined by a positive integral of a^{-1} carries one equation into the other. In the example, $\dot{z}_1 = -z_1$ and $\dot{z}_2 = -2z_2$ keep z_2/z_1^2 constant, which is (7.42), while ordinary gradient descent keeps z_2/z_1 constant, and for $z_1^0 z_2^0 \neq 0$ the images differ; the same equations are the Euclidean natural-gradient flow of $\frac{1}{2}(z_1^2 + 2z_2^2)$. □

A rate ratio between coordinate blocks, such as fast beliefs and slow models, is an anisotropic mobility (the example is the ratio $1 : 2$), so the selected orbit can depend on it; the rate-separated flow is in section 8.2.

Proposition 7.32 (Noncritical free-energy orbits are embedded). Let $\mathcal{Q}$ be finite-dimensional, $\mathcal{F}$ of class C^2, and M a C^1 field of positive-definite symmetric forms on the cotangent spaces, for instance $\mathsf{M} = \mathsf{a}\mathsf{G}^{-1}$ for a C^1 metric and a C^1 function $\mathsf{a} > 0$. Let γ be a maximal integral curve of $\mathscr{V} = -\mathsf{M}(d\mathcal{F})$ through a noncritical point. Then γ meets no critical point, $\mathcal{F} \circ \gamma$ is strictly decreasing, and γ is an injective immersion that is a homeomorphism onto its image, an embedded C^1 curve on which $\mathcal{F}$ is a strictly monotone continuous coordinate, independent of parameterization and gauge invariant whenever $\mathcal{F}$ is. In particular the multiplicity clause of definition 7.30 is vacuous for noncritical histories, and for the natural gradient the Fisher clock of definition 7.35 satisfies $d\mathfrak{t} = -d\mathcal{F}/\|\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}\|_{\mathsf{G}}$ along the orbit. [ESTABLISHED]

Proof. The field $\mathscr{V}$ is C^1, so its integral curves are unique, and its zeros are the critical points of $\mathcal{F}$; since the constant curve at a zero is an integral curve, a curve through a nonzero of $\mathscr{V}$ never reaches one. Along γ, $\frac{d}{dt}\mathcal{F}(\gamma) = -\mathsf{M}(d\mathcal{F}, d\mathcal{F}) < 0$, which makes $\mathcal{F} \circ \gamma$ strictly decreasing,

γ injective, and $\dot\gamma \neq 0$. On the image the inverse of γ is $Q \mapsto (\mathcal{F}\circ\gamma)^{-1}(\mathcal{F}(Q))$, a composite of continuous maps, so γ is an embedding. The clock relation is (7.52). □

Proposition 7.33 (A semidefinite tensor cannot be silently inverted). If the proposed metric on a configuration space is only semidefinite, the natural-gradient equation can have no solution or several. [ESTABLISHED]

Proof. On $\mathbb{R}^2$ let $\mathsf{G} = dz_1^2$. For $\mathcal{F} = z_2$ the equation $\mathsf{G}(\operatorname{grad}\mathcal{F}, \cdot) = dz_2$ has no solution because the left side annihilates ∂_{z_2}, and for $\mathcal{F} = z_1$ every $\partial_{z_1} + \mathrm{c}\partial_{z_2}$ solves it. □

A quotient constructed as in theorem 7.14, or a separate regularization, must precede any use of an inverse or a pseudoinverse.

Remark 7.34 (Discrete histories). In a chart where the metric has a constant coefficient matrix G, the explicit Euler step of (7.41) with $\mathsf{a} \equiv 1$ and step $\Delta t > 0$ is $Q_{n+1} = Q_n - \Delta t \mathsf{G}^{-1}\nabla\mathcal{F}(Q_n)$. If $\nabla^2\mathcal{F} \preceq \mathrm{c}_\mathsf{G}\mathsf{G}$ on the segment $[Q_n, Q_{n+1}]$, with $\mathrm{c}_\mathsf{G} > 0$, then

$$\mathcal{F}(Q_{n+1}) \leq \mathcal{F}(Q_n) - \Delta t\left(1 - \frac{\mathrm{c}_\mathsf{G}\Delta t}{2}\right)\left\|\operatorname{grad}^\mathsf{G}\mathcal{F}(Q_n)\right\|_\mathsf{G}^2, \tag{7.43}$$

so every step with $0 < \Delta t < 2/\mathrm{c}_\mathsf{G}$ decreases the objective off the critical set, and the bound is sharp: for a quadratic objective whose Hessian has largest generalized eigenvalue c_G relative to G, an iterate on that eigendirection is multiplied by $1 - \Delta t\mathrm{c}_\mathsf{G}$, so $\Delta t > 2/\mathrm{c}_\mathsf{G}$ increases the objective there. If, in the same chart, $\mathscr{V} = -\mathsf{G}^{-1}\nabla\mathcal{F}$ is Lipschitz with constant $\mathrm{c}_\mathrm{L} > 0$ on the closed δ-neighborhood of an exact arc $Q([0,t_1])$ and $\|\ddot Q\| \leq \mathrm{c}_2$ on the arc, then

$$\|Q_n - Q(n\Delta t)\| \leq \frac{\mathrm{c}_2\Delta t}{2\mathrm{c}_\mathrm{L}}\left(e^{\mathrm{c}_\mathrm{L} n\Delta t} - 1\right), \qquad n\Delta t \leq t_1, \tag{7.44}$$

for every Δt small enough that the right side stays below δ at $n\Delta t = t_1$. Euler iterates follow the selected orbit to first order in Δt on compact parameter intervals without lying on it. [ESTABLISHED]

Proof. Taylor's formula with integral remainder along the segment gives $\mathcal{F}(Q_{n+1}) \leq \mathcal{F}(Q_n) + \nabla\mathcal{F}^\top\Delta Q + \frac{1}{2}\mathrm{c}_\mathsf{G}\Delta Q^\top\mathsf{G}\Delta Q$ with $\Delta Q = -\Delta t\mathsf{G}^{-1}\nabla\mathcal{F}$, which is (7.43); sharpness is the stated eigendirection computation. The local truncation error is at most $\mathrm{c}_2\Delta t^2/2$, so $e_n = \|Q_n - Q(n\Delta t)\|$ obeys $e_{n+1} \leq (1 + \Delta t\mathrm{c}_\mathrm{L})e_n + \mathrm{c}_2\Delta t^2/2$ while the iterates stay in the neighborhood, the discrete Grönwall inequality gives (7.44), and induction on n keeps the iterates in the neighborhood. □

7.7 Fisher duration

Definition 7.35 (Fisher length and information clock). Let $Q : \mathcal{T} \to \mathcal{Q}$ be a C^1 curve in a configuration manifold with metric G. At c_* its velocity lies in the fibers, $\operatorname{ver}(\dot Q) = \dot Q \in T_Q\mathcal{Q}$, so no connection enters. Its Fisher speed, its length on $[t_0, t_1]$, its cumulative clock from a chosen origin

t_0, and, for a curve Γ in one law fiber with its Fisher metric, the vertical Fisher length are

$$\dot{\mathfrak{t}}(t) = \sqrt{\mathsf{G}\big(\dot{Q}(t), \dot{Q}(t)\big)}, \tag{7.45}$$

$$\mathrm{len}_{\mathsf{G}}(Q; [t_0, t_1]) = \int_{t_0}^{t_1} \dot{\mathfrak{t}}(t)\, dt, \tag{7.46}$$

$$\mathfrak{t}_{Q,t_0}(t) = \int_{t_0}^{t} \dot{\mathfrak{t}}(t')\, dt', \tag{7.47}$$

$$\mathrm{len}_{g^F}(\Gamma) = \int \sqrt{g^F\big(\mathrm{ver}(\dot{\Gamma}), \mathrm{ver}(\dot{\Gamma})\big)}\, dt. \tag{7.48}$$

The belief clock and the model clock of agent i are the Fisher lengths of $t \mapsto q_i^b(t)$ and $t \mapsto q_i^m(t)$, combined by the channel-weighted speed (7.18). Along a history of $\mathcal{Q}_V$ with block-diagonal metric $\bigoplus_i \mathsf{G}_i$, the clock $\mathfrak{t}_i$ of agent i is the G_i-length of $t \mapsto Q_i(t)$. The origin is part of the clock, and the parameter t does not enter the length.

Theorem 7.36 (Reparameterization invariance and unit-speed form). Let $\widetilde{Q} = Q \circ \phi$ with $\phi : \widetilde{\mathcal{T}} \to \mathcal{T}$ an orientation-preserving C^1 diffeomorphism. Corresponding subarcs have the same length, and

$$\mathfrak{t}_{\widetilde{Q},\widetilde{t}_0}(\widetilde{t}) = \mathfrak{t}_{Q,t_0}(\phi(\widetilde{t})), \qquad t_0 = \phi(\widetilde{t}_0). \tag{7.49}$$

If $\dot{\mathfrak{t}} > 0$, the clock is strictly increasing and gives the unit-speed representative of the oriented history; on a subinterval where $\dot{\mathfrak{t}}$ vanishes, as on a critical or null segment, the clock is constant and is not a coordinate. If Q satisfies (7.41), then on its noncritical part

$$\frac{d\mathfrak{t}}{dt} = \mathsf{a} \left\| \mathrm{grad}^{\mathsf{G}} \mathcal{F} \right\|_{\mathsf{G}}, \tag{7.50}$$

$$\frac{dQ}{d\mathfrak{t}} = -\frac{\mathrm{grad}^{\mathsf{G}} \mathcal{F}}{\left\| \mathrm{grad}^{\mathsf{G}} \mathcal{F} \right\|_{\mathsf{G}}}, \tag{7.51}$$

$$\frac{d\mathcal{F}}{d\mathfrak{t}} = -\left\| \mathrm{grad}^{\mathsf{G}} \mathcal{F} \right\|_{\mathsf{G}}. \tag{7.52}$$

If the clock is read with a second metric G', for instance the unweighted Fisher metric along a flow whose blocks carry rate weights, then along any C^1 history with velocity $Z \neq 0$

$$\left| \frac{d\mathcal{F}}{d\mathfrak{t}'} \right| = \frac{|d\mathcal{F}(Z)|}{\|Z\|_{\mathsf{G}'}} \leq \left\| \mathrm{grad}^{\mathsf{G}'} \mathcal{F} \right\|_{\mathsf{G}'}, \tag{7.53}$$

with equality exactly when $d\mathcal{F} = 0$ or Z is parallel to $\mathrm{grad}^{\mathsf{G}'} \mathcal{F}$. Along a history of $\mathcal{Q}_V$ with block-diagonal metric the squared speeds add, $\dot{\mathfrak{t}}^2 = \sum_i \dot{\mathfrak{t}}_i^2$, and over any common parameter interval $\max_i \mathfrak{t}_i \leq (\sum_i \mathfrak{t}_i^2)^{1/2} \leq \mathfrak{t} \leq \sum_i \mathfrak{t}_i$. [ESTABLISHED]

The proof is in appendix B.2.

Remark 7.37 (Endpoints do not determine an update duration). For an update Q from $Q(t_0) = Q'$ to $Q(t_1) = Q''$ in one path component of $\mathcal{Q}$, the distance $\mathrm{dist}_{\mathsf{G}}(Q', Q'')$, the infimum of

$\mathrm{len}_{\mathsf{G}}(\Gamma;[t_0,t_1])$ over curves Γ with these endpoints, is at most $\mathrm{len}_{\mathsf{G}}(Q;[t_0,t_1])$, because the realized update is one such curve, and the inequality is strict for a curved path in the Euclidean Fisher geometry of the two-parameter normal location family. Duration is also distinct from the amount of curve parameter: in the unit-variance normal location family the path $\mu(t)=e^{-t}$ has length $\int_0^\infty|\dot\mu(t)|dt=\int_0^\infty e^{-t}dt=1$ over an unbounded parameter interval. [ESTABLISHED]

Proposition 7.38 (Finite duration for analytic objectives). Let $\mathcal{Q}$ be finite-dimensional with a continuous positive-definite metric G, and let $\gamma:[0,\infty)\to\mathcal{Q}$ satisfy $\dot\gamma=-\operatorname{grad}^{\mathsf{G}}\mathcal{F}$. If $\gamma(t)$ has an accumulation point Q_* as $t\to\infty$ and $\mathcal{F}$ is real analytic in a chart around Q_*, then $\gamma(t)\to Q_*$, Q_* is critical, and $\mathrm{len}_{\mathsf{G}}(\gamma;[0,\infty))$ is finite. This holds for every forward history with relatively compact image when $\mathcal{Q}$ has a real-analytic atlas in whose charts $\mathcal{F}$ is real analytic. Analyticity cannot be weakened to smoothness: for the C^∞ function of Absil et al. [1, Section 2, Eq. (8)] on the Euclidean plane, the gradient history from the initial data given there is bounded, has the unit circle as its ω-limit set, and has infinite duration. [ESTABLISHED]

Proposition 7.39 (Invariant free-energy histories are horizontal). Let a Lie group $\mathbb{G}$ act smoothly on a finite-dimensional configuration manifold $\mathcal{Q}$ with a positive-definite metric G, write $\xi_{\mathcal{Q}}(Q)=\frac{d}{dt}\big|_{t=0}\exp(t\xi)\cdot Q$ and $V_Q=\{\xi_{\mathcal{Q}}(Q):\xi\in\operatorname{Lie}\mathbb{G}\}$, and let $\mathcal{F}$ be C^1 and $\mathbb{G}$-invariant. Then $\operatorname{grad}^{\mathsf{G}}\mathcal{F}(Q)$ is G-orthogonal to V_Q at every Q, every natural-gradient history has velocity orthogonal to the orbits, and its gauge-quotient speed $\min_\xi\|\dot Q-\xi_{\mathcal{Q}}(Q)\|_{\mathsf{G}}$ equals its Fisher speed. If G is C^∞ and the action is free, proper, and isometric, the quotient map $\pi:\mathcal{Q}\to\mathcal{Q}/\mathbb{G}$ is a Riemannian submersion for a unique metric $\bar{\mathsf{G}}$ on the quotient, $\mathcal{F}=\bar{\mathcal{F}}\circ\pi$, and $T\pi\operatorname{grad}^{\mathsf{G}}\mathcal{F}=\operatorname{grad}^{\bar{\mathsf{G}}}\bar{\mathcal{F}}\circ\pi$; each natural-gradient history descends to a natural-gradient history of $\bar{\mathcal{F}}$ with the same Fisher duration. [ESTABLISHED]

The proofs of propositions 7.38 and 7.39 are in appendix B.2.

7.8 Orthogonal clocks

Theorem 7.40 (Exact criterion for an orthogonal clock). Let $\mathcal{U}\subseteq\mathcal{Q}$ be a connected open set without critical points, with $\mathcal{F}$ of class C^2 and G of class C^1. Define n, $\mathbf{e}_{\mathcal{F}}$, and $\boldsymbol{\alpha}$ by the first line below, and call a C^1 function $\mathfrak{t}^\perp:\mathcal{U}\to\mathbb{R}$ that satisfies the second an orthogonal clock:

$$\mathsf{n}=\left\|\operatorname{grad}^{\mathsf{G}}\mathcal{F}\right\|_{\mathsf{G}},\qquad \mathbf{e}_{\mathcal{F}}=-\frac{\operatorname{grad}^{\mathsf{G}}\mathcal{F}}{\mathsf{n}},\qquad \boldsymbol{\alpha}=\mathbf{e}_{\mathcal{F}}^{\flat}=-\frac{d\mathcal{F}}{\mathsf{n}}, \tag{7.54}$$

$$d\mathfrak{t}^\perp=\boldsymbol{\alpha}. \tag{7.55}$$

An orthogonal clock exists if and only if

$$d\boldsymbol{\alpha}=0,\qquad \oint_\gamma\boldsymbol{\alpha}=0\quad\text{for every piecewise } C^1 \text{ closed curve } \gamma\subset\mathcal{U}, \tag{7.56}$$

and on a simply connected $\mathcal{U}$ closedness suffices. The local obstruction is

$$d\boldsymbol{\alpha} = \frac{1}{\mathsf{n}^2} d\mathsf{n} \wedge d\mathcal{F}, \tag{7.57}$$

so $\boldsymbol{\alpha}$ is closed exactly when n is locally a function of $\mathcal{F}$, and an orthogonal clock is then constant on every connected piece of every level set of $\mathcal{F}$ and locally a strictly decreasing function of $\mathcal{F}$. Whenever $\mathsf{t}^{\perp}$ exists, every unit-speed natural-gradient history in $\mathcal{U}$ obeys $d\mathsf{t}^{\perp}/d\mathsf{t} = 1$, and the level sets of $\mathsf{t}^{\perp}$ are G-orthogonal to $\mathbf{e}_{\mathcal{F}}$. For $\mathcal{F} = z_1 z_2$ on the open first quadrant with the Euclidean metric,

$$d\boldsymbol{\alpha} = \frac{z_1^2 - z_2^2}{(z_1^2 + z_2^2)^{3/2}} dz_1 \wedge dz_2, \tag{7.58}$$

which vanishes identically on no open set, so no orthogonal clock exists on any open subset of the quadrant. [ESTABLISHED]

Proposition 7.41 (Unit-speed clocks from a global cross-section). Let $\mathcal{U}$ be as in theorem 7.40, and let $\mathcal{S} \subset \mathcal{U}$ be a C^1 hypersurface that every maximal integral curve of $\mathbf{e}_{\mathcal{F}}$ in $\mathcal{U}$ meets exactly once and transversally. Then the signed Fisher arc length along each orbit, measured from its crossing with $\mathcal{S}$, is a C^1 function $\mathsf{t}^{\mathcal{S}} : \mathcal{U} \to \mathbb{R}$ with $d\mathsf{t}^{\mathcal{S}}(\mathbf{e}_{\mathcal{F}}) = 1$, so every unit-speed natural-gradient history obeys $d\mathsf{t}^{\mathcal{S}}/d\mathsf{t} = 1$. A level set of $\mathcal{F}$ that every orbit meets is such a cross-section, and the level sets of $\mathsf{t}^{\mathcal{S}}$ are G-orthogonal to $\mathbf{e}_{\mathcal{F}}$ only if $\boldsymbol{\alpha}$ is closed. For $\mathcal{F} = z_1 z_2$ on the open first quadrant with the Euclidean metric, the hyperbola $\{z_1 z_2 = 1\}$ is a global cross-section, so a unit-speed clock exists on the whole quadrant although no orthogonal one exists on any open subset. [ESTABLISHED]

The proofs of theorem 7.40 and proposition 7.41 are in appendix B.2.

7.9 Record clocks

Corollary 7.42 (Same-path record and meta-agent clock contraction). Let $t \mapsto P_t$ be a C^1 history in a family of laws on Y that is differentiable in quadratic mean along it, with directional score $\mathbf{s}_t$, and let $K : \mathsf{Y} \rightsquigarrow \bar{\mathsf{Y}}$ be a Markov kernel that does not depend on t and satisfies $K(y, \bar{\mathsf{Y}}) = 1$ for every y. Then the record history $\bar{P}_t = P_t K$, the image of the same fine path, satisfies

$$\bar{\mathbf{s}}_t(\bar{Y}) = \mathbb{E}_t \left[\mathbf{s}_t(Y) \mid \bar{Y} \right], \tag{7.59}$$

$$\dot{\mathsf{t}}_P^2 - \dot{\mathsf{t}}_{\bar{P}}^2 = \mathbb{E}_t \operatorname{Var}_t \left(\mathbf{s}_t(Y) \mid \bar{Y} \right) \geq 0, \tag{7.60}$$

and on every common parameter interval

$$\operatorname{len}(\bar{P}) \leq \operatorname{len}(P), \qquad \mathsf{t}_{\bar{P}}(t_1) - \mathsf{t}_{\bar{P}}(t_0) \leq \mathsf{t}_P(t_1) - \mathsf{t}_P(t_0), \tag{7.61}$$

with equality, when the fine length on the interval is finite, exactly when the fine score is $\bar{Y}$-measurable for almost every t in the interval. Equality of two infinite lengths gives no such conclusion. The inequality applies to a meta-agent exactly when its law is the image of the same fine history under one such kernel. Parameter independence of the kernel cannot be dropped: for a

dominated channel K_t with density $\mathsf{k}_t(\bar{y} \mid y)$, assume the joint family $P_t(dy)K_t(y, d\bar{y})$ is differentiable in quadratic mean with square-integrable score $\mathbf{s}_t(y) + \partial_t \log \mathsf{k}_t(\bar{y} \mid y)$. Its record score is

$$\bar{\mathbf{s}}_t(\bar{Y}) = \mathbb{E}_t \left[\mathbf{s}_t(Y) + \partial_t \log \mathsf{k}_t(\bar{Y} \mid Y) \,\middle|\, \bar{Y} \right], \tag{7.62}$$

and a constant input followed by a channel that emits a Bernoulli bit of success probability $(1+e^{-t})^{-1}$ has zero fine speed and record speed $(2\cosh(t/2))^{-1} > 0$. [ESTABLISHED]

The proof is in appendix B.2.

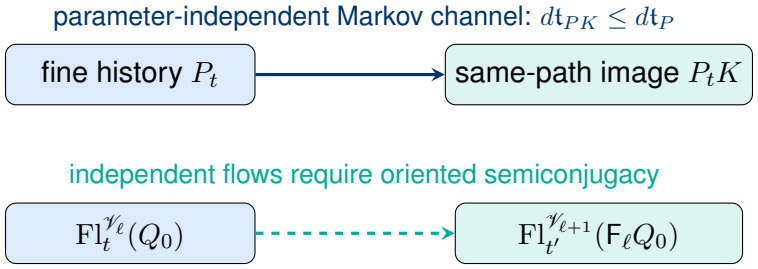


Figure 7.1: The upper arrow is the same-path comparison of corollary 7.42, with its conditional-variance defect. The lower arrow compares independently generated flows and exists only under the semiconjugacy condition of definition 7.45.

Open Problem 7.43 (Quantitative meta-agent clock ratio). Corollary 7.42 shows that a meta-agent whose law is a Markov image of its constituents' joint history accumulates no more Fisher duration than that history, with the exact defect (7.60). Open is the ratio between the meta-agent's duration and the constituents' collective duration for an N-constituent meta-agent of chapter 12, as a function of N and of the correlation between the constituents' velocities, for coherent histories (aligned velocities) and incoherent ones (independent random velocities). The expected ratio of arc lengths is not the square root of the expected ratio of relative entropies, so a divergence computation does not settle it; a theorem that computes or bounds the ratio for a stated coarse map and a stated law of constituent velocities, including its dependence on N in both regimes, does.

7.10 The coarse configuration map and oriented semiconjugacy

Definition 7.44 (Coarse configuration map). A coarse configuration map at depth ℓ is a smooth map

$$\mathsf{F}_\ell : \mathcal{Q}_\ell \longrightarrow \mathcal{Q}_{\ell+1} \tag{7.63}$$

from the configuration manifold of the fine agents at c_* to that of their meta-agents, specified separately from the sample kernels that may realize it. The fine and coarse natural-gradient fields are $\mathscr{V}_\ell = -\operatorname{grad}^{\mathsf{G}_\ell} \mathcal{F}_\ell$ and $\mathscr{V}_{\ell+1} = -\operatorname{grad}^{\mathsf{G}_{\ell+1}} \mathcal{F}_{\ell+1}$, with local flows $\mathrm{Fl}_t^{\mathscr{V}_\ell}$ and $\mathrm{Fl}_{t'}^{\mathscr{V}_{\ell+1}}$.

Definition 7.45 (Oriented semiconjugacy). For an open $\mathcal{U} \subseteq \mathcal{Q}_\ell$, the pair $(\mathsf{F}_\ell, \mathsf{a}_\ell)$ with $\mathsf{a}_\ell : \mathcal{U} \to (0, \infty)$ continuous is an oriented semiconjugacy of $\mathscr{V}_\ell$ onto $\mathscr{V}_{\ell+1}$ over $\mathcal{U}$ when

$$T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q) = \mathsf{a}_\ell(Q) \mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q) \quad \text{for every } Q \in \mathcal{U}. \tag{7.64}$$

Lemma 7.46 (The factor is determined and regular off the coarse critical set). If $\mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q) \neq 0$, then $\mathsf{a}_\ell(Q)$ is unique and

$$\mathsf{a}_\ell(Q) = \frac{\mathsf{G}_{\ell+1}\left(T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q), \mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q)\right)}{\|\mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q)\|^2_{\mathsf{G}_{\ell+1}}}, \tag{7.65}$$

so a_ℓ is C^k wherever $Q \mapsto T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q)$, $Q \mapsto \mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q)$, and the pulled-back coefficients of $\mathsf{G}_{\ell+1}$ are C^k, for instance when $\mathsf{F}_\ell \in C^{k+1}$ and $\mathscr{V}_\ell, \mathscr{V}_{\ell+1}, \mathsf{G}_{\ell+1}$ are C^k; continuity of a_ℓ is therefore automatic where $\mathscr{V}_{\ell+1} \neq 0$ on $\mathsf{F}_\ell(\mathcal{U})$. If $\mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q) = 0$, then (7.64) forces $T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q) = 0$ and fixes no value of $\mathsf{a}_\ell(Q)$; limits from the noncritical set may determine it or may obstruct any continuous positive extension. [ESTABLISHED]

Proof. Pair (7.64) with $\mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q)$ and divide by its positive squared norm. □

Theorem 7.47 (Oriented semiconjugacy and maximal intervals). Assume (7.64) on an open $\mathcal{U}$ with $\mathscr{V}_{\ell+1}$ locally Lipschitz. Fix $Q \in \mathcal{U}$, let $\mathcal{T}_Q$ be the connected component of 0 in the set of t for which $\mathrm{Fl}^{\mathscr{V}_\ell}_{t''}(Q) \in \mathcal{U}$ for every t'' between 0 and t, and set $\psi_Q(t) = \int_0^t \mathsf{a}_\ell(\mathrm{Fl}^{\mathscr{V}_\ell}_{t''} Q) dt''$. Then ψ_Q is a strictly increasing C^1 diffeomorphism of $\mathcal{T}_Q$ onto an open interval $\mathcal{T}'_Q \ni 0$, and

$$\mathcal{T}'_Q \subseteq \bar{\mathcal{T}}^{\max}_{\mathsf{F}_\ell Q}, \qquad \mathsf{F}_\ell\left(\mathrm{Fl}^{\mathscr{V}_\ell}_t(Q)\right) = \mathrm{Fl}^{\mathscr{V}_{\ell+1}}_{\psi_Q(t)}(\mathsf{F}_\ell Q) \quad \text{for every } t \in \mathcal{T}_Q, \tag{7.66}$$

where $\bar{\mathcal{T}}^{\max}_{\mathsf{F}_\ell Q} = (\bar{t}^-, \bar{t}^+)$ is the maximal interval of the coarse flow through $\mathsf{F}_\ell Q$. With $\mathcal{T}_Q = (t^-, t^+)$ and endpoints in the extended real line,

$$\mathcal{T}'_Q = \left(\lim_{t \downarrow t^-} \psi_Q(t), \lim_{t \uparrow t^+} \psi_Q(t)\right), \tag{7.67}$$

and the whole maximal coarse orbit is traversed if and only if the two limits equal $\bar{t}^-$ and $\bar{t}^+$; when the fine orbit is complete in $\mathcal{U}$ and the coarse maximal interval is $\mathbb{R}$, this means $\int_0^{+\infty} \mathsf{a}_\ell(\mathrm{Fl}^{\mathscr{V}_\ell}_t Q) dt = \int_{-\infty}^0 \mathsf{a}_\ell(\mathrm{Fl}^{\mathscr{V}_\ell}_t Q) dt = +\infty$. A positive lower bound on a_ℓ suffices and is not necessary: on $\mathbb{R}$ with $\mathscr{V}_\ell = \mathscr{V}_{\ell+1} = \partial_z$ and $\mathsf{F}_\ell = \operatorname{arsinh}$, $\mathsf{a}_\ell = (1+z^2)^{-1/2}$ has infimum zero while $\psi_0 = \operatorname{arsinh}$ maps $\mathbb{R}$ onto $\mathbb{R}$; with $\mathsf{F}_\ell = \arctan$, $\mathsf{a}_\ell = (1+z^2)^{-1}$ is positive while $\mathcal{T}'_0 = (-\pi/2, \pi/2)$ although both flows are complete. Conversely, differentiating a flow relation of the form (7.66) at $t = 0$ gives (7.64). [ESTABLISHED]

The proof is in appendix B.2.

Proposition 7.48 (Noncollapse is a separate hypothesis). Condition (7.64) alone permits total collapse: with F_ℓ constant and $\mathscr{V}_{\ell+1} \equiv 0$ it holds for every positive a_ℓ, while every fine orbit maps to a

point and every coarse duration is zero. A nonconstant shared oriented history additionally requires $T_Q \mathsf{F}_\ell \mathscr{V}_\ell(Q) \neq 0$ on every nontrivial subarc, equivalently $\mathscr{V}_{\ell+1}(\mathsf{F}_\ell Q) \neq 0$ there, and full traversal requires the endpoint equalities of (7.67); neither follows from (7.64). [ESTABLISHED]

Proof. Both sides of (7.64) vanish in the example, and the arctan example of theorem 7.47 satisfies it without full traversal. □

Theorem 7.49 (The semiconjugacy rate cancels from geometric length). Under (7.64), let $Q(t)$ be a fine orbit. The image speed is

$$\left\| \tfrac{d}{dt} \mathsf{F}_\ell Q(t) \right\|_{\mathsf{G}_{\ell+1}} = \mathsf{a}_\ell \left(Q(t) \right) \left\| \mathscr{V}_{\ell+1} \left(\mathsf{F}_\ell Q(t) \right) \right\|_{\mathsf{G}_{\ell+1}}, \tag{7.68}$$

and for $t_0 < t_1$ in $\mathcal{T}_Q$ the accumulated coarse duration is

$$\mathfrak{t}^{(\ell+1)}(t_1) - \mathfrak{t}^{(\ell+1)}(t_0) = \int_{\psi_Q(t_0)}^{\psi_Q(t_1)} \left\| \mathscr{V}_{\ell+1} \left(\mathrm{Fl}_{t'}^{\mathscr{V}_{\ell+1}} (\mathsf{F}_\ell Q) \right) \right\|_{\mathsf{G}_{\ell+1}} dt', \tag{7.69}$$

the intrinsic $\mathsf{G}_{\ell+1}$-length of the coarse orbit arc, which does not depend on a_ℓ. [ESTABLISHED]

Proof. Equation (7.68) is (7.64) with positive homogeneity of the norm and $\mathsf{a}_\ell > 0$. Substituting it and changing variables $t' = \psi_Q(t)$, whose Jacobian is $\mathsf{a}_\ell(\mathrm{Fl}_t^{\mathscr{V}_\ell} Q)$, cancels the factor. □

Remark 7.50 (Exact duration comparison). Under (7.64), along one orbit and from a common origin, the coarse and fine durations agree identically if and only if $\|T\mathsf{F}_\ell \mathscr{V}_\ell\|_{\mathsf{G}_{\ell+1}} = \|\mathscr{V}_\ell\|_{\mathsf{G}_\ell}$ almost everywhere along the orbit, they agree up to a factor $\mathrm{c} > 0$ if and only if the same holds with that factor, and the coarse duration has no larger increments on every subinterval if and only if

$$\|T\mathsf{F}_\ell \mathscr{V}_\ell\|_{\mathsf{G}_{\ell+1}} \leq \|\mathscr{V}_\ell\|_{\mathsf{G}_\ell} \quad \text{along the orbit,} \tag{7.70}$$

because both durations are integrals of continuous nonnegative densities from a common origin, and two such integrals agree for every upper limit exactly when the densities agree almost everywhere. The condition is scalar and concerns the single direction $\mathscr{V}_\ell$; the Loewner condition $\mathsf{F}_\ell^* \mathsf{G}_{\ell+1} \preceq \mathsf{G}_\ell$ is sufficient and stronger, and corollary 7.51 supplies it for channelwise Markov maps. Fiberwise Fisher contraction alone gives neither, for two independent reasons. The configuration metrics are separate data: on $\mathbb{R}$ with $\mathscr{V}_\ell = \mathscr{V}_{\ell+1} = \partial_z$ and F_ℓ the identity, the semiconjugacy holds with $\mathsf{a}_\ell \equiv 1$, and the metrics dz^2 and $4dz^2$ double the duration, whether realized with the identity fiber map and unequal channel weights 1 and 4 (zero Fisher defect) or with fibers $\mathcal{N}(\mu, 1)$ and $\mathcal{N}(\mu, \frac{1}{4})$ and the identity parameter map (Fisher defect $1 - 4 = -3$, so no Markov morphism). And contraction compares the image of the fine orbit, not a recomputed coarse orbit: on $\mathbb{R}^2$ with F_ℓ the identity and the Euclidean metric at both depths, $\mathcal{F}_\ell = \frac{1}{2}(z_1^2 + z_2^2)$ and $\mathcal{F}_{\ell+1} = \frac{1}{2}(z_1^2 + 4z_2^2)$ give the fine history (e^{-t}, e^{-t}) of duration $\sqrt{2}$ and the recomputed coarse history (e^{-t}, e^{-4t}) of duration $\int_0^1 \sqrt{1 + 16z^6} dz > \sqrt{2}$, and (7.64) fails because $(z_1, z_2) = \mathsf{a}(z_1, 4z_2)$ forces $\mathsf{a} = 1$ and $z_2 = 0$. [ESTABLISHED]

Corollary 7.51 (When fiberwise contraction does lift). Let $\mathcal{Q}_\ell$ and $\mathcal{Q}_{\ell+1}$ carry the channel-weighted metrics $\mathsf{G}_\ell = \sum_x w_x g_x^F$ and $\mathsf{G}_{\ell+1} = \sum_x w_x \bar{g}_x^F$ with the same positive weights, and let F_ℓ act channelwise by Markov coarse morphisms, $(\mathsf{F}_\ell Q)_x = \mathscr{K}_x(Q_x)$. Then for every $Z \in T_Q\mathcal{Q}_\ell$,

$$\|Z\|^2_{\mathsf{G}_\ell} - \|T\mathsf{F}_\ell Z\|^2_{\mathsf{G}_{\ell+1}} = \sum_x w_x \Delta_F^{\mathscr{K}_x}(Z_x, Z_x) \geq 0, \tag{7.71}$$

so the image of every C^1 fine curve has no greater speed and no greater duration on every subinterval, and (7.70) holds along every orbit of an oriented semiconjugacy. On a subinterval of finite fine duration, equality holds exactly when, for every channel and almost every parameter value, the fine score in the direction Z_x is measurable with respect to the coarse output. [ESTABLISHED]

Proof. Differentiate the channelwise action, insert it into $\mathsf{G}_{\ell+1}$, and subtract from G_ℓ; the difference is the weighted sum of the vertical defects, which are positive semidefinite with the stated equality case by corollary 7.11. Monotonicity of the square root and of the integral gives the speed and duration statements. □

7.11 Semiconjugacy on symmetry quotients

Throughout this section $\mathscr{V}_\ell$ and $\mathscr{V}_{\ell+1}$ are natural-gradient fields of C^2 objectives for positive-definite metrics.

Definition 7.52 (Functional compatibility). A coarse configuration map F_ℓ is functionally compatible on an open set $\mathcal{U} \subseteq \mathcal{Q}_\ell$ when there is a C^1 function χ_ℓ with $\chi_\ell' > 0$ such that $\mathcal{F}_\ell = \chi_\ell \circ \mathcal{F}_{\ell+1} \circ \mathsf{F}_\ell$ on $\mathcal{U}$.

Proposition 7.53 (Semiconjugacy on symmetry quotients). Equal objectives do not intertwine natural gradients: with F_ℓ the identity of $\mathbb{R}^2$, $\mathcal{F}_\ell = \mathcal{F}_{\ell+1} = \frac{1}{2}(z_1^2 + 2z_2^2)$, and metrics $\mathrm{diag}(1,1)$ and $\mathrm{diag}(1,\mathrm{c})$ with $\mathrm{c} \neq 1$, (7.64) fails wherever $z_1 z_2 \neq 0$. Functional compatibility on $\mathcal{U}$ forces $d\mathcal{F}_\ell$ to vanish on $\ker T\mathsf{F}_\ell$ at every point of $\mathcal{U}$, so when F_ℓ is a submersion, $\mathcal{F}_\ell$ is constant on each connected piece of each fiber of F_ℓ in $\mathcal{U}$. If F_ℓ is a surjective submersion, functionally compatible on $\mathcal{U}$, and horizontally conformal with dilation $\mathsf{d}_\ell > 0$, meaning that $\mathsf{G}_{\ell+1}(T\mathsf{F}_\ell Z_1, T\mathsf{F}_\ell Z_2) = \mathsf{d}_\ell^2 \mathsf{G}_\ell(Z_1, Z_2)$ for $Z_1, Z_2 \in \mathrm{Hor}_Q = (\ker T_Q\mathsf{F}_\ell)^\perp$, then (7.64) holds on $\mathcal{U}$ with

$$\mathsf{a}_\ell(Q) = \chi_\ell'\left(\mathcal{F}_{\ell+1}(\mathsf{F}_\ell Q)\right) \mathsf{d}_\ell(Q)^2 > 0. \tag{7.72}$$

In particular the quotient map of a free, proper, isometric action that preserves $\mathcal{F}_\ell$, with the quotient metric and descended objective of proposition 7.39, is an oriented semiconjugacy with $\mathsf{a}_\ell \equiv 1$; and at c_* the identity map from the mean coordinate of the fiber $\{\mathcal{N}(\mu,1)\}$ to that of $\{\mathcal{N}(\mu,2)\}$, realized by the kernel $y \mapsto \mathcal{N}(y,1)$, is horizontally conformal with $\mathsf{d}_\ell^2 = \frac{1}{2}$ and gives $\mathsf{a}_\ell = \frac{1}{2}$ for $\mathcal{F}_\ell = \mathcal{F}_{\ell+1} = \mu^2/2$, an injective case that does not exercise the fiber condition. The route does not reach maps that discard directions on which the fine objective depends: if $\mathcal{F}_{\ell+1}(Q') = \inf\{\mathcal{F}_\ell(Q) : \mathsf{F}_\ell Q = Q'\}$, F_ℓ is a submersion with positive-dimensional fibers, and each fiber contains at most one minimizer of $\mathcal{F}_\ell$,

then compatibility with $\chi_\ell = \mathrm{id}$ holds on no open set; for $\mathsf{F}_\ell(f,s) = s$ and $\mathcal{F}_\ell(f,s) = \frac{1}{2}(f-s)^2 + \frac{1}{2}s^2$ it holds for no χ_ℓ on any open set. [ESTABLISHED]

The proof is in appendix B.2.

For maps that discard directions on which the fine objective depends, the exact route runs through the reduced slow flow, which on the set of conditional minimizers is conjugate to the coarse natural-gradient flow (proposition 8.4, (8.8)); whether the fine or rate-separated flow follows it is open problem 8.7.

7.12 Coarse-graining depth, orbit parameter, and duration

A tower of coarse configurations $Q^{(\ell+1)} = \mathsf{F}_\ell(Q^{(\ell)})$, $\ell = 0, 1, 2, \ldots$, is indexed by the coarse-graining depth ℓ, which is not a parameter on a history. A history can be present at every depth, as $Q^{(\ell)}(t)$ with accumulated duration $\mathfrak{t}^{(\ell)}(t)$; when $Q^{(\ell+1)}(t) = \mathsf{F}_\ell(Q^{(\ell)}(t))$ and F_ℓ acts by Markov coarse morphisms as in corollary 7.51, the record-clock comparison applies between adjacent depths, while for a general oriented semiconjugacy the exact comparison is remark 7.50, and independently recomputed histories also need the noncollapse conditions of proposition 7.48. The Markov property of a configuration arrow can fail even for natural meta-agent maps: no parameter-independent Markov kernel sends every product of two normal laws to their joint mean-and-variance barycenter, as Supplement G proves.

Proposition 7.54 (Typed non-identification of depth, orbit parameter, and duration). Coarse-graining depth ℓ, an orbit parameter t, and Fisher duration $\mathfrak{t}^{(\ell)}$ are distinct: ℓ is a discrete index, t a chosen oriented parameter on a selected orbit, and $\mathfrak{t}^{(\ell)}$ the arc length for a chosen metric from a chosen origin. Once the metric, origin, orientation, and parameterized path are chosen, $\mathfrak{t}^{(\ell)}$ is a function of t, but it is not determined by the unparameterized orbit or by ℓ: changing the metric rescales it and changing the origin shifts it. Strict monotonicity of $\mathfrak{t}^{(\ell)}$ does not make it a regular coordinate: for $\mathsf{G} = dz^2$ on $\mathbb{R}$ and the path $z(t) = t^2$ on $[0,1]$, $\mathfrak{t}(t) = t^2$, whose inverse is not differentiable at the origin, and a zero-speed subinterval destroys strict monotonicity. [ESTABLISHED]

Proof. Only the metric, origin, orientation, and parameterized path produce a duration function. Metric dependence is homogeneity of the norm in (7.45), and the example is $\int_0^t 2t''dt'' = t^2$ with the square root not differentiable at zero. □

Physicist's summary

The Fisher metric on an agent's belief and model law fibers is a gauge-invariant ruler and the second-order term of the relative entropy, so a loop holonomy near the identity appears as a quadratic form on the Lie algebra that vanishes on the isotropy algebra of the law. A Markov channel loses exactly the conditional variance of the score, so a meta-agent that is a channel image of its constituents' history has an information clock no faster than theirs, by a ratio that is open. A

chosen metric turns the free energy into an orbit, Fisher arc length removes its parameterization, and a belief-model rate ratio, unlike a scalar mobility, can bend the orbit; an orthogonal clock needs the gradient norm to be locally a function of the free energy, a unit-speed clock only a global cross-section. By the parity witness, joint laws, Fisher metrics, and free energies are not functions of marginals. A recomputed coarse flow is the image of the fine flow only under an oriented semiconjugacy, which holds with rate one on symmetry quotients; information-discarding maps are compared through the reduced slow flow of the next chapter.

Chapter 8

Two-Timescale Dynamics

This chapter asks what the collective free energy of chapter 5 supplies for separating fast inference of beliefs from slower evolution of models, priors, and parents: descent for every positive-definite block mobility, local in a product family (proposition 8.1); a reduced slow flow conjugate to its tangent lift on the graph of the conditional optimum (lemma 8.3 and proposition 8.4), the exact route of chapter 7 for information-discarding maps; and a Markov parent history under instantaneous within-parent refresh (theorem 8.11), whose failure under finite refresh (remark 8.12) is measured in chapter 12. The slow block (definition 8.2) moves by accepted M-coordinates (definition 5.28) or by the rate-separated flow; the formation family of chapter 14 realizes one scalar slow parameter (14.25), not slow per-agent model laws, while on the four-cycle of that chapter, whose link translations this chapter writes $\check{\omega}$, slow model means are a supplied rate, and the model channel lags the belief channel exactly when the links are slow as well (proposition 8.6, the held-sheet instance of open problem 8.7). The finite model of the last section uses fine states $\xi \in \Xi$, parent values $z \in \mathsf{Z}$, a parent map par, a prior p^0, record laws ψ_ξ, a refresh fraction ω, and sans-serif kernels acting on row vectors from the right.

Scope. The collective free energy does not choose the fast–slow split, the rate ratio, or the block mobilities, and a rate ratio is an anisotropic mobility that can change the selected orbit (proposition 7.31). The chapter does not claim that the rate-separated flow tracks the reduced flow for a recognition family of the program beyond the held matching sheets of proposition 8.6, that the sampled dynamics with source jumps track it, that a sampled episode with source jumps at the temperature of definition 14.21 keeps a matched pair belief-coherent and model-incoherent on an interval of order $1/\varepsilon_{\mathrm{m}}$ with probability bounded below, that the discrete alternating scheme converges, that the parent sweep converges at a rate uniform in the number of fine states or in the smallest posterior weight, or that a static parent datum has closed dynamics (open problems 5.32, 8.7 and 12.40). [NOT-CLAIMED]

8.1 Block descent at fixed parameter

Proposition 8.1 (Block descent of the collective free energy). Let $\lambda \mapsto Q_\lambda$ be a family of recognition laws on an open domain $\Lambda = \prod_{i \in V} \Lambda_i$ with block coordinates $\lambda = (\lambda_i)_i$, such that $F(\lambda) := \mathcal{F}_o^{\mathrm{ext}}(Q_\lambda)$ is real and C^1 on Λ, and let $\mathsf{G}_i(\lambda)$ be continuous fields of symmetric positive-definite block metrics, with positive block rates absorbed into the metrics and blocks of zero rate

held fixed. Let $\lambda(t)$ solve

$$\dot{\lambda}_i = -\mathsf{G}_i^{-1}\nabla_{\lambda_i}F \tag{8.1}$$

for the moving blocks $i \in V' \subseteq V$ and remain in Λ.

(i) Along the solution

$$\frac{d}{dt}F = -\sum_{i\in V'}(\nabla_{\lambda_i}F)^{\top}\mathsf{G}_i^{-1}(\nabla_{\lambda_i}F) \le 0. \tag{8.2}$$

(ii) If the family is a regular product family $Q_\lambda = \bigotimes_i Q_{i,\lambda_i}$ with $\mathbb{E}_{Q_{-i}}|\log Z_{\{i\}}(Y_{-i})| < \infty$ for every i and λ, then $\nabla_{\lambda_i}F = \nabla_{\lambda_i}\overline{\mathcal{F}}_i$ for the outside-averaged local free energy (5.53), so an agent that descends its own local free energy moves along a block of a descent flow of the collective one. The Fisher metric of such a family is block diagonal, so with G_i equal to the block Fisher metrics the flow is the Fisher natural gradient of F.

(iii) Outside product families moving λ_i can change Q_{-i}, and the identification in (ii) needs the conditional lift of definition 7.17, whose outside-averaged objective (7.21) has the differential of the collective free energy restricted to fixed outside marginal. The Fisher metric then need not be block diagonal: for the bivariate normal location family with unit variances and correlation c_2, $0 < |\mathrm{c}_2| < 1$, its off-diagonal entry is $-\mathrm{c}_2/(1-\mathrm{c}_2^2)$.

[ESTABLISHED]

Proof. Part (i) is the chain rule for F along the solution. For (ii), a change of λ_i leaves Q_{-i} fixed, and theorem 5.23 with $B = \{i\}$, whose finiteness hypotheses hold because F is real, shows that F and $\overline{\mathcal{F}}_i$ differ by a quantity independent of λ_i. Scores of distinct factors of a product law are independent with mean zero, so their cross covariances vanish. For (iii), the Fisher metric of the location family is the inverse covariance matrix. □

8.2 Fast recognition and slow model coordinates

Definition 8.2 (Fast and slow coordinate blocks). Let $\mathcal{U}_{\mathrm{f}} \subseteq \mathbb{R}^{n_{\mathrm{f}}}$ and $\mathcal{U}_{\mathrm{s}} \subseteq \mathbb{R}^{n_{\mathrm{s}}}$ be open, and write $s = (s_{\mathrm{m}}, \theta) \in \mathcal{U}_{\mathrm{s}}$. A fast–slow coordinatization is a map $(f, s) \mapsto (Q_{f,s_{\mathrm{m}}}, \theta)$ into population recognition laws and generative parameters, together with the real-valued objective

$$F(f,s) := \mathcal{F}_o^{\mathrm{ext}}(Q_{f,s_{\mathrm{m}}};\theta) = -\log p_{\theta,V}(o \mid X) + D_{\mathrm{KL}}\big(Q_{f,s_{\mathrm{m}}} \| \mathbf{\Pi}_{\theta,V,o,X}\big), \tag{8.3}$$

in which the generative argument of (5.17) is explicit. The fast coordinate f carries the belief-channel recognition coordinates and, when present, the source-label rows of section 6.1; the slow coordinate carries the model-channel recognition coordinates s_{m}, which move a recognition law, and the generative parameter θ, which moves the fixed joint between evaluations and never inside one, so a composite $\theta(Q)$ inside one evaluation of (8.3) stays excluded by hypothesis 4.3. The assignment of coordinates to the two blocks is a declaration.

Lemma 8.3 (Envelope identity and reduced slow descent). Let $F \in C^1(\mathcal{U}_\mathrm{f} \times \mathcal{U}_\mathrm{s})$, let $\mathcal{U}_0 \subseteq \mathcal{U}_\mathrm{s}$ be open, and let $f^\star : \mathcal{U}_0 \to \mathcal{U}_\mathrm{f}$ be a C^1 map with $F(f^\star(s), s) = \min_{f \in \mathcal{U}_\mathrm{f}} F(f, s)$ for every $s \in \mathcal{U}_0$. The reduced functional

$$F_\mathrm{red}(s) := \min_{f \in \mathcal{U}_\mathrm{f}} F(f, s) = F(f^\star(s), s) \tag{8.4}$$

satisfies the envelope identity

$$dF_\mathrm{red}(s) = \partial_s F(f^\star(s), s), \qquad s \in \mathcal{U}_0. \tag{8.5}$$

For every continuous field G_s of symmetric positive-definite metrics on $\mathcal{U}_\mathrm{s}$, the reduced slow flow

$$\dot{s} = -\mathsf{G}_\mathrm{s}(s)^{-1} \partial_s F(f^\star(s), s) \tag{8.6}$$

is the G_s natural-gradient flow of F_red, and along the lifted curve $(f^\star(s(t)), s(t))$ the free energy decreases at the rate $dF_\mathrm{red} \cdot \mathsf{G}_\mathrm{s}^{-1} dF_\mathrm{red}$. A C^1 branch exists near $s_0 \in \mathcal{U}_0$ whenever $F \in C^2$, $\partial_{ff}^2 F(f^\star(s_0), s_0)$ is positive definite, and $f^\star$ is continuous at s_0. If $\mathcal{U}_\mathrm{f}$ is replaced by a compact set on which F and $\partial_s F$ are continuous and the conditional minimizer is unique, (8.5) holds without differentiability of $f^\star$. [ESTABLISHED]

Proof. At the minimizer $\partial_f F(f^\star(s), s) = 0$, so the chain rule gives $dF_\mathrm{red} = \partial_f F \cdot Df^\star + \partial_s F = \partial_s F$, and substitution in (8.6) gives the flow and its rate. For the branch, the implicit function theorem applied to $\partial_f F = 0$ at $(f^\star(s_0), s_0)$ gives a C^1 map whose graph is the whole critical set near that point, and continuity places $f^\star(s)$ on it for s near s_0. In the compact case, uniqueness and compactness make $f^\star$ continuous, since a limit point f' of $f^\star(s_n)$ with $s_n \to s$ satisfies $F(f', s) \le F(f^\star(s), s)$. With $s' = s + \Delta s$, the difference $F_\mathrm{red}(s') - F_\mathrm{red}(s)$ lies between $F(f^\star(s'), s') - F(f^\star(s'), s)$ and $F(f^\star(s), s') - F(f^\star(s), s)$, and by the mean value theorem and uniform continuity of $\partial_s F$ on a compact neighborhood both equal $\partial_s F(f^\star(s), s)\Delta s + o(|\Delta s|)$; this is the theorem of Danskin [30, Ch. III] in the unique-minimizer case. □

Proposition 8.4 (Rate-separated natural gradient and its reduced flow). Let $F \in C^1(\mathcal{U}_\mathrm{f} \times \mathcal{U}_\mathrm{s})$, let $\mathsf{G}_\mathrm{f}(f, s)$ and $\mathsf{G}_\mathrm{s}(s)$ be continuous symmetric positive-definite block metrics with G_s independent of f, and for $\varepsilon > 0$ set $\mathsf{G}_\varepsilon = \varepsilon \mathsf{G}_\mathrm{f} \oplus \mathsf{G}_\mathrm{s}$. The G_ε natural gradient of F is the rate-separated flow

$$\varepsilon \dot{f} = -\mathsf{G}_\mathrm{f}^{-1} \partial_f F, \qquad \dot{s} = -\mathsf{G}_\mathrm{s}^{-1} \partial_s F, \tag{8.7}$$

which decreases F at the rate $\varepsilon^{-1} \partial_f F \cdot \mathsf{G}_\mathrm{f}^{-1} \partial_f F + \partial_s F \cdot \mathsf{G}_\mathrm{s}^{-1} \partial_s F$ for every $\varepsilon > 0$; its orbit can depend on ε. Under the hypotheses of lemma 8.3, let $\mathcal{S}^\star = \{(f^\star(s), s) : s \in \mathcal{U}_0\}$. At every point of $\mathcal{S}^\star$ and for every $\varepsilon > 0$, the fast component of (8.7) vanishes and the slow component equals $-\mathsf{G}_\mathrm{s}^{-1} dF_\mathrm{red}$. The reduced field $\mathscr{V}^\star = (Df^\star v, v)$ with $v = -\mathsf{G}_\mathrm{s}^{-1} dF_\mathrm{red}$ is tangent to $\mathcal{S}^\star$, and

$$d\,\mathrm{pr}_\mathrm{s}\, \mathscr{V}^\star = -\operatorname{grad}^{\mathsf{G}_\mathrm{s}} F_\mathrm{red} \circ \mathrm{pr}_\mathrm{s} \quad \text{on } \mathcal{S}^\star. \tag{8.8}$$

Thus $(\mathrm{pr}_\mathrm{s}\,|_{\mathcal{S}^\star}, 1)$ is an oriented semiconjugacy in the sense of definition 7.45, with the manifold $\mathcal{S}^\star$ in place of an open fine domain, and a conjugacy, because $\mathrm{pr}_\mathrm{s}\,|_{\mathcal{S}^\star}$ is a diffeomorphism onto $\mathcal{U}_0$.

The speeds satisfy $\|\mathscr{V}^\star\|^2_{\mathsf{G}_\varepsilon} = \varepsilon\|Df^\star v\|^2_{\mathsf{G}_\mathrm{f}} + \|v\|^2_{\mathsf{G}_\mathrm{s}}$, so the G_ε-length of a reduced orbit is at least the G_s-length of its image. These lengths decrease to the latter as $\varepsilon \downarrow 0$ on every compact time interval of the orbit, and on any interval where the lifted length is finite at some positive ε_0. [ESTABLISHED]

Proof. Because $\mathsf{G}_\varepsilon^{-1} = \varepsilon^{-1}\mathsf{G}_\mathrm{f}^{-1} \oplus \mathsf{G}_\mathrm{s}^{-1}$, the flow is (8.7) with the stated dissipation, and the orbit statement is proposition 7.31, since G_ε differs from G_1 by an anisotropic rescaling. On $\mathcal{S}^\star$, $\partial_f F = 0$ and $\partial_s F = dF_{\mathrm{red}}$ by (8.5), and G_s does not depend on f. The field $\mathscr{V}^\star$ is the differential of $s \mapsto (f^\star(s), s)$ applied to v, hence tangent to $\mathcal{S}^\star$ with slow component v. The length identity is the block form of G_ε. For $0 < \varepsilon \le \varepsilon_0$ the lifted speed is bounded by its ε_0 value, which is integrable under the stated hypothesis and on compact time intervals by continuity. Dominated convergence gives the length limit. □

Remark 8.5 (The slow graph off the optimum and under correlated metrics). Off $\mathcal{S}^\star$ the slow component $-\mathsf{G}_\mathrm{s}^{-1}\partial_s F(f,s)$ need not be a positive multiple of $-\operatorname{grad}^{\mathsf{G}_\mathrm{s}} F_{\mathrm{red}}(s)$: for $F = [(f - s_1)^2 + s_1^2 + s_2^2]/2$ with $\mathsf{G}_\mathrm{s} = I$, $\partial_s F = (2s_1 - f, s_2)$ and $\operatorname{grad} F_{\mathrm{red}} = (s_1, s_2)$ are not parallel when $f \neq s_1$ and $s_2 \neq 0$, so the projection semiconjugates the finite-ε flow to the reduced flow only where that proportionality holds. At points of $\mathcal{S}^\star$ the finite-ε velocity $(0, v)$ is tangent to $\mathcal{S}^\star$ only where $Df^\star v = 0$, so $\mathcal{S}^\star$ is not invariant in general. A correlated belief–model metric requires the rate declaration to name its slow manifold. For $F(f,s) = [(f-s)^2 + s^2]/2$, whose conditional minimizer is $f = s$, and the metric G with unit diagonal and off-diagonal entry $\mathrm{c}_2 \in (0,1)$, the row-rated field $\varepsilon\dot{f} = -(\mathsf{G}^{-1}dF)_f$, $\dot{s} = -(\mathsf{G}^{-1}dF)_s$ has, in the fast time t/ε, the critical set

$$\{(\mathsf{G}^{-1}dF)_f = 0\} = \Big\{f = \frac{1+2\mathrm{c}_2}{1+\mathrm{c}_2}s\Big\}, \tag{8.9}$$

which attracts at rate $1/(1-\mathrm{c}_2)$ and differs from the conditional minimizer. This field need not descend F: at $\mathrm{c}_2 = 1/2$, every $\varepsilon \in (0, 7-4\sqrt{3})$, and $(f,s) = (1+2\delta, 1+\delta)$ with $\delta = (1+\varepsilon)/4$, it gives $dF/dt = (1+\varepsilon)^2/(12\varepsilon) - 4/3 > 0$, so for those rates it is the natural gradient of no metric. The congruence rescaling $\mathsf{G}_\varepsilon = \Delta_\varepsilon \mathsf{G} \Delta_\varepsilon$ with $\Delta_\varepsilon = \operatorname{diag}(\sqrt{\varepsilon}, 1)$ gives a natural gradient that descends F for every ε, and the critical set of its $\varepsilon = 0$ fast subsystem is the conditional-minimizer set $\{\partial_f F = 0\}$. [ESTABLISHED]

Proof. The first two statements are direct differentiation. For the correlated metric, G^{-1} has diagonal entries $1/(1-\mathrm{c}_2^2)$ and off-diagonal entries $-\mathrm{c}_2/(1-\mathrm{c}_2^2)$, and $dF = (f-s, 2s-f)$, so $(\mathsf{G}^{-1}dF)_f$ vanishes on the displayed line, and its derivative in f is $(1+\mathrm{c}_2)/(1-\mathrm{c}_2^2) = 1/(1-\mathrm{c}_2)$. Substituting the stated point into $dF/dt = -\varepsilon^{-1}\partial_f F(\mathsf{G}^{-1}dF)_f - \partial_s F(\mathsf{G}^{-1}dF)_s$ at $\mathrm{c}_2 = 1/2$ gives the displayed value, which is positive exactly when $\varepsilon^2 - 14\varepsilon + 1 > 0$. Under congruence, G_ε is positive definite for every $\varepsilon > 0$, and in the fast time the fast component of its natural gradient is $-(1-\mathrm{c}_2^2)^{-1}(\partial_f F - \mathrm{c}_2\sqrt{\varepsilon}\partial_s F)$, whose zero set at $\varepsilon = 0$ is $\{\partial_f F = 0\}$. □

Proposition 8.6 (Slow model means on the four-cycle and the model-link dichotomy). In definition 14.21 let the belief means $\bar{k}$ move at rate one, the model means $\bar{m}$ at rate $\varepsilon_\mathrm{m} \in [0,1]$, and

the link translations $\check{\omega}$ at rate $\varepsilon_{\check{\omega}} \in [0,1]$: between clock times

$$d\mathbf{x} = -\big(\partial_{\bar{k}} U_S, \varepsilon_{\mathrm{m}} \partial_{\bar{m}} U_S, \varepsilon_{\check{\omega}} \partial_{\check{\omega}} U_S\big) dt + \sqrt{2\mathrm{T}}\big(dW^{\bar{k}}, \varepsilon_{\mathrm{m}}^{1/2} dW^{\bar{m}}, \varepsilon_{\check{\omega}}^{1/2} dW^{\check{\omega}}\big) \tag{8.10}$$

with one standard Brownian motion W in $\mathbb{R}^{12}$, and the clock redraws the assignment as before, leaving $\mathbf{x}$ unchanged. The rates are supplied, since the Fisher metric of $\mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2)$ in $\bar{m}_i$ is $1/\sigma_{\mathrm{ev}}^2$ and a Fisher natural gradient would move the model means at rate $1/2$. For $i \in \{0,2\}$ write $\bar{k}_i^{\pm} = (\bar{k}_{i+1} \pm \bar{k}_i)/\sqrt{2}$ and $\bar{m}_i^{\pm} = (\bar{m}_{i+1} \pm \bar{m}_i)/\sqrt{2}$, so that $r_i^b = \sqrt{2}\bar{k}_i^- - \check{\omega}_i$ and $r_i^m = \sqrt{2}\bar{m}_i^- - \check{\omega}_i$.

(i) *Held matching.* On the held assignment $S^{(0)}$ a Gaussian law of $\mathbf{x}$ stays Gaussian and its mean follows (8.10) without the noise; for $\varepsilon_{\mathrm{m}}, \varepsilon_{\check{\omega}} > 0$, in the slow time $\tau = \varepsilon_{\mathrm{m}} t$, this mean flow is (8.7) for $F = U_{S^{(0)}}$, which is the collective free energy of proposition 14.22 at the Gaussian law up to terms independent of the mean, with $\varepsilon = \varepsilon_{\mathrm{m}}$, $f = \bar{k}$, $s = (\bar{m}, \check{\omega})$, $\mathsf{G}_{\mathrm{f}} = I_4$ and $\mathsf{G}_{\mathrm{s}} = I_4 \oplus (\varepsilon_{\mathrm{m}}/\varepsilon_{\check{\omega}}) I_4$. In the orthonormal coordinates $(\bar{k}_i^-, \bar{m}_i^-, \check{\omega}_i)$ and $(\bar{k}_i^+, \bar{m}_i^+)$ for $i \in \{0,2\}$, $\check{\omega}_1$, and $\check{\omega}_3$, the Hessian of $U_{S^{(0)}}$ and the mobility are block diagonal, with the blocks

$$\begin{pmatrix} 4+3\mathrm{T} & -2\mathrm{T} & -2\sqrt{2} \\ -2\mathrm{T} & 4+3\mathrm{T} & -2\sqrt{2} \\ -2\sqrt{2} & -2\sqrt{2} & 4+\mathrm{T} \end{pmatrix}, \qquad \mathrm{T}\begin{pmatrix} 3 & -2 \\ -2 & 3 \end{pmatrix}, \qquad \mathrm{T}, \tag{8.11}$$

so the mean flow is a direct sum of six flows and r_i^b, r_i^m read only the block of e_i. With $\varepsilon_{\check{\omega}} = \varepsilon_{\mathrm{m}}$ the split by mobility, $f = \bar{k}$ and $s = (\bar{m}, \check{\omega})$, is the only split for which the metric has the form G_ε with blocks independent of ε_{m}; restricted to a selected-edge block it satisfies the three hypotheses of open problem 8.7 with $\mathrm{c}_1 = 4$ for every $\mathrm{T} > 0$, on every closed ball whose interior contains the reduced orbit, while on the full sheet its fast-block curvature along the pair means $\bar{k}_i^+$ is $3\mathrm{T}$.

(ii) *Dichotomy at zero temperature.* Hold $S^{(0)}$, drop the temperature terms, and start the mean flow $\dot{\mathbf{x}} = -(\partial_{\bar{k}} V, \varepsilon_{\mathrm{m}} \partial_{\bar{m}} V, \varepsilon_{\check{\omega}} \partial_{\check{\omega}} V)$, $V = V_{S^{(0)}}$, from $\mathbf{x}^\circ$, where every residual equals -1. Unless $(\varepsilon_{\mathrm{m}}, \varepsilon_{\check{\omega}}) \in \{(0,0),(1,0)\}$, the flow matrix of a selected-edge block has the simple eigenvalue 0 on $(1,1,\sqrt{2})$, which both residual rows annihilate, and the two rates

$$\varsigma_\pm = 2(1 + \varepsilon_{\mathrm{m}} + \varepsilon_{\check{\omega}}) \pm 2\sqrt{(1-\varepsilon_{\mathrm{m}})^2 + \varepsilon_{\check{\omega}}^2}, \tag{8.12}$$

and each internal residual is the combination of $e^{-\varsigma_- t}$ and $e^{-\varsigma_+ t}$ with value -1 and derivative $4(\varepsilon_{\mathrm{m}} + \varepsilon_{\check{\omega}})$ (model) or $4(1+\varepsilon_{\check{\omega}})$ (belief) at $t = 0$. With a fast link, $\varepsilon_{\check{\omega}} = 1$, one has $\varsigma_- \geq 4 - 2\sqrt{2}$ for every $\varepsilon_{\mathrm{m}} \in [0,1]$ and $|r_i^m(t)| \leq e^{-(4-2\sqrt{2})t}$: the link removes the model disagreement on the belief clock even when the model means are frozen. With fast model means, $\varepsilon_{\mathrm{m}} = 1$ and $\varepsilon_{\check{\omega}} \in (0,1]$, $r_i^m(t) = -e^{-(4+4\varepsilon_{\check{\omega}})t}$. With slow model means and links, $\varepsilon_{\check{\omega}} = \varepsilon_{\mathrm{m}} \leq 1/2$, one has $6\varepsilon_{\mathrm{m}} - 2\varepsilon_{\mathrm{m}}^2 \leq \varsigma_- \leq 6\varepsilon_{\mathrm{m}}$, $\varsigma_+ \geq 4$, and

$$|r_i^b(t)| \leq (1+\varepsilon_{\mathrm{m}})e^{-4t} + \varepsilon_{\mathrm{m}}, \qquad \big(1 - \tfrac{3}{2}\varepsilon_{\mathrm{m}}\big)e^{-6\varepsilon_{\mathrm{m}} t} - \tfrac{3}{2}\varepsilon_{\mathrm{m}} \leq |r_i^m(t)| \leq e^{-(6\varepsilon_{\mathrm{m}} - 2\varepsilon_{\mathrm{m}}^2)t} + \tfrac{3}{2}\varepsilon_{\mathrm{m}}, \tag{8.13}$$

so for $\varepsilon_{\mathrm{m}} \leq 1/25$ each matched pair has $|r_i^b| \leq 2\varepsilon_{\mathrm{m}}$ and $|r_i^m| \geq 1/2$ at every time from $\frac{1}{4}\log(4/\varepsilon_{\mathrm{m}})$ to $1/(12\varepsilon_{\mathrm{m}})$.

(iii) *Persistence along every history.* Let $\mathrm{T} > 0$, let $\mathbf{x}_t$ be the sampled process from any initial configuration and assignment, and let $\bar{\varepsilon} = \max(\varepsilon_{\mathrm{m}}, \varepsilon_{\breve{\omega}}) > 0$. For an edge e_i with $|r_i^m(\mathbf{x}_0)| \geq 1$, numbers $\mathrm{R} \geq 1$ and $\mathrm{R}' > 0$, and $0 < t \leq [4\bar{\varepsilon}(12\mathrm{R} + 9\mathrm{T}\mathrm{R}')]^{-1}$, the event that $|r_i^m(\mathbf{x}_u)| < 1/2$ for some $u \leq t$ while every residual stays bounded by R and $|\mathbf{x}_u| \leq \mathrm{R}'$ on $[0, t]$ has probability at most $2\exp[-1/(192\mathrm{T}\bar{\varepsilon}t)]$, whatever the assignment path, and at most $2e^{-1/(4\mathrm{T})}$ at the largest such t.

In these arms a belief-agreeing, model-disagreeing interval of order $1/\varepsilon_{\mathrm{m}}$ exists exactly when the model means and the links are both slow: with both slow it is the interval of (ii), whose model disagreement (iii) protects along every assignment path up to a time of order $1/\varepsilon_{\mathrm{m}}$ while residuals and configuration stay bounded, and a fast link or fast model means remove the disagreement at a rate at least $4 - 2\sqrt{2}$ whatever the other rate. [ESTABLISHED]

Proof. For (i), both endpoints of e_0 and of e_2 attend them under $S^{(0)}$ and no receiver attends e_1 or e_3, so $V_{S^{(0)}} = 2\mathcal{D}_{e_0} + 2\mathcal{D}_{e_2}$ with $2\mathcal{D}_{e_i} = (\sqrt{2}\bar{k}_i^- - \breve{\omega}_i)^2 + (\sqrt{2}\bar{m}_i^- - \breve{\omega}_i)^2$, the reference term is invariant under the orthonormal change of coordinates, and the evaluator term is $\mathrm{T}\sum_{i\in\{0,2\}}[(\bar{k}_i^- - \bar{m}_i^-)^2 + (\bar{k}_i^+ - \bar{m}_i^+)^2]$; collecting second derivatives gives (8.11), and the mobility stays diagonal because the new coordinates mix one channel at a time. On a held assignment (8.10) is linear with constant coefficients, so a Gaussian law stays Gaussian and its mean solves the noiseless equation. Since $U_S(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top\nabla^2 U_S\mathbf{x}$, the identity $\mathcal{F}_{\mathrm{T}}(q) = \mathrm{T}D_{\mathrm{KL}}(q\|q^\star) + \mathrm{B}_{\mathrm{T}} - \mathrm{T}\log Z_{\mathrm{opt}}$ of proposition 14.22 at $q = \mathcal{N}(\mu, \Sigma) \otimes \delta_S$ gives $\mathcal{F}_{\mathrm{T}} = U_S(\mu) + \frac{1}{2}\operatorname{Tr}(\nabla^2 U_S\Sigma) - \frac{\mathrm{T}}{2}\log\det\Sigma$ up to a constant. Dividing the mean equation by ε_{m} gives the slow-time form with the metric $\varepsilon_{\mathrm{m}}I_4 \oplus I_4 \oplus (\varepsilon_{\mathrm{m}}/\varepsilon_{\breve{\omega}})I_4$, whose blocks are independent of ε_{m} exactly when f consists of the unit-rate coordinates. On a selected-edge block with $\varepsilon_{\breve{\omega}} = \varepsilon_{\mathrm{m}}$, F is the quadratic form of the first block of (8.11), positive definite for $\mathrm{T} > 0$ since its leading minors are $4 + 3\mathrm{T}$, $(4 + \mathrm{T})(4 + 5\mathrm{T})$ and $\mathrm{T}(8 + \mathrm{T})(4 + 5\mathrm{T})$; $\partial_{ff}^2 F = 4 + 3\mathrm{T} \geq 4\mathsf{G}_{\mathrm{f}}$; the reduced matrix is a Schur complement, hence positive definite, so $|s|$ is nonincreasing along the reduced flow and every closed ball whose interior contains the initial slow state serves as $\mathcal{K}$; and the frozen fast flow is linear with coefficient $4 + 3\mathrm{T}$, so every fast state lies in its basin. The pair-mean block gives the curvature $3\mathrm{T}$ along $\bar{k}_i^+$.

For (ii), the block flow matrix is $\operatorname{diag}(1, \varepsilon_{\mathrm{m}}, \varepsilon_{\breve{\omega}})$ times the first block of (8.11) at $\mathrm{T} = 0$, whose kernel is spanned by $(1, 1, \sqrt{2})$, on which $r_i^b = r_i^m = 0$. Expanding the determinant, its characteristic polynomial is $\varsigma[\varsigma^2 - 4(1 + \varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}})\varsigma + 8(2\varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}} + \varepsilon_{\mathrm{m}}\varepsilon_{\breve{\omega}})]$, whose quadratic factor has discriminant $16[(1 - \varepsilon_{\mathrm{m}})^2 + \varepsilon_{\breve{\omega}}^2]$ and constant term vanishing only at $(0, 0)$, so its roots are (8.12) and the eigenvalues are distinct outside the two excluded pairs. The matrix is then diagonalizable and the residual rows annihilate the kernel, so each internal residual is $\mathrm{c}_2 e^{-\varsigma_- t} + \mathrm{c}_3 e^{-\varsigma_+ t}$ with $\mathrm{c}_2 + \mathrm{c}_3 = -1$ and $\varsigma_-\mathrm{c}_2 + \varsigma_+\mathrm{c}_3$ equal to minus the derivative at $t = 0$, which is $a \cdot \operatorname{diag}(1, \varepsilon_{\mathrm{m}}, \varepsilon_{\breve{\omega}})(-2\sqrt{2}, -2\sqrt{2}, 4)^\top$ for the residual row a; hence $\mathrm{c}_3 = (\varsigma_- - \mathrm{c}_5)/(\varsigma_+ - \varsigma_-)$ with $\mathrm{c}_5 = 4(\varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}})$ for r_i^m and $\mathrm{c}_5 = 4(1 + \varepsilon_{\breve{\omega}})$ for r_i^b. For $\varepsilon_{\breve{\omega}} = 1$ put $\mathrm{c}_4 = \sqrt{(1 - \varepsilon_{\mathrm{m}})^2 + 1}$, so $\varsigma_\pm = 4 + 2\varepsilon_{\mathrm{m}} \pm 2\mathrm{c}_4$; ς_- increases in ε_{m} from $4 - 2\sqrt{2}$, and $\varsigma_- \leq 4 + 4\varepsilon_{\mathrm{m}} \leq \varsigma_+$ because $\mathrm{c}_4 \geq \varepsilon_{\mathrm{m}}$, so both coefficients of r_i^m are nonpositive with sum -1. For $\varepsilon_{\mathrm{m}} = 1$, $\varsigma_- = 4 = \mathrm{c}_5 - 4\varepsilon_{\breve{\omega}}$ gives $\mathrm{c}_3 = -1$. For $\varepsilon_{\breve{\omega}} = \varepsilon_{\mathrm{m}} \leq 1/2$ put $\mathrm{c}_4 = \sqrt{(1 - \varepsilon_{\mathrm{m}})^2 + \varepsilon_{\mathrm{m}}^2}$, so that $1 - \varepsilon_{\mathrm{m}} \leq \mathrm{c}_4 \leq 1 - \varepsilon_{\mathrm{m}} + \varepsilon_{\mathrm{m}}^2$ by squaring, $\varsigma_\pm = 2 + 4\varepsilon_{\mathrm{m}} \pm 2\mathrm{c}_4$ and $\varsigma_+ - \varsigma_- = 4\mathrm{c}_4 \geq 4(1 - \varepsilon_{\mathrm{m}})$, which gives the stated brackets; the coefficient of $e^{-\varsigma_+ t}$ in r_i^m is $(\varsigma_- - 8\varepsilon_{\mathrm{m}})/(4\mathrm{c}_4)$, of absolute value at

most $\varepsilon_{\mathrm{m}}(1+\varepsilon_{\mathrm{m}})/(2(1-\varepsilon_{\mathrm{m}})) \le \frac{3}{2}\varepsilon_{\mathrm{m}}$, and the coefficient of $e^{-\varsigma_- t}$ in r_i^b is $(\varsigma_- - 4\varepsilon_{\mathrm{m}})/(4\mathrm{c}_4) \in [0, \varepsilon_{\mathrm{m}}]$. Since each pair of coefficients sums to -1, (8.13) follows, and the last claim is (8.13) at the two times, using $(1+\varepsilon_{\mathrm{m}})/4 \le 1$ and $e^{-1/2}(1-\frac{3}{50}) - \frac{3}{50} > \frac{1}{2}$.

For (iii), write a for the row of r_i^m, with entries -1 and 1 at $\bar{m}_i$ and $\bar{m}_{i+1}$ and -1 at $\breve{\omega}_i$. Since the clock leaves $\mathbf{x}$ unchanged, (8.10) gives, along every assignment path, $r_i^m(\mathbf{x}_t) - r_i^m(\mathbf{x}_0) = -\int_0^t a \cdot (\partial_{\bar{k}} U_{S_u}, \varepsilon_{\mathrm{m}} \partial_{\bar{m}} U_{S_u}, \varepsilon_{\breve{\omega}} \partial_{\breve{\omega}} U_{S_u})(\mathbf{x}_u) du + \sqrt{2\mathrm{T}(2\varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}})} B_t$ with B a standard Brownian motion. With $\mathrm{n}_j(S) \in \{0,1,2\}$ the number of receivers attending e_j under S, $\partial_{\bar{m}_j} U_S = \mathrm{n}_{j-1}(S) r_{j-1}^m - \mathrm{n}_j(S) r_j^m + 3\mathrm{T}\bar{m}_j - 2\mathrm{T}\bar{k}_j$ and $\partial_{\breve{\omega}_j} U_S = -\mathrm{n}_j(S)(r_j^b + r_j^m) + \mathrm{T}\breve{\omega}_j$, so on the stated event the integrand is bounded by $\varepsilon_{\mathrm{m}}(8\mathrm{R} + 8\mathrm{TR}') + \varepsilon_{\breve{\omega}}(4\mathrm{R} + \mathrm{TR}') \le \bar{\varepsilon}(12\mathrm{R} + 9\mathrm{TR}')$, using $3|\bar{m}_j| + 2|\bar{k}_j| \le \sqrt{13}|\mathbf{x}|$. For t as stated the drift moves the residual by at most $1/4$, so a crossing below $1/2$ requires $\sup_{u \le t} |B_u| \ge [4\sqrt{2\mathrm{T}(2\varepsilon_{\mathrm{m}} + \varepsilon_{\breve{\omega}})}]^{-1} \ge [4\sqrt{6\mathrm{T}\bar{\varepsilon}}]^{-1}$. Doob's maximal inequality applied to the martingale $\exp(\theta B_u - \theta^2 u/2)$ with $\theta = \mathrm{y}/t$ gives $\Pr(\sup_{u \le t} B_u \ge \mathrm{y}) \le e^{-\mathrm{y}^2/(2t)}$, and with $\mathrm{y}^2 = 1/(96\mathrm{T}\bar{\varepsilon})$ the two-sided bound is $2e^{-1/(192\mathrm{T}\bar{\varepsilon}t)}$; at the largest t the exponent is $-(12\mathrm{R} + 9\mathrm{TR}')/(48\mathrm{T}) \le -1/(4\mathrm{T})$. The closing sentence collects (ii) and (iii), the fast-means case having rate $4 + 4\varepsilon_{\breve{\omega}} \ge 4$. □

In the latent reading of chapter 9, the link translations are generators of the translation instance with anchor zero and no loop term, and the supplied rate $\varepsilon_{\breve{\omega}}$ places their recognition means in the slow block on the held matching sheet; the natural-gradient recognition flow of the generators assigns belief means, model means, and link means one common rate on an invariant slice, so the separation used here comes from the supplied mobilities (remark 9.30).

Open Problem 8.7 (Slow-manifold reduction of the collective free energy). Consider the rate-separated flow (8.7) under three hypotheses. First, $F \in C^2$ and the block metric $\mathsf{G}_{\mathrm{f}}(f,s) \oplus \mathsf{G}_{\mathrm{s}}(s)$ has C^1 symmetric positive-definite blocks with G_{s} independent of f. Second, there are a compact set $\mathcal{K} \subset \mathcal{U}_0$, a C^1 branch $f^\star$ of conditional minimizers, and $\mathrm{c}_1 > 0$ with $\partial_{ff}^2 F(f^\star(s), s) \succeq \mathrm{c}_1 \mathsf{G}_{\mathrm{f}}(f^\star(s), s)$ for all $s \in \mathcal{K}$. Third, the reduced flow (8.6) from s_0 stays in the interior of $\mathcal{K}$ on $[0, t_1]$, and the initial fast state lies in the basin of $f^\star(s_0)$ for the frozen fast flow $\dot{f} = -\mathsf{G}_{\mathrm{f}}^{-1} \partial_f F(f, s_0)$. Under the second hypothesis the linearized frozen fast flow has spectrum in $(-\infty, -\mathrm{c}_1]$, so $f^\star(s)$ is isolated and uniformly asymptotically stable on $\mathcal{K}$, and Tikhonov's theorem gives $s_\varepsilon(t) \to s(t)$ for $0 \le t \le t_1$ and $f_\varepsilon(t) \to f^\star(s(t))$ for $0 < t \le t_1$ as $\varepsilon \to 0$ [97]; for smooth data Fenichel's theory gives a locally invariant slow manifold $O(\varepsilon)$-close to a compact normally hyperbolic piece of $\mathcal{S}^\star$ [36]. Their application at c_* is open beyond the held matching sheets of proposition 8.6(i), whose selected-edge blocks meet the three hypotheses with a curvature constant uniform in the temperature, so that Tikhonov's theorem applies to them, while the split by mobility on the full sheet has a constant of order T: proposition 6.8 shows that the second hypothesis can fail after the source rows are optimized even when every source energy and every frozen-weight curvature is positive, the sampled dynamics with source jumps of chapter 14 lie outside this deterministic statement, and no explicit bound on the slow orbit or its Fisher duration in terms of ε is available. It is settled in the class of the sampled dynamics of definition 14.21 under the mobility of proposition 8.6 with $\varepsilon_{\breve{\omega}} = \varepsilon_{\mathrm{m}}$ and no parameter learning, where the split by mobility, $f = \bar{k}$ and $s = (\bar{m}, \breve{\omega})$, and (8.6) for $F = U_S$ under the current assignment S define the reduced slow flow of an assignment path: positively, by a bound on the distance between the slow coordinates of the sampled process and that reduced

flow over a bounded slow-time interval, explicit in ε_{m}, T, and the number of assignment changes, tending to zero with ε_{m} and T, and holding, for every positive integer n, with probability tending to one on the event that the assignment path has at most n changes per unit slow time; negatively, by a positive integer n, a sequence of mobility ratios and temperatures tending to zero, and initial states from which, with probability bounded below, the assignment path has at most n changes per unit slow time and that distance stays bounded below over a bounded slow-time interval. The learning case, in which the generative parameter is the slow coordinate and the reduced object is the recognition law at the frozen parameter, lies outside this class.

8.3 Parent-sensitive inference and the prepared parent

Definition 8.8 (Parent-sensitive sweep). Let Ξ be a nonempty finite set of fine states with a parent map $\mathrm{par} : \Xi \to \mathsf{Z}$ that retains the complete decorated parent, keeping only parent values with nonempty fibers. Take a prior p^0 on Ξ with $p^0(\xi) > 0$, normalized conditional laws ψ_ξ of an unobserved record r in a finite set O', and record likelihoods $L_o(\xi) > 0$ at the observation, none of which reads recognition. The generative law of (ξ, r, o) has density $p^0(\xi)\psi_\xi(r)L_o(\xi)$, and its evidence and posterior are

$$\mathcal{Z}_o = \sum_\xi p^0(\xi)L_o(\xi), \qquad p(\xi) = \frac{p^0(\xi)L_o(\xi)}{\mathcal{Z}_o}. \tag{8.14}$$

On an undirected graph of allowed moves take symmetric proposal weights $\mathsf{a}_{\xi\xi'} = \mathsf{a}_{\xi'\xi} \geq 0$ with off-diagonal row sums at most $1/2$. The Metropolis kernel, the parent projection, the posterior lift, the refresh, and the sweep are

$$\begin{aligned} &\mathsf{M}_{\xi\xi'} = \mathsf{a}_{\xi\xi'} \min\bigl(1, p(\xi')/p(\xi)\bigr) \quad (\xi \neq \xi'), \quad \mathsf{M}_{\xi\xi} = 1 - \sum_{\xi' \neq \xi} \mathsf{M}_{\xi\xi'}, \\ &\mathsf{C}_{\xi z} = \mathbf{1}_{\mathrm{par}(\xi)=z}, \qquad \mathsf{J}_{z\xi} = \frac{p(\xi)\mathbf{1}_{\mathrm{par}(\xi)=z}}{p_Z(z)}, \qquad p_Z = p\mathsf{C}, \\ &\mathsf{D} = \mathsf{CJ}, \qquad \mathsf{T} = \mathsf{MD}. \end{aligned} \tag{8.15}$$

The sweep depends on the parent through the conditional laws $p(\cdot \mid z)$. The marked version of a kernel K on Ξ is $\widetilde{\mathsf{K}}((\xi, r), (\xi', r')) = \mathsf{K}_{\xi\xi'}\psi_{\xi'}(r')$. The proposal and the update order are declared choices.

Hypothesis 8.9 (Instantaneous within-parent refresh). In this section the refresh is the exact kernel $\mathsf{D} = \mathsf{CJ}$: each application resamples the fine state from its exact conditional law given the current complete parent. This is the adiabatic elimination of the within-parent fine state, whose timescale is infinitely shorter than the parent's. It excludes finite within-fiber mixing, for example the partial refresh $\mathsf{D}_\omega = (1-\omega)I + \omega\mathsf{CJ}$ with $0 \leq \omega < 1$, which is equally normalized, p-reversible, and parent preserving.

Theorem 8.10 (Stationarity, exact descent, and finite-component convergence).

(i) For a recognition law $q(\xi)\psi'_\xi(r)$ the free energy is

$$\mathcal{F}(q \otimes \psi'; o) = D_{\mathrm{KL}}(q\|p) + \mathbb{E}_q D_{\mathrm{KL}}(\psi'_\xi\|\psi_\xi) - \log \mathcal{Z}_o, \tag{8.16}$$

so matching the record conditionals, $\psi' = \psi$, is a specialization of the joint objective and not the removal of its dependence terms.

(ii) The kernels M, D, T are normalized and preserve p, and M and D are p-reversible; their marked versions are normalized and preserve $p(\xi)\psi_\xi(r)$.

(iii) qD minimizes $D_{\mathrm{KL}}(\cdot\|p)$ among laws with parent marginal qC, with $D_{\mathrm{KL}}(q\mathsf{D}\|p) = D_{\mathrm{KL}}(q\mathsf{C}\|p_Z)$.

(iv) Call a component a connected component of the union of the positive off-diagonal edges of M and D. For every initial q, T preserves component masses, and for every p-stationary kernel K on Ξ, with $q' = q\mathsf{K}$ and reversal $\mathsf{K}^*_{\xi'\xi} = p(\xi)\mathsf{K}_{\xi\xi'}/p(\xi')$,

$$D_{\mathrm{KL}}(q\|p) - D_{\mathrm{KL}}(q\mathsf{K}\|p) = \sum_{\xi':q'(\xi')>0} q'(\xi') D_{\mathrm{KL}}\big(q(\cdot \mid \xi')\|\mathsf{K}^*_{\xi'\cdot}\big), \tag{8.17}$$

where $q(\xi \mid \xi') = q(\xi)\mathsf{K}_{\xi\xi'}/q'(\xi')$. Hence T decreases (8.16), strictly unless $q(\xi) = q(\Xi')p(\xi \mid \Xi')$ on each component Ξ'.

(v) On each component, $q\mathsf{T}^n$ converges geometrically to that conditional target.

[ESTABLISHED]

Proof. For (i), the posterior of (ξ, r) given o is $p(\xi)\psi_\xi(r)$ because r is unobserved and conditionally independent of o, and the evidence is $\mathcal{Z}_o$; definition 5.6 and the chain rule lemma 3.12 give the display. For (ii), $p(\xi)\mathsf{M}_{\xi\xi'} = \mathsf{a}_{\xi\xi'} \min(p(\xi), p(\xi')) = p(\xi')\mathsf{M}_{\xi'\xi}$; JC $= I$ gives $p\mathsf{D} = p$, and $p(\xi)\mathsf{D}_{\xi\xi'} = p(\xi)p(\xi')/p_Z(z)$ on a common parent fiber z. Stationarity survives composition; summing the terminal record of a marked kernel proves normalization and summing the initial record proves invariance. Part (iii) is corollary 5.22 for the block of fine coordinates given the parent: qD has parent marginal $q\mathsf{CJC} = q\mathsf{C}$ and conditionals $p(\cdot \mid z)$, and the chain rule over the parent gives the value. For (iv), the input–output laws $q(\xi)\mathsf{K}_{\xi\xi'}$ and $p(\xi)\mathsf{K}_{\xi\xi'}$ have divergence $D_{\mathrm{KL}}(q\|p)$, and disintegrating them over the output with lemma 3.12 gives (8.17). With positive holding, equality forces q/p to be constant across every positive edge; equality for M makes q stationary under M, and equality for the subsequent D makes the ratio constant on parent fibers, which is constancy on components. For (v), positive holding makes every M or D edge a positive T edge, so on a component Ξ' of n states $\mathsf{T}^k > 0$ for $k = \max(1, n-1)$. With target $p_{\Xi'} = p(\cdot \mid \Xi')$ and $\mathrm{c}_1 = \min_{\xi\xi'}(\mathsf{T}^k)_{\xi\xi'}/p_{\Xi'}(\xi') \in (0, 1]$, one has $\mathsf{T}^k = \mathrm{c}_1 \mathbf{1} p_{\Xi'} + (1 - \mathrm{c}_1)\mathsf{K}'$ on Ξ' with a $p_{\Xi'}$-stationary stochastic K' (or $\mathsf{T}^k = \mathbf{1}p_{\Xi'}$ when $\mathrm{c}_1 = 1$), and convexity with data processing give $D_{\mathrm{KL}}(q\mathsf{T}^k\|p_{\Xi'}) \le (1 - \mathrm{c}_1) D_{\mathrm{KL}}(q\|p_{\Xi'})$ for laws q on Ξ'. □

Theorem 8.11 (The prepared parent is Markov). Under hypothesis 8.9, the reduced sweep $\overline{\mathsf{T}} := \mathsf{JMC}$ is reversible under p_Z and

$$\mathsf{JT} = \overline{\mathsf{T}}\mathsf{J}, \qquad (\nu\mathsf{J})\mathsf{T}^n = \nu\overline{\mathsf{T}}^n\mathsf{J}. \tag{8.18}$$

For a start $q_0 = \nu\mathsf{J}$ the parent history $Z_t = \mathrm{par}(\xi_t)$ of the chain with kernel T is Markov with transition $\overline{\mathsf{T}}$, and for an arbitrary start it is so after the first completed refresh. [ESTABLISHED]

Proof. With $\mathsf{T} = \mathsf{MCJ}$ and $\mathsf{JC} = I$, $\mathsf{JT} = \mathsf{JMCJ} = \overline{\mathsf{T}}\mathsf{J}$, and induction gives the second identity. Moreover $p_Z(z)\overline{\mathsf{T}}_{zz'}$ is the sum of $p(\xi)\mathsf{M}_{\xi\xi'}$ over $\mathrm{par}(\xi) = z$ and $\mathrm{par}(\xi') = z'$, which is symmetric by reversibility of M. The final refresh makes the fine law given the whole parent history $\mathsf{J}_{z\cdot}$, with z the last parent, so the next parent row is $\mathsf{J}_{z\cdot}\mathsf{MC} = \overline{\mathsf{T}}_{z\cdot}$, and induction over histories gives the Markov property. □

The refresh holds the fast fine state at its conditional optimum given the slow parent (theorem 8.10(iii)), so $\mathsf{JT} = \overline{\mathsf{T}}\mathsf{J}$ is the discrete instantaneous-equilibration case of (8.8).

Remark 8.12 (Finite refresh leaves parent memory). Hypothesis 8.9 cannot be dropped. Take $\Xi = \{1, 2, 3\}$ with $\mathrm{par}(1) = \mathrm{par}(2) = z_1$ and $\mathrm{par}(3) = z_2$, uniform p, the single proposal $\mathsf{a}_{23} = 1/2$, the partial sweep $\mathsf{T}_\omega = \mathsf{MD}_\omega$, and a start $\nu\mathsf{J}$ with $\nu(z_1), \nu(z_2) > 0$. Then

$$\mathbb{P}(Z_2 = z_2 \mid Z_1 = z_1, Z_0 = z_1) = 1/6 + \omega/12, \qquad \mathbb{P}(Z_2 = z_2 \mid Z_1 = z_1, Z_0 = z_2) = 1/2 - \omega/4. \tag{8.19}$$

The dependence on Z_0 is $(1-\omega)/3$, equal to $7/30$ at $\omega = 3/10$, and it vanishes only at $\omega = 1$, where both values equal $\overline{\mathsf{T}}_{z_1 z_2} = 1/4$. For $\omega < 1$ the parent history is not Markov. It remains a deterministic function of the fine Markov chain with kernel T_ω, so the hidden-history law of section 12.8.1 retains it exactly, and the memory cost of a first-order parent closure is the defect $D_N^{(1)}$ of theorem 12.46, whose value $I(Z_2; Z_0 \mid Z_1)$ at $N = 3$ is positive. [ESTABLISHED]

Proof. Only state 2 moves under M, to 3 with probability $1/2$, and D_ω keeps a state of $\{1, 2\}$ with probability $1 - \omega/2$. Given $Z_1 = z_1$, the state before the second sweep is 2 with probability $1 - \omega/2$ if $Z_0 = z_2$, since it moved from 3, and with probability $1/3 + \omega/6$ if $Z_0 = z_1$, since after M it is 2 with probability $1/3$ and the refresh redraws it uniformly with probability ω; multiplying by $1/2$ gives (8.19). Both conditioning events have positive probability, so $I(Z_2; Z_0 \mid Z_1) > 0$. □

Physicist's summary

Fast and slow are declared, not derived. The collective free energy is then a potential for every positive-definite mobility. The reduced gradient flow is conjugate to its tangent lift on the graph of the conditional optimum; the finite-rate flow shares its slow component there but can leave that graph. Instantaneous within-parent refresh gives the Markov parent proved here. On the four-cycle, slow model means lag the beliefs only if the links are slow too; a fast link carries the model agreement on the belief clock.

Chapter 9

Lie-Algebra Links as Latent Fields

This chapter asks what becomes of the independent links of Definition 2.11 when they are inferred jointly with the agents' states instead of being held as structural data or moved as parameters. Written in the exponential chart of a reference point, every twist is the exponential of a generator in the Lie algebra, on which passive re-expressions do not act (Definition 9.2 and proposition 9.3). A Gaussian base law on the generators multiplied by the exponential of minus a nonnegative loop action is a normalized prior for every structure group whose Lie algebra carries an inner product, compact or not (Proposition 9.12); on the general linear group the loop action measures the reference holonomy with a fiber metric, because no continuous class function separates unipotent holonomy from the identity (Corollary 9.7 and proposition 9.10). With the generators latent, the collective bound of Proposition 5.26 specializes to a joint that has unit mass on every record hypergraph and to a free energy whose link sector is the relative entropy from the generator prior (Propositions 9.16 and 9.18). Its stationarity condition is a link field equation in which the stress of the prior balances the expected matter current transported by the adjoint derivative of the chart (Proposition 9.21); for a comparison record the current has the outer-product form of a delta rule (Corollary 9.22), and without the prior the link problem degenerates (Corollary 9.24). Held links are the zero-width limit of the latent model, and statements proved at every held link transfer conditionally on the generators (Theorem 9.25 and proposition 9.26). Matter attached by record extension induces no link action, while matter normalized jointly with the generators induces minus the logarithm of its normalizer (Proposition 9.27). Recognition of the generators is a first-order natural-gradient history whose explicit Euler step is the Gaussian Bayesian learning rule (Proposition 9.28); across learning episodes a Gibbs prior on generator histories gives the most probable history an inertia whose mass is the precision of the generator increment (Propositions 9.32 and 9.33); and near zero generators the stationarity condition is a linear equation whose positive operator collects the anchor, the loops, the episodes, and the matter (Proposition 9.34). These results are used for the coarse links and the holonomy-conditioned parents of Chapters 11 and 13, whose link statements gain a version with latent generators, and the link translations of Chapter 14 are their translation-group instance (Proposition 9.13); Supplement G realizes them for Gaussian matter. Longer proofs are deferred to appendix B.

The symbols of table 9.1 are declared for this chapter and its proof section, and those that recur in Chapters 11 and 13 are also listed in table D.1; $\mathbf{B}$, $\mathbf{S}$, $\mathbf{T}$, and $\mathbf{H}$ denote generic square matrices, $\mathbf{b}$ and $\mathbf{c}$ generic vectors, and $\mathbf{h}$ a generic real function, each bound in the statement or proof where it

appears, and a ring accent marks an object of the generator sector written in the coordinates of the reference point.

Table 9.1: Symbols of the latent-link construction.

Symbol	Meaning
$\mathfrak{g}$, $\exp$, Ad_g, $\mathrm{d}\rho_x$	Lie algebra of G, its exponential map, the adjoint action, the differential of ρ_x at e_G
E_+, $\mathscr{T} = \mathfrak{g}^{E_+}$	one orientation of each reversal pair; link space of one channel, $\mathscr{T} \times \mathscr{T}$ for two independent channels
$\mathring{\omega}_e^x$, $\exp_\sigma$, $\mathring{\Theta}_p^x$	generator of the twist of e; exponential chart at σ; reference path product, the reference holonomy of a closed walk
$\mathsf{G}^{\mathrm{fib}}$, $\langle\cdot,\cdot\rangle_{\mathsf{G}}$, $\|\cdot\|_{\mathsf{G}}$	fiber metric at the reference point; trace form on $\mathfrak{gl}(\mathbb{V})$ and the inner product on $\mathfrak{g}$
$\mathfrak{Q}_p^x$, $\mathbb{L}$, c_p, $\mathbf{C}$	loop defect of a closed walk; family of closed walks and their coefficients; cycle map
$\mathbb{L}_+$, $\mathbf{C}_+$, z_p, $r_{\mathbb{L}}$, κ	loops with positive coefficient and their cycle map; net traversal vector of a loop; rank of these vectors; number of components of the multigraph
d, Z, $\mathscr{T}_{\mathrm{ex}}$, $\mathscr{T}_{\mathrm{har}}$	vertex-difference map and cycle space of $\mathbb{R}^{E_+}$; exact and harmonic generators
$\nu^{\mathscr{T}}$, $\mathscr{S}$, $Z_{\mathscr{S}}$, $\mathbb{P}_\theta^{\mathscr{T}}$	base law, link action, normalizer, generator prior
$\mathring{\alpha}_e^x$, $\mathring{\Sigma}_e^x$; $\mathring{\mu}$, $\mathring{\Lambda}$	anchor and prior covariance; mean and precision of a Gaussian recognition law R of the generators
$\mathbf{L}(\mathbf{B}, \mathbf{T})$	Fréchet derivative of the matrix exponential
$\mathbf{H}_o$, $\mathbf{U}$, $\mathbf{V}$, $\mathbf{J}_e^x$	matter energy, expected action plus matter energy, most-probable-link objective, link current
$\mathbf{M}$, $\mathbf{M}^\star$, $\mathbf{l}$	transport matrix of a comparison record, its rank-one minimizer, expected record energy
$\mathbf{F}_Q$, $\mathbf{F}$	held-link free energy as a function of the generators, and its infimum over a class
$\mathscr{S}^{\mathrm{ind}}$, $\mathfrak{Z}$	induced link action, joint normalizer of generators and undirected matter
$\mathfrak{i}_\lambda$, $\mathring{\Xi}$, $\mathring{\vartheta}$, $\mathring{\varpi}$, $\mathring{\Gamma}$	generator integrand, space of admissible precisions, natural and expectation parameters, log-partition function
$\mathbf{m}$, $\mathfrak{U}$, $\mathbb{P}^{\mathrm{hist}}$, Z_{hist}, $\mathring{Z}_n^{\leftarrow}$	increment precision, kinetic term, generator history prior, its normalizer, backward functions
$\mathscr{S}^{\mathrm{hist}}$, $\mathfrak{V}_n$	history action and episode potentials
$\mathbf{K}$, $\mathbf{N}_0$, $\mathring{\Phi}$, $\mathbf{j}$	linearized link operator, current correction of the Hessian, matter operator, source

Scope. The chapter does not claim that latent links improve the evidence, the prediction of any record, or any learning outcome relative to held links; that the generator prior is invariant under reference changes that do not act by multiples of isometries of the fiber metric, or that the loop defect is independent of the base vertex of a closed walk; that every local minimizer of the most-probable-link objective is global; that the recognition flow of the generators converges, or that stationary generator histories are unique outside the ball of proposition 9.34; that the joint M-coordinate of the anchors and the prior covariances exists when the action has loop terms; that frames are dynamical, so that time-like links would carry variables; that the linearized link operator

is the operator of a physical field; or that the outer-product form of the link current identifies the model with a fast-weight or delta-rule architecture. [NOT-CLAIMED] The interaction multigraph stays structural, and edge creation and deletion is open problem 6.9; a finite-dimensional family of link laws that survives the coarse step is open problem 13.21; and a slow-manifold reduction with the generator recognition means in the slow block is part of open problem 8.7.

9.1 The exponential chart of the links

This section writes the independent links of Definition 2.11 in the exponential chart of the reference point σ of Proposition 2.12. The twist of an edge copy is unchanged by every passive re-expression, so a generator of the twist carries no vertex gauge: the frames U_i carry the passive action, while a change of reference point or an active automorphism moves all generators by one adjoint map (Proposition 9.3). The chart covers exactly the link families whose twists are exponentials. For $G = \mathrm{GL}(K, \mathbb{R})$ this excludes some twists of positive determinant, while on a closed walk that traverses two reversal pairs once each the reference holonomies of chart links still reach every element of $\mathrm{GL}^+(K)$ (Proposition 9.5).

Lemma 9.1 (Exponential maps and homomorphisms). Let $\varphi : G \to \mathbb{Y}$ be a smooth homomorphism of Lie groups with differential $\mathrm{d}\varphi : \mathfrak{g} \to \mathrm{Lie}(\mathbb{Y})$ at e_G. Then $\varphi(\exp \mathring{\omega}) = \exp(\mathrm{d}\varphi \mathring{\omega})$ for every $\mathring{\omega} \in \mathfrak{g}$. In particular, let Ad_g be the differential at e_G of the conjugation $g' \mapsto g g' g^{-1}$, and let $\rho_x : G \to \mathrm{GL}(\mathbb{V})$ be a smooth homomorphism into the general linear group of a finite-dimensional real vector space $\mathbb{V}$, with differential $\mathrm{d}\rho_x$ at e_G. Then for all $g \in G$ and $\mathring{\omega} \in \mathfrak{g}$, with the matrix exponential as the exponential map of $\mathrm{GL}(\mathbb{V})$,

$$g \exp(\mathring{\omega}) g^{-1} = \exp(\mathrm{Ad}_g \mathring{\omega}), \qquad \rho_x(\exp \mathring{\omega}) = \exp(\mathrm{d}\rho_x \mathring{\omega}), \qquad \mathrm{d}\rho_x(\mathrm{Ad}_g \mathring{\omega}) = \rho_x(g) \mathrm{d}\rho_x(\mathring{\omega}) \rho_x(g)^{-1}. \tag{9.1}$$

[ESTABLISHED]

Proof. The exponential map sends $\mathring{\omega}$ to the value at time one of the integral curve through the identity of the left-invariant vector field whose value at the identity is $\mathring{\omega}$, and $t \mapsto \exp(t\mathring{\omega})$ is a smooth homomorphism from $(\mathbb{R}, +)$. Hence $\gamma(t) = \varphi(\exp(t\mathring{\omega}))$ is a smooth homomorphism from $(\mathbb{R}, +)$ into $\mathbb{Y}$. Differentiating $\gamma(t + t') = \gamma(t)\gamma(t')$ in t' at $t' = 0$ shows that $\gamma'(t)$ is the image of $\gamma'(0)$ under the differential at the identity of the left multiplication by $\gamma(t)$, so γ is the integral curve through the identity of the left-invariant vector field with value $\gamma'(0) = \mathrm{d}\varphi \mathring{\omega}$, and uniqueness of integral curves gives $\gamma(t) = \exp(t \mathrm{d}\varphi \mathring{\omega})$; the first claim is the case $t = 1$. Conjugation by g is a smooth homomorphism with differential Ad_g, which gives the first identity of eq. (9.1). On $\mathrm{GL}(\mathbb{V})$ the left-invariant vector field with value $\mathbf{H}$ at the identity is $y \mapsto y\mathbf{H}$, and the matrix exponential satisfies $\frac{\mathrm{d}}{\mathrm{d}t} e^{t\mathbf{H}} = e^{t\mathbf{H}}\mathbf{H}$ with $e^0 = I_{\mathbb{V}}$, so it is the exponential map of $\mathrm{GL}(\mathbb{V})$; this gives the second identity. Applying ρ_x to the first identity for $t\mathring{\omega}$ and using the second gives $\rho_x(g) \exp(t \mathrm{d}\rho_x \mathring{\omega}) \rho_x(g)^{-1} = \exp(t \mathrm{d}\rho_x(\mathrm{Ad}_g \mathring{\omega}))$, whose derivative at $t = 0$ is the third identity. □

Definition 9.2 (Exponential chart of the links). Let $\mathfrak{g}$ be the Lie algebra of G and $\exp : \mathfrak{g} \to G$ its exponential map, let $E_+ \subseteq E$ contain exactly one edge copy of each reversal pair of $\mathfrak{I}$, and let the reference point σ and the frames $u_i = \sigma \cdot U_i^{-1}$ be those of Proposition 2.12. For a channel x, a generator family is a point $\mathring{\omega}^x = (\mathring{\omega}^x_e)_{e\in E_+}$ of the link space $\mathscr{T} = \mathfrak{g}^{E_+}$, extended to the reversed copies by the first line below. Its twists and links are those of the second line, and for a walk $p = (e_0, \dots, e_{r-1})$ with $e_a : i_{a+1} \to i_a$, as in Definition 2.11, its reference path product is that of the third line, called the reference holonomy of p when p is closed:

$$\mathring{\omega}^x_{\bar{e}} = -\mathring{\omega}^x_e, \tag{9.2}$$

$$\mathring{\Theta}^x_e = \exp(\mathring{\omega}^x_e), \qquad \Theta^x_e = U_i \exp(\mathring{\omega}^x_e) U_j^{-1} \quad (e : j \to i), \tag{9.3}$$

$$\mathring{\Theta}^x_p = \mathring{\Theta}^x_{e_0}\mathring{\Theta}^x_{e_1}\cdots\mathring{\Theta}^x_{e_{r-1}} = \exp(\mathring{\omega}^x_{e_0})\exp(\mathring{\omega}^x_{e_1})\cdots\exp(\mathring{\omega}^x_{e_{r-1}}). \tag{9.4}$$

The links satisfy eq. (2.27) because $\exp(-\mathring{\omega}) = \exp(\mathring{\omega})^{-1}$, and $\mathring{\Theta}^x_e$ is their twist in the sense of eq. (2.32). The exponential chart at σ is the continuous map $\exp_\sigma : \mathscr{T} \to G^{E_+}$, $\mathring{\omega}^x \mapsto (U_i \exp(\mathring{\omega}^x_e) U_j^{-1})_{e\in E_+}$. The link space is a real vector space of dimension $|E_+| \dim G$, standard Borel with its Borel sets, and the chart covers exactly the link families whose twists lie in $\exp(\mathfrak{g})$ (Proposition 9.5). When ρ_x is a smooth homomorphism $G \to \mathrm{GL}(\mathbb{V})$ on a real vector sample fiber $\mathbb{V} = \mathbb{R}^{\dim \mathbb{V}}$, the represented generators are $\mathrm{d}\rho_x(\mathring{\omega}^x_e) \in \mathfrak{gl}(\mathbb{V})$, and $\rho_x(\mathring{\Theta}^x_e) = \exp(\mathrm{d}\rho_x \mathring{\omega}^x_e)$ by Lemma 9.1. The two channels carry independent generator families, with link space $\mathscr{T} \times \mathscr{T}$; under the common-link specialization eq. (2.31) one generator family $\mathring{\omega} = \mathring{\omega}^b = \mathring{\omega}^m$ serves both channels and the index x is dropped.

Proposition 9.3 (Gauge action on generators). In Definition 9.2, let the generator family $\mathring{\omega}^x$ have the links $\Theta^x = \exp_\sigma(\mathring{\omega}^x)$. (a) After a passive re-expression $(g_i)_{i\in V}$ the links $g_i \Theta^x_e g_j^{-1}$ are the links of the same $\mathring{\omega}^x$ relative to σ and the re-expressed frames $g_i U_i$, so every generator is unchanged. (b) Relative to the reference point $\sigma' = \sigma \cdot g_0$ the frames are $U_i g_0$, and the links Θ^x are those of the generator family $\mathrm{Ad}_{g_0^{-1}} \mathring{\omega}^x = (\mathrm{Ad}_{g_0^{-1}} \mathring{\omega}^x_e)_{e\in E_+}$, so that $\exp_{\sigma'} \circ \mathrm{Ad}_{g_0^{-1}} = \exp_\sigma$; every reference path product becomes $g_0^{-1}\mathring{\Theta}^x_p g_0$. (c) An active F with $F(\sigma) = \sigma \cdot \kappa$, as in Proposition 2.12, carries Θ^x to the links of $\mathrm{Ad}_\kappa \mathring{\omega}^x$ relative to σ and the same frames, and every reference path product to $\kappa \mathring{\Theta}^x_p \kappa^{-1}$. (d) The holonomy of a closed walk p based at i_0 and, for a smooth homomorphism $\rho_x : G \to \mathrm{GL}(\mathbb{V})$, its represented reference holonomy are

$$\mathrm{Hol}^x(p) = U_{i_0} \mathring{\Theta}^x_p U_{i_0}^{-1}, \qquad \rho_x(\mathring{\Theta}^x_p) = \exp(\mathrm{d}\rho_x \mathring{\omega}^x_{e_0}) \exp(\mathrm{d}\rho_x \mathring{\omega}^x_{e_1}) \cdots \exp(\mathrm{d}\rho_x \mathring{\omega}^x_{e_{r-1}}). \tag{9.5}$$

Each of (a) to (c) moves all generators by one linear map of $\mathfrak{g}$, namely the identity, $\mathrm{Ad}_{g_0^{-1}}$, or Ad_κ, and the passive vertex group acts on the frames alone. [ESTABLISHED]

Proof. (a) The vertex law eq. (2.28) and $U_i' = g_i U_i$ from eq. (2.11) give

$$g_i \Theta^x_e g_j^{-1} = (g_i U_i) \exp(\mathring{\omega}^x_e) (g_j U_j)^{-1}.$$

(b) From $\sigma = \sigma' \cdot g_0^{-1}$ we get $u_i = \sigma' \cdot (U_i g_0)^{-1}$, and Lemma 9.1 gives $(U_i g_0) \exp(\mathrm{Ad}_{g_0^{-1}} \mathring{\omega}^x_e)(U_j g_0)^{-1} = U_i g_0 g_0^{-1} \exp(\mathring{\omega}^x_e) g_0 g_0^{-1} U_j^{-1} = \Theta^x_e$; the same identity applied to each factor of eq. (9.4) gives the path

products. (c) By Proposition 2.12 the image twists are $\kappa \exp(\mathring{\omega}_e^x)\kappa^{-1} = \exp(\mathrm{Ad}_\kappa \mathring{\omega}_e^x)$, and a twist determines its link given the frames through $\Theta_e^x = U_i \mathring{\Theta}_e^x U_j^{-1}$. (d) The first identity is eq. (2.33), and the second follows from the homomorphism property of ρ_x and Lemma 9.1. □

Proposition 9.4 (Real logarithms). A real square matrix is the exponential of a real square matrix exactly when it is nonsingular and each Jordan block belonging to a negative eigenvalue occurs an even number of times [29, Theorem 1]. [STANDARD]

Proposition 9.5 (Image of the exponential chart). In Definition 9.2 the following hold. (a) The image of $\exp_\sigma$ is the set of link families of channel x whose twists all lie in $\exp(\mathfrak{g})$; it does not depend on σ, and passive re-expressions and active automorphisms map it onto itself. Link families with a twist outside $\exp(\mathfrak{g})$ have no generator family. (b) For $G = \mathrm{GL}(K,\mathbb{R})$ or $G = \mathrm{GL}^+(K)$ the set $\exp(\mathfrak{gl}(K,\mathbb{R}))$ is the set of matrices described in Proposition 9.4 and lies in $\mathrm{GL}^+(K)$; for $K \geq 2$ it is a proper subset of $\mathrm{GL}^+(K)$, since $\mathrm{diag}(-1,-2,1,\ldots,1)$ is not the exponential of a real matrix, so the chart misses every link family with such a twist, and for $G = \mathrm{GL}(K,\mathbb{R})$ it also misses every link family with a twist of negative determinant. (c) Every element of $\mathrm{GL}^+(K)$ equals $\exp(\mathbf{S})\exp(\mathbf{T})$ with $\mathbf{S}$ symmetric and $\mathbf{T}$ skew-symmetric, and every element of $SO(K)$ equals $\exp(\mathbf{T})$ with $\mathbf{T}$ skew-symmetric; hence $\mathrm{GL}^+(K)$ is connected and generated by $\exp(\mathfrak{gl}(K,\mathbb{R}))$, and for $G = SO(K)$ the chart covers every link family. (d) For every connected Lie group G, $\exp(\mathfrak{g})$ generates G. (e) Let G be $\mathrm{GL}(K,\mathbb{R})$ or $\mathrm{GL}^+(K)$, and let the closed walk p traverse two distinct reversal pairs exactly once each. Then every element of $\mathrm{GL}^+(K)$ is the reference holonomy $\mathring{\Theta}_p$ of some generator family, so for $K \geq 2$ the reference holonomies of chart links leave $\exp(\mathfrak{gl}(K,\mathbb{R}))$. On a triangle walk (e_0,e_1,e_2) with every $e_a \in E_+$ and $K = 2$, the generators $\mathrm{diag}(0,\log 2)$, $\pi(E_{21} - E_{12})$, and 0 give $\mathring{\Theta}_p = \mathrm{diag}(-1,-2)$. [ESTABLISHED]

The proof is in appendix B.3.

9.2 Curvature on a noncompact structure group

A loop penalty should vanish exactly on the closed walks with trivial represented holonomy. For compact G and unitary ρ_x the class function of Remark 2.16 does so; on the general linear group no continuous class function can (Corollary 9.7), because conjugation contracts every unipotent element to the identity (Proposition 9.6). The loop defect defined below measures the reference holonomy with a fiber metric placed at the reference point. It separates every holonomy from the identity; with the metric held as a matrix at the reference point, it is invariant under the reference changes that act by multiples of isometries of that metric, and for the general linear group under no others (Proposition 9.10).

Proposition 9.6 (Continuous class functions are blind to unipotent holonomy). Let $K \geq 2$, let $\mathbb{H}$ be $\mathrm{GL}(K,\mathbb{R})$ or $\mathrm{GL}^+(K)$, and let $\mathbf{h} : \mathbb{H} \to \mathbb{R}$ be continuous and constant on the conjugacy classes of $\mathbb{H}$. Then $\mathbf{h}(I_K + \mathbf{T}) = \mathbf{h}(I_K)$ for every nilpotent $\mathbf{T}$, that is, at every unipotent matrix $I_K + \mathbf{T}$, which lies in $\mathrm{GL}^+(K)$ because $\det(I_K + \mathbf{T}) = 1$. [ESTABLISHED]

Proof. Since $\mathbf{T}^K = 0$, the kernels $\ker \mathbf{T} \subseteq \ker \mathbf{T}^2 \subseteq \cdots \subseteq \ker \mathbf{T}^K = \mathbb{R}^K$ increase, and $\mathbf{T}$ maps $\ker \mathbf{T}^{k+1}$ into $\ker \mathbf{T}^k$. Listing a basis of $\ker \mathbf{T}$, extending it to a basis of $\ker \mathbf{T}^2$, and so on, gives the columns of an invertible $\mathbf{B}$ with $\mathbf{B}^{-1}\mathbf{T}\mathbf{B} = \mathbf{T}_0$ strictly upper triangular. Replacing the first column of $\mathbf{B}$ by its negative conjugates $\mathbf{T}_0$ by a diagonal matrix, which keeps it strictly upper triangular, so $\det \mathbf{B} > 0$ may be assumed. For $t > 0$ let $\mathbf{S}_t = \mathrm{diag}(1, t, \ldots, t^{K-1})$; then $\mathbf{B}\mathbf{S}_t \in \mathrm{GL}^+(K) \subseteq \mathbb{H}$ and $(\mathbf{S}_t^{-1}\mathbf{T}_0\mathbf{S}_t)_{kl} = t^{l-k}(\mathbf{T}_0)_{kl}$, which tends to zero as $t \to 0$ because $(\mathbf{T}_0)_{kl} = 0$ unless $l > k$. Hence $\mathbf{h}(I_K + \mathbf{T}) = \mathbf{h}\big((\mathbf{B}\mathbf{S}_t)^{-1}(I_K + \mathbf{T})\mathbf{B}\mathbf{S}_t\big) = \mathbf{h}(I_K + \mathbf{S}_t^{-1}\mathbf{T}_0\mathbf{S}_t)$ for every $t > 0$, and continuity gives $\mathbf{h}(I_K + \mathbf{T}) = \mathbf{h}(I_K)$. □

Corollary 9.7 (No invariant loop defect on the general linear group). Let $K \geq 2$ and $\mathbb{H}$ be as in Proposition 9.6. No continuous function on $\mathbb{H}$ that is constant on its conjugacy classes vanishes exactly at I_K: every such function that vanishes at I_K vanishes at the shear $I_K + E_{12} \neq I_K$, and so does every function of the eigenvalues, counted with multiplicity, that vanishes at I_K. Consequently, for $G = \mathrm{GL}(K, \mathbb{R})$ or $G = \mathrm{GL}^+(K)$ and a closed walk p that traverses some reversal pair exactly once, a loop penalty that is a continuous function of $\mathrm{Hol}^x(p)$ on G and is unchanged by every passive re-expression of every link family takes at every unipotent holonomy the value it takes at the trivial holonomy, so it vanishes at every unipotent holonomy whenever it vanishes at the trivial one. In particular no norm on $\mathfrak{gl}(K, \mathbb{R})$ is invariant under conjugation by $\mathbb{H}$, which for $\mathbb{H} = \mathrm{GL}(K, \mathbb{R})$ is the observation of Remark 2.16; the corollary is stronger, since it excludes every continuous conjugation-invariant penalty and not only those of the form $g' \mapsto \|g' - I_K\|$. [ESTABLISHED]

Proof. The first claim is Proposition 9.6 at $\mathbf{T} = E_{12}$, which is nilpotent because $E_{12}^2 = 0$; the shear is upper triangular with unit diagonal, so its eigenvalues are those of I_K. If p traverses the pair of e exactly once, the link family with Θ_e^x equal to a given $g' \in G$ or to g'^{-1}, according to the direction of the traversal, and with all other links equal to e_G has $\mathrm{Hol}^x(p) = g'$, and the passive re-expression with $g_{i_0} = g$ replaces it by $gg'g^{-1}$ by eq. (2.33); the penalty is therefore a continuous class function on G. For a norm $\|\cdot\|_\star$ invariant under conjugation by $\mathbb{H}$, the function $g' \mapsto \|g' - I_K\|_\star$ is continuous, constant on conjugacy classes because $gg'g^{-1} - I_K = g(g' - I_K)g^{-1}$, and vanishes exactly at I_K. □

Definition 9.8 (Loop defect and cycle map). In Definition 9.2, let $\rho_x : G \to \mathrm{GL}(\mathbb{V})$ be a smooth homomorphism on a real vector sample fiber $\mathbb{V} = \mathbb{R}^{\dim \mathbb{V}}$, and let $\mathsf{G}^{\mathrm{fib}}$ be a symmetric positive definite matrix at the reference point: the matrix, in the coordinates of σ, of an inner product on the fiber at c_* of the vector bundle associated with ρ_x. It plays the role of the fiber metric of Remark 2.16, placed at σ instead of at a vertex. On $\mathfrak{gl}(\mathbb{V})$ put

$$\langle \mathbf{H}, \mathbf{H}' \rangle_{\mathsf{G}} = \mathrm{Tr}\big((\mathsf{G}^{\mathrm{fib}})^{-1}\mathbf{H}^\top \mathsf{G}^{\mathrm{fib}}\mathbf{H}'\big), \qquad \|\mathbf{H}\|_{\mathsf{G}} = \langle \mathbf{H}, \mathbf{H} \rangle_{\mathsf{G}}^{1/2}, \tag{9.6}$$

and, when $\mathrm{d}\rho_x$ is injective, write $\langle \mathring{\omega}, \mathring{\omega}' \rangle_{\mathsf{G}} = \langle \mathrm{d}\rho_x \mathring{\omega}, \mathrm{d}\rho_x \mathring{\omega}' \rangle_{\mathsf{G}}$ for the pulled-back inner product on $\mathfrak{g}$. The loop defect of a closed walk p based at i_0 is the first quantity below, with $\mathring{\Theta}_p^x$ the reference holonomy eq. (9.4). For a finite family $\mathbb{L}$ of closed walks, the cycle map $\mathbf{C} : \mathscr{T} \to \mathfrak{g}^{\mathbb{L}}$ is the second,

with $p = (e_0, \dots, e_{r-1})$ and the reversal convention eq. (9.2):

$$\mathfrak{Q}_p^x(\mathring{\omega}^x) = \tfrac{1}{2}\big\|I_{\mathbb{V}} - \rho_x(\mathring{\Theta}_p^x)\big\|_{\mathsf{G}}^2, \tag{9.7}$$

$$(\mathbf{C}\mathring{\omega}^x)_p = \mathring{\omega}_{e_0}^x + \mathring{\omega}_{e_1}^x + \cdots + \mathring{\omega}_{e_{r-1}}^x. \tag{9.8}$$

Each reversal pair thus enters $(\mathbf{C}\mathring{\omega}^x)_p$ with its generator on E_+ times the number of traversals of p along the orientation in E_+ minus the number against it, and $\mathbf{C}$ is linear.

Lemma 9.9 (The trace form determines the metric up to scale). For symmetric positive definite matrices G' and G'' on $\mathbb{V}$, write $\|\cdot\|_{\mathsf{G}'}$ and $\|\cdot\|_{\mathsf{G}''}$ for eq. (9.6) with G' and G'' in place of $\mathsf{G}^{\mathrm{fib}}$. Then $\|\mathbf{H}\|_{\mathsf{G}'} = \|\mathbf{H}\|_{\mathsf{G}''}$ for every rank-one $\mathbf{H}$ exactly when G'' is a positive multiple of G', and in that case the two norms agree on all of $\mathfrak{gl}(\mathbb{V})$. [ESTABLISHED]

Proof. For $\mathbf{H} = \mathbf{b}\mathbf{c}^\top$, cyclicity of the trace gives $\|\mathbf{H}\|_{\mathsf{G}'}^2 = (\mathbf{b}^\top\mathsf{G}'\mathbf{b})(\mathbf{c}^\top\mathsf{G}'^{-1}\mathbf{c})$, and likewise for G''. If $\mathsf{G}'' = \mathrm{c_s}\mathsf{G}'$ with $\mathrm{c_s} > 0$, the factors $\mathrm{c_s}^{-1}$ and $\mathrm{c_s}$ in eq. (9.6) cancel for every $\mathbf{H}$. Conversely, fixing $\mathbf{c} \neq 0$ gives $\mathbf{b}^\top\mathsf{G}''\mathbf{b} = \mathrm{c_s}\mathbf{b}^\top\mathsf{G}'\mathbf{b}$ for every $\mathbf{b}$, with $\mathrm{c_s} = (\mathbf{c}^\top\mathsf{G}'^{-1}\mathbf{c})/(\mathbf{c}^\top\mathsf{G}''^{-1}\mathbf{c}) > 0$, and symmetric matrices with proportional quadratic forms are proportional. □

Proposition 9.10 (Properties of the loop defect). In Definition 9.8, let $p = (e_0, \dots, e_{r-1})$ be a closed walk based at i_0.

(a) The form eq. (9.6) is an inner product on $\mathfrak{gl}(\mathbb{V})$ with $\|\mathbf{H}\mathbf{H}'\|_{\mathsf{G}} \le \|\mathbf{H}\|_{\mathsf{G}}\|\mathbf{H}'\|_{\mathsf{G}}$ and $\|I_{\mathbb{V}}\|_{\mathsf{G}} = (\dim\mathbb{V})^{1/2}$. The defect $\mathfrak{Q}_p^x$ is a continuous function on $\mathscr{T}$ that depends on $\mathring{\omega}^x$ only through the links; it is nonnegative and vanishes exactly when $\rho_x(\mathring{\Theta}_p^x) = I_{\mathbb{V}}$, equivalently when $\rho_x(\mathrm{Hol}^x(p)) = I_{\mathbb{V}}$.

(b) Passive re-expressions leave $\mathfrak{Q}_p^x$ unchanged. Under the reference change $\sigma' = \sigma \cdot g_0$, with $\mathbf{R} = \rho_x(g_0)$ and the generators $\mathrm{Ad}_{g_0^{-1}}\mathring{\omega}^x$ of Proposition 9.3, the defect computed at σ' with the matrix $\mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R}$, which represents the same inner product in the coordinates of σ', equals $\mathfrak{Q}_p^x$ for every link family, while the defect computed at σ' with the matrix $\mathsf{G}^{\mathrm{fib}}$ equals the defect at σ computed with $\mathbf{R}^{-\top}\mathsf{G}^{\mathrm{fib}}\mathbf{R}^{-1}$. The latter equals $\mathfrak{Q}_p^x$ for every generator family whenever $\mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R}$ is a positive multiple of $\mathsf{G}^{\mathrm{fib}}$, that is, whenever $\mathbf{R}$ is a nonzero multiple of a $\mathsf{G}^{\mathrm{fib}}$-isometry, and, when $\mathrm{d}\rho_x(\mathfrak{g}) = \mathfrak{gl}(\mathbb{V})$ and p traverses some reversal pair exactly once, only then. Likewise, the defect of the rotated walk $(e_1, \dots, e_{r-1}, e_0)$ based at i_1 is that of p computed with $\rho_x(\mathring{\Theta}_{e_0}^x)^{-\top}\mathsf{G}^{\mathrm{fib}}\rho_x(\mathring{\Theta}_{e_0}^x)^{-1}$.

(c) For each agent i let $\mathsf{G}_i^{\mathrm{fib}} = \rho_x(U_i)^{-\top}\mathsf{G}^{\mathrm{fib}}\rho_x(U_i)^{-1}$, the matrix of the same inner product in the coordinates of u_i. Under passive re-expression $\mathsf{G}_i^{\mathrm{fib}}$ becomes $(R_i^x)^{-\top}\mathsf{G}_i^{\mathrm{fib}}(R_i^x)^{-1}$, the law of Remark 2.16; the matrices of different agents satisfy $\mathsf{G}_i^{\mathrm{fib}} = \rho_x(T_{ij})^{-\top}\mathsf{G}_j^{\mathrm{fib}}\rho_x(T_{ij})^{-1}$; and, with $\mathsf{U} = \rho_x(\mathrm{Hol}^x(p))$,

$$\mathfrak{Q}_p^x = \tfrac{1}{2}\,\mathrm{Tr}\big((\mathsf{G}_{i_0}^{\mathrm{fib}})^{-1}(I_{\mathbb{V}} - \mathsf{U})^\top\mathsf{G}_{i_0}^{\mathrm{fib}}(I_{\mathbb{V}} - \mathsf{U})\big), \tag{9.9}$$

which is one half of the invariant penalty of Remark 2.16 at i_0. Conversely, if matrices $\widetilde{\mathsf{G}}_{i_0}$ assigned to the frames of agent i_0 transform by the law of that remark, then $\rho_x(U_{i_0})^\top\widetilde{\mathsf{G}}_{i_0}\rho_x(U_{i_0})$ is unchanged by passive re-expression, and $\widetilde{\mathsf{G}}_{i_0}$ is its frame image.

(d) If $\rho_x(\mathring{\Theta}_p^x)$ is a $\mathsf{G}^{\mathrm{fib}}$-isometry, then $\mathfrak{Q}_p^x = \dim \mathbb{V} - \operatorname{Tr} \rho_x(\mathring{\Theta}_p^x) = \dim \mathbb{V} - \operatorname{Tr} \mathsf{U} \in [0, 2 \dim \mathbb{V}]$, which is $\dim \mathbb{V}$ times the plaquette term $1 - \operatorname{Tr} \mathsf{U} / \dim \mathbb{V}$ of Wilson's lattice action [106, Eq. (3.29)], in the normalization of Creutz [25, Ch. 7, Eq. (7.6)].

(e) Let $\mathring{\Omega}_a = \mathrm{d}\rho_x(\mathring{\omega}_{e_a}^x)$ and $\mathring{\varsigma}_p = \sum_{a=0}^{r-1} \|\mathring{\Omega}_a\|_{\mathsf{G}}$. Then

$$\left| \mathfrak{Q}_p^x - \tfrac{1}{2} \left\| \mathrm{d}\rho_x(\mathbf{C}\mathring{\omega}^x)_p \right\|_{\mathsf{G}}^2 \right| \leq \mathring{\varsigma}_p \big(\exp(\mathring{\varsigma}_p) - 1 - \mathring{\varsigma}_p \big) + \tfrac{1}{2} \big(\exp(\mathring{\varsigma}_p) - 1 - \mathring{\varsigma}_p \big)^2, \tag{9.10}$$

and the right side is at most $(\frac{1}{2}\exp(1) + \frac{1}{8}\exp(2))\mathring{\varsigma}_p^3 < \frac{23}{10}\mathring{\varsigma}_p^3$ when $\mathring{\varsigma}_p \leq 1$. When $\mathrm{d}\rho_x$ is injective and $\mathscr{T}$ carries the norm $\|\mathring{\omega}\|^2 = \sum_{e \in E_+} \|\mathring{\omega}_e\|_{\mathsf{G}}^2$, one has $\mathring{\varsigma}_p \leq r\|\mathring{\omega}^x\|$, so the bound is $\frac{23}{10} r^3 \|\mathring{\omega}^x\|^3$ on the ball $r\|\mathring{\omega}^x\| \leq 1$. [ESTABLISHED]

The proof is in appendix B.3.

The matrix at the reference point is thus the one datum beyond the group that the defect uses. Carried along with the reference point, it leaves the defect invariant under every reference change; held as a matrix, it leaves the defect invariant under the reference changes acting by nonzero multiples of its isometries, and, when $\mathrm{d}\rho_x(\mathfrak{g}) = \mathfrak{gl}(\mathbb{V})$ and the walk traverses some reversal pair exactly once, under no other (Proposition 9.10(b)). The symmetry of the measurement is reduced in this way from all reference changes to those acting by multiples of isometries of $\mathsf{G}^{\mathrm{fib}}$; on holonomies that are isometries the defect is Wilson's trace density (part (d)), and one reference matrix supplies the metric of Remark 2.16 at every vertex and for every closed walk (part (c)). A cyclic rotation of a closed walk replaces the matrix by its image under the twist moved across (part (b)), so the base vertex does not affect the defect when that twist acts by a multiple of an isometry. The quadratic part of the defect is carried by the cycle map (part (e)), which reappears in Proposition 9.34.

9.3 The generator prior

The generator chart supplies a link prior for every Lie group: a Gaussian base law on the generators, which exists whether or not G is compact, multiplied by the exponential of minus a nonnegative loop action (Proposition 9.12). It replaces the Haar reference, whose normalizability the scope of Chapter 2 declines for noncompact and general linear G, by a reference on the Lie algebra, and the abelian links of Chapter 14 are its translation-group instance (Proposition 9.13).

Definition 9.11 (Generator prior). In Definitions 9.2 and 9.8, equip $\mathfrak{g}$ with an inner product, written $\langle \cdot, \cdot \rangle_{\mathsf{G}}$ (the pulled-back inner product of Definition 9.8 when $\mathrm{d}\rho_x$ is injective), and $\mathscr{T}$ with the product inner product and the Lebesgue measure $\mathrm{d}\mathring{\omega}$ normalized by it, under which the cube spanned by an orthonormal basis has volume one. For each $e \in E_+$ let the anchor $\mathring{\alpha}_e^x \in \mathfrak{g}$ and the covariance $\mathring{\Sigma}_e^x$, a positive definite self-adjoint operator on $(\mathfrak{g}, \langle \cdot, \cdot \rangle_{\mathsf{G}})$, be generative parameters, that is, coordinates of θ, and let $\mathbb{L}$ be a finite family of closed walks with coefficients $\mathrm{c}_p \geq 0$. The base

law, link action, normalizer, and generator prior are

$$\nu^{\mathscr{T}} = \bigotimes_{e\in E_+} \mathcal{N}(\mathring{\alpha}_e^x, \mathring{\Sigma}_e^x), \qquad \mathscr{S}(\mathring{\omega}^x) = \sum_{p\in\mathbb{L}} \mathrm{c}_p \mathfrak{Q}_p^x(\mathring{\omega}^x), \tag{9.11}$$

$$Z_{\mathscr{S}} = \int_{\mathscr{T}} e^{-\mathscr{S}} d\nu^{\mathscr{T}}, \qquad Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}} \nu^{\mathscr{T}}, \tag{9.12}$$

where $\mathcal{N}(\mathring{\alpha}, \mathring{\Sigma})$ has the density $(2\pi)^{-\dim \mathfrak{g}/2} (\det \mathring{\Sigma})^{-1/2} \exp\big(-\frac{1}{2}\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\rangle_{\mathsf{G}}\big)$ with respect to the normalized Lebesgue measure of $\mathfrak{g}$. With independent links in the two channels the link space is $\mathscr{T} \times \mathscr{T}$, the base law is the product of the two base laws, and the action is the sum of the two channel actions; under the common-link specialization one base law on $\mathscr{T}$ serves both channels and the action is the sum over channels.

Proposition 9.12 (Normalization and covariance of the generator prior). Let G be a Lie group, compact or not, and let the data of Definition 9.11 be given.

(a) The action $\mathscr{S}$ is finite, continuous, and nonnegative on $\mathscr{T}$ and has finite $\nu^{\mathscr{T}}$-mean, and

$$0 < \exp\big(-\mathbb{E}_{\nu^{\mathscr{T}}} \mathscr{S}\big) \le Z_{\mathscr{S}} \le 1; \tag{9.13}$$

when $\rho_x(G)$ consists of $\mathsf{G}^{\mathrm{fib}}$-isometries, also $Z_{\mathscr{S}} \ge \exp(-2 \dim \mathbb{V} \sum_p \mathrm{c}_p)$. The generator prior is a probability law on the standard Borel space $\mathscr{T}$ with a strictly positive continuous density with respect to $\mathrm{d}\mathring{\omega}$. Hence $\nu^{\mathscr{T}}$, $\mathscr{S}$, and $Z_{\mathscr{S}}$ satisfy the hypotheses that Proposition 5.26 places on its base law, action, and normalizer, with the generator family as the latent link variable and the links $\exp_\sigma(\mathring{\omega}^x)$ as a continuous function of it. With $\widetilde{\mathscr{S}}$ the action written as a function of the link family through $\mathring{\Theta}_p^x = U_{i_0}^{-1} \mathrm{Hol}^x(p) U_{i_0}$, so that $\mathscr{S} = \widetilde{\mathscr{S}} \circ \exp_\sigma$, the image of the prior under $\exp_\sigma$ is $Z_{\mathscr{S}}^{-1} e^{-\widetilde{\mathscr{S}}} (\exp_\sigma)_{\#} \nu^{\mathscr{T}}$, with the same normalizer.

(b) Passive re-expressions leave the generator prior unchanged and carry its image on link families by the vertex law eq. (2.28).

(c) Let $\sigma' = \sigma \cdot g_0$ and $\mathbf{R} = \rho_x(g_0)$, and let $\mathrm{Ad}_{g_0^{-1}}$ act on every component of $\mathscr{T}$. The image of the generator prior under $\mathrm{Ad}_{g_0^{-1}}$ is the generator prior relative to σ' with the anchors $\mathrm{Ad}_{g_0^{-1}} \mathring{\alpha}_e^x$, the covariances $\mathrm{Ad}_{g_0^{-1}} \mathring{\Sigma}_e^x \mathrm{Ad}_{g_0}$, the fiber matrix $\mathbf{R}^\top \mathsf{G}^{\mathrm{fib}} \mathbf{R}$, the inner product $\langle \mathrm{Ad}_{g_0} \cdot, \mathrm{Ad}_{g_0} \cdot \rangle_{\mathsf{G}}$ on $\mathfrak{g}$, which is the pullback for $\mathbf{R}^\top \mathsf{G}^{\mathrm{fib}} \mathbf{R}$ when $\mathrm{d}\rho_x$ is injective, and the same walks and coefficients; both priors have the same image on link families.

(d) If Ad_{g_0} is an isometry of $\langle \cdot, \cdot \rangle_{\mathsf{G}}$, fixes every anchor, and commutes with every covariance, and $\mathbf{R}$ is a nonzero multiple of a $\mathsf{G}^{\mathrm{fib}}$-isometry, then the image of the generator prior under $\mathrm{Ad}_{g_0^{-1}}$ is the generator prior itself, so the reference change by g_0 leaves the anchors, covariances, and coefficients unchanged and multiplies the fiber matrix by a positive scalar, which changes neither the inner product nor the defect. For the pulled-back inner product the isometry condition follows from the condition on $\mathbf{R}$, and when $\mathrm{d}\rho_x(\mathfrak{g}) = \mathfrak{gl}(\mathbb{V})$ it is equivalent to it. [ESTABLISHED]

The proof is in appendix B.3.

The Haar-referenced density of Remark 2.16 needs the finite invariant measure of a compact group, whereas the base law of Definition 9.11 lives on the Lie algebra, so eq. (9.13) holds for every Lie group; the price is a set of anchors and covariances that a reference change transforms (Proposition 9.12).

Proposition 9.13 (Translation links). Let $G = (\mathbb{R}^K, +)$ act on the sample fiber $\mathbb{R}^K$ by translation, and write the group additively.

(a) The affine representation and its differential,

$$\rho(\mathring{\omega}) = \begin{pmatrix} I_K & \mathring{\omega} \\ 0 & 1 \end{pmatrix}, \qquad \mathrm{d}\rho(\mathring{\omega}) = \begin{pmatrix} 0 & \mathring{\omega} \\ 0 & 0 \end{pmatrix}, \tag{9.14}$$

form a faithful smooth linear representation on $\mathbb{R}^{K+1}$ with $\mathfrak{g} = \mathbb{R}^K$ and $\exp$ the identity, and $\rho(\mathring{\omega})(k, 1) = (k + \mathring{\omega}, 1)$, so the translations of the sample fiber act on the affine slice $\mathbb{R}^K \times \{1\}$. A link $\Theta_e \in \mathbb{R}^K$ on $e : j \to i$, acting by $k \mapsto k + \Theta_e$, has the generator $\mathring{\omega}_e = \Theta_e - U_i + U_j$, and the reference holonomy of a closed walk is the loop sum $\mathring{\Theta}_p = (\mathbf{C}\mathring{\omega})_p = \sum_a \Theta_{e_a}$, in which the frames cancel.

(b) For every symmetric positive definite $\mathsf{G}^{\mathrm{fib}}$ on $\mathbb{R}^{K+1}$, the pulled-back norm and the loop defect are

$$\|\mathring{\omega}\|_{\mathsf{G}}^2 = \big((\mathsf{G}^{\mathrm{fib}})^{-1}\big)_{K+1,K+1}(\mathring{\omega}, 0)^\top \mathsf{G}^{\mathrm{fib}}(\mathring{\omega}, 0), \qquad \mathfrak{Q}_p = \tfrac{1}{2}\big\|(\mathbf{C}\mathring{\omega})_p\big\|_{\mathsf{G}}^2, \tag{9.15}$$

so the left side of eq. (9.10) vanishes; for $\mathsf{G}^{\mathrm{fib}} = I_{K+1}$, $\mathfrak{Q}_p = \frac{1}{2}|\sum_a \Theta_{e_a}|^2$.

(c) In the four-cycle experiment of Definition 14.21 and the gated translation family of Definition 14.23, where $K = 1$ and the link translations ω_e are the links Θ_e, the configuration prior is $\mathcal{N}(0, I)$, so the link translations are independent of the means and have the law $\prod_e \mathcal{N}(0, 1)$. In generator coordinates this is the base law of Definition 9.11 for $\mathsf{G}^{\mathrm{fib}} = I_2$ with unit covariances and the anchors $\mathring{\alpha}_e = U_j - U_i$, which vanish when every frame equals σ and always have zero loop sums, and the action is $\mathscr{S} = 0$. The loop $p = (e_3, e_2, e_1, e_0)$ of the four-cycle, based at agent 0, has $(\mathbf{C}\mathring{\omega})_p = \sum_i \omega_i$ and $\mathfrak{Q}_p = \frac{1}{2}(\sum_i \omega_i)^2$; in these models it enters only through the edge costs, whose residuals satisfy $\sum_i r_i^b = \sum_i r_i^m = -\sum_i \omega_i$.

(d) In the triangle-chain experiment of Definition 14.3, with the symbols of table 14.1 and $\mathsf{G}^{\mathrm{fib}} = I_2$, every cyclically oriented triangle q has $\mathfrak{h}_q = (\mathbf{C}\mathring{\omega})_q$ and loop energy $\mathrm{J}_1 h_q^2 = \frac{2}{3}\mathrm{J}_1\mathfrak{Q}_q$, and for every configuration

$$\frac{\mathbf{x}^\top \Omega_0(\vartheta)\mathbf{x}}{2\mathrm{T}} = \sum_q \frac{\vartheta \mathrm{J}_1}{3\mathrm{T}}\mathfrak{Q}_q + \frac{\mathbf{x}^\top \Omega_0^\perp(\vartheta)\mathbf{x}}{2\mathrm{T}}, \qquad \Omega_0^\perp(\vartheta) = \vartheta B^\top \operatorname{diag}(\mathrm{J}_\ell I) B + P_{\mathcal{K}} - \mathrm{T}R_{\mathrm{ev}}, \tag{9.16}$$

where the kernel of $\Omega_0^\perp(\vartheta)$ is the loop space $\mathcal{H}$ and $\Omega_0^\perp(\vartheta)$ is positive definite on $\mathcal{U} \oplus \mathcal{K}$. Under the configuration prior the loop sums $\mathfrak{h}_q$ are therefore independent with law $\mathcal{N}(0, 3\mathrm{T}/(\vartheta \mathrm{J}_1))$ and independent of the projection of $\mathbf{x}$ onto $\mathcal{U} \oplus \mathcal{K}$, and the quadratic energies E_ϑ and $\mathrm{E}_\vartheta^{\mathrm{phys}}$ of eq. (14.13) are $\frac{1}{3}\vartheta \mathrm{J}_1 \sum_q \mathfrak{Q}_q$ plus a function of that projection. This prior differs from Definition 9.11 in that its

second factor is a Gaussian energy of means and links jointly that is constant along $\mathcal{H}$, so it is not integrable alone, and the loop defect with coefficient $\vartheta \mathrm{J}_1/(3\mathrm{T})$ supplies all of the prior's precision along loop coordinates. [ESTABLISHED]

The proof is in appendix B.3.

Remark 9.14 (The running example in the chart). In the running example of Section 2.5 take $E_+ = \{e_1, e_2, e_3\}$, $\sigma = u_1$, and $U_i = e_G$, and write $R_x(\phi) = \exp(\phi(E_{32} - E_{23}))$ and $R_z(\phi) = \exp(\phi(E_{21} - E_{12}))$ for the rotations about the first and third axes. The chart covers every link family of $SO(3)$ (Proposition 9.5), and the generators of rotation angle at most π are $\mathring{\omega}^b_{e_1} = \frac{\pi}{2}(E_{21} - E_{12})$, $\mathring{\omega}^b_{e_2} = \frac{\pi}{2}(E_{32} - E_{23})$, and $\mathring{\omega}^b_{e_3} = 0$. The reference holonomy of $p = (e_1, e_2, e_3)$ is $\mathring{\Theta}^b_p = R_z(\frac{\pi}{2})R_x(\frac{\pi}{2})$, the cyclic permutation matrix that sends each standard basis vector to the next one, and it equals $\mathrm{Hol}^b(p)$, a rotation by $2\pi/3$ about the axis spanned by $(1,1,1)$. With $\mathsf{G}^{\mathrm{fib}} = I_3$ every rotation is an isometry, a rotation by ϕ has defect $4\sin^2(\phi/2) \in [0,4]$, and $\mathfrak{Q}^b_p = 3 - \operatorname{Tr}\mathring{\Theta}^b_p = 3$, while the trivial model representation gives $\mathfrak{Q}^m_p = 0$. The cycle map gives $(\mathbf{C}\mathring{\omega}^b)_p = \frac{\pi}{2}(E_{21} - E_{12} + E_{32} - E_{23})$, whose quadratic term $\frac{1}{2}\|(\mathbf{C}\mathring{\omega}^b)_p\|^2_{\mathsf{G}} = \pi^2/2$ exceeds the defect; here $\mathring{\varsigma}_p = \pi\sqrt{2} > 1$, outside the ball of Proposition 9.10(e). A passive re-expression g_2 changes $\Theta^b_{e_1}$ and $\Theta^b_{e_2}$ but no generator. [ESTABLISHED]

Proof. The series gives $R_z(\frac{\pi}{2})$ and $R_x(\frac{\pi}{2})$ as the quarter turns of eq. (2.38), whose product is the displayed permutation matrix; it fixes $(1,1,1)$ and has trace $0 = 1 + 2\cos\phi$ with $\phi = 2\pi/3$. A rotation by ϕ has eigenvalues 1 and $e^{\pm i\phi}$, so Proposition 9.10(d) gives the defect $3-(1+2\cos\phi) = 4\sin^2(\phi/2)$, and $\mathfrak{Q}^b_p = 3 - 0$. The four nonzero entries of $(\mathbf{C}\mathring{\omega}^b)_p$ have modulus $\pi/2$, so its squared Frobenius norm is π^2, and each nonzero generator has norm $\pi/\sqrt{2}$. Generators are unchanged by passive re-expression by Proposition 9.3(a). □

9.4 Latent generators in the collective evidence bound

What joint law and what collective free energy result when the link values of a channel leave the structural data and the generators of the exponential chart become latent coordinates? The joint has unit mass on every finite record hypergraph, and its free energy splits into a link sector and the conditional free energies of the held-link model (proposition 9.16); for a Gaussian recognition law of the generators every term is finite and the link sector is the relative entropy from the generator prior (proposition 9.18). Throughout, the generator prior is that of definition 9.11 in the following form: the base law $\nu^{\mathscr{T}}$ is the product, over the copies $e \in E_+$, of the Gaussian laws $\mathcal{N}(\mathring{\alpha}^x_e, \mathring{\Sigma}^x_e)$ on $(\mathfrak{g}, \langle\cdot,\cdot\rangle_{\mathsf{G}})$, written $\mathcal{N}(\mathring{\alpha}, \mathring{\Sigma})$ with a block-diagonal covariance $\mathring{\Sigma}$; the action $\mathscr{S}$ is a finite nonnegative combination of the loop defects $\mathfrak{Q}^x_p$ of definition 9.8; and $0 < Z_{\mathscr{S}} \leq 1$ by proposition 9.12. The definitions and statements of this section and of sections 9.5 to 9.7 are written for the link space $\mathscr{T}$ of one latent channel x, which covers the common-link specialization. When both channels carry independent latent generator families, the link space is $\mathscr{T} \times \mathscr{T}$ with the product of the two base laws and the sum of the two actions (definitions 9.2 and 9.11); every definition and statement then holds with these data in place of $\mathscr{T}$, $\nu^{\mathscr{T}}$, and $\mathscr{S}$, with the summands indexed by the pairs (x, e), by

the same proof, since the proofs use only that the link space is a finite-dimensional inner product space carrying a Gaussian base law that factors over orthogonal summands and an action that is a finite nonnegative combination of loop defects. The space $\mathscr{T}$ carries the orthogonal sum of the inner products of its summands, its norm $\|\cdot\|$, and the Lebesgue measure normalized by it; gradients and Hessians on $\mathscr{T}$ refer to this inner product, $\|\cdot\|$ is the operator norm on matrices, and upright $\mathrm{c}_1, \mathrm{c}_2, \dots$ are constants local to one statement.

Definition 9.15 (Joint law with latent generators). Let the structural data X contain the interaction multigraph $\mathfrak{I}$, an orientation set E_+ with one copy of each reversal pair, the reference point σ with the frames U_i, $u_i = \sigma \cdot U_i^{-1}$, and the acyclic graph Γ of definition 4.1, but no link values of the channels x whose links are latent. For those channels the generator family $\mathring{\omega} = (\mathring{\omega}_e^x)$ is a latent coordinate with values in $\mathscr{T}$, or in $\mathscr{T} \times \mathscr{T}$ when both channels are latent, and it determines the links of the exponential chart of definition 9.2,

$$\Theta_e^x(\mathring{\omega}) = U_i \exp(\mathring{\omega}_e^x) U_j^{-1}, \qquad \Theta_{\bar{e}}^x(\mathring{\omega}) = \Theta_e^x(\mathring{\omega})^{-1}, \qquad e : j \to i \text{ in } E_+, \tag{9.17}$$

a continuous map $\mathring{\omega} \mapsto \Theta(\mathring{\omega})$ into the link families. Write (X, Θ) for X with a link family Θ adjoined, and let the factors (4.1) and the record kernels of hypothesis 5.5 be measurable in the link values, so that for each Θ the held-link model of chapters 4 and 5 at (X, Θ) has the joint $\mathbb{P}_{\theta,V}(dy, do \mid X, \Theta) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X, \Theta) \prod_{a \in \mathfrak{A}} K_a(X, \Theta, y_{\partial a}, do_a)$, the evidence $p_{\theta,V}(o \mid X, \Theta)$, the posterior $\mathbf{\Pi}_{\theta,V,o,X,\Theta}$, and, for laws Q on Y, the free energy $\mathcal{F}_V[Q; X, \Theta, o] = -\log p_{\theta,V}(o \mid X, \Theta) + D_{\mathrm{KL}}(Q \| \mathbf{\Pi}_{\theta,V,o,X,\Theta})$ of definition 5.6. With the generator prior $\mathbb{P}^{\mathscr{T}}_\theta(d\mathring{\omega} \mid X) = Z_{\mathscr{S}}^{-1} e^{-\mathscr{S}(\mathring{\omega})} \nu^{\mathscr{T}}(d\mathring{\omega})$, the joint law with latent generators is the law on $\mathscr{T} \times \mathsf{Y} \times \mathsf{O}$

$$\mathbb{P}_{\theta,V}(d\mathring{\omega}, dy, do \mid X) = \mathbb{P}^{\mathscr{T}}_\theta(d\mathring{\omega} \mid X) \mathbb{P}^{\mathsf{Y}}_{\theta,V}\big(dy \mid X, \Theta(\mathring{\omega})\big) \prod_{a \in \mathfrak{A}} K_a\big(X, \Theta(\mathring{\omega}), y_{\partial a}, do_a\big), \tag{9.18}$$

whose evidence $p_{\theta,V}(o \mid X)$, posterior $\mathbf{\Pi}_{\theta,V,o,X}(d\mathring{\omega}, dy)$, and extended free energy $\mathcal{F}_V[Q; X, o]$ for laws Q on $\mathscr{T} \times \mathsf{Y}$ are those of definition 5.6 with the latent space $\mathscr{T} \times \mathsf{Y}$ and the baseline $\mathbb{P}^{\mathscr{T}}_\theta(d\mathring{\omega} \mid X) \mathbb{P}^{\mathsf{Y}}_{\theta,V}(dy \mid X, \Theta(\mathring{\omega}))$. With θ, X and o held, write $p(o \mid \mathring{\omega}) = p_{\theta,V}(o \mid X, \Theta(\mathring{\omega}))$, $\mathbf{\Pi}_{\mathring{\omega}} = \mathbf{\Pi}_{\theta,V,o,X,\Theta(\mathring{\omega})}$, $\mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega}) = \mathbb{P}^{\mathsf{Y}}_{\theta,V}(\cdot \mid X, \Theta(\mathring{\omega}))$, and, for records with densities (5.8), $L_o(\mathring{\omega}, y) = \prod_a e^{-E_{a,o}(X, \Theta(\mathring{\omega}), y_{\partial a})}$ and $H_o(\mathring{\omega}, y) = \sum_a E_{a,o}(X, \Theta(\mathring{\omega}), y_{\partial a})$. The anchors $\mathring{\alpha}_e^x$, the prior covariances $\mathring{\Sigma}_e^x$, and the loop coefficients are coordinates of θ, and the walks of $\mathscr{S}$ belong to X. The prior reads θ and X; each agent factor reads θ, X, its parents and $\mathring{\omega}$; each record reads θ, X, $y_{\partial a}$ and $\mathring{\omega}$. No factor reads a recognition law, a variational parameter, or the posterior, so hypothesis 4.3 holds: moving an anchor or a prior covariance is an M-coordinate (definition 5.28), while the generators are moved by recognition. Passive re-expression changes the frames and no generator (proposition 2.12); a change of reference point acts on the generator prior as in proposition 9.3.

Proposition 9.16 (Normalization, evidence, and the conditional decomposition). Assume definition 9.15. (a) For every finite record hypergraph, cyclic or not, and every (θ, X), (9.18) is a probability law on $\mathscr{T} \times \mathsf{Y} \times \mathsf{O}$, and the generator family is a root block of the directed latent law. (b)

If the records have densities (5.8), then $\mathring{\omega} \mapsto p(o \mid \mathring{\omega}) = \int L_o(\mathring{\omega}, y)\mathbb{P}^{\mathsf{Y}}(dy \mid \mathring{\omega})$ is measurable and

$$p_{\theta,V}(o \mid X) = \int_{\mathscr{T}} p(o \mid \mathring{\omega})\mathbb{P}^{\mathscr{T}}_{\theta}(d\mathring{\omega} \mid X); \tag{9.19}$$

the evidence is positive exactly when $p(o \mid \cdot)$ is positive on a set of positive Lebesgue measure, and if every record density is bounded, $e^{-E_{a,o}} \le \mathrm{c}_a$, then $p(o \mid \mathring{\omega}) \le \mathrm{c}_{\mathfrak{A}} := \prod_a \mathrm{c}_a$ for every $\mathring{\omega}$ and the evidence is at most $\mathrm{c}_{\mathfrak{A}}$; binary success records have $\mathrm{c}_a = 1$. (c) Suppose further that the record densities are bounded and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$. Then the posterior disintegrates as $\mathbf{\Pi}_{\theta,V,o,X}(d\mathring{\omega}, dy) = \mathbf{\Pi}^{\mathscr{T}}(d\mathring{\omega})\mathbf{\Pi}_{\mathring{\omega}}(dy)$ with link marginal $\mathbf{\Pi}^{\mathscr{T}}(d\mathring{\omega}) = p_{\theta,V}(o \mid X)^{-1}p(o \mid \mathring{\omega})\mathbb{P}^{\mathscr{T}}_{\theta}(d\mathring{\omega} \mid X)$, and every law Q on $\mathscr{T} \times \mathsf{Y}$, disintegrated as $Q(d\mathring{\omega}, dy) = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$, satisfies

$$\mathcal{F}_V[Q; X, o] = D_{\mathrm{KL}}\big(R \| \mathbb{P}^{\mathscr{T}}_{\theta}(\cdot \mid X)\big) + \int_{\mathscr{T}} \mathcal{F}_V\big[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o\big] R(d\mathring{\omega}) \qquad \text{in } (-\infty, +\infty], \tag{9.20}$$

with a measurable integrand bounded below by $-\log \mathrm{c}_{\mathfrak{A}}$. When $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) < \infty$ and $\mathbb{E}_R|\mathscr{S}| < \infty$, the first term equals $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \mathbb{E}_R\mathscr{S} + \log Z_{\mathscr{S}}$, and (9.20) is (5.57) for the latent variable $\mathring{\omega}$. [ESTABLISHED]

The proof is in appendix B.3.

Lemma 9.17 (Exponential moments and growth of loop actions). (a) For a Gaussian law $\mathcal{N}(\mu, \Sigma)$ on $\mathscr{T}$ and every $\mathrm{c} \ge 0$, $\mathbb{E}_{\mathcal{N}(\mu,\Sigma)} e^{\mathrm{c}\|\mathring{\omega}\|} \le e^{\mathrm{c}\|\mu\|}\big(2e^{\mathrm{c}^2\|\Sigma\|/2}\big)^{\dim \mathscr{T}}$, and the same bound holds on each summand of $\mathscr{T}$ with its dimension. (b) For a representation ρ_x on a real vector space of finite dimension, the map $\mathring{\xi} \mapsto \rho_x(\exp \mathring{\xi}) = \exp(\mathrm{d}\rho_x \mathring{\xi})$ on $(\mathfrak{g}, \langle\cdot,\cdot\rangle_{\mathsf{G}})$ is smooth, and its value and its first and second derivatives at $\mathring{\xi}$ have norms at most $e^{\|\mathrm{d}\rho_x\|\|\mathring{\xi}\|}$, $\|\mathrm{d}\rho_x\| e^{\|\mathrm{d}\rho_x\|\|\mathring{\xi}\|}$ and $\|\mathrm{d}\rho_x\|^2 e^{\|\mathrm{d}\rho_x\|\|\mathring{\xi}\|}$. (c) Every real polynomial $\mathbf{h}$ in the matrix entries of the represented twists $\rho_x(\mathring{\Theta}^x_e)^{\pm 1} = \exp(\pm \mathrm{d}\rho_x \mathring{\omega}^x_e)$, $e \in E_+$, is smooth on $\mathscr{T}$ and satisfies, for constants depending on $\mathbf{h}$,

$$|\mathbf{h}(\mathring{\omega})| + \|\nabla \mathbf{h}(\mathring{\omega})\| + \|\nabla^2 \mathbf{h}(\mathring{\omega})\| \le \mathrm{c}_1 e^{\mathrm{c}_2\|\mathring{\omega}\|}; \tag{9.21}$$

each loop defect of definition 9.8, and hence the action $\mathscr{S}$, is such a polynomial. [ESTABLISHED]

Proof. (a) Write $\mathring{\omega} = \mu + \Sigma^{1/2}\mathbf{b}$ with $\mathbf{b}$ standard Gaussian in an orthonormal basis, so that $\|\mathring{\omega}\| \le \|\mu\| + \|\Sigma\|^{1/2}\sum_l |\mathbf{b}_l|$; the coordinates are independent and, for $\mathrm{c}' \ge 0$, $\mathbb{E}e^{\mathrm{c}'|\mathbf{b}_l|} \le \mathbb{E}e^{\mathrm{c}'\mathbf{b}_l} + \mathbb{E}e^{-\mathrm{c}'\mathbf{b}_l} = 2e^{\mathrm{c}'^2/2}$. (b) The identity $\rho_x(\exp \mathring{\xi}) = \exp(\mathrm{d}\rho_x \mathring{\xi})$ is lemma 9.1. The series $\exp \mathbf{B} = \sum_l \mathbf{B}^l/l!$ converges with all its derivatives. The first derivative of $\mathbf{B} \mapsto \mathbf{B}^l$ in the direction $\mathbf{T}$ is a sum of l products of $l-1$ factors $\mathbf{B}$ and one factor $\mathbf{T}$, and its second derivative in the directions $\mathbf{T}$ and $\mathbf{T}'$ is a sum of $l(l-1)$ products of $l-2$ factors $\mathbf{B}$ and one factor each of $\mathbf{T}$ and $\mathbf{T}'$. Summing the series, the value and the first and second derivatives of the exponential have norms at most $e^{\|\mathbf{B}\|}$, $e^{\|\mathbf{B}\|}\|\mathbf{T}\|$ and $e^{\|\mathbf{B}\|}\|\mathbf{T}\|\|\mathbf{T}'\|$; composing with the linear map $\mathrm{d}\rho_x$ gives (b). (c) By (b) every entry of $\exp(\pm \mathrm{d}\rho_x \mathring{\omega}^x_e)$ and its first two derivatives are bounded by $\mathrm{c}e^{\|\mathrm{d}\rho_x\|\|\mathring{\omega}\|}$, and the Leibniz rule bounds a monomial of degree c_0 and its derivatives by $\mathrm{c}'e^{\mathrm{c}_0\|\mathrm{d}\rho_x\|\|\mathring{\omega}\|}$. The reference-coordinate holonomy of a closed walk is the product of the matrices $\rho_x(\mathring{\Theta}^x_e)^{\pm 1}$ along the walk (proposition 2.12), and a loop defect is a quadratic polynomial in its entries (definition 9.8). □

Proposition 9.18 (Gaussian recognition of the generators). Let $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ be a Gaussian law on $\mathscr{T}$ with mean $\mathring{\mu}$ and positive-definite self-adjoint precision $\mathring{\Lambda}$; in the mean-field case $R = \bigotimes \mathcal{N}(\mathring{\mu}_e^x, (\mathring{\Lambda}_e^x)^{-1})$ over the summands of $\mathscr{T}$. (a) The relative entropy from the base law is finite and equals

$$D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) = \tfrac{1}{2}\Big[\mathrm{Tr}\big(\mathring{\Sigma}^{-1}\mathring{\Lambda}^{-1}\big) + \big\langle \mathring{\mu} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha})\big\rangle - \dim \mathscr{T} + \log\det \mathring{\Sigma} + \log\det \mathring{\Lambda}\Big]; \quad (9.22)$$

in the mean-field case it is the sum of the same expressions over the summands, and in general it equals $\mathrm{TC}(R) + \sum D_{\mathrm{KL}}(R_e^x \| \mathcal{N}(\mathring{\alpha}_e^x, \mathring{\Sigma}_e^x))$, with R_e^x the summand marginals and $\mathrm{TC}(R) = \frac{1}{2}[\sum \log\det(\mathring{\Lambda}^{-1})_e^x + \log\det\mathring{\Lambda}]$, where $(\mathring{\Lambda}^{-1})_e^x$ are the diagonal blocks of $\mathring{\Lambda}^{-1}$. (b) $\mathbb{E}_R e^{\mathrm{c}\|\mathring{\omega}\|} < \infty$ for every $\mathrm{c} \geq 0$ and $\mathbb{E}_R \mathscr{S} < \infty$. Consequently, if the record densities are bounded and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$, then for every Markov kernel $\mathring{\omega} \mapsto Q_{\mathring{\omega}}$ the free energy of $Q = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ is (5.57) for the latent variable $\mathring{\omega}$, with the finite link sector

$$D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \mathbb{E}_R \mathscr{S} + \log Z_{\mathscr{S}} = D_{\mathrm{KL}}\big(R\|\mathbb{P}_\theta^{\mathscr{T}}(\cdot \mid X)\big). \quad (9.23)$$

(c) If moreover $Q_{\mathring{\omega}} \ll \mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega})$ for R-almost every $\mathring{\omega}$, $\int D_{\mathrm{KL}}(Q_{\mathring{\omega}}\|\mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega}))R(d\mathring{\omega}) < \infty$, and $\mathbb{E}_Q|H_o| < \infty$, then the free energy is finite and equals

$$\mathcal{F}_V[Q; X, o] = D_{\mathrm{KL}}\big(R\|\mathbb{P}_\theta^{\mathscr{T}}(\cdot \mid X)\big) + \int D_{\mathrm{KL}}\big(Q_{\mathring{\omega}}\|\mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega})\big)R(d\mathring{\omega}) + \mathbb{E}_Q H_o. \quad (9.24)$$

The condition on H_o holds for a product law $Q = R \otimes Q^{\mathsf{Y}}$ when Y is a real vector space of finite dimension, Q^{Y} has finite second moments, and every record satisfies $|E_{a,o}(X, \Theta(\mathring{\omega}), y_{\partial a})| \leq \mathrm{c}_1(1 + \|y\|^2)e^{\mathrm{c}_2\|\mathring{\omega}\|}$. [ESTABLISHED]

Proof. (a) Both laws have densities with respect to Lebesgue measure on $\mathscr{T}$, and for a self-adjoint operator $\mathbf{B}$, $\mathbb{E}_R\langle\mathring{\omega} - \mathring{\alpha}, \mathbf{B}(\mathring{\omega} - \mathring{\alpha})\rangle = \mathrm{Tr}(\mathbf{B}\mathring{\Lambda}^{-1}) + \langle\mathring{\mu} - \mathring{\alpha}, \mathbf{B}(\mathring{\mu} - \mathring{\alpha})\rangle$; taking the R-expectation of the difference of the two log densities gives (9.22). In the mean-field case all operators are block diagonal, so traces and log-determinants are sums over blocks. The general identity is corollary 5.1 with the product law $\nu^{\mathscr{T}}$, and the total correlation of a Gaussian law is the sum of the summand entropies minus the joint entropy, in which the factors $2\pi e$ cancel because the summand dimensions add up to $\dim \mathscr{T}$. (b) The first claim is lemma 9.17(a), and lemma 9.17(c) gives $\mathbb{E}_R \mathscr{S} \leq \mathrm{c}_1 \mathbb{E}_R e^{\mathrm{c}_2\|\mathring{\omega}\|} < \infty$. With (a), the hypotheses of the last sentence of proposition 9.16(c) hold, and (9.23) is $\log(dR/d\mathbb{P}_\theta^{\mathscr{T}}) = \log(dR/d\nu^{\mathscr{T}}) + \mathscr{S} + \log Z_{\mathscr{S}}$ integrated against R. (c) The baseline of the latent space is $\mathbb{P}_\theta^{\mathscr{T}}(d\mathring{\omega} \mid X)\mathbb{P}^{\mathsf{Y}}(dy \mid \mathring{\omega})$, and lemma 3.12(a) writes its divergence from Q as the first two terms of (9.24), which are finite, so Q is absolutely continuous with respect to it; the negative part of H_o is Q-integrable, and (5.19) of corollary 5.7, applied on the latent space $\mathscr{T} \times \mathsf{Y}$, gives (9.24). For a product law, $\mathbb{E}_Q|E_{a,o}| \leq \mathrm{c}_1 \mathbb{E}_R e^{\mathrm{c}_2\|\mathring{\omega}\|}\mathbb{E}_{Q^{\mathsf{Y}}}(1 + \|y\|^2) < \infty$ by (b) and Tonelli's theorem. □

The relative entropy (9.23) is the whole link sector: a loop energy placed in the prior enters with its normalizer, as remark 4.7 requires of any reciprocal energy that is to be part of an evidence bound,

and in the mean-field case the sector is a sum over edge copies of Gaussian divergences from the anchors plus the recognition expectation of the loop action.

9.5 The link field equation

Which condition does the recognition law of the generators satisfy where the collective free energy is stationary? For a Gaussian recognition law and a given matter recognition law, the stress of the anchor plus the expected gradient of the loop action equals the expected matter current, filtered through the adjoint derivative of the exponential chart (proposition 9.21); for a comparison record the current is a precision-weighted prediction error times the source mean, minus the transported source covariance (corollary 9.22). Without the generator prior the stationary problem degenerates, at the sample level and at the law level (proposition 9.23 and corollary 9.24). The recognition flow that relaxes to these conditions is proposition 9.28. Matrices are real, $\langle \mathbf{S}, \mathbf{T}\rangle = \mathrm{Tr}(\mathbf{S}^\top \mathbf{T})$ is the trace pairing of square matrices of one size, gradients of functions of matrices refer to it, and for a representation ρ_x on a sample fiber written in reference coordinates, $(\mathrm{d}\rho_x)^\top$ is the adjoint of $\mathrm{d}\rho_x$ for $\langle\cdot,\cdot\rangle_{\mathsf{G}}$ on $\mathfrak{g}$ and the trace pairing.

Lemma 9.19 (Derivative of the exponential chart). For square matrices $\mathbf{B}, \mathbf{T}$ of one size put

$$\mathbf{L}(\mathbf{B}, \mathbf{T}) = \int_0^1 e^{t\mathbf{B}} \mathbf{T} e^{(1-t)\mathbf{B}} dt. \tag{9.25}$$

(a) For all $\mathbf{B}, \mathbf{T}$, in the operator norm, $\|e^{\mathbf{B}+\mathbf{T}} - e^{\mathbf{B}} - \mathbf{L}(\mathbf{B}, \mathbf{T})\| \le \frac{1}{2}\|\mathbf{T}\|^2 e^{\|\mathbf{B}\|+\|\mathbf{T}\|}$, so the matrix exponential is Fréchet differentiable at $\mathbf{B}$ with derivative $\mathbf{T} \mapsto \mathbf{L}(\mathbf{B}, \mathbf{T})$, and $\|\mathbf{L}(\mathbf{B}, \mathbf{T})\| \le e^{\|\mathbf{B}\|}\|\mathbf{T}\|$. (b) For every $\mathbf{S}$,

$$\langle \mathbf{S}, \mathbf{L}(\mathbf{B}, \mathbf{T})\rangle = \langle \mathbf{L}(\mathbf{B}^\top, \mathbf{S}), \mathbf{T}\rangle. \tag{9.26}$$

(c) If a function $\mathbf{h}$ is differentiable at $e^{\mathbf{B}}$ with gradient $\nabla\mathbf{h}(e^{\mathbf{B}})$, then $\mathbf{B} \mapsto \mathbf{h}(e^{\mathbf{B}})$ has gradient $\mathbf{L}(\mathbf{B}^\top, \nabla\mathbf{h}(e^{\mathbf{B}}))$ at $\mathbf{B}$, and $\mathring{\xi} \mapsto \mathbf{h}(\rho_x(\exp \mathring{\xi}))$ has gradient $(\mathrm{d}\rho_x)^\top \mathbf{L}\big((\mathrm{d}\rho_x \mathring{\xi})^\top, \nabla\mathbf{h}(\rho_x(\exp \mathring{\xi}))\big)$ at $\mathring{\xi}$ for $\langle\cdot,\cdot\rangle_{\mathsf{G}}$. [ESTABLISHED]

Proof. For square matrices $\mathbf{B}$ and $\mathbf{B}'$, Duhamel's formula $e^{\mathbf{B}'} - e^{\mathbf{B}} = \int_0^1 e^{t\mathbf{B}'}(\mathbf{B}' - \mathbf{B})e^{(1-t)\mathbf{B}} dt$ holds because the integrand is the t-derivative of $e^{t\mathbf{B}'}e^{(1-t)\mathbf{B}}$, and with $\|e^{t\mathbf{B}}\| \le e^{t\|\mathbf{B}\|}$ it gives $\|e^{\mathbf{B}'} - e^{\mathbf{B}}\| \le \|\mathbf{B}' - \mathbf{B}\| e^{\max(\|\mathbf{B}'\|, \|\mathbf{B}\|)}$. With $\mathbf{B}' = \mathbf{B} + \mathbf{T}$, $e^{\mathbf{B}+\mathbf{T}} - e^{\mathbf{B}} - \mathbf{L}(\mathbf{B}, \mathbf{T}) = \int_0^1 (e^{t(\mathbf{B}+\mathbf{T})} - e^{t\mathbf{B}})\mathbf{T} e^{(1-t)\mathbf{B}} dt$, and the same bound applied to $t(\mathbf{B} + \mathbf{T})$ and $t\mathbf{B}$ bounds the integrand by $t\|\mathbf{T}\|^2 e^{t(\|\mathbf{B}\|+\|\mathbf{T}\|)+(1-t)\|\mathbf{B}\|} \le t\|\mathbf{T}\|^2 e^{\|\mathbf{B}\|+\|\mathbf{T}\|}$, whose integral gives (a); the bound on $\mathbf{L}$ follows from $\|e^{t\mathbf{B}}\|\|e^{(1-t)\mathbf{B}}\| \le e^{\|\mathbf{B}\|}$. For (b), cyclicity of the trace and $(e^{t\mathbf{B}})^\top = e^{t\mathbf{B}^\top}$ give $\mathrm{Tr}(\mathbf{S}^\top e^{t\mathbf{B}} \mathbf{T} e^{(1-t)\mathbf{B}}) = \mathrm{Tr}\big((e^{t\mathbf{B}^\top} \mathbf{S} e^{(1-t)\mathbf{B}^\top})^\top \mathbf{T}\big)$, and integration over t gives (9.26). Part (c) is the chain rule with (a) and (b); for the representation, $\rho_x(\exp \mathring{\xi}) = \exp(\mathrm{d}\rho_x \mathring{\xi})$ (lemma 9.1), its derivative in the direction $\mathring{\eta}$ is $\mathbf{L}(\mathrm{d}\rho_x \mathring{\xi}, \mathrm{d}\rho_x \mathring{\eta})$, and (9.26) moves $\mathbf{L}$ onto the gradient before the adjoint $(\mathrm{d}\rho_x)^\top$ is applied. □

The integral form of the Fréchet derivative of the matrix exponential is classical [52, Ch. 10].

Lemma 9.20 (Gaussian expectations and their derivatives). Let $\mathbf{h}$ be a twice continuously differentiable function on $\mathscr{T}$, or on one of its summands, with $|\mathbf{h}(\mathring{\omega})| + \|\nabla\mathbf{h}(\mathring{\omega})\| + \|\nabla^2\mathbf{h}(\mathring{\omega})\| \le \mathrm{c}_1 e^{\mathrm{c}_2\|\mathring{\omega}\|}$. Then $(\mu,\Sigma) \mapsto \mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\mathbf{h}$ is continuously differentiable on the product of the space and the open cone of positive-definite self-adjoint operators, and for every self-adjoint $\mathring{\Delta}$

$$\nabla_\mu \mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\mathbf{h} = \mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\nabla\mathbf{h}, \qquad \frac{d}{dt}\mathbb{E}_{\mathcal{N}(\mu,\Sigma+t\mathring{\Delta})}\mathbf{h}\Big|_{t=0} = \tfrac{1}{2}\operatorname{Tr}\big(\mathring{\Delta}\mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\nabla^2\mathbf{h}\big); \tag{9.27}$$

that is, for the trace pairing of self-adjoint operators the gradient in Σ is one half of $\mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\nabla^2\mathbf{h}$. [ESTABLISHED]

The proof is in appendix B.3.

The two identities are the Gaussian derivative identities used for variational Gaussian approximations in Opper and Archambeau [84, Eqs. (18)–(19)].

Proposition 9.21 (Link field equation). Assume definition 9.15 with bounded record densities and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$, a σ-finite reference ν^{Y} with respect to which every $\mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega})$ has a density, and a matter recognition law Q^{Y} with density q^{Y} and $\mathbb{E}_{Q^{\mathsf{Y}}}|\log q^{\mathsf{Y}}| < \infty$. Let the matter energy $\mathbf{H}_o(\mathring{\omega}, y) = -\log p_{\theta,V}(o, y \mid X, \Theta(\mathring{\omega}))$, the negative logarithm of the density (4.12) of the held-link joint at $\Theta(\mathring{\omega})$, be twice continuously differentiable in $\mathring{\omega}$ for every y with

$$\mathbb{E}_{Q^{\mathsf{Y}}} \sup_{\|\mathring{\omega}\| \le \mathring{r}} \Big(|\mathbf{H}_o| + \|\nabla_{\mathring{\omega}}\mathbf{H}_o\| + \|\nabla^2_{\mathring{\omega}}\mathbf{H}_o\|\Big)(\mathring{\omega}, Y) \le \mathrm{c}_1 e^{\mathrm{c}_2 \mathring{r}} \qquad \text{for every } \mathring{r} \ge 0. \tag{9.28}$$

Restrict recognition to the laws $R \otimes Q^{\mathsf{Y}}$ with $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ Gaussian on $\mathscr{T}$, general or mean-field over the summands, and put $\mathbf{U}(\mathring{\omega}) = \mathscr{S}(\mathring{\omega}) + \mathbb{E}_{Q^{\mathsf{Y}}}\mathbf{H}_o(\mathring{\omega}, Y)$. Then:

(a) $\mathbf{U}$ is twice continuously differentiable, with $|\mathbf{U}| + \|\nabla\mathbf{U}\| + \|\nabla^2\mathbf{U}\| \le \mathrm{c}_3 e^{\mathrm{c}_4\|\mathring{\omega}\|}$, and the free energy

$$\mathcal{F}_V[R \otimes Q^{\mathsf{Y}}; X, o] = D_{\mathrm{KL}}(R \| \nu^{\mathscr{T}}) + \mathbb{E}_R\mathbf{U} + \log Z_{\mathscr{S}} + \mathbb{E}_{Q^{\mathsf{Y}}}\log q^{\mathsf{Y}} \tag{9.29}$$

is finite and continuously differentiable in $(\mathring{\mu}, \mathring{\Lambda})$.

(b) A pair $(\mathring{\mu}, \mathring{\Lambda})$ is stationary for (9.29) exactly when

$$\mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha}) + \mathbb{E}_R\nabla\mathscr{S} = -\mathbb{E}_{R\otimes Q^{\mathsf{Y}}}\nabla_{\mathring{\omega}}\mathbf{H}_o, \tag{9.30}$$

$$\mathring{\Lambda} = \mathring{\Sigma}^{-1} + \mathbb{E}_R\nabla^2\mathscr{S} + \mathbb{E}_{R\otimes Q^{\mathsf{Y}}}\nabla^2_{\mathring{\omega}}\mathbf{H}_o, \tag{9.31}$$

and in the mean-field case exactly when the summand components of (9.30) and the diagonal blocks of (9.31) hold. When the agent latent law does not read the links, $\nabla_{\mathring{\omega}}\mathbf{H}_o = \sum_a \nabla_{\mathring{\omega}} E_{a,o}(X, \Theta(\mathring{\omega}), y_{\partial a})$, the sum over the records whose kernels read a link, so the right side of (9.30) is minus the expected gradient of the incident record energies.

(c) If $\mathbf{H}_o$ reads $\mathring{\omega}^x_e$ only through the represented twist $\rho_x(\mathring{\Theta}^x_e) = \exp(\mathrm{d}\rho_x\mathring{\omega}^x_e)$, as a differentiable function of this matrix on an open set of matrices that contains the represented twists, and its

gradient $\partial \mathbf{H}_o / \partial \rho_x(\mathring{\Theta}_e^x)$ is Q^{Y}-integrable at every $\mathring{\omega}$, then with the link current

$$\mathbf{J}_e^x(\mathring{\omega}) = -\mathbb{E}_{Q^{\mathsf{Y}}} \frac{\partial \mathbf{H}_o}{\partial \rho_x(\mathring{\Theta}_e^x)}(\mathring{\omega}, Y) \tag{9.32}$$

the (x, e) component of (9.30) reads

$$(\mathring{\Sigma}_e^x)^{-1}(\mathring{\mu}_e^x - \mathring{\alpha}_e^x) + \mathbb{E}_R \nabla_{(x,e)} \mathscr{S} = \mathbb{E}_R \Big[(\mathrm{d}\rho_x)^\top \mathbf{L}\big((\mathrm{d}\rho_x \mathring{\omega}_e^x)^\top, \mathbf{J}_e^x(\mathring{\omega})\big) \Big] : \tag{9.33}$$

the expected stress of the generator prior equals the expected matter current transported by the adjoint derivative of the exponential chart.

(d) Put $\mathbf{V}(\mathring{\omega}) = \frac{1}{2}\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\rangle + \mathbf{U}(\mathring{\omega})$. Then (9.30) is $\mathbb{E}_R \nabla \mathbf{V} = 0$, and $\mathbb{E}_{\mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})} \nabla \mathbf{V} \to \nabla \mathbf{V}(\mathring{\mu})$ as $\mathring{\Lambda}^{-1} \to 0$, uniformly for $\mathring{\mu}$ in compact sets. Every limit point $\mathring{\mu}_*$, as $\mathring{\Lambda}^{-1} \to 0$, of solutions of (9.30) satisfies $\nabla \mathbf{V}(\mathring{\mu}_*) = 0$; and if $\nabla \mathbf{V}(\mathring{\mu}_*) = 0$ with $\nabla^2 \mathbf{V}(\mathring{\mu}_*)$ invertible, then for every positive-definite $\mathring{\Lambda}_0$ there are $t_0 > 0$ and a continuously differentiable curve $t \mapsto \mathring{\mu}(t)$ on $(-t_0, t_0)$ with $\mathring{\mu}(0) = \mathring{\mu}_*$ such that $(\mathring{\mu}(t), t^{-2}\mathring{\Lambda}_0)$ solves (9.30) for $0 < |t| < t_0$. When $\mathbf{U}$ is bounded below, the law $R^\star(d\mathring{\omega}) \propto e^{-\mathbf{U}(\mathring{\omega})} \nu^{\mathscr{T}}(d\mathring{\omega})$ is the unique minimizer of $\mathcal{F}_V[R \otimes Q^{\mathsf{Y}}; X, o]$ over all laws R on $\mathscr{T}$, and $-\mathbf{V}$ is its log density up to a constant, so the limit condition $\nabla \mathbf{V} = 0$ is the stationarity condition of the most probable link given the matter recognition.

(e) No point mass is admissible: $D_{\mathrm{KL}}(\delta_{\mathring{\mu}} \| \nu^{\mathscr{T}}) = +\infty$, so the free energy of $\delta_{\mathring{\mu}} \otimes Q^{\mathsf{Y}}$ is $+\infty$, and (9.29) tends to $+\infty$ as $\mathring{\Lambda}^{-1} \to 0$ at a given $\mathring{\mu}$. [ESTABLISHED]

The proof is in appendix B.3.

At a point law the matter current is evaluated at one link value and the condition $\nabla \mathbf{V} = 0$ is the most-probable-link equation; part (e) types it as a generative quantity at given matter recognition, in the sense of proposition 3.21, and not as a recognition optimum.

Corollary 9.22 (The current of a comparison record). In the setting of proposition 9.21(c), let $x = b$, let $e : j \to i$ lie in E_+, and write samples and links in the reference gauge, in which every frame is the identity, reached by the passive re-expression by the inverse frames, which changes no twist and hence no generator (proposition 2.12). Suppose that the only term of $\mathbf{H}_o$ that reads $\mathring{\omega}_e^b$ is either (a) a binary success record a with scope $\{i, j\}$, observed as a success, with $E_{a,o} = \frac{1}{2}(k_i - \rho_b(\mathring{\Theta}_e^b)k_j)^\top W_a (k_i - \rho_b(\mathring{\Theta}_e^b)k_j)$ and $W_a \succ 0$, or (b) the comparison record (6.41) of row i at $o_i = 0$ in the per-receiver source-label model of definition 6.1, with query map S_i^Q, key map $S_e^K = S_{\mathrm{ref}}^K \rho_b(\mathring{\Theta}_e^b)$ as in definition 6.15, tempered weight $\widetilde{W}_i = W_i/\tau_i$, the source labels counted among the matter coordinates, and a matter recognition law of the factorization (6.5) whose row β_i^Q does not depend on $(\mathring{\omega}, y)$. Let the belief marginals of Q^{Y} have finite second moments, with means μ_i, μ_j, covariance Σ_j of k_j, and cross covariance $\Sigma_{ij} = \mathbb{E}_{Q^{\mathsf{Y}}}[(k_i - \mu_i)(k_j - \mu_j)^\top]$, which vanishes for a product law. Then the current (9.32) is, in the two cases,

$$\mathbf{J}_e^b(\mathring{\omega}) = W_a \big[(\mu_i - \rho_b(\mathring{\Theta}_e^b)\mu_j)\mu_j^\top + \Sigma_{ij} - \rho_b(\mathring{\Theta}_e^b)\Sigma_j \big], \tag{9.34}$$

$$\mathbf{J}_e^b(\mathring{\omega}) = \beta_{ij}^Q (S_{\mathrm{ref}}^K)^\top \widetilde{W}_i \big[(S_i^Q \mu_i - S_{\mathrm{ref}}^K \rho_b(\mathring{\Theta}_e^b)\mu_j)\mu_j^\top + S_i^Q \Sigma_{ij} - S_{\mathrm{ref}}^K \rho_b(\mathring{\Theta}_e^b)\Sigma_j \big], \tag{9.35}$$

and (9.33) holds for e with this current. For a product law in case (a) the current is the precision-weighted prediction error of the receiver times the source mean, minus the precision-weighted transported source covariance. [ESTABLISHED]

Proof. For a matrix $\mathbf{M}$ in place of $\rho_b(\mathring{\Theta}_e^b)$, the gradient of $\frac{1}{2}(k_i - \mathbf{M}k_j)^\top W_a(k_i - \mathbf{M}k_j)$ in $\mathbf{M}$ for the trace pairing is $-W_a(k_i - \mathbf{M}k_j)k_j^\top$, which is Q^{Y}-integrable by the second moments, and $\mathbb{E}_{Q^{\mathsf{Y}}}[(k_i - \mathbf{M}k_j)k_j^\top] = \mu_i\mu_j^\top + \Sigma_{ij} - \mathbf{M}(\Sigma_j + \mu_j\mu_j^\top)$, which gives (9.34). In case (b), proposition 6.2(b) with the latent state $(\mathring{\omega}, y)$ shows that row i contributes $\tau_i^{-1}\sum_j \beta_{ij}^Q \mathbb{E}D_{ij}$ together with terms that do not read the links, because c_i of proposition 6.19 is constant; the source j reached through e has $D_{ij} = V_e$ of definition 6.15, whose gradient in $\mathbf{M}$ is $-(S_{\mathrm{ref}}^K)^\top W_i(S_i^Q k_i - S_{\mathrm{ref}}^K \mathbf{M}k_j)k_j^\top$, and the same expectation gives (9.35). The last sentence is proposition 9.21(c). □

By corollary 9.22, the source of the link field equation (9.33) has the outer-product form of the delta rule that Schlag et al. [89, Sec. 4.2, Eq. (24)] write for fast-weight programmers, a line of work they trace in their Sec. 2 to Schmidhuber [90]: a weighted prediction error of the receiver times the source mean. Three features have no counterpart there and are consequences of the joint: the transported source covariance, the transport of the current by the adjoint derivative of the exponential chart, and the average over the recognition law of the generators, balanced by the stress of the generator prior. The correspondence is one of form between update rules; no learning architecture is identified with the model.

Proposition 9.23 (Law-level link energies and their stabilizers). Let q_i, q_j be laws in the law fiber $\mathcal{B}_x$ of a channel, let $D_e(\Theta) = D_{\mathrm{KL}}(q_i \| \widehat{\rho}_x(\Theta)q_j)$ for $\Theta \in G$ be the transported divergence (13.12) of a copy $e : j \to i$ as a function of its link value, and let $G_q = \{g \in G : \widehat{\rho}_x(g)q = q\}$ be the stabilizer of q, a closed subgroup when $g \mapsto \widehat{\rho}_x(g)q$ is continuous for the weak topology on laws, with Lie algebra $\mathfrak{g}_q$; under hypothesis 7.1 it is the isotropy algebra of (7.9).

(a) $D_e(h\Theta h') = D_e(\Theta)$ for all $\Theta \in G$, $h \in G_{q_i}$, $h' \in G_{q_j}$. Hence the set of minimizers of D_e, and the set of its stationary points when D_e is finite and continuously differentiable on G, are unions of double cosets $G_{q_i}\Theta G_{q_j}$, and a minimizer or stationary point is not isolated when $\dim \mathfrak{g}_{q_i} > 0$ or $\dim \mathfrak{g}_{q_j} > 0$. At every generator at which the exponential map of $\mathfrak{g}$ is a local diffeomorphism, a minimizer or stationary point of the chart energy $\mathring{\omega}_e^x \mapsto D_e(U_i \exp(\mathring{\omega}_e^x)U_j^{-1})$ is likewise not isolated when $\dim \mathfrak{g}_{q_i} > 0$ or $\dim \mathfrak{g}_{q_j} > 0$.

(b) Let $G = \mathrm{GL}^+(K)$ act on $\mathbb{R}^K$ by its defining representation, and let $q = \mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$. Then $G_q = \Sigma^{1/2}\{O \in \mathrm{SO}(K) : O\Sigma^{-1/2}\mu = \Sigma^{-1/2}\mu\}\Sigma^{-1/2}$, of dimension $(K-1)(K-2)/2$ if $\mu \neq 0$ and $K(K-1)/2$ if $\mu = 0$. The Mahalanobis norm $|\mu|_\Sigma = (\mu^\top\Sigma^{-1}\mu)^{1/2}$ satisfies $|g\mu|_{g\Sigma g^\top} = |\mu|_\Sigma$ for every g, and for $K \geq 2$ a link Θ_0 with $\widehat{\rho}_x(\Theta_0)q_j = q_i$ exists exactly when $|\mu_i|_{\Sigma_i} = |\mu_j|_{\Sigma_j}$; the zero set of D_e is then the double coset $G_{q_i}\Theta_0 G_{q_j} = \Theta_0 G_{q_j}$. For $K \geq 3$ every minimizer and stationary point of D_e is non-isolated, and for $K = 2$ this holds when $\mu_i = 0$ or $\mu_j = 0$.

(c) If $\widehat{\rho}_x(\Theta_0)q_j = q_i$ and hypothesis 7.1 holds in the channel, then D_e is C^3 near Θ_0 with zero value and differential at Θ_0, and its Hessian at Θ_0 on the tangent vector of $t \mapsto \exp(t\mathring{\xi})\Theta_0$ is the orbit Fisher form $\mathfrak{t}_{q_i}(\mathring{\xi}, \mathring{\xi})$ of corollary 7.7; it is positive semidefinite with kernel the tangent space of the

zero set. For Gaussian laws in (b),

$$\mathfrak{k}_q(\mathring{\xi},\mathring{\xi}) = \left|\Sigma^{-1/2}\mathring{\xi}\mu\right|^2 + \tfrac{1}{2}\left\|\Sigma^{-1/2}\mathring{\xi}\Sigma^{1/2} + (\Sigma^{-1/2}\mathring{\xi}\Sigma^{1/2})^\top\right\|_{\mathrm{F}}^2, \qquad \mathring{\xi} \in \mathfrak{gl}(K,\mathbb{R}). \tag{9.36}$$

[ESTABLISHED]

The proof is in appendix B.3.

Corollary 9.24 (Without the generator prior the link problem degenerates). (a) Let $G = \mathrm{GL}^+(K)$, $K \geq 2$, act on the belief fiber by its defining representation, and in the reference gauge let the record of corollary 9.22(a) be the only factor reading the link of $e : j \to i$. For a matter recognition law with finite second moments, $\Sigma_j \succ 0$, and $\Sigma_{ij} = 0$, let the link be moved as a parameter without a generator prior, with expected record energy $\mathbf{l}(\mathbf{M}) = \mathbb{E}_{Q^{\mathsf{Y}}}\frac{1}{2}(k_i - \mathbf{M}k_j)^\top W_a(k_i - \mathbf{M}k_j)$ for real $K \times K$ matrices $\mathbf{M}$. Then: (i) $\mathbf{l}$ is a strictly convex quadratic whose unique stationary point and minimizer is

$$\mathbf{M}^\star = \mu_i\mu_j^\top(\Sigma_j + \mu_j\mu_j^\top)^{-1} = \frac{\mu_i\mu_j^\top\Sigma_j^{-1}}{1 + \mu_j^\top\Sigma_j^{-1}\mu_j}, \tag{9.37}$$

of rank at most one and therefore singular; with a cross covariance it is instead the second-moment regression $\mathbb{E}[k_ik_j^\top]\mathbb{E}[k_jk_j^\top]^{-1}$, which can be invertible. (ii) $\mathbf{l}$ has no stationary point on $\mathrm{GL}(K)$; its infimum over $\mathrm{GL}^+(K)$ equals $\mathbf{l}(\mathbf{M}^\star)$, is not attained, and every minimizing sequence in $\mathrm{GL}^+(K)$ converges to $\mathbf{M}^\star$. (iii) A generator $\mathring{\omega}_e^b$ is stationary for $\mathring{\omega}_e^b \mapsto \mathbf{l}(\exp\mathring{\omega}_e^b)$ only where $\mathbf{L}(\mathring{\omega}_e^b,\cdot)$ is singular, and such generators exist: for $K = 2$, $\mu_i = \mu_j = \left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$, $\Sigma_j = W_a = I_2$, the generator $\mathring{\omega}_e^b = -(\log 3)I_2 + 2\pi\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ is stationary, with link $\exp\mathring{\omega}_e^b = \frac{1}{3}I_2 \neq \mathbf{M}^\star$. (iv) If $\mu_j^\top\Sigma_j^{-1}\mu_i \geq 0$, then $\mathbf{M}^\star$ is a limit of chart links $\exp\mathring{\omega}_e^b$, so the infimum over chart links equals $\mathbf{l}(\mathbf{M}^\star)$, is not attained, and is approached only along links converging to $\mathbf{M}^\star$; if $\mu_j^\top\Sigma_j^{-1}\mu_i < 0$, no sequence of chart links converges to $\mathbf{M}^\star$, the infimum over chart links exceeds $\mathbf{l}(\mathbf{M}^\star)$, and for $K = 2$ it is not attained.

(b) If the free energy depends on a link value only through a law-point energy D_e of proposition 9.23, finite and continuously differentiable on G, and the link is moved as a parameter without a generator prior, its stationary link values form a union of double cosets, none isolated when $\dim\mathfrak{g}_{q_i} + \dim\mathfrak{g}_{q_j} > 0$; for nondegenerate Gaussian laws under $\mathrm{GL}^+(K)$, where D_e is finite and smooth, this holds for every pair of laws when $K \geq 3$.

(c) With the generator prior: (i) under the hypotheses of proposition 9.21 with $\mathbf{U}$ bounded below, $\mathbf{V}$ attains its minimum on $\mathscr{T}$, and $(\mathring{\mu},\mathring{\Lambda}) \mapsto \mathcal{F}_V[\mathcal{N}(\mathring{\mu},\mathring{\Lambda}^{-1}) \otimes Q^{\mathsf{Y}}; X, o]$ has compact sublevel sets in the product of $\mathscr{T}$ and the open cone of positive-definite precisions, or of block-diagonal ones in the mean-field case, and attains its minimum there; (ii) if $\mathbb{P}^{\mathsf{Y}}$ is a nondegenerate Gaussian law on a real vector space Y of finite dimension that does not read the links, and every record energy is nonnegative, lower semicontinuous in $(\mathring{\omega}, y)$, and bounded by $\mathrm{c}_1(1 + \|y\|^2)e^{\mathrm{c}_2\|\mathring{\omega}\|}$, then the free energy restricted to the product Gaussian laws $\mathcal{N}(\mathring{\mu},\mathring{\Lambda}^{-1}) \otimes \mathcal{N}(\mu^{\mathsf{Y}},\Sigma^{\mathsf{Y}})$ has compact sublevel sets in $(\mathring{\mu},\mathring{\Lambda},\mu^{\mathsf{Y}},\Sigma^{\mathsf{Y}})$ and attains its minimum. [ESTABLISHED]

The proof is in appendix B.3.

At the sample level the degeneracy of (a) is tied to a product matter recognition, since a correlated one can make the stationary matrix invertible, while the coercivity of (c) comes from the anchor term of the prior at every width $\mathring{\Sigma}$. Part (a)(iii) shows that the chart itself creates stationary generators at its folds: a gradient flow in generator coordinates started at such a generator does not move, although its link value is not stationary for the link energy.

9.6 Held links as a limit, and conditional statements

How does the theory with latent generators relate to the theory in which links are structural data or parameters? As the prior covariance shrinks at a held anchor, the infimum of the latent-link free energy converges to the infimum of the held-link free energy at the anchor links, and near-minimizing link recognition laws concentrate at the anchor (theorem 9.25); at every width, the conditional laws given the generators are held-link laws, so statements proved for every link value transfer conditionally, while statements about mixtures over the generators need not (proposition 9.26).

Theorem 9.25 (Held links are the zero-width limit). Hold θ except for the prior covariances, which are $\mathring{\Sigma} = \mathring{\sigma}^2\mathring{\Sigma}_0$ for $\mathring{\sigma} > 0$ and a block-diagonal $\mathring{\Sigma}_0 \succ 0$, and hold X, o and the anchor $\mathring{\alpha}$. Write $\nu^{\mathscr{T}}_{\mathring{\sigma}}$, $\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}$, $Z_{\mathscr{S},\mathring{\sigma}}$ and $\mathcal{F}_{\mathring{\sigma}}$ for the base law, the generator prior, its normalizer, and the free energy of the latent-link model at width $\mathring{\sigma}$. Let $\mathcal{Q}'$ be a nonempty class of laws on Y, put $\mathbf{F}_Q(\mathring{\omega}) = \mathcal{F}_V[Q; X, \Theta(\mathring{\omega}), o]$ for $Q \in \mathcal{Q}'$ and $\mathbf{F}(\mathring{\omega}) = \inf_{Q\in\mathcal{Q}'} \mathbf{F}_Q(\mathring{\omega})$, and assume: (i) the record densities are bounded, $L_o \le \mathrm{c}_{\mathfrak{A}}$, and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$; (ii) $\mathbf{F}$ is finite and lower semicontinuous at $\mathring{\alpha}$; (iii) a subclass $\mathcal{Q}'_0 \subseteq \mathcal{Q}'$ with $\inf_{Q\in\mathcal{Q}'_0} \mathbf{F}_Q(\mathring{\alpha}) = \mathbf{F}(\mathring{\alpha})$ consists of laws for which $\mathbf{F}_Q$ is continuous at $\mathring{\alpha}$ and bounded by $\mathrm{c}_Q e^{\mathrm{c}_Q\|\mathring{\omega}\|}$. Then:

(a) As $\mathring{\sigma} \to 0$, the infimum of $\mathcal{F}_{\mathring{\sigma}}$ over the product laws $R \otimes Q$ with $Q \in \mathcal{Q}'$ and R in any class of laws on $\mathscr{T}$ that contains $\nu^{\mathscr{T}}_{\mathring{\sigma}}$, and its infimum over all laws $R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ whose kernel takes values in $\mathcal{Q}'$, both converge to

$$\mathbf{F}(\mathring{\alpha}) = \inf_{Q\in\mathcal{Q}'} \mathcal{F}_V\big[Q; X, \Theta(\mathring{\alpha}), o\big], \qquad \Theta^x_e(\mathring{\alpha}) = U_i \exp(\mathring{\alpha}^x_e) U_j^{-1}, \tag{9.38}$$

the infimum over $\mathcal{Q}'$ of the held-link free energy at the anchor links.

(b) Under (i) alone, every family of laws $R_{\mathring{\sigma}}(d\mathring{\omega})Q_{\mathring{\sigma},\mathring{\omega}}(dy)$ with $\sup_{\mathring{\sigma}} \mathcal{F}_{\mathring{\sigma}} < \infty$ satisfies $R_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \ge \mathring{\epsilon}\} \to 0$ for every $\mathring{\epsilon} > 0$; under (i) and (ii), every such family with kernels in $\mathcal{Q}'$ and $\mathcal{F}_{\mathring{\sigma}} \to \mathbf{F}(\mathring{\alpha})$ satisfies in addition $D_{\mathrm{KL}}(R_{\mathring{\sigma}} \| \mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}) \to 0$.

(c) Under (i) alone, if $p(o \mid \cdot)$ is continuous at $\mathring{\alpha}$, then the evidence of the latent-link model converges to $p_{\theta,V}(o \mid X, \Theta(\mathring{\alpha}))$, and the link marginal $\mathbf{\Pi}^{\mathscr{T}}_{\mathring{\sigma}}$ of its posterior satisfies $\mathbf{\Pi}^{\mathscr{T}}_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \ge \mathring{\epsilon}\} \to 0$ for every $\mathring{\epsilon} > 0$.

(d) Under (i), hypothesis (ii) holds when $\mathcal{Q}' = \{Q_\lambda : \lambda \in \mathbb{U}\}$ for a subset $\mathbb{U}$ of a Euclidean space, $(\mathring{\omega}, \lambda) \mapsto \mathbf{F}_{Q_\lambda}(\mathring{\omega})$ is lower semicontinuous, $\mathbf{F}(\mathring{\alpha}) < \infty$, and for some $\mathring{\epsilon}_0 > 0$ and some $\mathrm{c} > \mathbf{F}(\mathring{\alpha})$ the set of λ with $\mathbf{F}_{Q_\lambda}(\mathring{\omega}) \le \mathrm{c}$ for some $\|\mathring{\omega} - \mathring{\alpha}\| < \mathring{\epsilon}_0$ has compact closure in $\mathbb{U}$. [ESTABLISHED]

The proof is in appendix B.3.

The present held-link theory is therefore the zero-width limit of the latent one in value: the prior and the recognition law narrow together, so the link sector costs nothing in the limit, in contrast with a recognition law that narrows under a prior of constant width, whose cost diverges (proposition 9.21(e)).

Proposition 9.26 (Conditional statements transfer; link-marginal statements need not). Assume the hypotheses of proposition 9.16(c). (a) For every $\mathring{\omega}$ the conditional law of the posterior given $\mathring{\omega}$ is the held-link posterior $\mathbf{\Pi}_{\theta,V,o,X,\Theta(\mathring{\omega})}$, and for every law $Q = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ the conditional free energy $-\log p(o \mid \mathring{\omega}) + D_{\mathrm{KL}}(Q_{\mathring{\omega}} \| \mathbf{\Pi}_{\mathring{\omega}})$ is the held-link free energy $\mathcal{F}_V[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o]$. If $Q' = R(d\mathring{\omega})Q'_{\mathring{\omega}}(dy)$ has the same link marginal and both free energies are finite, then

$$\mathcal{F}_V[Q'; X, o] - \mathcal{F}_V[Q; X, o] = \int \Big(\mathcal{F}_V\big[Q'_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o\big] - \mathcal{F}_V\big[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o\big] \Big) R(d\mathring{\omega}). \tag{9.39}$$

(b) Let $\mathbb{O}$ be a measurable set of link families, and consider a property of a pair consisting of held-link structural data and a law on Y, expressed through the held-link joint, evidence, posterior and free energy. If the property holds for $((X, \Theta), Q'')$ for every $\Theta \in \mathbb{O}$ and every Q'' in a class $\mathcal{Q}''$, then it holds for $((X, \Theta(\mathring{\omega})), Q_{\mathring{\omega}})$ at every $\mathring{\omega}$ with $\Theta(\mathring{\omega}) \in \mathbb{O}$ and $Q_{\mathring{\omega}} \in \mathcal{Q}''$, hence for R-almost every $\mathring{\omega} \in \Theta^{-1}(\mathbb{O})$ when the kernel lies in $\mathcal{Q}''$ R-almost surely. For instance, for $G = \mathrm{GL}^+(K)$ in its defining representation and a closed walk p in which the copies of some reversal pair occur with nonzero net count, the generators at which $|\det \rho_x(\mathrm{Hol}^x(p))| = 1$ form a hyperplane of $\mathscr{T}$, so for every R absolutely continuous with respect to Lebesgue measure the conclusion of corollary 11.12 for a block containing p holds conditionally on $\mathring{\omega}$ for R-almost every $\mathring{\omega}$. (c) A property of laws on Y transfers from the conditional laws to the link-marginal laws, namely the matter marginal $\int Q_{\mathring{\omega}} R(d\mathring{\omega})$ of a recognition law, the matter marginal $\int \mathbf{\Pi}_{\mathring{\omega}} \mathbf{\Pi}^{\mathscr{T}}(d\mathring{\omega})$ of the posterior, and their images under a coarse channel that does not read the links, when the set of laws with the property is preserved by mixtures, and need not otherwise. For $G = \mathrm{GL}^+(1)$ acting on $\mathbb{R}$ by multiplication, two agents with $k_j \sim \mathcal{N}(0,1)$ and $k_i \mid k_j \sim \mathcal{N}(\rho_b(\Theta^b_e)k_j, 1)$ for the copy $e : j \to i$, trivial frames, $\mathscr{S} = 0$, $\nu^{\mathscr{T}} = \mathcal{N}(0,1)$, and no record, every held-link posterior is Gaussian, while the k_i-marginal of the latent-link posterior has $\mathbb{E}k_i^4 = 3(1 + 2e^2 + e^8) > 3(1 + e^2)^2 = 3(\mathbb{E}k_i^2)^2$, so it is not Gaussian. [ESTABLISHED]

Proof. (a) The first claim is the disintegration of proposition 9.16(c), and the second is definition 5.6 at $(X, \Theta(\mathring{\omega}))$. Finite free energies make both integrals in (9.20) finite, because the integrands are bounded below and the divergence term is common, so subtraction gives (9.39). (b) By (a) the conditional objects at $\mathring{\omega}$ are the held-link objects at $(X, \Theta(\mathring{\omega}))$, and kernels are determined up to R-null sets. For the example, $\det e^{\mathbf{B}} = e^{\mathrm{tr}\,\mathbf{B}}$ (triangularize $\mathbf{B}$ over $\mathbb{C}$), so by (2.33) $\det \rho_x(\mathrm{Hol}^x(p))$ is the exponential of the linear functional $\mathring{\omega} \mapsto \sum_{e \in E_+} \mathrm{tr}\, \mathring{\omega}^x_e$ weighted by the net count of each e in p, which is nonzero by assumption, and the set where the determinant has modulus one is the kernel of this functional, which is Lebesgue null. Off it the holonomy has $|\det| \neq 1$, and corollary 11.12 applies at every such link value. (c) The marginals are mixtures over $\mathring{\omega}$ of the conditional laws, and a link-independent kernel commutes with mixing, which gives the positive clause. In the example there are no records, so the posterior is the prior; given the generator,

(k_i, k_j) is Gaussian with $\mathbb{E}[k_i^2 \mid \mathring{\omega}] = 1 + e^{2\mathring{\omega}}$, and a centered Gaussian has $\mathbb{E}k^4 = 3(\mathbb{E}k^2)^2$, so $\mathbb{E}k_i^4 = 3\mathbb{E}(1+e^{2\mathring{\omega}})^2 = 3(1+2e^2+e^8)$ and $\mathbb{E}k_i^2 = 1+e^2$, by $\mathbb{E}e^{2\mathring{\omega}} = e^2$ and $\mathbb{E}e^{4\mathring{\omega}} = e^8$; strict inequality is strict Jensen, since $e^{2\mathring{\omega}}$ is not almost surely constant. □

A coarse channel that reads the links, such as the tree transports of definition 13.17, is a kernel on $(\mathring{\omega}, y)$ and is covered by (b) conditionally on $\mathring{\omega}$; its link-marginal image is again a mixture.

9.7 Directed and undirected matter

Does matter generate a link action of its own? In the record extension it does not: the generator marginal is the generator prior, and matter reaches the links only through the evidence. When generators and matter are normalized jointly as an undirected model, the generator marginal acquires minus the logarithm of the matter normalizer (proposition 9.27).

Proposition 9.27 (Induced link action). (a) For the joint (9.18), $\int_{\mathsf{Y}\times\mathsf{O}} \mathbb{P}_{\theta,V}(d\mathring{\omega}, dy, do \mid X) = \mathbb{P}_\theta^{\mathscr{T}}(d\mathring{\omega} \mid X)$, and with record densities the link marginal of the posterior is proportional to $p(o \mid \mathring{\omega})\mathbb{P}_\theta^{\mathscr{T}}(d\mathring{\omega} \mid X)$. (b) Let the potentials $\psi_i(\cdot; X, \Theta)$, $\psi_{ij}(\cdot; X, \Theta)$, $\psi_a(\cdot; X, \Theta)$ of definition 4.5 depend jointly measurably on the link family, write $\widetilde{p}_{X,\Theta}$ and $Z_{X,\Theta}$ for (4.13) and (4.14) at (X, Θ), and normalize generators and matter jointly: with

$$\mathfrak{Z} = \int_{\mathscr{T}} e^{-\mathscr{S}(\mathring{\omega})} Z_{X,\Theta(\mathring{\omega})} \nu^{\mathscr{T}}(d\mathring{\omega}), \tag{9.40}$$

assume $0 < \mathfrak{Z} < \infty$, and take the law with density $\mathfrak{Z}^{-1} e^{-\mathscr{S}(\mathring{\omega})} \widetilde{p}_{X,\Theta(\mathring{\omega})}(o, y)$ with respect to $\nu^{\mathscr{T}} \otimes \nu^{\mathsf{Y}} \otimes \nu^{\mathsf{O}}$. It is a probability law, $Z_{X,\Theta(\mathring{\omega})} < \infty$ for $\nu^{\mathscr{T}}$-almost every $\mathring{\omega}$, its generator marginal is

$$\mathfrak{Z}^{-1} e^{-\mathscr{S}(\mathring{\omega}) - \mathscr{S}^{\mathrm{ind}}(\mathring{\omega})} \nu^{\mathscr{T}}(d\mathring{\omega}), \qquad \mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) = -\log Z_{X,\Theta(\mathring{\omega})}, \tag{9.41}$$

with $\mathscr{S}^{\mathrm{ind}}(\mathring{\omega}) \in (-\infty, +\infty]$ for $\nu^{\mathscr{T}}$-almost every $\mathring{\omega}$, and on $\{0 < Z_{X,\Theta(\mathring{\omega})} < \infty\}$ its conditional law given $\mathring{\omega}$ is the undirected model (4.15) at $(X, \Theta(\mathring{\omega}))$. A record extension with densities is the case $Z_{X,\Theta} = 1$ for every Θ, in which $\mathscr{S}^{\mathrm{ind}} = 0$. The condition $\mathfrak{Z} < \infty$ is not implied by $Z_{\mathscr{S}} < \infty$: for $G = \mathrm{GL}^+(1)$, trivial frames, $\mathscr{S} = 0$, two scalar agents with standard Gaussian self potentials, and the edge potential $\exp(\rho_b(\Theta_e)k_1k_2)$, $Z_{X,\Theta} = (1-\rho_b(\Theta_e)^2)^{-1/2}$ when $\rho_b(\Theta_e) < 1$ and $Z_{X,\Theta} = +\infty$ when $\rho_b(\Theta_e) = e^{\mathring{\omega}_e} \geq 1$, a set of positive $\nu^{\mathscr{T}}$-measure, so $\mathfrak{Z} = +\infty$. [ESTABLISHED]

Proof. (a) By Tonelli's theorem, integrate the records, each a kernel of unit mass, and then y against the probability kernel $\mathbb{P}^{\mathsf{Y}}(\cdot \mid \mathring{\omega})$; the posterior marginal is that of proposition 9.16(c). (b) Tonelli's theorem gives $\int e^{-\mathscr{S}} \widetilde{p}_{X,\Theta(\mathring{\omega})} d(\nu^{\mathsf{Y}} \otimes \nu^{\mathsf{O}}) = e^{-\mathscr{S}(\mathring{\omega})} Z_{X,\Theta(\mathring{\omega})}$, whose $\nu^{\mathscr{T}}$-integral is $\mathfrak{Z}$, so the density integrates to one; finiteness of $\mathfrak{Z}$ with $e^{-\mathscr{S}} > 0$ forces $Z_{X,\Theta(\mathring{\omega})} < \infty$ almost everywhere, the generator marginal is $\mathfrak{Z}^{-1} e^{-\mathscr{S}} Z_{X,\Theta(\mathring{\omega})} \nu^{\mathscr{T}}$, which is (9.41) with the value $+\infty$ of $\mathscr{S}^{\mathrm{ind}}$ where $Z_{X,\Theta(\mathring{\omega})} = 0$, and dividing the joint density by the marginal density gives (4.15). The densities of a record extension are conditional densities of unit mass in (y, o) for every link family (proposition 4.4), so as potentials they have $Z_{X,\Theta} = 1$. In the example, $\iint (2\pi)^{-1} e^{-(k_1^2+k_2^2)/2 + \rho_b(\Theta_e)k_1k_2} dk_1 dk_2 = \det \begin{pmatrix} 1 & -\rho_b(\Theta_e) \\ -\rho_b(\Theta_e) & 1 \end{pmatrix}^{-1/2}$

for $\rho_b(\Theta_e) < 1$, the integral diverges for $\rho_b(\Theta_e) \geq 1$ by proposition 4.6, and the Gaussian law $\nu^{\mathscr{T}}$ on $\mathfrak{g} = \mathbb{R}$ charges $\{\mathring{\omega}_e \geq 0\}$. □

For Gaussian self potentials and pairwise potentials of the connection-Laplacian form (11.24), the matter normalizer is a Gaussian determinant, so the induced action is one half the log-determinant of the connection Laplacian plus the self term, and its expansion in closed walks produces loop products of the links; Supplement G develops both.

9.8 Inference dynamics of the links

The generators are latent coordinates, so they move by the dynamics of every recognition coordinate of the program, descent of the collective free energy. This section writes that descent for a Gaussian recognition law of the generators jointly with the agents' recognition coordinates, shows that it is a natural-gradient history whose explicit Euler step is the Gaussian Bayesian learning rule (proposition 9.28), gives the exact M-coordinate of the anchor (proposition 9.29), and reads the supplied link rate of proposition 8.6 in this setting (remark 9.30). In this section and the next two, the links of one channel are latent, or both channels share them under the common-link specialization (2.31), and the channel index is dropped. The link space $\mathscr{T} = \mathfrak{g}^{E_+}$ carries the product inner product $\langle \mathring{\omega}, \mathring{\omega}' \rangle = \sum_{e \in E_+} \langle \mathring{\omega}_e, \mathring{\omega}'_e \rangle_{\mathsf{G}}$ of definition 9.11, its norm $\|\cdot\|$, and the Lebesgue measure of that definition; self-adjointness, traces, and determinants of operators on $\mathscr{T}$ refer to this inner product. Where a representation ρ enters, $\mathrm{d}\rho$ is injective and $\mathfrak{g}$ carries the pulled-back inner product of definition 9.8, so that $\|\mathring{\omega}_e\|_{\mathsf{G}} = \|\mathrm{d}\rho \mathring{\omega}_e\|_{\mathsf{G}}$. For a matrix $\mathbf{H}$ on the represented fiber, $\mathbf{H}^* = (\mathsf{G}^{\mathrm{fib}})^{-1} \mathbf{H}^\top \mathsf{G}^{\mathrm{fib}}$ is its adjoint for the fiber metric, so that $\langle \mathbf{H}, \mathbf{H}' \rangle_{\mathsf{G}} = \mathrm{Tr}(\mathbf{H}^* \mathbf{H}')$ in (9.6) and left or right multiplication by $\mathbf{H}$ has as adjoint the left or right multiplication by $\mathbf{H}^*$; $\mathrm{pr}_{\mathfrak{g}}$ is the $\langle \cdot, \cdot \rangle_{\mathsf{G}}$-orthogonal projection of matrices onto $\mathrm{d}\rho(\mathfrak{g})$, read in $\mathfrak{g}$.

Proposition 9.28 (Natural-gradient recognition flow of the generators). In the latent-link model of definition 9.15, with bounded record densities and $p(o \mid \mathring{\omega}) > 0$ for every $\mathring{\omega}$, let the recognition law be $R \otimes Q_\lambda$. Here $R = \mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ is a Gaussian law on $\mathscr{T}$ with mean $\mathring{\mu} \in \mathscr{T}$ and precision $\mathring{\Lambda}$ in the open cone $\mathring{\Xi}_{\succ 0}$ of positive-definite elements of the space $\mathring{\Xi}$ of self-adjoint operators that preserve each summand of a given orthogonal decomposition of $\mathscr{T}$: one summand admits every positive-definite precision, the summands $\mathfrak{g}$ of the edge copies give the products $\bigotimes_{e \in E_+} \mathcal{N}(\mathring{\mu}_e, \mathring{\Lambda}_e^{-1})$, and the lines of an orthonormal basis give the diagonal precisions. The law Q_λ is the agents' recognition law, with coordinates λ in a collective configuration manifold $\mathcal{Q}_V$ (definition 7.13) that satisfies hypothesis 7.16 with metric G_V. By (9.20) and proposition 9.18(b) with $Q_{\mathring{\omega}} = Q_\lambda$ for every $\mathring{\omega}$, the collective free energy at $R \otimes Q_\lambda$ is

$$F(\mathring{\mu}, \mathring{\Lambda}, \lambda) = \mathbb{E}_R\big[\mathfrak{i}_\lambda\big] + \tfrac{1}{2} \log\det \mathring{\Lambda} + \log Z_{\mathscr{I}} - \tfrac{1}{2} \dim \mathscr{T} \log(2\pi e), \tag{9.42}$$

with the generator integrand

$$\mathfrak{i}_\lambda(\mathring{\omega}) = \mathcal{F}_V\big[Q_\lambda; X, \Theta(\mathring{\omega}), o\big] + \mathscr{I}(\mathring{\omega}) + \tfrac{1}{2} \big\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha}) \big\rangle + \tfrac{1}{2} \log\det\big(2\pi \mathring{\Sigma}\big). \tag{9.43}$$

Assume that F is C^2 on $\mathscr{T} \times \mathring{\Xi}_{\succ 0} \times \mathcal{Q}_V$ and that each i_λ is C^2 with $|\mathrm{i}_\lambda| + \|\nabla \mathrm{i}_\lambda\| + \|\nabla^2 \mathrm{i}_\lambda\| \le \mathrm{c}_1 e^{\mathrm{c}_2 \|\mathring{\omega}\|}$ for constants $\mathrm{c}_1, \mathrm{c}_2$. Write $\mathring{\mu} \otimes \mathring{\mu}$ for the operator $\mathring{\omega} \mapsto \langle \mathring{\mu}, \mathring{\omega} \rangle \mathring{\mu}$ and $\mathrm{pr}_{\mathring{\Xi}}$ for the orthogonal projection onto $\mathring{\Xi}$ in the trace inner product.

(a) The laws $\mathcal{N}(\mathring{\mu}, \mathring{\Lambda}^{-1})$ with $\mathring{\Lambda} \in \mathring{\Xi}_{\succ 0}$ form a minimal regular exponential family with natural parameter $\mathring{\vartheta} = (\mathring{\Lambda}\mathring{\mu}, -\frac{1}{2}\mathring{\Lambda})$, expectation parameter $\mathring{\varpi} = (\mathring{\mu}, \mathring{\Lambda}^{-1} + \mathrm{pr}_{\mathring{\Xi}}(\mathring{\mu} \otimes \mathring{\mu}))$, and log-partition function $\mathring{\Gamma}$, hence a regular statistical manifold. Its Fisher metric is smooth and positive definite, and in the coordinates $(\mathring{\mu}, \mathring{\Lambda})$ it is

$$g^F\big((\delta\mathring{\mu}, \delta\mathring{\Lambda}), (\delta\mathring{\mu}, \delta\mathring{\Lambda})\big) = \langle \delta\mathring{\mu}, \mathring{\Lambda}\delta\mathring{\mu} \rangle + \tfrac{1}{2} \operatorname{Tr}\big(\mathring{\Lambda}^{-1}\delta\mathring{\Lambda}\mathring{\Lambda}^{-1}\delta\mathring{\Lambda}\big). \tag{9.44}$$

The product $\mathcal{Q}$ of this family with $\mathcal{Q}_V$, with the metric $\mathsf{G} = g^F \oplus \mathsf{G}_V$, satisfies hypothesis 7.16.

(b) The natural-gradient field $-\operatorname{grad}^{\mathsf{G}} F$ of (7.40) on $\mathcal{Q}$ is C^1, so its maximal integral curves exist and are unique. Along each of them $\frac{d}{dt}F = -\|\operatorname{grad}^{\mathsf{G}} F\|_{\mathsf{G}}^2 \le 0$, and each one through a noncritical point has nowhere-vanishing velocity and satisfies (7.41) with $\mathsf{a} \equiv 1$, so it represents a natural-gradient history (definition 7.30).

(c) The flow is of first order. Its velocity is a function of the configuration alone and vanishes exactly at the critical points of F. It obeys the force balance $\mathsf{G}(\dot{Q}, \cdot) = -dF$, in which G is the friction tensor, and $\frac{d}{dt}F = -\mathsf{G}(\dot{Q}, \dot{Q})$. The free energy strictly decreases along every nonconstant solution, so no nonconstant solution is periodic. At a critical point $Q_\star$ with invertible Hessian the linearized field is $-\mathsf{G}(Q_\star)^{-1}\nabla^2 F(Q_\star)$, which is diagonalizable with real eigenvalues, so no linearized mode oscillates.

(d) In the chart $(\mathring{\vartheta}, \lambda)$, the generator block of the explicit Euler step of the flow with step $\Delta t > 0$ is

$$\mathring{\Lambda}' = (1 - \Delta t)\mathring{\Lambda} + \Delta t\, \mathrm{pr}_{\mathring{\Xi}}\, \mathbb{E}_R\big[\nabla^2 \mathrm{i}_\lambda\big], \qquad \mathring{\mu}' = \mathring{\mu} - \Delta t \mathring{\Lambda}'^{-1} \mathbb{E}_R\big[\nabla \mathrm{i}_\lambda\big], \tag{9.45}$$

whenever $\mathring{\Lambda}'$ is positive definite, for instance when $0 < \Delta t < 1$ and $\mathrm{pr}_{\mathring{\Xi}}\, \mathbb{E}_R[\nabla^2 \mathrm{i}_\lambda]$ is positive semidefinite. For one summand this is the Gaussian form of the Bayesian learning rule [66, Eq. (12)] applied to i_λ, and for the lines of an orthonormal basis it is its diagonal-precision form, which Khan and Rue [66, Sec. 4.2, Eq. (37)] state with the expected gradient and Hessian replaced by their values at the mean.

(e) Let $R' = \mathcal{N}(\mathring{\mu}', \mathring{\Lambda}'^{-1})$ with expectation parameter $\mathring{\varpi}'$, and let $\widetilde{F}(\mathring{\varpi})$ denote F at fixed λ in the expectation chart. If $\mathring{\Lambda}'$ is positive definite and $\nabla^2 \widetilde{F} \preceq \mathrm{c}_{\mathsf{G}} \nabla^2 \mathring{\Gamma}^\star$ on the segment from $\mathring{\varpi}$ to $\mathring{\varpi}'$ for some $\mathrm{c}_{\mathsf{G}} > 0$, where $\mathring{\Gamma}^\star$ is the convex conjugate of $\mathring{\Gamma}$ and $\nabla^2 \mathring{\Gamma}^\star$ is the Fisher metric in the expectation chart, then

$$F(R') \le F(R) - \Delta t^{-1} D_{\mathrm{KL}}(R \| R') - \big(\Delta t^{-1} - \mathrm{c}_{\mathsf{G}}\big) D_{\mathrm{KL}}(R' \| R), \tag{9.46}$$

so a step with $0 < \Delta t \le 1/\mathrm{c}_{\mathsf{G}}$ decreases F unless $R' = R$. For one summand and a quadratic integrand $\mathrm{i}_\lambda(\mathring{\omega}) = \frac{1}{2}\langle \mathring{\omega}, \mathbf{Q}\mathring{\omega} \rangle - \langle \mathbf{b}, \mathring{\omega} \rangle + \mathrm{c}_3$ with $\mathbf{Q}$ positive definite, the hypothesis holds with

$\mathrm{c_G} = 1$ everywhere, and the step $\Delta t = 1$ gives $R' = \mathcal{N}(\mathbf{Q}^{-1}\mathbf{b}, \mathbf{Q}^{-1})$, the minimizer of F over Gaussian R.

[ESTABLISHED]

The proof is in appendix B.3.

For links, the latent option of definition 6.23 asks of a random relation coordinate a prior that is a probability law, dynamics in the augmented state, and a regular configuration metric or a justified quotient, and the present construction discharges these requirements. The generator prior of definition 9.11, in the joint of definition 9.15, supplies the first for every Lie group G, compact or not (propositions 9.12 and 9.16), with directed matter, and with undirected matter normalized jointly whenever the normalizer (9.40) is finite (proposition 9.27); the dynamics and the metric are those of proposition 9.28.

Proposition 9.29 (Exact M-coordinate of the anchor). Let the generator prior of definition 9.11 have base law $\nu^{\mathscr{T}}_{\mathring{\alpha}} = \bigotimes_{e\in E_+} \mathcal{N}(\mathring{\alpha}_e, \mathring{\Sigma}_e)$ and an action $\mathscr{S} \geq 0$ that does not depend on $\mathring{\alpha}$, so that the prior is $\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}} = Z_{\mathscr{S}}(\mathring{\alpha})^{-1} e^{-\mathscr{S}} \nu^{\mathscr{T}}_{\mathring{\alpha}}$ with $Z_{\mathscr{S}}(\mathring{\alpha}) = \int e^{-\mathscr{S}} d\nu^{\mathscr{T}}_{\mathring{\alpha}}$, and let no other factor of the joint depend on $\mathring{\alpha}$. Hold the recognition coupling $\mathbb{Q}_{V,o,X}$ fixed, let its generator marginal R have finite second moments, and assume the finiteness hypotheses of proposition 5.27 for every anchor.

(a) Up to terms that do not depend on $\mathring{\alpha}$, the objective of (5.58) is

$$\mathcal{L}^{\mathrm{anc}}(\mathring{\alpha}) = -\tfrac{1}{2}\mathbb{E}_R\big\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\big\rangle - \log Z_{\mathscr{S}}(\mathring{\alpha}), \tag{9.47}$$

which contains the normalizer of the prior, and

$$\nabla_{\mathring{\alpha}_e} \mathcal{L}^{\mathrm{anc}} = \mathring{\Sigma}_e^{-1}\big(\mathbb{E}_R[\mathring{\omega}_e] - \mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}}}[\mathring{\omega}_e]\big). \tag{9.48}$$

(b) The function $\mathcal{L}^{\mathrm{anc}}$ is strictly concave and has exactly one maximizer $\mathring{\alpha}^{\mathrm{new}}$, the unique anchor at which the prior mean equals the recognition mean, $\mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}^{\mathrm{new}}}}[\mathring{\omega}] = \mathbb{E}_R[\mathring{\omega}]$. By proposition 5.27 it is the exact M-coordinate on the anchor at the given covariances.

(c) If $\mathscr{S} = 0$, as when every loop coefficient c_p vanishes, then $\mathring{\alpha}^{\mathrm{new}} = \mathbb{E}_R[\mathring{\omega}]$, which is $\mathring{\mu}$ for a Gaussian generator recognition law. If $\mathscr{S}(\mathring{\omega}) = \frac{1}{2}\sum_p \mathrm{c}_p \|\mathbf{C}_p\mathring{\omega}\|^2_{\mathsf{G}}$, where $\mathbf{C}_p\mathring{\omega} = (\mathbf{C}\mathring{\omega})_p$ is the loop sum of the cycle map (9.8) and $\mathbf{C}_p^{\top}$ is the adjoint of $\mathbf{C}_p : \mathscr{T} \to \mathfrak{g}$, as for the loop defects of the translation instance in its affine representation (proposition 9.13), then

$$\mathring{\alpha}^{\mathrm{new}} = \Big(I + \mathring{\Sigma}\sum_p \mathrm{c}_p \mathbf{C}_p^{\top}\mathbf{C}_p\Big)\mathbb{E}_R[\mathring{\omega}], \tag{9.49}$$

which differs from $\mathbb{E}_R[\mathring{\omega}]$ exactly when $\mathbf{C}_p\mathbb{E}_R[\mathring{\omega}] \neq 0$ for some loop p with $\mathrm{c}_p > 0$.

(d) If the covariances $\mathring{\Sigma}_e$ range as well over the positive-definite self-adjoint operators on $\mathfrak{g}$, the objective is strictly concave in the coordinates $(\mathring{\Sigma}_e^{-1}\mathring{\alpha}_e, -\frac{1}{2}\mathring{\Sigma}_e^{-1})_{e\in E_+}$, it has at most one maximizer, and a maximizer matches the mean and the covariance of every edge generator under the

prior and under R. If $\mathscr{S} = 0$ and every $\mathrm{Cov}_R(\mathring{\omega}_e)$ is positive definite, the maximizer exists and is $(\mathbb{E}_R[\mathring{\omega}_e], \mathrm{Cov}_R(\mathring{\omega}_e))_{e\in E_+}$. With a loop term it need not exist: in the translation instance with $\mathsf{G}^{\mathrm{fib}} = I_2$, on one agent i with one reversal pair of edge copies $e, \bar{e} : i \to i$, the loop $p = (e)$ with $\mathrm{c}_p = \mathrm{c} > 0$, and $R = \mathcal{N}(\mathring{\mu}, \mathrm{v})$ with $\mathrm{v} \geq 1/\mathrm{c}$, the objective has no maximizer over $(\mathring{\alpha}_e, \mathring{\Sigma}_e) \in \mathbb{R} \times (0, \infty)$, and its supremum is approached as $\mathring{\Sigma}_e \to \infty$.

[ESTABLISHED]

The proof is in appendix B.3.

Remark 9.30 (The supplied link rate in the latent reading). Read definition 14.21 with every frame at the reference point, so that each twist equals its link (proposition 2.12). (i) Its link translations, written $\check{\omega}_i$ in proposition 8.6, are generators of the translation instance (proposition 9.13): $(\mathbb{R}, +)$ acting on the scalar fibers by translation, in its affine representation with $\mathsf{G}^{\mathrm{fib}} = I_2$, for which $\mathfrak{g} = \mathbb{R}$, $\langle \mathrm{x}, \mathrm{y} \rangle_{\mathsf{G}} = \mathrm{xy}$, and every loop defect is the quadratic $\frac{1}{2}(\sum_a \check{\omega}_{e_a})^2$. The link block $\mathcal{N}(0, I_4)$ of the configuration prior is the generator prior of definition 9.11 with anchor 0, $\mathring{\Sigma}_e = 1$, and no loop term; the loop energy of the experiment enters through its records. For a law of the configuration, the generator recognition law R is its $\check{\omega}$-marginal, and its mean is the generator recognition mean $\mathring{\mu}$. (ii) Moving the link translations at the supplied rate $\varepsilon_{\check{\omega}}$ is therefore, on the held matching sheet and for $\varepsilon_{\mathrm{m}}, \varepsilon_{\check{\omega}} > 0$, the rate-separated flow (8.7) with the generator recognition means in the slow block, $s = (\bar{m}, \mathring{\mu})$ and $\mathsf{G}_{\mathrm{s}} = I_4 \oplus (\varepsilon_{\mathrm{m}}/\varepsilon_{\check{\omega}}) I_4$, by proposition 8.6(i). (iii) The anchor $\mathring{\alpha}$ is a coordinate of the generative parameter θ, so it belongs to the slow block $s = (s_{\mathrm{m}}, \theta)$ of definition 8.2: it moves the joint between evaluations, by accepted M-coordinates (definition 5.28), of which proposition 9.29 gives the exact one, or by the rate-separated flow, and never inside an evaluation (hypothesis 4.3). In the experiment no generative parameter is learned and the anchor stays at 0. (iv) The recognition flow of proposition 9.28 does not supply the separation of (ii). In the notation of definition 14.21, take the held sheet $S^{(0)}$, the Gaussian configuration laws $\mathcal{N}(\mu, \Sigma) \otimes \delta_{S^{(0)}}$ on $\mathbb{R}^{12}$ with their Fisher metric, of the form (9.44), and the free energy $\mathcal{F}_{\mathrm{T}}/\mathrm{T}$ of proposition 14.22, at the temperature T of that proposition. Then $\nabla^2 U_{S^{(0)}} \succeq \mathrm{T} I$, the natural-gradient flow leaves the slice $\Sigma = \mathrm{T}(\nabla^2 U_{S^{(0)}})^{-1}$ invariant, and on it $\mu(t) = e^{-t}\mu(0)$: belief means, model means, and link means relax at one common rate. From $\mu(0) = \mathbf{x}^\circ$, every residual is $r_i^b(t) = r_i^m(t) = -e^{-t}$, so at no time is $|r_i^b| \leq 2\varepsilon_{\mathrm{m}} < \frac{1}{2} \leq |r_i^m|$, and the belief-agreeing, model-disagreeing interval of proposition 8.6(ii) does not occur. [ESTABLISHED]

The proof is in appendix B.3.

On the slice of remark 9.30(iv) the Fisher geometry of the recognition family assigns every coordinate the same rate, so there the separation in proposition 8.6 comes from its supplied mobilities alone, and a slow link block is a declaration of the kind that definition 8.2 allows.

9.9 Link histories across learning episodes

Across learning episodes the generators form a history, and the latent route of definition 13.26 gives that history one joint law. This section puts a Gibbs prior on generator histories whose kinetic

term is the loop defect of a time-like loop (definition 9.31), proves that it is a latent route with the generators in the parent readout (proposition 9.32), and shows that its increment precision is the inertial mass of the most probable history (proposition 9.33). Throughout, $\mathfrak{Q}(\mathbf{H}) = \frac{1}{2}\|\mathbf{H} - I\|_{\mathsf{G}}^2$ for a matrix $\mathbf{H}$ on the represented fiber, so that the loop defect (9.7) of a closed walk p is $\mathfrak{Q}_p = \mathfrak{Q}(\rho(\mathring{\Theta}_p))$.

Definition 9.31 (Generator history prior). Index learning episodes by $n = 0, \dots, n_1$ as in definition 13.26, let $\mathring{\omega}^{(n)} = (\mathring{\omega}_e^{(n)})_{e \in E_+} \in \mathscr{T}$ be the generators of episode n, with $\mathring{\omega}_{\bar{e}}^{(n)} = -\mathring{\omega}_e^{(n)}$, and write $\mathring{\omega}^{(0:n_1)}$ for a history. Fix the generator prior of definition 9.11, with base law $\nu^{\mathscr{T}} = \bigotimes_{e \in E_+} \mathcal{N}(\mathring{\alpha}_e, \mathring{\Sigma}_e)$ and action $\mathscr{S} = \sum_p \mathrm{c}_p \mathfrak{Q}_p$ (the statements below use only that $\mathscr{S}$ is finite, measurable, and nonnegative), and a mass $\mathbf{m} > 0$. Frames are static at c_*: the copies of agent i at episodes n and $n+1$ are joined by the identity link. For an edge copy $e : j \to i$, the time-like loop of e between episodes n and $n+1$ is the closed walk that runs from j at episode $n+1$ to j at episode n, along e at episode n, from i at episode n to i at episode $n+1$, and along $\bar{e}$ at episode $n+1$. Its reference holonomy (9.4), with the identity twist for each time-like step, is $\mathring{\Theta}_e^{(n+1 \leftarrow n)} = \exp(-\mathring{\omega}_e^{(n+1)}) \exp(\mathring{\omega}_e^{(n)})$, with the represented holonomy

$$\rho\big(\mathring{\Theta}_e^{(n+1 \leftarrow n)}\big) = e^{-\mathrm{d}\rho \mathring{\omega}_e^{(n+1)}} e^{\mathrm{d}\rho \mathring{\omega}_e^{(n)}}. \tag{9.50}$$

The kinetic term of e between episodes n and $n+1$ is either the time-like loop defect $\mathfrak{U}_e^{(n)} = \mathfrak{Q}(\rho(\mathring{\Theta}_e^{(n+1 \leftarrow n)}))$ or the Gaussian increment $\mathfrak{U}_e^{(n)} = \frac{1}{2}\|\mathring{\omega}_e^{(n+1)} - \mathring{\omega}_e^{(n)}\|_{\mathsf{G}}^2$, and $\mathfrak{U}(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}) = \sum_{e \in E_+} \mathfrak{U}_e^{(n)}$. The generator history prior is the law on $\mathscr{T}^{n_1+1}$

$$\mathbb{P}^{\mathrm{hist}}\big(d\mathring{\omega}^{(0:n_1)}\big) = Z_{\mathrm{hist}}^{-1} \exp\Big[-\sum_{n=0}^{n_1} \mathscr{S}\big(\mathring{\omega}^{(n)}\big) - \mathbf{m} \sum_{n=0}^{n_1-1} \mathfrak{U}\big(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}\big)\Big] \prod_{n=0}^{n_1} \nu^{\mathscr{T}}\big(d\mathring{\omega}^{(n)}\big), \tag{9.51}$$

with Z_{hist} the integral of the exponential against $\bigotimes_n \nu^{\mathscr{T}}$. The anchor, the prior covariances, the loop coefficients, and the mass are coordinates of the generative parameter θ.

Proposition 9.32 (The generator history prior). In definition 9.31, with either kinetic term:

(a) $0 < Z_{\mathrm{hist}} \le Z_{\mathscr{S}}^{n_1+1} \le 1$, so $\mathbb{P}^{\mathrm{hist}}$ is a probability law, absolutely continuous with respect to $\bigotimes_n \nu^{\mathscr{T}}$ with density at most Z_{hist}^{-1}; every function integrable under $\bigotimes_n \nu^{\mathscr{T}}$, in particular every function bounded by $\mathrm{c}_1 e^{\mathrm{c}_2 \|\mathring{\omega}^{(0:n_1)}\|}$, is integrable under it.

(b) With the backward functions $\mathring{Z}_{n_1}^{\leftarrow} \equiv 1$ and

$$\mathring{Z}_n^{\leftarrow}(\mathring{\omega}) = \int \exp\big[-\mathscr{S}(\mathring{\omega}') - \mathbf{m}\mathfrak{U}(\mathring{\omega}', \mathring{\omega})\big] \mathring{Z}_{n+1}^{\leftarrow}(\mathring{\omega}') \nu^{\mathscr{T}}(d\mathring{\omega}'), \qquad n < n_1, \tag{9.52}$$

which are measurable with values in $(0,1]$, $\mathbb{P}^{\mathrm{hist}}$ is the law of the Markov chain with root and transition kernels

$$\begin{aligned}\mathbb{P}^{(0)}_{\mathrm{hist}}(d\mathring{\omega}) &= Z^{-1}_{\mathrm{hist}} e^{-\mathscr{S}(\mathring{\omega})} \mathring{Z}^{\leftarrow}_0(\mathring{\omega}) \nu^{\mathscr{T}}(d\mathring{\omega}),\\ \mathbb{P}^{(n+1\leftarrow n)}_{\mathrm{hist}}(d\mathring{\omega}' \mid \mathring{\omega}) &= \frac{e^{-\mathscr{S}(\mathring{\omega}') - \mathbf{m}\mathfrak{U}(\mathring{\omega}',\mathring{\omega})} \mathring{Z}^{\leftarrow}_{n+1}(\mathring{\omega}')}{\mathring{Z}^{\leftarrow}_n(\mathring{\omega})} \nu^{\mathscr{T}}(d\mathring{\omega}').\end{aligned} \tag{9.53}$$

These kernels are the conditional laws of the Gibbs law (9.51) and depend on n and on the horizon n_1 through $\mathring{Z}^{\leftarrow}_{n+1}$. The chain whose kernels normalize each increment factor separately, started from the law proportional to $e^{-\mathscr{S}}\nu^{\mathscr{T}}$, has the law (9.51) exactly when $\mathring{Z}^{\leftarrow}_{n_1-1}$ is constant $\nu^{\mathscr{T}}$-almost everywhere, and this fails for one scalar generator with $\nu^{\mathscr{T}} = \mathcal{N}(0,1)$, $\mathscr{S} = 0$, the Gaussian increment, and $n_1 = 1$. With tower states $\mathbf{Y}^{(n)} = (\mathring{\omega}^{(n)}, y^{(n)})$ at depth zero, parent readout $\chi^{\mathrm{ep}}(\mathbf{Y}^{(n)}) = \mathring{\omega}^{(n)}$, the agents' latent law of each episode given by a Markov kernel $\mathbb{P}^{\mathrm{ag}}_n(dy^{(n)} \mid \mathring{\omega}^{(n)}, X)$, and records of episode n that read $\mathbf{Y}^{(n)}$ only, the joint is a latent route of definition 13.26, with root $\mathbb{P}^{(0)}_{\mathrm{hist}}(d\mathring{\omega}^{(0)})\mathbb{P}^{\mathrm{ag}}_0(dy^{(0)} \mid \mathring{\omega}^{(0)}, X)$ and transitions $\mathbb{P}^{(n\leftarrow n-1)}_{\mathrm{hist}}(d\mathring{\omega}^{(n)} \mid \mathring{\omega}^{(n-1)})\mathbb{P}^{\mathrm{ag}}_n(dy^{(n)} \mid \mathring{\omega}^{(n)}, X)$, each a composition of kernels. Hence proposition 13.27 applies: the joint has unit mass and is acyclic with forward cross-episode edges, the root is required, the predictive prior of the generators at episode $n+1$ is the filtered law of $\mathring{\omega}^{(n)}$ pushed through $\mathbb{P}^{(n+1\leftarrow n)}_{\mathrm{hist}}$, and, on the finite-density domain with observation densities, the free energy decomposes along n as in (13.43).

(c) The time-like links carry no variable. The twist of the identity link of agent i between two episodes is e_G, so $\mathbb{P}^{\mathrm{hist}}$ is a law on the spatial generator histories alone, and each kinetic term is a function of the generators of one edge at two consecutive episodes. In lattice gauge theory this configuration is the temporal gauge, in which a time-like plaquette couples two space-like links at consecutive times [25, Ch. 15, Eq. (15.9)].

[ESTABLISHED]

The proof is in appendix B.3.

Proposition 9.33 (Inertia of the most probable history). Let $\mathfrak{V}_n : \mathscr{T} \to \mathbb{R}$, $n = 0, \dots, n_1$, be differentiable, let $\mathbf{m} > 0$ and $\mathfrak{U}$ be either kinetic term of definition 9.31, and let

$$\mathscr{S}^{\mathrm{hist}}\big(\mathring{\omega}^{(0:n_1)}\big) = \sum_{n=0}^{n_1} \mathfrak{V}_n\big(\mathring{\omega}^{(n)}\big) + \mathbf{m} \sum_{n=0}^{n_1-1} \mathfrak{U}\big(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}\big). \tag{9.54}$$

For the history prior, $\mathfrak{V}_n = \mathscr{S} + \frac{1}{2}\langle \cdot - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\cdot - \mathring{\alpha})\rangle$ makes $\mathscr{S}^{\mathrm{hist}}$ the negative log density of (9.51) relative to Lebesgue measure up to a constant; adding to $\mathfrak{V}_n$ a matter term of episode n, such as the negative log evidence of its records given the generators or their expected energy under a matter recognition law, gives the corresponding posterior or conditional objective. A most probable history is a minimizer of $\mathscr{S}^{\mathrm{hist}}$, with free ends or with ends held at given values.

(a) For the Gaussian increment, every critical point satisfies

$$\mathbf{m}\big(2\mathring{\omega}^{(n)} - \mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n-1)}\big) + \nabla\mathfrak{V}_n\big(\mathring{\omega}^{(n)}\big) = 0, \qquad 0 < n < n_1, \tag{9.55}$$

and, with free ends, $\mathbf{m}(\mathring{\omega}^{(0)} - \mathring{\omega}^{(1)}) + \nabla\mathfrak{V}_0(\mathring{\omega}^{(0)}) = 0$ and $\mathbf{m}(\mathring{\omega}^{(n_1)} - \mathring{\omega}^{(n_1-1)}) + \nabla\mathfrak{V}_{n_1}(\mathring{\omega}^{(n_1)}) = 0$. With unit step in n, (9.55) is the central-difference form of $\mathbf{m}\ddot{\mathring{\omega}} = \nabla\mathfrak{V}$, Newton's equation with mass $\mathbf{m}$ in the potential $-\mathfrak{V}$, the Euclidean sign. If each $\mathfrak{V}_n$ is continuous and $\sum_n \mathfrak{V}_n(\mathring{\omega}^{(n)}) \to \infty$ as $\|\mathring{\omega}^{(0:n_1)}\| \to \infty$, as when each $\mathfrak{V}_n$ is at least the anchor quadratic minus a constant, a most probable history exists.

(b) For the time-like loop defect, write $\mathring{\Theta}_e^{\leftarrow} = \exp(-\mathring{\omega}'_e)\exp(\mathring{\omega}_e)$ for a later family $\mathring{\omega}'$ and an earlier family $\mathring{\omega}$, so that $\rho(\mathring{\Theta}_e^{\leftarrow})$ is (9.50) with $\mathring{\omega}'$ and $\mathring{\omega}$ in place of $\mathring{\omega}^{(n+1)}$ and $\mathring{\omega}^{(n)}$, and let $\mathbf{L}$ be the Fréchet derivative (9.25). The gradients of $\mathfrak{U}(\mathring{\omega}', \mathring{\omega})$ in its earlier and later arguments are

$$\begin{aligned}\big(\partial_{\mathring{\omega}}\mathfrak{U}\big)_e &= \operatorname{pr}_{\mathfrak{g}}\mathbf{L}\big((\mathrm{d}\rho\mathring{\omega}_e)^*, (e^{-\mathrm{d}\rho\mathring{\omega}'_e})^*\big(\rho(\mathring{\Theta}_e^{\leftarrow}) - I\big)\big),\\ \big(\partial_{\mathring{\omega}'}\mathfrak{U}\big)_e &= -\operatorname{pr}_{\mathfrak{g}}\mathbf{L}\big(-(\mathrm{d}\rho\mathring{\omega}'_e)^*, \big(\rho(\mathring{\Theta}_e^{\leftarrow}) - I\big)(e^{\mathrm{d}\rho\mathring{\omega}_e})^*\big),\end{aligned} \tag{9.56}$$

and every critical point satisfies, for $0 < n < n_1$,

$$\mathbf{m}\big[\partial_{\mathring{\omega}}\mathfrak{U}\big(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}\big) + \partial_{\mathring{\omega}'}\mathfrak{U}\big(\mathring{\omega}^{(n)}, \mathring{\omega}^{(n-1)}\big)\big] + \nabla\mathfrak{V}_n\big(\mathring{\omega}^{(n)}\big) = 0. \tag{9.57}$$

For $\mathrm{r} > 0$ let $\mathrm{c}_\mathrm{r} = \sqrt{2}\sup\|D^3\mathfrak{U}(\mathring{\omega}', \mathring{\omega})\|$ over $\|\mathring{\omega}\|, \|\mathring{\omega}'\| \le \mathrm{r}$. On that set $\|\partial_{\mathring{\omega}}\mathfrak{U} - (\mathring{\omega} - \mathring{\omega}')\|$ and $\|\partial_{\mathring{\omega}'}\mathfrak{U} - (\mathring{\omega}' - \mathring{\omega})\|$ are at most $\mathrm{c}_\mathrm{r}\mathrm{r}\|\mathring{\omega}' - \mathring{\omega}\|$, so at a history in the ball of radius r the left side of (9.57) differs from that of (9.55) by at most $\mathbf{m}\mathrm{c}_\mathrm{r}\mathrm{r}(\|\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)}\| + \|\mathring{\omega}^{(n)} - \mathring{\omega}^{(n-1)}\|)$. At a given $\mathring{\omega}$ and small increments $\delta\mathring{\omega}$, $\mathfrak{U}(\mathring{\omega} + \delta\mathring{\omega}, \mathring{\omega}) = \frac{1}{2}\sum_e \|\mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)e^{\mathrm{d}\rho\mathring{\omega}_e}\|_{\mathsf{G}}^2 + O(\|\delta\mathring{\omega}\|^3)$, so the time-like defect gives the increment the mass tensor $\delta\mathring{\omega} \mapsto \mathbf{m}\sum_e \|\mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)e^{\mathrm{d}\rho\mathring{\omega}_e}\|_{\mathsf{G}}^2$, which depends on the generators and reduces to $\mathbf{m}\|\delta\mathring{\omega}\|^2$ at $\mathring{\omega} = 0$.

(c) Let $\mathscr{T}$ be one-dimensional, as for one reversal pair of edge copies of the translation instance, identified with $\mathbb{R}$ isometrically, take the Gaussian increment, hold the ends at $\mathring{\omega}^{(0)} = \mathrm{a} < \mathrm{b} = \mathring{\omega}^{(n_1)}$, and let $\mathfrak{V}_n = \mathfrak{V}$ for $0 < n < n_1$ be continuous with $\mathfrak{V} \ge \mathfrak{V}(\mathrm{a})$ on $(-\infty, \mathrm{a}]$ and $\mathfrak{V} \ge \mathfrak{V}(\mathrm{b})$ on $[\mathrm{b}, \infty)$, as for a double well with minima at a and b. Then a most probable history exists, and every most probable history is nondecreasing in n; if moreover $\mathfrak{V}$ is differentiable with $\mathfrak{V}'(\mathrm{a}) = \mathfrak{V}'(\mathrm{b}) = 0$, every most probable history is strictly increasing. The most probable path between the wells is a monotone kink and does not oscillate.

[ESTABLISHED]

The proof is in appendix B.3.

The inertia of proposition 9.33 belongs to the smoothed history, a joint optimization over n, while the recognition flow of proposition 9.28 is of first order in its parameter. Recognition dynamics with velocity variables arise when the generative model itself declares the motion of its states, as in the generalized coordinates of motion of Friston et al. [38, p. 852, Eq. (13)], whose ensemble dynamics couple each state to its motion; the history prior declares no such variables. The Gaussian

increment term of (9.54) is, up to a constant, the negative log transition density of a random walk with increment covariance $\mathbf{m}^{-1}I$, the discrete-time form of the path weight of a driftless diffusion.

9.10 The linearized link equation

Near zero generators the stationarity condition of the generator history becomes a linear equation whose operator collects every term of the construction: the anchor, the loops, the episodes, and the matter. Its positivity makes the linearized history unique, and the neglected terms are of second order in the sources (proposition 9.34).

Proposition 9.34 (The linearized link equation). In the latent-link model of definition 9.15, let the generators carry the history prior of definition 9.31 with zero anchor, prior covariance $\mathring{\Sigma}$, action $\mathscr{S} = \sum_p \mathrm{c}_p \mathfrak{Q}_p$, mass $\mathbf{m}$, and either kinetic term, and write samples in the reference gauge of corollary 9.22. At each episode n let $\mathfrak{A}_n$ be a finite set of binary success records (hypothesis 5.5), observed as successes; record a has a scope $\{i, j\}$ joined by an edge copy $e : j \to i$ of E_+ or of its reversal, with $\mathring{\omega}_{\bar{e}} = -\mathring{\omega}_e$, and the energy of corollary 9.22(a),

$$E_{a,o}(k; \mathring{\omega}) = \tfrac{1}{2}\big(k_i - e^{\mathrm{d}\rho\mathring{\omega}_e} k_j\big)^\top W_a \big(k_i - e^{\mathrm{d}\rho\mathring{\omega}_e} k_j\big), \qquad W_a \succ 0, \tag{9.58}$$

where i, j, and e are those of a. Record a enters the free energy of its episode with a weight $\beta_a \in [0, 1]$ that does not depend on the generators: $\beta_a = 1$ for a record that is always read, and a weight below one arises when a record enters through a recognized source label, as in corollary 9.22(b). Hold a product Gaussian matter recognition under which k_i at episode n has mean $\mu_i^{(n)}$ and covariance $\Sigma_i^{(n)}$, and let $\mathscr{S}^{\mathrm{hist}}$ be the history action (9.54) with $\mathfrak{V}_n = \frac{1}{2}\langle \cdot, \mathring{\Sigma}^{-1} \cdot \rangle + \mathscr{S} + \sum_{a \in \mathfrak{A}_n} \beta_a \mathbb{E}[E_{a,o}(k; \cdot)]$, that is,

$$\begin{aligned} \mathscr{S}^{\mathrm{hist}}(\mathring{\omega}) = \sum_{n=0}^{n_1} \Big[\tfrac{1}{2} \big\langle \mathring{\omega}^{(n)}, \mathring{\Sigma}^{-1} \mathring{\omega}^{(n)} \big\rangle + \sum_p \mathrm{c}_p \mathfrak{Q}_p\big(\mathring{\omega}^{(n)}\big) + \sum_{a \in \mathfrak{A}_n} \beta_a \mathbb{E}\big[E_{a,o}\big(k; \mathring{\omega}^{(n)}\big)\big] \Big] \\ + \mathbf{m} \sum_{n=0}^{n_1 - 1} \mathfrak{U}\big(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)}\big), \end{aligned} \tag{9.59}$$

which, when the agents' latent law does not read the links, is the generator part of the expected negative log joint density. Its minimizers are the most probable generator histories given the matter recognition, in the sense of proposition 9.33. For one episode with every $\beta_a = 1$, and an agent latent law that does not read the links, $\mathscr{S}^{\mathrm{hist}}$ is the function $\mathbf{V}$ of proposition 9.21(d) up to a constant, so $\nabla \mathscr{S}^{\mathrm{hist}} = 0$ is the history form of the most-probable-link condition $\nabla \mathbf{V} = 0$. On $\mathscr{T}^{n_1+1}$, with

$\langle \mathring{\omega}, \mathring{\omega}' \rangle = \sum_n \langle \mathring{\omega}^{(n)}, \mathring{\omega}'^{(n)} \rangle$, define self-adjoint operators $\mathbf{K}$ and $\mathbf{N}_0$ and a source $\mathbf{j}$ by

$$\begin{aligned}\langle \mathring{\omega}, \mathbf{K}\mathring{\omega}\rangle &= \sum_n \Big[\langle \mathring{\omega}^{(n)}, \mathring{\Sigma}^{-1}\mathring{\omega}^{(n)}\rangle + \sum_p \mathrm{c}_p \big\| \mathbf{C}_p \mathring{\omega}^{(n)} \big\|_{\mathsf{G}}^2 + \sum_{a\in\mathfrak{A}_n} \beta_a \operatorname{Tr}\big(W_a \mathrm{d}\rho \mathring{\omega}_e^{(n)} \mathbb{E}\big[k_j k_j^\top\big] (\mathrm{d}\rho \mathring{\omega}_e^{(n)})^\top \big) \Big] \\ &\quad + \mathbf{m} \sum_{n<n_1} \big\| \mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)} \big\|^2, \\ \langle \mathring{\omega}, \mathbf{N}_0 \mathring{\omega}\rangle &= -\sum_n \sum_{a\in\mathfrak{A}_n} \beta_a \operatorname{Tr}\big(\mathbf{J}_a^\top (\mathrm{d}\rho \mathring{\omega}_e^{(n)})^2 \big), \qquad \langle \mathbf{j}, \mathring{\omega}\rangle = \sum_n \sum_{a\in\mathfrak{A}_n} \beta_a \operatorname{Tr}\big(\mathbf{J}_a^\top \mathrm{d}\rho \mathring{\omega}_e^{(n)} \big),\end{aligned} \tag{9.60}$$

where e and j are those of a, $\mathbb{E}[k_j k_j^\top] = \Sigma_j^{(n)} + \mu_j^{(n)} \mu_j^{(n)\top}$, $\mathbf{C}_p \mathring{\omega} = (\mathbf{C}\mathring{\omega})_p$ is the loop sum of the cycle map (9.8), and, for $a \in \mathfrak{A}_n$,

$$\mathbf{J}_a = W_a \big[(\mu_i^{(n)} - \mu_j^{(n)}) \mu_j^{(n)\top} - \Sigma_j^{(n)} \big] \tag{9.61}$$

is the current (9.34) of record a at zero generators for a product law. Thus $\mathbf{K} = \mathring{\Sigma}^{-1} + \sum_p \mathrm{c}_p \mathbf{C}_p^\top \mathbf{C}_p + \mathbf{m}(-\Delta_n) + \mathring{\Phi}$, where $\mathring{\Sigma}^{-1}$ and the loop term act episode by episode, $(-\Delta_n \mathring{\omega})^{(n)} = 2\mathring{\omega}^{(n)} - \mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n-1)}$ in the interior, $\mathring{\omega}^{(0)} - \mathring{\omega}^{(1)}$ and $\mathring{\omega}^{(n_1)} - \mathring{\omega}^{(n_1-1)}$ at the ends, and $\mathring{\Phi}$ is the matter operator of the last term of the bracket. Let $\mathrm{c}_{\mathring{\Sigma}}$ be the smallest eigenvalue of $\mathring{\Sigma}^{-1}$, and let $\mathrm{cond}(\mathsf{G}^{\mathrm{fib}})$ be the ratio of the extreme eigenvalues of $\mathsf{G}^{\mathrm{fib}}$.

(a) $\nabla \mathscr{S}^{\mathrm{hist}}(0) = -\mathbf{j}$ and $\nabla^2 \mathscr{S}^{\mathrm{hist}}(0) = \mathbf{K} + \mathbf{N}_0$, so to first order in the generators and the currents the stationarity condition is

$$\mathbf{K}\mathring{\omega} = \mathbf{j}. \tag{9.62}$$

(b) $\mathbf{K} \succeq \mathrm{c}_{\mathring{\Sigma}} I$. Hence (9.62) has the unique solution $\mathring{\omega}_{\mathrm{lin}} = \mathbf{K}^{-1}\mathbf{j}$, with $\|\mathring{\omega}_{\mathrm{lin}}\| \le \|\mathbf{j}\|/\mathrm{c}_{\mathring{\Sigma}}$, and $\|\mathbf{K}^{-1}\mathbf{j} - \mathbf{K}^{-1}\mathbf{j}'\| \le \|\mathbf{j} - \mathbf{j}'\|/\mathrm{c}_{\mathring{\Sigma}}$.

(c) Let $\mathrm{c}_0 = \|\mathbf{N}_0\|$ and, for $\mathrm{r} > 0$, $\mathrm{c}_{\mathrm{r}} = \sup_{\|\mathring{\omega}\| \le \mathrm{r}} \|D^3 \mathscr{S}^{\mathrm{hist}}(\mathring{\omega})\|$. Then $\mathrm{c}_0 \le \mathrm{cond}(\mathsf{G}^{\mathrm{fib}}) \max_{n, e\in E_+} \mathrm{j}_e^{(n)}$ and $\|\mathbf{j}\| \le \mathrm{cond}(\mathsf{G}^{\mathrm{fib}})^{1/2} (\sum_{n,e\in E_+} (\mathrm{j}_e^{(n)})^2)^{1/2}$, where $\mathrm{j}_e^{(n)}$ is the sum of $\beta_a \|\mathbf{J}_a\|_{\mathrm{F}}$ over the records $a \in \mathfrak{A}_n$ on e or $\bar{e}$. If $\mathrm{c}_0 + \mathrm{c}_{\mathrm{r}}\mathrm{r} \le \mathrm{c}_{\mathring{\Sigma}}/2$ and $\|\mathbf{j}\| \le \mathrm{c}_{\mathring{\Sigma}}\mathrm{r}/4$, then $\mathscr{S}^{\mathrm{hist}}$ has exactly one critical point $\mathring{\omega}_\star$ in the ball of radius r about 0; it is the unique minimizer of $\mathscr{S}^{\mathrm{hist}}$ on that ball, $\|\mathring{\omega}_\star\| \le 2\|\mathbf{j}\|/\mathrm{c}_{\mathring{\Sigma}}$, and

$$\|\mathring{\omega}_\star - \mathring{\omega}_{\mathrm{lin}}\| \le \frac{2\|\mathbf{j}\|}{\mathrm{c}_{\mathring{\Sigma}}^2} \Big(\mathrm{c}_0 + \frac{\mathrm{c}_{\mathrm{r}} \|\mathbf{j}\|}{\mathrm{c}_{\mathring{\Sigma}}} \Big), \tag{9.63}$$

which, at fixed c_{r}, is of second order in the currents.

[ESTABLISHED]

The proof is in appendix B.3.

In the language of lattice gauge theory, proposition 9.34 reads as follows, and the reading adds nothing to it. The spatial term $\sum_p \mathrm{c}_p \mathbf{C}_p^\top \mathbf{C}_p$ is a discrete curl-curl operator: $\mathbf{C}_p$ sums the generators around the loop p, so the term penalizes the circulation of the generator field on the cycle space of the multigraph, the magnetic part of a lattice action. The episode term $\mathbf{m}(-\Delta_n)$ has the quadratic form $\mathbf{m} \sum_n \|\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)}\|^2$ of a forward difference in the episode index; by proposition 9.32(c) and

proposition 9.33(b) it is the second-order part of the time-like loop defect with identity time-like links, the temporal gauge, in which the time-like plaquettes give the electric term of the Hamiltonian of Kogut and Susskind [68], as derived in Creutz [25, Ch. 15, Eq. (15.41)]. The anchor precision $\mathring{\Sigma}^{-1}$ is a bare mass and the matter operator $\mathring{\Phi}$ a state-dependent screening mass, so at linear order $\mathbf{K}$ has the form of a massive Euclidean vector-field operator of Proca type, a mass term plus curl-curl plus a squared episode difference, acting on the spatial components alone, since the history carries no time-like variable. Its curl-curl and episode terms act as scalar graph operators on each component of $\mathfrak{g}$ in an orthonormal basis, and the bracket of $\mathfrak{g}$ enters only through the terms of order three and higher, which proposition 9.34(c) bounds. The episode index is an inference index in the sense of section 1.4, and the signature is Euclidean because $\mathbf{K}$ is positive (proposition 9.34(b)).

The loop term sees only part of the link space, and the declared family $\mathbb{L}$ fixes which part. The next statement makes this exact for the quadratic loop action of proposition 9.29(c) and for the loop term of $\mathbf{K}$.

Corollary 9.35 (Cycle-space splitting of the loop sector). Let $\mathfrak{I} = (V, E, {}^{-})$ be the interaction multigraph of definition 2.11, with κ connected components, write $\mathscr{T} = \mathbb{R}^{E_+} \otimes \mathfrak{g}$, let $\mathbb{L}_+ = \{p \in \mathbb{L} : \mathrm{c}_p > 0\}$, and let $\mathbf{C}_+ : \mathscr{T} \to \mathfrak{g}^{\mathbb{L}_+}$ be the cycle map (9.8) of $\mathbb{L}_+$. Let $d : \mathbb{R}^V \to \mathbb{R}^{E_+}$ be $(da)_e = a_i - a_j$ for $e : j \to i$, let $Z = (\operatorname{im} d)^{\perp}$ be the cycle space, and for $p \in \mathbb{L}_+$ let $z_p \in \mathbb{R}^{E_+}$ count the traversals of each $e \in E_+$ along its orientation minus those against it, so that $(\mathbf{C}\mathring{\omega})_p = \sum_{e \in E_+} z_{p,e} \mathring{\omega}_e$. Put $r_{\mathbb{L}} = \dim \operatorname{span}\{z_p : p \in \mathbb{L}_+\} \le |\mathbb{L}_+|$, and call

$$\mathscr{T}_{\mathrm{ex}} = \operatorname{im} d \otimes \mathfrak{g}, \qquad \mathscr{T}_{\mathrm{har}} = \big(Z \cap \{z_p : p \in \mathbb{L}_+\}^{\perp}\big) \otimes \mathfrak{g} \tag{9.64}$$

the exact and the harmonic generators; $\mathring{\omega}$ is exact when $\mathring{\omega}_e = a_i - a_j$ for some $a \in \mathfrak{g}^V$.

(a) $\ker \mathbf{C}_+ = \mathscr{T}_{\mathrm{ex}} \oplus \mathscr{T}_{\mathrm{har}}$, a sum orthogonal for the inner product of $\mathscr{T}$, with

$$\dim \mathscr{T}_{\mathrm{ex}} = \dim \mathfrak{g}\,(|V| - \kappa), \qquad \dim \mathscr{T}_{\mathrm{har}} = \dim \mathfrak{g}\,(|E_+| - |V| + \kappa - r_{\mathbb{L}}) \ge \dim \mathfrak{g}\,(|E_+| - |V| + \kappa - |\mathbb{L}_+|). \tag{9.65}$$

In the translation instance (proposition 9.13) an exact generator shift by $a_i - a_j$ is the link change $\Theta_e \mapsto \Theta_e + a_i - a_j$ with the frames held, and every loop defect $\mathfrak{Q}_p$ with $p \in \mathbb{L}_+$ is constant along $\ker \mathbf{C}_+$.

(b) Under the quadratic action of proposition 9.29(c), with $\mathring{\Sigma} = \bigoplus_{e \in E_+} \mathring{\Sigma}_e$,

$$\mathring{\alpha}^{\mathrm{new}} - \mathbb{E}_R[\mathring{\omega}] \in \mathring{\Sigma} \operatorname{im} \mathbf{C}_+^{\top}, \qquad \mathscr{T} = \mathscr{T}_{\mathrm{ex}} \oplus \mathscr{T}_{\mathrm{har}} \oplus \mathring{\Sigma} \operatorname{im} \mathbf{C}_+^{\top}, \tag{9.66}$$

and the last summand is orthogonal to the first two for $\langle \cdot, \mathring{\Sigma}^{-1} \cdot \rangle$. Hence $\mathring{\alpha}^{\mathrm{new}}$ and $\mathbb{E}_R[\mathring{\omega}]$ have the same components in $\mathscr{T}_{\mathrm{ex}}$ and in $\mathscr{T}_{\mathrm{har}}$, and the anchor update corrects only the component in $\mathring{\Sigma} \operatorname{im} \mathbf{C}_+^{\top}$.

(c) For the operator $\mathbf{K}$ of proposition 9.34, if every $\mathring{\omega}^{(n)}$ lies in $\ker \mathbf{C}_+$, the loop term vanishes and $\langle \mathring{\omega}, \mathbf{K}\mathring{\omega} \rangle = \sum_n \langle \mathring{\omega}^{(n)}, \mathring{\Sigma}^{-1} \mathring{\omega}^{(n)} \rangle + \mathbf{m} \sum_{n < n_1} \|\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)}\|^2 + \langle \mathring{\omega}, \mathring{\Phi}\mathring{\omega} \rangle$.

[ESTABLISHED]

Proof. (a) The kernel of d is the functions constant on components, so $\dim \operatorname{im} d = |V| - \kappa$ and $\dim Z = |E_+| - |V| + \kappa$. Along a closed walk every arrival at a vertex is followed by a departure, so the net traversals at each vertex cancel, $d^\top z_p = 0$, and $z_p \in Z$. For the matrix Z_+ with rows z_p, $p \in \mathbb{L}_+$, this gives $\ker \mathsf{Z}_+ = (\operatorname{span}\{z_p\})^\perp = \operatorname{im} d \oplus \big(Z \cap \{z_p\}^\perp\big)$, orthogonally, and the second summand has dimension $\dim Z - r_{\mathbb{L}}$ because $\operatorname{span}\{z_p\} \subset Z$. By (9.8) and the reversal convention (9.2), $\mathbf{C}_+ = \mathsf{Z}_+ \otimes \mathrm{id}_{\mathfrak{g}}$, and the inner product of $\mathscr{T}$ is the Euclidean one on $\mathbb{R}^{E_+}$ times $\langle \cdot, \cdot \rangle_{\mathsf{G}}$, so tensoring with $\mathfrak{g}$ carries the decomposition, its orthogonality, and the dimensions over. In the translation instance $\mathring{\omega}_e = \Theta_e - U_i + U_j$ by proposition 9.13(a), so with the frames held an exact shift of $\mathring{\omega}$ is the stated shift of Θ, and by (9.15) $\mathfrak{Q}_p = \frac{1}{2}\|(\mathbf{C}\mathring{\omega})_p\|^2_{\mathsf{G}}$ depends on $\mathring{\omega}$ only through $\mathbf{C}_+\mathring{\omega}$ when $p \in \mathbb{L}_+$. (b) Terms with $\mathrm{c}_p = 0$ vanish in (9.49), so $\mathring{\alpha}^{\mathrm{new}} - \mathbb{E}_R[\mathring{\omega}] = \mathring{\Sigma} \sum_{p \in \mathbb{L}_+} \mathrm{c}_p \mathbf{C}_p^\top \mathbf{C}_p \mathbb{E}_R[\mathring{\omega}] \in \mathring{\Sigma} \operatorname{im} \mathbf{C}_+^\top$. Each $\mathring{\Sigma}_e$ is self-adjoint and positive, so for $h \in \ker \mathbf{C}_+$ and $v \in \operatorname{im} \mathbf{C}_+^\top$, $\langle \mathring{\Sigma} v, \mathring{\Sigma}^{-1} h \rangle = \langle v, h \rangle = 0$. The subspace $\mathring{\Sigma} \operatorname{im} \mathbf{C}_+^\top$ therefore meets $\ker \mathbf{C}_+$ only in zero, and its dimension $\operatorname{rank} \mathbf{C}_+ = \dim \mathscr{T} - \dim \ker \mathbf{C}_+$ makes the sum all of $\mathscr{T}$; with (a) this gives (9.66), and the components of the difference in the first two summands vanish. (c) $\mathbf{C}_p \mathring{\omega}^{(n)} = 0$ for $p \in \mathbb{L}_+$, and the other loop terms carry $\mathrm{c}_p = 0$, so only the remaining terms of (9.60) survive. □

Attaching a two-cell along each walk of $\mathbb{L}_+$ makes the declared loop family a two-complex on the agents, and corollary 9.35 is the Hodge decomposition of its one-cochains with values in $\mathfrak{g}$ [47, Theorem 3.1], applied to the constant sheaf: exact generators are its gradient part, $\mathring{\Sigma} \operatorname{im} \mathbf{C}_+^\top$ its circulating part in the metric of the prior, and $\mathscr{T}_{\mathrm{har}}$ the circulation around cycles of $\mathfrak{I}$ that bound no declared walk. By (9.65), a family of short loops on a multigraph whose cycle space is mostly spanned by longer cycles leaves most of that space to the anchor, the episodes, and the matter, so the choice of $\mathbb{L}$ is a structural declaration on the agent set of the same kind as the support of the source prior. Away from zero generators the loop defects of a nonabelian group are not functions of $\mathbf{C}\mathring{\omega}$, and the splitting describes only their quadratic part (proposition 9.10(e)).

Physicist's summary

The links between agents are treated here as a field that is inferred, not as a background that is held. Each link is the exponential of a Lie-algebra generator measured in the frame of a reference point; per-agent frame changes do not touch the generators, and a change of reference point rotates all of them by one adjoint map. A Gaussian law on the generators, reweighted by a Wilson-type loop penalty, is a normalized prior for any structure group, so no finite Haar measure is needed; on the general linear group the loop penalty is measured with a fiber metric, because conjugation-invariant penalties cannot see unipotent holonomy. The links enter the free energy through one relative entropy from this prior. Stationarity is a field equation: the restoring force of the prior balances a matter current, which for a comparison record is a precision-weighted prediction error times the source state, the form of a delta rule, transported by the derivative of the exponential map. Shrinking the prior to its anchor recovers the free energy and the evidence of the held links at the anchor. Integrating out matter that is normalized jointly with the links generates an induced link action, as in

induced lattice gauge theory, while matter attached by record extension generates none. Inference of the generators is first-order gradient flow in the Fisher geometry of their Gaussian law, with the Bayesian learning rule as its Euler step; inertia appears only in the most probable history across learning episodes, where the precision of the increment acts as a mass. Linearized about zero generators, the stationarity operator is a bare mass from the anchor, a curl-curl term from the loops, a squared episode difference, and a screening mass from the matter, which is the form of a massive Euclidean vector operator in the episode index. On the cycle space the loop term acts only on the circulating part of the generators: exact generators and circulation around cycles that bound no declared loop are fixed by the anchor, the episodes, and the matter alone, and the anchor update corrects only the circulating part.

Part II

Exact Coarse-Graining at One Base Point

Chapter 10

Coarse Channels

Part II asks what a coarse description of the agent network at c_* keeps of the joint of Chapter 4, of its posterior, and of a recognition law. For one coarse channel the answer is Theorem 10.7: a channel that pushes all three laws preserves the evidence and the posterior version, and splits the free energy into the coarse free energy plus the expected relative entropy of the discarded conditional laws. Corollary 10.11 specializes it to a candidate parent of a block of agents, and the exact results of Chapters 11 to 14 rest on both. The chapter also separates Markov pushforward, precomposition, and Galerkin restriction (Section 10.4) and closes with finite labeled families, in which a retained label carries information of order N^2 at a tie among N agents (Section 10.8). Fisher contraction under a channel is Theorem 7.8 and is not restated. The longer proofs of the chapter are deferred to appendix B.

Scope. The common-channel identity concerns one channel applied to all three laws with the observation unchanged. It does not compare different generative and recognition channels, a coarse joint fitted rather than pushed forward, or a channel that also coarsens the observation, even an invertible or evidence-sufficient one, because each changes the comparison target or the evidence event; nor is a Fisher tensor of a labeled family used as the metric of a dynamics. [NOT-CLAIMED] A natural selector of partitions or membership laws is Open Problem 10.20, and equivariant Bayes versions beyond the finite and compact cases are Open Problem 10.19.

10.1 Markov coarse maps

Definition 10.1 (Coarse channel). A coarse channel between standard Borel spaces X and Y is a Markov kernel $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ in the sense of Section 1.4. It sends a law P on X to $(PK)(B) = \int_{\mathsf{X}} K(x, B)P(dx)$ and a statistical experiment $(\mathsf{X}, \{P_\theta\}_{\theta\in\Theta})$ to $(\mathsf{Y}, \{P_\theta K\}_{\theta\in\Theta})$. A measurable statistic $c : \mathsf{X} \to \mathsf{Y}$ is the deterministic kernel $K_c(x, B) = \mathbf{1}_B(c(x))$. A family K_θ that varies with the parameter is a parameter-dependent model transformation, not one channel between experiments.

Proposition 10.2 (Markov kernels form a category). If $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ and $L : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ are Markov kernels, then $(KL)(x, D) = \int_{\mathsf{Y}} L(y, D)K(x, dy)$ is a Markov kernel, composition is associative with the deterministic identity as unit, and

$$P(KL) = (PK)L \tag{10.1}$$

for every law P on X. [ESTABLISHED]

Proof. Kernel integration gives measurability in x, and monotone convergence gives countable additivity in D. By Tonelli's theorem $[P(KL)](D) = \int_{\mathsf{X}} \int_{\mathsf{Y}} L(y,D)K(x,dy)P(dx)$, which equals $\int_{\mathsf{Y}} L(y,D)(PK)(dy)$; this is (10.1), and applied to point masses and a third kernel it proves associativity. These are the morphisms and the composition of the Kleisli category of the Giry probability monad [42]. □

10.2 KL contraction, equality, and recovery

Every Markov kernel contracts relative entropy in $[0,+\infty]$ (Lemma 3.12); the equality case needs finiteness.

Theorem 10.3 (Data processing with its exact equality condition). Let P, Q be laws on X with $P \ll Q$ and $D_{\mathrm{KL}}(P\|Q) < \infty$, let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a Markov kernel, and put $\mathbb{J}(dx,dy) = Q(dx)K(x,dy)$, $r = dP/dQ$, and $\bar{r}(y) = \mathbb{E}_{\mathbb{J}}[r(X) \mid Y = y]$. Then $d(PK)/d(QK) = \bar{r}$ and $D_{\mathrm{KL}}(PK\|QK) \le D_{\mathrm{KL}}(P\|Q)$, with equality exactly when $r(X) = \bar{r}(Y)$ $\mathbb{J}$-almost surely. [ESTABLISHED]

Proof. For bounded measurable f on Y, $\int f d(PK) = \mathbb{E}_{\mathbb{J}}[f(Y)r(X)] = \mathbb{E}_{\mathbb{J}}[f(Y)\bar{r}(Y)]$, which is the Radon-Nikodym formula. Conditional Jensen for the strictly convex function $t \mapsto t\log t$ gives $\mathbb{E}_{\mathbb{J}}[\bar{r}(Y)\log\bar{r}(Y)] \le \mathbb{E}_{\mathbb{J}}[r(X)\log r(X)]$, and finite divergence and strict convexity make equality equivalent to $r(X) = \bar{r}(Y)$ almost surely. This is the information-loss theorem of Csiszár [28], Kullback and Leibler [69] in kernel form. □

Corollary 10.4 (Pairwise Bayes recovery). In the setting of Theorem 10.3, let $R_Q(y,dx)$ be a regular conditional law of X given $Y = y$ under $\mathbb{J}$. Then $QKR_Q = Q$, and at equality in the data-processing inequality also $PKR_Q = P$. Conversely, any one Markov kernel R that recovers both P and Q forces that equality. For a dominated experiment $\{P_\theta\}$ with one common dominating law P_0, simultaneous finite equalities $D_{\mathrm{KL}}(P_\theta K\|P_0K) = D_{\mathrm{KL}}(P_\theta\|P_0)$ for every θ make the single kernel R_{P_0} recover every P_θ. [ESTABLISHED]

Proof. Disintegration gives $QKR_Q = Q$. At equality, $(PKR_Q)(A) = \int \bar{r}(y)R_Q(y,A)(QK)(dy) = \iint \mathbf{1}_A(x)r(x)\mathbb{J}(dx,dy) = P(A)$. The converse follows by data processing through K and then through R, and the family statement is the pairwise statement with $Q = P_0$ for each θ. This is the commutative case of the channel-sufficiency theorem of Petz [85], whose reference-dependent reverse map is here the Bayes kernel R_Q. □

Proposition 10.5 (Common recovery by likelihood descent). Let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be a coarse channel, let $\{P_\theta\}_{\theta\in\Theta}$ be laws on X dominated by a law P_0, with likelihood ratios $r_\theta = dP_\theta/dP_0$, and let R_{P_0} be a regular conditional law of X given Y under $\mathbb{J}_{P_0}(dx,dy) = P_0(dx)K(x,dy)$; no relative entropy is assumed finite. The density $r = dP/dP_0$ of a law $P \ll P_0$ descends when it agrees $\mathbb{J}_{P_0}$-almost everywhere with a measurable function of the coarse coordinate, which is then

$\bar{r} = d(PK)/d(P_0K)$:

$$r(x) = \bar{r}(y) \qquad \text{for } \mathbb{J}_{P_0}\text{-almost every } (x, y). \tag{10.2}$$

(i) One Markov kernel $R : \mathsf{Y} \rightsquigarrow \mathsf{X}$ recovers P_0 and every P_θ, that is, $P_0KR = P_0$ and $P_\theta KR = P_\theta$, exactly when every r_θ descends, and R_{P_0} is then such a kernel. The laws $P \ll P_0$ that R_{P_0} recovers are those whose density descends; when every r_θ descends, they include every law whose density lies in the $L^1(P_0)$-closed linear span of the constant 1 and the r_θ. For the records of Definition 12.1, with $P_0 = \varrho$, the family $\{\bar{\mathsf{m}}_o\}$ indexed by records o, and $K = C$, the likelihood ratios are L_o/M_o and their pushed densities are L^c_o/M_o, so one kernel recovers the reference and every $\bar{\mathsf{m}}_o$ exactly when each record likelihood L_o agrees with its effective likelihood L^c_o almost everywhere under the reference joint. (ii) For the deterministic kernel K_c of a statistic $c : \mathsf{X} \to \mathsf{Y}$, descent reads $r = \bar{r} \circ c$ P_0-almost everywhere, and a Markov kernel R supported on fibers, $R(y, c^{-1}(\{y\})) = 1$ for $c_\# P_0$-almost every y, recovers P_0 exactly when it is a version of R_{P_0}. A kernel supported on fibers therefore recovers P_0 and every P_θ exactly when it is a version of R_{P_0} and every r_θ descends, and it is then unique up to a $c_\# P_0$-null set. (iii) Let $\mathsf{X} = \mathsf{X}_1 \times \mathsf{X}_2$ with coordinates X_1, X_2, let $c : \mathsf{X}_1 \to \mathsf{Y}$ be a statistic, and let the channel be the deterministic kernel of $c \times \mathrm{id}_{\mathsf{X}_2} : \mathsf{X} \to \mathsf{Y} \times \mathsf{X}_2$, which coarsens X_1 and retains X_2. A kernel $R((y, x_2), \cdot) = R'(y, \cdot) \otimes \delta_{x_2}$, which does not read the retained coordinate, with $R' : \mathsf{Y} \rightsquigarrow \mathsf{X}_1$ satisfying $R'(y, c^{-1}(\{y\})) = 1$ for $(c(X_1))_\# P_0$-almost every y, recovers P_0 and every P_θ exactly when R' is a regular conditional law of X_1 given $c(X_1)$ under P_0, X_1 and X_2 are conditionally independent given $c(X_1)$ under P_0, and every r_θ descends. Descent does not imply the conditional independence: for $\mathsf{X}_1 = \mathsf{X}_2 = \{0, 1\}$, P_0 uniform on $\{(0, 0), (1, 1)\}$, the family $\{P_0\}$, and Y a point, R_{P_0} recovers P_0, while every kernel that does not read X_2 sends P_0 to a law giving mass $1/2$ to the P_0-null set $\{x_1 \neq x_2\}$. [ESTABLISHED]

The proof is in appendix B.4.

Part (i) replaces the simultaneous finite equalities in the family clause of Corollary 10.4 by descent, a condition on likelihood ratios rather than on relative entropies; under finite divergence the two agree by Theorem 10.3, while Corollary 10.6 shows that the relative-entropy form fails without finiteness. For a channel that coarsens a block and leaves the other agents unchanged, part (iii) separates reconstruction of the block from its coarse coordinate alone from reconstruction that also reads the other agents.

For a deterministic channel, the equivalence in (ii) between descent of every likelihood ratio and recovery of the family by R_{P_0} is the factorization theorem of Halmos and Savage [46, Sec. 5, Theorem 1] applied to the set $\{P_0\} \cup \{P_\theta\}$, which is equivalent to P_0. For finitely many θ, part (i) characterizes when the coarse experiment is sufficient for the fine one in the sense of Blackwell [15, Sec. 2], and the conditions of part (iii) on P_0 and the r_θ are equivalent to adequacy [94, Sec. 2, Definition 1] of the σ-algebra of $c(X_1)$ for that of X_1 with respect to that of X_2 and the family $\{P_0\} \cup \{P_\theta\}$.

Corollary 10.6 (Infinite equality carries no recovery conclusion). The finiteness hypothesis of Theorem 10.3 cannot be removed from its equality characterization: $D_{\mathrm{KL}}(PK\|QK) = D_{\mathrm{KL}}(P\|Q) = +\infty$ does not imply that one reverse kernel recovers P and Q. [ESTABLISHED]

Proof. Let $\mathsf{X} = \{a, b, c\}$ and $\mathsf{Y} = \{u, v\}$, let the deterministic channel send a to u and b, c to v, and set $P = \frac{1}{2}\delta_a + \frac{1}{2}\delta_b$ and $Q = \frac{1}{2}\delta_b + \frac{1}{2}\delta_c$. Both divergences are $+\infty$, because $P(a) > 0 = Q(a)$ and $(PK)(u) > 0 = (QK)(u)$. A reverse kernel R recovering Q has $R(v, \cdot) = Q$, since $QK = \delta_v$; then $(PK)R = \frac{1}{2}R(u, \cdot) + \frac{1}{2}Q$ assigns at least $1/4$ to c, while $P(c) = 0$. □

10.3 The common-channel VFE identity and the pointwise parent datum

Fix the joint $\mathbb{P}(do, dy)$ on $\mathsf{O} \times \mathsf{Y}$ of Chapter 4 with (θ, V, X) suppressed, an admitted observation o with evidence density representative $p(o)$, the selected posterior-version kernel $o \mapsto \mathbf{\Pi}_o$, a recognition law $\mathbb{Q}_o$ on Y, and a coarse channel $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ on the latent coordinates, which reads neither $\mathbb{Q}_o$ nor the observation. The common pushforward of the three laws is

$$\mathbb{P}^c(do, dz) = \int_{\mathsf{Y}} C(y, dz)\mathbb{P}(do, dy), \qquad \mathbf{\Pi}^c_o = \mathbf{\Pi}_o C, \qquad \mathbb{Q}^c_o = \mathbb{Q}_o C, \tag{10.3}$$

and free energies are the extended values $\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o) = -\log p(o) + D_{\mathrm{KL}}(\mathbb{Q}_o \| \mathbf{\Pi}_o)$ of Definition 5.6, defined for every law on the latent space.

Theorem 10.7 (Exact coarse VFE and discarded conditional information). In the setting of (10.3), the following hold. (i) The law $\mathbb{P}^c$ is normalized and has the observation marginal of $\mathbb{P}$, so the same evidence representative $p(o)$ serves both scales. The kernel $o \mapsto \mathbf{\Pi}_o C$ is a selected posterior version of $\mathbb{P}^c$, including at the exceptional observations of the fine version. If $\mathbb{Q}_o \ll \mathbf{\Pi}_o$, then $\mathbb{Q}^c_o \ll \mathbf{\Pi}^c_o$. (ii) With the bridge lifts $\widehat{\mathbb{Q}}_o(dy, dz) = \mathbb{Q}_o(dy)C(y, dz)$ and $\widehat{\mathbf{\Pi}}_o(dy, dz) = \mathbf{\Pi}_o(dy)C(y, dz)$ and their disintegrations over z, the free energies satisfy, in $(-\infty, +\infty]$ and with no absolute-continuity hypothesis,

$$\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o) = \mathcal{F}_{\mathbb{P}^c}(\mathbb{Q}^c_o) + \Delta(o), \qquad \Delta(o) = \int_{\mathsf{Z}} D_{\mathrm{KL}}\big(\widehat{\mathbb{Q}}_o(dy \mid z) \| \widehat{\mathbf{\Pi}}_o(dy \mid z)\big)\mathbb{Q}^c_o(dz) \in [0, +\infty], \tag{10.4}$$

so the fine free energy is at least the coarse one. (iii) $\Delta(o) = 0$ exactly when the two discarded conditional laws in (10.4) agree $\mathbb{Q}^c_o$-almost surely. (iv) If $D_{\mathrm{KL}}(\mathbb{Q}_o \| \mathbf{\Pi}_o) < \infty$, then $\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_o) - \mathcal{F}_{\mathbb{P}^c}(\mathbb{Q}^c_o) = \Delta(o)$ is an ordinary difference of real numbers, and $\Delta(o) = 0$ exactly when one Markov kernel $R : \mathsf{Z} \rightsquigarrow \mathsf{Y}$ recovers both laws, $\mathbb{Q}^c_o R = \mathbb{Q}_o$ and $\mathbf{\Pi}^c_o R = \mathbf{\Pi}_o$. This recovery is pairwise; one kernel recovering every member of a family of laws requires the hypothesis for each member simultaneously. [ESTABLISHED]

The proof is in appendix B.4.

Corollary 10.8 (Evidence-preserving coarse channel). In the setting of Theorem 10.7, $\mathbb{P}^c(do \times \mathsf{Z}) = \mathbb{P}(do \times \mathsf{Y})$, $p^c(o) = p(o)$, and $\mathbf{\Pi}^c_o = \mathbf{\Pi}_o C$ for every declared o, and the extended ELBO of Definition 5.6 satisfies, in $[-\infty, \log p(o)]$,

$$\mathcal{L}^{\mathrm{ext}}_{\mathbb{P}^c}(\mathbb{Q}_o C) = \log p(o) - D_{\mathrm{KL}}(\mathbb{Q}_o C \| \mathbf{\Pi}_o C) \geq \log p(o) - D_{\mathrm{KL}}(\mathbb{Q}_o \| \mathbf{\Pi}_o) = \mathcal{L}^{\mathrm{ext}}_{\mathbb{P}}(\mathbb{Q}_o). \tag{10.5}$$

When the fine ELBO is finite, the difference of the two ELBOs is $\Delta(o)$: the rise is the discarded conditional information at unchanged evidence, neither model improvement nor evidence ascent. For two separately declared models with evidences $p_1(o), p_2(o)$ and finite gaps, the difference of their ELBOs is $\log p_1(o) - \log p_2(o)$ minus the difference of their gaps, so it compares variational tightness only when the evidences agree, as they do under one evidence-preserving common pushforward. [ESTABLISHED]

Proof. The first three identities are part (i) of the theorem. The inequality is data processing in $[0, +\infty]$ (Lemma 3.12), and the difference is part (ii) written for $\mathcal{L}^{\mathrm{ext}} = -\mathcal{F}$. The statement on separately declared models is the gap identity of Corollary 5.7 for each model. □

Remark 10.9 (Support boundaries of the discarded term). If $\mathbb{Q}_o \not\ll \mathbf{\Pi}_o$ while $D_{\mathrm{KL}}(\mathbb{Q}_o^c \| \mathbf{\Pi}_o^c) < \infty$, then the fine free energy is $+\infty$, the coarse one is finite, and $\Delta(o) = +\infty$. If $\mathbb{Q}_o^c \not\ll \mathbf{\Pi}_o^c$, both free energies are $+\infty$; the conditional law $\widehat{\mathbf{\Pi}}_o(dy \mid z)$ is then undetermined on a $\mathbb{Q}_o^c$-positive set of targets, so $\Delta(o)$ depends there on an arbitrary version and the difference of the two infinite free energies is not a numerical loss. [ESTABLISHED]

Proof. In the first case (10.4) reads $+\infty = \text{finite} + \Delta(o)$. In the second, a $\mathbf{\Pi}_o^c$-null set B with $\mathbb{Q}_o^c(B) > 0$ gives $\widehat{\mathbb{Q}}_o(\mathsf{Y} \times B) > 0 = \widehat{\mathbf{\Pi}}_o(\mathsf{Y} \times B)$, so both divergences are infinite, and a disintegration of $\widehat{\mathbf{\Pi}}_o$ over z is determined only $\mathbf{\Pi}_o^c$-almost surely. □

Proposition 10.10 (Stochastic loss and deterministic minimization). Let Y and Z be finite and write $\mathcal{F}^c(q') = -\log p(o) + D_{\mathrm{KL}}(q' \| \mathbf{\Pi}_o C)$ for a law q' on Z. (i) For a deterministic statistic $c : \mathsf{Y} \to \mathsf{Z}$ and $q' \ll c_{\#}\mathbf{\Pi}_o$, the fine law $\mathbb{Q}_*(y) = q'(c(y))\mathbf{\Pi}_o(y \mid c(y))$, defined to be zero where $q'(c(y)) = 0$ without evaluating a posterior fiber at a null target, has image q' and satisfies $\mathcal{F}_{\mathbb{P}}(\mathbb{Q}_*) = \mathcal{F}^c(q') = \min\{\mathcal{F}_{\mathbb{P}}(\mathbb{Q}) : \mathbb{Q}K_c = q'\}$. (ii) For a stochastic channel the statement $\mathcal{F}^c(q') = \inf_{\mathbb{Q}:\mathbb{Q}C=q'} \mathcal{F}_{\mathbb{P}}(\mathbb{Q})$ can fail: with a one-point observation alphabet, $\mathbf{\Pi}_o = (1/2, 1/2)$, $C = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$, $\mathbb{Q} = (1, 0)$, and $q' = \mathbb{Q}C = (3/4, 1/4)$, the law $\mathbb{Q}$ is the only fine law with image q', its free energy is $\log 2$, the coarse free energy is $\frac{3}{4}\log\frac{3}{2} + \frac{1}{4}\log\frac{1}{2}$, and the gap is $2\log 2 - \frac{3}{4}\log 3 > 0$. [ESTABLISHED]

Proof. (i) Summing $\mathbb{Q}_*$ over a fiber $c^{-1}(z)$ with $q'(z) > 0$ gives $q'(z)$, because the posterior fiber $\mathbf{\Pi}_o(\cdot \mid z)$ is a law on $c^{-1}(z)$; so $\mathbb{Q}_*$ is a probability law with image q'. By (10.4), every $\mathbb{Q}$ with image q' has $\mathcal{F}_{\mathbb{P}}(\mathbb{Q}) = \mathcal{F}^c(q') + \Delta \geq \mathcal{F}^c(q')$, and $\Delta = 0$ for $\mathbb{Q}_*$ because its occupied reverse fibers are the posterior fibers. (ii) The matrix C has determinant $1/2$, so $\mathbb{Q}$ is the unique probability vector with image q'. The evidence is one, $\mathbf{\Pi}_o C = (1/2, 1/2)$, and direct evaluation gives the two free energies, whose difference $2\log 2 - \frac{3}{4}\log 3$ is positive because $2^8 > 3^3$. □

Corollary 10.11 (Full pointwise probabilistic datum for a candidate parent). Fix a nonempty finite block $I \subseteq V$ of agents with $c_* \in \bigcap_{i \in I} \mathcal{C}_i$, a parent label A, and structural data X with $X_A = \chi_A(X)$ kept outside the random channel. Let the parent state space $\mathsf{Z}_A = \mathsf{B}_A \times \mathsf{M}_A \times \mathbf{\Xi}_A \times \mathsf{H}_A$ have nonempty standard Borel factors for parent belief samples, parent model presentations, the inputs read by the parent evaluator, and retained marks such as the rooted-holonomy and dressed-boundary data of Section 11.2. Supply the joint $\mathbb{P}_I(do, dy \mid X)$ of the block with its selected

posterior-version kernel $o \mapsto \mathbf{\Pi}_{I,o,X}$, a recognition law $\mathbb{Q}_{I,o,X} \ll \mathbf{\Pi}_{I,o,X}$ at an admitted o, and one parent coarse kernel $C_A : \mathsf{Y}_I \rightsquigarrow \mathsf{Z}_A$, a declared input like I, not a canonical selection. Define

$$\mathbb{P}_A(do, dz \mid X) = \int_{\mathsf{Y}_I} C_A(y, dz)\mathbb{P}_I(do, dy \mid X), \tag{10.6}$$

$$\mathbf{\Pi}_{A,o,X} = \mathbf{\Pi}_{I,o,X}C_A, \qquad \mathbb{Q}_{A,o,X} = \mathbb{Q}_{I,o,X}C_A. \tag{10.7}$$

Then the following hold. (i) All three parent laws are normalized, the parent observation marginal and evidence equal the fine ones, $o \mapsto \mathbf{\Pi}_{A,o,X}$ is the selected parent posterior version, $\mathbb{Q}_{A,o,X} \ll \mathbf{\Pi}_{A,o,X}$, and in $[0, +\infty]$ $D_{\mathrm{KL}}(\mathbb{Q}_{I,o,X} \| \mathbf{\Pi}_{I,o,X}) = D_{\mathrm{KL}}(\mathbb{Q}_{A,o,X} \| \mathbf{\Pi}_{A,o,X}) + \Delta_A(o, X)$, where $\Delta_A(o, X) \geq 0$ is the discarded term $\Delta(o)$ of (10.4) for $\mathbb{P} = \mathbb{P}_I$ and $C = C_A$. Adding $-\log p_X(o)$ gives $\mathcal{F}_I = \mathcal{F}_A + \Delta_A$ in the extended reals, where a finite free energy may be negative. The zero-defect and recovery statements of Theorem 10.7 hold with $C = C_A$; a bare equality $+\infty = +\infty$ supplies neither. (ii) The parent model coordinate is generative only through a jointly measurable normalized evaluator $\mathrm{ev}_A : \mathsf{M}_A \to \mathrm{Kern}(\mathbf{\Xi}_A, \mathsf{B}_A \times \mathsf{O} \times \mathsf{H}_A)$, $m_A \mapsto K^{X_A}_{A,m_A}$. A selected standard-Borel disintegration of $\mathbb{P}_A$ over (m_A, ξ_A) supplies one such evaluator; an evaluator declared independently of $\mathbb{P}_A$ is compatible with it only under the separate hypothesis that $\mathbb{P}_A(db_A, do, dh_A \mid \xi_A, m_A, X) = K^{X_A}_{A,m_A}(\xi_A; db_A, do, dh_A)$ almost surely for the parent (m_A, ξ_A) marginal. Disintegration neither validates an arbitrary declared evaluator nor selects its null extension or makes evaluation injective. (iii) With $\mathbb{P}^Z_A(D \mid X) = \mathbb{P}_A(\mathsf{O} \times D \mid X)$, the parent recognition, prior, and posterior marginals in channel $x \in \{b, m\}$ are the coordinate pushforwards $q^x_A = (\mathrm{pr}_x)_\# \mathbb{Q}_{A,o,X}$, $p^x_A = (\mathrm{pr}_x)_\# \mathbb{P}^Z_A(\cdot \mid X)$, and $\mathbf{\Pi}^x_{A,o,X} = (\mathrm{pr}_x)_\# \mathbf{\Pi}_{A,o,X}$. If both laws of a marginal pair are non-Dirac, or if $\mathbf{\Xi}_A \times \mathsf{H}_A$ has at least two points, the pair does not determine the full law on Z_A. If $\mathbf{\Xi}_A \times \mathsf{H}_A$ is a point and one marginal is a point mass, the coupling class of the pair is a single law. [ESTABLISHED]

The proof is in appendix B.4.

10.4 Pushforward, precomposition, and Galerkin restriction

Three operations share the name coarse-graining. A Markov pushforward (Definition 10.1) moves laws and obeys data processing; precomposition identifies variables inside an energy and defines a new model; Galerkin restriction compresses an operator and needs further data before it propagates signals. A proof about one of them does not transfer to another.

Definition 10.12 (Energy precomposition). For a measurable injection $\iota : \mathsf{Z} \to \mathsf{X}$ and an energy E on X, the precomposed model is

$$\bar{E}(z) = E(\iota z), \qquad \bar{P}(dz) = \frac{e^{-\bar{E}(z)}}{\bar{Z}}\bar{\nu}(dz), \tag{10.8}$$

defined once a reference $\bar{\nu}$ on Z and a finite normalizer $\bar{Z}$ are declared; the reference, its Radon-Nikodym or Jacobian rule, and the normalizer are part of the construction.

Proposition 10.13 (Hard identification is conditioning). (i) On a finite or countable X, with counting references on Z and X and $E = -\log p$ for the mass function of a law P, the precomposed law is $\bar{P}(z) = p(\iota z)/P(\iota(\mathsf{Z}))$: the law of X conditioned on $X \in \iota(\mathsf{Z})$, transported by ι^{-1}. (ii) Conditioning can increase relative entropy. On $\mathsf{X} = \{1,2,3\}$ with $\iota(\mathsf{Z}) = \{1,2\}$, the laws $P = (1,1,98)/100$ and $Q = (1,2,97)/100$ have $D_{\mathrm{KL}}(P\|Q) \approx 0.0031$, while their conditioned laws $(1/2,1/2)$ and $(1/3,2/3)$ have relative entropy ≈ 0.0589. Hence no Markov kernel sends P and Q to their precomposed laws, and data processing, evidence preservation, and (10.5) do not transfer to precomposition. (iii) When $\iota(\mathsf{Z})$ is null for the fine reference, as for block-constant configurations of a continuous fiber, the fine law defines no conditional law on it, and different references give different precomposed laws: for $\bar{E}(z) = z^2$ on $\mathsf{Z} = \mathbb{R}$, Lebesgue reference gives $\mathcal{N}(0,1/2)$ and the reference $e^{-z^2}dz$ gives $\mathcal{N}(0,1/4)$. Exact marginalization is a Markov pushforward, and restricting a recognition family keeps the joint and optimizes the same ELBO over a subset, which is neither pushforward nor precomposition. [ESTABLISHED]

Proof. (i) Substitution in (10.8) gives $\bar{P}(z) \propto p(\iota z)$, with normalizer $P(\iota(\mathsf{Z}))$. (ii) The values are direct evaluation; every Markov kernel contracts relative entropy (Lemma 3.12), and the conditioned pair has larger relative entropy. (iii) Conditioning on a null event is not defined by the law, and the densities e^{-z^2} and e^{-2z^2} are proportional to the stated Gaussians. The last sentence restates the definitions. □

10.4.1 Galerkin operators and message passing

Definition 10.14 (Galerkin data). Let $S : V_c \to V_f$ be an injective linear map of finite-dimensional real spaces, let $M \succ 0$ be a fine reference form, and let L be a fine bilinear form. Put $M_c = S^\top M S$, $S^+ = M_c^{-1} S^\top M$, $H = M^{-1}L$, and $H_c = S^+ H S = M_c^{-1} S^\top L S$. The form $S^\top L S$ is the Galerkin coarse form of aggregation-based algebraic multigrid [82, Eqs. (2.1) and (2.2)], [75, Sec. 2], here for an arbitrary injective prolongation as in smoothed aggregation [100]; the map H_c is the coarse propagator.

Proposition 10.15 (Exact message-passing residual). The data of Definition 10.14 satisfy $S^+S = I$ and $S^+H - H_cS^+ = S^+H(I - SS^+)$. The idempotent SS^+ is the M-orthogonal projector onto $\operatorname{range} S$, with kernel $\ker S^+ = (\operatorname{range} S)^{\perp_M}$. Hence exact coarse message passing, $S^+H = H_cS^+$ on all fine signals, holds exactly when

$$H\big((\operatorname{range} S)^{\perp_M}\big) \subseteq (\operatorname{range} S)^{\perp_M}. \tag{10.9}$$

The operator H is self-adjoint in the M-inner product exactly when L is symmetric, and then (10.9) is equivalent to H-invariance of $\operatorname{range} S$. Without symmetry the two invariance conditions are independent: with $M = I$, $S = e_1$, and $L = \left(\begin{smallmatrix} 2 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{smallmatrix}\right)$, the subspace $\operatorname{range} S$ is invariant but the residual $S^+H(I - SS^+) = (0,1,0)$ is nonzero, while $L^\top$ leaves the complement invariant and gives zero residual although $\operatorname{range} S$ is not invariant. [ESTABLISHED]

Proof. $S^+S = M_c^{-1}S^\top M S = I$, and substituting $H_c = S^+HS$ gives the residual identity. The map SS^+ is idempotent with range $\operatorname{range} S$ and kernel $\ker S^+$; since M_c is invertible, $S^+y = 0$ exactly

when $S^\top My = 0$, that is, when y is M-orthogonal to $\operatorname{range} S$. The residual vanishes exactly when $S^+Hy = 0$ for every $y \in \operatorname{range}(I - SS^+) = \ker S^+$, which is (10.9). Next $\langle Hx, y\rangle_M = x^\top L^\top y$ and $\langle x, Hy\rangle_M = x^\top Ly$, and for an M-self-adjoint operator a subspace is invariant exactly when its M-orthogonal complement is. The two witnesses are direct computations. □

10.5 Gauge and permutation equivariance

Proposition 10.16 (Equivariant Markov channels). Let a group $\mathbb{G}$ act bimeasurably on X and Y, and let $K : \mathsf{X} \rightsquigarrow \mathsf{Y}$ be equivariant, $K(gx, B) = K(x, g^{-1}B)$. Then $(g_\# P)K = g_\#(PK)$ for every law P on X, and composites of equivariant kernels are equivariant. [ESTABLISHED]

Proof. For measurable B the law $(g_\# P)K$ gives B the mass $\int K(gx, B)P(dx)$, which by equivariance equals $\int K(x, g^{-1}B)P(dx) = [g_\#(PK)](B)$. Substitution in the composition of Proposition 10.2 proves closure under composition. □

At c_* a principal gauge transformation acts on belief and model coordinates through $\rho_b(g)$ and $\rho_m(g)$ of one $g \in G$ (Proposition 2.6), and each passive re-expression g_i acts on both channels; a coarse channel on joint belief and model coordinates is gauge covariant when it is equivariant for these actions, and for agent relabelings the acting group is a permutation group.

Proposition 10.17 (Equivariant transport of Bayes recovery kernels). Let $\mathbb{G}$ act on X and Y by bimeasurable bijections, let K be equivariant as in Proposition 10.16, let Q be a law on X, let R_Q be a regular conditional law of X given Y under $\mathbb{J}_Q(dx, dy) = Q(dx)K(x, dy)$, and put $R^g_Q(y, A) = R_Q(g^{-1}y, g^{-1}A)$ for $g \in \mathbb{G}$. (i) R^g_Q is a regular conditional law of X given Y under $\mathbb{J}_{g_\# Q}$ for every g. (ii) If $PKR_Q = P$, then $(g_\# P)KR^g_Q = g_\# P$, and $D_{\mathrm{KL}}((g_\# P)K\|(g_\# Q)K) = D_{\mathrm{KL}}(PK\|QK)$ and $D_{\mathrm{KL}}(g_\# P\|g_\# Q) = D_{\mathrm{KL}}(P\|Q)$. (iii) If X and Y are finite and $\mathbb{G}$ acts by permutations, the rule $R_Q(y, \{x\}) = Q(\{x\})K(x, \{y\})/(QK)(\{y\})$ where $(QK)(\{y\}) > 0$ and $R_Q(y, \{x\}) = 1/|\mathsf{X}|$ where $(QK)(\{y\}) = 0$ selects a version for every Q, depends Borel measurably on (Q, y), and satisfies $R_{g_\# Q}(gy, \{gx\}) = R_Q(y, \{x\})$ for all g, Q, y, x. (iv) If $\mathbb{G}$ is a topological group acting jointly measurably and the stabilizer $\mathbb{G}_Q = \{h : h_\# Q = Q\}$ is a compact metrizable group with Haar probability dh, then $\bar{R}_Q(y, A) = \int_{\mathbb{G}_Q} R^h_Q(y, A)dh$ is a version of R_Q with $\bar{R}_Q(hy, hA) = \bar{R}_Q(y, A)$ for $h \in \mathbb{G}_Q$, and $R_{g_\# Q} := \bar{R}^g_Q$ is well defined on the orbit of Q and satisfies $R_{k_\# Q'}(ky, kA) = R_{Q'}(y, A)$ for every Q' in that orbit and every $k \in \mathbb{G}$. [ESTABLISHED]

The proof is in appendix B.4.

Corollary 10.18 (Continuous posterior versions are equivariant). In the setting of Proposition 10.17, let X and Y be Polish, let every $h \in \mathbb{G}_Q$ act on both by homeomorphisms, and let S be the support of QK. If some version of R_Q is weakly continuous on S, then $R(y, \cdot) = R_Q(y, \cdot)$ for $y \in S$ and $R(y, \cdot) = Q$ for $y \notin S$ is a version of R_Q with $R(hy, hA) = R(y, A)$ for every $h \in \mathbb{G}_Q$, every $y \in \mathsf{Y}$, and every measurable A, whether or not the stabilizer is compact. [ESTABLISHED]

Proof. By Proposition 10.16, $h_\#(QK) = (h_\# Q)K = QK$ for $h \in \mathbb{G}_Q$, so S is a closed $\mathbb{G}_Q$-invariant set, and R is a Markov kernel that agrees with R_Q on the full-measure set S. For $h \in \mathbb{G}_Q$ the

kernel R_Q^h is a version of R_Q by part (i) of the proposition, and $y \mapsto R_Q^h(y,\cdot) = h_\# R_Q(h^{-1}y,\cdot)$ is weakly continuous on S, because h^{-1} is a homeomorphism of Y preserving S and pushforward by the homeomorphism h of X is weakly continuous. Two versions agree QK-almost everywhere, and the set where two weakly continuous maps from S into the Hausdorff space of laws agree is closed in S; its complement in S is an open QK-null subset of the support, hence empty. So $R_Q^h = R_Q$ on S, which is the equivariance identity at $y \in S$; for $y \notin S$ also $h^{-1}y \notin S$, and $R(h^{-1}y, h^{-1}A) = Q(h^{-1}A) = Q(A) = R(y,A)$. □

Open Problem 10.19 (Equivariant Bayes versions beyond the finite and compact cases). Proposition 10.17 settles equivariant Bayes versions for finite spaces and along orbits with compact stabilizer, and Corollary 10.18 settles them for every stabilizer when the posterior has a weakly continuous version on the support of QK. Two cases remain open in the class of a locally compact second countable group acting continuously on Polish spaces, an equivariant channel K, and laws Q whose stabilizer is noncompact, so that no Haar probability averages versions, and whose posterior has no weakly continuous version on the support of QK: an everywhere-defined version equivariant under the stabilizer; and a version that is simultaneously equivariant and jointly measurable across a measurable family of such laws meeting several orbits, which also needs a measurable choice of orbit representatives. The problem is settled by an equivariant measurable selection theorem giving, for every member of that class, an everywhere-defined version equivariant under the stabilizer, jointly measurable across every measurable family of such laws meeting several orbits, or by a pair (Q,K) of the class with no everywhere-defined version equivariant under the stabilizer.

Open Problem 10.20 (A natural selector of partitions and membership laws). A data-dependent hard partition selector $S(X)$ must be natural under agent relabeling: for every fine permutation P there must exist a coarse relabeling $Q = Q(P,X)$ with $S(P\cdot X) = PS(X)Q^\top$, that is, the unlabeled partition is equivariant. The quantifier on Q is existential because coarse labels are arbitrary; a universal one would force $S(X) = S(X)Q^\top$ for every Q at $P = \mathrm{id}$, which no partition into two or more blocks satisfies. Gauge-carrying inputs must enter through invariant assignment scores, while represented messages transform equivariantly. These conditions type an admissible selector of a partition, a membership law, or a structural channel, and select none. The costs analyzed so far do not select one: an unpenalized sum of nonnegative within-block divergences is minimized by the singleton partition (Remark 11.22), the Gaussian recognition costs of the Gaussian supplement select trivial endpoints or need an external coefficient, and one common recovery kernel for a whole statistical experiment, (11.23), is available only under simultaneous family hypotheses. The problem is settled by a relabeling-natural, gauge-compatible functional of the laws, not monotone under refinement, with a nondegenerate optimizer, or by a theorem that no such selector exists in a stated class.

10.6 Scale diagrams

Definition 10.21 (Scale diagram). Let S be a category whose objects are resolutions ℓ, and choose one target category $\mathscr{K}$: sets and maps for a deterministic state, standard Borel spaces and Markov kernels for a stochastic state, or a declared category of Banach spaces and bounded linear maps, or a declared product of such categories, for an operator or mixed state. A scale diagram is a functor $\mathfrak{X} : \mathsf{S} \to \mathscr{K}$, with objects $\mathfrak{X}_\ell$ and arrows $C_{\ell k} \in \mathrm{Hom}_{\mathscr{K}}(\mathfrak{X}_\ell, \mathfrak{X}_k)$ satisfying $C_{kr} \circ C_{\ell k} = C_{\ell r}$, where $\circ$ is composition in $\mathscr{K}$, componentwise in a product. For a stochastic component, kernels act on the right,

$$K_{\ell r} := K_{\ell k} K_{kr}, \qquad (\mu K_{\ell k}) K_{kr} = \mu K_{\ell r}, \tag{10.10}$$

so that $K_{\ell k} K_{kr}$ represents the abstract composite $K_{kr} \circ K_{\ell k}$. One symbol is never at once a map, a kernel, and an operator. An expression on a reference space needs declared isomorphisms, and transformations of reference measures stay explicit.

Proposition 10.22 (When functoriality has content). (i) On a chain of resolutions $0 < 1 < \cdots < L$, any family of adjacent arrows $C_{\ell,\ell+1}$ extends uniquely to a scale diagram, by $C_{\ell k} = C_{k-1,k} \circ \cdots \circ C_{\ell,\ell+1}$ and $C_{\ell\ell} = \mathrm{id}$; the composition law imposes no condition. (ii) On a poset of resolutions that is not a chain, functoriality requires any two routes between the same resolutions to agree. For a family of kernels generated by one blocking rule indexed by block factors, functoriality is the semigroup law $K_{b_1 b_2} = K_{b_1} K_{b_2}$. (iii) Deterministic arrows that keep the complete block configuration, or an additive block statistic such as block sums, satisfy both conditions. Majority with ties resolved by the first block member does not: on six agents it sends $(0, 0, 1, 0, 1, 0)$ to 1 by pairs and then triples, and to 0 by triples and then pairs. (iv) Mutually inverse Markov kernels between finite spaces are permutation matrices, so no stochastic isomorphism exists between finite levels with different numbers of configurations. [ESTABLISHED]

The proof is in appendix B.4.

10.7 Labeled families and retained label information

A finite mixture that keeps its label in the sample space is the elementary model of a random partition or random link pattern among agents at c_*; its Fisher tensor splits into a label part and a weighted branch part.

Definition 10.23 (Finite labeled family). Let $\Theta \subseteq \mathbb{R}^d$ be open, $1 \le d < \infty$, and let $\mathfrak{G}$ be a finite nonempty set of labels γ, for instance partitions of a block of agents or link patterns. Each label has a finite nonempty carrier X_γ, and the sample space is the disjoint union of the $\{\gamma\} \times \mathsf{X}_\gamma$, so that labels stay distinct when branch laws coincide. A finite labeled family on Θ consists of effective log masses $\lambda_\gamma \in C^3(\Theta)$ and branch laws $p_{\gamma,\theta}$ on X_γ, with

$$P_\theta(\gamma, x) = W_\gamma(\theta) p_{\gamma,\theta}(x), \qquad W_\gamma = \frac{e^{\lambda_\gamma}}{\sum_{\delta \in \mathfrak{G}} e^{\lambda_\delta}}, \tag{10.11}$$

on a fixed support stratum: every branch mass is either identically zero on Θ, a structural zero that carries no score, or positive and C^3 in θ. A common additive function of θ in all λ_γ is a change of representative. When the family comes from a normalization model, the effective log masses absorb its label weights and branch partitions (Proposition 13.13). For a fixed-support finite family $\theta \mapsto R_\theta$ the score is $\mathbf{s}_R = \nabla_\theta \log R_\theta$ on positive atoms and the Fisher tensor is $g^F(R) = \sum_{z:R_\theta(z)>0} R_\theta(z)\mathbf{s}_R(z)\mathbf{s}_R(z)^\top$; write $\xi_\gamma = \nabla\lambda_\gamma$ and $\bar{\xi} = \sum_\gamma W_\gamma \xi_\gamma$. An observation model is a record kernel $K^{\mathrm{obs}}(o \mid \gamma, x)$ on a finite alphabet O, of the same fixed-support C^3 type and independent of recognition; it defines the complete observation law $P_\theta^{\mathrm{obs}}(\gamma, x, o) = P_\theta(\gamma, x)K^{\mathrm{obs}}(o \mid \gamma, x)$ and the evidence $p_\theta(o) = \sum_{\gamma,x} P_\theta^{\mathrm{obs}}(\gamma, x, o)$. The complete latent, complete observation, and observed-data experiments are $\theta \mapsto P_\theta$, $\theta \mapsto P_\theta^{\mathrm{obs}}$, and $\theta \mapsto p_\theta$. An extensive labeled family is a sequence $P_N = (W_{\gamma,N}p_{\gamma,N})$ of finite labeled families indexed by an extensive size $N \geq 1$, such as the number of agents, with carriers that may depend on N. On a compact $\mathsf{K} \subset \Theta$ it has extensive label growth with bulk log-mass densities s_γ if, on an open neighborhood of K,

$$\|\lambda_{\gamma,N}\|_{C^3(\mathsf{K})} \leq \mathrm{C}_\mathsf{K} N, \qquad \lambda_{\gamma,N}/N \longrightarrow \mathsf{s}_\gamma \quad \text{in } C^3(\mathsf{K}), \tag{10.12}$$

where $\|f\|_{C^k(\mathsf{K})}$ is the largest supremum over K of the ordinary coordinate derivatives of f through order k, including order zero, with fixed norms on vector and matrix entries.

Proposition 10.24 (Complete labeled Fisher decomposition). For a finite labeled family, the branch score $\mathbf{s}_{p_\gamma} = \nabla \log p_\gamma$ on the positive atoms of X_γ satisfies $\sum_x p_\gamma(x)\mathbf{s}_{p_\gamma}(x) = 0$, and the latent score, the Fisher tensor, and, for every multi-index α with $|\alpha| \leq 2$, the derivatives of its weighted branch term satisfy

$$\mathbf{s}_P(\gamma, x) = \xi_\gamma - \bar{\xi} + \mathbf{s}_{p_\gamma}(x), \tag{10.13}$$

$$g^F(P) = \mathrm{Cov}_W(\xi_\gamma) + \sum_\gamma W_\gamma g^F(p_\gamma), \tag{10.14}$$

$$\partial^\alpha\Big(\sum_\gamma W_\gamma g^F(p_\gamma)\Big) = \sum_\gamma \sum_{\beta\leq\alpha} \binom{\alpha}{\beta} (\partial^\beta W_\gamma)(\partial^{\alpha-\beta} g^F(p_\gamma)). \tag{10.15}$$

The first term of $g^F(P)$ is the Fisher tensor of the label marginal W. A common additive function in every λ_γ leaves these identities and their derivatives unchanged, and $g^F(P) \succeq 0$ with no rank assumption. [ESTABLISHED]

The proof is in appendix B.4.

Proposition 10.25 (Selected posterior and its complete score). For a finite labeled family with an observation model and a record o, let $e_\gamma(o) = \sum_x p_\gamma(x)K^{\mathrm{obs}}(o \mid \gamma, x)$ be the branch evidence, so that $p_\theta(o) = \sum_\gamma W_\gamma e_\gamma(o)$, and where $p_\theta(o) > 0$ define the selected posterior $\mathbf{\Pi}_o(\gamma, x) = W_\gamma p_\gamma(x)K^{\mathrm{obs}}(o \mid \gamma, x)/p_\theta(o)$ and its label weights $W_\gamma^o = W_\gamma e_\gamma(o)/p_\theta(o)$. A label with $e_\gamma(o) = 0$ stays on the carrier with zero posterior mass and has no conditional branch; zero evidence leaves the posterior undefined. On occupied labels W^o is the softmax of $\lambda_\gamma^o = \lambda_\gamma + \log e_\gamma(o)$. On the positive posterior support the complete score and, within an occupied branch $p_\gamma^o(x) = p_\gamma(x)K^{\mathrm{obs}}(o \mid$

$\gamma, x)/e_\gamma(o)$, the branch score are

$$\mathbf{s}_{\mathbf{\Pi}_o}(\gamma, x) = \mathbf{s}_P(\gamma, x) + \nabla \log K^{\mathrm{obs}}(o \mid \gamma, x) - \nabla \log p_\theta(o), \tag{10.16}$$

$$\nabla \log p^o_\gamma(x) = \mathbf{s}_{p_\gamma}(x) + \nabla \log K^{\mathrm{obs}}(o \mid \gamma, x) - \nabla \log e_\gamma(o), \tag{10.17}$$

and the decomposition (10.14) holds for the posterior family $\theta \mapsto \mathbf{\Pi}_{o,\theta}$, with the weights W^o and the branches p^o_γ. [ESTABLISHED]

Proof. Summing the posterior over all labeled atoms gives one, and the label weights are proportional to $e^{\lambda_\gamma} e_\gamma(o)$. Differentiating the logarithm of the positive posterior mass gives (10.16), and the same computation within one branch gives (10.17). The fixed zero patterns of branch laws and record kernel keep the occupied labels and atoms fixed on the stratum, so no division by zero occurs, and the proof of Proposition 10.24 applies to the posterior family. □

Proposition 10.26 (Observed-data information and missing information). For a finite labeled family with an observation model, let $\mathsf{O}^+ = \{o : p_\theta(o) > 0\}$, fixed on the stratum, let $\mathbf{s}^c = \nabla \log P^{\mathrm{obs}}$ be the complete score on the positive support of the complete observation law, and let $g^F(p) = \sum_{o \in \mathsf{O}^+} p_\theta(o) \nabla \log p_\theta(o) \nabla \log p_\theta(o)^\top$ be the observed-data Fisher tensor. Then

$$g^F(P^{\mathrm{obs}}) = g^F(P) + \sum_\gamma W_\gamma \sum_x p_\gamma(x) g^F\big(K^{\mathrm{obs}}(\cdot \mid \gamma, x)\big), \tag{10.18}$$

$$\nabla \log p_\theta(o) = \mathbb{E}_{\mathbf{\Pi}_o}\big[\mathbf{s}^c(\gamma, X, o)\big], \qquad g^F(p) = g^F(P^{\mathrm{obs}}) - \sum_{o \in \mathsf{O}^+} p_\theta(o)\, \mathrm{Cov}_{\mathbf{\Pi}_o}(\mathbf{s}^c). \tag{10.19}$$

In particular $g^F(p) \preceq g^F(P^{\mathrm{obs}})$, and positive definiteness of $g^F(P)$ or of $g^F(P^{\mathrm{obs}})$ does not imply that of $g^F(p)$: for a one-point alphabet $p_\theta(o) = 1$ identically and $g^F(p) = 0$ for all branch data. [ESTABLISHED]

The proof is in appendix B.4.

Fisher contraction requires a channel that does not depend on the parameter; the constant-input Bernoulli witness of Corollary 7.42 shows that a parameter-dependent channel can create information that its input does not carry.

Corollary 10.27 (Stochastic score projection and retained-label loss). A label-preserving channel is given by row-stochastic matrices $C_\gamma(x, y)$ from X_γ to finite nonempty targets Y_γ, independent of θ and of recognition, and lifted as zero between different labels, so that $P^C(\gamma, y) = \sum_x P(\gamma, x) C_\gamma(x, y)$. The source law may be the latent family or a positive selected posterior; on the complete observation law the channel acts on x and keeps o. For such a channel, on positive targets, the score projection $\mathbf{s}_{P^C}(\gamma, y) = \mathbb{E}_{P \otimes C}[\mathbf{s}_P(\gamma, X) \mid \gamma, y]$ and the Fisher loss $g^F(P) - g^F(P^C) = \mathbb{E}_{P \otimes C}[(\mathbf{s}_P - \mathbf{s}_{P^C})(\mathbf{s}_P - \mathbf{s}_{P^C})^\top] \succeq 0$ hold, and for $P = (W_\gamma p_\gamma)$

$$g^F(P) - g^F(P^C) = \sum_{\gamma : W_\gamma > 0} W_\gamma \big[g^F(p_\gamma) - g^F(p_\gamma C_\gamma)\big], \qquad 0 \preceq g^F(P) - g^F(P^C) \preceq \sum_\gamma W_\gamma g^F(p_\gamma), \tag{10.20}$$

while the label marginal and all its derivatives are unchanged. [ESTABLISHED]

Proof. The score projection and the Fisher loss are Theorem 7.8 for the finite fixed-support family, or directly: differentiating P^C at a positive target and dividing by its mass gives the conditional expectation, and the residual $\mathbf{s}_P - \mathbf{s}_{P^C}$ has zero conditional mean given the target, so the cross terms vanish. Row sums give $\sum_y P^C(\gamma, y) = W_\gamma$ as functions of θ. Subtracting the two decompositions (10.14) gives the first part of (10.20), and $g^F(p_\gamma C_\gamma) \succeq 0$ gives the upper bound. A zero-mass target receives no incidence from a positive atom and is a structural zero. □

Proposition 10.28 (Finite-word score and VFE telescopes). Let $C_1, \dots, C_m$ be parameter-independent Markov kernels between finite spaces, let P_0 be a fixed-support finite family, and put $P_j = P_0 C_1 \cdots C_j$ on its fixed positive support, with score $\mathbf{s}_j$ and Fisher tensor $g_j = g^F(P_j)$. Then $g_0 - g_m = \sum_{j=1}^m D_j$ with $D_j = g_{j-1} - g_j \succeq 0$, and on the path law $P_0(x_0)C_1(x_0, x_1) \cdots C_m(x_{m-1}, x_m)$ the increments $\varepsilon_j^{\mathrm{sc}} = \mathbf{s}_{j-1}(X_{j-1}) - \mathbf{s}_j(X_j)$ are orthogonal with $\mathbb{E}[\varepsilon_j^{\mathrm{sc}}(\varepsilon_j^{\mathrm{sc}})^\top] = D_j$, so $\mathbb{E}[(\mathbf{s}_0(X_0) - \mathbf{s}_m(X_m))(\mathbf{s}_0(X_0) - \mathbf{s}_m(X_m))^\top] = \sum_j D_j$. For a record with positive evidence, a selected posterior $\mathbf{\Pi}_0$, and a recognition law $\mathbb{Q}_0 \ll \mathbf{\Pi}_0$, push both laws and the joint by the same word and set $\mathcal{F}_j = -\log p(o) + D_{\mathrm{KL}}(\mathbb{Q}_j \| \mathbf{\Pi}_j)$. Then $\mathcal{F}_0 - \mathcal{F}_m = \sum_{j=1}^m K_j^{\mathrm{KL}}$, where $K_j^{\mathrm{KL}} \geq 0$ is the discarded conditional relative entropy of step j. If the fine and coarse families have extensive sizes N_0 and N_m, the per-agent bookkeeping reads $\frac{N_0}{N_m} \cdot \frac{\mathcal{F}_0}{N_0} = \frac{\mathcal{F}_m}{N_m} + \frac{1}{N_m} \sum_{j=1}^m K_j^{\mathrm{KL}}$. [ESTABLISHED]

The proof is in appendix B.4.

10.8 Retained-label Fisher asymptotics

When log masses grow like the number of agents N, the softmax calculus of Proposition 10.24 has three regimes with different hypotheses and coordinates: a strict winner, a tie with converging weights, and a window of width $1/N$ at a tie. Fix an extensive labeled family with weights $W_{\gamma,N}$, log masses $\lambda_{\gamma,N}$, gradients $\xi_{\gamma,N} = \nabla \lambda_{\gamma,N}$, Fisher density $g_N = N^{-1} g^F(P_N)$, and branch densities $\mathcal{I}_{\gamma,N} = N^{-1} g^F(p_{\gamma,N})$ with limits $\mathcal{I}_\gamma$ when they converge; C_K depends on K and the family and may change from line to line, and limits are taken at a fixed label set.

Theorem 10.29 (Strict-phase convergence with both metric derivatives). Assume extensive label growth (10.12) on a neighborhood of a compact K, and $\mathcal{I}_{\gamma,N} \to \mathcal{I}_\gamma$ in $C^2(\mathsf{K})$ for every label. Suppose a label γ_* satisfies $\mathsf{s}_{\gamma_*} - \mathsf{s}_\gamma \geq \delta_\mathsf{K} > 0$ on K for every $\gamma \neq \gamma_*$. Then $\|g_N - \mathcal{I}_{\gamma_*}\|_{C^2(\mathsf{K})} \to 0$. More precisely, for $j = 0, 1, 2$ and all large N,

$$\left\| D^j \left[N^{-1} \operatorname{Cov}_{W_N}(\xi_{\gamma,N}) \right] \right\|_\mathsf{K} \leq \mathrm{C}_\mathsf{K} N^{j+1} e^{-\delta_\mathsf{K} N/2}, \tag{10.21}$$

and the weighted branch error $\sum_\gamma W_{\gamma,N} \mathcal{I}_{\gamma,N} - \mathcal{I}_{\gamma_*,N}$ has $C^2(\mathsf{K})$ norm at most $\mathrm{C}_\mathsf{K} N^2 e^{-\delta_\mathsf{K} N/2}$. Only three derivatives of the effective log masses are used. If moreover $\mathcal{I}_{\gamma_*} \succeq \mathrm{c}_{\min} \mathrm{Id}_d$ on K for a constant $\mathrm{c}_{\min} > 0$, then $g_N(\theta) \succeq \frac{1}{2} \mathrm{c}_{\min} \mathrm{Id}_d$ for all $\theta \in \mathsf{K}$ and all large N. [ESTABLISHED]

The proof is in appendix B.4.

Proposition 10.30 (Squared-size Fisher tensor at a finite-weight tie). Fix θ_0 and assume (10.12) on a neighborhood of θ_0. Let S be the nonempty set of labels maximizing $\mathsf{s}_\gamma(\theta_0)$, suppose the weights satisfy $W_{\gamma,N}(\theta_0) \to \bar{\mathsf{p}}_\gamma$, and assume $g^F(p_{\gamma,N})(\theta_0) = O(N)$ for every label. Write $\mathsf{v}_\gamma = \nabla \mathsf{s}_\gamma(\theta_0)$ and $\bar{\mathsf{v}} = \sum_\gamma \bar{\mathsf{p}}_\gamma \mathsf{v}_\gamma$. Then $\bar{\mathsf{p}}$ is supported on S and $N^{-2} g^F(P_N)(\theta_0) \to \mathrm{Cov}_{\bar{\mathsf{p}}}(\mathsf{v}_\gamma) = \sum_\gamma \bar{\mathsf{p}}_\gamma (\mathsf{v}_\gamma - \bar{\mathsf{v}})(\mathsf{v}_\gamma - \bar{\mathsf{v}})^\top$. Its rank is the dimension of the span of the differences $\mathsf{v}_\gamma - \mathsf{v}_\eta$ over pairs with positive limiting weights, at most $\min(d, |\operatorname{supp} \bar{\mathsf{p}}| - 1)$. A direction with positive variance has Fisher information asymptotic to a positive multiple of N^2; zero variance gives only $o(N^2)$ in that direction. [ESTABLISHED]

The proof is in appendix B.4.

Theorem 10.31 (Center ratios, smooth weights, and the full Fisher limit). Assume (10.12) and $\mathcal{I}_{\gamma,N} \to \mathcal{I}_\gamma$ in C^2 on a neighborhood of θ_0. Let S be the maximizing set of $\mathsf{s}_\gamma(\theta_0)$, choose $\gamma_0 \in S$, and assume the center-ratio condition

$$\exp\big[\lambda_{\gamma,N}(\theta_0) - \lambda_{\gamma_0,N}(\theta_0)\big] \longrightarrow \mathrm{c}_\gamma \in (0,\infty) \quad (\gamma \in S), \qquad \mathrm{c}_{\gamma_0} = 1. \tag{10.22}$$

Set $\mathsf{v}_\gamma = \nabla \mathsf{s}_\gamma(\theta_0)$ and define the window experiment on the original labeled carrier by $\theta_N(\mathsf{w}) = \theta_0 + \mathsf{w}/N$ and $\mathcal{E}_N(\mathsf{w}) = P_{N,\theta_N(\mathsf{w})}$. On every compact set of w its label law converges in C^3 to

$$\bar{\mathsf{p}}_\gamma(\mathsf{w}) = \frac{\mathrm{c}_\gamma e^{\mathsf{w}\cdot\mathsf{v}_\gamma}}{\sum_{\eta\in S} \mathrm{c}_\eta e^{\mathsf{w}\cdot\mathsf{v}_\eta}} \quad (\gamma \in S), \qquad \bar{\mathsf{p}}_\gamma(\mathsf{w}) = 0 \quad (\gamma \notin S), \tag{10.23}$$

and its Fisher tensor in the coordinate w converges in C^2:

$$g^F_{\mathsf{w}}(\mathcal{E}_N)(\mathsf{w}) = N^{-2} g^F(P_N)(\theta_N(\mathsf{w})) \longrightarrow \mathrm{Cov}_{\bar{\mathsf{p}}(\mathsf{w})}(\mathsf{v}_\gamma) = D^2_{\mathsf{w}} \log \sum_{\gamma\in S} \mathrm{c}_\gamma e^{\mathsf{w}\cdot\mathsf{v}_\gamma}. \tag{10.24}$$

The center-ratio condition is additional data at finite N; equality of the bulk densities supplies no such condition. [ESTABLISHED]

The proof is in appendix B.4.

Proposition 10.32 (Ties that diverge or do not converge). Extensive label growth, even with identical parameter-independent branches, implies neither a finite extensive Fisher density at a tie, nor convergence of the label law, nor C^2 convergence of a uniformly convergent Fisher density. For two labels $0, 1$ with one common branch law independent of θ and log-odds $\omega_N = \lambda_{1,N} - \lambda_{0,N}$,

$$W_{1,N} = \frac{e^{\omega_N}}{1+e^{\omega_N}}, \qquad g^F(P_N) = \mathrm{bv}(\omega_N)(\omega_N')^2, \qquad \mathrm{bv}(x) = \frac{e^x}{(1+e^x)^2} = \frac{1}{4\cosh^2(x/2)}, \tag{10.25}$$

and each of the following families has extensive label growth on every compact interval. (a) For $\omega_N = N\theta$ the tie at $\theta = 0$ has $g^F(P_N)(\theta) = N^2/[4\cosh^2(N\theta/2)]$ and $g^F(P_N)(0)/N = N/4$, so the extensive density diverges, and $N^{-2} g^F(P_N)$ tends to $1/4$ at zero and to zero elsewhere. (b) For ω_N equal to 0 at even N and $\log 2$ at odd N, the Fisher tensor vanishes identically while $W_{1,N}$

alternates between $1/2$ and $2/3$. (c) For $\omega_N = N^{\mathrm{c}}\theta$ with $0 < \mathrm{c} < 1$, both bulk densities vanish and

$$\frac{g^F(P_N)(0)}{N} = \frac{1}{4}N^{2\mathrm{c}-1}, \tag{10.26}$$

which is below, at, or above the extensive scale according as $\mathrm{c} < 1/2$, $\mathrm{c} = 1/2$, or $\mathrm{c} > 1/2$; for $\omega_N = \mathrm{c}_N + \sqrt{N}\theta$ with c_N alternating between 0 and $\log 2$, the density at zero alternates between $1/4$ and $2/9$. (d) For $\omega_N = N\theta - \mathrm{c}\log N$ with $\mathrm{c} > 0$, the tied label is rare, $W_{1,N}(0) = (1 + N^{\mathrm{c}})^{-1} \to 0$, yet

$$\frac{g^F(P_N)(0)}{N} = \frac{N^{1+\mathrm{c}}}{(1+N^{\mathrm{c}})^2} \sim N^{1-\mathrm{c}}, \tag{10.27}$$

so the limiting N^2 coefficient is zero while the density diverges for $\mathrm{c} < 1$, tends to one for $\mathrm{c} = 1$, and vanishes for $\mathrm{c} > 1$. (e) For $\omega_N(\theta) = N^{-1/6}\sin(N^{1/3}\theta)$ both weights converge uniformly to $1/2$ and $F_N = N^{-1}g^F(P_N)$ converges uniformly to zero, while

$$F_N'' = \frac{1}{N}\Big[\mathrm{bv}''(\omega_N)(\omega_N')^4 + 5\,\mathrm{bv}'(\omega_N)(\omega_N')^2\omega_N'' + 2\,\mathrm{bv}(\omega_N)(\omega_N'')^2 + 2\,\mathrm{bv}(\omega_N)\omega_N'\omega_N'''\Big] \tag{10.28}$$

satisfies $F_N''(0) = -\frac{1}{2} - \frac{1}{8}N^{-1/3} \to -\frac{1}{2}$. [ESTABLISHED]

The proof is in appendix B.4.

Remark 10.33 (The retained label is observed information). The covariance term of (10.14) is the Fisher tensor of the label marginal, so the N^2 scales of Proposition 10.30 and theorem 10.31 are information about the label. When all carriers are one common set, the experiment that observes only the configuration, through $(\gamma, x) \mapsto x$, loses exactly the expected conditional covariance of s_P given X, which vanishes when X determines the label; for a record the loss is given by Proposition 10.26. In the binary families of Proposition 10.32 the configuration alone has a law independent of θ and zero Fisher information, while for $\omega_N = N\theta$ the labeled window experiment has Fisher tensor $\mathrm{bv}(\mathrm{w})$ in w for every N. [ESTABLISHED]

Proof. The label marginal has score $\xi_\gamma - \bar{\xi}$. The map $(\gamma, x) \mapsto x$ is a parameter-independent statistic, so Theorem 7.8 gives the loss as the expected conditional covariance, which vanishes when the score is a function of X, in particular when X determines (γ, x). With one common θ-independent branch law p, the configuration has law $\sum_\gamma W_\gamma p = p$. For $\omega_N = N\theta$ at $\theta = \mathrm{w}/N$, (10.25) gives $N^{-2}\,\mathrm{bv}(\mathrm{w})N^2$. □

Physicist's summary. A coarse channel is a noisy detector on latent coordinates, under which relative entropy and Fisher information can only drop. Applied to the generative joint, the posterior, and the recognition law at once, it leaves the evidence untouched, and the free energy drops by exactly what it discarded about the disagreement of recognition and posterior inside each coarse cell, which is bookkeeping, not learning. A retained discrete label, such as which partition is present, carries its own Fisher information: with N agents a strict winner hides it, a tie exposes information of order N^2, and near a tie the natural coordinate is $N(\theta - \theta_0)$, as in the finite-volume rounding of a first-order transition.

Chapter 11

Gauge-Retained Network State and Fixed Sectors

Grouping the agents at c_* into blocks does not average away their gauge data: links are independent group elements and closed walks carry holonomy. This chapter asks what a coarse description of a block must retain of that data and which part of the block's state survives exact aggregation. A rooted spanning tree retains tree transports, the based holonomy, and block-dressed boundary edges as one root-framed datum (Proposition 11.5). The fixed sector of a block is the kernel of its connection Laplacian, with positive-definite weights the holonomy-fixed root values (Proposition 11.19), and transported marginal laws admit a parent law exactly when they occupy one holonomy-stabilized mode (Theorem 11.10). Cut links compress exactly into rectangular endpoint maps (Theorem 11.27), and into one group-valued link with new self terms exactly when an explicit matrix is invertible (Lemma 11.24), with one balanced choice of link and weight (Proposition 11.25) that no presentation at the level of laws reproduces once the cut has excess (Proposition 11.26) and that admits no continuous extension to all singular cuts (Proposition 11.28); the coarse frame is invisible to the constituents and cannot be computed from their frames by a symmetric equivariant rule (Proposition 11.29); and a record at c_* distinguishes loop holonomies at the level of laws (Theorem 11.30). These results supply the retained gauge data of Chapters 12 and 13, the formation witnesses of Chapter 14, and the testability discussion of Chapter 15; Supplement G realizes them for Gaussian families.

Scope. The fixed-sector rank is an admissibility criterion: it assigns no partition cost, keeps singletons admissible, and does not require constituent beliefs to coincide. The marginal-law score takes its block and weights as inputs, selects no partition, supplies no Markov kernel, and does not recover the dependence structure of a joint law. A cross map parallel for the links neither identifies the belief and model bundles nor orders their two scores, and the operational record, which sees a loop only through its represented holonomy at the probe, recovers no link values. [NOT-CLAIMED] The obstruction to a continuous extension of balanced cut links is Proposition 11.28.

11.1 Levelwise fiber state and scale intertwiners at c_*

Definition 11.1 (Levelwise fiber state). At depth ℓ fix a Lie group G_ℓ and a right G_ℓ-torsor $\mathscr{P}_\ell$, the fiber over c_* of the principal bundle of that depth, containing one frame of each agent or meta-agent

of depth ℓ, used for both channels. Law fibers $\mathcal{B}_{\ell,x}$, $x \in \{b,m\}$, carry left actions $\widehat{\rho}_{\ell,x}$ of G_ℓ, which may be inequivalent, with associated fibers $\mathcal{E}_{\ell,x} = \mathscr{P}_\ell \times_{\widehat{\rho}_{\ell,x}} \mathcal{B}_{\ell,x}$. The depth-$\ell$ interaction multigraph $\mathfrak{I}_\ell$ carries links $\Theta_e^x \in G_\ell$ with $\Theta_{\bar{e}}^x = (\Theta_e^x)^{-1}$ (Definition 2.11), every block I of depth ℓ carries its based holonomy groups Hol_I^x (Definition 11.4), and declared cross maps $\Phi_\ell : \mathcal{E}_{\ell,b} \to \mathcal{E}_{\ell,m}$ and $\widetilde{\Phi}_\ell : \mathcal{E}_{\ell,m} \to \mathcal{E}_{\ell,b}$ need not be invertible or mutually inverse. The fiber state of depth ℓ is

$$\mathfrak{X}_\ell = \Big(G_\ell, \mathscr{P}_\ell;\ \mathcal{E}_{\ell,b}, \mathcal{E}_{\ell,m};\ \mathfrak{I}_\ell, (\Theta_e^b, \Theta_e^m)_e, (\mathrm{Hol}_I^b, \mathrm{Hol}_I^m)_I;\ \Phi_\ell, \widetilde{\Phi}_\ell;\ \ldots\Big), \tag{11.1}$$

where the ellipsis stands for law, kernel, or operator components declared by a realization.

Proposition 11.2 (Scale maps and intertwiners at c_*). (i) A map $\Phi_\ell : \mathcal{E}_{\ell,b} \to \mathcal{E}_{\ell,m}$, expressed in a frame u by the fiber map f_u with $\Phi_\ell[u,w] = [u, f_u(w)]$, has one expression f_ℓ^{bm} in every frame exactly when $f_\ell^{bm} \circ \widehat{\rho}_{\ell,b}(g) = \widehat{\rho}_{\ell,m}(g) \circ f_\ell^{bm}$ for every $g \in G_\ell$, which is also the condition that Φ_ℓ commutes with every automorphism of $\mathscr{P}_\ell$ and constrains the covariance of a cross map, not its existence; the reverse direction is analogous. For a homomorphism $\kappa_\ell : G_\ell \to G_{\ell+1}$, points $p_0 \in \mathscr{P}_\ell$ and $p_1 \in \mathscr{P}_{\ell+1}$, and fiber maps $q_{\ell,x} : \mathcal{B}_{\ell,x} \to \mathcal{B}_{\ell+1,x}$, $x \in \{b,m\}$, consider

$$\mathsf{P}_\ell(p \cdot g) = \mathsf{P}_\ell(p) \cdot \kappa_\ell(g), \tag{11.2}$$

$$\widehat{q}_{\ell,x}[p,w] = [\mathsf{P}_\ell(p), q_{\ell,x}(w)], \tag{11.3}$$

$$q_{\ell,x} \circ \widehat{\rho}_{\ell,x}(g) = \widehat{\rho}_{\ell+1,x}(\kappa_\ell(g)) \circ q_{\ell,x} \qquad \text{for every } g \in G_\ell. \tag{11.4}$$

(ii) Exactly one map $\mathsf{P}_\ell : \mathscr{P}_\ell \to \mathscr{P}_{\ell+1}$ satisfies $\mathsf{P}_\ell(p_0) = p_1$ and (11.2). (iii) The formula (11.3) defines a map $\widehat{q}_{\ell,x} : \mathcal{E}_{\ell,x} \to \mathcal{E}_{\ell+1,x}$ exactly when (11.4) holds. (iv) For composable steps the homomorphisms, torsor maps, fiber maps, and induced maps compose, the rightmost factor acting first, and on law fibers given by pushforward the composite is the pushforward of the composed kernel, by (10.1). [ESTABLISHED]

Proof. (i) Each point of $\mathcal{E}_{\ell,b}$ has one representative with first entry u, and $[u \cdot g, w] = [u, \widehat{\rho}_{\ell,b}(g)w]$ gives $f_{u\cdot g} = \widehat{\rho}_{\ell,m}(g)^{-1} \circ f_u \circ \widehat{\rho}_{\ell,b}(g)$, as in (2.22). An automorphism of the torsor, $u \mapsto u \cdot \kappa$, acts by $[u,w] \mapsto [u, \widehat{\rho}(\kappa)w]$, so commuting with all of them is the same condition. (ii) The action is free and transitive, so $\mathsf{P}_\ell(p_0 \cdot g) := p_1 \cdot \kappa_\ell(g)$ is well defined, satisfies (11.2), and is forced by it. (iii) The representatives (p,w) and $(p \cdot g, \widehat{\rho}_{\ell,x}(g)^{-1}w)$ of one point have images $[\mathsf{P}_\ell(p), q_{\ell,x}(w)]$ and $[\mathsf{P}_\ell(p), \widehat{\rho}_{\ell+1,x}(\kappa_\ell(g))q_{\ell,x}(\widehat{\rho}_{\ell,x}(g)^{-1}w)]$, which agree for every (p,w,g) exactly under (11.4). (iv) The composites satisfy the defining identities by substitution. □

On a chain of depths these composites form a scale diagram (Definition 10.21 and proposition 10.22); at the information-geometric tier an induced map also needs the pushed laws to stay in the coarse model and $q_{\ell,x}$ to be smooth. For an edge copy $e : j \to i$ of $\mathfrak{I}_\ell$, the cross-scale morphism defects and the link-intertwining defects are the ordered pairs

$$\begin{aligned} \mathbf{D}_\ell^\Phi &= \big(\widehat{q}_{\ell,m} \circ \Phi_\ell,\ \Phi_{\ell+1} \circ \widehat{q}_{\ell,b}\big), & \mathbf{D}_\ell^{\widetilde{\Phi}} &= \big(\widehat{q}_{\ell,b} \circ \widetilde{\Phi}_\ell,\ \widetilde{\Phi}_{\ell+1} \circ \widehat{q}_{\ell,m}\big), \\ \mathbf{D}_e^\Phi &= \big(\widehat{\mathsf{t}}_e^m \circ \Phi_\ell,\ \Phi_\ell \circ \widehat{\mathsf{t}}_e^b\big), & \mathbf{D}_e^{\widetilde{\Phi}} &= \big(\widehat{\mathsf{t}}_e^b \circ \widetilde{\Phi}_\ell,\ \widetilde{\Phi}_\ell \circ \widehat{\mathsf{t}}_e^m\big), \end{aligned} \tag{11.5}$$

where $\widehat{\mathsf{t}}^x_e[p,w] = [\mathsf{t}^x_e(p), w]$ is induced by the link map t^x_e of Proposition 2.12. A defect vanishes when its two components are equal; subtracting them needs a vector or affine realization.

Proposition 11.3 (Covariance of the defect diagrams). The maps $\widehat{\mathsf{t}}^x_e$ are well defined and act from the frame of j to the frame of i as $\widehat{\rho}_x(\Theta^x_e)$. Under passive re-expressions at depths ℓ and $\ell+1$, both components of every pair in (11.5) acquire the same coordinate changes at source and target, so the equality locus of each pair is gauge invariant; in a fiberwise-linear realization the difference of the components transforms by the same left and right laws. [ESTABLISHED]

Proof. Right equivariance of t^x_e gives $[\mathsf{t}^x_e(p\cdot g), \widehat{\rho}_x(g)^{-1}w] = [\mathsf{t}^x_e(p), w]$, and $\mathsf{t}^x_e(u_j\cdot g) = u_i\cdot\Theta^x_e g$ gives the coordinate form. Both components of each pair are frame-free maps between the same two associated fibers, so their frame expressions are conjugated by the same coordinate changes, and equality of two maps does not depend on the frames. □

11.2 Rooted trees, based holonomy, and dressed edges

Fix a depth and suppress its index. Let $\mathcal{P}$ be a partition of the agents into blocks and $\mathfrak{I}_I$ the internal multigraph of a block I, with its loops and parallel copies. A passive re-expression $g=(g_i)$ acts by $\Theta^x_e \mapsto g_i\Theta^x_e g_j^{-1}$ for $e: j\to i$ (2.28) and by $z^x_i \mapsto \rho_x(g_i)z^x_i$ on sample coordinates (2.9).

Definition 11.4 (Rooted spanning-tree data). The component-indexed meta-agent set of $\mathcal{P}$ is

$$\widehat{\mathcal{P}} = \{(I,\mathsf{c}) : I\in\mathcal{P},\ \mathsf{c}\in\pi_0(\mathfrak{I}_I)\}. \tag{11.6}$$

From here on I, J denote elements of $\widehat{\mathcal{P}}$, each connected with one root r_I; the state of a disconnected block is the product over its components, and collapsing that product to one root, row, or feature is an additional coarse channel. Choose a rooted spanning tree in each I, nested across depths. With the path products Θ^x_p and holonomies $\mathrm{Hol}^x(p)$ of (2.30), the tree transport of $i\in I$ along the tree path $p_{I,i}$ from i to r_I, the holonomy of the fundamental cycle of a non-tree copy $e: j\to i$ of $\mathfrak{I}_I$, the based holonomy group of I at a vertex r, and the block-dressed edge of a boundary copy $e: j\to i$ from J to I are

$$\Theta^x_{I\leftarrow i} = \Theta^x_{p_{I,i}}, \qquad \mathrm{Hol}^x(p_e) = \Theta^x_{I\leftarrow i}\Theta^x_e(\Theta^x_{I\leftarrow j})^{-1}, \tag{11.7}$$

$$\mathrm{Hol}^x_I(r) = \{\Theta^x_p : p \text{ a closed walk of } \mathfrak{I}_I \text{ at } r\}\subseteq G, \qquad \mathrm{Hol}^x_I = \mathrm{Hol}^x_I(r_I), \tag{11.8}$$

$$\widehat{\Theta}^x_e = \Theta^x_{I\leftarrow i}\Theta^x_e(\Theta^x_{J\leftarrow j})^{-1}. \tag{11.9}$$

For an action ρ_x of G on the channel-x sample fiber, the root-framed child tuple $\bar{z}^x_I = (\rho_x(\Theta^x_{I\leftarrow i})z^x_i)_{i\in I}$ collects the member coordinates in the root frame, and a law Q on the root sample fiber of I is holonomy stabilized when

$$\widehat{\rho}_x(\mathsf{k})Q = Q \qquad \text{for every } \mathsf{k}\in\mathrm{Hol}^x_I, \tag{11.10}$$

which is law coherence up to holonomy stabilizers on the block (Definition 2.17).

Proposition 11.5 (Gauge laws of the rooted datum). The map $p \mapsto \Theta^x_p$ on closed walks at r_I is multiplicative under concatenation and invariant under insertion or removal of backtracks, so it is a homomorphism on the fundamental group of $\mathfrak{I}_I$ at r_I, with image Hol^x_I generated by the fundamental-cycle holonomies $\mathrm{Hol}^x(p_e)$. Under a passive re-expression g the tree transports, this homomorphism, and, for a boundary copy $e : j \to i$ from J to I, the link and the dressed edge transform as

$$\Theta^x_{I\leftarrow i} \longmapsto g_{r_I}\Theta^x_{I\leftarrow i}g_i^{-1}, \tag{11.11}$$

$$\pi_1(\mathfrak{I}_I, r_I) \longrightarrow G, \quad p \longmapsto \Theta^x_p, \qquad \Theta^x_p \longmapsto g_{r_I}\Theta^x_p g_{r_I}^{-1}, \tag{11.12}$$

$$\Theta^x_e \longmapsto g_i\Theta^x_e g_j^{-1}, \qquad \widehat{\Theta}^x_e \longmapsto g_{r_I}\widehat{\Theta}^x_e g_{r_J}^{-1}, \tag{11.13}$$

and $\bar{z}^x_I \mapsto \rho_x(g_{r_I})\bar{z}^x_I$, one root element acting on every entry. The datum consisting of $\bar{z}^x_I$, the homomorphism (11.12), and the dressed edges is therefore root framed and equivariant under one root re-expression per block. The homomorphism does not depend on the trees, and a change of tree multiplies each tree transport on the left by a based holonomy. The tree links can be set to the identity by a passive re-expression unique up to one element per block, after which each non-tree internal link equals the conjugated fundamental-cycle holonomy it closes. [ESTABLISHED]

Proof. Substitution cancels interior factors of path products and gives the law of $\bar{z}^x_I$, and a backtrack contributes $\Theta^x_e\Theta^x_{\bar{e}} = e$. Every internal copy satisfies $\Theta^x_e = (\Theta^x_{I\leftarrow i})^{-1}\mathrm{Hol}^x(p_e)\Theta^x_{I\leftarrow j}$, with $\mathrm{Hol}^x(p_e) = e$ on tree edges, so the product along a closed walk at r_I telescopes to a product of fundamental-cycle holonomies and their inverses; walks and holonomy do not involve the trees, and for a second tree $\widetilde{\Theta}^x_{I\leftarrow i}(\Theta^x_{I\leftarrow i})^{-1}$ is the product along a closed walk at r_I. Finally $g_i = g_{r_I}\Theta^x_{I\leftarrow i}$ makes every tree transport, hence every tree link, the identity, is forced up to the choice of g_{r_I}, and turns a non-tree link into $g_{r_I}\mathrm{Hol}^x(p_e)g_{r_I}^{-1}$. □

The last statement is the maximal-tree gauge fixing of lattice gauge theory [26], in the spanning-tree form of Delcamp and Dittrich [31, Sec. IV A, Eq. (4.1)]. The measure tier retains the raw root-framed datum, since quotienting each holonomy separately would lose its orientation relative to the root features and boundary legs, and an orbit space would need a quotient theorem not supplied here.

Proposition 11.6 (Two full-law treatments of holonomy for a candidate parent). In the setting of Corollary 10.11 two alternatives are available, which need not exclude each other at different retained-coordinate or quotient levels. (i) Groupoid-covariant parent: supply bimeasurable actions of a typed groupoid on the fine random space, on Z_A, on the observation space, and on the structural data; let the fine generative and recognition laws be covariant and the selected posterior versions compatible with the actions; and require $C_A(g\cdot y, g\cdot D) = C_A(y, D)$ for every admitted arrow g and measurable $D \subseteq \mathsf{Z}_A$, together with covariance of the evaluator for the actions on (m_A, ξ_A) and on (b_A, o, h_A). Then the parent generative, posterior, and recognition laws are covariant. Invariance at one slice (o, X) holds for the arrows that fix (o, X) and the selected posterior versions; invariance of q^b_A and q^m_A separately is only a projection consequence and does not replace these hypotheses. (ii) Retention: the component roots, the based holonomy homomorphisms, and the dressed boundary

edges of Proposition 11.5 are placed in H_A, and their joint law is pushed forward with no invariance claim. Neither alternative selects I, A, a partition, or a membership kernel. [ESTABLISHED]

Proof. (i) Substituting the invariance of C_A and the covariant fine laws into (10.6) and (10.7) gives covariance of the three parent laws, and evaluator covariance carries it to the evaluator tier; at a fixed slice only arrows fixing (o, X) and the versions map the slice to itself. (ii) defines the retained coordinates, and the pushforward is that of Corollary 10.11. □

11.3 Linear feature tier and tree dependence

Proposition 11.7 (Linear cross-scale maps and their tree dependence). Let $\rho_x^{\mathbb{W}} : G \to \mathrm{GL}(\mathbb{W}_x)$ be a linear representation on a feature fiber, choose weights $w_{Ii} \geq 0$ with $\sum_{i\in I} w_{Ii} = 1$, and set

$$(\mathsf{C}_x z)_I = \sum_{i\in I} w_{Ii}\rho_x^{\mathbb{W}}(\Theta^x_{I\leftarrow i})z_i, \qquad (\mathsf{S}_x \hat{z})_i = \rho_x^{\mathbb{W}}(\Theta^x_{I\leftarrow i})^{-1}\hat{z}_I \tag{11.14}$$

for a fine feature z and a coarse feature $\hat{z} = (\hat{z}_I)_I$, with $\rho^{\mathbb{W}}_{x,f}(g) = \bigoplus_i \rho_x^{\mathbb{W}}(g_i)$ and $\rho^{\mathbb{W}}_{x,c}(g) = \bigoplus_I \rho_x^{\mathbb{W}}(g_{r_I})$. (i) $\mathsf{C}_x\mathsf{S}_x = I$, the averaged feature $(\mathsf{C}_x z)_I$ is a weighted mean of the entries of the root-framed tuple, which it compresses, and $\mathsf{C}'_x = \rho^{\mathbb{W}}_{x,c}(g)\mathsf{C}_x\rho^{\mathbb{W}}_{x,f}(g)^{-1}$, $\mathsf{S}'_x = \rho^{\mathbb{W}}_{x,f}(g)\mathsf{S}_x\rho^{\mathbb{W}}_{x,c}(g)^{-1}$. (ii) At nested depths $0 \to 1 \to 2$, with $\Theta^{x,12}_{A\leftarrow I}$ the tree transport of the coarse vertex I to the root of its depth-two block A, the conditions $\Theta^{x,02}_{A\leftarrow i} = \Theta^{x,12}_{A\leftarrow I}\Theta^{x,01}_{I\leftarrow i}$ and $w^{02}_{Ai} = w^{12}_{AI}w^{01}_{Ii}$ give $\mathsf{C}^{02}_x = \mathsf{C}^{12}_x\mathsf{C}^{01}_x$ and $\mathsf{S}^{02}_x = \mathsf{S}^{01}_x\mathsf{S}^{12}_x$; nested forests alone give neither equality. (iii) For a fine message operator $\mathcal{A}_x$ with $\mathcal{A}'_x = \rho^{\mathbb{W}}_{x,f}(g)\mathcal{A}_x\rho^{\mathbb{W}}_{x,f}(g)^{-1}$, the fine-to-meta, meta-to-fine, and meta-to-meta operators $\mathcal{A}^x_{I\leftarrow j} = (\mathsf{C}_x\mathcal{A}_x)_{Ij}$, $\mathcal{A}^x_{j\leftarrow I} = (\mathcal{A}_x\mathsf{S}_x)_{jI}$, and $\mathcal{A}^x_{I\leftarrow J} = (\mathsf{C}_x\mathcal{A}_x\mathsf{S}_x)_{IJ}$ transform between the corresponding endpoint frames, and for a linear cross map $\Phi_f : \mathbb{W}_{b,f} \to \mathbb{W}_{m,f}$ with $\Phi'_f = \rho^{\mathbb{W}}_{m,f}(g)\Phi_f\rho^{\mathbb{W}}_{b,f}(g)^{-1}$ the compressed map $\Phi^c = \mathsf{C}_m\Phi_f\mathsf{S}_b$ satisfies $\Phi^{c\prime} = \rho^{\mathbb{W}}_{m,c}(g)\Phi^c\rho^{\mathbb{W}}_{b,c}(g)^{-1}$. (iv) A second family of trees gives transports $\widetilde{\Theta}^x_{I\leftarrow i} = \mathrm{Hol}^x(\gamma_i)\Theta^x_{I\leftarrow i}$, with γ_i the closed walk from r_I to i along the first tree and back along the second, and

$$\begin{aligned}(\widetilde{\mathsf{C}}_x z)_I - (\mathsf{C}_x z)_I &= \sum_{i\in I} w_{Ii}\big(\rho_x^{\mathbb{W}}(\mathrm{Hol}^x(\gamma_i)) - I\big)\rho_x^{\mathbb{W}}(\Theta^x_{I\leftarrow i})z_i,\\ (\widetilde{\mathsf{S}}_x\hat{z})_i &= \rho_x^{\mathbb{W}}(\Theta^x_{I\leftarrow i})^{-1}\rho_x^{\mathbb{W}}(\mathrm{Hol}^x(\gamma_i))^{-1}\hat{z}_I.\end{aligned} \tag{11.15}$$

The difference is not a root re-expression, so C_x, S_x, and the message operators depend on the trees through gauge-invariant quantities: for two vertices joined by two edge copies with $\mathrm{SO}(2)$ links the identity and the rotation by ϑ, the defining representation, equal weights, equal features e_1, and root at the first vertex, the two trees give averaged features of norms 1 and $|\cos(\vartheta/2)|$, although these norms are invariant under every root re-expression. Tree independence holds for inputs whose root-framed features lie in the fixed space $\mathrm{Fix}(\rho_x^{\mathbb{W}} \circ \mathrm{Hol}^x_I) = \{v \in \mathbb{W}_x : \rho_x^{\mathbb{W}}(\mathsf{k})v = v \text{ for every } \mathsf{k} \in \mathrm{Hol}^x_I\}$, for S_x on coarse features in that subspace, and for all inputs when $\rho_x^{\mathbb{W}} \circ \mathrm{Hol}^x_I$ is trivial. [ESTABLISHED]

Proof. (i) $(\mathsf{C}_x\mathsf{S}_x\hat{z})_I = \sum_i w_{Ii}\hat{z}_I = \hat{z}_I$, and substituting (11.11) gives the covariance laws. (ii) Substitute the two conditions into the composites. (iii) follows from (i). (iv) The closed walk γ_i has path product $\widetilde{\Theta}^x_{I\leftarrow i}(\Theta^x_{I\leftarrow i})^{-1}$, which gives (11.15). In the example the two trees give e_1 and

$\frac{1}{2}(e_1 + R_\vartheta e_1)$, of norm $|\cos(\vartheta/2)|$. If every $\rho_x^{\mathbb{W}}(\Theta^x_{I\leftarrow i})z_i$ is fixed by the represented holonomy, the difference vanishes, and likewise for S_x. □

This fixed space is the feature-level counterpart of (11.10); a tree choice is a gauge fixing for the retained datum but part of the model for the averaged feature $\mathsf{C}_x z$.

Proposition 11.8 (Composite trees and one rooted forest compose the tree transports). At nested depths $0 \to 1 \to 2$ as in Proposition 11.7(ii), fix a channel x with links $\Theta^x_e \in G$, let the blocks I of depth one partition V, let the depth-one multigraph $\mathfrak{I}_1$ of Definition 13.17 carry the dressed marks $\Theta^{x,1}_e = \widehat{\Theta}^x_e$ as links, and let the blocks A of depth two partition the vertices of $\mathfrak{I}_1$, every block connected; A also denotes the set of agents of its children $I \subseteq A$, with internal multigraph $\mathfrak{I}_A$ in $\mathfrak{I}$. The transports $\Theta^{x,01}_{I\leftarrow i}$, $\Theta^{x,12}_{A\leftarrow I}$, and $\Theta^{x,02}_{A\leftarrow i}$ are taken along rooted spanning trees of $\mathfrak{I}_I$, of the internal multigraph of A in $\mathfrak{I}_1$, and of $\mathfrak{I}_A$. (i) For any such trees of the children, with roots r_I, and of A in $\mathfrak{I}_1$, with root child I_A, the copies of these trees form a spanning tree of $\mathfrak{I}_A$, the composite tree of A. With the composite tree and the root r_{I_A}, every agent $i \in I \subseteq A$ and every copy $e : j \to i$ with $j \in J \subseteq A'$ satisfy

$$\begin{aligned} \Theta^{x,02}_{A\leftarrow i} &= \Theta^{x,12}_{A\leftarrow I}\Theta^{x,01}_{I\leftarrow i}, \\ \Theta^{x,12}_{A\leftarrow I}\widehat{\Theta}^x_e\big(\Theta^{x,12}_{A'\leftarrow J}\big)^{-1} &= \Theta^{x,02}_{A\leftarrow i}\Theta^x_e\big(\Theta^{x,02}_{A'\leftarrow j}\big)^{-1}, \end{aligned} \tag{11.16}$$

so the depth-two link obtained by lifting twice with (13.31) is the one-step lift through the composite trees: for $A' \neq A$ a dressed edge (11.9), and for $A' = A$ the fundamental-cycle holonomy $\mathrm{Hol}^x(p_e)$ of (11.7) for the composite tree, which is e_G on its copies. The transport condition of Proposition 11.7(ii) therefore holds, and with the weights $w^{02}_{Ai} = w^{12}_{AI}w^{01}_{Ii}$, which sum to one over A, both equalities of that part hold. (ii) Choose, before the partitions and independently of them, a rooted spanning forest of $\mathfrak{I}$, that is, a rooted spanning tree of each connected component of $\mathfrak{I}$, and let every block be a connected subtree of it, meaning that the forest copies joining agents of the block connect the block. Each block then has exactly one agent nearest to the root of its component, measured by the number of copies on the forest path; call it r_I for a child and r_A for a block of depth two. Give each child the forest copies joining its agents, rooted at r_I, and give A in $\mathfrak{I}_1$ the forest copies joining distinct children of A, rooted at the child I_A that contains r_A. These are rooted spanning trees, $r_{I_A} = r_A$, and the composite tree of A consists of the forest copies joining agents of A, so (i) applies to the forest tree of A rooted at r_A, and

$$\Theta^{x,12}_{A\leftarrow I} = \Theta^{x,02}_{A\leftarrow r_I}, \qquad \Theta^{x,02}_{A\leftarrow i} = \Theta^{x,02}_{A\leftarrow r_I}\Theta^{x,01}_{I\leftarrow i}. \tag{11.17}$$

For a longer chain of nested partitions whose blocks are all connected subtrees of the forest, the trees induced at every depth ℓ, with the agents as the blocks of depth zero, formed by the forest copies inside a block of depth $\ell + 1$ that join distinct blocks of depth ℓ and rooted at the block of depth ℓ that contains its nearest agent, satisfy the hypotheses of (i) at the depths $\ell \to \ell + 1 \to \ell + 2$ for every ℓ. (iii) If the links of channel x are graph flat on every block of depth two (Definition 2.14), in particular if they are frame aligned, (11.16) holds with any rooted spanning tree of $\mathfrak{I}_A$ with root r_{I_A} in place of the composite tree. The transports in (i) and (ii) are path products of the group-valued links, whose reversed copies carry the inverse links; a matrix communication map L_e of Definition 6.12,

whose reversed copy is specified separately, is not a transport, and on the vector tier, where ρ_x acts linearly on the sample fibers, its dressings by the represented transports of (i) satisfy

$$\begin{aligned}\rho_x(\Theta^{x,12}_{A\leftarrow I})\big[\rho_x(\Theta^{x,01}_{I\leftarrow i})\mathsf{L}_e\rho_x(\Theta^{x,01}_{J\leftarrow j})^{-1}\big]\rho_x(\Theta^{x,12}_{A'\leftarrow J})^{-1}\\ = \rho_x(\Theta^{x,02}_{A\leftarrow i})\mathsf{L}_e\rho_x(\Theta^{x,02}_{A'\leftarrow j})^{-1},\end{aligned} \tag{11.18}$$

which inverts only represented transports, so L_e may be singular. [ESTABLISHED]

Proof. Deleting a backtrack leaves a path product unchanged, since $\Theta^x_e\Theta^x_{\bar e} = e_G$, and a walk in a tree reduces in this way to the tree path between its endpoints, because a closed walk of positive length in a tree backtracks at a vertex farthest from its base point. The argument for (i) uses only that the links are elements of G whose reversed copies carry the inverses, which holds at every depth by Proposition 13.18. (i) Counting reversal pairs, the trees of the children contain $|A|$ less the number of children, each pair joining agents of one child, and the tree of A in $\mathfrak{I}_1$ contains the number of children less one, each pair joining distinct children; together they connect the agents of A with $|A|-1$ pairs, so they form a spanning tree. Let the tree path p from I to I_A in $\mathfrak{I}_1$ traverse the copies $e_a : j_a \to i_a$ from I_{a-1} to I_a for $a = 1,\dots,|p|$, oriented along the path, with $I_0 = I$, $I_{|p|} = I_A$, and $i_0 = i$. By (13.31),

$$\Theta^{x,12}_{A\leftarrow I}\Theta^{x,01}_{I\leftarrow i} = \Theta^{x,01}_{I_{|p|}\leftarrow i_{|p|}}\Theta^x_{e_{|p|}}\big[(\Theta^{x,01}_{I_{|p|-1}\leftarrow j_{|p|}})^{-1}\Theta^{x,01}_{I_{|p|-1}\leftarrow i_{|p|-1}}\big]\Theta^x_{e_{|p|-1}}\cdots\Theta^x_{e_1}\big[(\Theta^{x,01}_{I_0\leftarrow j_1})^{-1}\Theta^{x,01}_{I_0\leftarrow i_0}\big].$$

Each bracket is the path product of a walk in the tree of I_a from i_a through r_{I_a} to j_{a+1}, and the leftmost factor is that of the tree path from $i_{|p|}$ to r_{I_A}, so the right side is the path product of a walk in the composite tree from i to r_{I_A}, which is $\Theta^{x,02}_{A\leftarrow i}$. Substituting this at both endpoints of $\widehat\Theta^x_e = \Theta^{x,01}_{I\leftarrow i}\Theta^x_e(\Theta^{x,01}_{J\leftarrow j})^{-1}$ gives the second equality, whose right side is (11.9) for $A' \neq A$ and (11.7) for $A' = A$, equal to e_G on tree copies as in Proposition 13.18. The weights are nonnegative with $\sum_{I\subseteq A} w^{12}_{AI}\sum_{i\in I} w^{01}_{Ii} = 1$, so Proposition 11.7(ii) applies. (ii) If two distinct agents of a block were nearest to the root, the forest path between them would lie in the block, since the forest copies inside the block connect it and paths in a tree are unique, and it would pass through their last common ancestor, which is nearer to the root. The forest copies inside a block connect it and contain no cycle, so they form a spanning tree of its internal multigraph. Inside A lie $|A|-1$ forest pairs, of which $|A|$ less the number of children lie inside children, so the number of children less one join distinct children; these pairs connect the children, because the forest copies inside A connect A, so they form a spanning tree of A in $\mathfrak{I}_1$, and with the trees of the children they make up the forest copies inside A. The agent r_A lies in $I_A \subseteq A$ and is nearest in A, hence in I_A, so $r_{I_A} = r_A$. Then (i) applies; putting $i = r_I$, where $\Theta^{x,01}_{I\leftarrow r_I} = e_G$, gives the first equality of (11.17), and substituting it into (i) gives the second. For a longer chain the same count shows that each induced tree is a spanning tree. For a block of depth $\ell+2$, its induced tree at depth ℓ is the union of the induced trees at depth ℓ of its children and of the forest copies joining distinct children, which form its induced tree at depth $\ell+1$, and the block of depth ℓ that contains its nearest agent is the root at depth ℓ of the child containing that agent, whose nearest agent it is. (iii) Graph-flat links have path product e_G on every closed walk in the block (Proposition 2.13), and another tree with the same root multiplies each transport on the left by a based holonomy (Proposition 11.5), which leaves

$\Theta^{x,02}_{A\leftarrow i}$ unchanged; frame-aligned links are graph flat (Remark 2.15). The matrix identity follows from the first equality of (11.16) at i and at j and the multiplicativity of ρ_x. □

Part (i) supplies what nesting alone leaves open in Proposition 11.7(ii): a depth-zero tree of A that contains the depth-one tree and is rooted at the root of the root child; another tree with that root multiplies the direct transports $\Theta^{x,02}_{A\leftarrow i}$ by based holonomies (Proposition 11.5). In (i) the trees may be chosen after each partition is realized, and the composite tree then depends on the intermediate partition. In (ii) the forest is part of the blocking scheme and is chosen before the partitions, so the direct transports of A depend only on A and the forest; a random membership mark R_ℓ (Definition 13.14) enters (ii) only through realized partitions whose blocks are connected subtrees of that forest, a realized block that is not such a subtree is outside (ii), and overlapping membership kernels are outside the proposition, as in Definition 13.17.

Setting the tree links to the identity is the spanning-tree gauge fixing of lattice gauge theory in the form of Delcamp and Dittrich [31, Sec. IV A, Eq. (4.1)], whose change of spanning tree at a fixed root is their Sec. IV C, Eq. (4.4); the coarse link as an ordered product of fine links along a composite path is the macroscopic holonomy of Bahr [6, Sec. 4, Eq. (4.1)]. Neither source composes trees across nested partitions.

11.4 Holonomy-conditioned marginal-law modes

Definition 11.9 (Holonomy-conditioned marginal-law score). Fix a channel x, a connected block I with root r, weights $w_i > 0$ with $\sum_{i\in I} w_i = 1$, and marginal laws q^x_i of the agents of I on the sample fiber Z^x, on which G acts by bimeasurable bijections $\rho_x(g)$ with pushforward $\widehat{\rho}_x(g)$. Let parent-law families $\mathscr{M}^x_i \subseteq \mathrm{Prob}(\mathsf{Z}^x)$ be equivariant: $\widehat{\rho}_x(\Theta^x_p)\mathscr{M}^x_j = \mathscr{M}^x_i$ for every walk p of $\mathfrak{I}_I$ from j to i, and $\mathscr{M}^{x\prime}_i = \widehat{\rho}_x(g_i)\mathscr{M}^x_i$ under a passive re-expression. For walks p_i from i to r put $\bar{q}^x_i = \widehat{\rho}_x(\Theta^x_{p_i})q^x_i$, and with the based holonomy group $\mathrm{Hol}^x_I(r)$ of (11.8) define the fixed parent family and the score

$$\mathscr{Q}^x_{I,\mathrm{fix}}(r) = \big\{Q \in \mathscr{M}^x_r : \widehat{\rho}_x(\mathsf{k})Q = Q \text{ for every } \mathsf{k} \in \mathrm{Hol}^x_I(r)\big\}, \tag{11.19}$$

$$\mathfrak{D}^x_I = \inf_{Q\in\mathscr{Q}^x_{I,\mathrm{fix}}(r)} \sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}^x_i \| Q), \qquad \inf\varnothing := +\infty. \tag{11.20}$$

A minimizer, when one exists, is a holonomy-conditioned marginal-law mode.

Theorem 11.10 (Invariant holonomy-conditioned marginal-law mode). The score $\mathfrak{D}^x_I$ does not depend on the walks, the root, or the passive frames. If the infimum is attained, then $\mathfrak{D}^x_I = 0$ exactly when some $Q_\star \in \mathscr{Q}^x_{I,\mathrm{fix}}(r)$ satisfies $\bar{q}^x_i = Q_\star$ for every $i \in I$, and the laws $\widehat{\rho}_x(\Theta^x_{p_i})^{-1}Q_\star$ then form a holonomy-stabilized parallel marginal-law section. If $\mathscr{Q}^x_{I,\mathrm{fix}}(r) = \varnothing$, then $\mathfrak{D}^x_I = +\infty$. If $\mathfrak{D}^x_I = 0$ without attainment, the conclusion is only forward-KL approximability by fixed parent laws. [ESTABLISHED]

Proof. Relative entropy is invariant under bimeasurable bijections (3.24). Another walk p'_i from i to r replaces $\bar{q}^x_i$ by $\widehat{\rho}_x(\mathsf{k})\bar{q}^x_i$ with $\mathsf{k} = \Theta^x_{p'_i}(\Theta^x_{p_i})^{-1} \in \mathrm{Hol}^x_I(r)$, and for every fixed parent Q,

$D_{\mathrm{KL}}(\widehat{\rho}_x(\mathsf{k})\bar{q}_i^x\|Q) = D_{\mathrm{KL}}(\widehat{\rho}_x(\mathsf{k})\bar{q}_i^x\|\widehat{\rho}_x(\mathsf{k})Q) = D_{\mathrm{KL}}(\bar{q}_i^x\|Q)$, so every objective value is path independent. A walk δ from r to r' conjugates $\mathrm{Hol}_I^x(r)$ onto $\mathrm{Hol}_I^x(r')$, and pushforward by Θ_δ^x maps the two fixed families bijectively, by equivariance, and the transported laws of r to those of r', preserving every term. A passive re-expression sends Θ_p^x to $g_i\Theta_p^x g_j^{-1}$ for p from j to i and q_i^x to $\widehat{\rho}_x(g_i)q_i^x$, so it pushes every transported law by $\widehat{\rho}_x(g_r)$, conjugates the holonomy group by g_r, and carries the fixed family bijectively, and invariance of relative entropy gives gauge invariance. Relative entropy vanishes exactly between equal laws and every w_i is positive, which gives the zero case; since $Q_\star$ is holonomy fixed, the reconstructed laws do not depend on the walks and are law coherent by Proposition 2.18. The empty case is the convention, and without attainment the infimum supplies Q_n with score below $1/n$ and no minimizer. □

Proposition 11.11 (Compact holonomy: closed form and attainment). In the setting of Theorem 11.10, let the represented holonomy group $\mathfrak{H}_r = \rho_x(\mathrm{Hol}_I^x(r))$ be a compact metrizable group acting jointly measurably on Z^x, with Haar probability $d\mathsf{k}$, and put

$$\bar{P} = \sum_{i\in I} w_i\bar{q}_i^x, \qquad \bar{M}(B) = \int_{\mathfrak{H}_r} \bar{P}(\mathsf{k}^{-1}B)d\mathsf{k}. \tag{11.21}$$

Then $\bar{M}$ is $\mathfrak{H}_r$-invariant, and every $\mathfrak{H}_r$-invariant law Q satisfies, in $[0,+\infty]$,

$$\sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}_i^x\|Q) = \sum_{i\in I} w_i D_{\mathrm{KL}}(\bar{q}_i^x\|\bar{M}) + D_{\mathrm{KL}}(\bar{M}\|Q). \tag{11.22}$$

Consequently $\mathfrak{D}_I^x = \sum_i w_i D_{\mathrm{KL}}(\bar{q}_i^x\|\bar{M}) + \inf_{Q\in\mathscr{Q}_{I,\mathrm{fix}}^x(r)} D_{\mathrm{KL}}(\bar{M}\|Q)$. If $\bar{M}\in\mathscr{M}_r^x$, for instance when $\mathscr{M}_r^x = \mathrm{Prob}(\mathsf{Z}^x)$, the infimum is attained at $\bar{M}$, the score is the holonomy-symmetrized information radius $\sum_i w_i D_{\mathrm{KL}}(\bar{q}_i^x\|\bar{M})$, and $\bar{M}$ is the unique minimizer when that value is finite. If instead $\mathsf{Z}^x = \mathbb{R}^K$, the holonomy acts linearly, $\mathscr{M}_r^x$ is the nondegenerate Gaussian family, and $\bar{M}$ has finite second moments, mean $\bar{m}$, nonsingular covariance $\bar{S}$, and $D_{\mathrm{KL}}(\bar{M}\|\mathcal{N}(\bar{m},\bar{S})) < \infty$, then $\mathcal{N}(\bar{m},\bar{S})$ is holonomy invariant and attains the infimum, uniquely when $\mathfrak{D}_I^x < \infty$. For a finite holonomy group $d\mathsf{k}$ is uniform; a holonomy group that is only relatively compact in a group acting jointly continuously on a Polish fiber has the same fixed laws as its closure, to which the statement applies; and when every $\bar{q}_i^x$ is a nondegenerate Gaussian and the holonomy acts linearly with compact closure, the Gaussian hypotheses hold and the score is finite. [ESTABLISHED]

Proof. Left invariance of $d\mathsf{k}$ gives invariance of $\bar{M}$. Since $d\bar{q}_i^x/d\bar{P} \le w_i^{-1}$, splitting $\log(d\bar{q}_i^x/dQ)$ through $\bar{P}$ gives, for every Q, the compensation identity $\sum_i w_i D_{\mathrm{KL}}(\bar{q}_i^x\|Q) = \sum_i w_i D_{\mathrm{KL}}(\bar{q}_i^x\|\bar{P}) + D_{\mathrm{KL}}(\bar{P}\|Q)$ in $[0,+\infty]$: the negative part of the second term is integrable because $t(\log t)^- \le e^{-1}$, and both sides are infinite unless $\bar{P}\ll Q$. Applied with Q and with $\bar{M}$, it reduces (11.22) to $D_{\mathrm{KL}}(\bar{P}\|Q) = D_{\mathrm{KL}}(\bar{P}\|\bar{M}) + D_{\mathrm{KL}}(\bar{M}\|Q)$ for invariant Q, where both sides are infinite unless $\bar{P}\ll Q$. If $\bar{P}\ll Q$ with density f, invariance of Q gives $\bar{M}$ the invariant density $\bar{f}(z) = \int f(\mathsf{k}^{-1}z)d\mathsf{k}$; the invariant set $\{\bar{f}=0\}$ is null for $\bar{M}$ and hence for $\bar{P}$, so $\bar{P}\ll\bar{M}\ll Q$. Averaging the invariant function $\log\bar{f}$ over the group by Tonelli's theorem gives $\int\log\bar{f}d\bar{P} = \int\log\bar{f}d\bar{M} = D_{\mathrm{KL}}(\bar{M}\|Q)$ with finite negative parts, and integrating $\log f = \log(f/\bar{f}) + \log\bar{f}$ against $\bar{P}$ proves the identity. Attainment and uniqueness follow because relative entropy vanishes only between equal laws. In the Gaussian

case, invariance of $\bar{M}$ and linearity give $\mathsf{k}\bar{m} = \bar{m}$ and $\mathsf{k}\bar{S}\mathsf{k}^\top = \bar{S}$, and $\log[d\mathcal{N}(\bar{m}, \bar{S})/d\mathcal{N}(\mu, \Sigma)]$ is a quadratic polynomial whose $\bar{M}$- and $\mathcal{N}(\bar{m}, \bar{S})$-expectations agree, which gives $D_{\mathrm{KL}}(\bar{M}\|\mathcal{N}(\mu, \Sigma)) = D_{\mathrm{KL}}(\bar{M}\|\mathcal{N}(\bar{m}, \bar{S})) + D_{\mathrm{KL}}(\mathcal{N}(\bar{m}, \bar{S})\|\mathcal{N}(\mu, \Sigma))$. For a relatively compact group, $\mathsf{k} \mapsto \int \phi(\mathsf{k}z)Q(dz)$ is continuous for bounded continuous ϕ by dominated convergence, so fixed laws of the group and of its closure agree. For Gaussian constituents with compact linear holonomy, $\bar{M}$ is a mixture of Gaussians with covariances in a compact set of positive-definite matrices, so its density is bounded and its second moments are finite, and convexity of relative entropy in its second argument bounds each $D_{\mathrm{KL}}(\bar{q}_i^x\|\bar{M})$ by an average of Gaussian relative entropies continuous on the compact group. □

Corollary 11.12 (Noncompact Gaussian holonomy leaves no admissible parent). Let $\mathsf{Z}^x = \mathbb{R}^K$, let the holonomy act by invertible linear maps, and let $\mathscr{M}_r^x$ be the nondegenerate Gaussian family. Then $\mathscr{Q}_{I,\mathrm{fix}}^x(r) \neq \varnothing$ if and only if the represented holonomy group is conjugate into $\mathrm{O}(K)$. If one loop holonomy has an eigenvalue off the unit circle, for instance $|\det| \neq 1$, or is not diagonalizable, then $\mathfrak{D}_I^x = +\infty$ for every configuration of marginals. For a block containing a loop with independent $\mathrm{GL}(K)$ links, the link configurations with $|\det| = 1$ on that loop form a Lebesgue-null set, so the score is $+\infty$ off a null set of configurations; it distinguishes configurations only after the links are restricted to a group conjugate into $\mathrm{O}(K)$ or a parent family with a nonempty fixed set is declared. [ESTABLISHED]

Proof. $\mathcal{N}(\mu, \Sigma)$ with $\Sigma \succ 0$ is fixed by k exactly when $\mathsf{k}\mu = \mu$ and $\mathsf{k}\Sigma\mathsf{k}^\top = \Sigma$. If one such Σ is fixed by the whole group, then $\Sigma^{-1/2}\mathsf{k}\Sigma^{1/2}$ is orthogonal for every element; conversely, if $S^{-1}\mathsf{k}S \in \mathrm{O}(K)$ for every element, then $\mathcal{N}(0, SS^\top)$ is fixed. An element conjugate into $\mathrm{O}(K)$ is diagonalizable with eigenvalues of modulus one, and $\det(\mathsf{k}\Sigma\mathsf{k}^\top) = (\det \mathsf{k})^2 \det \Sigma$; an empty fixed family gives $+\infty$ by convention. The holonomy of a simple loop is a product of link matrices in which one chosen link Θ enters once, as $A_1\Theta^{\pm 1}A_2$ with A_1, A_2 fixed by the other links; this is a diffeomorphism of $\mathrm{GL}(K)$, so the preimage of the null set $\{|\det| = 1\}$ is null, and Fubini's theorem over the remaining links completes the argument. □

With latent links (Chapter 9), the generators carry a law, and the score of Definition 11.9 becomes a function of them.

Corollary 11.13 (Latent noncompact links empty the Gaussian parent family almost surely). Let $G = \mathrm{GL}^+(K)$ with $K \geq 1$ act on the sample fiber $\mathsf{Z}^x = \mathbb{R}^K$ of a channel x with latent links by its defining representation, let the parent family be the nondegenerate Gaussian family as in Corollary 11.12, and let I be a connected block, as in Definition 11.9, whose internal multigraph $\mathfrak{I}_I$ is not a forest, that is, has a loop copy, two parallel copies, or a cycle through three or more agents. Then $\mathfrak{I}_I$ has a closed walk $p = (e_0, \ldots, e_{r-1})$ that traverses some reversal pair a different number of times in its two orientations, and for every such walk the following hold. (i) For every choice of frames, $\det \mathrm{Hol}^x(p) = \exp\big(\sum_{a=0}^{r-1} \mathrm{Tr}\, \mathring{\omega}_{e_a}^x\big)$, with $\mathring{\omega}_{\bar{e}}^x = -\mathring{\omega}_e^x$. (ii) The generator families with $\sum_a \mathrm{Tr}\, \mathring{\omega}_{e_a}^x = 0$ form a hyperplane of $\mathscr{T}$, which is Lebesgue-null, and for every generator family off this hyperplane the fixed parent family (11.19) is empty and $\mathfrak{D}_I^x = +\infty$ for every configuration of marginals. (iii) Under every law of the generators that is absolutely continuous with respect

to Lebesgue measure on $\mathscr{T}$, in particular the generator prior of Definition 9.11 whenever it has a density, as it does when its base law is Gaussian with positive-definite covariances, and every recognition law absolutely continuous with respect to such a prior, $\mathfrak{D}_I^x = +\infty$ almost surely, and its expectation is $+\infty$. (iv) A law of the generators under which $\mathfrak{D}_I^x$ is finite with positive probability charges the hyperplane of (ii); the point mass at $\mathring{\omega} = 0$ is such a law for nondegenerate Gaussian marginals. (v) For $G = \mathrm{SL}(2,\mathbb{R})$ acting on $\mathbb{R}^2$ by its defining representation and a block whose internal multigraph has cycle rank one, the generator families at which the represented holonomy group is conjugate into $\mathrm{SO}(2)$ contain a nonempty open subset of $\mathscr{T}$; on it the fixed parent family is nonempty, and every law of the generators with an everywhere positive density gives it positive probability. [ESTABLISHED]

Proof. If $\mathfrak{I}_I$ is a forest, every copy is a bridge, and a closed walk crosses a bridge as often in one orientation as in the other, because it returns to the side it started from. A loop copy traversed once, the walk $(e, \bar{e}')$ through two parallel copies e, e', and a cycle traversed once each traverse some reversal pair once in one orientation and never in the other. (i) By (2.33), $\mathrm{Hol}^x(p) = U_{i_0} e^{\mathring{\omega}^x_{e_0}} \cdots e^{\mathring{\omega}^x_{e_{r-1}}} U_{i_0}^{-1}$, and $\det e^{\mathbf{B}} = e^{\mathrm{Tr}\,\mathbf{B}}$ by Jacobi's formula. (ii) Collecting terms, $\sum_a \mathrm{Tr}\,\mathring{\omega}^x_{e_a} = \sum_{e \in E_+} \mathrm{n}_e\,\mathrm{Tr}\,\mathring{\omega}^x_e$, where n_e is the number of traversals of e minus that of $\bar{e}$. Some n_e is nonzero, and the family with $\mathring{\omega}^x_e = I_K$ and every other generator zero gives the value $\mathrm{n}_e K \neq 0$, so the functional is a nonzero linear form and its zero set is a hyperplane. Off the hyperplane $|\det \mathrm{Hol}^x(p)| \neq 1$ by (i). For the root r of I, the closed walk at r that follows a walk from r to i_0, then p, then the reverse walk has a holonomy conjugate to $\mathrm{Hol}^x(p)$, with the same determinant, so a loop holonomy of the block has an eigenvalue off the unit circle, and Corollary 11.12 gives an empty fixed family and $\mathfrak{D}_I^x = +\infty$ for every configuration of marginals. (iii) Such a law gives the null hyperplane probability zero, and a law absolutely continuous with respect to one that is absolutely continuous with respect to Lebesgue measure is itself absolutely continuous with respect to Lebesgue measure. The generator prior $Z_{\mathscr{J}}^{-1} e^{-\mathscr{J}} \nu^{\mathscr{T}}$ of Definition 9.11 is absolutely continuous with respect to its base law, which has a Lebesgue density when it is Gaussian with positive-definite covariances. (iv) The first statement is (ii). At $\mathring{\omega} = 0$ every twist is the identity, so every holonomy is the identity by (2.33); the trivial group is compact, and Proposition 11.11 gives a finite score for nondegenerate Gaussian marginals. (v) With cycle rank one, Hol_I^x is generated by one fundamental-cycle holonomy k (Proposition 11.5), which by (2.33) is conjugate to an ordered product of exponentials of the generators along the cycle, so $\mathrm{Tr}\,\mathsf{k}$ is a continuous function of $\mathring{\omega}$. At the generator family with $\frac{\pi}{2}\mathfrak{J}$ on the non-tree copy, $\mathfrak{J} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and zero elsewhere, k is conjugate to the rotation by $\frac{\pi}{2}$, so the set where $|\mathrm{Tr}\,\mathsf{k}| < 2$ is open and nonempty. There $\mathsf{k} \in \mathrm{SL}(2,\mathbb{R})$ has the eigenvalues $e^{\pm \mathrm{i}\phi}$ with $\sin\phi \neq 0$; the real and imaginary parts of an eigenvector for $e^{\mathrm{i}\phi}$ form a basis of $\mathbb{R}^2$ in which k acts as a rotation, so the group generated by k is conjugate into $\mathrm{SO}(2)$, and Corollary 11.12 gives a nonempty fixed family. A positive density has positive integral over a nonempty open set. □

The null-set sentence of Corollary 11.12 measures link configurations in $\mathrm{GL}(K)$. With latent generators the exceptional set is the explicit hyperplane of (ii), linear in the generators because $\log\det e^{\mathbf{B}} = \mathrm{Tr}\,\mathbf{B}$, and it depends neither on the frames, nor on the trace-free components of the generators, nor on whether they commute. The score is then not a statistic of the recognition law:

it is $+\infty$ under every law with a density, and only a law that charges the hyperplane, such as a point mass at a held link family, can make it finite. By (v), on $\mathrm{SL}(2,\mathbb{R})$, which has no determinant direction, a set of generator families of positive probability keeps a nonempty fixed family.

Remark 11.14 (Holonomy score on a three-agent loop). On the triangle of the running example (Section 2.5), now with belief sample fiber $\mathbb{R}^2$ and links in $\mathrm{SO}(2)$ acting by rotation, root the block at the first agent, take the tree edges joining it to the other two, and let the loop holonomy be a rotation by ϕ with $\sin\phi \neq 0$. For transported marginals $\bar{q}_i^b = \mathcal{N}(m_i, \sigma^2 I_2)$ with weights w_i and the nondegenerate Gaussian parent family, the fixed family is $\{\mathcal{N}(0, sI_2) : s > 0\}$ and $\mathfrak{D}_I^b = \log(1 + \frac{1}{2\sigma^2}\sum_i w_i |m_i|^2)$, attained at $s = \sigma^2 + \frac{1}{2}\sum_i w_i |m_i|^2$, the moment-matched Gaussian of the Haar-symmetrized mixture; rotations preserve norms, so the score does not depend on the walks. With frame-aligned links on the same triangle the holonomy is trivial, every nondegenerate Gaussian is fixed, and the score is $\frac{1}{2}\log\det(I_2 + \mathrm{Cov}_w(m)/\sigma^2)$, which never exceeds the value above. Replacing one link by a matrix of determinant 2 empties the fixed family and makes the score $+\infty$ (Corollary 11.12). [ESTABLISHED]

Proof. A fixed Gaussian has mean fixed by R_ϕ, hence zero because $R_\phi - I$ is invertible, and a symmetric covariance commuting with R_ϕ; the linear conditions on its off-diagonal entry and the difference of its diagonal entries have determinant $2\sin^2\phi \neq 0$, so the covariance is scalar. In dimension two, $D_{\mathrm{KL}}(\mathcal{N}(m, \sigma^2 I_2) \| \mathcal{N}(0, sI_2)) = \sigma^2/s + |m|^2/(2s) - 1 + \log(s/\sigma^2)$, whose weighted sum is minimized at the stated s with the stated value. In the flat case moment matching gives $\mathcal{N}(\bar{m}, \sigma^2 I_2 + \mathrm{Cov}_w(m))$ and the stated value. With c_1, c_2 the eigenvalues of $\mathrm{Cov}_w(m)/\sigma^2$ and $\sum_i w_i |m_i|^2 = |\bar{m}|^2 + \mathrm{tr}\,\mathrm{Cov}_w(m)$, the inequality of arithmetic and geometric means gives $\frac{1}{2}\log((1 + c_1)(1 + c_2)) \leq \log(1 + \frac{1}{2}(c_1 + c_2))$, which is at most the holonomy value. A link of determinant 2 gives the loop holonomy a determinant of modulus 2. □

Definition 11.15 (Exact common recovery). A statistical experiment $\{P_\theta\}_{\theta\in\Theta}$ admits exact common recovery through a parameter-independent Markov kernel $K_\downarrow$ when a parameter-independent Markov kernel $K_\uparrow$ satisfies

$$P_\theta K_\downarrow K_\uparrow = P_\theta \qquad \text{for every } \theta \in \Theta. \tag{11.23}$$

Remark 11.16 (A zero score is not common recovery). The score (11.20) depends on the laws of the block only through their transported marginals. When both marginals of a pair of agents are non-Dirac, there are joint laws $P_1 \neq P_2$ with the same marginals (Proposition 3.17), and every kernel $K_\downarrow$ for which $PK_\downarrow$ depends on P only through its marginals sends them to one law, so the experiment $\{P_1, P_2\}$ admits no common recovery through it. [ESTABLISHED]

Proof. If $P_1 K_\downarrow = P_2 K_\downarrow$, a kernel $K_\uparrow$ recovering both would give $P_1 = P_1 K_\downarrow K_\uparrow = P_2 K_\downarrow K_\uparrow = P_2$. □

Corollary 11.17 (Parallel cross morphisms preserve fixed marginal-law sections). Let $\Phi : \mathcal{E}_b \to \mathcal{E}_m$ be a cross map, expressed in the frame of agent i by $\Phi_i : \mathcal{B}_b \to \mathcal{B}_m$, whose link-intertwining defects (11.5) vanish on the internal copies of the block, so that $\widehat{\rho}_m(\Theta_p^m) \circ \Phi_j = \Phi_i \circ \widehat{\rho}_b(\Theta_p^b)$ for

every walk p from j to i, and assume $\Phi_r(\mathscr{M}_r^b) \subseteq \mathscr{M}_r^m$. Then $\Phi_r(\mathscr{Q}_{I,\mathrm{fix}}^b(r)) \subseteq \mathscr{Q}_{I,\mathrm{fix}}^m(r)$, and applying Φ_i agentwise to a holonomy-stabilized parallel belief-law section gives such a model-law section. If $\widetilde{\Phi} : \mathcal{E}_m \to \mathcal{E}_b$ likewise has vanishing link-intertwining defects and $\widetilde{\Phi}_r(\mathscr{M}_r^m) \subseteq \mathscr{M}_r^b$, then $\widetilde{\Phi}_r(\mathscr{Q}_{I,\mathrm{fix}}^m(r)) \subseteq \mathscr{Q}_{I,\mathrm{fix}}^b(r)$, with the analogous statement for sections. [ESTABLISHED]

Proof. Vanishing defects on the edges give the intertwining for walks by composition, using the coordinate form of Proposition 11.3. If $Q \in \mathscr{Q}_{I,\mathrm{fix}}^b(r)$ and p is a closed walk at r, then $\widehat{\rho}_m(\Theta_p^m)\Phi_r(Q) = \Phi_r(\widehat{\rho}_b(\Theta_p^b)Q) = \Phi_r(Q)$, and family compatibility gives the first inclusion. If the belief marginals transport to Q, the same square shows that their images transport to $\Phi_r(Q)$. Interchanging the channels and replacing Φ by $\widetilde{\Phi}$ gives the rest. □

11.5 Structural fixed sectors

Definition 11.18 (Connection Laplacian and fixed fiber). Fix a channel and a block I with internal multigraph $\mathfrak{I}_I$, let the channel's sample fiber be a real vector space $\mathbb{V}$ of dimension K with a linear representation ρ, and write Θ_e for the represented link $\rho(\Theta_e^x) \in \mathrm{GL}(\mathbb{V})$. For each unoriented internal copy choose an orientation $e : j \to i$ and a positive-semidefinite form W_e on $\mathbb{V}$. The connection Laplacian of the block is the positive-semidefinite form L_I on $\mathbb{V}^I$ with

$$z^\top L_I z = \sum_{e:j\to i} (z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j), \tag{11.24}$$

in which loops and parallel copies each contribute one term. Its fixed fiber and rank are $\mathbb{V}_{I,\mathrm{fix}} = \ker L_I$ and $f_I = \dim \mathbb{V}_{I,\mathrm{fix}}$. For a root r the represented based holonomy group $\mathfrak{H}_r = \rho(\mathrm{Hol}_I^x(r))$ has the fixed space $\mathrm{Fix}(\mathfrak{H}_r) = \{v \in \mathbb{V} : \mathsf{k}v = v \text{ for every } \mathsf{k} \in \mathfrak{H}_r\}$.

Proposition 11.19 (Kernel-holonomy fixed-space isomorphism). (i) The energy (11.24) does not depend on the chosen orientations when the reversed copy carries $W_{\bar{e}} = \Theta_e^\top W_e \Theta_e$. For positive-semidefinite weights, $z \in \ker L_I$ exactly when

$$W_e^{1/2}(z_i - \Theta_e z_j) = 0 \qquad \text{for every internal copy } e : j \to i, \tag{11.25}$$

and $\ker L_I$ is the space H^0 of global sections of the cellular sheaf with vertex stalks $\mathbb{V}$ and restriction maps $W_e^{1/2}$ at i and $W_e^{1/2}\Theta_e$ at j, of which L_I is the degree-zero Laplacian [47, Sec. 3.2 and Thm. 3.1]; for orthogonal transports and scalar weights L_I is the vector-diffusion operator of Singer and Wu [93, Sec. 3]. A passive re-expression, with the weights re-expressed as $W_e \mapsto \rho(g_i)^{-\top} W_e \rho(g_i)^{-1}$, carries $\ker L_I$ to $\rho_f(g) \ker L_I$, with $\rho_f(g) = \bigoplus_{i\in I} \rho(g_i)$. (ii) If $W_e \succ 0$ on every internal copy, then on every connected component I_α with root r_α root evaluation $z \mapsto z_{r_\alpha}$ is a linear isomorphism $\ker L_{I_\alpha} \to \mathrm{Fix}(\mathfrak{H}_{r_\alpha})$, so $\ker L_I \cong \bigoplus_\alpha \mathrm{Fix}(\mathfrak{H}_{r_\alpha})$ and $f_I = \sum_\alpha \dim \mathrm{Fix}(\mathfrak{H}_{r_\alpha})$. (iii) With positive-definite weights a connected block has $f_I = K$ exactly when every represented holonomy element is the identity, equivalently when the represented links are graph flat on I (Proposition 2.13); this is the full fixed sector, $0 < f_I < K$ is a partial fixed sector, and $f_I = 0$ leaves no retained degree. A nonfaithful representation can make the represented holonomy trivial

when the principal holonomy is not. (iv) With positive-semidefinite weights holonomy may survive in weight-null directions: on a triangle with $W_{12} = \mathrm{diag}(1,0)$, $\Theta_{12} = \mathrm{diag}(1,2)$, $W_{23} = \mathrm{diag}(0,1)$, $\Theta_{23} = \mathrm{diag}(2,1)$, and $W_{31} = \Theta_{31} = I$, the loop product is $2I$ and fixes no nonzero vector, while $\ker L_I$ is the two-dimensional space of constant configurations. [ESTABLISHED]

Proof. (i) Substituting $\Theta_{\bar{e}} = \Theta_e^{-1}$ and $W_{\bar{e}}$ reproduces the term of e. Since $L_I \succeq 0$, $z \in \ker L_I$ exactly when $z^\top L_I z = \sum_e |W_e^{1/2}(z_i - \Theta_e z_j)|^2 = 0$. The diagonal blocks of L_I are sums of W_e and $\Theta_e^\top W_e \Theta_e$ and its off-diagonal blocks are $-W_e\Theta_e$, the blocks of the sheaf Laplacian with the stated restriction maps, whose kernel is H^0 by the discrete Hodge theorem. The energy is unchanged under $z_i \mapsto \rho(g_i)z_i$, $\Theta_e \mapsto \rho(g_i)\Theta_e\rho(g_j)^{-1}$, and the weight law, so $L_I' = \rho_f(g)^{-\top} L_I \rho_f(g)^{-1}$. (ii) With positive-definite weights, $z \in \ker L_{I_\alpha}$ exactly when $z_i = \Theta_e z_j$ on every copy. Such a section is determined by its root value, and transporting that value around a closed walk returns it, so root evaluation is injective with image in $\mathrm{Fix}(\mathfrak{H}_{r_\alpha})$. Conversely, for $u \in \mathrm{Fix}(\mathfrak{H}_{r_\alpha})$ put $z_v = \rho(\Theta^x_{I_\alpha \leftarrow v})^{-1} u$. Tree constraints hold by construction, and for a non-tree copy $e : j \to i$ the represented fundamental-cycle holonomy $\rho(\mathrm{Hol}^x(p_e)) = \rho(\Theta^x_{I_\alpha \leftarrow i})\Theta_e \rho(\Theta^x_{I_\alpha \leftarrow j})^{-1}$ of (11.7) fixes u, which gives $\Theta_e z_j = z_i$. The Laplacian is the direct sum over components. (iii) $\mathrm{Fix}(\mathfrak{H}_r) = \mathbb{V}$ exactly when every element is the identity, the fundamental-cycle holonomies generate the group (Proposition 11.5), and elements of the kernel of ρ are represented by the identity. (iv) The loop product is $\mathrm{diag}(1,2)\,\mathrm{diag}(2,1) = 2I$. The copy 31 forces $z_3 = z_1$, the copy 12 equal first coordinates of z_1 and z_2, and the copy 23 equal second coordinates of z_2 and z_3, so the kernel consists of the constant configurations. □

Proposition 11.20 (Fixed-section compression). Let Λ be a positive-semidefinite form on the fine configuration space, for instance the connection Laplacian of the network plus positive-semidefinite self terms. For a partition $\mathcal{P}$ into blocks with positive-semidefinite internal weights, let ι_I be a full-column matrix whose range is $\ker L_I$, and $\iota_{\mathcal{P}} = \mathrm{blkdiag}_{I \in \mathcal{P}}\, \iota_I$. Then

$$L_I \iota_I = 0, \qquad \Lambda_c = \iota_{\mathcal{P}}^\top \Lambda \iota_{\mathcal{P}} \succeq 0, \qquad \Lambda_c \succ 0 \iff \mathrm{range}(\iota_{\mathcal{P}}) \cap \ker \Lambda = \{0\}. \tag{11.26}$$

Internal edges contribute nothing to Λ_c, and cut edges become the rectangular endpoint data of Theorem 11.27. With positive-definite weights, the tree construction in the proof of Proposition 11.19 extends a basis of root values to one such matrix. [ESTABLISHED]

Proof. The first identity holds by construction and the second by congruence; $y^\top \Lambda_c y = 0$ exactly when $\iota_{\mathcal{P}} y \in \ker \Lambda$, and $\iota_{\mathcal{P}}$ is injective. An internal term vanishes on $\mathrm{range}\, \iota_I = \ker L_I$, and a cut term restricted to the ranges is the energy of Theorem 11.27 with the blocks of ι as injections. □

Proposition 11.21 (Nested fixed sections compose exactly). Let a larger block A be a union of first-level blocks with positive-semidefinite internal weights, and let $\iota_{A \leftarrow \mathcal{P}} = \mathrm{blkdiag}_{I \subset A}\, \iota_I$ and $L_A^{(1)} = \iota_{A \leftarrow \mathcal{P}}^\top L_A \iota_{A \leftarrow \mathcal{P}}$. Then $\ker L_A = \iota_{A \leftarrow \mathcal{P}} \ker L_A^{(1)}$, and for every injection $\iota_A^{(2)} : \mathbb{V}_{A,\mathrm{fix}} \hookrightarrow \bigoplus_{I \subset A} \mathbb{V}_{I,\mathrm{fix}}$ with range $\ker L_A^{(1)}$ the fine fixed-section injection is $\iota_A^{\mathrm{fine}} = \iota_{A \leftarrow \mathcal{P}} \iota_A^{(2)}$. [ESTABLISHED]

Proof. A zero mode of L_A has zero energy in every positive-semidefinite edge summand, so its restriction to each first-level block lies in $\ker L_I$ and it equals $\iota_{A \leftarrow \mathcal{P}} y$ for some y. Since $L_A \succeq 0$, $y \in \ker L_A^{(1)}$ exactly when $y^\top L_A^{(1)} y = 0$, that is, exactly when $\iota_{A \leftarrow \mathcal{P}} y \in \ker L_A$, which proves the first

identity; substituting the range of $\iota_A^{(2)}$ gives the second, and associativity of matrix composition iterates the statement. □

Remark 11.22 (Structural fixed sector versus marginal law occupation). Relative entropy between the marginals of two agents needs one of them transported into the frame of the other and depends on the walk: a second walk changes the transported law by a loop-holonomy action, and the two values agree for every pair of laws only under stabilization; with $\mathsf{k} = \mathrm{diag}(1,-1,-1) \in \mathrm{SO}(3)$ and $q_i = q_j = \mathcal{N}(e_2, I_3)$, the identity walk gives zero and a walk differing by k gives 2. An unpenalized sum of nonnegative within-block pairwise divergences vanishes at, and is minimized by, the singleton partition; it contains only marginal information and does not test common recovery (Remark 11.16). The structural fixed sector and the marginal-law score are independent: on a two-vertex tree with identity transport and $W = I_K$ the represented holonomy is trivial, yet $D_{\mathrm{KL}}(\mathcal{N}(-ae_1, I_K) \| \mathcal{N}(ae_1, I_K)) = 2a^2$ is arbitrarily large; on a triangle with $K = 3$, unit weights, identity transports on two edges, and closing transport k, the represented holonomy is nontrivial and $\mathrm{Fix}(\mathsf{k}) = \mathrm{span}(e_1)$, so $f_I = 1 < K$, while k preserves $\mathcal{N}(0, \sigma^2 I_3)$ and assigning that law to every vertex makes every transported marginal divergence zero. Transported divergence thus measures the occupation of the available law modes, not the dimension of the structural fixed sector. [ESTABLISHED]

Proof. For equal covariances $D_{\mathrm{KL}}(\mathcal{N}(m_1, I) \| \mathcal{N}(m_2, I)) = \frac{1}{2}|m_1 - m_2|^2$, which gives $\frac{1}{2}|e_2 - \mathsf{k}e_2|^2 = 2$ and $2a^2$. A sum of nonnegative terms over pairs inside blocks is zero for singleton blocks. The fixed space of k is $\mathrm{span}(e_1)$, and $\mathsf{k}\sigma^2 I_3 \mathsf{k}^\top = \sigma^2 I_3$. □

11.6 Cut links and rectangular sheaf closure

Let I and J be blocks reduced to root coordinates $z_I, z_J \in \mathbb{V}$, with represented dressed boundary edges $\widehat{\Theta}_e \in \mathrm{GL}(\mathbb{V})$ (Definition 11.4) and positive-definite weights W_e on the cut copies e from J to I. The cut energy $\mathcal{E}_{\mathrm{cut}}(z_I, z_J) = \sum_e (z_I - \widehat{\Theta}_e z_J)^\top W_e (z_I - \widehat{\Theta}_e z_J)$ equals $z_I^\top \Lambda_{II} z_I - 2 z_I^\top \Lambda_{IJ} z_J + z_J^\top \Lambda_{JJ} z_J$ with $\Lambda_{II} = \sum_e W_e$, $\Lambda_{IJ} = \sum_e W_e \widehat{\Theta}_e$, and $\Lambda_{JJ} = \sum_e \widehat{\Theta}_e^\top W_e \widehat{\Theta}_e$. Coarse families admit newly generated positive-semidefinite self terms.

Definition 11.23 (Anchored and no-new-anchor presentations of a cut). An anchored one-link presentation of the cut is a tuple $(\Theta, W, \Lambda_I^{\mathrm{s}}, \Lambda_J^{\mathrm{s}})$ with $W \succ 0$, $\Theta \in \mathrm{GL}(\mathbb{V})$, and $\Lambda_I^{\mathrm{s}}, \Lambda_J^{\mathrm{s}} \succeq 0$ such that, for all z_I, z_J,

$$\mathcal{E}_{\mathrm{cut}}(z_I, z_J) = (z_I - \Theta z_J)^\top W (z_I - \Theta z_J) + z_I^\top \Lambda_I^{\mathrm{s}} z_I + z_J^\top \Lambda_J^{\mathrm{s}} z_J. \tag{11.27}$$

The no-new-anchor pair subfamily consists of the presentations with $\Lambda_I^{\mathrm{s}} = \Lambda_J^{\mathrm{s}} = 0$: one link and one weight, with no new self terms.

Lemma 11.24 (Completed square for a cut). One has $\Lambda_{JJ} \succ 0$ and, with $\bar{\Theta} = \Lambda_{II}^{-1}\Lambda_{IJ}$,

$$\mathcal{E}_{\mathrm{cut}}(z_I, z_J) = (z_I - \bar{\Theta} z_J)^{\top} \Lambda_{II} (z_I - \bar{\Theta} z_J) + z_J^{\top} \Delta z_J, \tag{11.28}$$

$$\Delta = \Lambda_{JJ} - \Lambda_{IJ}^{\top}\Lambda_{II}^{-1}\Lambda_{IJ} = \sum_e (\widehat{\Theta}_e - \bar{\Theta})^{\top} W_e (\widehat{\Theta}_e - \bar{\Theta}) \succeq 0. \tag{11.29}$$

(i) An anchored presentation exists if and only if $\det \Lambda_{IJ} \neq 0$; with links restricted to $\mathrm{GL}^+(\mathbb{V})$ the condition is $\det \Lambda_{IJ} > 0$. (ii) When $\det \Lambda_{IJ} \neq 0$, the anchored presentations are exactly $\Theta = W^{-1}\Lambda_{IJ}$, $\Lambda_I^{\mathrm{s}} = \Lambda_{II} - W$, and $\Lambda_J^{\mathrm{s}} = \Lambda_{JJ} - \Lambda_{IJ}^{\top} W^{-1}\Lambda_{IJ}$, with W in the Loewner interval

$$\mathfrak{W}_{\mathrm{cut}} = \{W : \Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top} \preceq W \preceq \Lambda_{II}\} = \{W \succ 0 : W \preceq \Lambda_{II},\ \Lambda_{IJ}^{\top} W^{-1} \Lambda_{IJ} \preceq \Lambda_{JJ}\}. \tag{11.30}$$

The choice $W = \Lambda_{II}$ gives $\Lambda_I^{\mathrm{s}} = 0$ and $\Lambda_J^{\mathrm{s}} = \Delta$, and $W = \Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top}$ gives $\Lambda_J^{\mathrm{s}} = 0$, the reversed completion with link $\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top}$ from I to J and weight Λ_{JJ}. (iii) When $\det \Lambda_{IJ} \neq 0$, the anchored presentation is unique if and only if $\Delta = 0$, and it is then the bare pair $(\bar{\Theta}, \Lambda_{II}, 0, 0)$; otherwise there are infinitely many. The condition $\Delta = 0$ itself forces $\det \Lambda_{IJ} \neq 0$. (iv) The no-new-anchor pair subfamily contains a presentation of the cut if and only if $\Delta = 0$, that is, if and only if all cut twists $\widehat{\Theta}_e$ coincide. (v) The weighted mean $\bar{\Theta}$ has the gauge law of a dressed edge, $\bar{\Theta} \mapsto \rho(g_{r_I})\bar{\Theta}\rho(g_{r_J})^{-1}$, but it can be singular: equal weights with twists I and $-I$ give $\bar{\Theta} = 0$. [ESTABLISHED]

Proof. Each $\widehat{\Theta}_e$ is invertible and $W_e \succ 0$, so $\Lambda_{JJ} \succ 0$. Completing the square in z_I gives (11.28), and expanding the sum in (11.29) gives $\Lambda_{JJ} - \Lambda_{IJ}^{\top}\Lambda_{II}^{-1}\Lambda_{IJ}$. Matching the three blocks of (11.27) gives $W\Theta = \Lambda_{IJ}$, $W + \Lambda_I^{\mathrm{s}} = \Lambda_{II}$, and $\Theta^{\top} W \Theta + \Lambda_J^{\mathrm{s}} = \Lambda_{JJ}$, hence the formulas of (ii); both self terms are positive semidefinite exactly on the second description in (11.30). The link is invertible exactly when Λ_{IJ} is, and $\det \Theta = \det \Lambda_{IJ} / \det W$ has its sign. If $\det \Lambda_{IJ} \neq 0$, then $\Lambda_{IJ}^{\top} W^{-1} \Lambda_{IJ} \preceq \Lambda_{JJ}$ is equivalent to $W^{-1} \preceq \Lambda_{IJ}^{-\top}\Lambda_{JJ}\Lambda_{IJ}^{-1}$ and, since inversion reverses the Loewner order on positive-definite matrices, to $W \succeq \Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top}$; this is the first description, which contains $W = \Lambda_{II}$ because $\Delta \succeq 0$, and proves (i). The reversed completion is the same computation with z_I and z_J exchanged. Expanding the right side shows $\Lambda_{II} - \Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top} = \Lambda_{II}\Lambda_{IJ}^{-\top}\Delta\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top}$, so the interval is a single point exactly when $\Delta = 0$; otherwise every point of the segment between its endpoints is admissible and distinct weights give distinct presentations. If $\Delta = 0$ then $\Lambda_{IJ}^{\top}\Lambda_{II}^{-1}\Lambda_{IJ} = \Lambda_{JJ} \succ 0$, so Λ_{IJ} is invertible. For (iv), a presentation without self terms needs $W = \Lambda_{II}$, $W\Theta = \Lambda_{IJ}$, and $\Theta^{\top} W \Theta = \Lambda_{JJ}$, that is, $\Delta = 0$, and the sum in (11.29) with $W_e \succ 0$ vanishes exactly when every $\widehat{\Theta}_e = \bar{\Theta}$. For (v), invariance of the energy under $z_I \mapsto \rho(g_{r_I})z_I$ and $z_J \mapsto \rho(g_{r_J})z_J$ transforms W_e at the receiving end and $\widehat{\Theta}_e$ as in (11.13), and substitution gives the law of $\bar{\Theta}$; the example is direct. □

Proposition 11.25 (Balanced presentation of a cut). In the setting of Lemma 11.24 with $\det \Lambda_{IJ} \neq 0$, write $W_{\mathrm{lo}} = \Lambda_{IJ}\Lambda_{JJ}^{-1}\Lambda_{IJ}^{\top}$ and $W_{\mathrm{hi}} = \Lambda_{II}$ for the endpoints of $\mathfrak{W}_{\mathrm{cut}}$, and call an anchored presentation balanced when the self term at J, carried to I along the presentation's own link, equals the self term at I,

$$\Theta^{-\top}\Lambda_J^{\mathrm{s}}\Theta^{-1} = \Lambda_I^{\mathrm{s}}, \qquad \text{equivalently} \qquad \Theta^{\top}\Lambda_{II}\Theta = \Lambda_{JJ}, \tag{11.31}$$

so that the balanced link carries the form Λ_{JJ} to the form Λ_{II}. (i) Exactly one anchored presentation is balanced. Its weight is the matrix geometric mean of the endpoints,

$$W_{\mathrm{bal}} = W_{\mathrm{lo}}^{1/2}\big(W_{\mathrm{lo}}^{-1/2} W_{\mathrm{hi}} W_{\mathrm{lo}}^{-1/2}\big)^{1/2} W_{\mathrm{lo}}^{1/2}, \tag{11.32}$$

the unique positive-definite solution of $W W_{\mathrm{lo}}^{-1} W = W_{\mathrm{hi}}$, and its link is $\Theta_{\mathrm{bal}} = W_{\mathrm{bal}}^{-1}\Lambda_{IJ}$. (ii) Under root re-expressions the weight transforms as a weight at I, $W_{\mathrm{bal}} \mapsto \rho(g_{r_I})^{-\top} W_{\mathrm{bal}} \rho(g_{r_I})^{-1}$, the link as a dressed edge, $\Theta_{\mathrm{bal}} \mapsto \rho(g_{r_I})\Theta_{\mathrm{bal}}\rho(g_{r_J})^{-1}$, and each self term as a form at its own block. The presentation is end symmetric: read from J, with link $\Theta_{\mathrm{bal}}^{-1}$ and weight $\Theta_{\mathrm{bal}}^{\top} W_{\mathrm{bal}} \Theta_{\mathrm{bal}}$, it is the balanced presentation of the reversed cut, so that weight is the geometric mean of the endpoints $\Lambda_{IJ}^{\top}\Lambda_{II}^{-1}\Lambda_{IJ}$ and Λ_{JJ} of J. (iii) For scalar weights $W_e = w_e I$ and orthogonal twists, an anchored presentation is balanced exactly when its link is orthogonal; then $W_{\mathrm{bal}} = (\Lambda_{IJ}\Lambda_{IJ}^{\top})^{1/2}$ and Θ_{bal} is the orthogonal polar factor of $\bar{\Theta}$. [ESTABLISHED]

Proof. By part (ii) of the lemma an anchored presentation has $\Theta = W^{-1}\Lambda_{IJ}$, $\Lambda_I^{\mathrm{s}} = \Lambda_{II} - W$, and $\Lambda_J^{\mathrm{s}} = \Lambda_{JJ} - \Lambda_{IJ}^{\top} W^{-1}\Lambda_{IJ}$, so $\Theta^{-\top}\Lambda_J^{\mathrm{s}}\Theta^{-1} = W\Lambda_{IJ}^{-\top}\Lambda_{JJ}\Lambda_{IJ}^{-1}W - W = W W_{\mathrm{lo}}^{-1} W - W$, and the first form of (11.31) reads $W W_{\mathrm{lo}}^{-1} W = W_{\mathrm{hi}}$. The second form is $\Lambda_J^{\mathrm{s}} = \Theta^{\top}\Lambda_I^{\mathrm{s}}\Theta$ with the common term $\Theta^{\top} W \Theta$ removed from both sides, and after the substitution $\Theta = W^{-1}\Lambda_{IJ}$ it reads $W^{-1} W_{\mathrm{hi}} W^{-1} = W_{\mathrm{lo}}^{-1}$, the same equation. (i) The substitution $W = W_{\mathrm{lo}}^{1/2} Y W_{\mathrm{lo}}^{1/2}$ is a bijection of positive-definite matrices and turns the equation into $Y^2 = M$ with $M = W_{\mathrm{lo}}^{-1/2} W_{\mathrm{hi}} W_{\mathrm{lo}}^{-1/2} \succ 0$. A positive-definite Y with $Y^2 = M$ commutes with M, so it is diagonal in an eigenbasis of M with positive eigenvalues whose squares are those of M; hence $Y = M^{1/2}$, which is (11.32). Since $\Delta \succeq 0$ gives $W_{\mathrm{lo}} \preceq W_{\mathrm{hi}}$, one has $M \succeq I$, hence $I \preceq M^{1/2} \preceq M$, and congruence by $W_{\mathrm{lo}}^{1/2}$ gives $W_{\mathrm{lo}} \preceq W_{\mathrm{bal}} \preceq W_{\mathrm{hi}}$; the solution lies in $\mathfrak{W}_{\mathrm{cut}}$ and is the weight of exactly one anchored presentation. (ii) As in the proof of part (v) of the lemma, a root re-expression acts by $W_e \mapsto g_I^{-\top} W_e g_I^{-1}$ and $\widehat{\Theta}_e \mapsto g_I \widehat{\Theta}_e g_J^{-1}$ with $g_I = \rho(g_{r_I})$ and $g_J = \rho(g_{r_J})$, so Λ_{II}, Λ_{IJ}, and Λ_{JJ} transform by $g_I^{-\top}(\cdot)g_I^{-1}$, $g_I^{-\top}(\cdot)g_J^{-1}$, and $g_J^{-\top}(\cdot)g_J^{-1}$, and both endpoints by $g_I^{-\top}(\cdot)g_I^{-1}$. If W solves $W W_{\mathrm{lo}}^{-1} W = W_{\mathrm{hi}}$, then $g_I^{-\top} W g_I^{-1}$ solves the transformed equation, so uniqueness gives the law of W_{bal}, and substitution gives the laws of the link and of the self terms. For end symmetry, $(z_I - \Theta z_J)^{\top} W (z_I - \Theta z_J) = (z_J - \Theta^{-1} z_I)^{\top}\Theta^{\top} W \Theta (z_J - \Theta^{-1} z_I)$, so an anchored presentation of the cut, read from J, is the anchored presentation $(\Theta^{-1}, \Theta^{\top} W \Theta, \Lambda_J^{\mathrm{s}}, \Lambda_I^{\mathrm{s}})$ of the reversed cut, whose blocks are Λ_{JJ}, $\Lambda_{IJ}^{\top}$, and Λ_{II}. The second form of (11.31) is equivalent to $\Theta^{-\top}\Lambda_{JJ}\Theta^{-1} = \Lambda_{II}$, the balance condition of the reversed cut, so the balanced presentation read from J is balanced for the reversed cut, and (i) applied to that cut identifies its weight. (iii) Here $\Lambda_{II} = \Lambda_{JJ} = (\sum_e w_e) I$, so the second form of (11.31) says that Θ is orthogonal. For orthogonal Θ the weight $W = \Lambda_{IJ}\Theta^{\top}$ is positive definite exactly when $\Lambda_{IJ} = W\Theta$ is a polar decomposition of Λ_{IJ}, and then $W^2 = \Lambda_{IJ}\Lambda_{IJ}^{\top}$ forces $W = (\Lambda_{IJ}\Lambda_{IJ}^{\top})^{1/2}$ and $\Theta = W^{-1}\Lambda_{IJ}$, the polar factor of Λ_{IJ} and of $\bar{\Theta} = \Lambda_{IJ}/\sum_e w_e$. □

Proposition 11.26 (No balanced presentation of a cut at the level of laws). Let $\mathrm{Gau}(\mathbb{V})$ be the fiber of nondegenerate Gaussian laws on $\mathbb{V}$, and let k copies e from J to I carry dressed edges $\widehat{\Theta}_e \in \mathrm{GL}(\mathbb{V})$ and weights $w_e > 0$ with $\sum_e w_e \widehat{\Theta}_e^{-1}$ invertible. For $q_J = \mathcal{N}(m_J, \Sigma_J) \in \mathrm{Gau}(\mathbb{V})$ the transported laws $(\widehat{\Theta}_e)_{\#} q_J = \mathcal{N}(\widehat{\Theta}_e m_J, \Sigma_e)$, $\Sigma_e = \widehat{\Theta}_e \Sigma_J \widehat{\Theta}_e^{\top}$, form the cut of Lemma 11.24 with

weights $W_e = \frac{1}{2} w_e \Sigma_e^{-1}$ and twists $\widehat{\Theta}_e$; its blocks $\Lambda_{II}(\Sigma_J)$, $\Lambda_{IJ}(\Sigma_J)$, $\Lambda_{JJ}(\Sigma_J)$, its excess $\Delta(\Sigma_J)$ (11.29), and its balanced weight and link $W_{\mathrm{bal}}(\Sigma_J)$ and $\Theta_{\mathrm{bal}}(\Sigma_J)$ (11.32) are functions of Σ_J, with $\Lambda_{IJ}(\Sigma_J)$ invertible for every Σ_J. The cut has excess, $\Delta(\Sigma_J) \neq 0$ for one and hence for every Σ_J, exactly when the dressed edges are not all equal. Let $\mathscr{C} \subseteq \mathrm{Gau}(\mathbb{V})$ be nonempty and closed under rescaling of the covariance, so that $\mathcal{N}(m, \Sigma) \in \mathscr{C}$ implies $\mathcal{N}(m, t\Sigma) \in \mathscr{C}$ for every $t > 0$. A presentation of the copies at the level of laws on $\mathscr{C}$ is a weight $W' > 0$, an image map $\mathfrak{T} : \mathscr{C} \to \mathrm{Gau}(\mathbb{V})$, and self terms $\mathfrak{S}_I : \mathrm{Gau}(\mathbb{V}) \to \mathbb{R}$ and $\mathfrak{S}_J : \mathscr{C} \to \mathbb{R}$ with

$$\sum_e w_e D_{\mathrm{KL}}\big(q_I \| (\widehat{\Theta}_e)_{\#} q_J\big) = W' D_{\mathrm{KL}}\big(q_I \| \mathfrak{T}[q_J]\big) + \mathfrak{S}_I(q_I) + \mathfrak{S}_J(q_J) \quad \text{for all } q_I \in \mathrm{Gau}(\mathbb{V}) \text{ and } q_J \in \mathscr{C}, \tag{11.33}$$

and it is balanced when its Gaussian mean part is the link term of the balanced presentation of the cut of Σ_J, that is, writing $\mathfrak{T}[q_J] = \mathcal{N}(\mu_{\mathfrak{T}}, \Sigma_{\mathfrak{T}})$,

$$\tfrac{1}{2} W' \Sigma_{\mathfrak{T}}^{-1} = W_{\mathrm{bal}}(\Sigma_J) \quad \text{and} \quad \mu_{\mathfrak{T}} = \Theta_{\mathrm{bal}}(\Sigma_J) m_J \qquad \text{for every } q_J = \mathcal{N}(m_J, \Sigma_J) \in \mathscr{C}. \tag{11.34}$$

If the cut has excess, no presentation at the level of laws on $\mathscr{C}$ is balanced. Two copies with dressed edges I and $2I$ and weights w_1, w_2 have $\Delta(\Sigma_J) = \frac{w_1 w_2}{2(4w_1 + w_2)} \Sigma_J^{-1}$, so they admit no balanced presentation on any such $\mathscr{C}$. [ESTABLISHED]

Proof. Since $\Sigma_e^{-1} = \widehat{\Theta}_e^{-\top} \Sigma_J^{-1} \widehat{\Theta}_e^{-1}$, the blocks are $\Lambda_{II}(\Sigma_J) = \frac{1}{2} \sum_e w_e \widehat{\Theta}_e^{-\top} \Sigma_J^{-1} \widehat{\Theta}_e^{-1}$, $\Lambda_{IJ}(\Sigma_J) = \frac{1}{2} \big(\sum_e w_e \widehat{\Theta}_e^{-\top}\big) \Sigma_J^{-1}$, and $\Lambda_{JJ}(\Sigma_J) = \frac{1}{2} \big(\sum_e w_e\big) \Sigma_J^{-1}$, so $\Lambda_{IJ}(\Sigma_J)$ is invertible exactly when $\sum_e w_e \widehat{\Theta}_e^{-1}$ is, and each block is homogeneous of degree -1 in Σ_J. Hence so are the endpoints $W_{\mathrm{lo}} = \Lambda_{IJ} \Lambda_{JJ}^{-1} \Lambda_{IJ}^{\top}$ and $W_{\mathrm{hi}} = \Lambda_{II}$, the excess Δ, and, since (11.32) is jointly homogeneous of degree one in its two arguments, the balanced weight, $W_{\mathrm{bal}}(t\Sigma_J) = t^{-1} W_{\mathrm{bal}}(\Sigma_J)$. By part (iv) of the lemma, $\Delta(\Sigma_J) = 0$ exactly when all $\widehat{\Theta}_e$ coincide, a condition that does not involve Σ_J.

Fix a presentation, a law $q_J = \mathcal{N}(m_J, \Sigma_J) \in \mathscr{C}$, and $q_I = \mathcal{N}(m_I, \Sigma_I) \in \mathrm{Gau}(\mathbb{V})$; write $\mathfrak{T}[q_J] = \mathcal{N}(\mu_{\mathfrak{T}}, \Sigma_{\mathfrak{T}})$, $B = \frac{1}{2} W' \Sigma_{\mathfrak{T}}^{-1}$, and abbreviate $\Lambda_{II} = \Lambda_{II}(\Sigma_J)$, $\Lambda_{IJ} = \Lambda_{IJ}(\Sigma_J)$. For Gaussian laws $D_{\mathrm{KL}}(\mathcal{N}(m_1, \Sigma_1) \| \mathcal{N}(m_2, \Sigma_2)) = \frac{1}{2}(m_1 - m_2)^{\top} \Sigma_2^{-1} (m_1 - m_2) + \frac{1}{2}[\mathrm{Tr}(\Sigma_2^{-1} \Sigma_1) - \dim \mathbb{V} + \log\det \Sigma_2 - \log\det \Sigma_1]$, so both sides of (11.33) are quadratic polynomials in m_I, and their difference is

$$\begin{aligned} 0 = {} & m_I^{\top} (\Lambda_{II} - B) m_I - 2 m_I^{\top} (\Lambda_{IJ} m_J - B \mu_{\mathfrak{T}}) + \mathrm{Tr}[(\Lambda_{II} - B) \Sigma_I] \\ & + \tfrac{1}{2} \big(W' - \textstyle\sum_e w_e\big) \log\det \Sigma_I - \mathfrak{S}_I(q_I) + \mathrm{c}(q_J), \end{aligned}$$

where $\mathrm{c}(q_J)$ collects the terms that read q_J alone. Subtracting this identity for a second law $q_J' \in \mathscr{C}$ removes $\mathfrak{S}_I(q_I)$ and the $\log\det \Sigma_I$ term and leaves a quadratic polynomial in m_I that vanishes for every $m_I \in \mathbb{V}$ and every $\Sigma_I \succ 0$, so its quadratic and linear coefficients vanish: $M = \Lambda_{II}(\Sigma_J) - B$ and $v = \Lambda_{IJ}(\Sigma_J) m_J - B \mu_{\mathfrak{T}}$ take the same value at every $q_J \in \mathscr{C}$. If the presentation is balanced, (11.34) gives $B = W_{\mathrm{bal}}(\Sigma_J)$ and $B \mu_{\mathfrak{T}} = W_{\mathrm{bal}}(\Sigma_J) \Theta_{\mathrm{bal}}(\Sigma_J) m_J = \Lambda_{IJ}(\Sigma_J) m_J$, so $v = 0$ and $M = \Lambda_{II}(\Sigma_J) - W_{\mathrm{bal}}(\Sigma_J)$ for every $q_J \in \mathscr{C}$. Comparing q_J with its rescaling $\mathcal{N}(m_J, t\Sigma_J) \in \mathscr{C}$, the homogeneity gives $M = t^{-1} M$ for every $t > 0$, hence $M = 0$ and $W_{\mathrm{bal}}(\Sigma_J) = W_{\mathrm{hi}}(\Sigma_J)$. Substituting into $W W_{\mathrm{lo}}^{-1} W = W_{\mathrm{hi}}$ gives $W_{\mathrm{hi}} W_{\mathrm{lo}}^{-1} W_{\mathrm{hi}} = W_{\mathrm{hi}}$, so $W_{\mathrm{lo}} = W_{\mathrm{hi}}$, and the identity $\Lambda_{II} - \Lambda_{IJ} \Lambda_{JJ}^{-1} \Lambda_{IJ}^{\top} = \Lambda_{II} \Lambda_{IJ}^{-\top} \Delta \Lambda_{JJ}^{-1} \Lambda_{IJ}^{\top}$ of the proof of the lemma, whose outer factors are invertible,

gives $\Delta(\Sigma_J) = 0$: the cut has no excess. For the two copies I and $2I$, $\Sigma_1 = \Sigma_J$ and $\Sigma_2 = 4\Sigma_J$, so $\Lambda_{II} = \frac{1}{8}(4w_1 + w_2)\Sigma_J^{-1}$, $\Lambda_{IJ} = \frac{1}{4}(2w_1 + w_2)\Sigma_J^{-1}$, and $\Lambda_{JJ} = \frac{1}{2}(w_1 + w_2)\Sigma_J^{-1}$, whence $\Delta = \big[\frac{1}{2}(w_1 + w_2) - \frac{(2w_1+w_2)^2}{2(4w_1+w_2)}\big]\Sigma_J^{-1} = \frac{w_1 w_2}{2(4w_1+w_2)}\Sigma_J^{-1}$. □

The balance condition is imposed only on the mean part of the link term, so the conclusion does not depend on whether the image map is gauge covariant or the self terms are end symmetric: a self term that reads one law cannot carry $\Lambda_{II} - W_{\mathrm{bal}}$, which is homogeneous of degree -1 in the covariance of the other law, and the excess of the cut is exactly what keeps that difference from vanishing.

The exact closed category is larger than one invertible link; its endpoint maps and stalks use the notation of the comparison maps and spaces of Chapter 6, which are an instance.

Theorem 11.27 (Rectangular endpoint-map closure). Choose a minimal factor $W_e = (S_e^Q)^\top S_e^Q$ with $S_e^Q : \mathbb{V} \to H_e$ of full row rank, and put $S_e^K = S_e^Q \Theta_e$. Then $(z_i - \Theta_e z_j)^\top W_e (z_i - \Theta_e z_j) = \|S_e^Q z_i - S_e^K z_j\|_{H_e}^2$. For linear maps $\iota_{iI} : \mathbb{V}_I \to \mathbb{V}$ from block coordinates, the coarse endpoint maps $S_{I,e}^{Q,c} = S_e^Q \iota_{iI}$ and $S_{J,e}^{K,c} = S_e^K \iota_{jJ}$ reproduce the Galerkin compression of the cut energy exactly, parallel cut copies compress losslessly into the direct sum of their stalks H_e, and nested blocking is associative. Parallel copies between the same two coarse vertices also admit a minimal exact compression: with coboundary rows $\delta_e = (S_{I,e}^{Q,c}, -S_{J,e}^{K,c})$, one edge with stalk H and coboundary δ reproduces their summed energy exactly when $\delta^\top \delta = \sum_e \delta_e^\top \delta_e$; the minimal stalk dimension is $\operatorname{rank} \sum_e \delta_e^\top \delta_e \le \dim \mathbb{V}_I + \dim \mathbb{V}_J$, attained by any full-row-rank factor, and two full-row-rank factors differ by an orthogonal change of edge basis. [ESTABLISHED]

Proof. Expanding the square gives the pair form. Precomposition gives the coarse maps, whose cross block $-(S_{I,e}^{Q,c})^\top S_{J,e}^{K,c} = -\iota_{iI}^\top W_e \Theta_e \iota_{jJ}$ and diagonal blocks agree with the compressed energy. Stacking in $\bigoplus_e H_e$ sums the energies, and a second injection ι' gives $(S\iota)\iota' = S(\iota\iota')$. Two quadratic forms agree exactly when their symmetric matrices agree, which gives the compression criterion and the minimal stalk dimension; for full-row-rank factors δ_1, δ_2 of one matrix, $O = \delta_2 \delta_1^+$ satisfies $O^\top O = I$ and $\delta_2 = O\delta_1$. □

This is the degree-zero cellular-sheaf Laplacian category [47], in which vertex spaces may have unequal dimensions and automorphism groups $\mathrm{GL}(\mathbb{V}_I)$ and edge basis changes act on both endpoint maps and the edge metric. Twice the quadratic disagreement V_e of Definition 6.15 is the edge term of such an energy once its comparison precision is absorbed into the endpoint maps, as in the discourse sheaves of Hansen and Ghrist [48]; the closure selects no decoder and no law, and the normalized realization for fixed maps is (6.41).

Proposition 11.28 (No continuous extension of balanced cut links). There is no rule assigning a matrix-valued link and a positive-semidefinite weight and self terms to every cut that is continuous in the constituent weights and twists and agrees with Proposition 11.25 whenever Λ_{IJ} is invertible. This already fails for two parallel copies with identity weights and twists in $\mathrm{SO}(2) \subset \mathrm{GL}^+(2)$. Consequently adding gauge covariance, invariance under reversal, or a rank requirement on singular links cannot produce such a rule on all singular and rectangular cuts. This obstruction

concerns extension of the balanced presentation; the exact rectangular endpoint-map closure of Theorem 11.27 remains available. [ESTABLISHED]

Proof. For $-1 < t < 1$, take $W_1 = W_2 = I_2$ and

$$\widehat{\Theta}_1(t) = \begin{pmatrix} t & -\sqrt{1-t^2} \\ \sqrt{1-t^2} & t \end{pmatrix}, \qquad \widehat{\Theta}_2(t) = \widehat{\Theta}_1(t)^\top. \tag{11.35}$$

The twists lie in $\mathrm{SO}(2)$ and depend continuously on t, while $\Lambda_{II} = \Lambda_{JJ} = 2I_2$ and $\Lambda_{IJ} = 2tI_2$. For $t \neq 0$, the interval endpoints are $W_{\mathrm{lo}} = 2t^2 I_2$ and $W_{\mathrm{hi}} = 2I_2$, so the balanced weight and link are $W_{\mathrm{bal}} = 2|t|I_2$ and $\Theta_{\mathrm{bal}} = \mathrm{sign}(t)I_2$. The same limiting cut at $t = 0$ is approached from both sides, but the links tend to I_2 and $-I_2$. A matrix-valued continuous extension would have to equal both limits. □

11.7 The frame sector

Proposition 11.29 (The frame sector). Fix a block I whose constituent frames are $u_i = \sigma \cdot U_i^{-1}$ relative to a reference point σ (Proposition 2.5), and a coarse datum (u_I, q_I) with $q_I \in \mathcal{B}_x$ and $u_I = \sigma \cdot U_I^{-1}$ a point of the same torsor, so that $u_I = u_i \cdot T_{iI}$ with $T_{iI} = U_i U_I^{-1}$. (i) The constituent expressions $\widehat{\rho}_x(T_{iI})q_I = \widehat{\rho}_x(U_i)\bar{q}_I$ of the coarse law, with $\bar{q}_I = \widehat{\rho}_x(U_I)^{-1}q_I$ the coordinate in σ of the point $[u_I, q_I]$ of the associated fiber, depend on (u_I, q_I) only through that point; for the Gaussian law fiber of the defining representation of $\mathrm{GL}^+(K)$ the constituent expression of $q_I = \mathcal{N}(\mu_I, \Sigma_I)$ is $\mathcal{N}(U_i\bar{m}, U_i\bar{S}U_i^\top)$ with $\bar{m} = U_I^{-1}\mu_I$ and $\bar{S} = U_I^{-1}\Sigma_I U_I^{-\top}$. The data with the same constituent expressions form exactly one orbit $\{(u_I \cdot g, \widehat{\rho}_x(g)^{-1}q_I) : g \in G\}$, on which G acts freely, so the coarse frame occupies a copy of G, of dimension K^2 for $G = \mathrm{GL}^+(K)$, that the constituents do not see. If only the represented frame $\rho_x(U_I)$ is recorded, a nonfaithful representation determines the principal coarse frame only modulo its kernel. (ii) If G contains an element $v \neq e$ with $v^n = e$, no map $F : G^n \to G$ is both left equivariant, $F(gg_1, \ldots, gg_n) = gF(g_1, \ldots, g_n)$, and invariant under permutations of its arguments. For $G = \mathrm{GL}^+(K)$ with $K \geq 2$ and every $n \geq 2$ such an element exists, a rotation of order n in one coordinate plane. The torsion hypothesis cannot be dropped: on the additive group $\mathbb{R}^d$ the arithmetic mean is left equivariant and symmetric. [ESTABLISHED]

Proof. (i) By the quotient convention (2.7), $[u_I, q_I] = [u_i \cdot T_{iI}, q_I] = [u_i, \widehat{\rho}_x(T_{iI})q_I]$, so the constituent expression is the coordinate of $[u_I, q_I]$ in the frame u_i and determines that point. Likewise $[u_I, q_I] = [\sigma, \widehat{\rho}_x(U_I)^{-1}q_I] = [u_i \cdot U_i, \bar{q}_I] = [u_i, \widehat{\rho}_x(U_i)\bar{q}_I]$, and the Gaussian pushforward by the linear map U_i gives the Gaussian form. Two data define the same point exactly when they differ by $(u_I, q_I) \mapsto (u_I \cdot g, \widehat{\rho}_x(g)^{-1}q_I)$, and $u_I \cdot g = u_I$ forces $g = e$. Elements of the kernel of ρ_x give equal represented frames. (ii) Left multiplication by v permutes the tuple $(e, v, \ldots, v^{n-1})$ cyclically, because $v \cdot v^{n-1} = e$. Symmetry gives F the same value at the permuted tuple, while equivariance multiplies it by v; hence $vF = F$ and $v = e$, a contradiction. A rotation by $2\pi/n$ in one coordinate plane lies in $\mathrm{GL}^+(K)$ and has order n. The mean on $\mathbb{R}^d$ satisfies $\frac{1}{n}\sum_i(g + g_i) = g + \frac{1}{n}\sum_i g_i$. □

A rule $u_I = \sigma \cdot F(a_1, \dots, a_n)$ with $u_i = \sigma \cdot a_i$ is independent of σ exactly when F is left equivariant, because $\sigma' = \sigma \cdot g_0$ replaces every a_i by $g_0^{-1} a_i$; (ii) therefore excludes every symmetric rule of this kind for a meta-agent's frame.

11.8 An operational record at c_* sensitive to loop holonomy

Theorem 11.30 (An operational record sensitive to network-loop holonomy). Let p be a closed walk of the interaction multigraph based at a probe agent i_0 whose belief and model sample fibers are real vector spaces $\mathbb{V}_b, \mathbb{V}_m$ with linear representations ρ_b, ρ_m, and write $z_{i_0}(y) \in \mathbb{V}_{\mathrm{pr}} = \mathbb{V}_b \oplus \mathbb{V}_m$ for the probe coordinates of a latent configuration y and $\rho_{\mathrm{pr}} = \rho_b \oplus \rho_m$. Let Ω be a finite set of labels ω with prior masses $\psi_\omega > 0$, each prescribing link values on the edges of p, with represented loop holonomy $\mathrm{Hol}^{\mathrm{pr}}_\omega = \rho_b(\mathrm{Hol}^b_\omega(p)) \oplus \rho_m(\mathrm{Hol}^m_\omega(p))$. Keep the joint $\mathbb{P}(dy, do)$ independent of ω, with selected posterior version $o \mapsto \mathbf{\Pi}_o$, the slice $\mathbf{\Pi}_{\theta,V,o,X}$ at fixed parameters and structural data; fix $\Sigma_{\mathrm{r}} \succ 0$ on $\mathbb{V}_{\mathrm{pr}}$, and declare the record joint

$$P^{\mathrm{rec}}(\omega, dy, do, dr) = \psi_\omega \mathbb{P}(dy, do) \mathcal{N}\big(r; (\mathrm{Hol}^{\mathrm{pr}}_\omega - I) z_{i_0}(y), \Sigma_{\mathrm{r}}\big) dr.$$

(i) P^{rec} is a normalized joint. At an admitted o and given ω, the record is distributed as $(\mathrm{Hol}^{\mathrm{pr}}_\omega - I)Z + \epsilon$, with $Z \sim (z_{i_0})_{\#}\mathbf{\Pi}_o$ and $\epsilon \sim \mathcal{N}(0, \Sigma_{\mathrm{r}})$ independent of Z. Two labels ω, ω' have the same record law exactly when $(\mathrm{Hol}^{\mathrm{pr}}_\omega - I)Z$ and $(\mathrm{Hol}^{\mathrm{pr}}_{\omega'} - I)Z$ have the same law. (ii) If $\mathrm{Hol}^{\mathrm{pr}}_{\omega_0} = I$ and $\mathrm{Hol}^{\mathrm{pr}}_{\omega_1} - I$ is invertible, as for $\mathrm{SO}(2)$ links in both channels whose loop rotation angles are not multiples of 2π, the record laws of ω_0 and ω_1 differ exactly when $(z_{i_0})_{\#}\mathbf{\Pi}_o$ is not the point mass at 0. Under the same condition the statistic $\mathsf{q} = r^\top \Sigma_{\mathrm{r}}^{-1} r$ has different laws: its mean is $\dim \mathbb{V}_{\mathrm{pr}}$ under ω_0 and $\dim \mathbb{V}_{\mathrm{pr}} + \mathbb{E}\big[Z^\top (\mathrm{Hol}^{\mathrm{pr}}_{\omega_1} - I)^\top \Sigma_{\mathrm{r}}^{-1} (\mathrm{Hol}^{\mathrm{pr}}_{\omega_1} - I) Z\big] \in (\dim \mathbb{V}_{\mathrm{pr}}, +\infty]$ under ω_1. (iii) Under a passive re-expression g, with the record expressed in the probe frame, $r \mapsto \rho_{\mathrm{pr}}(g_{i_0}) r$ and $\Sigma_{\mathrm{r}} \mapsto \rho_{\mathrm{pr}}(g_{i_0}) \Sigma_{\mathrm{r}} \rho_{\mathrm{pr}}(g_{i_0})^\top$, the record joint is the image of the original under a bimeasurable bijection; the posterior over ω given (o, r) and the law of q are unchanged. [ESTABLISHED]

Proof. (i) The Gaussian factor integrates to one in r, and ψ and $\mathbb{P}$ are normalized. The label is independent of (y, o), so given ω and o the configuration has law $\mathbf{\Pi}_o$ and the record kernel is $\mathcal{N}((\mathrm{Hol}^{\mathrm{pr}}_\omega - I) z_{i_0}(y), \Sigma_{\mathrm{r}})$. The characteristic function of the record is $\mathbb{E}[e^{is^\top (\mathrm{Hol}^{\mathrm{pr}}_\omega - I)Z}] e^{-s^\top \Sigma_{\mathrm{r}} s/2}$, whose Gaussian factor never vanishes, so two record laws agree exactly when the laws of the two images of Z agree. (ii) Under ω_0 the image is 0. Under ω_1 invertibility gives $(\mathrm{Hol}^{\mathrm{pr}}_{\omega_1} - I)Z = 0$ almost surely exactly when $Z = 0$ almost surely; a rotation by α in the plane has $\det(R_\alpha - I) = 2 - 2\cos\alpha$. Given Z, the conditional mean of q is $|\Sigma_{\mathrm{r}}^{-1/2}(\mathrm{Hol}^{\mathrm{pr}}_\omega - I)Z|^2 + \dim \mathbb{V}_{\mathrm{pr}}$, and Tonelli's theorem gives the two means; the second exceeds the first exactly when the image of Z is nonzero with positive probability. (iii) The loop product telescopes under $\Theta^x_e \mapsto g_i \Theta^x_e g_j^{-1}$, so with $U = \rho_{\mathrm{pr}}(g_{i_0})$ one has $\mathrm{Hol}^{\mathrm{pr}}_\omega \mapsto U \mathrm{Hol}^{\mathrm{pr}}_\omega U^{-1}$ and $z_{i_0} \mapsto U z_{i_0}$, hence the new mean is $U(\mathrm{Hol}^{\mathrm{pr}}_\omega - I) z_{i_0}$ and the new record kernel is the image of the old one under $r \mapsto Ur$. The joint is thus the image under the bijection that changes the latent coordinates and multiplies r by U, which leaves the posterior over ω unchanged, and $r^\top U^\top (U \Sigma_{\mathrm{r}} U^\top)^{-1} U r = r^\top \Sigma_{\mathrm{r}}^{-1} r$. □

Physicist's summary. A block of agents at one base point is a small lattice gauge system whose invariants are loop products and edge twists modulo one global conjugation. A coarse description keeps a root frame, the tree transports to it, the holonomy of every independent loop, and the tree-dressed boundary links; the tree is a gauge fixing, as in lattice gauge theory, but averaged member features depend on it whenever the holonomy acts on them. The configurations that survive exact aggregation form the kernel of the connection Laplacian, for positive-definite couplings the holonomy-fixed vectors, and the members' marginal laws have one parent exactly when their transported laws coincide in a holonomy-invariant law, so a planar rotation loop forces a zero parent mean and a volume-stretching loop allows no Gaussian parent. Links between blocks compress exactly into rectangular endpoint maps, and into one link with new self-energies when an explicit matrix is invertible, uniquely once the self-energies are required to balance across the link; at the level of the laws themselves no such balanced form exists once the copies disagree. The parent frame is pure gauge for the constituents, and a noisy record of how the loop rotates the probe's state distinguishes holonomies unless that state is pinned at zero; Supplement B computes that holonomy from a connection along a base loop.

Chapter 12

Exact Agent-Network Coarse-Graining

This chapter asks what one exact coarse-graining step does to the interacting agent network of Chapter 5 at c_*, and which state must be kept so that the step closes. The effective action of a coarse agent network is a conditional partition function (Theorem 12.3), real analytic in bounded perturbations (Theorem 12.4); in Hoeffding interaction coordinates the step is an exact nonlinear map between spaces of different dimension whose finite-step multipliers are bounded by $3^{|V_n|}-1$ at every depth (Proposition 12.21); and exact closure keeps induced hyperedges, bridge kernels, joint attention-event laws, gauge-retained marks, and path laws with their memory (Table 12.1). The multiscale free energy of Chapter 13 is built from these objects, and the formation witnesses of Chapter 14 are exact marked parents in the sense fixed here.

At depth ℓ a coarse channel $C_\ell : \mathsf{Y}_\ell \rightsquigarrow \mathsf{Y}_{\ell+1}$ (Definition 10.1) maps the product state space of the finite agent set V_ℓ to the next depth; for one arrow the spaces are Y and Z with samples y and z. Local symbols are $\varrho_\ell, \mathsf{m}_\ell, M_\ell, \bar{\mathsf{m}}_\ell, H_\ell$ for the measure pair; $C^\dagger, \mathsf{U}, \mathcal{A}$ for the reverse kernel, its conditional expectation, and the action increment; $\mathcal{G}_\ell, \mathsf{g}, \mathsf{E}_{\ell,B}, \mathsf{P}_{\ell,B}, \mathrm{asm}_\ell, \mathrm{hd}_\ell$ for interaction coordinates; $\mathcal{T}_\ell, D_\ell, D_{n\leftarrow\ell}$ for the exact step and its multipliers; χ and T for a deterministic coarse map and a transition kernel; and a superscript c for coarse objects. Longer proofs are deferred to appendix B.

Scope. The retained-state table of Section 12.9 catalogs exact operations under their admission hypotheses and is not a completeness theorem; the coarse channel and the child blocks are inputs, not canonical selections; and the auxiliary path experiment of Section 12.8.1 has its own evidence and free energy, not identified with those of the original sensor model. [NOT-CLAIMED] Open items are the invariant retained subspaces of the interaction step (Open Problem 12.27), the dynamic closure of the pointwise parent (Open Problem 12.40), and jointly measurable equivariant conditional versions beyond the compact and finite cases (Open Problem 10.19).

12.1 Exact effective action

Definition 12.1 (Scale-indexed measure pair). For each depth ℓ let ϱ_ℓ be a probability measure on Y_ℓ, let $\mathsf{m}_\ell \ll \varrho_\ell$ be a finite positive measure, and let a depth arrow be a parameter-independent

Markov kernel $C_\ell : \mathsf{Y}_\ell \rightsquigarrow \mathsf{Y}_{\ell+1}$ acting on both components:

$$\mathsf{m}_\ell(dy) = \exp[-H_\ell(y)]\varrho_\ell(dy), \qquad 0 < M_\ell := \mathsf{m}_\ell(\mathsf{Y}_\ell) < \infty, \qquad \bar{\mathsf{m}}_\ell = \mathsf{m}_\ell/M_\ell, \tag{12.1}$$

$$\varrho_{\ell+1} = \varrho_\ell C_\ell, \qquad \mathsf{m}_{\ell+1} = \mathsf{m}_\ell C_\ell, \qquad H_{\ell+1} = -\log \frac{d\mathsf{m}_{\ell+1}}{d\varrho_{\ell+1}} \quad \varrho_{\ell+1}\text{-almost everywhere}, \tag{12.2}$$

with $\exp(-\infty) = 0$, so that each action is an extended real function defined almost everywhere. At a record o and for one arrow $C : \mathsf{Y} \rightsquigarrow \mathsf{Z}$ with its index suppressed, the fine pair is (ϱ, m_o) with $\mathsf{m}_o(dy) = L_o(y)\varrho(dy)$, $L_o = \exp(-H_o)$, $0 < M_o := \mathsf{m}_o(\mathsf{Y}) < \infty$, and $\bar{\mathsf{m}}_o = M_o^{-1}\mathsf{m}_o$, and the pushed pair, effective likelihood, and effective action are

$$\varrho^c = \varrho C, \quad \mathsf{m}_o^c = \mathsf{m}_o C, \quad \bar{\mathsf{m}}_o^c = \bar{\mathsf{m}}_o C = M_o^{-1}\mathsf{m}_o^c, \quad L_o^c = \frac{d\mathsf{m}_o^c}{d\varrho^c}, \quad H_o^c = -\log L_o^c. \tag{12.3}$$

When ϱ is the latent prior and L_o the record likelihood, $\bar{\mathsf{m}}_o$ is the posterior slice and M_o the evidence.

Proposition 12.2 (Exact composition and normalized law). Composable arrows act on the pair componentwise: $(\varrho_\ell, \mathsf{m}_\ell)(C_\ell C_{\ell+1}) = ((\varrho_\ell C_\ell)C_{\ell+1}, (\mathsf{m}_\ell C_\ell)C_{\ell+1})$. Moreover $\mathsf{m}_{\ell+1} \ll \varrho_{\ell+1}$, so that (12.2) is defined, $\mathsf{m}_{\ell+1}(\mathsf{Y}_{\ell+1}) = M_\ell$, and $\bar{\mathsf{m}}_{\ell+1} = \bar{\mathsf{m}}_\ell C_\ell$; the same statements hold for the single-arrow pair (12.3). [ESTABLISHED]

Proof. Composition on each component is Proposition 10.2. If $\varrho_{\ell+1}(A) = 0$, then $C_\ell(y, A) = 0$ for ϱ_ℓ-almost every y, hence for m_ℓ-almost every y, and $\mathsf{m}_{\ell+1}(A) = 0$. A Markov kernel preserves total mass, and dividing by that mass gives the normalized law. □

Theorem 12.3 (Conditional-partition formula and composition). Let $C_\varrho^\dagger(z, dy)$ be a reverse conditional of Y given $Z = z$ under the joint law $\varrho(dy)C(y, dz)$. Then for ϱ^c-almost every z

$$\exp[-H_o^c(z)] = \int \exp[-H_o(y)]C_\varrho^\dagger(z, dy), \qquad M_o = \int \exp[-H_o^c(z)]\varrho^c(dz). \tag{12.4}$$

For a second channel C' on Z, the direct effective likelihood of CC' and the staged one agree $\varrho CC'$-almost everywhere when their reverse conditionals are chosen from one joint tower. If $0 \le L_o \le 1$, as for a binary success record, then $0 \le L_o^c \le 1$, so one effective binary record realizes the whole coarse action; for a general record density L_o^c need not be bounded by one. [ESTABLISHED]

Proof. For bounded measurable f, $\int f(z)\mathsf{m}_o^c(dz) = \mathbb{E}_\varrho[f(Z)L_o(Y)] = \int f(z)\mathbb{E}_\varrho[L_o(Y) \mid Z = z]\varrho^c(dz)$ under the joint law, which identifies the Radon–Nikodym derivative, as in the density formula of Theorem 10.3 with $L_o = d\mathsf{m}_o/d\varrho$; $f = 1$ gives the evidence. The tower property proves composition, and a conditional expectation of a function with values in $[0, 1]$ takes values in $[0, 1]$. □

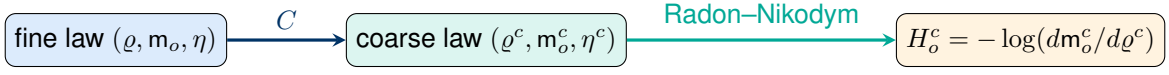


Figure 12.1: One exact step acts on the reference probability, the evidence measure, and the joint attention-event law; the effective action is the Radon–Nikodym derivative of the pushed evidence measure with respect to the pushed reference.

12.2 Bounded action calculus and contraction

Theorem 12.4 (Exact bounded action calculus). Fix a jointly measurable reverse kernel $C^\dagger(z, dy)$ of the normalized joint law and its reverse conditional expectation,

$$\bar{\mathsf{m}}_o(dy)C(y, dz) = \bar{\mathsf{m}}_o^c(dz)C^\dagger(z, dy), \qquad (\mathsf{U}h)(z) = \int h(y)C^\dagger(z, dy), \tag{12.5}$$

whose values on $\bar{\mathsf{m}}_o^c$-null sets are not used. A bounded ψ perturbs the evidence measure to $e^{-\psi}\mathsf{m}_o$; on $\mathfrak{B} = L^\infty(\bar{\mathsf{m}}_o; \mathbb{R})$ the exact coarse action increment $\mathcal{A} : \mathfrak{B} \to L^\infty(\bar{\mathsf{m}}_o^c; \mathbb{R})$ is

$$\mathcal{A}(\psi)(z) = -\log \frac{d\big((e^{-\psi}\mathsf{m}_o)C\big)}{d(\mathsf{m}_o C)}(z) = -\log \int e^{-\psi(y)}C^\dagger(z, dy), \tag{12.6}$$

the difference of the coarse actions of m_o and $e^{-\psi}\mathsf{m}_o$ where both are finite; elsewhere the calculus supplies no action difference and the Radon–Nikodym ratio is the definition.

(a) $e^{-\|\psi\|_\infty} \le \mathsf{U}(e^{-\psi}) \le e^{\|\psi\|_\infty}$, so $\|\mathcal{A}(\psi)\|_\infty \le \|\psi\|_\infty$, and $\mathcal{A}$ is real analytic on the whole of $\mathfrak{B}$.

(b) For $\psi \in \mathfrak{B}$ the tilted reverse kernel and its operator

$$C_\psi^\dagger(z, dy) = \frac{e^{-\psi(y)}C^\dagger(z, dy)}{\int e^{-\psi(u)}C^\dagger(z, du)}, \qquad \mathsf{U}^\psi h = \frac{\mathsf{U}(e^{-\psi}h)}{\mathsf{U}(e^{-\psi})}, \tag{12.7}$$

are positive, unital, and bounded from $L^\infty(\bar{\mathsf{m}}_o)$ to $L^\infty(\bar{\mathsf{m}}_o^c)$, and for bounded h and k

$$\mathcal{A}(\psi + h) - \mathcal{A}(\psi) = -\log \mathsf{U}^\psi(e^{-h}), \tag{12.8}$$

$$D\mathcal{A}(\psi)[h](z) = \int h(y)C_\psi^\dagger(z, dy) = (\mathsf{U}^\psi h)(z), \tag{12.9}$$

$$D^2\mathcal{A}(\psi)[h, k](z) = -\operatorname{Cov}_{C_\psi^\dagger(z,\cdot)}(h, k), \tag{12.10}$$

which at the origin read $D\mathcal{A}(0)[h] = \mathsf{U}h$ and $D^2\mathcal{A}(0)[h, k] = -\operatorname{Cov}_{C^\dagger(z,\cdot)}(h, k)$.

(c) The map is order preserving and additively homogeneous: $\psi \le \psi' + c$ implies $\mathcal{A}(\psi) \le \mathcal{A}(\psi') + c$, and $\mathcal{A}(\psi + c) = \mathcal{A}(\psi) + c$ for constants c. Consequently

$$\|\mathcal{A}(\psi) - \mathcal{A}(\psi')\|_\infty \le \|\psi - \psi'\|_\infty. \tag{12.11}$$

(d) Both $\mathcal{A}$ and U descend to the projective action space $\overline{\mathfrak{B}} = \mathfrak{B}/\mathbb{R}1$, which forgets the perturbed mass $(e^{-\psi}\mathsf{m}_o)(\mathsf{Y})$.

(e) Normalizing the perturbed fine law instead gives

$$\widehat{\mathsf{m}}^\psi = \frac{e^{-\psi}\bar{\mathsf{m}}_o}{\bar{\mathsf{m}}_o(e^{-\psi})}, \qquad \widetilde{\mathcal{A}}(\psi) := -\log \frac{d(\widehat{\mathsf{m}}^\psi C)}{d\bar{\mathsf{m}}_o^c} = \mathcal{A}(\psi) + \log \bar{\mathsf{m}}_o(e^{-\psi}), \tag{12.12}$$

which differs from $\mathcal{A}$ by a constant, with derivatives $D\widetilde{\mathcal{A}}(0)[h] = \mathsf{U}h - \bar{\mathsf{m}}_o(h)$ and $D^2\widetilde{\mathcal{A}}(0)[h,k] = -\operatorname{Cov}_{C^\dagger(z,\cdot)}(h,k) + \operatorname{Cov}_{\bar{\mathsf{m}}_o}(h,k)$. [ESTABLISHED]

Theorem 12.5 (Conditional-expectation contraction and exact defect). For every $1 \le p \le \infty$ the operator U of (12.5) extends uniquely to a mean-preserving contraction $\mathsf{U} : L^p(\bar{\mathsf{m}}_o;\mathbb{R}) \to L^p(\bar{\mathsf{m}}_o^c;\mathbb{R})$, and at $p = 2$ the loss is a conditional variance:

$$\|\mathsf{U}h\|_{L^p(\bar{\mathsf{m}}_o^c)} \le \|h\|_{L^p(\bar{\mathsf{m}}_o)}, \tag{12.13}$$

$$\|h\|^2_{L^2(\bar{\mathsf{m}}_o)} - \|\mathsf{U}h\|^2_{L^2(\bar{\mathsf{m}}_o^c)} = \int \operatorname{Var}_{C^\dagger(z,\cdot)}(h)\,\bar{\mathsf{m}}_o^c(dz) \ge 0, \tag{12.14}$$

with equality exactly when $h(y) = f(z)$ for $\bar{\mathsf{m}}_o(dy)C(y,dz)$-almost every (y,z) and some measurable f on Z; for a deterministic coarse map $z = \chi(y)$ this means $h = f \circ \chi$ $\bar{\mathsf{m}}_o$-almost everywhere. With the oscillation norm on $L^\infty(\bar{\mathsf{m}}_o)/\mathbb{R}1$ and the contraction coefficient of Dobrushin [33] of the reverse kernel,

$$\|[h]\|_{\mathrm{osc}} = \inf_{c\in\mathbb{R}} \|h - c\|_\infty = \tfrac{1}{2}\big(\operatorname{ess\,sup} h - \operatorname{ess\,inf} h\big), \tag{12.15}$$

$$\delta = \operatorname*{ess\,sup}_{(z,z')\sim(\bar{\mathsf{m}}_o^c)^{\otimes 2}} \sup_A \big|C^\dagger(z,A) - C^\dagger(z',A)\big|, \tag{12.16}$$

the inner supremum running over Borel sets, one has $\|[\mathsf{U}h]\|_{\mathrm{osc}} \le \delta\|[h]\|_{\mathrm{osc}}$, and along a finite composition of arrows the oscillation norm of the composed reverse operators is at most the product of the one-step coefficients. [ESTABLISHED]

The proofs of Theorems 12.4 and 12.5 are in appendix B.5.

Corollary 12.6 (Divergence bounds for a bounded action perturbation). In the setting of Theorem 12.4, which is the bounded tier of Remark 12.9, let $\psi \in \mathfrak{B}$ perturb the evidence measure to $e^{-\psi}\mathsf{m}_o$, whose action on the same reference is $H_o + \psi$. Then $\widehat{\mathsf{m}}^\psi$ of (12.12) is equivalent to $\bar{\mathsf{m}}_o$, and

$$D_{\mathrm{KL}}\big(\bar{\mathsf{m}}_o\|\widehat{\mathsf{m}}^\psi\big), \quad D_{\mathrm{KL}}\big(\widehat{\mathsf{m}}^\psi\|\bar{\mathsf{m}}_o\big) \le \min\big\{2\|[\psi]\|_{\mathrm{osc}}, \|[\psi]\|^2_{\mathrm{osc}}/2\big\}, \tag{12.17}$$

where both sides read ψ only through its class in the projective action space $\overline{\mathfrak{B}}$ of part (d) of Theorem 12.4. The offsets read the essential bounds of ψ instead: the log evidence ratio $\log \bar{\mathsf{m}}_o(e^{-\psi}) = \log\big((e^{-\psi}\mathsf{m}_o)(\Upsilon)/M_o\big)$, the constant of part (e), lies in $[-\operatorname{ess\,sup}\psi, -\operatorname{ess\,inf}\psi]$, and for every law $Q \ll \bar{\mathsf{m}}_o$ the free energies $\mathcal{F}_{\mathsf{m}_o}(Q) = -\log M_o + D_{\mathrm{KL}}(Q\|\bar{\mathsf{m}}_o)$ and $\mathcal{F}_{e^{-\psi}\mathsf{m}_o}(Q)$, of the form of Definition 5.6, satisfy $\mathcal{F}_{e^{-\psi}\mathsf{m}_o}(Q) = \mathcal{F}_{\mathsf{m}_o}(Q) + \mathbb{E}_Q\psi$ in $(-\infty,+\infty]$ with $\mathbb{E}_Q\psi \in [\operatorname{ess\,inf}\psi, \operatorname{ess\,sup}\psi]$, while both are $+\infty$ for every other law Q. Along arrows $C_0,\dots,C_{L-1}$ with pairs $(\varrho_\ell, \mathsf{m}_\ell)$ as in Definition 12.1, let $\widetilde{\mathsf{m}}_0 = e^{-\psi_0}\mathsf{m}_0$ and $\widetilde{\mathsf{m}}_{\ell+1} = e^{-\psi_{\ell+1}}(\widetilde{\mathsf{m}}_\ell C_\ell)$ with $\psi_\ell \in L^\infty(\bar{\mathsf{m}}_\ell)$, let $\widetilde{H}_\ell = -\log(d\widetilde{\mathsf{m}}_\ell/d\varrho_\ell)$, and let $\mathcal{A}_\ell$ be the action increment (12.6) of C_ℓ at m_ℓ. Then each $\widetilde{H}_\ell - H_\ell$ is essentially bounded, and

$$\widetilde{H}_{\ell+1} - H_{\ell+1} = \mathcal{A}_\ell(\widetilde{H}_\ell - H_\ell) + \psi_{\ell+1}, \qquad \sum_{j=0}^{L} \operatorname{ess\,inf}\psi_j \le \widetilde{H}_L - H_L \le \sum_{j=0}^{L} \operatorname{ess\,sup}\psi_j, \tag{12.18}$$

the recurrence holding $\mathsf{m}_{\ell+1}$-almost everywhere and the bounds m_L-almost everywhere. Hence the preceding statements hold with m_L in place of m_o and $\psi = \widetilde{H}_L - H_L$, with $\|[\widetilde{H}_L - H_L]\|_{\mathrm{osc}} \le \sum_{j=0}^{L} \|[\psi_j]\|_{\mathrm{osc}}$ and both offsets in the summed interval. [ESTABLISHED]

The proof is in appendix B.5.

The divergence bound sees the width of a perturbation and not its offset, which the evidence and the free energy carry. Outside $\mathfrak{B}$, for instance for the polynomial couplings on $\mathbb{R}^K$ described in Remark 12.9, the corollary gives no bound.

Corollary 12.7 (Contracted widths and transmitted offsets along a perturbed tower). In the setting of the tower of Corollary 12.6, write $\mathfrak{r}_\ell = \widetilde{H}_\ell - H_\ell$ for the propagated residual, let $C_\ell^\dagger$ be a reverse kernel of the normalized joint law $\bar{\mathsf{m}}_\ell(dy)C_\ell(y,dz)$ as in (12.5), so that $\mathcal{A}_\ell(\psi)(z) = -\log \int e^{-\psi(y)} C_\ell^\dagger(z,dy)$, and measure widths by the full essential range under $\bar{\mathsf{m}}_\ell$,

$$\omega_\ell = 2\|[\mathfrak{r}_\ell]\|_{\mathrm{osc}} = \operatorname{ess\,sup} \mathfrak{r}_\ell - \operatorname{ess\,inf} \mathfrak{r}_\ell, \qquad \omega_\ell^{\mathrm{st}} = 2\|[\psi_\ell]\|_{\mathrm{osc}}, \tag{12.19}$$

so that $\omega_0 = \omega_0^{\mathrm{st}}$. Let δ_ℓ be the coefficient (12.16) of $C_\ell^\dagger$, and let its projective diameter in the sense of Birkhoff, $\Delta_\ell \in [0,\infty]$, be the infimum of the $t \ge 0$ for which

$$C_\ell^\dagger(z,A)C_\ell^\dagger(z',B) \le e^t C_\ell^\dagger(z',A)C_\ell^\dagger(z,B) \tag{12.20}$$

holds, for $(\bar{\mathsf{m}}_{\ell+1})^{\otimes 2}$-almost every (z,z'), simultaneously for all Borel $A, B \subseteq \mathsf{Y}_\ell$, with $\Delta_\ell = \infty$ when no t qualifies and $\tanh(\infty) = 1$.

(a) For $0 \le \ell < L$,

$$\omega_{\ell+1} \le \min\Big\{\log\big(1 + \delta_\ell(e^{\omega_\ell} - 1)\big), \tanh(\Delta_\ell/4)\omega_\ell\Big\} + \omega_{\ell+1}^{\mathrm{st}}. \tag{12.21}$$

When $\omega_\ell > 0$, the first term of the minimum is smaller than ω_ℓ exactly when $\delta_\ell < 1$, and the second exactly when $\Delta_\ell < \infty$.

(b) The first term is the exact supremum at given coefficient and width: for every $\delta \in [0,1]$ and $x \ge 0$ there are a normalized law on a two-point space, a channel into a two-point space whose reverse kernel has coefficient δ, and a ψ with $\operatorname{ess\,sup}\psi - \operatorname{ess\,inf}\psi = x$, for which $\operatorname{ess\,sup}\mathcal{A}(\psi) - \operatorname{ess\,inf}\mathcal{A}(\psi) = \log(1+\delta(e^x-1))$, and for $\delta > 0$ this reverse kernel has infinite projective diameter. Hence no bound on the width of $\mathcal{A}_\ell(\psi)$ by a function of δ_ℓ and the width of ψ alone is smaller than the first term, and $\delta_\ell < 1$ does not imply $\Delta_\ell < \infty$.

(c) If $\delta_\ell \le \delta < 1$ for $0 \le \ell < L$ and $\omega_\ell^{\mathrm{st}} \le s$ for $1 \le \ell \le L$ with $s < \log(1/\delta)$, read as $+\infty$ when $\delta = 0$, then $\omega_\ell \le \max\{\omega_0, x_{\delta,s}\}$ for $0 \le \ell \le L$, where $x_{\delta,s}$ is the unique $x \ge 0$ with $x - \log(1+\delta(e^x-1)) = s$. If instead $\Delta_\ell \le \Delta < \infty$ for $0 \le \ell < L$ and $\omega_\ell^{\mathrm{st}} \le s$ for $1 \le \ell \le L$, with no condition on $s \ge 0$, then $\omega_\ell \le \max\{\omega_0, s/(1-\tanh(\Delta/4))\}$ for $0 \le \ell \le L$. Neither bound depends on L, and by Corollary 12.6 both divergences between $\bar{\mathsf{m}}_L$ and the normalization of $\widetilde{\mathsf{m}}_L$ are at most $\min\{\omega_L, \omega_L^2/8\}$.

(d) If $C_\ell(y,A) = 1_A(\chi_\ell(y))$ for a Borel map $\chi_\ell : \mathsf{Y}_\ell \to \mathsf{Y}_{\ell+1}$ and $\bar{\mathsf{m}}_{\ell+1}$ is not a point mass, then $\delta_\ell = 1$ and $\Delta_\ell = \infty$, and (12.21) reduces to the additive recursion $\omega_{\ell+1} \le \omega_\ell + \omega_{\ell+1}^{\mathrm{st}}$ behind

(12.18). If every arrow has this form, F is a bounded Borel function on Y_L, $c_0, \dots, c_L \geq 0$, and $\psi_j = c_j F \circ \chi_{L-1} \circ \cdots \circ \chi_j$ (with $\psi_L = c_L F$), then $\omega_\ell = \sum_{j=0}^{\ell} \omega_j^{\mathrm{st}}$ for $0 \leq \ell \leq L$, so the additive bound is attained at every depth.

(e) For every $\psi \in L^\infty(\bar{\mathsf{m}}_\ell)$, $\log \bar{\mathsf{m}}_{\ell+1}(e^{-\mathcal{A}_\ell(\psi)}) = \log \bar{\mathsf{m}}_\ell(e^{-\psi})$. With $\widehat{\mathsf{n}}_j = \widetilde{\mathsf{m}}_{j-1} C_{j-1} / \widetilde{\mathsf{m}}_{j-1}(\mathsf{Y}_{j-1})$ for $1 \leq j \leq L$, the log evidence ratio decomposes as

$$\log \frac{\widetilde{\mathsf{m}}_L(\mathsf{Y}_L)}{M_L} = \log \bar{\mathsf{m}}_L(e^{-\mathfrak{r}_L}) = \log \bar{\mathsf{m}}_0(e^{-\psi_0}) + \sum_{j=1}^{L} \log \widehat{\mathsf{n}}_j(e^{-\psi_j}), \tag{12.22}$$

each summand lying in $[-\operatorname{ess\,sup} \psi_j, -\operatorname{ess\,inf} \psi_j]$. Replacing one ψ_j by $\psi_j + c$ for a constant c leaves every ω_ℓ and the normalization of every $\widetilde{\mathsf{m}}_\ell$ unchanged and lowers the log evidence ratio by exactly c, whatever the arrows, including arrows with $\delta_\ell = 0$. [ESTABLISHED]

The proof is in appendix B.5.

The factor $\tanh(\Delta_\ell/4)$ in (a) has the form of the contraction ratio $\tanh(\Delta/4)$, in Hilbert's projective metric, of a positive linear map whose image cone has projective diameter Δ [13, Sec. 4, Lemma 1, Eq. (9a)], stated for nonnegative kernels acting on nonnegative measures by Le Gland and Oudjane [73, Lemma 3.8, Eq. (11)]. The proof does not invoke that lemma: it bounds the width directly for rows defined almost everywhere, with Δ_ℓ defined on rows by the cross-ratio (12.20).

Along a stochastic tower the widths, which control the divergences, contract through (12.21), while the offsets, which the evidence and the free energy carry, pass every arrow with coefficient one.

12.3 Finite-network interaction coordinates

Definition 12.8 (Product-referenced network scale). At depth ℓ agent $i \in V_\ell$ has state space $\mathsf{Y}_{\ell i}$ and a probability law $\nu_{\ell i}$, and the normalized law $\bar{\mathsf{m}}_\ell$ of Definition 12.1 on $\mathsf{Y}_\ell = \prod_{i \in V_\ell} \mathsf{Y}_{\ell i}$ is mutually absolutely continuous, $\bar{\mathsf{m}}_\ell \sim \nu_\ell$, with the product reference $\nu_\ell = \bigotimes_{i \in V_\ell} \nu_{\ell i}$. Equivalence identifies $L^\infty(\bar{\mathsf{m}}_\ell)$ with $L^\infty(\nu_\ell)$ isometrically, with no bound on the Radon–Nikodym ratio, and the product structure of ν_ℓ makes the coordinate expectations below well defined on equivalence classes by Fubini's theorem. For an arrow C_ℓ, equivalence of $\bar{\mathsf{m}}_{\ell+1} = \bar{\mathsf{m}}_\ell C_\ell$ to a product reference $\nu_{\ell+1}$ is a separate premise (Proposition 12.10). If $V_\ell = \varnothing$, then Y_ℓ is a point, $\mathcal{G}_\ell = \{0\}$, and every map below is zero; estimates involving $3^{|V_\ell|} - 1$ assume $V_\ell \neq \varnothing$.

Remark 12.9 (The bounded tier). Every interaction component below is essentially bounded. The bounded tier contains every interaction on finite agent spaces and every continuous interaction on compact ones, hence the natural couplings of categorical agents and of compact group-valued states. On $\mathsf{Y}_{\ell i} = \mathbb{R}^K$ it excludes the polynomial couplings: for a matrix $\mathsf{N} \neq 0$ and every $M > 0$ the set where $|y_i^\top \mathsf{N} y_j| > M$ is a nonempty open subset of $\mathbb{R}^K \times \mathbb{R}^K$, so it has positive mass under every reference with full support, and $y_i^\top \mathsf{N} y_j \notin L^\infty(\nu_\ell)$. Such couplings need a separate tier, exact Gaussian marginalization or a weighted action space, which this chapter does not construct. [ESTABLISHED]

Proposition 12.10 (Product equivalence is an admitted, not an automatic, scale premise). Let $\mathsf{Y}_0 = \{0,1\}$ with its uniform law $\bar{\mathsf{m}}_0 = \nu_0$, and let C be the deterministic diagonal-cloning channel $y \mapsto (y,y)$ from $\{0,1\}$ to $\{0,1\}^2$. Then $\bar{\mathsf{m}}_1 = \bar{\mathsf{m}}_0 C$ is supported on $\{(0,0),(1,1)\}$, no product probability $\nu_{11} \otimes \nu_{12}$ on $\{0,1\}^2$ is equivalent to it, and no target Hoeffding decomposition relative to a product reference is admitted at this target. [ESTABLISHED]

Proof. Equivalence would make both diagonal atoms positive under the product, so each marginal would charge both 0 and 1 and both off-diagonal atoms would have positive product mass, while $\bar{\mathsf{m}}_1$ assigns them mass zero. □

Definition 12.11 (Hoeffding coordinates). Under Definition 12.8, for $B \subseteq V_\ell$ let $(\mathbb{E}_{\ell,B} f)(y_B) = \int f(y_B, y'_{B^c}) \nu_{\ell,B^c}(dy'_{B^c})$ integrate out the complementary coordinates against the product reference, with $\mathbb{E}_{\ell,\varnothing} f = \nu_\ell(f)$. The Hoeffding projectors are the Möbius inversions

$$\mathsf{P}_{\ell,B} = \sum_{B' \subseteq B} (-1)^{|B|-|B'|} \mathbb{E}_{\ell,B'} = \prod_{i \in B} \big(I - \mathbb{E}_{\ell,V_\ell \setminus \{i\}}\big) \mathbb{E}_{\ell,B}, \qquad B \subseteq V_\ell, \tag{12.23}$$

where the alternating sum is the definition and the product is the identity of Theorem 12.12. The component spaces $\mathcal{H}_{\ell,B} = \mathsf{P}_{\ell,B} L^\infty(\nu_\ell)$ form the interaction space $\mathcal{G}_\ell = \bigoplus^{\ell^1}_{\varnothing \neq B \subseteq V_\ell} \mathcal{H}_{\ell,B}$ with norm $\|\mathsf{g}\|_{\mathcal{G}_\ell} = \sum_{\varnothing \neq B \subseteq V_\ell} \|\mathsf{g}_B\|_\infty$, so every nonempty hyperedge has its component and no bound on $|V_\ell|$ is imposed. With the action quotient $\overline{\mathfrak{B}}_\ell = L^\infty(\nu_\ell)/\mathbb{R}1$, normed by (12.15), the assembly and extraction maps are

$$\psi_{\mathsf{g}} = \sum_{\varnothing \neq B \subseteq V_\ell} \mathsf{g}_B, \qquad \operatorname{asm}_\ell \mathsf{g} = [\psi_{\mathsf{g}}], \qquad \operatorname{hd}_\ell[f] = (\mathsf{P}_{\ell,B} f)_{\varnothing \neq B \subseteq V_\ell}. \tag{12.24}$$

Theorem 12.12 (Hoeffding assembly is an exact finite-network action isomorphism). For every finite V_ℓ the coordinate expectations commute and satisfy $\mathbb{E}_{\ell,B}\mathbb{E}_{\ell,B'} = \mathbb{E}_{\ell,B\cap B'}$, the product formula in (12.23) holds, and $\mathsf{P}_{\ell,B}\mathsf{P}_{\ell,B'} = \mathbf{1}_{B=B'}\mathsf{P}_{\ell,B}$ with $\sum_{B \subseteq V_\ell} \mathsf{P}_{\ell,B} = I$. Each component g_B depends only on y_B and has zero ν_ℓ-mean in every coordinate of B. The maps of (12.24) are mutually inverse bounded linear isomorphisms,

$$\operatorname{hd}_\ell \operatorname{asm}_\ell = I_{\mathcal{G}_\ell}, \qquad \operatorname{asm}_\ell \operatorname{hd}_\ell = I_{\overline{\mathfrak{B}}_\ell}, \qquad \|\operatorname{asm}_\ell\| \le 1, \qquad \|\operatorname{hd}_\ell\| \le 3^{|V_\ell|} - 1. \tag{12.25}$$

The last bound is not dimension free, and its exponential dependence is sharp: for independent $\{-1,1\}$-valued coordinates with $\Pr(Y_i = 1) = p$ with $1/2 \le p < 1$ and $f = \prod_i Y_i$ on $n = |V_\ell|$ agents, the sum of the component norms is $(4p-1)^n - (2p-1)^n$, which tends to $3^n - 1$ as $p \uparrow 1$. [ESTABLISHED]

The proof is in appendix B.5.

Corollary 12.13 (Gauge covariance of interaction components). Let a passive change $g = (g_i)$ of Chapter 2, one element per agent acting on both the belief and the model components of that agent's state, be realized at depth ℓ by coordinate Borel isomorphisms and a Borel permutation ς of

the agents, whose product map $\kappa_\ell : \mathsf{Y}_\ell \to \widehat{\mathsf{Y}}_\ell$ sends ν_ℓ to a product reference $\widehat{\nu}_\ell$. With $\mathsf{K}_\ell f = f \circ \kappa_\ell^{-1}$, its action $\overline{\mathsf{K}}_\ell$ on the quotient by constants, and its action $\mathsf{K}_\ell^{\mathcal{G}}$ on components,

$$\mathsf{K}_\ell \mathbb{E}^{\nu}_{\ell,B} = \mathbb{E}^{\widehat{\nu}}_{\ell,\varsigma B} \mathsf{K}_\ell, \qquad \mathsf{K}_\ell \mathsf{P}^{\nu}_{\ell,B} = \mathsf{P}^{\widehat{\nu}}_{\ell,\varsigma B} \mathsf{K}_\ell, \qquad \overline{\mathsf{K}}_\ell \operatorname{asm}_\ell = \widehat{\operatorname{asm}}_\ell \mathsf{K}^{\mathcal{G}}_\ell, \qquad \mathsf{K}^{\mathcal{G}}_\ell \operatorname{hd}_\ell = \widehat{\operatorname{hd}}_\ell \overline{\mathsf{K}}_\ell. \tag{12.26}$$

Preservation of the product reference alone is insufficient: the product-Haar-preserving shear $(y_1, y_2) \mapsto (y_1, y_1 + y_2)$ of the torus $\mathbb{T}^2$ sends a function of y_2 to one depending on both coordinates and moves a singleton component into a pair component. If only almost-everywhere reverse-kernel equivariance is supplied, the covariance holds for L^∞ equivalence classes and not for a selected pointwise version. [ESTABLISHED]

Proof. Fubini's theorem proves the first identity, and the others follow by the finite Möbius sums. The shear maps $f(y_2)$ to $f(y_2 - y_1)$, which is not a function of one coordinate when f is nonconstant. □

12.4 The exact interaction step, its residual, and finite-step multipliers

Definition 12.14 (Exact interaction step). Assume Definition 12.8 at depths ℓ and $\ell+1$, let $\mathcal{A}_\ell$ be the action increment of Theorem 12.4 for the arrow C_ℓ at the base law $\bar{\mathsf{m}}_\ell$, and let $\overline{\mathcal{A}}_\ell : \overline{\mathfrak{B}}_\ell \to \overline{\mathfrak{B}}_{\ell+1}$ be its quotient map, defined on the whole bounded space by additive homogeneity. The exact interaction step is

$$\mathcal{T}_\ell = \operatorname{hd}_{\ell+1} \overline{\mathcal{A}}_\ell \operatorname{asm}_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}, \qquad \mathsf{g}^{\mathrm{ex}}_{\ell+1} = \mathcal{T}_\ell(\mathsf{g}_\ell). \tag{12.27}$$

Lemma 12.15 (Mass, origin, and derivative of the exact step). The sum ψ_{g} is the representative of $\operatorname{asm}_\ell \mathsf{g}$ with zero ν_ℓ-mean, and the perturbed evidence mass is carried separately and preserved by the arrow,

$$M_\ell(\mathsf{g}) := M_\ell \bar{\mathsf{m}}_\ell\big(e^{-\psi_{\mathsf{g}}}\big) = \big((e^{-\psi_{\mathsf{g}}} \mathsf{m}_\ell) C_\ell\big)(\mathsf{Y}_{\ell+1}) =: M_{\ell+1}(\mathsf{g}). \tag{12.28}$$

The step fixes the origin, $\mathcal{T}_\ell(0) = 0$, for every model and every channel. The tilted operator $\mathsf{U}_\ell^{\psi_{\mathsf{g}}}$ of (12.7) is unchanged when a constant is added to ψ_{g}, descends to the action quotient, and gives the derivative

$$D\mathcal{T}_\ell(\mathsf{g}) = \operatorname{hd}_{\ell+1} \overline{\mathsf{U}_\ell^{\psi_{\mathsf{g}}}} \operatorname{asm}_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}; \tag{12.29}$$

only at $\mathsf{g} = 0$ may the tilted operator be replaced by U_ℓ. [ESTABLISHED]

Proof. A Markov kernel preserves total mass, which gives (12.28), and $\mathcal{A}_\ell(0) = -\log \mathsf{U}_\ell(1) = 0$. A constant added to ψ_{g} multiplies numerator and denominator of $\mathsf{U}_\ell^{\psi_{\mathsf{g}}} h$ by one positive number, and the chain rule with part (b) of Theorem 12.4 gives (12.29). □

Remark 12.16 (Two agents and one exact step). Let $V_0 = \{1, 2\}$ with $\mathsf{Y}_{0i} = \{-1, 1\}$ and uniform product base $\bar{\mathsf{m}}_0 = \nu_0$, and perturb by $\psi(y) = -\xi_1 y_1 - \xi_2 y_2 - \xi_{12} y_1 y_2$, whose Hoeffding components are $-\xi_1 y_1$, $-\xi_2 y_2$, and $-\xi_{12} y_1 y_2$, so $\dim \mathcal{G}_0 = 3$. The channel $C_0(y, dz) = \delta_{y_1}(dz)$ keeps the first agent, so $V_1 = \{I\}$, $\mathsf{Y}_1 = \{-1, 1\}$, the pushed base is uniform, and $\dim \mathcal{G}_1 = 1$. The reverse kernel

sets $y_1 = z$ and leaves y_2 uniform, so $\mathcal{A}_0(\psi)(z) = -\xi_1 z - \log\cosh(\xi_2 + \xi_{12} z)$, and the exact step is $\mathcal{T}_0(\mathsf{g})_{\{I\}}(z) = -\xi_1' z$ with

$$\xi_1' = \xi_1 + \tfrac{1}{2}\log\frac{\cosh(\xi_2+\xi_{12})}{\cosh(\xi_2-\xi_{12})} = \xi_1 + \operatorname{artanh}\big(\tanh\xi_2\tanh\xi_{12}\big).$$

The pair coupling enters the coarse field only through the polarization of the eliminated agent, $\partial\xi_1'/\partial\xi_{12} = \tanh\xi_2$ at $\xi_{12} = 0$. At the origin $D\mathcal{T}_0(0)$ keeps the field of agent 1 and sends the field of agent 2 and the pair direction to zero, since $\mathsf{U}_0(y_2) = \mathsf{U}_0(y_1y_2) = 0$; at a point with $\xi_2 \neq 0$ the pair direction has the nonzero image $-\tanh(\xi_2)z$. The drop from dimension 3 to 1 is the obstruction of Proposition 12.22, and the multipliers obey the bound $3^2 - 1 = 8$ of Proposition 12.21. [ESTABLISHED]

Proof. Each monomial has zero mean in each of its coordinates, which gives the components by (12.23). Averaging $e^{-\psi}$ over y_2 at $y_1 = z$ gives $e^{\xi_1 z}\cosh(\xi_2 + \xi_{12}z)$, and subtracting the mean of $\mathcal{A}_0(\psi)$ over $z = \pm 1$ gives the coefficient $-\xi_1'$; the artanh form follows from the addition formula for $\cosh$. Differentiating at $\xi_{12} = 0$ gives $\tanh\xi_2$, and (12.29) at $\psi = -\xi_2 y_2$ gives $\mathsf{U}_0^{\psi}(y_1y_2)(z) = z\tanh\xi_2$. □

Proposition 12.17 (A product base and a block-product channel preserve the intra-block sector). Suppose that $\bar{\mathsf{m}}_\ell = \nu_\ell$ is a product law, that blocks $I \in V_{\ell+1}$ partition V_ℓ, each identified with its coarse agent, and that the channel factors as

$$C_\ell(y, dz) = \prod_{I\in V_{\ell+1}} C_I(y_I, dz_I) \tag{12.30}$$

with Markov kernels C_I. Then $\bar{\mathsf{m}}_{\ell+1} = \bar{\mathsf{m}}_\ell C_\ell$ is a product law, so the convention $\bar{\mathsf{m}}_{\ell+1} = \nu_{\ell+1}$ is admissible at the next depth. Let $\mathcal{G}_\ell^{\mathrm{intra}} = \{\mathsf{g} \in \mathcal{G}_\ell : \mathsf{g}_B = 0 \text{ unless } B \subseteq I \text{ for some } I\}$. For every $\mathsf{g} \in \mathcal{G}_\ell^{\mathrm{intra}}$ and every product reference $\nu_{\ell+1} \sim \bar{\mathsf{m}}_{\ell+1}$, the components of $\mathcal{T}_\ell(\mathsf{g})$ vanish on every $B \subseteq V_{\ell+1}$ with $|B| \geq 2$; in particular the singleton sector $\{\mathsf{g} : \mathsf{g}_B = 0 \text{ for } |B| \geq 2\}$, which lies in $\mathcal{G}_\ell^{\mathrm{intra}}$, is mapped into the singleton sector of the next depth. [ESTABLISHED]

The proof is in appendix B.5.

Definition 12.18 (Scale-indexed evolution and its derivative cocycle). Let $(\mathscr{X}_\ell, \|\cdot\|_\ell)_{\ell\geq 0}$ be Banach spaces and let $\mathcal{T}_\ell : \mathscr{O}_\ell \to \mathscr{X}_{\ell+1}$ be continuously Fréchet differentiable on an open set $\mathscr{O}_\ell \subseteq \mathscr{X}_\ell$. The ordered evolution $\mathcal{T}_{n\leftarrow\ell} = \mathcal{T}_{n-1}\circ\cdots\circ\mathcal{T}_\ell$, with $\mathcal{T}_{\ell\leftarrow\ell} = \mathrm{id}_{\mathscr{X}_\ell}$, is defined at the points whose successive images stay in the successive domains. Along an exact orbit $\xi_{\ell+1} = \mathcal{T}_\ell(\xi_\ell)$ in those domains, the finite-step multipliers are

$$D_\ell = D\mathcal{T}_\ell(\xi_\ell) \in \mathcal{L}(\mathscr{X}_\ell, \mathscr{X}_{\ell+1}), \qquad D_{n\leftarrow\ell} = D_{n-1}\cdots D_\ell, \qquad D_{\ell\leftarrow\ell} = I_{\mathscr{X}_\ell}, \tag{12.31}$$

with the rightmost factor acting first.

Proposition 12.19 (Ordered composition of the evolution and its derivative). For $\ell \le n \le r$ and an exact orbit as in Definition 12.18,

$$\mathcal{T}_{r\leftarrow n} \circ \mathcal{T}_{n\leftarrow \ell} = \mathcal{T}_{r\leftarrow \ell}, \qquad D_{n\leftarrow \ell} = D\mathcal{T}_{n\leftarrow \ell}(\xi_\ell), \qquad D_{r\leftarrow n} D_{n\leftarrow \ell} = D_{r\leftarrow \ell}. \tag{12.32}$$

[ESTABLISHED]

Proof. The first and third identities are associativity. The second follows by induction on $n - \ell$ from the Fréchet chain rule applied to $\mathcal{T}_{n+1\leftarrow \ell} = \mathcal{T}_n \circ \mathcal{T}_{n\leftarrow \ell}$ at ξ_ℓ. □

Definition 12.20 (Finite-step multipliers of the interaction step). Instantiate Definition 12.18 with $\mathscr{X}_\ell = \mathcal{G}_\ell$, $\mathcal{T}_\ell$ the exact interaction step, and an exact interaction orbit $\mathrm{g}_{\ell+1} = \mathcal{T}_\ell(\mathrm{g}_\ell)$. By Lemma 12.15 the one-step multipliers are

$$D_\ell = D\mathcal{T}_\ell(\mathrm{g}_\ell) = \mathrm{hd}_{\ell+1}\, \overline{\mathsf{U}_\ell^{\psi_{\mathrm{g}_\ell}}}\, \mathrm{asm}_\ell : \mathcal{G}_\ell \longrightarrow \mathcal{G}_{\ell+1}, \tag{12.33}$$

and $D_{n\leftarrow \ell}$ are their ordered products over the finite depth.

Proposition 12.21 (Native growth bound for the exact interaction cocycle). Equip every action quotient $\overline{\mathfrak{B}}_k$ with the oscillation norm (12.15). Along an exact interaction orbit with multipliers (12.33) and for $n > \ell$, the ordered product telescopes, and its norm is bounded by the target size:

$$D_{n\leftarrow \ell} = \mathrm{hd}_n\, \overline{\mathsf{U}_{n-1}^{\psi_{\mathrm{g}_{n-1}}}} \cdots \overline{\mathsf{U}_\ell^{\psi_{\mathrm{g}_\ell}}}\, \mathrm{asm}_\ell, \tag{12.34}$$

$$\|D_{n\leftarrow \ell}\|_{\mathcal{G}_\ell \to \mathcal{G}_n} \le 3^{|V_n|} - 1. \tag{12.35}$$

The bound does not depend on the number of steps, and it is zero when $V_n = \varnothing$. [ESTABLISHED]

Proof. By (12.25), $\mathrm{asm}_{k+1}\, \mathrm{hd}_{k+1} = I$ on $\overline{\mathfrak{B}}_{k+1}$, so every interior pair cancels in $D_{n-1} \cdots D_\ell$, which gives (12.34). Each tilted operator is positive and unital by Theorem 12.4, so $\mathsf{U}_k^\psi(f - c) = \mathsf{U}_k^\psi f - c$ and $\|\mathsf{U}_k^\psi(f - c)\|_\infty \le \|f - c\|_\infty$ for every constant c, and the infimum over c shows that its quotient has norm at most one. Multiplying by the bounds on asm_ℓ and hd_n in (12.25) gives (12.35). □

Proposition 12.22 (Finite networks admit no typed interaction comparison across a reducing step). Let every agent space be finite. Then $\mathcal{G}_\ell \cong L^\infty(\nu_\ell)/\mathbb{R}1$ has dimension $|\mathrm{supp}\, \nu_\ell| - 1 = \prod_{i\in V_\ell} |\mathrm{supp}\, \nu_{\ell i}| - 1$. If a step reduces this number, no space admits linear isomorphisms onto both $\mathcal{G}_\ell$ and $\mathcal{G}_{\ell+1}$, so no comparison identification of the interaction spaces exists across that step; six binary agents blocked into three binary blocks give $\dim \mathcal{G}_\ell = 63$ and $\dim \mathcal{G}_{\ell+1} = 7$. If the blocking ratios are the vertex ratios $b_k = |V_k|/|V_{k+1}| \ge b > 1$, then $|V_n| \le |V_\ell| b^{-(n-\ell)}$, and a nonempty V_n forces $n - \ell \le \log_b |V_\ell|$, so a finite network admits only finitely many such steps. [ESTABLISHED]

Proof. By Theorem 12.12, $\mathcal{G}_\ell$ is linearly isomorphic to $L^\infty(\nu_\ell)/\mathbb{R}1$. On a finite space $L^\infty(\nu_\ell)$ is the space of functions on the support of ν_ℓ, the product of the coordinate supports, and the quotient lowers the dimension by one; linear isomorphisms preserve dimension. The vertex ratios telescope to $|V_\ell|/|V_n| = \prod_{k=\ell}^{n-1} b_k \ge b^{n-\ell}$, and $|V_n| \ge 1$ bounds $n - \ell$. □

On finite networks the depth objects are therefore the exact step, its residual (Proposition 12.24), and the multipliers, none of which needs a comparison isomorphism; blocking sequences with independently declared ratios, and infinite networks, are treated in Supplement B.

Definition 12.23 (Retained projection and closure residual). Let $\mathsf{P}_\ell^{\mathrm{ret}} : \mathcal{G}_\ell \to \mathcal{G}_\ell$ be a bounded idempotent that intertwines the gauge action of Corollary 12.13. The retained output, the coordinate residual, and the action residual are

$$\mathsf{g}_{\ell+1}^{\mathrm{ret}} = \mathsf{P}_{\ell+1}^{\mathrm{ret}} \mathcal{T}_\ell(\mathsf{g}_\ell), \qquad r_{\ell+1}^{\mathcal{G}}(\mathsf{g}_\ell) = \big(I - \mathsf{P}_{\ell+1}^{\mathrm{ret}}\big)\mathcal{T}_\ell(\mathsf{g}_\ell), \tag{12.36}$$

$$\overline{r}_{\ell+1}^{\mathcal{A}}(\mathsf{g}_\ell) = \overline{\mathcal{A}}_\ell \,\mathrm{asm}_\ell\, \mathsf{g}_\ell - \mathrm{asm}_{\ell+1}\, \mathsf{P}_{\ell+1}^{\mathrm{ret}}\, \mathrm{hd}_{\ell+1}\, \overline{\mathcal{A}}_\ell \,\mathrm{asm}_\ell\, \mathsf{g}_\ell. \tag{12.37}$$

Proposition 12.24 (Residual identity and exact-image invariance). The residuals satisfy $\overline{r}_{\ell+1}^{\mathcal{A}} = \mathrm{asm}_{\ell+1}\, r_{\ell+1}^{\mathcal{G}}$, so one vanishes exactly when the other does, and

$$\mathcal{T}_\ell\big(\mathrm{Ran}\,\mathsf{P}_\ell^{\mathrm{ret}}\big) \subseteq \mathrm{Ran}\,\mathsf{P}_{\ell+1}^{\mathrm{ret}} \quad \Longleftrightarrow \quad r_{\ell+1}^{\mathcal{G}}(\mathsf{g}) = 0 \quad \text{for every } \mathsf{g} \in \mathrm{Ran}\,\mathsf{P}_\ell^{\mathrm{ret}}. \tag{12.38}$$

A retained update is exact precisely under this exact-image invariance; otherwise it is a projection scheme with the displayed residual. For $V_{\ell+1} \neq \varnothing$,

$$\mathrm{hd}_{\ell+1}\, \overline{r}_{\ell+1}^{\mathcal{A}} = r_{\ell+1}^{\mathcal{G}}, \qquad \frac{\|r_{\ell+1}^{\mathcal{G}}\|_{\mathcal{G}_{\ell+1}}}{3^{|V_{\ell+1}|} - 1} \le \|\overline{r}_{\ell+1}^{\mathcal{A}}\|_{\overline{\mathfrak{B}}_{\ell+1}} \le \|r_{\ell+1}^{\mathcal{G}}\|_{\mathcal{G}_{\ell+1}}. \tag{12.39}$$

[ESTABLISHED]

Proof. Since $\mathrm{asm}_{\ell+1}\, \mathrm{hd}_{\ell+1} = I$, the first term of (12.37) is $\mathrm{asm}_{\ell+1}\, \mathcal{T}_\ell(\mathsf{g}_\ell)$, so the action residual is $\mathrm{asm}_{\ell+1}$ applied to $(I - \mathsf{P}_{\ell+1}^{\mathrm{ret}})\mathcal{T}_\ell(\mathsf{g}_\ell)$, and $\mathrm{asm}_{\ell+1}$ is injective. An element lies in the range of an idempotent exactly when the complementary idempotent annihilates it, which is (12.38). Applying $\mathrm{hd}_{\ell+1}$ gives the coordinate identity, and the norm bounds are (12.25). □

Proposition 12.25 (Radon–Nikodym covariance of the nonlinear interaction step). At adjacent depths let κ_ℓ and $\kappa_{\ell+1}$ be realizations as in Corollary 12.13, transform the measure pair and the normalized law by pushforward, $\widehat{\varrho}_\ell = (\kappa_\ell)_{\#}\varrho_\ell$, $\widehat{\mathsf{m}}_\ell = (\kappa_\ell)_{\#}\mathsf{m}_\ell$, and $\widehat{\bar{\mathsf{m}}}_\ell = (\kappa_\ell)_{\#}\bar{\mathsf{m}}_\ell$, and suppose that the transformed channel satisfies $\widehat{C}_\ell(\kappa_\ell y, \kappa_{\ell+1} A) = C_\ell(y, A)$ for every Borel set A. Then, as Radon–Nikodym equivalence classes,

$$\widehat{\mathcal{A}}_\ell(\mathsf{K}_\ell \psi) = \mathsf{K}_{\ell+1} \mathcal{A}_\ell(\psi), \tag{12.40}$$

$$\widehat{\mathsf{U}}_\ell^{\mathsf{K}_\ell\psi} \mathsf{K}_\ell h = \mathsf{K}_{\ell+1} \mathsf{U}_\ell^{\psi} h, \tag{12.41}$$

$$\widehat{\mathcal{T}}_\ell \mathsf{K}_\ell^{\mathcal{G}} = \mathsf{K}_{\ell+1}^{\mathcal{G}} \mathcal{T}_\ell. \tag{12.42}$$

If the retained projections also intertwine, $\widehat{\mathsf{P}}^{\mathrm{ret}}_k \mathsf{K}^{\mathcal{G}}_k = \mathsf{K}^{\mathcal{G}}_k \mathsf{P}^{\mathrm{ret}}_k$ for $k = \ell, \ell+1$, then both residuals are covariant,

$$\widehat{r}^{\mathcal{G}}_{\ell+1}(\mathsf{K}^{\mathcal{G}}_\ell \mathrm{g}) = \mathsf{K}^{\mathcal{G}}_{\ell+1} r^{\mathcal{G}}_{\ell+1}(\mathrm{g}), \tag{12.43}$$

$$\widehat{\overline{r}}^{\mathcal{A}}_{\ell+1}(\mathsf{K}^{\mathcal{G}}_\ell \mathrm{g}) = \overline{\mathsf{K}}_{\ell+1} \overline{r}^{\mathcal{A}}_{\ell+1}(\mathrm{g}). \tag{12.44}$$

[ESTABLISHED]

The proof is in appendix B.5.

Remark 12.26 (Grouping, symmetry, and pullback pairs). Under Definition 12.8 at depths ℓ and $\ell+1$, three families of retained projections satisfy (12.38). (a) Let blocks $I \in V_{\ell+1}$ partition V_ℓ, let the blocks be grouped into at least two groups, let the channel be block product (12.30), and let $\bar{\mathsf{m}}_\ell$ be a product over the groups. The grouping sector at either depth consists of the elements whose components vanish on every set that meets two groups; the two grouping sectors form an invariant pair, and when a group contains two blocks the depth-ℓ sector contains components coupling agents of different blocks. (b) Let a finite group act at both depths by product maps as in Corollary 12.13 that preserve the references, the measure pair, and the channel. The fixed sectors of its two actions on components form an invariant pair onto which the averaging operators are bounded idempotents; the base law may couple agents of different blocks, as for a law on $\{-1,1\}^{V_\ell}$ whose density is even under $y \mapsto -y$ with a channel that keeps one agent of each block, where the fixed sector at depth ℓ is spanned by the components of even order. (c) Let the channel be a deterministic Borel surjection χ and put $\chi^*[h] = [h \circ \chi]$ on $\overline{\mathfrak{B}}_{\ell+1}$. Then $\overline{\mathcal{A}}_\ell \chi^* = I$ and $\overline{\mathsf{U}}_\ell \chi^* = I$ for every base law, so for every retained projection $\mathsf{P}^{\mathrm{ret}}_{\ell+1}$ the operator

$$\mathsf{P}^{\mathrm{ret}}_\ell = \mathrm{hd}_\ell \, \chi^* \, \mathrm{asm}_{\ell+1} \, \mathsf{P}^{\mathrm{ret}}_{\ell+1} \, \mathrm{hd}_{\ell+1} \, \overline{\mathsf{U}_\ell} \, \mathrm{asm}_\ell \tag{12.45}$$

is a bounded idempotent whose range, the pullback sector $\mathrm{hd}_\ell \, \chi^* \, \mathrm{asm}_{\ell+1} \, \mathrm{Ran} \, \mathsf{P}^{\mathrm{ret}}_{\ell+1}$, is mapped by $\mathcal{T}_\ell$ onto $\mathrm{Ran} \, \mathsf{P}^{\mathrm{ret}}_{\ell+1}$. Transported with the gauge together with the grouping, the group action, and the channel, each projection intertwines the gauge action. [ESTABLISHED]

Proof. (a) The joint law $\bar{\mathsf{m}}_\ell(dy) C_\ell(y, dz)$ is a product over the groups, so a product of group reverse kernels is a version of the reverse kernel, and for g in the depth-ℓ sector ψ_{g} is a sum over groups of functions of the agents of one group, so that $\mathcal{A}_\ell(\psi_{\mathrm{g}})$ is by Fubini's theorem a sum over groups of functions of the coarse agents of one group. If B meets two groups, choose $j \in B$ outside the group of one summand; $\mathbb{E}_{\ell+1,B}$ of that summand does not depend on z_j, so the factor $I - \mathbb{E}_{\ell+1, V_{\ell+1} \setminus \{j\}}$ of (12.23) annihilates it, as in the proof of Proposition 12.17. (b) Proposition 12.25 with every hatted object equal to the original gives $\mathcal{T}_\ell \mathsf{K}^{\mathcal{G}}_\ell = \mathsf{K}^{\mathcal{G}}_{\ell+1} \mathcal{T}_\ell$ for each group element, so a fixed g has a fixed image, and the average of a representation of a finite group by bounded operators is a bounded idempotent onto its fixed space. In the example the components of a set B of binary agents are the multiples of $\prod_{i \in B} y_i$, on which the flip acts by $(-1)^{|B|}$, and the channel commutes with the flips at the two depths. (c) The reverse kernel of a deterministic channel is carried by the fibers of χ, so $\int e^{-h(\chi(y))} C^\dagger(z, dy) = e^{-h(z)}$ and $\mathsf{U}_\ell(h \circ \chi) = h$ for $\bar{\mathsf{m}}_{\ell+1}$-almost every z. Idempotence follows

from $\overline{\mathrm{U}_\ell}\chi^* = I$ and $\mathrm{asm}_{\ell+1}\,\mathrm{hd}_{\ell+1} = I$, the same identities show that $\mathsf{P}^{\mathrm{ret}}_\ell$ fixes the pullback sector, and $\mathcal{T}_\ell\,\mathrm{hd}_\ell\,\chi^*\,\mathrm{asm}_{\ell+1} = \mathrm{hd}_{\ell+1}\,\overline{\mathcal{A}}_\ell\chi^*\,\mathrm{asm}_{\ell+1} = I$ on $\mathcal{G}_{\ell+1}$. A gauge change permutes the sets that index components, conjugates the group action, and turns χ into $\kappa_{\ell+1}\chi\kappa_\ell^{-1}$, so the transported projections intertwine by (12.26) and, for (c), (12.41) at $\psi = 0$. □

Open Problem 12.27 (Invariant retained subspaces). Under a product base law and a block-product channel, Proposition 12.17 gives (12.38) for the retained projections onto the intra-block sector and the singleton sector of the next depth, so the residual vanishes on them, and Remark 12.26 adds the grouping sectors of a base law that is a product over groups of blocks, the fixed sectors of a symmetry of the measure pair and the channel, and the pullback sectors of a deterministic channel under every base law. Unproved is an invariant pair beyond these. The problem is posed for finite agent spaces with fixed product references of full support, at least two blocks, and deterministic block-product channels each of whose block maps depends on every coordinate of its block, in each of two classes of base laws of full support, the product laws and all laws. It is settled by a proper invariant pair, with fixed retained projections, that is invariant under the step of every base law in a nonempty open subset of the class and whose depth-ℓ range is not contained in the sum of the pullback sector of $\mathcal{G}_{\ell+1}$ and the grouping sector of any partition of the blocks into at least two groups; or by a proof that, for base laws outside a closed nowhere dense subset of the class, every proper invariant pair has its depth-ℓ range in the sum of the pullback sector of $\mathcal{G}_{\ell+1}$ and the grouping sector of some such partition.

12.5 Exact closure generically requires hyperedges

Proposition 12.28 (Elimination cliques and hyperedge generation). (a) Suppose the complete nonnegative density of the law, baseline included, is factorized against a product or block-product reference. Marginalizing an internal variable multiplies every incident factor and integrates their product, giving one factor on the union of the remaining incident scopes. Iterating represents the normalized law by generally unnormalized factors on the elimination cliques, whose product is normalized by the inherited total normalizer; the joint law does not depend on the elimination order, although the displayed factorization does. A correlated baseline without such a factorization is kept as one global factor and pushed at the level of laws.

(b) For a finite coarse agent set V^c, a finite-valued coarse action H^c_o, and an anchor z°, the anchored components satisfy, for $B \subseteq V^c$,

$$H^{c,\circ}_{o,B}(z_B) = \sum_{B' \subseteq B} (-1)^{|B|-|B'|} H^c_o(z_{B'}, z^\circ_{V^c \setminus B'}), \qquad H^c_o(z) = \sum_{B \subseteq V^c} H^{c,\circ}_{o,B}(z_B). \tag{12.46}$$

The expansion exists for every finite-valued function on a finite product, so closure of the full hypergraph family is automatic. The anchored components are indexed by the same subsets as the Hoeffding components (12.23) and also depend only on z_B, but their values differ, and only the Hoeffding components enter the exact step. An extended-valued coarse action is kept as one top-order factor, which closes without forming undefined alternating sums.

(c) The anchor is a covariant background: if the action is gauge invariant, each component satisfies $H^{c,\circ}_{o,B}(gz_B; gz^\circ) = H^{c,\circ}_{o,B}(z_B; z^\circ)$, while an anchor held fixed in one coordinate chart gives covariance of the sum only.

(d) Pairwise closure fails in general. Eliminating the center y_0 of a star of $\{-1, 1\}$-valued agents with field ξ_0 and couplings ξ_1, ξ_2, ξ_3 gives the factor $\sum_{y_0=\pm1} \exp[y_0(\xi_0 + \sum_{r=1}^3 \xi_r y_r)] = 2\cosh(\xi_0 + \sum_{r=1}^3 \xi_r y_r)$, whose negative logarithm has the coefficient $2\,\mathrm{sech}^2(\xi_0)\tanh(\xi_0)\xi_1\xi_2\xi_3 + O(\xi^5)$ on $y_1y_2y_3$, nonzero for all sufficiently small nonzero couplings when $\xi_0 \neq 0$. At $\xi_0 = 0$ this coefficient vanishes identically, because $\log 2\cosh(\sum_r \xi_r y_r)$ is even under $y \mapsto -y$, so hyperedge generation is generic, not universal. [ESTABLISHED]

The proof is in appendix B.5.

12.6 Fine–meta bridge kernels

Definition 12.29 (Posterior bridge and bridge kernels). At a record o with posterior slice $\mathbf{\Pi}_o$ on the fine latent space and a coarse channel C, the posterior bridge is the joint law $\mathbb{B}_o(dy, dz) = \mathbf{\Pi}_o(dy)C(y, dz)$, with coarse marginal $\mathbf{\Pi}^c_o = \mathbf{\Pi}_o C$ and full reverse kernel $\mathbb{B}_o(dy \mid z)$. For a fine agent j and a meta-agent I, the (Y_j, Z_I) marginal of $\mathbb{B}_o$ disintegrated in both directions,

$$\mathbf{\Pi}_{o,j}(dy_j)K_{I\leftarrow j}(y_j, dz_I) = \mathbf{\Pi}^c_{o,I}(dz_I)K_{j\leftarrow I}(z_I, dy_j), \tag{12.47}$$

defines the agent-to-meta and meta-to-agent bridge kernels. For a deterministic statistic $z = \chi(y)$, the lift and the posterior conditional expectation are

$$(\mathsf{J}f)(y) = f(\chi(y)), \qquad (\mathsf{C}g)(z) = \mathbb{E}_{\mathbf{\Pi}_o}[g(Y) \mid \chi(Y) = z], \tag{12.48}$$

and for a randomized channel $(\mathsf{J}f)(y) = \int f(z)C(y, dz)$ and $(\mathsf{C}g)(z) = \int g(y)\mathbb{B}_o(dy \mid z)$.

Proposition 12.30 (Bridge-kernel identities). For bounded observables f and g the bridge kernels are adjoint, $\int f(y_j)(K_{I\leftarrow j}g)(y_j)\mathbf{\Pi}_{o,j}(dy_j) = \int (K_{j\leftarrow I}f)(z_I)g(z_I)\mathbf{\Pi}^c_{o,I}(dz_I)$. The kernel $K_{j\leftarrow I}$ does not reconstruct the fine posterior, whose full reverse kernel generally reads all meta variables: a product baseline and a block-product channel make the baseline reverse kernel factor over blocks, but a likelihood that couples blocks can destroy this factorization in the posterior, and posterior-local refinement requires $Y_I \perp Z_{-I} \mid Z_I$ under $\mathbb{B}_o$. The operators of (12.48) are adjoint in L^2; for a deterministic statistic $\mathsf{CJ} = I$ in $L^p(\mathbf{\Pi}^c_o)$, and the operators compose by the tower property along a nested filtration, while for a randomized channel $\mathsf{CJ} = I$ fails in general. [ESTABLISHED]

The proof is in appendix B.5.

12.7 Exact attention between meta-agents

A row-stochastic attention matrix is a conditional law of a source given a receiver and cannot be coarse-grained without the receiver law. The recursive attention state is therefore the joint edge-event law η of (6.13) over the oriented edge copies $e : j \to i$, with receiver occupancy α supplied by a structural receiver-selection law or by the single-edge model of Definition 6.5.

Proposition 12.31 (Exact meta-attention pushforward). For a posterior bridge $\mathbb{B}_o$ and a partition of the agents into blocks, put

$$\eta^c_{IJ}(z) = \mathbb{E}_{\mathbb{B}_o}\Big[\sum_{\substack{e:j\to i\\ i\in I,\ j\in J}} \eta_e(Y)\ \Big|\ Z = z\Big], \qquad \alpha^c_I(z) = \sum_J \eta^c_{IJ}(z), \qquad \beta^c_{IJ}(z) = \frac{\eta^c_{IJ}(z)}{\alpha^c_I(z)}, \tag{12.49}$$

the last ratio on $\{\alpha^c_I > 0\}$. The coarse event law is normalized, pushing then disintegrating is associative under nested partitions, and the next depth pushes η^c, not β^c. If the channel and the selected posterior versions are gauge equivariant, η^c, α^c, and β^c are invariant as values at corresponding coarse states and covariant as functions of z. Events with both endpoints in I become meta-self events in η^c_{II}, so the coarse source set of I contains I whenever $\eta^c_{II} > 0$, even when the fine rows exclude self-edges; for rows that do not depend on the latent state, η^c is the block sum, and corollary 6.6 gives the uniform case. [ESTABLISHED]

Proof. Conditional expectation preserves $\sum_e \eta_e = 1$, and the tower property gives associativity. Equivariance transports $\mathbb{B}_o$ and leaves the scalar event probabilities unchanged. When the rows do not depend on the latent state the conditional expectation is the block sum. □

Proposition 12.32 (Evidence-weighted meta-attention in the single-edge model). Assume Definition 6.5, fix the latent state y, let $\mathcal{Z}_i(y) = \sum_j \pi_{ij} e^{-D_{ij}(y)/\tau_i}$ be the normalizer of the row $\beta^P_{ij}(y)$ of (6.4), and put $\overline{\mathcal{Z}}_I(y) = \sum_{k\in I} a_{k|I}\widetilde{c}_k(o,y)\mathcal{Z}_k(y)$ for a receiver block I with $a_I = \sum_{k\in I} a_k > 0$ and $a_{i|I} = a_i/a_I$. On $\{\overline{\mathcal{Z}}_I(y) > 0\}$ the posterior receiver law conditional on the receiver lying in I and the posterior block row are

$$\alpha'_i(y) = \frac{a_{i|I}\widetilde{c}_i(o,y)\mathcal{Z}_i(y)}{\overline{\mathcal{Z}}_I(y)}, \qquad \beta^c_{IJ}(y) = \sum_{i\in I} \alpha'_i(y) \sum_{j\in J} \beta^P_{ij}(y). \tag{12.50}$$

At that y they coincide with the block sums and disintegration of (12.49) applied to the posterior edge-label law $\eta_{ij}(y) = \alpha^P_i(y)\beta^P_{ij}(y)$, whose receiver occupancy is $\alpha^P_i(y) \propto a_i\widetilde{c}_i(o,y)\mathcal{Z}_i(y)$. The formula holds posterior-almost everywhere; on a zero-evidence block the row is an arbitrary conditional version and is removed from the effective support. The equality $\alpha'_i = a_{i|I}$ holds exactly when $\widetilde{c}_i(o,y)\mathcal{Z}_i(y)$ is constant on the conditional support in I, which guarantees equality with the pre-observation coarse mixture, although the block rows can also agree accidentally after source coarsening. [ESTABLISHED]

Proof. Bayes' formula gives $P(i,j \mid y,o) \propto a_i \pi_{ij} \widetilde{c}_i(o,y) \exp[-D_{ij}(y)/\tau_i]$, because the label-free likelihood factor cancels. Summing over j gives $P(i \mid y,o) \propto a_i \widetilde{c}_i(o,y) \mathcal{Z}_i(y)$, and conditioning on $i \in I$ gives α'_i. Given i the source posterior is $\beta^P_{ij}(y)$, and the block sums of $P(i,j \mid y,o)$ over $I \times J$ and over $I \times V$ have the stated ratio. □

Proposition 12.33 (Prior-weighted collapse in the per-receiver model). In the per-receiver label model of Section 6.1, under its exclusivity hypothesis (6.3), let I be a set of receivers and let a structural coarse channel draw a receiver $i^\star \in I$ with the law $a_{\cdot|I}$, independently of the latent state, the labels, and the record, and output the source label of $i^\star$. Then the posterior law of $i^\star$ given (y,o) is $a_{\cdot|I}$, and the posterior block row is

$$\Pr(\text{source of } i^\star \in J \mid y,o) = \sum_{i \in I} a_{i|I} \sum_{j \in J} \beta^P_{ij}(y), \tag{12.51}$$

with β^P_{ij} of (6.4); its receiver occupancy is the selection law, and (12.50) does not apply to this model. The two block rows agree when $\widetilde{c}_i \mathcal{Z}_i$ is constant on the conditional support in I and otherwise can differ: for two receivers with $a_{\cdot|I} = (1/2, 1/2)$, block-source masses $\sum_{j \in J} \beta^P_{ij}$ equal to 1 and 0, and $\widetilde{c}_i \mathcal{Z}_i$ equal to 3 and 1, the evidence-weighted mass of J is $3/4$ and the prior-weighted mass is $1/2$. If all receiver labels are kept, the exact meta-label is their vector; the collapse to one categorical row is a further coarse channel, and its conditional relative entropy enters (10.4). [ESTABLISHED]

Proof. The selector enters neither the baseline nor the likelihood, so its posterior equals its prior and it is conditionally independent of the labels given (y,o). By Proposition 6.2 the label posterior factors by receiver with row i equal to $\beta^P_i(y)$, which gives (12.51). In the example the evidence weights are $3/4$ and $1/4$. □

Remark 12.34 (Coarse prior and log-sum-exp energy of one row). For one categorical row with prior π_j, row energies D_j, temperature τ, and a source partition $j \mapsto J$, the block posterior is the Gibbs row of the coarse prior and coarse energy

$$\pi^c_J = \sum_{j \in J} \pi_j, \qquad D^c_J = -\tau \log\Big[\frac{1}{\pi^c_J} \sum_{j \in J} \pi_j \exp(-D_j/\tau)\Big], \tag{12.52}$$

for every block with $\pi^c_J > 0$; a block of zero prior has zero posterior mass and leaves the effective support. The rule associates for nested partitions of one row. Merging independently normalized receiver rows needs a receiver law: the evidence-weighted occupancy of Proposition 12.32 in the single-edge model and the selection law of Proposition 12.33 in the per-receiver model. [ESTABLISHED]

Proof. By construction $\pi^c_J \exp(-D^c_J/\tau) = \sum_{j \in J} \pi_j \exp(-D_j/\tau)$, so the block sums of the Gibbs row equal the Gibbs row of (π^c, D^c) with the same normalizer. Applying the rule to a coarser partition of the blocks gives the same sums, which is associativity. □

Proposition 12.35 (Transported meta-messages as marked moments). Let each edge copy $e : j \to i$ with $i \in I$ and $j \in J$ carry its dressed link $\widehat{\Theta}^x_e$ of (11.13), let $\rho^{\mathbb{W}}_x : G \to \mathrm{GL}(\mathbb{W}_x)$ be the representation of the linear feature tier of (11.14), and assume the conditional Bochner-integrability condition $\mathbb{E}_{\mathbb{B}_o}[\sum_e \eta_e(Y)\|\rho^{\mathbb{W}}_x(\widehat{\Theta}^x_e)\| \mid Z] < \infty$ almost everywhere, the sum running over these edge copies. Define

$$\begin{aligned} \mathcal{M}^x_{IJ}(z) &= \mathbb{E}_{\mathbb{B}_o}\Big[\sum_{\substack{e:j\to i \\ i\in I,\ j\in J}} \eta_e(Y)\rho^{\mathbb{W}}_x(\widehat{\Theta}^x_e) \;\Big|\; Z = z\Big], \\ \overline{\mathcal{M}}^x_{IJ} &= \frac{\mathcal{M}^x_{IJ}}{\eta^c_{IJ}} \quad \text{on } \{\eta^c_{IJ} > 0\}, \qquad \mathcal{M}^{x,\mathrm{att}}_{IJ} = \frac{\mathcal{M}^x_{IJ}}{\alpha^c_I} \quad \text{on } \{\alpha^c_I > 0\}. \end{aligned} \tag{12.53}$$

Then $\mathcal{M}^{x,\mathrm{att}}_{IJ} = \beta^c_{IJ}\overline{\mathcal{M}}^x_{IJ}$ on $\{\eta^c_{IJ} > 0\}$, and the attention-weighted block vanishes on $\{\alpha^c_I > 0, \eta^c_{IJ} = 0\}$, where $\overline{\mathcal{M}}^x_{IJ}$ is a null-event conditional left undefined. Under a passive root-frame change with no relabeling of agents, corresponding conditional versions, and the transformed coarse state z', $\mathcal{M}^{x'}_{IJ}(z') = \rho^{\mathbb{W}}_x(g_{r_I})\mathcal{M}^x_{IJ}(z)\rho^{\mathbb{W}}_x(g_{r_J})^{-1}$ for $\mathbf{\Pi}^c_o$-almost every z, so these operators map the J root frame to the I root frame. The conditional mean need not lie in $\rho^{\mathbb{W}}_x(G)$: the average of the rotations by $\pi/2$ and $-\pi/2$ is zero. The moment is exact only as the first operator moment on a common source input: for equally likely marks and features (I, v) and $(-I, -v)$ the separate means vanish while the mean product is v, so exact closure keeps the marked operator–feature kernel, with (12.53) as a derived moment. Since neither the definition nor the covariance uses invertibility, both hold for any measurable edge mark with values in $\mathrm{End}(\mathbb{W}_x)$ that obeys the two-endpoint law, under the corresponding integrability condition. In particular, when $\mathbb{W}_x$ is the vector sample fiber of Section 6.3 and $\rho^{\mathbb{W}}_x$ is its representation ρ_x, the selective links of (6.28) obey that law by (6.30), and their dressed event-weighted first moment is a covariant operator between root frames that need not be invertible. [ESTABLISHED]

Proof. The first two statements are the definitions of β^c and of the ratios. The dressed link transforms as $\widehat{\Theta}^{x\prime}_e = g_{r_I}\widehat{\Theta}^x_e g^{-1}_{r_J}$ by (11.13), the event probabilities are invariant, and conditional expectation commutes with the fixed linear maps $\rho^{\mathbb{W}}_x(g_{r_I})$ and $\rho^{\mathbb{W}}_x(g_{r_J})^{-1}$. For a general mark, substituting its two-endpoint law and the tree transport law (11.11) into the dressed mark gives the same root-frame law. The two witnesses are direct computations. □

Proposition 12.36 (Meta-attention and meta-agent kernels split one conditional divergence). At a fixed outside context z_{-I} let $\beta^{\mathbb{Q}}_I(J \mid z_{-I}) = \mathbb{Q}(J \mid z_{-I})$ and $K^{\mathbb{Q}}_{I\leftarrow J}(dz_I \mid z_{-I}) = \mathbb{Q}(dz_I \mid J, z_{-I})$, and define the generative counterparts under $\mathbb{P}(\cdot \mid z_{-I}, o)$. If the recognition row and kernels are absolutely continuous with respect to their generative counterparts, then, in $[0, +\infty]$,

$$\begin{aligned} D_{\mathrm{KL}}\big(\mathbb{Q}(J, Z_I \mid z_{-I})\|\mathbb{P}(J, Z_I \mid z_{-I}, o)\big) &= D_{\mathrm{KL}}\big(\beta^{\mathbb{Q}}_I(\cdot \mid z_{-I})\|\beta^{\mathbb{P}}_I(\cdot \mid z_{-I}, o)\big) \\ &+ \sum_J \beta^{\mathbb{Q}}_I(J \mid z_{-I}) D_{\mathrm{KL}}\big(K^{\mathbb{Q}}_{I\leftarrow J}(\cdot \mid z_{-I})\|K^{\mathbb{P}}_{I\leftarrow J}(\cdot \mid z_{-I}, o)\big). \end{aligned} \tag{12.54}$$

[ESTABLISHED]

Proof. This is the chain rule Lemma 3.12 for the categorical coordinate J followed by Z_I. □

12.8 Path laws and memory

On a finite ordered path interval the pointwise coarse channel applied to the joint path law is again a coarse channel, so Theorem 10.7 applies on path space; the order parameter is an inference index, not the Fisher duration of Chapter 7. The coarse path law need not be first-order Markov (Remark 12.38); this section says when it is and gives the exact non-Markov dynamics and the cost of finite memory.

Theorem 12.37 (Standard-Borel strong lumpability). Let $\chi : \mathsf{Y} \to \mathsf{Z}$ be Borel and surjective, and let T be a Markov kernel on Y. Consider

$$\mathsf{T}(y, \chi^{-1}A) = \mathsf{T}(y', \chi^{-1}A) \quad \text{whenever } \chi(y) = \chi(y'), \qquad \text{for every Borel } A \subseteq \mathsf{Z}. \tag{12.55}$$

If a Markov kernel T^χ on Z satisfies

$$\chi_\#(\mu\mathsf{T}) = (\chi_\#\mu)\mathsf{T}^\chi \qquad \text{for every probability } \mu \text{ on } \mathsf{Y}, \tag{12.56}$$

then (12.55) holds and T^χ is unique. Conversely, if (12.55) holds and χ has a Borel right inverse ι, then $\mathsf{T}^\chi(z, A) = \mathsf{T}(\iota(z), \chi^{-1}A)$ is a Markov kernel satisfying (12.56). A Borel right inverse exists for a countable coarse space and for a product projection $\mathsf{Y} = \mathsf{Z} \times \mathsf{Y}'$ with Y' nonempty. For finite spaces the biconditional is the lumpability criterion of Kemeny and Snell [64, Thm. 6.3.2]. [ESTABLISHED]

The proof is in appendix B.5.

Remark 12.38 (Two memory witnesses). Strong lumpability is the every-initial-law condition and is not necessary for a single initial law. On $\mathsf{Y} = \{1, 2, 3\}$ with $\chi(1) = \chi(2) = 0$ and $\chi(3) = 1$, let T send 1 to 3, send 2 to 1 or 2 with equal probability, and send 3 to 1 or 3 with equal probability. Then $\mathsf{T}(1, \chi^{-1}\{0\}) = 0$ while $\mathsf{T}(2, \chi^{-1}\{0\}) = 1$, so (12.55) fails, but started at δ_3 the chain stays in $\{1, 3\}$, on which χ is injective, and the coarse process is Markov. A stationary coarse process can also have memory. On the deterministic cycle $1 \to 2 \to 3 \to 1$ with uniform stationary law and labels $\chi = (0, 0, 1)$, the observed histories $(0, 0)$ and $(1, 0)$ both have positive probability and the same last symbol, but predict the next symbols 1 and 0 respectively, so first-order Markov closure fails under this stationary preparation. [ESTABLISHED]

Proof. In the first witness the reachable set from 3 is $\{1, 3\}$, and the coarse transitions are $0 \mapsto 1$ surely and $1 \mapsto 0$ or 1 with equal probability. In the second, the history $(0, 0)$ is the state pair $(1, 2)$, followed by 3, and the history $(1, 0)$ is $(3, 1)$, followed by 2. □

Definition 12.39 (Regular finite-dimensional filter class). For the dynamic-closure problem below, an admitted filter is a deterministic functional of the retained path taking values in a finite-dimensional smooth manifold, with dimension bounded independently of ϵ and the horizon. Between assignment jumps it has a time-homogeneous finite-dimensional stochastic realization with locally Lipschitz coefficients; its initialization and jump-update maps are continuous in their declared native

inputs. A decoded conditional law of the hidden coordinates has a density with respect to one declared sigma-finite measure on the hidden-coordinate space, and the decoder is continuously Fréchet differentiable from the filter manifold into that L^1 space. Discrete assignment values are treated as separate strata. This is an additional regularity restriction for this problem: dimension alone does not rule out Borel encodings of an entire history or of a probability-valued filter into one real coordinate. The restriction excludes such arbitrary encodings and singular decoded families without a common dominating measure, while allowing exact finite-dimensional filters that satisfy these conditions. Changing the regularity class changes the closure question.

Open Problem 12.40 (Dynamic closure of the pointwise parent). The static construction of Corollary 10.11 supplies no fine evolution and no parent evolution. In a differentiable special case with fine vector field $\mathscr{V}_t$, moving deterministic coarse map F_t, and candidate parent field $\overline{\mathscr{V}}_t$, the exact semiconjugacy defect is

$$\delta_t = \partial_t \mathsf{F}_t + D\mathsf{F}_t \mathscr{V}_t - \overline{\mathscr{V}}_t \circ \mathsf{F}_t. \tag{12.57}$$

For a Markov evolution the corresponding exact condition is semigroup intertwining; a generator identity is an alternative criterion only on an invariant domain or core, with generation and uniqueness hypotheses sufficient to lift it to the semigroups. Neither condition is proved for the static datum together with any autonomy, moving-membership, path-law, stochastic, delayed, memory, or nonequilibrium hypothesis. Without a class the question is decided in both directions by statements already present: strongly lumpable kernels lump (Theorem 12.37), the prepared parent intertwines under exact refresh (Theorem 8.11), a parent carrying the terminal posterior of Theorem 12.44 reproduces every path law, and the stationary cycle of Remark 12.38 and the finite refresh of Remark 8.12 admit no first-order parent on their coarse labels. The problem is therefore posed for switching diffusions at c_* of the type of Definition 14.21, indexed by the regime parameter ϵ: finitely many source assignments, a positive-definite quadratic sheet energy for each assignment with Hessian bounded uniformly in ϵ, diffusion of the configuration with identity mobility on the sheet of the current assignment, and one rate-one clock at which the assignment is redrawn from its conditional law under the restricted optimizer. The channels keep the assignment and an orthogonal projection of the configuration, fixed or depending on the assignment, as in part (b) of Proposition 14.22, and on some assignment whose weight under the restricted optimizer is bounded below in ϵ they hide coordinates on which the receiver rows of that assignment depend. The parents are time-homogeneous Markov evolutions of the retained assignment and coordinates, or of these together with a filter in the explicitly restricted class of Definition 12.39, each activated from the pushforward of the recognition law at the activation time. Two levels are asked: the retained and assignment path law on windows of fixed length, uniformly over windows placed up to the switching time, the time after activation by which the expected number of assignment changes reaches one; and the complete retained and assignment history over horizons on which the expected number of assignment changes diverges as $\epsilon \to 0$. At either level the problem is settled by a proof that every member of the class admits such a parent whose retained and assignment path law is within a total variation tending to zero with ϵ of the native one, or by a member for which every such parent keeps that distance bounded below.

Theorem 12.41 (Exact discrete projection-memory recurrence). Let X and W be vector spaces, let $\mathsf{T} : \mathsf{X} \to \mathsf{X}$, $\mathsf{C} : \mathsf{X} \to \mathsf{W}$, and $\mathsf{J} : \mathsf{W} \to \mathsf{X}$ be linear with $\mathsf{CJ} = I_{\mathsf{W}}$, and set $\mathsf{P}^{\mathrm{res}} = \mathsf{JC}$ and $\mathsf{Q} = I - \mathsf{P}^{\mathrm{res}}$, so that $\mathsf{P}^{\mathrm{res}}$ is idempotent. For the fine recursion $\xi_{n+1} = \mathsf{T}\xi_n$ write $w_n = \mathsf{C}\xi_n$, so that $\mathsf{P}^{\mathrm{res}}\xi_n = \mathsf{J}w_n$. Then for every $n \geq 0$ the unresolved part and the resolved coordinate obey

$$\mathsf{Q}\xi_n = (\mathsf{QTQ})^n \mathsf{Q}\xi_0 + \sum_{k=0}^{n-1} (\mathsf{QTQ})^{n-1-k} \mathsf{QTJ} w_k, \tag{12.58}$$

$$w_{n+1} = \mathsf{CTJ} w_n + \sum_{k=0}^{n-1} \mathsf{CTQ}(\mathsf{QTQ})^{n-1-k} \mathsf{QTJ} w_k + \mathsf{CTQ}(\mathsf{QTQ})^n \mathsf{Q}\xi_0, \tag{12.59}$$

an exact closed recurrence for w_n. The operators $\mathsf{CTQ}(\mathsf{QTQ})^m\mathsf{QTJ}$, $m \geq 0$, are the memory kernels and the last term of (12.59) is the unresolved initial-state term; no closure or Markov assumption is used. [ESTABLISHED]

Proof. Applying $\mathsf{P}^{\mathrm{res}}$ and Q to $\xi_{n+1} = \mathsf{T}\xi_n$ with $\xi_n = \mathsf{J}w_n + \mathsf{Q}\xi_n$ and $\mathsf{Q}^2 = \mathsf{Q}$ gives $w_{n+1} = \mathsf{CTJ}w_n + \mathsf{CTQQ}\xi_n$ and $\mathsf{Q}\xi_{n+1} = \mathsf{QTQQ}\xi_n + \mathsf{QTJ}w_n$. The second relation gives (12.58) by induction on n, and substituting it into the first gives (12.59). □

For the slow block of a fast–slow coordinatization (Definition 8.2) of a linear evolution, (12.59) is the exact discrete slow dynamics; the projection-operator method goes back to Nakajima [80] and Zwanzig [110], in continuous time on ensemble densities.

Corollary 12.42 (Autonomy of the resolved recursion). Let $\mathscr{S}_0 \subseteq \mathsf{X}$ be a class of initial states. The resolved recursion is autonomous on $\mathscr{S}_0$, meaning $w_{n+1} = \mathsf{CTJ}w_n$ for every $n \geq 0$ and every $\xi_0 \in \mathscr{S}_0$, exactly when the total correction in (12.59) vanishes for every such n and ξ_0. It suffices that all memory kernels vanish and that $\mathsf{CTQ}(\mathsf{QTQ})^n\mathsf{Q}\xi_0 = 0$ for $n \geq 0$ throughout $\mathscr{S}_0$. On the resolved class $\mathsf{Q}\xi_0 = 0$ the second family is vacuous, and there $\mathsf{QTJ} = 0$ is sufficient but not necessary: on $\mathsf{X} = \mathbb{R}^2$ with $\mathsf{W} = \mathbb{R}$, $\mathsf{C}(u, v) = u$, $\mathsf{J}(w) = (w, 0)$, and $\mathsf{T}(u, v) = (u, u)$, one has $\mathsf{QTJ}w = (0, w) \neq 0$ for $w \neq 0$ while $\mathsf{CTQ} = 0$, so every memory kernel vanishes and $w_{n+1} = w_n$ is exactly autonomous. [ESTABLISHED]

Proof. The biconditional is (12.59) read as an identity, and the two families are its two correction terms. In the witness $\mathsf{Q}(u, v) = (0, v)$ and $\mathsf{T}(0, v) = (0, 0)$ give $\mathsf{CTQ} = 0$, while $\mathsf{T}(w, 0) = (w, w)$ gives $\mathsf{QTJ}w = (0, w)$. □

12.8.1 The hidden history and its filter

Definition 12.43 (Hidden path experiment). Let Y_t take values in a finite set Y with initial law p_0 and transition matrix T, and let $Z_t = \chi(Y_t)$ take values in a finite set Z, with P_z the diagonal projector onto $\chi^{-1}(z)$. The joint path law over N steps and its observed marginal are

$$p_0(y_0) \prod_{t=0}^{N-2} \mathsf{T}_{y_t y_{t+1}} \prod_{t=0}^{N-1} \mathbf{1}_{\chi(y_t)=z_t}, \qquad \mathbb{P}_N^{\mathrm{path}}(z_{0:N-1}) = p_0 \mathsf{P}_{z_0} \mathsf{T} \mathsf{P}_{z_1} \cdots \mathsf{T} \mathsf{P}_{z_{N-1}} \mathbf{1}. \tag{12.60}$$

This auxiliary model, the law-level counterpart of Theorem 12.41, is fixed before a recognition law on its paths is varied.

Theorem 12.44 (Exact filtering without a projected Markov assumption). For an observed word $z_{0:n}$ of positive probability put

$$\mathsf{f}_{z_{0:n}} = p_0 \mathsf{P}_{z_0} \mathsf{T}\mathsf{P}_{z_1} \cdots \mathsf{T}\mathsf{P}_{z_n}, \qquad \widehat{p}_{z_{0:n}} = \frac{\mathsf{f}_{z_{0:n}}}{\mathsf{f}_{z_{0:n}}\mathbf{1}}, \qquad \mathbb{P}(Z_{n+1} = z \mid z_{0:n}) = \widehat{p}_{z_{0:n}} \mathsf{T}\mathsf{P}_z \mathbf{1}. \tag{12.61}$$

These are the exact terminal posterior and next-symbol prediction. A word of zero probability has zero messages for all its extensions and undefined conditional readouts. The terminal posterior determines the law of every future suffix, but its attainable values need not form a finite set. [ESTABLISHED]

Proof. Summing the joint path law with the final hidden state fixed gives $\mathsf{f}_{z_{0:n}}$. Replacing P_z by I and using $\mathsf{T}\mathbf{1} = \mathbf{1}$ proves normalization and prefix consistency, division by the positive word mass gives the posterior, and a further matrix product gives any suffix. Nonnegative zero messages stay zero under multiplication. A posterior is a probability vector on Y, and sufficiency does not assert that its attainable subset is finite. □

12.8.2 Finite-memory projection

Definition 12.45 (Finite-memory projection of a path law). Let $\mathbb{P}_N^{\mathrm{path}}$ be any law on Z^N, stationary or not. For an integer $k \geq 0$ and a position t let $\mathbf{z}_t = z_{\max(0,t-k):t-1}$ be the retained context. On contexts of positive mass define the rows and the projected law

$$\mathsf{T}_t^{(k)}(z \mid \mathbf{z}) = \mathbb{P}_N^{\mathrm{path}}(Z_t = z \mid \mathbf{z}_t = \mathbf{z}), \qquad \widehat{\mathbb{P}}_N^{(k)}(z_{0:N-1}) = \prod_{t=0}^{N-1} \mathsf{T}_t^{(k)}(z_t \mid \mathbf{z}_t), \tag{12.62}$$

recursively, so that a prefix of zero mass has zero extensions and no row is chosen on a null context.

Theorem 12.46 (Forward-KL optimality and its exact information defect). The projected law (12.62) is normalized, has the same context-plus-symbol marginals as $\mathbb{P}_N^{\mathrm{path}}$, and is positive on every word of positive $\mathbb{P}_N^{\mathrm{path}}$ probability. Among all laws with time-dependent k-context rows it uniquely minimizes the forward path divergence, and the minimum is

$$D_N^{(k)} := D_{\mathrm{KL}}\big(\mathbb{P}_N^{\mathrm{path}} \| \widehat{\mathbb{P}}_N^{(k)}\big) = \sum_{t=0}^{N-1} I\big(Z_t; Z_{0:t-k-1} \mid Z_{\max(0,t-k):t-1}\big), \tag{12.63}$$

an empty discarded history contributing zero. For a competing law $\widetilde{\mathbb{P}}$ with rows $\widetilde{\mathsf{T}}_t$,

$$D_{\mathrm{KL}}\big(\mathbb{P}_N^{\mathrm{path}} \| \widetilde{\mathbb{P}}\big) - D_N^{(k)} = \sum_{t,\mathbf{z}:\ \mathbb{P}_N^{\mathrm{path}}(\mathbf{z}_t = \mathbf{z}) > 0} \mathbb{P}_N^{\mathrm{path}}(\mathbf{z}_t = \mathbf{z}) D_{\mathrm{KL}}\big(\mathsf{T}_t^{(k)}(\cdot \mid \mathbf{z}) \| \widetilde{\mathsf{T}}_t(\cdot \mid \mathbf{z})\big). \tag{12.64}$$

The defect decreases in k and vanishes for $k \geq N-1$; the projection is idempotent and commutes with restriction to a shorter prefix; and $2\,\mathrm{TV}(\mathbb{P}_N^{\mathrm{path}}, \widehat{\mathbb{P}}_N^{(k)})^2 \leq D_N^{(k)}$. The reverse divergence is not controlled: the stationary cycle of Remark 12.38 has $\mathbb{P}_3^{\mathrm{path}}(000) = 0$ but $\widehat{\mathbb{P}}_3^{(1)}(000) = 1/6$. For a path recognition law $\mathbb{Q}^{\mathrm{path}}$ supported on the support of $\mathbb{P}_N^{\mathrm{path}}$,

$$D_{\mathrm{KL}}\big(\mathbb{Q}^{\mathrm{path}} \| \widehat{\mathbb{P}}_N^{(k)}\big) - D_{\mathrm{KL}}\big(\mathbb{Q}^{\mathrm{path}} \| \mathbb{P}_N^{\mathrm{path}}\big) = \mathbb{E}_{\mathbb{Q}^{\mathrm{path}}} \log \frac{\mathbb{P}_N^{\mathrm{path}}}{\widehat{\mathbb{P}}_N^{(k)}}, \tag{12.65}$$

whose worst case over such laws is set by the extrema of the log ratio, not by its $\mathbb{P}_N^{\mathrm{path}}$-average. [ESTABLISHED]

Theorem 12.47 (Memory identity on standard Borel records). Let $\mathbb{P}_N^{\mathrm{path}}$ be any law on Z^N for a record space Z that need not be finite, and for each position t let $\mathbf{f}_t$ be a Borel map from Z^t, a point when $t = 0$, to a statistic space. Let $\mathsf{T}_t^{\mathbf{f}}$ be a regular conditional law of Z_t given $\mathbf{f}_t(Z_{0:t-1})$ under $\mathbb{P}_N^{\mathrm{path}}$, let $\widetilde{\mathsf{T}}_t$ be any Markov kernel from the statistic space to Z, and let $\widetilde{\mathbb{P}}$ be the law on Z^N under which Z_t given $Z_{0:t-1} = z_{0:t-1}$ has law $\widetilde{\mathsf{T}}_t(\mathbf{f}_t(z_{0:t-1}), \cdot)$. A conditional mutual information is, as in Proposition 5.18, the average over the conditioning variable of the divergence between the conditional joint law and the product of the conditional marginals. Then, in $[0, +\infty]$, with no absolute-continuity or finiteness hypothesis, and whatever versions are chosen,

$$\begin{aligned} D_{\mathrm{KL}}\big(\mathbb{P}_N^{\mathrm{path}} \| \widetilde{\mathbb{P}}\big) &= D_N^{\mathbf{f}} + \sum_{t=0}^{N-1} \mathbb{E}_{\mathbb{P}_N^{\mathrm{path}}} D_{\mathrm{KL}}\big(\mathsf{T}_t^{\mathbf{f}}(\mathbf{f}_t(Z_{0:t-1}), \cdot) \big\| \widetilde{\mathsf{T}}_t(\mathbf{f}_t(Z_{0:t-1}), \cdot)\big), \\ D_N^{\mathbf{f}} &= \sum_{t=0}^{N-1} I\big(Z_t; Z_{0:t-1} \mid \mathbf{f}_t(Z_{0:t-1})\big). \end{aligned} \tag{12.66}$$

Hence $D_N^{\mathbf{f}}$ is the minimum of the forward path divergence over all such laws $\widetilde{\mathbb{P}}$, attained when $\widetilde{\mathsf{T}}_t = \mathsf{T}_t^{\mathbf{f}}$ for every t; if $D_N^{\mathbf{f}} < \infty$, subtracting it leaves the sum of expected row divergences as the excess risk; and the t-th term of $D_N^{\mathbf{f}}$ vanishes exactly when the law of Z_t given $Z_{0:t-1}$ has a version that depends on the history only through $\mathbf{f}_t$. For $\mathbf{f}_t(z_{0:t-1}) = \mathbf{z}_t$ of Definition 12.45, $I(Z_t; Z_{0:t-1} \mid \mathbf{z}_t) = I(Z_t; Z_{0:t-k-1} \mid \mathbf{z}_t)$, zero when the discarded history is empty, and on a finite Z the identity is (12.63) with (12.64). If at $\mathbb{P}_N^{\mathrm{path}}$-almost every history the competitor row at $\mathbf{f}_t(z_{0:t-1})$ is $e^{-\psi}$ times the source row there for a bounded ψ, that row divergence is at most $\min\{2\|[\psi]\|_{\mathrm{osc}}, \|[\psi]\|_{\mathrm{osc}}^2/2\}$, with the oscillation norm taken under the source row, as in Corollary 12.6. [ESTABLISHED]

The defect is what a predictor that reads the history only through $\mathbf{f}_t$ loses by forgetting, and the row sum is what it loses by predicting badly from what it keeps. Neither the finite hidden chain of Definition 12.43 nor a finite record set enters, so the identity covers record laws with continuous values. For the window statistic $\mathbf{f}_t(z_{0:t-1}) = \mathbf{z}_t$ and a competitor that dominates $\mathbb{P}_N^{\mathrm{path}}$, (12.66) is Gray [43, Sec. 7.3, Theorem 7.3.1, Eqs. (7.13) and (7.16)] on standard alphabets; the theorem removes the domination hypothesis and allows any Borel statistic of the history.

Theorem 12.48 (Stationary projection and the finite hidden information budget). Let the hidden chain of Definition 12.43 start from a stationary law $p_0 = \bar{p}$, $\bar{p}\mathsf{T} = \bar{p}$, restricted to its positive support, which is closed under T; arbitrary initial preparations are excluded. Write $\mathrm{h}_j = H(Z_j \mid Z_{0:j-1})$ in natural logarithms, with $\mathrm{h}_0 = H(Z_0)$, and $\mathrm{c}_Y = H(Y_0)$. Then the stationary projection is normalized, stationary, and prefix consistent, and agrees with the source on every window of length at most $k+1$. The sequence h_j decreases to a limit h_∞, and with $\mathrm{h}_k^{\mathrm{ex}} = \mathrm{h}_k - \mathrm{h}_\infty$,

$$
\begin{aligned}
D_N^{(k)} &= \sum_{t=k+1}^{N-1} (\mathrm{h}_k - \mathrm{h}_t), & \lim_{N\to\infty} \frac{D_N^{(k)}}{N} &= \mathrm{h}_k^{\mathrm{ex}}, \\
\sum_{j=0}^{\infty} \mathrm{h}_j^{\mathrm{ex}} &\le \mathrm{c}_Y, & \mathrm{h}_k^{\mathrm{ex}} &\le \frac{\mathrm{c}_Y}{k+1}, \\
0 \le \mathrm{h}_k^{\mathrm{ex}} - \frac{D_N^{(k)}}{N} &\le \frac{\mathrm{c}_Y}{N}, & D_N^{(k)} &\le (N-k-1)_+ \mathrm{h}_k^{\mathrm{ex}},
\end{aligned}
\tag{12.67}
$$

an empty sum being zero. The series $\sum_j \mathrm{h}_j^{\mathrm{ex}}$ is the excess entropy of the observed process in the sense of Crutchfield and Feldman [27], so the second line bounds it by the hidden-state entropy. No irreducibility or forgetting assumption is used. [ESTABLISHED]

The proofs of Theorems 12.46 to 12.48 are in appendix B.5.

The memory cost of a prepared parent whose within-parent refresh is only partial (Section 8.3) is the defect $D_N^{(1)}$ of Theorem 12.46 for the parent process.

12.9 The exact retained state

Exact closure keeps every object of Table 12.1, listed with its exact operation, source statement, and admission hypotheses, uniformly in the number and arrangement of the agents.

The blocks and the channel enter the table as inputs, and the smallest case in which three two-dimensional blocks are asked for shows that the law cannot supply them.

Proposition 12.49 (Law and Fisher geometry do not canonically agentize blocks). At the uniform law on seven outcomes, no rule that depends only on the unlabeled law and its Fisher metric and is natural under every relabeling of the outcomes can select an unordered decomposition of the six-dimensional tangent space into three two-dimensional blocks. Declared typing or other symmetry-breaking data can supply blocks; the law and its Fisher tensor alone do not. [ESTABLISHED]

The proof is in appendix B.5.

Table 12.1: The retained state of exact agent-network coarse-graining at c_*.

Retained object	Exact operation and standing	Admission hypotheses
Normalized generative law, posterior versions, recognition law	One channel pushes all three; the variational free energy decomposes by the chain rule. Derived: Theorem 10.7, Proposition 12.2.	Recognition-independent channel; finite positive evidence; recognition law absolutely continuous with respect to the posterior.
Reference probability, evidence measure, effective action	Radon–Nikodym action by the conditional partition function, composing along towers. Derived: Theorem 12.3.	Probability reference pushed with the evidence measure; reverse conditionals from one joint tower.
Bounded action calculus	Analytic action increment, tilted derivatives, contraction and oscillation bounds. Derived: Theorem 12.4, Theorem 12.5.	Bounded perturbations; jointly measurable reverse kernel.
Interaction coordinates and the exact step	Hoeffding isomorphism and exact nonlinear step with its closure residual. Derived: Theorem 12.12, Lemma 12.15, Proposition 12.24. Retained projection: declared.	Product-equivalent normalized laws at both depths (Definition 12.8).
Finite-step multipliers	Ordered products of tilted reverse operators, bounded by $3^{\lvert V_n\rvert}-1$. Derived: Definition 12.20, Proposition 12.21.	An exact orbit in the bounded tier.
Induced hyperedges	Elimination cliques and Möbius components; generically nonpairwise. Derived: Proposition 12.28.	Finite-valued coarse action, or one top-order factor.
Bridge kernels	Pair-marginal disintegrations of the posterior bridge. Derived: Proposition 12.30. Gauge-covariant versions: declared beyond the compact and finite cases (Open Problem 10.19).	Regular disintegration on standard Borel spaces.
Roots, based holonomy, dressed boundary marks	Raw root-framed marks and their joint law, pushed without a holonomy-blind claim. Derived: Section 11.2. Trees: declared.	Rooted spanning forests of the blocks.
Joint attention-event law	Event pushforward and disintegration on positive occupancy. Derived: Proposition 12.31, Proposition 12.32, Proposition 12.33.	Receiver law of the label model.
Marked operator–feature kernels	Event-weighted first operator moments. Derived: Proposition 12.35.	Conditional Bochner integrability.
Path laws and memory	Lumped kernel when strongly lumpable; otherwise memory kernels, filters, and finite-memory projections with their defects. Derived: Theorem 12.37, Theorem 12.41, Theorem 12.46, Theorem 12.48.	The full path law, or the memory kernels and initial-state term.

Summary for physicists

One exact coarse-graining step pushes normalized laws through a Markov channel; the effective action is the Radon–Nikodym derivative of the pushed evidence measure with respect to a pushed reference probability, the conditional partition function of the fine action over each coarse state,

analytic in bounded changes of the fine action and never expanding the supremum norm. In Hoeffding coordinates, one component per set of coupled agents, the step is an exact nonlinear map whose finite-step multipliers telescope to one extraction, a product of tilted conditional expectations, and one assembly, so they are bounded by $3^{|V_n|} - 1$ at every depth. Exactness costs state: eliminated agents generate hyperedges unless a symmetry forbids them, attention is carried as the joint event law, transported messages are moments of marked kernels, and the coarse path law generally has memory, with an exact memory-kernel recurrence and, for memory length k, a finite-memory cost of at most the hidden-state entropy divided by $k + 1$ per step.

Chapter 13

The Multiscale Graph VFE and Closing the Tower

This chapter asks what probabilistic structure places adaptive event laws and links on the interaction multigraph, random partitions, and the influence of parents on their children inside one variational free energy, and how a tower of such depths closes. The answer is the free energy of one normalized generative law and one recognition law over the whole tower (Definition 13.1). For an adaptive tower both laws factor top-down, each depth conditioned on the whole next-depth state and the structural mark (Lemma 13.5), and the free energy splits into observation, top-prior, partition, graph, and parent–child terms (Theorem 13.7). A finite acyclic tower needs a declared latent root (Proposition 13.23); participation across depths enters through the inference loop, parameter learning (Proposition 13.25), and delayed feedback (Propositions 13.27 and 13.28). With latent links, the product Gaussian generator laws are not closed under the coarse links (Proposition 13.20). The formation witnesses of Chapter 14 are adaptive towers, and Chapter 15 reads the three loci of participation.

The depth ℓ runs from 0 to a finite L. At depth ℓ, Z_ℓ are the coordinates, $\mathfrak{N}_\ell$ the graph datum, $Y_\ell = (Z_\ell, \mathfrak{N}_\ell)$ the state on a standard Borel space Y_ℓ that here includes the graph datum, $R_\ell \in \mathsf{R}_\ell$ the structural mark with realized value γ, and $\mathfrak{h}_\ell$ the retained root, holonomy, and boundary marks. The whole tower is $\mathbf{Y}$, row normalizers are $\mathcal{Z}$, and $\mathbb{M}$ is an input law: generative, posterior, or recognition. In the episode-indexed models of Section 13.7.2, $\widetilde{o}_n$ is a latent copy of the record o_n, ϵ_{e} is the constant step of the aggregate parameter update, and T is the temperature of the descent kinetics, the symbol of Chapter 14. With latent links the generator symbols are those of Table 9.1, and the direction $\mathring{v}$, the noise $\mathring{\zeta}$, the width $\mathring{\tau}$, the cumulant forms $\mathring{\kappa}_3$ and $\mathring{\kappa}_4$, and the set $\mathbb{X}_K$ are declared in Definition 13.19 and proposition 13.20.

Scope. In an adaptive tower the downward kernel is a reversal kernel, and the coupling it carries is not read as causal downward influence; declining that reading does not refute a causal reading of a separately declared top-down hierarchy. The composite potential of Definition 13.10 is not claimed to be an evidence bound, and delayed feedback is not claimed to support a sustained nonequilibrium reading. [NOT-CLAIMED] Open items are the persistent hierarchy (Open Problem 13.30), the accounting across episodes (Open Problem 13.29), the dynamic closure of the pointwise parent (Open Problem 12.40), a natural partition selector (Open Problem 10.20), and a finite-dimensional description of coarse link laws (Open Problem 13.21).

13.1 The full answer before decomposition

Definition 13.1 (Tower variational free energy). Write $X_* = X(c_*)$, fix an admitted observation $o \in \mathsf{O}^{\mathrm{reg}}_{\theta,X_*}$ and a finite depth L, and let $\mathbf{Y}$ contain every random object of the multiscale model: belief and model coordinates at depths $0,\dots,L$, links, event laws, retained marks, structural marks, and auxiliary latents. Declare one normalized generative kernel $\mathbb{P}_\theta(do, d\mathbf{Y} \mid X_*)$ and one recognition kernel $\mathbb{Q}_\lambda(d\mathbf{Y} \mid o, X_*)$. With posterior slice $\mathbf{\Pi}_{\theta,o,X_*}$ and evidence density $p_\theta(o \mid X_*) \in (0,\infty)$, the tower variational free energy is

$$\boxed{\mathcal{F}^{\mathrm{ext}}_{\mathrm{tower}}[\mathbb{Q}_\lambda; X_*, o] = -\log p_\theta(o \mid X_*) + D_{\mathrm{KL}}\big(\mathbb{Q}_\lambda(\cdot \mid o, X_*) \| \mathbf{\Pi}_{\theta,o,X_*}\big).} \tag{13.1}$$

By the evidence identity of Chapter 5, on the finite-density domain it equals

$$\mathcal{F}_{\mathrm{tower}} = \mathbb{E}_{\mathbb{Q}_\lambda}\Big[\log \frac{q_\lambda(\mathbf{Y} \mid o, X_*)}{p_\theta(o, \mathbf{Y} \mid X_*)}\Big]. \tag{13.2}$$

Every later term is an accounting decomposition of this object or is named a composite potential.

13.2 A normalized hierarchical graph model

Definition 13.2 (Closed scale datum). At depth ℓ let V_ℓ be the vertex set and let $Z_\ell = (Z_{i,\ell})_{i \in V_\ell}$ contain the full probabilistic coordinates of that depth, belief and model presentations rather than only their marginals. The graph datum lives on a finite interaction multigraph $\mathfrak{I}_\ell = (V_\ell, E_\ell, {}^-)$ (Definition 2.11), at depth 0 the multigraph of section 1.4 and at deeper depths induced by the link lift of Definition 13.17. Put

$$\eta_\ell = (\eta^{b,\ell}, \eta^{m,\ell}), \qquad \Theta^{x,\ell} = (\Theta^{x,\ell}_e)_{e \in E_\ell}, \qquad \mathfrak{N}_\ell = (\eta_\ell, \Theta^{b,\ell}, \Theta^{m,\ell}, \mathfrak{h}_\ell), \tag{13.3}$$

where $\eta^{x,\ell} = (\eta^{x,\ell}_e)_{e \in E_\ell}$ is a normalized event law on the copies, $\Theta^{x,\ell}_e \in G$ is the channel-x link of the copy e with $\Theta^{x,\ell}_{\bar e} = (\Theta^{x,\ell}_e)^{-1}$, and $\mathfrak{h}_\ell$ contains the root, based-holonomy, and dressed boundary marks of Section 11.2. Each copy carries one link, one event mass, and one divergence per channel; endpoint notation abbreviates a copy only when it is unique, and $\eta^{x,\ell}_{ij}$ is the total mass of the copies $j \to i$. Occupancies and rows are derived from η_ℓ as in (6.13) and do not replace it in the recursive state. The observation and X_* stay outside every coarse channel.

Definition 13.3 (Common structural channel). A structural step is one recognition-independent Markov channel

$$C^{\mathrm{struct}}_{\ell+1\leftarrow\ell} : \mathsf{Y}_\ell \rightsquigarrow \mathsf{R}_\ell \times \mathsf{Y}_{\ell+1}, \tag{13.4}$$

whose output includes $(R_\ell, \eta_{\ell+1}, \mathfrak{h}_{\ell+1})$, with augmented transition state

$$\widehat{\mathsf{Y}}_{\ell+1\leftarrow\ell} := (\mathsf{Y}_\ell, R_\ell, \mathsf{Y}_{\ell+1}), \tag{13.5}$$

which does not duplicate R_ℓ. The same channel is applied to the fine generative marginal, the selected posterior, and the recognition law,

$$\widehat{\mathbb{M}}^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(dy_\ell, d\gamma, dy_{\ell+1}) := \mathbb{M}_\ell(dy_\ell) C^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(y_\ell; d\gamma, dy_{\ell+1}), \qquad \mathbb{M} \in \{\mathbb{P}, \mathbf{\Pi}, \mathbb{Q}\}, \tag{13.6}$$

so the three laws of R_ℓ may differ while the channel does not. If the channel is realized by an auxiliary seed, the same measurable map and seed law define all three pushforwards; a blocking rule that depends on the input law is not admitted.

Hypothesis 13.4 (Depth-Markov hierarchy). The generative law factors top-down as

$$\begin{aligned}\mathbb{P}_\theta(do, d\mathbf{Y} \mid X_*) = \mathbb{P}^{\mathrm{obs}}_\theta(do \mid Z_0, X_*)\mathbb{P}_L(dY_L \mid X_*) \\ \times \prod_{\ell=0}^{L-1} \mathbb{P}^\ell_R(dR_\ell \mid Y_{\ell+1}, X_*)\mathbb{P}^\ell_{\mathfrak{N}}(d\mathfrak{N}_\ell \mid Y_{\ell+1}, R_\ell, X_*)K^\ell_\downarrow(dZ_\ell \mid Y_{\ell+1}, R_\ell, \mathfrak{N}_\ell, X_*),\end{aligned} \tag{13.7}$$

with the kernels composed from depth L downward, and recognition factors along the same order,

$$\begin{aligned}\mathbb{Q}_\lambda(d\mathbf{Y} \mid o, X_*) = \mathbb{Q}_L(dY_L \mid o, X_*) \prod_{\ell=0}^{L-1} \mathbb{Q}^\ell_R(dR_\ell \mid Y_{\ell+1}, o, X_*) \\ \times \mathbb{Q}^\ell_{\mathfrak{N}}(d\mathfrak{N}_\ell \mid Y_{\ell+1}, R_\ell, o, X_*)\mathbb{Q}^\ell_Z(dZ_\ell \mid Y_{\ell+1}, R_\ell, \mathfrak{N}_\ell, o, X_*).\end{aligned} \tag{13.8}$$

Each depth-ℓ factor reads the complete next-depth state $Y_{\ell+1} = (Z_{\ell+1}, \mathfrak{N}_{\ell+1})$ and no deeper variable; this excludes direct dependence on depths $\ell+2, \dots, L$ given $(Y_{\ell+1}, R_\ell)$, leaves arbitrary dependence within a depth and between adjacent depths, and is not a mean-field assumption. In a declared top-down hierarchy $K^\ell_\downarrow$ is where downward influence enters, and the bottom-up coarse map belongs to recognition or to an induced pushforward, never to a second generative arrow at the same depth.

Lemma 13.5 (Depth-Markov reversal of an adaptive tower). Let $\mathbb{M}_0$ be a probability law on Y_0, and generate the adaptive tower bottom-up through the common structural channels,

$$\mathbb{M}_0(dy_0) \prod_{\ell=0}^{L-1} C^{\mathrm{struct}}_{\ell+1\leftarrow\ell}(y_\ell; d\gamma_\ell, dy_{\ell+1}). \tag{13.9}$$

For every ℓ, the conditional law of R_ℓ given all variables $(Y_{\ell+1}, R_{\ell+1}, \dots, Y_L)$ of greater depth has a version that depends only on $Y_{\ell+1}$, and the conditional law of Y_ℓ given R_ℓ and those variables has a version that depends only on $(R_\ell, Y_{\ell+1})$. Hence (13.9) factors as in (13.7) without its observation factor, with $\mathbb{P}^\ell_R$, $\mathbb{P}^\ell_{\mathfrak{N}}$, and $K^\ell_\downarrow$ selected versions of these conditional laws. The same holds for the recognition input $\mathbb{Q}_0(dy_0 \mid o, X_*)$, for the selected posterior slice, and for the generative input $\mathbb{P}_0(do, dy_0 \mid X_*)$ whenever its observation kernel reads only Z_0, so Hypothesis 13.4 holds automatically for an adaptive tower. These factors are reversal, or bridge, kernels in the sense

of (12.47), not independently chosen structural channels, and when the parent coordinates are a function of the child coordinates and the structural mark, $K_{\downarrow}^{\ell}$ is supported on the child configurations compatible with the realized parent. [ESTABLISHED]

Proof. For each ℓ, the variables $(R_{\ell+1}, Y_{\ell+2}, \dots, Y_L)$ are produced from $Y_{\ell+1}$ by kernels that read neither the observation nor any of $(Y_0, R_0, \dots, Y_\ell, R_\ell)$, so they are conditionally independent of the latter given $Y_{\ell+1}$, which gives both version statements. Regular conditional laws exist on standard Borel spaces [61], and successive disintegration from depth L downward gives the factorization. A conditional law of Z_ℓ given a value of a function of (Z_ℓ, R_ℓ) charges only the preimage of that value. □

Proposition 13.6 (The next-depth graph datum cannot be dropped). If the depth-ℓ factors of an adaptive tower read only $(Z_{\ell+1}, R_\ell)$, the product $\mathbb{P}^{\mathrm{red}}$ of the corresponding selected conditional laws of the tower law $\mathbb{P}^{\mathrm{tow}}$ of (13.9) differs from it by

$$D_{\mathrm{KL}}\big(\mathbb{P}^{\mathrm{tow}} \| \mathbb{P}^{\mathrm{red}}\big) = \sum_{\ell=0}^{L-1} I_{\mathbb{P}^{\mathrm{tow}}}\big(Y_\ell; \mathfrak{N}_{\ell+1} \mid Z_{\ell+1}, R_\ell\big). \tag{13.10}$$

The cost vanishes exactly when every $\mathfrak{N}_{\ell+1}$ is conditionally independent of Y_ℓ given $(Z_{\ell+1}, R_\ell)$. For a deterministic event pushforward it is positive as soon as two event laws in the support, at the same $(Z_{\ell+1}, R_\ell)$, are pushed to different coarse event laws: for four fine vertices with the partition $\{0,1\} \mapsto I$, $\{2,3\} \mapsto J$, constant coordinates, trivial links and marks, and η_0 equal with probability $1/2$ each to the unit mass on a copy $2 \to 0$ or on a copy $1 \to 0$, the pushed η_1 is the unit mass on (I, J) or on (I, I), and the divergence is $\log 2$. [ESTABLISHED]

Proof. By Lemma 13.5 the reduced product replaces each conditional law of Y_ℓ given $(R_\ell, Y_{\ell+1})$ by its conditional law given $(R_\ell, Z_{\ell+1})$, and the chain rule Lemma 3.12 turns each replacement into the displayed conditional mutual information. In the witness the joint law of $(\mathfrak{N}_0, \mathfrak{N}_1)$ has two atoms of mass $1/2$ while the reduced product is the product of their marginals, which puts mass $1/2$ on pairs that violate the pushforward. □

Theorem 13.7 (Exact hierarchical decomposition). Assume Hypothesis 13.4, which holds for an adaptive tower by Lemma 13.5, together with common domination and the integrability needed for the finite log-density form. Then

$$\begin{aligned}\mathcal{F}_{\mathrm{tower}} = \; & -\mathbb{E}_{\mathbb{Q}_\lambda} \log p_\theta^{\mathrm{obs}}(o \mid Z_0, X_*) + D_{\mathrm{KL}}(\mathbb{Q}_L \| \mathbb{P}_L) \\ & + \sum_{\ell=0}^{L-1} \mathbb{E}_{\mathbb{Q}_\lambda}\big[D_{\mathrm{KL}}(\mathbb{Q}_R^\ell \| \mathbb{P}_R^\ell) + D_{\mathrm{KL}}(\mathbb{Q}_{\mathfrak{N}}^\ell \| \mathbb{P}_{\mathfrak{N}}^\ell) + D_{\mathrm{KL}}(\mathbb{Q}_Z^\ell \| K_{\downarrow}^\ell)\big],\end{aligned} \tag{13.11}$$

where each divergence compares the recognition and generative factors of (13.8) and (13.7) at their common conditioning variables, the recognition factor also reading o, and is averaged over those variables under $\mathbb{Q}_\lambda$. These are the observation, top-prior, partition, graph, and parent–child contributions. [ESTABLISHED]

Proof. Substitute the two factorizations into the Radon–Nikodym ratio of (13.2) and apply the chain rule for conditional relative entropy, Lemma 3.12 and Cover and Thomas [24, Sec. 2.5], in its measure-level form, which subtracts no infinities. □

13.3 The scale-local two-channel graph energy

Definition 13.8 (Transported divergences and row free energies). For every copy $e : j \to i$ in E_ℓ and $x \in \{b, m\}$ put

$$D_e^{x,\ell} := D_{\mathrm{KL}}\big(q_{i,\ell}^x \| (\Theta_e^{x,\ell})_\# q_{j,\ell}^x\big), \tag{13.12}$$

where $(\Theta)_\#$ is the pushforward action $\widehat{\rho}_x(\Theta)$ on channel-x laws; parallel copies carry separate divergences, and these are law-point energies in the sense of Section 3.8. A selectively communicating variant uses the role divergences (6.35) or a message channel instead, which changes the energy, and its recursive state keeps the interface and noise data and any private conditional law or memory that later kernels read (Section 6.5, Proposition 6.26). With $E_\ell(i)$ the copies received by i, the occupancies and rows of an event law are

$$\alpha_i^{x,\ell} := \sum_{e \in E_\ell(i)} \eta_e^{x,\ell}, \qquad \beta_e^{x,\ell} := \frac{\eta_e^{x,\ell}}{\alpha_i^{x,\ell}} \quad (e \in E_\ell(i),\ \alpha_i^{x,\ell} > 0), \tag{13.13}$$

with pair rows $\beta_{ij}^{x,\ell} = \sum_{e:j\to i} \beta_e^{x,\ell}$. The rows are the conditional source laws of whichever event law is displayed, generative, posterior, or recognition; a row of zero occupancy is not determined by the joint law. With normalized copy priors $\pi_i^{x,\ell}$ on $E_\ell(i)$ at every vertex i that carries a row (a vertex with an empty source set carries no row and has zero occupancy, although $E_\ell(i)$ may contain the reversals of copies it sends) and temperatures $\tau_{i,\ell}^x > 0$, the row free energies in energy units are

$$\mathcal{F}_{i,\ell}^{\mathrm{att},x} = \sum_{e \in E_\ell(i)} \beta_e^{x,\ell} D_e^{x,\ell} + \tau_{i,\ell}^x D_{\mathrm{KL}}\big(\beta_i^{x,\ell} \| \pi_i^{x,\ell}\big), \tag{13.14}$$

and their dimensionless form is $\mathcal{F}_{i,\ell}^{\mathrm{att},x}/\tau_{i,\ell}^x$.

Proposition 13.9 (Gibbs rows). With $\mathcal{Z}_i^{x,\ell} = \sum_{e\in E_\ell(i)} \pi_e^{x,\ell} e^{-D_e^{x,\ell}/\tau_{i,\ell}^x}$, assume $\mathcal{Z}_i^{x,\ell} > 0$, equivalently that at least one edge of positive prior mass has finite energy. On the closed simplex of rows supported on $\{e : \pi_e^{x,\ell} > 0\}$, with $0 \cdot (+\infty) = 0$, the extended-real identity is $\mathcal{F}_{i,\ell}^{\mathrm{att},x} = \tau_{i,\ell}^x D_{\mathrm{KL}}(\beta_i^{x,\ell} \| \beta_i^{x,\ell,*}) - \tau_{i,\ell}^x \log \mathcal{Z}_i^{x,\ell}$ with

$$\beta_e^{x,\ell,*} = \frac{\pi_e^{x,\ell} e^{-D_e^{x,\ell}/\tau_{i,\ell}^x}}{\mathcal{Z}_i^{x,\ell}}, \tag{13.15}$$

so the minimizer is unique and the minimum is $-\tau_{i,\ell}^x \log \mathcal{Z}_i^{x,\ell}$, with zero optimal mass on every infinite-energy edge. If $\mathcal{Z}_i^{x,\ell} = 0$, every prior-supported row has infinite free energy, and the Gibbs formula defines no probability row or finite minimizer. When all prior-supported energies

are finite, the optimum has positive mass on every prior-supported edge. The optimal rows move with the transported divergences, and omitting the row relative entropy changes the objective. The dimensionless form is the row free energy (6.7) with prior $\pi_i^{x,\ell}$ and the fixed energies $D_e^{x,\ell}$ in place of the expected divergences, with the same minimizer (6.8) and minimum $-\log \mathcal{Z}_i^{x,\ell}$. The energy-unit form with $\tau_{i,\ell}^x \neq 1$ is not an independently weighted sector of one evidence bound (Proposition 6.2), and the composite potential below inherits this scaling. [ESTABLISHED]

Proof. Expanding $D_{\mathrm{KL}}(\beta\|\beta^*)$ with (13.15) gives $\sum_e \beta_e \log(\beta_e/\pi_e) + \sum_e \beta_e D_e/\tau + \log \mathcal{Z}$, which is the stated identity after multiplication by τ. Relative entropy is nonnegative and vanishes only at $\beta = \beta^*$. A row charging an infinite-energy edge has both sides equal to $+\infty$; otherwise the expansion uses only finite-energy edges. If $\mathcal{Z} = 0$, every edge of positive prior mass has infinite energy, giving the stated exceptional case. □

Definition 13.10 (Composite multiscale potential). If a scale-local prior is a normalized Gibbs law containing the self terms and $\sum_{e\in E_\ell}(\eta_e^{b,\ell} D_e^{b,\ell} + \eta_e^{m,\ell} D_e^{m,\ell})$, those terms and the global normalizer belong to the tower variational free energy. If they are appended to a fixed population free energy $\mathcal{F}_{\mathrm{pop}}$ of Chapter 5 instead, the resulting objective is the composite potential

$$\mathcal{J}_{\mathrm{comp}} = \mathcal{F}_{\mathrm{pop}} + \sum_\ell \sum_i \left(\alpha_i^{b,\ell} \mathcal{F}_{i,\ell}^{\mathrm{att},b} + \alpha_i^{m,\ell} \mathcal{F}_{i,\ell}^{\mathrm{att},m}\right) + \sum_\ell \mathcal{F}_{\mathrm{part}}^\ell + \sum_\ell \mathcal{F}_{\mathrm{cross}}^\ell + \sum_\ell \mathcal{J}_{\mathrm{link}}^\ell, \tag{13.16}$$

where the sum over i runs over the vertices that carry a row, a term of zero occupancy is zero, the partition and cross-scale terms are those of (13.17) and (13.38), and $\mathcal{J}_{\mathrm{link}}^\ell$ is a declared link regularizer. It is a descent potential and carries no evidence interpretation of its own. With latent links, a loop action placed in the generator prior of Definition 9.11 is the in-bound counterpart of a link regularizer: it enters the free energy with its normalizer through the link sector (9.23) (Proposition 9.18), so it belongs to the bound instead of being appended to it.

13.4 Partition variables and event pushforwards

Attention rows do not create a parent random variable. A hierarchy meant to emerge under the same variational principle puts the structural mark R_ℓ inside both the generative and the recognition law.

Definition 13.11 (Partition factor). A finite-temperature partition term has the form

$$\mathcal{F}_{\mathrm{part}}^\ell = \mathbb{E}_{\mathbb{Q}_R^\ell}\left[H_{\mathrm{block}}^\ell(R_\ell; Z_\ell, \mathfrak{N}_\ell)\right] + \tau_R^\ell D_{\mathrm{KL}}\left(\mathbb{Q}_R^\ell \| \mathbb{P}_R^{\ell,\circ}\right), \tag{13.17}$$

and enters the variational free energy only when the normalized Gibbs factor that it represents is part of the generative law. The block action H_{block}^ℓ may trade internal transported mismatch against boundary retention, evaluator closure, and residual cost. No canonical partition selector follows from the global free energy alone (Open Problem 10.20).

13.4.1 A random partition catalog

The structural mark may be random with a finite catalog of partitions, or membership configurations, of V_ℓ; each label fixes the branch carrier of the next depth, X stays outside the channel, no edge is created, and the two normalizations of the catalog are distinct generative models.

Definition 13.12 (Finite partition catalog). Let $\mathfrak{G}$ be a finite nonempty catalog of values of R_ℓ. For each $\gamma \in \mathfrak{G}$ declare a finite nonempty carrier Y_γ, a strictly positive probability reference ν_γ, a positive raw factor $\widetilde{p}_\gamma$, a positive weight a_γ, and a normalized, possibly vanishing, likelihood $L_\gamma(o \mid y)$ on a common finite observation alphabet, all fixed generative data independent of recognition. On the disjoint union of the carriers put

$$\mathcal{Z}_\gamma = \sum_y \nu_\gamma(y)\widetilde{p}_\gamma(y), \qquad p_\gamma(y) = \frac{\nu_\gamma(y)\widetilde{p}_\gamma(y)}{\mathcal{Z}_\gamma}, \qquad \bar{\mathsf{a}}_\gamma = \frac{\mathsf{a}_\gamma}{\sum_\delta \mathsf{a}_\delta}. \tag{13.18}$$

The hierarchically normalized and the globally normalized joints are

$$\mathbb{P}_{\mathrm{h}}(\gamma, y, o) = \bar{\mathsf{a}}_\gamma p_\gamma(y) L_\gamma(o \mid y), \tag{13.19}$$

$$\mathbb{P}_{\mathrm{f}}(\gamma, y, o) = \frac{\mathsf{a}_\gamma \nu_\gamma(y)\widetilde{p}_\gamma(y) L_\gamma(o \mid y)}{\mathcal{Z}_{\mathrm{all}}}, \qquad \mathcal{Z}_{\mathrm{all}} = \sum_\delta \mathsf{a}_\delta \mathcal{Z}_\delta. \tag{13.20}$$

Proposition 13.13 (Branch normalization changes label weights). Both joints are normalized and have the same conditional branch laws, while the label weight of the globally normalized joint is $\mathsf{a}_\gamma \mathcal{Z}_\gamma / \mathcal{Z}_{\mathrm{all}}$. The label weights of the two joints are the softmax weights of Definition 10.23 with effective log masses $\log \mathsf{a}_\gamma$ and $\log \mathsf{a}_\gamma + \log \mathcal{Z}_\gamma$ respectively. For the same positive weights the joints coincide exactly when all $\mathcal{Z}_\gamma$ coincide. Replacing $\widetilde{p}_\gamma$ by $\mathrm{c}_\gamma \widetilde{p}_\gamma$ with $\mathrm{c}_\gamma > 0$ leaves the hierarchical joint unchanged and reweights the global one by c_γ, and a common factor cancels in both. For a strictly positive probability reference $\nu_{\mathfrak{G}}$ on the catalog write $\mathsf{a}_\gamma = \nu_{\mathfrak{G}}(\gamma) e^{-H_\gamma^{\mathrm{lab}}}$ and $\widetilde{p}_\gamma = e^{-H_\gamma}$; every positive family of weights has this form. Fix positive evidence $M_o = \sum_{\gamma,y} \mathbb{P}(\gamma, y, o)$ and a recognition law $\mathbb{Q}(\gamma, y) = \mathbb{Q}^{\mathrm{lab}}(\gamma)\mathbb{Q}_\gamma(y)$, with $\mathbb{Q}_\gamma$ defined where $\mathbb{Q}^{\mathrm{lab}}(\gamma) > 0$. Then, with $\mathcal{F}_o = D_{\mathrm{KL}}(\mathbb{Q} \| \mathbf{\Pi}_o) - \log M_o$,

$$\begin{aligned} \mathcal{F}_{\mathrm{f},o}(\mathbb{Q}) &= D_{\mathrm{KL}}(\mathbb{Q}^{\mathrm{lab}} \| \nu_{\mathfrak{G}}) + \mathbb{E}_{\mathbb{Q}^{\mathrm{lab}}} H_\gamma^{\mathrm{lab}} + \log \mathcal{Z}_{\mathrm{all}} \\ &\quad + \sum_\gamma \mathbb{Q}^{\mathrm{lab}}(\gamma) \big[D_{\mathrm{KL}}(\mathbb{Q}_\gamma \| \nu_\gamma) + \mathbb{E}_{\mathbb{Q}_\gamma} H_\gamma - \mathbb{E}_{\mathbb{Q}_\gamma} \log L_\gamma(o \mid Y) \big], \end{aligned} \tag{13.21}$$

$$\mathcal{F}_{\mathrm{h},o}(\mathbb{Q}) = \mathcal{F}_{\mathrm{f},o}(\mathbb{Q}) - \log \mathcal{Z}_{\mathrm{all}} + \log \sum_\delta \mathsf{a}_\delta + \sum_\gamma \mathbb{Q}^{\mathrm{lab}}(\gamma) \log \mathcal{Z}_\gamma, \tag{13.22}$$

with the convention $0 \log 0 = 0$; a recognition-positive atom with zero likelihood gives $+\infty$. Each expansion contains one full recognition entropy, decomposed by its conditional chain, and no further correlation-entropy term. [ESTABLISHED]

Proof. Sum either joint over o, then y, then γ. Equality of the joints implies equality of the label marginals, hence $\mathcal{Z}_\gamma = \mathcal{Z}_{\mathrm{all}}/\sum_\delta \mathsf{a}_\delta$ for every γ; the converse is substitution. A branch rescaling changes $\mathcal{Z}_\gamma$ to $\mathrm{c}_\gamma \mathcal{Z}_\gamma$ and cancels in p_γ. The softmax statement is the label marginal of each joint written as the exponential of its log mass divided by the sum of these exponentials. The expansions follow by writing $\log \mathbb{P}$ with the displayed factors and taking the expectation under $\mathbb{Q}$; the two differ only in the normalizers, $\log \sum_\delta \mathsf{a}_\delta + \log \mathcal{Z}_\gamma$ against $\log \mathcal{Z}_{\mathrm{all}}$. □

Reference, raw factor, label weight, evidence, and posterior are distinct objects, and the score of either joint in a parameter is the complete labeled score of Proposition 10.24, which contains every varying generative factor, the $\mathcal{Z}_\gamma$ of the global model included.

13.4.2 Membership kernels and event pushforwards

Definition 13.14 (Membership kernels). Overlapping assignment is a membership Markov kernel

$$\mathfrak{m}_\ell(I \mid i, R_\ell) \geq 0, \qquad \sum_{I \in V_{\ell+1}} \mathfrak{m}_\ell(I \mid i, R_\ell) = 1, \tag{13.23}$$

of which a hard partition is the deterministic case $\mathfrak{m}_\ell(I \mid i, R_\ell) = \mathbf{1}\{R_\ell(i) = I\}$. Correlated endpoints use one joint endpoint kernel

$$\mathfrak{M}_\ell(I, J \mid i, j, R_\ell) \geq 0, \qquad \sum_{I,J} \mathfrak{M}_\ell(I, J \mid i, j, R_\ell) = 1, \tag{13.24}$$

whose product form requires a declared conditional independence of the two draws.

Proposition 13.15 (Event pushforwards). Fix $y \in \mathsf{Y}_\ell$ and a realized mark $\gamma \in \mathsf{R}_\ell$. If every vertex receives one membership draw and the draws of distinct vertices are conditionally independent at (y, γ), the conditional coarse event law is

$$\eta^{x,\ell+1}_{IJ}(y,\gamma) = \sum_{\substack{i,j \in V_\ell \\ i \neq j}} \eta^{x,\ell}_{ij}(y)\mathfrak{m}_\ell(I \mid i,\gamma)\mathfrak{m}_\ell(J \mid j,\gamma) + \delta_{IJ} \sum_{i \in V_\ell} \eta^{x,\ell}_{ii}(y)\mathfrak{m}_\ell(I \mid i,\gamma), \tag{13.25}$$

where the loop term uses the diagonal endpoint kernel, since both endpoints of a loop follow one draw. With correlated endpoints,

$$\eta^{x,\ell+1}_{IJ}(y,\gamma) = \sum_{i,j \in V_\ell} \eta^{x,\ell}_{ij}(y)\mathfrak{M}_\ell(I, J \mid i, j, \gamma). \tag{13.26}$$

With the assignment marginal $C^R_\ell(y, d\gamma) = C^{\mathrm{struct}}_{\ell+1 \leftarrow \ell}(y; d\gamma \times \mathsf{Y}_{\ell+1})$ of the structural channel and an input law $\mathbb{M}$, the unconditional event mass averages the whole conditional expression,

$$\overline{\eta}^{x,\ell+1}_{IJ}[\mathbb{M}] = \sum_{i,j \in V_\ell} \int \eta^{x,\ell}_{ij}(y)\mathfrak{M}_\ell(I, J \mid i, j, \gamma) C^R_\ell(y, d\gamma)\mathbb{M}(dy). \tag{13.27}$$

All three coarse event laws sum to one. Conditional endpoint independence permits the product in (13.25) but not the factoring of the expectation of a random event mass and a random assignment, so (13.27) applies to all three input laws with no independence assumed between event and membership marks. The random $\eta_{\ell+1}(y,\gamma)$ is part of $\mathfrak{N}_{\ell+1}$ with rows recovered by (13.13), and its unconditional mean does not replace it. Loop copies arise at every depth after the first link lift (13.31); they are induced by blocking and are not a new support declaration, as in Proposition 12.31. For a hard partition the loop and product terms agree, and under the posterior bridge the pushforward is (12.49). Literal unit-weight membership in several parents has total mass greater than one; it is finite-measure replication, not a probability pushforward. [ESTABLISHED]

Proof. Summing the first term of (13.25) over I, J gives $\sum_{i\neq j}\eta_{ij}$ and the loop term gives $\sum_i \eta_{ii}$, which add to one; the correlated and unconditional cases use normalization of $\mathfrak{M}_\ell$, η_ℓ, C_ℓ^R, and $\mathbb{M}$. The expectation of a product of dependent random quantities is not the product of expectations. A vertex with unit weight in k parents contributes its mass k times. □

Definition 13.16 (Chapman–Kolmogorov composition). Nested assignments compose by

$$\mathfrak{m}_{\ell+2\leftarrow\ell}(A \mid i, R_\ell, R_{\ell+1}) = \sum_I \mathfrak{m}_{\ell+1}(A \mid I, R_{\ell+1})\mathfrak{m}_\ell(I \mid i, R_\ell), \tag{13.28}$$

$$\mathfrak{M}_{\ell+2\leftarrow\ell}(A, A' \mid i, j, R_\ell, R_{\ell+1}) = \sum_{I,J} \mathfrak{M}_{\ell+1}(A, A' \mid I, J, R_{\ell+1})\mathfrak{M}_\ell(I, J \mid i, j, R_\ell), \tag{13.29}$$

and the augmented structural channels by

$$C^{\mathrm{struct}}_{\ell+2\leftarrow\ell} = C^{\mathrm{struct}}_{\ell+2\leftarrow\ell+1} \circ C^{\mathrm{struct}}_{\ell+1\leftarrow\ell}, \tag{13.30}$$

where composition integrates every intermediate assignment, event law, and mark, and the root, holonomy, and boundary marks stay in the target of each composite. An independently declared multi-depth channel is admitted only when it coincides with the composite, and the composite is the exact multi-depth step only when the augmented state carries every mark that the next structural kernel reads.

13.5 Coarse links and what the parent retains

The links are not assumed flat: $\mathfrak{h}_\ell$ keeps the root-framed datum of Proposition 11.5 in every connected block with its joint law, and on a non-flat graph a reduced mean of the dressed marks must carry a declared residual, whose penalization to zero is a modeling preference, not a consequence of gauge covariance.

Definition 13.17 (Coarse links under a hard partition). Let the realized mark $R_\ell = \gamma$ be a hard partition whose blocks are replaced by their component-indexed meta-agents (11.6), so that each coarse vertex I has one root r_I and one rooted spanning tree. This excludes overlapping membership kernels, for which a copy would need a root transport into every block that contains

one of its endpoints with positive membership; the event pushforwards of Proposition 13.15 hold for general membership kernels. The depth-$(\ell+1)$ multigraph $\mathfrak{I}_{\ell+1} = (V_{\ell+1}, E_{\ell+1}, {}^{-})$ has the coarse vertices, the same copies $E_{\ell+1} := E_\ell$ with the same reversal, and endpoints $e : J \to I$ whenever $e : j \to i$ with $j \in J$ and $i \in I$; its event law is $\eta_e^{x,\ell+1} := \eta_e^{x,\ell}$, and its links are the dressed marks of all copies,

$$\Theta_e^{x,\ell+1} := \widehat{\Theta}_e^{x,\ell} = \Theta_{I\leftarrow i}^{x,\ell}\Theta_e^{x,\ell}\big(\Theta_{J\leftarrow j}^{x,\ell}\big)^{-1}, \qquad e : j \to i,\ j \in J,\ i \in I, \tag{13.31}$$

internal as well as boundary, so that internal copies become loops at their block. The parent laws $q_{I,\ell+1}^x$ are expressed in the frame of the root r_I. The coarse multigraph is induced structure; no edge is created.

Proposition 13.18 (Properties of the coarse links). The coarse links are well typed and keep the whole link content of depth ℓ. An internal tree copy has the identity link 1_G. An internal non-tree copy carries the based holonomy of its fundamental loop, and for every closed walk p in I based at r_I that traverses the copies $e_0, \dots, e_{n-1}$ in this order, $\mathrm{Hol}_I^{x,\ell}(p) = \Theta_{e_{n-1}}^{x,\ell+1} \cdots \Theta_{e_0}^{x,\ell+1}$. Inverse reversal persists, and the dressed law is the vertex law (2.28) with vertex coordinates g_{r_I}, so the divergences, rows, and row free energies of Definition 13.8 are defined and gauge invariant at depth $\ell+1$. On a loop the divergence $D_{\mathrm{KL}}(q_{I,\ell+1}^x \| (\Theta_e^{x,\ell+1})_\# q_{I,\ell+1}^x)$ vanishes on tree copies and, on a non-tree copy, exactly when the parent law is fixed by that holonomy, as in (11.10). The pair marginal $\sum_{e:J\to I} \eta_e^{x,\ell+1}$ is (13.25) for the hard partition, and no averaged group element replaces the parallel copies. The links depend on the trees, not only on the roots: changing the root of a fixed tree is a vertex gauge transformation at depth $\ell+1$, while replacing the tree changes $\Theta_{I\leftarrow i}^{x,\ell}$ to $\mathrm{Hol}_I^{x,\ell}(p_i)\Theta_{I\leftarrow i}^{x,\ell}$ for a based loop p_i that depends on i, and a copy that leaves the tree changes its loop link from 1_G to the holonomy $\mathrm{Hol}_I^{x,\ell}(p)$ of its new fundamental loop p and its loop divergence from 0 to $D_{\mathrm{KL}}(q_{I,\ell+1}^x \| \mathrm{Hol}_I^{x,\ell}(p)_\# q_{I,\ell+1}^x)$. Whenever such a holonomy is not the identity, the trees are part of the coarse channel rather than a gauge choice. When $q_{I,\ell+1}^x$ and $q_{J,\ell+1}^x$ lie in the stabilized sector, the divergences $D_e^{x,\ell+1}$ do not depend on the trees. This link-level tree dependence complements the feature-level one of (11.15). [ESTABLISHED]

Proof. The dressed link of an internal copy is 1_G on a tree copy, since there $\Theta_e = \Theta_{I\leftarrow i}^{-1}\Theta_{I\leftarrow j}$, and is the fundamental-cycle holonomy (11.7) otherwise. Inserting $(\Theta_{I\leftarrow k})^{-1}\Theta_{I\leftarrow k}$ at each intermediate vertex k of a closed walk turns the ordered product of the fine links into that of the coarse links. Reversal and the vertex law follow from (11.11), and relative entropy is invariant under a common bijective pushforward, which gives gauge invariance; a loop divergence vanishes exactly when $q_I = \mathrm{Hol}_I(p)_\# q_I$. A root change multiplies every transport of the block by one element, and a tree change multiplies each tree transport on the left by a based holonomy (Proposition 11.5), so the link of $e : j \to i$ becomes $\mathrm{Hol}_I(p_i)\Theta_e^{x,\ell+1}\mathrm{Hol}_J(p_j)^{-1}$, and when $\mathrm{Hol}_I(p_i)$ fixes q_I and $\mathrm{Hol}_J(p_j)$ fixes q_J, invariance under the pushforward by $\mathrm{Hol}_I(p_i)^{-1}$ gives $D_{\mathrm{KL}}(q_I \| (\mathrm{Hol}_I(p_i)\Theta\mathrm{Hol}_J(p_j)^{-1})_\# q_J) = D_{\mathrm{KL}}(q_I \| \Theta_\# q_J)$. □

When the links are latent generators (Chapter 9), the coarse links of Definition 13.17 are random elements of G: measurable functions of the fine generators given the partition and the trees, whose law is the pushforward of the fine generator law. Two questions follow: whether the exponential

chart and its Gaussian laws survive the coarse step, and how the holonomy-conditioned score of Definition 11.9 behaves once the links carry a law.

Definition 13.19 (Series coarse twist and cumulant forms). Let G be a closed subgroup of $\mathrm{GL}(K,\mathbb{R})$ with Lie algebra $\mathfrak{g} \subseteq \mathfrak{gl}(K,\mathbb{R})$, acting on the channel by the inclusion, so that the exponential is the matrix exponential, and let $\|\cdot\|$ be the operator norm. Fix one channel and drop its superscript. In Definition 13.17 at depth ℓ, let $e' : i \to r_I$ be a copy of the rooted spanning tree of a block I that joins an agent i to the root, and let $e : j \to i$ be a copy from the root $j = r_J$ of a block $J \neq I$. Then $\Theta_{I\leftarrow i} = \Theta_{e'}$ and $\Theta_{J\leftarrow j} = 1_G$, so the coarse link of e is $\Theta_{e'}\Theta_e = U_{r_I} e^{\mathring{\omega}_{e'}} e^{\mathring{\omega}_e} U_{r_J}^{-1}$, and its twist relative to the root frames, in which the parent laws are expressed, is

$$\mathring{\Theta}_e^{\ell+1} = e^{\mathring{\omega}_{e'}} e^{\mathring{\omega}_e}. \tag{13.32}$$

A coarse generator is a $\mathfrak{g}$-valued random variable $\mathring{\omega}_e^{\ell+1}$ with $e^{\mathring{\omega}_e^{\ell+1}} = \mathring{\Theta}_e^{\ell+1}$ almost surely; where the principal logarithm $\mathrm{Log}\,\mathring{\Theta}_e^{\ell+1}$ exists, it is the natural candidate. Extend $\langle\cdot,\cdot\rangle_{\mathsf{G}}$ to $\mathfrak{gl}(K,\mathbb{R})$ by the same formula. For a direction $\mathring{v} \in \mathfrak{g}$, positive-definite self-adjoint operators $\mathring{\Lambda}_{e'}, \mathring{\Lambda}_e$ on $(\mathfrak{g}, \langle\cdot,\cdot\rangle_{\mathsf{G}})$, the directions $\mathring{v}_{e'} = \mathring{\Lambda}_{e'}^{-1}\mathring{v}$ and $\mathring{v}_e = \mathring{\Lambda}_e^{-1}\mathring{v}$, and independent centered Gaussian random elements $\mathring{\zeta}_{e'} \sim \mathcal{N}(0, \mathring{\Lambda}_{e'}^{-1})$ and $\mathring{\zeta}_e \sim \mathcal{N}(0, \mathring{\Lambda}_e^{-1})$ of $\mathfrak{g}$, put

$$\begin{aligned}
\mathring{\kappa}_3(\mathring{v}) &= \big\langle \mathring{v}, [\mathring{v}_{e'}, \mathring{v}_e] \big\rangle_{\mathsf{G}}, \\
\mathring{\kappa}_4(\mathring{v}) &= 3\mathbb{E}\big\langle \mathring{v}, [\mathring{v}_{e'}, \mathring{\zeta}_e] \big\rangle_{\mathsf{G}}^2 + 3\mathbb{E}\big\langle \mathring{v}, [\mathring{\zeta}_{e'}, \mathring{v}_e] \big\rangle_{\mathsf{G}}^2 + 2\big\langle \mathring{v}, [\mathring{v}_{e'}, [\mathring{v}_{e'}, \mathring{v}_e]] + [\mathring{v}_e, [\mathring{v}_e, \mathring{v}_{e'}]] \big\rangle_{\mathsf{G}}.
\end{aligned} \tag{13.33}$$

Proposition 13.20 (Gaussian generator laws are not closed under coarse links). In Definition 13.19 let $\mathring{\omega}_{e'}$ and $\mathring{\omega}_e$ be independent Gaussian random elements of $\mathfrak{g}$ with means $\mathring{\mu}_{e'}, \mathring{\mu}_e$ and positive-definite precisions $\mathring{\Lambda}_{e'}, \mathring{\Lambda}_e$, as under a product Gaussian recognition law of the generators (Proposition 9.18). (i) If $\mathfrak{g}$ is abelian, $\mathring{\omega}_{e'} + \mathring{\omega}_e$ is a Gaussian coarse generator. (ii) Let $G = \mathrm{GL}^+(K)$ with $K \geq 2$, and let $\mathbb{X}_K$ be the set of real $K \times K$ matrices with K distinct real eigenvalues, exactly two of them negative and the others positive. Then $\mathbb{X}_K$ is a nonempty open subset of $\mathrm{GL}^+(K)$, no element of $\mathbb{X}_K$ has a real logarithm, and $\Pr(\mathring{\Theta}_e^{\ell+1} \in \mathbb{X}_K) > 0$. Consequently no coarse generator exists, and the law of $\mathring{\Theta}_e^{\ell+1}$ is the image under the exponential of no law on $\mathfrak{g}$. (iii) Let the laws depend on a width $\mathring{\tau} \in (0,1]$ through the means $\mathring{\tau}\mathring{\mu}_{e'}$ and $\mathring{\tau}\mathring{\mu}_e$ and the precisions $\mathring{\tau}^{-2}\mathring{\Lambda}_{e'}$ and $\mathring{\tau}^{-2}\mathring{\Lambda}_e$, with $\mathring{\mu}_{e'}, \mathring{\mu}_e, \mathring{\Lambda}_{e'}, \mathring{\Lambda}_e$ held. On the event $\|\mathring{\omega}_{e'}\| + \|\mathring{\omega}_e\| \leq \frac{1}{2}$ the principal logarithm of $\mathring{\Theta}_e^{\ell+1}$ exists and lies in $\mathfrak{g}$, and the event has probability at least $1 - \mathrm{c}_1 e^{-\mathrm{c}_2/\mathring{\tau}^2}$ for constants $\mathrm{c}_1, \mathrm{c}_2 > 0$. If the cubic form $\mathring{\kappa}_3$ of (13.33) is not identically zero, or if $\mathring{\mu}_{e'} = \mathring{\mu}_e = 0$ and the quartic form $\mathring{\kappa}_4$ is not identically zero, there is $\mathring{\tau}_0 > 0$ such that for every $\mathring{\tau} \in (0, \mathring{\tau}_0)$ no Gaussian random element of $\mathfrak{gl}(K,\mathbb{R})$ coincides with $\mathrm{Log}\,\mathring{\Theta}_e^{\ell+1}$ almost surely on that event. For $\mathfrak{g} = \mathfrak{gl}(K,\mathbb{R})$ with $K \geq 2$, $\mathsf{G}^{\mathrm{fib}} = I_K$, and $\mathring{\Lambda}_{e'} = \mathring{\Lambda}_e = \mathrm{Id}$, the form $\mathring{\kappa}_3$ vanishes identically and $\mathring{\kappa}_4(\mathring{v}) = 12$ at $\mathring{v} = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \oplus 0_{K-2}$; with $\mathring{\Lambda}_{e'} = \mathrm{Id}$ and $\mathring{\Lambda}_e^{-1}$ the identity plus the orthogonal projection onto the line of that matrix, $\mathring{\kappa}_3(\mathring{v}) = 1$ at $\mathring{v} = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right) \oplus 0_{K-2}$. For an inner product on $\mathfrak{g}$ invariant under the adjoint action of $\mathfrak{g}$ and precisions that are multiples of the identity, both forms vanish identically. (iv) Consequently the product Gaussian generator laws are not closed under the coarse links of Definition 13.17: for $G = \mathrm{GL}^+(K)$ with $K \geq 2$ and

every pair of nondegenerate laws by (ii), and for every closed subgroup under the conditions of (iii) at small widths, in the sense that the principal-logarithm coarse generator is not Gaussian. [ESTABLISHED]

Proof. (i) Commuting matrices satisfy $e^{\mathring{\omega}_{e'}}e^{\mathring{\omega}_e}=e^{\mathring{\omega}_{e'}+\mathring{\omega}_e}$, and a sum of independent Gaussian random elements is Gaussian.

(ii) A matrix in $\mathbb{X}_K$ has determinant equal to the product of its eigenvalues, which is positive. Near a matrix of $\mathbb{X}_K$ the eigenvalues depend continuously on the entries, a simple real eigenvalue stays real because a nonreal eigenvalue nearby would bring its complex conjugate into the same small disk, and signs and distinctness persist, so $\mathbb{X}_K$ is open. Each negative eigenvalue of a matrix in $\mathbb{X}_K$ is simple, so its single Jordan block occurs once, and by Culver [29, Theorem 1] no element of $\mathbb{X}_K$ has a real logarithm. With $\mathfrak{J}=\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$, $\exp(\pi\mathfrak{J}\oplus 0_{K-2})\exp(\operatorname{diag}(0,\log 2,\log 3,\dots,\log K))=\operatorname{diag}(-1,-2,3,\dots,K)\in\mathbb{X}_K$, so the preimage of $\mathbb{X}_K$ under the continuous map $(\mathring{\xi}_1,\mathring{\xi}_2)\mapsto e^{\mathring{\xi}_1}e^{\mathring{\xi}_2}$ is a nonempty open subset of $\mathfrak{g}\times\mathfrak{g}$, and the joint law of $(\mathring{\omega}_{e'},\mathring{\omega}_e)$, which has an everywhere positive density, charges it. A $\mathfrak{g}$-valued $\mathring{\omega}_e^{\ell+1}$ with $e^{\mathring{\omega}_e^{\ell+1}}=\mathring{\Theta}_e^{\ell+1}$ almost surely would place $\mathring{\Theta}_e^{\ell+1}$ in $\exp(\mathfrak{g})$, which misses $\mathbb{X}_K$, almost surely, and the image under the exponential of any law on $\mathfrak{g}$ gives $\mathbb{X}_K$ mass zero.

(iii) Expansion of the logarithm. Let $\mathring{\xi}_1,\mathring{\xi}_2\in\mathfrak{gl}(K,\mathbb{R})$ with $\|\mathring{\xi}_1\|+\|\mathring{\xi}_2\|\le\frac12$, and let $\mathring{\Pi}_l=\sum_{a+b=l}\mathring{\xi}_1^a\mathring{\xi}_2^b/(a!b!)$ be the part of degree l of $e^{\mathring{\xi}_1}e^{\mathring{\xi}_2}$. Then $\|e^{\mathring{\xi}_1}e^{\mathring{\xi}_2}-I\|\le e^{1/2}-1<1$, so the series $\sum_{k\ge1}(-1)^{k+1}(e^{\mathring{\xi}_1}e^{\mathring{\xi}_2}-I)^k/k$ converges; its value is the principal logarithm, the logarithm whose eigenvalues have imaginary parts in $(-\pi,\pi)$. Substituting $\sum_{l\ge1}\mathring{\Pi}_l$ and grouping by total degree in $(\mathring{\xi}_1,\mathring{\xi}_2)$ is legitimate, because replacing every coefficient by its absolute value and every word in the letters by the product of the norms of its letters gives the expansion of $-\log(2-e^{\|\mathring{\xi}_1\|+\|\mathring{\xi}_2\|})$, which converges since $\frac12<\log 2$. Hence $\operatorname{Log}(e^{\mathring{\xi}_1}e^{\mathring{\xi}_2})=\sum_{l\ge1}\operatorname{BCH}_l(\mathring{\xi}_1,\mathring{\xi}_2)$ with homogeneous polynomials BCH_l of degree l satisfying $\|\operatorname{BCH}_l(\mathring{\xi}_1,\mathring{\xi}_2)\|\le \mathrm{b}_l(\|\mathring{\xi}_1\|+\|\mathring{\xi}_2\|)^l$, where $\mathrm{b}_l\ge0$ are the Taylor coefficients at zero of $\mathrm{u}\mapsto-\log(2-e^{\mathrm{u}})$. The components of degree one, two, and three are $\operatorname{BCH}_1=\mathring{\Pi}_1$, $\operatorname{BCH}_2=\mathring{\Pi}_2-\frac12\mathring{\Pi}_1^2=\frac12[\mathring{\xi}_1,\mathring{\xi}_2]$, and $\operatorname{BCH}_3=\mathring{\Pi}_3-\frac12(\mathring{\Pi}_1\mathring{\Pi}_2+\mathring{\Pi}_2\mathring{\Pi}_1)+\frac13\mathring{\Pi}_1^3=\frac{1}{12}([\mathring{\xi}_1,[\mathring{\xi}_1,\mathring{\xi}_2]]+[\mathring{\xi}_2,[\mathring{\xi}_2,\mathring{\xi}_1]])$, the last by expanding the products and collecting the eight words of degree three, so

$$\begin{gathered}\operatorname{Log}(e^{\mathring{\xi}_1}e^{\mathring{\xi}_2})=\mathring{\xi}_1+\mathring{\xi}_2+\tfrac12[\mathring{\xi}_1,\mathring{\xi}_2]+\tfrac{1}{12}\big([\mathring{\xi}_1,[\mathring{\xi}_1,\mathring{\xi}_2]]+[\mathring{\xi}_2,[\mathring{\xi}_2,\mathring{\xi}_1]]\big)+\operatorname{BCH}_{\ge4}(\mathring{\xi}_1,\mathring{\xi}_2),\\ \|\operatorname{BCH}_{\ge4}(\mathring{\xi}_1,\mathring{\xi}_2)\|\le \mathrm{c}_3\big(\|\mathring{\xi}_1\|+\|\mathring{\xi}_2\|\big)^4,\end{gathered}\tag{13.34}$$

with $\mathrm{c}_3=\sum_{l\ge4}\mathrm{b}_l2^{4-l}<\infty$.

The logarithm lies in $\mathfrak{g}$ when $\mathring{\xi}_1,\mathring{\xi}_2\in\mathfrak{g}$. For $t\in[0,1]$ put $\mathring{\gamma}(t)=\operatorname{Log}(e^{t\mathring{\xi}_1}e^{\mathring{\xi}_2})$. By the bound above, $\mathring{\gamma}$ is real-analytic with $\|\mathring{\gamma}(t)\|\le-\log(2-e^{1/2})<\frac{\pi}{2}$, and $\mathring{\gamma}(0)=\mathring{\xi}_2$, because the grouped series of $\operatorname{Log}(e^{\mathring{\xi}_2})$ is that of the formal identity $\log(e^{\mathrm{u}})=\mathrm{u}$. Differentiating $e^{\mathring{\gamma}(t)}=e^{t\mathring{\xi}_1}e^{\mathring{\xi}_2}$ with Lemma 9.19 gives $\mathbf{L}(\mathring{\gamma},\mathring{\gamma}')=\mathring{\xi}_1e^{\mathring{\gamma}}$, and $e^{\mathrm{u}\mathring{\gamma}}\mathbf{T}e^{-\mathrm{u}\mathring{\gamma}}=e^{\mathrm{u}\operatorname{ad}_{\mathring{\gamma}}}\mathbf{T}$ (Lemma 9.1 for the homomorphism $g\mapsto\operatorname{Ad}_g$, whose differential is ad), with $\operatorname{ad}_{\mathring{\gamma}}\mathbf{T}=[\mathring{\gamma},\mathbf{T}]$, writes $\mathbf{L}(\mathring{\gamma},\mathbf{T})=\big(\int_0^1 e^{\mathrm{u}\operatorname{ad}_{\mathring{\gamma}}}d\mathrm{u}\big)(\mathbf{T})e^{\mathring{\gamma}}$. Since $\|\operatorname{ad}_{\mathring{\gamma}}\|\le2\|\mathring{\gamma}\|<\pi$, the operator $\int_0^1 e^{\mathrm{u}\operatorname{ad}_{\mathring{\gamma}}}d\mathrm{u}=\sum_{k\ge0}\operatorname{ad}_{\mathring{\gamma}}^k/(k+1)!$ is inverted by the power series of $\mathrm{w}\mapsto\mathrm{w}/(e^{\mathrm{w}}-1)$, which converges for $|\mathrm{w}|<2\pi$. Hence $\mathring{\gamma}$ solves the differential equation whose

right side at $\mathbf{T}$ is that power series in $\mathrm{ad}_{\mathbf{T}}$ applied to $\mathring{\xi}_1$, an analytic vector field on the ball $\|\mathbf{T}\| < \pi$ that is tangent to $\mathfrak{g}$ at the points of $\mathfrak{g}$, because $\mathfrak{g}$ is a closed Lie subalgebra. The solution of this equation in $\mathfrak{g}$ from $\mathring{\xi}_2$ coincides with $\mathring{\gamma}$ by uniqueness while it exists, and it exists on $[0,1]$ because $\mathring{\gamma}$ stays in a compact subset of the ball; so $\mathrm{Log}(e^{\mathring{\xi}_1}e^{\mathring{\xi}_2}) = \mathring{\gamma}(1) \in \mathfrak{g}$.

Probability of the event. Write $\mathring{\omega}_{e'} = \mathring{\tau}\mathring{\chi}_{e'}$ and $\mathring{\omega}_e = \mathring{\tau}\mathring{\chi}_e$, where $\mathring{\chi}_{e'} = \mathring{\mu}_{e'} + \mathring{\zeta}_{e'}$ and $\mathring{\chi}_e = \mathring{\mu}_e + \mathring{\zeta}_e$ are independent with laws that do not depend on $\mathring{\tau}$. A Gaussian random element $\mathring{\chi}$ of a space of finite dimension satisfies $\mathbb{E}e^{\mathrm{c}\|\mathring{\chi}\|^2} < \infty$ for some $\mathrm{c} > 0$, by a Gaussian integral, and Markov's inequality for $e^{\mathrm{c}\|\mathring{\chi}\|^2}$ bounds $\Pr(\|\mathring{\chi}\| > 1/(4\mathring{\tau}))$ by a multiple of $e^{-\mathrm{c}/(16\mathring{\tau}^2)}$, which bounds the probability of the complement of the event.

Polynomial part. Put $\mathring{\psi}_1 = \langle \mathring{v}, \mathring{\chi}_{e'} + \mathring{\chi}_e\rangle_{\mathsf{G}}$, $\mathring{\psi}_2 = \frac{1}{2}\langle \mathring{v}, [\mathring{\chi}_{e'}, \mathring{\chi}_e]\rangle_{\mathsf{G}}$, $\mathring{\psi}_3 = \frac{1}{12}\langle \mathring{v}, [\mathring{\chi}_{e'}, [\mathring{\chi}_{e'}, \mathring{\chi}_e]] + [\mathring{\chi}_e, [\mathring{\chi}_e, \mathring{\chi}_{e'}]]\rangle_{\mathsf{G}}$, $\mathring{\psi}_4 = \mathring{\tau}^{-4}\langle \mathring{v}, \mathrm{BCH}_{\geq 4}(\mathring{\omega}_{e'}, \mathring{\omega}_e)\rangle_{\mathsf{G}}$ on the event and $\mathring{\psi}_4 = 0$ off it, and

$$\mathring{\Psi}_{\mathring{\tau}} = \mathring{\psi}_1 + \mathring{\tau}\mathring{\psi}_2 + \mathring{\tau}^2\mathring{\psi}_3 + \mathring{\tau}^3\mathring{\psi}_4. \tag{13.35}$$

By (13.34) and homogeneity, $\mathring{\Psi}_{\mathring{\tau}} = \mathring{\tau}^{-1}\langle \mathring{v}, \mathrm{Log}\,\mathring{\Theta}_e^{\ell+1}\rangle_{\mathsf{G}}$ on the event, and $|\mathring{\psi}_4| \leq \mathrm{c}_4(\|\mathring{\chi}_{e'}\| + \|\mathring{\chi}_e\|)^4$ with c_4 independent of $\mathring{\tau}$, so every moment of $\mathring{\psi}_1, \dots, \mathring{\psi}_4$ is bounded uniformly in $\mathring{\tau}$. Joint cumulants are multilinear and are polynomials in joint moments, and the cumulants of order three and four of the Gaussian variable $\mathring{\psi}_1$ vanish, so, with cum_l the cumulant of order l and cum the joint cumulant,

$$\begin{aligned}
&\mathrm{cum}_3\,\mathring{\Psi}_{\mathring{\tau}} = 3\mathring{\tau}\,\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2) + O(\mathring{\tau}^2),\\
&\mathrm{cum}_4\,\mathring{\Psi}_{\mathring{\tau}} = 4\mathring{\tau}\,\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2) + \mathring{\tau}^2\big[6\,\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2, \mathring{\psi}_2) + 4\,\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_3)\big] + O(\mathring{\tau}^3).
\end{aligned} \tag{13.36}$$

Gaussian integration by parts. Let $\partial_{\mathring{v}}$ be the derivative of a function of $(\mathring{\zeta}_{e'}, \mathring{\zeta}_e)$ along $(\mathring{v}_{e'}, \mathring{v}_e)$. Every polynomial function $\mathring{\varphi}$ of $(\mathring{\zeta}_{e'}, \mathring{\zeta}_e)$ satisfies $\mathbb{E}[(\mathring{\psi}_1 - \mathbb{E}\mathring{\psi}_1)\mathring{\varphi}] = \mathbb{E}[\partial_{\mathring{v}}\mathring{\varphi}]$: here $\mathring{\psi}_1 - \mathbb{E}\mathring{\psi}_1 = \langle \mathring{v}, \mathring{\zeta}_{e'}\rangle_{\mathsf{G}} + \langle \mathring{v}, \mathring{\zeta}_e\rangle_{\mathsf{G}}$, and integration by parts against the Gaussian density, whose gradient in $\mathring{\zeta}_{e'}$ for $\langle\cdot,\cdot\rangle_{\mathsf{G}}$ is $-\mathring{\Lambda}_{e'}\mathring{\zeta}_{e'}$ times the density and likewise in $\mathring{\zeta}_e$, turns each term into the derivative along $\mathring{\Lambda}_{e'}^{-1}\mathring{v}$, respectively $\mathring{\Lambda}_e^{-1}\mathring{v}$, as in the proof of Lemma 9.20. Two applications give $\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2) = \mathbb{E}[(\mathring{\psi}_1 - \mathbb{E}\mathring{\psi}_1)^2(\mathring{\psi}_2 - \mathbb{E}\mathring{\psi}_2)] = \partial_{\mathring{v}}^2\mathring{\psi}_2 = \mathring{\kappa}_3(\mathring{v})$ for all means, because $\mathring{\psi}_2$ is a quadratic polynomial whose second derivative along $(\mathring{v}_{e'}, \mathring{v}_e)$ is $\langle \mathring{v}, [\mathring{v}_{e'}, \mathring{v}_e]\rangle_{\mathsf{G}}$. If the means vanish, the map $(\mathring{\zeta}_{e'}, \mathring{\zeta}_e) \mapsto (-\mathring{\zeta}_{e'}, -\mathring{\zeta}_e)$ preserves the law and changes the signs of $\mathring{\psi}_1$ and $\mathring{\psi}_3$ but not of $\mathring{\psi}_2$, so every joint cumulant and every joint moment with an odd number of entries from $\{\mathring{\psi}_1, \mathring{\psi}_3\}$ vanishes; in particular $\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2) = 0$ and $\mathbb{E}[\mathring{\psi}_1\mathring{\psi}_2] = 0$, while $\mathbb{E}\mathring{\psi}_2 = 0$ because the two generators are independent and centered. Repeated integration by parts then gives $\mathbb{E}[\mathring{\psi}_1^2\mathring{\psi}_2^2] = \mathbb{E}[\mathring{\psi}_1^2]\mathbb{E}[\mathring{\psi}_2^2] + 2\mathbb{E}[(\partial_{\mathring{v}}\mathring{\psi}_2)^2]$ and $\mathbb{E}[\mathring{\psi}_1^3\mathring{\psi}_3] = 3\mathbb{E}[\mathring{\psi}_1^2]\mathbb{E}[\mathring{\psi}_1\mathring{\psi}_3] + \partial_{\mathring{v}}^3\mathring{\psi}_3$, hence $\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_2, \mathring{\psi}_2) = 2\mathbb{E}[(\partial_{\mathring{v}}\mathring{\psi}_2)^2]$ and $\mathrm{cum}(\mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_1, \mathring{\psi}_3) = \partial_{\mathring{v}}^3\mathring{\psi}_3$, which is six times the value of the homogeneous cubic $\mathring{\psi}_3$ at $(\mathring{v}_{e'}, \mathring{v}_e)$. Since $\partial_{\mathring{v}}\mathring{\psi}_2 = \frac{1}{2}\langle \mathring{v}, [\mathring{v}_{e'}, \mathring{\zeta}_e]\rangle_{\mathsf{G}} + \frac{1}{2}\langle \mathring{v}, [\mathring{\zeta}_{e'}, \mathring{v}_e]\rangle_{\mathsf{G}}$ with independent centered summands, the coefficient of $\mathring{\tau}^2$ in (13.36) is $\mathring{\kappa}_4(\mathring{v})$ in the centered case.

Gaussian rigidity. Fix a direction $\mathring{v}$ with $\mathring{\kappa}_3(\mathring{v}) \neq 0$, or, in the centered case, with $\mathring{\kappa}_4(\mathring{v}) \neq 0$, and suppose that along a sequence of widths tending to zero some Gaussian random element of $\mathfrak{gl}(K, \mathbb{R})$ coincides with $\mathrm{Log}\,\mathring{\Theta}_e^{\ell+1}$ almost surely on the event. Pairing it with $\mathring{v}$ and dividing by $\mathring{\tau}$ gives real Gaussian variables that equal $\mathring{\Psi}_{\mathring{\tau}}$ outside an event of probability at most $\mathrm{c}_1 e^{-\mathrm{c}_2/\mathring{\tau}^2}$. Since

$\mathring{\Psi}_{\mathring{\tau}} \to \mathring{\psi}_1$ in mean square, these Gaussian variables converge in law to $\mathring{\psi}_1$; the moduli of their characteristic functions converge pointwise, which gives convergence of their variances, and then their phase factors converge pointwise, which by integration against a standard normal density forces convergence of their means. Their moments of order eight are therefore bounded along the sequence, and by the Cauchy–Schwarz inequality on the exceptional event their moments of order at most four differ from those of $\mathring{\Psi}_{\mathring{\tau}}$ by $O(e^{-c_2/(2\mathring{\tau}^2)})$, and so do their cumulants of order three and four. Those of a Gaussian variable vanish, so by (13.36) $3\mathring{\tau}\mathring{\kappa}_3(\mathring{v}) + O(\mathring{\tau}^2)$, and in the centered case $\mathring{\tau}^2\mathring{\kappa}_4(\mathring{v}) + O(\mathring{\tau}^3)$, tend to zero faster than every power of $\mathring{\tau}$, which is impossible. Hence such a sequence does not exist, and $\mathring{\tau}_0$ does.

Instances. For $\mathsf{G}^{\mathrm{fib}} = I_K$ the inner product is $\mathrm{Tr}(\mathbf{S}^\top\mathbf{T})$. With unit precisions $\mathring{v}_{e'} = \mathring{v}_e = \mathring{v}$, so $\mathring{\kappa}_3(\mathring{v}) = \langle \mathring{v}, [\mathring{v}, \mathring{v}]\rangle_{\mathsf{G}} = 0$ and the double brackets of $\mathring{\kappa}_4$ vanish. For the stated $\mathring{v}$, $\langle \mathring{v}, [\mathring{v}, \mathring{\zeta}]\rangle_{\mathsf{G}} = \mathring{\zeta}_{22} - \mathring{\zeta}_{11}$, which has variance 2 when $\mathring{\zeta}$ has independent standard normal entries, and $\langle \mathring{v}, [\mathring{\zeta}, \mathring{v}]\rangle_{\mathsf{G}}$ is its negative, so $\mathring{\kappa}_4(\mathring{v}) = 3 \cdot 2 + 3 \cdot 2$. In the second instance $\mathring{v}_{e'} = \mathring{v}$ and $\mathring{v}_e = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 0 \end{smallmatrix}\right) \oplus 0_{K-2}$, whose bracket $[\mathring{v}_{e'}, \mathring{v}_e] = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \oplus 0_{K-2}$ pairs with $\mathring{v}$ to 1. For an invariant inner product, $\langle \mathring{v}, [\mathring{v}, \mathbf{T}]\rangle_{\mathsf{G}} = -\langle [\mathring{v}, \mathring{v}], \mathbf{T}\rangle_{\mathsf{G}} = 0$ for every $\mathbf{T} \in \mathfrak{g}$, and with precisions that are multiples of the identity $\mathring{v}_{e'}$ and $\mathring{v}_e$ are multiples of $\mathring{v}$, so every term of both forms vanishes.

(iv) is (ii) and (iii). □

For $\mathrm{GL}^+(K)$ the exponential chart does not survive the coarse step: the product of two independent links with full-support generator laws leaves the image of the exponential with positive probability, so the coarse links stay typed as group elements, as in Definition 13.17, and their law, not a generator law, is what the next depth receives. Where the logarithm exists, the commutator term of the expansion makes the coarse generator non-Gaussian at first order in the width whenever $\mathring{\kappa}_3$ is not identically zero, and, for centered laws, at second order whenever $\mathring{\kappa}_4$ is not; for an invariant inner product with isotropic precisions both leading coefficients vanish and this test is silent. The expansion (13.34) is the Baker–Campbell–Hausdorff series through third order [45, Ch. 5], proved above with the remainder bound used here.

Open Problem 13.21 (A finite-dimensional description of coarse link laws). By Proposition 13.20 the product Gaussian generator laws are not closed under the coarse links of Definition 13.17, and for $G = \mathrm{GL}^+(K)$ with $K \geq 2$ no law of generators represents a coarse link. The law of the coarse links always exists, as the pushforward of the fine law through the dressed products, which are deterministic given the partition and the trees (Proposition 13.18), and Theorem 10.7 applies to the coarse step. What is open is a finite-dimensional family of link laws that survives the step. For the two copies in series of Proposition 13.20, the step sends independent links with laws ν' and ν on G to the law $\nu' * \nu$ of their product, so such a family must be closed under convolution. It is settled, for $G = \mathrm{GL}^+(K)$ with $K \geq 2$, by a weakly continuous injective map from a connected open subset of some $\mathbb{R}^d$ into the Borel probability laws on $\mathrm{GL}^+(K)$ whose image contains the image under the exponential of every nondegenerate Gaussian law on $\mathfrak{gl}(K, \mathbb{R})$ and contains $\nu' * \nu$ whenever it contains ν' and ν; or by a proof that for some $K \geq 2$ no such map exists.

13.6 How a parent influences its children

Proposition 13.22 (Downward coupling in a declared hierarchy). Under Hypothesis 13.4 for a declared top-down hierarchy, a factorization such as

$$K^{\ell}_{\downarrow}(dZ_{\ell} \mid Y_{\ell+1}, R_{\ell}, \mathfrak{N}_{\ell}, X_{*}) = \prod_{I \in V_{\ell+1}} K^{\ell}_{\downarrow,I}\big(dZ_{I,\ell} \mid Z_{I,\ell+1}, \mathfrak{N}_{\ell+1}, R_{\ell}, \mathfrak{N}_{\ell}, X_{*}\big) \tag{13.37}$$

makes the parent belief and model presentations arguments of the children's conditional generative prior, and the corresponding term of (13.11) is

$$\mathcal{F}^{\ell}_{\text{cross}} = \mathbb{E}_{\mathbb{Q}_{\lambda}} D_{\mathrm{KL}}\big(\mathbb{Q}^{\ell}_{Z}(\cdot \mid Y_{\ell+1}, R_{\ell}, \mathfrak{N}_{\ell}, o, X_{*}) \| K^{\ell}_{\downarrow}(\cdot \mid Y_{\ell+1}, R_{\ell}, \mathfrak{N}_{\ell}, X_{*})\big). \tag{13.38}$$

The generative denominator in this cross-scale divergence reads the parent coordinates through $K^{\ell}_{\downarrow}$. The full tower objective can also couple parent coordinates to child structural marks through $\mathbb{P}^{\ell}_{R}$ and $\mathbb{P}^{\ell}_{N}$ in Hypothesis 13.4. A downward kernel that does not depend on the parent coordinates contributes no parent–child cross-derivative through this generative factor, and normalization of the downward kernel alone implies no coupling. When $K^{\ell}_{\downarrow}$ depends on the parent coordinates and the recognition parameterization transmits that dependence, the mixed derivative in a parent and a child parameter can be nonzero, so that optimization is coupled in both directions although the generative arrow is top-down; a universal claim of such coupling for a family requires a proved nonzero cross-derivative in that family. For a binary parent b and child y with product recognition factors $q_b(1) = u$ and $q_y(1) = v$, the mixed derivative of $\mathbb{E}_{q_b q_y}[-\log K_{\downarrow}(y \mid b)]$ in (u, v) is $-\log[K_{\downarrow}(1 \mid 1)K_{\downarrow}(0 \mid 0)/(K_{\downarrow}(0 \mid 1)K_{\downarrow}(1 \mid 0))]$, which vanishes exactly when the odds ratio of the kernel is one. [ESTABLISHED]

Proof. The generative denominator in (13.38) is $K^{\ell}_{\downarrow}$. If that kernel does not read the parent coordinates, its negative log density has no parent argument; the expectation and recognition parameterization may still depend on the parent. For the binary case the expectation is bilinear in (u, v), and its uv coefficient is the displayed alternating sum of $-\log K_{\downarrow}(y \mid b)$ over the four configurations. □

This is the static sense in which a meta-agent influences its sub-agents without a generative factor reading a posterior; in an adaptive tower the same kernel is the reversal kernel of Lemma 13.5.

13.7 Closing the tower

A tower in which each depth's prior is supplied by the depth above must terminate, and within one finite acyclic joint its top cannot be constituted by the agents it organizes (Proposition 13.23). A same-time reciprocal structure among latents is admissible only as a globally normalized joint (Section 4.2, Proposition 4.6): a flat unanchored energy is improper, an anchored or non-flat energy with its global normalizer is a valid undirected model, and an energy appended without its normalizer is not an evidence bound. A same-time deterministic parent $Z_{\ell+1} = \chi_{\ell}(Z_{\ell})$ and an

independently sampled parent that generates Z_ℓ are therefore not two free descriptions of one joint: one uses the hierarchy above, a deterministic pushforward, or a normalized undirected compatibility factor, and reciprocal same-time influence requires the last with its partition function. Supplement G works out the anchored Gaussian star and the convergence of its coordinate ascent.

13.7.1 A declared root

A truncated tower with a declared top prior $\mathbb{P}_L$ satisfies the hypotheses of Chapter 5 without amendment; its top is given, not constituted.

Proposition 13.23 (A declared root is unavoidable). Let a finite directed acyclic generative model be typed as in Chapter 4, so that every observed record is a child of at least one latent vertex. Then at least one latent has no parents, and its factor in the joint depends only on the parameters and the structural data. No finite acyclic construction of this type has every prior constituted by other latents. [ESTABLISHED]

Proof. A finite directed acyclic graph admits a topological order, whose first element has no incoming edge. That element is not a record, because every record has a latent parent, so it is a latent, and its factor is conditioned on no other vertex. The typing hypothesis is needed: in the graph $o \to y$ with observed root o, the only latent has a parent. □

Anchored, globally normalized cyclic models remain available, but their anchors and partition functions are declared; parameter learning fits only their parameters.

13.7.2 Three loci of participation

What a cycle in the model was reaching for is partly available without one, at three loci, each a sequence of evaluations of fixed normalized joints.

Remark 13.24 (The inference loop is block coordinate ascent). In a mean-field recognition family, the coordinate update of a factor reads the current factors of every variable that shares a factor of the joint with it [14, Sec. 10.1.1]. In every connected hierarchical model in which the children's conditionals depend on their parents, parent and child updates therefore read each other; at a stationary point of the schedule the factor that acts as a child's prior in its own update is $\exp \mathbb{E}[\log K_\downarrow(z_{\text{child}} \mid \cdot)]$ under the parent's current factor, which is itself formed from the factors of all its children. The effective prior is thus formed from the population's current beliefs through a latent inside the joint. This is a schedule of updates against one fixed, acyclic, normalized joint, not participation in time, and no kernel reads a posterior, so Hypothesis 4.3 holds. [STANDARD]

Proposition 13.25 (The aggregate recognition law is the optimal top prior). Let episodes $n = 1, \dots, N_{\text{e}}$ be conditionally independent given θ, each with its own top latent in a common space carrying the prior density p_θ relative to a fixed reference measure, let $\mathbb{Q}_n$ be the recognition law of the top latent of episode n, and let every other factor be free of θ. The θ-dependent part of the

summed bound is

$$\sum_{n=1}^{N_{\mathrm{e}}} \mathbb{E}_{\mathbb{Q}_n}[\log p_\theta] = N_{\mathrm{e}} \mathbb{E}_{\bar{\mathbb{Q}}}[\log p_\theta], \qquad \bar{\mathbb{Q}} = \frac{1}{N_{\mathrm{e}}} \sum_{n=1}^{N_{\mathrm{e}}} \mathbb{Q}_n. \tag{13.39}$$

If $\bar{\mathbb{Q}}$ has a density $\bar{q}$ with finite entropy relative to the same reference, then $\mathbb{E}_{\bar{\mathbb{Q}}}[\log p] = \mathbb{E}_{\bar{\mathbb{Q}}}[\log \bar{q}] - D_{\mathrm{KL}}(\bar{\mathbb{Q}} \| p)$ for every prior density p. Over an unrestricted family of prior densities the bound is maximized exactly at the aggregate recognition law $p = \bar{q}$, and within a restricted family its maximizers are the minimizers of $D_{\mathrm{KL}}(\bar{\mathbb{Q}} \| p)$ over the family, when they exist. Each evaluation uses one fixed normalized joint, and updating θ at fixed recognition is a coordinate step of the two-argument objective [81], so Hypothesis 4.3 holds. [ESTABLISHED]

Proof. The first identity is linearity of expectation in the recognition law. The second is $\log p = \log \bar{q} - \log(\bar{q}/p)$ integrated against $\bar{\mathbb{Q}}$, and Gibbs' inequality gives the maximizers. □

This is the variational counterpart of empirical Bayes [88], whose optimum is the aggregate-posterior prior of Hoffman and Johnson [56] and Tomczak and Welling [98]; only the parameter of the declared prior family is fitted, by a slow-block update in the typing of Section 8.2.

Definition 13.26 (Episode-indexed joint). Episodes are indexed by $n = 0, 1, \ldots$ with records o_n. In the parameter route each episode has its own normalized joint $\mathbb{P}_{\theta_n, V}$, and the parameter is updated by

$$\theta_{n+1} = \mathcal{U}_n\big(\theta_n, \mathbb{Q}^{(n)}_{V, o_n, X}\big), \tag{13.40}$$

where $\mathbb{Q}^{(n)}_{V, o_n, X}$ is the recognition coupling selected at episode n and $\mathcal{U}_n$ is a declared map; within each episode the joint is fixed, and its generative kernels read θ_n and never a recognition law of the same episode. In the latent route one joint spans episodes $0, \ldots, n_1$: episode n has a tower state $\mathbf{Y}^{(n)}$ with parent coordinates $Z^{(n)}_{\mathrm{par}} = \chi^{\mathrm{ep}}(\mathbf{Y}^{(n)})$ for a measurable read-out χ^{ep}, and

$$\begin{aligned} \mathbb{P}^{\mathrm{ep}}\big(d\mathbf{Y}^{(0:n_1)}, do_{0:n_1} \mid X\big) = \mathbb{P}^{(0)}(d\mathbf{Y}^{(0)} \mid X) \prod_{n=1}^{n_1} \mathbb{P}^{(n \leftarrow n-1)}\big(d\mathbf{Y}^{(n)} \mid Z^{(n-1)}_{\mathrm{par}}, X\big) \\ \times \prod_{n=0}^{n_1} \mathbb{P}^{\mathrm{obs}}\big(do_n \mid \mathbf{Y}^{(n)}, X\big), \end{aligned} \tag{13.41}$$

with a declared root in $\mathbb{P}^{(0)}$; the parent latent of episode $n-1$ parameterizes the conditional law of the tower of episode n. The observation kernel is any Markov kernel and need not have a density: the tower state may carry a latent copy $\widetilde{o}_n$ of the record, observed through $\mathbb{P}^{\mathrm{obs}}(do_n \mid \mathbf{Y}^{(n)}, X) = \delta_{\widetilde{o}_n}(do_n)$, in which case the free energy of the joint is its extended form (13.1) and the decomposition (13.43) applies as displayed only for discrete records.

Proposition 13.27 (Delayed feedback in the episode-indexed joint). In the latent route, (13.41) is a normalized joint whose kernels read structural data and earlier latents only, so Hypothesis 4.3 holds; its graph is acyclic with every cross-episode edge pointing forward in n, so Proposition 13.23 still requires a declared root in episode 0. The prior of the tower at episode $n+1$ given the records

so far is the filtered parent law pushed through the transition,

$$\mathbb{P}^{\mathrm{ep}}\big(\mathbf{Y}^{(n+1)} \in \cdot \mid o_{0:n}\big) = \int \mathbb{P}^{(n+1\leftarrow n)}(\cdot \mid z, X)\mathbb{P}^{\mathrm{ep}}\big(Z^{(n)}_{\mathrm{par}} \in dz \mid o_{0:n}\big), \tag{13.42}$$

so the children's prior at episode $n+1$ is set by the parent inferred from the population at episode n. For every recognition law $\mathbb{Q}$ on $\mathbf{Y}^{(0:n_1)}$, on the finite-density domain with observation densities p^{obs},

$$\begin{aligned}\mathcal{F}^{\mathrm{ep}}[\mathbb{Q}] = &\sum_{n=0}^{n_1} \mathbb{E}_{\mathbb{Q}}\big[-\log p^{\mathrm{obs}}(o_n \mid \mathbf{Y}^{(n)}, X)\big] + D_{\mathrm{KL}}\big(\mathbb{Q}^{(0)} \| \mathbb{P}^{(0)}\big) \\ &+ \sum_{n=1}^{n_1} \mathbb{E}_{\mathbb{Q}} D_{\mathrm{KL}}\big(\mathbb{Q}(\cdot \mid \mathbf{Y}^{(0:n-1)}) \| \mathbb{P}^{(n\leftarrow n-1)}(\cdot \mid Z^{(n-1)}_{\mathrm{par}}, X)\big),\end{aligned} \tag{13.43}$$

where $\mathbb{Q}^{(0)}$ is the law of $\mathbf{Y}^{(0)}$ under $\mathbb{Q}$ and $\mathbb{Q}(\cdot \mid \mathbf{Y}^{(0:n-1)})$ the conditional law of $\mathbf{Y}^{(n)}$. In the parameter route, if $\mathcal{U}_n$ maximizes the bound of episode n over the parameter at fixed $\mathbb{Q}^{(n)}_{V,o_n,X}$, the bound of episode n at θ_{n+1} is at least its value at θ_n. [ESTABLISHED]

Proof. Normalization is composition of Markov kernels, as in Proposition 4.4, and acyclicity follows because every kernel reads only earlier episodes. Given $Z^{(n)}_{\mathrm{par}}$, the tower of episode $n+1$ is independent of $(\mathbf{Y}^{(0:n)}, o_{0:n})$, because its kernel reads only $Z^{(n)}_{\mathrm{par}}$ and the records up to n are generated before it; conditioning on $o_{0:n}$ and integrating out $Z^{(n)}_{\mathrm{par}}$ gives (13.42). The decomposition is the chain rule Lemma 3.12 applied along n to the Radon–Nikodym ratio of $\mathbb{Q}$ against the joint. The last statement is the definition of a maximizer. □

With latent links, Definition 9.31 puts a Gibbs prior on the generator histories across episodes, and Proposition 9.32 shows that it is a latent route of Definition 13.26 with the generators in the parent read-out, to which Proposition 13.27 applies.

Proposition 13.28 (One joint for a declared parameter route). Let the episode joints of the parameter route form a Markov kernel $\theta \mapsto \mathbb{P}_{\theta,V}(dy, do \mid X)$ on the parameter space, with evidence densities $p_{\theta,V}(o \mid X)$ relative to one reference ν on O and posterior slices $\mathbf{\Pi}_{\theta,V,o,X}$ jointly measurable in (θ, o); let the coupling $\mathbb{Q}^{\theta}_{V,o,X}$ selected by Definition 3.15 for the joint at θ, so that $\mathbb{Q}^{(n)}_{V,o_n,X} = \mathbb{Q}^{\theta_n}_{V,o_n,X}$, be jointly measurable in (θ, o); and let each update $\mathcal{U}_n$ of (13.40) be measurable, so that $\mathcal{U}^{\mathrm{sel}}_n(\theta, o) := \mathcal{U}_n(\theta, \mathbb{Q}^{\theta}_{V,o,X})$ is measurable and the route reads $\theta_{n+1} = \mathcal{U}^{\mathrm{sel}}_n(\theta_n, o_n)$ from a declared θ_0. Write $\theta_n(o_{0:n-1})$ for the parameter the route reaches after the records $o_{0:n-1}$, and let each realized record be admitted at that parameter. The lifted joint on the tower states $\mathbf{Y}^{(n)} = (\theta_n, y_n, \widetilde{o}_n)$, $n = 0, \dots, n_1$, and the records,

$$\mathbb{P}^{\mathrm{lift}}\big(d\mathbf{Y}^{(0:n_1)}, do_{0:n_1} \mid X\big) = \delta_{\theta_0}(d\theta_0) \prod_{n=0}^{n_1} \mathbb{P}_{\theta_n,V}(dy_n, d\widetilde{o}_n \mid X)\delta_{\widetilde{o}_n}(do_n) \prod_{n=1}^{n_1} \delta_{\mathcal{U}^{\mathrm{sel}}_{n-1}(\theta_{n-1},\widetilde{o}_{n-1})}(d\theta_n), \tag{13.44}$$

has four properties. (i) It is a latent route of Definition 13.26 with parent coordinates $Z^{(n)}_{\mathrm{par}} = (\theta_n, \widetilde{o}_n)$, transition $\mathbb{P}^{(n\leftarrow n-1)}(d\mathbf{Y}^{(n)} \mid z, X) = \delta_{\mathcal{U}^{\mathrm{sel}}_{n-1}(z)}(d\theta_n)\mathbb{P}_{\theta_n,V}(dy_n, d\widetilde{o}_n \mid X)$, observation kernel $\delta_{\widetilde{o}_n}(do_n)$, and root $\delta_{\theta_0} \otimes \mathbb{P}_{\theta_0,V}$, so Proposition 13.27 applies to it. (ii) Given $o_{0:n-1}$, the conditional law of

$(y_n, \widetilde{o}_n)$ is the episode joint $\mathbb{P}_{\theta_n(o_{0:n-1}),V}$ of the parameter route, and the evidence density of the lifted joint relative to $\nu^{\otimes(n_1+1)}$ is the product of the episode evidences,

$$p^{\mathrm{lift}}(o_{0:n_1} \mid X) = \prod_{n=0}^{n_1} p_{\theta_n(o_{0:n-1}),V}(o_n \mid X). \tag{13.45}$$

(iii) Its posterior slice fixes $\widetilde{o}_n = o_n$ and $\theta_n = \theta_n(o_{0:n-1})$ and is the product $\bigotimes_n \mathbf{\Pi}_{\theta_n(o_{0:n-1}),V,o_n,X}$ on $y_{0:n_1}$, so the posterior law of y_n reads $o_{0:n}$ only. (iv) For every recognition law $\mathbb{Q}$ on the tower states, the free energy (13.1) of the lifted joint is

$$\mathcal{F}^{\mathrm{lift}}[\mathbb{Q}] = \sum_{n=0}^{n_1} \Big[-\log p_{\theta_n,V}(o_n \mid X) + D_{\mathrm{KL}}\big(\mathbb{Q}_n^y \| \mathbf{\Pi}_{\theta_n,V,o_n,X}\big) \Big] + \mathrm{TC}(\mathbb{Q}^y), \qquad \theta_n = \theta_n(o_{0:n-1}), \tag{13.46}$$

when $\mathbb{Q}$ fixes $\widetilde{o}_n = o_n$ and $\theta_n = \theta_n(o_{0:n-1})$ almost surely, and $+\infty$ otherwise, where $\mathbb{Q}^y$ is the law of $y_{0:n_1}$ under $\mathbb{Q}$, $\mathbb{Q}_n^y$ its marginal on y_n, each bracket is the extended free energy (5.16) of episode n at the parameter the route reaches, and TC is the total correlation (5.1) across episodes. At product laws, and in particular at the selected couplings $\bigotimes_n \mathbb{Q}^{\theta_n}_{V,o_n,X}$, the free energy is the sum of the episode bounds of the parameter route. [ESTABLISHED]

Proof. Each factor of (13.44) is a Markov kernel in the coordinates to its left: $\theta \mapsto \mathbb{P}_{\theta,V}$ by hypothesis, and the Dirac factors as kernels of measurable maps, $\mathcal{U}_{n-1}^{\mathrm{sel}}$ being the composition of $\mathcal{U}_{n-1}$ with the jointly measurable selection. Finite composition normalizes the product (Proposition 4.4), and reading the factors in the order of (13.41) gives (i); every kernel reads structural data and latent coordinates only, because the selection is evaluated at the latent copy $\widetilde{o}_{n-1}$ and not at a recognition law of the joint being scored. For (ii), integrate (13.44) against a bounded measurable $f(o_{0:n-1}, y_n, \widetilde{o}_n)$: the factors of episodes $n+1, \dots, n_1$ have unit mass, the Dirac factors identify $o_k = \widetilde{o}_k$ for $k < n$ and, by induction along the recursion, $\theta_n = \theta_n(o_{0:n-1})$, and Tonelli's theorem leaves the integral of $\int f(o_{0:n-1}, y, \widetilde{o}) \mathbb{P}_{\theta_n(o_{0:n-1}),V}(dy, d\widetilde{o} \mid X)$ against the law of $o_{0:n-1}$, which is the stated conditional law. Its record marginal is $p_{\theta_n(o_{0:n-1}),V}(o_n \mid X)\nu(do_n)$, and multiplying the conditional densities along n gives (13.45); the product is measurable because the evidence density is jointly measurable and $o_{0:n-1} \mapsto \theta_n(o_{0:n-1})$ is a composition of measurable maps, and it is finite and positive because each factor is. For (iii), substitute the disintegration $\mathbb{P}_{\theta,V}(dy, d\widetilde{o} \mid X) = \mathbf{\Pi}_{\theta,V,\widetilde{o},X}(dy) p_{\theta,V}(\widetilde{o} \mid X)\nu(d\widetilde{o})$ into each factor and integrate out the Dirac coordinates: for bounded measurable f of the tower states and g of the records,

$$\begin{aligned} \mathbb{E}_{\mathbb{P}^{\mathrm{lift}}}[fg] = \int g(o) \Big[\int f\big((\theta_n(o_{0:n-1}), y_n, o_n)_{n \le n_1}\big) \bigotimes_{n=0}^{n_1} \mathbf{\Pi}_{\theta_n(o_{0:n-1}),V,o_n,X}(dy_n) \Big] \\ \times\, p^{\mathrm{lift}}(o \mid X) \nu^{\otimes(n_1+1)}(do), \end{aligned} \tag{13.47}$$

and the bracket, measurable in o by the joint measurability of the slices, is a version of the conditional expectation of f given the records. For (iv), the posterior is carried by the set on which $\widetilde{o}_n = o_n$ and $\theta_n = \theta_n(o_{0:n-1})$ for every n; a $\mathbb{Q}$ that charges its complement is not absolutely continuous with respect to the posterior, and its relative entropy is $+\infty$. Otherwise disintegrate both laws over $y_{0:n_1}$

with Lemma 3.12(a): the conditional laws of the remaining coordinates given $y_{0:n_1}$ are the same Dirac mass under both, so $D_{\mathrm{KL}}(\mathbb{Q}\|\mathbf{\Pi}^{\mathrm{lift}}) = D_{\mathrm{KL}}(\mathbb{Q}^y\|\bigotimes_n \mathbf{\Pi}_{\theta_n,V,o_n,X})$, and the product-reference identity (5.2) splits the right side into $\mathrm{TC}(\mathbb{Q}^y) + \sum_n D_{\mathrm{KL}}(\mathbb{Q}^y_n\|\mathbf{\Pi}_{\theta_n,V,o_n,X})$. Adding $-\log p^{\mathrm{lift}}(o \mid X)$ from (13.45) gives (13.46), and the total correlation of a product law is zero. □

The route's bookkeeping with the parameter read from the law being scored is the composite that Hypothesis 4.3 excludes; the lifted joint evaluates the selection at the latent copy of the record instead, and its evidence is the sequential empirical-Bayes evidence of the route.

Open Problem 13.29 (Accounting across episodes). In the parameter route the bounds of different episodes refer to different joints and records. Within an episode a maximizing update is monotone (Proposition 13.27), and for an update that reads a coupling selected by a declared rule the lifted joint of Proposition 13.28 is one joint whose evidence is the product of the episode evidences and whose free energy is the sum of the episode bounds plus the cross-episode total correlation; an update that reads the recognition law being scored is excluded by Hypothesis 4.3. The lifted joint informs the parent of episode $n+1$ by the record of episode n through its latent copy. What is unproved is whether the label coordinates alone can carry the feedback. The problem is posed for label-sufficient latent routes of Definition 13.26: the episode model, with a finite label coordinate y^{lab} whose prior is the parameter, its private latents, and its record kernel, is that of the parameter route with the posterior slice as its recognition coupling and the aggregate update of constant step, $\theta_{n+1} = (1-\epsilon_{\mathrm{e}})\theta_n + \epsilon_{\mathrm{e}}\mathbf{\Pi}^{\mathrm{lab}}_{\theta_n,V,o_n,X}$ with $\mathbf{\Pi}^{\mathrm{lab}}$ the label marginal of the posterior slice; the tower state of episode n consists of the episode model's latents and an auxiliary preference coordinate; each record is emitted from its own episode's latents by the record kernel; and the read-out χ^{ep} reads the label and the preference coordinate only. The records either have one value whose density $p_V(o \mid y^{\mathrm{lab}}, X)$ given the label does not depend on the label, while the expected squared total-variation displacement of the label posterior from the prior, under the evidence at that prior, is bounded below uniformly over priors whose smallest weight is at least half that of θ_0; or their likelihood vectors $(p_V(o \mid y^{\mathrm{lab}}, X))_{y^{\mathrm{lab}}}$ span a cone on which no nonzero polynomial vanishes. For such routes, with the per-episode discrepancy the total variation between the route's filtered label law $\mathbb{P}^{\mathrm{ep}}(y^{\mathrm{lab}}_{n+1} \in \cdot \mid o_{0:n})$ and $(1-\epsilon_{\mathrm{e}})\mathbb{P}^{\mathrm{ep}}(y^{\mathrm{lab}}_n \in \cdot \mid o_{0:n-1}) + \epsilon_{\mathrm{e}}\mathbb{P}^{\mathrm{ep}}(y^{\mathrm{lab}}_n \in \cdot \mid o_{0:n})$, the aggregate update applied to the route's own filtered law and label posterior, the problem is settled by a route, defined for every step, whose discrepancy is $o(\epsilon_{\mathrm{e}})$ uniformly over record histories and over episodes n up to any fixed multiple of $1/\epsilon_{\mathrm{e}}$, or by a lower bound of order ϵ_{e} on the supremum of the discrepancy over such histories and episodes for every such route. Whether a declared update intertwines the fine and parent evolutions is part of Open Problem 12.40; θ_n belongs to the slow block of Definition 8.2.

13.7.3 The remaining hierarchy question

The prepared parent of Section 8.3, the memory cost of partial refresh in Section 12.8.2, and the successive formation of Theorem 14.17 are constructions at stated hypotheses, not consequences of arbitrary descent.

Open Problem 13.30 (Persistent multiscale hierarchy). Starting from c_* with a finite multigraph datum $\mathfrak{N}_0$, construct a normalized partition law and parent–child kernels such that descent of (13.1) produces persistent nested blocks, the pushforward rules (13.25) to (13.30) and the link lift (13.31) compose across depths, the parent family has a controlled closure residual in the sense of Proposition 12.24, and non-flat holonomy is kept or stabilized without imposing flatness. What holds: every candidate hierarchy has the exact accounting of Theorem 13.7; at one depth the fast–slow typing of Definition 8.2 and its conditional slow-manifold limit, Open Problem 8.7, are available; Theorem 14.17 forms nested blocks under a supplied energy hierarchy whose strengths are tied to the temperature; and Proposition 14.22 forms one level. What is unproved is that descent produces a second level that depends on the first. The problem is posed in the class of models at c_* whose structural marks are drawn from a finite partition catalog with a partition factor inside the generative and the recognition law (Definitions 13.11 and 13.12), whose recognition coordinates descend (13.1) by a reversible diffusion at temperature T with rate-one redraws of the discrete marks from their conditional laws, as in Definition 14.21, and whose energies, precisions, prior ratios, hyperparameters, and mobilities do not depend on T, as $\mathrm{T} \to 0$. It is settled by three results together, or by a proof that the first of them cannot hold in this class. The first is a lineage of at least two levels, formed with probability bounded below from every initial law with a density and a finite second moment, whose second level meets two conditions. It depends on the first level: for a declared handoff time, a declared coalescence event whose block is a union of at least two first-level blocks and whose windows are measured from the handoff, one declared statistic of the handoff state (the probability of the event, of its occurrence by declared times, or of a declared block), and a handoff class fixed before the comparison that contains every first-level handoff and, with probability bounded below, the handoffs of histories without first-level formation, the event has probability bounded below after first-level formation, does not have probability tending to one uniformly over the handoff class at the declared windows, and the statistic exceeds its value after a counterfactual handoff in the class by a margin bounded below. It is a coherence episode rather than restored equilibrium or coincidental coherence: a restoring interaction, evaluated on the law of the coordinates given the realized marks or on their marginal over the marks and not on a conditional law held fixed by hand, holds the second-level block's coherence at a declared resolution, with the restoring rate times the squared resolution divided by T tending to infinity, on horizons short compared with the stationary time of entrance into the event from a set of stationary mass bounded below, or with a probability exceeding its stationary value by a margin bounded below. The second result is a nondegenerate intrinsic partition selector (Open Problem 10.20), and the third is a separated slow manifold on which depth-$(\ell+1)$ variables feed back consistently into depth-ℓ kernels. The negative resolution is a proof that in this class no second level produced by descent, apart from a supplied-scale sequence whose event occurs with probability tending to one uniformly over the handoff class at the declared windows, both meets the restoring and horizon conditions and exceeds its counterfactual statistic by such a margin.

Summary for physicists

The multiscale free energy is that of one generative and one recognition law over the whole tower, not a sum of energies chosen scale by scale. Bayes reversal of a bottom-up structural channel conditions every depth on the whole next-depth state, graph datum included. Partitions enter as a normalized factor, normalized branch by branch or globally, and coarse links are the dressed links of all copies, so internal loops keep their holonomy. A parent can couple to child coordinates through a downward kernel that reads it and to child structural marks through the partition and graph factors. A finite acyclic tower has a declared root, and the agents participate in their own top through coordinate ascent, a learned aggregate prior, and delayed feedback. A sequence of episodes whose parameter update reads a coupling selected by a declared rule is itself one joint, with the product of the episode evidences as its evidence (Proposition 13.28).

Part III

Formation and Interpretation

Chapter 14

Formation Witnesses

Exact parents are normalized pushforwards of the fine law (Theorem 10.7) but in general not autonomous reduced dynamics (Theorem 12.41). This chapter asks whether, in one normalized experiment at c_*, coherent components can form in succession while every parent is activated from the complete preceding recognition law, retains every loop coordinate and source assignment, and learns its own copy of a model parameter from its own law. Two witnesses answer yes under explicit restrictions. Theorem 14.17 treats a triangle chain with a supplied hierarchy of interaction energies: at each finite depth L the coherent components pass through $3 \cdot 2^{L-1} \to 2^{L-1} \to \cdots \to 1$ with probability tending to one, while the discarded conditional losses, parameter covectors, assignment histories, and continuous paths obey separate vanishing bounds. Proposition 14.22 treats a homogeneous four-cycle without supplied hierarchy, in which the source assignment decides which pairs form, with probability bounded below independently of the small parameter. Proposition 14.24 bounds what that mechanism can do on any finite graph: in the gated family of Definition 14.23, the stationary coherent-block partition is, up to the error of eq. (14.59), the partition into the components of a minimum edge cover, and each merge beyond such a cover costs a factor of the temperature. The proposition and the conjecture of the interpretation chapter (Proposition 15.4 and theorem 15.5) are posed on the four-cycle family of the second witness. Both witnesses are approximation statements under a stated preparation and regime, weaker than the exact intertwining that an autonomous parent dynamics would require (Open Problem 12.40). Both realize the general construction for the translation group $(\mathbb{R}, +)$ on scalar sample fibers: agents are local sections at c_* (Definition 2.8), one common frame per agent lets one independent link translate both channels (Definition 2.11), and the free energy is that of Definition 5.6. In generator coordinates the link prior of the four-cycle experiment and of the gated family is the Gaussian base law of the generator prior of Chapter 9 with no loop action, while the configuration prior of the triangle chain places all of its precision along loop coordinates in a loop defect (Proposition 9.13).

Letters of the notation table that would collide with chapter-local objects are replaced by the symbols of table 14.1; upright $\mathrm{J}, \mathrm{Q}, \mathrm{A}, \mathrm{T}, \mathrm{c}_L, \mathrm{C}_L$ are local scalars, the configuration is bold, and $\ell' < \ell$ whenever two levels occur.

Scope. For the finite-depth family this chapter declines the following claims: that attention or learning drives formation, since coherence is linear relaxation of the prepared mean, the source record saturates at the coherence scale (Remark 14.18), and learning is calibration dominated up to $\epsilon^{2\mathrm{Q}_1+20}$; that per-agent models are slow, since means and links have unit mobility and $\mathfrak{d}_i$

Table 14.1: Chapter-local symbols of the formation families, shared with the proof appendix.

Symbol	Meaning
L, $N_{\triangle} = 2^{L-1}$, $N = 3N_{\triangle}$, ℓ, ℓ'	depth, number of triangles, number of agents, level indices
$\mathbf{x} = (\bar{k}, \bar{m}, \omega)$, $\mathbf{X}_t$	configuration of law presentations (belief means, model means, link translations) and its process
$\mathfrak{d}_i = \bar{k}_i - \bar{m}_i$, $\mathfrak{D}$, r_e	belief–model discrepancy, its row matrix, edge residual
$\mathfrak{h}_q$, h_q, $O_{\triangle}$	oriented triangle holonomy, normalized loop coordinate, contrast rows
$\mathcal{U}, \mathcal{H}, \mathcal{K}$; $W_\ell, \mathcal{U}_\ell, M_\ell, P_\ell$	normal, loop, and gauge spaces; retained spaces and projections
H, H_ℓ, Q_ℓ	stiffness, its Schur complement on M_ℓ, orthonormal flag columns
J_ℓ, Q_ℓ, A, T, t_*	energy strengths, their exponents, prepared amplitude, temperature, horizon
σ^2_{ev}, R_{ev}, ν_i^*	evaluator variance, precision of the evaluator restriction cost, optimal evaluator coupling
ϑ, $\Omega_0(\vartheta)$, M, $\mathsf{m} = \mathrm{TM}$	learned scalar, prior precision, complete Fisher metric and its scaled form
κ, $z_\star$, $o_{\mathcal{K}}$, o_{cal}, o	root observation precision, root record, root and calibration observations, observed records
$S \in \mathcal{J}$, $\mathcal{N}_i^{\triangle}$, i_0, $\pi_{ij}(\vartheta)$, $\pi_S(\vartheta)$	complete source assignment, internal source palette, distinguished receiver, source priors
χ, ϕ_S, δ, B_0, B_*	source function, source potential, record coefficient, and its two bounds
β_S, $\beta^{\ell}_{SS'}$, $\bar{\beta}_{jk}$	complete posterior source row, inherited proposal row, averaged fast row
$q^{\star}_{\vartheta}$, $q^{\star,\ell}_{\vartheta}$, $\mathbf{\Pi}_{\vartheta,\mathsf{o}}$, q^{ℓ}_t	restricted optimizer and its level-ℓ push, physical posterior, recognition density
$C_{\ell \leftarrow \ell'}$, $\mathcal{D}^{\ell,\mu}_e$, $\Delta_{\ell':\ell}$	marked channel, inherited edge statistic, discarded conditional loss
$\mathcal{F}_{\mathrm{T}}$, C_{T}, E_{ϑ}, $U_{\vartheta,S}$, $V_{\ell,S}$, $\psi_{\ell,S}$, g_ℓ	scaled restricted free energy, its offset, quadratic and sheet energies, parent potential and tilt, parameter covector
$\mathscr{L}^{\ell}_{\vartheta}$	inference generator at level ℓ
t_ℓ, I_ℓ, t^{act}_ℓ, k_L, η_L, ς_ϵ, r_ϵ	window start, window, activation time, entrance constant, tolerance, $1 + \log(1/\epsilon)$, parameter remainder
c_L, C_L, j_m	positive constants depending on L and the fixed data; proposal marks
B_{sr}, j_ℓ, $\mathfrak{Y}_\ell$, $\mathfrak{y}_\ell$, $\mathfrak{x}_\ell$, ϱ_s, Q_{T}, Q_{Δ}	flag blocks of the residual rows, least eigenvalue of a diagonal Gram block, forcing map and core coefficient of an interface, current-layer transient, strength ratio, exponents of the thermal ratio and of a loss target
(V, E), $\mathfrak{n}$, F, $\mathrm{cc}(F)$, $\mathrm{att}(S)$	graph of the gated family, its matching number, an edge set, the number of components of (V, F), the edge set attended under S
$\mathbf{a}_e$, $\mathbf{a}_F$, W_F, s_F	residual pair map of an edge, its stack over F, the kernel of the stack, the least positive singular value of the stack
H_S, k_S, h_S, Ω, Z_S, z_S, V_π, $V_{\min}$	sheet stiffness, its rank and least positive eigenvalue, reference precision, sheet weight and its constant, source-marginal and gated energies
$\mathrm{Coh}_{\mathrm{a}}(\mathbf{x})$, $\mathfrak{P}_{\mathrm{a}}(\mathbf{x})$, a	coherent edges and coherent blocks of a configuration; resolution

is a level-one row; that loop holonomy is of order one or nonabelian, since the group is $(\mathbb{R},+)$ and the retained loops are thermally negligible; that there is an inter-agent model-channel record, since only the belief channel is observed; that parents are coarse agents with their own belief and model fibers, since a parent removes only normal coordinates; that source learning creates the supplied contrasts; that the doubling recursion $\mathrm{Q}_\ell = 2\mathrm{Q}_{\ell+1} + 20$ is required, as distinct from the doubly exponential separation of inner base $4/3$ that Proposition 14.15 shows the Gaussian core to require; that the continuous path laws converge in total variation or are exactly lumpable; and that the construction saves computation or uses depth-uniform resources. For the four-cycle it declines generic formation, formation of a matching from every initial assignment, a unique block chosen by the invariant full law, all-slow parents, exact closure under arbitrary source switching, permanent membership, repeated coalescence, and a physical posterior conditioned on the episode. The corresponding constructions are Open Problems 14.16, 14.20 and 14.25. [NOT-CLAIMED]

14.1 The pointwise carrier and its restoring flag

Definition 14.1 (Triangle-chain carrier). Fix an integer $L \geq 2$, set $N_\triangle = 2^{L-1}$, and arrange $N_\triangle$ disjoint triangles in a chain, with one bridge between each consecutive pair. Each triangle is cyclically oriented and every bridge has one fixed orientation, so the graph has $N = 3N_\triangle$ vertices (the agents) and $4N_\triangle - 1$ oriented edge copies. Bridge number j, between triangles j and $j+1$, has level $\ell(j) = 2 + v_2(j)$, where $v_2(j)$ is the exponent of two dividing j; triangle edges have level one. Energy strengths $\mathrm{J}_1 \geq \cdots \geq \mathrm{J}_L > 0$ are attached to the levels. The group $(\mathbb{R},+)$ acts by translation on two scalar sample fibers at c_*; fix $0 < \sigma_{\mathrm{ev}}^2 < 1$. A configuration and its law presentations are

$$\begin{aligned} \mathbf{x} = (\bar{k}, \bar{m}, \omega) \in \mathbb{R}^{2N+4N_\triangle-1} = \mathbb{R}^{10N_\triangle-1}, \qquad & \Theta_e(k) = k + \omega_e, \\ q_i^b(\cdot \mid \mathbf{x}) = \mathcal{N}(\bar{k}_i, 1), \qquad & q_i^m(\cdot \mid \mathbf{x}) = \mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2), \\ \mathfrak{d}_i(\mathbf{x}) = \bar{k}_i - \bar{m}_i, \qquad & r_e(\mathbf{x}) = \bar{k}_i - \bar{k}_j - \omega_e \quad (e : j \to i), \end{aligned} \tag{14.1}$$

with $\omega_{\bar{e}} = -\omega_e$. A passive frame change g_i sends $\bar{k}_i, \bar{m}_i, \omega_e$ to $\bar{k}_i + g_i$, $\bar{m}_i + g_i$, $\omega_e + g_i - g_j$, and re-expresses reference centers and observation records in the same way. For triangle q put $\mathfrak{h}_q = \sum_{e \in q} \omega_e$ and $h_q = \mathfrak{h}_q/\sqrt{3}$, and let $O_\triangle = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 1/\sqrt{6} & 1/\sqrt{6} & -2/\sqrt{6} \end{pmatrix}$. Let B_1 stack the N rows $\mathfrak{d}_i$ and the $2N_\triangle$ internal contrast rows $O_\triangle r_q$, where r_q is the vector of the three residuals of triangle q; for $\ell \geq 2$ let B_ℓ stack the bridge residuals of level ℓ, and let B stack all of them. With the oriented incidence operator $(\partial t)_e = t_i - t_j$,

$$\begin{aligned} \mathcal{K} &= \{(t, t, \partial t) : t \in \mathbb{R}^N\}, \quad \mathcal{H} = \operatorname{span}\{\text{normalized loop covectors}\}, \\ \mathcal{U} &= \operatorname{range} B^\top, \qquad\qquad H = B^\top \operatorname{diag}(\mathrm{J}_\ell I) B + \mathrm{J}_1 P_{\mathcal{H}}. \end{aligned} \tag{14.2}$$

The retained spaces and their Euclidean orthogonal projections are

$$W_\ell = \ker(B_1, \ldots, B_\ell), \qquad \mathcal{U}_\ell = W_\ell \cap \mathcal{U}, \qquad M_\ell = \mathcal{U}_\ell \oplus \mathcal{H}, \qquad P_\ell : \mathbb{R}^{10N_\triangle-1} \to W_\ell, \qquad P_0 = I. \tag{14.3}$$

Lemma 14.2 (Carrier algebra). In Definition 14.1, $\mathfrak{d}_i$ and r_e are invariant under passive frame changes, and every triangle satisfies

$$|r_q|^2 = |O_\triangle r_q|^2 + h_q^2, \tag{14.4}$$

so the loop energy belongs to the original sum of edge mismatches. The rows of B are independent, $\operatorname{rank} B = 6N_\triangle - 1$, the normalized loop covectors are orthonormal and orthogonal to every row of B, and $\mathbb{R}^{10N_\triangle - 1} = \mathcal{U} \oplus \mathcal{H} \oplus \mathcal{K}$ is an orthogonal direct sum. The stiffness has kernel $\mathcal{K}$ and $\mathbf{x}^\top H \mathbf{x} = \mathrm{J}_1 \sum_i \mathfrak{d}_i(\mathbf{x})^2 + \sum_e \mathrm{J}_{\ell(e)} r_e(\mathbf{x})^2$. Finally $P_\ell P_{\ell'} = P_\ell$ for $\ell' < \ell$, every W_ℓ contains $\mathcal{H} \oplus \mathcal{K}$, and

$$\dim W_\ell = 4N_\triangle - 1 + \frac{N_\triangle}{2^{\ell-1}} \qquad (1 \le \ell \le L). \tag{14.5}$$

[ESTABLISHED]

Proof. A passive change adds $g_i - g_i$ to $\mathfrak{d}_i$ and $g_i - g_j - (g_i - g_j)$ to r_e. Around an oriented triangle the vertex means telescope, so $\sum_{e \in q} r_e = -\mathfrak{h}_q$; the rows of $O_\triangle$ are orthonormal and orthogonal to $(1,1,1)/\sqrt{3}$, which gives eq. (14.4). In a linear relation among the rows of B, the link columns occur only in contrast and bridge rows, with the independent patterns $-O_\triangle$ on the triangle edges and -1 on each bridge, so every contrast and bridge coefficient vanishes; the model-mean columns then force the discrepancy coefficients to vanish. A normalized loop covector has link entries $1/\sqrt{3}$ on one triangle and nothing else, so it is orthogonal to the discrepancy rows, to the contrast rows because $O_\triangle(1,1,1)^\top = 0$, and to the bridges. Every row of B and every loop covector annihilates $(t, t, \partial t)$, and $(6N_\triangle - 1) + N_\triangle + N = 10N_\triangle - 1$. The quadratic identity is the sum of $\mathrm{J}_\ell |B_\ell \mathbf{x}|^2$ and $\mathrm{J}_1 |P_{\mathcal{H}} \mathbf{x}|^2$ combined with eq. (14.4); it vanishes exactly when $B\mathbf{x} = 0$ and $P_{\mathcal{H}} \mathbf{x} = 0$, that is on $\mathcal{K}$. The spaces W_ℓ decrease, which gives $P_\ell P_{\ell'} = P_\ell$, and $\mathcal{H} \oplus \mathcal{K}$ lies in each of them. There are $N_\triangle / 2^{\ell'' - 1}$ bridges of level $\ell'' \ge 2$, so $\operatorname{rank}(B_1, \dots, B_\ell) = 3N_\triangle + 2N_\triangle + N_\triangle(1 - 2^{1-\ell})$, which gives eq. (14.5). □

The restoring estimates of the proof appendix rest on the bounds eq. (C.2) of H by weighted energies $\sum_\ell \mathrm{J}_\ell |w_\ell|^2$ along orthonormal flag columns Q_ℓ spanning $\mathcal{U}_{\ell-1} \cap \mathcal{U}_\ell^\perp$.

14.2 One normalized experiment with a nonlinear source record

Definition 14.3 (Normalized formation experiment). At each vertex the physical model sample is $m_i \sim \mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2)$ and the evaluator sends m_i to $\mathcal{N}(m_i, 1 - \sigma_{\mathrm{ev}}^2)$ on the belief sample k_i. Among couplings with the prescribed marginals $\mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2)$ and $\mathcal{N}(\bar{k}_i, 1)$, the recognition coupling at a fixed configuration is

$$m_i = \bar{m}_i + \xi_i, \qquad k_i = \bar{k}_i + \xi_i + \xi_i', \qquad \xi_i \sim \mathcal{N}(0, \sigma_{\mathrm{ev}}^2), \quad \xi_i' \sim \mathcal{N}(0, 1 - \sigma_{\mathrm{ev}}^2), \tag{14.6}$$

with independent noises; write ν_i^* for its law and $R_{\mathrm{ev}} = \mathfrak{D}^\top \mathfrak{D} / (1 - \sigma_{\mathrm{ev}}^2)$, where $\mathfrak{D}\mathbf{x} = \bar{k} - \bar{m}$. Let $\vartheta \in [1, 21/10]$ be the learned scalar and $0 < \mathrm{T} \le \mathrm{J}_1(1 - \sigma_{\mathrm{ev}}^2)/2$. The configuration prior and its

Lebesgue normalizer are

$$\begin{aligned}\Omega_0(\vartheta) &= \vartheta H + P_{\mathcal{K}} - \mathrm{T}R_{\mathrm{ev}},\\ \mathbb{P}^0_\vartheta(d\mathbf{x}) &= Z_0(\vartheta)^{-1}\exp\left[-\frac{\mathbf{x}^\top\Omega_0(\vartheta)\mathbf{x}}{2\mathrm{T}}\right]d\mathbf{x},\\ Z_0(\vartheta) &= (2\pi\mathrm{T})^{(10N_\triangle-1)/2}\det\Omega_0(\vartheta)^{-1/2}.\end{aligned} \tag{14.7}$$

Fix $0<\kappa\le 1$ and a nonzero $z_\star\in\mathcal{K}$. The observation kernels are

$$o_{\mathcal{K}}\mid\mathbf{x}\sim\mathcal{N}_{\mathcal{K}}\big(P_{\mathcal{K}}\mathbf{x},(\mathrm{T}/\kappa)I_{\mathcal{K}}\big),\qquad o_{\mathrm{cal}}\mid\vartheta\sim\mathcal{N}(\vartheta,\mathrm{T}), \tag{14.8}$$

with the records $o_{\mathcal{K}}=z_\star$ and $o_{\mathrm{cal}}=2$; the first is a density on the N-dimensional carrier $\mathcal{K}$. Each receiver i has the two candidate sources $\mathcal{N}_i^{\triangle}$ inside its own triangle, and the complete assignment is $S=(S_i)_i\in\mathcal{J}=\prod_i\mathcal{N}_i^{\triangle}$, with $|\mathcal{J}|=2^N$; every bridge is an added reversal pair of copies that keeps its link and interaction energy but lies outside the source palettes. A distinguished receiver i_0 has sources labeled $+$ and $-$, and the independent source priors are

$$\begin{aligned}\pi_{i_0,+}(\vartheta)&=\mathrm{sig}\big(4(\vartheta-3/2)\big),\quad \pi_{i_0,-}(\vartheta)=1-\pi_{i_0,+}(\vartheta),\\ \pi_{ij}(\vartheta)&=\tfrac12\quad(i\neq i_0),\qquad\qquad \pi_S(\vartheta)=\prod_i\pi_{iS_i}(\vartheta),\end{aligned} \tag{14.9}$$

where $\mathrm{sig}(u)=(1+e^{-u})^{-1}$. Set $B_0=10^{-4}$, $\delta=B_0/N$, and $\chi(r)=1-e^{-r^2/2}$. At each receiver a binary record E_i, conditionally independent given $(\mathbf{x},S)$, has

$$\begin{aligned}\Pr(E_i=1\mid\mathbf{x},S_i=j)&=q_0\exp\left[-\delta\chi\left(\sqrt{\mathrm{J}_1/\mathrm{T}}r_{ij}(\mathbf{x})\right)\right],\qquad q_0=\tfrac12,\\ \phi_S(\mathbf{x})&=\delta\sum_i\chi\left(\sqrt{\mathrm{J}_1/\mathrm{T}}r_{iS_i}(\mathbf{x})\right),\qquad 0\le\phi_S\le B_0,\end{aligned} \tag{14.10}$$

where r_{ij} is the residual of the edge from source j to receiver i, and the observed record is $E_i=1$ for every i. Since $\mathrm{J}_1r_{ij}^2/(2\mathrm{T})=(\mathrm{J}_1/\mathrm{T})D_{\mathrm{KL}}(\mathcal{N}(\bar k_i,1)\|\mathcal{N}(\bar k_j+\omega_{ij},1))$, the record is a bounded observation of the standardized edge mismatch. The complete joint is

$$\begin{aligned}\mathbb{P}_\vartheta=\mathbb{P}^0_\vartheta(d\mathbf{x})L_{\mathcal{K}}(do_{\mathcal{K}}\mid\mathbf{x})L_{\mathrm{cal}}(do_{\mathrm{cal}}\mid\vartheta)\\ \times\prod_i\left[\pi_i(dS_i\mid\vartheta)L_i(dE_i\mid\mathbf{x},S_i)\mathcal{N}(dm_i;\bar m_i,\sigma_{\mathrm{ev}}^2)\mathcal{N}(dk_i;m_i,1-\sigma_{\mathrm{ev}}^2)\right].\end{aligned} \tag{14.11}$$

Proposition 14.4 (Normalization and the evaluator optimum). In Definition 14.3, for every admissible coupling ν_i of the two prescribed marginals,

$$D_{\mathrm{KL}}\big(\nu_i\|\mathcal{N}(\bar m_i,\sigma_{\mathrm{ev}}^2)\,\mathrm{ev}_i\big)=D_{\mathrm{KL}}(\nu_i\|\nu_i^*)+\frac{\mathfrak{d}_i^2}{2(1-\sigma_{\mathrm{ev}}^2)}, \tag{14.12}$$

so ν_i^* is the optimal coupling and the total restriction cost is $\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2$. Moreover $\mathrm{T}R_{\mathrm{ev}}\preceq H/2$, hence $\Omega_0(\vartheta)\succeq(\vartheta/2)H+P_{\mathcal{K}}\succ 0$ and the prior is a normalized correlated Gaussian, and the joint

eq. (14.11) has total mass one, contains no kernel that takes a recognition law as input, and gives the selected observation a strictly positive density. [ESTABLISHED]

The term $-\mathrm{T}R_{\mathrm{ev}}$ of the prior cancels the restriction cost $\frac{\mathrm{T}}{2}\mathbf{x}^{\top}R_{\mathrm{ev}}\mathbf{x}$ that eq. (14.12) adds, so the restricted optimizer below has a temperature-independent quadratic energy.

14.3 Physical posterior, restricted optimizer, and recognition

Definition 14.5 (Three laws of the formation experiment). Write $\mathsf{o} = (z_\star, 2, \mathbf{1})$ for the observed records. The recognition family consists of the laws with a smooth positive density $q(\mathbf{x}, S)$ relative to Lebesgue measure times counting measure and conditional evaluator coupling $\prod_i \nu_i^*$; assignments and configurations may be correlated, and the conditional assignment law need not be the source posterior. With

$$\begin{aligned} \widehat{K}_\vartheta &= \vartheta H + (1+\kappa)P_{\mathcal{K}}, & \mathrm{E}_\vartheta(\mathbf{x}) &= \tfrac{1}{2}\mathbf{x}^{\top}\widehat{K}_\vartheta\mathbf{x} - \kappa z_\star^{\top}\mathbf{x}, \\ \mathrm{E}_\vartheta^{\mathrm{phys}}(\mathbf{x}) &= \mathrm{E}_\vartheta(\mathbf{x}) - \tfrac{\mathrm{T}}{2}\mathbf{x}^{\top}R_{\mathrm{ev}}\mathbf{x}, & U_{\vartheta,S}(\mathbf{x}) &= \mathrm{E}_\vartheta(\mathbf{x}) + \mathrm{T}\phi_S(\mathbf{x}), \\ Z_{\mathrm{opt}}(\vartheta) &= \sum_S \pi_S(\vartheta)\int e^{-\mathrm{E}_\vartheta(\mathbf{x})/\mathrm{T}-\phi_S(\mathbf{x})}d\mathbf{x}, & Z_{\mathrm{phys}}(\vartheta) &= \sum_S \pi_S(\vartheta)\int e^{-\mathrm{E}_\vartheta^{\mathrm{phys}}(\mathbf{x})/\mathrm{T}-\phi_S(\mathbf{x})}d\mathbf{x}, \end{aligned} \tag{14.13}$$

the restricted optimizer is $q_\vartheta^\star(\mathbf{x}, S) = \pi_S(\vartheta)e^{-U_{\vartheta,S}(\mathbf{x})/\mathrm{T}}/Z_{\mathrm{opt}}(\vartheta)$, and the physical posterior $\mathbf{\Pi}_{\vartheta,\mathsf{o}}$ on $(\mathbf{x}, S)$ is the same expression with $\mathrm{E}_\vartheta^{\mathrm{phys}}$ and Z_{phys}. The scaled restricted free energy $\mathcal{F}_{\mathrm{T}}(q,\vartheta)$ is T times the negative ELBO of the evaluator restriction, evaluated at the records o.

Proposition 14.6 (Laws, restricted free energy, and the physical gap). The integrals eq. (14.13) are finite and positive, both laws have full support, and their common conditional assignment row at fixed $\mathbf{x}$ is the product of the receiver posterior rows,

$$\beta_S(\mathbf{x},\vartheta) = \frac{\pi_S(\vartheta)e^{-\phi_S(\mathbf{x})}}{\sum_{S'}\pi_{S'}(\vartheta)e^{-\phi_{S'}(\mathbf{x})}}. \tag{14.14}$$

With the scalars B_{T}, C_{T}, and $p_\vartheta(\mathsf{o})$ of eq. (C.1),

$$\begin{aligned} \mathcal{F}_{\mathrm{T}}(q,\vartheta) &= \mathrm{T}\sum_S \int q(\mathbf{x},S)\log q(\mathbf{x},S)d\mathbf{x} + \mathbb{E}_q\big[\mathrm{E}_\vartheta(\mathbf{x}) + \mathrm{T}\phi_S(\mathbf{x}) - \mathrm{T}\log\pi_S(\vartheta)\big] + \mathrm{B}_{\mathrm{T}}(\vartheta) \\ &= \mathrm{T}D_{\mathrm{KL}}(q\|q_\vartheta^\star) + \mathrm{C}_{\mathrm{T}}(\vartheta), \end{aligned} \tag{14.15}$$

and the excess over the unconstrained physical optimum is $\mathrm{C}_{\mathrm{T}} + \mathrm{T}\log p_\vartheta(\mathsf{o}) = \mathrm{T}\log(Z_{\mathrm{phys}}/Z_{\mathrm{opt}}) \geq 0$. Whenever $N\mathrm{T} \leq (1-\sigma_{\mathrm{ev}}^2)\mathrm{J}_1$,

$$0 \leq \mathrm{T}\log\frac{Z_{\mathrm{phys}}(\vartheta)}{Z_{\mathrm{opt}}(\vartheta)} \leq \frac{2e^{B_0}N\mathrm{T}^2}{(1-\sigma_{\mathrm{ev}}^2)\mathrm{J}_1}, \qquad D_{\mathrm{KL}}(q_\vartheta^\star\|\mathbf{\Pi}_{\vartheta,\mathsf{o}}) \leq \frac{2e^{B_0}N\mathrm{T}}{(1-\sigma_{\mathrm{ev}}^2)\mathrm{J}_1}. \tag{14.16}$$

[ESTABLISHED]

14.4 Exact marked parents and their inherited inference

Definition 14.7 (Marked parents and inherited edge statistics). The parent carrier at level ℓ is $W_\ell \times \mathcal{J}$. For $\ell' < \ell$ the deterministic Markov kernel

$$C_{\ell\leftarrow\ell'}((y,S),\cdot) = \delta_{(P_{\ell'}y,S)}, \qquad C_\ell = C_{\ell\leftarrow 0}, \tag{14.17}$$

extended by the identity on observations, pushes the generative joint, the physical posterior, the restricted optimizer, and the recognition law, giving $q_\vartheta^{\star,\ell} = (C_\ell)_\# q_\vartheta^\star$, $\mathbf{\Pi}^\ell_{\vartheta,\mathrm{o}}$, and $(C_\ell)_\# q$, and erases no assignment. The evaluator samples $(m_i,k_i)_i$, the records $(E_i)_i$, and the discarded normal coordinates leave the parent carrier through the disintegration of eq. (14.11) given $(P_\ell \mathbf{x}, S)$, so their correlations are retained and no mark is replaced by its value at $P_\ell\mathbf{x}$. For an original edge e and each $\mu_\vartheta \in \{\mathbf{\Pi}_{\vartheta,\mathrm{o}}, q_\vartheta^\star\}$, the inherited edge statistic is

$$\mathcal{D}_e^{\ell,\mu}(y,S) = \mathbb{E}_{\mu_\vartheta}\left[\tfrac{1}{2} r_e(\mathbf{X})^2 \mid P_\ell \mathbf{X} = y, S\right], \tag{14.18}$$

the discrepancy observables for $\mathfrak{d}_i$ are defined in the same way, and at $\ell = 0$ both reduce to the configuration residuals. For a density q^ℓ on the marked parent carrier, the parent free energy is

$$\mathcal{F}^\ell_{\mathrm{T}}(q^\ell,\vartheta) = \mathrm{T} D_{\mathrm{KL}}(q^\ell \| q_\vartheta^{\star,\ell}) + \mathrm{C}_{\mathrm{T}}(\vartheta). \tag{14.19}$$

Proposition 14.8 (Exact parent potential and discarded conditional cost). The kernels eq. (14.17) compose, $C_{\ell''\leftarrow\ell} \circ C_{\ell\leftarrow\ell'} = C_{\ell''\leftarrow\ell'}$, and positive smooth versions of every pushed law and conditional exist. The root factor of $q_\vartheta^{\star,\ell}$ is independent of the assignment and of ϑ. On M_ℓ let H_ℓ be the Schur complement of $H|_{\mathcal{K}^\perp}$ and $\gamma^{\mathrm{hid}}_{\ell,\vartheta}(dz \mid y)$ the cumulative hidden Gaussian conditional of the quadratic optimizer, whose mean is independent of ϑ and whose covariance is T/ϑ times a fixed inverse precision. Integration at fixed S gives the sheet potential

$$\begin{aligned} V_{\ell,S}(y,\vartheta) &= \tfrac{1}{2}\vartheta y^\top H_\ell y + \mathrm{C}^{\mathrm{G}}_\ell(\vartheta) + \mathrm{T}\psi_{\ell,S}(y,\vartheta), \\ \psi_{\ell,S}(y,\vartheta) &= -\log \mathbb{E}_{\gamma^{\mathrm{hid}}_{\ell,\vartheta}(\cdot|y)} e^{-\phi_S}, \qquad 0 \le \psi_{\ell,S} \le B_0, \\ \frac{d}{d\vartheta}\mathrm{C}^{\mathrm{G}}_\ell(\vartheta) &= \frac{\mathrm{T}\mathrm{d}_\ell}{2\vartheta}, \end{aligned} \tag{14.20}$$

where $\mathrm{d}_\ell = \dim W_\ell^\perp$ is the cumulative eliminated dimension, $\mathrm{C}^{\mathrm{G}}_0 = 0$, $\psi_{0,S} = \phi_S$, and the unchanged root potential is appended on W_ℓ. At one common parameter and one common time,

$$\begin{aligned} \Delta_{\ell':\ell}(q^{\ell'},\vartheta) &= \mathbb{E}_{(C_{\ell\leftarrow\ell'})_\# q^{\ell'}} D_{\mathrm{KL}}\left(q^{\ell'}(dz \mid y,S) \middle\| q_\vartheta^{\star,\ell'}(dz \mid y,S)\right), \\ \mathcal{F}^{\ell'}_{\mathrm{T}}(q^{\ell'},\vartheta) - \mathcal{F}^{\ell}_{\mathrm{T}}\big((C_{\ell\leftarrow\ell'})_\# q^{\ell'},\vartheta\big) &= \mathrm{T}\Delta_{\ell':\ell}(q^{\ell'},\vartheta), \end{aligned} \tag{14.21}$$

where z consists only of the discarded normal coordinates, loops and S staying in the retained state. [ESTABLISHED]

Proof. Composition follows from $P_{\ell''}P_{\ell} = P_{\ell''}$, and finite Gaussian domination supplies the positive smooth versions. The potential is the Gaussian integral computed in Lemma C.1. The cost identity is Theorem 10.7 for the channel $C_{\ell\leftarrow\ell'}$, since the scalar $\mathrm{C_T}$ is common to both sides and the channel leaves the observations unchanged. □

Definition 14.9 (Declared inference kinetics). Every microscopic coordinate has unit continuous mobility, and a rate-one Poisson proposal clock redraws the whole assignment from $\beta(\mathbf{x}, \vartheta)$ without moving any continuous coordinate, a proposal being allowed to return the current assignment. The microscopic generator on global Euclidean space times the finite set $\mathcal{J}$ is

$$\mathscr{L}^0_\vartheta \mathsf{f}(\mathbf{x}, S) = \mathrm{T}\Delta_{\mathbf{x}}\mathsf{f} - \nabla_{\mathbf{x}} U_{\vartheta,S} \cdot \nabla_{\mathbf{x}}\mathsf{f} + \sum_{S'} \beta_{S'}(\mathbf{x}, \vartheta)[\mathsf{f}(\mathbf{x}, S') - \mathsf{f}(\mathbf{x}, S)], \tag{14.22}$$

with diffusion covariance $2\mathrm{T}I\,dt$. The inherited proposal row and parent generator are

$$\begin{aligned} \beta^\ell_{SS'}(y, \vartheta) &= \mathbb{E}_{q^\star_\vartheta}[\beta_{S'}(\mathbf{X}, \vartheta) \mid P_\ell \mathbf{X} = y, S], \\ \mathscr{L}^\ell_\vartheta \mathsf{f}(y, S) &= \mathrm{T}\Delta_y \mathsf{f} - \nabla_y V_{\ell,S} \cdot \nabla_y \mathsf{f} + \sum_{S'} \beta^\ell_{SS'}(y, \vartheta)[\mathsf{f}(y, S') - \mathsf{f}(y, S)], \end{aligned} \tag{14.23}$$

with the unchanged root drift included when y denotes all of W_ℓ. Self-proposals are kept, so every row sums to one and every exit rate is at most one; these rows declare the parent kinetics rather than project the fine process in time.

Proposition 14.10 (Reversible inherited rows). The rows β^ℓ satisfy detailed balance with $q^{\star,\ell}_\vartheta$, and for $\ell' < \ell$

$$\beta^{\ell'}_{SS'}(y, \vartheta) \geq e^{-B_0}\pi_{S'}(\vartheta) > 0, \qquad \mathbb{E}_{q^{\star,\ell'}_\vartheta}[\beta^{\ell'}_{SS'}(Y, \vartheta) \mid P_\ell Y = z, S] = \beta^\ell_{SS'}(z, \vartheta). \tag{14.24}$$

[ESTABLISHED]

Proof. The product $q^\star_\vartheta(\mathbf{x}, S)\beta_{S'}(\mathbf{x}, \vartheta)$ equals the continuous marginal of $q^\star_\vartheta$ times $\beta_S\beta_{S'}$ and is symmetric in S, S'; conditional integration over a retained fiber gives detailed balance. The lower bound follows from eq. (14.14) and $0 \leq \phi_S \leq B_0$, and the second identity is the tower property of conditional expectation. □

14.5 Complete model scores and dissipation

Definition 14.11 (Own-law parameter dynamics). Let $\mathsf{M}(\vartheta)$ be the Fisher information of the complete normalized experiment eq. (14.11) in ϑ, before conditioning on the records, and $\mathsf{m}(\vartheta) = \mathrm{T}\mathsf{M}(\vartheta)$. With the model metric $\mathsf{m}(\vartheta)/\epsilon$, $\epsilon > 0$, each level ℓ evolves its own recognition density and its own parameter by

$$\partial_t q^\ell_t = (\mathscr{L}^\ell_{\vartheta_\ell(t)})^* q^\ell_t, \qquad \dot\vartheta_\ell(t) = -\epsilon \mathsf{m}(\vartheta_\ell(t))^{-1} g_\ell(q^\ell_t, \vartheta_\ell(t)), \qquad g_\ell = \partial_\vartheta \mathcal{F}^\ell_{\mathrm{T}}\Big|_{q^\ell}. \tag{14.25}$$

The parameter path is deterministic, since the feedback is an expectation under the law of its level. Parents inherit this metric, with no new rate multiplier at activation.

Proposition 14.12 (Complete metric, covector, and dissipation). The complete Fisher metric is

$$\mathrm{M}(\vartheta) = \frac{1}{\mathrm{T}} + \frac{1}{2}\operatorname{Tr}\big[(\Omega_0(\vartheta)^{-1}H)^2\big] + 16\pi_{i_0,+}(\vartheta)\big[1-\pi_{i_0,+}(\vartheta)\big], \tag{14.26}$$

and the full parameter covector of level ℓ is

$$\begin{aligned} g_\ell(q^\ell,\vartheta) = \vartheta - 2 + \tfrac{1}{2}\mathbb{E}_{q^\ell} y^\top H_\ell y + \frac{\mathrm{T}\mathrm{d}_\ell}{2\vartheta} + \mathrm{T}\mathbb{E}_{q^\ell}\partial_\vartheta \psi_{\ell,S}(y,\vartheta) \\ - \frac{\mathrm{T}}{2}\operatorname{Tr}\big(\Omega_0(\vartheta)^{-1}H\big) - \mathrm{T}\mathbb{E}_{q^\ell}\partial_\vartheta \log \pi_S(\vartheta), \end{aligned} \tag{14.27}$$

a partial derivative at fixed recognition law, which at $\ell = 0$, where $\mathrm{d}_0 = 0$ and $\partial_\vartheta \phi_S = 0$, is the microscopic score. Along eq. (14.25), with $\mathfrak{r}_S = q^\ell_S / q^{\star,\ell}_{\vartheta_\ell}(\cdot, S)$,

$$\begin{aligned} \frac{d}{dt}\mathcal{F}^\ell_{\mathrm{T}} = & -\mathrm{T}^2 \sum_S \int q^\ell_S \left|\nabla \log \mathfrak{r}_S\right|^2 dy - \frac{\mathrm{T}}{2}\int \sum_{S,S'} q^{\star,\ell}_{\vartheta_\ell}(y,S)\beta^\ell_{SS'}(y,\vartheta_\ell)(\mathfrak{r}_S - \mathfrak{r}_{S'})(\log \mathfrak{r}_S - \log \mathfrak{r}_{S'})dy \\ & - \epsilon \mathrm{m}(\vartheta_\ell)^{-1} g_\ell(q^\ell,\vartheta_\ell)^2 \le 0. \end{aligned} \tag{14.28}$$

At the microscopic level the jump term equals $-\mathrm{T}\int q_{\mathbf{X}}[D_{\mathrm{KL}}(q(\cdot \mid \mathbf{x})\|\beta) + D_{\mathrm{KL}}(\beta\|q(\cdot \mid \mathbf{x}))]d\mathbf{x}$. [ESTABLISHED]

The proofs are in appendix C.1; the identities hold for the solutions admitted by Lemma C.3, whose cutoff passage is Lemma C.6.

14.6 Preparation and successive activation

Definition 14.13 (Regime, preparation, and activation schedule). For $0 < \epsilon < 1$ the regime is

$$\mathrm{Q}_L = 0, \qquad \mathrm{Q}_\ell = 2\mathrm{Q}_{\ell+1} + 20, \qquad \mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}, \qquad \mathrm{A}^2 = \epsilon^{3\mathrm{Q}_1+20}, \qquad \mathrm{T} = \mathrm{A}^2\epsilon^6, \qquad t_* = \frac{2}{\epsilon}, \tag{14.29}$$

with unit mobility and proposal rate and the record coefficient δ of eq. (14.10); the weakest energy is exactly $\mathrm{J}_L = 1$. Put $\varsigma_\epsilon = 1 + \log(1/\epsilon)$ and

$$r_\epsilon = \mathrm{A}^2\mathrm{J}_1 + \mathrm{T}\varsigma_\epsilon^2 = \epsilon^{2\mathrm{Q}_1+20} + \epsilon^{3\mathrm{Q}_1+26}\varsigma_\epsilon^2. \tag{14.30}$$

Let $\mathbf{v}_\star \in \mathcal{U}$ be the unit vector of Lemma C.2, with margin $\mathrm{m}_L > 0$, and $\mathbf{v}_0$ any vector of the neighborhood of $\mathbf{v}_\star$ on which the margin is at least $\mathrm{m}_L/2$. The initial parameter is $\vartheta(0) = 1$. The root is independent of the non-root coordinates and the assignments, with law $\mathcal{N}_{\mathcal{K}}(\kappa z_\star/(1+\kappa), \mathrm{T}(1+\kappa)^{-1} I_{\mathcal{K}})$. On every assignment sheet the reference non-root law has common mean $\mathrm{A}\mathbf{v}_0$, zero loop mean, and covariance $\mathrm{T}(H|_{\mathcal{K}^\perp})^{-1}$; the positive assignment masses are near the product prior at $\vartheta = 1$, and the distinguished source mass $\lambda_{i_0,+}(0) = \Pr\{S_{i_0}(0) = +\}$ satisfies

$|\lambda_{i_0,+}(0) - \mathrm{sig}(-2)| < 10^{-2}$. The preparation neighborhood consists of the smooth within-sheet transports of this reference with bounded log-density ratio, retaining the root factor, whose coupling satisfies

$$\left|\widehat{K}_1^{1/2}(\mathbf{X}_0 - \mathbf{X}_0^{\mathrm{G}})\right| < \epsilon\sqrt{\mathrm{T}} \quad \text{almost surely;} \tag{14.31}$$

small compactly supported diffeomorphisms in standardized non-root coordinates supply such transports, the neighborhood may depend on ϵ and on L, and later laws need not stay Gaussian or factor across sheets. Given an entrance constant k_L, the windows and activation times are

$$t_\ell = \frac{k_L \log(1/\epsilon)}{\mathrm{J}_\ell}, \qquad I_\ell = [t_\ell, 2t_\ell], \qquad t_0^{\mathrm{act}} = 0, \quad t_\ell^{\mathrm{act}} = 2t_\ell. \tag{14.32}$$

At t_ℓ^{act} the complete level-$(\ell-1)$ law is pushed through $C_{\ell\leftarrow\ell-1}$, its parameter is copied, and the new law then evolves by eq. (14.25); an independently continued source process keeps its own law and parameter.

Remark 14.14 (The regime in closed form). Solving the recursion gives $\mathrm{Q}_\ell = 20(2^{L-\ell} - 1)$, so

$$\mathrm{Q}_1 = 20(2^{L-1} - 1), \qquad \frac{\mathrm{J}_1}{\mathrm{J}_L} = \epsilon^{-\mathrm{Q}_1}, \qquad \mathrm{A}^2 = \epsilon^{3\mathrm{Q}_1+20}, \qquad \mathrm{T} = \epsilon^{3\mathrm{Q}_1+26} = \epsilon^{60\cdot 2^{L-1}-34}. \tag{14.33}$$

The regime is a low-temperature limit with doubly exponential energy separation in the depth: $\mathrm{T} = \epsilon^{86}, \epsilon^{206}, \epsilon^{446}$ for $L = 2, 3, 4$, every ratio $\mathrm{J}_\ell/\mathrm{T}$ diverges, $\mathrm{J}_{\ell+1}/\mathrm{J}_\ell = \epsilon^{\mathrm{Q}_{\ell+1}+20}$ and $\mathrm{J}_{\ell+1}^2/\mathrm{J}_\ell = \epsilon^{20}$ for $\ell < L$, and the relative thermal scale is $\sqrt{\mathrm{T}}/\mathrm{A} = \epsilon^3$. There is no common acceleration of every energy, no vanishing source rate, and no decrease of δ. [ESTABLISHED]

The recursion comes from the early estimate of Lemma C.7, which bounds the retained Fisher information by the full budget eq. (C.25) and hence the discarded loss on I_ℓ by the term $(\mathrm{A}^2/\mathrm{T})\mathrm{J}_{\ell+1}^2/\mathrm{J}_\ell$ of eq. (C.41). The linear-Gaussian core shows that this term is not sharp.

Proposition 14.15 (Gaussian core of the discarded conditional loss). Take $\delta = 0$, $\vartheta \equiv 1$, and an active non-root normal law $\mathcal{N}(\bar{\mathbf{x}}(t), \mathrm{T}H_{\ell'}^{-1})$ at level $\ell' < \ell$, whose covariance is stationary for the Ornstein–Uhlenbeck flow, so $\bar{\mathbf{x}}(t) = e^{-H_{\ell'}t}\bar{\mathbf{x}}(0)$. Split the coordinates into the discarded z and retained y, with blocks H_{zz}, H_{zy}, H_{yy} of $H_{\ell'}$ and Schur complement $H_\ell = H_{yy} - H_{zy}^\top H_{zz}^{-1} H_{zy}$.

(a) *Loss formula and upper bound.* Then

$$\begin{aligned} \Delta_{\ell':\ell}(t) &= \frac{1}{2\mathrm{T}}\mathfrak{v}^\top H_{zz}^{-1}\mathfrak{v}, & \mathfrak{v} &= (H_{\ell'}\bar{\mathbf{x}})_z, \\ \dot{\mathfrak{v}} &= -H_{zz}\mathfrak{v} - H_{zy}(H_{\ell'}\bar{\mathbf{x}})_y, & (H_{\ell'}\bar{\mathbf{x}})_y &= H_\ell \bar{\mathbf{x}}_y + H_{zy}^\top H_{zz}^{-1}\mathfrak{v}, \end{aligned} \tag{14.34}$$

and, for $|\bar{\mathbf{x}}| \le \mathrm{C}_L\mathrm{A}$, $\Delta_{\ell':\ell} \le \mathrm{C}_L(\mathrm{A}^2/\mathrm{T})[\mathrm{J}_{\ell'+1}e^{-\mathrm{c}_L\mathrm{J}_\ell(t-t_{\ell'}^{\mathrm{act}})} + \mathrm{J}_{\ell+1}^4/\mathrm{J}_\ell^3]$.

(b) *Two-sided asymptotic across an early interface.* Let the strengths $\mathrm{J}_1 \ge \cdots \ge \mathrm{J}_L = 1$ depend on ϵ with ratios $\varrho_s = \mathrm{J}_{s+1}/\mathrm{J}_s \to 0$ for $s < L$, put $\varrho_0 = \varrho_L = 0$, and let $1 \le \ell \le L-1$ and $\ell' = \ell - 1$. Write $\mathsf{B}_{sr} = B_s\mathsf{Q}_r$ for the blocks of the matrix B of eq. (C.2), which vanish for $r > s$ and are square and invertible for $r = s$ (Lemma C.1), let $\mathrm{j}_\ell = \lambda_{\min}(\mathsf{B}_{\ell\ell}^\top\mathsf{B}_{\ell\ell})$, let $\mathfrak{Y}_\ell = \mathsf{B}_{\ell+1,\ell}^\top\mathsf{B}_{\ell+1,\ell+1}\mathsf{B}_{\ell+1,\ell+1}^\top$ be the

forcing map of the interface, and for $\mathbf{v} \in \mathbb{R}^{10N_\triangle - 1}$ let

$$\mathfrak{y}_\ell(\mathbf{v}) = \tfrac{1}{2}\big|(\mathrm{B}_{\ell\ell}^{\top}\mathrm{B}_{\ell\ell})^{-3/2}\mathfrak{Y}_\ell B_{\ell+1}P_\ell\mathbf{v}\big|^2, \qquad \mathfrak{x}_\ell(t) = \varrho_\ell^{-2}\exp\big[-\tfrac{1}{2}\mathrm{j}_\ell\mathrm{J}_\ell(t - t^{\mathrm{act}}_{\ell-1})\big]. \tag{14.35}$$

Then $\mathrm{B}_{\ell+1,\ell}$ has full row rank for every choice of the bridge endpoints, so $\mathfrak{Y}_\ell$ is injective. There are $\mathrm{c}_0, \mathrm{C} > 0$ depending only on L and the carrier such that, for the fine process started from the reference non-root law of Definition 14.13 with mean $\mathrm{A}\mathbf{v}_0$ and activated at the times eq. (14.32), whenever $\varrho_{\ell-1} + \varrho_\ell + \varrho_{\ell+1} \le \mathrm{c}_0$, the active level $\ell - 1$ satisfies, for $t^{\mathrm{act}}_{\ell-1} \le t \le t^{\mathrm{act}}_\ell$,

$$\begin{aligned} &\Big|\frac{\mathrm{T}\mathrm{J}_\ell^3}{\mathrm{A}^2\mathrm{J}_{\ell+1}^4}\Delta_{\ell-1:\ell}(t) - \mathfrak{y}_\ell\big(\bar{\mathbf{x}}(t)/\mathrm{A}\big)\Big| \le \mathrm{C}\big[\varrho_{\ell-1} + \varrho_\ell + \varrho_{\ell+1} + \mathfrak{x}_\ell(t) + \mathfrak{x}_\ell(t)^2\big], \\ &\qquad |B_{\ell+1}P_\ell\bar{\mathbf{x}}(t) - \mathrm{A}B_{\ell+1}P_\ell\mathbf{v}_0| \le \mathrm{C}\mathrm{J}_{\ell+1}\mathrm{A}t. \end{aligned} \tag{14.36}$$

If moreover $k_L\log(1/\epsilon)\varrho_\ell \to 0$ and $\frac{1}{2}\mathrm{j}_\ell k_L\log(1/\epsilon)(1 - 2\varrho_{\ell-1}) - 2\log(1/\varrho_\ell) \to \infty$, then, uniformly on I_ℓ,

$$\Delta_{\ell-1:\ell}(t) = \frac{\mathrm{A}^2}{\mathrm{T}}\frac{\mathrm{J}_{\ell+1}^4}{\mathrm{J}_\ell^3}\big[\mathfrak{y}_\ell(\mathbf{v}_0) + o(1)\big], \qquad \mathfrak{y}_\ell(\mathbf{v}_0) \ge \frac{\sigma_{\min}(\mathfrak{Y}_\ell)^2 N_\triangle \mathrm{m}_L^2}{2^{\ell+3}\|\mathrm{B}_{\ell\ell}\|^6} > 0, \tag{14.37}$$

where $\sigma_{\min}$ denotes the least singular value.

(c) *Separation criterion.* With power laws $\mathrm{J}_s = \epsilon^{-\mathrm{Q}_s}$, $\mathrm{Q}_1 > \cdots > \mathrm{Q}_L = 0$, with $\mathrm{A}^2/\mathrm{T} = \epsilon^{-\mathrm{Q}_\mathrm{T}}$ for some $\mathrm{Q}_\mathrm{T} > 0$, which is $\mathrm{Q}_\mathrm{T} = 6$ in eq. (14.29), and with an entrance constant $k_L > 4(\mathrm{Q}_\ell - \mathrm{Q}_{\ell+1})/\mathrm{j}_\ell$ for every $\ell < L$, both $\inf_{I_\ell}\Delta_{\ell-1:\ell}$ and $\sup_{I_\ell}\Delta_{\ell-1:\ell}$ are of exact order $\epsilon^{3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} - \mathrm{Q}_\mathrm{T}}$. The core loss is $O(\epsilon^{\mathrm{Q}_\triangle})$ on every early window, $\mathrm{Q}_\triangle \ge 0$, exactly when $3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} \ge \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle$ for every $\ell \le L - 1$, which forces $\mathrm{Q}_1 \ge (\mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle)[(4/3)^{L-1} - 1]$, with equality along the recursion $3\mathrm{Q}_\ell = 4\mathrm{Q}_{\ell+1} + \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle$; it vanishes on every early window exactly when $3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} > \mathrm{Q}_\mathrm{T}$ for every $\ell \le L - 1$. [ESTABLISHED]

The proof of (b) and (c) is in appendix C.6.

Proof of (a). Both conditional laws of z given y have precision H_{zz}/T, and their means differ by $H_{zz}^{-1}(H_{\ell'}\bar{\mathbf{x}})_z$ independently of y, which gives the first formula. Differentiating $\mathfrak{v} = (H_{\ell'}\bar{\mathbf{x}})_z$ along $\dot{\bar{\mathbf{x}}} = -H_{\ell'}\bar{\mathbf{x}}$ and eliminating $\bar{\mathbf{x}}_z$ give the other two. Put $\mathfrak{u} = H_{zz}^{-1/2}\mathfrak{v}$. Its equation is

$$\dot{\mathfrak{u}} = -\big(H_{zz} + H_{zz}^{-1/2}H_{zy}H_{zy}^{\top}H_{zz}^{-1/2}\big)\mathfrak{u} - H_{zz}^{-1/2}H_{zy}H_\ell\bar{\mathbf{x}}_y.$$

The matrix multiplying $\mathfrak{u}$ is symmetric and bounded below by $\mathrm{c}_L\mathrm{J}_\ell I$. The retained loop block is orthogonal to H_{zy} and contributes no forcing; on the future normal coordinates, eq. (C.71) bounds the forcing by $\mathrm{C}_L\mathrm{A}\mathrm{J}_{\ell+1}^2/\sqrt{\mathrm{J}_\ell}$. At $t_0 = t^{\mathrm{act}}_{\ell'}$, the weighted flag gives $|\mathfrak{u}(t_0)| \le \mathrm{C}_L\mathrm{A}\sqrt{\mathrm{J}_{\ell'+1}}$. Variation of constants therefore gives

$$|\mathfrak{u}(t)| \le \mathrm{C}_L\mathrm{A}\left[\sqrt{\mathrm{J}_{\ell'+1}}e^{-\mathrm{c}_L\mathrm{J}_\ell(t-t_0)} + \frac{\mathrm{J}_{\ell+1}^2}{\mathrm{J}_\ell^{3/2}}\right].$$

Squaring and using $\Delta_{\ell':\ell} = |\mathfrak{u}|^2/(2\mathrm{T})$ proves the bound, with the decay constant adjusted. The forcing vanishes for $\ell = L$. □

Open Problem 14.16 (Sharpness of the doubly exponential regime). Decide whether the recursion $\mathrm{Q}_\ell = 2\mathrm{Q}_{\ell+1}+20$ is required for Theorem 14.17, and whether its constants and preparation neighborhoods can be made uniform in the depth. What holds is the theorem at every finite L with constants depending on L, and, by Proposition 14.15, that in the Gaussian core the discarded loss on every early window is of exact order $(\mathrm{A}^2/\mathrm{T})\mathrm{J}_{\ell+1}^4/\mathrm{J}_\ell^3$, smaller than the term of the general estimate by $(\mathrm{J}_{\ell+1}/\mathrm{J}_\ell)^2$, so that the early interfaces need exactly $3\mathrm{Q}_\ell \geq 4\mathrm{Q}_{\ell+1}+20$ for the core loss at the scale ϵ^{14}, a separation still doubly exponential in L but of inner base $4/3$, with $\mathrm{Q}_\ell \geq 14$ for the term $\mathrm{C}_{\mathcal{J}}/\mathrm{J}_\ell$ of the jump remainder eq. (C.39). The doubling enters the present proof through the full Fisher budget eq. (C.25) in the early estimate of Lemma C.7 and through the history estimate of Lemma C.11, which uses it at the top interface in eq. (C.66). The depth enters through Q_1 in the regime and every entrance condition, through $\log|\mathcal{J}| = N\log 2$ in the jump remainder, and through the Frobenius constants of eq. (C.2), whose squares grow linearly in $N_\triangle$. The problem is posed in the family of the theorem under (H1), (H3), and (H4), with power-law strengths $\mathrm{J}_\ell = \epsilon^{-\mathrm{Q}_\ell}$, $\mathrm{Q}_L = 0$, satisfying $3\mathrm{Q}_\ell \geq 4\mathrm{Q}_{\ell+1}+20$ and $\mathrm{Q}_\ell \geq 14$ for $\ell < L$, with the amplitude, temperature, horizon, and record coefficient of eq. (14.29) formed from Q_1, and with an entrance constant above the thresholds of the theorem and of Proposition 14.15. It is settled by conclusions (a) to (e) for every schedule of this class, with constants and preparation neighborhoods independent of L, through an early estimate of the retained Fisher information by the future-layer energy, a history-environment bound without the top-interface recursion, and depth-uniform flag and jump constants; or by an admitted preparation and a schedule of the class under which one of the conclusions fails, or a lower bound showing that one of the conclusions cannot hold with constants independent of L.

14.7 The finite-depth formation theorem

Formation in this family is hierarchical relaxation of the prepared mean. The non-root configuration starts near $\mathrm{A}\mathbf{v}_0$, where every residual has order A and every bridge residual $r_e(P_{\ell'}\mathrm{A}\mathbf{v}_0)$ with $\ell' < \ell(e)$ stays at least $\mathrm{Am}_L/2$ away from zero. The weighted flag makes residuals of level at most ℓ relax at rate of order J_ℓ, while bridges of higher level feel only couplings of order $\mathrm{J}_{\ell+1}$. On I_ℓ the first have decayed by $\epsilon^{\mathrm{c}_L k_L}$ relative to A, whereas the second have moved by at most $\mathrm{C}_L\mathrm{A}\mathrm{J}_{\ell+1}t_\ell \to 0$ and keep their margin, so each window joins pairs of coherent components along the bridges of level ℓ and halves their number. The source record reads residuals at the thermal resolution $\sqrt{\mathrm{T}/\mathrm{J}_1}$, far below the coherence scale A, so it is saturated there. Source jumps, the record, and learning are therefore perturbations, and the content of the theorem is that exact marked parents, activation from the complete law, and own-law learning can be carried through this relaxation with separately vanishing errors.

Theorem 14.17 (Finite-depth formation under a supplied energy hierarchy). Assume (H1) the experiment of Definitions 14.1, 14.3, 14.5 and 14.7 with the kinetics of Definitions 14.9 and 14.11, at a fixed depth $L \geq 2$; (H2) the regime eq. (14.29); (H3) an initial law in the preparation neighborhood eq. (14.31); (H4) parents activated at the times eq. (14.32). There exist constants $k_{\mathrm{geom},L}, \mathrm{c}_{\mathrm{early},L}, \mathrm{c}_{\mathrm{joint},L}, \mathrm{c}_{\mathrm{fast},L}, \eta_L^{\max} > 0$, depending only on L and the fixed data and constructed

respectively in Lemma C.12 (together with $\eta_L^{\max}$, a fixed fraction of the margin m_L), Lemma C.7, Lemma C.8, and Lemma C.10, with the following property. For every entrance constant k_L with

$$\begin{aligned} k_L > k_{\mathrm{geom},L}, \quad \tfrac{1}{2}\mathrm{c}_{\mathrm{early},L}k_L > \mathrm{Q}_1 + 20, \\ \mathrm{c}_{\mathrm{joint},L}k_L > \mathrm{Q}_1 + 8, \qquad \mathrm{c}_{\mathrm{fast},L}k_L > \mathrm{Q}_1/2 + 12, \end{aligned} \tag{14.38}$$

and every tolerance $0 < \eta_L \le \eta_L^{\max}$, there exist $\epsilon_L, \mathrm{c}_L, \mathrm{C}_L > 0$, depending on L, the fixed data, k_L, and η_L, such that for every $0 < \epsilon < \epsilon_L$ conclusions (a) to (e) hold for the fine process and for the lineage of activated parents. The lineage is said to admit all L successive activations when the fine process and every activated parent have a unique nonexplosive solution of eq. (14.25) through t_*, with positive smooth densities and parameter in $[1, 21/10]$.

(a) *Well-posedness and parameter.* The lineage admits all L successive activations, and every process satisfies, in absolute time,

$$1 \le \vartheta_\ell(t) \le 21/10, \qquad |\dot{\vartheta}_\ell(t)| \le \mathrm{C}_L\epsilon, \qquad |\vartheta_\ell(t) - (2 - e^{-\epsilon t})| \le \mathrm{C}_L r_\epsilon, \tag{14.39}$$

with $r_\epsilon = O(\epsilon^{2\mathrm{Q}_1+20})$ of eq. (14.30); the comparison curve solves the calibration part of eq. (14.25).

(b) *Coherence.* For a level-ℓ description at time t, an original edge e is coherent when $\sqrt{2\mathcal{D}_e^{\ell,\mu}} \le \eta_L\mathrm{A}$, with eq. (14.18) evaluated at the current retained state and parameter of that description, and a vertex is internally coherent when the corresponding observable for $\mathfrak{d}_i$ is at most $\eta_L\mathrm{A}$; the coherent components are the connected components of the graph on the N vertices whose edges are the coherent edges. For each of $\mu_\vartheta = \mathbf{\Pi}_{\vartheta,\mathrm{o}}$ and $\mu_\vartheta = q^\star_\vartheta$, except on an event of probability at most

$$\mathrm{C}_L(1 + t_*)\exp(-\mathrm{c}_L/\epsilon^6), \tag{14.40}$$

the microscopic process and its sequential parent descriptions have the successive coherent components

$$3 \cdot 2^{L-1} \longrightarrow 2^{L-1} \longrightarrow 2^{L-2} \longrightarrow \cdots \longrightarrow 1. \tag{14.41}$$

On I_ℓ every edge of level at most ℓ is coherent, every vertex is internally coherent, and every bridge of level greater than ℓ has residual observable at least a fixed positive multiple of A, so the count on I_ℓ is $2^{L-\ell}$; the single final component persists through t_*. Loop coordinates remain random and retained, with no identity-holonomy constraint.

(c) *Conditional loss and score.* On each window I_ℓ, at the current law and parameter,

$$\Delta_{\ell-1:\ell} \le \begin{cases} \mathrm{C}_L\epsilon^{14}, & \ell < L, \\ \mathrm{C}_L\epsilon^2, & \ell = L, \end{cases} \qquad \big|g_{\ell-1}(q,\vartheta) - g_\ell\big((C_{\ell\leftarrow\ell-1})_\#q, \vartheta\big)\big| \le \begin{cases} \mathrm{C}_L\mathrm{T}\epsilon^7, & \ell < L, \\ \mathrm{C}_L\mathrm{T}\epsilon, & \ell = L, \end{cases} \tag{14.42}$$

and $\mathrm{T}\Delta_{\ell-1:\ell}$ is the discarded restricted free energy.

(d) *Source and kernel movement.* Let $\lambda^{\ell}_{i_0,+}(t)$ be the distinguished receiver's source mass in a fine, parent, or sequential process. From each activation time at which that process is continued,

$$\lambda^{\ell}_{i_0,+}(t_*) - \lambda^{\ell}_{i_0,+}(t^{\mathrm{act}}_{\ell'}) > \tfrac{1}{2}, \tag{14.43}$$

so the binary recognition source row changes by total variation greater than one half. Between $\vartheta = 1$ and the final parameter, the first hidden physical conditional and the first hidden optimizing conditional each change by relative entropy greater than $1/200$, at every fixed retained input and assignment. Both changes are driven by the learned parameter through eq. (14.9) and ϑH, the record moving the distinguished log odds by at most δ, and both take place after formation, since $\vartheta \le 1 + \mathrm{C}_L \epsilon \log(1/\epsilon)$ until the last window closes.

(e) *Histories and paths.* For any source process and its inherited parent begun from a common marked activation sample at t^{act}_{ℓ}, there is a deterministic path $\bar{\mathbf{x}}_{\ell} : [t^{\mathrm{act}}_{\ell}, t_*] \to W_{\ell}$ with

$$\Pr\Big\{ \sup_{t^{\mathrm{act}}_{\ell} \le t \le t_*} |P_{\ell}\mathbf{X}_t - \bar{\mathbf{x}}_{\ell}(t)| > \mathrm{C}_L \mathrm{A}\epsilon^2 \Big\} + \Pr\Big\{ \sup_{t^{\mathrm{act}}_{\ell} \le t \le t_*} |\overline{\mathbf{X}}_t - \bar{\mathbf{x}}_{\ell}(t)| > \mathrm{C}_L \mathrm{A}\epsilon^2 \Big\} \le \mathrm{C}_L (1 + t_*) e^{-\mathrm{c}_L/\epsilon^2}. \tag{14.44}$$

There is a coupling preserving the activation sample under which the first bound below holds, and the second holds under every coupling of the two processes:

$$\begin{aligned} &\Pr\big\{ S_t \ne \overline{S}_t \text{ for some } t \in [t^{\mathrm{act}}_{\ell}, t_*] \big\} \le \mathrm{C}_L \epsilon^6, \\ &\Pr\Big\{ \sup_{t^{\mathrm{act}}_{\ell} \le t \le t_*} |P_{\ell}\mathbf{X}_t - \overline{\mathbf{X}}_t| > \mathrm{C}_L \mathrm{A}\epsilon^2 \Big\} \le \mathrm{C}_L (1 + t_*) e^{-\mathrm{c}_L/\epsilon^2}. \end{aligned} \tag{14.45}$$

The resolution $\mathrm{C}_L \mathrm{A}\epsilon^2 = \mathrm{C}_L \sqrt{\mathrm{T}}/\epsilon$ is coarser than the thermal scale, and a finite number of successive replacements changes only the constants. [ESTABLISHED]

Proof. The exact statements are Propositions 14.4, 14.6, 14.8 and 14.10, and Table C.1 lists the lemmas of appendix C with their inputs and exponents. Conclusion (a) is Lemmas C.3 and C.4, and (b) is Lemma C.12, which uses neither conditional equilibration nor history averaging. For (c), Lemma C.7 gives the early bound from $\mathrm{J}^2_{\ell+1}/\mathrm{J}_{\ell} = \epsilon^{20}$ and the second entrance condition, and Lemma C.8 the terminal bound by joint diffusion-plus-jump relaxation; conditional relative entropy is bounded by the full one through the chain rule, and the complete hidden score bounds the covector difference by $\mathrm{C}_L \mathrm{T}(\sqrt{\Delta} + \Delta)$. Conclusion (d) is Lemma C.13. For (e), Lemma C.11 couples the assignment histories and Lemma C.14 gives eq. (14.44); each of its events concerns the law of one process, so the union bound gives the second line of eq. (14.45) under every coupling. The four entrance conditions are equivalent to the single threshold

$$k_L > \max\left\{ k_{\mathrm{geom},L},\ \frac{2(\mathrm{Q}_1 + 20)}{\mathrm{c}_{\mathrm{early},L}},\ \frac{\mathrm{Q}_1 + 8}{\mathrm{c}_{\mathrm{joint},L}},\ \frac{\mathrm{Q}_1/2 + 12}{\mathrm{c}_{\mathrm{fast},L}} \right\}, \tag{14.46}$$

whose terms supply, in order, the geometric windows, the early entrance, terminal joint relaxation, and the source-history estimate; every lemma uses the same preparation, regime, and entrance choice. The final window has order $k_L \log(1/\epsilon)$ and lies before $t_* - 10 = 2/\epsilon - 10$ once ϵ is small;

this, the ordering of the windows, and the smallness conditions inside the lemmas fix ϵ_L after k_L and η_L. □

Remark 14.18 (Saturated record, calibrated learning, and thermally negligible loops). Uniformly in the configuration, the assignment, and the parameter,

$$\begin{aligned} &|\log \beta_S(\mathbf{x},\vartheta) - \log \pi_S(\vartheta)| \le B_0 = 10^{-4}, \\ &\qquad |H^{-1/2}\mathrm{T}\nabla_{\mathbf{x}}\phi_S(\mathbf{x})| \le B_*\sqrt{\mathrm{T}}, \qquad B_* = \delta\sqrt{2N} = 10^{-4}\sqrt{2/N} \le 5.8\times 10^{-5}, \end{aligned} \tag{14.47}$$

with $H^{-1/2}$ acting on $\mathcal{K}^{\perp}$, by eq. (14.14) with $0 \le \phi_S \le B_0$ and by eq. (C.9). At a residual of order A the argument of χ has order $\epsilon^{-(3+\mathrm{Q}_1/2)}$, so the record equals its saturated value up to an exponentially small error. The proof of Lemma C.12 uses the record and the assignments only through $0 \le \phi_S \le B_0$ and the pathwise force bound eq. (C.17), valid for every adapted assignment path, and the parameter only through $1 \le \vartheta \le 21/10$, so with $\delta = 0$ and $\vartheta \equiv 1$ it gives the same components. By Lemmas C.3 and C.4 every law-dependent term of the covector is at most $\mathrm{C}_L r_\epsilon$, which gives the parameter bounds of Theorem 14.17. The loop energy $\mathrm{J}_1|h|^2$ belongs to the strongest term, so $\mathrm{J}_1\mathbb{E}|h|^2 \le \mathrm{C}_L\mathrm{T}$ by eq. (C.15), and the loops have root-mean-square size at most $\mathrm{C}_L\sqrt{\mathrm{T}/\mathrm{J}_1} = \mathrm{C}_L\mathrm{A}\epsilon^{3+\mathrm{Q}_1/2}$. [ESTABLISHED]

14.8 A two-triangle instance and the remaining scope

Remark 14.19 (The two-triangle instance). For $L = 2$, take triangles $(1,2,3)$ and $(4,5,6)$ with the single bridge $(3,4)$. The retained carriers are

$$\mathbb{R}^{19}\times\mathcal{J} \longrightarrow \mathbb{R}^{9}\times\mathcal{J} \longrightarrow \mathbb{R}^{8}\times\mathcal{J}, \qquad |\mathcal{J}| = 64, \tag{14.48}$$

the coherent component counts are $6 \to 2 \to 1$, both loops survive both maps, and the regime becomes

$$\mathrm{J}_1 = \epsilon^{-20}, \qquad \mathrm{J}_2 = 1, \qquad \mathrm{A} = \epsilon^{40}, \qquad \mathrm{T} = \epsilon^{86}. \tag{14.49}$$

The first window has order $\epsilon^{20}\log(1/\epsilon)$, the second $\log(1/\epsilon)$, and the source and model changes occur over the longer horizon $2/\epsilon$. The source-dependent term is not quadratic: along a direction with selected internal residual slope $\lambda \neq 0$, $\chi(\lambda t) = \lambda^2 t^2/2 - \lambda^4 t^4/8 + O(t^6)$, and every assignment selects an internal edge with nonzero first-normal projection, so the hidden log density has a nonzero fourth derivative in some first-normal direction; exact marginalization keeps the full conditional integral and the negative covariance term of its Hessian, and neither first hidden conditional is a Gaussian. [ESTABLISHED]

Open Problem 14.20 (A family in which learning or holonomy selects the components). Construct a normalized family at c_* in which parameter learning or loop holonomy changes, at the level of the law, which components form, with exact marked parents and activation from the complete law. What holds is Theorem 14.17, whose components are fixed by the supplied contrasts

and internal source support, and Proposition 14.22, whose pairs are seeded by the random initial assignment without learning, so that in it the source record selects only along sampled paths; Proposition 15.4 shows that its seeded bound is unchanged by learning and that from an initial law invariant under the cycle automorphisms learning leaves the rows uniform, so a law-level selection needs an initial law without that invariance. Such a family needs at least one of a source record standardized at the coherence scale, for example through $\chi(r/(\eta_L \mathrm{A}))$, model means and model links both slow relative to the belief mobility, since a fast link or fast model means remove the model disagreement on the belief clock whatever the other rate (Proposition 8.6), loop coordinates of order A or nonabelian links, or interacting model-channel records, together with new fast-marginal and history estimates; sources on different bridge levels and source learning that creates the contrasts belong to the same problem. It is posed on the four-cycle of Proposition 15.4, with its temperature, learning mobility, and horizon, for Gaussian initial laws $\mathcal{N}(\mathrm{m}_0, I_{12})$ with m_0 outside a Lebesgue-null set, and for parents of the type of part (b) of Proposition 14.22 activated from the recognition law of the episode prefix; the law of the formed block means the conditional law, given contrast formation, of the first block for which it occurs, the conditioning event having probability bounded below in every arm by part (b) of Proposition 15.4, so that a difference in whether formation occurs is not selection. It is settled by such parents with separately vanishing conditional and dynamical errors together with a total-variation difference, bounded below as $\epsilon \to 0$, between the conditional laws of the formed block in the learned and in the frozen-row arm, or between those of two initial means that differ only in their loop coordinate $\sum_i \omega_i$, the amplified datum being the initial law or its loop coordinate and the other arm its null; or by a proof that, for almost every m_0, each of these two pairs of arms gives conditional laws of the formed block whose difference vanishes as $\epsilon \to 0$.

A separate slower-mobility regime is resolved by proposition C.17: keeping the same contrast event and horizon but taking $\epsilon_{\mathrm{L}} = o(\mathrm{T}^{1/4})$, the learned and frozen conditional first-block laws have vanishing total variation for every fixed Gaussian mean. The original coupled mobility $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$ and the loop-coordinate comparison remain outside that estimate. The late coherent-partition contrast in proposition C.16 concerns restored equilibrium coherence and does not close this first-block problem. For the late coherence event on the joint slow schedule, propositions C.18 and C.20 establish actual conditioned activation of the existing parent with separately vanishing conditional-information and dynamical errors. The activation event is a completed coherence prefix after selection. It is not the first-contrast event in the original regime posed above. For the seeded sub-event of that original contrast, at every mobility $\epsilon_{\mathrm{L}} \in [0, 1]$ including $\mathrm{T}^{1/8}$, propositions C.25 and C.27 give separately vanishing conditional and dynamical errors when the parent is activated after hidden thermalization, at a deterministic activation time θ with $\theta - \log(1/\epsilon) \to \infty$ and $\epsilon^4 \theta \to 0$, on following windows short compared with T^{-1}, while proposition C.26 shows that activation at the first qualifying completion has diverging conditional error, also under conditioning on the whole contrast event. The learned and frozen first-block comparison at $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$, the loop-coordinate comparison, and conditioning on contrast histories outside the seeds remain open.

14.9 An endogenous four-cycle witness

Definition 14.21 (Homogeneous four-cycle experiment). Index four agents by $i \in \mathbb{Z}/4$ and orient edge e_i from i to $i+1$. A configuration is $\mathbf{x} = (\bar{k}, \bar{m}, \omega) \in \mathbb{R}^{12}$, with $q_i^b = \mathcal{N}(\bar{k}_i, 1)$, $q_i^m = \mathcal{N}(\bar{m}_i, \sigma_{\mathrm{ev}}^2)$, $\sigma_{\mathrm{ev}}^2 = 1/2$, four independent link translations ω_i, residuals $r_i^b = \bar{k}_{i+1} - \bar{k}_i - \omega_i$ and $r_i^m = \bar{m}_{i+1} - \bar{m}_i - \omega_i$, and the edge cost

$$\mathcal{D}_{e_i}(\mathbf{x}) = \tfrac{1}{2}\big[(r_i^b)^2 + (r_i^m)^2\big] = D_{\mathrm{KL}}\big(q_{i+1}^b \| (\Theta_{e_i})_{\#} q_i^b\big) + \sigma_{\mathrm{ev}}^2 D_{\mathrm{KL}}\big(q_{i+1}^m \| (\Theta_{e_i})_{\#} q_i^m\big), \tag{14.50}$$

which is the same for both orientations. Every receiver has both neighbors as sources, $S \in \mathcal{J} = \prod_i \{i-1, i+1\}$ with $|\mathcal{J}| = 16$ and uniform prior $\pi_S = 1/16$, and $V_S = \sum_i \mathcal{D}_{e(i,S_i)}$, where $e(i,j)$ joins i and j. The configuration prior is $\mathcal{N}(0, I_{12})$; given $(\mathbf{x}, S)$, independent binary records have $\Pr(E_i = 1 \mid \mathbf{x}, S) = \frac{1}{2}\exp[-\mathcal{D}_{e(i,S_i)}(\mathbf{x})/\mathrm{T}]$, and $E = \mathbf{1}$ is observed; the physical evaluator and recognition coupling are those of Definition 14.3. With $\mathfrak{D}\mathbf{x} = \bar{k} - \bar{m}$ and $R_{\mathrm{ev}} = 2\mathfrak{D}^\top \mathfrak{D}$, the restricted optimizer is $q^\star(\mathbf{x}, S) \propto \pi_S e^{-U_S(\mathbf{x})/\mathrm{T}}$ with $U_S = V_S + \frac{\mathrm{T}}{2}|\mathbf{x}|^2 + \frac{\mathrm{T}}{2}\mathbf{x}^\top R_{\mathrm{ev}} \mathbf{x}$, and the physical posterior omits the last term. The kinetics are identity mobility with diffusion covariance $2\mathrm{T} I dt$ and one rate-one clock at which the complete assignment is redrawn from the product of the receiver posterior rows $\beta_S(\mathbf{x})$, self-proposals allowed; no generative parameter is learned. The two perfect-matching assignments are $S^{(0)} = (1, 0, 3, 2)$, which selects e_0, e_2 from both ends, and $S^{(1)} = (3, 2, 1, 0)$. For $0 < \epsilon \le 1/100$ put

$$\mathrm{T} = \epsilon^8, \quad t_\epsilon = 2\log(1/\epsilon), \quad t_*^\circ = t_\epsilon + 1, \quad \varrho_\circ = \tfrac{1}{100}, \quad \sigma_\circ = \varrho_\circ/\sqrt{48}, \quad \mathbf{x}^\circ = (0, 0, \mathbf{1}_4). \tag{14.51}$$

The preparation class consists of the laws $\frac{1}{16}\mathfrak{p}_\circ(\mathbf{x})(1 + \mathfrak{z}(\mathbf{x}))d\mathbf{x}$ on $(\mathbf{x}, S)$, where $\mathfrak{p}_\circ$ is the equal mixture of $\mathcal{N}(\mathbf{x}^\circ, \sigma_\circ^2 I)$ and $\mathcal{N}(-\mathbf{x}^\circ, \sigma_\circ^2 I)$ and $\mathfrak{z}$ is smooth, invariant under the cycle automorphisms (with link signs transformed under reflections), of $\mathfrak{p}_\circ$-mean zero, and with $\|\mathfrak{z}\|_\infty < 1/2$.

The preparation names no block: its assignment is uniform and independent of the configuration, reflections exchange its two centers, which carry opposite loop orientations, and all initial edge mismatch second moments agree. Under the source gating an edge that neither endpoint selects carries no interaction energy, although every edge stays in the palettes of both its endpoints and every attention weight stays positive.

Proposition 14.22 (Endogenous pairing on a homogeneous four-cycle). In Definition 14.21, let $0 < \epsilon \le 1/100$ and let the initial law belong to the preparation class. The joint is normalized, the scaled negative ELBO is $\mathcal{F}_{\mathrm{T}}(q) = \mathrm{T} D_{\mathrm{KL}}(q \| q^\star) + \mathrm{B}_{\mathrm{T}} - \mathrm{T}\log Z_{\mathrm{opt}}$ with $\mathrm{B}_{\mathrm{T}} = \mathrm{T}[6\log(2\pi) + 4\log 2]$, and the recognition law of the kinetics has nonincreasing $\mathcal{F}_{\mathrm{T}}$ and is invariant under the cycle automorphisms at every time. Put

$$p_\epsilon = 73728\epsilon^4 t_*^\circ + 4 t_*^\circ e^{-1/\mathrm{T}} < \tfrac{1}{2}, \qquad \eta_\epsilon = \tfrac{9}{2}\varrho_\circ^2 \epsilon^8 + \frac{3}{4(1 - e^{-8})}\epsilon^{32}, \qquad \bar{\eta}_\epsilon = \frac{\eta_\epsilon + \mathcal{S}_2(p_\epsilon)}{1 - p_\epsilon}, \tag{14.52}$$

with $\mathcal{S}_2(p) = -p\log p - (1-p)\log(1-p)$. Then:

(a) *Episode.* With probability at least $p_0 = 3e^{-1/4}/512$, the complete assignment equals one perfect matching throughout $[0, t_*^\circ]$; at time zero the internal belief transported relative entropy of each matched pair exceeds $1/4$, so that the contrast is entered during the episode and is not present at its start; on $[t_\epsilon, t_*^\circ]$ both internal belief and model transported relative entropies of each matched pair are at most ϵ^4; and on $[1/4, t_*^\circ]$ every exterior belief relative entropy exceeds $1/2$, the model ones being larger. This is a coherence episode held by the two attended edge costs: on the held matching sheet the normal restoring rates of the kinetics are $8 + \mathrm{T}$ and $4 + 5\mathrm{T}$, those of the physical law $8 + \mathrm{T}$ and $4 + \mathrm{T}$, so at the residual resolution ϵ^2 the least restoring rate times the squared resolution, divided by the temperature, is $4\epsilon^{-4}$, and the sheet is selected by the posterior proposals on the event. The kinetics leave $q^\star$ invariant, and started from $q^\star$ they realize the same event with probability at most $4e^{-1/(4\mathrm{T})}$, so the episode exceeds its stationary probability by at least $p_0 - 4e^{-1/(4\mathrm{T})} > p_0/2$.

(b) *Parent channel.* For each matching M, let Z_M be the span of the three orthonormal directions $(n_e^+ - n_f^+)/\sqrt{2}$, n_e^-, n_f^-, where e, f are its two edges, $n_e^\pm$ the unit normal covectors of eq. (C.94), and $W_M = Z_M^\perp$. The channel $C(\mathbf{x}, S) = (S, P_{W_S}\mathbf{x})$, with $W_S = W_M$ on the two matchings and $W_S = \mathbb{R}^{12}$ otherwise, extended by the identity on observations and evaluator samples, keeps every assignment, the loop $\sum_i \omega_i$, and a fast common-normal coordinate, and pushes the joint, the physical posterior, the restricted optimizer, and the recognition law to normalized parent laws.

(c) *Discarded conditional free energy.* For every $t \in [t_\epsilon, t_*^\circ]$, the recognition law q_t of the episode has mean discarded optimizing conditional relative entropy at most $\bar{\eta}_\epsilon$ and mean discarded physical conditional relative entropy, with the evaluator samples retained, at most $\bar{\eta}_\epsilon + \mathrm{T}$.

(d) *Parent dynamics.* The inherited parent of C, activated at t_ϵ from the complete recognition law of the episode prefix, has retained path law on $[t_\epsilon, t_*^\circ]$, including the complete assignment path, within total variation $2p_\epsilon/(1 - p_\epsilon) + 4\sqrt{8/3}e^{-1/\mathrm{T}}$ of the retained native path conditioned on the same prefix.

The estimates hold under the stated source gating, random matching seed, symmetric circulated preparation, temperature ϵ^8, Euclidean kinetic metric, duration-one episode, and fixed generative parameters. [ESTABLISHED]

The proof is in appendix C.7. On the center path eq. (C.95), which holds $S^{(0)}$ from $\mathbf{x}^\circ$, where every residual equals -1, the selected residuals decay as $-e^{-(8+\mathrm{T})t}$ while the exterior ones grow toward $-3/2$, so a source-cost gap greater than one develops by time $1/4$, and the posterior proposals then keep the matching except with probability at most $4t_*^\circ e^{-1/\mathrm{T}}$. Since the full recognition law is invariant, the pairing is an event of sampled paths seeded by the initial assignment, not a block chosen by the law.

The witness raises the question whether its mechanism, an edge that carries energy only while an endpoint attends to it, produces a stationary law that favors coherence beyond one level, or a second coalescence of the pairs it forms. The next statement answers the stationary part negatively in the class to which the four-cycle belongs, the family of Definition 14.23 on an arbitrary finite graph; the second coalescence is posed for models outside that family in Open Problem 14.25. Under exclusive gating the source label enters only the weight of a sheet and never its stiffness, so

every sheet restores its attended residual directions at rates of order one and leaves every other direction, including every direction that translates one coherent block relative to another, to the thermal reference at rates of order T. This has a spectral and a stationary consequence.

Definition 14.23 (Gated translation family on a graph). Let (V, E) be a finite simple graph without isolated vertices, with $N = |V|$ agents and matching number $\mathfrak{n}$, the largest number of pairwise disjoint edges. Orient each edge $e \in E$ once as $e : j \to i$; the interaction multigraph at c_* consists of the copies e and $\bar{e}$ of the edges of (V, E), and the source set $\mathcal{N}_i$ of receiver i is its set of neighbors in (V, E), with $e(i, j)$ the edge joining i and j. The agents, links, laws, evaluator, and recognition coupling are those of Definition 14.21 on this graph: a configuration is $\mathbf{x} = (\bar{k}, \bar{m}, \omega) \in \mathbb{R}^{2N+|E|}$ with one link translation ω_e per edge, $q_i^b = \mathcal{N}(\bar{k}_i, 1)$, $q_i^m = \mathcal{N}(\bar{m}_i, 1/2)$, and edge $e : j \to i$ carries the residual pair and the cost

$$\mathbf{a}_e\mathbf{x} = (r_e^b, r_e^m) = (\bar{k}_i - \bar{k}_j - \omega_e,\ \bar{m}_i - \bar{m}_j - \omega_e), \qquad \mathcal{D}_e(\mathbf{x}) = \tfrac{1}{2}|\mathbf{a}_e\mathbf{x}|^2, \tag{14.53}$$

so $\mathbf{a}_e : \mathbb{R}^{2N+|E|} \to \mathbb{R}^2$ is linear and invariant under passive frame changes as in Lemma 14.2, and $\mathcal{D}_e$ is the transported relative entropy eq. (14.50). Each receiver carries one source label of Definition 6.1 with source set $\mathcal{N}_i$, so a complete assignment is $S \in \mathcal{J} = \prod_i \mathcal{N}_i$; the labels are exclusive, the prior is the product $\pi_S = \prod_i \pi_{iS_i}$ with every $\pi_{ij} > 0$ independent of T, and given $(\mathbf{x}, S)$ independent binary records have $\Pr(E_i = 1 \mid \mathbf{x}, S) = \frac{1}{2}\exp[-\mathcal{D}_{e(i,S_i)}(\mathbf{x})/\mathrm{T}]$, with $E = \mathbf{1}$ observed. The attended edge set, sheet energy, and sheet stiffness of S are

$$\mathrm{att}(S) = \{e(i, S_i) : i \in V\}, \qquad V_S = \sum_i \mathcal{D}_{e(i,S_i)} = \tfrac{1}{2}\mathbf{x}^\top H_S \mathbf{x}, \qquad H_S = \sum_i \mathbf{a}_{e(i,S_i)}^\top \mathbf{a}_{e(i,S_i)}. \tag{14.54}$$

The configuration prior is $\mathcal{N}(0, I_{2N+|E|})$, and with $R_{\mathrm{ev}} = 2\mathfrak{D}^\top\mathfrak{D}$ the restricted optimizer and the physical posterior are the laws

$$q_\Omega(\mathbf{x}, S) = \frac{\pi_S}{\sum_{S'} \pi_{S'} Z_{S'}} \exp\Big[-\frac{V_S(\mathbf{x})}{\mathrm{T}} - \tfrac{1}{2}\mathbf{x}^\top\Omega\mathbf{x}\Big], \qquad Z_S = \int \exp\Big[-\frac{V_S(\mathbf{x})}{\mathrm{T}} - \tfrac{1}{2}\mathbf{x}^\top\Omega\mathbf{x}\Big]d\mathbf{x}, \tag{14.55}$$

with the reference precision $\Omega = I + R_{\mathrm{ev}}$ for the optimizer and $\Omega = I$ for the physical posterior; Z_S is the sheet weight. For either Ω the kinetics have identity mobility, diffusion covariance $2\mathrm{T}I\,dt$, drift $-\nabla_{\mathbf{x}} U_S$ with $U_S = V_S + \frac{\mathrm{T}}{2}\mathbf{x}^\top\Omega\mathbf{x}$, and one rate-one clock at which the complete assignment is redrawn from the product of the receiver posterior rows $\beta_i(j \mid \mathbf{x}) = \pi_{ij} e^{-\mathcal{D}_{e(i,j)}(\mathbf{x})/\mathrm{T}} \big/ \sum_{j' \in \mathcal{N}_i} \pi_{ij'} e^{-\mathcal{D}_{e(i,j')}(\mathbf{x})/\mathrm{T}}$, self-proposals allowed; no generative parameter is learned. For an edge set $F \subseteq E$, $\mathbf{a}_F$ stacks the maps $\mathbf{a}_e$, $e \in F$, $W_F = \ker \mathbf{a}_F$, s_F is the least positive singular value of $\mathbf{a}_F$, and $\mathrm{cc}(F)$ is the number of connected components of the graph (V, F), isolated vertices included. At resolution $\mathrm{a} > 0$ the coherent edges of a configuration are $\mathrm{Coh}_\mathrm{a}(\mathbf{x}) = \{e \in E : |\mathbf{a}_e\mathbf{x}| \le \mathrm{a}\}$, and $\mathfrak{P}_\mathrm{a}(\mathbf{x})$ is the partition of V into the vertex sets of the components of $(V, \mathrm{Coh}_\mathrm{a}(\mathbf{x}))$, the coherent blocks.

The four-cycle of Definition 14.21 is the member with (V, E) the four-cycle and the uniform prior, and its kinetics are those with $\Omega = I + R_{\mathrm{ev}}$. A minimum edge cover of (V, E) is an edge set covering

every vertex with the fewest edges; the proposition identifies these covers with the spanning forests of stars with $\mathfrak{n}$ components.

Proposition 14.24 (Depth one under exclusive source gating). In Definition 14.23, for either reference precision Ω, let $\mathrm{k}_S = \operatorname{rank} H_S$, let h_S be the least positive eigenvalue of H_S, $\mathrm{h}_* = \min_S \mathrm{h}_S$, and $\mathrm{z}_S = [\operatorname{pdet} H_S \cdot \det(\Omega|_{W_{\mathrm{att}(S)}})]^{-1/2}$, where pdet is the product of the positive eigenvalues and $\Omega|_W = \mathsf{Q}_W^{\top}\Omega\mathsf{Q}_W$ for a matrix Q_W whose columns form an orthonormal basis of W; let $\mathcal{M}$ be the set of assignments of minimum rank k_S.

(a) *Two restoring bands.* For every S and $\mathrm{T} > 0$, the eigenvalues $\lambda_1 \le \cdots \le \lambda_{2N+|E|}$ of $H_S + \mathrm{T}\Omega$ satisfy $\lambda_k \in [\mathrm{T}\|\Omega^{-1}\|^{-1}, \mathrm{T}\|\Omega\|]$ for $k \le 2N+|E|-\mathrm{k}_S$ and $\lambda_k \in [\mathrm{h}_S, \|H_S\|+\mathrm{T}\|\Omega\|]$ for $k > 2N+|E|-\mathrm{k}_S$. The range of H_S is $W^{\perp}_{\mathrm{att}(S)}$, the span of the attended residual covectors, and the prior enters q_Ω only through the sheet weights $\pi_S Z_S$, never through H_S.

(b) *Gating wells.* With

$$V_\pi(\mathbf{x}) = -\mathrm{T}\sum_i \log \sum_{j\in\mathcal{N}_i} \pi_{ij} e^{-\mathcal{D}_{e(i,j)}(\mathbf{x})/\mathrm{T}}, \qquad V_{\min}(\mathbf{x}) = \sum_i \min_{j\in\mathcal{N}_i} \mathcal{D}_{e(i,j)}(\mathbf{x}),$$

one has

$$0 \le V_\pi(\mathbf{x}) - V_{\min}(\mathbf{x}) \le \mathrm{T}\sum_i \log \frac{1}{\min_{j\in\mathcal{N}_i}\pi_{ij}} \qquad \text{for every } \mathbf{x}, \tag{14.56}$$

the configuration marginal of q_Ω has density proportional to $\exp[-V_\pi(\mathbf{x})/\mathrm{T} - \tfrac{1}{2}\mathbf{x}^{\top}\Omega\mathbf{x}]$, and if every edge of an edge set F covering V has zero residual at $\mathbf{x}$, then $V_{\min}$ vanishes at $\mathbf{x}$ and at every configuration obtained from it by translating the belief and model means of the agents of one component of (V, F) by a common amount with the links unchanged.

(c) *Rank, minimum edge covers, and sheet weights.* For every $F \subseteq E$, $\operatorname{rank}\mathbf{a}_F = |F| + N - \mathrm{cc}(F)$. Hence $\mathrm{k}_S = |\operatorname{att}(S)| + N - \mathrm{cc}(\operatorname{att}(S)) \ge 2(N - \mathfrak{n})$, with equality exactly when $\operatorname{att}(S)$ is a minimum edge cover of (V, E); the minimum edge covers are the spanning forests of stars with $\mathfrak{n}$ components, every one of them is attended by some assignment, and an edge outside $\operatorname{att}(S)$ joining two of its components raises the rank by two. The sheet weight satisfies

$$\Big(1 + \frac{\mathrm{T}\|\Omega\|}{\mathrm{h}_S}\Big)^{-\mathrm{k}_S/2} \le \frac{Z_S}{(2\pi)^{(2N+|E|)/2}\mathrm{z}_S\mathrm{T}^{\mathrm{k}_S/2}} \le 1, \tag{14.57}$$

and consequently, for $0 < \mathrm{T} \le 1$ and every integer $m \ge 1$,

$$q_\Omega\{S : \mathrm{cc}(\operatorname{att}(S)) \le \mathfrak{n} - m\} \le \mathrm{K}_m\mathrm{T}^m, \qquad \mathrm{K}_m = \Big(1 + \frac{\|\Omega\|}{\mathrm{h}_*}\Big)^{N-\mathfrak{n}} \frac{\sum_{S:\mathrm{cc}(\operatorname{att}(S))\le\mathfrak{n}-m}\pi_S\mathrm{z}_S}{\sum_{S\in\mathcal{M}}\pi_S\mathrm{z}_S}. \tag{14.58}$$

(d) *Stationary block law.* Let s_* be the least, over all S and all edges $e \notin \operatorname{att}(S)$ joining two components of $\operatorname{att}(S)$, of the least positive singular value of $\mathbf{a}_e P_{W_{\mathrm{att}(S)}}$, where P_W is the orthogonal projection onto W. When no such pair (S, e) exists, put $\mathrm{s}_* = +\infty$ and $1/\mathrm{s}_*^2 = 0$. For $\mathrm{a} > 0$ and $0 < \mathrm{T} \le 1$, the probability under q_Ω that $\mathfrak{P}_{\mathrm{a}}(\mathbf{x})$ is not the partition of V into the components of a

minimum edge cover is at most

$$\mathrm{K}_1\mathrm{T} + \frac{|E|\|\Omega\|}{2\mathrm{s}_*^2}\mathrm{a}^2 + 4|E|e^{-\mathrm{a}^2/(4\mathrm{T})}. \tag{14.59}$$

[ESTABLISHED]

The proof is in appendix C.8. In the four-cycle, $\mathfrak{n} = 2$ and the minimum edge covers are the two perfect matchings, so (c) gives them all but $O(\mathrm{T})$ of the stationary mass of either law and (d) makes coherence of an unattended bridge at resolution a cost $O(\mathrm{a}^2)$ per bridge. A second coalescence that is held by an attended bridge at stationarity with probability bounded below as $\mathrm{T} \to 0$ therefore needs a model outside the family.

Open Problem 14.25 (Repeated endogenous formation). Construct, at c_*, a second endogenous coalescence with inherited source interactions and retained marks, in a family where a matching or partition forms from initially nonmatching source assignments, generative parameters are learned, and the episode length is not fixed in advance. What holds is Proposition 14.22, one duration-one episode seeded by the initial assignment, Theorem 14.17, whose hierarchy is supplied, and Proposition 14.24, by which in the gated translation family of Definition 14.23 the stationary coherent-block partition is, up to the error of eq. (14.59), the partition into the components of a minimum edge cover, and every further merge at stationarity costs a factor T. The question includes arbitrary families and initial laws, mixed-level source support, permanent membership under later forcing, and control as the depth tends to infinity; the intrinsic selector and the hierarchy question are Open Problems 10.20 and 13.30. It is posed for models at c_* with source-gated attention on a finite connected graph whose energies, precisions, prior ratios, hyperparameters, and mobilities do not depend on the temperature T, as $\mathrm{T} \to 0$, outside the family of Definition 14.23. A second coalescence in such a model is declared by a handoff time, deterministic or a stopping time; a first-stage event of the history up to the handoff, with its block partition; a second-stage event, with windows measured from the handoff, whose block is a union of at least two first-stage blocks; one statistic of the handoff state, namely the probability of the second-stage event, the probability that it has occurred by declared times, or the probability of a declared block; and a handoff class, fixed before the comparison, that contains every first-stage handoff and, with probability bounded below, the handoffs of histories without first-stage formation. The problem is settled by such a coalescence, with learned generative parameters and from initially nonmatching assignments, whose second-stage event does not occur with probability tending to one uniformly over the handoff class, whose statistic after first-stage formation exceeds its value after a counterfactual handoff in the class by a margin bounded below, whose block is held at a declared resolution by a restoring interaction whose rate times the squared resolution, divided by T, tends to infinity on the sheets the kinetics select, on horizons short compared with the stationary recurrence time of the event measured by its rate of entrance from a set of stationary mass bounded below, and with parents whose conditional and dynamical errors vanish separately; or by a proof that in this class every coalescence of first-stage blocks on such horizons fails the margin or the restoring clause or has a second-stage event occurring with probability tending to one uniformly over the handoff class.

Physicist's summary

In the triangle chain a supplied hierarchy of stiffnesses $\mathrm{J}_1 \gg \cdots \gg \mathrm{J}_L = 1$ makes the prepared displacement relax like overdamped springs of very different stiffness: the stiffest bonds equilibrate first and glue their endpoints, and each softer level glues the clusters already formed (Theorem 14.17). Every parent keeps every loop and source assignment, starts from the complete current law, and learns its own parameter; the price of each reduction vanishes in an extreme regime (temperature ϵ^{86} at depth two) in which nothing adaptive decides the outcome. In the homogeneous four-cycle, without a supplied hierarchy, an edge carries energy only when an endpoint attends to it; attended edges contract, the prepared circulation stretches the unattended ones, and the posterior keeps attending to the contracted edges, so a pair forms with probability bounded away from zero (Proposition 14.22). The same gating is why the story stops there in that family (Proposition 14.24): an unattended edge between two formed blocks feels only the thermal reference, so at stationarity the blocks are the components of a minimum edge cover and each further merge costs a factor T.

Chapter 15

Interpretation and Its Limits

What do the preceding chapters commit to, and which readings of them could an observation or a proof count against? The chapter's readings are definitions, hypotheses, and open problems, and it proves a proposition and a theorem. Proposition 15.4 shows that on the four-agent cycle of section 14.9 a collective forms from every Gaussian initial law with probability bounded below uniformly in the regime parameter, whether or not the source prior learns, because the random initial assignment seeds the pair; its proof is in appendix C.7. The main statement, theorem 15.5, sharpens the participatory reading: at a sufficiently small fixed temperature, slow learning of the source prior selects one pairing at the level of the law from almost every Gaussian initial law, while a frozen prior leaves the long-time pairing law symmetric. It feeds open problem 15.8.

Beyond section 1.4, the chapter declines the claims that a structural-realist, Neo-Kantian, or noumenal reading follows from the invariance results or is selected by the mathematics; that the lineage citations below prove the probability construction or establish its interpretation; that definition 15.1 follows from constructive empiricism; that participation entails consciousness or that variational free energy is a physical action principle; and that a constructed record sensitive to link holonomy is a canonical observable or an empirical realization. Its open items are open problems 15.7 and 15.8. [NOT-CLAIMED]

15.1 Scaffolding, idle posits, and structural readings

The verdict of the preceding chapters is that the architecture is local-first, not background-free. Agent i carries the law points

$$q_i^b \in (\mathcal{E}_b)_{c*}, \qquad q_i^m \in (\mathcal{E}_m)_{c*}, \tag{15.1}$$

which the cross maps Φ and $\widetilde{\Phi}$ do not identify, and the base, principal bundle, and measurable spaces are coordination scaffolds that make agent-relative interfaces comparable and that local inference laws do not construct. The verdict would change if edges became random structural coordinates updated by inference (open problem 6.9) or links were recovered from population records, with single-point part open problem 15.8. A Neo-Kantian reading takes local frames, model evaluations, and access maps as constitutive but revisable conditions of a record's inferential meaning, in the lineage of Kant, Reichenbach, Cassirer, and Friedman [20, 37, 62, 87]; a noumenal reading takes the base and bundles as structure reached only through agent-side representations. As scaffolding is no inferential target, the base's lack of a law decides neither reading.

Definition 15.1 (Idle-wheel criterion). A posit is idle when changing it alters the law of no record and of no statistic of a record in the model. The idle-wheel criterion removes idle posits by parsimony. It is a declared rule of reading.

The criterion is close in spirit to constructive empiricism, on which accepting a theory involves belief only in its empirical adequacy [99, 11–12], but it is declared, not derived. A closed walk p and an edge copy e determine the classes

$$[\mathrm{Hol}^b(p)]_{\mathrm{conj}}, \qquad [\mathrm{Hol}^m(p)]_{\mathrm{conj}}, \qquad [\mathring{\Theta}^x_e]_{\mathrm{conj}} \tag{15.2}$$

of the link holonomies (2.30) and the per-edge twists, invariant under passive and active changes (proposition 2.12). On a graph-flat family the loop classes are trivial (proposition 2.13) and give a noumenal reading no work; a nontrivial one is a referent only within the interaction complex, made operational at c_* by theorem 11.30.

The interface theory of perception permits an arbitrary measurable relation between representation and world [54, 55]. Here the principal fiber at c_* is a G-torsor, a frame is the choice of a reference point, and what survives a change of frame is structural data,

$$\mathcal{I}_{\mathrm{struct}} = \Big\{\text{ELBO and relative-entropy values},\ [\mathrm{Hol}^b(p)]_{\mathrm{conj}},\ [\mathrm{Hol}^m(p)]_{\mathrm{conj}},\ \big[(\mathring{\Theta}^b_e, \mathring{\Theta}^m_e)_e\big]\Big\}, \tag{15.3}$$

the last entry being the per-edge twists of both channels modulo one simultaneous conjugation, invariant by proposition 4.9, (3.24), and proposition 2.12. Of the three agreements of proposition 2.18, frame agreement is independent of the other two, law coherence implies mean agreement but not conversely, and none selects a partition (open problem 10.20). These invariants are compatible with epistemic structural realism [107], with moderate structural realism, on which objects share the ontological footing of relations and are characterized only by them [35], and with ontic structural realism, on which structure is all that exists [70, 71]. No argument from theory change is given, since these are invariants of passive frame change within one theory whereas Worrall's structure is retained across theories, and nothing here selects among these positions or the Neo-Kantian reading. Support would need, respectively, an invariant retained across a change of theory, a typed interpretation of the formalism's objects and relations in the moderate ontology, and an argument from formal unidentifiability to ontological elimination with discriminating evidence.

Chapter 9 changes the status of the links, not of the scaffolding. Inferred as Lie-algebra generators under a declared prior, the links take the formal role of a latent field: stationarity balances the stress of the prior against a matter current that, for a comparison record, has the outer-product form of a delta rule (proposition 9.21 and corollary 9.22); near zero generators the stationarity operator has the form of a massive Euclidean vector operator in the episode index, with the matter as a screening mass (proposition 9.34); and the held-link model is the zero-width limit of the latent one (theorem 9.25). These are correspondences of form, which that chapter declines to read as a physical field or as a learning architecture. The base, the principal bundle, the frames, and the reference point remain structural data that inference does not construct, so the verdict above is

unchanged, and links inferred within the model are not links recovered from the records of a target system, which remains part of open problem 15.8.

15.2 Participation, formation, and testability

Hypothesis 15.2 (Operational observational closure). Every datum directly available to an interior agent is a clamped state or a message of another interior agent or of a boundary environment node, an operational environment node in the sense of definition 5.16. The hypothesis excludes data that reach a record kernel other than through such a node. It does not remove the conditioning record, does not require a boundary node to be autonomous, and does not exclude an unobserved environment that affects all agents without becoming the argument of any agent's belief.

By theorem 5.17 the hypothesis is free for kernels but removes no conditioning: a statistic of the record is posterior-sufficient exactly when the conditional mutual information in (5.35) vanishes, and a latent-independent kernel is a zero-information boundary message (5.34), not an agent.

Hypothesis 15.3 (Every boundary node is called an agent). Every boundary environment node of hypothesis 15.2 is called an agent. The hypothesis is a naming decision. It adds to no node a persistent state, an action, a Markov blanket, or a local free energy, and it excludes reading the constructed joint $\mathbb{P}_{\theta,V}$ or the selected coupling $\mathbb{Q}_{V,o,X}$ as an agent: these are complete joint laws at the population level. A block conditional is a meta-agent only after its own coarse state, observation interface, evaluated generative mechanism, recognition law, and update rule have been constructed, as in chapters 12 and 13. A strong agent-only ontology would additionally need a criterion of agency for every boundary node, and the formalism supplies no unique one.

Participation is reciprocal inferential dependence through records inside one joint, with no live posterior in a same-step generative factor (hypothesis 4.3). Section 13.7 locates it in three sequences of fixed joints: the same-time inference loop, block coordinate ascent as in any connected hierarchical model under mean-field updates, which resembles Wheeler's participatory proposal [105] without being derived from it or evidence for it; parameter learning, whose optimal prior over an unrestricted family is the aggregate recognition law (13.39); and delayed feedback through (13.40), which for an update reading a declared recognition kernel is the free energy of one joint (proposition 13.28) and whose dynamic closure is part of open problem 12.40. None of these separates the program from a hierarchical model with learned priors; formation could. Theorem 14.17 forms successive collectives under a supplied energy hierarchy and source support, without showing that source learning creates the contrasts, and proposition 14.22 forms one collective on a homogeneous four-cycle with probability bounded below uniformly in its regime parameter, from a symmetric preparation whose random initial source assignment seeds the pair, and without parameter learning. The same seed forms a collective from every Gaussian initial law, whether or not the source prior learns.

Proposition 15.4 (Seeded contrast formation with idle source learning). Take the homogeneous four-cycle experiment of definition 14.21, with its regime parameter $0 < \epsilon \leq 1/100$, the constants of eq. (14.51), and its records, evaluator, reference law, and kinetics, and modify it in two ways. First, the uniform prior on assignments becomes a product $\pi_S = \prod_i \pi_{i,S_i}$ of receiver rows, each a generative parameter that starts uniform and learns from the recognition law q_t of the kinetics by

$$\dot{\pi}_{ij} = \epsilon\big(q_t(S_i = j) - \pi_{ij}\big), \tag{15.4}$$

the natural-gradient flow of the free energy in the prior row under the Fisher metric of the simplex at mobility ϵ; its rest point is the recognition label marginal, as for the aggregate prior (13.39), and the assignment is redrawn from posterior rows computed with the current prior. Second, the initial law is $\mathcal{N}(\mathrm{m}_0, I_{12})$ on configurations, with an arbitrary mean $\mathrm{m}_0 \in \mathbb{R}^{12}$, times the uniform law on assignments. Call contrast formation the event that some connected block I with $2 \leq |I| \leq |V| - 1$ has, at time zero, an internal edge whose belief transported relative entropy exceeds $1/4$, and, on some time interval of length one inside $[0, 2/\epsilon]$, all its internal belief and model transported relative entropies at most ϵ^4 while every belief transported relative entropy on an edge joining it to its complement exceeds $1/2$, the bounds of part (a) of proposition 14.22; the first clause makes the event an entrance into the contrast and excludes a contrast present at time zero that persists. Then:

(a) *Rows.* The kinetics and eq. (15.4) have exactly one joint solution on $[0, \infty)$, with continuous deterministic rows, and $|\pi_{ij}(t) - 1/2| \leq \epsilon t/2$ for all $t \geq 0$.

(b) *Formation.* For every $\mathrm{m}_0 \in \mathbb{R}^{12}$ and every $\epsilon \in (0, 1/100]$,

$$\begin{aligned}\Pr(\text{contrast formation}) &\geq \frac{e^{-1/4}}{16} \sum_{\pm} \mathcal{N}(\mathrm{m}_0, I_{12})\big\{|\mathbf{x} \mp \mathbf{x}^\circ| < \varrho_\circ\big\} \\ &\geq \frac{e^{-1/4}\varrho_\circ^{12}}{737280} \exp\Big[-\tfrac{1}{2}\big(|\mathbf{x}^\circ - \mathrm{m}_0| + \varrho_\circ\big)^2\Big] > 0,\end{aligned}$$

and the same bound holds when the rows are frozen at $1/2$. The bound is carried by the sub-event in which I is a matched pair of the initial assignment, held throughout $[0, t_*^\circ]$, and the contrast is that of part (a) of proposition 14.22. This sub-event is a coherence episode with the restoring interaction and rates of that part, and for every fixed row configuration π the kinetics at rows π leave $q_\pi^\star(\mathbf{x}, S) \propto \pi_S e^{-U_S(\mathbf{x})/\mathrm{T}}$ invariant and, started from it, realize the sub-event with probability at most $4e^{-1/(4\mathrm{T})}$; so for every m_0, in either arm, the sub-event exceeds every such stationary value by a margin bounded below as $\epsilon \to 0$, the first display not depending on ϵ.

(c) *Invariant initial laws.* The initial law is invariant under the cycle automorphisms, with link signs transformed under reflections, exactly when $\mathrm{m}_0 = (\mathrm{c}_1\mathbf{1}_4, \mathrm{c}_2\mathbf{1}_4, 0)$ for some $\mathrm{c}_1, \mathrm{c}_2 \in \mathbb{R}$, and then $\pi_{ij}(t) = 1/2$ for every receiver, source, and $t \geq 0$. [ESTABLISHED]

The proof is in appendix C.7. The probability in part (b) is supplied by the preparation's matching seed, an initial assignment drawn uniformly and independently of a configuration law of full support, and source learning is idle in it. A statement in which learning creates a contrast must therefore

compare learned and frozen priors and concern what the law selects, not the probability of a sample-path episode.

Theorem 15.5 (Pairing selection by slow source learning). Take the homogeneous four-cycle experiment of definition 14.21, with its records, evaluator, reference law, and kinetics, at a fixed temperature $\mathrm{T} > 0$ in place of ϵ^8; let the product receiver rows of proposition 15.4 start uniform and learn by eq. (15.4) with a mobility $\epsilon_{\mathrm{L}} > 0$ in place of ϵ; and let the initial law be $\mathcal{N}(\mathrm{m}_0, I_{12})$ on configurations times the uniform law on assignments. Write $q_t^{\epsilon_{\mathrm{L}}}$ for the recognition law of the kinetics and q_t^0 for that of the frozen-prior arm, whose rows stay at $1/2$. First, there is $\mathrm{T}_0 > 0$ such that for every $\mathrm{T} \in (0, \mathrm{T}_0]$ there is $\mathrm{C}_{\mathrm{T}} < \infty$ for which, for Lebesgue-almost every $\mathrm{m}_0 \in \mathbb{R}^{12}$, one of the two perfect matchings $M \in \{S^{(0)}, S^{(1)}\}$ satisfies

$$\lim_{\epsilon_{\mathrm{L}} \to 0} \sup_{t_{\mathrm{L}} \le t \le 2t_{\mathrm{L}}} q_t^{\epsilon_{\mathrm{L}}}(S \neq M) = 0, \qquad t_{\mathrm{L}} = \mathrm{C}_{\mathrm{T}} \epsilon_{\mathrm{L}}^{-1} \log(1/\epsilon_{\mathrm{L}}).$$

Second, for every $\mathrm{T} > 0$ and every m_0, $q_t^0(S = S^{(0)}) - q_t^0(S = S^{(1)}) \to 0$ as $t \to \infty$. [ESTABLISHED]

The proof is in appendix C.9.

Corollary 15.6 (Joint selection along a temperature schedule). In the model of theorem 15.5, fix any sequence $\mathrm{T}_n \downarrow 0$ within its temperature interval and any $\eta_n \in (0, 1)$ tending to zero. There are thresholds $d_n \downarrow 0$, independent of the initial mean, such that for Lebesgue-almost every fixed m_0, all sufficiently large n, and every $0 < \epsilon_{\mathrm{L}} \le d_n$, a matching $M_n(\mathrm{m}_0)$ satisfies

$$\sup_{t_{\mathrm{L}} \le t \le 2t_{\mathrm{L}}} q_t^{\epsilon_{\mathrm{L}}, \mathrm{T}_n}(S \neq M_n) \le \eta_n, \qquad t_{\mathrm{L}} = 2\epsilon_{\mathrm{L}}^{-1} \log(1/\epsilon_{\mathrm{L}}).$$

The same thresholds can make the learned law's relative entropy to its instantaneous invariant law, and the frozen law's relative entropy to the uniform-row invariant law, at most η_n throughout this window. Consequently the frozen matching probabilities tend to $1/2$ each. For any prescribed $a > 7/2$, the thresholds can also satisfy $d_n \le \mathrm{T}_n^a$. They are existence thresholds, rather than a proved power-law selection condition; the selected matching may depend on n. [ESTABLISHED]

The proof and an explicit temperature-dependent tracking bound are in appendix C.10. Setting $\mathrm{T}(\epsilon_{\mathrm{L}}) = \mathrm{T}_n$ for $d_{n+1} < \epsilon_{\mathrm{L}} \le d_n$ gives a joint limit with a schedule common to almost every initial mean. Proposition C.16 identifies what persists on this late window. At residual resolution $\mathrm{a} = \mathrm{T}^{1/4}$, the selected pairing remains the coherent partition for a following window of length $\mathrm{T}^{-1/4}$, with probability tending to one when the schedule tolerances satisfy $\eta_n \le \mathrm{T}_n$. The frozen arm approaches the equal mixture of the two persistent pairings. The statement controls the source redraws of the kinetics; its comparison with a held-sheet Gaussian process is an estimate on path laws. This is restored equilibrium coherence.

The first contrast block is a different statistic. Proposition C.17 keeps the contrast event and the early horizon of proposition 15.4, writing $\xi = \mathrm{T}^{1/8}$ and separating the learning mobility ϵ_{L} from ξ. If $\epsilon_{\mathrm{L}} = o(\mathrm{T}^{1/4})$, the learned and frozen laws of the first block, each conditioned on contrast formation,

have total variation tending to zero for every fixed Gaussian initial mean. Thus the slow schedules can select different late pairing laws while leaving the early first-block comparison unchanged. The original regime $\epsilon_{\mathrm{L}} = \xi$ is outside this estimate.

The retained channel in part (b) of proposition 14.22 also applies after this late coherence. Complete a duration-one coherence prefix at $\theta = t_{\mathrm{L}} + 1$, and initialize the parent from the full recognition law conditioned on that prefix, pushed through the channel. Propositions C.18 and C.20 give, along corollary 15.6 with tolerances $\eta, \delta \leq \mathrm{T}$ and mobility cap $\epsilon_{\mathrm{L}} \leq \mathrm{T}^2$, separately vanishing discarded conditional relative entropy and retained-path total variation over a following interval $\mathrm{T}^{-1/4}$. The parent inherits the rows at activation and then learns from its own actual law. It retains the assignment, the loop, and the common fast normal coordinate; no independent Gaussian preparation is substituted for its activation law. These are averaged guarantees for this declared past event. They do not assert closure conditional on every realized history or activate a parent at the first contrast.

The same channel can be activated early, on the seeded sub-event of the original contrast, at every mobility $\epsilon_{\mathrm{L}} \in [0, 1]$ and so at $\epsilon_{\mathrm{L}} = \xi$. By proposition C.25, the discarded conditional relative entropy at a deterministic activation time θ vanishes when $\theta - \log(1/\xi) \to \infty$ and $\xi^4\theta \to 0$, and diverges when $\log(1/\xi) - \theta \to \infty$; proposition C.27 adds a vanishing retained-path error on following windows short compared with T^{-1}. By proposition C.26, activation at the first qualifying completion instead has diverging conditional error, also when conditioned on the whole contrast event. These results concern the seeded sub-event and leave the comparison of learned and frozen first blocks at $\epsilon_{\mathrm{L}} = \xi$ open.

The limit $\epsilon_{\mathrm{L}} \to 0$ at fixed T places the mobility of the rows below any positive mixing rate of the kinetics at fixed rows, and t_{L} is, up to its constant, the time in which an imprint of order ϵ_{L} on the rows that grows at a rate of order ϵ_{L} reaches order one. The asymmetry that learning would amplify is supplied by the initial law, not by the generative model or the kinetics: by part (c) of proposition 15.4, learning from a Gaussian initial law that is invariant under the cycle automorphisms leaves the rows uniform for all time. The theorem amplifies the asymmetry of $\mathcal{N}(\mathrm{m}_0, I_{12})$ and excludes a null set of means. The frozen arm is the null against which selection is measured. This is a result about the one-time assignment law at fixed temperature. The conditional block events, a power-law vanishing-temperature regime, and conditional and dynamical parent errors of open problem 14.20 remain separate obligations. The participatory reading requires collectives constituted by the coupled inference, with a restoring interaction and a dependence on the preceding state. Assignment selection alone does not establish that recursive claim. The noumenal reading buys structure behind the agents' representations with posits that definition 15.1 removes unless an observable such as a loop holonomy carries them.

Open Problem 15.7 (Physical-law identification). Identifying an object produced by the coarse-graining of chapter 12, or by its limits, with a physical law requires a named target system, an observable and an estimator, a validation and uncertainty protocol set before the data, a baseline margin, probability-level closure, and held-out evidence that is robust across admissible blocking

schemes. None is supplied here, and the mathematics neither proves a physical law nor supports ontological closure. For a named target system whose data the model does not generate, observed through a declared record kernel that, under the model's own law, separates two objects of the coarse-graining that differ in a gauge-invariant datum, with an observable, an estimator, and a validation and uncertainty protocol set before the data, the problem is settled by held-out evidence, robust across admissible blocking schemes, that an object of the coarse-graining meets a baseline margin with probability-level closure, or by held-out evidence under the same protocol that the baseline margin is not met.

Graph flatness (proposition 2.13) and the fixed-sector condition of proposition 11.19 check declared data but detect no bundle topology or base curvature, so an interpretation must keep graph-link holonomy, base-connection holonomy, bundle topology, and gauge redundancy apart.

Open Problem 15.8 (Single-point testability). State an observation at c_* that could count against a claim of the program. What holds is the following: theorem 11.30 constructs a population record and a frame-invariant statistic whose law separates two independent-link configurations that differ only in the conjugacy class of a loop holonomy, and theorem 15.5 establishes a learned pairing selection against a symmetric frozen-prior arm in one declared family. What is missing is a target system: sampling units, a noise model, an estimator of gauge-typed parameters with its uncertainty, a held-out statistic, an acceptance tolerance set in advance, and baselines without link holonomy or without formation. A cross-scale test additionally needs two or more admissible partitions, coarse channels, gauge alignment, and held-out coarse statistics predicted from the fine fit without coarse refitting, and its failure would falsify the added cross-scale hypothesis, not the finite-dimensional theorems. For a named target system at c_* whose data the model does not generate, observed through a declared record kernel, with sampling units and a noise model, that reads the links of a closed walk or the coordinates of a candidate block, the problem is settled by a test fixed before the data, with an estimator of gauge-typed parameters, a held-out statistic, an acceptance tolerance, and baselines without link holonomy or without formation, whose failure would falsify the presence of nontrivial holonomy on that walk or of formation of that block by the inference in the target, together with a proof that under the declared kernel the law of the tested statistic depends on that hypothesis; or, negatively, by a proof that under the declared kernel every statistic of the records has the same law for two link configurations that differ only in the conjugacy class of the walk's holonomy, or the same law whether or not the block is coherent at the declared resolution.

Physicist's summary. Beyond scaffolding, frame-change invariants select no realist position and environment messages create no agents; coordinate ascent, learning, and delayed feedback do not distinguish the program from hierarchical modeling, but formation by the inference would.

Chapter 16

Related Work

Where does this construction sit among existing work, and what does it owe to it? The central comparison, with Bayesian renormalization in section 16.5, rests on proposition 5.11 and corollary 10.8, and the open bridge problem of Supplement G builds on it.

No construction below is identified with this one or adopted as authority beyond its cited statement, and qualifications of other works are this manuscript's reading. [NOT-CLAIMED]

16.1 Gauge theory, free energy, and multi-agent active inference

Sengupta et al. [91] propose, in an essay, a gauge theory of neuronal dynamics in which variational free energy plays the role of a Lagrangian, gauge fields arise from the information geometry of the brain's probability distributions, and attention, the cognitive counterpart of precision weighting, is a force produced by the curvature of that geometry. Here the group acts on the sample fibers of one principal bundle (chapter 2), links are independent group elements on edge copies at one base point (definition 2.11), and attention is a latent source label with a derived posterior (proposition 6.2) whose precision is that of a record (proposition 6.19).

In federated inference [39], agents share a generative model of a shared world and communicate beliefs through a likelihood mapping to overt expressions, and language emerges when agents share an econiche. Heins et al. [51] obtain cohesion, milling, and directed motion from agents minimizing surprise under generative models of their neighbors, recovering social forces as suppressed prediction errors without a group-level model. Thestrup Waade et al. [96] treat a collective whose group-level Markov blanket consists of its sensory agents' observations and its active agents' actions as a group-level agent, fit a model of the same structure to simulated collective behavior, and find the individual and group models nontrivially related. Here agents may carry inequivalent representations and independent links, their encoder, transmission, and decoder kernels enter one joint with one evidence bound (definition 6.10 and chapter 5), parent laws follow from the fine joint by exact coarse channels (theorem 10.7 and corollary 10.11), formation is proved for a stated family (chapter 14), and a low-rank interface does not by itself give a Markov blanket (proposition 6.26).

16.2 Connection Laplacians, sheaves, and equivariant networks

Singer and Wu [93, Section 3 and Theorem 5.1] build a graph connection Laplacian from edge weights and orthogonal transformations aligning the tangent spaces of neighboring sample points and prove its convergence to the connection Laplacian of the underlying manifold; Hansen and Ghrist [47, Section 3.6] identify it as the sheaf Laplacian of an orthogonal vector bundle over a graph, in a spectral theory of sheaves whose discrete Hodge theorem identifies harmonic cochains with sheaf cohomology [47, Theorem 3.1]. Our quadratic energies with represented links are of this type, with kernel the holonomy-fixed sector (proposition 11.19), but with links in G acting through representations, semidefinite weights, and normalized laws; the quadratic disagreement of definition 6.15 is the edge term of a weighted sheaf energy with rectangular restriction maps, as in the discourse sheaves of selective expression [48]. Hansen and Ghrist [47, Section 4.2] show that Kron reduction is in general impossible for sheaves, since an eliminated vertex can impose constraints between boundary vertices that no pairwise restriction maps express, and possible for vertex stalks of dimension at most one, where the Laplacians have factor width at most two [47, Theorem 4.3]; this is prior art for the Kron obstructions of matrix-weighted Gaussian energies that Supplement G classifies, whereas Galerkin closure (theorem 11.27) restricts a form to a subspace.

Group-equivariant convolutional networks share weights across the orbits of a global symmetry group [22], gauge-equivariant ones make a network on a manifold independent of the choice of local frames [23], and a gauge-equivariant transformer applies multi-head self-attention to feature fields on meshes with a parallel transport that makes the layer agnostic to the orientation of local coordinate systems [50]. These constrain learned maps over a base manifold with a connection. Here links are independent dynamic data at one base point, a nonidentity per-edge twist has no counterpart in an architecture with one shared key map (table 6.1), and corollary 6.11 and proposition 6.13 describe how declared data transform, not constraints on a learned layer.

16.3 Lattice gauge theory, induced actions, and learning rules

Lattice gauge theory places group elements on the links of a lattice and weights them by traces of plaquette holonomies [106, Eq. (3.29)]; its Hamiltonian form in the temporal gauge is that of Kogut and Susskind [68], derived from the transfer matrix of the Wilson action in Creutz [25, Ch. 15, Eq. (15.41)]. On holonomies that are isometries of the fiber metric, the loop defect of definition 9.8 is the Wilson plaquette term in the normalization of Creutz [25, Ch. 7, Eq. (7.6)] (proposition 9.10), and the time-like loop of definition 9.31 with identity time-like links is the temporal-gauge configuration of Creutz [25, Ch. 15, Eq. (15.9)]. Here the links live on the edge copies of an interaction multigraph at one base point, the episode index is an inference index, and the reference measure is a Gaussian law on the Lie algebra instead of the Haar measure, which normalizes the prior for noncompact and general linear groups at the cost of anchors and covariances that a change of reference point transforms (proposition 9.12); on the general linear group the defect needs a fiber metric, since no continuous class function separates unipotent holonomy from the identity (corollary 9.7).

In induced gauge theory the action of the links is generated by integrating out matter: Kazakov and Migdal [63, Sec. 2, Eqs. (2)–(4)] induce the Yang–Mills interaction of a lattice model by a heavy scalar field in the adjoint representation. The induced link action of proposition 9.27 arises in the same way from undirected matter normalized jointly with the generators, and it vanishes for matter attached by record extension. For Gaussian matter with pairwise potentials of connection-Laplacian form it is one half the log-determinant of a connection Laplacian plus a self term (section 9.7), and Kenyon [65, Theorems 1 and 9] expands the determinant of a line-bundle Laplacian, and the quaternion determinant of the Laplacian of an $\mathrm{SL}(2,\mathbb{C})$ connection, as sums over cycle-rooted spanning forests with the weight $2 - w - w^{-1}$ or $2 - \operatorname{Tr} w$ for each cycle, w the monodromy of the cycle.

The current of a comparison record (corollary 9.22) has the outer-product form of the delta rule that Schlag et al. [89, Sec. 4.2, Eq. (24)] write for fast-weight programmers, a line of work they trace in their Sec. 2 to Schmidhuber [90]; here it is the source of a stationarity condition for a latent generator, transported by the derivative of the exponential map and averaged over the recognition law of the generators, and the correspondence is one of form. The recognition flow of the generators is natural-gradient flow in the Fisher geometry of their Gaussian recognition family, and its explicit Euler step is the Gaussian form of the Bayesian learning rule [66, Eq. (12)] (proposition 9.28); the mean and precision conditions of the link field equation use the Gaussian derivative identities that Opper and Archambeau [84, Eqs. (18)–(19)] use for variational Gaussian approximations (lemma 9.20 and proposition 9.21).

16.4 Averaging and path-space coarse-graining

Hartmann et al. [49, Section 2.1.4, Theorem 2.9] bound path-space relative entropy for an effective diffusion under regularity, conditional functional inequalities, and Novikov's condition, with the resolved diffusion coefficient independent of unresolved coordinates. Their fixed-time estimates in Theorems 2.4 and 2.7 have a different target. Neither estimate supplies the growing-history, switching-source comparison requested by open problem 12.40. The one-time selection result of theorem 15.5 uses the separate entropy, transient imprint, and saddle-escape estimates in appendix C.9. A change in diffusion covariance can instead make continuous-path laws singular. Backhoff-Veraguas and Unterberger [5, Remark 1 and Theorem 1] study a partition-refinement relative entropy for martingale diffusions; that quantity is distinct from ordinary path-space relative entropy. The filter and memory constructions of section 8.3 retain the dynamical information that a stationary marginal alone omits.

16.5 Information-geometric and Bayesian renormalization

Bény and Osborne [9] formulate the renormalization group as a family of quantum channels connecting the microscopic world to the observable world at a given scale, endow the space of Hamiltonians with a metric induced by an information geometry on states, measure the information

lost along a flow by the distinguishability of thermal states, and introduce functions of two-point correlations that do not increase along the flow; theorem 7.8 is only the classical Fisher-information statement for one parameter-independent kernel at one parameter value.

Bayesian renormalization, as formulated by Berman, Klinger, and Stapleton, builds on the dynamical Bayesian inference of Berman et al. [11] and uses Fisher distinguishability on model-parameter space to define an information scale and to separate stiff from sloppy directions. In the late-data regime the posterior is a Gaussian whose covariance is the inverse Fisher information times the scale parameter [12, Eq. (38)]. For a differentiable forward map $\mathfrak{O}$ from parameters to data they write the pushforward of that posterior as a Gaussian [12, Eq. (43)] whose covariance is the scale parameter times the pushed inverse Fisher metric

$$D_\theta^{\mathfrak{O}} = \partial\mathfrak{O}(\theta) g^F(\theta)^{-1} \partial\mathfrak{O}(\theta)^\top \tag{16.1}$$

at the posterior center [12, Eq. (44)], the diffusion tensor of a Fokker–Planck equation and a stochastic flow on data space [12, Eqs. (45) and (46)]. In our reading the delta method makes their Gaussian form the leading local approximation of the pushforward, exact when $\mathfrak{O}$ is affine, and its continuation beyond the late-data regime is a renormalization construction, not an exact nonlinear pushforward or a Bayesian asymptotic theorem. It has been inverted into a statistical-inference reading of exact renormalization flows [10] and applied to neural-network field theories [58].

The information scale ranks which directions of a parent model presentation stay distinguishable at a declared precision, but by itself it selects no graph blocks, coarse-grains no nonflat links, and leaves the two-channel interaction law open. Letting a posterior flow does not make a generative kernel read its own posterior, so hypothesis 4.3 does not refute Bayesian renormalization, but inserting a posterior from one scale as the generative top factor at another declares a new model or an explicit transition between models, whose bounds refer to different joints unless a common target and a comparison map are supplied. By proposition 5.11, pointwise identities designate no one member or evidence, and comparing two bounds shows an improvement in one quantity only on an evidence level set; for two independently declared joints with no evidence-preserving channel between them, the difference of optimized bounds mixes evidence and tightness differences and supplies no order, whereas an evidence-preserving recognition-independent Markov kernel applied to the posterior and the recognition law orders the bounds by data processing (corollary 10.8 and lemma 3.12). Relating the widening of the posterior along its scale to the addition of precision under Gaussian aggregation would need a common state space, a map between the scale parameters, and a transported monotone; Supplement G records this bridge as open.

Mehta and Schwab [78] map Kadanoff's variational renormalization group exactly onto deep restricted Boltzmann machines; a deterministic identification of coarse and fine coordinates does not supply its kernel, so the Galerkin coarse form of section 10.4.1 is not thereby a deep network.

16.6 Network renormalization, graph coarsening, and multigrid

Network renormalization addresses heterogeneous graphs, which have no lattice, momentum, or geometric length [40]. The Laplacian renormalization group of Villegas et al. [103] uses the diffusion density operator $e^{-tL}/\operatorname{Tr} e^{-tL}$ of a graph Laplacian L, its spectral entropy $\mathcal{S}(t)$ normalized by the logarithm of the number of nodes, and the specific heat

$$\chi(t) = -\frac{d\mathcal{S}}{d\log t}, \tag{16.2}$$

whose peaks they use to identify characteristic resolution scales [103, Eqs. (1)–(2)].

Remark 16.1 (Gauge behavior of heat-kernel diagnostics). Let $R \succ 0$ and L be block matrices on $\bigoplus_{i\in V} \mathbb{R}^{n_i}$, and let a vertex reframing $P = \operatorname{diag}(P_i)$ with invertible blocks act by congruence, $R \mapsto P^\top RP$ and $L \mapsto P^\top LP$. Then the generator $R^{-1}L$ becomes $P^{-1}R^{-1}LP$, so its spectrum, its heat trace $\operatorname{Tr} e^{-tR^{-1}L}$, and every spectral entropy built from them are invariant, whereas the off-diagonal blocks of $e^{-tR^{-1}L}$ become $P_i^{-1}(e^{-tR^{-1}L})_{ij}P_j$ and are not invariant scalar node affinities. [ESTABLISHED]

Proof. $(P^\top RP)^{-1}P^\top LP = P^{-1}R^{-1}LP$, and the exponential of a similar matrix is similar by the same matrix. □

A gauge-invariant block selector built from heat-kernel data therefore needs invariant statistics of the blocks, and a scale diagnostic alone selects no unique partition (open problem 10.20). Garuccio et al. [41] give a renormalization scheme for any hierarchy of heterogeneous coarse-grainings and identify a class of scale-invariant random graphs with a necessary dependence on additive node hidden variables, a characterized family rather than arbitrary hidden-variable networks; for random graphs whose link probabilities are conditioned on other links, Catanzaro et al. [21] derive a closed-form renormalization transformation for maximum coordination number two and show that higher coordination numbers admit none. These candidate partition and closure mechanisms do not by themselves handle directed row-normalized attention, two gauge representations, or nonflat link holonomy, and the aggregation of chapter 11 does not integrate out edge variables. A combined scheme would infer a partition from gauge-invariant multiplex diagnostics of edge-event laws through a structural channel with a named law and push the complete law exactly.

Graph coarsening and pooling in machine learning add hypotheses of their own: restricted spectral approximation controls spectral and cut properties [77], coarsening can carry message-passing guarantees [59], differentiable pooling learns soft cluster assignments end to end [109], and deep graph convolutional networks whose weights satisfy spectral conditions lose all information except connected components and node degrees exponentially in depth [83]. The Galerkin identity preserves the restricted quadratic energy, while spectral approximation, message-passing commutation, and a graph-neural-network realization need the further controls of section 10.4.1 (proposition 10.15). That coarse form is the one of algebraic multigrid: aggregation-based multigrid forms $P^\top AP$ with a piecewise-constant prolongation defined by disjoint aggregates [82, Section 2,

Eqs. (2.1)–(2.2)], smoothed aggregation smooths that tentative prolongation [100], and lean algebraic multigrid for graph Laplacians combines piecewise-constant interpolation, node aggregation by an affinity measure, and an energy correction of the coarse systems [75]. What is added here is the placement of that operator inside normalized laws with gauge covariance and exact evidence accounting.

Physicist's summary. Here the group acts on sample fibers with independent links at one point, and parent laws are computed from the fine joint rather than shared, omitted, or fitted as in multi-agent active inference. Sheaf Laplacians supply the linear algebra and the Kron obstruction, and renormalization flows meet the exact channels only through a common comparison target. The latent links take the Wilson plaquette and the temporal gauge from lattice gauge theory and the induced action from induced gauge theory, with a Gaussian reference on the Lie algebra in place of the Haar measure.

Appendix A

Measure-Theoretic Lemmas and Integrability Witnesses

This appendix proves the measure-theoretic statements of Chapters 3 and 5 and separates the hypotheses of hypothesis 5.3 from factorwise log-integrability (proposition 5.15) by witnesses.

A.1 Kernel measurability and proofs for the probability chapter

Theorem A.1 (Jointly measurable Radon–Nikodym version for a dominated kernel). Let $(\mathsf{E}, \mathscr{E})$ be a measurable space, and let K and L be probability kernels from E to $(\mathsf{F}, \mathscr{F})$ with $K(x,\cdot) \ll L(x,\cdot)$ for every x. Then there is a finite $\mathscr{E} \otimes \mathscr{F}$-measurable $r : \mathsf{E} \times \mathsf{F} \to [0,\infty)$ with $K(x,B) = \int_B r(x,y) L(x,dy)$ for every $x \in \mathsf{E}$ and $B \in \mathscr{F}$. Selecting a derivative separately for each x does not give joint measurability; the theorem supplies one simultaneous version. [ESTABLISHED]

Proof. Standard Borelness gives finite refining measurable partitions $(\mathcal{P}_n)_{n\geq 1}$ whose generated sigma-algebras increase to $\mathscr{F}$. With $0/0 := 0$ put

$$r_n(x,y) = \sum_{A \in \mathcal{P}_n} \frac{K(x,A)}{L(x,A)} \mathbf{1}_A(y), \tag{A.1}$$

and let $r = \limsup_n r_n$ where this is finite and $r = 0$ elsewhere; like every r_n, r is jointly measurable. For fixed x, $r_n(x,\cdot)$ is the conditional expectation of $h_x = dK(x,\cdot)/dL(x,\cdot)$ given $\sigma(\mathcal{P}_n)$ under $L(x,\cdot)$, so martingale convergence gives $r_n(x,\cdot) \to h_x < \infty$ $L(x,\cdot)$-almost everywhere, and $r(x,\cdot)$ is a derivative for every x. □

Corollary A.2 (Joint densities against one reference). Let ν be a sigma-finite measure on $(\mathsf{F}, \mathscr{F})$ and K a probability kernel from $(\mathsf{E}, \mathscr{E})$ to F with $K(x,\cdot) \ll \nu$ for every x. Then K has a finite jointly measurable density against ν, and every finite or countable collection of kernels dominated by the same ν has jointly measurable densities against that common reference; in particular this applies to the generative and recognition kernels (3.6)–(3.7) when they are dominated by the product references. [ESTABLISHED]

Proof. Sigma-finiteness supplies measurable w with $0 < w < \infty$ everywhere and $\int w(y)\nu(dy) = 1$, so the constant kernel $L(x, dy) = w(y)\nu(dy)$ is equivalent to ν. Theorem A.1 gives a finite jointly measurable $r = dK/dL$, and $k = rw$ is the density. Apply this to each member of the collection. □

Proposition A.3 (Kernel integration preserves measurability). Let κ be an s-finite kernel from $(\mathsf{E}, \mathscr{E})$ to $(\mathsf{F}, \mathscr{F})$, that is, a countable sum of finite kernels; probability kernels and fixed sigma-finite measures are instances. For every $\mathscr{E} \otimes \mathscr{F}$-measurable $f : \mathsf{E} \times \mathsf{F} \to [0, \infty]$,

$$x \longmapsto \int_{\mathsf{F}} f(x, y)\kappa(x, dy) \tag{A.2}$$

is $\mathscr{E}$-measurable. For an extended signed integrand the same holds on the measurable set where the positive-part and negative-part integrals are not both infinite, and no expression $+\infty - \infty$ is formed. Consequently the evidence (3.12) and the $\mathbb{Q}_{V,o,X}$-expectations of Chapter 5 are measurable in (o, X) whenever their integrands are jointly measurable. [ESTABLISHED]

Proof. For finite κ, the class of $C \in \mathscr{E} \otimes \mathscr{F}$ for which $x \mapsto \kappa(x, C_x)$ is measurable, with $C_x = \{y : (x, y) \in C\}$, contains the rectangles, since $\kappa(x, (A \times B)_x) = \mathbf{1}_A(x)\kappa(x, B)$, and is closed under proper differences and increasing unions because each $\kappa(x, \cdot)$ is finite. Rectangles form a π-system generating $\mathscr{E} \otimes \mathscr{F}$, so the π–λ theorem gives every C. Linearity and monotone convergence pass to nonnegative measurable f, and countable sums to s-finite κ; a sigma-finite ν with a measurable partition into sets F_m of finite measure is the constant s-finite kernel $\sum_m \nu(\cdot \cap F_m)$. A signed integrand is split into f^+ and f^-, whose difference is measurable where they are not both infinite. □

Corollary A.4 (Measurability of kernel relative entropy). Let κ and η be probability kernels from $(\mathsf{E}, \mathscr{E})$ to $(\mathsf{F}, \mathscr{F})$. Then $x \mapsto D_{\mathrm{KL}}(\kappa(x, \cdot)\|\eta(x, \cdot))$ is $\mathscr{E}$-measurable as a $[0, +\infty]$-valued map, and for finite refining partitions $(\mathcal{P}_n)$ generating $\mathscr{F}$,

$$D_{\mathrm{KL}}(\kappa(x, \cdot)\|\eta(x, \cdot)) = \sup_{n \geq 1} \sum_{A \in \mathcal{P}_n} \kappa(x, A) \log \frac{\kappa(x, A)}{\eta(x, A)}, \tag{A.3}$$

with $0 \log(0/b) = 0$ and $a \log(a/0) = +\infty$ for $a > 0$. [ESTABLISHED]

Proof. Put $L = (\kappa + \eta)/2$. Theorem A.1 applied to (κ, L) and (η, L) gives finite jointly measurable densities a and b. Fix x, write $L_x = L(x, \cdot)$, and let a_n, b_n be the conditional expectations of $a(x, \cdot), b(x, \cdot)$ under L_x given $\sigma(\mathcal{P}_n)$, whose cell values are the cellwise ratios in (A.3). The perspective $\phi_{\mathrm{KL}}(u, v) = u \log(u/v) - u + v$, with $\phi_{\mathrm{KL}}(0, v) = v$, $\phi_{\mathrm{KL}}(u, 0) = +\infty$ for $u > 0$, and $\phi_{\mathrm{KL}}(0, 0) = 0$, is nonnegative, convex, and lower semicontinuous. Since both kernels have unit mass, the n-th partition sum is $D_n(x) = \int \phi_{\mathrm{KL}}(a_n, b_n)dL_x$ and the divergence is $\int \phi_{\mathrm{KL}}(a, b)dL_x$. Conditional Jensen makes $D_n(x)$ nondecreasing in n and at most the divergence, and martingale convergence $(a_n, b_n) \to (a, b)$ L_x-almost everywhere, with Fatou's lemma and lower semicontinuity, gives $\int \phi_{\mathrm{KL}}(a, b)dL_x \leq \liminf_n D_n(x)$. Each D_n is measurable in x, so the countable supremum is measurable. □

Proof of Theorem 3.5. If A_* is countable, ν_{mix} is sigma-finite, and a ν_{mix}-null set is Lebesgue null and contains no point of A_*, so every term of every μ_t vanishes on it. Conversely, let a sigma-finite ξ dominate every μ_t. Each $a \in A_*$ has $\mu_t(\{a\}) > 0$ for some t, whence $\xi(\{a\}) > 0$, and a sigma-finite measure has at most countably many points of positive mass, as in the proof of Remark 3.6. The moving-atom family has $A_* = \mathbb{R}$. □

Remark A.5 (A regular-conditional version is not determined at a point). On $\mathbb{R}^2$ with coordinates (o, y), let ϕ be the standard normal density and $\psi \neq \phi$ another probability density. Then $p(o, y) = \phi(o)\phi(y)$ and $\widetilde{p}$, equal to p for $o \neq 0$ and to $\phi(0)\psi(y)$ for $o = 0$, are versions of the same joint density; both evidence representatives equal $\phi(0)$ at $o = 0$, but their density-ratio conditionals there are $\phi(y)dy$ and $\psi(y)dy$. The joint measure therefore determines neither pointwise posterior at $o = 0$. [ESTABLISHED]

Proof. The two functions differ only on the Lebesgue-null line $\{0\} \times \mathbb{R}$, and integrating out y at $o = 0$ gives $\phi(0)$ in both cases. □

Proof of Lemma 3.12. (a) Measurability of the integrand is Corollary A.4 with $\mathsf{E} = \mathsf{Y}_1$. Suppose first $Q \ll P$. Then $Q_1 \ll P_1$, since $P_1(A) = 0$ gives $P(A \times \mathsf{Y}_2) = 0$. Let $r = dQ/dP$ and $r_1 = dQ_1/dP_1$. For measurable $A \subseteq \mathsf{Y}_1$ and $B \subseteq \mathsf{Y}_2$,

$$\int_A Q_{2|1}(y_1, B) r_1(y_1) P_1(dy_1) = Q(A \times B) = \int_A \int_B r(y_1, y_2) P_{2|1}(y_1, dy_2) P_1(dy_1),$$

so for each B the integrands agree P_1-almost everywhere; over a countable generating algebra of Y_2 the exceptional sets form one P_1-null set N, off which both sides are finite measures in B that agree on the algebra and hence everywhere. Off the Q_1-null set $N \cup \{r_1 = 0\}$, $Q_{2|1}(y_1, \cdot)$ therefore has the density $r(y_1, \cdot)/r_1(y_1)$ against $P_{2|1}(y_1, \cdot)$, and $\log r$ splits Q-almost surely into $\log r_1$ and the logarithm of that density. Both summands have negative parts of Q-integral at most $1/e$, as in Definition 3.10, so expectations add in $(-\infty, +\infty]$; the first gives $D_{\mathrm{KL}}(Q_1 \| P_1)$, and Tonelli's theorem for the positive and negative parts along $Q = Q_1 \otimes Q_{2|1}$ gives the integral of the conditional divergences. Suppose next $Q \not\ll P$. If $Q_1 \not\ll P_1$, the first term on the right is $+\infty$. If $Q_1 \ll P_1$ and the conditional term were finite, then $Q_{2|1}(y_1, \cdot) \ll P_{2|1}(y_1, \cdot)$ for Q_1-almost every y_1, so every P-null C, whose sections are $P_{2|1}(y_1, \cdot)$-null for P_1-, hence Q_1-almost every y_1, would be Q-null. Conditional versions are unique Q_1-almost everywhere when $Q_1 \ll P_1$, and otherwise the right side is $+\infty$ for every version.

(b) Quantize every block by finite refining partitions generating its sigma-algebra. On the finite product alphabet a cell of positive Q-mass has marginal cells of positive Q_b-mass, so the first term on the right is finite; if every marginal term is finite, every such cell has positive reference mass and the logarithm splits cellwise, and otherwise a marginal cell of positive Q_b-mass and zero ρ_b-mass makes both sides infinite. By (A.3) for constant kernels, each of the three divergences is the nondecreasing limit of its quantized values, which gives (3.21).

(c) By (a), every Q with $Q_1 = \mu$ has $D_{\mathrm{KL}}(Q\|P) \geq D_{\mathrm{KL}}(\mu\|P_1)$. When $D_{\mathrm{KL}}(\mu\|P_1) < \infty$, equality holds exactly when $Q_{2|1} = P_{2|1}$ μ-almost everywhere, which determines Q. When the marginal divergence is infinite, every member of the class has divergence $+\infty$.

(e) If $Q \ll P$ with $r = dQ/dP$, then $A_{\#}Q(B) = \int_{A^{-1}B} r(y)P(dy) = \int_B r(A^{-1}y')(A_{\#}P)(dy')$ for measurable B, which is (3.23), and integrating $\log(r \circ A^{-1})$ against $A_{\#}Q$ gives $\int \log r(y)Q(dy)$. If $Q(S) > 0 = P(S)$, then $A(S)$ is measurable because A^{-1} is, and $A_{\#}Q(A(S)) > 0 = A_{\#}P(A(S))$, so both divergences are $+\infty$.

(d) Form the laws $Q(dy)K(y, dz)$ and $P(dy)K(y, dz)$ on $\mathsf{Y} \times \mathsf{Z}$. By (a) with y as the first coordinate their divergence is $D_{\mathrm{KL}}(Q\|P) + \int D_{\mathrm{KL}}(K(y,\cdot)\|K(y,\cdot))Q(dy) = D_{\mathrm{KL}}(Q\|P)$. By (e) for the coordinate swap and (a) with z as the first coordinate, the same divergence is $D_{\mathrm{KL}}(QK\|PK)$ plus a term in $[0, +\infty]$. □

A.2 The relative-log representation and integrability witnesses

Proposition A.6 (Relative-log representation). Under (H1)–(H2) of Hypothesis 5.3, let $M_{\theta,V,o,X}$ be the slice measure (5.14). If $Q \ll M_{\theta,V,o,X}$ and $r = dQ/dM_{\theta,V,o,X}$, then

$$\begin{aligned} \mathcal{L}_V^{\mathrm{ext}}(Q; X, o) &= -\int_{\mathsf{Y}} \log r(y)Q(dy), \\ \int_{\{r<1\}} -\log r(y)Q(dy) &= \int_{\{r<1\}} -r(y)\log r(y)M_{\theta,V,o,X}(dy) \leq \frac{p_{\theta,V}(o \mid X)}{e}, \end{aligned} \tag{A.4}$$

so the integral is well defined in $(-\infty, +\infty]$ and its negative lies in $[-\infty, +\infty)$. If $Q \not\ll M_{\theta,V,o,X}$, then $\mathcal{L}_V^{\mathrm{ext}}(Q; X, o) = -\infty$. [ESTABLISHED]

Proof. Since $\mathbf{\Pi}_{\theta,V,o,X} = M_{\theta,V,o,X}/p_{\theta,V}(o \mid X)$, the Radon–Nikodym chain rule gives $dQ/d\mathbf{\Pi}_{\theta,V,o,X} = p_{\theta,V}(o \mid X)r$. The negative part of $\log r$ is Q-integrable by the displayed bound, because $-u\log u \leq 1/e$ on $[0, 1]$ and $M_{\theta,V,o,X}(\mathsf{Y}) = p_{\theta,V}(o \mid X)$. Hence $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) = \log p_{\theta,V}(o \mid X) + \int \log r(y)Q(dy)$ is a legitimate extended identity, and substitution in (5.15) gives the first branch. The slice measure and the posterior have the same null sets, which gives the second. □

Proposition A.7 (Integrability witnesses). The integrability hypotheses of Chapter 5 are separated by the following witnesses.

(a) Clauses (H1)–(H3) of Hypothesis 5.3 do not imply (H4). On $\mathbb{R}$ with a standard Gaussian posterior and a standard Cauchy recognition law, (H1)–(H3) hold, $\mathbb{E}_Q \log q = -\log 4\pi$ is finite, $\mathbb{E}_Q|\log p_{\theta,V}(o, Y \mid X)| = +\infty$, the divergence is $+\infty$, and the extended bound is $-\infty$.

(b) Finite relative entropy does not imply (H4). On $\{2, 3, \ldots\}$ with a trivial record, let the posterior and the recognition law both equal $P(\{n\}) = \mathrm{c}/(n(\log n)^2)$, with c normalizing. The divergence is 0, while $\sum_n P(\{n\})|\log P(\{n\})| = +\infty$, so both expectations in the classical split equal $-\infty$ and their difference is undefined.

(c) Factorwise log-integrability (proposition 5.15) does not imply (H4). With one latent $k \in \mathbb{R}$, generative factor $\mathcal{N}(0,1)$, and a record whose log density is bounded for $k \in [0,1]$, let the recognition law spread mass $w_n = 2^{-n}$ uniformly over disjoint intervals in $[0,1]$ of widths $w_n e^{-1/w_n}$, whose total is below e^{-2}. Every factor logarithm is bounded on the support, while $\log q = 1/w_n$ on the n-th interval and $\mathbb{E}_Q \log q = \sum_n w_n/w_n = +\infty$.

(d) Factorwise log-integrability is strictly stronger than the generative half of (H4). Take a root model coordinate $m \in \{2,3,\ldots\}$ with $p(m) = \mathrm{c}/(m(\log m)^2)$, a bridge $k \mid m \sim \mathcal{N}(0,\sigma_m^2)$ with $\sigma_m^2 = p(m)^2/(2\pi)$, a trivial record, and the recognition law equal to this law. Then $\log p(m) + \log p(k \mid m) = -k^2/(2\sigma_m^2)$ has expectation $-\frac{1}{2}$ and is also the recognition log density, so (H4) holds, while $\mathbb{E}|\log p(m)| = +\infty$.

[ESTABLISHED]

Proof. (a) Both laws have positive Lebesgue densities, so (H1)–(H3) hold. The log joint differs from $-y^2/2$ by a constant and the Cauchy law has no second moment; the Cauchy differential entropy is $\log 4\pi$; hence the divergence $\mathbb{E}_Q \log q - \mathbb{E}_Q \log p$ is $+\infty$. (b) For large n, $P(\{n\}) \le 1/n$, so $|\log P(\{n\})| \ge \log n$ and the sum dominates $\mathrm{c} \sum_n 1/(n \log n) = +\infty$. (c) The widths sum to $\sum_n 2^{-n} e^{-2^n} < e^{-2}$, so the intervals fit in $[0,1]$; the density on the n-th interval is $w_n/(w_n e^{-1/w_n}) = e^{1/w_n}$. (d) The bridge density at k is $(2\pi\sigma_m^2)^{-1/2} e^{-k^2/(2\sigma_m^2)} = p(m)^{-1} e^{-k^2/(2\sigma_m^2)}$, which gives the identity and $\mathbb{E}|\log p(m) + \log p(k \mid m)| = \frac{1}{2}$; and $\mathbb{E}|\log p(m)| = +\infty$ as in (b). □

Appendix B

Deferred Proofs

This appendix collects the longer proofs of chapters 6, 7, 9, 10 and 12, in the order of the statements they prove.

B.1 Proofs for Attention and Selective Communication

Proof of Proposition 6.2. (a) The product label prior and exclusivity make the posterior of J factor by rows, and Bayes' formula gives (6.4); a weight $\pi_{ij}e^{-D_{ij}/\tau_i}$ is positive exactly when $\pi_{ij} > 0$, since the energies are finite, and with equal prior weights the largest weight has the smallest energy, $D_{ii} = 0$. (b) With $Z_i^{\mathrm{att}}(y) = \sum_k \pi_{ik}e^{-D_{ik}(y)/\tau_i}$, $D_{\mathrm{KL}}(\beta_i^Q \| \beta_i^P(y)) = D_{\mathrm{KL}}(\beta_i^Q \| \pi_i) + \tau_i^{-1}\sum_j \beta_{ij}^Q D_{ij}(y) + \log Z_i^{\mathrm{att}}(y)$. Subtracting $\log Z_i^{\mathrm{att}}(y)$ and the source-independent $\log \mathrm{c}_i(o_i, y)$ and taking the recognition expectation proves (6.6) and (6.7); Lagrange multiplication under $\sum_j \beta_{ij}^Q = 1$ gives (6.8), and strict convexity of $D_{\mathrm{KL}}(\cdot\|\pi_i)$ gives uniqueness, which the factor $\tau_i > 0$ does not change. (c) The label part of the VFE is $\mathbb{E}_{\mathbb{Q}}[\log \mathbb{Q}_{J|Y} - \sum_i \log \pi_{iJ_i} - \sum_i \log L_i(o_i \mid Y, J_i)]$. The product-reference case of lemma 3.12 splits $D_{\mathrm{KL}}(\mathbb{Q}_{J|Y=y} \| \prod_i \pi_i)$ into the total correlation and the row relative entropies, which with (6.2) gives (6.9). At fixed y the expansion of (b), applied to the product reference, equals the label terms other than $-\sum_i \log \mathrm{c}_i$ to $D_{\mathrm{KL}}(\mathbb{Q}_{J|Y=y} \| \prod_i \beta_i^P(y)) - \sum_i \log Z_i^{\mathrm{att}}(y)$, which the product of the posterior rows minimizes. □

Proof of Proposition 6.3. Substituting (6.8) into (6.7) gives R_i^Q, the minimum by the expansion in the proof of proposition 6.2. At fixed y the same expansion gives $D_{\mathrm{KL}}(\mathbb{Q}_{J_i|Y=y} \| \pi_i) + \tau_i^{-1}\mathbb{E}_{\mathbb{Q}_{J_i|Y=y}} D_{iJ_i}(y) = D_{\mathrm{KL}}(\mathbb{Q}_{J_i|Y=y} \| \beta_i^P(y)) + R_i^P(y)$, minimized by $\beta_i^P(y)$, and the conditional total correlation in (6.9) is nonnegative and vanishes on the product of these rows. Normalization of π_i and concavity of the logarithm give $\min_j D_{ij}/\tau_i \le R_i^P \le \sum_j \pi_{ij} D_{ij}/\tau_i$, so $R_i^P(Y)$ is integrable. The softmax formulas give $\log \beta_{ij}^{Q\star} - \log \beta_{ij}^P(y) = (D_{ij}(y) - d_{ij})/\tau_i + R_i^Q - R_i^P(y)$, and averaging over $\beta_i^{Q\star}$ and then over $\mathbb{Q}_Y$ removes the first term, since $\mathbb{E}_{\mathbb{Q}_Y} D_{ij} = d_{ij}$, which proves (6.11). The relative entropy vanishes at y exactly when $D_{ij}(y) - d_{ij}$ does not depend on j, and almost-sure constancy of the differences forces those constants to equal $d_{ij} - d_{ik}$. Under the affine hypothesis $d_{ij} - d_{ik} = (\widetilde{D}_{ij} - \widetilde{D}_{ik})(\bar{\psi}_i)$, and a softmax row depends only on logit differences, which gives (6.12); template differences are affine in y_i. For the random source, uncorrelatedness gives $\mathbb{E} r_{ij} = \bar{y}_i - \rho_b(\Theta_{ij}^b)\bar{y}_j$ and $\operatorname{Cov} r_{ij} = \operatorname{Cov} Y_i + \rho_b(\Theta_{ij}^b) \operatorname{Cov}(Y_j) \rho_b(\Theta_{ij}^b)^\top$, so d_{ij} exceeds D_{ij} at the recognition mean by $\frac{1}{2}\operatorname{Tr}(W_i \operatorname{Cov} Y_i) + \frac{1}{2}\operatorname{Tr}(W_i \rho_b(\Theta_{ij}^b) \operatorname{Cov}(Y_j) \rho_b(\Theta_{ij}^b)^\top)$, whose first trace is common to the row and cancels from the softmax. □

Proof of Corollary 6.7. On the open simplex the Fisher metric is the Shahshahani metric, whose natural gradient of a function with Euclidean gradient g is $-\beta_j(g_j - \sum_k \beta_k g_k)$. The Euclidean gradient of (6.7) is $\mathsf{c}_{ij} + 1$, and the constant cancels on the simplex, which gives (6.17). Pairing the gradient with the flow gives $-\mathsf{r}_i \sum_j \beta_{ij}\mathsf{c}_{ij}(\mathsf{c}_{ij} - \bar{\mathsf{c}}_i) = -\mathsf{r}_i \operatorname{Var}$, since $\sum_j \dot\beta_{ij} = 0$; with moving λ the chain rule adds the explicit λ-derivative, the second term of (6.19). The quotient rule for $\beta = \eta/\alpha$ gives (6.20), and differentiating $\log \beta_j = \log \pi_j - E_j/\tau - \log Z$ gives $\dot\beta_j/\beta_j = \chi_j - \dot Z/Z$ with $\dot Z/Z = \sum_k \beta_k \chi_k$, which is (6.21). □

Proof of Proposition 6.8. Differentiating the finite sum gives $\nabla\beta_{ij} = -\tau_i^{-1}\beta_{ij}(\nabla E_{ij} - \bar g)$ and the gradient; once more, $\nabla^2 R_i = \tau_i^{-1}\sum_j(\beta_{ij}\nabla^2 E_{ij} + \nabla E_{ij}\nabla\beta_{ij}^\top)$, and $\sum_j \beta_{ij}\bar g(\nabla E_{ij} - \bar g)^\top = 0$ turns the second sum into $-\tau_i^{-2}\operatorname{Cov}_{\beta_i}(\nabla E_i)$. The joint Hessian of $\mathscr{J}_i$ has blocks $\tau_i^{-1}\sum_j \beta_{ij}\nabla^2 E_{ij}$, the mixed form $(v, w) \mapsto \tau_i^{-1}\sum_j w_j \nabla E_{ij}\cdot v$, and $w \mapsto \sum_j w_j^2/\beta_{ij}$ on the tangent space; minimizing the joint quadratic form over tangent w at fixed v gives $w_j = -\tau_i^{-1}\beta_{ij}(\nabla E_{ij} - \bar g)\cdot v$, which satisfies $\sum_j w_j = 0$, with minimum $v^\top(\tau_i^{-1}\sum_j \beta_{ij}\nabla^2 E_{ij})v - \tau_i^{-2}\operatorname{Var}_{\beta_i}(\nabla E_{ij}\cdot v)$. The constant-row identification is proposition 6.3; for exact label integration, the product label prior and exclusivity give $\sum_j \pi_{ij}\mathrm{c}_i(o_i, y)e^{-D_{ij}(y)/\tau_i} = \mathrm{c}_i(o_i, y)e^{-R_i(y)}$. In the witness, O given $J = j$ is $\mathcal{N}(-j\delta, \tau(1 + \omega^{-1}))$, whose density at zero is the displayed evidence for both labels. At $o = 0$, $\frac{1}{2}\sum_{j=\pm1} e^{-(x-j\delta)^2/(2\tau)} = e^{-(x^2+\delta^2)/(2\tau)}\cosh(\delta x/\tau)$, which with the baseline gives (6.24), and $\frac{d^2}{dx^2}\log\cosh(\delta x/\tau) = \delta^2/\tau^2$ at $x = 0$. The constant-row value follows from (6.23) with the affine energies, whose weights are equal at $\mathsf{m} = 0$, together with the recognition entropy and the baseline term, whose m-curvature is ω/τ. □

Proof of Corollary 6.11. (a) Integrating o, then ξ^+, then ξ^- gives total mass one, and measurability follows from measurable kernel integration (chapter 3). (b) Substituting $\xi^- = (\phi_e^-)^{-1}\xi'$ and $\xi^+ = (\phi_e^+)^{-1}\xi''$ in the composition cancels the message recoordinatizations. (c) The first identity is the definition of F_e^v, and the represented link law $\rho_x(\Theta_e^x)' = \mathsf{g}_i\rho_x(\Theta_e^x)\mathsf{g}_j^{-1}$ combined with (b) gives the last. □

Proof of Proposition 6.13. (a) The two internal basis changes cancel in $\mathsf{L}_e^{\mathrm{dec}\prime}\mathsf{L}_e^{\mathrm{tr}\prime}\mathsf{L}_e^{\mathrm{enc}\prime}$. (b) The inclusions and the rank bound follow from composition. For $k \in \ker \mathsf{L}_e^{\mathrm{enc}}$ and any u, $(\mathsf{G}_j^{-1}(\mathsf{L}_e^{\mathrm{enc}})^\top \mathsf{u})^\top \mathsf{G}_j k = \mathsf{u}^\top \mathsf{L}_e^{\mathrm{enc}} k = 0$, and $\dim \operatorname{im}(\mathsf{L}_e^{\mathrm{enc}})^\top = \dim E_j - \dim\ker \mathsf{L}_e^{\mathrm{enc}}$, which identifies the complement; the metric law is the congruence induced by g_j. Coordinates on $\operatorname{im}\mathsf{L}_e$, the induced quotient as encoder, the identity as transmission, and the inclusion of the image as decoder factor L_e through its rank, which proves existence, not canonicity of an interface. (c) Rank is subadditive and bounded by the target dimension, and the example is direct. □

Proof of Proposition 6.14. (a) The factorization holds by definition of F_e, and $\mathsf{L}_e' = \mathsf{g}_i\mathsf{L}_e\mathsf{g}_j^{-1}$ with $\Theta_e' = \mathsf{g}_i\Theta_e\mathsf{g}_j^{-1}$ gives the law of F_e; for every choice of Θ_e some F_e exists, so the link is not determined, and a common F_i exists exactly when all $\mathsf{L}_e\Theta_e^{-1}$ coincide. (b) Consecutive factors $\mathsf{g}_{i_{a+1}}^{-1}\mathsf{g}_{i_{a+1}}$ cancel, which uses no inverse of any L_e. (c) and (d) are direct matrix computations; the holonomy of the loop through b and $\bar a$ is $\Theta_a^{-1}\Theta_b = \Theta_b \neq I_2$. □

Proof of Corollary 6.17. (a) Expanding $D_{\mathrm{KL}}(\beta_i \| \beta_i^P(y))$ with (6.4) and $D_{ij} = d_{ij}$ gives the identity, and the relative entropy vanishes only at $\beta_i^P(y)$; the point mass is singular with respect to the posterior slice when the baseline gives $\{y\}$ zero mass. (b) The terms of $D_{\mathrm{KL}}(\mathcal{N}(\mu_i^Q, \Sigma^Q)\|\mathcal{N}(\mu_j^K, \Sigma^K))$

depending on j are those of $\frac{1}{2}(\mu_j^K - \mu_i^Q)^\top(\Sigma^K)^{-1}(\mu_j^K - \mu_i^Q)$, and expanding the quadratic form gives (6.39). □

Proof of Proposition 6.19. (a) The product label prior and exclusivity make J_i independent of k_i and of every other record given y_{-i}, and the records are conditionally independent given the latent state, so the joint posterior density of (k_i, j) is proportional to $\pi_{ij}\mathcal{N}(k_i;\mu_i,\Sigma_i)\mathcal{N}(0;S_i^Q k_i - S_e^K k_j, \widetilde{W}_i^{-1})$. For fixed j, adding the two quadratic forms in k_i gives the precision $(\Sigma_i^+)^{-1}$ and the component mean. Integrating k_i instead gives the density at zero of o_i given $J_i = j$, an affine image of independent Gaussian variables with mean r_{ij} and covariance Σ_i^{pred}, whose normalizer does not depend on j; Bayes' formula gives w_{ij}. The mean follows from the tower property, and multiplying $(\Sigma_i^+)^{-1} = \Sigma_i^{-1} + S_i^{Q\top}\widetilde{W}_i S_i^Q$ by Σ_i^+ gives the last identity. (b) Here $\Sigma_i^{\mathrm{pred}} = 2$, $r_{i2} = -2$, and $\Sigma_i^+ = 1/2$; substitution gives the displayed values. (c) The identity follows from $\Sigma_i^+ S_i^{Q\top}\widetilde{W}_i\Sigma_i^{\mathrm{pred}} = \Sigma_i^+(S_i^{Q\top} + S_i^{Q\top}\widetilde{W}_i S_i^Q\Sigma_i S_i^{Q\top}) = \Sigma_i^+(\Sigma_i^+)^{-1}\Sigma_i S_i^{Q\top}$. Since $S_i^Q\Sigma_i S_i^{Q\top}$ is invertible, $(\Sigma_i^{\mathrm{pred}})^{-1} \to (S_i^Q\Sigma_i S_i^{Q\top})^{-1}$ as $\widetilde{W}_i^{-1} \to 0$, which gives the limit. Directly $S_i^Q R_e = I$, and any δ with $S_i^Q\delta = \mathsf{y}$ is $R_e\mathsf{y} + w$ with $S_i^Q w = 0$, for which the cross term $(R_e\mathsf{y})^\top\Sigma_i^{-1}w = \mathsf{y}^\top(S_i^Q\Sigma_i S_i^{Q\top})^{-1}S_i^Q w$ vanishes, so $R_e\mathsf{y}$ uniquely minimizes $\delta^\top\Sigma_i^{-1}\delta$. (d) Repeat (a) with the copy-dependent data. □

Proof of Proposition 6.20. At fixed k_i the proof of proposition 6.2 applies verbatim to the rows $(i,1)$ and $(i,2)$. Given the labels, the two records at zero are a Gaussian vector with mean $-(k_j, k_l)$ and covariance $I_2 + \mathbf{1}\mathbf{1}^\top$, whose inverse is $I_2 - \frac{1}{3}\mathbf{1}\mathbf{1}^\top$; completing the square gives (6.45). The masses are 1 for $(1,1)$ and $e^{-a^2/3}$ for each of $(1,2)$, $(2,1)$, and $(2,2)$, so the odds ratio is $e^{-a^2/3}/e^{-2a^2/3} = e^{a^2/3}$. □

Proof of Proposition 6.24. The chain rule of lemma 3.12, applied to mutual information as a relative entropy, gives $I(S_j;\xi^+, O \mid \Upsilon) = I(S_j;\xi^+ \mid \Upsilon) = I(S_j;O \mid \Upsilon) + I(S_j;\xi^+ \mid O,\Upsilon)$, because $I(S_j;O \mid \xi^+,\Upsilon) = 0$ by the conditional Markov property; nonnegativity proves the first inequality, including infinite values, and the second is identical. For finite alphabets without side information this is the data-processing inequality [24, Section 2.8, Theorem 2.8.1]. □

Proof of Proposition 6.26. (a) Necessity follows by evaluating the commuting identity on $\ker M_j$. For sufficiency define $\overline{\mathsf{L}}_e(M_j\mathsf{u}) = M_i\mathsf{L}_e\mathsf{u}$; the condition makes this independent of the representative, and surjectivity of M_j gives its domain and its uniqueness. (b) If $M\mathcal{A} = \overline{\mathcal{A}}M$, then $M\mathcal{A}^n = \overline{\mathcal{A}}^n M$ for every n, and summing the exponential series gives $Me^{t\mathcal{A}} = e^{t\overline{\mathcal{A}}}M$; conversely, differentiating at $t = 0$ returns (6.50), and $M\mathcal{A}$ factors through M exactly when it vanishes on $\ker M$, which is (a) with $\mathsf{L}_e = \mathcal{A}$ and $M_j = M_i = M$. (c) Along any solution $\frac{d}{dt}(M_t\mathsf{u}) = (\dot{M}_t + M_t\mathcal{A}_t)\mathsf{u}$, so (6.51) is sufficient. For necessity fix t and $\mathsf{v} \in E$; the linear equation with continuous coefficients has a solution through $\mathsf{u}(t) = \mathsf{v}$, so $(\dot{M}_t + M_t\mathcal{A}_t)\mathsf{v} = \overline{\mathcal{A}}_t M_t\mathsf{v}$ for every v, and a linear map $E \to \overline{E}$ factors through the surjection M_t exactly when it vanishes on $\ker M_t$, with a unique factor. Multiplying (6.51) on the right by M_t^+ and using $M_tM_t^+ = I$ gives the formula; in fixed inner products $M_t^+ = M_t^\top(M_tM_t^\top)^{-1}$ is continuous. (d) For the swap, $\ker M_j$ is spanned by the second coordinate, which the swap sends outside $\ker M_i$. For the flow, choose $\mathsf{u} \in \ker M$ with $M\mathcal{A}\mathsf{u} \neq 0$; the solutions from u and from 0 have retained value 0 at $t = 0$ and retained velocities $M\mathcal{A}\mathsf{u}$ and 0. (e) The conditional law is the image of the law of z^{in} under the diagonal map, which is a product only for a degenerate law. □

B.2 Proofs for Fiber Information Geometry and Inference Histories

Proof of Proposition 7.3. Each $\widehat{\rho}_x(g)$ is pushforward by the bimeasurable sample map $\rho_x(g)$, which does not depend on the law parameter. Pushforward is a unitary map $L^2(p) \to L^2(\rho_x(g)_{\#}p)$, $f \mapsto f \circ \rho_x(g)^{-1}$, differentiability in quadratic mean makes $\mathbf{s}_Z \circ \rho_x(g)^{-1}$ the score of the pushed direction, and the change of variables preserves (7.1) and (7.2); since $\iota_{u\cdot g} = \iota_u \circ \widehat{\rho}_x(g)$, $(\iota_u^{-1})^*\tau$ is independent of u exactly when $\widehat{\rho}_x(g)^*\tau = \tau$ for every g. For the counterexample, $\mathrm{X} = \mu/\sqrt{2}$ and $\mathrm{S} = \varsigma$ turn g^F into $2(d\mathrm{X}^2 + d\mathrm{S}^2)/\mathrm{S}^2$, twice the hyperbolic metric of the upper half-plane, of which the real Möbius map $z \mapsto (1+z)/(1-z)$ of determinant $2 > 0$ is an isometry. It fixes $z = \mathrm{i}$ with derivative $2/(1-\mathrm{i})^2 = \mathrm{i}$, a quarter turn, so it sends the g^F-unit vector ∂_μ at $(0,1)$ to $\partial_\varsigma/\sqrt{2}$. There the only nonzero components of T^F are $T^F(\partial_\mu, \partial_\mu, \partial_\varsigma) = 2$ and $T^F(\partial_\varsigma, \partial_\varsigma, \partial_\varsigma) = 8$, so the cubic values of the two unit vectors are 0 and $8/2^{3/2} = 2\sqrt{2}$. □

Proof of Proposition 7.6. Invariance is (3.24) for the sample maps $\rho(g)$. With local parameters λ and λ' of the two arguments, primed derivatives acting on λ', and $\mathbf{s}_i = \partial_i \log p_\lambda$, the identities $\mathbb{E}_\lambda[\mathbf{s}_i] = 0$, $\partial_i p_\lambda = p_\lambda \mathbf{s}_i$, and $\partial_i\partial_j p_\lambda = p_\lambda(\mathbf{s}_i\mathbf{s}_j + \partial_i \mathbf{s}_j)$ and differentiation at $\lambda' = \lambda$ under hypothesis 7.1 give

$$\begin{aligned} -\partial_i\partial'_j D_{\mathrm{KL}}(p_\lambda \| p_{\lambda'}) &= \mathbb{E}_\lambda[\mathbf{s}_i\mathbf{s}_j], \\ -\partial_i\partial_j\partial'_k D_{\mathrm{KL}}(p_\lambda \| p_{\lambda'}) &= \mathbb{E}_\lambda[(\mathbf{s}_i\mathbf{s}_j + \partial_i\mathbf{s}_j)\mathbf{s}_k], \\ -\partial'_i\partial'_j\partial_k D_{\mathrm{KL}}(p_\lambda \| p_{\lambda'}) &= \mathbb{E}_\lambda[\mathbf{s}_k\partial_i\mathbf{s}_j]. \end{aligned} \tag{B.1}$$

The second derivatives of the score are symmetric, so the last two lines differ by $\mathbb{E}_\lambda[\mathbf{s}_i\mathbf{s}_j\mathbf{s}_k]$, and positivity of the metric jet is positivity of g^F. The same dualistic structure is derived from contrast functions in Amari and Nagaoka [2], Ay et al. [4]. □

Proof of Corollary 7.7. Positive definiteness of g^F identifies the radical of $\mathfrak{k}_q$ with the kernel of $\xi \mapsto \xi_{\mathcal{B}}(q)$, which is $\mathfrak{g}_q$. The curve $\gamma(t) = \widehat{\rho}(\exp t\xi)q$ is the integral curve of $\xi_{\mathcal{B}}$ through q, so by uniqueness of integral curves $\xi_{\mathcal{B}}(q) = 0$ exactly when $\gamma \equiv q$, that is, when $\exp t\xi \in G_q$ for every t; this gives the vanishing clause, and since G_q is closed, its Lie algebra is the set of ξ with $\exp t\xi \in G_q$ for every t, which is $\mathfrak{g}_q$. The function $\phi(t) = \mathscr{D}(q, \gamma(t))$ is C^3 with $\phi(0) = \phi'(0) = 0$ because $\mathscr{D}(q,\cdot)$ is minimal at q, and differentiating $\partial'_j\mathscr{D}(\lambda,\lambda) = 0$ along the diagonal gives $\partial_i\partial'_j\mathscr{D} = -\partial_i\partial'_j\mathscr{D} = (g_{\mathscr{D}})_{ij}$ there, so $\phi''(0) = g_{\mathscr{D}}(\xi_{\mathcal{B}}(q), \xi_{\mathcal{B}}(q))$ and Taylor's theorem gives the expansion; for relative entropy $g_{\mathscr{D}} = g^F$ by proposition 7.6. □

Proof of Theorem 7.8. Pollard's Le Cam decomposition at λ_0 is $P_{\lambda_0+\delta} = \mathsf{p}_\delta P_{\lambda_0} + P_\delta^\perp$ with $P_\delta^\perp \perp P_{\lambda_0}$, $P_\delta^\perp(\mathrm{Y}) = o(\|\delta\|^2)$, and $\int(\sqrt{\mathsf{p}_\delta} - 1 - \frac{1}{2}\delta^\top \mathbf{s}_{\lambda_0})^2 dP_{\lambda_0} = o(\|\delta\|^2)$. Hence $P_{\lambda_0+\delta} \otimes K = \mathsf{p}_\delta(y)(P_{\lambda_0} \otimes K) + P_\delta^\perp \otimes K$, where the second term has mass $o(\|\delta\|^2)$ and is singular, because a P_{λ_0}-null set carrying $P_\delta^\perp$, multiplied by $\bar{\mathrm{Y}}$, carries it and is $P_{\lambda_0} \otimes K$-null since each $K(y,\cdot)$ is a probability, and the square-root remainder of the first term is unchanged because the kernel integrates to one. The joint experiment is therefore differentiable in quadratic mean with score $\mathbf{s}_{\lambda_0}(Y)$, without a common dominating measure, and Pollard's preservation theorem for measurable statistics [86, Theorem 7], applied to $(y, \bar{y}) \mapsto \bar{y}$, gives differentiability of $\{P_\lambda K\}$ with the score (7.11) in $L^2(P_{\lambda_0}K; \mathbb{R}^d)$; the projected singular term may acquire an absolutely continuous part, of mass still $o(\|\delta\|^2)$. The law of

total covariance gives (7.12), and since conditional expectation is an orthogonal projection, equality in the direction z holds exactly when $z^\top \mathbf{s}_{\lambda_0}(Y)$ is $\bar{Y}$-measurable. □

Proof of Theorem 7.14. Since $h_Q(Z,\cdot) = g(d\,\mathrm{ret}_Q\, Z, d\,\mathrm{ret}_Q\,\cdot)$, the Cauchy–Schwarz inequality for the semidefinite g makes $h_Q(Z,Z) = 0$, $h_Q(Z,\cdot) = 0$, and $d\,\mathrm{ret}_Q\, Z \in \operatorname{rad} g$ equivalent, which is (7.19) and the transversality criterion. The radical is the kernel of the constant-rank C^1 bundle map $h^\flat : T\mathcal{Q} \to T^*\mathcal{Q}$, hence a C^1 subbundle, and h descends to the quotient because it annihilates its radical. When $\operatorname{rad} h = \ker d\,\mathrm{ret}$, ret has zero derivative along, hence is constant on, each connected leaf and factors as $\bar{r}\circ\pi$ through the leaf space, with π a submersion, $\ker d\pi = \ker d\,\mathrm{ret}$, and $\bar{r} = \mathrm{ret}\circ\iota$ smooth for smooth local sections ι of π; then $h = \pi^*(\bar{r}^* g)$, $d\bar{r}$ is injective, and $\operatorname{im} d\bar{r} = \operatorname{im} d\,\mathrm{ret}$ meets $\operatorname{rad} g$ only in zero, so $\bar{r}^* g$ is positive definite. For a finite family, $h = \sum_y \partial p_y \partial p_y / p_y$ is the Gram matrix of the Jacobian of ret weighted by $1/p_y$, hence C^2 and positive definite exactly where the Jacobian has full column rank; an immersion is locally injective because the inverse function theorem [74, Ch. 4] makes a coordinate projection of ret with invertible square Jacobian block a local diffeomorphism, and the Levi-Civita connection of a C^2 metric is C^1. For softmax, $Z^\top h(z) Z = \mathrm{Var}_p(Z)$ vanishes exactly on constant vectors, so $\operatorname{rad} h = \mathbb{R}\mathbf{1} = \ker d\,\mathrm{ret}$ with constant rank $k-1$, and an objective of p has $d\mathcal{F}(\mathbf{1}) = 0$, so $\nabla\mathcal{F} \in \mathbf{1}^\perp = \operatorname{im} h$, which gives the solution class and $h^+\nabla\mathcal{F} \in \operatorname{im} h = \mathbf{1}^\perp$. □

Proof of Proposition 7.18. The tower property gives $\mathbb{E}[\mathbf{s}\mathbf{s}_i^\top] = \mathsf{G}_i$ and its transpose, so expanding $\mathfrak{K}$ gives $\mathfrak{K} = \mathsf{G}_\mathrm{J} - \sum_i \mathsf{G}_i + \mathfrak{C}$, which rearranges to (7.24); the criterion is its restriction, and (7.25) is its quadratic form for $n = 2$ and unit weights. Under independence the joint score is the sum of the block scores, so $\mathbf{r} = 0$, $\mathfrak{C} = 0$, and the weight condition is $(1-w_1)\mathsf{G}_1 + (1-w_2)\mathsf{G}_2 = 0$ in the stated direction. For the normal location family with precision Λ the joint Fisher metric is Λ and each marginal is a location family of variance $1/(1-\mathrm{c}^2)$, so $\mathsf{G}_1 + \mathsf{G}_2 = (1-\mathrm{c}^2)\mathrm{I}_2$, which gives the displayed difference and its two values. The lift metrics are the pullbacks of $d\lambda_1^2 + d\lambda_2^2$ along the three maps. □

Proof of Theorem 7.23. Each atom has $p_0/\mathrm{D} = \prod_{y_{i,x}=1}(1-\lambda_{i,x})^{-1} \prod_{y_{i,x}=0} \lambda_{i,x}^{-1} > 1$, so $p_\varkappa \geq p_0 - |\varkappa|\mathrm{D} > (1-|\varkappa|)\mathrm{D} > 0$. Summing out any one coordinate cancels the parity term, because D does not depend on y and par changes sign with that coordinate; this gives normalization and the product proper marginals, hence $\mathrm{mar}\circ\iota_\varkappa = \mathrm{id}$, which with polynomial dependence gives smoothness and injectivity of $d\iota_\varkappa$. The marginal map is the restriction to the open 63-dimensional simplex of a linear map of rank six, so its fibers are open subsets of 57-dimensional affine spaces, and $p_\varkappa$ is affine in $\varkappa$. A paired complement changes two bits, so it preserves par and D while p_0 transforms by Bernoulli relabeling, and a typed permutation preserves all three expressions, which proves (7.28); a single complement changes the sign of par and sends $\lambda_{i,x}$ to $1-\lambda_{i,x}$, which preserves D and proves (7.29). □

Proof of Corollary 7.25. Two-bit marginals are products by theorem 7.23, so centered scores of distinct bits are uncorrelated. With $\mathbf{s}_{i,x}$ the one-bit Bernoulli score, differentiation gives the residual components

$$\mathbf{r}_{i,x} = \frac{\varkappa\,\mathrm{par}(y)\mathrm{D}(\lambda)}{p_\varkappa(y;\lambda)} \left(\frac{1-y_{i,x}}{\lambda_{i,x}} - \frac{y_{i,x}}{1-\lambda_{i,x}} \right). \tag{B.2}$$

If $Z^{\top}\mathbf{r}$ vanished on every atom, positivity and $\varkappa \mathrm{D} \neq 0$ would make the Z-weighted sum of the six bracketed one-bit functions vanish on the cube, and comparing two atoms that differ in one coordinate forces each component of Z to vanish; so $\mathfrak{K} \succ 0$. At the center $\mathrm{D} = 4^{-6}$, its derivative vanishes, and the law is (7.36), so, each one-bit score having square 4, averaging $(1+\bar{\varkappa}\,\mathrm{par})^{-1} = (1-\bar{\varkappa}\,\mathrm{par})/(1-\bar{\varkappa}^2)$ over the cube gives (7.32), and $\bar{\varkappa} = 1/128$ gives (7.33). The one-bit tensors at the center are $4e_{i,x}e_{i,x}^{\top}$, which gives (7.34), and the last sentence compares $\varkappa = 0$ with $\varkappa \neq 0$. □

Proof of Proposition 7.28. Half of the atoms have each parity, so averaging (7.37) under Unif gives evidence $\frac{1}{2}$, Bayes' rule gives the posterior $64^{-1}(1+\bar{\varkappa}\,\mathrm{par})$, which is (7.36), and the gap identity of the free energy gives the tight value $-\log\frac{1}{2}$. Every posterior one-bit marginal is uniform, so its product law is Unif, and since the two parities have equal mass, $-\mathbb{E}_{\mathrm{Unif}}\log K_{\mathrm{hyp}}(1 \mid Y) = \log 2 - \frac{1}{2}\log(1-\bar{\varkappa}^2)$, which is (7.38). The posterior and Unif have the same one-bit marginals, the posterior input and the evidence are unchanged, and the two free-energy values differ when $\varkappa \neq 0$. □

Proof of Theorem 7.36. The chain rule gives $\dot{\widetilde{Q}} = (\dot{Q}\circ\phi)\phi'$ with $\phi' > 0$, and the substitution $t = \phi(\widetilde{t})$ proves length invariance and (7.49). Positive speed makes the clock invertible, and dividing by $d\mathfrak{t}/dt$ gives unit speed. Substituting (7.41) into (7.45) gives (7.50), and the other two identities follow from (7.40). For a second metric, $d\mathcal{F}(Z) = \mathsf{G}'(\mathrm{grad}^{\mathsf{G}'}\mathcal{F}, Z)$, and the Cauchy–Schwarz inequality gives (7.53). For a block-diagonal metric the squared speeds add, so $\dot{\mathfrak{l}}_i \le \dot{\mathfrak{l}} \le \sum_i \dot{\mathfrak{l}}_i$, and Minkowski's integral inequality for the vector $(\dot{\mathfrak{l}}_i)_i$ gives $\big(\sum_i(\int \dot{\mathfrak{l}}_i)^2\big)^{1/2} \le \int \dot{\mathfrak{l}}$. □

Proof of Proposition 7.38. Along the history $\frac{d}{dt}\mathcal{F}(\gamma) = -\|\dot{\gamma}\|_{\mathsf{G}}^2 \le 0$, so $\mathcal{F}(\gamma(t))$ decreases to $\mathcal{F}_* = \mathcal{F}(Q_*)$; if it reaches $\mathcal{F}_*$ at some t_1, then $\dot{\gamma} = 0$ on $[t_1,\infty)$ and the duration is that of a compact segment. Otherwise choose a closed Euclidean ball $\bar{B}$ of radius δ_0 around Q_* in the chart on which $\mathrm{c}_1\mathrm{I} \preceq \mathsf{G} \preceq \mathrm{c}_2\mathrm{I}$ with $0 < \mathrm{c}_1 \le \mathrm{c}_2$ and the Łojasiewicz gradient inequality $\|\nabla\mathcal{F}\| \ge \mathrm{c}_0|\mathcal{F}-\mathcal{F}_*|^{\mathrm{q}}$ holds with $\mathrm{c}_0 > 0$ and $\mathrm{q} \in [0,1)$ [1, Lemma 2.1]. While γ stays in $\bar{B}$, $\|\dot{\gamma}\|_{\mathsf{G}}^2 = \nabla\mathcal{F}^{\top}\mathsf{G}^{-1}\nabla\mathcal{F} \ge \mathrm{c}_2^{-1}\|\nabla\mathcal{F}\|^2$, hence

$$\frac{d}{dt}\,(\mathcal{F}(\gamma)-\mathcal{F}_*)^{1-\mathrm{q}} = -(1-\mathrm{q})\,(\mathcal{F}(\gamma)-\mathcal{F}_*)^{-\mathrm{q}}\,\|\dot{\gamma}\|_{\mathsf{G}}^2 \le -(1-\mathrm{q})\mathrm{c}_0\mathrm{c}_2^{-1/2}\|\dot{\gamma}\|_{\mathsf{G}}, \tag{B.3}$$

so, with $\mathrm{C} = ((1-\mathrm{q})\mathrm{c}_0\mathrm{c}_2^{-1/2})^{-1}$, a segment $\gamma([t_1,t_2]) \subseteq \bar{B}$ has Fisher length at most $\mathrm{C}(\mathcal{F}(\gamma(t_1)) - \mathcal{F}_*)^{1-\mathrm{q}}$ and Euclidean length at most $\mathrm{c}_1^{-1/2}$ times that. For t_1 with $\|\gamma(t_1)-Q_*\| < \delta_0/2$ and $\mathrm{c}_1^{-1/2}\mathrm{C}(\mathcal{F}(\gamma(t_1))-\mathcal{F}_*)^{1-\mathrm{q}} < \delta_0/2$, a first exit from $\bar{B}$ after t_1 would give a segment in $\bar{B}$ of Euclidean length both at least and less than $\delta_0/2$; so γ stays in $\bar{B}$, has finite length on $[t_1,\infty)$, and is Cauchy, hence converges to Q_*. If Q_* were noncritical, $\frac{d}{dt}\mathcal{F}(\gamma)$ would tend to $-\|\mathrm{grad}^{\mathsf{G}}\mathcal{F}(Q_*)\|_{\mathsf{G}}^2 < 0$, contradicting convergence of $\mathcal{F}(\gamma)$. This is the argument of Absil et al. [1, Theorem 2.2], with its angle condition supplied by the metric bounds. In the smooth counterexample the Euclidean plane is the Fisher geometry of the unit-variance normal location family, and a curve of finite length in the plane converges, so a history that does not converge has infinite duration. □

Proof of Proposition 7.39. Differentiating $\mathcal{F}(\exp(t\xi)\cdot Q) = \mathcal{F}(Q)$ at $t=0$ gives $\mathsf{G}(\mathrm{grad}^{\mathsf{G}}\mathcal{F}, \xi_{\mathcal{Q}}(Q)) = 0$, and for a velocity $Z \perp V_Q$ the quantity $\|Z-\xi_{\mathcal{Q}}(Q)\|^2 = \|Z\|^2 + \|\xi_{\mathcal{Q}}(Q)\|^2$ is minimal where $\xi_{\mathcal{Q}}(Q) = 0$. In the free and proper case the orbit space is a smooth manifold and π a surjective submersion [74, Theorem 21.10]. With $\mathrm{Hor}_Q = (\ker T_Q\pi)^{\perp}$, $T_Q\pi$ maps Hor_Q isomorphically onto

$T_{\pi(Q)}(\mathcal{Q}/\mathbb{G})$; define $\bar{\mathsf{G}}$ by transporting $\mathsf{G}|_{\mathrm{Hor}_Q}$ through this isomorphism. Each $L_g : Q \mapsto g \cdot Q$ is an isometry with $\pi \circ L_g = \pi$, so TL_g maps Hor_Q onto Hor_{gQ} with $T_{gQ}\pi \circ TL_g = T_Q\pi$ and the transported inner products agree along orbits; $\bar{\mathsf{G}}$ is smooth because, through a smooth local section ι_π of π given by the rank theorem, its coefficients are those of G on the horizontal projections of the pushed coordinate fields, and it is the unique metric for which π is a Riemannian submersion. The invariant $\mathcal{F}$ is constant on the fibers of π, so $\bar{\mathcal{F}} = \mathcal{F} \circ \iota_\pi$ is well defined and C^1 with $\mathcal{F} = \bar{\mathcal{F}} \circ \pi$; hence $d\mathcal{F}$ vanishes on $\ker T\pi$, the gradient is horizontal, and for horizontal Z, $\mathsf{G}(\mathrm{grad}^{\mathsf{G}}\,\mathcal{F}, Z) = d\bar{\mathcal{F}}(T\pi Z) = \bar{\mathsf{G}}(\mathrm{grad}^{\bar{\mathsf{G}}}\,\bar{\mathcal{F}}, T\pi Z)$, which gives the gradient identity. Applying $T\pi$ to (7.41) shows that $\pi \circ Q$ is a natural-gradient history of $\bar{\mathcal{F}}$, and since $T\pi$ is isometric on horizontal vectors and the velocity is horizontal, the two Fisher speeds agree. □

Proof of Theorem 7.40. Exact forms are closed and have zero periods; conversely, under (7.56), integrating $\boldsymbol{\alpha}$ from a base point along curves in $\mathcal{U}$ is path independent and defines a function with $d\mathfrak{t}^\perp = \boldsymbol{\alpha}$. Differentiating $\boldsymbol{\alpha} = -\mathsf{n}^{-1} d\mathcal{F}$ gives (7.57). Since $d\mathcal{F} \neq 0$, the form $d\mathsf{n} \wedge d\mathcal{F}$ vanishes exactly when $d\mathsf{n}$ is locally a multiple of $d\mathcal{F}$, that is, when n does not depend on the other coordinates of a chart whose first coordinate is $\mathcal{F}$. Then $d\mathfrak{t}^\perp \wedge d\mathcal{F} = -\mathsf{n}^{-1} d\mathcal{F} \wedge d\mathcal{F} = 0$, so $\mathfrak{t}^\perp$ is constant on connected pieces of level sets and locally $\mathfrak{t}^\perp = \tau \circ \mathcal{F}$ with $\tau' = -1/\mathsf{n} < 0$. Since $\boldsymbol{\alpha}(\mathbf{e}_\mathcal{F}) = \mathsf{G}(\mathbf{e}_\mathcal{F}, \mathbf{e}_\mathcal{F}) = 1$, the unit rate follows, and the kernel of $\boldsymbol{\alpha} = \mathbf{e}_\mathcal{F}^\flat$ is the orthogonal complement of $\mathbf{e}_\mathcal{F}$. In the example $\mathsf{n} = (z_1^2 + z_2^2)^{1/2}$ and $\boldsymbol{\alpha} = -(z_2 dz_1 + z_1 dz_2)/\mathsf{n}$, whose exterior derivative is (7.58), with coefficient vanishing only on the diagonal. □

Proof of Proposition 7.41. The unit field $\mathbf{e}_\mathcal{F}$ is C^1 because $\mathsf{n} > 0$. Let $\mathcal{W} \subseteq \mathbb{R} \times \mathcal{S}$ be the open set of pairs (t, q) for which $\mathrm{Fl}_t^{\mathbf{e}_\mathcal{F}}(q)$ is defined in $\mathcal{U}$, and put $\Xi(t, q) = \mathrm{Fl}_t^{\mathbf{e}_\mathcal{F}}(q)$, a C^1 map. Each orbit meets $\mathcal{S}$, so Ξ is onto; $\mathcal{F}$ decreases strictly along orbits, so no orbit is closed, and each orbit meets $\mathcal{S}$ once, so Ξ is injective. At (t, q), $T\Xi$ sends ∂_t to $\mathbf{e}_\mathcal{F}$ and $T_q\mathcal{S}$ onto the image of a complement of $\mathbf{e}_\mathcal{F}(q)$ under the invertible map $T\,\mathrm{Fl}_t^{\mathbf{e}_\mathcal{F}}$, which also carries $\mathbf{e}_\mathcal{F}(q)$ to $\mathbf{e}_\mathcal{F}$ at the image point, so transversality makes $T\Xi$ invertible. Hence Ξ is a C^1 diffeomorphism, and $\mathfrak{t}^\mathcal{S} = \mathrm{pr}_1 \circ \Xi^{-1}$ satisfies $d\mathfrak{t}^\mathcal{S}(\mathbf{e}_\mathcal{F}) = 1$, with t the Fisher arc length because $\mathbf{e}_\mathcal{F}$ has unit speed. A level set is transversal and met once because $d\mathcal{F}(\mathbf{e}_\mathcal{F}) = -\mathsf{n} < 0$. If the level sets of $\mathfrak{t}^\mathcal{S}$ are orthogonal to $\mathbf{e}_\mathcal{F}$, then $d\mathfrak{t}^\mathcal{S}$ vanishes on the orthogonal complement and equals one on $\mathbf{e}_\mathcal{F}$, so it equals $\boldsymbol{\alpha}$. In the example the Euclidean descent orbits $\dot{z}_1 = -z_2$, $\dot{z}_2 = -z_1$ in the quadrant are $z_1 = \mathrm{c}_1 e^t + \mathrm{c}_2 e^{-t}$, $z_2 = -\mathrm{c}_1 e^t + \mathrm{c}_2 e^{-t}$ with $\mathrm{c}_2 > |\mathrm{c}_1|$, along which $\mathcal{F} = \mathrm{c}_2^2 e^{-2t} - \mathrm{c}_1^2 e^{2t}$ decreases from $+\infty$ toward 0, so each meets $\{z_1 z_2 = 1\}$ exactly once. □

Proof of Corollary 7.42. At each t, theorem 7.8 applied to the one-parameter family $\{P_{t'}\}$ gives (7.59) and the defect identity (7.60) with its equality clause; taking nonnegative square roots and integrating gives (7.61). For a dominated channel, differentiating the joint density $p_t(y)\mathsf{k}_t(\bar{y} \mid y)$ gives the joint score $\mathbf{s}_t(y) + \partial_t \log \mathsf{k}_t(\bar{y} \mid y)$, whose projection onto the record is (7.62). In the example the fine law does not move, and a Bernoulli law of success probability $\mathrm{b}(t) = (1 + e^{-t})^{-1}$ has Fisher speed $|\dot{\mathrm{b}}|/\sqrt{\mathrm{b}(1-\mathrm{b})} = \sqrt{\mathrm{b}(1-\mathrm{b})} = (2\cosh(t/2))^{-1}$. □

Proof of Theorem 7.47. Put $\gamma(t) = \mathsf{F}_\ell(\mathrm{Fl}_t^{\mathscr{V}_\ell} Q)$, so $\dot{\gamma} = \mathsf{a}_\ell(\mathrm{Fl}_t^{\mathscr{V}_\ell} Q)\mathscr{V}_{\ell+1}(\gamma)$. The integrand of ψ_Q is continuous and positive, so ψ_Q is a C^1 diffeomorphism onto an open interval. The curve $\gamma \circ \psi_Q^{-1}$ satisfies $(\gamma \circ \psi_Q^{-1})' = \mathscr{V}_{\ell+1}(\gamma \circ \psi_Q^{-1})$ on $\mathcal{T}_Q'$ and starts at $\mathsf{F}_\ell Q$, so by local Lipschitz continuity it is

contained in the maximal integral curve through $\mathsf{F}_\ell Q$; this gives the inclusion and the flow identity, with the curve constructed before the coarse flow is evaluated at $\psi_Q(t)$. The endpoint formula is monotonicity of ψ_Q, and in the complete case the limits are the two integrals. In the examples $\mathsf{a}_\ell = \mathsf{F}'_\ell$ because both fields are ∂_z. The converse is the chain rule. □

Proof of Proposition 7.53. In the first example, (7.64) reads $(z_1, 2z_2) = \mathsf{a}(z_1, 2z_2/\mathrm{c})$, which forces $\mathsf{a} = 1$ and $\mathrm{c} = 1$ where $z_1 z_2 \neq 0$. Differentiating the compatibility identity gives $d\mathcal{F}_\ell(Z) = \chi'_\ell d\mathcal{F}_{\ell+1}(T\mathsf{F}_\ell Z) = 0$ for $Z \in \ker T\mathsf{F}_\ell$, so $\mathcal{F}_\ell$ is constant along curves tangent to the fibers, $Z_0 = \operatorname{grad}^{\mathsf{G}_\ell} \mathcal{F}_\ell$ lies in Hor_Q, and for $Z \in \mathrm{Hor}_Q$

$$\mathsf{G}_{\ell+1}(T\mathsf{F}_\ell Z_0, T\mathsf{F}_\ell Z) = \mathsf{d}_\ell^2 d\mathcal{F}_\ell[Z] = \mathsf{d}_\ell^2 \chi'_\ell \mathsf{G}_{\ell+1}\left(\operatorname{grad}^{\mathsf{G}_{\ell+1}} \mathcal{F}_{\ell+1} \circ \mathsf{F}_\ell, T\mathsf{F}_\ell Z\right).$$

Since $T\mathsf{F}_\ell(\mathrm{Hor}_Q)$ is the whole tangent space, $T\mathsf{F}_\ell Z_0 = \mathsf{d}_\ell^2 \chi'_\ell \operatorname{grad}^{\mathsf{G}_{\ell+1}} \mathcal{F}_{\ell+1} \circ \mathsf{F}_\ell$, and negating gives the claim. The quotient case is $\mathsf{d}_\ell \equiv 1$ and $\chi_\ell = \mathrm{id}$. In the normal example the kernel sends $\mathcal{N}(\mu, 1)$ to $\mathcal{N}(\mu, 2)$, the two Fisher metrics are $d\mu^2$ and $\frac{1}{2}d\mu^2$, and $\mathscr{V}_\ell = -\mu$, $\mathscr{V}_{\ell+1} = -2\mu$. For the exact-contraction objective, $\mathcal{F}_{\ell+1} \circ \mathsf{F}_\ell \leq \mathcal{F}_\ell$ with equality exactly on the set $\mathcal{A}$ of fiberwise minimizers, so compatibility with $\chi_\ell = \mathrm{id}$ holds exactly on $\mathcal{A}$; an open subset of $\mathcal{A}$ would contain, with each of its points, a neighborhood of that point in its fiber, which has more than one point, so $\mathcal{A}$ has empty interior. In the last example $\mathcal{A} = \{f = s\}$ and $d\mathcal{F}_\ell(\partial_f) = f - s$, which vanishes on no open set, so the fiber condition fails for every χ_ℓ. □

B.3 Proofs for Lie-Algebra Links

Proof of Proposition 9.5. (a) A link family lies in the image exactly when $\mathring{\Theta}_e^x = U_i^{-1}\Theta_e^x U_j \in \exp(\mathfrak{g})$ for every $e \in E_+$, and then for every $e \in E$ because $\exp(\mathring{\omega})^{-1} = \exp(-\mathring{\omega})$. Passive re-expressions leave twists unchanged, reference changes and active automorphisms conjugate them (Proposition 2.12), and $\exp(\mathfrak{g})$ is closed under conjugation by Lemma 9.1. (b) The first statement is Proposition 9.4. Jacobi's formula $\frac{\mathrm{d}}{\mathrm{d}t} \det e^{t\mathbf{H}} = \operatorname{Tr}(\mathbf{H}) \det e^{t\mathbf{H}}$ gives $\det e^{\mathbf{H}} = e^{\operatorname{Tr}\mathbf{H}} > 0$. If $\operatorname{diag}(-1, -2, 1, \ldots, 1)$ were $e^{\mathbf{H}}$ with $\mathbf{H}$ real, then $\mathbf{H}$ would commute with it, since it is a power series in $\mathbf{H}$, and would therefore preserve its eigenspace for the eigenvalue -1, which is spanned by the first standard basis vector $\mathbf{b}$; with $\mathbf{H}\mathbf{b} = \mathrm{c}\mathbf{b}$ for a real c, this gives $e^{\mathbf{H}}\mathbf{b} = e^{\mathrm{c}}\mathbf{b} \neq -\mathbf{b}$, a contradiction. (c) For $\mathbf{B} \in \mathrm{GL}^+(K)$ the spectral theorem writes the positive definite matrix $\mathbf{B}\mathbf{B}^\top$ as $\mathbf{O}_0 \operatorname{diag}(d_1, \ldots, d_K)\mathbf{O}_0^\top$ with $\mathbf{O}_0$ orthogonal and every $d_k > 0$. The symmetric matrix $\mathbf{S} = \mathbf{O}_0 \operatorname{diag}(\frac{1}{2}\log d_1, \ldots, \frac{1}{2}\log d_K)\mathbf{O}_0^\top$ has $e^{2\mathbf{S}} = \mathbf{B}\mathbf{B}^\top$, and $\mathbf{O}_1 = e^{-\mathbf{S}}\mathbf{B}$ satisfies $\mathbf{O}_1\mathbf{O}_1^\top = e^{-\mathbf{S}}\mathbf{B}\mathbf{B}^\top e^{-\mathbf{S}} = I_K$ and $\det \mathbf{O}_1 = \det \mathbf{B} / \det e^{\mathbf{S}} > 0$, so $\mathbf{O}_1 \in SO(K)$. By the real normal form of an orthogonal matrix there is an orthogonal $\mathbf{O}_2$ such that $\mathbf{O}_2^\top \mathbf{O}_1 \mathbf{O}_2$ is block diagonal with diagonal entries ± 1 and 2×2 blocks $R(\phi) = \exp(\phi(E_{21} - E_{12}))$; since $\det \mathbf{O}_1 = 1$ the entries -1 occur an even number of times and pair into blocks $R(\pi)$, so $\mathbf{O}_2^\top \mathbf{O}_1 \mathbf{O}_2 = \exp(\mathbf{T}_0)$ with $\mathbf{T}_0$ block diagonal and skew-symmetric. Hence $\mathbf{O}_1 = \exp(\mathbf{T})$ with $\mathbf{T} = \mathbf{O}_2 \mathbf{T}_0 \mathbf{O}_2^\top$ skew-symmetric, $\mathbf{B} = \exp(\mathbf{S})\exp(\mathbf{T})$, and the path $t \mapsto \exp(t\mathbf{S})\exp(t\mathbf{T})$ joins I_K to $\mathbf{B}$ inside $\mathrm{GL}^+(K)$. The Lie algebra of $SO(K)$ consists of the skew-symmetric matrices, so for $G = SO(K)$ every twist is an exponential. (d) The differential of $\exp$ at 0 is the identity of $\mathfrak{g}$, since $\exp(t\mathring{\omega})$ has velocity $\mathring{\omega}$ at $t = 0$, so by the inverse function theorem $\exp(\mathfrak{g})$ contains an open

neighborhood of e_G. The subgroup generated by $\exp(\mathfrak{g})$ is therefore a union of translates of that neighborhood, hence open; its complement is a union of its cosets, hence open; and a nonempty subset of a connected group that is open and closed is the whole group. (e) Let the two pairs be traversed at the positions $a < a'$ of p. Given $\mathbf{B} \in \mathrm{GL}^+(K)$, write $\mathbf{B} = \exp(\mathbf{S})\exp(\mathbf{T})$ by (c), give the pair traversed at a the generator $\pm\mathbf{S}$ and the pair traversed at a' the generator $\pm\mathbf{T}$, with the signs chosen by eq. (9.2) so that the traversed copies carry $\mathbf{S}$ and $\mathbf{T}$, and give every other pair the generator 0. Every other twist along p is then I_K, and $\mathring{\Theta}_p = \mathbf{B}$. The displayed case is $\mathrm{diag}(1,2)R(\pi) = \mathrm{diag}(-1,-2)$, which is not an exponential by (b). □

Proof of Proposition 9.10. Write $\mathsf{G} = \mathsf{G}^{\mathrm{fib}}$ and $\Delta = I_{\mathbb{V}} - \rho_x(\mathring{\Theta}_p^x)$, and note that, for invertible $\mathbf{R}$ and symmetric positive definite G', cyclicity of the trace gives

$$\|\mathbf{R}^{-1}\Delta\mathbf{R}\|^2_{\mathsf{G}'} = \mathrm{Tr}\big(\mathbf{R}\mathsf{G}'^{-1}\mathbf{R}^\top\Delta^\top\mathbf{R}^{-\top}\mathsf{G}'\mathbf{R}^{-1}\Delta\big) = \|\Delta\|^2_{\mathbf{R}^{-\top}\mathsf{G}'\mathbf{R}^{-1}}, \tag{B.4}$$

where a subscript names the matrix used in eq. (9.6).

(a) Cyclicity gives $\langle\mathbf{H},\mathbf{H}'\rangle_{\mathsf{G}} = \mathrm{Tr}\big((\mathsf{G}^{1/2}\mathbf{H}\mathsf{G}^{-1/2})^\top(\mathsf{G}^{1/2}\mathbf{H}'\mathsf{G}^{-1/2})\big)$, the Frobenius inner product of the images under the algebra isomorphism $\mathbf{H} \mapsto \mathsf{G}^{1/2}\mathbf{H}\mathsf{G}^{-1/2}$, which yields the inner product, submultiplicativity, and $\|I_{\mathbb{V}}\|_{\mathsf{G}} = \|I_{\mathbb{V}}\|_{\mathrm{F}}$. The defect is continuous because $\exp$, products, and ρ_x are; it depends only on the twists along p, which are functions of the links; its zero set is that of a norm; and $\rho_x(\mathrm{Hol}^x(p)) = \rho_x(U_{i_0})\rho_x(\mathring{\Theta}_p^x)\rho_x(U_{i_0})^{-1}$ by eq. (9.5).

(b) A passive re-expression changes neither the generators relative to σ (Proposition 9.3) nor σ and G. Relative to σ' the reference holonomy is $g_0^{-1}\mathring{\Theta}_p^x g_0$ (Proposition 9.3), whose defect matrix is $\mathbf{R}^{-1}\Delta\mathbf{R}$. The quotient convention eq. (2.7) gives $[\sigma, v] = [\sigma'\cdot g_0^{-1}, v] = [\sigma', \mathbf{R}^{-1}v]$, so the coordinates of σ' are $\mathbf{R}^{-1}$ times those of σ, and the inner product with matrix G at σ has the matrix $\mathbf{R}^\top\mathsf{G}\mathbf{R}$ at σ'. Taking $\mathsf{G}' = \mathbf{R}^\top\mathsf{G}\mathbf{R}$ in eq. (B.4) returns $\|\Delta\|^2_{\mathsf{G}}$, and $\mathsf{G}' = \mathsf{G}$ gives the second statement. If $\mathbf{R}^\top\mathsf{G}\mathbf{R} = \mathrm{c_o}^2\mathsf{G}$ with $\mathrm{c_o} \neq 0$, then $\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1} = \mathrm{c_o}^{-2}\mathsf{G}$, and Lemma 9.9 gives equality. Conversely, let p traverse the pair of e exactly once, let $\mathbf{H} \in \mathfrak{gl}(\mathbb{V})$, choose $\mathring{\omega} \in \mathfrak{g}$ with $\mathrm{d}\rho_x\mathring{\omega} = \mathbf{H}$, and give the pair of e the generator $t\mathring{\omega}$ and every other pair the generator 0. Then $\rho_x(\mathring{\Theta}_p^x) = \exp(\pm t\mathbf{H})$ by Lemma 9.1, so $\Delta = \mp t\mathbf{H} + O(t^2)$. Dividing the assumed equality $\|\Delta\|^2_{\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1}} = \|\Delta\|^2_{\mathsf{G}}$ by t^2 and letting $t \to 0$ gives $\|\mathbf{H}\|_{\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1}} = \|\mathbf{H}\|_{\mathsf{G}}$ for every $\mathbf{H}$, so Lemma 9.9 gives $\mathbf{R}^{-\top}\mathsf{G}\mathbf{R}^{-1} = \mathrm{c_s}\mathsf{G}$ with $\mathrm{c_s} > 0$, and $\mathrm{c_s}^{1/2}\mathbf{R}$ is a G-isometry. For the rotated walk, $\mathring{\Theta}^x_{(e_1,\ldots,e_{r-1},e_0)} = (\mathring{\Theta}^x_{e_0})^{-1}\mathring{\Theta}_p^x\mathring{\Theta}^x_{e_0}$, and eq. (B.4) applies with $\mathbf{R} = \rho_x(\mathring{\Theta}^x_{e_0})$ and $\mathsf{G}' = \mathsf{G}$.

(c) By the quotient convention, $[\sigma, v] = [u_i \cdot U_i, v] = [u_i, \rho_x(U_i)v]$, so the coordinates of u_i are $\rho_x(U_i)$ times those of σ, which gives the matrix $\mathsf{G}_i^{\mathrm{fib}}$. The law under passive re-expression follows from $U_i' = g_i U_i$ in eq. (2.11), and the relation between agents from $T_{ij} = U_i U_j^{-1}$. Since $I_{\mathbb{V}} - \mathsf{U} = \rho_x(U_{i_0})\Delta\rho_x(U_{i_0})^{-1}$, eq. (B.4) with $\mathbf{R} = \rho_x(U_{i_0})^{-1}$ and $\mathsf{G}' = \mathsf{G}^{\mathrm{fib}}_{i_0}$ gives $\|I_{\mathbb{V}} - \mathsf{U}\|^2_{\mathsf{G}^{\mathrm{fib}}_{i_0}} = \|\Delta\|^2_{\mathsf{G}}$, which is eq. (9.9). For the converse, $\rho_x(g_{i_0}U_{i_0})^\top(R^x_{i_0})^{-\top}\widetilde{\mathsf{G}}_{i_0}(R^x_{i_0})^{-1}\rho_x(g_{i_0}U_{i_0}) = \rho_x(U_{i_0})^\top\widetilde{\mathsf{G}}_{i_0}\rho_x(U_{i_0})$.

(d) For $\mathbf{B} = \rho_x(\mathring{\Theta}_p^x)$ with $\mathbf{B}^\top\mathsf{G}\mathbf{B} = \mathsf{G}$, $\Delta^\top\mathsf{G}\Delta = 2\mathsf{G} - \mathbf{B}^\top\mathsf{G} - \mathsf{G}\mathbf{B}$ and $\mathrm{Tr}(\mathsf{G}^{-1}\mathbf{B}^\top\mathsf{G}) = \mathrm{Tr}\,\mathbf{B}$, so $2\mathfrak{Q}_p^x = 2\dim\mathbb{V} - 2\,\mathrm{Tr}\,\mathbf{B}$. The matrix $\mathsf{G}^{1/2}\mathbf{B}\mathsf{G}^{-1/2}$ is orthogonal, so the eigenvalues of $\mathbf{B}$ have modulus one and $|\mathrm{Tr}\,\mathbf{B}| \leq \dim\mathbb{V}$; $\mathrm{Tr}\,\mathsf{U} = \mathrm{Tr}\,\mathbf{B}$ because the two are similar.

(e) By linearity of $\mathrm{d}\rho_x$ and eq. (9.2), $\sum_a \mathring{\Omega}_a = \mathrm{d}\rho_x(\mathbf{C}\mathring{\omega}^x)_p$. Expanding each $\exp(\mathring{\Omega}_a)$ in its series and multiplying, $\rho_x(\mathring{\Theta}_p^x) - I_{\mathbb{V}} - \sum_a \mathring{\Omega}_a$ is the absolutely convergent sum $\mathbf{T}_p$ of the ordered products $\mathring{\Omega}_0^{l_0}\cdots\mathring{\Omega}_{r-1}^{l_{r-1}}/(l_0!\cdots l_{r-1}!)$ over the multi-indices with $l_0+\cdots+l_{r-1}\geq 2$, and by (a) $\|\mathbf{T}_p\|_{\mathsf{G}} \leq \sum_{l_0+\cdots+l_{r-1}\geq 2}\prod_a \|\mathring{\Omega}_a\|_{\mathsf{G}}^{l_a}/l_a! = \exp(\mathring{\varsigma}_p) - 1 - \mathring{\varsigma}_p$. Since $2\mathfrak{Q}_p^x = \|\sum_a \mathring{\Omega}_a + \mathbf{T}_p\|_{\mathsf{G}}^2$ and $\|\sum_a \mathring{\Omega}_a\|_{\mathsf{G}} \leq \mathring{\varsigma}_p$, expanding the square gives eq. (9.10). From $\exp(\mathring{\varsigma}) - 1 - \mathring{\varsigma} = \mathring{\varsigma}^2\sum_{k\geq 0}\mathring{\varsigma}^k/(k+2)! \leq \frac{1}{2}\mathring{\varsigma}^2\exp(\mathring{\varsigma})$, the right side of eq. (9.10) is at most $\frac{1}{2}\mathring{\varsigma}_p^3\exp(\mathring{\varsigma}_p) + \frac{1}{8}\mathring{\varsigma}_p^4\exp(2\mathring{\varsigma}_p)$, hence at most $(\frac{1}{2}\exp(1) + \frac{1}{8}\exp(2))\mathring{\varsigma}_p^3$ for $\mathring{\varsigma}_p \leq 1$, and $\frac{1}{2}\exp(1) + \frac{1}{8}\exp(2) < 2.283$. With the pulled-back norm, $\|\mathring{\Omega}_a\|_{\mathsf{G}} = \|\mathring{\omega}_{e_a}^x\|_{\mathsf{G}} \leq \|\mathring{\omega}^x\|$. □

Proof of Proposition 9.12. (a) By Proposition 9.10(a) each $\mathfrak{Q}_p^x$, and hence $\mathscr{S}$, is finite, continuous, and nonnegative. For the mean, the series of the exponential and Proposition 9.10(a) give $\|\exp\mathbf{H}\|_{\mathsf{G}} \leq \|I_{\mathbb{V}}\|_{\mathsf{G}} + e^{\|\mathbf{H}\|_{\mathsf{G}}} - 1 \leq (\dim\mathbb{V})^{1/2}e^{\|\mathbf{H}\|_{\mathsf{G}}}$, so for a walk p of length r, with $\mathring{\varsigma}_p$ as in Proposition 9.10(e),

$$\mathfrak{Q}_p^x \leq \tfrac{1}{2}\big((\dim\mathbb{V})^{1/2} + (\dim\mathbb{V})^{r/2}e^{\mathring{\varsigma}_p}\big)^2 \leq 2(\dim\mathbb{V})^r e^{2\mathring{\varsigma}_p}.$$

The quantity $\mathring{\varsigma}_p$ is at most the operator norm of $\mathrm{d}\rho_x$ times the sum of the norms of the generators on the pairs that p traverses, each counted as often as it is traversed. Under $\nu^{\mathscr{T}}$ these generators are independent Gaussian vectors, and the norm of a Gaussian vector has finite exponential moments of every order, so $e^{2\mathring{\varsigma}_p}$ and therefore $\mathscr{S}$ have finite means. Since $0 < e^{-\mathscr{S}} \leq 1$ and $\nu^{\mathscr{T}}$ is a probability law, $0 < Z_{\mathscr{S}} \leq 1$, and convexity of $u \mapsto e^{-u}$ gives $Z_{\mathscr{S}} = \mathbb{E}_{\nu^{\mathscr{T}}}e^{-\mathscr{S}} \geq \exp(-\mathbb{E}_{\nu^{\mathscr{T}}}\mathscr{S})$ by Jensen's inequality. When $\rho_x(G)$ consists of isometries, Proposition 9.10(d) gives $\mathscr{S} \leq 2\dim\mathbb{V}\sum_p \mathrm{c}_p$. The density is $Z_{\mathscr{S}}^{-1}e^{-\mathscr{S}}$ times the product of the Gaussian densities, which are positive and continuous because every covariance is positive definite. The hypotheses of Proposition 5.26 on base law, action, and normalizer are that $\nu^{\mathscr{T}}$ is a probability law on a standard Borel space, that $\mathscr{S}$ is measurable with values in $(-\infty,+\infty]$, and that $0 < Z_{\mathscr{S}} < \infty$. The defect depends on the generators only through the links (Proposition 9.10(a)), and eq. (9.5) gives $\mathring{\Theta}_p^x = U_{i_0}^{-1}\mathrm{Hol}^x(p)U_{i_0}$, so $\widetilde{\mathscr{S}}$ is a continuous function on G^{E_+} with $\mathscr{S} = \widetilde{\mathscr{S}}\circ\exp_\sigma$, and the change of variables for image measures gives $\int e^{-\widetilde{\mathscr{S}}}d(\exp_\sigma)_{\#}\nu^{\mathscr{T}} = Z_{\mathscr{S}}$.

(b) The generators, anchors, covariances, inner product, walks, and the matrix at σ are unchanged by passive re-expression, and the links move by the vertex law (Proposition 9.3(a)).

(c) By eq. (9.1), $\mathrm{d}\rho_x(\mathrm{Ad}_{g_0^{-1}}\mathring{\omega}) = \mathbf{R}^{-1}\mathrm{d}\rho_x(\mathring{\omega})\mathbf{R}$, so polarizing eq. (B.4) with $\mathsf{G}' = \mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R}$ shows that $\mathrm{Ad}_{g_0^{-1}}$ maps the pullback for $\mathsf{G}^{\mathrm{fib}}$ isometrically onto the pullback for $\mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R}$; in general $\mathrm{Ad}_{g_0^{-1}}$ is an isometry from $\langle\cdot,\cdot\rangle_{\mathsf{G}}$ onto $\langle\mathrm{Ad}_{g_0}\cdot,\mathrm{Ad}_{g_0}\cdot\rangle_{\mathsf{G}}$ by construction. A linear isometry between inner product spaces carries the normalized Lebesgue measure to the normalized Lebesgue measure and $\mathcal{N}(\mathring{\alpha},\mathring{\Sigma})$ to the Gaussian law with mean $\mathrm{Ad}_{g_0^{-1}}\mathring{\alpha}$ and covariance operator $\mathrm{Ad}_{g_0^{-1}}\mathring{\Sigma}\,\mathrm{Ad}_{g_0}$: in the new inner product $\langle\mathrm{Ad}_{g_0}\cdot,\mathrm{Ad}_{g_0}\cdot\rangle_{\mathsf{G}}$, the pairing of $\mathring{\beta}\in\mathfrak{g}$ with $\mathrm{Ad}_{g_0^{-1}}\mathring{\omega}$ is $\langle\mathrm{Ad}_{g_0}\mathring{\beta},\mathring{\omega}\rangle_{\mathsf{G}}$, whose variance is $\langle\mathrm{Ad}_{g_0}\mathring{\beta},\mathring{\Sigma}\,\mathrm{Ad}_{g_0}\mathring{\beta}\rangle_{\mathsf{G}}$. By Proposition 9.10(b) the defect at σ' with $\mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R}$ of $\mathrm{Ad}_{g_0^{-1}}\mathring{\omega}^x$ equals $\mathfrak{Q}_p^x(\mathring{\omega}^x)$, so the transformed action composed with $\mathrm{Ad}_{g_0^{-1}}$ is $\mathscr{S}$ and the normalizer is unchanged. Finally $\exp_{\sigma'}\circ\mathrm{Ad}_{g_0^{-1}} = \exp_\sigma$ by Proposition 9.3(b).

(d) Under the hypotheses, the data of (c) are the anchors, covariances, inner product, walks, and coefficients of the prior together with the fiber matrix $\mathbf{R}^\top\mathsf{G}^{\mathrm{fib}}\mathbf{R} = \mathrm{c_o}^2\mathsf{G}^{\mathrm{fib}}$ for a real $\mathrm{c_o} \neq 0$, and the

trace form eq. (9.6), hence the defect, is unchanged when its matrix is multiplied by a positive scalar (Lemma 9.9). For the pulled-back inner product, $\|\mathrm{d}\rho_x(\mathrm{Ad}_{g_0^{-1}}\mathring{\omega})\|_{\mathsf{G}} = \|\mathrm{d}\rho_x\mathring{\omega}\|_{\mathbf{R}^{-\top}\mathsf{G}^{\mathrm{fib}}\mathbf{R}^{-1}}$ by eq. (B.4), which equals $\|\mathrm{d}\rho_x\mathring{\omega}\|_{\mathsf{G}}$ when $\mathbf{R}^{-\top}\mathsf{G}^{\mathrm{fib}}\mathbf{R}^{-1} = \mathrm{c_o}^{-2}\mathsf{G}^{\mathrm{fib}}$; when $\mathrm{d}\rho_x(\mathfrak{g}) = \mathfrak{gl}(\mathbb{V})$, equality for every $\mathring{\omega}$ forces $\mathbf{R}^{-\top}\mathsf{G}^{\mathrm{fib}}\mathbf{R}^{-1}$ to be a positive multiple of $\mathsf{G}^{\mathrm{fib}}$ by Lemma 9.9. □

Proof of Proposition 9.13. (a) Block multiplication gives $\rho(\mathring{\omega})\rho(\mathring{\omega}') = \rho(\mathring{\omega}+\mathring{\omega}')$, and ρ is injective and smooth. The one-parameter subgroups of $(\mathbb{R}^K,+)$ are $t\mapsto t\mathring{\omega}$, so $\exp$ is the identity, and $\mathrm{d}\rho(\mathring{\omega})^2 = 0$ gives $\exp(\mathrm{d}\rho\mathring{\omega}) = I_{K+1} + \mathrm{d}\rho(\mathring{\omega}) = \rho(\mathring{\omega})$. The twist eq. (2.32) reads $\mathring{\omega}_e = -U_i + \Theta_e + U_j$ additively, and along a closed walk the frame terms $U_{i_{a+1}} - U_{i_a}$ sum to zero.

(b) With $\mathbf{b}$ the last standard basis vector of $\mathbb{R}^{K+1}$ and $\mathring{\xi} = (\mathbf{C}\mathring{\omega})_p$, the matrix $I_{K+1} - \rho(\mathring{\xi}) = -\mathrm{d}\rho(\mathring{\xi}) = -(\mathring{\xi},0)\mathbf{b}^\top$ has rank at most one, and the rank-one formula in the proof of Lemma 9.9 gives $\|(\mathring{\xi},0)\mathbf{b}^\top\|_{\mathsf{G}}^2 = (\mathring{\xi},0)^\top\mathsf{G}^{\mathrm{fib}}(\mathring{\xi},0)\cdot\mathbf{b}^\top(\mathsf{G}^{\mathrm{fib}})^{-1}\mathbf{b}$, which is both $2\mathfrak{Q}_p$ and $\|\mathring{\xi}\|_{\mathsf{G}}^2$.

(c) The configuration priors are stated in Definitions 14.21 and 14.23, and a standard normal vector has independent coordinates. In generator coordinates $\mathring{\omega}_e = \omega_e - U_i + U_j$ is normal with mean $U_j - U_i$ and unit variance, and around a closed walk the means telescope to zero. The edge e_i points from i to $i+1$, so $e_a : i_{a+1}\to i_a$ holds along (e_3,e_2,e_1,e_0) with the vertices $0,3,2,1,0$, and part (b) gives the defect. The residual sums telescope the means around the cycle.

(d) The normalized loop covector of q has the entries $1/\sqrt{3}$ on the links of q and zero elsewhere (Lemma 14.2), so $h_q = \mathfrak{h}_q/\sqrt{3}$, and part (b) gives $\mathfrak{Q}_q = \frac{1}{2}\mathfrak{h}_q^2$, hence $\mathrm{J}_1 h_q^2 = \frac{2}{3}\mathrm{J}_1\mathfrak{Q}_q$. By eqs. (14.2) and (14.7), $\Omega_0(\vartheta) = \Omega_0^{\perp}(\vartheta) + \vartheta\mathrm{J}_1 P_{\mathcal{H}}$, and $\mathbf{x}^\top P_{\mathcal{H}}\mathbf{x} = \sum_q h_q^2$ because the normalized loop covectors are orthonormal, which gives eq. (9.16). By Lemma 14.2 the rows of B and the loop covectors are orthogonal and the configuration space is the orthogonal sum $\mathcal{U}\oplus\mathcal{H}\oplus\mathcal{K}$. The matrices $B^\top\mathrm{diag}(\mathrm{J}_\ell I)B$, $P_{\mathcal{K}}$, and $R_{\mathrm{ev}} = \mathfrak{D}^\top\mathfrak{D}/(1-\sigma_{\mathrm{ev}}^2)$, whose rows $\mathfrak{d}_i$ are rows of B, annihilate $\mathcal{H}$ and map into $\mathcal{U}\oplus\mathcal{K}$, so $\Omega_0^{\perp}(\vartheta)$ annihilates $\mathcal{H}$ and agrees with $\Omega_0(\vartheta)$ on $\mathcal{U}\oplus\mathcal{K}$, where it is positive definite by Proposition 14.4. Hence $\Omega_0(\vartheta)$ is block diagonal for $\mathcal{H}\oplus(\mathcal{U}\oplus\mathcal{K})$ with the block $\vartheta\mathrm{J}_1$ on $\mathcal{H}$, and the Gaussian prior with precision $\Omega_0(\vartheta)/\mathrm{T}$ makes $P_{\mathcal{H}}\mathbf{x}$ independent of the other component, with covariance $(\mathrm{T}/(\vartheta\mathrm{J}_1))P_{\mathcal{H}}$; since $\mathfrak{h}_q = \sqrt{3}h_q$ and the loop covectors are orthonormal, the $\mathfrak{h}_q$ are independent with variance $3\mathrm{T}/(\vartheta\mathrm{J}_1)$. In eq. (14.13) the matrix of E_ϑ differs from ϑH by a multiple of $P_{\mathcal{K}}$, its linear term is along $z_\star\in\mathcal{K}$, and $\mathrm{E}_\vartheta^{\mathrm{phys}}$ differs from E_ϑ by a multiple of $\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}$, so the same block decomposition gives the energy statement. Finally, $\Omega_0^{\perp}(\vartheta)$ has the nonzero kernel $\mathcal{H}$, so $e^{-\mathbf{x}^\top\Omega_0^{\perp}(\vartheta)\mathbf{x}/(2\mathrm{T})}$ is constant along $\mathcal{H}$ and not integrable, and it depends on means and links through B. □

Proof of Proposition 9.16. (a) The prior is a probability law by proposition 9.12. Composing the measurable map $\mathring{\omega}\mapsto\Theta(\mathring{\omega})$ with the probability kernel $\Theta\mapsto\mathbb{P}^{\mathsf{Y}}_{\theta,V}(\cdot\mid X,\Theta)$ of proposition 4.4 and with the record kernels gives probability kernels in $\mathring{\omega}$. Placing the generator family before every agent in a topological order of Γ, as a parent of every factor that reads a link, makes (9.18) the iterated integral (4.7) of a directed latent law on $\mathscr{T}\times\mathsf{Y}$ with records conditionally independent given $(\mathring{\omega},y)$, and proposition 4.4 gives unit mass on every finite record hypergraph. (b) The function L_o is jointly measurable, so Tonelli's theorem gives the measurability of $p(o\mid\cdot)$ and, after integrating y before $\mathring{\omega}$, (9.19). Since $\mathscr{S}$ is finite, the prior has a strictly positive density with respect to Lebesgue measure on $\mathscr{T}$, so the integral of the nonnegative function $p(o\mid\cdot)$ is

positive exactly when that function is positive on a set of positive Lebesgue measure. Bounded densities give $L_o \le \mathrm{c}_{\mathfrak{A}}$, and integration against probability kernels gives both bounds; a binary success record has densities $e^{-E} \le 1$ and $1 - e^{-E} \le 1$ because $E \ge 0$. (c) By (5.13) at $(X, \Theta(\mathring{\omega}))$, the measure $L_o(\mathring{\omega}, y)\mathbb{P}^{\mathsf{Y}}(dy \mid \mathring{\omega})\mathbb{P}_\theta^{\mathscr{T}}(d\mathring{\omega} \mid X)$ equals $p(o \mid \mathring{\omega})\mathbf{\Pi}_{\mathring{\omega}}(dy)\mathbb{P}_\theta^{\mathscr{T}}(d\mathring{\omega} \mid X)$, and dividing by the evidence gives the disintegration. Lemma 3.12(a) gives $D_{\mathrm{KL}}(Q\|\mathbf{\Pi}_{\theta,V,o,X}) = D_{\mathrm{KL}}(R\|\mathbf{\Pi}^{\mathscr{T}}) + \int D_{\mathrm{KL}}(Q_{\mathring{\omega}}\|\mathbf{\Pi}_{\mathring{\omega}})R(d\mathring{\omega})$ in $[0, +\infty]$ with a measurable integrand, and the evidence lies in $(0, \mathrm{c}_{\mathfrak{A}}]$ by (b). If $D_{\mathrm{KL}}(R\|\mathbb{P}_\theta^{\mathscr{T}}) < \infty$, then $R \ll \mathbb{P}_\theta^{\mathscr{T}}$, and since $d\mathbf{\Pi}^{\mathscr{T}}/d\mathbb{P}_\theta^{\mathscr{T}} = p(o \mid \cdot)/p_{\theta,V}(o \mid X)$ is positive and finite, $R \ll \mathbf{\Pi}^{\mathscr{T}}$ with $\log(dR/d\mathbf{\Pi}^{\mathscr{T}}) = \log(dR/d\mathbb{P}_\theta^{\mathscr{T}}) - \log p(o \mid \mathring{\omega}) + \log p_{\theta,V}(o \mid X)$. The first term has an R-integrable negative part (definition 3.10) and the second is bounded below by $-\log \mathrm{c}_{\mathfrak{A}}$, so the integral splits, and adding the conditional term and $-\log p_{\theta,V}(o \mid X)$ gives (9.20), because $\mathcal{F}_V[Q_{\mathring{\omega}}; X, \Theta(\mathring{\omega}), o] = -\log p(o \mid \mathring{\omega}) + D_{\mathrm{KL}}(Q_{\mathring{\omega}}\|\mathbf{\Pi}_{\mathring{\omega}})$. If $D_{\mathrm{KL}}(R\|\mathbb{P}_\theta^{\mathscr{T}}) = +\infty$, the bound $\mathbf{\Pi}^{\mathscr{T}} \le (\mathrm{c}_{\mathfrak{A}}/p_{\theta,V}(o \mid X))\mathbb{P}_\theta^{\mathscr{T}}$ gives $dR/d\mathbf{\Pi}^{\mathscr{T}} \ge (p_{\theta,V}(o \mid X)/\mathrm{c}_{\mathfrak{A}})dR/d\mathbb{P}_\theta^{\mathscr{T}}$ whenever $R \ll \mathbf{\Pi}^{\mathscr{T}}$, so $D_{\mathrm{KL}}(R\|\mathbf{\Pi}^{\mathscr{T}}) = +\infty$ and both sides of (9.20) are $+\infty$. The last sentence is $\log(dR/d\mathbb{P}_\theta^{\mathscr{T}}) = \log(dR/d\nu^{\mathscr{T}}) + \mathscr{S} + \log Z_{\mathscr{S}}$ integrated under the stated integrability. □

Proof of Lemma 9.20. All expectations below are finite by lemma 9.17(a). First, for a continuously differentiable $\mathbf{t}$ with $|\mathbf{t}| + \|\nabla\mathbf{t}\| \le \mathrm{c}_1 e^{\mathrm{c}_2\|\mathring{\omega}\|}$,

$$\mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\big[(\mathring{\omega} - \mu)\mathbf{t}(\mathring{\omega})\big] = \Sigma\mathbb{E}_{\mathcal{N}(\mu,\Sigma)}\nabla\mathbf{t}. \tag{B.5}$$

Indeed, for a standard Gaussian vector $\mathbf{b}$ in an orthonormal basis and a continuously differentiable $\mathbf{t}'$ of exponential growth with its gradient, integration by parts in the l-th coordinate against the standard normal density, whose derivative is minus the coordinate times the density, has vanishing boundary terms and gives $\mathbb{E}[\mathbf{b}_l\mathbf{t}'(\mathbf{b})] = \mathbb{E}[\partial_l\mathbf{t}'(\mathbf{b})]$ after Fubini's theorem; with $\mathbf{t}'(\mathbf{b}) = \mathbf{t}(\mu + \Sigma^{1/2}\mathbf{b})$, whose gradient is $\Sigma^{1/2}\nabla\mathbf{t}$, this is (B.5). The mean derivative follows by differentiating $\mathbb{E}\mathbf{h}(\mu + \Sigma^{1/2}\mathbf{b})$ under the expectation, with domination by $\mathrm{c}_1 e^{\mathrm{c}_2(\|\mu\| + \|\mathring{\eta}\| + \|\Sigma\|^{1/2}\|\mathbf{b}\|)}$ along $\mu + t\mathring{\eta}$, $|t| \le 1$, for each direction $\mathring{\eta}$. For the covariance, $\Sigma_t = \Sigma + t\mathring{\Delta}$ stays between half the smallest eigenvalue of Σ and $\|\Sigma\|$ plus that amount for small $|t|$, so the t-derivative of the Gaussian density, $\frac{1}{2}[\langle\mathring{\omega} - \mu, \Sigma_t^{-1}\mathring{\Delta}\Sigma_t^{-1}(\mathring{\omega} - \mu)\rangle - \operatorname{Tr}(\Sigma_t^{-1}\mathring{\Delta})]$ times the density, is dominated uniformly by a Gaussian times a quadratic, and against $|\mathbf{h}| \le \mathrm{c}_1 e^{\mathrm{c}_2\|\mathring{\omega}\|}$ the derivative passes under the integral: $\frac{d}{dt}\mathbb{E}\mathbf{h}|_{t=0} = \frac{1}{2}\mathbb{E}[\mathbf{h}(\mathring{\omega})\langle\mathring{\omega} - \mu, \Sigma^{-1}\mathring{\Delta}\Sigma^{-1}(\mathring{\omega} - \mu)\rangle] - \frac{1}{2}\operatorname{Tr}(\Sigma^{-1}\mathring{\Delta})\mathbb{E}\mathbf{h}$. Applying (B.5) to $\mathbf{t} = \mathbf{h}\langle\mathring{\xi}, \mathring{\omega} - \mu\rangle$ for a given vector $\mathring{\xi}$, and then to the directional derivatives of $\mathbf{h}$, gives $\mathbb{E}[\mathbf{h}(\mathring{\omega} - \mu) \otimes (\mathring{\omega} - \mu)] = \Sigma(\mathbb{E}\nabla^2\mathbf{h})\Sigma + \Sigma\mathbb{E}\mathbf{h}$, so the first term equals $\frac{1}{2}\operatorname{Tr}(\mathring{\Delta}\mathbb{E}\nabla^2\mathbf{h}) + \frac{1}{2}\operatorname{Tr}(\Sigma^{-1}\mathring{\Delta})\mathbb{E}\mathbf{h}$, which is (9.27). Both derivatives are continuous in (μ, Σ) by dominated convergence in the representation $\mathring{\omega} = \mu + \Sigma^{1/2}\mathbf{b}$, since $\Sigma \mapsto \Sigma^{1/2}$ is continuous on the cone. □

Proof of Proposition 9.21. (a) The action satisfies (9.21) by lemma 9.17(c). On each ball $\|\mathring{\omega}\| \le \mathring{r}$, (9.28) dominates $\mathbf{H}_o$ and its first two $\mathring{\omega}$-derivatives by an integrable function of y, so differentiation passes under $\mathbb{E}_{Q^{\mathsf{Y}}}$ and $\mathbb{E}_{Q^{\mathsf{Y}}}\mathbf{H}_o$ is twice continuously differentiable with derivatives bounded by $\mathrm{c}_1 e^{\mathrm{c}_2\mathring{r}}$ on that ball. At every $\mathring{\omega}$ the held-link model at $\Theta(\mathring{\omega})$ satisfies hypothesis 5.3 for Q^{Y}: its posterior has the density proportional to $e^{-\mathbf{H}_o(\mathring{\omega}, \cdot)}$, which is positive wherever $\mathbf{H}_o$ is finite, hence Q^{Y}-almost everywhere by (9.28), and both halves of (5.6)–(5.7) are finite. So $\mathcal{F}_V[Q^{\mathsf{Y}}; X, \Theta(\mathring{\omega}), o] = \mathbb{E}_{Q^{\mathsf{Y}}}[\log q^{\mathsf{Y}} + \mathbf{H}_o(\mathring{\omega}, Y)]$ by theorem 5.8, and proposition 9.16(c) with proposition 9.18(a,b) gives

(9.29), finite because $\mathbb{E}_R|\mathbf{U}| < \infty$ by lemma 9.17(a). The closed form (9.22) is smooth, and $\mathbb{E}_R\mathbf{U}$ is continuously differentiable in the mean and the covariance $\mathring{\Lambda}^{-1}$ by lemma 9.20; the inversion $\mathring{\Lambda} \mapsto \mathring{\Lambda}^{-1}$ is a diffeomorphism of the cone.

(b) By (9.22) and lemma 9.20, the derivative of (9.29) in the mean along $\mathring{\eta}$ is $\langle \mathring{\Sigma}^{-1}(\mathring{\mu}-\mathring{\alpha})+\mathbb{E}_R\nabla\mathbf{U}, \mathring{\eta}\rangle$, and its derivative in the covariance along a self-adjoint $\mathring{\Delta}$ is $\frac{1}{2}\operatorname{Tr}\big(\mathring{\Delta}(\mathring{\Sigma}^{-1}-\mathring{\Lambda}+\mathbb{E}_R\nabla^2\mathbf{U})\big)$. Stationarity in $(\mathring{\mu}, \mathring{\Lambda})$ is stationarity in $(\mathring{\mu}, \mathring{\Lambda}^{-1})$, and a self-adjoint operator orthogonal to every self-adjoint $\mathring{\Delta}$ vanishes; differentiation under $\mathbb{E}_{Q^\Upsilon}$ gives $\mathbb{E}_R\nabla\mathbf{U} = \mathbb{E}_R\nabla\mathscr{S} + \mathbb{E}_{R\otimes Q^\Upsilon}\nabla_{\mathring{\omega}}\mathbf{H}_o$ and the same for Hessians. In the mean-field case the admissible $\mathring{\Delta}$ are block diagonal, and $\mathring{\Sigma}$ and $\mathring{\Lambda}$ are block diagonal, so only the diagonal blocks are constrained.

(c) For given y and given remaining generators, lemma 9.19(c) applied to $\mathbf{H}_o$ as a function of the matrix $\rho_x(\mathring{\Theta}_e^x)$ gives $\nabla_{(x,e)}\mathbf{H}_o = (\mathrm{d}\rho_x)^\top\mathbf{L}\big((\mathrm{d}\rho_x\mathring{\omega}_e^x)^\top, \partial\mathbf{H}_o/\partial\rho_x(\mathring{\Theta}_e^x)\big)$; the map $\mathbf{S} \mapsto (\mathrm{d}\rho_x)^\top\mathbf{L}(\mathbf{B}, \mathbf{S})$ is linear and continuous, so it commutes with $\mathbb{E}_{Q^\Upsilon}$, and the (x, e) component of (9.30) becomes (9.33).

(d) Since $\mathbb{E}_R(\mathring{\omega} - \mathring{\alpha}) = \mathring{\mu} - \mathring{\alpha}$, (9.30) is $\mathbb{E}_R\nabla\mathbf{V} = 0$. With $\mathring{\omega} = \mathring{\mu} + \mathring{\Lambda}^{-1/2}\mathbf{b}$, $\nabla\mathbf{V}$ is uniformly continuous on bounded sets and dominated by $\mathrm{c}(1 + \|\mathring{\mu}\| + \|\mathbf{b}\|)e^{\mathrm{c}(\|\mathring{\mu}\|+\|\mathbf{b}\|)}$ for $\|\mathring{\Lambda}^{-1}\| \le 1$; splitting the expectation at $\|\mathbf{b}\| = \mathrm{c}'$ bounds the tail uniformly for $\mathring{\mu}$ in a compact set, and the modulus of continuity controls the rest, which gives the uniform convergence. A limit point then satisfies $\|\nabla\mathbf{V}(\mathring{\mu}_*)\| \le \|\nabla\mathbf{V}(\mathring{\mu}_*) - \nabla\mathbf{V}(\mathring{\mu})\| + \|\nabla\mathbf{V}(\mathring{\mu}) - \mathbb{E}_R\nabla\mathbf{V}\| \to 0$ along the solutions. For the curve, the map $(\mathring{\mu}, t) \mapsto \mathbb{E}\nabla\mathbf{V}(\mathring{\mu}+t\mathring{\Lambda}_0^{-1/2}\mathbf{b})$ is continuously differentiable on $\mathscr{T}\times\mathbb{R}$ by dominated differentiation, equals $\nabla\mathbf{V}(\mathring{\mu})$ at $t = 0$, and has $\mathring{\mu}$-derivative $\nabla^2\mathbf{V}(\mathring{\mu}_*)$ at $(\mathring{\mu}_*, 0)$, so the implicit function theorem gives $t \mapsto \mathring{\mu}(t)$. For the last claim, every law R with $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) = +\infty$ has $D_{\mathrm{KL}}(R\|\mathbb{P}_\theta^{\mathscr{T}}) = +\infty$, because $\mathbb{P}_\theta^{\mathscr{T}} \le Z_{\mathscr{T}}^{-1}\nu^{\mathscr{T}}$, and hence infinite free energy by (9.20); for the others, (9.20) and the split used in (a) give (9.29) in $(-\infty, +\infty]$, because the matter integrand is bounded below by $-\log \mathrm{c}_{\mathfrak{A}}$ and $\mathscr{S} \ge 0$, and with $0 < \int e^{-\mathbf{U}}d\nu^{\mathscr{T}} < \infty$, $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \mathbb{E}_R\mathbf{U} = D_{\mathrm{KL}}(R\|R^\star) - \log\int e^{-\mathbf{U}}d\nu^{\mathscr{T}}$, which is minimal exactly at $R = R^\star$. The density of $\nu^{\mathscr{T}}$ is proportional to $e^{-\frac{1}{2}\langle\mathring{\omega}-\mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega}-\mathring{\alpha})\rangle}$.

(e) A singleton is null for $\nu^{\mathscr{T}}$, which has a Lebesgue density, so (3.19) of proposition 3.11 gives the infinite divergence, and the free energy is infinite as in (d). At a given $\mathring{\mu}$, (9.22) is at least $\frac{1}{2}(\log\det\mathring{\Lambda} - \log\det\mathring{\Sigma}^{-1} - \dim\mathscr{T})$, which tends to $+\infty$ as $\mathring{\Lambda}^{-1} \to 0$, while $\mathbb{E}_R\mathbf{U} \to \mathbf{U}(\mathring{\mu})$ by the argument of (d) applied to $\mathbf{U}$. □

Proof of Proposition 9.23. (a) Pushforward by the bimeasurable map $\rho_x(h)$ preserves relative entropy (lemma 3.12(e)), so $D_e(h\Theta h') = D_{\mathrm{KL}}(q_i\|\widehat{\rho}_x(h)\widehat{\rho}_x(\Theta)q_j) = D_{\mathrm{KL}}(\widehat{\rho}_x(h)^{-1}q_i\|\widehat{\rho}_x(\Theta)q_j) = D_e(\Theta)$. The maps $\Theta \mapsto h\Theta h'$ are diffeomorphisms of G that preserve D_e, so they preserve its minimizers and, where it is continuously differentiable, its stationary points. If $0 \ne \mathring{\xi} \in \mathfrak{g}_{q_i}$, the curve $t \mapsto \exp(t\mathring{\xi})\Theta$ lies in $G_{q_i}\Theta$ and is not constant near $t = 0$, since the exponential map is injective near 0; the case of $\mathfrak{g}_{q_j}$ uses $\Theta\exp(t\mathring{\xi})$. If the exponential map is a local diffeomorphism at the generator, so is the chart map $\mathring{\omega}_e^x \mapsto U_i\exp(\mathring{\omega}_e^x)U_j^{-1}$, and its local inverse carries such a curve to a non-constant curve of minimizers or stationary points of the chart energy. (b) The pushforward of $\mathcal{N}(\mu, \Sigma)$ by g is $\mathcal{N}(g\mu, g\Sigma g^\top)$. Writing $g = \Sigma^{1/2}O\Sigma^{-1/2}$, the condition $g\Sigma g^\top = \Sigma$ is $OO^\top = I_K$, $\det g = \det O > 0$ gives $O \in \mathrm{SO}(K)$, and $g\mu = \mu$ is $O\Sigma^{-1/2}\mu = \Sigma^{-1/2}\mu$; the stabilizer in $\mathrm{SO}(K)$ of a nonzero vector is a copy of $\mathrm{SO}(K-1)$. Invariance of the Mahalanobis norm is

$(g\mu)^\top(g\Sigma g^\top)^{-1}g\mu = \mu^\top\Sigma^{-1}\mu$, which proves necessity. For sufficiency, $\mathrm{SO}(K)$ acts transitively on spheres of $\mathbb{R}^K$ when $K \geq 2$, so some O carries $\Sigma_j^{-1/2}\mu_j$ to $\Sigma_i^{-1/2}\mu_i$, and $\Theta_0 = \Sigma_i^{1/2}O\Sigma_j^{-1/2}$ transports q_j to q_i. The zero set of D_e is $\{\Theta : \widehat{\rho}_x(\Theta)q_j = q_i\} = \Theta_0 G_{q_j}$, and $\Theta_0^{-1}G_{q_i}\Theta_0 = G_{q_j}$. The dimension count and (a) give the last sentence. (c) Under hypothesis 7.1 relative entropy is C^3 near the diagonal (proposition 7.6) and $\Theta \mapsto \widehat{\rho}_x(\Theta)q_j$ is smooth, so D_e is C^3 near Θ_0; it is nonnegative and vanishes at Θ_0, so its differential vanishes there and its Hessian is a well-defined form. Along $t \mapsto \exp(t\mathring{\xi})\Theta_0$, $D_e = D_{\mathrm{KL}}(q_i\|\widehat{\rho}_x(\exp t\mathring{\xi})q_i) = \frac{t^2}{2}\mathfrak{k}_{q_i}(\mathring{\xi},\mathring{\xi}) + O(t^3)$ by (7.10), so the Hessian on the velocity $\mathring{\xi}\Theta_0$ is $\mathfrak{k}_{q_i}(\mathring{\xi},\mathring{\xi})$; every tangent vector at Θ_0 has this form, the radical of $\mathfrak{k}_{q_i}$ is $\mathfrak{g}_{q_i}$ (corollary 7.7), and $\mathfrak{g}_{q_i}\Theta_0$ is the tangent space of $G_{q_i}\Theta_0$. For Gaussian laws the expansion can be computed directly: with $\mu_t = e^{t\mathring{\xi}}\mu = \mu + t\mathring{\xi}\mu + O(t^2)$, $\Sigma_t = e^{t\mathring{\xi}}\Sigma e^{t\mathring{\xi}^\top}$, and $\Sigma^{-1/2}\Sigma_t\Sigma^{-1/2} = I_K + \mathbf{T}_t$, where $\mathbf{T}_t = t\mathbf{B} + O(t^2)$ is self-adjoint and $\mathbf{B} = \Sigma^{-1/2}\mathring{\xi}\Sigma^{1/2} + (\Sigma^{-1/2}\mathring{\xi}\Sigma^{1/2})^\top$, the Gaussian divergence $D_{\mathrm{KL}}(\mathcal{N}(\mu,\Sigma)\|\mathcal{N}(\mu_t,\Sigma_t))$ is $\frac{1}{2}[\mathrm{Tr}((I_K+\mathbf{T}_t)^{-1}) - K + \log\det(I_K+\mathbf{T}_t)] + \frac{1}{2}(\mu_t-\mu)^\top\Sigma_t^{-1}(\mu_t-\mu)$. The power series of $(I_K+\mathbf{T}_t)^{-1}$ and of $\log\det(I_K+\mathbf{T}_t) = \mathrm{Tr}\log(I_K+\mathbf{T}_t)$, valid for $\|\mathbf{T}_t\| < 1$, make the first term $\frac{1}{4}\mathrm{Tr}(\mathbf{T}_t^2) + O(t^3) = \frac{t^2}{4}\mathrm{Tr}(\mathbf{B}^2) + O(t^3)$, since its terms of first order cancel, and the second term is $\frac{t^2}{2}|\Sigma^{-1/2}\mathring{\xi}\mu|^2 + O(t^3)$, which gives (9.36). The form vanishes exactly when $\mathring{\xi}\mu = 0$ and $\Sigma^{-1/2}\mathring{\xi}\Sigma^{1/2}$ is skew, the Lie algebra of G_q in (b). □

Proof of Corollary 9.24. (a)(i) With the second moment $\Sigma_j + \mu_j\mu_j^\top \succ 0$ and $\mathbb{E}[k_ik_j^\top] = \mu_i\mu_j^\top$, the gradient is $\nabla\mathbf{l}(\mathbf{M}) = -W_a\big(\mu_i\mu_j^\top - \mathbf{M}(\Sigma_j+\mu_j\mu_j^\top)\big)$, which vanishes exactly at $\mathbf{M}^\star$, and

$$\mathbf{l}(\mathbf{M}) - \mathbf{l}(\mathbf{M}^\star) = \tfrac{1}{2}\big\|W_a^{1/2}(\mathbf{M}-\mathbf{M}^\star)(\Sigma_j+\mu_j\mu_j^\top)^{1/2}\big\|_{\mathrm{F}}^2, \tag{B.6}$$

so $\mathbf{l}$ is strictly convex; the second form of (9.37) is the Sherman–Morrison identity, and the rank bound is that of $\mu_i\mu_j^\top$. With a cross covariance the same computation uses $\mathbb{E}[k_ik_j^\top] = \mu_i\mu_j^\top + \Sigma_{ij}$. (ii) A rank at most one matrix is not invertible for $K \geq 2$. Every singular matrix is a limit of matrices of positive determinant: in a singular value decomposition, replace each zero singular value by a small positive number, and change the sign of one of them if the determinant would be negative. Continuity gives the infimum, uniqueness of the minimizer excludes attainment, and (B.6) forces minimizing sequences to converge. (iii) By lemma 9.19(c) the gradient of $\mathring{\omega} \mapsto \mathbf{l}(e^{\mathring{\omega}})$ for the trace pairing is $\mathbf{L}(\mathring{\omega}^\top, \nabla\mathbf{l}(e^{\mathring{\omega}}))$, and $\nabla\mathbf{l}(e^{\mathring{\omega}}) \neq 0$ by (ii), so stationarity forces $\mathbf{L}(\mathring{\omega}^\top,\cdot)$, and with it its adjoint $\mathbf{L}(\mathring{\omega},\cdot)$, to be singular. In the example, with $\mathbf{R}_{\mathring{\phi}} = \exp\big(\mathring{\phi}\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)\big)$ the rotation by the angle $\mathring{\phi}$, $e^{\mathring{\omega}} = \frac{1}{3}\mathbf{R}_{2\pi} = \frac{1}{3}I_2$ and $\nabla\mathbf{l}(\frac{1}{3}I_2) = -\big(\mu_i\mu_j^\top - \frac{1}{3}(I_2+\mu_j\mu_j^\top)\big) = \frac{1}{3}\,\mathrm{diag}(-1,1)$; this matrix composed with $\mathbf{R}_{\mathring{\phi}}$ equals $\mathbf{R}_{-\mathring{\phi}}$ composed with it, and since $e^{t\mathring{\omega}^\top} = 3^{-t}\mathbf{R}_{-2\pi t}$ and $e^{(1-t)\mathring{\omega}^\top} = 3^{t-1}\mathbf{R}_{2\pi t}$, $\mathbf{L}\big(\mathring{\omega}^\top, \mathrm{diag}(-1,1)\big) = \frac{1}{3}\int_0^1 \mathbf{R}_{-4\pi t}dt\,\mathrm{diag}(-1,1) = 0$. (iv) If $\mu_j^\top\Sigma_j^{-1}\mu_i > 0$, then $\mathbf{M}^\star$ has the simple eigenvalue $\mathrm{tr}\,\mathbf{M}^\star > 0$ and the eigenvalue 0 with a $(K-1)$-dimensional eigenspace, so $\mathbf{M}^\star = \mathbf{S}\,\mathrm{diag}(\mathrm{tr}\,\mathbf{M}^\star, 0, \dots, 0)\mathbf{S}^{-1}$ for a real invertible $\mathbf{S}$, and for $\mathrm{c}_0 > 0$ the chart links $\mathbf{S}\,\mathrm{diag}(\mathrm{tr}\,\mathbf{M}^\star, \mathrm{c}_0, \dots, \mathrm{c}_0)\mathbf{S}^{-1} = \exp\big(\mathbf{S}\,\mathrm{diag}(\log\mathrm{tr}\,\mathbf{M}^\star, \log\mathrm{c}_0, \dots, \log\mathrm{c}_0)\mathbf{S}^{-1}\big)$ converge to it as $\mathrm{c}_0 \to 0$. If $\mu_j^\top\Sigma_j^{-1}\mu_i = 0$, then $(\mathbf{M}^\star)^2 = 0$, and $\mathrm{c}_0 I_K + \mathbf{M}^\star = \exp\big((\log\mathrm{c}_0)I_K + \mathrm{c}_0^{-1}\mathbf{M}^\star\big)$. Then (B.6) gives the infimum and the convergence of minimizing sequences, and the infimum is not attained because chart links are invertible and $\mathbf{M}^\star$ is not. If $\mu_j^\top\Sigma_j^{-1}\mu_i < 0$, then $\mathrm{tr}\,\mathbf{M}^\star < 0$ is a simple eigenvalue of $\mathbf{M}^\star$; for $\mathbf{M}$ near $\mathbf{M}^\star$ exactly one eigenvalue lies in the disk of radius $\frac{1}{2}|\mathrm{tr}\,\mathbf{M}^\star|$ about $\mathrm{tr}\,\mathbf{M}^\star$, by continuity of the roots of the characteristic polynomial, and it is real, negative and simple because the disk is

symmetric under conjugation. A real matrix $e^{\mathbf{B}}$ has no simple negative eigenvalue: triangularizing $\mathbf{B}$ over $\mathbb{C}$ shows that the eigenvalues of $e^{\mathbf{B}}$ with multiplicity are the e^{β} for the eigenvalues β of $\mathbf{B}$, a negative value requires $\beta \notin \mathbb{R}$, and then $\bar{\beta} \neq \beta$ is an eigenvalue with the same image; this is a case of the criterion of Culver [29, Theorem 1]: a real invertible matrix has a real logarithm exactly when every Jordan block belonging to a negative eigenvalue occurs an even number of times. So a neighborhood of $\mathbf{M}^\star$ contains no chart link, and (B.6) bounds l on its complement away from $\mathrm{l}(\mathbf{M}^\star)$. For $K = 2$, the chart links are the matrices of $\exp(\mathfrak{gl}(2,\mathbb{R}))$. By (B.6), l attains its minimum over the closure of this set, that minimum is the infimum over chart links, and a chart link attaining the infimum would minimize l over the closure. No chart link $\mathbf{M}$ does, because such an $\mathbf{M}$ would satisfy $\nabla \mathrm{l}(\mathbf{M}) = 0$, that is $\mathbf{M} = \mathbf{M}^\star$, which is not in the closure. If $\mathbf{M}$ has no eigenvalue in $(-\infty, 0]$, then neither has any matrix near it, and these matrices are chart links by the criterion, so $\mathbf{M}$ is an interior point of the closure and the gradient of l vanishes there. Otherwise the criterion leaves only $\mathbf{M} = \mathrm{c}' I_2$ with $\mathrm{c}' < 0$, since the second eigenvalue of a real 2×2 matrix with a negative eigenvalue is real. For every $\mathbf{E}$ with $(\mathbf{E}_{11} - \mathbf{E}_{22})^2 + 4\mathbf{E}_{12}\mathbf{E}_{21} < 0$ and every $t > 0$ the matrix $\mathrm{c}' I_2 + t\mathbf{E}$ has non-real eigenvalues and is a chart link, so minimality gives $\langle \nabla \mathrm{l}(\mathrm{c}' I_2), \mathbf{E} \rangle \geq 0$; these $\mathbf{E}$ form a nonempty open set invariant under $\mathbf{E} \mapsto -\mathbf{E}$, so $\nabla \mathrm{l}(\mathrm{c}' I_2)$ is orthogonal to an open set and vanishes. (b) This is proposition 9.23(a,b), since $(K-1)(K-2)/2 \geq 1$ for $K \geq 3$. (c)(i) $\mathbf{V}$ is continuous and at least $\frac{1}{2}\langle \mathring{\omega} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega} - \mathring{\alpha})\rangle + \inf \mathbf{U}$, which tends to $+\infty$ with $\|\mathring{\omega}\|$. By (9.29) the free energy is at least $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \inf \mathbf{U} + \log Z_{\mathscr{S}} + \mathbb{E}_{Q^{\mathsf{Y}}} \log q^{\mathsf{Y}}$. With $\mathring{\lambda}_l$ the eigenvalues of $\mathring{\Sigma}^{-1/2}\mathring{\Lambda}^{-1}\mathring{\Sigma}^{-1/2}$, (9.22) is $\frac{1}{2}\sum_l(\mathring{\lambda}_l - 1 - \log \mathring{\lambda}_l) + \frac{1}{2}\langle \mathring{\mu} - \mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\mu} - \mathring{\alpha})\rangle$, so a bound on it confines the $\mathring{\lambda}_l$ to a compact subset of $(0,\infty)$ and $\mathring{\mu}$ to a bounded set: the sublevel sets are compact in the open domain, and the free energy is continuous there by proposition 9.21(a). (ii) The record densities are at most one and positive, and by proposition 9.18(c) the free energy is $D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_\theta) + D_{\mathrm{KL}}(Q^{\mathsf{Y}}\|\mathbb{P}^{\mathsf{Y}}) + \mathbb{E}_{R\otimes Q^{\mathsf{Y}}} H_o$, finite everywhere by the growth bound, and at least $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \log Z_{\mathscr{S}} + D_{\mathrm{KL}}(Q^{\mathsf{Y}}\|\mathbb{P}^{\mathsf{Y}})$ because $\mathscr{S} \geq 0$ and $H_o \geq 0$; both divergences have compact sublevel sets in their parameters by the argument of (i). The divergences and $\mathbb{E}_R \mathscr{S}$ are continuous, and $\mathbb{E}_{R\otimes Q^{\mathsf{Y}}} H_o$ is lower semicontinuous by Fatou's lemma in the representation by standard Gaussian vectors, so the minimum is attained. □

Proof of Theorem 9.25. Write $\mathring{\omega} = \mathring{\alpha} + \mathring{\sigma}\mathring{\Sigma}_0^{1/2}\mathbf{b}$ under $\nu^{\mathscr{T}}_{\mathring{\sigma}}$, with $\mathbf{b}$ standard Gaussian. Since $0 \leq e^{-\mathscr{S}} \leq 1$ is continuous, $Z_{\mathscr{S},\mathring{\sigma}} = \mathbb{E} e^{-\mathscr{S}(\mathring{\alpha} + \mathring{\sigma}\mathring{\Sigma}_0^{1/2}\mathbf{b})} \to e^{-\mathscr{S}(\mathring{\alpha})} > 0$, and $\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \geq \mathring{\epsilon}\} \leq \nu^{\mathscr{T}}_{\mathring{\sigma}}\{\|\mathring{\omega} - \mathring{\alpha}\| \geq \mathring{\epsilon}\}/Z_{\mathscr{S},\mathring{\sigma}} \to 0$ for every $\mathring{\epsilon} > 0$. By (9.20), every law $Q = R(d\mathring{\omega})Q_{\mathring{\omega}}(dy)$ has $\mathcal{F}_{\mathring{\sigma}}[Q] = D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}) + \int \mathbf{F}_{Q_{\mathring{\omega}}}(\mathring{\omega}) R(d\mathring{\omega})$, with $\mathbf{F}_{Q_{\mathring{\omega}}} \geq -\log p(o \mid \cdot) \geq -\log \mathrm{c}_{\mathfrak{A}}$.

Upper bound. Let $Q \in \mathcal{Q}'_0$ with $\mathbf{F}_Q(\mathring{\alpha}) < \infty$. The law $\nu^{\mathscr{T}}_{\mathring{\sigma}} \otimes Q$ is admissible, and $D_{\mathrm{KL}}(\nu^{\mathscr{T}}_{\mathring{\sigma}}\|\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}) = \mathbb{E}_{\nu^{\mathscr{T}}_{\mathring{\sigma}}} \mathscr{S} + \log Z_{\mathscr{S},\mathring{\sigma}}$. For $\mathring{\sigma} \leq 1$, lemma 9.17(c) dominates $\mathscr{S}(\mathring{\alpha} + \mathring{\sigma}\mathring{\Sigma}_0^{1/2}\mathbf{b})$ by $\mathrm{c}_1 e^{\mathrm{c}_2(\|\mathring{\alpha}\| + \|\mathring{\Sigma}_0\|^{1/2}\|\mathbf{b}\|)}$, which is integrable by lemma 9.17(a), so $\mathbb{E}_{\nu^{\mathscr{T}}_{\mathring{\sigma}}} \mathscr{S} \to \mathscr{S}(\mathring{\alpha})$ and $D_{\mathrm{KL}}(\nu^{\mathscr{T}}_{\mathring{\sigma}}\|\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}) \to 0$. Hypothesis (iii) and the same domination give $\mathbb{E}_{\nu^{\mathscr{T}}_{\mathring{\sigma}}} \mathbf{F}_Q \to \mathbf{F}_Q(\mathring{\alpha})$. Hence the product infimum has $\limsup$ at most $\mathbf{F}_Q(\mathring{\alpha})$, and taking the infimum over $\mathcal{Q}'_0$ gives $\limsup \leq \mathbf{F}(\mathring{\alpha})$.

Lower bound. Let $\mathrm{c}_0 > 0$. By (ii) there is $\mathring{\epsilon} > 0$ with $\mathbf{F} > \mathbf{F}(\mathring{\alpha}) - \mathrm{c}_0$ on $\{\|\mathring{\omega} - \mathring{\alpha}\| < \mathring{\epsilon}\}$, so the bounded simple function $\mathbf{q} = (\mathbf{F}(\mathring{\alpha}) - \mathrm{c}_0)\mathbf{1}_{\{\|\mathring{\omega}-\mathring{\alpha}\|<\mathring{\epsilon}\}} - (\log \mathrm{c}_{\mathfrak{A}})\mathbf{1}_{\{\|\mathring{\omega}-\mathring{\alpha}\|\geq\mathring{\epsilon}\}}$ satisfies $\mathbf{q} \leq \mathbf{F}_{Q_{\mathring{\omega}}}$ for every kernel with values in $\mathcal{Q}'$. For every law R with $D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}) < \infty$, the law $\mathbb{P}^{\mathbf{q}} \propto e^{-\mathbf{q}}\mathbb{P}^{\mathscr{T}}_{\mathring{\sigma}}$ has a density

bounded above and below, so $0 \le D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathbf{q}}) = D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}) + \mathbb{E}_R\mathbf{q} + \log \mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}} e^{-\mathbf{q}}$, and therefore $\mathcal{F}_{\mathring\sigma}[Q] \ge D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}) + \mathbb{E}_R\mathbf{q} \ge -\log \mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}} e^{-\mathbf{q}}$; the bound holds trivially when $D_{\mathrm{KL}}(R\|\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}) = +\infty$. Since $\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\} \to 0$, the right side tends to $\mathbf{F}(\mathring\alpha) - \mathrm{c}_0$. The kernel infimum is at most the product infimum, so both converge to $\mathbf{F}(\mathring\alpha)$.

(b) The decomposition gives $D_{\mathrm{KL}}(R_{\mathring\sigma}\|\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}) \le \mathcal{F}_{\mathring\sigma} + \log \mathrm{c}_{\mathfrak{A}} \le \mathrm{c}$. Data processing (lemma 3.12(d)) through the indicator of $\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}$ bounds c below by the relative entropy between the Bernoulli laws with success probabilities $R_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}$ and $\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}$. Since the entropy of a Bernoulli law is at most $\log 2$ and the failure outcome contributes a nonnegative cross-entropy term, that relative entropy is at least $R_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\} \log\bigl(1/\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}\bigr) - \log 2$, so $R_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\} \le (\mathrm{c} + \log 2)/\log\bigl(1/\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}\bigr) \to 0$. For the second claim, with $\mathbf{q}$ as above, $D_{\mathrm{KL}}(R_{\mathring\sigma}\|\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}) = \mathcal{F}_{\mathring\sigma} - \int \mathbf{F}_{Q_{\mathring\sigma,\mathring\omega}} dR_{\mathring\sigma} \le \mathcal{F}_{\mathring\sigma} - \mathbb{E}_{R_{\mathring\sigma}}\mathbf{q}$, and $\mathbb{E}_{R_{\mathring\sigma}}\mathbf{q} \to \mathbf{F}(\mathring\alpha) - \mathrm{c}_0$ by the first claim, so the $\limsup$ of the divergence is at most c_0.

(c) By (9.19), $|p_{\theta,V}(o \mid X) - p(o \mid \mathring\alpha)| \le \sup_{\|\mathring\omega - \mathring\alpha\| < \mathring\epsilon} |p(o \mid \cdot) - p(o \mid \mathring\alpha)| + \mathrm{c}_{\mathfrak{A}}\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}$, which tends to zero when $\mathring\epsilon$ is chosen first. Then $\mathbf{\Pi}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\} \le \mathrm{c}_{\mathfrak{A}}\mathbb{P}^{\mathscr{T}}_{\mathring\sigma}\{\|\mathring\omega - \mathring\alpha\| \ge \mathring\epsilon\}/p_{\theta,V}(o \mid X) \to 0$, because the evidence tends to $p(o \mid \mathring\alpha) > 0$. For the class of all laws on Y the infimum of the free energy is minus the log evidence (corollary 5.7), so the first claim is the value statement of (a) for that class, here obtained without (ii) and (iii).

(d) Suppose $\mathring\omega_l \to \mathring\alpha$ with $\mathbf{F}(\mathring\omega_l) \to \mathrm{c}' < \mathbf{F}(\mathring\alpha)$ along a subsequence. For large l choose λ_l with $\mathbf{F}_{Q_{\lambda_l}}(\mathring\omega_l) \le \min\{\mathbf{F}(\mathring\omega_l) + 1/l, \mathrm{c}\}$; the λ_l lie in a compact subset of $\mathbb{U}$, a further subsequence converges to some $\lambda \in \mathbb{U}$, and lower semicontinuity gives $\mathbf{F}_{Q_\lambda}(\mathring\alpha) \le \mathrm{c}' < \mathbf{F}(\mathring\alpha)$, which contradicts the definition of $\mathbf{F}$. □

Proof of Proposition 9.28. For (9.42), the link sector is $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) + \mathbb{E}_R\mathscr{S} + \log Z_{\mathscr{S}}$ by (9.23). With r the Lebesgue density of R, $D_{\mathrm{KL}}(R\|\nu^{\mathscr{T}}) = \mathbb{E}_R[\log r] - \mathbb{E}_R[\log(d\nu^{\mathscr{T}}/d\mathring\omega)]$, where $\mathbb{E}_R[\log r] = \frac{1}{2}\log\det\mathring\Lambda - \frac{1}{2}\dim\mathscr{T}\log(2\pi e)$ and $-\log(d\nu^{\mathscr{T}}/d\mathring\omega)$ is the sum of the last two terms of (9.43); substituting in (9.20) gives the display.

(a) The density of R is $\exp(\langle\mathring\vartheta_1, \mathring\omega\rangle + \langle\mathring\omega, \mathring\vartheta_2\mathring\omega\rangle - \mathring\Gamma(\mathring\vartheta))$ with $\mathring\Gamma(\mathring\vartheta) = -\frac{1}{4}\langle\mathring\vartheta_1, \mathring\vartheta_2^{-1}\mathring\vartheta_1\rangle - \frac{1}{2}\log\det(-2\mathring\vartheta_2) + \frac{1}{2}\dim\mathscr{T}\log(2\pi)$. For $\mathring\vartheta_2 \in \mathring\Xi$, $\langle\mathring\omega, \mathring\vartheta_2\mathring\omega\rangle = \mathrm{Tr}(\mathring\vartheta_2\,\mathrm{pr}_{\mathring\Xi}(\mathring\omega\otimes\mathring\omega))$, so the sufficient statistic is $(\mathring\omega, \mathrm{pr}_{\mathring\Xi}(\mathring\omega\otimes\mathring\omega))$. The integral of $\exp(\langle\mathring\vartheta_1, \mathring\omega\rangle + \langle\mathring\omega, \mathring\vartheta_2\mathring\omega\rangle)$ is finite exactly when $\mathring\vartheta_2$ is negative definite, an open condition in $\mathscr{T}\times\mathring\Xi$, so the family is regular, and it is minimal because a nonzero polynomial of degree at most two in $\mathring\omega$ vanishes only on a Lebesgue-null set. A positive-definite operator that preserves each summand has an inverse that preserves each summand, so $\mathring\Lambda^{-1} \in \mathring\Xi$, and $\nabla\mathring\Gamma$, the R-mean of the sufficient statistic, is $\mathring{\varpi}$ as stated. Under R the centered generator $\mathring\omega - \mathring\mu$ has law $\mathcal{N}(0, \mathring\Lambda^{-1})$, and the score of R in the direction $(\delta\mathring\mu, \delta\mathring\Lambda)$ is $\langle\delta\mathring\mu, \mathring\Lambda(\mathring\omega - \mathring\mu)\rangle - \frac{1}{2}(\langle\mathring\omega - \mathring\mu, \delta\mathring\Lambda(\mathring\omega - \mathring\mu)\rangle - \mathrm{Tr}(\mathring\Lambda^{-1}\delta\mathring\Lambda))$. The two parts are uncorrelated because the odd central moments vanish, the first has variance $\langle\delta\mathring\mu, \mathring\Lambda\delta\mathring\mu\rangle$, and Isserlis' identity, by which $\langle\mathring\omega - \mathring\mu, \mathring\Delta(\mathring\omega - \mathring\mu)\rangle$ has variance $2\,\mathrm{Tr}(\mathring\Delta\mathring\Lambda^{-1}\mathring\Delta\mathring\Lambda^{-1})$ for self-adjoint $\mathring\Delta$, gives the second part of (9.44). The form vanishes only when $\delta\mathring\mu = 0$ and $\mathring\Lambda^{-1/2}\delta\mathring\Lambda\mathring\Lambda^{-1/2} = 0$, and its coefficients are smooth. Every minimal regular exponential family with its Fisher metric is an instance of hypothesis 7.16, and a finite product of instances with the direct-sum metric, C^1 and positive definite, is again one.

(b) The metric G is C^1 and F is C^2, so $\operatorname{grad}^{\mathsf{G}} F$ is C^1 and the Picard–Lindelöf theorem gives existence and uniqueness of maximal integral curves. Along one, $\frac{d}{dt}F = dF[\dot{Q}] = -\mathsf{G}(\operatorname{grad}^{\mathsf{G}} F, \operatorname{grad}^{\mathsf{G}} F)$ by (7.40). By proposition 7.32, applied with $\mathsf{M} = \mathsf{G}^{-1}$, a curve through a noncritical point meets no critical point, so its velocity never vanishes, and it satisfies (7.41) with $\mathsf{a} \equiv 1$.

(c) The field depends on the configuration only and its zeros are the zeros of dF. The force balance is (7.40) with Z arbitrary and $\dot{Q} = -\operatorname{grad}^{\mathsf{G}} F$, and evaluating it on $\dot{Q}$ gives the dissipation. Strict decrease along nonconstant solutions is proposition 7.32, and a periodic nonconstant solution would return to a value of F it has left. In a chart, the derivative of $Q \mapsto -\mathsf{G}(Q)^{-1}\nabla F(Q)$ at $Q_\star$ is $-\mathsf{G}(Q_\star)^{-1}\nabla^2 F(Q_\star)$, because the term containing the derivative of G^{-1} is multiplied by $\nabla F(Q_\star) = 0$; this matrix is similar to the symmetric matrix $-\mathsf{G}(Q_\star)^{-1/2}\nabla^2 F(Q_\star)\mathsf{G}(Q_\star)^{-1/2}$, hence diagonalizable with real eigenvalues, and every solution of the linearized equation is a combination of real exponentials.

(d) In the natural chart the Fisher metric is $\nabla^2\mathring{\Gamma}$, and the chain rule through $\mathring{\varpi} = \nabla\mathring{\Gamma}(\mathring{\vartheta})$ gives $\nabla_{\mathring{\vartheta}} F = \nabla^2\mathring{\Gamma}\nabla_{\mathring{\varpi}} F$, so the generator block of the flow is $\dot{\mathring{\vartheta}} = -\nabla_{\mathring{\varpi}} F$ and its Euler step is $\mathring{\vartheta}' = \mathring{\vartheta} - \Delta t\nabla_{\mathring{\varpi}} F$. The growth bound and the Gaussian tails of R justify differentiation under the integral and two integrations by parts, which give the Gaussian gradient identities of lemma 9.20: $\nabla_{\mathring{\mu}}\mathbb{E}_R[\mathrm{i}_\lambda] = \mathbb{E}_R[\nabla\mathrm{i}_\lambda]$, and the derivative of $\mathbb{E}_R[\mathrm{i}_\lambda]$ in the covariance $\mathring{\Lambda}^{-1}$ along $\mathring{\Delta} \in \mathring{\Xi}$ is $\frac{1}{2}\operatorname{Tr}(\mathring{\Delta}\mathbb{E}_R[\nabla^2\mathrm{i}_\lambda])$, so its gradient in $\mathring{\Xi}$ is $\frac{1}{2}\operatorname{pr}_{\mathring{\Xi}}\mathbb{E}_R[\nabla^2\mathrm{i}_\lambda]$. The covariance gradient of $\frac{1}{2}\log\det\mathring{\Lambda} = -\frac{1}{2}\log\det\mathring{\Lambda}^{-1}$ is $-\frac{1}{2}\mathring{\Lambda}$. Write $\mathring{\Upsilon} = \frac{1}{2}\operatorname{pr}_{\mathring{\Xi}}\mathbb{E}_R[\nabla^2\mathrm{i}_\lambda] - \frac{1}{2}\mathring{\Lambda} \in \mathring{\Xi}$ for the covariance gradient of F. Since $\mathring{\mu} = \mathring{\varpi}_1$ and $\mathring{\Lambda}^{-1} = \mathring{\varpi}_2 - \operatorname{pr}_{\mathring{\Xi}}(\mathring{\varpi}_1 \otimes \mathring{\varpi}_1)$, and $\operatorname{Tr}(\mathring{\Upsilon}\operatorname{pr}_{\mathring{\Xi}}(\mathring{\Delta})) = \operatorname{Tr}(\mathring{\Upsilon}\mathring{\Delta})$ for every self-adjoint $\mathring{\Delta}$, the chain rule gives $\nabla_{\mathring{\varpi}_2} F = \mathring{\Upsilon}$ and $\nabla_{\mathring{\varpi}_1} F = \mathbb{E}_R[\nabla\mathrm{i}_\lambda] - 2\mathring{\Upsilon}\mathring{\mu}$. The second component of the Euler step is $-\frac{1}{2}\mathring{\Lambda}' = -\frac{1}{2}\mathring{\Lambda} - \Delta t\mathring{\Upsilon}$, which is the first equation of (9.45), and the first component is $\mathring{\Lambda}'\mathring{\mu}' = \mathring{\Lambda}\mathring{\mu} - \Delta t\mathbb{E}_R[\nabla\mathrm{i}_\lambda] + 2\Delta t\mathring{\Upsilon}\mathring{\mu} = \mathring{\Lambda}'\mathring{\mu} - \Delta t\mathbb{E}_R[\nabla\mathrm{i}_\lambda]$, which is the second. When $0 < \Delta t < 1$ and the projected expected Hessian is positive semidefinite, $\mathring{\Lambda}' \succeq (1 - \Delta t)\mathring{\Lambda}$ is positive definite. For one summand $\operatorname{pr}_{\mathring{\Xi}}$ is the identity and (9.45) has the form of Khan and Rue [66, Eq. (12)] with their loss replaced by i_λ, and for the lines of an orthonormal basis $\operatorname{pr}_{\mathring{\Xi}}$ keeps the diagonal, as in their Eq. (37), where the expected gradient and Hessian are evaluated at the mean.

(e) The Lebesgue density of R gives $\mathbb{E}_R[\log r] = \langle\mathring{\vartheta}, \mathring{\varpi}\rangle - \mathring{\Gamma}(\mathring{\vartheta}) = \mathring{\Gamma}^\star(\mathring{\varpi})$, so $\widetilde{F} = \mathbb{E}_R[\mathrm{i}_\lambda] + \mathring{\Gamma}^\star + \log Z_{\mathscr{F}}$. The relative entropy of two members is $D_{\mathrm{KL}}(R\|R') = \mathring{\Gamma}(\mathring{\vartheta}') - \mathring{\Gamma}(\mathring{\vartheta}) - \langle\mathring{\varpi}, \mathring{\vartheta}' - \mathring{\vartheta}\rangle$, and the Bregman divergence of $\mathring{\Gamma}^\star$ is the relative entropy with the arguments exchanged, $\mathring{\Gamma}^\star(\mathring{\varpi}') - \mathring{\Gamma}^\star(\mathring{\varpi}) - \langle\mathring{\vartheta}, \mathring{\varpi}' - \mathring{\varpi}\rangle = D_{\mathrm{KL}}(R'\|R)$. The expectation domain $\{(\mathring{\varpi}_1, \mathring{\varpi}_2) : \mathring{\varpi}_2 - \operatorname{pr}_{\mathring{\Xi}}(\mathring{\varpi}_1 \otimes \mathring{\varpi}_1) \succ 0\}$ is convex, because along a segment the difference of convex combinations is the convex combination of the differences plus a nonnegative multiple of $\operatorname{pr}_{\mathring{\Xi}}$ of a positive-semidefinite operator, which is positive semidefinite because $\operatorname{pr}_{\mathring{\Xi}}$ keeps the diagonal blocks of the decomposition. Taylor's formula with integral remainder along the segment and the curvature hypothesis give $\widetilde{F}(\mathring{\varpi}') \le \widetilde{F}(\mathring{\varpi}) + \langle\nabla\widetilde{F}(\mathring{\varpi}), \mathring{\varpi}' - \mathring{\varpi}\rangle + \mathrm{c}_{\mathsf{G}} D_{\mathrm{KL}}(R'\|R)$, since the same formula applied to $\mathring{\Gamma}^\star$ produces the Bregman divergence. By (d), $\nabla\widetilde{F}(\mathring{\varpi}) = \Delta t^{-1}(\mathring{\vartheta} - \mathring{\vartheta}')$, and adding the two expressions for the divergences gives $\langle\mathring{\vartheta}' - \mathring{\vartheta}, \mathring{\varpi}' - \mathring{\varpi}\rangle = D_{\mathrm{KL}}(R\|R') + D_{\mathrm{KL}}(R'\|R)$; substituting proves (9.46). Both divergences vanish only when $R' = R$. For the quadratic integrand, $\mathbb{E}_R[\mathrm{i}_\lambda] = \frac{1}{2}\operatorname{Tr}(\mathbf{Q}\mathring{\varpi}_2) - \langle\mathbf{b}, \mathring{\varpi}_1\rangle + \mathrm{c}_3$ is affine in $\mathring{\varpi}$, so $\nabla^2\widetilde{F} = \nabla^2\mathring{\Gamma}^\star$; moreover $\mathbb{E}_R[\nabla^2\mathrm{i}_\lambda] = \mathbf{Q}$ and $\mathbb{E}_R[\nabla\mathrm{i}_\lambda] = \mathbf{Q}\mathring{\mu} - \mathbf{b}$, so (9.45) with

$\Delta t = 1$ gives $\mathring{\Lambda}' = \mathbf{Q}$ and $\mathring{\mu}' = \mathbf{Q}^{-1}\mathbf{b}$, and R' is the law proportional to $e^{-\mathrm{i}_\lambda}$, which minimizes $F = D_{\mathrm{KL}}(R\|R') + \mathrm{const}$ over R. □

Proof of Proposition 9.29. (a) The joint log density is the prior log density plus terms free of $\mathring{\alpha}$, and $\log(d\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}}/d\mathring{\omega}) = -\mathscr{S}(\mathring{\omega}) - \log Z_{\mathscr{S}}(\mathring{\alpha}) - \frac{1}{2}\langle\mathring{\omega}-\mathring{\alpha}, \mathring{\Sigma}^{-1}(\mathring{\omega}-\mathring{\alpha})\rangle - \frac{1}{2}\log\det(2\pi\mathring{\Sigma})$, which gives (9.47). Differentiating under the integral, which the factor $e^{-\mathscr{S}} \le 1$ and the Gaussian tails permit locally uniformly in $\mathring{\alpha}$, $\nabla_{\mathring{\alpha}_e}\log Z_{\mathscr{S}} = \mathring{\Sigma}_e^{-1}(\mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}}}[\mathring{\omega}_e] - \mathring{\alpha}_e)$, and subtracting it from the gradient of the first term of (9.47) gives (9.48).

(b) With $\mathring{\pi} = \mathring{\Sigma}^{-1}\mathring{\alpha}$, the prior density with respect to the measure $\exp(-\frac{1}{2}\langle\mathring{\omega}, \mathring{\Sigma}^{-1}\mathring{\omega}\rangle - \mathscr{S}(\mathring{\omega}))d\mathring{\omega}$ is $\exp(\langle\mathring{\pi},\mathring{\omega}\rangle - \mathrm{A}(\mathring{\pi}))$, with $\mathrm{A}(\mathring{\pi}) = \log\int\exp(\langle\mathring{\pi},\mathring{\omega}\rangle - \frac{1}{2}\langle\mathring{\omega}, \mathring{\Sigma}^{-1}\mathring{\omega}\rangle - \mathscr{S}(\mathring{\omega}))d\mathring{\omega}$, finite on all of $\mathscr{T}$ because $\mathscr{S} \ge 0$. Thus $\mathcal{L}^{\mathrm{anc}} = \langle\mathring{\pi}, \mathbb{E}_R[\mathring{\omega}]\rangle - \mathrm{A}(\mathring{\pi}) + \mathrm{const}$, and A is smooth with Hessian $\mathrm{Cov}_{\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}}}(\mathring{\omega})$, which is positive definite because the prior has a positive density; so $\mathcal{L}^{\mathrm{anc}}$ is strictly concave in $\mathring{\pi}$, hence in $\mathring{\alpha}$. For a unit vector $\mathring{\delta}$, the set $\{\langle\mathring{\delta},\mathring{\omega}\rangle \ge \langle\mathring{\delta}, \mathbb{E}_R[\mathring{\omega}]\rangle + 1\}$ has positive measure $\mathrm{c}_{\mathring{\delta}}$ under that measure, so the convex function $\mathrm{f} = \mathrm{A} - \langle\cdot, \mathbb{E}_R[\mathring{\omega}]\rangle$ satisfies $\mathrm{f}(t\mathring{\delta}) \ge t + \log\mathrm{c}_{\mathring{\delta}} \to \infty$ as $t \to \infty$. A convex function on $\mathscr{T}$ that tends to infinity along every ray from the origin has bounded sublevel sets: if $\mathrm{f}(\mathring{\pi}_1) \le \mathrm{c}$ with $\|\mathring{\pi}_1\| \to \infty$ and, after passing to a subsequence, $\mathring{\pi}_1/\|\mathring{\pi}_1\| \to \mathring{\delta}$, convexity gives $\mathrm{f}(t\mathring{\pi}_1/\|\mathring{\pi}_1\|) \le \max(\mathrm{f}(0), \mathrm{c})$ for $\|\mathring{\pi}_1\| \ge t$, and continuity bounds $\mathrm{f}(t\mathring{\delta})$ for every t. Hence f attains its minimum, at exactly one point, where $\nabla\mathrm{A} = \mathbb{E}_{\mathbb{P}^{\mathscr{T}}_{\mathring{\alpha}}}[\mathring{\omega}] = \mathbb{E}_R[\mathring{\omega}]$. Since the other factors of the joint do not depend on $\mathring{\alpha}$, the maximizers of (5.58) over the anchor are those of $\mathcal{L}^{\mathrm{anc}}$, and proposition 5.27 applies.

(c) If $\mathscr{S} = 0$ the prior is $\nu^{\mathscr{T}}_{\mathring{\alpha}}$, with mean $\mathring{\alpha}$, and the moment condition reads $\mathring{\alpha} = \mathbb{E}_R[\mathring{\omega}]$. In the affine representation $\rho(\mathring{\omega}) = \left(\begin{smallmatrix}1 & \mathring{\omega}\\ 0 & 1\end{smallmatrix}\right)$ of the translations, products of represented translations are represented sums, so the represented holonomy of p minus the identity is $\mathrm{d}\rho(\mathbf{C}_p\mathring{\omega})$ and the loop defect is exactly $\frac{1}{2}\|\mathbf{C}_p\mathring{\omega}\|^2_{\mathsf{G}}$, as in proposition 9.13. For a quadratic action, the prior is Gaussian with precision $\mathring{\Sigma}^{-1} + \sum_p \mathrm{c}_p\mathbf{C}_p^\top\mathbf{C}_p$ and mean $(\mathring{\Sigma}^{-1} + \sum_p \mathrm{c}_p\mathbf{C}_p^\top\mathbf{C}_p)^{-1}\mathring{\Sigma}^{-1}\mathring{\alpha}$, and the moment condition gives (9.49). The operator $\sum_p \mathrm{c}_p\mathbf{C}_p^\top\mathbf{C}_p$ is positive semidefinite with quadratic form $\sum_p \mathrm{c}_p\|\mathbf{C}_p\mathring{\omega}\|^2_{\mathsf{G}}$, so it annihilates $\mathbb{E}_R[\mathring{\omega}]$ exactly when every $\mathbf{C}_p\mathbb{E}_R[\mathring{\omega}]$ with $\mathrm{c}_p > 0$ vanishes, and $\mathring{\Sigma}$ is invertible.

(d) In the coordinates $(\mathring{\Sigma}_e^{-1}\mathring{\alpha}_e, -\frac{1}{2}\mathring{\Sigma}_e^{-1})_{e\in E_+}$ the prior is the exponential family with sufficient statistic $(\mathring{\omega}_e, \mathring{\omega}_e\otimes\mathring{\omega}_e)_{e\in E_+}$ and base measure $e^{-\mathscr{S}(\mathring{\omega})}d\mathring{\omega}$; the Gaussian normalizers of the base law and $Z_{\mathscr{S}}$ combine into its log-partition function, which is strictly convex because the components of the statistic are affinely independent on a set of positive Lebesgue measure. The range of these coordinates is convex, so the strictly concave objective has at most one maximizer, and at an interior maximizer the gradient of the log-partition function, the vector of prior moments of the statistic, equals its expectation under R, which is the stated matching. If $\mathscr{S} = 0$, the prior is $\bigotimes_e \mathcal{N}(\mathring{\alpha}_e, \mathring{\Sigma}_e)$, the matching is solved by the stated pair, which lies in the range when every $\mathrm{Cov}_R(\mathring{\omega}_e)$ is positive definite, and a critical point of a concave function maximizes it. For the witness, with $\mathsf{G}^{\mathrm{fib}} = I_2$ the inner product on $\mathfrak{g} = \mathbb{R}$ is the product of reals, the loop defect is $\frac{1}{2}\mathring{\omega}_e^2$, and with $\mathrm{s} = \mathring{\Sigma}_e$ the prior is $\mathcal{N}(\mathring{\alpha}_e/(1+\mathrm{cs}), \mathrm{s}/(1+\mathrm{cs}))$. Its variance $\mathrm{s}' = \mathrm{s}/(1+\mathrm{cs})$ ranges over $(0, 1/\mathrm{c})$, and for each s the best anchor makes the prior mean $\mathring{\mu}$, leaving $-\frac{1}{2}\log(2\pi\mathrm{s}') - \mathrm{v}/(2\mathrm{s}')$, whose derivative in s' is $(\mathrm{v}-\mathrm{s}')/(2\mathrm{s}'^2) > 0$ for $\mathrm{s}' < 1/\mathrm{c} \le \mathrm{v}$. The objective therefore increases strictly as $\mathrm{s} \to \infty$ and no pair attains its supremum. □

Proof of Remark 9.30. (i) A frame at the reference point is $U_i = e_G$, and then $\mathring{\Theta}_e = \Theta_e$ by (2.32). In the affine representation a product of represented translations is the represented sum, so a represented holonomy minus the identity is $\mathrm{d}\rho$ of the loop sum, and for $\mathsf{G}^{\mathrm{fib}} = I_2$, $\langle \mathrm{d}\rho\mathrm{x}, \mathrm{d}\rho\mathrm{y}\rangle_{\mathsf{G}} = \mathrm{Tr}\big((\mathrm{x}E_{12})^{\top}\mathrm{y}E_{12}\big) = \mathrm{xy}$. The configuration prior of definition 14.21 is $\mathcal{N}(0, I_{12})$, whose $\check{\omega}$-block is $\mathcal{N}(0, I_4)$ and independent of $(\bar{k}, \bar{m})$, and the records are the binary records with success probability $\frac{1}{2}\exp[-\mathcal{D}_{e(i,S_i)}/\mathrm{T}]$, whose edge costs carry the loop structure. Disintegrating a configuration law over $\check{\omega}$ gives the form $R(d\check{\omega})Q_{\check{\omega}}$ of proposition 5.26. (ii) is proposition 8.6(i) with the link block of the mean read as $\mathring{\mu}$ by (i). (iii) is definition 8.2, whose slow coordinate carries θ, together with the statement of definition 14.21 that no generative parameter is learned.

(iv) Write $U = U_{S^{(0)}}$. It is the quadratic form $U(\mathbf{x}) = \frac{1}{2}\mathbf{x}^{\top}\nabla^2 U\mathbf{x}$, and $\nabla^2 U \succeq \mathrm{T}I$ because $U - \frac{\mathrm{T}}{2}|\mathbf{x}|^2$ is a sum of squares. By proposition 14.22, $\mathcal{F}_{\mathrm{T}}(q) = \mathrm{T}D_{\mathrm{KL}}(q\|q^{\star}) + \mathrm{const}$ with $q^{\star} \propto \pi_S e^{-U_S/\mathrm{T}}$, so at $q = \mathcal{N}(\mu, \Sigma) \otimes \delta_{S^{(0)}}$, $F = \mathcal{F}_{\mathrm{T}}/\mathrm{T} = \mathrm{T}^{-1}U(\mu) + \frac{1}{2}\mathrm{T}^{-1}\,\mathrm{Tr}(\nabla^2 U\Sigma) - \frac{1}{2}\log\det\Sigma + \mathrm{const}$. In covariance coordinates, where $\delta\Lambda = -\Lambda\delta\Sigma\Lambda$ turns (9.44) into $\langle\delta\mu, \Sigma^{-1}\delta\mu\rangle + \frac{1}{2}\,\mathrm{Tr}(\Sigma^{-1}\delta\Sigma\Sigma^{-1}\delta\Sigma)$, the natural gradient of F is $\dot{\mu} = -\mathrm{T}^{-1}\Sigma\nabla^2 U\mu$ and $\dot{\Sigma} = -2\Sigma\big(\frac{1}{2}\mathrm{T}^{-1}\nabla^2 U - \frac{1}{2}\Sigma^{-1}\big)\Sigma = \Sigma - \mathrm{T}^{-1}\Sigma\nabla^2 U\Sigma$. At $\Sigma = \mathrm{T}(\nabla^2 U)^{-1}$ the second right side vanishes and the first is $-\mu$; the field is smooth, so by uniqueness the slice is invariant and $\mu(t) = e^{-t}\mu(0)$. The residuals are linear functionals of the mean that equal -1 at $\mathbf{x}^{\circ}$, which gives the stated residuals, and $|r_i^b| = |r_i^m|$ excludes $|r_i^b| \le 2\varepsilon_{\mathrm{m}} < \frac{1}{2} \le |r_i^m|$; the interval of proposition 8.6(ii) is stated for $\varepsilon_{\mathrm{m}} \le 1/25$, where $2\varepsilon_{\mathrm{m}} < \frac{1}{2}$. □

Proof of Proposition 9.32. (a) Since $\mathscr{S} \ge 0$, $\mathfrak{U} \ge 0$, and $\mathbf{m} > 0$, the exponent in (9.51) is finite and at most $-\sum_n \mathscr{S}(\mathring{\omega}^{(n)}) \le 0$. The integrand is therefore positive everywhere and at most $\prod_n e^{-\mathscr{S}(\mathring{\omega}^{(n)})}$, whose integral is $Z_{\mathscr{S}}^{n_1+1} \le 1$, so $0 < Z_{\mathrm{hist}} \le Z_{\mathscr{S}}^{n_1+1}$ and the density bound follows. A Gaussian law integrates $e^{\mathrm{c}_2\|\cdot\|}$.

(b) By Tonelli's theorem each $\mathring{Z}_n^{\leftarrow}$ is measurable; by induction from n_1, its integrand is positive and at most 1 and $\nu^{\mathscr{T}}$ is a probability law, so $\mathring{Z}_n^{\leftarrow} \in (0, 1]$. Each kernel of (9.53) has total mass one by (9.52), and $\int e^{-\mathscr{S}}\mathring{Z}_0^{\leftarrow}\,d\nu^{\mathscr{T}} = Z_{\mathrm{hist}}$ by iterated integration of (9.51). At a history, the root density is $Z_{\mathrm{hist}}^{-1}e^{-\mathscr{S}(\mathring{\omega}^{(0)})}\mathring{Z}_0^{\leftarrow}(\mathring{\omega}^{(0)})$ and the kernel density of the step from n to $n+1$ is $e^{-\mathscr{S}(\mathring{\omega}^{(n+1)}) - \mathbf{m}\mathfrak{U}(\mathring{\omega}^{(n+1)}, \mathring{\omega}^{(n)})}$ times $\mathring{Z}_{n+1}^{\leftarrow}(\mathring{\omega}^{(n+1)})/\mathring{Z}_n^{\leftarrow}(\mathring{\omega}^{(n)})$; the ratios telescope, and since $\mathring{Z}_{n_1}^{\leftarrow} \equiv 1$ the product is the density of (9.51). So $\mathbb{P}^{\mathrm{hist}}$ is the law of this chain, the conditional law of $\mathring{\omega}^{(n+1)}$ given $\mathring{\omega}^{(0:n)}$ is $\mathbb{P}_{\mathrm{hist}}^{(n+1\leftarrow n)}(\cdot \mid \mathring{\omega}^{(n)})$, and the kernels are the conditionals. The chain that normalizes each increment factor separately has as kernels the factors $e^{-\mathscr{S}(\mathring{\omega}') - \mathbf{m}\mathfrak{U}(\mathring{\omega}', \mathring{\omega})}$ against $\nu^{\mathscr{T}}(d\mathring{\omega}')$ divided by $\mathring{Z}_{n_1-1}^{\leftarrow}(\mathring{\omega})$, which is the normalizer of one increment factor, so the density of that chain relative to $\mathbb{P}^{\mathrm{hist}}$ is proportional to $\prod_{n<n_1}\mathring{Z}_{n_1-1}^{\leftarrow}(\mathring{\omega}^{(n)})^{-1}$, and by Fubini's theorem this product is constant almost everywhere exactly when $\mathring{Z}_{n_1-1}^{\leftarrow}$ is. In the scalar case, $\mathring{Z}_0^{\leftarrow}(\mathring{\omega}) = (1+\mathbf{m})^{-1/2}\exp[-\mathbf{m}\mathring{\omega}^2/(2(1+\mathbf{m}))]$ is not constant; equivalently, the Gibbs law has precision matrix $\left(\begin{smallmatrix} 1+\mathbf{m} & -\mathbf{m} \\ -\mathbf{m} & 1+\mathbf{m}\end{smallmatrix}\right)$, so its first marginal has precision $(1+2\mathbf{m})/(1+\mathbf{m}) \ne 1$, whereas the other chain starts from $\nu^{\mathscr{T}}$, of precision 1. For the latent route, the joint $\mathbb{P}^{\mathrm{hist}}(d\mathring{\omega}^{(0:n_1)})\prod_n \mathbb{P}_n^{\mathrm{ag}}(dy^{(n)} \mid \mathring{\omega}^{(n)}, X)\prod_n \mathbb{P}^{\mathrm{obs}}(do_n \mid \mathbf{Y}^{(n)}, X)$, with the chain factorization substituted, is (13.41) for the stated root, transitions, and readout: each transition is a Markov kernel from $Z_{\mathrm{par}}^{(n-1)} = \mathring{\omega}^{(n-1)}$ to $\mathbf{Y}^{(n)}$ composed of two Markov kernels, and the root is a law of $\mathbf{Y}^{(0)}$. Every kernel reads structural data and earlier latent coordinates, as hypothesis 4.3 requires, and proposition 13.27 gives the listed consequences, the predictive prior being (13.42) with $Z_{\mathrm{par}}^{(n)} = \mathring{\omega}^{(n)}$.

(c) With a static frame U_i, the twist of the identity link of agent i is $U_i^{-1} e_G U_i = e_G$ by (2.32), independent of every variable, so it is the identity twist of each time-like step in (9.50), and (9.51) contains no other link. □

Proof of Proposition 9.33. (a) Differentiating (9.54) in $\mathring{\omega}^{(n)}$ gives the equations, since $\mathring{\omega}^{(n)}$ enters the increments to $n-1$ and $n+1$ in the interior and one increment at each end. Under the growth condition $\mathscr{S}^{\mathrm{hist}}$ is continuous and coercive on a finite-dimensional space, so it attains its minimum.

(b) By lemma 9.19(a), $\frac{d}{dt} e^{\mathbf{B}+t\mathbf{T}}|_{t=0} = \mathbf{L}(\mathbf{B},\mathbf{T})$. For each $t \in [0,1]$ the map $\mathbf{X} \mapsto e^{t\mathbf{B}} \mathbf{X} e^{(1-t)\mathbf{B}}$ has the adjoint $\mathbf{X} \mapsto e^{t\mathbf{B}^*} \mathbf{X} e^{(1-t)\mathbf{B}^*}$ for $\langle \cdot,\cdot \rangle_{\mathsf{G}}$, because $(e^{t\mathbf{B}})^* = e^{t\mathbf{B}^*}$, so the adjoint of $\mathbf{L}(\mathbf{B},\cdot)$ is $\mathbf{L}(\mathbf{B}^*,\cdot)$. The differential of $\mathfrak{Q}$ at $\mathbf{H}$ is $\mathbf{T} \mapsto \langle \mathbf{H} - I, \mathbf{T} \rangle_{\mathsf{G}}$, and the differential of $\rho(\mathring{\Theta}_e^{\leftarrow})$ in the direction $\delta\mathring{\omega}_e$ is $e^{-\mathrm{d}\rho\mathring{\omega}_e'} \mathbf{L}(\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e)$ in the earlier argument and $-\mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e', \mathrm{d}\rho\delta\mathring{\omega}_e) e^{\mathrm{d}\rho\mathring{\omega}_e}$ in the later one; moving the factors across the inner product and projecting onto $\mathrm{d}\rho(\mathfrak{g})$ gives (9.56), and (9.57) is stationarity of (9.54). The function $\mathfrak{U}$ is smooth, nonnegative, and zero on the diagonal, so its gradient vanishes there; and $\rho(\mathring{\Theta}_e^{\leftarrow}) = I + \mathrm{d}\rho(\mathring{\omega}_e - \mathring{\omega}_e') + O(\|(\mathring{\omega},\mathring{\omega}')\|^2)$ gives $\mathfrak{U} = \frac{1}{2}\|\mathring{\omega} - \mathring{\omega}'\|^2 + O(\|(\mathring{\omega},\mathring{\omega}')\|^3)$, so the mixed second derivative at the origin is $-I$. Let $\mathrm{R}_1(\mathring{\omega}',\mathring{\omega}) = \partial_{\mathring{\omega}}\mathfrak{U} - (\mathring{\omega} - \mathring{\omega}')$. Then $\mathrm{R}_1(\mathring{\omega},\mathring{\omega}) = 0$ and $\mathrm{R}_1(\mathring{\omega}',\mathring{\omega}) = \int_0^1 D_{\mathring{\omega}'}\mathrm{R}_1(\mathring{\omega} + t(\mathring{\omega}' - \mathring{\omega}),\mathring{\omega})[\mathring{\omega}' - \mathring{\omega}]dt$, where $D_{\mathring{\omega}'}\mathrm{R}_1$ vanishes at the origin and has derivative bounded by $\sup\|D^3\mathfrak{U}\|$ on the convex set $\{\|\mathring{\omega}\|, \|\mathring{\omega}'\| \le \mathrm{r}\}$; since $\|(\mathring{\omega},\mathring{\omega}')\| \le \sqrt{2}\mathrm{r}$ there, the bound follows, and the same argument applies to $\partial_{\mathring{\omega}'}\mathfrak{U} - (\mathring{\omega}' - \mathring{\omega})$. For the quadratic form at a given $\mathring{\omega}$, with $\mathring{\omega}' = \mathring{\omega} + \delta\mathring{\omega}$, the differential of the later argument gives $\rho(\mathring{\Theta}_e^{\leftarrow}) = I - \mathbf{L}(-\mathrm{d}\rho\mathring{\omega}_e, \mathrm{d}\rho\delta\mathring{\omega}_e) e^{\mathrm{d}\rho\mathring{\omega}_e} + O(\|\delta\mathring{\omega}\|^2)$, and $\mathbf{L}(0,\mathbf{T})e^0 = \mathbf{T}$ at the origin.

(c) Write $\mathrm{E}(\mathring{\omega}) = \frac{\mathbf{m}}{2}\sum_{n<n_1}(\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)})^2 + \sum_{0<n<n_1}\mathfrak{V}(\mathring{\omega}^{(n)})$ for histories with the given ends. Replacing every value by its clamp to $[\mathrm{a},\mathrm{b}]$ does not increase E: clamping is 1-Lipschitz, and by the hypothesis on $\mathfrak{V}$ outside $[\mathrm{a},\mathrm{b}]$ it does not raise the potential terms. If some value leaves $[\mathrm{a},\mathrm{b}]$, the step that enters a maximal run of such values from a value in $[\mathrm{a},\mathrm{b}]$ strictly shortens, so the decrease is strict. Hence minimizers over $[\mathrm{a},\mathrm{b}]^{n_1-1}$, which exist by compactness and continuity, are global, and every global minimizer takes its values in $[\mathrm{a},\mathrm{b}]$. Now let $\mathrm{z}_0 \le \cdots \le \mathrm{z}_{n_1}$ be the values of such a history in increasing order, so $\mathrm{z}_0 = \mathrm{a}$ and $\mathrm{z}_{n_1} = \mathrm{b}$, and let $\mathrm{g}_\mathrm{l} = \mathrm{z}_{\mathrm{l}+1} - \mathrm{z}_\mathrm{l}$. No value lies strictly inside a gap, so each step from $\mathring{\omega}^{(n)}$ to $\mathring{\omega}^{(n+1)}$ covers a set of whole gaps and its square is at least the sum of their squares. A history from a to b crosses each gap of positive length an odd number of times, so summing over steps gives $\sum_n(\mathring{\omega}^{(n+1)} - \mathring{\omega}^{(n)})^2 \ge \sum_\mathrm{l} \mathrm{g}_\mathrm{l}^2$, the kinetic sum of the increasing rearrangement; a decreasing step crosses some gap of positive length downward, which forces at least three crossings of it and a strict inequality. The rearrangement keeps a and b at the ends and the multiset of interior values, so it leaves the potential sum unchanged. A history that is not nondecreasing therefore has a strictly smaller competitor, and every most probable history is nondecreasing. For strictness, let $[\mathrm{p},\mathrm{q}]$ be a maximal run of equal values x with $\mathrm{q} \ge \mathrm{p}+1$. If $0 < \mathrm{p}$ and $\mathrm{q} < n_1$, (9.55) at p and at q gives $\mathbf{m}(\mathrm{x} - \mathring{\omega}^{(\mathrm{p}-1)}) = -\mathfrak{V}'(\mathrm{x}) = \mathbf{m}(\mathrm{x} - \mathring{\omega}^{(\mathrm{q}+1)})$, which contradicts $\mathring{\omega}^{(\mathrm{p}-1)} < \mathrm{x} < \mathring{\omega}^{(\mathrm{q}+1)}$. If $\mathrm{p} = 0$, then $\mathrm{x} = \mathrm{a}$, $\mathrm{q} < n_1$, and (9.55) at q with $\mathfrak{V}'(\mathrm{a}) = 0$ gives $\mathring{\omega}^{(\mathrm{q}+1)} = \mathrm{a}$, against maximality; the case $\mathrm{q} = n_1$ is symmetric, and $\mathrm{p} = 0$ with $\mathrm{q} = n_1$ would give $\mathrm{a} = \mathrm{b}$. □

Proof of Proposition 9.34. (a) The anchor term has gradient 0 and Hessian $\mathring{\Sigma}^{-1}$ at 0. Each $\mathfrak{Q}_p$ is smooth, a polynomial in the entries of a product of matrix exponentials, and $\mathfrak{Q}_p(\mathring{\omega}) = \frac{1}{2}\|\mathbf{C}_p\mathring{\omega}\|_{\mathsf{G}}^2 + O(\|\mathring{\omega}\|^3)$ by proposition 9.10(e), so by uniqueness of Taylor coefficients its gradient vanishes at 0 and

its Hessian there is $\mathbf{C}_p^\top \mathbf{C}_p$. The Gaussian increment is the quadratic form of $\mathbf{m}(-\Delta_n)$, and the time-like defect has the same second-order part by the expansion in the proof of proposition 9.33(b), with vanishing gradient at 0. For a record a with scope $\{i, j\}$ on $e : j \to i$ and $\mathring{\Omega} = \mathrm{d}\rho\mathring{\omega}_e$, independence of k_i and k_j gives $\mathbb{E}[E_{a,o}] = \frac{1}{2}[\|\mu_i - e^{\mathring{\Omega}}\mu_j\|^2_{W_a} + \mathrm{Tr}(W_a\Sigma_i) + \mathrm{Tr}(W_a e^{\mathring{\Omega}}\Sigma_j(e^{\mathring{\Omega}})^\top)]$, dropping the episode index. With $e^{\mathring{\Omega}} = I + \mathring{\Omega} + \frac{1}{2}\mathring{\Omega}^2 + O(\|\mathring{\Omega}\|^3)$, the first-order part is $\mathrm{Tr}(W_a\mathring{\Omega}\Sigma_j) - (\mu_i - \mu_j)^\top W_a\mathring{\Omega}\mu_j = -\mathrm{Tr}(\mathbf{J}_a^\top\mathring{\Omega})$ and the second-order part is $\frac{1}{2}[\mathrm{Tr}(W_a\mathring{\Omega}\mathbb{E}[k_jk_j^\top]\mathring{\Omega}^\top) + \mathrm{Tr}(W_a\mathring{\Omega}^2\Sigma_j) - (\mu_i - \mu_j)^\top W_a\mathring{\Omega}^2\mu_j] = \frac{1}{2}[\mathrm{Tr}(W_a\mathring{\Omega}\mathbb{E}[k_jk_j^\top]\mathring{\Omega}^\top) - \mathrm{Tr}(\mathbf{J}_a^\top\mathring{\Omega}^2)]$, because $\mathbf{J}_a^\top = (\mu_j(\mu_i - \mu_j)^\top - \Sigma_j)W_a$. For a record on a reversed copy, $\mathring{\Omega} = -\mathrm{d}\rho\mathring{\omega}_{\bar{e}}$ with $\bar{e} \in E_+$, which changes the sign of the first-order part only. Summing with the weights gives $\nabla\mathscr{S}^{\mathrm{hist}}(0) = -\mathbf{j}$ and $\nabla^2\mathscr{S}^{\mathrm{hist}}(0) = \mathbf{K} + \mathbf{N}_0$.

(b) The loop, episode, and matter forms are nonnegative, the last because $\mathrm{Tr}(W_a\mathring{\Omega}\mathbb{E}[k_jk_j^\top]\mathring{\Omega}^\top) = \mathbb{E}\|W_a^{1/2}\mathring{\Omega}k_j\|^2$, so $\langle\mathring{\omega}, \mathbf{K}\mathring{\omega}\rangle \geq \mathrm{c}_{\mathring{\Sigma}}\|\mathring{\omega}\|^2$, and the bounds follow from $\|\mathbf{K}^{-1}\| \leq 1/\mathrm{c}_{\mathring{\Sigma}}$.

(c) A represented generator $\mathring{\Omega}$ has $\|\mathring{\Omega}\|_{\mathsf{G}} = \|(\mathsf{G}^{\mathrm{fib}})^{1/2}\mathring{\Omega}(\mathsf{G}^{\mathrm{fib}})^{-1/2}\|_{\mathrm{F}}$, so that $\|\mathring{\Omega}\|^2_{\mathrm{F}} \leq \mathrm{cond}(\mathsf{G}^{\mathrm{fib}})\|\mathring{\Omega}\|^2_{\mathsf{G}}$. Then $|\mathrm{Tr}(\mathbf{J}_a^\top\mathring{\Omega}^2)| \leq \|\mathbf{J}_a\|_{\mathrm{F}}\|\mathring{\Omega}\|^2_{\mathrm{F}}$ and $|\mathrm{Tr}(\mathbf{J}_a^\top\mathring{\Omega})| \leq \|\mathbf{J}_a\|_{\mathrm{F}}\|\mathring{\Omega}\|_{\mathrm{F}}$, summed over records and combined with the Cauchy–Schwarz inequality, give the bounds on c_0 and $\|\mathbf{j}\|$, the first because $\mathbf{N}_0$ is self-adjoint. Write $\nabla\mathscr{S}^{\mathrm{hist}}(\mathring{\omega}) = -\mathbf{j} + \mathbf{K}\mathring{\omega} + \nabla\mathrm{R}(\mathring{\omega})$, so that $\nabla\mathrm{R}(0) = 0$, $\nabla^2\mathrm{R}(0) = \mathbf{N}_0$, and $\|\nabla^2\mathrm{R}(\mathring{\omega})\| \leq \mathrm{c}_0 + \mathrm{c}_{\mathrm{r}}\|\mathring{\omega}\|$ for $\|\mathring{\omega}\| \leq \mathrm{r}$. The critical points of $\mathscr{S}^{\mathrm{hist}}$ are the fixed points of the map $\mathring{\omega} \mapsto \mathbf{K}^{-1}(\mathbf{j} - \nabla\mathrm{R}(\mathring{\omega}))$. On the ball of radius $\mathrm{r}_1 = 2\|\mathbf{j}\|/\mathrm{c}_{\mathring{\Sigma}} \leq \mathrm{r}/2$, $\|\nabla\mathrm{R}(\mathring{\omega})\| \leq (\mathrm{c}_0 + \mathrm{c}_{\mathrm{r}}\mathrm{r})\|\mathring{\omega}\| \leq \frac{1}{2}\mathrm{c}_{\mathring{\Sigma}}\mathrm{r}_1$, so the map sends this ball into itself, its values having norm at most $\|\mathbf{j}\|/\mathrm{c}_{\mathring{\Sigma}} + \mathrm{r}_1/2 = \mathrm{r}_1$, and it has Lipschitz constant at most $(\mathrm{c}_0 + \mathrm{c}_{\mathrm{r}}\mathrm{r})/\mathrm{c}_{\mathring{\Sigma}} \leq \frac{1}{2}$ there, so it has exactly one fixed point $\mathring{\omega}_\star$ in that ball. On the ball of radius r, $\nabla^2\mathscr{S}^{\mathrm{hist}} \succeq (\mathrm{c}_{\mathring{\Sigma}} - \mathrm{c}_0 - \mathrm{c}_{\mathrm{r}}\mathrm{r})I \succeq \frac{1}{2}\mathrm{c}_{\mathring{\Sigma}}I$, so $\mathscr{S}^{\mathrm{hist}}$ is strongly convex on that convex set and has there at most one critical point, which is its unique minimizer on the set. Finally $\mathring{\omega}_\star - \mathring{\omega}_{\mathrm{lin}} = -\mathbf{K}^{-1}\nabla\mathrm{R}(\mathring{\omega}_\star)$ and $\|\nabla\mathrm{R}(\mathring{\omega}_\star)\| \leq (\mathrm{c}_0 + \frac{1}{2}\mathrm{c}_{\mathrm{r}}\|\mathring{\omega}_\star\|)\|\mathring{\omega}_\star\|$, which with $\|\mathring{\omega}_\star\| \leq \mathrm{r}_1$ gives (9.63). Since c_0 and $\|\mathbf{j}\|$ are bounded by multiples of the currents, the right side is of second order in them. □

B.4 Proofs for Coarse Channels

The proofs for section 10.8 use the notation fixed at the head of that section, including its convention on the constant C_{K}.

Proof of Proposition 10.5. For a law $P \ll P_0$ with density r and bounded measurable g on Y, $\int g\,d(PK) = \mathbb{E}_{\mathbb{J}_{P_0}}[g(Y)r(X)] = \mathbb{E}_{\mathbb{J}_{P_0}}[g(Y)\bar{r}(Y)]$ with $\bar{r}(Y) = \mathbb{E}_{\mathbb{J}_{P_0}}[r(X) \mid Y]$, because r is P_0-integrable. This is the computation in the proof of Theorem 10.3, which uses no finiteness of relative entropy, and it shows that a measurable function of the coarse coordinate that agrees with r almost everywhere is $\bar{r} = d(PK)/d(P_0K)$. (i) If r descends, disintegrating $\mathbb{J}_{P_0}$ over y gives $(PKR_{P_0})(A) = \int \bar{r}(y)R_{P_0}(y, A)(P_0K)(dy) = \mathbb{E}_{\mathbb{J}_{P_0}}[\mathbf{1}_A(X)\bar{r}(Y)] = \mathbb{E}_{\mathbb{J}_{P_0}}[\mathbf{1}_A(X)r(X)] = P(A)$, and $r = 1$ gives $P_0KR_{P_0} = P_0$. Conversely, let one Markov kernel R recover P_0 and P. Conditional Jensen for the concave square root gives $\mathbb{E}_{\mathbb{J}_{P_0}}[\sqrt{r(X)} \mid Y] \leq \sqrt{\bar{r}(Y)}$ almost surely, so the affinity $\mathbb{E}_{P_0}[\sqrt{r}]$, which is at most 1 by the Cauchy-Schwarz inequality, does not decrease through K: $\mathbb{E}_{P_0}[\sqrt{r}] \leq \mathbb{E}_{P_0K}[\sqrt{\bar{r}}]$. The kernel R sends the pair (PK, P_0K), of density $\bar{r}$, to (P, P_0), of density r, so the same inequality through R gives $\mathbb{E}_{P_0K}[\sqrt{\bar{r}}] \leq \mathbb{E}_{P_0}[\sqrt{r}]$. The two finite affinities are therefore

equal, the conditional inequality is an almost sure equality, and

$$\mathbb{E}_{\mathbb{J}_{P_0}}\big[(\sqrt{r(X)}-\sqrt{\bar r(Y)})^2\big]=2-2\mathbb{E}_{\mathbb{J}_{P_0}}\big[\sqrt{\bar r(Y)}\mathbb{E}_{\mathbb{J}_{P_0}}[\sqrt{r(X)}\mid Y]\big]=2-2\mathbb{E}_{\mathbb{J}_{P_0}}[\bar r(Y)]=0,$$

where $\sqrt{r(X)\bar r(Y)}$ is integrable because it is at most $(r(X)+\bar r(Y))/2$. Hence r descends. Applied to each P_θ, and to a single law $P\ll P_0$, this proves the first sentence of (i) and the characterization of the laws that R_{P_0} recovers. The functions in $L^1(P_0)$ that agree $\mathbb{J}_{P_0}$-almost everywhere with a measurable function of the coarse coordinate form a linear subspace containing 1 and, when every r_θ descends, every r_θ; it is closed, so it then contains their closed linear span: if $r^{(n)}\to r$ in $L^1(P_0)$ and $r^{(n)}(X)=\bar r^{(n)}(Y)$ almost surely, then, since X has law P_0 under $\mathbb{J}_{P_0}$, a subsequence converges $\mathbb{J}_{P_0}$-almost surely, and the limit superior of $\bar r^{(n)}$ along it, set to zero where it is infinite, is a measurable function of y equal to $r(x)$ almost everywhere. For records, $d\bar{\mathsf{m}}_o/d\varrho=L_o/M_o$ by Definition 12.1, and $d(\bar{\mathsf{m}}_oC)/d(\varrho C)=L_o^c/M_o$ by (12.3). (ii) The graph $\operatorname{gr}c=\{(x,y):c(x)=y\}$ is measurable because the diagonal of the standard Borel space Y is, so $y\mapsto R(y,c^{-1}(\{y\}))=\int\mathbf{1}_{\operatorname{gr}c}(x,y)R(y,dx)$ is measurable. Under K_c the joint $\mathbb{J}_{P_0}$ is the law of $(X,c(X))$ and is carried by $\operatorname{gr}c$, which gives the stated form of descent, and $\int R_{P_0}(y,c^{-1}(\{y\}))(c_\#P_0)(dy)=\mathbb{J}_{P_0}(\operatorname{gr}c)=1$, so every version of R_{P_0} is supported on fibers and recovers P_0. If R is supported on fibers and recovers P_0, the measure on $\mathsf{X}\times\mathsf{Y}$ that gives $A\times B$ the mass $\int_B R(y,A)(c_\#P_0)(dy)$ is carried by $\operatorname{gr}c$ and has first marginal $P_0K_cR=P_0$, so its value on $A\times B$ equals its value on $(A\cap c^{-1}(B))\times\mathsf{Y}$, which is $P_0(A\cap c^{-1}(B))=\mathbb{J}_{P_0}(A\times B)$. Hence R disintegrates $\mathbb{J}_{P_0}$ over y and is a version of R_{P_0}, and two versions agree $c_\#P_0$-almost everywhere because X is countably generated. Since $P_\theta K_c\ll P_0K_c$, all versions give the same $P_\theta K_cR$, and (i) gives the rest. (iii) The fiber of $c\times\mathrm{id}_{\mathsf{X}_2}$ over (y,x_2) is $c^{-1}(\{y\})\times\{x_2\}$, and the first marginal of $(c\times\mathrm{id}_{\mathsf{X}_2})_\#P_0$ is the law of $c(X_1)$, so R is supported on fibers and, by (ii), recovers P_0 and every P_θ exactly when it recovers P_0 and every r_θ descends. For measurable $A_1\subseteq\mathsf{X}_1$, $A_2\subseteq\mathsf{X}_2$, and $B\subseteq\mathsf{Y}$, fiber support gives $R'(c(X_1),A_1\cap c^{-1}(B))=R'(c(X_1),A_1)\mathbf{1}_B(c(X_1))$ almost surely, and measurable rectangles generate the product σ-algebra, so R recovers P_0 exactly when

$$P_0\big(X_1\in A_1,c(X_1)\in B,X_2\in A_2\big)=\mathbb{E}_{P_0}\big[R'(c(X_1),A_1)\mathbf{1}_B(c(X_1))\mathbf{1}_{A_2}(X_2)\big]$$

for all such sets. With $A_2=\mathsf{X}_2$ this says that R' is a regular conditional law of X_1 given $c(X_1)$. Granted that, conditioning both sides on $c(X_1)$ shows that the identity is equivalent to $P_0(X_1\in A_1,X_2\in A_2\mid c(X_1))=P_0(X_1\in A_1\mid c(X_1))P_0(X_2\in A_2\mid c(X_1))$ almost surely for all A_1,A_2, which is the conditional independence. These conditions on P_0 and the r_θ are equivalent to adequacy in the sense named after the proposition, that is, to a conditional law of X_1 given $c(X_1)$ common to P_0 and every P_θ together with the conditional independence of X_1 and X_2 given $c(X_1)$ under each of these laws. Conditional independence under P_0 with descent $r_\theta(x_1,x_2)=\bar r_\theta(c(x_1),x_2)$ makes the conditional law of X_1 given $(c(X_1),X_2)$ under P_θ, whose density with respect to P_0 is a function of $(c(X_1),X_2)$, equal to that under P_0, which is the conditional law of X_1 given $c(X_1)$ under P_0, so the conditional independence holds under every P_θ with this common conditional law; conversely, adequacy writes each member of the family as the law of $(c(X_1),X_2)$ followed by the common kernel to X_1, so every r_θ is the density between these laws of $(c(X_1),X_2)$ evaluated at $(c(x_1),x_2)$

and descends. In the witness every R' is supported on the one fiber, the density 1 of the family descends, $R_{P_0}((y,x_2),\cdot)=\delta_{(x_2,x_2)}$ recovers P_0, and $R'(y,\cdot)\otimes\delta_{x_2}$ sends P_0 to the product of $R'(y,\cdot)$ with the uniform law on $\{0,1\}$, which gives $\{x_1\neq x_2\}$ the mass $\frac{1}{2}R'(y,\{1\})+\frac{1}{2}R'(y,\{0\})=\frac{1}{2}$. □

Proof of Theorem 10.7. (i) Since $C(y,\mathsf{Z})=1$, a bounded function of o alone has the same integral against $\mathbb{P}^c$ and against $\mathbb{P}$, which gives normalization and the observation marginal. For bounded measurable f on Z and g on O put $h(y)=\int f(z)C(y,dz)$. Then

$$\int g\Big(\int f\,d(\mathbf{\Pi}_o C)\Big)\mathbb{P}^c(do\times\mathsf{Z})=\int g\Big(\int h\,d\mathbf{\Pi}_o\Big)\mathbb{P}(do\times\mathsf{Y})$$
$$=\int g(o)h(y)\mathbb{P}(do,dy)=\int g(o)f(z)\mathbb{P}^c(do,dz),$$

using the preserved marginal, the defining property of the fine version, and (10.3). Hence $o\mapsto\mathbf{\Pi}_oC$ is a posterior version of $\mathbb{P}^c$ at every observation where it is declared, because C acts on the conditioned variable only. If $\mathbf{\Pi}^c_o(N)=0$, then $C(\cdot,N)=0$ $\mathbf{\Pi}_o$-almost surely, hence $\mathbb{Q}_o$-almost surely under $\mathbb{Q}_o\ll\mathbf{\Pi}_o$, and $\mathbb{Q}^c_o(N)=0$. (ii) Disintegrated over y, both lifts have the marginals $\mathbb{Q}_o$ and $\mathbf{\Pi}_o$ and the common conditional $C(y,\cdot)$, so the chain rule of Lemma 3.12, in $[0,+\infty]$, gives $D_{\mathrm{KL}}(\widehat{\mathbb{Q}}_o\|\widehat{\mathbf{\Pi}}_o)=D_{\mathrm{KL}}(\mathbb{Q}_o\|\mathbf{\Pi}_o)$; disintegrated over z, it gives $D_{\mathrm{KL}}(\widehat{\mathbb{Q}}_o\|\widehat{\mathbf{\Pi}}_o)=D_{\mathrm{KL}}(\mathbb{Q}^c_o\|\mathbf{\Pi}^c_o)+\Delta(o)$. Adding the finite real $-\log p(o)$, the same at both scales by (i), and using that $\mathbf{\Pi}^c_o$ is the coarse posterior version gives (10.4); every term lies in $[0,+\infty]$ before the finite shift, so no infinite quantity is subtracted. (iii) $\Delta(o)$ is the integral of a nonnegative measurable function, which vanishes exactly when the integrand vanishes $\mathbb{Q}^c_o$-almost surely, and a relative entropy vanishes exactly when its arguments agree. (iv) Finite fine divergence makes both terms on the right of (10.4) finite. If $\Delta(o)=0$, the two divergences are equal and finite, and Theorem 10.3 and corollary 10.4 give recovery of both laws by the Bayes kernel of $\mathbf{\Pi}_o$ under C. Conversely, a kernel R recovering both laws gives $D_{\mathrm{KL}}(\mathbb{Q}_o\|\mathbf{\Pi}_o)\geq D_{\mathrm{KL}}(\mathbb{Q}^c_o\|\mathbf{\Pi}^c_o)\geq D_{\mathrm{KL}}(\mathbb{Q}^c_oR\|\mathbf{\Pi}^c_oR)=D_{\mathrm{KL}}(\mathbb{Q}_o\|\mathbf{\Pi}_o)$ by data processing through C and then R, so $\Delta(o)=0$. □

Proof of Corollary 10.11. (i) is Theorem 10.7 with $\mathbb{P}=\mathbb{P}_I$, $C=C_A$, and the admitted recognition law. (ii) On standard Borel spaces a disintegration of $\mathbb{P}_A(\cdot\mid X)$ over (m_A,ξ_A) exists and is jointly measurable (Chapter 3); it satisfies the compatibility identity by construction, while a family declared independently of $\mathbb{P}_A$ satisfies it only if assumed. (iii) The pushforward formulas hold by testing each parent law with functions of one coordinate. For two non-Dirac marginals, Proposition 3.17 gives distinct couplings with the same pair, and tensoring each with one fixed law on $\mathbf{\Xi}_A\times\mathsf{H}_A$ gives distinct laws on Z_A. For two points $h\neq h'$ of $\mathbf{\Xi}_A\times\mathsf{H}_A$ and the (b_A,m_A) marginal R of the given law, $R\otimes\delta_h$ and $R\otimes\delta_{h'}$ share the pair and differ, so one of them differs from the given law. If that space is a point and one marginal is Dirac, every coupling assigns mass one to the Dirac coordinate. □

Proof of Proposition 10.17. (i) By equivariance, $\mathbb{J}_{g_\#Q}(A\times B)=\mathbb{J}_Q(g^{-1}A\times g^{-1}B)$. Bimeasurability makes R^g_Q a Markov kernel, the Y-marginal of $\mathbb{J}_{g_\#Q}$ is $g_\#(QK)$ by Proposition 10.16, and the change of variables $y=gy'$ gives $\int\mathbf{1}_B(y)R^g_Q(y,A)(g_\#(QK))(dy)=\mathbb{J}_{g_\#Q}(A\times B)$. (ii) With $(g_\#P)K=g_\#(PK)$, $((g_\#P)KR^g_Q)(A)=(PKR_Q)(g^{-1}A)=(g_\#P)(A)$; the relative-entropy identities are invariance under bimeasurable bijections, (3.23) and (3.24). (iii) Where $(QK)(\{y\})>0$ the first

value is the value of every version; the remaining points are QK-null, both branches transform as stated because $(g_{\#}Q)(\{gx\}) = Q(\{x\})$ and $K(gx, \{gy\}) = K(x, \{y\})$, and the first branch is a rational function of Q on a Borel set. (iv) Each R_Q^h with $h \in \mathbb{G}_Q$ is a version of R_Q by (i). Joint measurability makes $(h, y) \mapsto R_Q^h(y, A)$ measurable, so $\bar{R}_Q$ is a Markov kernel by Tonelli's theorem and monotone convergence, and the defining identity, linear in the kernel, passes to the mixture. Left invariance of dh gives $\bar{R}_Q(ky, kA) = \bar{R}_Q(y, A)$ for $k \in \mathbb{G}_Q$. If $g_{\#}Q = g'_{\#}Q$ then $g^{-1}g' \in \mathbb{G}_Q$, so $\bar{R}_Q^{g'} = \bar{R}_Q^g$; finally $\bar{R}_Q^{kg}(ky, kA) = \bar{R}_Q^g(y, A)$. □

Proof of Proposition 10.22. (i) Functoriality forces the stated composites, and associativity gives $C_{kr} \circ C_{\ell k} = C_{\ell r}$. (ii) is the functor axiom on two routes, and for a rule family it is the axiom on the arrows indexed by b_1, b_2, and $b_1 b_2$. (iii) Each composite of configuration-keeping or additive arrows is the same statistic of the final partition, and the witness is direct evaluation: by pairs $(0, 0, 1, 0, 1, 0) \mapsto (0, 1, 1) \mapsto 1$, by triples $(0, 0, 1, 0, 1, 0) \mapsto (0, 0) \mapsto 0$. (iv) If stochastic matrices $P, Q \geq 0$ are mutually inverse, rank gives them equal sizes. From $PQ = I$, every positive entry P_{ik} forces row k of Q to vanish off column i, so each column of P has at most one positive entry, and unit row sums make P a permutation matrix. □

Proof of Proposition 10.24. Differentiating $\sum_x p_\gamma(x) = 1$ on the fixed support gives the branch-score identity, and differentiating the softmax gives $\nabla \log W_\gamma = \xi_\gamma - \bar{\xi}$, hence (10.13). In the outer product of (10.13), summation over x within each branch removes both cross terms by the branch-score identity, which gives (10.14); the label marginal has score $\xi_\gamma - \bar{\xi}$, so its Fisher tensor is $\mathrm{Cov}_W(\xi_\gamma)$. Positive denominators on a finite support make the C^3 masses give a C^2 Fisher tensor, and the Leibniz rule gives (10.15). A common shift cancels from the weights and adds one vector to every ξ_γ, which leaves the covariance unchanged as a function of θ. Finally $v^\top g^F(P) v = \sum_z P(z)(v^\top \mathbf{s}_P(z))^2 \geq 0$. □

Proof of Proposition 10.26. On its positive support $\mathbf{s}^c = \mathbf{s}_P + \nabla \log K^{\mathrm{obs}}(o \mid \gamma, x)$, and each record row has fixed support and sums to one, so $\sum_o K^{\mathrm{obs}}(o \mid \gamma, x) \nabla \log K^{\mathrm{obs}}(o \mid \gamma, x) = 0$ and the cross terms of the second moment vanish; this is (10.18). The record is the image of the complete sample under the parameter-independent map $(\gamma, x, o) \mapsto o$, with conditional law $\mathbf{\Pi}_o$ given $o \in \mathsf{O}^+$. A finite fixed-support family with C^3 masses is differentiable in quadratic mean, so Theorem 7.8 gives the score projection and the Fisher loss in (10.19). For a one-point alphabet every record row equals one, so $p_\theta(o) = \sum_{\gamma, x} P_\theta(\gamma, x) = 1$. The pointwise observed-information form of this identity is due to Louis [76]; its average over the record is (10.19). □

Proof of Proposition 10.28. The score telescope comes first. Differentiating $P_j = P_{j-1} C_j$ at positive atoms gives $\mathbf{s}_j(X_j) = \mathbb{E}[\mathbf{s}_{j-1}(X_{j-1}) \mid X_j]$, so the single-step loss is $D_j \succeq 0$ by Theorem 7.8, and the sum telescopes. The reversed path is Markov, so $\mathbb{E}[\varepsilon_j^{\mathrm{sc}} \mid X_j, \ldots, X_m] = 0$; for $j < k$ the increment $\varepsilon_k^{\mathrm{sc}}$ is measurable with respect to $X_j, \ldots, X_m$, which gives orthogonality, and $\mathbb{E}[\varepsilon_j^{\mathrm{sc}} (\varepsilon_j^{\mathrm{sc}})^\top] = g_{j-1} - g_j$ because $\mathbb{E}[\mathbf{s}_{j-1} \mathbf{s}_j^\top] = g_j$. Domination persists under each kernel on finite spaces, so every divergence is finite, and Theorem 10.7 and corollary 10.8 at each step give $\mathcal{F}_{j-1} - \mathcal{F}_j = K_j^{\mathrm{KL}}$ at unchanged evidence; summing proves the free-energy telescope, and division by N_m gives the per-agent form. □

Proof of Theorem 10.29. Uniform convergence of λ/N gives $\lambda_{\gamma,N} - \lambda_{\gamma_*,N} \le -\delta_{\mathsf{K}} N/2$ for each $\gamma \neq \gamma_*$ and large N; the softmax denominator contains the winning mass, so $W_{\gamma,N} \le e^{-\delta_{\mathsf{K}} N/2}$. Suppress N and write $\bar{\xi}_i = \sum_\eta W_\eta \xi_{\eta,i}$. The softmax derivatives $\partial_i W_\gamma = W_\gamma(\xi_{\gamma,i} - \bar{\xi}_i)$ and $\partial_j \bar{\xi}_i = \sum_\eta W_\eta \partial_j \xi_{\eta,i} + \mathrm{Cov}_W(\xi_i, \xi_j)$ show that, for $|\beta| \le 2$, $\partial^\beta W_\gamma$ is W_γ times a polynomial in the weights and in the components of the ξ_η and $\partial \xi_\eta$, of degree at most $|\beta|$ in the latter, each of which is $O(N)$ by (10.12). Hence $\|\partial^\beta W_{\gamma,N}\|_{\mathsf{K}} \le \mathrm{C}_{\mathsf{K}} N^{|\beta|} e^{-\delta_{\mathsf{K}} N/2}$ for $\gamma \neq \gamma_*$ and $|\beta| \le 2$, and the same bound holds for $W_{\gamma_*,N} - 1$, which is minus the sum of the losing weights. With $\delta\xi_\gamma = \xi_\gamma - \xi_{\gamma_*}$ and $\bar{d} = \sum_{\gamma \neq \gamma_*} W_\gamma \delta\xi_\gamma$, exact centering gives $\mathrm{Cov}_W(\xi) = \sum_{\gamma \neq \gamma_*} W_\gamma \delta\xi_\gamma \delta\xi_\gamma^\top - \bar{d}\bar{d}^\top$. Each derivative of $\delta\xi_\gamma$ through order two is $O(N)$, so a term with j derivatives in the sum is bounded by $\mathrm{C}_{\mathsf{K}} N^{j+2} e^{-\delta_{\mathsf{K}} N/2}$: at most j derivatives fall on a weight and the two score factors contribute at most N each. The product $\bar{d}\bar{d}^\top$ carries two exponential factors and obeys the same bound. Dividing by N proves (10.21). The branch error equals $\sum_{\gamma \neq \gamma_*} W_\gamma (\mathcal{I}_{\gamma,N} - \mathcal{I}_{\gamma_*,N})$, whose branch differences have bounded C^2 norms, so the Leibniz sum with the weight bound gives the stated bound. Adding the C^2 error of the winning branch and the covariance error, via (10.14), proves the C^2 convergence, since exponentials dominate the powers of N. For the floor, uniform convergence in operator norm gives $\|g_N - \mathcal{I}_{\gamma_*}\| \le \frac{1}{2}\mathrm{c}_{\min}$ on K for large N, so $z^\top g_N z \ge z^\top \mathcal{I}_{\gamma_*} z - \frac{1}{2}\mathrm{c}_{\min}|z|^2 \ge \frac{1}{2}\mathrm{c}_{\min}|z|^2$. □

Proof of Proposition 10.30. The label set is finite, so there is a positive bulk gap from S to every other label at θ_0; comparison with one maximizing label and convergence of λ/N make every nonmaximizing weight exponentially small, so its limit is zero. C^1 convergence gives $\xi_{\gamma,N}/N \to \mathsf{v}_\gamma$. Dividing (10.14) by N^2 makes the branch term $O(1/N)$ and leaves $\mathrm{Cov}_{W_N}(\xi_{\gamma,N}/N)$, which converges by finite summation. For every z, $z^\top \mathrm{Cov}_{\bar{\mathsf{p}}}(\mathsf{v}_\gamma) z = \sum_\gamma \bar{\mathsf{p}}_\gamma [z \cdot (\mathsf{v}_\gamma - \bar{\mathsf{v}})]^2$, which vanishes exactly when $z \cdot \mathsf{v}_\gamma$ is constant on the positive-weight labels, so the kernel is the orthogonal complement of the stated span. A positive limiting quadratic form gives the asymptotic after multiplication by N^2, and a zero one gives only the little-o conclusion. □

Proof of Theorem 10.31. A compact set of w is bounded, so its image under θ_N and the segments to θ_0 lie in the chosen neighborhood for large N. Put $\lambda^{\mathrm{inc}}_{\gamma,N}(\mathsf{w}) = \lambda_{\gamma,N}(\theta_0 + \mathsf{w}/N) - \lambda_{\gamma,N}(\theta_0)$. The fundamental theorem of calculus gives

$$\lambda^{\mathrm{inc}}_{\gamma,N}(\mathsf{w}) = \int_0^1 N^{-1} \nabla \lambda_{\gamma,N}(\theta_0 + \tau \mathsf{w}/N) \cdot \mathsf{w} \, d\tau, \tag{B.7}$$

whose integrand tends uniformly to $\mathsf{v}_\gamma \cdot \mathsf{w}$ by the C^1 part of (10.12) and continuity of $\nabla \mathsf{s}_\gamma$. Differentiation in w gives

$$D_{\mathsf{w}} \lambda^{\mathrm{inc}}_{\gamma,N} \to \mathsf{v}_\gamma, \qquad \|D^2_{\mathsf{w}} \lambda^{\mathrm{inc}}_{\gamma,N}\| = O(N^{-1}), \qquad \|D^3_{\mathsf{w}} \lambda^{\mathrm{inc}}_{\gamma,N}\| = O(N^{-2}), \tag{B.8}$$

so $\lambda^{\mathrm{inc}}_{\gamma,N} \to \mathsf{w} \cdot \mathsf{v}_\gamma$ in C^3. For tied labels, the relative log masses $\lambda_{\gamma,N} - \lambda_{\gamma_0,N}$ at $\theta_N(\mathsf{w})$ are the logarithms of the center ratios plus the difference of increments, and converge in C^3 to $\log \mathrm{c}_\gamma + \mathsf{w} \cdot (\mathsf{v}_\gamma - \mathsf{v}_{\gamma_0})$; exponentiation and the finite softmax are smooth on the resulting compact ranges, with denominator at least the reference mass one. For a nonmaximizing label the relative center log mass is at most $-\mathrm{c}N$ for some constant $\mathrm{c} > 0$, while its increment and w-derivatives through order three are bounded, so its relative exponential and those derivatives are $O(e^{-\mathrm{c}N})$.

Normalizing proves (10.23) in C^3, including the zero coordinates outside S. With $\hat{\xi}_{\gamma,N}(\mathsf{w}) = N^{-1}\xi_{\gamma,N}(\theta_N(\mathsf{w}))$, (B.8) gives $\hat{\xi}_{\gamma,N} \to \mathsf{v}_\gamma$ in C^2, and the affine change θ_N has derivative $N^{-1}\mathrm{Id}$, so (10.14) becomes

$$g^F_{\mathsf{w}}(\mathcal{E}_N)(\mathsf{w}) = \mathrm{Cov}_{W_N(\theta_N(\mathsf{w}))}\big(\hat{\xi}_{\gamma,N}(\mathsf{w})\big) + \frac{1}{N}\sum_\gamma W_{\gamma,N}(\theta_N(\mathsf{w}))\mathcal{I}_{\gamma,N}(\theta_N(\mathsf{w})). \tag{B.9}$$

The covariance converges in C^2 by finite sums and products, without dividing by a weight that tends to zero. The branch densities and their first two w-derivatives are bounded, since a w-derivative is N^{-1} times a θ-derivative, and the weights have bounded C^2 norms, so every term of the Leibniz sum for the last term is $O(N^{-1})$. Finally the log-sum-exponential has gradient $\sum_\gamma \bar{\mathsf{p}}_\gamma \mathsf{v}_\gamma$ and Hessian $\mathrm{Cov}_{\bar{\mathsf{p}}}(\mathsf{v}_\gamma)$. The window is the finite-mixture counterpart of the finite-volume rounding of a first-order transition, which Borgs and Kotecký [16] derive from a lattice model as a hyperbolic-tangent crossover of width of order the inverse volume. □

Proof of Proposition 10.32. The common branch law cancels from the relative log mass and has zero score, so the label is the only source of information. Then $W_1' = W_1(1-W_1)\omega_N'$, the centered label scores are $(1-W_1)\omega_N'$ and $-W_1\omega_N'$, and averaging their squares gives (10.25), with $W_1(1-W_1) = \mathrm{bv}(\omega_N)$. In (a) the C^3 norm of ω_N on a compact interval is at most $\mathrm{C}N$ and $\omega_N/N = \theta$; the formulas follow by substitution. In (b) the derivatives vanish and $\omega_N/N \to 0$ in C^3. In (c) the C^3 norms are $O(N^{\mathrm{c}})$ and $O(\sqrt{N})$, both $o(N)$, and substitution gives the two statements, with $\mathrm{bv}(0) = 1/4$ and $\mathrm{bv}(\log 2) = 2/9$. In (d) $\omega_N/N = \theta - \mathrm{c}N^{-1}\log N \to \theta$ in C^3 and the C^3 norm is $O(N)$; at zero, $\mathrm{bv}(-\mathrm{c}\log N)N^2/N = N^{1+\mathrm{c}}/(1+N^{\mathrm{c}})^2$. In (e) derivatives through order three are bounded by $N^{-1/6}N = N^{5/6}$, so $\omega_N/N \to 0$ in C^3; $|\omega_N| \le N^{-1/6}$ gives uniform convergence of the weights, and $0 \le F_N \le N^{-1}\frac{1}{4}N^{1/3}$. The Leibniz rule gives (10.28). At zero $\omega_N = 0$, $\omega_N' = N^{1/6}$, $\omega_N'' = 0$, $\omega_N''' = -N^{5/6}$, and $\mathrm{bv}(0) = 1/4$, $\mathrm{bv}'(0) = 0$, $\mathrm{bv}''(0) = -1/8$, which gives the value of $F_N''(0)$; the uniform limit has second derivative zero. □

B.5 Proofs for Exact Agent-Network Coarse-Graining

Proof of Theorem 12.4. Positivity and unitality of U give the bounds in (a), the bound on $\mathcal{A}(\psi)$, and the boundedness and positivity of U^ψ; substitution in (12.6) gives (12.8). If $\|h\|_\infty < \log 2$, then $\|\mathsf{U}^\psi(e^{-h}) - 1\|_\infty \le e^{\|h\|_\infty} - 1 < 1$, so the exponential series in $L^\infty(\bar{\mathsf{m}}_o)$ and the series of $\log(1+u)$ in the commutative Banach algebra $L^\infty(\bar{\mathsf{m}}_o^c)$ converge, and $h \mapsto \mathcal{A}(\psi+h)$ is real analytic on the ball of radius $\log 2$ about every ψ, which proves (a). The Fréchet chain rule for the exponential, U^ψ, and the logarithm gives (12.9); differentiating $\int h(y)C^\dagger_{\psi+tk}(z,dy)$ at $t=0$ gives (12.10); and $\psi = 0$ gives the origin. For (c), $\psi \le \psi' + c$ implies $e^{-\psi} \ge e^{-c}e^{-\psi'}$, and positivity of U gives the order statement; applying it in both directions with $c = \|\psi - \psi'\|_\infty$ gives (12.11), and $\mathsf{U}(e^{-\psi-c}) = e^{-c}\mathsf{U}(e^{-\psi})$ gives homogeneity, which with $\mathsf{U}1 = 1$ gives (d). For (e), $\widehat{\mathsf{m}}^\psi C = (e^{-\psi}\mathsf{m}_o)C/\mathsf{m}_o(e^{-\psi})$ and $\bar{\mathsf{m}}_o^c = \mathsf{m}_o C/M_o$ give the density $e^{-\mathcal{A}(\psi)}/\bar{\mathsf{m}}_o(e^{-\psi})$; the first two derivatives of $t \mapsto \log \bar{\mathsf{m}}_o(e^{-th})$ at $t=0$ are $-\bar{\mathsf{m}}_o(h)$ and $\mathrm{Var}_{\bar{\mathsf{m}}_o}(h)$, and polarization gives the covariance. □

Proof of Theorem 12.5. Conditional Jensen gives $|\mathsf{U}h|^p \le \mathsf{U}|h|^p$ for finite p, and integrating the disintegration (12.5) proves (12.13) and preservation of the mean; $p=\infty$ follows from the pointwise bound. The conditional variance identity

$$\int h^2 d\bar{\mathsf{m}}_o = \int (\mathsf{U}h)^2 d\bar{\mathsf{m}}_o^c + \int \mathrm{Var}_{C^\dagger(z,\cdot)}(h)\bar{\mathsf{m}}_o^c(dz), \tag{B.10}$$

the decomposition behind Theorem 7.8 applied to h, proves (12.14), and a nonnegative conditional variance integrates to zero exactly when it vanishes almost everywhere. For the oscillation bound, fix a representative of h and let N be the $\bar{\mathsf{m}}_o$-null set on which it lies outside $[\operatorname{ess\,inf} h, \operatorname{ess\,sup} h]$. Disintegration gives $\int C^\dagger(z,N)\bar{\mathsf{m}}_o^c(dz) = \bar{\mathsf{m}}_o(N) = 0$, so $C^\dagger(z,N)=0$ for almost every z. Thus this fixed h lies between its essential bounds almost surely under those kernels. For such a pair (z,z') the signed measure $\mu = C^\dagger(z,\cdot) - C^\dagger(z',\cdot)$ has total mass zero and positive and negative parts of equal mass $t \le \delta$, so $(\mathsf{U}h)(z) - (\mathsf{U}h)(z') = \int h(y)\mu^+(dy) - \int h(y)\mu^-(dy) \le t(\operatorname{ess\,sup} h - \operatorname{ess\,inf} h)$. The sets where $\mathsf{U}h$ lies within any positive distance of its essential supremum or infimum have positive $\bar{\mathsf{m}}_o^c$ measure, so the bound passes to $\operatorname{ess\,sup} \mathsf{U}h - \operatorname{ess\,inf} \mathsf{U}h$; applying it factor by factor gives the composition statement. □

Proof of Corollary 12.6. Fix a representative of ψ with values in $[\mathrm{a},\mathrm{b}]$, where $\mathrm{a} = \operatorname{ess\,inf}\psi$ and $\mathrm{b} = \operatorname{ess\,sup}\psi$, so that $\mathrm{b}-\mathrm{a} = 2\|[\psi]\|_{\mathrm{osc}}$. Then $e^{-\mathrm{b}} \le \bar{\mathsf{m}}_o(e^{-\psi}) \le e^{-\mathrm{a}}$, which is the bracket of the log evidence ratio, and the density $e^{-\psi}/\bar{\mathsf{m}}_o(e^{-\psi})$ of $\widehat{\mathsf{m}}^\psi$ with respect to $\bar{\mathsf{m}}_o$ is positive with a logarithm in $[\mathrm{a}-\mathrm{b}, \mathrm{b}-\mathrm{a}]$. This gives equivalence and, since each divergence is the mean of a logarithmic density ratio bounded by $\mathrm{b}-\mathrm{a}$ in absolute value, the linear branch of (12.17). For the quadratic branch, the density of $\widehat{\mathsf{m}}^\psi$ gives $\widehat{\mathsf{m}}^\psi(e^{\psi}) = 1/\bar{\mathsf{m}}_o(e^{-\psi})$, so

$$D_{\mathrm{KL}}\big(\bar{\mathsf{m}}_o \| \widehat{\mathsf{m}}^\psi\big) = \bar{\mathsf{m}}_o(\psi) + \log \bar{\mathsf{m}}_o(e^{-\psi}) = \log \bar{\mathsf{m}}_o\big(e^{-(\psi - \bar{\mathsf{m}}_o(\psi))}\big),$$
$$D_{\mathrm{KL}}\big(\widehat{\mathsf{m}}^\psi \| \bar{\mathsf{m}}_o\big) = -\widehat{\mathsf{m}}^\psi(\psi) - \log \bar{\mathsf{m}}_o(e^{-\psi}) = \log \widehat{\mathsf{m}}^\psi\big(e^{\psi - \widehat{\mathsf{m}}^\psi(\psi)}\big).$$

Since $\widehat{\mathsf{m}}^\psi$ is equivalent to $\bar{\mathsf{m}}_o$, ψ takes values in $[\mathrm{a},\mathrm{b}]$ almost surely under both laws, and Hoeffding [53, Sec. 4, Eq. (4.16)], applied under $\bar{\mathsf{m}}_o$ to $-\psi$ and under $\widehat{\mathsf{m}}^\psi$ to ψ with $h=1$, bounds both divergences by $(\mathrm{b}-\mathrm{a})^2/8 = \|[\psi]\|_{\mathrm{osc}}^2/2$. Adding a constant to ψ changes neither $\widehat{\mathsf{m}}^\psi$ nor $\|[\psi]\|_{\mathrm{osc}}$. For $Q \ll \bar{\mathsf{m}}_o$ one has $\log(dQ/d\widehat{\mathsf{m}}^\psi) = \log(dQ/d\bar{\mathsf{m}}_o) + \psi + \log \bar{\mathsf{m}}_o(e^{-\psi})$ Q-almost surely; the last two terms are bounded and the first has a Q-integrable negative part (Definition 3.10), so $D_{\mathrm{KL}}(Q\|\widehat{\mathsf{m}}^\psi) = D_{\mathrm{KL}}(Q\|\bar{\mathsf{m}}_o) + \mathbb{E}_Q\psi + \log \bar{\mathsf{m}}_o(e^{-\psi})$ in $(-\infty,+\infty]$, and the mass $(e^{-\psi}\mathsf{m}_o)(\mathsf{Y}) = M_o\bar{\mathsf{m}}_o(e^{-\psi})$ turns this into the free-energy identity. Any other Q charges a set that is null for both equivalent laws, so both divergences are $+\infty$. Along the arrows, suppose $\widetilde{H}_\ell - H_\ell$ is essentially bounded, as it is for $\ell=0$, where it equals ψ_0. Then $\widetilde{\mathsf{m}}_\ell = e^{-(\widetilde{H}_\ell - H_\ell)}\mathsf{m}_\ell$, so (12.6) gives $\widetilde{\mathsf{m}}_\ell C_\ell = e^{-\mathcal{A}_\ell(\widetilde{H}_\ell - H_\ell)}\mathsf{m}_{\ell+1}$ and $\widetilde{\mathsf{m}}_{\ell+1} = e^{-\mathcal{A}_\ell(\widetilde{H}_\ell - H_\ell) - \psi_{\ell+1}}\mathsf{m}_{\ell+1}$; taking densities with respect to $\varrho_{\ell+1}$ where $H_{\ell+1}$ is finite, which is $\mathsf{m}_{\ell+1}$-almost everywhere, gives the recurrence in (12.18). Since $\mathcal{A}_\ell(0)=0$, part (c) of Theorem 12.4 gives $\mathcal{A}_\ell(c)=c$ for constants c, and applied to the constant bounds of $\widetilde{H}_\ell - H_\ell$ it keeps $\mathcal{A}_\ell(\widetilde{H}_\ell - H_\ell)$ between the essential infimum and supremum of $\widetilde{H}_\ell - H_\ell$. Adding $\psi_{\ell+1}$ and inducting gives essential boundedness at every depth and the bounds in (12.18), and the first part of the proof applies at depth L. □

Proof of Corollary 12.7. Fix $\ell < L$ and $\psi \in L^\infty(\bar{\mathfrak{m}}_\ell)$ with $\mathrm{a} = \operatorname{ess\,inf}\psi$, $\mathrm{b} = \operatorname{ess\,sup}\psi$, and $x = \mathrm{b} - \mathrm{a}$, and fix a representative with values in $[\mathrm{a}, \mathrm{b}]$ off a $\bar{\mathfrak{m}}_\ell$-null set N. As in the proof of Theorem 12.5, $C_\ell^\dagger(z, N) = 0$ for $\bar{\mathfrak{m}}_{\ell+1}$-almost every z. When $\Delta_\ell < \infty$, the sets of full $(\bar{\mathfrak{m}}_{\ell+1})^{\otimes 2}$ measure on which (12.20) holds for a sequence $t_n \downarrow \Delta_\ell$ intersect in a set of full measure on which it holds with $t = \Delta_\ell$. Hence the pairs (z, z') whose rows $\mu = C_\ell^\dagger(z, \cdot)$ and $\mu' = C_\ell^\dagger(z', \cdot)$ both vanish on N, satisfy $\sup_A |\mu(A) - \mu'(A)| \le \delta_\ell$, and, when $\Delta_\ell < \infty$, satisfy (12.20) with $t = \Delta_\ell$, form a set of full measure. For such a pair put $g = e^{\mathrm{b}-\psi}$, which takes values in $[1, e^x]$ almost surely under both rows, so that $\mathcal{A}_\ell(\psi)(z') - \mathcal{A}_\ell(\psi)(z) = \log(\mu(g)/\mu'(g))$. Since $\mu - \mu'$ has total mass zero and its positive part has mass at most δ_ℓ, $\mu(g) - \mu'(g) = (\mu - \mu')(g - 1) \le \delta_\ell(e^x - 1)$, and $\mu'(g) \ge 1$, so

$$\log \frac{\mu(g)}{\mu'(g)} \le \log\bigl(1 + \delta_\ell(e^x - 1)\bigr).$$

For the second bound let $\Delta_\ell < \infty$, put $\lambda = \mu(g)/\mu'(g)$, let P be a positive set of a Hahn decomposition of the signed measure $\nu = \mu - \lambda\mu'$, and put $h = 1 + (e^x - 1)1_P$. Because $0 \le g - 1 \le e^x - 1$ almost surely under both rows, $0 = \nu(g) = \nu(1) + \nu(g - 1) \le \nu(1) + (e^x - 1)\nu(P) = \nu(h)$, so $\mu(h) \ge \lambda\mu'(h)$. With $p = \mu(P)$ and $q = \mu'(P)$,

$$\log \frac{\mu(h)}{\mu'(h)} = \log\bigl((1 - p) + pe^x\bigr) - \log\bigl((1 - q) + qe^x\bigr),$$

which is positive only if $p > q$; otherwise $\log\lambda \le 0$. If $p > q$, (12.20) with $A = P$ and $B = \mathrm{Y}_\ell \setminus P$ reads $p(1 - q) \le e^{\Delta_\ell} q(1 - p)$, which excludes $q = 0$ and $p = 1$, so $p, q \in (0, 1)$ and $\gamma = \log\bigl(p(1-q)/(q(1-p))\bigr) \in (0, \Delta_\ell]$. Put $\beta = \log(q/(1-q))$ and $\mathrm{sg}(u) = (1+e^{-u})^{-1}$, the derivative of $u \mapsto \log(1 + e^u)$. Since $1 - q = (1 + e^\beta)^{-1}$ and $1 - p = (1 + e^{\beta+\gamma})^{-1}$,

$$\log \frac{\mu(h)}{\mu'(h)} = \int_0^x \bigl[\mathrm{sg}(\beta + \gamma + u) - \mathrm{sg}(\beta + u)\bigr]\,du.$$

The function $u \mapsto \mathrm{sg}(u + \gamma) - \mathrm{sg}(u)$ has derivative $\mathrm{sg}'(u + \gamma) - \mathrm{sg}'(u)$, and $\mathrm{sg}' = \mathrm{sg}(1 - \mathrm{sg})$ is even and decreasing in $|u|$, so the derivative is positive for $u < -\gamma/2$ and negative for $u > -\gamma/2$; the maximum is $\mathrm{sg}(\gamma/2) - \mathrm{sg}(-\gamma/2) = \tanh(\gamma/4) \le \tanh(\Delta_\ell/4)$. Hence $\log(\mu(g)/\mu'(g)) = \log\lambda \le \log(\mu(h)/\mu'(h)) \le \tanh(\Delta_\ell/4)x$, the form of the contraction ratio of Birkhoff's lemma on Hilbert's projective metric [13, Sec. 4, Lemma 1, Eq. (9a)]. The sets where $\mathcal{A}_\ell(\psi)$ lies within any positive distance of its essential supremum or infimum have positive $\bar{\mathfrak{m}}_{\ell+1}$ measure, so their product meets the full-measure set of pairs, and letting the distance tend to zero gives

$$\operatorname{ess\,sup}\mathcal{A}_\ell(\psi) - \operatorname{ess\,inf}\mathcal{A}_\ell(\psi) \le \min\bigl\{\log\bigl(1 + \delta_\ell(e^x - 1)\bigr), \tanh(\Delta_\ell/4)x\bigr\},$$

the second term being x when $\Delta_\ell = \infty$ by part (c) of Theorem 12.4.

(a) Take $\psi = \mathfrak{r}_\ell$, so $x = \omega_\ell$. The recurrence $\mathfrak{r}_{\ell+1} = \mathcal{A}_\ell(\mathfrak{r}_\ell) + \psi_{\ell+1}$ of (12.18), subadditivity of the essential supremum, and superadditivity of the essential infimum give (12.21). For $\omega_\ell > 0$ one has $\delta_\ell(e^{\omega_\ell} - 1) < e^{\omega_\ell} - 1$ exactly when $\delta_\ell < 1$, and $\tanh(\Delta_\ell/4) < 1$ exactly when $\Delta_\ell < \infty$.

(b) For $\delta = 0$ take two points y_0, y_1 of mass $1/2$, a channel sending each to the uniform law on $\{z, z'\}$, and $\psi = x1_{\{y_1\}}$; both reverse rows equal the fine law, so $\mathcal{A}(\psi)$ is constant. For $\delta \in (0, 1]$ give (y_1, z), (y_0, z'), and (y_1, z') the joint masses $1/2$, $\delta/2$, and $(1-\delta)/2$; the fine law gives y_0 and y_1 the positive masses $\delta/2$ and $1 - \delta/2$, the coarse law is uniform on $\{z, z'\}$, the channel is the forward conditional law, and the reverse rows are given by $C^\dagger(z, \{y_1\}) = 1$, $C^\dagger(z', \{y_0\}) = \delta$, and $C^\dagger(z', \{y_1\}) = 1 - \delta$. Their total variation distance, and hence the coefficient, is δ. For $\psi = x1_{\{y_1\}}$ one has $\mathcal{A}(\psi)(z) = x$ and $\mathcal{A}(\psi)(z') = -\log(\delta + (1-\delta)e^{-x})$, whose difference is $x + \log(\delta + (1-\delta)e^{-x}) = \log(1 + \delta(e^x - 1))$. The pair (z', z) carries product mass $1/4$, and (12.20) with $A = \{y_0\}$ and $B = \{y_1\}$ would require $\delta \le e^t \cdot 0$, so no t qualifies and the diameter is infinite.

(c) The map $x \mapsto \log(1 + \delta(e^x - 1))$ is increasing in x and in δ, so $\omega_{\ell+1} \le \log(1 + \delta(e^{\omega_\ell} - 1)) + s$ for $\ell < L$. The function $\mathrm{k}(x) = x - \log(1 + \delta(e^x - 1))$ has $\mathrm{k}(0) = 0$ and $\mathrm{k}'(x) = (1-\delta)/(1 + \delta(e^x - 1)) > 0$, and it tends to $\log(1/\delta)$ as $x \to \infty$, since $\mathrm{k}(x) = -\log(\delta + (1-\delta)e^{-x})$; so it is a continuous increasing bijection of $[0, \infty)$ onto $[0, \log(1/\delta))$, and $x_{\delta,s}$ exists and is unique. Put $\mathrm{B} = \max\{\omega_0, x_{\delta,s}\}$. If $\omega_\ell \le \mathrm{B}$, then $\omega_{\ell+1} \le \log(1 + \delta(e^{\mathrm{B}} - 1)) + s = \mathrm{B} - \mathrm{k}(\mathrm{B}) + s \le \mathrm{B}$, because $\mathrm{k}(\mathrm{B}) \ge \mathrm{k}(x_{\delta,s}) = s$; induction from $\ell = 0$ gives the first bound. For the second put $\mathrm{B}' = \max\{\omega_0, s/(1 - \tanh(\Delta/4))\}$; if $\omega_\ell \le \mathrm{B}'$, then $\omega_{\ell+1} \le \tanh(\Delta/4)\mathrm{B}' + s \le \mathrm{B}'$, because $s \le (1 - \tanh(\Delta/4))\mathrm{B}'$. The divergence bound is (12.17) at depth L with $\|[\mathfrak{r}_L]\|_{\mathrm{osc}} = \omega_L/2$.

(d) The joint law $\bar{\mathsf{m}}_\ell(dy)C_\ell(y, dz)$ is concentrated on the graph of χ_ℓ, a Borel set, so disintegration gives $\int C_\ell^\dagger(z, \chi_\ell^{-1}\{z\})\bar{\mathsf{m}}_{\ell+1}(dz) = 1$ and $C_\ell^\dagger(z, \chi_\ell^{-1}\{z\}) = 1$ for almost every z. Because $\bar{\mathsf{m}}_{\ell+1}$ is not a point mass, the diagonal has $(\bar{\mathsf{m}}_{\ell+1})^{\otimes 2}$ measure equal to the sum of the squared atom masses, at most the largest atom mass, which is below one; so pairs $z \ne z'$ whose rows are concentrated on the disjoint fibers $\chi_\ell^{-1}\{z\}$ and $\chi_\ell^{-1}\{z'\}$ have positive measure. For such a pair, $A = \chi_\ell^{-1}\{z\}$ gives $C_\ell^\dagger(z, A) - C_\ell^\dagger(z', A) = 1$, so $\delta_\ell = 1$, and with $B = \mathsf{Y}_\ell \setminus A$ the left side of (12.20) is one and the right side zero, so $\Delta_\ell = \infty$. Both terms of the minimum in (12.21) then equal ω_ℓ. For a bounded Borel f on $\mathsf{Y}_{\ell+1}$ the same concentration gives $\mathcal{A}_\ell(f \circ \chi_\ell)(z) = -\log \int e^{-f(\chi_\ell(y))} C_\ell^\dagger(z, dy) = f(z)$ for almost every z. Put $F_L = F$, $F_\ell = F_{\ell+1} \circ \chi_\ell$, and $\bar{c}_\ell = \sum_{j=0}^{\ell} c_j$. Then $\mathfrak{r}_0 = \psi_0 = \bar{c}_0 F_0$, and if $\mathfrak{r}_\ell = \bar{c}_\ell F_\ell$, the recurrence gives $\mathfrak{r}_{\ell+1} = \bar{c}_\ell F_{\ell+1} + c_{\ell+1}F_{\ell+1} = \bar{c}_{\ell+1}F_{\ell+1}$. By Proposition 12.2, $\bar{\mathsf{m}}_L$ is the image of $\bar{\mathsf{m}}_\ell$ under $\chi_{L-1} \circ \cdots \circ \chi_\ell$, so F_ℓ has under $\bar{\mathsf{m}}_\ell$ the essential range of F under $\bar{\mathsf{m}}_L$, of some length W. Since the c_j are nonnegative, $\omega_j^{\mathrm{st}} = c_j \mathrm{W}$ and $\omega_\ell = \bar{c}_\ell \mathrm{W} = \sum_{j=0}^{\ell} \omega_j^{\mathrm{st}}$.

(e) By (12.6), $e^{-\mathcal{A}_\ell(\psi)}$ is the density of $(e^{-\psi}\bar{\mathsf{m}}_\ell)C_\ell$ with respect to $\bar{\mathsf{m}}_\ell C_\ell = \bar{\mathsf{m}}_{\ell+1}$, and a Markov kernel preserves total mass, so $\bar{\mathsf{m}}_{\ell+1}(e^{-\mathcal{A}_\ell(\psi)}) = \bar{\mathsf{m}}_\ell(e^{-\psi})$. Since $\widetilde{\mathsf{m}}_L = e^{-\mathfrak{r}_L}\mathsf{m}_L$, the first equality of (12.22) holds. Mass preservation also gives $\widetilde{\mathsf{m}}_j(\mathsf{Y}_j) = \widetilde{\mathsf{m}}_{j-1}(\mathsf{Y}_{j-1})\widehat{\mathsf{n}}_j(e^{-\psi_j})$ for $j \ge 1$, $\widetilde{\mathsf{m}}_0(\mathsf{Y}_0) = M_0 \bar{\mathsf{m}}_0(e^{-\psi_0})$, and $M_L = M_0$ by Proposition 12.2; the product of these factors gives the second equality. The proof of Corollary 12.6 gives $\widetilde{\mathsf{m}}_{j-1}C_{j-1} = e^{-\mathcal{A}_{j-1}(\mathfrak{r}_{j-1})}\mathsf{m}_j$ with a bounded exponent, so $\widehat{\mathsf{n}}_j$ is equivalent to $\bar{\mathsf{m}}_j$, ψ_j has the same essential bounds under both, and each summand lies in the stated interval. Finally, by part (c) of Theorem 12.4, $\mathcal{A}_{\ell'}(\mathfrak{r}_{\ell'} + c) = \mathcal{A}_{\ell'}(\mathfrak{r}_{\ell'}) + c$ for $\ell' \ge j$, so replacing ψ_j by $\psi_j + c$ replaces $\mathfrak{r}_\ell$ by $\mathfrak{r}_\ell + c$ and $\widetilde{\mathsf{m}}_\ell$ by $e^{-c}\widetilde{\mathsf{m}}_\ell$ for every $\ell \ge j$ and changes nothing at smaller depths, for every choice of arrows. Widths and normalizations are therefore unchanged, and $\log \bar{\mathsf{m}}_L(e^{-\mathfrak{r}_L})$ decreases by c. □

Proof of Theorem 12.12. Fubini's theorem for the product ν_ℓ gives commutation and $\mathbb{E}_{\ell,B}\mathbb{E}_{\ell,B'} = \mathbb{E}_{\ell,B\cap B'}$; expanding the product in (12.23) with this rule gives the alternating sum, and Möbius inversion on the Boolean lattice gives the orthogonality relations and the resolution of the identity. Every term of $\mathsf{P}_{\ell,B}f$ is a function of y_B, and the factor $I - \mathbb{E}_{\ell,V_\ell\setminus\{i\}}$ removes its mean in coordinate i; hence the nonempty components recover g and sum to $f - \mathbb{E}_{\ell,\varnothing}f$, the class $[f]$. The triangle inequality and $\|[\psi]\|_{\mathrm{osc}} \le \|\psi\|_\infty$ give $\|\mathrm{asm}_\ell\| \le 1$. For nonempty B the projector $\mathsf{P}_{\ell,B}$ annihilates constants and is an alternating sum of $2^{|B|}$ contractions, so $\|\mathsf{P}_{\ell,B}f\|_\infty \le 2^{|B|}\|f - c\|_\infty$ for every constant c, and $\sum_{B\neq\varnothing} 2^{|B|} = 3^{|V_\ell|} - 1$. For the biased coordinates, expanding $\prod_i\big((Y_i - m) + m\big)$ with $m = \mathbb{E}Y_i = 2p - 1$ gives the components $m^{n-|B|}\prod_{i\in B}(Y_i - m)$, whose supremum norms $m^{n-|B|}(1+m)^{|B|}$ sum to $(1+2m)^n - m^n = (4p-1)^n - (2p-1)^n$. □

Proof of Proposition 12.17. Because $\bar{\mathsf{m}}_\ell$ is a product over agents and hence over blocks, the joint law $\bar{\mathsf{m}}_\ell(dy)C_\ell(y,dz)$ is the product of the block laws $\bar{\mathsf{m}}_\ell^I(dy_I)C_I(y_I,dz_I)$; its z-marginal is the product of the block marginals, and the product of the block reverse kernels $C_I^\dagger(z_I,dy_I)$ is a version of the reverse kernel. For $\mathrm{g}\in\mathcal{G}_\ell^{\mathrm{intra}}$, grouping components by block writes $\psi_{\mathrm{g}} = \sum_I \psi_I$ with ψ_I a function of y_I, so Fubini's theorem gives $\mathcal{A}_\ell(\psi_{\mathrm{g}}) = \sum_I f_I(z_I)$ with $f_I = -\log \mathsf{U}_I(e^{-\psi_I})$. For $|B|\ge 2$ choose $j\in B\setminus\{I\}$; the function $\mathbb{E}_{\ell+1,B}f_I$ does not depend on z_j, so the factor $I - \mathbb{E}_{\ell+1,V_{\ell+1}\setminus\{j\}}$ of the product form of (12.23) annihilates it. Hence $\mathsf{P}_{\ell+1,B}f_I = 0$, and linearity finishes the proof. □

Proof of Proposition 12.25. Channel covariance gives $\widehat{\mathsf{m}}_\ell\widehat{C}_\ell = (\kappa_{\ell+1})_\#(\mathsf{m}_\ell C_\ell)$ and $(e^{-\mathsf{K}_\ell\psi}\widehat{\mathsf{m}}_\ell)\widehat{C}_\ell = (\kappa_{\ell+1})_\#((e^{-\psi}\mathsf{m}_\ell)C_\ell)$. The Radon–Nikodym derivative of two measures pushed through one Borel isomorphism is the transported derivative, and applying $-\log$ gives (12.40) without selecting a pointwise reverse kernel. Differentiating at an arbitrary bounded ψ with part (b) of Theorem 12.4 gives (12.41). Composing with the intertwinings (12.26) gives (12.42), and inserting the intertwining of the retained projections into $I - \mathsf{P}_{\ell+1}^{\mathrm{ret}}$ and into Proposition 12.24 gives the residual formulas. □

Proof of Proposition 12.28. (a) is Tonelli's theorem applied to the product of nonnegative factors. (b) is Möbius inversion on the Boolean lattice of subsets of V^c, and the component indexed by B reads only z_B because every term does. (c) follows by substituting gz and gz° in (12.46). For (d), write $f(u) = -\log 2\cosh u$, so $f''' = 2\,\mathrm{sech}^2\tanh$. With $x = \sum_r \xi_r y_r$ and $y_r^2 = 1$, only odd powers of x contain the monomial $y_1y_2y_3$; the cubic term $f'''(\xi_0)x^3/6$ contributes $f'''(\xi_0)\xi_1\xi_2\xi_3$, and higher odd powers are of order five in the couplings. At $\xi_0 = 0$ the function $f(x)$ is even in y, so every odd Walsh coefficient vanishes. □

Proof of Proposition 12.30. Adjointness is (12.47) integrated against $f(y_j)g(z_I)$. For the factorization witness take two blocks with one $\{-1,1\}$-valued agent each, a uniform product baseline, a channel that flips each coordinate independently with probability $1/5$, and the likelihood $e^{Jy_1y_2}$. The baseline reverse probability of $y_1 = 1$ given $z_1 = 1$ is $4/5$ whatever z_2 is, while with $x = e^{2J}$ the posterior reverse probability is $4(x+4)/(8x+17)$ at $z_2 = -1$ and $4(4x+1)/(17x+8)$ at $z_2 = 1$, which differ exactly when $J\neq 0$. For a deterministic statistic, $\mathsf{CJ}f(z) = \mathbb{E}[f(\chi(Y)) \mid \chi(Y) = z] = f(z)$ almost everywhere. A channel whose output does not depend on y gives $\mathsf{CJ}f = \mathbf{\Pi}_o^c(f)$, which differs from f for nonconstant f. □

Proof of Theorem 12.37. Assume (12.56) and take $\mu = \delta_y$. Then $\mathsf{T}(y,\chi^{-1}A) = \mathsf{T}^\chi(\chi(y),A)$ for every y and Borel A, whose right side depends on y only through $\chi(y)$, and surjectivity determines

T^{χ} at every z. Conversely, for fixed z the map $A \mapsto \mathsf{T}(\iota(z), \chi^{-1}A)$ is a probability measure because χ^{-1} commutes with countable disjoint unions, and for fixed A it is Borel in z as the composition of ι with $y \mapsto \mathsf{T}(y, \chi^{-1}A)$. Since $\chi(\iota(\chi(y))) = \chi(y)$, hypothesis (12.55) gives the same pointwise identity, and integrating it against μ gives (12.56). □

Proof of Theorem 12.46. At $t = 0$ the empty context has mass one. If the context at t has its $\mathbb{P}_N^{\mathrm{path}}$ marginal, multiplying by its row gives the correct context-plus-symbol marginal, and marginalizing the oldest symbol gives the next context marginal; by induction no positive prefix enters a null context, and successive row sums give normalization. Expanding the logarithm of the products and regrouping by contexts gives (12.64); a competitor that excludes a positive word has infinite divergence. Equality forces every occupied row to agree, and recursive reachability forces the same path law. Subtracting the source chain-rule logarithm from the projected product leaves, in expectation, the difference between entropy given the context and entropy given the full past, which is the conditional mutual information (12.63) by the chain rules of Cover and Thomas [24, Sec. 2.5]. Conditioning on more history does not increase entropy, so the defect decreases in k, and $k \geq N - 1$ keeps the whole past. The projection keeps its own defining marginals, and a shorter prefix only clips the available history. The total-variation bound is Pinsker's inequality [24, Lemma 11.6.1]. For the cycle, $\mathbb{P}(Z_0 = 0) = 2/3$ and both one-step rows from 0 put mass $1/2$ on 0, so the projection gives 000 the mass $1/6$ although the cycle never produces it. Subtracting the two finite log ratios gives (12.65). □

Proof of Theorem 12.47. Write $\mathbb{P}_t^{\mathrm{path}}$ and $\widetilde{\mathbb{P}}_t$ for the laws of $Z_{0:t-1}$ under $\mathbb{P}_N^{\mathrm{path}}$ and $\widetilde{\mathbb{P}}$, the case $t = 0$ living on a point, abbreviate $\mathbf{f}_t(Z_{0:t-1})$ to $\mathbf{f}_t$, and let $\mathbb{P}_{t+1}^{\mathbf{f}}$ and $\widetilde{\mathbb{P}}_{t+1}^{\mathbf{f}}$ be the laws $\mathbb{P}_t^{\mathrm{path}}(dz_{0:t-1})\mathsf{T}_t^{\mathbf{f}}(\mathbf{f}_t(z_{0:t-1}), dz_t)$ and $\mathbb{P}_t^{\mathrm{path}}(dz_{0:t-1})\widetilde{\mathsf{T}}_t(\mathbf{f}_t(z_{0:t-1}), dz_t)$ on Z^{t+1}. The kernel $\widetilde{\mathsf{T}}_t(\mathbf{f}_t(\cdot), \cdot)$ is a version of the law of Z_t given $Z_{0:t-1}$ both under $\widetilde{\mathbb{P}}$ and under $\widetilde{\mathbb{P}}_{t+1}^{\mathbf{f}}$, whose first marginal is $\mathbb{P}_t^{\mathrm{path}}$, so part (a) of Lemma 3.12 with first coordinate $Z_{0:t-1}$, applied to both pairs, gives

$$D_{\mathrm{KL}}\big(\mathbb{P}_{t+1}^{\mathrm{path}} \| \widetilde{\mathbb{P}}_{t+1}\big) = D_{\mathrm{KL}}\big(\mathbb{P}_t^{\mathrm{path}} \| \widetilde{\mathbb{P}}_t\big) + D_{\mathrm{KL}}\big(\mathbb{P}_{t+1}^{\mathrm{path}} \| \widetilde{\mathbb{P}}_{t+1}^{\mathbf{f}}\big)$$

in $[0, +\infty]$, and summing over t writes $D_{\mathrm{KL}}(\mathbb{P}_N^{\mathrm{path}} \| \widetilde{\mathbb{P}})$ as the sum of the last terms. Fix t and let $\mathbb{P}_t^{\mathrm{path}}(\cdot \mid \mathbf{w})$ be a regular conditional law of $Z_{0:t-1}$ given $\mathbf{f}_t = \mathbf{w}$. The map $(z_{0:t-1}, z_t) \mapsto (\mathbf{f}_t(z_{0:t-1}), z_t, z_{0:t-1})$ is Borel and the Borel projection onto its last two components inverts it, so part (d) of the lemma, applied to the map and to the projection, shows that pushing two laws through it preserves their divergence; equivalently, the map is a bimeasurable bijection onto its image, which is Borel as the preimage of the diagonal of the standard Borel statistic space under a Borel map, and part (e) applies there. After the push, Fubini's theorem, and the fact that under the two laws just defined Z_t reads the history only through $\mathbf{f}_t$, give the triple $(\mathbf{f}_t, Z_t, Z_{0:t-1})$ the law $(\mathbf{f}_t)_{\#}\mathbb{P}_t^{\mathrm{path}}(d\mathbf{w})\mathsf{T}_t^{\mathbf{f}}(\mathbf{w}, dz_t)\mathbb{P}_t^{\mathrm{path}}(dz_{0:t-1} \mid \mathbf{w})$ under $\mathbb{P}_{t+1}^{\mathbf{f}}$ and the same law with $\widetilde{\mathsf{T}}_t$ in place of $\mathsf{T}_t^{\mathbf{f}}$ under $\widetilde{\mathbb{P}}_{t+1}^{\mathbf{f}}$, while under $\mathbb{P}_{t+1}^{\mathrm{path}}$ the pair $(\mathbf{f}_t, Z_t)$ has the law $(\mathbf{f}_t)_{\#}\mathbb{P}_t^{\mathrm{path}}(d\mathbf{w})\mathsf{T}_t^{\mathbf{f}}(\mathbf{w}, dz_t)$ by the definition of $\mathsf{T}_t^{\mathbf{f}}$. Part (a) with first coordinate $(\mathbf{f}_t, Z_t)$ therefore writes $D_{\mathrm{KL}}(\mathbb{P}_{t+1}^{\mathrm{path}} \| \widetilde{\mathbb{P}}_{t+1}^{\mathbf{f}})$ as the divergence of the $(\mathbf{f}_t, Z_t)$-marginals plus an average, under the $(\mathbf{f}_t, Z_t)$-marginal of $\mathbb{P}_{t+1}^{\mathrm{path}}$, of the divergence from the conditional history law of $\mathbb{P}_{t+1}^{\mathrm{path}}$ to $\mathbb{P}_t^{\mathrm{path}}(\cdot \mid \mathbf{w})$; the same application to $\mathbb{P}_{t+1}^{\mathrm{path}}$ and $\mathbb{P}_{t+1}^{\mathbf{f}}$, whose $(\mathbf{f}_t, Z_t)$-marginals coincide, shows that this average is $D_{\mathrm{KL}}(\mathbb{P}_{t+1}^{\mathrm{path}} \| \mathbb{P}_{t+1}^{\mathbf{f}})$. Part (a)

along $\mathbf{f}_t$ turns the marginal term into $\int D_{\mathrm{KL}}(\mathsf{T}^{\mathbf{f}}_t(\mathbf{w},\cdot)\|\widetilde{\mathsf{T}}_t(\mathbf{w},\cdot))(\mathbf{f}_t)_{\#}\mathbb{P}^{\mathrm{path}}_t(d\mathbf{w})$, the t-th row term of (12.66) after the change of variables $\mathbf{w} = \mathbf{f}_t(z_{0:t-1})$. Pushing $\mathbb{P}^{\mathrm{path}}_{t+1}$ and $\mathbb{P}^{\mathbf{f}}_{t+1}$ instead through $(z_{0:t-1}, z_t) \mapsto (\mathbf{f}_t(z_{0:t-1}), (z_t, z_{0:t-1}))$, inverted in the same way, and applying (a) along $\mathbf{f}_t$, whose laws agree, writes $D_{\mathrm{KL}}(\mathbb{P}^{\mathrm{path}}_{t+1}\|\mathbb{P}^{\mathbf{f}}_{t+1})$ as the average of the divergence between the conditional law of $(Z_t, Z_{0:t-1})$ given $\mathbf{f}_t$ and the product $\mathsf{T}^{\mathbf{f}}_t(\mathbf{f}_t,\cdot) \otimes \mathbb{P}^{\mathrm{path}}_t(\cdot \mid \mathbf{f}_t)$ of its conditional marginals, which is $I(Z_t; Z_{0:t-1} \mid \mathbf{f}_t)$. Summing over t proves (12.66), each term being independent of the chosen versions by (a).

The row terms are nonnegative and vanish for $\widetilde{\mathsf{T}}_t = \mathsf{T}^{\mathbf{f}}_t$, which gives the minimum and, when $D^{\mathbf{f}}_N$ is finite, the difference. The t-th defect term vanishes exactly when $\mathbb{P}^{\mathrm{path}}_{t+1} = \mathbb{P}^{\mathbf{f}}_{t+1}$, that is, by uniqueness of disintegrations, when $\mathsf{T}^{\mathbf{f}}_t(\mathbf{f}_t(\cdot),\cdot)$ is a version of the law of Z_t given $Z_{0:t-1}$; conversely, a version that depends on the history only through $\mathbf{f}_t$ is a conditional law of Z_t given $\mathbf{f}_t$ and so agrees with $\mathsf{T}^{\mathbf{f}}_t$ at almost every value. For $\mathbf{f}_t(z_{0:t-1}) = \mathbf{z}_t$ the history is the concatenation of $Z_{0:t-k-1}$ with $\mathbf{z}_t$, so at almost every value $\mathbf{z}$ of $\mathbf{z}_t$ the two laws whose divergence enters $I(Z_t; Z_{0:t-1} \mid \mathbf{z}_t)$ are the images of those entering $I(Z_t; Z_{0:t-k-1} \mid \mathbf{z}_t)$ under $(z_t, z_{0:t-k-1}) \mapsto (z_t, (z_{0:t-k-1}, \mathbf{z}))$, which a coordinate projection inverts; part (d) twice equates the two divergences, and an empty discarded history is a point and contributes zero. On a finite Z the rows at contexts of positive mass are those of (12.62), the expectations are the sums of (12.64), and $\widetilde{\mathsf{T}}_t = \mathsf{T}^{(k)}_t$ gives (12.63). Finally, where the competitor row is $e^{-\psi}$ times the source row, the two rows are the laws $\bar{\mathsf{m}}_o$ and $\widehat{\mathsf{m}}^{\psi}$ of Corollary 12.6 with $\bar{\mathsf{m}}_o$ the source row and $\bar{\mathsf{m}}_o(e^{-\psi}) = 1$, and (12.17) bounds the row divergence. □

Proof of Theorem 12.48. Stationarity and nonnegativity force $\mathsf{T}_{yy'} = 0$ when $\bar{p}_y > 0 = \bar{p}_{y'}$, so the positive support is closed. For $k \geq 1$ the positive k-context chain shifts a context to its last k symbols after appending a symbol, with row $\mathbb{P}^{\mathrm{path}}_{k+1}(\mathbf{z}z)/\mathbb{P}^{\mathrm{path}}_k(\mathbf{z})$; its incoming mass at a context is the corresponding marginal of $\mathbb{P}^{\mathrm{path}}_{k+1}$, which equals $\mathbb{P}^{\mathrm{path}}_k$ of that context by stationarity, and this gives stationarity and exact $(k+1)$-windows; for $k = 0$ the projection is the stationary product of the one-symbol marginal. Conditioning and stationarity give $0 \leq \mathrm{h}_{j+1} \leq \mathrm{h}_j$. The chain rule gives $H(Z_{0:N-1}) = \sum_{j<N} \mathrm{h}_j$, and for $N > k$ the projected cross-entropy is $H(Z_{0:k-1}) + (N-k)\mathrm{h}_k$, which gives the sum and its rate. For two adjacent observed blocks of lengths n_1 and n_2, past and future are conditionally independent given the intervening hidden state, so their mutual information is at most c_Y; expanding it by stationary entropies gives $H(Z_{0:n_2-1}) - \sum_{j=0}^{n_2-1} \mathrm{h}_{n_1+j} \leq \mathrm{c}_Y$, and letting n_1 and then n_2 grow gives the series bound, whose monotone terms give $(k+1)\mathrm{h}^{\mathrm{ex}}_k \leq \mathrm{c}_Y$. Finally, for $N > k$, $N\mathrm{h}^{\mathrm{ex}}_k - D^{(k)}_N = k\mathrm{h}^{\mathrm{ex}}_k + \sum_{t=k}^{N-1} \mathrm{h}^{\mathrm{ex}}_t \leq \mathrm{c}_Y$, for $N \leq k$ the bound $N\mathrm{h}^{\mathrm{ex}}_k \leq \mathrm{c}_Y$ holds, and bounding each nonzero summand of $D^{(k)}_N$ by $\mathrm{h}^{\mathrm{ex}}_k$ gives the last inequality. □

Proof of Proposition 12.49. Relabelings fix the uniform law and its Fisher metric, so a natural rule would be permuted by them and would induce a homomorphism from the symmetric group S_7 to the symmetric group S_3 of the three blocks. Its restriction to the alternating group A_7, which is perfect, is trivial, because S_3 is solvable, so every block would be A_7-invariant. The tangent space is the sum-zero representation of A_7. Since A_7 is doubly transitive on the seven outcomes, the commutant of its permutation representation is spanned by the identity and the all-ones matrix, so every equivariant endomorphism of the sum-zero space is scalar, the representation is irreducible, and it has no invariant two-dimensional subspace. □

Appendix C

Proofs for Finite-Depth Formation

This appendix proves the propositions of chapter 14, the lemmas behind Theorem 14.17, Proposition 14.22, and its extension Proposition 15.4 of chapter 15, in the symbols of table 14.1; Table C.1 orders the lemmas. The depth $L \geq 2$ is fixed. Constants $\mathrm{c}_L, \mathrm{C}_L > 0$ may change between displays; they depend on L, on the fixed data, and, once chosen, on k_L and η_L, but not on ϵ, A, T, or the strengths unless the dependence is displayed, and all limits have $\epsilon, \mathrm{T} > 0$. Densities are relative to Lebesgue measure times counting measure, values of the complete assignment are written $j, k \in \mathcal{J}$, and the independent equilibrium root factor of every law is omitted from non-root integrals and appended unchanged. On M_ℓ, H_ℓ is the Gaussian Schur precision and $\mathrm{d}_\ell = \dim M_0 - \dim M_\ell$ the number of eliminated coordinates.

Table C.1: Lemma dependencies of the finite-depth formation theorem.

Lemma	Inputs	Output	Exponent or constant
Lemma C.1	Lemma 14.2	parent potentials, rows, covector; force, Hessian, metric bounds	$B_* = \delta\sqrt{2N}$, $\delta_* < 1/2$
Lemma C.2	Lemma 14.2	prepared direction, strict margin	$\mathrm{m}_L > 0$
Lemma C.3	Lemma C.1	well-posedness, moments, parameter bounds	$\mathrm{A}^2\mathrm{J}_{\ell+1} + \mathrm{T}\varsigma_\epsilon^2$
Lemma C.4	Lemma C.3	calibration-dominated parameter	$r_\epsilon = O(\epsilon^{2\mathrm{Q}_1+20})$
Lemma C.5	cited inequalities	log-Sobolev, Herbst, duality, transport, Pinsker	curvature, T explicit
Lemma C.6	Lemmas C.1, C.3 and C.5	activation and Fisher budgets; score moments	$\mathrm{A}^2\mathrm{J}_{\ell+1} + \mathrm{T}\varsigma_\epsilon^2$
Lemma C.7	Lemmas C.5 and C.6	early discarded loss; covector error	ϵ^{14}; $\frac{1}{2}\mathrm{c}_{\mathrm{early},L}k_L > \mathrm{Q}_1 + 20$
Lemma C.8	Lemmas C.1 and C.6	terminal loss by joint relaxation	ϵ^2; $\mathrm{c}_{\mathrm{joint},L}k_L > \mathrm{Q}_1 + 8$
Lemma C.9	Lemma C.1	averaged row; environment errors	$\mathrm{e}_\epsilon(y)$
Lemma C.10	Lemmas C.3 and C.9	mixing to the moving fast equilibrium	rate $\mathrm{c}_{\mathrm{fast},L}\mathrm{J}_1$
Lemma C.11	Lemmas C.4, C.9 and C.10	assignment histories	ϵ^6; $\mathrm{c}_{\mathrm{fast},L}k_L > \mathrm{Q}_1/2 + 12$
Lemma C.12	Lemmas C.2 and C.3	coherent components	$e^{-\mathrm{c}_L/\epsilon^6}$; $k_{\mathrm{geom},L}$, $\eta_L^{\max}$
Lemma C.13	Lemma C.4	source movement; kernel change	$57/100$; $1/200$
Lemma C.14	Lemmas C.4 and C.12	common-path concentration	$\mathrm{A}\epsilon^2$; $e^{-\mathrm{c}_L/\epsilon^2}$

C.1 Exact marginals, rates, and parameter derivatives

Proof of Proposition 14.4. Under ν_i^*, $k_i - m_i = \mathfrak{d}_i + \xi_i'$, whereas the physical evaluator gives $k_i - m_i = \xi_i'$ with the same law of m_i. The log-density ratio $[\mathfrak{d}_i(k_i - m_i) - \mathfrak{d}_i^2/2]/(1 - \sigma_{\mathrm{ev}}^2)$ therefore

has expectation $\mathfrak{d}_i^2/(2(1-\sigma_{\mathrm{ev}}^2))$ under every coupling of the two marginals, since $\mathbb{E}[k_i-m_i]=\bar{k}_i-\bar{m}_i$; this proves eq. (14.12), and summation gives $\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2$. The rows $\mathfrak{d}_i$ belong to B_1, so $\mathrm{J}_1\mathfrak{D}^\top\mathfrak{D}\preceq H$, and $\mathrm{T}\le \mathrm{J}_1(1-\sigma_{\mathrm{ev}}^2)/2$ gives $\mathrm{T}R_{\mathrm{ev}}\preceq H/2$; with $\vartheta\ge 1$ and the orthogonal split of Lemma 14.2, $\Omega_0(\vartheta)\succeq(\vartheta/2)H+P_{\mathcal{K}}\succ 0$. Every factor of eq. (14.11) is a positive normalized kernel of inputs none of which is a recognition law, so iterated integration gives total mass one and the records a positive density. □

Proof of Proposition 14.6. Since $\widehat{K}_\vartheta-\mathrm{T}R_{\mathrm{ev}}\succeq(\vartheta/2)H+(1+\kappa)P_{\mathcal{K}}\succ 0$ and $0\le\phi_S\le B_0$, both integrals are finite and positive. At fixed $\mathbf{x}$ both laws are proportional to $\pi_S(\vartheta)e^{-\phi_S(\mathbf{x})}$ in S, which gives eq. (14.14), a product of receiver rows because ϕ_S is a sum and π_S a product over receivers. Define

$$\begin{aligned}
\mathrm{n}_{\mathcal{K}}&=\big(\kappa/(2\pi\mathrm{T})\big)^{N/2}, & \mathrm{n}_{\mathrm{cal}}&=(2\pi\mathrm{T})^{-1/2},\\
\mathrm{B}_{\mathrm{T}}(\vartheta)&=\mathrm{T}\log Z_0(\vartheta)-\mathrm{T}\log\big(\mathrm{n}_{\mathcal{K}}\mathrm{n}_{\mathrm{cal}}q_0^N\big)+\tfrac12\kappa|z_\star|^2+\tfrac12(\vartheta-2)^2, & &\\
\mathrm{C}_{\mathrm{T}}(\vartheta)&=\mathrm{B}_{\mathrm{T}}(\vartheta)-\mathrm{T}\log Z_{\mathrm{opt}}(\vartheta), & p_\vartheta(\mathsf{o})&=e^{-\mathrm{B}_{\mathrm{T}}(\vartheta)/\mathrm{T}}Z_{\mathrm{phys}}(\vartheta).
\end{aligned}\tag{C.1}$$

As $z_\star\in\mathcal{K}$, the log density of the joint at the records is $-[\mathrm{E}_\vartheta^{\mathrm{phys}}+\mathrm{T}\phi_S+\tfrac12\kappa|z_\star|^2+\tfrac12(\vartheta-2)^2]/\mathrm{T}-\log Z_0+\log(\mathrm{n}_{\mathcal{K}}\mathrm{n}_{\mathrm{cal}}q_0^N)+\log\pi_S$ plus the physical evaluator density. Hence $\mathrm{T}\mathbb{E}\log(q\prod_i\nu_i^*/p)$, in which eq. (14.12) adds $\frac{\mathrm{T}}{2}\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}=\mathrm{E}_\vartheta-\mathrm{E}_\vartheta^{\mathrm{phys}}$, is the first line of eq. (14.15), and completing the relative entropy to $q_\vartheta^\star$ gives the second. Integrating the joint over $(\mathbf{x},S)$ at the records gives $p_\vartheta(\mathsf{o})$, so $\mathrm{C}_{\mathrm{T}}+\mathrm{T}\log p_\vartheta(\mathsf{o})=\mathrm{T}\log(Z_{\mathrm{phys}}/Z_{\mathrm{opt}})\ge 0$, as $\mathrm{E}_\vartheta^{\mathrm{phys}}\le\mathrm{E}_\vartheta$.

For the gap, $Z_{\mathrm{phys}}/Z_{\mathrm{opt}}=\mathbb{E}_{q_\vartheta^\star}e^{\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2}$. The Gaussian law $\gamma_\vartheta^{\mathrm{E}}\propto e^{-\mathrm{E}_\vartheta/\mathrm{T}}$ has its mean in $\mathcal{K}$, which R_{ev} annihilates, and since R_{ev} has rank N the eigenvalues λ_k of $\mathrm{T}\widehat{K}_\vartheta^{-1}R_{\mathrm{ev}}$ satisfy $\lambda_k\le\mathrm{T}/(\mathrm{J}_1(1-\sigma_{\mathrm{ev}}^2))\le 1/2$ and $\sum_k\lambda_k\le N\mathrm{T}/(\mathrm{J}_1(1-\sigma_{\mathrm{ev}}^2))\le 1$, so $\mathbb{E}_{\gamma_\vartheta^{\mathrm{E}}}e^{\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2}=\prod_k(1-\lambda_k)^{-1/2}\le e^{\sum_k\lambda_k}$. The density of $q_\vartheta^\star$ relative to $\gamma_\vartheta^{\mathrm{E}}\otimes\pi(\vartheta)$ is at most e^{B_0} and $e^{\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2}\ge 1$, so $\mathbb{E}_{q_\vartheta^\star}[e^{\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}/2}-1]\le e^{B_0}(e^{\sum_k\lambda_k}-1)\le 2e^{B_0}\sum_k\lambda_k$, and $\log(1+y)\le y$ gives the first bound of eq. (14.16). The second follows from $D_{\mathrm{KL}}(q_\vartheta^\star\|\mathbf{\Pi}_{\vartheta,\mathsf{o}})=\log(Z_{\mathrm{phys}}/Z_{\mathrm{opt}})-\frac12\mathbb{E}_{q_\vartheta^\star}\mathbf{x}^\top R_{\mathrm{ev}}\mathbf{x}$. □

Lemma C.1 (Weighted geometry and exact parent score). For the triangle-chain experiment, all marked marginal laws and inherited proposal rows are normalized and compose exactly. With orthonormal flag columns Q_ℓ spanning $\mathcal{U}_{\ell-1}\cap\mathcal{U}_\ell^\perp$, $\mathsf{B}=B[\mathsf{Q}_1\cdots\mathsf{Q}_L]$, and $\mathbf{x}_{\mathcal{U}}=\sum_\ell\mathsf{Q}_\ell w_\ell$,

$$\|\mathsf{B}^{-1}\|_F^{-2}\sum_\ell \mathrm{J}_\ell|w_\ell|^2\le\mathbf{x}_{\mathcal{U}}^\top H\mathbf{x}_{\mathcal{U}}\le\|\mathsf{B}\|_F^2\sum_\ell\mathrm{J}_\ell|w_\ell|^2,\tag{C.2}$$

and the same bounds hold for every Schur complement obtained by minimizing over an initial group of normal layers. On $1\le\vartheta\le 21/10$, the parent potentials are eq. (14.20) with the tilt and eliminated Gaussian scalar of eqs. (C.6) and (C.7) below, the inherited rows are those of eq. (14.23) and satisfy the detailed balance of Proposition 14.10, and the complete covector, including the source-prior score, is eq. (14.27). With $B_*=\delta\sqrt{2N}$ and $\delta_*=2\delta+2N\delta^2<1/2$, the parent forces,

Hessians, and scaled metric obey

$$|H_\ell^{-1/2}\mathrm{T}\nabla\psi_{\ell,j}| \le B_*\sqrt{\mathrm{T}}, \qquad (\vartheta-\delta_*)H_\ell \preceq \nabla^2 V_{\ell,j} \preceq (\vartheta+2\delta)H_\ell, \tag{C.3}$$

$$1 \le \mathsf{m}(\vartheta) \le 1+\mathrm{C}_L\mathrm{T}, \qquad |\mathsf{m}'(\vartheta)| \le \mathrm{C}_L\mathrm{T}, \tag{C.4}$$

for every fixed depth and all sufficiently small ϵ. [ESTABLISHED]

Proof. By Lemma 14.2, rows of level at most ℓ annihilate $\mathcal{U}_\ell$, so B and B^{-1} are block lower triangular with invertible diagonal blocks. With $\mathsf{D}_\mathrm{J} = \mathrm{diag}(\sqrt{\mathrm{J}_\ell} I)$, each nonzero block (ℓ, r) of $\mathsf{D}_\mathrm{J}\mathsf{B}^{\pm1}\mathsf{D}_\mathrm{J}^{-1}$ is that of $\mathsf{B}^{\pm1}$ times $\sqrt{\mathrm{J}_\ell/\mathrm{J}_r} \le 1$, which proves eq. (C.2), and $H|_{M_0} = H_\mathcal{U} \oplus \mathrm{J}_1 I_\mathcal{H}$. The Schur quadratic is the infimum over the eliminated coordinates, so minimizing both bounds proves the retained versions; in particular $H_\ell \succeq \mathrm{h}_L I$ for a fixed $\mathrm{h}_L > 0$, since $\mathrm{J}_L = 1$, the normal part of H_ℓ is at most $\mathrm{C}_L\mathrm{J}_{\ell+1} I$ with $\mathrm{J}_{L+1} = 0$, and the loop part keeps the weight J_1.

For a cumulative hidden/retained split (z, y) of M_0 with blocks $H^{(\ell)}_{zz}$, $H^{(\ell)}_{zy}$ of H, completing the square gives the lift in the first line below, with z embedded in the hidden subspace. The centered hidden Gaussian $\gamma^{\mathrm{hid}}_{\ell,\vartheta}$, of covariance $(\mathrm{T}/\vartheta)(H^{(\ell)}_{zz})^{-1}$, has a mean independent of ϑ, and integrating the source factor gives eq. (14.20) with the tilt, the scalar, and the normalized optimizer of the other lines, $\mathrm{C}^\mathrm{G}_0 = 0$:

$$\Xi_\ell y = \big(-(H^{(\ell)}_{zz})^{-1}H^{(\ell)}_{zy}y,\ y\big), \qquad \Xi_\ell^\top H \Xi_\ell = H_\ell, \qquad \Xi_\ell^\top H z = 0, \tag{C.5}$$

$$\psi_{\ell,j}(y,\vartheta) = -\log \int e^{-\phi_j(\Xi_\ell y + z)} \gamma^{\mathrm{hid}}_{\ell,\vartheta}(dz), \qquad 0 \le \psi_{\ell,j} \le B_0, \tag{C.6}$$

$$\mathrm{C}^\mathrm{G}_\ell(\vartheta) = -\frac{\mathrm{T}\mathrm{d}_\ell}{2}\log\frac{2\pi\mathrm{T}}{\vartheta} + \frac{\mathrm{T}}{2}\log\det H^{(\ell)}_{zz}, \tag{C.7}$$

$$q^{\star,\ell}_\vartheta(y,j) = \frac{e^{-\psi_{\ell,j}(y,\vartheta)}\gamma_{\ell,\vartheta}(y)\pi_j(\vartheta)}{Z_\ell(\vartheta)}, \quad \gamma_{\ell,\vartheta} = \mathcal{N}\big(0, (\mathrm{T}/\vartheta)H_\ell^{-1}\big), \quad e^{-B_0} \le Z_\ell \le 1. \tag{C.8}$$

Positivity and Gaussian domination justify iterated integration and differentiation on compact parameter intervals. Proposition 14.10 gives detailed balance, composition, and $\beta^\ell_{jk} \ge e^{-B_0}\pi_k(\vartheta)$ for the rows of eq. (14.23), and the physical, optimizing, and recognition laws compose by separate integration through the same channel.

Let $\mathfrak{B}_\mathrm{int}$ have rows $\sqrt{\mathrm{J}_1}\mathbf{r}_e$ on internal edges, $\mathbf{r}_e$ the covector of r_e, so $\mathfrak{B}_\mathrm{int}^\top\mathfrak{B}_\mathrm{int} \preceq H$. An edge is selected $\mathrm{n}_e \in \{0,1,2\}$ times, $\sum_e \mathrm{n}_e = N$, so the force of $\phi_j = \delta\sum_i \chi(\sqrt{\mathrm{J}_1/\mathrm{T}}\, r_{ij_i})$ is $\delta\sqrt{\mathrm{T}}\mathfrak{B}_\mathrm{int}^\top\mathfrak{c}$ with $\mathfrak{c}_e = \mathrm{n}_e\chi'(\sqrt{\mathrm{J}_1/\mathrm{T}}\, r_e)$ and, since $|\chi'|, |\chi''| \le 1$, $|\mathfrak{c}|^2 \le 2N$. This gives the first line below on M_0, and with the centered Gaussian variable of eq. (C.6) held fixed, the normalized tilted law $\nu_{\ell,y,j}$ gives the other two:

$$|H^{-1/2}\mathrm{T}\nabla\phi_j| \le B_*\sqrt{\mathrm{T}}, \qquad -2\delta H \preceq \mathrm{T}\nabla^2\phi_j \preceq 2\delta H, \qquad B_* = \delta\sqrt{2N}, \tag{C.9}$$

$$\mathrm{T}\nabla\psi_{\ell,j} = \mathbb{E}_{\nu_{\ell,y,j}}\Xi_\ell^\top(\mathrm{T}\nabla\phi_j), \tag{C.10}$$

$$\mathrm{T}\nabla^2\psi_{\ell,j} = \mathbb{E}_{\nu_{\ell,y,j}}\Xi_\ell^\top(\mathrm{T}\nabla^2\phi_j)\Xi_\ell - \mathrm{T}^{-1}\,\mathrm{Cov}_{\nu_{\ell,y,j}}(\Xi_\ell^\top\mathrm{T}\nabla\phi_j). \tag{C.11}$$

Through the isometry $H^{1/2}\Xi_\ell H_\ell^{-1/2}$, the first line bounds the force by $B_*\sqrt{\mathrm{T}}$ in the H_ℓ norm, the covariance by $B_*^2 H_\ell = 2N\delta^2 H_\ell$, and the first Hessian term between $\pm 2\delta H_\ell$, which proves eq. (C.3). The source factor is even, so centered Gaussian integration makes $\psi_{\ell,j}$ even in y, and every non-root sheet has its unique minimum at zero.

The hidden Gaussian scale score $\mathfrak{s}_\ell = (\mathrm{d}_\ell - |\mathbf{Z}|^2)/(2\vartheta)$, with $\mathbf{Z}$ standard Gaussian, has mean zero and second moment $\mathrm{d}_\ell/(2\vartheta^2)$, and the tilted density ratio lies in $[e^{-B_0}, e^{B_0}]$; as the Gaussian conditional center is independent of ϑ, uniformly in the retained input,

$$\partial_\vartheta \psi_{\ell,j} = -\mathbb{E}_{\nu_{\ell,y,j}}\mathfrak{s}_\ell, \qquad |\partial_\vartheta \psi_{\ell,j}| \le (e^{B_0}-1)\frac{\sqrt{\mathrm{d}_\ell}}{\sqrt{2}\vartheta}. \tag{C.12}$$

By eq. (14.20) the parent free energy is $\mathrm{T}\mathbb{E}_q \log q + \mathbb{E}_q[V_{\ell,S} - \mathrm{T}\log\pi_S] + \mathrm{B}_\mathrm{T}$, with $\mathrm{B}'_\mathrm{T} = \vartheta - 2 - \frac{\mathrm{T}}{2}\operatorname{Tr}(\Omega_0^{-1}H)$, and differentiation at a fixed recognition law, which does not differentiate assignment masses, gives eq. (14.27), whose source-prior term, present although it has no continuous gradient, has absolute unscaled score at most four for the distinguished binary prior. In the normalized joint the prior, calibration, and source-prior scores have zero mean and zero cross expectations, the other kernels have no explicit parameter dependence at fixed inputs, and the three scores have variances $\frac{1}{2}\operatorname{Tr}[(\Omega_0^{-1}H)^2]$, $1/\mathrm{T}$, and, for $4(\mathbf{1}_{\{S_{i_0}=+\}} - \pi_{i_0,+})$, $16\pi_{i_0,+}(1-\pi_{i_0,+})$, which gives eq. (14.26). As R_{ev} is fixed, annihilates $\mathcal{K}$, and is supported on the first normal layer, Ω_0 is positive for small T, and conjugation by the quadratic core bounds the trace terms and their derivatives uniformly; calibration contributes exactly one to m and the other two terms are nonnegative and $O(\mathrm{T})$, which proves eq. (C.4). □

Proof of Proposition 14.12. The metric and the covector are derived in Lemma C.1. The diffusion part of the generator is reversible with respect to $e^{-V_{\ell,S}/\mathrm{T}}$ on each sheet, which gives the first term of the dissipation after integration by parts. For the jump part, detailed balance (Proposition 14.10) turns $\mathrm{T}\int\sum_S \partial_t q_S \log\mathfrak{r}_S$ into $\mathrm{T}\int\sum_{S,S'} q^{\star,\ell}(y,S)\beta^\ell_{SS'}(\mathfrak{r}_{S'} - \mathfrak{r}_S)\log\mathfrak{r}_S$, and symmetrizing in S, S' gives the second term, whose integrand is nonnegative because the logarithm is increasing. The parameter term is $g_\ell\dot\vartheta_\ell = -\epsilon\mathrm{m}^{-1}g_\ell^2$. At $\ell = 0$ the row $\beta_{S'}$ does not depend on S, and with $\mathfrak{r}_S = (q_\mathbf{X}/q^\star_\mathbf{X})q(S\mid\mathbf{x})/\beta_S$ the jump integrand sums to $2[D_{\mathrm{KL}}(q(\cdot\mid\mathbf{x})\|\beta) + D_{\mathrm{KL}}(\beta\|q(\cdot\mid\mathbf{x}))]$. These identities hold for positive smooth densities with finite entropy and moments; the moment bounds of Lemma C.3 and the cutoff passage of Lemma C.6 justify them for the admitted solutions. □

C.2 Prepared laws and coupled continuation

Lemma C.2 (Prepared direction with a strict margin). Consider the list of linear functionals on $\mathcal{U}$ consisting of each internal edge covector and, for each bridge e and each $\ell < \ell(e)$, the functional $r_e \circ P_\ell|_\mathcal{U}$. Every functional in the list is nonzero, and there is a unit vector $\mathbf{v}_\star \in \mathcal{U}$ with

$$\mathrm{m}_L = \min_{e,\ \ell<\ell(e)} |r_e(P_\ell \mathbf{v}_\star)| > 0, \tag{C.13}$$

where internal edges enter only at $\ell = 0$. A fixed neighborhood of $\mathbf{v}_\star$ keeps a margin of at least $\mathrm{m}_L/2$. [ESTABLISHED]

Proof. For a bridge e and $\ell < \ell(e)$, r_e is independent of the rows of $B_1, \dots, B_\ell$ (Lemma 14.2), so it does not vanish on W_ℓ; it annihilates $\mathcal{H} \oplus \mathcal{K}$, hence does not vanish on $\mathcal{U}_\ell$, which P_ℓ fixes. An internal residual of triangle q is a nonzero combination of the contrast rows $O_{\triangle} r_q$, which lie in $\mathcal{U}$, plus a multiple of the loop covector, so its restriction to $\mathcal{U}$ is nonzero. Evaluated in an orthonormal basis of $\mathcal{U}$ on $(1, t, \dots, t^{\dim\mathcal{U}-1})$, each of the n_0 functionals is a nonzero polynomial of degree less than $\dim\mathcal{U}$, so some integer $t \in [0, \mathrm{n}_0(\dim\mathcal{U} - 1)]$ is a simultaneous nonroot; normalizing gives $\mathbf{v}_\star$, and continuity gives the neighborhood. □

Lemma C.3 (Prepared continuation). For the preparation of Definition 14.13 and every finite sequence of activations, the coupled diffusion, source proposals, and own-law parameter equation have a unique nonexplosive solution through $t_* = 2/\epsilon$, with positive smooth joint densities. On every active level ℓ and for all $t \le t_*$,

$$\mathbb{E}_{q_t^\ell} y^\top H_\ell y \le \mathrm{C}_L[\mathrm{A}^2 \mathrm{J}_{\ell+1} + \mathrm{T}\varsigma_\epsilon^2], \tag{C.14}$$

$$\mathbb{E}_{q_t^\ell} |y|^2 \le \mathrm{C}_L[\mathrm{A}^2 + \mathrm{T}\varsigma_\epsilon^2], \qquad \mathrm{J}_1 \mathbb{E}_{q_t^\ell} |h|^2 \le \mathrm{C}_L \mathrm{T}, \tag{C.15}$$

and the parameter satisfies $1 \le \vartheta_\ell(t) \le 21/10$ and $|\dot\vartheta_\ell(t)| \le \mathrm{C}_L\epsilon$; these bounds use no conditional-entropy or history estimate. [ESTABLISHED]

Proof. For a fixed positive instance and a prescribed continuous parameter path near $[1, 21/10]$, each sheet drift has a bounded Hessian and linear growth by eq. (C.3), and the proposal rows are smooth with bounded derivatives and a positive lower bound, their Gaussian-integral denominator being at least e^{-B_0}; interlacing the sheet diffusions on an independent rate-one Poisson grid gives a unique nonexplosive process with all polynomial moments. For the parameter feedback let $\mathfrak{u}_j^\vartheta(t', \mathbf{x}) = \mathbb{E}_\vartheta[\mathsf{f}(\mathbf{X}_t, S_t) \mid \mathbf{X}_{t'} = \mathbf{x}, S_{t'} = j]$, for smooth f with polynomially growing derivatives. Given the proposal times and labels, it composes Lipschitz sheet flows with products of positive row weights; flow derivatives are bounded on compact intervals, a differentiated weight is at most a constant times the weight, and the proposal count has all exponential moments, so $\mathfrak{u}$ and $\nabla\mathfrak{u}$ have polynomial bounds. Itô's formula for $\mathfrak{u}^{\widetilde\vartheta}$ along the ϑ process, stopped in a ball and then using the moments, gives

$$|\mathbb{E}_\vartheta \mathsf{f}(\mathbf{X}_t, S_t) - \mathbb{E}_{\widetilde\vartheta} \mathsf{f}(\mathbf{X}_t, S_t)| \le \mathrm{C} \int_0^t |\vartheta(t') - \widetilde\vartheta(t')| dt', \tag{C.16}$$

since the second-order coefficients agree and the drift and row differences are at most $\mathrm{C}|\vartheta - \widetilde\vartheta|(1 + |\mathbf{x}|)$ and $\mathrm{C}|\vartheta - \widetilde\vartheta|$. The observable terms of eq. (14.27) meet these hypotheses, and their explicit terminal parameter dependence adds a local Lipschitz term, so the integral map for $\dot\vartheta = -\epsilon \mathrm{m}(\vartheta)^{-1} g_\ell(q, \vartheta)$ is a contraction on a short interval, which gives local coupled existence and uniqueness. The step length is uniform: for parameter paths in $[3/4, 5/2]$, $\vartheta - \delta_* \ge 1/4$ and $H_\ell \succeq \mathrm{h}_L I$ make every non-root sheet drift uniformly confining toward zero, the root is a fixed Ornstein–Uhlenbeck factor, proposals fix and projections contract the configuration, and so the generator maps $|\mathbf{x}|^{2k}$ below $\mathrm{C}_k - \mathrm{c}_k|\mathbf{x}|^{2k}$; all moments on $[0, t_*]$, hence the constants of eq. (C.16)

and of the covector, are uniform over such paths, and finitely many steps reach t_* unless ϑ leaves $[1, 21/10]$, which the stopped argument below excludes.

Stop at the first possible exit of ϑ from $[1, 21/10]$. The Gaussian comparison $d\mathbf{X}^{\mathrm{G}} = -\vartheta(t)H_\ell \mathbf{X}^{\mathrm{G}}dt + \sqrt{2\mathrm{T}}dW$, with the same parameter and Brownian increments on the active level, starts from the reference of Definition 14.13 coupled by eq. (14.31); at an activation its projected reference continues while the parent starts at the full retained sample. For $\mathbf{e} = \mathbf{X} - \mathbf{X}^{\mathrm{G}}$ the noise cancels and proposals leave $\mathbf{e}$ continuous, so eq. (C.3) gives, for every adapted label history,

$$\frac{d}{dt}|\mathbf{e}|^2 \le -\mathbf{e}^\top H_\ell \mathbf{e} + \mathrm{C}_L \mathrm{T}, \qquad \sup_{t \le t_*} |\mathbf{e}_t| \le |\mathbf{e}_0| + \mathrm{C}_L \sqrt{\mathrm{T} t_*}. \tag{C.17}$$

For the energy, variation of constants with the commuting propagator $\exp[-H_\ell \int_{t'}^{t} \vartheta]$ uses the first bound below, obtained by splitting the integral at the inverse extreme eigenvalues, where the integrand is at most $\lambda_{\max}(H_\ell)$, $1/(e\sigma)$, and $\lambda_{\min}(H_\ell)e^{-\lambda_{\min}(H_\ell)\sigma}$; Euclidean projection and the Schur variational identity contract the two bounds at each activation, and finitely many stages give the second:

$$\int_0^\infty \|H_\ell e^{-H_\ell \sigma}\| d\sigma \le \mathrm{C}_L \left[1 + \log \frac{\lambda_{\max}(H_\ell)}{\lambda_{\min}(H_\ell)}\right], \qquad \sup_t |H_\ell^{1/2} \mathbf{e}_t| \le \mathrm{C}_L \sqrt{\mathrm{T}} \varsigma_\epsilon. \tag{C.18}$$

The Gaussian covariance barrier $\Sigma^{\mathrm{G}} \preceq \mathrm{C}_L \mathrm{T} H_\ell^{-1}$ needs only $\vartheta \ge 1$, by substitution into $\dot\Sigma^{\mathrm{G}} = -\vartheta(H_\ell \Sigma^{\mathrm{G}} + \Sigma^{\mathrm{G}} H_\ell) + 2\mathrm{T} I$, and survives retention because the retained block of H_ℓ^{-1} is the inverse Schur complement; the deterministic mean contracts, has norm $O_L(\mathrm{A})$, and has zero loop part. This gives eqs. (C.14) and (C.15), the loop bound following from the core rate $\vartheta \mathrm{J}_1$ and a force of at most $\mathrm{C}_L\sqrt{\mathrm{T}\mathrm{J}_1}$, whose forced error is at most $\mathrm{C}_L\sqrt{\mathrm{T}/\mathrm{J}_1}$.

Substituting these moments in the covector and metric bounds gives $\dot\vartheta = \epsilon(2 - \vartheta) + O_L(\epsilon r_\epsilon)$, with r_ϵ of eq. (14.30), before any bound on $\dot\vartheta$ is used. The field points inward at both endpoints of $[1, 21/10]$ for small ϵ, which excludes the stopped exit; the moments exclude explosion, so the solution reaches t_* with $|\dot\vartheta| \le \mathrm{C}_L \epsilon$. Positive-temperature elliptic smoothing and positive finite rates keep densities positive and smooth, and the construction restarts after an activation from the full retained law. □

Lemma C.4 (Comparison of independently learned parameters). Every process or finite replacement lineage of Lemma C.3 satisfies eq. (14.39) through t_*, where every independently evolving descendant uses its own law and copies only the starting parameter. [ESTABLISHED]

Proof. The first two bounds are in Lemma C.3. Subtracting $\dot\vartheta_\circ = \epsilon(2 - \vartheta_\circ)$, $\vartheta_\circ(0) = 1$, from the equation of the process and integrating with the factor $e^{\epsilon t}$ bounds the forcing contribution by $\mathrm{C}_L r_\epsilon$ through the horizon, where $r_\epsilon = \epsilon^{2\mathrm{Q}_1+20}(1 + \epsilon^{\mathrm{Q}_1+6}\varsigma_\epsilon^2) \to 0$ by eq. (14.30), and copying the parameter at finitely many activations adds a finite multiple of that error. The comparison trajectory solves the calibration part of the own-law equation, since the calibration record contributes $\vartheta - 2$ to the covector and $1/\mathrm{T}$ to M, and every law-dependent term is of order $\epsilon^{2\mathrm{Q}_1+20}$. As $2 - e^{-2} > 7/4$, the coefficient exceeds $7/4$ on the last ten units of the horizon for small ϵ. □

C.3 Entropy estimates

Lemma C.5 (Continuous entropy and score tools). Let $\mu(d\mathbf{x}) \propto e^{-\mathcal{V}(\mathbf{x})/\mathrm{T}}d\mathbf{x}$ on a finite-dimensional Euclidean space, where $\mathcal{V}$ is smooth with a globally bounded Hessian and $\nabla^2\mathcal{V} \succeq \lambda I$, $\lambda > 0$. For a positive density f relative to μ, with $\mathrm{Ent}_\mu(\mathsf{f}) = \int \mathsf{f}\log(\mathsf{f}/\int \mathsf{f}d\mu)d\mu$, for $\mathfrak{F}$ with $\mathrm{Lip}(\mathfrak{F}) \le \mathrm{K}$, for every probability law ν, and whenever the right sides are finite,

$$\mathrm{Ent}_\mu(\mathsf{f}) \le \frac{\mathrm{T}}{2\lambda}\int \frac{|\nabla \mathsf{f}|^2}{\mathsf{f}}d\mu, \tag{C.19}$$

$$\log \mathbb{E}_\mu e^{\upsilon(\mathfrak{F}-\mathbb{E}_\mu\mathfrak{F})} \le \frac{\mathrm{TK}^2\upsilon^2}{2\lambda}, \qquad \upsilon \in \mathbb{R}, \tag{C.20}$$

$$\upsilon\mathbb{E}_\nu\mathfrak{F} \le D_{\mathrm{KL}}(\nu\|\mu) + \log \mathbb{E}_\mu e^{\upsilon\mathfrak{F}}, \tag{C.21}$$

so $|\mathbb{E}_\nu\mathfrak{F} - \mathbb{E}_\mu\mathfrak{F}| \le \mathrm{K}\sqrt{2\mathrm{T}D_{\mathrm{KL}}(\nu\|\mu)/\lambda}$. A centered observable ξ that obeys the first line below obeys the second:

$$\log \mathbb{E}_\mu e^{\upsilon\xi} \le \frac{\mathrm{v}\upsilon^2}{2(1-\mathrm{b}|\upsilon|)} \quad \text{for } |\upsilon| < \mathrm{b}^{-1}, \tag{C.22}$$

$$|\mathbb{E}_\nu\xi| \le \sqrt{2\mathrm{v}D_{\mathrm{KL}}(\nu\|\mu)} + \mathrm{b}D_{\mathrm{KL}}(\nu\|\mu). \tag{C.23}$$

Moreover $\|\nu-\mu\|_{\mathrm{TV}} \le \sqrt{D_{\mathrm{KL}}(\nu\|\mu)/2}$. These statements apply to every conditional sheet potential of the construction. [ESTABLISHED]

Proof. For the semigroup P_t generated by $\mathrm{T}\Delta - \nabla\mathcal{V}\cdot\nabla$, the derivative of the synchronous stochastic flow has operator norm at most $e^{-\lambda t}$, so weighted Cauchy–Schwarz gives $|\nabla P_t\mathsf{f}|^2/P_t\mathsf{f} \le e^{-2\lambda t}P_t(|\nabla\mathsf{f}|^2/\mathsf{f})$ and invariance of μ gives $\mathcal{I}_\mu(P_t\mathsf{f}) \le e^{-2\lambda t}\mathcal{I}_\mu(\mathsf{f})$ for the Fisher information $\mathcal{I}_\mu$. Integration by parts gives $d\,\mathrm{Ent}_\mu(P_t\mathsf{f})/dt = -\mathrm{T}\mathcal{I}_\mu(P_t\mathsf{f})$; for bounded positive Lipschitz f, contraction against a stationary initial sample gives $P_t\mathsf{f} \to \int \mathsf{f}d\mu$ and, by bounded convergence, entropy tending to zero, so integrating the Fisher bound proves eq. (C.19). Cutoff, positive truncation, and short semigroup smoothing extend it by lower semicontinuity, and Gaussian tails from strong convexity justify the integrations. This is the curvature argument of Bakry and Émery [7, pp. 187, 194–200] with the diffusion coefficient made explicit; its Gaussian case is Boucheron et al. [17, Thm. 5.4]. Applied to $e^{\upsilon\mathfrak{F}}$, eq. (C.19) gives $\mathrm{Ent}_\mu(e^{\upsilon\mathfrak{F}}) \le (\mathrm{TK}^2\upsilon^2/(2\lambda))\mathbb{E}_\mu e^{\upsilon\mathfrak{F}}$, the hypothesis of Herbst's argument [17, Prop. 6.1] with variance factor TK^2/λ; applied to $\pm\mathfrak{F}$ it gives eq. (C.20). The inequality eq. (C.21) is the variational formula for relative entropy [17, Cor. 4.15], and optimizing it over υ with eq. (C.20) gives the Lipschitz consequence. Under eq. (C.22), the transportation lemma [17, Lemma 4.18] applied to $\pm\xi$, with the inverse convex conjugate $\sqrt{2\mathrm{v}t} + \mathrm{b}t$ of the sub-gamma function [17, Sec. 2.4], gives eq. (C.23); the case $D_{\mathrm{KL}} = \infty$ is trivial. The last inequality is Pinsker's inequality [17, Thm. 4.19], with the constant $1/2$ due to Csiszár [28]. □

Lemma C.6 (Entropy at activations). For the law of the active level ℓ, let $\mathfrak{E}_\ell(t) = \mathrm{T} D_{\mathrm{KL}}(q_t^\ell \| q^{\star,\ell}_{\vartheta_\ell(t)})$, with the root factor canceled. At an activation time and on each ensuing active interval,

$$\mathfrak{E}_\ell(t_\ell^{\mathrm{act}}) \le \mathrm{C}_L[\mathrm{A}^2 \mathrm{J}_{\ell+1} + \mathrm{T}\varsigma_\epsilon^2], \tag{C.24}$$

$$\mathrm{T}^2 \int_{t_\ell^{\mathrm{act}}}^{t} \mathcal{I}_\ell(\sigma) d\sigma \le \mathrm{C}_L[\mathfrak{E}_\ell(t_\ell^{\mathrm{act}}) + \mathrm{T}], \tag{C.25}$$

where $\mathcal{I}_\ell$ is the full continuous relative Fisher information of the joint law. The normalized parameter scores of the hidden Gaussian conditionals and of the joint satisfy eq. (C.22) with constants $\mathrm{v}_L, \mathrm{b}_L$ independent of the retained input, of T, of the strengths, and of the source masses. [ESTABLISHED]

Proof. The normalized hidden Gaussian conditionals have centers independent of ϑ, and their logarithmic parameter scores, bounded marginal tilt included, are centered versions of $-|\mathbf{Z}|^2/(2\vartheta) - \partial_\vartheta \psi_{\ell,j}$; the joint score adds $\partial_\vartheta \log \pi_j(\vartheta)$, of absolute value at most four. A bounded tilt of a standard Gaussian satisfies

$$\mathbb{E}|\mathbf{Z}|^{2m} \le e^{B_0} 2^{\dim M_0/2} 4^m m!, \qquad m \ge 0, \tag{C.26}$$

because $r^m \le m! 4^m e^{r/4}$ and $\mathbb{E} e^{|\mathbf{Z}|^2/4} \le 2^{\dim M_0/2}$, and the other score terms are bounded by eq. (C.12). Centering at most doubles a geometric moment constant, and summing the exponential series without its zero linear term, with $\log(1+x) \le x$, gives eq. (C.22), which conditional expectations keep by Jensen's inequality.

For the differential and counting entropy $\mathcal{S}(q) = -\sum_j \int q_j \log q_j$ across changes of dimension, write $q_j = \lambda_j q_{|j}$, omitting terms of zero mass. The entropy–Fisher bound $\mathcal{I}(\mathrm{f}) \ge 2\pi e \mathrm{d} e^{-2\mathcal{S}(\mathrm{f})/\mathrm{d}}$ of Stam [95], equivalent to the Gaussian logarithmic Sobolev inequality [19], follows from the Gaussian case of eq. (C.19) by expanding both sides relative to $\mathcal{N}(0, \sigma^2 I)$, integrating $\int \mathbf{x} \cdot \nabla \mathrm{f} = -\mathrm{d}$, canceling the second moments, and minimizing at $\sigma^2 = \mathrm{d}/\mathcal{I}(\mathrm{f})$; Jensen's inequality and $\mathcal{S}(q) = \sum_j \lambda_j \mathcal{S}(q_{|j}) + \mathcal{S}(\lambda)$, with $\mathcal{S}(\lambda) \ge 0$, extend it to $\sum_j \lambda_j \mathcal{I}(q_{|j}) \ge 2\pi e \mathrm{d} e^{-2\mathcal{S}(q)/\mathrm{d}}$. At fixed $\mathbf{x}$, a proposal moves the conditional label row $\mathfrak{a}$ in the direction $\mathfrak{a}\beta^\ell - \mathfrak{a}$ and leaves the continuous marginal unchanged, and concavity of categorical entropy bounds the directional derivative below by $\mathcal{S}(\mathfrak{a}\beta^\ell) - \mathcal{S}(\mathfrak{a}) \ge -\log|\mathcal{J}|$. On a d-dimensional sheet, diffusion therefore gives the first line below, and the two bounds together give the second:

$$\mathcal{S}'(q)_{\mathrm{diff}} = \mathrm{T} \sum_j \lambda_j \mathcal{I}(q_{|j}) - \mathbb{E}_q \Delta V_{\ell,S} \ge \mathrm{T} \sum_j \lambda_j \mathcal{I}(q_{|j}) - \mathrm{d}\Lambda_\ell, \qquad \Lambda_\ell \le \mathrm{C}_L \mathrm{J}_1, \tag{C.27}$$

$$\mathcal{S}'(q) \ge 2\pi e \mathrm{d} \mathrm{T} e^{-2\mathcal{S}(q)/\mathrm{d}} - \mathrm{d}\Lambda_\ell - \log|\mathcal{J}|. \tag{C.28}$$

Scalar comparison bounds $\mathcal{S}(q)$ below by the lesser of its initial value and $\frac{\mathrm{d}}{2}\log[2\pi e \mathrm{d}\mathrm{T}/(\mathrm{d}\Lambda_\ell + \log|\mathcal{J}|)]$. At a projection dropping k coordinates z, the chain rule and the maximum-entropy Gaussian bound, applied conditionally on (Y, S) and then with Jensen's inequality, give

$$\mathcal{S}(Y,S) \ge \mathcal{S}(\mathbf{X},S) - \frac{k}{2}\log\frac{2\pi e \mathbb{E}|z-\bar{z}|^2}{k}, \tag{C.29}$$

where $\bar{z}$ can be the projected comparison mean, whose centered covariance the moment proof of Lemma C.3 bounds by $\mathrm{C}_L \mathrm{T} \varsigma_\epsilon^2 H_\ell^{-1}$. Initial entropies and all logarithms in these estimates are $O_L(\varsigma_\epsilon^2)$, because the covariance eigenvalues, A, and T are fixed powers of ϵ, the admitted initial density ratios are bounded, and there are finitely many projections. Expanding $\mathrm{T} D_{\mathrm{KL}}(q \| \gamma_{\ell,\vartheta} \otimes \pi(\vartheta))$ and using eq. (C.14) proves eq. (C.24); the bounded optimizer tilt adds at most $\mathrm{T} B_0$, and the source prior has a positive minimum on $[1, 21/10]$, so its logarithmic contribution does not depend on the masses λ_j.

For the time-integrated estimate put $\mathcal{V} = y^\top H_\ell y/2$ and $\mathrm{R}_L = e^{B_0} 2^{\dim M_0/2}$. The reference satisfies the first bound below. Entropy duality gives $\vartheta \mathbb{E}_q \mathcal{V} \le 2\mathfrak{E}_\ell + 2\mathrm{T} \log \mathrm{R}_L$, Jensen's inequality gives $\vartheta \mathbb{E}_{q_\vartheta^{\star,\ell}} \mathcal{V} \le 2\mathrm{T} \log \mathrm{R}_L$, and the other centered terms of the normalized full score are at most $2\mathrm{C}_{\mathrm{sc},L}$, which gives the second. Diffusion dissipates $\mathrm{T}^2 \mathcal{I}_\ell$ and detailed balance makes the source contribution nonpositive, which gives the third for a fixed $\mathrm{C}_{0,L}$:

$$\mathbb{E}_{q_\vartheta^{\star,\ell}} \exp \frac{\vartheta \mathcal{V}}{2\mathrm{T}} \le \mathrm{R}_L, \tag{C.30}$$

$$|\mathrm{T} \mathbb{E}_q \partial_\vartheta \log q_\vartheta^{\star,\ell}| \le \frac{2\mathfrak{E}_\ell + 4\mathrm{T} \log \mathrm{R}_L}{\vartheta} + 2\mathrm{T}\mathrm{C}_{\mathrm{sc},L}, \tag{C.31}$$

$$\mathfrak{E}_\ell' \le -\mathrm{T}^2 \mathcal{I}_\ell + \frac{2|\dot{\vartheta}|}{\vartheta} \big(\mathfrak{E}_\ell + \mathrm{C}_{0,L} \mathrm{T} \big). \tag{C.32}$$

The integrating factor $\exp[-2 \int |\dot{\vartheta}|/\vartheta]$ has exponent bounded over t_* by the speed bound, and integration proves eq. (C.25). At an activation the law and the reference are pushed through the same channel at the same parameter, so the chain rule makes $\mathfrak{E}$ decrease along the finite lineage. All entropy derivatives are first taken for positive truncated densities on bounded domains, and the moment and entropy bounds remove the cutoffs. □

Lemma C.7 (Conditional entropy and complete score error). There is a constant $\mathrm{c}_{\mathrm{early},L} > 0$ such that, for every retention $\ell - 1 \to \ell$, the conditional relative entropy Δ_ℓ with respect to the optimizing hidden conditional satisfies

$$\sup_{t \in I_\ell \text{ and later active times}} \Delta_\ell(t) \le \begin{cases} \mathrm{C}_L \epsilon^{14}, & \ell < L, \\ \mathrm{C}_L \epsilon^2, & \ell = L, \end{cases} \tag{C.33}$$

where the early bound also applies, until the next activation, to the corresponding unreplaced comparison process, and the terminal bound is Lemma C.8. At every such time the complete fine/parent covector difference is at most $\mathrm{C}_L \mathrm{T}(\sqrt{\Delta_\ell} + \Delta_\ell)$. [ESTABLISHED]

Proof of the early bound and the covector estimate. For an early retention use the auxiliary continuous split of the first line below. The true parent retains the loops, so the chain rule bounds its hidden conditional defect Δ_ℓ by the auxiliary defect $\widehat{\Delta}_\ell$, which also integrates h; the independent root cancels, and conditional strong convexity in Z is at least $\lambda_\ell = \mathrm{c}_{\mathrm{early},L} \mathrm{J}_\ell$ after reducing the

constant if necessary. The mixed Hessian obeys the second line:

$$Z = (\mathcal{U}_{\ell-1} \cap \mathcal{U}_\ell^\perp) \oplus \mathcal{H}, \qquad Y = \mathcal{U}_\ell, \qquad S \text{ retained}, \tag{C.34}$$

$$\|\nabla_Z \nabla_Y V_{\ell-1,j}\| \le \Gamma_\ell = \mathrm{C}_L \mathrm{J}_{\ell+1}, \qquad \Gamma_L = 0. \tag{C.35}$$

The nonlinear residuals lie in the first normal layer and the loop space, and at $\ell - 1 = 0$ the nonlinear potential has no future normal dependence. For $\ell - 1 \ge 1$, in coordinates $r_E = B_E w_E$ on the eliminated normal rows, the Gaussian conditional precision $\Omega_E = \mathrm{J}_E + \mathcal{C}^\top \mathrm{J}_{\mathrm{fut}} \mathcal{C}$, with the diagonal strengths J_E and $\mathrm{J}_{\mathrm{fut}}$ of the eliminated and future rows, has $\|\Omega_E - \mathrm{J}_E\| \le \mathrm{C}_L \mathrm{J}_\ell$ and $\|\Omega_E^{-1}\| \le \mathrm{C}_L/\mathrm{J}_{\ell-1}$, so the identity in the first line below bounds the first block row of Ω_E^{-1} by $\mathrm{C}_L/\mathrm{J}_1$. A future force column of level r has size $\mathrm{C}_L \mathrm{J}_r$, so the first conditional mean responds by $\mathrm{C}_L \mathrm{J}_r/\mathrm{J}_1$, with conditional covariance comparable to T/J_1. In the derivatives of the normalized Gaussian integral defining $\psi_{\ell-1,j}$, a first-layer or loop derivative of ϕ_j has size $\mathrm{C}_L\sqrt{\mathrm{J}_1/\mathrm{T}}$ and a second derivative size $\mathrm{C}_L \mathrm{J}_1/\mathrm{T}$, each future derivative brings that mean response, and the covariance term is bounded by products of first-derivative bounds, which gives the second line:

$$\Omega_E^{-1} = \mathrm{J}_E^{-1} - \mathrm{J}_E^{-1}(\Omega_E - \mathrm{J}_E)\Omega_E^{-1}, \tag{C.36}$$

$$\mathrm{T}|\partial_h \partial_{w_r} \psi_{\ell-1,j}| \le \mathrm{C}_L \mathrm{J}_r, \qquad \mathrm{T}|\partial_{w_i} \partial_{w_r} \psi_{\ell-1,j}| \le \mathrm{C}_L \frac{\mathrm{J}_i \mathrm{J}_r}{\mathrm{J}_1}. \tag{C.37}$$

The Gaussian Schur cross block is at most $\mathrm{C}_L \mathrm{J}_{\ell+1}$ by the lower triangular residual form and its variational bound, and summing finitely many blocks, generated conditional covariance included, proves eq. (C.35).

Write $q = q_Y(y,j) q_{Z|Y}(z \mid y,j)$ and $q^\star = q_Y^\star(y,j) q_{Z|Y}^\star(z \mid y,j)$, and let $\mathfrak{w}_Y = \nabla_Y \log(q/q^\star)$, $\mathfrak{v}_Y = \nabla_Y \log(q_Y/q_Y^\star)$, and $\mathfrak{e}_Y = \mathbb{E}_{q_{Z|Y}} \nabla_Y \log q_{Z|Y}^\star$. Normalization gives $\mathbb{E}_{q_{Z|Y}^\star} \nabla_Y \log q_{Z|Y}^\star = 0$ and $\mathbb{E}_{q_{Z|Y}} \mathfrak{w}_Y = \mathfrak{v}_Y - \mathfrak{e}_Y$, the marginal flux is $\mathrm{T} q_Y (\mathfrak{v}_Y - \mathfrak{e}_Y)$, and subtracting the marginal entropy derivative from the full one yields

$$\begin{aligned} \widehat{\Delta}'_\ell|_{\mathrm{cont}} = & -\mathrm{T}\mathcal{I}_Z - \mathrm{T}\mathbb{E}_{q_Y} \mathrm{Var}_{q_{Z|Y}}(\mathfrak{w}_Y) + \mathrm{T}\mathbb{E}_{q_Y}\left[\mathfrak{e}_Y \cdot \mathbb{E}_{q_{Z|Y}} \mathfrak{w}_Y\right] \\ & - \dot{\vartheta} \mathbb{E}_q \partial_\vartheta \log q_{Z|Y}^\star. \end{aligned} \tag{C.38}$$

The conditional logarithmic Sobolev inequality gives $\mathrm{T}\mathcal{I}_Z \ge 2\lambda_\ell \widehat{\Delta}_\ell$. Applying eq. (C.20) to each unit-vector component of $\nabla_Y \log q_{Z|Y}^\star$, whose Lipschitz constant in Z is at most Γ_ℓ/T, gives $|\mathfrak{e}_Y| \le \Gamma_\ell \sqrt{2 D_{\mathrm{KL}}(q_{Z|Y} \| q_{Z|Y}^\star)/(\mathrm{T}\lambda_\ell)}$, so Cauchy–Schwarz and Young's inequality bound the mixed term by $\lambda_\ell \widehat{\Delta}_\ell/2 + \mathrm{T}\Gamma_\ell^2 \mathcal{I}_Y/\lambda_\ell^2$, with $\mathcal{I}_Y$ the Y-component of the full relative Fisher information. The normalized conditional parameter score, sub-gamma by Lemma C.6, contributes at most $\lambda_\ell \widehat{\Delta}_\ell/2 + \mathrm{v}_L |\dot{\vartheta}|^2/\lambda_\ell + \mathrm{b}_L |\dot{\vartheta}| \widehat{\Delta}_\ell$.

For the source updates, at fixed $Y = y$ each $\log(\gamma^{\mathrm{hid}}/q_{Z|Y,j}^\star)$, with γ^{hid} the common label-independent hidden Gaussian, lies in $[-B_0, B_0]$, and the conditional entropy relative to γ^{hid} splits as $D_{\mathrm{KL}}(q_Z \| \gamma^{\mathrm{hid}}) + \mathbb{I}(Z;S)$. A proposal leaves q_Z unchanged, and the mutual information is convex in the conditional label channel at fixed q_Z by joint convexity of relative entropy, so its directional

derivative toward the new channel is at most $\log|\mathcal{J}|$; the difference from the optimizing target, a linear expectation of $\log(\gamma^{\mathrm{hid}}/q^{\star}_{Z|Y,j})$, changes at rate at most $2B_0$. This divides by no source mass and extends to vanishing masses by convex approximation. With $\mathrm{b}_L|\dot{\vartheta}| \le \lambda_\ell/2$ for small ϵ and the negative variance term dropped,

$$\widehat{\Delta}'_\ell|_{\mathrm{jump}} \le \mathrm{C}_{\mathcal{J}} := \log|\mathcal{J}| + 2B_0, \tag{C.39}$$

$$\widehat{\Delta}'_\ell \le -\frac{\lambda_\ell}{2}\widehat{\Delta}_\ell + \frac{\mathrm{T}\Gamma_\ell^2}{\lambda_\ell^2}\mathcal{I}_Y + \frac{\mathrm{v}_L|\dot{\vartheta}|^2}{\lambda_\ell} + \mathrm{C}_{\mathcal{J}}. \tag{C.40}$$

The initial conditional entropy is at most $\mathfrak{E}_{\ell-1}(t^{\mathrm{act}}_{\ell-1})/\mathrm{T}$. Integrating with eqs. (C.24) and (C.25) and $\mathrm{A}^2/\mathrm{T} = \epsilon^{-6}$ gives

$$\widehat{\Delta}_\ell(t) \le \mathrm{C}_L\Big[e^{-\mathrm{c}_{\mathrm{early},L}\mathrm{J}_\ell(t-t^{\mathrm{act}}_{\ell-1})}(\epsilon^{-6}\mathrm{J}_\ell + \varsigma_\epsilon^2) + \epsilon^{-6}\frac{\mathrm{J}^2_{\ell+1}}{\mathrm{J}_\ell} + \varsigma_\epsilon^2\frac{\mathrm{J}^2_{\ell+1}}{\mathrm{J}^2_\ell} + \frac{\epsilon^2}{\mathrm{J}^2_\ell} + \frac{\mathrm{C}_{\mathcal{J}}}{\mathrm{J}_\ell}\Big], \tag{C.41}$$

with $\mathrm{c}_{\mathrm{early},L}$ absorbing the factor $1/2$. For $\ell < L$, $\mathrm{J}^2_{\ell+1}/\mathrm{J}_\ell = \epsilon^{20}$ and $\mathrm{J}_\ell \ge \epsilon^{-20}$; on I_ℓ the entrance time and $\mathrm{c}_{\mathrm{early},L}k_L/2 > \mathrm{Q}_1 + 20$ bound the first term by $\mathrm{C}_L\epsilon^{14}$, and the other terms meet the same bound; at $\ell = L$ the last term would be $\mathrm{C}_{\mathcal{J}}$.

For the true retention, eq. (14.21) with $\ell' = \ell - 1$, differentiated at fixed recognition law and channel, gives $g_{\ell-1} - g_\ell = -\mathrm{T}\mathbb{E}_q\partial_\vartheta \log q^{\star}_{Z|Y,\mathrm{true}}$, the common scalar and the source-prior derivative canceling. The normalized conditional score has the same sub-gamma bound, so eq. (C.23), applied conditionally, with Jensen's inequality gives the covector bound $|g_{\ell-1}-g_\ell| \le \mathrm{C}_L\mathrm{T}(\sqrt{\Delta_\ell}+\Delta_\ell)$. □

Lemma C.8 (Joint relaxation at the weakest strength). There is a constant $\mathrm{c}_{\mathrm{joint},L} > 0$, independent of ϵ, such that every lineage satisfies

$$D_{\mathrm{KL}}\big(q^\ell_t \| q^{\star,\ell}_{\vartheta_\ell(t)}\big) \le \mathrm{C}_L e^{-\mathrm{c}_{\mathrm{joint},L}t}[\epsilon^{-(\mathrm{Q}_1+6)} + \varsigma_\epsilon^2] + \mathrm{C}_L\epsilon^2. \tag{C.42}$$

Consequently the terminal part of eq. (C.33) holds once $t \ge t_L = k_L\log(1/\epsilon)$. [ESTABLISHED]

Proof. Fix a level and a parameter, write $q^\star = q^{\star,\ell}_\vartheta$ and $\mathfrak{r}_j = q_j/q^\star_j$. The bounded tilt gives $e^{-B_0} \le q^\star/(\gamma_{\ell,\vartheta}\otimes\pi) \le e^{B_0}$, and eq. (14.23) gives the conductance comparison in the first line below. For a probability measure μ, $\mathrm{Ent}_\mu(\mathfrak{r}) = \inf_{r>0}\int[\mathfrak{r}\log(\mathfrak{r}/r) - \mathfrak{r} + r]d\mu$ with a nonnegative integrand, so $\mathrm{Ent}_{q^\star}(\mathfrak{r}) \le e^{B_0}\mathrm{Ent}_{\gamma\otimes\pi}(\mathfrak{r})$, and the product chain rule and convexity give the second line, whose first term the Gaussian entropy inequality bounds by $\mathrm{T}/(2\vartheta\mathrm{h}_L)$ times its continuous Fisher information:

$$q^\star_j(y)\beta^\ell_{jk}(y) \ge e^{-2B_0}\gamma_{\ell,\vartheta}(y)\pi_j(\vartheta)\pi_k(\vartheta), \tag{C.43}$$

$$\mathrm{Ent}_{\gamma\otimes\pi}(\mathfrak{r}) \le \sum_j \pi_j\,\mathrm{Ent}_\gamma(\mathfrak{r}_j) + \mathbb{E}_\gamma\,\mathrm{Ent}_\pi(\mathfrak{r}(y)). \tag{C.44}$$

For any positive finite vector $\mathbf{a}$, Jensen's inequality for the logarithm gives

$$\frac{1}{2}\sum_{j,k}\pi_j\pi_k(\mathbf{a}_j-\mathbf{a}_k)(\log\mathbf{a}_j-\log\mathbf{a}_k)=\sum_j\pi_j\mathbf{a}_j\log\mathbf{a}_j-\bar{\mathbf{a}}\sum_j\pi_j\log\mathbf{a}_j\geq\mathrm{Ent}_\pi(\mathbf{a}),\tag{C.45}$$

with $\bar{\mathbf{a}}=\sum_j\pi_j\mathbf{a}_j$. Comparing the Fisher measures and the conductances through their pointwise bounds, with the jump dissipation of the first line below, gives the second:

$$\mathcal{D}_{\mathrm{jump}}=\frac{1}{2}\int\sum_{j,k}q_j^\star\beta_{jk}^\ell(\mathfrak{r}_j-\mathfrak{r}_k)(\log\mathfrak{r}_j-\log\mathfrak{r}_k)dy,\tag{C.46}$$

$$D_{\mathrm{KL}}(q\|q^\star)\leq e^{2B_0}\frac{\mathrm{T}}{2\vartheta\mathrm{h}_L}\mathcal{I}_\ell+e^{3B_0}\mathcal{D}_{\mathrm{jump}}.\tag{C.47}$$

With $\mathrm{c}_{0,L}=\min\{2\mathrm{h}_Le^{-2B_0},e^{-3B_0}\}>0$, $\mathrm{T}\mathcal{I}_\ell+\mathcal{D}_{\mathrm{jump}}\geq\mathrm{c}_{0,L}D_{\mathrm{KL}}$. The time-dependent normalized full score satisfies eq. (C.22), hence, for small ϵ,

$$D_{\mathrm{KL}}'\leq-\mathrm{c}_{0,L}D_{\mathrm{KL}}+\mathrm{C}_L\epsilon(\sqrt{D_{\mathrm{KL}}}+D_{\mathrm{KL}})\leq-\frac{\mathrm{c}_{0,L}}{2}D_{\mathrm{KL}}+\mathrm{C}_L\epsilon^2,\tag{C.48}$$

and $\mathrm{c}_{\mathrm{joint},L}=\mathrm{c}_{0,L}/2$. Activations decrease the relative entropy at the unchanged parameter, and each descendant then obeys the same inequality with its own parameter speed. The initial relative entropy is at most $\mathrm{C}_L[\epsilon^{-(\mathrm{Q}_1+6)}+\varsigma_\epsilon^2]$, and integration through finitely many decreases proves eq. (C.42); the condition $\mathrm{c}_{\mathrm{joint},L}k_L>\mathrm{Q}_1+8$ makes its decaying term $O_L(\epsilon^2)$ at t_L. Both the auxiliary and the true hidden conditional entropies are bounded by the full joint relative entropy, which proves the terminal assertion. □

C.4 Exact fast marginals and complete source histories

Lemma C.9 (Common fast marginal and environmental error). In the flag coordinates $w=(w_1,\dots,w_L)$ of eq. (C.2) and the loop coordinate h, put $\Omega_*=\mathrm{J}_1\mathrm{B}_{11}^\top\mathrm{B}_{11}$ and

$$\mu_{\vartheta,j}^*(dw_1,dh)=\frac{1}{Z_{\vartheta,j}^*}\exp\left\{-\frac{\vartheta}{2\mathrm{T}}(w_1^\top\Omega_*w_1+\mathrm{J}_1|h|^2)-\phi_j(w_1,h)\right\}dw_1dh,\tag{C.49}$$

with loop marginal $\mu_{\vartheta,j}^{*,h}$, first-normal conditional $\mu_{\vartheta,j}^{*,1}(dw_1\mid h)$, and, with environment scales ϱ_ϵ, ϱ_ϵ',

$$\beta_{jk}^{*,h}(h,\vartheta)=\int\beta_k(w_1,h,\vartheta)\mu_{\vartheta,j}^{*,1}(dw_1\mid h),\tag{C.50}$$

$$\bar{\beta}_{jk}(\vartheta)=\int\beta_k(w_1,h,\vartheta)\mu_{\vartheta,j}^*(dw_1,dh)=\int\beta_{jk}^{*,h}(h,\vartheta)\mu_{\vartheta,j}^{*,h}(dh),\tag{C.51}$$

$$\varrho_\epsilon=\frac{\mathrm{J}_2}{\mathrm{J}_1},\qquad\varrho_\epsilon'=\frac{\mathrm{J}_2}{\sqrt{\mathrm{J}_1\mathrm{T}}},\qquad\mathrm{e}_\epsilon(y)=\varrho_\epsilon+\varrho_\epsilon'|y|.\tag{C.52}$$

Then eq. (C.49) is the exact (w_1, h)-marginal of the optimizer conditional on $S = j$; averaging the inherited row β^{ℓ}_{jk} against the optimizer conditional on j gives the same $\bar{\beta}_{jk}(\vartheta)$ for every ℓ; and the exact parent rows and loop forces satisfy eqs. (C.56) to (C.58) below. [ESTABLISHED]

Proof. In the flag coordinates

$$w^{\top} H_{\mathcal{U}} w = \mathrm{J}_1 |\mathrm{B}_{11} w_1|^2 + \sum_{\ell=2}^{L} \mathrm{J}_{\ell} \left| \sum_{r \le \ell} \mathrm{B}_{\ell r} w_r \right|^2 , \tag{C.53}$$

and ϕ_j depends only on (w_1, h). For fixed w_1, h, change the later w_ℓ to the residuals $r_\ell = \sum_{r \le \ell} \mathrm{B}_{\ell r} w_r$; the domain stays the whole Euclidean product and the Jacobian is the constant $\prod_{\ell \ge 2} |\det \mathrm{B}_{\ell\ell}|$, so integrating those Gaussian residuals contributes a scalar independent of w_1, h, j, which proves the first claim, the prior factor $\pi_j(\vartheta)$ canceling on conditioning. The change of variables and the tower property give the second.

For an intermediate $\ell \ge 1$, keep $y = (w_{\ell+1}, \dots, w_L)$ fixed and integrate $w_2, \dots, w_\ell$. The unperturbed conditional first-normal law $\gamma^1_{\ell,\vartheta,y}$ and its divergence from $\gamma^{1,*}_{\vartheta} = \mathcal{N}(0, (\mathrm{T}/\vartheta)\Omega_*^{-1})$ are

$$\gamma^1_{\ell,\vartheta,y} = \mathcal{N}\left(\mu_\ell(y), \frac{\mathrm{T}}{\vartheta} (\Omega^{(1)}_\ell)^{-1} \right), \qquad \Omega_* \preceq \Omega^{(1)}_\ell \preceq \Omega_* + \mathrm{C}_L \mathrm{J}_2 I, \qquad |\mu_\ell(y)| \le \mathrm{C}_L \frac{\mathrm{J}_2}{\mathrm{J}_1} |y|, \tag{C.54}$$

$$\frac{1}{2} \left[\frac{\vartheta}{\mathrm{T}} \mu_\ell^{\top} \Omega_* \mu_\ell + \mathrm{Tr}\big(\Omega_* (\Omega^{(1)}_\ell)^{-1}\big) - \dim w_1 - \log\det\big(\Omega_* (\Omega^{(1)}_\ell)^{-1}\big) \right] \le \mathrm{C}_L [\varrho_\epsilon^2 + \varrho_\epsilon'^2 |y|^2]. \tag{C.55}$$

For the lower precision bound, minimize the nonnegative later residual squares at $y = 0$; for the upper, set the eliminated later coordinates to zero and use $\mathrm{J}_\ell \le \mathrm{J}_2$ for $\ell \ge 2$; at $\ell = L$, $\Omega^{(1)}_L = \Omega_*$ and $\mu_L = 0$. The mean bound is the first-block-row estimate eq. (C.36), each future coordinate of level ℓ contributing at most $\mathrm{C}_L \mathrm{J}_\ell / \mathrm{J}_1$. The divergence bound holds because the whitened covariance eigenvalues lie between $(1 + \mathrm{C}_L \varrho_\epsilon)^{-1}$ and one, where $t - 1 - \log t = O((t-1)^2)$, and Pinsker's inequality turns it into a total variation at most $\mathrm{C}_L \mathrm{e}_\epsilon(y)$.

Tilting two probability measures by one weight $w \in [e^{-B_0}, 1]$ multiplies their total-variation distance by at most $2e^{B_0}$, since each term of $w\mu/\mu(w) - w\nu/\nu(w) = w(\mu - \nu)/\mu(w) + w\nu[1/\mu(w) - 1/\nu(w)]$ has total variation at most $e^{B_0} \|\mu - \nu\|_{\mathrm{TV}}$. Applied to $w = e^{-\phi_j(w_1,h)}$ uniformly in h, this gives the first bound below; the second follows through the common Gaussian force $\vartheta \mathrm{J}_1 h$ and $\mathrm{T}\mathbb{E}\nabla_h \phi_j$, and the third because $\gamma^{1,*}_{\vartheta}$ does not depend on h, so differentiation under its normalized tilt gives

$$\nabla_h \beta^{*,h}_{jk} = \mathbb{E}_{\mu^{*,1}_{\vartheta,j}(\cdot|h)} \nabla_h \beta_k - \mathrm{Cov}_{\mu^{*,1}_{\vartheta,j}(\cdot|h)} (\beta_k, \nabla_h \phi_j),$$

and both terms have the asserted Lipschitz bound: the standardized source potential has bounded first derivatives and $\sum_k \beta_k = 1$. Thus

$$\|\beta^{\ell}_{j\cdot}(y, h, \vartheta) - \beta^{*,h}_{j\cdot}(h, \vartheta)\|_{\mathrm{TV}} \le \mathrm{C}_L \mathrm{e}_\epsilon(y), \tag{C.56}$$

$$|\nabla_h V_{\ell,j}(y, h, \vartheta) - \nabla_h V^{*,h}_j(h, \vartheta)| \le \mathrm{C}_L \sqrt{\mathrm{T}\mathrm{J}_1}\, \mathrm{e}_\epsilon(y), \tag{C.57}$$

$$\mathrm{Lip}_{(w_1,h)} \beta_{j\cdot} + \mathrm{Lip}_h \beta^{*,h}_{j\cdot} \le \mathrm{C}_L \sqrt{\mathrm{J}_1/\mathrm{T}}. \tag{C.58}$$

□

Lemma C.10 (Fast mixing with a learned coefficient). There is a constant $\mathrm{c}_{\mathrm{fast},L} > 0$ with the following property. Let the auxiliary process have, on a proposal-free interval, fixed assignment j and either the full fast potential of eq. (C.49) or its exact loop marginal, and a deterministic coefficient with $1 \le \vartheta \le 21/10$ and $|\dot\vartheta| \le \mathrm{C}_L\epsilon$. For a point start ξ at time t' and $t - t' \ge (\mathrm{c}_{\mathrm{fast},L}\mathrm{J}_1)^{-1}$,

$$\|P^*_{t',t}(\xi,\cdot) - \mu^*_{\vartheta(t),j}\|_{\mathrm{TV}} \le \mathrm{C}_L\big(1 + \sqrt{\mathrm{J}_1/\mathrm{T}}|\xi|\big)e^{-\mathrm{c}_{\mathrm{fast},L}\mathrm{J}_1(t-t')} + \mathrm{C}_L\epsilon/\mathrm{J}_1, \tag{C.59}$$

where μ^* means its loop marginal in the loop case. Also $\bar\beta_{jk}(\vartheta)$ is uniformly Lipschitz in ϑ. [ESTABLISHED]

Proof. The relative Hessian bounds and the normalized marginal covariance calculation give $\mathrm{c}_L\mathrm{J}_1 I \preceq \nabla^2 V^* \preceq \mathrm{C}_L\mathrm{J}_1 I$ for both potentials, which are even, minimize at zero, and have equilibrium second moments at most $\mathrm{C}_L\mathrm{T}/\mathrm{J}_1$. For two point starts ξ, ξ', reflection coupling before meeting gives their distance a one-dimensional noise of size $2\sqrt{2\mathrm{T}}$ and a drift at most $-\lambda$ times the distance, $\lambda = \mathrm{c}_L\mathrm{J}_1$; multiplying by $e^{\lambda(t-t')}$ gives a time-changed Brownian motion of variance $4\mathrm{T}(e^{2\lambda(t-t')} - 1)/\lambda$, and the reflection principle bounds the probability of not meeting by

$$\mathrm{C}|\xi - \xi'|\sqrt{\lambda/\mathrm{T}}\big(e^{2\lambda(t-t')} - 1\big)^{-1/2}. \tag{C.60}$$

The coupling reflects one Brownian motion orthogonally and predictably, so both marginals have the required law; the drifts are globally Lipschitz for each instance, so the copies are constructed away from the diagonal and joined at meeting, the comparison and hitting bound following by stopping before the diagonal and letting the stopping radius tend to zero. Averaging ξ' against the equilibrium at time t' gives the first term of eq. (C.59).

For the second, compare that evolving equilibrium start with the instantaneous equilibrium at time t. After standardizing (w_1, h) by $\sqrt{\mathrm{J}_1/\mathrm{T}}$, the normalized parameter score is a centered quadratic with uniform exponential moments, and the loop-marginal score, its conditional expectation, has the same bound by Jensen's inequality. The moving-reference entropy calculation with the continuous entropy inequality at curvature $\mathrm{c}_L\mathrm{J}_1$ gives, for the relative entropy $\mathrm{D}_{\mathrm{lag}}$ of the evolved law to the current equilibrium,

$$\mathrm{D}'_{\mathrm{lag}} \le -\mathrm{c}_L\mathrm{J}_1\mathrm{D}_{\mathrm{lag}} + \mathrm{C}_L\epsilon^2/\mathrm{J}_1, \qquad \mathrm{D}_{\mathrm{lag}}(t') = 0, \tag{C.61}$$

the parameter term linear in $\mathrm{D}_{\mathrm{lag}}$ being absorbed for small ϵ; hence $\mathrm{D}_{\mathrm{lag}} \le \mathrm{C}_L\epsilon^2/\mathrm{J}_1^2$, and Pinsker's inequality gives the second term. Finally, differentiating $\bar\beta_{jk} = \mathbb{E}_{\mu^*_{\vartheta,j}}\beta_k$ gives the explicit row derivative, whose absolute values sum to at most eight, and a covariance with the centered scale score, whose standardized first moment is bounded, so $\sum_k |\partial_\vartheta \bar\beta_{jk}| \le \mathrm{C}_L$. □

Lemma C.11 (Coupling of complete source histories). The source-assignment histories of a process and of an activated exact parent can be coupled, together with their common initial marked sample, so that the probability of any disagreement through t_* is at most $\mathrm{C}_L\epsilon^6$. This holds through

the finite activation lineage, for the full finite-state path with arbitrary label correlations and the rate-one proposal clock. [ESTABLISHED]

Proof. First bound a proposal row on one proposal-free interval. For a microscopic process let $\xi = (w_1, h)$, and couple its auxiliary fast diffusion to the same initial fast point, Brownian increments, assignment, and parameter path. The omitted Gaussian environmental force $\vartheta[(H_{11} - \Omega_*)w_1 + H_{1,>1}w_{>1}]$ has norm at most $\mathrm{C}_L\mathrm{J}_2|w|$, and strong contraction from the most recent proposal or activation time t_{prev} gives the first bound below. For a parent at level $\ell \geq 1$, coupling its retained loop to the auxiliary loop process, eq. (C.57) gives the second:

$$|\xi_t - \xi_t^*| \leq \mathrm{C}_L\mathrm{J}_2 \int_{t_{\mathrm{prev}}}^{t} e^{-\mathrm{c}_L\mathrm{J}_1(t-\sigma)}|w_\sigma|d\sigma, \tag{C.62}$$

$$|h_t - h_t^*| \leq \mathrm{C}_L\sqrt{\mathrm{TJ}_1} \int_{t_{\mathrm{prev}}}^{t} e^{-\mathrm{c}_L\mathrm{J}_1(t-\sigma)}\mathrm{e}_\epsilon(y_\sigma)d\sigma, \tag{C.63}$$

with the adapted environment of the process. The row Lipschitz constant and the direct parent row error eq. (C.56) convert these into proposal-row errors.

At random proposal times, condition on the whole independent future Poisson grid, but not on its marks or on any future continuous path, so that the Brownian law is unchanged. Evenness and the uniform lower curvature on M_ℓ give the first bound below on every sheet and level; proposals preserve y, projections contract its norm, and the preparation has second moment at most $\mathrm{C}_L(\mathrm{A}^2 + \mathrm{T})$, which gives the second:

$$(\mathrm{T}\Delta - \nabla V_{\ell,j} \cdot \nabla)|y|^2 \leq 2\mathrm{T}\dim M_0 - 2\mathrm{c}_L|y|^2, \tag{C.64}$$

$$\sup_{t \leq t_*} \mathbb{E}\big[|\mathbf{X}_{M_\ell,t}|^2 \;\big|\; \text{Poisson grid}\big] \leq \mathrm{C}_L(\mathrm{A}^2 + \mathrm{T}). \tag{C.65}$$

If the initial sample is fixed as well, the right side first contains its squared norm, and integrating that sample returns the bound; the future grid after a deterministic activation is independent of the past, so the argument applies at each activation and at grid-selected times. Averaging the row errors of eqs. (C.62) and (C.63) with eq. (C.65), where the exponential convolution has integral at most $\mathrm{C}_L/\mathrm{J}_1$, over the distribution of every past prefix, and dividing by the probability of no rare prefix, gives the common expected per-proposal bound

$$\eta_\epsilon = \mathrm{C}_L\left[\frac{\mathrm{J}_2}{\mathrm{J}_1} + \frac{\mathrm{J}_2}{\sqrt{\mathrm{J}_1}}\left(1 + \frac{\mathrm{A}}{\sqrt{\mathrm{T}}}\right)\right] = O_L(\epsilon^7), \tag{C.66}$$

since $\mathrm{Q}_1 = 2\mathrm{Q}_2 + 20$ and $\mathrm{A}/\sqrt{\mathrm{T}} = \epsilon^{-3}$.

Define one comparison source process on the same proposal grid, started at the initial assignment and using the row $\bar{\beta}_{j\cdot}(\vartheta_\circ(t))$, $\vartheta_\circ(t) = 2 - e^{-\epsilon t}$, a single comparison law common to both processes while each process keeps its own law and coefficient. At a proposal time t, condition on the common starting sample, the full future grid, and all previous marks, self marks included; the assignment and level are fixed on (t_{prev}, t), and no future continuous trajectory is conditioned on. Starting the auxiliary diffusion from the fast point at t_{prev}, the environmental coupling, then eq. (C.59), then the

Lipschitz comparison of $\vartheta(t)$ with $\vartheta_\circ(t)$ bound the averaged conditional row error, for gaps at least t_{gap}, by the second line below:

$$t_{\mathrm{gap}} = \frac{k_L}{\mathrm{J}_1}\log(1/\epsilon), \tag{C.67}$$

$$\mathrm{C}_L\left[\left(1+\mathrm{A}\sqrt{\mathrm{J}_1/\mathrm{T}}\right)e^{-\mathrm{c}_{\mathrm{fast},L}\mathrm{J}_1 t_{\mathrm{gap}}} + \epsilon/\mathrm{J}_1 + \eta_\epsilon + r_\epsilon\right]; \tag{C.68}$$

the moment of the point-start mixing factor, which includes $\sqrt{\mathrm{J}_1/\mathrm{T}}(\mathrm{A}+\sqrt{\mathrm{T}})$, is absorbed because $\mathrm{A}/\sqrt{\mathrm{T}} \geq 1$, and averaging over the past marks returns the grid-conditioned moments exactly. Short gaps use the trivial bound one. For a rate-one Poisson grid the expected number of ordered pairs of proposal times less than t_{gap} apart is at most $t_* t_{\mathrm{gap}}$, by integrating the two-time intensity; proposals within t_{gap} after the initial time or one of the finitely many activations add at most $(L+1)t_{\mathrm{gap}}$, and the expected total count is t_*.

To pass from conditional row errors to the whole history, condition on a grid with n proposals, write the two joint mark laws as products of their conditional kernels K_m and $\overline{\mathsf{K}}_m$, and replace kernels in order, keeping the earlier prefix distributed according to the process and integrating the comparison tail, whose mass is one. With $\mathsf{j}_m \in \mathcal{J}$ the mth proposal mark,

$$\left\|\mathrm{Law}(\mathsf{j}_1,\dots,\mathsf{j}_{\mathrm{n}}) - \mathrm{Law}(\bar{\mathsf{j}}_1,\dots,\bar{\mathsf{j}}_{\mathrm{n}})\right\|_{\mathrm{TV}} \leq \sum_{m=1}^{\mathrm{n}} \mathbb{E}_{\mathsf{prefix}}\|\mathsf{K}_m(\cdot \mid \mathsf{j}_{<m}) - \overline{\mathsf{K}}_m(\cdot \mid \mathsf{j}_{<m})\|_{\mathrm{TV}}, \tag{C.69}$$

and integrating over the grid with the preceding bounds gives

$$\begin{aligned}\mathrm{Err}_{\mathrm{hist}} \leq \mathrm{C}_L\big[&(t_*+L+1)t_{\mathrm{gap}} + t_*\big(1+\mathrm{A}\sqrt{\mathrm{J}_1/\mathrm{T}}\big)e^{-\mathrm{c}_{\mathrm{fast},L}\mathrm{J}_1 t_{\mathrm{gap}}} \\ &+ t_*\epsilon/\mathrm{J}_1 + t_*\eta_\epsilon + t_* r_\epsilon\big].\end{aligned} \tag{C.70}$$

With $\mathrm{c}_{\mathrm{fast},L}k_L > \mathrm{Q}_1/2 + 12$, the five contributions are bounded by constants times $\epsilon^{19}\varsigma_\epsilon$, ϵ^8, ϵ^{Q_1}, ϵ^6, and $\epsilon^{2\mathrm{Q}_1+19} + \epsilon^{3\mathrm{Q}_1+25}\varsigma_\epsilon^2$, so $\mathrm{Err}_{\mathrm{hist}} \leq \mathrm{C}_L\epsilon^6$.

Fix the common initial sample and the grid, couple each mark sequence maximally with the comparison sequence, and glue the two couplings along it; the glued sequences disagree with probability at most $\mathrm{C}_L\epsilon^6$. The path spaces, continuous paths in a Euclidean space and right-continuous paths in $\mathcal{J}$, are standard Borel, so each continuous trajectory can be drawn from its regular conditional law given the initial sample, grid, and marks, which preserves both full-process marginals and the parent projection of the initial sample. Once self-proposals are discarded, each source path is a function of grid and marks, which proves the first bound of eq. (14.45); finitely many activations change only C_L. □

C.5 Coherence, source movement, and continuous paths

Lemma C.12 (Successive coherence with the original conditional marks). Under the preparation and regime of eqs. (C.13), (14.29) and (14.31), there are a finite $k_{\mathrm{geom},L}$ and a tolerance

bound $\eta_L^{\max} > 0$, a fixed fraction of m_L, both depending only on L and the fixed data, with the following property. For every $k_L > k_{\mathrm{geom},L}$ and every $0 < \eta_L \le \eta_L^{\max}$ there are $\epsilon_L, \mathrm{c}_L, \mathrm{C}_L > 0$ such that, for $0 < \epsilon < \epsilon_L$, conclusion (b) of Theorem 14.17 holds for the fine process and its successive parent descriptions on the windows $I_\ell = [t_\ell, 2t_\ell]$, $t_\ell = k_L \log(1/\epsilon)/\mathrm{J}_\ell$, with the probability bound eq. (14.40). On I_ℓ the discrepancies $\mathfrak{d}_i$ and the residuals of level at most ℓ are below $\eta_L \mathrm{A}$ and future bridges stay separated by a fixed positive multiple of A, in the conditional root-mean-square original residuals of eq. (14.18) for eliminated coordinates, separately for the physical posterior and the restricted optimizer. [ESTABLISHED]

Proof. On an active level $\ell' < \ell$, write the non-root Gaussian comparison of Lemma C.3 through a current normal component $z \in \mathcal{U}_{\ell'} \cap \mathcal{U}_\ell^\perp$, a future component $y \in \mathcal{U}_\ell$, and the separate loop component. The weighted flag gives, for the current and future blocks of $H_{\ell'}$,

$$H_{zz} \succeq \mathrm{c}_L \mathrm{J}_\ell I, \qquad \|H_{zy}\| \le \mathrm{C}_L \mathrm{J}_{\ell+1}, \qquad \|H_{yy}\| \le \mathrm{C}_L \mathrm{J}_{\ell+1}, \qquad \mathrm{J}_{L+1} = 0. \tag{C.71}$$

Indeed rows of levels at most ℓ annihilate $\mathcal{U}_\ell$, so the original cross and future blocks contain only strengths at most $\mathrm{J}_{\ell+1}$; in an earlier elimination through ℓ'', the eliminated-current block is $O_L(\mathrm{J}_{\ell''+1})$, the inverse hidden block $O_L(\mathrm{J}_{\ell''}^{-1})$, and the eliminated-future block $O_L(\mathrm{J}_{\ell+1})$, so the Schur correction is still $O_L(\mathrm{J}_{\ell+1})$; at level zero there is no correction, and loop/normal cross blocks vanish in the Gaussian core.

The deterministic reference mean μ^{G} has norm at most $\mathrm{C}_L \mathrm{A}$ and zero loop component. Integrating $\dot{\mu}_z^{\mathrm{G}} = -\vartheta(t)(H_{zz}\mu_z^{\mathrm{G}} + H_{zy}\mu_y^{\mathrm{G}})$ with $1 \le \vartheta \le 21/10$ gives the first bound below; integrating the full future row, of norm $O_L(\mathrm{J}_{\ell+1})$, from the common center, earlier projections preserving $P_\ell \mu^{\mathrm{G}}$, gives the second for a sequential process begun at $t_{\ell-1}^{\mathrm{act}}$:

$$|\mu_z^{\mathrm{G}}(t)| \le \mathrm{C}_L \mathrm{A} \left[e^{-\mathrm{c}_L \mathrm{J}_\ell (t - t_0)} + \frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell} \right], \tag{C.72}$$

$$|\mu^{\mathrm{G}}(t) - \mathrm{A} P_\ell \mathbf{v}_0| \le \mathrm{C}_L \mathrm{A} \left[e^{-\mathrm{c}_L \mathrm{J}_\ell (t - t_{\ell-1}^{\mathrm{act}})} + \frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell} + \mathrm{J}_{\ell+1} t \right]. \tag{C.73}$$

The future terms are absent when $\ell = L$, and a fine process uses $t_0 = 0$.

For the fluctuations, on each active non-root space $\mathbf{Z} = \mathbf{X}^{\mathrm{G}} - \mu^{\mathrm{G}}$ obeys $d\mathbf{Z} = -\vartheta(t) H_{\ell'} \mathbf{Z} dt + \sqrt{2\mathrm{T}} dW$, with $\vartheta H_{\ell'} \succeq \mathrm{h}_L I$ and initial covariance at most $\mathrm{C}_L \mathrm{T} H_{\ell'}^{-1}$. Choose $\mathrm{a}_* > 0$ below h_L and small enough for the initial Gaussian exponential moment. For $\mathcal{Y}(\mathbf{z}) = e^{\upsilon}$ with $\upsilon = \mathrm{a}_* |\mathbf{z}|^2/(4\mathrm{T})$, direct differentiation gives the first line below; the initial expectation of $\mathcal{Y}$ is bounded, orthogonal retention decreases $\mathcal{Y}$, and Itô's formula up to the first exit from a ball, localization, and the nonnegative stopped value of $\mathcal{Y}$ yield the second for every $\varrho > 0$, higher stiffnesses only strengthening the negative term:

$$\mathscr{L}_t \mathcal{Y} = \left[\frac{\mathrm{a}_* \mathrm{d}}{2} + \frac{\mathrm{a}_*^2 |\mathbf{z}|^2}{4\mathrm{T}} - \frac{\mathrm{a}_* \vartheta \mathbf{z}^\top H_{\ell'} \mathbf{z}}{2\mathrm{T}} \right] \mathcal{Y} \le \mathrm{a}_* (\mathrm{d}/2 - \upsilon) e^{\upsilon} \le \mathrm{C}_L, \tag{C.74}$$

$$\Pr\Big\{ \sup_{t \le t_*} |\mathbf{X}_t^{\mathrm{G}} - \mu^{\mathrm{G}}(t)| > \varrho \mathrm{A} \Big\} \le \mathrm{C}_L (1 + t_*) \exp\left[-\mathrm{c}_L \varrho^2 \mathrm{A}^2 / \mathrm{T} \right]. \tag{C.75}$$

Let $\bar{r}_L$ bound the norms of the finitely many residual and discrepancy covectors, let $\eta_L^{\max}$ be a fixed fraction of m_L, fix $\eta_L \in (0, \eta_L^{\max}]$, and choose a fluctuation radius $\varrho\mathrm{A}$ with $\bar{r}_L \varrho\mathrm{A}$ smaller than the reserved tolerance margin, c_L in the probability depending on ϱ. Take for example $k_{\mathrm{geom},L} = 2/\mathrm{c}_L$ for the restoring constant of eq. (C.72); increasing this finite constant is allowed. For small ϵ, $t_{\ell-1}^{\mathrm{act}} < t_\ell/2$, so the exponential term of eq. (C.73) tends to zero uniformly on I_ℓ, and

$$\frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell} \longrightarrow 0, \qquad \mathrm{J}_{\ell+1} t_\ell \longrightarrow 0 \quad (\ell < L), \qquad \frac{t_L^{\mathrm{act}}}{t_*} \longrightarrow 0. \tag{C.76}$$

The nonlinear correction of eq. (C.17) is at most $\mathrm{C}_L \mathrm{A} \epsilon^{5/2}$, including the admitted initial transport and finitely many inherited errors; it enters through the source force only via the uniform bound $B_* \sqrt{\mathrm{T}}$, so it holds for every adapted assignment path. Since $\mathrm{A}^2/\mathrm{T} = \epsilon^{-6}$, the strict prepared residual margins and eq. (C.75) give the asserted probability and separation on every window. The stable current-block equation keeps bounding formed residuals as their future forcing decreases, and after the last normal block there is no future forcing, so the final regime persists through t_* on the same event, with every loop retained.

For an eliminated edge and a cumulative elimination through $\ell' \geq 1$, write a fine non-root state as $\mathbf{z} + y$ with $\mathbf{z} \in W_{\ell'}^{\perp}$ and $y \in M_{\ell'}$. The optimizing and physical Gaussian precisions are ϑH and $\vartheta H - \mathrm{T} R_{\mathrm{ev}}$. Every row of $\mathfrak{D}$ lies in the first normal layer, so R_{ev} annihilates $\mathcal{U}_1$, $\mathcal{H}$, and $\mathcal{K}$, and with $\mathrm{T} R_{\mathrm{ev}} \preceq H/2$ both laws have hidden precision at least $\mathrm{c}_L \mathrm{J}_{\ell'} I$, hidden/future block $O_L(\mathrm{J}_{\ell'+1})$, and zero hidden/loop block. The Gaussian conditional mean therefore has norm at most $\mathrm{C}_L(\mathrm{J}_{\ell'+1}/\mathrm{J}_{\ell'})|y_{\mathcal{U}_{\ell'}}|$ and covariance trace at most $\mathrm{C}_L \mathrm{T}/\mathrm{J}_{\ell'}$, and, given the full assignment, each exact hidden density is a normalized tilt of its Gaussian conditional with log oscillation at most B_0, hence density ratio at most e^{B_0}. For $\mu_\vartheta \in \{\mathbf{\Pi}_{\vartheta,\mathrm{o}}, q_\vartheta^\star\}$ and every retained state with $|y_{\mathcal{U}_{\ell'}}| \leq \mathrm{C}_L \mathrm{A}$, this gives the first bound below, and for every original residual covector the triangle inequality and its reverse in conditional L^2 give the second:

$$\mathbb{E}_{\mu_\vartheta}[|\mathbf{z}|^2 \mid P_{\ell'}\mathbf{X} = y, S = j] \leq \mathrm{C}_L e^{B_0} \left[\mathrm{A}^2 \left(\frac{\mathrm{J}_{\ell'+1}}{\mathrm{J}_{\ell'}} \right)^2 + \frac{\mathrm{T}}{\mathrm{J}_{\ell'}} \right], \tag{C.77}$$

$$\left| \sqrt{\mathbb{E}_{\mu_\vartheta}[r_e(\mathbf{X})^2 \mid P_{\ell'}\mathbf{X} = y, S = j]} - |r_e(y)| \right| \leq \|\mathbf{r}_e\| \sqrt{\mathbb{E}_{\mu_\vartheta}[|\mathbf{z}|^2 \mid P_{\ell'}\mathbf{X} = y, S = j]}. \tag{C.78}$$

On the event already constructed the right side of eq. (C.78) is $o(\mathrm{A})$, uniformly in the assignment and separately for the two laws, and the same holds for $\mathfrak{d}_i$; this transfers the small formed and the separated future residuals to eq. (14.18), with no erased mark replaced by its value at $P_{\ell'}\mathbf{x}$. The bridge levels join successive pairs of already formed components, giving the stated counts. □

Lemma C.13 (Source envelope and changing hidden kernels). Every process or finite replacement lineage of Lemma C.3 satisfies eq. (14.43) for the distinguished binary source, from any of its activation times. Between $\vartheta = 1$ and its final parameter, the first hidden physical conditional and the first hidden optimizing conditional each change by relative entropy greater than $1/200$, uniformly at the same retained input and assignment. [ESTABLISHED]

Proof. The distinguished receiver's posterior row $\beta_{i_0,\pm}$, a factor of the product row eq. (14.14), has the log odds of the first line below. Since $0 \le \chi \le 1$, it lies between $\operatorname{sig}(4(\vartheta - 3/2) \mp \delta)$, and summing the inherited row over the assignments with distinguished label $+$ and taking the expectation preserves these bounds, so the generator applied to the indicator of that label gives the second line, with $\lambda(t) = \Pr\{S_{i_0}(t) = +\}$:

$$\log \frac{\beta_{i_0,+}(\mathbf{x},\vartheta)}{\beta_{i_0,-}(\mathbf{x},\vartheta)} = 4(\vartheta - 3/2) - \delta\left[\chi\big(\sqrt{\mathrm{J}_1/\mathrm{T}}\, r_{i_0,+}(\mathbf{x})\big) - \chi\big(\sqrt{\mathrm{J}_1/\mathrm{T}}\, r_{i_0,-}(\mathbf{x})\big)\right], \tag{C.79}$$

$$\lambda' = \bar{\beta}_\lambda - \lambda, \qquad \operatorname{sig}(4(\vartheta - 3/2) - \delta) \le \bar{\beta}_\lambda(t) \le \operatorname{sig}(4(\vartheta - 3/2) + \delta). \tag{C.80}$$

Neither diffusion nor a marked projection changes this indicator, so variation of constants remains valid across every activation, even when the inherited row depends on the current assignment and continuous state.

By Lemma C.4, uniformly before $t_L^{\mathrm{act}} = 2k_L \log(1/\epsilon)$, $\vartheta \le 1 + \mathrm{C}_L \epsilon \log(1/\epsilon) + \mathrm{C}_L r_\epsilon$. A solution of eq. (C.80) cannot exceed the larger of its starting value and the supremum forcing, so the preparation tolerance and $\|\operatorname{sig}'\|_\infty \le 1/4$ give the first bound below. For small ϵ, every activation precedes $t_* - 10$ and $\vartheta \ge 7/4$ throughout $[t_* - 10, t_*]$, so nonnegativity of the earlier contribution in variation of constants yields the second; for $\delta < 10^{-4}$, the elementary bounds $e^{1-\delta} > 249/100$, $e^2 > 7$, and $e^{10} > 2^{10}$ give the third:

$$\lambda(t_\ell^{\mathrm{act}}) \le \operatorname{sig}(-2) + 10^{-2} + \delta/4 + \mathrm{C}_L[\epsilon \log(1/\epsilon) + r_\epsilon], \qquad 0 \le \ell \le L, \tag{C.81}$$

$$\lambda(t_*) \ge (1 - e^{-10}) \operatorname{sig}(1 - \delta), \tag{C.82}$$

$$(1 - e^{-10}) \operatorname{sig}(1 - \delta) - \operatorname{sig}(-2) - 10^{-2} - \delta/4 > \frac{1023}{1024}\frac{249}{349} - \frac{1}{8} - \frac{1}{100} - \frac{1}{40000} > \frac{57}{100}. \tag{C.83}$$

The remaining error in eq. (C.81) tends to zero, hence $\lambda(t_*) - \lambda(t_\ell^{\mathrm{act}}) > 1/2$ for every activation, which for a binary row is its total-variation change.

For the hidden kernels, fix the retained input and the assignment, and first omit the bounded source factor. On the first hidden normal space let $\Omega_1, \Omega_{\mathrm{f}}$ be the initial and final Gaussian precisions, with covariances $\mathrm{T}\Omega_1^{-1}, \mathrm{T}\Omega_{\mathrm{f}}^{-1}$. For the optimizer the precision scales with ϑ; for the physical law, on the non-root space,

$$\Omega_1 = H - \mathrm{T}R_{\mathrm{ev}} \succ 0, \qquad \Omega_{\mathrm{f}} = \Omega_1 + (\vartheta_{\mathrm{f}} - 1)H \succeq \vartheta_{\mathrm{f}}\Omega_1, \qquad \vartheta_{\mathrm{f}} \ge 7/4. \tag{C.84}$$

Restriction to a conditional block preserves the order, and so does an earlier Gaussian marginal, whose Schur quadratic is the minimum over the discarded coordinates, so in either case $\Omega_{\mathrm{f}} \succeq (5/4)\Omega_1$. For the Gaussian laws $\gamma_1, \gamma_{\mathrm{f}}$ on a hidden space of dimension $\mathrm{d_h} \ge 1$, whitening Ω_1 and diagonalizing the conjugated final precision, with eigenvalues $\varrho_i \ge 5/4$, and completing the square in the Hellinger affinity give the first line below; each factor decreases for $\varrho_i \ge 1$, so the affinity is at most $[2\sqrt{5/4}/(1 + 5/4)]^{\mathrm{d_h}/2}$. Each exact conditional ν_1, ν_{f} is a normalized source tilt of its Gaussian conditional bounded above by e^{B_0}, also after exact hidden integration, so $\mathcal{A}(\nu_1, \nu_{\mathrm{f}}) \le e^{B_0}\mathcal{A}(\gamma_1, \gamma_{\mathrm{f}})$, and Jensen's inequality under ν_1 gives $-D_{\mathrm{KL}}(\nu_1 \| \nu_{\mathrm{f}})/2 \le \log \mathcal{A}(\nu_1, \nu_{\mathrm{f}})$. With $\log(1 + 1/80) \ge 1/81$

this gives the second line, separately for the physical and optimizing conditionals and uniformly in the retained input and assignment:

$$\mathcal{A}(\gamma_1,\gamma_{\mathrm{f}}) = \prod_{i=1}^{\mathrm{d_h}} \left(\frac{2\sqrt{\varrho_i}}{1+\varrho_i}\right)^{1/2} e^{-\mathrm{M}}, \qquad \mathrm{M}\geq 0, \tag{C.85}$$

$$D_{\mathrm{KL}}(\nu_1\|\nu_{\mathrm{f}}) \geq \frac{\mathrm{d_h}}{2}\log\frac{81}{80} - 2B_0 \geq \frac{1}{162} - \frac{1}{5000} > \frac{1}{200}. \tag{C.86}$$

□

Lemma C.14 (Concentration of continuous paths on a common deterministic path). Let a source process and its exact parent start from a common retained sample and parameter at t_ℓ^{act}. There is a deterministic path $\bar{\mathbf{x}}_\ell : [t_\ell^{\mathrm{act}}, t_*] \to W_\ell$, namely the parent's Gaussian reference mean on the non-root coordinates and the constant equilibrium root mean, such that

$$\Pr\Big\{\sup_{t_\ell^{\mathrm{act}}\leq t\leq t_*} |P_\ell \mathbf{X}_t - \bar{\mathbf{x}}_\ell(t)| > \mathrm{C}_L\mathrm{A}\epsilon^2\Big\} + \Pr\Big\{\sup_{t_\ell^{\mathrm{act}}\leq t\leq t_*} |\overline{\mathbf{X}}_t - \bar{\mathbf{x}}_\ell(t)| > \mathrm{C}_L\mathrm{A}\epsilon^2\Big\} \leq \mathrm{C}_L(1+t_*)e^{-\mathrm{c}_L/\epsilon^2}. \tag{C.87}$$

Consequently the displacement bound of eq. (14.45) holds under every coupling of the two processes, in particular under the coupling of Lemma C.11. A finite number of successive replacements changes only the constants. [ESTABLISHED]

Proof. First compare Gaussian means at one deterministic parameter path. In the split of eq. (C.71) the inherited normal precision is $H_\ell = H_{yy} - H_{zy}^\top H_{zz}^{-1} H_{zy}$. With $\mathfrak{w} = \mu_z^{\mathrm{G}} + H_{zz}^{-1}H_{zy}\mu_y^{\mathrm{G}}$, the future mean of the source process satisfies the first line below, the bound following from eq. (C.72) and $\|H_{zz}^{-1}H_{zy}\| \leq \mathrm{C}_L\mathrm{J}_{\ell+1}/\mathrm{J}_\ell$, and integrating the contractive future propagator from a matching retained mean gives the second:

$$\dot{\mu}_y^{\mathrm{G}} = -\vartheta H_\ell \mu_y^{\mathrm{G}} - \vartheta H_{zy}^\top \mathfrak{w}, \qquad |\mathfrak{w}(t)| \leq \mathrm{C}_L\mathrm{A}\left[e^{-\mathrm{c}_L\mathrm{J}_\ell(t-t_\ell^{\mathrm{act}})} + \frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell}\right], \tag{C.88}$$

$$\sup_{t_\ell^{\mathrm{act}}\leq t\leq t_*} |\mu_y^{\mathrm{G}}(t) - \bar{\mu}^{\mathrm{G}}(t)| \leq \mathrm{C}_L\mathrm{A}\left[\frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell} + t_*\frac{\mathrm{J}_{\ell+1}^2}{\mathrm{J}_\ell}\right]. \tag{C.89}$$

There is no future normal contribution when $\ell = L$, and the loop Gaussian core is unchanged by the reduction and has zero mean. If the independently learned parameters differ, with $\mathrm{e}_\vartheta = \sup|\vartheta - \bar{\vartheta}| \leq \mathrm{C}_L r_\epsilon$ by Lemma C.4, Young's inequality in the H_ℓ quadratic form gives, for homogeneous parent means $\mu, \bar{\mu}$ with difference $\mathfrak{e}$,

$$\frac{d}{dt}|\mathfrak{e}|^2 \leq -\vartheta \mathfrak{e}^\top H_\ell \mathfrak{e} + \mathrm{C}_L \mathrm{e}_\vartheta^2 \bar{\vartheta}\bar{\mu}^\top H_\ell \bar{\mu}, \tag{C.90}$$

and since $\frac{d}{dt}|\bar{\mu}|^2 = -2\bar{\vartheta}\bar{\mu}^\top H_\ell \bar{\mu}$, the time integral of the final energy is at most $\mathrm{C}_L\mathrm{A}^2$, so the extra displacement is at most $\mathrm{C}_L\mathrm{A}\mathrm{e}_\vartheta$.

Attach to each process its own Gaussian comparison from Lemma C.3; the discrepancy between the parent's projected starting reference and its full retained starting sample is included in eq. (C.17).

On the non-root coordinates let $\bar{\mathbf{x}}_\ell = \bar{\mu}^{\mathrm{G}}$ be the parent's reference mean. For the parent, eqs. (C.17) and (C.75) with $\varrho = \epsilon^2$ give $\sup_t |\overline{\mathbf{X}}_t - \bar{\mathbf{x}}_\ell(t)| \le \mathrm{C}_L\mathrm{A}(\epsilon^2 + \epsilon^{5/2})$ outside a set of probability $\mathrm{C}_L(1 + t_*)e^{-\mathrm{c}_L/\epsilon^2}$. For the source process the same two estimates for its own reference, then eq. (C.89) and the parameter comparison, give, outside a set of the same form,

$$\sup_{t_\ell^{\mathrm{act}} \le t \le t_*} |P_\ell \mathbf{X}_t - \bar{\mathbf{x}}_\ell(t)| \le \mathrm{C}_L\mathrm{A}\left[\epsilon^2 + \epsilon^{5/2} + r_\epsilon + \frac{\mathrm{J}_{\ell+1}}{\mathrm{J}_\ell} + t_* \frac{\mathrm{J}_{\ell+1}^2}{\mathrm{J}_\ell}\right]. \tag{C.91}$$

For $\ell < L$ the last term equals $2\epsilon^{19}$, $\mathrm{J}_{\ell+1}/\mathrm{J}_\ell = \epsilon^{\mathrm{Q}_{\ell+1}+20}$, and $r_\epsilon = o(\epsilon^2)$; both future terms vanish at $\ell = L$. The root coordinates of each process form the same autonomous Ornstein–Uhlenbeck process in law, with stiffness $1 + \kappa \ge 1$, started at its equilibrium and independent of the source history, and the argument of eq. (C.75) keeps each within $\epsilon^2\mathrm{A}$ of the constant mean $\kappa z_\star/(1+\kappa)$ outside a set of the same form. Taking that constant as the root component of $\bar{\mathbf{x}}_\ell$ proves eq. (C.87). Each of its events concerns the law of one process alone, so for any coupling the displacement event of eq. (14.45), after doubling C_L, lies in their union; a finite sequence of replacements uses a finite triangle inequality and union bound. □

C.6 The Gaussian core across an early interface

Proof of parts (b) and (c) of Proposition 14.15. Fix $1 \le \ell \le L-1$. In this section C and c denote positive constants depending only on L and the carrier, never on the strengths, A, T, t, or ϵ; put $\mathrm{J}_{L+1} = 0$ and $\bar{\varrho}_\ell = \varrho_{\ell-1} + \varrho_\ell + \varrho_{\ell+1}$.

Flag layers and block scales. In the flag coordinates $w = (w_1, \dots, w_L)$ of eq. (C.2), $\mathbf{x}_{\mathcal{U}} = \sum_s \mathsf{Q}_s w_s$, the quadratic form $\mathbf{x}_{\mathcal{U}}^\top H \mathbf{x}_{\mathcal{U}} = \sum_s \mathrm{J}_s |\sum_{r \le s} \mathsf{B}_{sr} w_r|^2$ has the blocks $(H_{\mathcal{U}})_{rr'} = \sum_{s \ge \max(r,r')} \mathrm{J}_s \mathsf{B}_{sr}^\top \mathsf{B}_{sr'}$, each of norm at most $\mathrm{C}\mathrm{J}_{\max(r,r')}$; the loop block $\mathrm{J}_1 I_{\mathcal{H}}$ is orthogonal to every layer and plays no part. On the active level $\ell - 1$ the normal part of $H_{\ell-1}$ is the Schur complement of $H_{\mathcal{U}}$ onto the layers $r \ge \ell$, obtained by eliminating the layers $\mathcal{E} = \{1, \dots, \ell-1\}$, with the discarded block $z = w_\ell$ and the retained block $y = (w_{\ell+1}, \dots, w_L)$. The lower bound of eq. (C.2) on $\mathcal{E}$ gives $H_{\mathcal{E}\mathcal{E}} \succeq \mathrm{c}\mathrm{J}_{\ell-1} I$, so the Schur correction to a block (r, r') with $r, r' \ge \ell$ has norm at most $\mathrm{C}\mathrm{J}_r\mathrm{J}_{r'}/\mathrm{J}_{\ell-1} \le \mathrm{C}\mathrm{J}_{\max(r,r')}\varrho_{\ell-1}$. Since in each block the summand with the largest admissible strength leads and the remainder carries the next strength,

$$\begin{aligned} H_{zz} &= \mathrm{J}_\ell\big[\mathsf{B}_{\ell\ell}^\top \mathsf{B}_{\ell\ell} + O(\varrho_\ell + \varrho_{\ell-1})\big], \\ H_{z,\ell+1} &= \mathrm{J}_{\ell+1}\big[\mathsf{B}_{\ell+1,\ell}^\top \mathsf{B}_{\ell+1,\ell+1} + O(\varrho_{\ell+1} + \varrho_{\ell-1})\big], \\ H_{\ell+1,\ell+1} &= \mathrm{J}_{\ell+1}\big[\mathsf{B}_{\ell+1,\ell+1}^\top \mathsf{B}_{\ell+1,\ell+1} + O(\varrho_{\ell+1} + \varrho_{\ell-1})\big], \end{aligned} \tag{C.92}$$

with the constants of the remainders depending only on L and the carrier, together with $\|H_{z,r}\| \le \mathrm{C}\mathrm{J}_r$ and $\|H_{\ell+1,r}\| \le \mathrm{C}\mathrm{J}_{\ell+1}(\varrho_{\ell+1} + \varrho_{\ell-1})$ for $r \ge \ell + 2$, and $\|H_{r,\cdot}\| \le \mathrm{C}\mathrm{J}_r$ for every row block $r \ge \ell + 1$, in particular $\|H_{zy}\| \le \mathrm{C}\mathrm{J}_{\ell+1}$. Hence, for $\bar{\varrho}_\ell \le \mathrm{c}_0$ with c_0 small, $\lambda_{\min}(H_{zz}) \ge \frac{3}{4}\mathrm{j}_\ell\mathrm{J}_\ell$, $\|H_{zz}^{-1}\| \le \mathrm{C}/\mathrm{J}_\ell$, and $\|H_{zz}^{-1}H_{zy}\| \le \mathrm{C}\varrho_\ell$. The same computation at every earlier active level $p \le \ell - 1$ bounds the rows of layer $\ell + 1$ of H_p by $\mathrm{C}\mathrm{J}_{\ell+1}$.

Full row rank. Write $\mathbf{r}_e$ for the covector of r_e. Since $\mathcal{U}_\ell = \mathcal{U}_{\ell-1} \cap \bigcap_e \ker \mathbf{r}_e$ over the rows e of B_ℓ, the layer spanned by Q_ℓ is the span of the projections $P_{\mathcal{U}_{\ell-1}}\mathbf{r}_e$, and $\mathsf{B}_{\ell+1,\ell}$ has full row rank exactly when the matrix $\mathrm{M}_{fe} = \mathbf{r}_f \cdot P_{\mathcal{U}_{\ell-1}}\mathbf{r}_e$, with f over the rows of $B_{\ell+1}$ and e over those of B_ℓ, has full row rank, because $\mathrm{M}_{fe} = (P_{\mathcal{U}_{\ell-1}}\mathbf{r}_f)\cdot(P_{\mathcal{U}_{\ell-1}}\mathbf{r}_e)$, the component of $P_{\mathcal{U}_{\ell-1}}\mathbf{r}_f$ in $\mathcal{U}_\ell$ pairs to zero with the layer, and the second factors span the layer. For $\ell = 1$, $\mathcal{U}_0 = \mathcal{U}$ contains every row of B, so $\mathrm{M}_{fe} = \mathbf{r}_f \cdot \mathbf{r}_e$; in the column of the discrepancy row of agent i the entry of a bridge f is $[i = \mathrm{head}(f)] - [i = \mathrm{tail}(f)]$, and the level-two bridges, those of odd index, join disjoint pairs of triangles and have pairwise disjoint endpoint sets, so the columns of the discrepancy rows of their heads form an identity submatrix. For $\ell \geq 2$ let S be the set of bridges of level at least ℓ, let $\boldsymbol{\partial}_f \in \mathbb{R}^V$ be the vector with entry 1 at the head of the bridge f, -1 at its tail, and 0 elsewhere, let ∂_S have the rows $\boldsymbol{\partial}_f^\top$, $f \in S$, and let ∂ be the oriented incidence operator of the whole graph, so that $\partial^\top\partial$ is its Laplacian. For a vector b on the bridges put $\mathbf{x}_{\mathrm{br}}(\mathsf{b}) = (0, 0, -\mathsf{b})$, the configuration with only the bridge links, and $\mathbf{x}(\mathsf{b}) = P_{\mathcal{K}^\perp}\mathbf{x}_{\mathrm{br}}(\mathsf{b})$. A configuration of $\mathcal{U}_1$ has $\bar{k} = \bar{m} =: t$ and, in each triangle, three equal residuals which sum to $-\mathfrak{h}_q = 0$ by orthogonality to $\mathcal{H}$, so its internal links are ∂t and its bridge links are $(\partial t)_f - \mathsf{b}_f$ with $\mathsf{b}_f = r_f$; it is therefore $(t, t, \partial t) + \mathbf{x}_{\mathrm{br}}(\mathsf{b})$ and, being orthogonal to $\mathcal{K}$, equals $\mathbf{x}(\mathsf{b})$. Conversely $\mathbf{x}_{\mathrm{br}}(\mathsf{b}) \in W_1 \cap \mathcal{H}^\perp$ and $\mathcal{K} \subseteq W_1 \cap \mathcal{H}^\perp$, so $\mathbf{x}(\mathsf{b}) \in \mathcal{U}_1$, and $r_f(\mathbf{x}(\mathsf{b})) = r_f(\mathbf{x}_{\mathrm{br}}(\mathsf{b})) = \mathsf{b}_f$ because residuals annihilate $\mathcal{K}$. Hence $\mathcal{U}_{\ell-1} = \{\mathbf{x}(\mathsf{b}) : \operatorname{supp}\mathsf{b} \subseteq S\}$ with $\mathsf{b} \mapsto \mathbf{x}(\mathsf{b})$ injective, and from $\langle (t,t,\partial t), (t',t',\partial t')\rangle = t^\top(2I + \partial^\top\partial)t'$ and $\langle (t,t,\partial t), \mathbf{x}_{\mathrm{br}}(\mathsf{b})\rangle = -t^\top\partial_S^\top\mathsf{b}$ the Gram matrix is $\langle \mathbf{x}(\mathsf{b}), \mathbf{x}(\mathsf{b}')\rangle = \mathsf{b}^\top \mathrm{G}\mathsf{b}'$ with $\mathrm{G} = I - \partial_S(2I + \partial^\top\partial)^{-1}\partial_S^\top$ on $\mathbb{R}^S$. The functional $\mathsf{b} \mapsto \mathsf{b}_f$, $f \in S$, has the Riesz representative $\mathbf{x}(\mathsf{b}^{(f)})$ with $\mathsf{b}^{(f)}$ the fth column of G^{-1}, so $P_{\mathcal{U}_{\ell-1}}\mathbf{r}_f = \mathbf{x}(\mathsf{b}^{(f)})$ and $\mathrm{M}_{fe} = (\mathrm{G}^{-1})_{fe}$. Since $2I + \partial^\top\partial - \partial_S^\top\partial_S = 2I + \partial_{\setminus S}^\top\partial_{\setminus S}$, with $\partial_{\setminus S}$ the incidence operator of the graph with the bridges of S removed, is positive definite, the Woodbury identity gives $\mathrm{M}_{fe} = \delta_{fe} + \boldsymbol{\partial}_f^\top(2I + \partial_{\setminus S}^\top\partial_{\setminus S})^{-1}\boldsymbol{\partial}_e$. The graph with the bridges of S removed is the disjoint union of $2^{L-\ell+1}$ connected blocks of $2^{\ell-2}$ consecutive triangles, and the inverse is block diagonal with entrywise strictly positive blocks: on a block B with Laplacian $\partial_B^\top\partial_B$ and maximal degree D_B, the matrix $(\mathrm{D}_B + 2)I - (2I + \partial_B^\top\partial_B)$ is entrywise nonnegative, irreducible, and of spectral radius at most $\mathrm{D}_B < \mathrm{D}_B + 2$, since the Laplacian's eigenvalues lie in $[0, 2\mathrm{D}_B]$, so the Neumann series of $(2I + \partial_B^\top\partial_B)^{-1}$ in it converges and is entrywise positive. The bridges of S are those of index $j = \mathrm{w}2^{\ell-2}$, $1 \leq \mathrm{w} \leq 2^{L-\ell+1} - 1$, of level $\ell + v_2(\mathrm{w})$, the bridge at position w joining block w to block $\mathrm{w}+1$; level ℓ means w odd and level $\ell+1$ means $\mathrm{w} \equiv 2 \pmod 4$. For $f \neq e$ in S, with positions $\mathrm{w}_f, \mathrm{w}_e$, the entry $\boldsymbol{\partial}_f^\top(2I + \partial_{\setminus S}^\top\partial_{\setminus S})^{-1}\boldsymbol{\partial}_e$ is a sum over the blocks both bridges touch, so it vanishes unless $|\mathrm{w}_f - \mathrm{w}_e| = 1$, and then it is plus or minus one entry of the positive inverse on the shared block, hence nonzero. Pair each level-$(\ell+1)$ bridge f with the level-ℓ bridge $e(f)$ at position $\mathrm{w}_f + 1$, which exists because 4 divides $2^{L-\ell+1}$; then $\mathrm{M}_{f,e(f')} \neq 0$ exactly when $|\mathrm{w}_f - \mathrm{w}_{f'} - 1| = 1$, that is $\mathrm{w}_f - \mathrm{w}_{f'} \in \{0, 2\}$, and as $\mathrm{w}_f - \mathrm{w}_{f'} \in 4\mathbb{Z}$ exactly when $f = f'$. The square submatrix $(\mathrm{M}_{f,e(f')})_{f,f'}$ is diagonal with nonzero entries, so the rows of M of level $\ell+1$ are independent and $\mathsf{B}_{\ell+1,\ell}$ has full row rank; as $\mathsf{B}_{\ell+1,\ell+1}\mathsf{B}_{\ell+1,\ell+1}^\top$ is invertible and $\mathsf{B}_{\ell+1,\ell}^\top$ is injective, $\mathfrak{Y}_\ell$ is injective. Nothing used the choice of bridge endpoints.

Slaving on the active interval. Let level $\ell - 1$ be active on $[t_0, t_\ell^{\mathrm{act}}]$ with $t_0 = t_{\ell-1}^{\mathrm{act}}$, with mean $\bar{\mathbf{x}}(t)$ in the layers $r \geq \ell$, $\dot{\bar{\mathbf{x}}} = -H_{\ell-1}\bar{\mathbf{x}}$, and $|\bar{\mathbf{x}}(t_0)| \leq \mathrm{CA}$, which holds because $|\bar{\mathbf{x}}(0)| = \mathrm{A}|\mathbf{v}_0|$, the flow contracts $|\bar{\mathbf{x}}|$, and the projections do not increase it. Write $\mathbf{g} = H_{\ell-1}\bar{\mathbf{x}}$, so that $\mathfrak{v} = \mathbf{g}_z$

and $\mathrm{T}\Delta_{\ell-1:\ell} = \frac{1}{2}\mathbf{g}_z^\top H_{zz}^{-1}\mathbf{g}_z$ by eq. (14.34), and $\mathrm{b}_{\ell+1}(t) = B_{\ell+1}P_\ell\bar{\mathbf{x}}(t) = \mathrm{B}_{\ell+1,\ell+1}\bar{\mathbf{x}}_{\ell+1}(t)$, since P_ℓ removes the layer ℓ and $\mathrm{B}_{\ell+1,r} = 0$ for $r > \ell+1$. The row bound gives $|\dot{\bar{\mathbf{x}}}_{\ell+1}| = |\mathbf{g}_{\ell+1}| \le \mathrm{C}\mathrm{J}_{\ell+1}\mathrm{A}$, and the same bound at the earlier levels, whose projections P_p, $p \le \ell$, leave the layer $\ell+1$ unchanged, gives the second line of eq. (14.36). Since $\dot{\mathbf{g}} = -H_{\ell-1}\mathbf{g}$, the variable $\mathbf{u} = \mathbf{g}_z + H_{zz}^{-1}H_{zy}\mathbf{g}_y$ satisfies $\dot{\mathbf{u}} = -H_{zz}\mathbf{u} - H_{zz}^{-1}H_{zy}(H_{\ell-1}\mathbf{g})_y$, where $|(H_{\ell-1}\mathbf{g})_y| \le \mathrm{C}\mathrm{J}_{\ell+1}(|\mathbf{g}_z| + |\mathbf{g}_y|)$, $|\mathbf{g}_y| \le \mathrm{C}\mathrm{J}_{\ell+1}\mathrm{A}$, and $|\mathbf{g}_z| \le |\mathbf{u}| + \mathrm{C}\varrho_\ell\mathrm{J}_{\ell+1}\mathrm{A}$. Variation of constants with $\|e^{-H_{zz}s}\| \le e^{-\lambda s}$, $\lambda = \lambda_{\min}(H_{zz})$, gives

$$|\mathbf{u}(t)| \le e^{-\lambda(t-t_0)}|\mathbf{u}(t_0)| + \int_{t_0}^{t} e^{-\lambda(t-s)}\big(\mathrm{C}\varrho_\ell^2\mathrm{J}_\ell|\mathbf{u}(s)| + \mathrm{C}\varrho_\ell\mathrm{J}_{\ell+1}^2\mathrm{A}\big)ds, \tag{C.93}$$

and iterating the positive Volterra operator of the right side compares $|\mathbf{u}|$ with the solution of the scalar equation with decay rate $\lambda - \mathrm{C}\varrho_\ell^2\mathrm{J}_\ell$, constant forcing $\mathrm{C}\varrho_\ell\mathrm{J}_{\ell+1}^2\mathrm{A}$, and initial value $|\mathbf{u}(t_0)|$, so that $|\mathbf{u}(t)| \le e^{-(\lambda-\mathrm{C}\varrho_\ell^2\mathrm{J}_\ell)(t-t_0)}|\mathbf{u}(t_0)| + \mathrm{C}\varrho_\ell\mathrm{J}_{\ell+1}^2\mathrm{A}/(\lambda - \mathrm{C}\varrho_\ell^2\mathrm{J}_\ell)$. For $\bar{\varrho}_\ell \le \mathrm{c}_0$, $\lambda - \mathrm{C}\varrho_\ell^2\mathrm{J}_\ell \ge \frac{1}{2}\mathrm{j}_\ell\mathrm{J}_\ell$, and $|\mathbf{u}(t_0)| \le \mathrm{C}\mathrm{J}_\ell\mathrm{A}$, so $|\mathbf{u}(t)| \le \mathrm{C}(\mathrm{J}_{\ell+1}^2\mathrm{A}/\mathrm{J}_\ell)[\varrho_\ell + \mathfrak{x}_\ell(t)]$. For the forcing, $H_{zy}\mathbf{g}_y = H_{z,\ell+1}\mathbf{g}_{\ell+1} + \sum_{r\ge\ell+2} H_{z,r}\mathbf{g}_r$, the sum being at most $\mathrm{C}\mathrm{J}_{\ell+2}^2\mathrm{A} \le \mathrm{C}\mathrm{J}_{\ell+1}^2\varrho_{\ell+1}^2\mathrm{A}$; from $\mathbf{g}_z = H_{zz}\bar{\mathbf{x}}_z + H_{zy}\bar{\mathbf{x}}_y$, $|\bar{\mathbf{x}}_z| \le \mathrm{C}\mathrm{A}(\varrho_\ell + \mathfrak{x}_\ell)$, and eq. (C.92) then gives $\mathbf{g}_{\ell+1} = \mathrm{J}_{\ell+1}[\mathrm{B}_{\ell+1,\ell+1}^\top\mathrm{b}_{\ell+1} + O(\mathrm{A}(\bar{\varrho}_\ell + \mathfrak{x}_\ell))]$ and $H_{zy}\mathbf{g}_y = \mathrm{J}_{\ell+1}^2[\mathfrak{Y}_\ell\mathrm{b}_{\ell+1} + O(\mathrm{A}(\bar{\varrho}_\ell + \mathfrak{x}_\ell))]$. Writing $\mathbf{g}_z = -H_{zz}^{-1}H_{zy}\mathbf{g}_y + \mathbf{u}$ with $H_{zz}^{-1} = \mathrm{J}_\ell^{-1}[(\mathrm{B}_{\ell\ell}^\top\mathrm{B}_{\ell\ell})^{-1} + O(\bar{\varrho}_\ell)]$ and inserting in $\frac{1}{2}\mathbf{g}_z^\top H_{zz}^{-1}\mathbf{g}_z$, with $|\mathrm{b}_{\ell+1}| \le \mathrm{C}\mathrm{A}$, gives $\mathrm{T}\Delta_{\ell-1:\ell} = (\mathrm{A}^2\mathrm{J}_{\ell+1}^4/\mathrm{J}_\ell^3)[\frac{1}{2}|(\mathrm{B}_{\ell\ell}^\top\mathrm{B}_{\ell\ell})^{-3/2}\mathfrak{Y}_\ell\mathrm{b}_{\ell+1}/\mathrm{A}|^2 + O(\bar{\varrho}_\ell + \mathfrak{x}_\ell + \mathfrak{x}_\ell^2)]$, which is the first line of eq. (14.36) by the definition of $\mathfrak{y}_\ell$.

Windows. On I_ℓ, $\mathrm{J}_\ell(t-t_0) \ge \mathrm{J}_\ell(t_\ell - 2t_{\ell-1}) = k_L\log(1/\epsilon)(1-2\varrho_{\ell-1})$, so $\mathfrak{x}_\ell \to 0$ uniformly on I_ℓ under the stated condition, and $\mathrm{J}_{\ell+1}t \le 2k_L\log(1/\epsilon)\varrho_\ell \to 0$ gives $\mathrm{b}_{\ell+1}(t)/\mathrm{A} = B_{\ell+1}P_\ell\mathbf{v}_0 + o(1)$, so the quadratic $\mathfrak{y}_\ell$ satisfies $\mathfrak{y}_\ell(\bar{\mathbf{x}}(t)/\mathrm{A}) = \mathfrak{y}_\ell(\mathbf{v}_0) + o(1)$. For the lower bound, $|(\mathrm{B}_{\ell\ell}^\top\mathrm{B}_{\ell\ell})^{-3/2}\mathfrak{Y}_\ell\mathrm{b}_{\ell+1}| \ge \|\mathrm{B}_{\ell\ell}\|^{-3}\sigma_{\min}(\mathfrak{Y}_\ell)|\mathrm{b}_{\ell+1}|$, and $|B_{\ell+1}P_\ell\mathbf{v}_0|^2 = \sum_e r_e(P_\ell\mathbf{v}_0)^2$ over the $N_\triangle/2^\ell$ bridges e of level $\ell+1$ is at least $N_\triangle\mathrm{m}_L^2/2^{\ell+2}$ by the margin of Lemma C.2, which gives eq. (14.37).

Power laws. With $\mathrm{J}_s = \epsilon^{-\mathrm{Q}_s}$, $\varrho_s = \epsilon^{\mathrm{Q}_s - \mathrm{Q}_{s+1}} \to 0$, $k_L\log(1/\epsilon)\varrho_\ell \to 0$, and on I_ℓ

$$\mathfrak{x}_\ell \le \epsilon^{\mathrm{j}_\ell k_L(1-2\varrho_{\ell-1})/2 - 2(\mathrm{Q}_\ell - \mathrm{Q}_{\ell+1})},$$

which tends to zero exactly under the stated threshold on k_L; the window statement then places $\Delta_{\ell-1:\ell}$ between $\frac{1}{2}\mathfrak{y}_\ell(\mathbf{v}_0)$ and $2\mathfrak{y}_\ell(\mathbf{v}_0)$ times $(\mathrm{A}^2/\mathrm{T})\mathrm{J}_{\ell+1}^4/\mathrm{J}_\ell^3 = \epsilon^{3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} - \mathrm{Q}_\mathrm{T}}$ on I_ℓ for small ϵ. It is $O(\epsilon^{\mathrm{Q}_\triangle})$ on every early window exactly when $3\mathrm{Q}_\ell - 4\mathrm{Q}_{\ell+1} - \mathrm{Q}_\mathrm{T} \ge \mathrm{Q}_\triangle$ for every $\ell \le L-1$, that is $\mathrm{Q}_\ell + \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle \ge \frac{4}{3}(\mathrm{Q}_{\ell+1} + \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle)$, and iterating from $\mathrm{Q}_L = 0$ gives $\mathrm{Q}_1 + \mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle \ge (4/3)^{L-1}(\mathrm{Q}_\mathrm{T} + \mathrm{Q}_\triangle)$, with equality along the recursion; vanishing is the case $\mathrm{Q}_\triangle = 0$ with strict inequality. □

C.7 The endogenous four-cycle

Proof of Proposition 14.22. Normalization, free energy, and descent. Every factor of the joint is a normalized kernel, so the joint has mass one, $E = \mathbf{1}$ has positive probability, and every candidate source has positive prior and posterior mass at every finite configuration. The reference density is $(2\pi)^{-6}e^{-|\mathbf{x}|^2/2}$, the four likelihood factors contribute $4\mathrm{T}\log 2$, and eq. (14.12) with $\sigma_{\mathrm{ev}}^2 = 1/2$ adds

$\frac{\mathrm{T}}{2}\mathbf{x}^{\top}R_{\mathrm{ev}}\mathbf{x}$, so the scaled negative ELBO is $\mathrm{T}\mathbb{E}_q\log(q/\pi_S)+\mathbb{E}_qU_S+\mathrm{B_T}=\mathrm{T}D_{\mathrm{KL}}(q\|q^{\star})+\mathrm{B_T}-\mathrm{T}\log Z_{\mathrm{opt}}$, and the evidence is $e^{-\mathrm{B_T}/\mathrm{T}}Z_{\mathrm{phys}}$, the physical energy omitting R_{ev}. Every U_S is a positive definite quadratic form with Hessian at least $\mathrm{T}I$, whose gradient contains both the sending and the receiving appearances of each agent. Between proposals the process is a linear diffusion with one of sixteen matrices, and proposals are finitely many on bounded intervals, so it exists, is pathwise unique with common random inputs, and does not explode. Since $q^{\star}(\mathbf{x},S)\beta_{S'}(\mathbf{x})=q^{\star}_{\mathbf{X}}(\mathbf{x})\beta_S\beta_{S'}$ is symmetric in S,S', the dissipation identity eq. (14.28) at level zero, without a parameter term, shows that $\mathcal{F}_{\mathrm{T}}$ does not increase. A cycle automorphism, with link signs transformed under reflections, preserves the reference, the kernels, the rows, the kinetics, and the preparation class, so the law at every time is invariant.

Matching sector. Fix $S^{(0)}$, with selected edges $e=e_0$ and $f=e_2$; $S^{(1)}$ follows by rotation. For each edge the covectors $\mathbf{a}^b_g,\mathbf{a}^m_g$ of r^b_g,r^m_g have Gram matrix $\begin{pmatrix}3&1\\1&3\end{pmatrix}$, and the two selected edges have disjoint supports. The interaction Hessian of $V_{S^{(0)}}$ is $H_M=2\sum_{g\in\{e,f\}}(\mathbf{a}^b_g\mathbf{a}^{b\top}_g+\mathbf{a}^m_g\mathbf{a}^{m\top}_g)$, and the unit covectors

$$n^{+}_g=\frac{\mathbf{a}^b_g+\mathbf{a}^m_g}{2\sqrt{2}},\qquad n^{-}_g=\frac{\mathbf{a}^b_g-\mathbf{a}^m_g}{2},\qquad g\in\{e,f\},\tag{C.94}$$

are orthonormal eigenvectors with eigenvalues 8 and 4; the kernel of H_M has dimension eight. With $\sigma^2_{\mathrm{ev}}=1/2$, $R_{\mathrm{ev}}n^{+}_g=0$, $R_{\mathrm{ev}}n^{-}_g=4n^{-}_g$, and R_{ev} leaves the kernel invariant, so the held-matching drift matrix $K_M=H_M+\mathrm{T}(I+R_{\mathrm{ev}})$ has normal rates $8+\mathrm{T}$ (twice) and $4+5\mathrm{T}$ (twice), all rates are at least T, and the physical matrix has normal rates $8+\mathrm{T}$ and $4+\mathrm{T}$. With $\mathbf{p}=\sum_{g\in\{e,f\}}(\mathbf{a}^b_g+\mathbf{a}^m_g)$, the vector $\mathbf{x}^{\circ}+\mathbf{p}/4$ lies in the kernel of H_M, $H_M\mathbf{p}=8\mathbf{p}$, and R_{ev} annihilates $\mathbf{x}^{\circ}$ and $\mathbf{p}$, so the deterministic held path from $\mathbf{x}^{\circ}$ is

$$\bar{\mathbf{x}}(t)=e^{-\mathrm{T}t}\Big[\mathbf{x}^{\circ}+\frac{1-e^{-8t}}{4}\mathbf{p}\Big],\qquad r_{\mathrm{int}}(\bar{\mathbf{x}}(t))=-e^{-(8+\mathrm{T})t},\qquad r_{\mathrm{ext}}(\bar{\mathbf{x}}(t))=e^{-\mathrm{T}t}\Big[-\frac{3}{2}+\frac{e^{-8t}}{2}\Big],\tag{C.95}$$

in both channels, where r_{int} is a selected and r_{ext} an exterior residual; along it $\bar{k}=\bar{m}$, so the evaluator force vanishes.

Episode. Let $\mathscr{B}$ be the event that $|\mathbf{X}_0-\mathbf{x}^{\circ}|<\varrho_{\circ}$, $S_0=S^{(0)}$, and no proposal occurs on $[0,1/4]$. The centered Gaussian component has expected squared norm $12\sigma^2_{\circ}=\varrho^2_{\circ}/4$, so it puts mass at least $3/4$ on the ball, and with the mixture weight $1/2$, the bound $1+\mathfrak{z}\geq 1/2$, the assignment mass $1/16$, and the independent clock, $\Pr(\mathscr{B})\geq\frac{3}{256}e^{-1/4}$. On $[0,1/4]$ the process is the held Ornstein–Uhlenbeck process $\widetilde{\mathbf{X}}_t=e^{-K_Mt}\mathbf{X}_0+\mathbf{Z}_t$, $\mathbf{Z}_t=\sqrt{2\mathrm{T}}\int_0^t e^{-K_M(t-\sigma)}dW_{\sigma}$, and afterwards the two coincide until the first proposal returning another assignment. In an eigenbasis of K_M, integration by parts gives $\sup_{t\leq t^{\circ}_*}|Z^k_t|\leq 2\sqrt{2\mathrm{T}}\sup_{t\leq t^{\circ}_*}|W^k_t|$, and $\mathbb{E}\sup_{t\leq t^{\circ}_*}|W^k_t|^2\leq 4t^{\circ}_*$ by Doob's L^2 maximal inequality: with $W^*=\sup|W^k|$, optional stopping at the hitting time of a level $\mathrm{y}>0$ gives $\mathrm{y}\Pr(W^*\geq\mathrm{y})\leq\mathbb{E}[|W^k_{t^{\circ}_*}|\mathbf{1}_{W^*\geq\mathrm{y}}]$, and integrating against $2d\mathrm{y}$ with truncation gives the bound. Summing twelve coordinates, $\mathbb{E}\sup|\mathbf{Z}|^2\leq 384\mathrm{T}t^{\circ}_*$, so with $\delta_{\epsilon}=\epsilon^2/(8\sqrt{3})$, Chebyshev's inequality gives $\Pr(\sup_{t\leq t^{\circ}_*}|\mathbf{Z}_t|>\delta_{\epsilon})\leq 73728\epsilon^4t^{\circ}_*$. On $\mathscr{B}$ and the complementary noise event, contraction gives $|\widetilde{\mathbf{X}}_t-\bar{\mathbf{x}}(t)|\leq\varrho_{\circ}+\delta_{\epsilon}$. Every residual covector has norm $\sqrt{3}$, so on $\mathscr{B}$ each residual lies within $\sqrt{3}\varrho_{\circ}$ of -1 at time zero and each internal belief relative entropy exceeds $(1-\sqrt{3}/100)^2/2>1/4$,

which is the entrance clause of (a); the restoring rates of (a) are those of K_M computed above. Since $\mathrm{T}t_*^\circ < 1/100$, for $1/4 \le t \le t_*^\circ$ the selected residuals have magnitude below $e^{-2} + \sqrt{3}(\varrho_\circ + \delta_\epsilon) < 1/5$ and the exterior ones above $e^{-\mathrm{T}t_*^\circ}(3/2 - e^{-2}/2) - \sqrt{3}(\varrho_\circ + \delta_\epsilon) > 13/10$. Each receiver's exterior cost therefore exceeds its partner cost by more than one, its posterior row gives its exterior source probability at most $e^{-1/\mathrm{T}}$, and a proposal leaves $S^{(0)}$ with probability at most $4e^{-1/\mathrm{T}}$; for the rate-one clock the expected number of departing proposals up to t_*° is at most $4t_*^\circ e^{-1/\mathrm{T}}$, with self-proposals unrestricted. For $t \ge t_\epsilon$, every residual annihilates the kernel of H_M, so a selected residual is at most $e^{-8t} + \sqrt{3}\varrho_\circ e^{-4t} + \sqrt{3}\delta_\epsilon \le \epsilon^{16} + \sqrt{3}\varrho_\circ \epsilon^8 + \epsilon^2/8 \le \epsilon^2$, the belief relative entropy $(r^b)^2/2$ and the model relative entropy $(r^m)^2/(2\sigma_{\mathrm{ev}}^2)$ are at most ϵ^4, and every exterior belief relative entropy exceeds $(13/10)^2/2 > 1/2$, the model ones being larger. Let $\mathscr{A}_t$ be the prefix event that the noise stays within δ_ϵ up to t and every proposal in $[1/4, t]$ returns $S^{(0)}$; then $\Pr(\mathscr{A}_{t_*^\circ}^c \mid \mathscr{B}) \le p_\epsilon$. Since $\epsilon^4 t_*^\circ$ increases on $(0, 1/100]$, the first term of p_ϵ is at most $73728 \cdot 10^{-8}(2\log 100 + 1) < 0.008$, and $e^{-1/\mathrm{T}} < \mathrm{T}$ makes the second below 10^{-14}, so $p_\epsilon < 1/2$ and $\Pr(\mathscr{B} \cap \mathscr{A}_{t_*^\circ}) \ge \frac{3}{256} e^{-1/4}(1 - p_\epsilon) \ge p_0$. The stopping rule is prefix-measurable, and the same argument holds for $S^{(1)}$ and at the reflected center, which proves the probability bound of (a).

Stationary comparison. On each sheet the diffusion with drift $-\nabla U_S$ and covariance $2\mathrm{T}I$ is reversible for $e^{-U_S/\mathrm{T}}d\mathbf{x}$, and the symmetric measure $q^\star(\mathbf{x}, S)\beta_{S'}(\mathbf{x})$ makes the proposal step reversible for $q^\star$, so $q^\star$ is invariant. Under $q^\star$, given $S_0 = M$, the configuration is $\mathcal{N}(0, \mathrm{T}K_M^{-1})$, and for a matched edge e with belief covector $\mathbf{a}_e^b$, $K_M \succeq H_M \succeq 2\mathbf{a}_e^b \mathbf{a}_e^{b\top}$, so $r_e^b = \mathbf{a}_e^b \cdot \mathbf{x}$ has variance $\mathrm{T}\mathbf{a}_e^{b\top} K_M^{-1} \mathbf{a}_e^b \le \mathrm{T}/2$ and $|r_e^b| > 1/\sqrt{2}$ has probability at most $2e^{-1/(2\mathrm{T})}$. The event of (a) requires S_0 to be a matching whose matched pairs carry internal belief relative entropy above $1/4$ at time zero, that is $|r_e^b| > 1/\sqrt{2}$ on a matched edge, so under the process started from $q^\star$ it has probability at most $2 \cdot 2e^{-1/(2\mathrm{T})} \le 4e^{-1/(4\mathrm{T})}$, and $4e^{-1/(4\mathrm{T})} < p_0/2$ because $\mathrm{T} \le 10^{-16}$. This completes (a).

Parent channel. The covector of $\sum_i \omega_i$ pairs to zero with n_g^- and to $-1/\sqrt{2}$ with n_g^+, so it annihilates Z_M and the loop is a function of $P_{W_M}\mathbf{x}$; W_M, of dimension nine, contains the kernel of H_M and the fast common-normal direction $(n_e^+ + n_f^+)/\sqrt{2}$. Both H_M and R_{ev} preserve W_M and Z_M, so on a matching fiber the hidden optimizing conditional is $\nu_{\mathrm{T}} = \mathcal{N}(0, \mathrm{T}K_Z^{-1})$ with $K_Z = \mathrm{diag}(8 + \mathrm{T}, 4 + 5\mathrm{T}, 4 + 5\mathrm{T})$, independent of the retained state, and the physical one has precision $\mathrm{diag}(8{+}\mathrm{T}, 4{+}\mathrm{T}, 4{+}\mathrm{T})$. The channel C is one deterministic measurable kernel onto the disjoint union $\coprod_S \{S\} \times W_S$, with Lebesgue measure on each W_S and counting measure on S; its pushforwards of the joint, the physical posterior, the optimizer, and the recognition law are normalized, it keeps every assignment, and the chain rule of Theorem 10.7 splits $D_{\mathrm{KL}}(q\|q^\star)$ into the retained term and the mean discarded conditional term. This proves (b).

Discarded conditional free energy. Under the held process the hidden coordinates are $z_t = e^{-K_Z t} z_0 + \xi_t$ with $\xi_t \sim \mathcal{N}(0, \mathrm{T}K_Z^{-1}(I - e^{-2K_Z t}))$ independent of the retained coordinates and of the initial data, and $|z_0| \le \varrho_\circ$ on $\mathscr{B}$ because $\mathbf{x}^\circ$ has zero hidden component. Given the retained state, the hidden law is a mixture over z_0 of these Gaussians, and for one coordinate with rate λ_k and $a = e^{-2\lambda_k t}$ the relative entropy to the stationary law is $\frac{1}{2}[-a - \log(1-a) + \lambda_k a z_0^2/\mathrm{T}]$. Convexity of relative entropy in its first argument, $4 \le \lambda_k \le 9$, and $-a - \log(1-a) \le a^2/(2(1-a))$ bound the mean conditional relative entropy of the held law conditioned on $\mathscr{B}$ by $\frac{9\varrho_\circ^2}{2\mathrm{T}} e^{-8t} + \frac{3}{4(1-e^{-8})} e^{-16t} \le \eta_\epsilon$ for $t \ge t_\epsilon$.

To pass to the episode, let $\widehat{Q}$ be this held law, $\mathscr{R} = \widehat{Q}_Y \otimes \nu_{\mathrm{T}}$, and $Q_{\mathscr{A}}$, $Q_{\mathscr{A}^c}$ its conditionals on $\mathscr{A}_t$ and its complement, of probabilities $1-p$ and $p \le p_\epsilon$. The relative entropy of the joint law of $(\mathbf{1}_{\mathscr{A}_t}, Y, Z)$ to $\mathrm{Bernoulli}(1-p) \otimes \mathscr{R}$, computed in both orders, gives $(1-p)D_{\mathrm{KL}}(Q_{\mathscr{A}}\|\mathscr{R}) + pD_{\mathrm{KL}}(Q_{\mathscr{A}^c}\|\mathscr{R}) = D_{\mathrm{KL}}(\widehat{Q}\|\mathscr{R}) + \mathbb{I}(\mathbf{1}_{\mathscr{A}_t}; (Y,Z))$, and the mutual information is at most $\mathcal{S}_2(p)$. Dropping the second term on the left and the nonnegative retained-marginal term of $D_{\mathrm{KL}}(Q_{\mathscr{A}}\|\mathscr{R})$, and using that $\mathcal{S}_2$ increases on $[0,1/2]$, bounds the mean discarded optimizing conditional relative entropy of $Q_{\mathscr{A}}$ by $\bar{\eta}_\epsilon$; on $\mathscr{B}\cap\mathscr{A}_t$ the held and native processes coincide, so $Q_{\mathscr{A}}$ is the episode law. The physical parent conditions the discarded coordinates also on the evaluator samples, so compare with the normalized law $\Pi(dz \mid y)P_{\mathrm{ev}}(dm, dk \mid \mathbf{x}) \exp[(\mathfrak{d}_y \cdot (k-m) - |\mathfrak{d}_y|^2/2)/(1-\sigma_{\mathrm{ev}}^2)]$, where $\mathfrak{d}_y = \mathfrak{D}y$ and $\mathfrak{d}_z = \mathfrak{D}z$ are orthogonal; the multiplier has expectation one and depends only on retained variables, so this law has the physical hidden conditional given (y,m,k). The recognition evaluator differs from it by the mean shift $\mathfrak{d}_z$, at relative entropy $|\mathfrak{d}_z|^2/(2(1-\sigma_{\mathrm{ev}}^2))$, and the chain rule in both orders bounds the mean discarded physical conditional relative entropy by $D_{\mathrm{KL}}(q(dz \mid y)\|\Pi(dz \mid y)) + \mathbb{E}_q|\mathfrak{d}_z|^2/(2(1-\sigma_{\mathrm{ev}}^2)) = D_{\mathrm{KL}}(q(dz \mid y)\|\nu_{\mathrm{T}}) + \log\frac{4+5\mathrm{T}}{4+\mathrm{T}}$, where $\log\frac{4+5\mathrm{T}}{4+\mathrm{T}} \le \frac{4\mathrm{T}}{4+\mathrm{T}} \le \mathrm{T}$. This proves (c).

Parent dynamics. The inherited proposal kernel

$$\beta_C((S,y),\cdot) = \int q^\star(d\mathbf{x} \mid C = (S,y)) \sum_{S'} \beta_{S'}(\mathbf{x}) \delta_{(S', P_{W_{S'}}\mathbf{x})}$$

is a probability kernel with the rate-one clock, reversible with respect to $C_\# q^\star$ because the symmetric measure $q^\star(d\mathbf{x}, S)\beta_{S'}(\mathbf{x})$ is pushed through (C,C); on a matching stratum the retained diffusion is the exact projection of the held process, and elsewhere C is the identity. On a matching fiber, with $z_+ = ((n_e^+ + n_f^+)/\sqrt{2}) \cdot y$ and the hidden coordinates z_1, z_2, z_3 of Z_M, the selected costs are $\mathcal{D}_e = (z_+ + z_1)^2 + z_2^2$ and $\mathcal{D}_f = (z_+ - z_1)^2 + z_3^2$, and each exterior common residual $(r_g^b + r_g^m)/2$ annihilates Z_M and satisfies $\mathcal{D}_g \ge ((r_g^b + r_g^m)/2)^2$. If both exterior common residuals have magnitude at least $\mathrm{r}_\circ$, each receiver chooses its exterior source with probability at most $\exp[(\mathcal{D}_{\mathrm{in}} - \mathcal{D}_{\mathrm{out}})/\mathrm{T}]$, and the Gaussian identity $\mathbb{E}e^{(a+Z)^2/\mathrm{T}} = (1-2\sigma^2/\mathrm{T})^{-1/2}\exp[a^2/(\mathrm{T}(1-2\sigma^2/\mathrm{T}))]$ with $\sigma^2 = \mathrm{T}/(8+\mathrm{T})$ and $\mathrm{T}/(4+5\mathrm{T})$, whose factors are at most $\sqrt{4/3}$ and $\sqrt{2}$, gives $\beta_C((S^{(0)},y), \{S' \ne S^{(0)}\}) \le 4\sqrt{8/3}\exp[-(\mathrm{r}_\circ^2 - 4z_+^2/3)/\mathrm{T}]$. On the episode $\mathrm{r}_\circ = 13/10$ and $|z_+| \le \epsilon^2$ on $[t_\epsilon, t_*^\circ]$, so this is at most $4\sqrt{8/3}e^{-1/\mathrm{T}}$. Condition on $\mathscr{B}\cap\mathscr{A}_{t_\epsilon}$ and let μ_{act} be the law of $C(\mathbf{X}_{t_\epsilon}, S_{t_\epsilon})$ under this prefix. The Brownian increments after t_ϵ are independent of the prefix, and the native future equals the held process until a departure or a tube exit, whose conditional probability is at most $\Pr(\mathscr{A}_{t_*^\circ}^c \mid \mathscr{B})/\Pr(\mathscr{A}_{t_\epsilon} \mid \mathscr{B}) \le p_\epsilon/(1-p_\epsilon)$. The inherited parent started from μ_{act} and driven by the same increments has the retained drift and diffusion of the projected held process until its first departure; the retained tube fails with conditional probability at most $p_\epsilon/(1-p_\epsilon)$, and on the tube the parent departs during the unit interval with probability at most $4\sqrt{8/3}e^{-1/\mathrm{T}}$. The union bound and the triangle inequality give (d). □

Proof of Proposition 15.4. Rows. For a continuous row path π whose entries lie in $[\mathrm{c}_0, 1-\mathrm{c}_0]$ for some $\mathrm{c}_0 \in (0,1/2]$, construct the kinetics pathwise as in the proof above: the configuration drift $-\nabla U_S$ does not involve the prior, and at a clock time τ receiver i takes source $i+1$ exactly when

a uniform variable on $[0,1]$, independent and fresh for each clock time and receiver, is at most its posterior row

$$\beta_{i,i+1}(\mathbf{x};\pi(\tau)) = \frac{\pi_{i,i+1}(\tau)e^{-\mathcal{D}_{e(i,i+1)}(\mathbf{x})/\mathrm{T}}}{\sum_{j=i\pm1}\pi_{ij}(\tau)e^{-\mathcal{D}_{e(i,j)}(\mathbf{x})/\mathrm{T}}}$$

at $\mathbf{x}=\mathbf{X}_{\tau-}$. Write q_t^{π} for the law at time t. Drive the constructions for two such row paths π,π' by the same initial data, Brownian motion, clock, and uniform variables. They coincide until the first clock time at which the uniform variable of some receiver falls between its posterior rows under π and π', and, given agreement before a clock time τ, this happens at τ with conditional probability at most $\sum_i|\beta_{i,i+1}(\mathbf{x};\pi(\tau))-\beta_{i,i+1}(\mathbf{x};\pi'(\tau))|$ at the common configuration. With $\mathrm{p}=\pi_{i,i+1}$ and $\pi_{i,i-1}=1-\mathrm{p}$, the derivative of a posterior row in p is $\beta_{i,i+1}(1-\beta_{i,i+1})/(\mathrm{p}(1-\mathrm{p}))\le 1/(4\mathrm{c}_0(1-\mathrm{c}_0))\le 1/(2\mathrm{c}_0)$, and the clock has rate one, so

$$\left|q_t^{\pi}(S_i=j)-q_t^{\pi'}(S_i=j)\right| \le \frac{2}{\mathrm{c}_0}\int_0^t \|\pi(s)-\pi'(s)\|_\infty ds,$$

where $\|\cdot\|_\infty$ is the maximum over receivers and sources. A solution of eq. (15.4) that starts uniform satisfies

$$\pi_{ij}(t)-\tfrac12 = \epsilon\int_0^t e^{-\epsilon(t-s)}\big(q_s(S_i=j)-\tfrac12\big)ds, \qquad \left|\pi_{ij}(t)-\tfrac12\right| \le \tfrac12\big(1-e^{-\epsilon t}\big)\le \tfrac{\epsilon t}{2},$$

because $0\le q_s(S_i=j)\le 1$; each row sums to one, and on $[0,t_1]$ its entries lie in $[\mathrm{c}_0,1-\mathrm{c}_0]$ with $\mathrm{c}_0=e^{-\epsilon t_1}/2$. Fix $t_1>0$ and let Γ send a row path π to the right-hand side of the first identity with $q_s=q_s^{\pi}$. It maps the closed set of continuous row paths on $[0,t_1]$ with $|\pi_{ij}(t)-1/2|\le(1-e^{-\epsilon t})/2$ into itself, and

$$\|\Gamma\pi(t)-\Gamma\pi'(t)\|_\infty \le \frac{2\epsilon t_1}{\mathrm{c}_0}\int_0^t\|\pi(s)-\pi'(s)\|_\infty ds, \qquad t\le t_1,$$

so the nth iterate of Γ contracts the supremum distance by the factor $(2\epsilon t_1^2/\mathrm{c}_0)^n/n!$, and Γ has exactly one fixed point in that set. Every solution lies in the set, solutions on nested intervals agree, and $s\mapsto q_s^{\pi}(S_i=j)$ is continuous because the clock has rate one, so the fixed point is differentiable and solves eq. (15.4). The rows are therefore a deterministic continuous function of time, and the row bound holds. This proves (a).

Seed events. The function $\epsilon t_*^{\circ}=\epsilon(2\log(1/\epsilon)+1)$ increases on $(0,1/100]$, with value $(1+4\log 10)/100<0.1022$ at $1/100$, so $t_*^{\circ}<2/\epsilon$ and, by (a), every row entry lies in $[0.4489,0.5511]$ on $[0,t_*^{\circ}]$, whence $\pi_{ik}(t)/\pi_{ij}(t)\le 0.5511/0.4489<1.23$ for the two entries of every row. For a sign $\sigma\in\{1,-1\}$ and a perfect matching $M\in\{S^{(0)},S^{(1)}\}$, let $\mathscr{B}_{\sigma,M}$ be the event that $|\mathbf{X}_0-\sigma\mathbf{x}^{\circ}|<\varrho_{\circ}$, $S_0=M$, and no proposal occurs on $[0,1/4]$. The initial configuration, the initial assignment, and the clock are independent, so

$$\Pr(\mathscr{B}_{\sigma,M}) = \frac{e^{-1/4}}{16}\mathcal{N}(\mathsf{m}_0,I_{12})\big\{|\mathbf{x}-\sigma\mathbf{x}^{\circ}|<\varrho_{\circ}\big\},$$

and the four events are disjoint because $|\mathbf{x}^{\circ}|=2>\varrho_{\circ}$.

Episode. Condition on $\mathscr{B}_{1,S^{(0)}}$. The configuration drift does not involve the rows, and the rows are deterministic, so the paragraph *Episode* of the proof above applies line by line from every starting point in the ball, whatever the law of $\mathbf{X}_0$ on it: the held Ornstein–Uhlenbeck process, whose noise bound involves only the Brownian motion; the contraction to the held path eq. (C.95); the residual bounds below $1/5$ and above $13/10$ on $[1/4, t_*^\circ]$, which make each receiver's exterior cost exceed its partner cost by more than one; and the relative-entropy bounds for $t \geq t_\epsilon$, all deterministic on the noise event. The rows enter only the exterior source probability: for a receiver i with partner j and exterior source k,

$$\beta_{ik}(\mathbf{x}; \pi(t)) \leq \frac{\pi_{ik}(t)}{\pi_{ij}(t)} \exp\Big[-\frac{\mathcal{D}_{e(i,k)}(\mathbf{x}) - \mathcal{D}_{e(i,j)}(\mathbf{x})}{\mathrm{T}}\Big] < 1.23 e^{-1/\mathrm{T}}$$

in place of $e^{-1/\mathrm{T}}$. A proposal therefore leaves $S^{(0)}$ with probability at most $5e^{-1/\mathrm{T}}$, the expected number of departing proposals up to t_*° is at most $5t_*^\circ e^{-1/\mathrm{T}}$, and the prefix event $\mathscr{A}_{t_*^\circ}$ of that paragraph fails with conditional probability at most

$$\tilde{p}_\epsilon = 73728\epsilon^4 t_*^\circ + 5t_*^\circ e^{-1/\mathrm{T}} < 0.008 + 10^{-14} < \tfrac{1}{2},$$

by the same monotonicity of $\epsilon^4 t_*^\circ$ and $e^{-1/\mathrm{T}} < \mathrm{T}$. On $\mathscr{B}_{1,S^{(0)}} \cap \mathscr{A}_{t_*^\circ}$, the block $\{0, 1\}$, of size two and joined to its complement by e_1 and e_3, carries on its internal edge e_0 a belief relative entropy above $(1 - \sqrt{3}\varrho_\circ)^2/2 > 1/4$ at time zero, because $|\mathbf{X}_0 - \mathbf{x}^\circ| < \varrho_\circ$ and every residual covector has norm $\sqrt{3}$; it has internal belief and model relative entropies at most ϵ^4 on $[t_\epsilon, t_*^\circ]$, an interval of length one inside $[0, 2/\epsilon]$, and both boundary belief relative entropies above $1/2$ on $[1/4, t_*^\circ]$, so contrast formation occurs.

The other seeds. A cycle automorphism g, with link signs transformed under reflections, acts on configurations by an orthogonal signed permutation O_g with $U_{gS}(\mathsf{O}_g\mathbf{x}) = U_S(\mathbf{x})$ and $\mathcal{D}_{e(gi,gj)}(\mathsf{O}_g\mathbf{x}) = \mathcal{D}_{e(i,j)}(\mathbf{x})$. With $(g\pi)_{gi,gj} = \pi_{ij}$, it therefore maps the kinetics with row path π from $(\mathbf{x}, S)$ to the kinetics with row path $g\pi$ from $(\mathsf{O}_g\mathbf{x}, gS)$, driven by the Brownian motion $\mathsf{O}_g W$ and by the receiver uniform variables relabeled by g, the variable of receiver i serving receiver gi and being replaced by one minus itself when g is a reflection. Indeed a reflection sends the source $i+1$ of receiver i to the source $gi-1$ of receiver gi, and the image receiver takes it exactly when the original variable is at most $\beta_{i,i+1}(\mathbf{x}; \pi) = \beta_{gi,gi-1}(\mathsf{O}_g\mathbf{x}; g\pi) = 1 - \beta_{gi,gi+1}(\mathsf{O}_g\mathbf{x}; g\pi)$, which is the rule of the image kinetics applied to one minus that variable, except on the null event that the variable equals its row; under the rotation the sources are carried along and the variable is kept. The relabeled family is again a family of independent uniform variables independent of the other inputs, so the image process has the law of the kinetics with rows $g\pi$, and $g\pi$ has the same ratio bound. The rotation $i \mapsto i+1$ fixes $\mathbf{x}^\circ$ and maps $S^{(0)}$ to $S^{(1)}$, the reflection $i \mapsto -i$ maps $(\mathbf{x}^\circ, S^{(0)})$ to $(-\mathbf{x}^\circ, S^{(1)})$, and their composition maps $(\mathbf{x}^\circ, S^{(0)})$ to $(-\mathbf{x}^\circ, S^{(0)})$, so the conditional bound holds on the other three seed events, with the images of the block. Summing over the four disjoint events,

$$\Pr(\text{contrast formation}) \geq (1 - \tilde{p}_\epsilon)\frac{2e^{-1/4}}{16} \sum_{\pm} \mathcal{N}(\mathsf{m}_0, I_{12})\big\{|\mathbf{x} \mp \mathbf{x}^\circ| < \varrho_\circ\big\},$$

which gives the first bound of (b) because $\tilde{p}_\epsilon < 1/2$. The Gaussian density on the ball about $\mathbf{x}^\circ$ is at least $(2\pi)^{-6}\exp[-(|\mathbf{x}^\circ - \mathrm{m}_0| + \varrho_\circ)^2/2]$, the ball has volume $\pi^6\varrho_\circ^{12}/6!$, and $(2\pi)^{-6}\pi^6/6! = 1/46080$, which gives the second bound after dropping the ball about $-\mathbf{x}^\circ$. Frozen rows have ratio one, so the argument and the bound hold without learning. The sub-event named in (b) is the event of part (a) of the proposition above with the block of the held matching. For fixed rows π the proposal step is reversible for $q^\star_\pi$ by the same symmetry $q^\star_\pi(\mathbf{x}, S)\beta_{S'}(\mathbf{x};\pi) = q^\star_{\pi,\mathbf{X}}(\mathbf{x})\beta_S(\mathbf{x};\pi)\beta_{S'}(\mathbf{x};\pi)$, the sheet diffusions do not involve the rows, and the conditional law $\mathcal{N}(0, \mathrm{T}K_M^{-1})$ of $q^\star_\pi$ given $S_0 = M$ does not depend on π, so the paragraph *Stationary comparison* above gives the invariance and the bound $4e^{-1/(4\mathrm{T})}$ for every π. This proves (b).

Invariant initial laws. Since O_g is orthogonal, g maps $\mathcal{N}(\mathrm{m}_0, I_{12})$ to $\mathcal{N}(\mathsf{O}_g\mathrm{m}_0, I_{12})$, so the initial law is invariant exactly when m_0 is fixed by the rotation and the reflection. The rotation fixes exactly the means whose belief, model, and link blocks are constant, and the reflection reverses every edge and hence the sign of a constant link block, which leaves $(\mathrm{c}_1\mathbf{1}_4, \mathrm{c}_2\mathbf{1}_4, 0)$. For such a mean, since the uniform assignment law and the uniform rows are also invariant, g maps the recognition law q_t of the rows π to the recognition law $g_\# q_t$ of the rows $g\pi$ from the same initial law, by the pathwise image of the paragraph *The other seeds*, whose relabeled driving inputs have the law of the original ones; and $g\pi$ solves eq. (15.4) with $g_\# q_t(S_{gi} = gj) = q_t(S_i = j)$, so $(g\pi, g_\# q)$ is a second solution from the same data and uniqueness in (a) gives $g\pi = \pi$. The reflection $j \mapsto 2i - j$ fixes receiver i and exchanges its sources $i \pm 1$, so $\pi_{i,i+1} = \pi_{i,i-1} = 1/2$. This proves (c). □

C.8 Depth one under exclusive source gating

Proof of Proposition 14.24. Throughout, $F = \mathrm{att}(S)$ and $W = W_F$ once a sheet S is fixed, P_W is the orthogonal projection onto W, and Q_W, $\mathsf{Q}_{W^\perp}$ are matrices whose columns form orthonormal bases of W and of $W^\perp$. Since H_S is the sum of the positive semidefinite maps $\mathbf{a}_{e(i,S_i)}^\top \mathbf{a}_{e(i,S_i)}$, its kernel is $\bigcap_{e\in F} \ker \mathbf{a}_e = W_F$ and its range is $W_F^\perp$, the span of the rows of $\mathbf{a}_F$, so $\mathrm{k}_S = \mathrm{rank}\,\mathbf{a}_F$. For every edge the Gram matrix of the two rows of $\mathbf{a}_e$ is $\left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, as in eq. (C.94), so $\|\mathbf{a}_e\| = 2$.

(a) Two bands. Let $M = H_S + \mathrm{T}\Omega$. By Weyl's inequality [57, Sec. 4.3], with eigenvalues in increasing order, $\lambda_k(H_S) + \mathrm{T}\lambda_{\min}(\Omega) \le \lambda_k(M) \le \lambda_k(H_S) + \mathrm{T}\|\Omega\|$ for every k. The first $2N + |E| - \mathrm{k}_S$ eigenvalues of H_S vanish, the remaining k_S lie in $[\mathrm{h}_S, \|H_S\|]$, and $\lambda_{\min}(\Omega) = \|\Omega^{-1}\|^{-1}$. The range was identified above, and π does not enter M.

(b) Gating wells. For each receiver, the row sums to one and every weight is at least $\min_j \pi_{ij}$, so $\min_j \pi_{ij} e^{-\min_j \mathcal{D}_{e(i,j)}/\mathrm{T}} \le \sum_j \pi_{ij} e^{-\mathcal{D}_{e(i,j)}/\mathrm{T}} \le e^{-\min_j \mathcal{D}_{e(i,j)}/\mathrm{T}}$; taking $-\mathrm{T}\log$ and summing over i gives eq. (14.56). Since V_S is a sum over receivers and π_S a product,

$$\sum_S \pi_S e^{-V_S/\mathrm{T}} = \prod_i \sum_{j\in\mathcal{N}_i} \pi_{ij} e^{-\mathcal{D}_{e(i,j)}/\mathrm{T}} = e^{-V_\pi/\mathrm{T}},$$

which is the configuration marginal up to the reference factor. If every edge of a covering set F has zero residual at $\mathbf{x}$, every receiver has an option of zero cost, so $V_{\min}(\mathbf{x}) = 0$; a common translation of the belief and model means of the agents of one component of (V, F), with the links

fixed, changes no residual of an edge of F, because the two endpoints of such an edge either both move or both stay, so $V_{\min}$ remains zero.

(c) Rank formula. In the coordinates $(\bar{k}, \mathfrak{d}, \omega)$ with $\mathfrak{d} = \bar{k} - \bar{m}$, the rows of $\mathbf{a}_e$ for $e : j \to i$ are $r_e^b = \bar{k}_i - \bar{k}_j - \omega_e$ and $r_e^m = r_e^b - (\mathfrak{d}_i - \mathfrak{d}_j)$; replacing the second by $r_e^m - r_e^b = -(\partial\mathfrak{d})_e$ is an invertible change within the rows of e. The rows r_e^b, $e \in F$, read only $(\bar{k}, \omega)$ and each has coefficient -1 on its own link coordinate and zero on every other, so they are linearly independent. The rows $(\partial\mathfrak{d})_e$, $e \in F$, read only $\mathfrak{d}$ and form the oriented incidence map of (V, F) on $\mathbb{R}^V$, whose kernel consists of the functions constant on each component, so their rank is $N - \mathrm{cc}(F)$. The two families act on disjoint coordinate blocks, so $\operatorname{rank} \mathbf{a}_F = |F| + N - \mathrm{cc}(F)$.

Minimum edge covers. Let $F = \mathrm{att}(S)$. Every vertex selects an incident edge, so F covers V and every component of (V, F) contains an edge; one edge from each component is a matching, so $\mathrm{cc}(F) \le \mathfrak{n}$, and a connected graph on c vertices has at least $\mathrm{c} - 1$ edges, so $|F| \ge N - \mathrm{cc}(F)$. Hence $\mathrm{k}_S \ge 2(N - \mathrm{cc}(F)) \ge 2(N - \mathfrak{n})$, with equality exactly when F is a forest with $\mathfrak{n}$ components and therefore $N - \mathfrak{n}$ edges. The same two inequalities show that every edge cover has at least $N - \mathfrak{n}$ edges, and a maximum matching together with one incident edge at each unmatched vertex, which exists because no vertex is isolated, is an edge cover with exactly $N - \mathfrak{n}$ edges, since the other endpoint of each added edge is matched, or else the matching could be enlarged. An edge cover with $N - \mathfrak{n}$ edges has at least $\mathfrak{n}$ components and hence exactly $\mathfrak{n}$, and is a forest; so the covering forests with $\mathfrak{n}$ components are exactly the minimum edge covers. In a minimum edge cover every edge has an endpoint of degree one, since otherwise removing it would leave a smaller cover, and a tree in which every edge has a leaf endpoint is a star, because two vertices of degree at least two would be joined by a path whose first edge has no leaf endpoint. Conversely, a star with center z is attended when every leaf selects z and z selects one leaf, which the source sets allow, so every minimum edge cover is some $\mathrm{att}(S)$. Adding to F an edge that joins two components raises $|F|$ by one and lowers $\mathrm{cc}(F)$ by one, hence raises the rank by two.

Sheet weights. Since $H_S/\mathrm{T} + \Omega \succ 0$, $Z_S = (2\pi)^{(2N+|E|)/2} \det(H_S/\mathrm{T} + \Omega)^{-1/2}$. In the basis $(\mathsf{Q}_{W^\perp}, \mathsf{Q}_W)$ the matrix $H_S/\mathrm{T} + \Omega$ has the blocks $H_1/\mathrm{T} + \Omega_{11}$, Ω_{10}, Ω_{01}, Ω_{00}, with $H_1 = \mathsf{Q}_{W^\perp}^\top H_S \mathsf{Q}_{W^\perp} \succeq \mathrm{h}_S I$ and $\Omega_{00} = \Omega|_W$. Eliminating the invertible block Ω_{00} by block row and column operations gives $\det(H_S/\mathrm{T} + \Omega) = \det \Omega_{00} \det(H_1/\mathrm{T} + \Sigma)$ with $\Sigma = \Omega_{11} - \Omega_{10}\Omega_{00}^{-1}\Omega_{01}$, and $0 \prec \Sigma \preceq \Omega_{11} \preceq \|\Omega\| I$. Then $\mathrm{T}\Sigma \preceq (\mathrm{T}\|\Omega\|/\mathrm{h}_S) H_1$, so $\det H_1 \le \det(H_1 + \mathrm{T}\Sigma) \le (1 + \mathrm{T}\|\Omega\|/\mathrm{h}_S)^{\mathrm{k}_S} \det H_1$, and $\det H_1 = \mathrm{pdet}\, H_S$. Since $\det(H_1/\mathrm{T} + \Sigma) = \mathrm{T}^{-\mathrm{k}_S} \det(H_1 + \mathrm{T}\Sigma)$, the determinant lies between $\mathrm{T}^{-\mathrm{k}_S}\mathrm{z}_S^{-2}$ and $(1 + \mathrm{T}\|\Omega\|/\mathrm{h}_S)^{\mathrm{k}_S}\mathrm{T}^{-\mathrm{k}_S}\mathrm{z}_S^{-2}$, which is eq. (14.57). The same block form shows that under $q_\Omega(\cdot \mid S)$, which is $\mathcal{N}(0, (H_S/\mathrm{T} + \Omega)^{-1})$, the coordinates $y = \mathsf{Q}_W^\top \mathbf{x}$ given $z = \mathsf{Q}_{W^\perp}^\top \mathbf{x}$ are Gaussian with precision Ω_{00}, the (W, W) block of the joint precision, which does not depend on T.

Merge cost. $q_\Omega(S) = \pi_S Z_S / \sum_{S'} \pi_{S'} Z_{S'} \le \pi_S Z_S / \sum_{S' \in \mathcal{M}} \pi_{S'} Z_{S'}$. The upper bound of eq. (14.57) for S and its lower bound for each $S' \in \mathcal{M}$, where $\mathrm{k}_{S'} = 2(N - \mathfrak{n})$ and $\mathrm{h}_{S'} \ge \mathrm{h}_*$, give $q_\Omega(S) \le \mathrm{T}^{(\mathrm{k}_S - 2(N-\mathfrak{n}))/2}(1 + \mathrm{T}\|\Omega\|/\mathrm{h}_*)^{N-\mathfrak{n}} \pi_S \mathrm{z}_S / \sum_{S' \in \mathcal{M}} \pi_{S'} \mathrm{z}_{S'}$. If $\mathrm{cc}(\mathrm{att}(S)) \le \mathfrak{n} - m$, then $\mathrm{k}_S \ge 2(N - \mathrm{cc}(\mathrm{att}(S))) \ge 2(N - \mathfrak{n}) + 2m$; with $\mathrm{T} \le 1$, summing over such S gives eq. (14.58).

(d) Block law. The sheets with $\mathrm{cc}(\mathrm{att}(S)) < \mathfrak{n}$ carry mass at most $\mathrm{K}_1\mathrm{T}$ by eq. (14.58). Fix a sheet with $\mathrm{cc}(F) = \mathfrak{n}$. No component of (V, F) contains two disjoint edges, since replacing that component's

edge in the matching of one edge per component would give $\mathfrak{n}+1$ disjoint edges. A connected graph in which no two edges are disjoint is a star or a triangle: fix an edge uv; every other edge meets $\{u,v\}$; if there are edges vw and uw' with $w\neq u$ and $w'\neq v$, then $w=w'$, and a further edge would have to meet $\{u,v\}$ and would then be disjoint from vw or from uw, so the graph is the triangle uvw; otherwise all edges share one vertex. Replacing each triangle component by a two-edge path on the same three vertices gives a covering forest of stars with $\mathfrak{n}$ components, a minimum edge cover by (c), so the partition of V into the components of F is a minimum-edge-cover partition. The coherent-block partition $\mathfrak{P}_{\mathrm{a}}(\mathbf{x})$ differs from it only if some attended edge has $|\mathbf{a}_e\mathbf{x}|>\mathrm{a}$ or some edge joining two components of F has $|\mathbf{a}_e\mathbf{x}|\le\mathrm{a}$. Under $q_\Omega(\cdot\mid S)$ the configuration is $\mathcal{N}(0,\mathrm{T}(H_S+\mathrm{T}\Omega)^{-1})$. For $e\in F$, $M=H_S+\mathrm{T}\Omega\succeq\mathbf{a}_e^\top\mathbf{a}_e$, so $\|\mathbf{a}_eM^{-1/2}\|\le1$ and the covariance $\mathrm{T}\mathbf{a}_eM^{-1}\mathbf{a}_e^\top$ of $\mathbf{a}_e\mathbf{x}$ is at most $\mathrm{T}I_2$; each of its two centered Gaussian coordinates has variance at most T, and $|\mathbf{a}_e\mathbf{x}|>\mathrm{a}$ forces one of them beyond $\mathrm{a}/\sqrt{2}$, an event of probability at most $2\cdot2e^{-\mathrm{a}^2/(4\mathrm{T})}$. For an edge $e\notin F$ joining two components, $\operatorname{rank}\mathbf{a}_{F\cup\{e\}}=\operatorname{rank}\mathbf{a}_F+\operatorname{rank}(\mathbf{a}_e|_W)$, because $\ker\mathbf{a}_{F\cup\{e\}}=W\cap\ker\mathbf{a}_e$; by (c) the restriction has rank two, so $\mathbf{a}_e^W=\mathbf{a}_e\mathsf{Q}_W$ has two nonzero singular values, the least of which is at least s_*. Writing $\mathbf{x}=\mathsf{Q}_{W^\perp}z+\mathsf{Q}_Wy$, the conditional law of y given z has precision $\Omega_{00}\preceq\|\Omega\|I$, so $\mathbf{a}_e^Wy$ is Gaussian in $\mathbb{R}^2$ with covariance $\mathbf{a}_e^W\Omega_{00}^{-1}\mathbf{a}_e^{W\top}\succeq\|\Omega\|^{-1}\mathbf{a}_e^W\mathbf{a}_e^{W\top}\succeq(\mathrm{s}_*^2/\|\Omega\|)I_2$ and density at most $(2\pi)^{-1}\|\Omega\|/\mathrm{s}_*^2$. The event $|\mathbf{a}_e\mathbf{x}|\le\mathrm{a}$ places $\mathbf{a}_e^Wy=\mathbf{a}_e\mathbf{x}-\mathbf{a}_e\mathsf{Q}_{W^\perp}z$ in a disc of radius a about a point determined by z, so its conditional probability is at most $\pi\mathrm{a}^2\cdot\|\Omega\|/(2\pi\mathrm{s}_*^2)$. A union bound over at most $|E|$ edges of each kind, integrated over the sheets, gives eq. (14.59). □

C.9 Proof of Pairing Selection

Proof of Theorem 15.5. Write $p_i=\pi_{i,i+1}$, encode the assignment by $\sigma_i=+1$ when receiver i selects $i+1$, and put $\varepsilon=\epsilon_{\mathrm{L}}$. The state remains $(\mathbf{x},\sigma)\in\mathbb{R}^{12}\times\{\pm1\}^4$, with the shared edge costs of eq. (14.50). The multiplicity of edge e is $m_e(\sigma)=1_{\{\sigma_e=+1\}}+1_{\{\sigma_{e+1}=-1\}}$. In this proof define

$$\Omega=I+R_{\mathrm{ev}},\qquad \mathsf{A}_\sigma=H_\sigma+\mathrm{T}\Omega,\qquad H_\sigma=\nabla^2V_\sigma,\qquad \mathsf{A}_\sigma\succeq\mathrm{T}I.$$

Let $\mathsf{w}_\sigma=\det(\mathsf{A}_\sigma/\mathrm{T})^{-1/2}$, $\mathsf{Z}(p)=\sum_\sigma\pi_\sigma(p)\mathsf{w}_\sigma$, and

$$\mu_p(\mathbf{x},\sigma)=\frac{\pi_\sigma(p)}{(2\pi)^6\mathsf{Z}(p)}\exp\left[-\frac{\mathbf{x}^\top\mathsf{A}_\sigma\mathbf{x}}{2\mathrm{T}}\right],\qquad \pi_\sigma(p)=\prod_i p_i^{1_{\{\sigma_i=+1\}}}(1-p_i)^{1_{\{\sigma_i=-1\}}}.$$

Each held-sheet diffusion preserves its Gaussian factor, and a clock ring resamples σ from its conditional law under μ_p. Thus μ_p is invariant for the frozen rows, including boundary rows on their supported sheets. Set

$$\bar\beta_i(p)=\mu_p(\sigma_i=+1),\qquad \mathfrak{g}_i(p)=\partial_{p_i}\log\mathsf{Z}(p),\qquad \mathscr{V}_i(p)=\bar\beta_i(p)-p_i=p_i(1-p_i)\mathfrak{g}_i(p).\tag{C.96}$$

The last identity follows by differentiating the product prior. Since Z is a positive multilinear polynomial on the closed cube, $\mathfrak{g}$ and $\mathscr{V}$ extend analytically to a neighborhood of that cube and have bounded derivatives there. The well-posedness argument for proposition 15.4 applies at every

fixed $\mathrm{T} > 0$; it gives deterministic continuously differentiable rows and

$$\min_i\{p_i(t), 1-p_i(t)\} \geq \tfrac{1}{2}e^{-\varepsilon t}, \qquad \|p(t) - \tfrac{1}{2}\|_\infty \leq \tfrac{1}{2}\varepsilon t. \tag{C.97}$$

All constants below may depend on the fixed temperature. Constants in bounds on a fixed initial law may also depend on its mean, whereas the window constant in the conclusion will depend only on the temperature.

Frozen convergence and the initial imprint

Here $\mathrm{Ent}_\nu(g) = \int g \log(g/\int g d\nu) d\nu$ for nonnegative g, and total variation is $\sup_A |q(A) - \nu(A)|$. For the time-one frozen kernel, the function $\mathsf{V}(\mathbf{x}) = |\mathbf{x}|^2$ satisfies

$$\mathbb{E}_{\mathbf{x}}|\mathbf{X}_t|^2 \leq e^{-2\mathrm{T}t}|\mathbf{x}|^2 + 12(1-e^{-2\mathrm{T}t}), \qquad P_1^p\mathsf{V} \leq e^{-2\mathrm{T}}\mathsf{V} + 12. \tag{C.98}$$

Indeed, jumps leave $\mathbf{X}$ unchanged, and the diffusion generator applied to its squared norm is at most $24\mathrm{T} - 2\mathrm{T}|\mathbf{X}|^2$. Choose $R > 24/(1-e^{-2\mathrm{T}})$. For any frozen p, some assignment $\sigma_*(p)$ has $\pi_{\sigma_*} \geq 1/16$. Restrict the rate-one clock to exactly one ring in $[0, 1/2]$ and none afterward. Its probability is $e^{-1}/2$. Starting with $|\mathbf{x}| \leq \sqrt{R}$, the first held Gaussian motion is in the ball of radius $r' = \sqrt{R} + \sqrt{24\mathrm{T}}$ at that ring with probability at least $1/2$. On the ball, the probability to redraw σ_* is at least $16^{-1}e^{-v(r')/\mathrm{T}}$, where $v(r') = \max_{|\mathbf{z}|\leq r',\sigma} V_\sigma(\mathbf{z}) < \infty$. The subsequent held Gaussian transition lasts between $1/2$ and 1. Its means are bounded and its covariance matrices have common positive lower and finite upper bounds. Its density consequently has a positive common lower bound on the unit ball. Hence, for some $\alpha_\mathrm{T} > 0$ independent of p,

$$P_1^p((\mathbf{x},\sigma),\cdot) \geq \alpha_\mathrm{T}\,\mathrm{Unif}(B_1) \otimes \delta_{\sigma_*(p)} \quad \text{when } |\mathbf{x}|^2 \leq R.$$

These are the drift and minorization hypotheses of Hairer and Mattingly [44, Assumptions 1–2 and Theorems 1.2–1.3]. The explicit dependence in their Theorem 1.3 makes the resulting constants uniform in the frozen rows, even though the minorizing measure depends on p. Thus μ_p is the unique invariant law and

$$|P_t^p\phi(\mathbf{x},\sigma) - \mu_p\phi| \leq \mathrm{K}_\mathrm{T}e^{-\lambda_\mathrm{T}t}(1+|\mathbf{x}|^2)\|\phi\|_\infty. \tag{C.99}$$

Intermediate times follow by applying the skeleton estimate to $P_r^p\phi$, $0 \leq r < 1$.

Put $\mathbf{v} = (1,-1,1,-1)$, $f_\mathbf{v}(\sigma) = \frac{1}{2}\sum_i \mathbf{v}_i\sigma_i$, and define

$$\mathsf{I}_i(\mathsf{m}_0) = \int_0^\infty [q_t^0(\sigma_i = +1) - \tfrac{1}{2}]dt, \qquad \mathsf{j}_\mathrm{T}(\mathsf{m}_0) = \mathbf{v}\cdot\mathsf{I}(\mathsf{m}_0). \tag{C.100}$$

At uniform rows, the stationary law is invariant under cycle rotations and reflections, so each stationary row is $1/2$ and $\mu_{1/2}f_\mathbf{v} = 0$. Estimate eq. (C.99) proves absolute convergence of these

integrals. It also gives a Poisson integral

$$\mathfrak{R}_{\mathrm{T}}(\mathbf{x},\sigma)=\int_0^\infty P_t^{1/2} f_{\mathbf{v}}(\mathbf{x},\sigma)dt, \qquad |\mathfrak{R}_{\mathrm{T}}(\mathbf{x},\sigma)|\le \mathrm{K}_{\mathrm{T}}(1+|\mathbf{x}|^2). \tag{C.101}$$

Convolution of this integral with the initial Gaussian expresses j_{T} as an entire function of its complexified mean: for every compact set of complex means, all differentiated integrands are bounded by an integrable polynomial times a Gaussian. The same holds for I. A rotation by one changes the sign of j_{T}.

For $t\le 1/(10\varepsilon)$, all row entries are in $[\mathrm{c}_0,1-\mathrm{c}_0]$, $\mathrm{c}_0=0.45$. A posterior receiver probability has derivative at most $1/[4\mathrm{c}_0(1-\mathrm{c}_0)]\le 1/(2\mathrm{c}_0)$ with respect to its prior entry. Coupling the common clocks and their four posterior rows therefore gives

$$\|q_t^{\varepsilon}-q_t^{0}\|_{\mathrm{TV}}\le \frac{2}{\mathrm{c}_0}\int_0^t \|p(s)-\tfrac12\|_\infty ds\le \frac{\varepsilon t^2}{2\mathrm{c}_0}.$$

Variation of constants in the learning equation, followed by eq. (C.99), gives

$$\left|p(t)-\tfrac12-\varepsilon \mathsf{I}(\mathsf{m}_0)\right|_\infty \le \mathrm{K}_{\mathrm{T},\mathsf{m}_0}(\varepsilon e^{-\lambda_{\mathrm{T}} t}+\varepsilon^2 t)+\frac{\varepsilon^2 t^3}{6\mathrm{c}_0}. \tag{C.102}$$

In particular, at $t_1=\varepsilon^{-1/4}$, $p(t_1)-1/2=\varepsilon \mathsf{I}(\mathsf{m}_0)+o(\varepsilon)$.

Nonvanishing of the matching imprint

Take $\mathbf{u}_\circ=(0_4,0_4,(0,1,0,-1))$, whose two residual vectors are $(0,-1,0,1)$. In the noiseless switched motion the subspace $\bar{k}=\bar{m}$ is invariant. After removing the scalar factor $e^{-\mathrm{T}t}$, its common residual vector satisfies

$$\dot\rho=-\mathsf{G}M_\sigma\rho, \qquad \mathsf{G}=4I-A_{C_4}, \qquad M_\sigma=\mathrm{diag}(m_e(\sigma)). \tag{C.103}$$

Here the cyclic difference matrix contributes $2I-A_{C_4}$ and the shared-link drift contributes $2I$. Of the sixteen assignments, six have $m_1=m_3$. Their first held motion is $e^{-ct}(0,-1,0,1)$, with $c=4m_1\in\{0,4,8\}$. At their first clock ring every receiver prefers the even-edge matching $S^{(0)}$. Holding that matching then leaves the odd residuals constant and the even residuals zero in the scalar-free coordinates.

Four further assignments have multiplicity patterns

$$(1,1,2,0),\quad (1,0,2,1),\quad (2,0,1,1),\quad (2,1,1,0).$$

The signed swap of coordinates $1,3$ and the rotation by two preserve the even-edge matching and carry the initial residual to itself or its negative. They reduce these four cases to $(1,1,2,0)$. Write its

scalar-free coordinates as $\rho = (-a, -b, -d, h)$. Then

$$\begin{pmatrix} \dot{a} \\ \dot{b} \\ \dot{d} \end{pmatrix} = \begin{pmatrix} -4 & 1 & 0 \\ 1 & -4 & 2 \\ 0 & 1 & -8 \end{pmatrix} \begin{pmatrix} a \\ b \\ d \end{pmatrix}, \qquad (a, b, d)(0) = (0, 1, 0), \qquad \dot{h} = -(a + 2d),\ h(0) = 1.$$

The nonnegative cone is invariant, and the negative row sums give $\max(a, b, d) \le 1$, hence $a \le 1/4$, $d \le 1/8$. The half-spaces $a - 3b/4 - 11d/25 \le 0$ and $d - b/4 \le 0$ are invariant: their outward derivatives on the corresponding boundary are $-b/400 - 7d/100$ and $-a/4 - b/8$, respectively. Consequently $a \le 43b/50$, $d \le b/4$, and $b \ge e^{-4t}$. The attended block is similar to a negative definite symmetric matrix because $\mathsf{G} \succ 0$. Solving $\rho(\infty) = \rho(0) - \mathsf{G}z$, with $z_3 = 0$ and the first three final coordinates zero, gives $h(\infty) = 6/7$. Thus $h \ge 6/7$.

If the first ring occurs at t_{r} and redraws $S^{(0)}$, the subsequent scalar-free held motion, in elapsed time r, is

$$\begin{aligned} a_r &= a_0 e^{-8r}, & d_r &= d_0 e^{-8r}, \\ b_r &= b_0 + \tfrac{1}{4}(a_0 + d_0)(1 - e^{-8r}), & h_r &= h_0 - \tfrac{1}{4}(a_0 + d_0)(1 - e^{-8r}) \ge \tfrac{171}{224}. \end{aligned}$$

Every adjacent odd-edge absolute residual exceeds its even-edge neighbor by at least $(7/50)e^{-4t_{\mathrm{r}}}$. Combining these four cases with the preceding six, the deterministic motion $\mathbf{Y}$ from $s\mathbf{u}_\circ$ has, throughout the hypothesized matching hold,

$$\sqrt{\mathcal{D}_{\mathrm{odd}}(\mathbf{Y}_t)} - \sqrt{\mathcal{D}_{\mathrm{even}}(\mathbf{Y}_t)} \ge \frac{7}{50} s e^{-\mathrm{T}t - 8t_{\mathrm{r}}}, \qquad t \ge t_{\mathrm{r}}. \tag{C.104}$$

There are therefore ten good initial assignments. For the remaining six only $f_{\mathbf{v}} \ge -2$ will be used.

Set $h_s = (\log s)/\mathrm{T}$, for $s > e$, and write the initial configuration as $s\mathbf{u}_\circ + \xi$, $\xi \sim \mathcal{N}(0, I)$. Along the actual assignment path decompose $\mathbf{X} = \mathbf{Y} + \mathbf{E} + \mathbf{Z}$, where the first two terms solve its switched deterministic equation from $s\mathbf{u}_\circ$ and ξ, respectively, while

$$d\mathbf{Z} = -\mathsf{A}_{\sigma_t}\mathbf{Z}dt + \sqrt{2\mathrm{T}}dW_t, \qquad \mathbf{Z}_0 = 0.$$

All three are adapted differential equations, and $|\mathbf{E}_t| \le e^{-\mathrm{T}t}|\xi|$. Itô's formula yields

$$\mathscr{L} e^{|\mathbf{Z}|^2/4} \le \mathrm{T}(6 - |\mathbf{Z}|^2/4) e^{|\mathbf{Z}|^2/4} \le \mathrm{T}e^5.$$

Stopping at radius $\log s$, and using $\mathbb{E}e^{|\xi|^2/4} = 64$, bounds the probability that either $|\xi| > \log s$ or $\sup_{t \le h_s} |\mathbf{Z}_t| > \log s$ by $(65 + \mathrm{T}e^5 h_s) e^{-(\log s)^2/4}$. On its complement each square-root edge cost changes from its $\mathbf{Y}$ value by at most $2\mathrm{C}_N \log s$, where C_N is the largest norm of a residual-pair map divided by $\sqrt{2}$.

Define

$$L_s(t_{\mathrm{r}}) = h_s - \frac{8}{\mathrm{T}} t_{\mathrm{r}} - \frac{2}{\mathrm{T}} \log\log s, \qquad B_s = (L_s(t_{\mathrm{r}}) - t_{\mathrm{r}})_+.$$

For a good initial assignment, eq. (C.104) gives a square-root gap at least $(7/50)(\log s)^2$ on $[t_{\mathrm{r}}, L_s]$. For sufficiently large s, while both noise bounds hold, the actual energy gap is at least $\kappa_0 (\log s)^4$,

$\kappa_0 = (7/50)^2/4$. Each receiver's wrong choice at a ring then has probability at most $e^{-\kappa_0(\log s)^4/\mathrm{T}}$. Apply this bound at the first ring, then use the rate-one compensator up to the first wrong redraw or exit from the noise bound, stopping at $L_s \wedge h_s$. The endpoint L_s is known after the first ring, and the gap estimate holds until that first failure. Thus the matching holds throughout the nonempty interval $[t_{\mathrm{r}}, L_s]$ except on an event of probability at most

$$\delta_s = (65 + \mathrm{T}e^5 h_s)e^{-(\log s)^2/4} + 4(1+h_s)e^{-\kappa_0(\log s)^4/\mathrm{T}}, \qquad h_s\delta_s \longrightarrow 0. \tag{C.105}$$

Since $\mathbb{E}t_{\mathrm{r}} = 1$, a good initial assignment contributes at least

$$-2h_s + 4\mathbb{E}B_s - 4h_s\delta_s \geq 2h_s - 4(1 + 8/\mathrm{T}) - (8/\mathrm{T})\log\log s - 4h_s\delta_s$$

to $\mathbb{E}\int_0^{h_s} f_{\mathbf{v}}(\sigma_t)dt$. The other six contribute at least $-2h_s$. Averaging the sixteen uniform initial assignments gives

$$\mathbb{E}\int_0^{h_s} f_{\mathbf{v}}(\sigma_t)dt \geq \tfrac{1}{2}h_s - \frac{5}{\mathrm{T}}\log\log s - O_{\mathrm{T}}(1).$$

By eq. (C.98), $\mathbb{E}|\mathbf{X}_{h_s}|^2 \leq 14$. The Markov property and eq. (C.101) bound the integral after h_s in absolute value by a constant independent of s. Therefore

$$\mathrm{j}_{\mathrm{T}}(s\mathbf{u}_\circ) \geq \frac{\log s}{2\mathrm{T}} - \frac{5}{\mathrm{T}}\log\log s - O_{\mathrm{T}}(1). \tag{C.106}$$

This proves that j_{T} is not identically zero at any fixed $\mathrm{T} > 0$. Its zero set is Lebesgue-null. For completeness, this last assertion follows by induction on dimension: a nonzero analytic function in one variable has isolated zeros; in several variables at least one coefficient $\partial_n^k f(\mathbf{x}', 0)$ is nonzero as an analytic function, or analyticity would make f vanish everywhere. Outside the null zero set of that coefficient, each one-dimensional slice has isolated zeros, and Fubini's theorem concludes the induction.

Entropy balance for the evolving recognition law

Fix a finite horizon h. All rows are bounded away from zero on this interval by eq. (C.97). Write $q_\sigma(t, \mathbf{x})$ for the evolving density, $Q = \sum_\sigma q_\sigma$, and $b_\sigma(t, \mathbf{x}) = \mu_{p(t)}(\mathbf{x}, \sigma)/\sum_\eta \mu_{p(t)}(\mathbf{x}, \eta)$. For the forward Ornstein–Uhlenbeck semigroup K_σ, the last-proposal formula is

$$q_\sigma(t) = e^{-t}K_\sigma(t)q_\sigma(0) + \int_0^t e^{-(t-r)}K_\sigma(t-r)[b_\sigma(r)Q(r)]dr. \tag{C.107}$$

Iterating this positive formula, and bounding each proposal factor by one, reduces each term to a Gaussian motion along a deterministic sequence of sheets. Put $M = \max_\sigma \|\mathsf{A}_\sigma\|$. Its means have norm at most $|\mathsf{m}_0|$, and its covariances lie between $e^{-2Mh}I$ and $(1 + 2\mathrm{T}h)I$. Summing the at most 16^n sequences on each time simplex, of volume $t^n/n!$, gives a finite upper Gaussian envelope. The no-proposal term gives a lower Gaussian envelope. Consequently, uniformly on $[0, h]$,

$$0 < \mathrm{c}_h e^{-\mathrm{C}_h(1+|\mathbf{x}|^2)} \leq q_\sigma(t, \mathbf{x}) \leq \mathrm{C}_h e^{-\mathrm{c}_h|\mathbf{x}|^2}. \tag{C.108}$$

This bound does not assert Gaussianity conditional on a state-dependent assignment history.

Differentiating one Ornstein–Uhlenbeck kernel against a function bounded by a polynomial times a Gaussian gives a bound of the same type with factor $(t-r)^{-1/2}$. To check it, use the Gaussian kernel covariance bounds proportional to $t-r$, absorb its linear derivative into a Gaussian with twice the covariance, and complete the square in the convolution. This singularity is integrable in eq. (C.107), proving a first-derivative Gaussian bound. Moreover

$$\nabla b_\sigma = -\frac{b_\sigma}{\mathrm{T}}\left(H_\sigma \mathbf{x} - \sum_\eta b_\eta H_\eta \mathbf{x}\right), \qquad |\nabla b_\sigma| \le \mathrm{K_T}|\mathbf{x}|.$$

The source $b_\sigma Q$ therefore has a first derivative with the same type of bound. For the second derivative, transfer one derivative to this source using $\nabla K_\sigma(r) g = e^{r\mathsf{A}_\sigma} K_\sigma(r) \nabla g$, and differentiate the other kernel factor. The singularity is again only $r^{-1/2}$. The explicit initial Gaussian term supplies the bounds at zero. The forward equation obtained by taking a time difference quotient in eq. (C.107) then bounds the time derivative too:

$$|\nabla q_\sigma| + |\nabla^2 q_\sigma| + |\partial_t q_\sigma| \le \mathrm{C}_h(1+|\mathbf{x}|)^k e^{-\mathrm{c}_h|\mathbf{x}|^2}. \tag{C.109}$$

The derivative transfer follows directly by integration by parts in the Gaussian transition density; its inverse mean map is uniformly bounded on this finite interval. These estimates justify the differentiations, without assuming finite Fisher information.

Set $f_\sigma = q_\sigma/\mu_{p(t),\sigma}$ and $\mathsf{H}(t) = D_{\mathrm{KL}}(q_t \| \mu_{p(t)})$. The logarithms of both densities grow at most quadratically by eq. (C.108), and $\partial_t \log \mu_p$ is bounded in $\mathbf{x}$, since only its prior weight and normalizer vary. Together with eq. (C.109), this proves that H is absolutely continuous on $[0,h]$ and permits its integral to be differentiated. To integrate the diffusion by parts, first multiply by a smooth radial cutoff χ_R, equal to one on B_R, zero outside B_{2R}, with $|\nabla \chi_R| \le \mathrm{C}/R$, increasing to one. On each compact cylinder the densities have positive lower bounds, so $\log f_\sigma$ is a legitimate test function. The boundary term is bounded by

$$\frac{\mathrm{C}_h}{R}\int_{R\le|\mathbf{x}|\le 2R} (1+|\mathbf{x}|)^{k+3} e^{-\mathrm{c}_h|\mathbf{x}|^2} d\mathbf{x} \longrightarrow 0.$$

Detailed balance $\mu_\sigma b_\eta = \mu_\eta b_\sigma$ symmetrizes the finite jump sum. Both resulting production integrands are nonnegative, so monotone convergence removes the cutoff and simultaneously proves finiteness and local time integrability of the production. The result is

$$\begin{aligned} \mathsf{H}'(t) &= -\mathscr{D}_{p(t)}(f) - \sum_\sigma \int q_\sigma \partial_t \log \mu_{p(t),\sigma}, \\ \mathscr{D}_p(f) &= \mathrm{T}\sum_\sigma \int \mu_{p,\sigma} \frac{|\nabla f_\sigma|^2}{f_\sigma} \\ &\quad + \frac{1}{2}\sum_{\sigma,\eta}\int \mu_{p,\sigma} b_\eta (f_\eta - f_\sigma)(\log f_\eta - \log f_\sigma). \end{aligned} \tag{C.110}$$

The equality holds almost everywhere and in integrated form on every finite interval. The jump term is integrable already from the Gaussian envelopes and quadratic logarithmic bounds; thus the cutoff argument does not presuppose a bound on the diffusion production. This also proves $\mathsf{H}(0) < \infty$.

Entropy contraction at frozen rows

In the original twelve-dimensional coordinates put $\mathsf{B}_\sigma = \mathsf{A}_\sigma/\mathrm{T}$, the Gaussian sheet precision, and set

$$\kappa = \min_\sigma \lambda_{\min}(\mathsf{B}_\sigma) > 0, \qquad M = \max_\sigma \|\mathsf{B}_\sigma\|, \qquad q^*_\sigma = \mu_p(\sigma) = \frac{\pi_\sigma \mathsf{w}_\sigma}{\mathsf{Z}}, \qquad q^*_{\min} = \min_\sigma q^*_\sigma.$$

Let $\gamma_\sigma = \mathcal{N}(0, \mathsf{B}_\sigma^{-1})$, and, for a real function u, define

$$\mathscr{E}_c(u) = \mathrm{T} \sum_\sigma \int \mu_{p,\sigma} |\nabla u_\sigma|^2, \qquad \mathscr{E}_j(u) = \frac{1}{2} \sum_{\sigma,\eta} \int \mu_{p,\sigma} b_\eta (u_\sigma - u_\eta)^2.$$

The Gaussian inequalities needed below follow from the explicit Ornstein–Uhlenbeck semigroup with invariant law γ_σ. Its gradient commutes as $\nabla P_t g = e^{-t\mathsf{B}_\sigma} P_t \nabla g$. Differentiating entropy and applying Cauchy–Schwarz inside this kernel gives

$$-\frac{d}{dt} \operatorname{Ent}_{\gamma_\sigma}(P_t g) = \int \frac{|\nabla P_t g|^2}{P_t g} d\gamma_\sigma \le e^{-2\kappa t} \int \frac{|\nabla g|^2}{g} d\gamma_\sigma.$$

For bounded positive smooth g, the explicit Gaussian formula and dominated convergence make the entropy tend to zero. Integration, with $g = u^2$, gives

$$\operatorname{Ent}_{\gamma_\sigma}(u^2) \le \frac{2}{\kappa} \int |\nabla u|^2 d\gamma_\sigma, \qquad \operatorname{Var}_{\gamma_\sigma}(u) \le \frac{1}{\kappa} \int |\nabla u|^2 d\gamma_\sigma. \tag{C.111}$$

The second estimate follows by differentiating variance in the same semigroup. Positive regularization, truncation, and smooth cutoffs extend both estimates to their form domains: the cutoff errors tend to zero by integrability of the squared function and its gradient, and the entropy passes to the limit by its variational formula.

Replace u by $|u|$, which decreases both energies, and normalize $\int u^2 d\mu_p = 1$. Put $z_\sigma = (\int u_\sigma^2 d\gamma_\sigma)^{1/2}$. The entropy chain rule gives

$$\operatorname{Ent}_{\mu_p}(u^2) = \sum_\sigma q^*_\sigma \operatorname{Ent}_{\gamma_\sigma}(u_\sigma^2) + \operatorname{Ent}_{q^*}(z^2).$$

Here $0 \le z_\sigma \le R = (q^*_{\min})^{-1/2}$. The function $\psi(z) = z^2 \log z^2 - z^2 + 1$ satisfies $\psi(1) = \psi'(1) = 0$ and $\psi''(z) = 4 + 4\log z$, whence $\psi(z) \le 2(1 + \log R)(z-1)^2$ on $[0, R]$. Since $\sum q^*_\sigma z_\sigma^2 = 1$,

$$\operatorname{Ent}_{q^*}(z^2) \le L_p \operatorname{Var}_{q^*}(z), \qquad L_p = 4 + 2\log(1/q^*_{\min}).$$

Indeed $\sum q^*_\sigma (z_\sigma - 1)^2 = 2(1 - \mathbb{E}_{q^*} z) \le 2 \operatorname{Var}_{q^*}(z)$.

Fix a ball $B = B_\rho$ in these coordinates, and define

$$m_\sigma = |B|^{-1}\int_B u_\sigma, \quad m_* = \sum_\sigma \pi_\sigma m_\sigma, \quad X_\sigma = \left[\frac{(2\pi)^6 \mathsf{w}_\sigma e^{\rho^2\|\mathsf{B}_\sigma\|/2}}{|B|} - 1\right]_+, \quad B_{\mathrm{T}} = \max_\sigma (1+\sqrt{X_\sigma})^2.$$

Variance about any center is an upper bound on the variance, so

$$\mathrm{Var}_{q^*}(z) \le 2\sum_\sigma q^*_\sigma (z_\sigma - m_\sigma)^2 + 2\sum_\sigma q^*_\sigma (m_\sigma - m_*)^2.$$

The first sum is at most $(B_{\mathrm{T}}/(\mathrm{T}\kappa))\mathscr{E}_c(u)$: $|z_\sigma - \gamma_\sigma u_\sigma| \le \sqrt{\mathrm{Var}_{\gamma_\sigma}(u_\sigma)}$, and Cauchy–Schwarz bounds $|m_\sigma - \gamma_\sigma u_\sigma|$ by the same standard deviation times $\sqrt{\chi^2(\mathrm{Unif}(B)\|\gamma_\sigma)} \le \sqrt{X_\sigma}$. For the second sum, Jensen's inequality gives

$$\sum_\sigma q^*_\sigma (m_\sigma - m_*)^2 \le \frac{\mathsf{w}_{\max}}{\mathsf{Z}|B|}\sum_{\sigma,\eta}\pi_\sigma\pi_\eta\int_B (u_\sigma - u_\eta)^2.$$

On B, the explicit Gaussian densities give

$$\mu_{p,\sigma} b_\eta \ge \frac{\pi_\sigma\pi_\eta e^{-\rho^2 M}}{(2\pi)^6 \mathsf{Z}}.$$

The normalizer therefore cancels, and the preceding sum is at most $2(2\pi)^6 \mathsf{w}_{\max} e^{\rho^2 M}\mathscr{E}_j(u)/|B|$. Combining these bounds proves

$$\mathrm{Ent}_{\mu_p}(u^2) \le \frac{2}{\mathrm{T}\kappa}(1 + L_p B_{\mathrm{T}})\mathscr{E}_c(u) + \frac{4L_p(2\pi)^6 \mathsf{w}_{\max} e^{\rho^2 M}}{|B|}\mathscr{E}_j(u).$$

For $u = \sqrt{f}$, the production is at least $4(\mathscr{E}_c(u) + \mathscr{E}_j(u))$. Its jump part uses $(a-b)(\log a - \log b) \ge 4(\sqrt{a} - \sqrt{b})^2$, which follows by Cauchy–Schwarz on the integral of $r^{-1/2}$ between a and b. The inequality extends to the form domain by the preceding approximations. Finally,

$$q^*_{\min} \ge p^4_{\min}\frac{\mathsf{w}_{\min}}{\mathsf{w}_{\max}}, \qquad p_{\min} = \min_i\{p_i, 1-p_i\}.$$

Thus there are finite positive constants depending only on T with

$$\mathscr{D}_p(f) \ge \lambda(\mathrm{T}, p)\,\mathrm{Ent}_{\mu_p}(f), \qquad \lambda(\mathrm{T}, p) \ge \frac{\mathrm{c}_{\mathrm{T}}}{\mathrm{K}_{\mathrm{T}} + 8\log(1/p_{\min})}. \tag{C.112}$$

The factor T in $\mathscr{E}_c$ is the same diffusion coefficient as in eq. (C.110).

The matching branch of the small-temperature landscape

The needed sheet weights can be computed using two four-dimensional determinants. Let D be the cyclic difference matrix, and $C_\sigma = \mathrm{diag}(m_e(\sigma))$. In the orthogonal coordinates $(\bar{k} + \bar{m})/\sqrt{2}$, $(\bar{k} - \bar{m})/\sqrt{2}$, and ω, the energy splits into $\frac{1}{2}(Du - \sqrt{2}\omega)^\top C_\sigma (Du - \sqrt{2}\omega) + \frac{1}{2}(Dv)^\top C_\sigma Dv$, while the

discrepancy coordinate has reference precision $5I$. The determinant identity for the first block then gives

$$\det \mathsf{A}_\sigma = \mathrm{T}^4 \det[\mathrm{T}I + C_\sigma(DD^\top + 2I)] \det[5\mathrm{T}I + D^\top C_\sigma D]. \tag{C.113}$$

The two matching assignments, eight end-path assignments, four middle-path assignments, and two oriented cycles have the respective determinants

$$\begin{aligned}
\mathsf{d}_M &= 25\mathrm{T}^8(\mathrm{T}+8)^2(5\mathrm{T}+4)^2,\\
\mathsf{d}_E &= 5\mathrm{T}^6(\mathrm{T}^3+16\mathrm{T}^2+77\mathrm{T}+112)(125\mathrm{T}^3+200\mathrm{T}^2+85\mathrm{T}+8),\\
\mathsf{d}_N &= 5\mathrm{T}^6(\mathrm{T}+4)(5\mathrm{T}+2)(\mathrm{T}^2+12\mathrm{T}+28)(25\mathrm{T}^2+30\mathrm{T}+4),\\
\mathsf{d}_C &= 5\mathrm{T}^5(\mathrm{T}+2)(\mathrm{T}+4)^2(\mathrm{T}+6)(5\mathrm{T}+2)^2(5\mathrm{T}+4).
\end{aligned}$$

These follow in eq. (C.113) from the multiplicity representatives $(2,0,2,0)$, $(2,1,1,0)$, $(1,2,1,0)$, and $(1,1,1,1)$, respectively. Normalize the matching weight to one and write

$$\mathsf{r}_E = \sqrt{\mathsf{d}_M/\mathsf{d}_E}, \qquad \mathsf{r}_N = \sqrt{\mathsf{d}_M/\mathsf{d}_N}, \qquad \mathsf{r}_C = \sqrt{\mathsf{d}_M/\mathsf{d}_C}.$$

Their leading terms are $\mathsf{r}_E, \mathsf{r}_N = \sqrt{40/7}\mathrm{T} + O(\mathrm{T}^2)$ and $\mathsf{r}_C = \sqrt{5/3}\mathrm{T}^{3/2}(1+O(\mathrm{T}))$. Choose $\mathrm{T}_0 > 0$ so that $0 < \mathsf{r}_E, \mathsf{r}_N, \mathsf{r}_C \le 1/8$ for $0 < \mathrm{T} \le \mathrm{T}_0$.

Use matching coordinates $x = (p_0, 1-p_1, p_2, 1-p_3)$. The normalized evidence is the multilinear polynomial whose vertex weights are one at $\mathbf{0}, \mathbf{1}$, r_E for subsets of cardinality one or three, r_N for the four adjacent pairs, and r_C for the two opposite pairs. Complementation preserves these weights, so the centered reduced field is odd. With $D_0 = 1 + 4\mathsf{r}_E + 2\mathsf{r}_N + \mathsf{r}_C$, its Jacobian at $x = \mathbf{1}/2$ is

$$\frac{1}{D_0}\begin{pmatrix}
0 & 1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C\\
1-\mathsf{r}_C & 0 & 1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C\\
1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C & 0 & 1-\mathsf{r}_C\\
1-\mathsf{r}_C & 1-2\mathsf{r}_N+\mathsf{r}_C & 1-\mathsf{r}_C & 0
\end{pmatrix}.$$

For example, the adjacent Hessian entry of the evidence is $(1-\mathsf{r}_C)/2$, its uniform value is $D_0/8$, and the row mobility is $1/4$, giving the displayed entry. The eigenvalues are

$$\alpha = \frac{3-2\mathsf{r}_N-\mathsf{r}_C}{D_0} > 0, \qquad \beta = \frac{-1-2\mathsf{r}_N+3\mathsf{r}_C}{D_0} < 0, \qquad \eta = \frac{-1+2\mathsf{r}_N-\mathsf{r}_C}{D_0} < 0 \quad \text{twice.} \tag{C.114}$$

The positive eigendirection is $\mathbf{1}$ in matching coordinates, or $\mathbf{v}$ in the original rows. In fact the chosen bounds give $\alpha \ge 7/5$, $\beta \le -1/3$, and $\eta \le -2/5$.

Cyclic invariance makes the matching diagonal $x = z\mathbf{1}$ invariant. Writing $w = z(1-z)$, its normalized evidence is

$$Z_{\mathrm{diag}}(z) = 1 + 4(\mathsf{r}_E - 1)w + (2 - 8\mathsf{r}_E + 4\mathsf{r}_N + 2\mathsf{r}_C)w^2, \qquad \dot{z} = \frac{z(1-z)}{4Z_{\mathrm{diag}}(z)} Z'_{\mathrm{diag}}(z).$$

The derivative is $(1-2z)[4(\mathsf{r}_E-1)+2(2-8\mathsf{r}_E+4\mathsf{r}_N+2\mathsf{r}_C)w]$. The bracket is negative at both endpoints $w=0,1/4$, where its values are $4(\mathsf{r}_E-1)$ and $-3+2\mathsf{r}_N+\mathsf{r}_C$. It is affine in w, hence negative throughout. Thus the diagonal motion tends to $\mathbf{1}$ from $z>1/2$, and to $\mathbf{0}$ from $z<1/2$, reaching every prescribed endpoint neighborhood in finite slow time. At either vertex the Jacobian is $(\mathsf{r}_E-1)I$: a single receiver flip changes weight one to r_E, the mobility vanishes, and its diagonal derivative supplies the factor $-(1-\mathsf{r}_E)$. Both vertices are therefore sinks with positive inward rate $\nu=1-\mathsf{r}_E$.

Tracking, escape, and selection

Let $a_i(t)=q_t(\sigma_i=+1)$ and $\Delta_i(t)=a_i(t)-\bar\beta_i(p(t))$. Differentiating the normalized product prior gives

$$\partial_{p_i}\log\mu_p=\frac{1_{\{\sigma_i=+1\}}-\bar\beta_i(p)}{p_i(1-p_i)}.$$

Inserting the learning equation and eq. (C.96) in eq. (C.110) yields

$$\mathsf{H}'=-\mathscr{D}_p(f)-\varepsilon\sum_i\mathfrak{g}_i(p)\Delta_i-\varepsilon\sum_i\frac{\Delta_i^2}{p_i(1-p_i)}\le-\lambda(\mathrm{T},p)\mathsf{H}+\varepsilon\|\mathfrak{g}(p)\|_1\sqrt{\mathsf{H}/2}. \tag{C.115}$$

The last bound uses $|\Delta_i|\le\mathrm{TV}(q_t,\mu_p)\le\sqrt{\mathsf{H}/2}$. The nonpositive square is kept until this step; it removes any inverse-row factor from the forcing. Writing $\Lambda(t)=\int_0^t\lambda(\mathrm{T},p(r))dr$, comparison for $\sqrt{\mathsf{H}+\delta^2}$, followed by $\delta\downarrow0$, gives

$$\sqrt{\mathsf{H}(t)}\le e^{-\Lambda(t)/2}\sqrt{\mathsf{H}(0)}+\frac{\varepsilon}{2\sqrt{2}}\int_0^t e^{-(\Lambda(t)-\Lambda(r))/2}\|\mathfrak{g}(p(r))\|_1dr. \tag{C.116}$$

The bounded fitness, eq. (C.97), and eq. (C.112) show that, on $t\le2\mathrm{C_T}\varepsilon^{-1}\log(1/\varepsilon)$,

$$\mathrm{TV}(q_t,\mu_{p(t)})\le e^{-\Lambda(t)/2}\sqrt{\mathsf{H}(0)/2}+O_{\mathrm{T}}(\varepsilon\log(1/\varepsilon)). \tag{C.117}$$

Indeed λ is bounded below on this horizon by a positive constant divided by $1+\log(1/\varepsilon)$.

For the saddle a finer bound is needed. Let $\|\mathfrak{g}(p)-\mathfrak{g}(p')\|_1\le L_{\mathrm{T}}\|p-p'\|_\infty$, and suppose $\lambda\ge\lambda_\flat>0$ on $[s_0,t]$. The estimate $|\dot p_i|\le\varepsilon$ in eq. (C.116) gives

$$\mathrm{TV}(q_t,\mu_{p(t)})\le e^{-\lambda_\flat(t-s_0)/2}\sqrt{\mathsf{H}(s_0)/2}+\frac{\varepsilon}{2\lambda_\flat}\|\mathfrak{g}(p(t))\|_1+\frac{L_{\mathrm{T}}\varepsilon^2}{\lambda_\flat^2}. \tag{C.118}$$

The constants follow by integrating 1 and $t-r$ against $e^{-\lambda_\flat(t-r)/2}$.

Fix $0<\mathrm{T}\le\mathrm{T}_0$ and a mean with $\mathsf{j}_{\mathrm{T}}(\mathsf{m}_0)\neq0$, and put $\mathbf{e}=\mathbf{v}/2$. In slow time $\mathfrak{s}=\varepsilon t$, write $y=p-1/2=a\mathbf{e}+z$, $z\perp\mathbf{e}$. By eq. (C.102), at $t_1=\varepsilon^{-1/4}$,

$$a(t_1)=\varepsilon(\mathsf{j}_{\mathrm{T}}(\mathsf{m}_0)/2+o(1)),\qquad|z(t_1)|=O(\varepsilon).$$

On $[0,t_1]$, symmetry gives $\mathfrak{g}(1/2)=0$, so $\|\mathfrak{g}(p(t))\|_1 \le L_{\mathrm{T}}\varepsilon t/2$. The bound in eq. (C.116) gives $\sqrt{\mathsf{H}(t_1)}=O(\varepsilon^{7/4})$. Consequently, until exit from any sufficiently small fixed ball, eq. (C.118) implies

$$|\Delta|_2 \le \kappa_\varepsilon |y| + b_\varepsilon, \qquad \kappa_\varepsilon = O(\varepsilon), \quad b_\varepsilon = O(\varepsilon^{7/4}) = o(|a(t_1)|). \tag{C.119}$$

The constants here may depend on the fixed mean.

Let $\beta_\perp = \max(\beta,\eta) < 0$ and $\gamma = \alpha - \beta_\perp > 0$. Oddness and diagonal invariance of the analytic reduced field imply, for $|y| \le r$,

$$|\langle \mathscr{V}(1/2+y), \mathbf{e}\rangle - \alpha a| \le \mathrm{C}|y|^3, \qquad |P_\perp \mathscr{V}(1/2+y) - D\mathscr{V}(1/2)z| \le \mathrm{C}|y|^2|z|.$$

For the second inequality, subtract the zero perpendicular field on $a\mathbf{e}$ and integrate its derivative along the transverse segment. The derivative differs from its value at zero by $O(|y|^2)$ because the field is odd. Writing $w = |z|/|a|$, the upper right derivatives of norms give

$$\begin{aligned} \frac{d|a|}{d\mathfrak{s}} &\ge |a|[\alpha - (\mathrm{C}r^2 + \kappa_\varepsilon)(1+w)] - b_\varepsilon, \\ D^+ w &\le -[\gamma - \mathrm{C}r^2 - (\mathrm{C}r^2 + \kappa_\varepsilon)(1+w)]w + (1+w)(\kappa_\varepsilon + b_\varepsilon/|a|). \end{aligned} \tag{C.120}$$

Choose W larger than twice the initial limiting ratio, then choose $r>0$ so small that the terms $\mathrm{C}r^2(1+W)$ consume less than one quarter of both positive rates. For small enough ε, eq. (C.119) controls the remaining terms. A bootstrap then keeps $w \le W$, makes $|a|$ increase without changing sign, and gives

$$w(\mathfrak{s}) \le w(\mathfrak{s}_1) e^{-\gamma(\mathfrak{s}-\mathfrak{s}_1)/2} + \frac{2(1+W)}{\gamma}(\kappa_\varepsilon + b_\varepsilon/|a(t_1)|).$$

The lower growth bound for $|a|$ yields an exit time no larger than

$$\frac{1+O(r^2)+o(1)}{\alpha}\log(1/\varepsilon)$$

in slow time. The corresponding upper norm-growth bound forces the exit time to tend to infinity. Hence $w \to 0$ at exit, and the exit point tends to $\mathrm{sign}(\mathrm{j_T}) r\mathbf{e}$.

The reduced diagonal orbit from that point enters any fixed neighborhood of its matching vertex in finite slow time. The tracking error in eq. (C.117) tends to zero after t_1; finite-time Grönwall comparison therefore carries the evolving rows into the same neighborhood. At its selected vertex let $\zeta_i = |p_i - p_i^M|$. By the derivative computation above and continuity, a sufficiently small neighborhood has $(1-\zeta_i)|\mathfrak{g}_i(p)| \ge \nu/2$, where $\nu = 1 - \mathsf{r}_E > 0$, and the fitness has the sign directed toward that vertex. Thus

$$D^+\zeta_i \le -\tfrac{\nu}{2}\zeta_i + \mathrm{TV}(q_t, \mu_{p(t)})$$

in slow time. The neighborhood remains invariant once the tracking floor is small enough, and ζ_i decays exponentially to a vanishing floor.

Choose any $\mathrm{C_T} > 1/\alpha$, independently of the mean, and then choose the local radius so that its escape coefficient is less than $\mathrm{C_T}$. The time remaining after the finite transit tends to infinity

uniformly on the slow-time window $[\mathrm{C_T}\log(1/\varepsilon), 2\mathrm{C_T}\log(1/\varepsilon)]$. The row distances therefore vanish uniformly on that window. Every normalized sheet weight is at most the matching weight one, so

$$\mu_p(M) \geq \pi_M \geq 1 - \sum_i \zeta_i, \qquad q_t(S \neq M) \leq \sum_i \zeta_i + \mathrm{TV}(q_t, \mu_{p(t)}).$$

The last total variation tends to zero by eq. (C.117); its integrated contraction exponent at the beginning of the window grows at least as a positive constant times ε^{-1}. This proves the first conclusion for every mean outside the null set in eq. (C.106). For the frozen arm, eq. (C.99) and rotation symmetry prove the second conclusion at every $\mathrm{T} > 0$ and every mean. The convergence is pointwise in the fixed mean; no uniform bound near the zero set of the imprint has been used. □

C.10 Proof of Joint Selection Along a Schedule

Proof of Corollary 15.6. The saddle estimate eq. (C.114) gives $\alpha(\mathrm{T}) \geq 7/5$ throughout the stated temperature range. The proof of Theorem 15.5 permits any window constant greater than $1/\alpha(\mathrm{T})$, independently of the initial mean. Thus the common choice 2 is valid. Write

$$\tau_\varepsilon = 2\varepsilon^{-1}\log(1/\varepsilon), \qquad W_\varepsilon = [\tau_\varepsilon, 2\tau_\varepsilon], \qquad 0 < \varepsilon < e^{-1}.$$

For a perfect matching M, define

$$\nu_{\mathrm{T}}^{M} = \mathcal{N}(0, \mathrm{T}\mathsf{A}_M^{-1}) \otimes \delta_M, \qquad \mu_{\mathrm{T}} = \mu_{\mathrm{T},1/2}.$$

Here $\mu_{\mathrm{T},p}$ is the frozen-row law used in the preceding proof, with its temperature now displayed explicitly. The inverse exists because $\mathsf{A}_M \succeq \mathrm{T}I$. Let $M_{\mathrm{T}}(\mathsf{m}_0)$ be the matching selected by the sign of $\mathsf{j}_{\mathrm{T}}(\mathsf{m}_0)$, choosing either matching on its null zero set.

First record the full one-time consequence of the fixed-temperature proof. Conditional on assignment M, the law $\mu_{\mathrm{T},p}$ has the Gaussian factor of ν_{T}^{M}, independently of p. The normalized sheet weights are at most the matching weight one. For $\zeta_i = |p_i - p_i^M|$, therefore,

$$\mu_{\mathrm{T},p}(M) \geq \pi_M \geq 1 - \sum_i \zeta_i, \qquad \mathrm{TV}(\mu_{\mathrm{T},p}, \nu_{\mathrm{T}}^{M}) = 1 - \mu_{\mathrm{T},p}(M) \leq \sum_i \zeta_i.$$

The row convergence and eq. (C.117) imply

$$\sup_{t \in W_\varepsilon} \mathrm{TV}(q_t^{\mathrm{T},\varepsilon,\mathsf{m}_0}, \nu_{\mathrm{T}}^{M_{\mathrm{T}}(\mathsf{m}_0)}) \longrightarrow 0$$

at fixed temperature outside the imprint's null zero set. Moreover eq. (C.116) itself gives convergence of the learned relative entropy to zero, uniformly on this window. For frozen rows, the entropy identity and contraction give

$$D_{\mathrm{KL}}(q_t^{\mathrm{T},0,\mathsf{m}_0} \| \mu_{\mathrm{T}}) \leq e^{-\lambda(\mathrm{T},1/2)t} D_{\mathrm{KL}}(q_0^{\mathrm{T},0,\mathsf{m}_0} \| \mu_{\mathrm{T}}).$$

The initial entropy is finite by the Gaussian density bounds, and $\lambda(\mathrm{T}, 1/2) > 0$. Frozen entropy and total variation therefore tend to zero on the window for every fixed mean.

Use the explicit entropy majorants as follows. Let

$$G_{\mathrm{T}} = \sup_{p\in[0,1]^4} \|\mathfrak{g}(p)\|_1, \qquad \ell_{\mathrm{T}}(\varepsilon) = \frac{\mathrm{c}_{\mathrm{T}}}{\mathrm{K}_{\mathrm{T}} + 8\log 2 + 32\log(1/\varepsilon)}, \qquad \lambda_u(\mathrm{T}) = \frac{\mathrm{c}_{\mathrm{T}}}{\mathrm{K}_{\mathrm{T}} + 8\log 2}.$$

The first constant is finite. The row lower bound and eq. (C.112) make $\ell_{\mathrm{T}}(\varepsilon)$ a rate lower bound throughout $t \le 2\tau_\varepsilon$, and $\lambda_u(\mathrm{T})$ a rate lower bound at uniform rows. The common initial entropy equals

$$H_0(\mathrm{T}, \mathsf{m}_0) = \log \mathsf{Z}(1/2) - 6 + \frac{\operatorname{tr} \overline{\mathsf{A}} + \mathsf{m}_0^\top \overline{\mathsf{A}} \mathsf{m}_0}{2\mathrm{T}}, \qquad \overline{\mathsf{A}} = \frac{1}{16}\sum_\sigma \mathsf{A}_\sigma.$$

Indeed the log density ratio is $\log \mathsf{Z}(1/2) - |\mathbf{x} - \mathsf{m}_0|^2/2 + \mathbf{x}^\top \mathsf{A}_\sigma \mathbf{x}/(2\mathrm{T})$; taking its initial expectation proves the formula. Thus H_0 is continuous and nonnegative. Uniformly on W_ε, the entropy tracking estimate gives

$$\begin{aligned} D_{\mathrm{KL}}(q_t^{\mathrm{T},\varepsilon,\mathsf{m}_0} \| \mu_{\mathrm{T},p(t)}) &\le B_L(\mathrm{T}, \varepsilon, \mathsf{m}_0) \\ &:= \left[e^{-\ell_{\mathrm{T}}(\varepsilon)\tau_\varepsilon/2} \sqrt{H_0} + \frac{\varepsilon G_{\mathrm{T}}}{\sqrt{2}\ell_{\mathrm{T}}(\varepsilon)} \right]^2, \\ D_{\mathrm{KL}}(q_t^{\mathrm{T},0,\mathsf{m}_0} \| \mu_{\mathrm{T}}) &\le B_F(\mathrm{T}, \varepsilon, \mathsf{m}_0) := e^{-\lambda_u(\mathrm{T})\tau_\varepsilon} H_0. \end{aligned}$$

These majorants are continuous in positive mobility and mean and tend to zero at fixed temperature and mean as $\varepsilon \downarrow 0$.

Let $\mathcal{E}_{\mathrm{T}}(\varepsilon, \mathsf{m}_0)$ be the minimum of one and the maximum of the following five nonnegative quantities:

$$\begin{aligned} &\sup_{t\in W_\varepsilon} \max_i |p_i(t) - p_i^{M_{\mathrm{T}}(\mathsf{m}_0)}|, \\ &\sup_{t\in W_\varepsilon} \mathrm{TV}(q_t^{\mathrm{T},\varepsilon,\mathsf{m}_0}, \nu_{\mathrm{T}}^{M_{\mathrm{T}}(\mathsf{m}_0)}), \\ &B_L(\mathrm{T}, \varepsilon, \mathsf{m}_0), \\ &\sup_{t\in W_\varepsilon} \mathrm{TV}(q_t^{\mathrm{T},0,\mathsf{m}_0}, \mu_{\mathrm{T}}), \\ &B_F(\mathrm{T}, \varepsilon, \mathsf{m}_0). \end{aligned}$$

The preceding estimates prove $\mathcal{E}_{\mathrm{T}}(\varepsilon, \mathsf{m}_0) \to 0$ for almost every fixed mean at each fixed temperature.

We will take a supremum over mobility, so measurability must be checked. Fix a temperature and restrict the positive mobility and the mean to compact sets, with the mobility bounded away from zero. The corresponding horizons are bounded, and eq. (C.97) bounds every row away from zero and one. On this row set the posterior redraw probabilities are uniformly Lipschitz in the rows, independently of the configuration. Maximally couple the initial laws and then use common Brownian motions and proposal clocks until the first different redraw. The law difference is bounded by the initial total variation plus a constant times the integral of row differences. Subtracting the learning equations bounds the row difference by integrals of row and law differences and by the

mobility difference. Grönwall's inequality proves total-variation continuity in the initial mean and positive mobility on each bounded time interval. Time continuity follows from eq. (C.107).

Parametrizing the window by $t = u\tau_\varepsilon$, $1 \le u \le 2$, proves continuity in positive mobility of each row or total-variation window supremum. The explicit entropy majorants are continuous by their formulas. The matching as a function of the mean is Borel because its imprint is analytic. Thus the error is measurable in the mean, and its supremum over an interval of positive mobilities equals its supremum over rational mobilities. No uniform continuity at zero mobility or separate entropy-continuity assertion is needed.

Fix the sequence $\mathrm{T}_n \downarrow 0$ in the stated temperature interval and $0 < \eta_n < 1$ with $\eta_n \to 0$. For the prescribed exponent $a > 7/2$, put $h_n = \mathrm{T}_n^a$. The following argument in fact permits arbitrary positive caps h_n. On the mean space $\mathbb{R}^{12}$, introduce the auxiliary probability measure $\rho = \mathcal{N}(0, I_{12})$. Its density is strictly positive; it introduces no assumption that the physical mean is random. Set

$$F_n(r, \mathsf{m}_0) = \sup_{0<\varepsilon\le r} \mathcal{E}_{\mathrm{T}_n}(\varepsilon, \mathsf{m}_0).$$

The measurable functions F_n are bounded by one. At fixed n, they tend to zero for ρ-almost every mean as $r \downarrow 0$. Dominated convergence yields

$$\int F_n(r, \mathsf{m}_0) d\rho(\mathsf{m}_0) \longrightarrow 0.$$

Choose $d_n = 2^{-k_n}$ recursively, where k_n is the smallest positive integer satisfying

$$d_n \le \min(e^{-1}, 2^{-n}, h_n, d_{n-1}/2), \qquad \int F_n(d_n, \mathsf{m}_0) d\rho(\mathsf{m}_0) \le \eta_n 2^{-n},$$

with the $d_{n-1}/2$ condition omitted for $n = 1$. The preceding limit proves existence. This defines deterministic thresholds independent of the mean; it does not give numerical values or an explicit power law.

Let $B_n = \{\mathsf{m}_0 : F_n(d_n, \mathsf{m}_0) > \eta_n\}$. Markov's inequality and countable subadditivity give

$$\rho(B_n) \le 2^{-n}, \qquad \rho\left(\bigcup_{n\ge N} B_n\right) \le 2^{1-N}.$$

The decreasing intersection of these tail unions consequently has measure zero. Add the countable union of the imprint zero sets. The resulting exceptional set is Lebesgue-null because ρ and Lebesgue measure have the same null sets. Outside it, for every sufficiently large n, all $0 < \varepsilon \le d_n$ satisfy $F_n(d_n, \mathsf{m}_0) \le \eta_n$. Since $\eta_n < 1$, clipping has no effect on this inequality: the row and total-variation errors and both entropy majorants are at most η_n. The majorants bound the actual entropies, giving all five stated conclusions.

The frozen matching probability is, exactly,

$$\mu_{\mathrm{T}}(M) = \frac{1}{2 + 8\mathsf{r}_E + 4\mathsf{r}_N + 2\mathsf{r}_C} = \frac{1}{2D_0(\mathrm{T})} = \frac{1}{2} + O(\mathrm{T}).$$

Indeed $\mathsf{r}_E, \mathsf{r}_N = O(\mathrm{T})$ and $\mathsf{r}_C = O(\mathrm{T}^{3/2})$, as proved from the sheet determinants. Pushforward cannot increase total variation. If q^S denotes the assignment marginal, the two established total-variation estimates therefore imply, uniformly on W_ε,

$$\left| \mathrm{TV}(q_t^{\mathrm{T}_n,\varepsilon,\mathsf{m}_0,S}, q_t^{\mathrm{T}_n,0,\mathsf{m}_0,S}) - \left(1 - \frac{1}{2D_0(\mathrm{T}_n)}\right) \right| \le 2\eta_n.$$

Here the comparison distance is exactly $\mathrm{TV}(\delta_M, \mu^S_{\mathrm{T}_n}) = 1 - \mu_{\mathrm{T}_n}(M)$, for either matching. The assignment-law contrast thus tends to $1/2$, even if the selected matching changes with n.

Finally, defining $\mathrm{T}(\varepsilon) = \mathrm{T}_n$ on $d_{n+1} < \varepsilon \le d_n$ gives a single stepwise path with both parameters tending to zero. The proved bounds hold for all sufficiently small mobilities on this path. Arbitrary mobility caps are admissible; for example $h_n = \mathrm{T}_n^n$ forces $\varepsilon_n/\mathrm{T}_n^a \to 0$ for each fixed $a > 0$ whenever $\varepsilon_n \le d_n$. This imposes an upper bound and does not certify a prescribed equality between temperature and mobility.

Nothing in this argument makes the imprint sign eventually constant as temperature vanishes. Its fixed-temperature nonvanishing and null zero set give no quantitative lower bound at a fixed mean in that limit. Accordingly the matching is allowed to depend on n, and the result concerns complete one-time laws, not complete histories or conditional first-formation events. □

An explicit temperature bound for tracking

Proposition C.15 (Temperature-dependent entropy tracking). In the four-cycle of theorem 15.5, let $0 < \mathrm{T} \le 1/100$, $p \in (0,1)^4$, and $p_{\min} = \min_i\{p_i, 1-p_i\}$. Define

$$\mathrm{K}_0 = 210\sqrt{2\pi/5}, \qquad \mathrm{c}_* = \min\left\{\frac{2}{1 + 4\mathrm{K}_0 e^{13/8}}, \frac{1}{\mathrm{K}_0 e^{13/4}}\right\}.$$

The entropy production of the full frozen law satisfies

$$\mathscr{D}_p(f) \ge \frac{\mathrm{c}_* \mathrm{T}^{5/2}}{4 + 3\log(1/\mathrm{T}) + 8\log(1/p_{\min})} \mathrm{Ent}_{\mu_p}(f).$$

Write $\varepsilon = \epsilon_{\mathrm{L}}$, $t_{\mathrm{L}} = 2\varepsilon^{-1}\log(1/\varepsilon)$, and

$$\mathrm{D}(\mathrm{T},\varepsilon) = 4 + 3\log(1/\mathrm{T}) + 8\log 2 + 32\log(1/\varepsilon), \qquad \ell = \frac{\mathrm{c}_* \mathrm{T}^{5/2}}{\mathrm{D}(\mathrm{T},\varepsilon)}.$$

For $0 < \varepsilon < e^{-1}$ and $0 \le t \le 2t_{\mathrm{L}}$,

$$\mathrm{TV}(q_t, \mu_{p(t)}) \le e^{-\ell t/2}\sqrt{\frac{13(12 + |\mathsf{m}_0|^2)}{4\mathrm{T}}} + \frac{2\varepsilon \mathrm{D}(\mathrm{T},\varepsilon)}{\mathrm{c}_* \mathrm{T}^{7/2}}. \tag{C.121}$$

In particular, for $\varepsilon = \mathrm{T}^{a}$ with any $a > 7/2$, this total variation tends to zero uniformly on $[t_{\mathrm{L}}, 2t_{\mathrm{L}}]$ and on bounded sets of initial means. This conclusion compares one-time laws; it supplies no imprint lower bound. [ESTABLISHED]

Proof. Use $u = (\bar{k} + \bar{m})/\sqrt{2}$, $v = (\bar{k} - \bar{m})/\sqrt{2}$, and ω, and put $R = [D, -\sqrt{2}I]$. The common kernel $K = \ker R \oplus \ker D$ has dimension five. Every H_{σ} vanishes there, and the reference precision preserves K and $N = K^{\perp}$. Its restrictions have eigenvalues one and five. The passive process on K is therefore an independent Gaussian Ornstein–Uhlenbeck process with entropy contraction at least $2\mathrm{T}$. The active process on N, together with the assignment, is autonomous: all redraw costs depend only on its residuals. Rescaling its coordinate by $\mathrm{T}^{-1/2}$ gives unit diffusion and drift matrix $a_{\sigma} = (\mathsf{H}_{\sigma} + \mathrm{T}\Omega)|_{N}$. Since every edge multiplicity is at most two,

$$\mathrm{T} I \preceq a_{\sigma} \preceq 13 I.$$

Indeed $RR^{\top} = DD^{\top} + 2I$ has eigenvalues $2, 4, 4, 6$, while $D^{\top}D$ has eigenvalues $0, 2, 2, 4$, so $\|\mathsf{H}_{\sigma}\| \le 12$.

Divide the four determinants in eq. (C.113) by the passive determinant $5\mathrm{T}^{5}$. Their positive coefficients, evaluated at $\mathrm{T} = 1/100$ for upper bounds, give the active bounds

$$5120\mathrm{T}^{3} \le d_{M} \le 5262\mathrm{T}^{3}, \quad 896\mathrm{T} \le d_{E} \le 1017\mathrm{T},$$
$$896\mathrm{T} \le d_{N} \le 2175\mathrm{T}, \qquad 3072 \le d_{C} < 5120.$$

Thus w_{M} and w_{C} are the largest and smallest active sheet weights, and $w_{M} \le \mathrm{T}^{-3/2}/\sqrt{5120}$, $w_{C}/w_{M} \ge \mathrm{T}^{3/2}$. The finite-label constant in the entropy proof consequently satisfies

$$L_{p} \le 4 + 3\log(1/\mathrm{T}) + 8\log(1/p_{\min}).$$

Apply the Gaussian-sheet and prior-weighted-ball argument of eqs. (C.111) and (C.112) in the seven active coordinates, using the ball of radius $1/2$, whose volume is $\pi^{3}/840$. Its Gaussian comparison factor is at most $4\mathrm{K}_{0}e^{13/8}\mathrm{T}^{-3/2}$, because

$$\frac{(2\pi)^{7/2}}{(\pi^{3}/840)\sqrt{5120}} = \mathrm{K}_{0}.$$

More explicitly, the two energy coefficients in that argument obey

$$C_{c} \le (2 + 8\mathrm{K}_{0}e^{13/8})L_{p}\mathrm{T}^{-5/2}, \qquad C_{j} \le 4\mathrm{K}_{0}e^{13/4}L_{p}\mathrm{T}^{-5/2}.$$

The continuous production equals four times its quadratic form and the jump production bounds four times its quadratic form. Hence the active rate is at least $\mathrm{c}_{*}\mathrm{T}^{5/2}/L_{p}$. Entropy decomposes over the independent passive and active factors. Cauchy–Schwarz bounds the energy of a marginal by the corresponding integrated energy, so the full rate is at least the smaller of the two rates. The

displayed active lower bound is less than $2\mathrm{T}$, proving the full-state inequality. The form-domain extension uses the positive regularization and cutoffs already used for eq. (C.112).

A one-receiver flip joins only the class pairs M–E, N–E, and C–E. The determinant bounds above make every ratio of such adjacent weights at most $1/\mathrm{T}$. Writing the evidence as $p_i Z_i^+ + (1-p_i)Z_i^-$, the ratio Z_i^+/Z_i^- is between the extremal adjacent ratios. Therefore $|\partial_{p_i} \log Z| \le 1/\mathrm{T}$, and $\|\mathsf{g}\|_1 \le 4/\mathrm{T}$. The row lower bound gives the rate floor ℓ on the stated horizon. The initial entropy is at most $13(12+|\mathsf{m}_0|^2)/(2\mathrm{T})$: in the Gaussian density ratio, $\mathsf{B}_\sigma \preceq 13I/\mathrm{T}$, $\mathsf{Z} \le 1$, and the remaining quadratic term has nonpositive expectation. Insert these bounds in eq. (C.116) and use Pinsker's inequality to obtain eq. (C.121). For $\varepsilon = \mathrm{T}^a$, its second term is $O(\mathrm{T}^{a-7/2}\log(1/\mathrm{T}))$, while ℓt_{L} grows as a positive constant times $\mathrm{T}^{5/2-a}$. The exponential therefore dominates the initial $\mathrm{T}^{-1/2}$ factor, proving the last assertion. □

C.11 Coherence and Persistence of the Selected Pairing

Proposition C.16 (Coherence and persistence after pairing selection). In the four-cycle of theorem 15.5, let $0 < \mathrm{T} \le 1/100$, let M be one matching, and suppose $[t_0, t_0+\mathsf{h}] \subseteq [t_{\mathrm{L}}, 2t_{\mathrm{L}}]$, with $\mathsf{h} > 0$. Write $\nu_{\mathrm{T}}^M = \mathcal{N}(0, \mathrm{T}\mathsf{A}_M^{-1}) \otimes \delta_M$. Assume the learned and frozen recognition laws satisfy

$$\sup_{t_0 \le t \le t_0+\mathsf{h}} \mathrm{TV}(q_t^{\mathrm{L}}, \nu_{\mathrm{T}}^M) \le \delta, \qquad \mathrm{TV}(q_{t_0}^0, \mu_{\mathrm{T},1/2}) \le \delta.$$

Let $\mathsf{O}_{\mathrm{T},\mathsf{h}}^M$ be the stationary Ornstein–Uhlenbeck path law with drift $-\mathsf{A}_M\mathbf{x}$, diffusion covariance $2\mathrm{T}I$, and constant assignment M. For the configuration and assignment path laws on this interval,

$$\mathrm{TV}(\mathsf{P}_{t_0,\mathsf{h}}^{\mathrm{L}}, \mathsf{O}_{\mathrm{T},\mathsf{h}}^M) \le (\mathsf{h}+2)\delta, \tag{C.122}$$

$$\mathrm{TV}\left(\mathsf{P}_{t_0,\mathsf{h}}^0, \tfrac{1}{2}\mathsf{O}_{\mathrm{T},\mathsf{h}}^{M_0} + \tfrac{1}{2}\mathsf{O}_{\mathrm{T},\mathsf{h}}^{M_1}\right) \le \delta + 16\mathrm{T} + 31\mathrm{T}\mathsf{h}. \tag{C.123}$$

The frozen arm keeps its original uniform rows.

For resolution $\mathsf{a} > 0$, let $\mathscr{E}_M(\mathsf{a},\mathsf{h})$ require throughout the interval that the assignment is M, the stacked residual norm $\sqrt{(r^b)^2+(r^m)^2}$ on its two edges is at most a, and that norm on each exterior edge exceeds a. Define

$$\mathsf{b}(\mathrm{T},\mathsf{a},\mathsf{h}) = 8\sqrt{2/\pi}\,\mathsf{a} + (8+3400\mathsf{h})\mathrm{T}/\mathsf{a}^2.$$

Then

$$\Pr_{\mathrm{L}}(\mathscr{E}_M^c) \le (\mathsf{h}+2)\delta + \mathsf{b}(\mathrm{T},\mathsf{a},\mathsf{h}). \tag{C.124}$$

In particular, the choice $\mathsf{a} = \mathrm{T}^{1/4}$, $\mathsf{h} = \mathrm{T}^{-1/4}$, and $\delta \le \mathrm{T}$ makes this probability tend to zero. The matching-window label, equal to M_j when $\mathscr{E}_{M_j}$ holds and to an additional symbol otherwise, converges to δ_M in the learned arm and to the equal matching mixture in the frozen arm.

At one time, define $\mathfrak{P}_{\mathsf{a}}(\mathbf{x})$ as the connected-component partition of edges whose stacked residual norm $\sqrt{(r^b)^2+(r^m)^2}$ is at most a. For times when both one-time hypotheses above hold, put

$$\mathsf{e}(\mathrm{T},\mathsf{a}) = 4\sqrt{2/\pi}\mathsf{a} + 4\mathrm{T}/\mathsf{a}^2.$$

The learned and frozen partition laws satisfy

$$\left|\mathrm{TV}(q_t^{\mathrm{L}}\mathfrak{P}_{\mathsf{a}}, q_t^{0}\mathfrak{P}_{\mathsf{a}}) - \tfrac{1}{2}\right| \le 2\delta + 16\mathrm{T} + 2\mathsf{e}(\mathrm{T},\mathsf{a}). \tag{C.125}$$

Thus the contrast persists for the coherent partition, not only the source label.

These conclusions apply along corollary 15.6 with tolerance $\delta \le \mathrm{T}$, mobility cap $\epsilon_{\mathrm{L}} \le \mathrm{T}^2$, and $t_0 = t_{\mathrm{L}}$. The matching can vary along that schedule. The internal restoring rates are $8+\mathrm{T}$ and $4+5\mathrm{T}$, on an event whose probability tends to one. The resolution therefore has restoring ratio at least $4\mathsf{a}^2/\mathrm{T} \to \infty$. If the rows are also within δ of their matching vertex, the stationary law with those rows frozen realizes the same persistence event with probability at least $1-4(\mathsf{h}+2)\delta-\mathsf{b}$. This late coherence is restored equilibrium coherence; it supplies no entrance or first-formation assertion. [ESTABLISHED]

Proof. First retain the proposal mechanism in the comparison. Put $r(t) = q_t^{\mathrm{L}}(S \neq M)$, and let $b_{\neq M}(t,\mathbf{x})$ be the probability of proposing a different assignment at a clock ring. Since the clock has rate one,

$$r'(t) = \mathbb{E}b_{\neq M}(t,\mathbf{X}_t) - r(t).$$

If $N_{\neq M}$ counts proposals to an assignment other than M on the interval, its compensator gives

$$\mathbb{E}N_{\neq M} = r(t_0+\mathsf{h}) - r(t_0) + \int_{t_0}^{t_0+\mathsf{h}} r(t)dt. \tag{C.126}$$

Consequently $\Pr(\exists t \in [t_0, t_0+\mathsf{h}] : S_t \neq M) \le (\mathsf{h}+1)\delta$. Couple the initial law maximally with ν_{T}^M, and use the same future Brownian motion for the native process and the stationary matching process. Their paths agree until an initial mismatch or a proposal to another assignment. The union bound and eq. (C.126) prove eq. (C.122). This argument counts proposals of the native process and does not condition its diffusion on an assignment history.

For the frozen comparison, use the normalized weights $1, \mathsf{r}_E, \mathsf{r}_N, \mathsf{r}_C$ from appendix C.9. The determinant bounds in the proof of proposition C.15 give

$$\mathsf{r}_E, \mathsf{r}_N \le \tfrac{5}{2}\mathrm{T}, \qquad \mathsf{r}_C \le \tfrac{3}{2}\mathrm{T}^{3/2}.$$

The invariant mass κ_{T} of nonmatching assignments is therefore at most $16\mathrm{T}$. Conditional on either matching, its configuration law is the Gaussian factor of ν_{T}^M. The invariant law is consequently within κ_{T} of the equal mixture of these two laws.

Detailed balance bounds the stationary flux from M to all nonmatching assignments by κ_{T}. For the other matching M', write $\mu_M(\mathbf{x})$ for the joint invariant density on the matching sheet. Pointwise,

$$\mu_M b_{M'} = \frac{\mu_M \mu_{M'}}{\sum_S \mu_S} \le \tfrac{1}{2}\sqrt{\mu_M \mu_{M'}}.$$

Since $(\mathsf{A}_M + \mathsf{A}_{M'})/2$ is the oriented-cycle matrix, Gaussian integration gives $\int \sqrt{\mu_M \mu_{M'}} = \mu_{\mathrm{T},1/2}(M)\mathsf{r}_C$. Dividing the two flux bounds by the matching mass yields

$$\mathbb{E}_{\nu_{\mathrm{T}}^M} b_{\neq M} \le 8\mathsf{r}_E + 4\mathsf{r}_N + \tfrac{5}{2}\mathsf{r}_C \le 31\mathrm{T}.$$

Thus the stationary flux out of the two matching sheets together is at most $31\mathrm{T}$. Couple the stationary native process at its initial time with the equal matching mixture and use common Brownian increments until the first departure from its coupled matching. Its failure probability is at most $16\mathrm{T} + 31\mathrm{Th}$. The frozen native process started at $q_{t_0}^0$ differs from its stationary path law by at most its initial total variation δ, because the frozen transition kernel is the same. This proves eq. (C.123).

It remains to bound residuals on the comparison path. The four normal coordinates $z_g^{\pm} = n_g^{\pm} \cdot \mathbf{X}$, $g \in M$, from eq. (C.94) have rates $8 + \mathrm{T}$ and $4 + 5\mathrm{T}$. Their stationary variances are at most $\mathrm{T}/4$, and

$$r_g^b = \sqrt{2} z_g^+ + z_g^-, \qquad r_g^m = \sqrt{2} z_g^+ - z_g^-.$$

For a scalar Ornstein–Uhlenbeck noise starting at zero, integration by parts in its stochastic convolution gives

$$Z_t = \sqrt{2\mathrm{T}}\left[W_t - \lambda \int_0^t e^{-\lambda(t-u)} W_u du\right], \qquad \mathbb{E}\sup_{t \le \mathsf{h}} |Z_t|^2 \le 32\mathrm{Th}.$$

The last inequality uses $\sup|Z_t| \le 2\sqrt{2\mathrm{T}}\sup|W_t|$ and the Brownian maximal estimate proved in the matching-episode argument of appendix C.7. For the four-dimensional stationary normal vector z, it follows that

$$\mathbb{E}\sup_{t \le \mathsf{h}} |z_t|^2 \le 2\mathrm{T} + 256\mathrm{Th}.$$

Each selected stacked residual norm is at most $2|z_t|$, since $(r_g^b)^2 + (r_g^m)^2 = 4(z_g^+)^2 + 2(z_g^-)^2$. Thus the probability that any internal stacked norm exceeds a at any time is at most $(8 + 1024\mathsf{h})\mathrm{T}/\mathsf{a}^2$.

Every exterior link coordinate is absent from H_M, and its block in A_M is the scalar T. It is therefore an independent $\mathcal{N}(0,1)$ coordinate in the stationary Gaussian. Each exterior belief or model residual has variance at least one. The probability that any of these four residuals initially has absolute value at most $2\mathsf{a}$ is at most $8\sqrt{2/\pi}\,\mathsf{a}$.

To control their increments, diagonalize the symmetric positive matrix A_M. In a scalar eigencoordinate with rate $\lambda > 0$, stationarity and the preceding noise estimate imply

$$\mathbb{E}\sup_{t \le \mathsf{h}} |X_t - X_0|^2 \le \frac{2\mathrm{T}}{\lambda}(1 - e^{-\lambda \mathsf{h}})^2 + 64\mathrm{Th} \le 66\mathrm{Th},$$

since $(1-e^{-v})^2 \le v$ for $v \ge 0$. Summing twelve coordinates gives

$$\mathbb{E}\sup_{t\le \mathsf{h}} |\mathbf{X}_t - \mathbf{X}_0|^2 \le 792\mathrm{Th}. \tag{C.127}$$

Every residual covector has norm $\sqrt{3}$. Except with probability $2376\mathrm{Th}/\mathsf{a}^2$, all exterior increments are therefore at most a in absolute value. The initial small-ball estimate and the two maximal estimates prove $\mathsf{O}^{M}_{\mathrm{T},\mathsf{h}}(\mathscr{E}^{c}_{M}) \le \mathsf{b}$. Together with eq. (C.122), this proves eq. (C.124). The frozen path comparison proves the stated matching-window label limit whenever $\mathsf{b} \to 0$, $\mathsf{h}\delta \to 0$, and $\mathrm{Th} \to 0$, in particular for the displayed choices.

For a single time no increment estimate is needed. The normal second moment is at most T, so the internal failure probability is at most $4\mathrm{T}/\mathsf{a}^2$. The exterior small-ball bound at radius a is $4\sqrt{2/\pi}\mathsf{a}$. Thus the partition induced by $\nu_{\mathrm{T}}^{M_j}$ differs from the matching partition $\mathfrak{P}_j$ with probability at most e. The learned partition law is within $\delta + \mathsf{e}$ of $\delta_{\mathfrak{P}_M}$; the frozen one is within $\delta + 16\mathrm{T} + \mathsf{e}$ of $(\delta_{\mathfrak{P}_0} + \delta_{\mathfrak{P}_1})/2$. These matching partitions are distinct, so the triangle inequality for total variation proves eq. (C.125).

The schedule in corollary 15.6 permits tolerance T and mobility cap T^2. Then $\mathrm{T}^{-1/4} \le t_{\mathrm{L}}$ for sufficiently small T, so the following persistence window lies within the selection window. All errors just bounded tend to zero. The normal rates establish the restoring ratio on the native event. Finally, if $\max_i |p_i - p_i^M| \le \delta$, $\mathrm{TV}(\mu_{\mathrm{T},p}, \nu_{\mathrm{T}}^{M}) = 1 - \mu_{\mathrm{T},p}(M) \le 4\delta$. Applying the same occupation and coupling argument to this stationary frozen-row process gives the stationary lower bound in the statement. The persistence event is therefore already typical at equilibrium for those rows and carries no first-entrance claim. □

C.11.1 The first contrast on the original episode window

Proposition C.17 (A first-block null under slower source learning). In the model of theorem 15.5, let $0 < \mathrm{T} \le 10^{-16}$, and let the initial law be $\mathcal{N}(\mathsf{m}_0, I_{12})$ times uniform assignments. The learned and frozen arms have the same initial mean and differ only in the source-learning mobility ϵ_{L}. Set

$$H_{\mathrm{T}} = 2\mathrm{T}^{-1/8}.$$

Keep the contrast event of proposition 15.4 with its regime parameter equal to $\mathrm{T}^{1/8}$: an eligible proper connected block has an internal belief relative entropy exceeding $1/4$ initially, and for a duration-one interval inside $[0, H_{\mathrm{T}}]$ all its internal belief and model relative entropies are at most $\sqrt{\mathrm{T}}$, while its boundary belief relative entropies exceed $1/2$. For each block take the infimum of its qualifying completion times, with value infinity when there is no such interval. Let K_{T} be the block with least such infimum, using one fixed tie rule in both arms, and give it a cemetery value when no block qualifies. Write E_{T} for the event that its value is not the cemetery value. The infimum need not itself be a qualifying completion time.

Define the positive, temperature-independent constant

$$\mathsf{c}(\mathsf{m}_0) = \frac{e^{-1/4}\varrho_\circ^{12}}{737280}\exp\left[-\frac{1}{2}(|\mathbf{x}^\circ - \mathsf{m}_0| + \varrho_\circ)^2\right], \qquad \mathsf{d}_{\mathrm{T}} = \min\{1, 8\epsilon_{\mathrm{L}}\mathrm{T}^{-1/4}\}.$$

Then

$$\Pr_0(E_{\mathrm{T}}) \geq \mathsf{c}(\mathsf{m}_0), \qquad \Pr_{\epsilon_{\mathrm{L}}}(E_{\mathrm{T}}) \geq \mathsf{c}(\mathsf{m}_0) - \mathsf{d}_{\mathrm{T}}.$$

If $\mathsf{d}_{\mathrm{T}} < \mathsf{c}(\mathsf{m}_0)$, the two conditional block laws satisfy

$$\mathrm{TV}\left(\mathrm{Law}_{\epsilon_{\mathrm{L}}}(K_{\mathrm{T}} \mid E_{\mathrm{T}}), \mathrm{Law}_0(K_{\mathrm{T}} \mid E_{\mathrm{T}})\right) \leq \frac{\mathsf{d}_{\mathrm{T}}}{\mathsf{c}(\mathsf{m}_0)}.$$

In particular their difference vanishes for every fixed Gaussian mean whenever $\mathrm{T} \downarrow 0$ and $\epsilon_{\mathrm{L}}\mathrm{T}^{-1/4} \to 0$. This includes the caps $\epsilon_{\mathrm{L}} \leq \mathrm{T}_n^a$, $a > 7/2$, in corollary 15.6. [ESTABLISHED]

Proof. First prove a uniform comparison of the complete native path laws on any finite horizon H. The learned rows are deterministic and satisfy

$$|p_i(t) - 1/2| \leq \epsilon_{\mathrm{L}} t/2.$$

At a common configuration, write $A, B > 0$ for a receiver's likelihood factors for its two sources. Its posterior positive-source probabilities in the two arms are

$$b(p) = \frac{pA}{pA + (1-p)B}, \qquad b(1/2) = \frac{A}{A+B}.$$

Direct subtraction gives

$$|b(p) - b(1/2)| = \frac{AB|2p-1|}{(pA + (1-p)B)(A+B)} \leq |2p-1|.$$

Indeed the denominator is $pA^2 + AB + (1-p)B^2 \geq AB$. This bound is independent of temperature and the likelihood ratio, without an interior-row restriction.

Use identical initial samples, a common Brownian motion, and the same rate-one proposal clock in the two arms. Until the first different redraw, their configurations and assignments coincide: a held-sheet drift and its noise do not depend on the prior rows. At each ring, couple each receiver's two Bernoulli laws maximally. The union bound and the row estimate bound the probability of any different receiver choice at time t by

$$\sum_i |b_i(p_i(t)) - b_i(1/2)| \leq 4\epsilon_{\mathrm{L}} t.$$

The compensator of the rate-one clock, stopped at the first mismatch, bounds its probability before H by $\int_0^H 4\epsilon_{\mathrm{L}} t dt = 2\epsilon_{\mathrm{L}} H^2$. Consequently,

$$\mathrm{TV}\left(\mathrm{Law}_{\epsilon_{\mathrm{L}}}(\mathbf{X}_{[0,H]}, S_{[0,H]}), \mathrm{Law}_0(\mathbf{X}_{[0,H]}, S_{[0,H]})\right) \leq \min\{1, 2\epsilon_{\mathrm{L}} H^2\}. \tag{C.128}$$

The bound follows because the complete paths are identical unless the coupling has had a mismatch. Taking $H = H_{\mathrm{T}}$ gives d_{T}.

The first-block mark is a measurable function of these paths. To see this directly, for a fixed block let $A_I(t)$ be the maximum of its internal belief and model relative entropies on $[t-1, t]$, and let $B_I(t)$ be the minimum boundary belief relative entropy there. Configuration paths are continuous, so these two window functions are continuous. Replace the strict condition $B_I(t) > 1/2$ by the countable union of conditions $B_I(t) \geq 1/2 + 1/k$, $k \geq 1$. For each k the qualifying end times form a closed subset of the compact interval $[1, H_{\mathrm{T}}]$. Its first time is measurable, as may be checked by minimizing $\max(A_I(t) - \sqrt{\mathrm{T}}, 1/2 + 1/k - B_I(t))$ on each compact initial subinterval. The infimum over k, the initial eligibility condition, and the finite comparison over blocks preserve measurability. This construction implements the strict boundary convention without asserting that its infimum is attained.

A measurable map of paths does not increase total variation. Let a_I, b_I be the probabilities of the non-cemetery marks in the learned and frozen arms, $p = \sum_I a_I$, $q = \sum_I b_I$, and d the total-variation distance of the two cemetery-augmented mark laws. Then

$$2d = \sum_I |a_I - b_I| + |p - q|, \qquad d \leq \mathsf{d}_{\mathrm{T}}.$$

If $p \geq q > 0$, the triangle inequality gives

$$\frac{1}{2} \sum_I \left| \frac{a_I}{p} - \frac{b_I}{q} \right| \leq \frac{\sum_I |a_I - b_I| + (p - q)}{2p} = \frac{d}{p}.$$

Interchanging the arms proves the bound $d/\max(p, q)$. The frozen matching-seed sub-event in proposition 15.4 belongs to E_{T} and has probability at least $\mathsf{c}(\mathsf{m}_0)$. Thus $q \geq \mathsf{c}(\mathsf{m}_0)$ and $p \geq q - d \geq \mathsf{c}(\mathsf{m}_0) - \mathsf{d}_{\mathrm{T}}$. The claimed conditional bound follows, and its limiting conclusion is immediate. □

The proposition keeps the original contrast event and uses its seeded sub-event only as a probability lower bound. That sub-event already has the restoring interaction and stationary comparison of proposition 14.22; the proof does not certify every history in the broader contrast event under the coherence-episode rule. Any common measurable admissibility restriction retaining the seeded sub-event has the same comparison.

The result does not cover the original coupled choice $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$, for which its bound is vacuous. It also does not compare different initial means or establish a conditional parent error. On the slower schedules, late assignment selection can coexist with a vanishing difference between the conditional laws of the first contrast on $[0, H_{\mathrm{T}}]$. These are different observables on different horizons.

C.12 Activation from the Actual Conditioned Recognition Law

Proposition C.18 (Conditional information after a completed coherence prefix). In the learned four-cycle, fix $0 < \mathrm{T} \leq 1/100$, a selected matching M, and $s = t_{\mathrm{L}} \geq 1$. Suppose, throughout

$[s, 2s]$,

$$D_{\mathrm{KL}}(q_t \| \mu_{\mathrm{T},p(t)}) \le \eta, \qquad \mathrm{TV}(q_t, \nu_{\mathrm{T}}^{M}) \le \delta.$$

Set $\mathsf{a} = \mathrm{T}^{1/4}$, $\theta = s + 1$, and let $E = \mathscr{E}_M(\mathsf{a}, 1)$ be the actual matching and coherence event of proposition C.16 on $[s, s+1]$. Assume

$$\rho = 3\delta + \mathsf{b}(\mathrm{T}, \mathsf{a}, 1) < \tfrac{1}{2}, \qquad h_b(r) = -r \log r - (1-r)\log(1-r), \qquad \kappa_E = \frac{\eta + h_b(\rho)}{1-\rho}.$$

For all $t \ge \theta$, write $q_t^E = \mathrm{Law}(\mathbf{X}_t, S_t \mid E)$, always conditioning on this same completed prefix. Keep the single marked channel C of proposition 14.22(b), including its identity action off the matching strata and its identity extension on evaluator samples. The parent activation law is exactly

$$\mu_{\mathrm{act}} = C_{\#} q_\theta^E.$$

Let $\mathcal{H}_{\mathrm{opt}}(q)$ be the mean discarded conditional relative entropy of q against the restricted optimizer under C. Let $\mathcal{H}_{\mathrm{phys}}(q)$ be the corresponding mean discarded conditional relative entropy against the full physical posterior, after lifting q with its recognition evaluator and retaining every evaluator sample. Then, for every $t \in [\theta, 2s]$,

$$D_{\mathrm{KL}}(q_t^E \| \mu_{\mathrm{T},p(t)}) \le \kappa_E, \tag{C.129}$$

$$\mathcal{H}_{\mathrm{opt}}(q_t^E) \le \kappa_E, \tag{C.130}$$

$$\mathcal{H}_{\mathrm{phys}}(q_t^E) \le \kappa_E + \log \frac{4 + 5\mathrm{T}}{4 + \mathrm{T}} \le \kappa_E + \mathrm{T}. \tag{C.131}$$

The two hidden-conditional bounds hold with any strictly positive alternative source rows in their reference laws; the full-law bound eq. (C.129) refers to the native rows $p(t)$.

These are unscaled, dimensionless relative entropies. The corresponding scaled optimizer free-energy loss is at most $\mathrm{T}\kappa_E$; the unscaled physical negative-ELBO loss is at most $\kappa_E + \mathrm{T}$, and its T-scaled value is at most $\mathrm{T}\kappa_E + \mathrm{T}^2$. Along the joint selection schedule with $\eta, \delta \le \mathrm{T}$,

$$\rho = O(\mathrm{T}^{1/4}), \qquad \kappa_E = O(\mathrm{T}^{1/4} \log(1/\mathrm{T})).$$

The assumption $\rho < 1/2$ then holds for all sufficiently small T, so these bounds are asymptotic. [ESTABLISHED]

Proof. The learned-arm persistence bound eq. (C.124), applied with $t_0 = s$ and $\mathsf{h} = 1$, gives $r := \Pr(E^c) \le \rho$; its proof uses only the learned-arm hypothesis. Thus E is a θ-measurable event of probability at least $1 - \rho$. The law q_θ^E is the actual conditional law and may have retained–hidden correlations and a non-Gaussian hidden conditional. Its marked pushforward is normalized by construction. At later times no further event is conditioned upon. In particular, the deterministic native row path is not recomputed from q_t^E.

For a fixed $t \in [\theta, 2s]$, put $B = \mathbf{1}_E$ and $R_t = (\mathbf{X}_t, S_t)$. Compute the relative entropy of the joint law of (B, R_t) against $\mathrm{Bernoulli}(1-r) \otimes \mu_{\mathrm{T},p(t)}$ in two orders:

$$\begin{aligned}(1-r)D_{\mathrm{KL}}(q_t^E \| \mu_{\mathrm{T},p(t)}) + rD_{\mathrm{KL}}(q_t^{E^c} \| \mu_{\mathrm{T},p(t)}) \\ = D_{\mathrm{KL}}(q_t \| \mu_{\mathrm{T},p(t)}) + \mathbb{I}(B; R_t) \le \eta + h_b(r).\end{aligned} \tag{C.132}$$

When $r = 0$, use just the first term. The second term is nonnegative, h_b is increasing on $[0, 1/2]$, and $1 - r \ge 1 - \rho$. This proves eq. (C.129). It also establishes the needed absolute continuity and finiteness; directly, $q_t^E \le q_t/(1-r)$.

The channel's fibers can be checked explicitly. On a matching with edges e, f, let F_M have the orthonormal columns

$$(n_e^+ - n_f^+)/\sqrt{2}, \qquad n_e^-, \qquad n_f^-.$$

Write $Z_M = \operatorname{im} F_M$, $W_M = Z_M^{\perp}$, and $\mathbf{x} = y + F_M z$. The normal identities in eq. (C.94) give

$$\nu_{\mathrm{T},Z} = \mathcal{N}(0, \mathrm{T}K_Z^{-1}), \qquad K_Z = \mathrm{diag}(8 + \mathrm{T}, 4 + 5\mathrm{T}, 4 + 5\mathrm{T})$$

as the hidden optimizer conditional. Both the interaction Hessian and the evaluator precision preserve W_M and Z_M. Consequently this conditional is independent of the retained coordinate and of the prior rows. It is a reference conditional, not an assertion that the actual conditioned recognition law has that form. The same statement holds on the other matching by rotation; off the two matchings the channel is the identity and has no discarded conditional.

The target of C is a finite disjoint union of Euclidean strata, so disintegration gives the exact entropy chain rule

$$D_{\mathrm{KL}}(q_t^E \| \mu_{\mathrm{T},p(t)}) = D_{\mathrm{KL}}(C_{\#} q_t^E \| C_{\#} \mu_{\mathrm{T},p(t)}) + \mathcal{H}_{\mathrm{opt}}(q_t^E). \tag{C.133}$$

Dropping its retained term proves eq. (C.130). This argument uses the complete actual law; it neither resets the hidden state nor assumes Gaussianity conditional on a source history.

For the physical bound, work first on one matching fiber. Its physical outer hidden Gaussian is

$$\Pi_{\mathrm{T},Z} = \mathcal{N}(0, \mathrm{T}(K_Z^{\mathrm{phys}})^{-1}), \qquad K_Z^{\mathrm{phys}} = \mathrm{diag}(8 + \mathrm{T}, 4 + \mathrm{T}, 4 + \mathrm{T}).$$

All displayed matrices are positive definite. Write $v = 1 - \sigma_{\mathrm{ev}}^2 = 1/2$, $d_y = \mathfrak{D}y$, and $d_z = \mathfrak{D}F_M z$. Since $R_{\mathrm{ev}} = \mathfrak{D}^{\top}\mathfrak{D}/v$ preserves the split, $d_y \cdot d_z = 0$; its hidden eigenvalues are $0, 4, 4$.

Let $U = (m, k)$ denote the full evaluator sample vectors. At configuration $\mathbf{x}$, the physical evaluator has $m \sim \mathcal{N}(\bar{m}, \sigma_{\mathrm{ev}}^2 I)$ and independent increment $k - m \sim \mathcal{N}(0, vI)$. Its recognition counterpart, eq. (14.6), has the same first law and increment mean $d_y + d_z$. At a fixed retained y, form the comparison probability law

$$\overline{P}_y(dz, dU) = \Pi_{\mathrm{T},Z}(dz) P_{\mathrm{ev}}(dU \mid y + F_M z) \exp\left[\frac{d_y \cdot (k - m) - |d_y|^2/2}{v}\right]. \tag{C.134}$$

Completing the Gaussian square in $k - m$ proves that the multiplier integrates to one for every z. It changes the increment mean to d_y and keeps $\Pi_{\mathrm{T},Z}$ as its z marginal. The multiplier is retained-measurable in (y, U), so $\overline{P}_y$ has the physical hidden conditional given these retained variables. At a fixed z, the recognition evaluator differs by the additional increment mean d_z, with relative entropy $|d_z|^2/(2v)$.

Compute the joint relative entropy first through z, then through U, and drop the nonnegative evaluator-marginal term. For the actual hidden law $q(dz \mid M, y)$, this gives

$$\mathcal{H}_{\mathrm{phys}}(q \mid M, y) \le D_{\mathrm{KL}}(q(dz \mid M, y) \| \Pi_{\mathrm{T},Z}) + \mathbb{E}_q[|d_z|^2/(2v) \mid M, y].$$

All terms are well-defined: the native process has finite second moments, conditioning increases the second-moment bound by a factor at most $(1 - \rho)^{-1}$, and the Gaussian reference logarithms are quadratic. The optimizer conditional is the exact tilt

$$\nu_{\mathrm{T},Z}(dz) = \frac{\Pi_{\mathrm{T},Z}(dz) e^{-|d_z|^2/(2v)}}{Z_{\mathrm{tilt}}}, \qquad Z_{\mathrm{tilt}} = \frac{4 + \mathrm{T}}{4 + 5\mathrm{T}}.$$

For its normalizer, the common coordinate is unchanged and each of the two discrepancy coordinates changes rate from $4 + \mathrm{T}$ to $4 + 5\mathrm{T}$; their two square-root Gaussian determinant factors multiply to Z_{tilt}. Therefore

$$D_{\mathrm{KL}}(q(dz \mid M, y) \| \Pi_{\mathrm{T},Z}) + \mathbb{E}_q[|d_z|^2/(2v) \mid M, y] = D_{\mathrm{KL}}(q(dz \mid M, y) \| \nu_{\mathrm{T},Z}) + \log \frac{4 + 5\mathrm{T}}{4 + \mathrm{T}}.$$

Integrate over the retained law. Matching strata have total probability at most one, and all other strata have zero discarded loss. Finally,

$$0 \le \log \frac{4 + 5\mathrm{T}}{4 + \mathrm{T}} = \log \left(1 + \frac{4\mathrm{T}}{4 + \mathrm{T}} \right) \le \frac{4\mathrm{T}}{4 + \mathrm{T}} \le \mathrm{T}.$$

This proves eq. (C.131). Only recognition has been conditioned on E; the physical target still uses the original observed records, with all evaluator samples retained.

A source-prior factor π_S cancels in any conditional law on a fiber because S is retained. The hidden optimizer and physical reference conditionals are consequently the same for every strictly positive choice of prior rows. The full-law entropy need not remain small for different rows, because its retained term may change. An imposed zero prior on an occupied stratum would remove reference support and is excluded; ordinary finite-time learning from positive rows preserves positivity.

The exact free-energy chain rules give the stated scaled and unscaled losses. With $\eta, \delta \le \mathrm{T}$ and $\mathsf{a} = \mathrm{T}^{1/4}$,

$$\rho \le 3\mathrm{T} + 8\sqrt{2/\pi}\mathrm{T}^{1/4} + 3408\mathrm{T}^{1/2}.$$

Use $h_b(\rho) \le \rho \log(e/\rho)$ and $1 - \rho \ge 1/2$ eventually to obtain the displayed order for κ_E. □

Remark C.19 (A high-probability prefix is needed). The high probability of the declared prefix has mathematical content. Under the exact matching Gaussian ν_{T}^{M}, a hidden coordinate z_1 is independent of the retained state and has zero unconditioned hidden relative entropy. Conditioning on $\{z_1 > 0\}$, of probability $1/2$, instead gives a density ratio two on that half-space and hidden relative entropy $\log 2$ at every temperature. Thus a probability merely bounded below does not suffice for a vanishing conditional loss. Conditioning on the entire realized fine history is different again: it fixes the hidden endpoint and gives a Dirac conditional, singular relative to the nondegenerate Gaussian fiber. The activation here uses the full joint law conditional on E, not a realized fine history. [ESTABLISHED]

The proposition supplies conditional information at activation and for the native future marginals under that same completed prefix. It does not yet compare an independently evolving parent path. Its prefix is the late restored coherence event, and carries no first-formation or recursive-coalescence conclusion.

Inherited dynamics from the conditioned activation law

Proposition C.20 (Dynamics of the conditioned parent). Use the selected matching, channel, and conditioned activation of proposition C.18. More explicitly, let $0 < \mathrm{T} \le 1/100$, $s = t_{\mathrm{L}}$, $\theta = s + 1$, and let $\mathscr{E}$ be the selected coherence event on $[s, \theta]$ at resolution a from proposition C.16. Assume $\Pr(\mathscr{E}^c) \le \rho < 1$, and, throughout $[s, \theta + \mathsf{h}] \subseteq [t_{\mathrm{L}}, 2t_{\mathrm{L}}]$,

$$\mathrm{TV}(q_t, \nu_{\mathrm{T}}^{M}) \le \delta, \qquad \max_i |p_i(t) - p_i^{M}| \le \delta.$$

Write $\varepsilon = \epsilon_{\mathrm{L}}$, $d = \delta + \varepsilon\mathsf{h} < 1/4$, and use relative future time $u = t - \theta$. Initialize the parent with

$$q_0^{C} = C_{\#}\,\mathrm{Law}(\mathbf{X}_\theta, S_\theta \mid \mathscr{E}), \qquad p^{C}(0) = p(\theta).$$

On the stratum $\{S\} \times W_S$, its continuous generator is $-\mathrm{A}_S y \cdot \nabla_{W_S} + \mathrm{T}\Delta_{W_S}$. Its rate-one proposal kernel is

$$J_p((S, y), \cdot) = \int \kappa_S(d\mathbf{x} \mid y) \sum_{S'} b_p(S' \mid \mathbf{x}) \delta_{(S', P_{W_{S'}}\mathbf{x})}, \tag{C.135}$$

where $b_p(S' \mid \mathbf{x}) = \mu_{\mathrm{T},p}(S' \mid \mathbf{x})$ is the product of receiver posterior rows at rows p, and κ_S is the optimizing hidden conditional: the three independent Gaussian coordinates on a matching stratum, and a point mass off the matchings. The parent rows evolve from its own law:

$$\frac{d}{du} p_i^{C}(u) = \varepsilon\big(q_u^{C}(\sigma_i = +1) - p_i^{C}(u)\big). \tag{C.136}$$

This coupled parent system has a unique global law with interior rows.

Let $\mathrm{P}_{\theta,\mathsf{h}}^{\mathrm{fine},C}$ be the law of $(C(\mathbf{X}_{\theta+u}, S_{\theta+u}))_{0 \le u \le \mathsf{h}}$ conditioned only on $\mathscr{E}$, with the fine process continuing its original row path. Let $\mathrm{P}_{\mathsf{h}}^{C}$ be the parent path law from the displayed activation and its own

learning equation. Then

$$\mathrm{TV}(\mathsf{P}^{\mathrm{fine},C}_{\theta,\mathsf{h}}, \mathsf{P}^{C}_{\mathsf{h}}) \le 2\rho + (\mathsf{h}+3)\delta + \frac{4\mathsf{h}d}{1-4d}. \tag{C.137}$$

The comparison includes all retained coordinates and the assignment history, hence also the loop history. Self-proposals remain part of the rate-one kernel.

Along corollary 15.6, choose $\delta \le \mathrm{T}$, $\varepsilon \le \mathrm{T}^2$, $\mathsf{a} = \mathrm{T}^{1/4}$, and $\mathsf{h} = \mathrm{T}^{-1/4}$. One may take $\rho = 3\delta + \mathsf{b}(\mathrm{T}, \mathsf{a}, 1)$, where b is from proposition C.16. The right side of eq. (C.137) is then $O(\mathrm{T}^{1/4})$. This is a conditional, finite-window approximation after the selected coherence event; the activation law is not replaced by a Gaussian law. [ESTABLISHED]

Proof. For a matching M, the decomposition $\mathbb{R}^{12} = W_M \oplus Z_M$ of proposition 14.22 is preserved by H_M, Ω, and A_M. The hidden drift matrix is $\mathrm{diag}(8+\mathrm{T}, 4+5\mathrm{T}, 4+5\mathrm{T})$. The prior weight $\pi_S(p)$ cancels from the conditional law at fixed (S, y), so κ_S is independent of p. Off the matchings no coordinate is discarded. Thus the displayed continuous generator is the projection of every held-sheet diffusion, and eq. (C.135) is a probability kernel. A self-proposal has $S' = S$ and $P_{W_S}\mathbf{x} = y$, so it leaves the parent state unchanged. At a different assignment it uses that assignment's projection, including its twelve-dimensional stratum when appropriate.

For fixed rows, the pushed optimizing law $C_{\#}\mu_{\mathrm{T},p}$ is invariant. Indeed each diffusion is reversible on its Gaussian stratum, and $\mu_{\mathrm{T},p}(d\mathbf{x}, S)b_p(S' \mid \mathbf{x})$ is symmetric in S, S'. Pushing both endpoints through C gives detailed balance for J_p. The loop covector annihilates each Z_M, so this construction retains the loop through the stratum changes as well.

We next check the existence assertion with evolving parent rows. For any continuous interior row path, concatenate the linear diffusions with the rate-one proposals, drawing κ_S at each proposal. On every finite interval the clock has finitely many rings and every inserted Gaussian is finite almost surely, so this construction is nonexplosive. Its second moments remain finite: projection cannot increase the norm, an inserted hidden draw adds expected squared norm at most $5\mathrm{T}/8$, and the diffusion generator applied to $|y|^2$ is at most $24\mathrm{T} - 2\mathrm{T}|y|^2$. The full generator is consequently bounded above by $25\mathrm{T} - 2\mathrm{T}|y|^2$. The activation law has finite second moment because it conditions a finite-moment fine law on an event of positive probability.

If two row paths have entries in $[\mathrm{c}_0, 1-\mathrm{c}_0]$, couple their parent processes by the same initial state, Brownian increments, clocks, hidden Gaussian draws, and receiver uniforms until their first different proposal. The derivative of a posterior receiver probability with respect to its prior entry is at most $1/[4\mathrm{c}_0(1-\mathrm{c}_0)]$. Therefore

$$\mathrm{TV}(q^{C,p}_u, q^{C,\widetilde{p}}_u) \le \frac{1}{\mathrm{c}_0(1-\mathrm{c}_0)} \int_0^u \|p(v) - \widetilde{p}(v)\|_\infty dv.$$

The variation-of-constants map for eq. (C.136) consequently has Lipschitz constant at most $\varepsilon u_*^2/[2\mathrm{c}_0(1-\mathrm{c}_0)]$ on an interval of length u_*. On a short enough interval it is a contraction on continuous row paths in an interior cube. Its image stays in that cube after further shortening the

interval, since the row velocity has absolute value at most ε. Moreover its solution satisfies

$$p_i^C(u) \geq p_i^C(0)e^{-\varepsilon u}, \qquad 1 - p_i^C(u) \geq (1 - p_i^C(0))e^{-\varepsilon u}.$$

On any prescribed finite horizon these lower bounds give a common interior cube and a positive contraction step length. Successive steps give existence and uniqueness on that horizon, and then on all finite horizons. The source marginal is continuous because its jump rate is bounded, so the resulting rows are continuously differentiable. This constructs learning from the parent's own law, without imposing the fine row path after activation.

Let $\overline{\nu}_{\mathrm{T}}^M$ be the retained Gaussian on W_M. Write $\mathsf{O}_{\mathrm{T},\mathsf{h}}^C$ for its stationary held-M path law. The projection of $\mathsf{O}_{\mathrm{T},\mathsf{h}}^M$ from proposition C.16 is exactly this law. Only the learned-arm conclusions of that proposition are used, and their proof does not use its frozen-arm hypothesis. Conditioning any probability law on an event of probability $1 - r$ changes its total variation by at most r after any measurable projection. Thus the conditioned native retained future obeys

$$\mathrm{TV}(\mathsf{P}_{\theta,\mathsf{h}}^{\mathrm{fine},C}, \mathsf{O}_{\mathrm{T},\mathsf{h}}^C) \leq \rho + (\mathsf{h}+2)\delta. \tag{C.138}$$

This applies to a prefix event through θ. No condition is placed on future coherence or future proposals. The same argument at the activation time gives $\mathrm{TV}(q_0^C, \overline{\nu}_{\mathrm{T}}^M \otimes \delta_M) \leq \rho + \delta$.

For the parent's own rows let ζ_i^C be the probability in row i assigned to the nonpartner of M. Its equation and the bound on any probability give

$$0 \leq \zeta_i^C(u) \leq \zeta_i^C(0) + \varepsilon u \leq \delta + \varepsilon u \leq d.$$

The row-independent normalized sheet weights are at most the matching weight one for $0 < \mathrm{T} \leq 1/100$ by proposition C.15; hence, for any fixed row vector with this bound,

$$\mu_{\mathrm{T},p}(M) \geq \pi_M(p) \geq 1 - \sum_i \zeta_i \geq 1 - 4d.$$

Define the parent departure probability at a proposal by $k_p(y) = J_p((M,y), \{S' \neq M\})$. Detailed balance, and the fact that the optimizing conditional retained law on M is $\overline{\nu}_{\mathrm{T}}^M$ independently of the rows, give

$$\int k_p(y)\overline{\nu}_{\mathrm{T}}^M(dy) \leq \frac{1 - \mu_{\mathrm{T},p}(M)}{\mu_{\mathrm{T},p}(M)} \leq \frac{4d}{1-4d}. \tag{C.139}$$

The first inequality bounds stationary flux from M to its complement by the complement's stationary mass.

Maximally couple the parent activation with the stationary retained matching law and use common Brownian increments, clocks, and parent proposal draws up to the first parent departure. Before departure, the continuous paths agree because the matching drift splits, and self-proposals do not move the retained coordinate. The row path $p^C(u)$ constructed above is deterministic, even though it was computed from the parent's own law. Under the stationary comparison path, its marginal

at every time is $\overline{v}_{\mathrm{T}}^{M}$. Bound the first-departure compensator with its stopped indicator by the compensator evaluated on that whole stationary path. Applying eq. (C.139) at each deterministic $p^{C}(u)$ gives

$$\Pr(\text{parent departure before } \mathsf{h} \text{ while initially coupled}) \le \frac{4\mathrm{h}d}{1-4d}.$$

No Gaussian assertion is made about the parent's law conditioned on survival or about the native hidden variables. We have proved

$$\mathrm{TV}(\mathsf{P}_{\mathsf{h}}^{C}, \mathsf{O}_{\mathrm{T},\mathsf{h}}^{C}) \le \rho + \delta + \frac{4\mathrm{h}d}{1-4d}.$$

Combining this with eq. (C.138) proves eq. (C.137).

For the displayed joint choices, $d \le \mathrm{T} + \mathrm{T}^{7/4} \le 2\mathrm{T}$, $\mathrm{h}\delta \le \mathrm{T}^{3/4}$, and $\rho = O(\mathrm{T}^{1/4})$. The activation and following window fit inside the selection window because $1 + \mathrm{T}^{-1/4} \le t_{\mathrm{L}}$ for sufficiently small temperature under the mobility cap. The departure term is $O(\mathrm{T}^{3/4})$, giving the stated total error. The Gaussian comparison laws served only to bound that error; the parent's activation and subsequent learning remain the laws in the statement. □

C.13 Early Activation on the Seeded Contrast

The two preceding sections activate the channel of proposition 14.22(b) after the late restored coherence of the joint slow schedule. This section returns to the early contrast of proposition 15.4 at every mobility $\epsilon_{\mathrm{L}} \in [0,1]$, a range that contains the original coupled mobility $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$, and conditions the recognition law on the seeded sub-event of that contrast. Proposition C.25 bounds the discarded conditional information at a deterministic activation time from above and from below, proposition C.26 treats activation at the first qualifying completion time of the first block, and proposition C.27 couples the retained path of the own-law parent with the conditioned native path. Remark C.28 records why the stationary comparison of proposition C.20 is not used.

Definition C.21 (Seeded early prefix). Take the model of theorem 15.5 at the temperature $\mathrm{T} = \epsilon^{8}$ of eq. (14.51), with $0 < \epsilon \le 1/100$, a learning mobility $\epsilon_{\mathrm{L}} \ge 0$, and the initial law $\mathcal{N}(\mathsf{m}_0, I_{12})$ times the uniform law on assignments; $\epsilon_{\mathrm{L}} = 0$ is the frozen arm and $\epsilon_{\mathrm{L}} = \epsilon$ is the regime of proposition 15.4. Write $\pi_{ij}(t)$ for the receiver rows, $p_i = \pi_{i,i+1}$, $K_S = \nabla^2 U_S$ for the drift matrix of sheet S, written A_S in proposition C.20, $\mu_{\mathrm{T},p} \propto \pi_S(p) e^{-U_S/\mathrm{T}}$ for the restricted optimizer at rows p, and

$$\Lambda_{\mathrm{L}}(t) = 2e^{\epsilon_{\mathrm{L}} t} - 1, \qquad \delta_\epsilon = \frac{\epsilon^2}{8\sqrt{3}}, \qquad \theta_\circ = \tfrac{1}{2}\log(1/\epsilon) + 1.$$

The kinetics are driven by mutually independent inputs: the initial state $(\mathbf{X}_0, S_0)$, a standard Brownian motion W on $\mathbb{R}^{12}$, a rate-one clock, and uniform variables on $[0,1]$, one for each ring and receiver. At a ring at time s, receiver i takes source $i+1$ exactly when its variable is at most

$\beta_{i,i+1}(\mathbf{X}_{s-};\pi(s))$, where

$$\beta_{ij}(\mathbf{x};\pi)=\frac{\pi_{ij}e^{-\mathcal{D}_{e(i,j)}(\mathbf{x})/\mathrm{T}}}{\sum_{j'=i\pm1}\pi_{ij'}e^{-\mathcal{D}_{e(i,j')}(\mathbf{x})/\mathrm{T}}},$$

and $\mathcal{F}_t$ is generated by the initial state and the inputs on $[0,t]$. For a perfect matching $M\in\{S^{(0)},S^{(1)}\}$ with edges e,f, let F_M have the orthonormal columns $(n_e^+-n_f^+)/\sqrt{2}$, n_e^-, n_f^- of eq. (C.94), and put $Z_M=\operatorname{im}F_M$, $W_M=Z_M^{\perp}$, $K_Z=\operatorname{diag}(8+\mathrm{T},4+5\mathrm{T},4+5\mathrm{T})$, and $\nu_{\mathrm{T},Z}=\mathcal{N}(0,\mathrm{T}K_Z^{-1})$. The hidden coordinates are $z=F_M^{\top}\mathbf{x}=(z_1,z_2,z_3)$, so that $z_2=n_e^-\cdot\mathbf{x}$ and $z_3=n_f^-\cdot\mathbf{x}$, and the retained common coordinate is $z_+=u_M\cdot\mathbf{x}$ with $u_M=(n_e^++n_f^+)/\sqrt{2}$. The held process, its noise, and the center path are

$$\widetilde{\mathbf{X}}_t^M=e^{-K_Mt}\mathbf{X}_0+\mathbf{Z}_t^M,\qquad \mathbf{Z}_t^M=\sqrt{2\mathrm{T}}\int_0^t e^{-K_M(t-s)}dW_s,\qquad \bar{\mathbf{x}}^{\sigma,M}(t)=\sigma e^{-K_Mt}\mathbf{x}^{\circ},$$

for $\sigma\in\{1,-1\}$. The seeds, remainders, and seeded prefix are

$$\begin{aligned}\mathscr{B}_{\sigma,M}&=\{|\mathbf{X}_0-\sigma\mathbf{x}^{\circ}|<\varrho_{\circ}\}\cap\{S_0=M\}\cap\{\text{no ring in }[0,1/4]\},\\ \mathscr{A}_t^M&=\Big\{\sup_{s\le t}|\mathbf{Z}_s^M|\le\delta_{\epsilon}\Big\}\cap\{S_s=M\text{ for all }s\le t\},\\ \mathscr{E}_{\theta}^{\circ}&=\bigcup_{\sigma,M}\big(\mathscr{B}_{\sigma,M}\cap\mathscr{A}_{\theta}^M\big),\end{aligned}$$

and $\mathscr{B}^{\circ}=\bigcup_{\sigma,M}\mathscr{B}_{\sigma,M}$. At a deterministic activation time θ the conditioned law is $q_{\theta}^{\circ}=\mathrm{Law}(\mathbf{X}_{\theta},S_{\theta}\mid\mathscr{E}_{\theta}^{\circ})$, the channel C is that of proposition 14.22(b), and $\mu_{\mathrm{act}}=C_{\#}q_{\theta}^{\circ}$. The functionals $\mathcal{H}_{\mathrm{opt}}$ and $\mathcal{H}_{\mathrm{phys}}$ are those of proposition C.18, and $\mathcal{S}_2$ is the binary entropy of eq. (14.52).

Lemma C.22 (Discarded information under conditioning and mixing). Let (Y,Z) be a random element of a product of standard Borel spaces, with law Q, let κ be a Markov kernel from the space of Y to that of Z, and put

$$\mathcal{H}_{\kappa}(Q)=\mathbb{E}D_{\mathrm{KL}}\big(\mathrm{Law}(Z\mid Y)\|\kappa_Y\big)\in[0,\infty].$$

(i) Let B be a random variable with finitely many values b, probabilities $w_b>0$, and conditional laws Q_b of (Y,Z) given $B=b$. Then

$$\sum_b w_b\mathcal{H}_{\kappa}(Q_b)=\mathcal{H}_{\kappa}(Q)+\mathbb{I}(B;Z\mid Y),\qquad 0\le\mathbb{I}(B;Z\mid Y)\le-\sum_b w_b\log w_b.$$

In particular $\mathcal{H}_{\kappa}$ is convex. If A is an event with $\Pr(A^c)\le\bar{r}<1/2$ and Q_A is the conditional law of (Y,Z) given A, then $\mathcal{H}_{\kappa}(Q_A)\le[\mathcal{H}_{\kappa}(Q)+\mathcal{S}_2(\bar{r})]/(1-\bar{r})$, and for every event A of positive probability, $\mathcal{H}_{\kappa}(Q)\ge\Pr(A)\mathcal{H}_{\kappa}(Q_A)-\log 2$.

(ii) If Υ is a random element of a standard Borel space defined jointly with (Y,Z), then $\mathcal{H}_{\kappa}(Q)\le\mathbb{E}D_{\mathrm{KL}}(\mathrm{Law}(Z\mid Y,\Upsilon)\|\kappa_Y)$.

(iii) If $\mathrm{g}\ge0$ is measurable and $\int e^{\mathrm{g}(y,z)}\kappa_y(dz)\le\mathrm{c}_1$ for every y, then $\mathcal{H}_{\kappa}(Q)\ge\mathbb{E}\mathrm{g}(Y,Z)-\log\mathrm{c}_1$.

For a law q of $(\mathbf{X}, S)$ in the four-cycle, $\mathcal{H}_{\mathrm{opt}}(q) = \mathcal{H}_\kappa$ of the law of $(C(\mathbf{X}, S), Z)$, where $Z = F_S^\top \mathbf{X}$ and $\kappa = \nu_{\mathrm{T},Z}$ on the matching strata and $Z = 0$, $\kappa = \delta_0$ elsewhere; $\mathcal{H}_{\mathrm{phys}}(q)$ is the same functional with the evaluator samples added to Y and κ the physical hidden conditional. [ESTABLISHED]

Proof. All identities below are the chain rule for relative entropy on standard Borel spaces and hold in $[0, \infty]$. For (i), let J be the joint law of (B, Y, Z), and let $\widehat{J}$ keep the joint law of (B, Y) and draw Z from κ_Y. Disintegrating in (B, Y) gives $D_{\mathrm{KL}}(J\|\widehat{J}) = \sum_b w_b \mathcal{H}_\kappa(Q_b)$, because the conditional law of Z given $(B = b, Y)$ is the hidden conditional of Q_b. The (Y, Z)-marginal of $\widehat{J}$ is $\mathrm{Law}(Y) \otimes \kappa$, and under $\widehat{J}$ the variable B is conditionally independent of Z given Y. Disintegrating in (Y, Z) therefore gives $D_{\mathrm{KL}}(J\|\widehat{J}) = \mathcal{H}_\kappa(Q) + \mathbb{E} D_{\mathrm{KL}}(\mathrm{Law}(B \mid Y, Z)\|\,\mathrm{Law}(B \mid Y))$, and the last term is $\mathbb{I}(B; Z \mid Y)$, the difference of the conditional Shannon entropies of B given Y and given (Y, Z); it lies between zero and the entropy of B. Convexity follows because the information term is nonnegative. For an event A with $r = \Pr(A^c)$, the case $r = 0$ is trivial; otherwise $B = \mathbf{1}_A$ gives $(1-r)\mathcal{H}_\kappa(Q_A) \le \mathcal{H}_\kappa(Q) + \mathcal{S}_2(r)$, and $\mathcal{S}_2$ increases on $[0, 1/2]$ while $1 - r \ge 1 - \bar{r}$. The same identity with the term of A^c dropped gives $\mathcal{H}_\kappa(Q) \ge \Pr(A)\mathcal{H}_\kappa(Q_A) - \mathcal{S}_2(\Pr(A))$.

For (ii), the chain rule gives $D_{\mathrm{KL}}(\mathrm{Law}(Y, \Upsilon, Z)\|\,\mathrm{Law}(Y, \Upsilon) \otimes \kappa) = \mathbb{E} D_{\mathrm{KL}}(\mathrm{Law}(Z \mid Y, \Upsilon)\|\kappa_Y)$. Forgetting Υ maps the two laws to $\mathrm{Law}(Y, Z)$ and $\mathrm{Law}(Y) \otimes \kappa$, whose relative entropy is $\mathcal{H}_\kappa(Q)$, and marginalization does not increase relative entropy.

For (iii), fix y and put $P = \mathrm{Law}(Z \mid Y = y)$, $R = \kappa_y$, and $\mathrm{g}_n = \mathrm{g}(y, \cdot) \wedge n$. It suffices to show $D_{\mathrm{KL}}(P\|R) \ge \int \mathrm{g}(y, z) P(dz) - \log \mathrm{c}_1$ and to average over Y. The case $D_{\mathrm{KL}}(P\|R) = \infty$ is trivial, so let $D_{\mathrm{KL}}(P\|R) < \infty$. With $R_n = e^{\mathrm{g}_n} R / \mathbb{E}_R e^{\mathrm{g}_n}$, $\log(dP/dR) = \log(dP/dR_n) + \mathrm{g}_n - \log \mathbb{E}_R e^{\mathrm{g}_n}$, and integration against P gives $D_{\mathrm{KL}}(P\|R) \ge \mathbb{E}_P \mathrm{g}_n - \log \mathbb{E}_R e^{\mathrm{g}_n} \ge \mathbb{E}_P \mathrm{g}_n - \log \mathrm{c}_1$, and monotone convergence as $n \to \infty$ proves (iii).

On a matching stratum $\mathbf{x} = y + F_S z$ is a linear isometry between $\mathbb{R}^{12}$ and $W_S \times \mathbb{R}^3$, so conditioning on $C(\mathbf{X}, S)$ is conditioning on Y, and off the matchings the reference is the point mass at the constant Z; this identifies $\mathcal{H}_{\mathrm{opt}}$. The physical functional is identified in the same way with the retained variables $(C(\mathbf{X}, S), U)$. □

Lemma C.23 (Physical and optimizing discarded information). In the four-cycle at any strictly positive rows, let q be a law of $(\mathbf{X}, S)$ with $m_{\mathrm{h}}(q) = \mathbb{E}_q[\mathbf{1}_{\{S^{(0)}, S^{(1)}\}}(S)|F_S^\top \mathbf{X}|^2] < \infty$. Then

$$\mathcal{H}_{\mathrm{opt}}(q) - 4m_{\mathrm{h}}(q) - 2\mathrm{T} \le \mathcal{H}_{\mathrm{phys}}(q) \le \mathcal{H}_{\mathrm{opt}}(q) + \log \frac{4 + 5\mathrm{T}}{4 + \mathrm{T}} \le \mathcal{H}_{\mathrm{opt}}(q) + \mathrm{T}.$$

[ESTABLISHED]

Proof. Off the matchings neither functional discards anything. Fix a matching M and a retained point y, write $q_y = q(dz \mid M, y)$, $v = 1/2$, $d_y = \mathfrak{D} y$, $d_z = \mathfrak{D} F_M z$, and let $\Pi_{\mathrm{T},Z}$ and the probability law $\overline{P}_y$ be those of eq. (C.134). The proof of proposition C.18 shows that $\overline{P}_y$ has z-marginal $\Pi_{\mathrm{T},Z}$, that given z its evaluator law is $\mathcal{N}(\bar{m}(y + F_M z), \sigma_{\mathrm{ev}}^2 I)$ for m times $\mathcal{N}(d_y, vI)$ for $k - m$, that its conditional law of z given the evaluator sample U is the physical hidden conditional, and that $D_{\mathrm{KL}}(q_y\|\Pi_{\mathrm{T},Z}) + \mathbb{E}_{q_y}|d_z|^2/(2v) = D_{\mathrm{KL}}(q_y\|\nu_{\mathrm{T},Z}) + \log[(4 + 5\mathrm{T})/(4 + \mathrm{T})]$. The recognition lift $Q_y(dz, dU) = q_y(dz)\nu^*_{y + F_M z}(dU)$ of eq. (14.6) has, given z, the same law of m and increment law

$\mathcal{N}(d_y + d_z, vI)$, at relative entropy $|d_z|^2/(2v)$. Computing $D_{\mathrm{KL}}(Q_y\|\overline{P}_y)$ through z and through U gives, in $[0,\infty]$,

$$\mathcal{H}_{\mathrm{phys}}(q \mid M, y) + D_{\mathrm{KL}}\big(Q_y(dU)\|\overline{P}_y(dU)\big) = D_{\mathrm{KL}}(q_y\|\nu_{\mathrm{T},Z}) + \log\frac{4+5\mathrm{T}}{4+\mathrm{T}}.$$

Dropping the second term on the left gives the upper bound. For the lower bound, the two evaluator marginals are the U-marginals of $q_y(dz)\Pi_{\mathrm{T},Z}(dz')\nu^*_{y+F_Mz}(dU)$ and $q_y(dz)\Pi_{\mathrm{T},Z}(dz')\overline{P}_y(dU \mid z')$, so marginalization gives

$$D_{\mathrm{KL}}\big(Q_y(dU)\|\overline{P}_y(dU)\big) \le \iint \big(|\bar{m}(F_M(z-z'))|^2 + |d_z|^2\big)q_y(dz)\Pi_{\mathrm{T},Z}(dz'),$$

the integrand being the relative entropy of two Gaussian laws with common covariance $I/2$, since $\sigma^2_{\mathrm{ev}} = v = 1/2$. Since $\bar{m}$ is a block of coordinates, F_M has orthonormal columns, and $\mathfrak{D}\mathfrak{D}^\top = 2I_4$, the integrand is at most $|z-z'|^2 + 2|z|^2 \le 4|z|^2 + 2|z'|^2$, and $\mathbb{E}_{\Pi_{\mathrm{T},Z}}|z'|^2 = \mathrm{T}[1/(8+\mathrm{T}) + 2/(4+\mathrm{T})] \le 5\mathrm{T}/8$. Since $m_{\mathrm{h}}(q) < \infty$, $\mathbb{E}_{q_y}|z|^2 < \infty$ for almost every retained point y, and for such y the relative entropy of the evaluator marginals is at most $4\mathbb{E}_{q_y}|z|^2 + 5\mathrm{T}/4 < \infty$; subtracting it from the identity gives $\mathcal{H}_{\mathrm{phys}}(q \mid M, y) \ge D_{\mathrm{KL}}(q_y\|\nu_{\mathrm{T},Z}) - 4\mathbb{E}_{q_y}|z|^2 - 5\mathrm{T}/4$. Integrate over the retained law, whose matching strata have total mass at most one. Finally $\log[(4+5\mathrm{T})/(4+\mathrm{T})] \le 4\mathrm{T}/(4+\mathrm{T}) \le \mathrm{T}$. □

Lemma C.24 (Rows and the held seed). In definition C.21 the following hold.

(i) For every $\epsilon_{\mathrm{L}} \ge 0$ the rows exist uniquely, are deterministic and continuous, and every entry lies in $[e^{-\epsilon_{\mathrm{L}}t}/2, 1 - e^{-\epsilon_{\mathrm{L}}t}/2]$. The two entries of a row have ratio at most $\Lambda_{\mathrm{L}}(t)$; a receiver i with sources j, k satisfies $\beta_{ik}(\mathbf{x}; \pi(t)) \le \Lambda_{\mathrm{L}}(t)\exp[-(\mathcal{D}_{e(i,k)}(\mathbf{x}) - \mathcal{D}_{e(i,j)}(\mathbf{x}))/\mathrm{T}]$; and each of its two choice probabilities is at most $\Lambda_{\mathrm{L}}(t)$ times its value at rows $1/2$.

(ii) For each matching M, $F_M^\top\mathbf{x}^\circ = 0$ and $z_+(\mathbf{x}^\circ) = -1$. Along $\bar{\mathbf{x}}^{\sigma,M}$ the hidden coordinates vanish, $z_+ = -\sigma e^{-(8+\mathrm{T})t}$, every selected residual equals $-\sigma e^{-(8+\mathrm{T})t}$, and every exterior residual equals $\sigma e^{-\mathrm{T}t}(-3/2 + e^{-8t}/2)$, in both channels.

(iii) Suppose $|\mathbf{X}_0 - \sigma\mathbf{x}^\circ| < \varrho_\circ$ and $\sup_{s\le t}|\mathbf{Z}^M_s| \le \delta_\epsilon$, and let $u \le t$. Then $\widetilde{\mathbf{x}} = \widetilde{\mathbf{X}}^M_u$ satisfies $|\widetilde{\mathbf{x}} - \bar{\mathbf{x}}^{\sigma,M}(u)| \le \varrho_\circ + \delta_\epsilon$ and $|F_M^\top\widetilde{\mathbf{x}}| \le \varrho_\circ + \delta_\epsilon$; every selected residual satisfies $|r(\widetilde{\mathbf{x}})| \le e^{-(8+\mathrm{T})u} + \sqrt{3}\varrho_\circ e^{-4u} + \epsilon^2/8$, and every exterior residual satisfies $|r(\widetilde{\mathbf{x}})| \ge e^{-\mathrm{T}u}(3/2 - e^{-8u}/2) - \sqrt{3}(\varrho_\circ + \delta_\epsilon)$. At $u = 0$ the internal belief relative entropy of each matched pair exceeds $1/4$. If $\mathrm{T}t \le 1/12$ and $u \in [1/4, t]$, each receiver's exterior cost exceeds its partner cost by more than one and every exterior belief relative entropy exceeds $1/2$. If $u \ge \frac{1}{2}\log(1/\epsilon)$, every internal belief and model relative entropy of the two matched pairs is at most ϵ^4.

(iv) Let $s \ge 0$, let $\mathbf{X}_s$ be independent of $(W_{s+r} - W_s)_{r\ge0}$ with $|F_M^\top\mathbf{X}_s| \le \mathrm{R}$, and let $\widehat{\mathbf{X}}_{s+r} = e^{-K_Mr}\mathbf{X}_s + \sqrt{2\mathrm{T}}\int_s^{s+r} e^{-K_M(s+r-v)}dW_v$. For $r > 0$,

$$\mathcal{H}_{\mathrm{opt}}\big(\mathrm{Law}(\widehat{\mathbf{X}}_{s+r}, M)\big) \le \frac{9\mathrm{R}^2e^{-8r}}{2\mathrm{T}} + \frac{3e^{-16r}}{4(1-e^{-8r})}.$$

[ESTABLISHED]

Proof. (i) The proof of proposition 15.4(a) uses the mobility only as a positive constant; with ϵ_{L} in place of ϵ it gives existence, uniqueness, continuity, and $|\pi_{ij}(t) - 1/2| \le (1 - e^{-\epsilon_{\mathrm{L}} t})/2$, and the case $\epsilon_{\mathrm{L}} = 0$ is trivial. The ratio of the endpoints of the entry interval is $\Lambda_{\mathrm{L}}(t)$. The bound on β_{ik} drops the partner term from its denominator. With likelihood factors $\mathrm{c}_1, \mathrm{c}_2 > 0$ and entry p, the ratio of $\mathrm{pc}_1/(\mathrm{pc}_1 + (1-\mathrm{p})\mathrm{c}_2)$ to its value at $\mathrm{p} = 1/2$ is $\mathrm{p}(\mathrm{c}_1 + \mathrm{c}_2)/(\mathrm{pc}_1 + (1-\mathrm{p})\mathrm{c}_2) \le \max\{1, \mathrm{p}/(1-\mathrm{p})\}$, and similarly for the other choice.

(ii) Every residual of $\mathbf{x}^\circ$ equals -1. Since $\mathbf{a}_g^b = \sqrt{2}n_g^+ + n_g^-$ and $\mathbf{a}_g^m = \sqrt{2}n_g^+ - n_g^-$, this gives $n_g^- \cdot \mathbf{x}^\circ = 0$ and $\sqrt{2}n_g^+ \cdot \mathbf{x}^\circ = -1$ for $g \in \{e, f\}$, hence $F_M^\top \mathbf{x}^\circ = 0$ and $z_+(\mathbf{x}^\circ) = (\sqrt{2}n_e^+ + \sqrt{2}n_f^+) \cdot \mathbf{x}^\circ/2 = -1$. The residuals along the center path are eq. (C.95) for $S^{(0)}$; the rotation $i \mapsto i+1$ fixes $\mathbf{x}^\circ$ and maps $S^{(0)}$ to $S^{(1)}$, and the sign σ enters by linearity. Since $K_M F_M = F_M K_Z$ and u_M is an eigenvector of K_M with eigenvalue $8 + \mathrm{T}$, the hidden coordinates stay zero and $z_+ = -\sigma e^{-(8+\mathrm{T})t}$.

(iii) Since $K_M \succeq \mathrm{T}I$, $e^{-K_M u}$ is a contraction, so $|\widetilde{\mathbf{x}} - \bar{\mathbf{x}}^{\sigma,M}(u)| \le |\mathbf{X}_0 - \sigma\mathbf{x}^\circ| + |\mathbf{Z}_u^M|$, and the hidden coordinates of the center path vanish. For a residual covector $\mathbf{a}$ of norm $\sqrt{3}$, $r(\widetilde{\mathbf{x}}) = r(\bar{\mathbf{x}}^{\sigma,M}(u)) + (e^{-K_M u}\mathbf{a}) \cdot (\mathbf{X}_0 - \sigma\mathbf{x}^\circ) + \mathbf{a} \cdot \mathbf{Z}_u^M$. A selected covector lies in the span of eigenvectors of K_M with eigenvalues at least 4, so $|e^{-K_M u}\mathbf{a}| \le \sqrt{3}e^{-4u}$; an exterior one has $|e^{-K_M u}\mathbf{a}| \le \sqrt{3}$; and $\sqrt{3}\delta_\epsilon = \epsilon^2/8$. With (ii) this gives the two residual bounds. At $u = 0$ a selected residual has magnitude at least $1 - \sqrt{3}\varrho_\circ$, and $(1 - \sqrt{3}\varrho_\circ)^2/2 > 0.48$. For $u \in [1/4, t]$ with $\mathrm{T}t \le 1/12$, the selected residuals are at most $e^{-2} + \sqrt{3}(\varrho_\circ + \delta_\epsilon) < 0.153 < 1/5$, and the exterior ones at least $e^{-1/12}(3/2 - e^{-2}/2) - \sqrt{3}(\varrho_\circ + \delta_\epsilon) > 1.3004 > 13/10$; the partner cost $[(r^b)^2 + (r^m)^2]/2$ is below $1/25$, the exterior cost above $169/100$, and every exterior belief relative entropy $(r^b)^2/2$ exceeds $169/200$. For $u \ge \frac{1}{2}\log(1/\epsilon)$ a selected residual is at most $\epsilon^4 + \sqrt{3}\varrho_\circ\epsilon^2 + \epsilon^2/8 < 0.15\epsilon^2$, so the belief relative entropy $(r^b)^2/2$ and the model relative entropy $(r^m)^2/(2\sigma_{\mathrm{ev}}^2) = (r^m)^2$ are at most ϵ^4.

(iv) The matrix K_M preserves W_M and Z_M, and the projections of the increments of W onto these orthogonal subspaces are independent Brownian motions. Hence the retained part of $\widehat{\mathbf{X}}_{s+r}$ is a function of $\mathbf{X}_s$ and the retained increments, while its hidden part is $e^{-K_Z r}z_s + \xi_r$, with $z_s = F_M^\top \mathbf{X}_s$ and $\xi_r \sim \mathcal{N}(0, \mathrm{T}K_Z^{-1}(I - e^{-2K_Z r}))$ independent of $\mathbf{X}_s$ and of the retained increments. By lemma C.22(ii) with $\Upsilon = \mathbf{X}_s$, the left side is at most the mean relative entropy of $\mathcal{N}(e^{-K_Z r}z_s, \mathrm{T}K_Z^{-1}(I - e^{-2K_Z r}))$ to $\nu_{\mathrm{T},Z}$. For a coordinate of rate λ_k and $a_k = e^{-2\lambda_k r}$, this relative entropy is $\frac{1}{2}[-a_k - \log(1-a_k) + \lambda_k a_k z_{s,k}^2/\mathrm{T}]$. The rates lie in $[4, 9]$, so $\lambda_k a_k \le 9e^{-8r}$ and $a_k \le e^{-8r}$; moreover $|z_s| \le \mathrm{R}$ and $-a - \log(1-a) = \sum_{n\ge 2} a^n/n \le a^2/(2(1-a))$. □

Proposition C.25 (The seeded early prefix at a deterministic activation time). In definition C.21, let $\theta \ge 1/4$ with $\mathrm{T}\theta \le 1/12$. Put

$$\mathsf{n}_\theta = \min\Big\{73728\epsilon^4\theta, 24\exp\Big[-\frac{\epsilon^{-4}}{36864\theta}\Big]\Big\}, \qquad p_\theta^\circ = \mathsf{n}_\theta + 4\Lambda_{\mathrm{L}}(\theta)\big(\theta - \tfrac{1}{4}\big)e^{-1/\mathrm{T}},$$

$$\eta^\circ(\theta) = \frac{9\varrho_\circ^2 e^{-8\theta}}{2\mathrm{T}} + \frac{3e^{-16\theta}}{4(1-e^{-8\theta})}, \qquad \bar{\eta}^\circ(\theta) = \frac{\eta^\circ(\theta) + \mathcal{S}_2(p_\theta^\circ)}{1 - p_\theta^\circ}, \qquad \mathrm{c}_{\mathsf{m}} = e^{-2\varrho_\circ(|\mathsf{m}_0|+2+\varrho_\circ)},$$

and assume $p_\theta^\circ < 1/2$. Then the following hold.

(a) *Prefix.* The four seeds are disjoint, $\Pr((\mathscr{A}_\theta^M)^c \mid \mathscr{B}_{\sigma,M}) \le p_\theta^\circ$ for each of them, and $\Pr(\mathscr{E}_\theta^\circ) \ge (1-p_\theta^\circ)\Pr(\mathscr{B}^\circ) \ge \mathsf{c}(\mathsf{m}_0)$, with c from proposition C.17. On $\mathscr{B}_{\sigma,M} \cap \mathscr{A}_\theta^M$ the native path equals $\widetilde{\mathbf{X}}^M$ on $[0,\theta]$, $S_\theta = M$, and $|\mathbf{X}_\theta - \bar{\mathbf{x}}^{\sigma,M}(\theta)| \le \varrho_\circ + \delta_\epsilon$; in particular q_θ° is carried by the matching strata with hidden coordinates of norm at most $\varrho_\circ + \delta_\epsilon$.

(b) *Upper bound.*

$$\mathcal{H}_{\mathrm{opt}}(q_\theta^\circ) \le \bar{\eta}^\circ(\theta), \qquad \mathcal{H}_{\mathrm{phys}}(q_\theta^\circ) \le \bar{\eta}^\circ(\theta) + \log\frac{4+5\mathrm{T}}{4+\mathrm{T}} \le \bar{\eta}^\circ(\theta) + \mathrm{T}.$$

The rows enter only through $\Lambda_{\mathrm{L}}(\theta)$, and the bound does not depend on m_0. For $\epsilon_{\mathrm{L}} \in [0,1]$, as $\epsilon \to 0$ with $\epsilon^4\theta \to 0$ and $\theta - \log(1/\epsilon)$ bounded below, $\bar{\eta}^\circ(\theta) = \frac{9}{2}\varrho_\circ^2 e^{-8(\theta-\log(1/\epsilon))} + o(1)$, which tends to zero when $\theta - \log(1/\epsilon) \to \infty$; uniformly on $[t_\epsilon, t_*^\circ]$, $\bar{\eta}^\circ(\theta) = \frac{9}{2}\varrho_\circ^2 e^{-8(\theta-t_\epsilon)}\mathrm{T} + O(\mathrm{T}^4)$.

(c) *Lower bound.* If $\theta \le \log(1/\epsilon)$, then

$$\mathcal{H}_{\mathrm{opt}}(q_\theta^\circ) \ge \frac{4+5\mathrm{T}}{4\mathrm{T}}\Big[\frac{\varrho_\circ^2}{7}\mathsf{c}_{\mathsf{m}} e^{-2(4+5\mathrm{T})\theta} - 3\varrho_\circ^2 (p_\theta^\circ)^{1/2}\Big] - \log 2,$$

and $\mathcal{H}_{\mathrm{phys}}(q_\theta^\circ) \ge \mathcal{H}_{\mathrm{opt}}(q_\theta^\circ) - 1/2000$. For each fixed m_0 and $\epsilon_{\mathrm{L}} \in [0,1]$, both tend to infinity as $\epsilon \to 0$ with $\log(1/\epsilon) - \theta \to \infty$, and at $\theta = \theta_\circ$ they are at least $(e^{-8}\varrho_\circ^2/7)\mathsf{c}_{\mathsf{m}}\mathrm{T}^{-1/2}(1+o(1))$.

(d) *Future marginals under the same prefix.* For $t \ge \theta$ with $\mathrm{T}t \le 1/12$ and $p_t^\circ < 1/2$, put $\mathfrak{L}_t = 17(|\mathsf{m}_0|^2+14)/(2\mathrm{T}) + 4(\log 2 + \epsilon_{\mathrm{L}} t)(t+t^2)^{1/2}$. Then

$$\mathcal{H}_{\mathrm{opt}}\big(\mathrm{Law}(\mathbf{X}_t, S_t \mid \mathscr{E}_\theta^\circ)\big) \le \bar{\eta}^\circ(t) + \frac{p_t^\circ \log\frac{1}{p_t^\circ \Pr(\mathscr{B}^\circ)} + \big(p_t^\circ/\Pr(\mathscr{B}^\circ)\big)^{1/2}\mathfrak{L}_t}{1-p_\theta^\circ},$$

and $\mathcal{H}_{\mathrm{phys}}$ of the same law is at most the right side plus T. For each fixed m_0 and $\epsilon_{\mathrm{L}} \in [0,1]$, the fraction tends to zero as $\epsilon \to 0$, uniformly for $\theta \le t \le \epsilon^{-4}/(663552\log(1/\epsilon))$.

(e) *A path-measurable prefix.* For $\theta > 1/4$, let $\mathscr{E}_\theta^{\mathrm{path}}$ be the union over seeds of the events $\{|\mathbf{X}_0 - \sigma\mathbf{x}^\circ| < \varrho_\circ\} \cap \{S_s = M \text{ for } s \le \theta\} \cap \{\sup_{s\le\theta}|\mathbf{X}_s - e^{-K_M s}\mathbf{X}_0| \le \delta_\epsilon\}$, a function of the configuration and assignment path on $[0,\theta]$ that contains $\mathscr{E}_\theta^\circ$. Put $p_\theta^{\mathrm{path}} = e^{1/4}\mathsf{n}_\theta/(1-\mathsf{n}_\theta) + 4\Lambda_{\mathrm{L}}(\theta)(\theta-\frac{1}{4})e^{-1/\mathrm{T}}$, and let $\eta^{\mathrm{path}}(\theta)$ be $\eta^\circ(\theta-\frac{1}{4})$ with $\varrho_\circ$ replaced by $\varrho_\circ + \delta_\epsilon$. If $p_\theta^{\mathrm{path}} < 1/2$, then $\mathcal{H}_{\mathrm{opt}}(\mathrm{Law}(\mathbf{X}_\theta, S_\theta \mid \mathscr{E}_\theta^{\mathrm{path}})) \le [\eta^{\mathrm{path}}(\theta) + \mathcal{S}_2(p_\theta^{\mathrm{path}})]/(1-p_\theta^{\mathrm{path}})$, and $\mathcal{H}_{\mathrm{phys}}$ of that law is at most this bound plus T. [ESTABLISHED]

Parts (b) and (c) fix the activation clock up to the window $\theta = \log(1/\epsilon) + O(1)$. On that window both bounds are of order one, and the estimates leave open whether the discarded information tends to zero there. The window $[t_\epsilon, t_*^\circ]$ of proposition 14.22 lies $\log(1/\epsilon)$ beyond the threshold. Only the discarded conditional term is bounded; the relative entropy of q_θ° to the restricted optimizer also contains a retained term, which is not bounded here.

Proof. (a) The balls about $\pm\mathbf{x}^\circ$ are disjoint because $|\mathbf{x}^\circ| = 2 > \varrho_\circ$, and S_0 separates the matchings. The initial state and the clock are independent, so $\Pr(\mathscr{B}_{\sigma,M}) = \frac{e^{-1/4}}{16}\mathcal{N}(\mathsf{m}_0, I_{12})\{|\mathbf{x} - \sigma\mathbf{x}^\circ| < \varrho_\circ\}$, and the proof of proposition 15.4(b) bounds this below by $\mathsf{c}(\mathsf{m}_0)$ for $\sigma = 1$; hence $\Pr(\mathscr{B}^\circ) \ge 2\mathsf{c}(\mathsf{m}_0)$. On $\mathscr{B}_{\sigma,M} \cap \{S_s = M \text{ for } s \le t\}$ the native configuration and $\widetilde{\mathbf{X}}^M$ solve the same linear equation

with the same Brownian path between rings, and a ring that keeps the assignment does not move the configuration, so they agree on $[0,t]$; with lemma C.24(iii) this gives the last assertions.

The tube event depends only on W, which is independent of the seed. Doob's L^2 maximal inequality gives $\mathbb{E}\sup_{s\le\theta}|\mathbf{Z}_s^M|^2 \le 384\mathrm{T}\theta$, as in the paragraph *Episode* of the proof of proposition 14.22, and Chebyshev's inequality bounds the tube failure by $384\mathrm{T}\theta/\delta_\epsilon^2 = 73728\epsilon^4\theta$. In an orthonormal eigenbasis of K_M the same paragraph gives $\sup_{s\le\theta}|Z_s^k| \le 2\sqrt{2\mathrm{T}}\sup_{s\le\theta}|W_s^k|$, so the tube fails only if some $|W^k|$ exceeds $\mathrm{a}_\delta = \delta_\epsilon/(4\sqrt{6\mathrm{T}})$ before θ, where $\mathrm{a}_\delta^2 = \epsilon^{-4}/18432$. The reflection principle and $\Pr(\mathcal{N}(0,1)\ge x) \le e^{-x^2/2}/2$ bound this by $2e^{-\mathrm{a}_\delta^2/(2\theta)}$ for each of the twelve coordinates. Hence the tube fails with probability at most n_θ.

For departures, let N count the rings in $(1/4,\theta]$ that are preceded by the good state, in which the seed holds and the assignment has equaled M and the noise has stayed in the tube up to the ring, and at which the redraw differs from M. In the good state the configuration is held, so by lemma C.24(iii) every exterior cost exceeds the partner cost by more than one, and by part (i) of that lemma the redraw leaves M with conditional probability at most $4\Lambda_{\mathrm{L}}(\theta)e^{-1/\mathrm{T}}$, the variables of the ring being independent of the past. The seed is $\mathcal{F}_{1/4}$-measurable, so the compensator of the clock gives $\mathbb{E}[N \mid \mathscr{B}_{\sigma,M}] \le 4\Lambda_{\mathrm{L}}(\theta)(\theta-\tfrac14)e^{-1/\mathrm{T}}$. If the tube holds on $[0,\theta]$ and the assignment leaves M by time θ, the first departure occurs after $1/4$ from the good state, so $N \ge 1$; Markov's inequality completes the remainder bound. Finally $\Pr(\mathscr{E}_\theta^\circ) = \sum \Pr(\mathscr{B}_{\sigma,M})\Pr(\mathscr{A}_\theta^M \mid \mathscr{B}_{\sigma,M}) \ge (1-p_\theta^\circ)\Pr(\mathscr{B}^\circ)$, which is at least $\mathsf{c}(\mathsf{m}_0)$ because $p_\theta^\circ < 1/2$.

(b) Fix a seed and write $\mathscr{B} = \mathscr{B}_{\sigma,M}$, $\mathscr{A} = \mathscr{A}_\theta^M$. Under $\Pr(\cdot \mid \mathscr{B})$ the Brownian motion keeps its law and is independent of $\mathbf{X}_0$, and $|F_M^\top \mathbf{X}_0| = |F_M^\top(\mathbf{X}_0 - \sigma\mathbf{x}^\circ)| < \varrho_\circ$ by lemma C.24(ii). By part (iv) of that lemma with $s=0$ and $\mathrm{R}=\varrho_\circ$, the law $\widehat{Q}$ of $(\widetilde{\mathbf{X}}_\theta^M, M)$ under $\Pr(\cdot \mid \mathscr{B})$ has $\mathcal{H}_{\mathrm{opt}}(\widehat{Q}) \le \eta^\circ(\theta)$. By (a), the law of $(\mathbf{X}_\theta, S_\theta)$ given $\mathscr{B}\cap\mathscr{A}$ is the conditional law of $\widehat{Q}$ given $\mathscr{A}$, and $\Pr(\mathscr{A}^c \mid \mathscr{B}) \le p_\theta^\circ < 1/2$, so lemma C.22(i) bounds its discarded information by $\bar\eta^\circ(\theta)$. The law q_θ° is the mixture of these four seed laws, and convexity gives the first bound. The physical bound follows from lemma C.23, the hidden coordinates being bounded by (a).

For the asymptotic statements, $\Lambda_{\mathrm{L}}(\theta) \le 2e^\theta$ when $\epsilon_{\mathrm{L}} \le 1$, so $p_\theta^\circ \le 24\exp[-\epsilon^{-4}/(36864\theta)] + 8\theta e^\theta e^{-1/\mathrm{T}}$. If $\epsilon^4\theta \to 0$, then $\theta \le \epsilon^{-4}$ eventually and both terms tend to zero, hence so do $\mathcal{S}_2(p_\theta^\circ)$ and $p_\theta^\circ\eta^\circ(\theta)$ whenever $\eta^\circ(\theta)$ stays bounded. Moreover $9\varrho_\circ^2 e^{-8\theta}/(2\mathrm{T}) = \frac92\varrho_\circ^2 e^{-8(\theta-\log(1/\epsilon))}$, and the second term of $\eta^\circ(\theta)$ tends to zero because $\theta\to\infty$. On $[t_\epsilon, t_*^\circ]$ one has $e^{-8\theta} = \mathrm{T}^2 e^{-8(\theta-t_\epsilon)}$, so the second term is at most $\mathrm{T}^4/(1-\mathrm{T}^2)$, and $\epsilon^{-4}/(36864 t_*^\circ)$ grows faster than every multiple of $\log(1/\epsilon)$, so p_θ° and $\mathcal{S}_2(p_\theta^\circ)$ are $O(\mathrm{T}^k)$ for every k.

(c) Put $z_\perp = (z_2, z_3)$ on each matching stratum. Under $\nu_{\mathrm{T},Z}$ the coordinates z_2, z_3 are independent with variance $\mathrm{T}/(4+5\mathrm{T})$, so $\mathrm{g} = (4+5\mathrm{T})|z_\perp|^2/(4\mathrm{T})$ has $\int e^{\mathrm{g}} d\nu_{\mathrm{T},Z} = 2$, and lemma C.22(iii) gives $\mathcal{H}_{\mathrm{opt}}(q) \ge \frac{4+5\mathrm{T}}{4\mathrm{T}}\mathbb{E}_q|z_\perp|^2 - \log 2$ for every law q carried by the matching strata. Since q_θ° is a mixture of the seed laws and $\Pr(\mathscr{A} \mid \mathscr{B}) \le 1$, $\mathbb{E}_{q_\theta^\circ}|z_\perp|^2$ is at least the least over seeds of $\mathbb{E}[|\widetilde{z}_\perp(\theta)|^2 \mathbf{1}_{\mathscr{A}} \mid \mathscr{B}]$, where $\widetilde{z}$ are the hidden coordinates of the held process. In these, $\widetilde{z}_\perp(\theta) = e^{-(4+5\mathrm{T})\theta} z_\perp(\mathbf{X}_0) + \xi_\perp$, with ξ centered and independent of $\mathbf{X}_0$ under $\Pr(\cdot \mid \mathscr{B})$, so $\mathbb{E}[|\widetilde{z}_\perp(\theta)|^2 \mid \mathscr{B}] \ge e^{-2(4+5\mathrm{T})\theta}\mathbb{E}[|z_\perp(\mathbf{X}_0)|^2 \mid \mathscr{B}]$. Given the seed, the offset $\mathbf{X}_0 - \sigma\mathbf{x}^\circ$ has a density with respect to the uniform law on the ball of radius $\varrho_\circ$ that is at least c_{m}: the squared distances from m_0 of two points of the ball

differ by at most $4\varrho_\circ(|\sigma\mathbf{x}^\circ - \mathrm{m}_0| + \varrho_\circ)$, so the Gaussian density on the ball varies by at most the factor $\exp[2\varrho_\circ(|\sigma\mathbf{x}^\circ - \mathrm{m}_0| + \varrho_\circ)]$, and $|\sigma\mathbf{x}^\circ - \mathrm{m}_0| \le |\mathrm{m}_0| + 2$. Under the uniform law each orthonormal coordinate has second moment $\varrho_\circ^2/14$, and $z_\perp(\mathbf{X}_0) = z_\perp(\mathbf{X}_0 - \sigma\mathbf{x}^\circ)$ consists of two of them, so $\mathbb{E}[|z_\perp(\mathbf{X}_0)|^2 \mid \mathscr{B}] \ge \mathrm{c_m}\varrho_\circ^2/7$. On the other side $|\widetilde{z}(\theta)| \le \varrho_\circ + |\xi|$ and $\operatorname{Cov}\xi \preceq (\mathrm{T}/4)I_3$, so $\mathbb{E}[|\widetilde{z}(\theta)|^4 \mid \mathscr{B}] \le 8\varrho_\circ^4 + 8\cdot\frac{15}{16}\mathrm{T}^2 \le 9\varrho_\circ^4$, and the Cauchy–Schwarz inequality with (a) gives $\mathbb{E}[|\widetilde{z}_\perp(\theta)|^2\mathbf{1}_{\mathscr{A}^c} \mid \mathscr{B}] \le 3\varrho_\circ^2(p_\theta^\circ)^{1/2}$. This proves the display. The physical bound is lemma C.23 with $4(\varrho_\circ + \delta_\epsilon)^2 + 2\mathrm{T} < 1/2000$.

For the limits, write $\theta = \log(1/\epsilon) - \mathrm{s}$. Then $e^{-2(4+5\mathrm{T})\theta}/\mathrm{T} = e^{8\mathrm{s}}e^{-10\mathrm{T}\theta}$ with $\mathrm{T}\theta \to 0$, while $p_\theta^\circ \le 24\exp[-\epsilon^{-4}/(36864\log(1/\epsilon))] + 8\epsilon^{-1}\log(1/\epsilon)e^{-1/\mathrm{T}}$, so $(p_\theta^\circ)^{1/2}/\mathrm{T} \to 0$. Hence the lower bound is $\frac{\varrho_\circ^2}{7}\mathrm{c_m}e^{8\mathrm{s}}(1+o(1)) - \log 2$, which tends to infinity with s; at $\theta = \theta_\circ$, $e^{8\mathrm{s}} = e^{-8}\mathrm{T}^{-1/2}$.

(d) Decompose $\mathscr{E}_\theta^\circ$ into $\mathscr{E}_t^\circ$ and $\mathscr{G}_t = \mathscr{E}_\theta^\circ \setminus \mathscr{E}_t^\circ$, since $\mathscr{A}_t^M \subseteq \mathscr{A}_\theta^M$. Convexity bounds the left side by $[\Pr(\mathscr{E}_t^\circ)\bar\eta^\circ(t) + \Pr(\mathscr{G}_t)\mathcal{H}_{\mathrm{opt}}(q_t^{\mathscr{G}_t})]/\Pr(\mathscr{E}_\theta^\circ)$, by (b) at time t, where $q_t^{\mathscr{G}_t}$ is the law of $(\mathbf{X}_t, S_t)$ given $\mathscr{G}_t$. Since the two seed balls are disjoint, $\Pr(\mathscr{B}^\circ) \le e^{-1/4}/8$, so (a) gives $\Pr(\mathscr{G}_t) \le p_t^\circ\Pr(\mathscr{B}^\circ) \le 1/16$ and $\Pr(\mathscr{E}_\theta^\circ) \ge (1-p_\theta^\circ)\Pr(\mathscr{B}^\circ)$.

To bound $\mathcal{H}_{\mathrm{opt}}(q_t^{\mathscr{G}_t})$, let $\mathcal{F}_t^{\mathrm{ch}}$ be generated by the initial state, by W and the clock on $[0,t]$, and by the source choices at the rings in $[0,t]$; the configuration and assignment path on $[0,t]$ is a measurable function of these data, and $\mathscr{G}_t \in \mathcal{F}_t^{\mathrm{ch}}$. Let Q be the law of the same data for the kinetics with rows frozen at $1/2$ started from $\mu_{\mathrm{T},1/2}$, which those kinetics leave invariant (proposition 14.22). Under both laws W and the clock have the same law and are independent of the initial state, and given the past the choices at a ring are independent Bernoulli variables with probabilities $\beta_{i,i+1}(\mathbf{X}_{s-};\pi(s))$ and $\beta_{i,i+1}(\mathbf{X}_{s-};1/2)$. Hence $\log(d\Pr/d\mathrm{Q})$ on $\mathcal{F}_t^{\mathrm{ch}}$ is the log density ratio of the initial laws plus the sum, over rings in $[0,t]$ and receivers, of the log ratios of the realized choice probabilities. The initial ratio is $(2\pi)^{-6}e^{-|\mathbf{x}-\mathrm{m}_0|^2/2}\mathrm{Z}e^{U_S(\mathbf{x})/\mathrm{T}}$ with $\mathrm{Z} = \frac{1}{16}\sum_S\int e^{-U_S/\mathrm{T}}d\mathbf{x} \le (2\pi)^6$, because $U_S \ge \mathrm{T}|\mathbf{x}|^2/2$; its logarithm is at most $U_S(\mathbf{x})/\mathrm{T} \le 17|\mathbf{x}|^2/(2\mathrm{T})$, since each of the four edge Hessians in H_S has norm 4 and $\|R_{\mathrm{ev}}\| = 4$. By lemma C.24(i) each choice ratio is at most $\Lambda_{\mathrm{L}}(t) \le 2e^{\epsilon_{\mathrm{L}}t}$. With N_t the number of rings in $[0,t]$, the positive part of $\log(d\Pr/d\mathrm{Q})$ is at most $17|\mathbf{X}_0|^2/(2\mathrm{T}) + 4N_t(\log 2 + \epsilon_{\mathrm{L}}t)$, whose $L^2(\Pr)$ norm is at most $\mathfrak{L}_t$ because $\mathbb{E}|\mathbf{X}_0|^4 = (|\mathrm{m}_0|^2+12)^2 + 4|\mathrm{m}_0|^2 + 24 \le (|\mathrm{m}_0|^2+14)^2$ and $\mathbb{E}N_t^2 = t + t^2$. The chain rule eq. (C.133) with the hidden conditional $\nu_{\mathrm{T},Z}$ of $\mu_{\mathrm{T},1/2}$, and marginalization to time t, give

$$\mathcal{H}_{\mathrm{opt}}(q_t^{\mathscr{G}_t}) \le D_{\mathrm{KL}}(q_t^{\mathscr{G}_t}\|\mu_{\mathrm{T},1/2}) \le \log\frac{1}{\Pr(\mathscr{G}_t)} + \frac{\mathbb{E}[\mathbf{1}_{\mathscr{G}_t}\log(d\Pr/d\mathrm{Q})]}{\Pr(\mathscr{G}_t)} \le \log\frac{1}{\Pr(\mathscr{G}_t)} + \frac{\mathfrak{L}_t}{\Pr(\mathscr{G}_t)^{1/2}},$$

the last step by the Cauchy–Schwarz inequality. Since $x\log(1/x)$ increases on $(0,1/e]$, the display of (d) follows. The physical bound uses lemma C.23; the native configuration has finite second moments, and conditioning multiplies them by at most $1/\Pr(\mathscr{E}_\theta^\circ)$.

For the limit, let $t \le \epsilon^{-4}/(663552\log(1/\epsilon))$. Then $\epsilon^{-4}/(36864t) \ge 18\log(1/\epsilon)$, so $\mathrm{n}_t \le 24\epsilon^{18}$, and $4\Lambda_{\mathrm{L}}(t)te^{-1/\mathrm{T}} \le 8te^te^{-1/\mathrm{T}}$ is smaller than any power of ϵ. Also $\mathfrak{L}_t = O(\epsilon^{-8})$ for fixed m_0, because $t \le \epsilon^{-4}$. Hence $(p_t^\circ)^{1/2}\mathfrak{L}_t = O(\epsilon)$ and $p_t^\circ\log(1/p_t^\circ) \to 0$, while $\Pr(\mathscr{B}^\circ) > 0$ does not depend on ϵ.

(e) On $\{S_s = M \text{ for } s \le \theta\}$ one has $\mathbf{X}_s - e^{-K_M s}\mathbf{X}_0 = \mathbf{Z}_s^M$, which gives the inclusion and shows that each piece of $\mathscr{E}_\theta^{\mathrm{path}}$ is the intersection of $\mathscr{B}^{\mathrm{path}} = \{|\mathbf{X}_0 - \sigma\mathbf{x}^\circ| < \varrho_\circ\} \cap \{S_s = M \text{ for } s \le \frac{1}{4}\} \cap$

$\{\sup_{s\le 1/4}|\mathbf{Z}_s^M|\le\delta_\epsilon\}$, which lies in $\mathcal{F}_{1/4}$, with $\mathscr{A}^{\mathrm{path}}=\{S_s=M \text{ for } \frac14<s\le\theta\}\cap\{\sup_{s\le\theta}|\mathbf{Z}_s^M|\le\delta_\epsilon\}$. On $\mathscr{B}^{\mathrm{path}}$, lemma C.24(iii) gives $|F_M^\top\mathbf{X}_{1/4}|\le\varrho_\circ+\delta_\epsilon$, and the Brownian increments after $1/4$ are independent of $\mathcal{F}_{1/4}$, so part (iv) of that lemma with $s=1/4$ bounds the discarded information of the held continuation at θ by $\eta^{\mathrm{path}}(\theta)$; on $\mathscr{B}^{\mathrm{path}}\cap\mathscr{A}^{\mathrm{path}}$ the native process equals that continuation. Let $\mathscr{O}$ be the event $\{|\mathbf{X}_0-\sigma\mathbf{x}^\circ|<\varrho_\circ\}\cap\{S_0=M\}$. Since W is independent of the initial state and of the clock, $\Pr(\mathscr{B}^{\mathrm{path}})\ge\Pr(\mathscr{O})e^{-1/4}(1-\mathsf{n}_\theta)$, while $\Pr(\mathscr{B}^{\mathrm{path}}\cap\{\text{tube fails on }[0,\theta]\})\le\Pr(\mathscr{O})\mathsf{n}_\theta$. Departures in $(1/4,\theta]$ are bounded as in (a), after the $\mathcal{F}_{1/4}$-measurable seed. Hence $\Pr((\mathscr{A}^{\mathrm{path}})^c\mid\mathscr{B}^{\mathrm{path}})\le p_\theta^{\mathrm{path}}$, and lemma C.22(i), convexity over the four pieces, and lemma C.23 finish the proof. □

Proposition C.26 (Activation at the first qualifying completion). In definition C.21, let $\epsilon_{\mathrm{L}}\in[0,1]$, and let E_{T} and K_{T} be the contrast event and the first-block mark of proposition C.17 with its regime parameter equal to $\mathrm{T}^{1/8}=\epsilon$. Let t_{c} be the least, over eligible blocks, of the infima of their qualifying completion times, so that $\mathsf{t}_{\mathrm{c}}<\infty$ exactly on E_{T}. With p° and c_{m} from proposition C.25, put

$$\mathscr{U}^\circ=\bigcup_{\sigma,M}\big(\mathscr{B}_{\sigma,M}\cap\mathscr{A}^M_{\mathsf{t}_{\mathrm{c}}\wedge\theta_\circ}\big),\qquad p_*=p^\circ_{t^\circ_*},\qquad \mathrm{a}_\circ=\min\{\epsilon,\varrho_\circ/4\}.$$

(a) *Completion on the seeded prefix.* For every $\theta\in[\theta_\circ,1/(12\mathrm{T})]$, $\mathscr{E}^\circ_\theta\subseteq\mathscr{U}^\circ\subseteq E_{\mathrm{T}}$. On $\mathscr{U}^\circ\cap\mathscr{B}_{\sigma,M}$, the mark K_{T} is one of the two matched pairs of M, $\mathsf{t}_{\mathrm{c}}\le\theta_\circ$, the assignment equals M and the native path equals $\widetilde{\mathbf{X}}^M$ on $[0,\mathsf{t}_{\mathrm{c}}]$, and if moreover $|n_e^-\cdot\mathbf{X}_0|\le\mathrm{a}_\circ$ for an edge e of M, then $\mathsf{t}_{\mathrm{c}}\le\frac14\log(1/\epsilon)+\frac{23}{20}$. Moreover $\Pr(\mathscr{U}^\circ)\ge\mathsf{c}(\mathsf{m}_0)$.

(b) *Divergence at completion.* Let $\mathscr{G}$ be $\mathscr{U}^\circ$ or $\mathscr{E}^\circ_\theta$ with $\theta\in[\theta_\circ,t^\circ_*]$, and let $q^{\mathscr{G}}=\mathrm{Law}(\mathbf{X}_{\mathsf{t}_{\mathrm{c}}},S_{\mathsf{t}_{\mathrm{c}}}\mid\mathscr{G})$. Then

$$\mathcal{H}_{\mathrm{opt}}(q^{\mathscr{G}})\ge e^{-10\mathrm{T}\theta_\circ}\max\Big\{\frac{e^{-8}\varrho_\circ^2}{14}(\mathrm{c}_{\mathsf{m}}-7p_*)\mathrm{T}^{-1/2},\frac{e^{-46/5}}{2}\big(0.1427\mathrm{c}_{\mathsf{m}}\varrho_\circ\mathrm{a}_\circ-\varrho_\circ^2p_*\big)\epsilon^{-6}\Big\}-65\theta_\circ-\log 2,$$

and $\mathcal{H}_{\mathrm{phys}}(q^{\mathscr{G}})\ge\mathcal{H}_{\mathrm{opt}}(q^{\mathscr{G}})-1/2000$. For each fixed m_0, both are at least $7.2\cdot10^{-8}\mathrm{c}_{\mathsf{m}}\mathrm{T}^{-5/8}(1+o(1))$ as $\epsilon\to0$.

(c) *The whole contrast event.*

$$\mathcal{H}_{\mathrm{opt}}\big(\mathrm{Law}(\mathbf{X}_{\mathsf{t}_{\mathrm{c}}},S_{\mathsf{t}_{\mathrm{c}}}\mid E_{\mathrm{T}})\big)\ge\mathsf{c}(\mathsf{m}_0)\mathcal{H}_{\mathrm{opt}}(q^{\mathscr{U}^\circ})-\log 2,$$

the same holds for $\mathcal{H}_{\mathrm{phys}}$, and both tend to infinity as $\epsilon\to0$ for each fixed m_0. [ESTABLISHED]

This is a refutation for the completion clock: activated at t_{c} from the recognition law conditioned on the seeded prefix, or on the whole contrast event, no parent of the type of proposition 14.22(b) has a vanishing conditional error, in either arm and at every mobility in $[0,1]$. The weight $\mathsf{c}(\mathsf{m}_0)$ is tiny, so the bound of part (c) becomes positive only at very small ϵ; it is a statement about the limit. The first completion occurs by $\theta_\circ$, well before the threshold $\log(1/\epsilon)$ of proposition C.25. The order $\mathrm{T}^{-5/8}$ comes from seeds whose first pair forms near $\frac14\log(1/\epsilon)$ while the other matched pair, whose

hidden discrepancy the channel also discards, still carries an imprint of order $\varrho_\circ\epsilon$. At $\mathtt{t}_\mathrm{c}$ only lower bounds are proved.

Proof. (a) The time $\mathtt{t}_\mathrm{c}$ is a measurable function of the path, by the measurability argument in the proof of proposition C.17, $\mathbf{X}$ is continuous, and S is right-continuous and piecewise constant, so $(\mathbf{X}_{\mathtt{t}_\mathrm{c}}, S_{\mathtt{t}_\mathrm{c}})$ is measurable on E_T. Fix a seed and let $s \geq 1$ satisfy $\mathrm{T}s \leq 1/12$. On $\mathscr{B}_{\sigma,M} \cap \mathscr{A}_s^M$ the native path is held on $[0, s]$, and every eligible block other than the two matched pairs of M, namely the other two pairs and the four triples, contains an edge exterior to M. For a completion time $t \in [1, s+1)$ the window $[t-1, t]$ meets $[1/4, s]$, where that edge's belief relative entropy exceeds $1/2 > \epsilon^4$ by lemma C.24(iii), so such a block has no qualifying completion time in $[1, s+1)$. If $s \geq \theta_\circ$, the same lemma shows that both matched pairs are eligible, have internal relative entropies at most ϵ^4 on $[\theta_\circ - 1, \theta_\circ]$, since $\theta_\circ - 1 = \frac{1}{2}\log(1/\epsilon) \geq 1/4$, and have boundary belief relative entropies above $1/2$ there; as $\theta_\circ \leq 2/\epsilon$, they qualify at $\theta_\circ$.

These facts give the assertions. The inclusion $\mathscr{E}_\theta^\circ \subseteq \mathscr{U}^\circ$ holds because $\mathscr{A}_\theta^M \subseteq \mathscr{A}_{\mathtt{t}_\mathrm{c} \wedge \theta_\circ}^M$. On $\mathscr{U}^\circ \cap \mathscr{B}_{\sigma,M}$, if $\mathtt{t}_\mathrm{c} > \theta_\circ$, then $\mathscr{A}_{\theta_\circ}^M$ holds and a matched pair qualifies at $\theta_\circ$, a contradiction; so $\mathtt{t}_\mathrm{c} \leq \theta_\circ$, E_T holds, the remainder holds up to $\mathtt{t}_\mathrm{c}$, and the path is held there by the pathwise identification in the proof of proposition C.25(a). Applying the first paragraph with $s = \mathtt{t}_\mathrm{c}$, which is at least 1 because a completion time ends a duration-one interval, every block containing an exterior edge has infimum at least $\mathtt{t}_\mathrm{c} + 1$, so the least infimum is attained by a matched pair of M, and K_T is one of them. On the subevent $|n_e^- \cdot \mathbf{X}_0| \leq \mathrm{a}_\circ$, write $\mathbf{d}_0 = \mathbf{X}_0 - \sigma\mathbf{x}^\circ$ for the offset. Then $\sqrt{2}n_e^+ \cdot \mathbf{X}_0 = -\sigma + \sqrt{2}n_e^+ \cdot \mathbf{d}_0$ and $n_e^- \cdot \mathbf{X}_0 = n_e^- \cdot \mathbf{d}_0$ by lemma C.24(ii), these coordinates decay at the rates $8 + \mathrm{T}$ and $4 + 5\mathrm{T}$ along the held flow, and both residuals of e satisfy $|r(\widetilde{\mathbf{X}}_u^M)| \leq (1 + \sqrt{2}\varrho_\circ)e^{-8u} + \mathrm{a}_\circ e^{-4u} + \epsilon^2/8$. For $u \geq \frac{1}{4}\log(1/\epsilon) + \frac{3}{20}$ this is at most $[(1 + \sqrt{2}\varrho_\circ)e^{-6/5} + e^{-3/5} + \frac{1}{8}]\epsilon^2 < 0.98\epsilon^2$, so the pair of e has internal relative entropies at most ϵ^4 from then on. Since $\frac{1}{4}\log(1/\epsilon) + \frac{23}{20} \leq \theta_\circ$, the same contradiction argument gives $\mathtt{t}_\mathrm{c} \leq \frac{1}{4}\log(1/\epsilon) + \frac{23}{20}$. Finally $\Pr(\mathscr{U}^\circ) \geq \Pr(\mathscr{E}_{\theta_\circ}^\circ) \geq (1 - p_{\theta_\circ}^\circ)2\mathrm{c}(\mathrm{m}_0)$ by proposition C.25(a), and $p_{\theta_\circ}^\circ \leq p_* < 1/100$, because $73728\epsilon^4 t_*^\circ < 0.008$ as in the proof of proposition 14.22 and $4\Lambda_\mathrm{L}(t_*^\circ)t_*^\circ e^{-1/\mathrm{T}} \leq 8e\epsilon^{-2}t_*^\circ e^{-1/\mathrm{T}} < 10^{-3}$.

(b) Fix a seed with edges e, f of M, write $\lambda_2 = 4 + 5\mathrm{T}$, and let $\xi_k(u)$ be the hidden coordinates of $\mathbf{Z}_u^M$. Under $\Pr(\cdot \mid \mathscr{B}_{\sigma,M})$ they are scalar Ornstein–Uhlenbeck noises driven by independent Brownian motions, and the maximal estimate of the paragraph *Episode* of the proof of proposition 14.22 gives $\mathbb{E}\sup_{u \leq \theta_\circ} \xi_k(u)^2 \leq 32\mathrm{T}\theta_\circ$. On $\mathscr{G} \cap \mathscr{B}_{\sigma,M}$, part (a) gives $z_k(\mathbf{X}_{\mathtt{t}_\mathrm{c}}) = e^{-\lambda_k \mathtt{t}_\mathrm{c}} z_k(\mathbf{X}_0) + \xi_k(\mathtt{t}_\mathrm{c})$ with $\mathtt{t}_\mathrm{c} \leq \theta_\circ$; also $\Pr(\mathscr{G}^c \mid \mathscr{B}_{\sigma,M}) \leq p_*$, by proposition C.25(a) and the monotonicity of p° in θ. With $(x+y)^2 \geq x^2/2 - y^2$, $|z_k(\mathbf{X}_0)| < \varrho_\circ$, and the moment bounds of the proof of proposition C.25(c),

$$\mathbb{E}\big[|z_\perp(\mathbf{X}_{\mathtt{t}_\mathrm{c}})|^2 \mathbf{1}_\mathscr{G} \,\big|\, \mathscr{B}_{\sigma,M}\big] \geq \tfrac{1}{2}e^{-2\lambda_2\theta_\circ}\varrho_\circ^2\big(\mathrm{c}_\mathsf{m}/7 - p_*\big) - 64\mathrm{T}\theta_\circ,$$

and $e^{-2\lambda_2\theta_\circ} = \epsilon^4 e^{-8} e^{-10\mathrm{T}\theta_\circ}$. For the second bound keep only z_3 on the subevent $\{|z_2(\mathbf{X}_0)| \leq \mathrm{a}_\circ\}$, where part (a) gives $\mathtt{t}_\mathrm{c} \leq \theta_\mathrm{r} = \frac{1}{4}\log(1/\epsilon) + \frac{23}{20}$ and $e^{-2\lambda_2\theta_\mathrm{r}} \geq \epsilon^2 e^{-46/5} e^{-10\mathrm{T}\theta_\circ}$:

$$\mathbb{E}\big[z_3(\mathbf{X}_{\mathtt{t}_\mathrm{c}})^2 \mathbf{1}_\mathscr{G} \mathbf{1}_{\{|z_2(\mathbf{X}_0)| \leq \mathrm{a}_\circ\}} \,\big|\, \mathscr{B}_{\sigma,M}\big] \geq \tfrac{1}{2}e^{-2\lambda_2\theta_\mathrm{r}}\big(\mathbb{E}[z_3(\mathbf{X}_0)^2 \mathbf{1}_{\{|z_2(\mathbf{X}_0)| \leq \mathrm{a}_\circ\}} \mid \mathscr{B}_{\sigma,M}] - \varrho_\circ^2 p_*\big) - 32\mathrm{T}\theta_\circ.$$

For the uniform law on the twelve-dimensional ball of radius $\varrho_\circ$, one orthonormal coordinate divided by $\varrho_\circ$ has density $\mathrm{c}_{12}(1-x^2)^{11/2}$ on $[-1,1]$, with $\mathrm{c}_{12}=\Gamma(7)/(\sqrt{\pi}\Gamma(13/2))=46080/(10395\pi)>1.411$, and given that coordinate the other eleven are uniform on a ball of radius $\varrho_\circ(1-x^2)^{1/2}$, so a second orthonormal coordinate has conditional second moment $\varrho_\circ^2(1-x^2)/13$. Write $\mathbb{E}_{\mathrm{unif}}$ for the expectation when the offset $\mathbf{d}_0=\mathbf{X}_0-\sigma\mathbf{x}^\circ$ of (a) has this uniform law. Since $\mathrm{a}_\circ/\varrho_\circ\le 1/4$,

$$\mathbb{E}_{\mathrm{unif}}\big[z_3(\mathbf{d}_0)^2\mathbf{1}_{\{|z_2(\mathbf{d}_0)|\le \mathrm{a}_\circ\}}\big]=\frac{\mathrm{c}_{12}\varrho_\circ^2}{13}\int_{-\mathrm{a}_\circ/\varrho_\circ}^{\mathrm{a}_\circ/\varrho_\circ}(1-x^2)^{13/2}dx\ge\frac{2\mathrm{c}_{12}}{13}\Big(\frac{15}{16}\Big)^{13/2}\varrho_\circ\mathrm{a}_\circ\ge 0.1427\varrho_\circ\mathrm{a}_\circ,$$

and the density bound c_{m} of the proof of proposition C.25(c) transfers it to the seed. The law $q^{\mathscr{G}}$ is a mixture of the four seed laws and is carried by the matching strata, so the Donsker–Varadhan bound of that proof, with the factor $(4+5\mathrm{T})/(4\mathrm{T})\ge 1/\mathrm{T}$ applied when the bracket is positive and $64\theta_\circ(1+5\mathrm{T}/4)\le 65\theta_\circ$, gives the display; when both brackets are negative the display is negative and holds trivially. On $\mathscr{G}$ the hidden coordinates have norm at most $\varrho_\circ+\delta_\epsilon$, and lemma C.23 gives the physical bound. For the limit, $\mathrm{a}_\circ=\epsilon$ once $\epsilon\le 1/400$, $p_*/\epsilon\to 0$, and $\theta_\circ=O(\log(1/\epsilon))$, so the second bracket gives $\frac{1}{2}e^{-46/5}0.1427\varrho_\circ\mathrm{c}_{\mathsf{m}}\epsilon^{-5}(1+o(1))$ with $\frac{1}{2}e^{-46/5}0.1427\varrho_\circ>7.2\cdot 10^{-8}$ and $\epsilon^{-5}=\mathrm{T}^{-5/8}$.

(c) By (a), $\mathscr{U}^\circ\subseteq E_{\mathrm{T}}$ and $\Pr(\mathscr{U}^\circ\mid E_{\mathrm{T}})\ge\Pr(\mathscr{U}^\circ)\ge\mathsf{c}(\mathsf{m}_0)$. The law given E_{T} is the mixture of the laws given $\mathscr{U}^\circ$ and given $E_{\mathrm{T}}\setminus\mathscr{U}^\circ$, and so are its evaluator lifts, so lemma C.22(i) gives both inequalities. The limits follow from (b). □

Proposition C.27 (Parent dynamics after the seeded early prefix). In definition C.21, let $\epsilon_{\mathrm{L}}\in[0,1]$, let $\theta\ge 1/4$ with $\mathrm{T}\theta\le 1/12$ and $p_\theta^\circ<1/2$ as in proposition C.25, and let $\mathsf{h}>0$ with $\mathrm{T}(\theta+\mathsf{h})\le 1/10$. Activate the parent of proposition C.20, with the kernel eq. (C.135) and own-law learning eq. (C.136) at mobility ϵ_{L}, from $q_0^C=\mu_{\mathrm{act}}=C_{\#}q_\theta^\circ$ and $p^C(0)=p(\theta)$.

(i) μ_{act} is carried by the union over seeds of $\{M\}\times\{y\in W_M:|y-P_{W_M}\bar{\mathbf{x}}^{\sigma,M}(\theta)|\le\varrho_\circ+\delta_\epsilon\}$.

(ii) The coupled parent system has exactly one solution on $[0,\infty)$, and its rows satisfy $p_i^C(u)\in[e^{-\epsilon_{\mathrm{L}}(\theta+u)}/2,1-e^{-\epsilon_{\mathrm{L}}(\theta+u)}/2]$.

(iii) Let $\mathsf{P}_{\theta,\mathsf{h}}^{\mathrm{fine}}$ be the law of $(C(\mathbf{X}_{\theta+u},S_{\theta+u}))_{0\le u\le\mathsf{h}}$ conditioned on $\mathscr{E}_\theta^\circ$, the native rows continuing on their own path, and $\mathsf{P}_{\mathsf{h}}^C$ the parent path law. Then

$$\mathrm{TV}\big(\mathsf{P}_{\theta,\mathsf{h}}^{\mathrm{fine}},\mathsf{P}_{\mathsf{h}}^C\big)\le 24\exp\Big[-\frac{1}{76800\mathrm{Th}}\Big]+4\big(1+\sqrt{8/3}\big)\Lambda_{\mathrm{L}}(\theta+\mathsf{h})\mathsf{h}e^{-1/\mathrm{T}}.$$

The comparison includes every retained coordinate and the assignment path, hence the loop. The right side does not depend on m_0 or on $\Pr(\mathscr{E}_\theta^\circ)$, and it tends to zero as $\epsilon\to 0$, uniformly in θ, $\epsilon_{\mathrm{L}}\in[0,1]$, and m_0, whenever $\mathrm{Th}\to 0$, for instance for $\mathsf{h}=\mathrm{T}^{-1+\mathrm{q}}$ with $\mathrm{q}\in(0,1)$.

In particular, if $\epsilon\to 0$ with $\theta-\log(1/\epsilon)\to\infty$, $\epsilon^4\theta\to 0$, and $\mathrm{Th}\to 0$, for instance for $\theta\in[t_\epsilon,t_*^\circ]$, then $\mathcal{H}_{\mathrm{opt}}(q_\theta^\circ)$, $\mathcal{H}_{\mathrm{phys}}(q_\theta^\circ)$, and the total variation of (iii) tend to zero separately, uniformly in m_0 and $\epsilon_{\mathrm{L}}\in[0,1]$. [ESTABLISHED]

Proof. (i) follows from proposition C.25(a), since P_{W_M} is a contraction.

(ii) The argument of the proof of proposition C.20, by coupled prescribed-row constructions, contraction of the variation-of-constants map on short intervals, and the lower bounds $p_i^C(u) \geq p_i^C(0)e^{-\epsilon_{\mathrm{L}}u}$ and $1-p_i^C(u) \geq (1-p_i^C(0))e^{-\epsilon_{\mathrm{L}}u}$, uses only an activation law with finite second moment and interior initial rows; neither the selection tolerances nor $d < 1/4$ enter it. Here μ_{act} has bounded support by (i), and $p(\theta)$ has entries in $[e^{-\epsilon_{\mathrm{L}}\theta}/2, 1-e^{-\epsilon_{\mathrm{L}}\theta}/2]$ by lemma C.24(i); the two lower bounds give the stated interval. For $\epsilon_{\mathrm{L}} = 0$ the rows stay at $p(\theta)$.

(iii) *Coupling.* Since the native rows are deterministic, the native path after θ is a measurable function of $(\mathbf{X}_\theta, S_\theta)$ and of the inputs after θ, which are independent of $\mathcal{F}_\theta \ni \mathscr{E}_\theta^{\circ\circ}$. On one probability space take independent objects: a state $(\mathbf{X}_0^N, S_0^N)$ with law q_θ°, a standard Brownian motion W', a rate-one clock, uniform variables for each ring and receiver, and draws $\zeta_k \sim \nu_{\mathrm{T},Z}$, one for each ring. The native copy starts at $(\mathbf{X}_0^N, S_0^N)$, solves $d\mathbf{X}^N = -K_{S^N}\mathbf{X}^N du + \sqrt{2\mathrm{T}}dW'$ between rings, and redraws with rows $p(\theta+u)$ at $\mathbf{X}_{u-}^N$; its retained path has law $\mathsf{P}_{\theta,\mathsf{h}}^{\mathrm{fine}}$. The parent copy starts at $C(\mathbf{X}_0^N, S_0^N)$, solves $dY^C = -K_S Y^C du + \sqrt{2\mathrm{T}}P_{W_S}dW'$ on stratum S, and at the kth ring redraws with rows $p^C(u)$ at $\widetilde{\mathbf{x}} = Y_{u-}^C + F_S\zeta_k$ on a matching stratum ($\widetilde{\mathbf{x}} = Y_{u-}^C$ otherwise), moving to $(S', P_{W_{S'}}\widetilde{\mathbf{x}})$; since $P_{W_S}W'$ is a standard Brownian motion on W_S and p^C is deterministic, its path law is $\mathsf{P}_{\mathsf{h}}^C$. Let $M_0 = S_0^N$ and let σ_{D} be the first ring at which either copy proposes an assignment other than M_0. Before σ_{D}, K_{M_0} commutes with $P_{W_{M_0}}$, so $P_{W_{M_0}}\mathbf{X}^N$ and Y^C solve the same linear equation with the same noise and the same initial value, and a ring that keeps M_0 moves neither copy, since $P_{W_{M_0}}(Y^C + F_{M_0}\zeta_k) = Y^C$. The retained paths therefore agree on $[0,\mathsf{h}]$ when $\sigma_{\mathrm{D}} > \mathsf{h}$, and the total variation is at most $\Pr(\sigma_{\mathrm{D}} \leq \mathsf{h})$.

Tube and gap. Let $\mathbf{Z}'_u = \sqrt{2\mathrm{T}}\int_0^u e^{-K_{M_0}(u-v)}dW'_v$ and $\mathsf{t}_\Gamma = \inf\{u : |\mathbf{Z}'_u| \geq 1/20\}$. Since W' is independent of the initial state, the argument of proposition C.25(a) with radius $1/20$ in place of δ_ϵ, and with the reflection principle for the closed event, gives $\Pr(\mathsf{t}_\Gamma \leq \mathsf{h}) \leq 24\exp[-(1/20)^2/(192\mathrm{T}\mathsf{h})] = 24\exp[-1/(76800\mathrm{T}\mathsf{h})]$. By proposition C.25(a), $|\mathbf{X}_0^N - \bar{\mathbf{x}}(\theta)| \leq \varrho_\circ + \delta_\epsilon$ for some sign σ, where $\bar{\mathbf{x}} = \bar{\mathbf{x}}^{\sigma,M_0}$. For $u \leq \sigma_{\mathrm{D}} \wedge \mathsf{t}_\Gamma$ the native configuration before the ring is $e^{-K_{M_0}u}\mathbf{X}_0^N + \mathbf{Z}'_u$, and $\bar{\mathbf{x}}(\theta+u) = e^{-K_{M_0}u}\bar{\mathbf{x}}(\theta)$, so $|\mathbf{X}_u^N - \bar{\mathbf{x}}(\theta+u)| \leq \mathrm{R} = \varrho_\circ + \delta_\epsilon + 1/20 < 0.0601$. For an exterior edge g, the common residual $c_g = (r_g^b + r_g^m)/2$ has a covector of norm $\sqrt{2}$ that annihilates Z_{M_0} (proof of proposition 14.22), $\mathcal{D}_g \geq c_g^2$, and along $\bar{\mathbf{x}}(t)$, $|c_g| = e^{-\mathrm{T}t}(3/2 - e^{-8t}/2)$ by lemma C.24(ii). For $t \in [\theta, \theta+\mathsf{h}]$, with $\theta \geq 1/4$ and $\mathrm{T}(\theta+\mathsf{h}) \leq 1/10$, this gives $|c_g(\mathbf{X}_u^N)| \geq e^{-1/10}(3/2 - e^{-2}/2) - \sqrt{2}\mathrm{R} > 1.21$. For a selected edge, $\mathcal{D}_e = (\sqrt{2}n_e^+ \cdot \mathbf{x})^2 + (n_e^- \cdot \mathbf{x})^2$ is the square of a seminorm of operator norm $\sqrt{2}$ that equals $e^{-(8+\mathrm{T})t} \leq e^{-2}$ at $\bar{\mathbf{x}}(t)$, so $\mathcal{D}_e(\mathbf{X}_u^N) \leq (e^{-2} + \sqrt{2}\mathrm{R})^2 \leq 2(e^{-2} + \mathrm{R})^2 < 0.08$. Likewise z_+ is 1-Lipschitz, retained, and equal to $-\sigma e^{-(8+\mathrm{T})t}$ on the center path, so $\frac{4}{3}z_+^2 \leq 2(e^{-2} + \mathrm{R})^2 < 0.08$. Since $1.21^2 - 0.08 > 1$, the energy gap exceeds one for both copies.

Hazards. At a ring at relative time $u \leq \sigma_{\mathrm{D}} \wedge \mathsf{t}_\Gamma$, the native copy departs with conditional probability at most $\sum_i \beta_{i,\mathrm{ext}} \leq 4\Lambda_{\mathrm{L}}(\theta+u)e^{-1/\mathrm{T}}$ by lemma C.24(i). For the parent, $\mathcal{D}_g(\widetilde{\mathbf{x}}) \geq c_g(Y^C)^2$ and $\mathcal{D}_e(\widetilde{\mathbf{x}}) = (z_+ + z_1)^2 + z_2^2$ with independent $z_1 \sim \mathcal{N}(0, \mathrm{T}/(8+\mathrm{T}))$ and $z_2 \sim \mathcal{N}(0, \mathrm{T}/(4+5\mathrm{T}))$, and similarly for f with $-z_1$ and z_3. The identity $\mathbb{E}e^{(\mathrm{c}_1+Z)^2/\mathrm{T}} = (1-2s^2/\mathrm{T})^{-1/2}\exp[\mathrm{c}_1^2/(\mathrm{T}(1-2s^2/\mathrm{T}))]$, valid for $Z \sim \mathcal{N}(0,s^2)$ with $s^2 < \mathrm{T}/2$, gives the factors $((8+\mathrm{T})/(6+\mathrm{T}))^{1/2} \leq \sqrt{4/3}$ and $((4+5\mathrm{T})/(2+5\mathrm{T}))^{1/2} \leq \sqrt{2}$ and an exponent at most $\frac{4}{3}z_+^2/\mathrm{T}$. With the row bound of (ii), the parent departs with conditional probability at most $4\sqrt{8/3}\Lambda_{\mathrm{L}}(\theta+u)e^{-1/\mathrm{T}}$. The variables of a ring are independent of its

past, so the number N_{D} of departing rings in $[0, \mathsf{h} \wedge \sigma_{\mathrm{D}} \wedge \mathsf{t}_{\Gamma}]$ has, by the compensator of the clock, $\mathbb{E}N_{\mathrm{D}} \le 4(1+\sqrt{8/3})\Lambda_{\mathrm{L}}(\theta+\mathsf{h})\mathsf{h}e^{-1/\mathrm{T}}$. If $\sigma_{\mathrm{D}} \le \mathsf{h}$, then $\mathsf{t}_{\Gamma} \le \mathsf{h}$ or $N_{\mathrm{D}} \ge 1$, which proves the bound.

Limits. The first term depends only on Th. For $\epsilon_{\mathrm{L}} \le 1$, $\Lambda_{\mathrm{L}}(\theta+\mathsf{h}) \le 2e^{1/(10\mathrm{T})}$ and $\mathsf{h} \le 1/(10\mathrm{T})$, so the second term is at most $3e^{-9/(10\mathrm{T})}/\mathrm{T}$. For $\mathsf{h} = \mathrm{T}^{-1+\mathrm{q}}$, $\mathrm{T}(\theta+\mathsf{h}) \le 1/10$ once ϵ is small. The last paragraph of the statement combines (iii) with proposition C.25(b), whose bound does not depend on m_0, and the window $[t_\epsilon, t_*^\circ]$ satisfies $\theta - \log(1/\epsilon) \ge \log(1/\epsilon)$. □

Remark C.28 (The stationary comparison does not apply to the early prefix). Let $\epsilon_{\mathrm{L}} \in [0, \epsilon]$ and $0 \le t \le t_*^\circ$. Every row entry $\pi_{ij}(t)$ lies in $[0.4489, 0.5511]$, so every row assigns more than $1/4$ to the nonpartner source of either matching and the hypothesis $d < 1/4$ of proposition C.20 fails; and for each matching M, $(1-\mu_{\mathrm{T},p(t)}(M))/\mu_{\mathrm{T},p(t)}(M) \ge 0.44$, so the averaged departure bound eq. (C.139) is at least 0.44 per unit time. Under the hypotheses of proposition C.27, for each matching M,

$$\mathrm{TV}\big(\mu_{\mathrm{act}}, \overline{\nu}_{\mathrm{T}}^{M} \otimes \delta_M\big) \ge 1 - 3\cdot 10^{-18},$$

where $\overline{\nu}_{\mathrm{T}}^{M} = (P_{W_M})_{\#}\mathcal{N}(0, \mathrm{T}K_M^{-1})$ is the retained Gaussian of that proposition. Consequently each leg of the triangle inequality through the stationary held-matching path law in its proof exceeds $1 - 3\cdot 10^{-18}$. [ESTABLISHED]

Proof. By lemma C.24(i), $|\pi_{ij}(t) - 1/2| \le (1-e^{-\epsilon t_*^\circ})/2$, and $\epsilon t_*^\circ < 0.1022$ as in the proof of proposition 15.4. The rotation $i \mapsto i+1$ acts by an orthogonal map carrying $U_{S^{(0)}}$ to $U_{S^{(1)}}$, so the two matchings have equal Gaussian masses $\int e^{-U_M/\mathrm{T}} d\mathbf{x}$. Dropping the other fourteen sheets from the normalizer gives $\mu_{\mathrm{T},p}(M) \le \pi_M/(\pi_M + \pi_{M'})$, where M' is the other matching, so $(1-\mu_{\mathrm{T},p}(M))/\mu_{\mathrm{T},p}(M) \ge \pi_{M'}/\pi_M = \prod_i \pi_{i,M_i'}/\pi_{i,M_i} \ge (0.4489/0.5511)^4 > 0.44$.

For the second claim, $R_{\mathrm{ev}} = 2\mathfrak{D}^{\top}\mathfrak{D}$ has eigenvalue 4 with multiplicity four and 0 with multiplicity eight, because $\mathfrak{D}\mathfrak{D}^{\top} = 2I_4$. It leaves $\ker H_M$ and the range of H_M invariant, and on the range it has eigenvalues $0, 0, 4, 4$ ($R_{\mathrm{ev}} n_g^+ = 0$, $R_{\mathrm{ev}} n_g^- = 4n_g^-$), so on $\ker H_M \subseteq W_M$ it has eigenvalue 4 twice and 0 six times. Under $\mathcal{N}(0, \mathrm{T}K_M^{-1})$ the component in $\ker H_M$ has covariance $(I + R_{\mathrm{ev}})^{-1}$ there, with eigenvalues 1 six times and $1/5$ twice, hence a density at most $5(2\pi)^{-4}$ on this eight-dimensional space. The projection onto $\ker H_M$ of a retained ball of radius $\mathrm{R}_0 = \varrho_\circ + \delta_\epsilon < 0.010008$ lies in an eight-dimensional ball of volume $\pi^4 \mathrm{R}_0^8/24$. By (i) of proposition C.27, μ_{act} is carried by a set whose part on the stratum of M consists of two such balls, of $\overline{\nu}_{\mathrm{T}}^{M}$-mass at most $2\cdot 5(2\pi)^{-4}\pi^4\mathrm{R}_0^8/24 = 10\mathrm{R}_0^8/384 < 3\cdot 10^{-18}$, and the rest of that set has $\overline{\nu}_{\mathrm{T}}^{M} \otimes \delta_M$-mass zero. Both $\mathsf{P}_{\theta,\mathsf{h}}^{\mathrm{fine}}$ and $\mathsf{P}_{\mathsf{h}}^{C}$ start from μ_{act}, the stationary path law starts from $\overline{\nu}_{\mathrm{T}}^{M} \otimes \delta_M$, and the total variation of path laws is at least that of their initial laws. □

The mismatch is structural. Early departures are rare because the exterior common residuals stay near $3/2$ while the partner costs are small, a pointwise property of the activation law; under the stationary retained law the slow coordinates are centered at zero with variance of order one, and the comparison charges the stationary mass of the other matching as potential flux. The proof of proposition C.27 therefore uses the energy gap along the conditioned path instead.

These statements concern the seeded sub-event of the contrast. The conditional law given the whole contrast event at a deterministic activation time, which would need control of contrast histories that enter a matching at a random positive time or start outside the seed balls, is not treated, and neither is the comparison of learned and frozen first blocks at $\epsilon_{\mathrm{L}} = \mathrm{T}^{1/8}$ or of two initial means differing in their loop coordinate; these remain part of open problem 14.20. The path coupling gives no conclusion on windows of order T^{-1} or longer, and the parent path over a following window describes the persistence of a formed block, not a second formation.

Appendix D

Notation

Table D.1 lists the symbols that recur across chapters, each with the one meaning it has in this manuscript and the statement or section that introduces it. A superscript $x \in \{b, m\}$ names the belief or the model channel. Symbols used inside a single chapter are declared at its opening, and those of the formation family are collected in Table 14.1. Metrics are sans-serif, statement-local constants are upright, and $\mathcal{N}(\mu, \Sigma)$ is the normal law with mean μ and covariance Σ.

Table D.1: Notation of the main manuscript.

Symbol	Meaning	Introduced
Base point, bundles, and frames		
$\mathcal{C}$, c, c_*	contextual base, a context, the fixed base point	definition 2.1
$\pi : \mathscr{P}_G \to \mathcal{C}$, G, e_G	principal right G-bundle, its structure group, the identity of G	definition 2.2
u_i, σ, U_i	common frame of agent i at c_* for both channels; a reference point of $(\mathscr{P}_G)_{c_*}$; the frame relative to it, $u_i = \sigma \cdot U_i^{-1}$	definition 2.2
g_i, $R_i^x = \rho_x(g_i)$	passive re-expression $u_i' = u_i \cdot g_i^{-1}$ (vertex gauge); the represented sample-coordinate map	section 2.2
φ_i	local function $F(u_i) = u_i \cdot \varphi_i$ of a fiber automorphism F	proposition 2.6
T_{ij}	pointwise frame comparison, $u_j = u_i \cdot T_{ij}$, $T_{ij} = U_i U_j^{-1}$	proposition 2.5
$(\mathsf{K}, \mathscr{K})$, $(\mathsf{M}, \mathscr{M})$	belief and model sample fibers	definition 2.2
ρ_b, ρ_m; $\widehat{\rho}_x$	representations of G on the sample fibers; the pushforward action $\widehat{\rho}_x(g)q = (\rho_x(g))_{\#}q$ on laws	definition 2.2
$\mathcal{B}_x$, $\mathcal{E}_x$, ϖ_x	law fiber, associated bundle $\mathscr{P}_G \times_{\widehat{\rho}_x} \mathcal{B}_x$, and its projection	definition 2.4
q_i^b, q_i^m	belief-law and model-law sections of agent i, and their coordinate laws at c_*	definition 2.8
Φ, $\widetilde{\Phi}$; Φ_i, $\widetilde{\Phi}_i$	cross morphisms $\mathcal{E}_b \to \mathcal{E}_m$ and $\mathcal{E}_m \to \mathcal{E}_b$; their representatives in the frame u_i	definition 2.9
Interaction multigraph and links		
V, N, i, j	agent set, number of agents (also the extensive size of a finite labeled family), agent indices	definition 2.8
$\mathfrak{I} = (V, E, {}^{-})$, $e : j \to i$, $\bar{e}$	interaction multigraph at c_*, containing the reversal closure of the source-prior supports; oriented edge copy; reversal	definition 2.11

continued on the next page

Symbol	Meaning	Introduced
Θ^x_e, t^x_e	independent link on e, with $\Theta^x_{\bar{e}} = (\Theta^x_e)^{-1}$; its link map $\mathsf{t}^x_e(u_j \cdot g) = u_i \cdot \Theta^x_e g$	definition 2.11
$\mathring{\Theta}^x_e$	per-edge twist $U_i^{-1}\Theta^x_e U_j$, whose conjugacy class is invariant	proposition 2.12
Θ^x_p, $\mathrm{Hol}^x(p)$	path product along a walk p; holonomy of a closed walk	eq. (2.30)
$\Theta^x_{I\leftarrow i}$, Hol^x_I, $\widehat{\Theta}^x_e$	tree transport to the root of a block I, based holonomy group of I, block-dressed boundary edge	definition 11.4
Samples, records, and the joint		
k_i, m_i; Y_i	belief sample and model presentation of agent i; its latent block $\mathsf{K}_i \times \mathsf{M}_i$	definition 3.1
$y \in \mathsf{Y}$, $o \in \mathsf{O}$	population latent sample and latent space; observation and observation space	definition 3.1
$X \in \mathsf{X}$, θ	structural data and their configuration space; generative parameters	definitions 3.2 and 4.1
$a \in \mathfrak{A}$, ∂a, K_a, $E_{a,o}$	record, its scope, its normalized record kernel, and its negative log density at the record o	definition 3.1 and hypothesis 5.5
ev_i, K^X_{i,m_i}	evaluator of agent i; the evaluated mechanism of the presentation m_i	definition 3.3
Acc_i, $\mathbb{Q}_{i,o,X}$, $\mathrm{Cpl}(\mathbb{Q}_{\bullet,o,X})$	access map, local recognition law, and the class of couplings of the local laws	definitions 3.2 and 3.13
$\mathfrak{q}^x_i$, $\mathsf{Y}^{\mathrm{law}}_i$; $\mathsf{Y}^{\mathrm{cfg}}$, pres, κ^{cfg}	law points and the law-point block of agent i; configuration space, presentation map, presentation kernel	definition 3.19
$\mathbb{P}_{\theta,V}$, $\mathbb{Q}_{V,o,X}$, $\mathbf{\Pi}_{\theta,V,o,X}$	constructed joint, selected recognition coupling, posterior slice	definitions 3.15 and 4.2
$\widetilde{p}_X$	unnormalized potential of an undirected model	definition 4.5
$\mathcal{L}_V$, $\mathcal{F}_V = -\mathcal{L}_V$; $\mathcal{L}^{\mathrm{ext}}$, $\mathcal{F}^{\mathrm{ext}}$	evidence bound and variational free energy; their extended versions, defined on every recognition law	definition 5.6
D_{KL}, TC	relative entropy with values in $[0,+\infty]$; total correlation	definition 3.10 and corollary 5.1
Attention and selective communication		
π_{ij}, $\mathcal{N}_i$, J_i	source prior of receiver i, its source set (the support of π_i), the source label	definition 6.1
(i,h)	receiving row of head h of agent i	definition 6.1
β^x_{ij}, α^x_i, η^x_{ij}	attention row of receiver i, receiver occupancy, and event law $\eta = \alpha\beta$, with the role mark P, Π, or Q of the law they are computed from (β^m names the model channel); the event law carries no head index	definition 6.4
τ_i, c_{ij}, r_i, $\mathcal{F}^{\mathrm{att},x}_i$	attention temperature, replicator cost, row relaxation rate, row free energy	corollary 6.7 and definition 13.8
S^Q_e, S^K_e, H^{cmp}_e; μ^Q_i, μ^K_j	query and key comparison maps into the comparison space; projected query and key means	definition 6.15
K^v_e; $\mathsf{L}^{\mathrm{enc}}_e$, $\mathsf{L}^{\mathrm{tr}}_e$, $\mathsf{L}^{\mathrm{dec}}_e$, L_e, $\mathsf{L}^{\mathrm{val}}_e$	composed value kernel of an edge copy; encoder, transmission, decoder, composite, and value linear maps	definitions 6.10 and 6.12
Σ^+_i, Σ^{pred}_i, $\widetilde{W}_i = W_i/\tau_i$	receiver posterior and predictive covariances; tempered receiver precision	proposition 6.19

continued on the next page

Symbol	Meaning	Introduced
S_j, Υ, v	source sample; random receiver side information and its values	proposition 6.24
Fiber geometry, histories, and time scales		
g^F, T^F, $\mathbf{s}$; λ	Fisher tensor, Amari–Chentsov tensor, score; coordinates of a recognition family	definition 7.2
$\mathfrak{k}_q$, $\mathfrak{g}_q$	orbit Fisher form and isotropy algebra at a law q	corollary 7.7
$\mathcal{Q}_i$, $\mathcal{Q}_V$, G	configuration manifolds of an agent and of the population at c_*; a declared configuration metric	definition 7.13
$\mathcal{Q}_\ell$, F_ℓ, $\mathscr{V}_\ell$, $\mathrm{Fl}_t^{\mathscr{V}_\ell}$	configuration manifold at depth ℓ, coarse configuration map, natural-gradient field, and its flow	definition 7.44
$\mathfrak{t}$	Fisher duration, the information clock of a selected history	definition 7.35
$\varkappa$; $\mathfrak{K}$, $\mathfrak{C}$; ret	parity amplitude; residual second-moment tensor and signed cross tensor of a joint score; retained-law map	definition 7.22, proposition 7.18, and theorem 7.14
f, $s = (s_{\mathrm{m}}, \theta)$	fast and slow coordinate blocks	definition 8.2
G_{f}, G_{s}, ε, G_ε, F_{red}	block metrics, rate ratio, rate-separated metric $\varepsilon\mathsf{G}_{\mathrm{f}} \oplus \mathsf{G}_{\mathrm{s}}$, reduced functional	lemma 8.3 and proposition 8.4
Lie-algebra links		
$\mathfrak{g}$, $\exp$, Ad_g, $\mathrm{d}\rho_x$	Lie algebra of G, its exponential map, the adjoint action, the differential of ρ_x at e_G	lemma 9.1
E_+, $\mathscr{T}$; $\mathring{\omega}_e^x$, $\mathring{\Theta}_p^x$	one orientation of each reversal pair, the link space $\mathfrak{g}^{E_+}$ of one latent channel; generator of the twist of e in the exponential chart, reference holonomy of a closed walk	definition 9.2
$\mathsf{G}^{\mathrm{fib}}$, $\langle\cdot,\cdot\rangle_{\mathsf{G}}$; $\mathfrak{Q}_p^x$, $\mathbf{C}$	fiber metric at the reference point and its trace form; loop defect of a closed walk, cycle map	definition 9.8
$\nu^{\mathscr{T}}$, $\mathscr{S}$, $Z_{\mathscr{S}}$, $\mathbb{P}_\theta^{\mathscr{T}}$; $\mathring{\alpha}_e^x$, $\mathring{\Sigma}_e^x$	base law, link action, normalizer, generator prior; anchor and prior covariance	definition 9.11
$\mathring{\mu}$, $\mathring{\Lambda}$; $\mathbf{L}(\mathbf{B}, \mathbf{T})$	mean and precision of a Gaussian recognition law of the generators; Fréchet derivative of the matrix exponential	proposition 9.18 and lemma 9.19
Coarse channels and the network state		
ℓ	depth of coarse-graining, and the level index of the formation family	definition 10.21
I, J; A	child blocks; the parent label of a block	corollary 10.11
C_A, Z_A, Δ_A	parent coarse kernel, parent state space, conditional-information defect of the coarse chain rule	corollary 10.11 and theorem 10.7
$\mathfrak{X}_\ell$, $C_{\ell k}$	objects and arrows of a scale diagram	definition 10.21
$\gamma \in \mathfrak{G}$, W_γ	label of a finite labeled family (a mixture or geometry label) and its weight	definition 10.23
s_γ, v_γ, $\mathcal{I}_\gamma$, $\bar{\mathsf{p}}$, w	bulk log-mass density, tie slope $\nabla\mathsf{s}_\gamma$, bulk Fisher density, limiting label law, window coordinate $\mathsf{w} = N(\theta - \theta_0)$	section 10.8
L_I, $\mathbb{V}_{I,\mathrm{fix}}$, f_I	connection Laplacian of a block, its fixed fiber $\ker L_I$, and the retained rank $\dim \mathbb{V}_{I,\mathrm{fix}}$	definition 11.18
Exact coarse-graining and the multiscale free energy		
V_ℓ, Y_ℓ, C_ℓ, $C^\dagger$	agents and state space at depth ℓ, the depth arrow, the reverse kernel of a channel	definition 12.1

continued on the next page

Symbol	Meaning	Introduced
$(\varrho_\ell, \mathsf{m}_\ell)$, M_ℓ, $\bar{\mathsf{m}}_\ell$, H_ℓ	measure pair of a reference probability and a finite evidence measure; its mass, normalized law, and action	definition 12.1
U, $\mathcal{A}$	reverse conditional expectation and the exact action increment of a channel	theorem 12.4
$\mathcal{G}_\ell$, g, $\mathcal{T}_\ell$	space of Hoeffding interaction coordinates, an element, and the exact interaction step	definitions 12.11 and 12.14
D_ℓ, $D_{n\leftarrow\ell}$	finite-step multipliers of the interaction step	definition 12.20
$\mathsf{P}_\ell^{\mathrm{ret}}$, $r_{\ell+1}^{\mathcal{G}}$	retained projection and closure residual	definition 12.23
$Y_\ell = (Z_\ell, \mathfrak{N}_\ell)$, $\mathbf{Y}$	state at depth ℓ, made of coordinates and the graph datum; the whole tower	definition 13.1
$R_\ell \in \mathsf{R}_\ell$; $C_{\ell+1\leftarrow\ell}^{\mathrm{struct}}$, $\widehat{\mathsf{Y}}_{\ell+1\leftarrow\ell}$	structural mark sampled by the structural channel, such as a random partition, and its space; the structural channel and its augmented transition state	definition 13.3
$\mathfrak{m}_\ell$, $\mathfrak{M}_\ell$	membership kernel and joint endpoint membership kernel	definition 13.14

Appendix E

Open Problems

Table E.1 collects the open problems of this manuscript in the order of the chapters. Each row names the item, the numbered statement where it arises, which also states what holds, and a condition that would settle it. Established results, refutations among them, stay at their statements and are not listed. Open problems of the base and Gaussian settings are collected in the two supplements.

Table E.1: Open problems of the main manuscript.

Item	Locator	Settling condition
Part I: single-point foundations		
Nondominated likelihood families	open problem 3.9	A treatment of the evidence bound, its local disintegrations, and the coarse channels at the level of joint measures and regular conditional laws, with a rule for record-null versions, in which the exact identities of chapters 5 and 10 are proved, or hypotheses that restore a jointly measurable likelihood; negatively, a record family for which those identities provably have no measure-level counterpart.
A normalized law-point prior	open problem 3.23	For a class of law points, a sigma-finite reference on the law-point or presentation space, a finite normalizer for the exponentiated law-point energies, the transformation law of the prior under the active gauge action, record kernels that read law points, and a compatibility relation between the law-point posterior and the sample-level recognition marginals.
An intrinsic criterion for recognition structure	open problem 5.14	A functional on admissible channels or partitions, expressible relative to one fixed joint and not monotone toward an endpoint, whose optimizer is nondegenerate; negatively, a proof of endpoint degeneracy for a precisely stated class of such functionals.

continued on the next page

Item	Locator	Settling condition
Convergence of the alternating scheme	open problem 5.32	For models of hypothesis 5.5 with finitely many latent states, parameters in an open cone, and joint densities twice continuously differentiable in the parameters, under a full exact E phase alternating with the exact M-coordinate with a unique maximizer: conditions checkable on the model, including compactness of the upper level sets of the evidence, met by some model of the class with parameter dimension at least two whose stationary set at the limit value of an orbit is not discrete, under which every orbit converges to one stationary point of the extended bound; or a model of that class with compact upper level sets whose orbit has more than one limit point.
Discrete creation and deletion of interaction edges	open problem 6.9	For the threshold rule of the statement (source sets in multigraphs of bounded multiplicity; deletion at a rate that is a continuous nonincreasing function of the recognition row weight and creation at a rate that is a continuous nonincreasing function of the averaged comparison energy, each positive below a threshold; rows on the current support following the replicator flow): for some pair of rate functions, a normalized joint at c_* with the source sets as a structural coordinate, declared with a named law and coupled to the link, interface, and label variables, in which the collective VFE decomposition, the event law, and its pushforward extend to the random support and the collective VFE is nonincreasing under the master equation of the rule from every initial support and recognition law; or a proof that for every pair of rate functions no joint of that class has this property.
Label-level multi-head attention	open problem 6.22	For heads of one receiver with exclusive labels and comparison records whose projected queries are correlated under the receiver's conditional Gaussian law and whose integrated positive label table does not factor (the factorizing subclass, including untied channel-split heads under a channel-product receiver law, is excluded): an exact recursion with bounded head count, a family of comparison-record models closed under the meta-attention pushforward composed with a declared head-merging channel, in which the parent's per-head rows are the disintegrations of the merged event law and its posterior mean is the coarse coordinate of its children's posterior means; or a member and a partition for which no comparison-record model at the block with fewer heads than the block has rows reproduces the posterior mean of the block's coarse coordinate.

continued on the next page

Item	Locator	Settling condition
Formation with evolving selective interfaces	open problem 6.27	For a connected vector-tier block with Gaussian belief means as fast coordinates, factored linear communication through surjective interfaces of one fixed rank below the fiber dimension, comparison records read through the receiver's interface, the kinetics of definition 14.21 at temperature T, and interfaces learned as generative parameters by a natural-gradient flow at mobility ϵ_{M}: a parent that is a retained mechanism of fixed dimension below the block's, independent of T, ϵ_{M}, and the horizon, whose path law of the communicated coordinates is within a relative entropy tending to zero as $\epsilon_{\mathrm{M}} \to 0$ on horizons of order $1/\epsilon_{\mathrm{M}}$, uniformly over $\mathrm{T} \in (0, \mathrm{T}_0]$; or a member whose learned interface path has a rest point at which no such parent is projectable, or in which every such parent keeps that relative entropy bounded below as ϵ_{M} and T tend to zero along a stated relation.
Global Fisher quotients and canonical block data	open problem 7.15	A criterion, checkable on the configuration manifolds at c_* and proved, under which the null foliation of a pulled-back Fisher tensor has a Hausdorff manifold leaf space, for instance through a free and proper group action, together with block data on the quotient constructed from a stated typing of agents and channels, or a counterexample showing that the typing does not determine them.
Scope beyond the finite parity witness	open problem 7.29	For a stated regular family of agent laws and a stated class of morphisms containing kernels that do not preserve products, a smooth right inverse of the marginal map that is natural under the class, and correlated lifts equivariant under a connected structure group acting on continuous law fibers, each by a construction with proof or a no-go theorem.
Quantitative meta-agent clock ratio	open problem 7.43	A theorem computing or bounding the ratio of a meta-agent's Fisher duration to its constituents' collective duration, for a stated coarse map and a stated law of constituent velocities, including its dependence on N for coherent and for incoherent histories.
Slow-manifold reduction of the collective free energy	open problem 8.7	For the sampled four-cycle dynamics of definition 14.21 with model means and links at the common slow rate ε_{m} of proposition 8.6, no parameter learning, and the split by mobility: a bound on the distance between the slow coordinates of the sampled process and the reduced slow flow of its assignment path over a bounded slow-time interval, explicit in ε_{m}, T, and the number of assignment changes, tending to zero with ε_{m} and T, for every positive integer n with probability tending to one on paths with at most n changes per unit slow time; or a positive integer n, mobility ratios and temperatures tending to zero, and initial states from which, with probability bounded below, the path has at most n changes per unit slow time and that distance stays bounded below.
Part II: exact coarse-graining at one base point		

continued on the next page

Item	Locator	Settling condition
Equivariant Bayes versions beyond the finite and compact cases	open problem 10.19	For a locally compact second countable group acting continuously on Polish spaces, an equivariant channel K, and laws Q with noncompact stabilizer whose posterior has no weakly continuous version on the support of QK: an equivariant measurable selection theorem giving, for every member, an everywhere-defined version equivariant under the stabilizer, jointly measurable across every measurable family of such laws meeting several orbits; or a pair (Q, K) of the class with no everywhere-defined version equivariant under the stabilizer.
A natural selector of partitions and membership laws	open problem 10.20	A relabeling-natural, gauge-compatible functional of the laws, not monotone under refinement, with a nondegenerate optimizer; or a theorem that no such selector exists in a stated class.
Invariant retained subspaces	open problem 12.27	For finite agent spaces with fixed product references of full support, at least two blocks, and deterministic block-product channels whose block maps depend on every coordinate of their block, with base laws of full support in the class of product laws or of all laws: a proper invariant pair with fixed retained projections, invariant under the step of every base law in a nonempty open subset of the class, whose depth-ℓ range is not contained in the sum of the pullback sector and a grouping sector of at least two groups; or a proof that off a closed nowhere dense subset of the class every proper invariant pair has its depth-ℓ range in such a sum.
Dynamic closure of the pointwise parent	open problem 12.40	For switching diffusions of the type of definition 14.21 indexed by ϵ, channels as in part (b) of proposition 14.22 that hide coordinates read by the receiver rows on an assignment of weight bounded below, and parents that are time-homogeneous Markov on the retained assignment and coordinates or carry in addition a filter in the explicitly declared regular class of definition 12.39, at two levels (windows of fixed length uniformly up to the switching time, and the complete history over horizons on which the expected number of assignment changes diverges): a proof that every member admits such a parent whose retained and assignment path law is within a total variation tending to zero with ϵ of the native one; or a member for which every such parent keeps that distance bounded below.
A finite-dimensional description of coarse link laws	open problem 13.21	For $G = \mathrm{GL}^+(K)$ with $K \geq 2$, a weakly continuous injective map from a connected open subset of some $\mathbb{R}^d$ into the Borel probability laws on $\mathrm{GL}^+(K)$ whose image contains the image under the exponential of every nondegenerate Gaussian law on $\mathfrak{gl}(K, \mathbb{R})$ and contains the convolution of any two of its laws; or a proof that for some $K \geq 2$ no such map exists.

continued on the next page

Item	Locator	Settling condition
Accounting across episodes	open problem 13.29	For label-sufficient latent routes of definition 13.26, whose episode model is that of the parameter route with a finite label coordinate, the posterior slice as recognition coupling, and the aggregate update of constant step ϵ_{e}, whose read-out reads the label and an auxiliary preference coordinate only, and whose records are silent and informative or have likelihood vectors spanning a cone on which no nonzero polynomial vanishes: a route whose per-episode discrepancy from the aggregate update applied to its own filtered label law and label posterior is $o(\epsilon_{\mathrm{e}})$ uniformly over record histories and over episodes up to any fixed multiple of $1/\epsilon_{\mathrm{e}}$; or a lower bound of order ϵ_{e} on that discrepancy for every such route.
Persistent multiscale hierarchy	open problem 13.30	For models at c_* with structural marks from a finite partition catalog inside both laws, descending the multiscale free energy by a reversible diffusion at temperature T with rate-one redraws of the marks, with energies, precisions, prior ratios, hyperparameters, and mobilities independent of T, as $\mathrm{T} \to 0$: three results together, a lineage of at least two levels, formed with probability bounded below from every initial law with a density and a finite second moment, whose second level depends on the first (a declared handoff comparison with a margin bounded below over counterfactual handoffs, and no occurrence with probability tending to one uniformly over the handoff class) and is a coherence episode (a restoring clause at a declared resolution and a horizon clause), a nondegenerate intrinsic partition selector, and a separated slow manifold on which depth-$(\ell+1)$ variables feed back consistently into depth-ℓ kernels; or a proof that in this class no second level produced by descent, apart from a supplied-scale sequence whose event occurs with probability tending to one uniformly over the handoff class, meets the restoring and horizon conditions and exceeds its counterfactual statistic by such a margin.
Part III: formation and interpretation		
Sharpness of the doubly exponential regime	open problem 14.16	In the family of theorem 14.17 under (H1), (H3), and (H4), with power-law strengths satisfying $3\mathrm{Q}_\ell \geq 4\mathrm{Q}_{\ell+1} + 20$ and $\mathrm{Q}_\ell \geq 14$ below the top, the regime formed from Q_1, and an entrance constant above the thresholds of the theorem and of proposition 14.15: conclusions (a) to (e) for every such schedule with constants and preparation neighborhoods independent of L, through an early estimate of the retained Fisher information by the future-layer energy, a history-environment bound without the top-interface recursion, and depth-uniform flag and jump constants; or an admitted preparation and a schedule of the class under which a conclusion fails, or a lower bound showing that a conclusion cannot hold with constants independent of L.

continued on the next page

Item	Locator	Settling condition
A family in which learning or holonomy selects the components	open problem 14.20	On the four-cycle of proposition 15.4, with its temperature, learning mobility, and horizon, Gaussian initial laws $\mathcal{N}(\mathrm{m}_0, I_{12})$ with m_0 outside a Lebesgue-null set, and parents of the type of part (b) of proposition 14.22: such parents with separately vanishing conditional and dynamical errors, together with a total-variation difference bounded below as $\epsilon \to 0$ between the conditional laws, given contrast formation, of the first formed block in the learned and the frozen-row arm, or between those of two initial means that differ only in their loop coordinate; or a proof that for almost every m_0 both conditional differences vanish as $\epsilon \to 0$.
Repeated endogenous formation	open problem 14.25	For models at c_* with source-gated attention on a finite connected graph, outside the family of definition 14.23, whose energies, precisions, prior ratios, hyperparameters, and mobilities do not depend on T, as $\mathrm{T} \to 0$, and a second coalescence declared by a handoff time, first- and second-stage events with their blocks, one statistic of the handoff state, and a handoff class fixed before the comparison: such a coalescence, with learned generative parameters and from initially nonmatching assignments, that does not occur with probability tending to one uniformly over the handoff class, whose statistic exceeds its counterfactual value by a margin bounded below, whose block is held at a declared resolution by a restoring interaction on horizons short compared with the stationary entrance time of the event, and whose parents have separately vanishing conditional and dynamical errors; or a proof that in this class every coalescence of first-stage blocks on such horizons fails the margin or the restoring clause or has a second-stage event occurring with probability tending to one uniformly over the handoff class.
Physical-law identification	open problem 15.7	For a named target system whose data the model does not generate, observed through a declared record kernel that under the model's own law separates two objects of the coarse-graining differing in a gauge-invariant datum, with an observable, an estimator, and a validation and uncertainty protocol set before the data: held-out evidence, robust across admissible blocking schemes, that an object of the coarse-graining meets a baseline margin with probability-level closure; or held-out evidence under the same protocol that the margin is not met.

continued on the next page

Item	Locator	Settling condition
Single-point testability	open problem 15.8	For a named target system at c_* whose data the model does not generate, observed with sampling units and a noise model through a declared record kernel that reads the links of a closed walk or the coordinates of a candidate block: a test fixed before the data, with an estimator of gauge-typed parameters, a held-out statistic, an acceptance tolerance, and baselines without link holonomy or without formation, whose failure would falsify nontrivial holonomy on that walk or formation of that block, with a proof that under the kernel the law of the tested statistic depends on that hypothesis; negatively, a proof that under the kernel every statistic of the records has the same law for two link configurations differing only in the conjugacy class of the walk's holonomy, or whether or not the block is coherent at the declared resolution.

Appendix F

Numerical Index

Table F.1 indexes the statements whose status is NUMERICAL in the sense of Table 1.1, each with its check, what the check tests, its control, and its seed. No statement of the main manuscript has that status: every lemma, proposition, theorem, corollary, and claim-bearing remark is proved here or cited to a primary source, and computations that corroborate its proved statements are not indexed here.

Table F.1: Numerical index of the main manuscript.

Claim	Check	What is tested	Control	Seed
No statement of the main manuscript has the status NUMERICAL.				

Bibliography

[1] P.-A. Absil, R. Mahony, and B. Andrews. Convergence of the iterates of descent methods for analytic cost functions. *SIAM Journal on Optimization*, 16(2):531–547, 2005. doi: 10.1137/040605266.

[2] Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, 2000.

[3] N. Ay, J. Jost, H.V. Lê, and L. Schwachhöfer. Information geometry and sufficient statistics. *Probability Theory and Related Fields*, 162:327–364, 2015.

[4] N. Ay, J. Jost, H. V. Lê, and L. Schwachhöfer. *Information Geometry*, volume 64 of *Ergebnisse der Mathematik und ihrer Grenzgebiete*. Springer, 2017.

[5] Julio Backhoff-Veraguas and Clara Unterberger. On the specific relative entropy between martingale diffusions on the line, 2023. URL https://arxiv.org/abs/2207.03312.

[6] Benjamin Bahr. Operator spin foams: holonomy formulation and coarse graining. *Journal of Physics: Conference Series*, 360:012042, 2012. doi: 10.1088/1742-6596/360/1/012042. Equation numbers cited refer to arXiv:1112.3567v1.

[7] Dominique Bakry and Michel Émery. Diffusions hypercontractives. *Séminaire de Probabilités*, 19:177–206, 1985. URL https://numdam.org/item/SPS_1985__19__177_0/.

[8] Martin Bauer, Martins Bruveris, and Peter W. Michor. Uniqueness of the Fisher–Rao metric on the space of smooth densities. *Bulletin of the London Mathematical Society*, 48(3):499–506, 2016. doi: 10.1112/blms/bdw020.

[9] Cédric Bény and Tobias J. Osborne. Information-geometric approach to the renormalization group. *Physical Review A*, 92(2):022330, 2015. doi: 10.1103/PhysRevA.92.022330.

[10] David S. Berman and Marc S. Klinger. The inverse of exact renormalization group flows as statistical inference. *Entropy*, 26(5):389, 2024. doi: 10.3390/e26050389.

[11] David S. Berman, Jonathan J. Heckman, and Marc Klinger. On the dynamics of inference and learning, 2022.

[12] David S. Berman, Marc S. Klinger, and Alexander G. Stapleton. Bayesian renormalization. *Machine Learning: Science and Technology*, 4(4):045011, 2023. doi: 10.1088/2632-2153/ad0102.

[13] Garrett Birkhoff. Extensions of Jentzsch's theorem. *Transactions of the American Mathematical Society*, 85(1):219–227, 1957. doi: 10.1090/S0002-9947-1957-0087058-6.

[14] C.M. Bishop. *Pattern Recognition and Machine Learning*. Springer, 2006.

[15] David Blackwell. Equivalent comparisons of experiments. *The Annals of Mathematical Statistics*, 24(2):265–272, 1953. doi: 10.1214/aoms/1177729032.

[16] Christian Borgs and Roman Kotecký. A rigorous theory of finite-size scaling at first-order phase transitions. *Journal of Statistical Physics*, 61(1–2):79–119, 1990. doi: 10.1007/BF01013955.

[17] Stéphane Boucheron, Gábor Lugosi, and Pascal Massart. *Concentration Inequalities: A Nonasymptotic Theory of Independence*. Oxford University Press, Oxford, 2013. ISBN 978-0-19-953525-5. doi: 10.1093/acprof:oso/9780199535255.001.0001.

[18] L. L. Campbell. An extended čencov characterization of the information metric. *Proceedings of the American Mathematical Society*, 98(1):135–141, 1986. doi: 10.1090/S0002-9939-1986-0848890-5.

[19] Eric A. Carlen. Superadditivity of Fisher's information and logarithmic Sobolev inequalities. *Journal of Functional Analysis*, 101(1):194–211, 1991. doi: 10.1016/0022-1236(91)90155-X.

[20] Ernst Cassirer. *Substance and Function and Einstein's Theory of Relativity*. Open Court, Chicago, 1923. Translated by W. C. Swabey and M. C. Swabey; originally published as Substanzbegriff und Funktionsbegriff, 1910.

[21] Alessio Catanzaro, Diego Garlaschelli, and Subodh P. Patil. Renormalization of interacting random graph models. *Physical Review E*, 113(2):024314, 2026. doi: 10.1103/34n8-pw8x.

[22] T. Cohen and M. Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning*, pages 2990–2999, 2016.

[23] Taco S. Cohen, Maurice Weiler, Berkay Kicanaoglu, and Max Welling. Gauge equivariant convolutional networks and the icosahedral CNN. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 2019. URL https://proceedings.mlr.press/v97/cohen19d.html.

[24] T.M. Cover and J.A. Thomas. *Elements of Information Theory*. Wiley-Interscience, Hoboken, 2nd edition, 2006.

[25] M. Creutz. *Quarks, Gluons and Lattices*. Cambridge University Press, Cambridge, 1983. ISBN 978-0521244053.

[26] Michael Creutz. Gauge fixing, the transfer matrix, and confinement on a lattice. *Physical Review D*, 15(4):1128–1136, 1977. doi: 10.1103/PhysRevD.15.1128.

[27] James P. Crutchfield and David P. Feldman. Regularities unseen, randomness observed: Levels of entropy convergence. *Chaos*, 13(1):25–54, 2003. doi: 10.1063/1.1530990.

[28] I. Csiszár. Information-type measures of difference of probability distributions and indirect observations. *Studia Scientiarum Mathematicarum Hungarica*, 2:299–318, 1967.

[29] Walter J. Culver. On the existence and uniqueness of the real logarithm of a matrix. *Proceedings of the American Mathematical Society*, 17(5):1146–1151, 1966.

[30] John M. Danskin. *The Theory of Max-Min and its Application to Weapons Allocation Problems*, volume 5 of *Ökonometrie und Unternehmensforschung / Econometrics and Operations Research*. Springer-Verlag, Berlin, Heidelberg, 1967. doi: 10.1007/978-3-642-46092-0. Chapter III, "The directional derivative", pp. 19–32.

[31] Clement Delcamp and Bianca Dittrich. Towards a phase diagram for spin foams. *Classical and Quantum Gravity*, 34(22):225006, 2017. doi: 10.1088/1361-6382/aa8f24. Equation and section numbers cited refer to arXiv:1612.04506v1.

[32] A.P. Dempster, N.M. Laird, and D.B. Rubin. Maximum likelihood from incomplete data via the EM algorithm. *Journal of the Royal Statistical Society Series B*, 39:1–38, 1977.

[33] R. L. Dobrushin. Central limit theorem for nonstationary Markov chains. I. *Theory of Probability and Its Applications*, 1(1):65–80, 1956. doi: 10.1137/1101006.

[34] James G. Dowty. Chentsov's theorem for exponential families. *Information Geometry*, 1(1): 117–135, 2018. doi: 10.1007/s41884-018-0006-4.

[35] Michael Esfeld and Vincent Lam. Moderate structural realism about space-time. *Synthese*, 160(1):27–46, 2008. doi: 10.1007/s11229-006-9076-2.

[36] N. Fenichel. Geometric singular perturbation theory for ordinary differential equations. *Journal of Differential Equations*, 31(1):53–98, 1979.

[37] Michael Friedman. *Dynamics of Reason*. CSLI Publications, Stanford, CA, 2001. ISBN 978-1575863467.

[38] K. J. Friston, N. Trujillo-Barreto, and J. Daunizeau. DEM: A variational treatment of dynamic systems. *NeuroImage*, 41(3):849–885, 2008.

[39] Karl J. Friston, Thomas Parr, Conor Heins, Axel Constant, Daniel Friedman, Takuya Isomura, Chris Fields, Tim Verbelen, Maxwell J. D. Ramstead, John Clippinger, and Christopher D. Frith. Federated inference and belief sharing. *Neuroscience & Biobehavioral Reviews*, 156: 105500, 2024. doi: 10.1016/j.neubiorev.2023.105500.

[40] Andrea Gabrielli, Diego Garlaschelli, Subodh P. Patil, and M. Ángeles Serrano. Network renormalization. *Nature Reviews Physics*, 7(4):203–219, 2025. doi: 10.1038/s42254-025-00817-5.

[41] Elena Garuccio, Margherita Lalli, and Diego Garlaschelli. Multiscale network renormalization: scale-invariance without geometry. *Physical Review Research*, 5(4):043101, 2023. doi: 10.1103/PhysRevResearch.5.043101.

[42] Michèle Giry. A categorical approach to probability theory. In *Categorical Aspects of Topology and Analysis*, volume 915 of *Lecture Notes in Mathematics*, pages 68–85. Springer, Berlin, Heidelberg, 1982. doi: 10.1007/BFb0092872.

[43] Robert M. Gray. *Entropy and Information Theory*. Springer-Verlag, New York, 1990. Corrected first edition, revised June 26, 2023, available at https://ee.stanford.edu/~gray/it.pdf; theorem and equation numbers cited refer to this version.

[44] Martin Hairer and Jonathan C. Mattingly. Yet another look at Harris' ergodic theorem for Markov chains. In *Seminar on Stochastic Analysis, Random Fields and Applications VI*, volume 63 of *Progress in Probability*, pages 109–117. Birkhäuser, 2011. doi: 10.1007/978-3-0348-0021-1_7. URL https://arxiv.org/abs/0810.2777.

[45] B.C. Hall. *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*. Springer, 2nd edition, 2015.

[46] Paul R. Halmos and L. J. Savage. Application of the Radon-Nikodym theorem to the theory of sufficient statistics. *The Annals of Mathematical Statistics*, 20(2):225–241, 1949. doi: 10.1214/aoms/1177730032.

[47] Jakob Hansen and Robert Ghrist. Toward a spectral theory of cellular sheaves. *Journal of Applied and Computational Topology*, 3:315–358, 2019. doi: 10.1007/s41468-019-00038-7.

[48] Jakob Hansen and Robert Ghrist. Opinion dynamics on discourse sheaves. *SIAM Journal on Applied Mathematics*, 2021. doi: 10.1137/20M1341088.

[49] Carsten Hartmann, Lara Neureither, and Upanshu Sharma. Coarse-graining of non-reversible stochastic differential equations: quantitative results and connections to averaging, 2020. URL https://arxiv.org/abs/1911.06081.

[50] Lingshen He, Yiming Dong, Yisen Wang, Dacheng Tao, and Zhouchen Lin. Gauge equivariant transformer. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://proceedings.neurips.cc/paper/2021/hash/e57c6b956a6521b28495f2886ca0977a-Abstract.html.

[51] Conor Heins, Beren Millidge, Lancelot Da Costa, Richard P. Mann, Karl J. Friston, and Iain D. Couzin. Collective behavior from surprise minimization. *Proceedings of the National Academy of Sciences*, 121(17):e2320239121, 2024. doi: 10.1073/pnas.2320239121.

[52] N.J. Higham. *Functions of Matrices: Theory and Computation*. SIAM, Philadelphia, 2008.

[53] Wassily Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. doi: 10.1080/01621459.1963.10500830.

[54] D.D. Hoffman, M. Singh, and C. Prakash. The interface theory of perception. *Psychonomic Bulletin & Review*, 22:1480–1506, 2015.

[55] Donald D. Hoffman. *The Case Against Reality: Why Evolution Hid the Truth from Our Eyes*. W. W. Norton & Company, New York, 2019. ISBN 978-0393254693.

[56] Matthew D. Hoffman and Matthew J. Johnson. ELBO surgery: Yet another way to carve up the variational evidence lower bound. In *Workshop in Advances in Approximate Bayesian Inference, 30th Conference on Neural Information Processing Systems (NIPS 2016)*, Barcelona, Spain, 2016. Checked against the author PDF; the average-KL decomposition is Eqs. (11)–(12).

[57] R.A. Horn and C.R. Johnson. *Matrix Analysis*. Cambridge University Press, 2nd edition, 2013.

[58] Jessica N. Howard, Marc S. Klinger, Anindita Maiti, and Alexander G. Stapleton. Bayesian RG flow in neural network field theories. *SciPost Physics Core*, 8(1):027, 2025. doi: $10.21468/\mathrm{SciPostPhysCore}.8.1.027$.

[59] Antonin Joly and Nicolas Keriven. Graph coarsening with message-passing guarantees. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: $10.52202/079017\text{-}3648$.

[60] Abram Kagan and Lawrence A. Shepp. A sufficiency paradox: An insufficient statistic preserving the Fisher information. *The American Statistician*, 59(1):54–56, 2005. doi: $10.1198/000313005\mathrm{X}21041$.

[61] Olav Kallenberg. *Foundations of Modern Probability*. Springer, Cham, 3 edition, 2021.

[62] Immanuel Kant. *Critique of Pure Reason*. Cambridge University Press, Cambridge, 1998. ISBN 978-0521657297. Originally published 1781/1787.

[63] V. A. Kazakov and A. A. Migdal. Induced gauge theory at large N. *Nuclear Physics B*, 397(1–2):214–238, 1993. doi: $10.1016/0550\text{-}3213(93)90342\text{-}\mathrm{M}$. The arXiv preprint is titled "Induced QCD at large N"; equation numbers cited refer to arXiv:hep-th/9206015v1.

[64] John G. Kemeny and J. Laurie Snell. *Finite Markov Chains*. Undergraduate Texts in Mathematics. Springer, New York, 1976. Reprint of the 1960 Van Nostrand edition.

[65] Richard Kenyon. Spanning forests and the vector bundle Laplacian. *The Annals of Probability*, 39(5):1983–2017, 2011. doi: $10.1214/10\text{-}\mathrm{AOP}596$. Theorem numbers cited refer to arXiv:1001.4028v3, which carries the published pagination.

[66] Mohammad Emtiyaz Khan and Håvard Rue. The Bayesian learning rule. *Journal of Machine Learning Research*, 24(281):1–46, 2023. URL http://jmlr.org/papers/v24/22-0291.html. Equation numbers cited refer to the JMLR version (arXiv:2107.04562v4).

[67] S. Kobayashi and K. Nomizu. *Foundations of Differential Geometry*, volume 1. Wiley, 1963.

[68] J. Kogut and L. Susskind. Hamiltonian formulation of Wilson's lattice gauge theories. *Physical Review D*, 11(2):395–408, 1975.

[69] S. Kullback and R.A. Leibler. On information and sufficiency. *Annals of Mathematical Statistics*, 22:79–86, 1951.

[70] James Ladyman. Structural realism. In Edward N. Zalta, editor, *The Stanford Encyclopedia of Philosophy*. Metaphysics Research Lab, Stanford University, winter 2014 edition, 2014. URL https://plato.stanford.edu/archives/win2014/entries/structural-realism/.

[71] James Ladyman, Don Ross, David Spurrett, and John Collier. *Every Thing Must Go: Metaphysics Naturalized*. Oxford University Press, Oxford, 2007. ISBN 978-0199276196. doi: 10.1093/acprof:oso/9780199276196.001.0001.

[72] Hông Vân Lê. The uniqueness of the Fisher metric as information metric. *Annals of the Institute of Statistical Mathematics*, 69:879–896, 2017.

[73] François Le Gland and Nadia Oudjane. Stability and uniform approximation of nonlinear filters using the Hilbert metric and application to particle filters. *The Annals of Applied Probability*, 14(1):144–187, 2004. doi: 10.1214/aoap/1075828050.

[74] John M. Lee. *Introduction to Smooth Manifolds*, volume 218 of *Graduate Texts in Mathematics*. Springer, New York, second edition, 2012. doi: 10.1007/978-1-4419-9982-5.

[75] Oren E. Livne and Achi Brandt. Lean algebraic multigrid (LAMG): Fast graph Laplacian linear solver. *SIAM Journal on Scientific Computing*, 34(4):B499–B522, 2012. doi: 10.1137/110843563.

[76] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society Series B*, 44(2):226–233, 1982. doi: 10.1111/j.2517-6161.1982.tb01203.x.

[77] Andreas Loukas. Graph reduction with spectral and cut guarantees. *Journal of Machine Learning Research*, 20(116):1–42, 2019.

[78] Pankaj Mehta and David J. Schwab. An exact mapping between the variational renormalization group and deep learning, 2014.

[79] Dov Monderer and Lloyd S. Shapley. Potential games. *Games and Economic Behavior*, 14 (1):124–143, 1996. doi: 10.1006/game.1996.0044.

[80] Sadao Nakajima. On quantum theory of transport phenomena. *Progress of Theoretical Physics*, 20(6):948–959, 1958. doi: 10.1143/PTP.20.948.

[81] Radford M. Neal and Geoffrey E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In Michael I. Jordan, editor, *Learning in Graphical Models*, volume 89 of *NATO ASI Series*, pages 355–368. Springer, Dordrecht, 1998. doi: 10.1007/978-94-011-5014-9_12. URL https://link.springer.com/chapter/10.1007/978-94-011-5014-9_12.

[82] Yvan Notay. An aggregation-based algebraic multigrid method. *Electronic Transactions on Numerical Analysis*, 37:123–146, 2010.

[83] Kenta Oono and Taiji Suzuki. Graph neural networks exponentially lose expressive power for node classification. In *International Conference on Learning Representations*, 2020.

[84] Manfred Opper and Cédric Archambeau. The variational Gaussian approximation revisited. *Neural Computation*, 21(3):786–792, 2009. doi: 10.1162/neco.2008.08-07-592. Equation numbers cited refer to the authors' web preprint dated 16 March 2009.

[85] Dénes Petz. Sufficiency of channels over von Neumann algebras. *The Quarterly Journal of Mathematics*, 39(1):97–108, 1988. doi: 10.1093/qmath/39.1.97.

[86] David Pollard. A note on insufficiency and the preservation of fisher information. In *From Probability to Statistics and Back: High-Dimensional Models and Processes*, volume 9 of *IMS Collections*, pages 266–275. Institute of Mathematical Statistics, Beachwood, OH, 2013. doi: 10.1214/12-IMSCOLL919. URL https://doi.org/10.1214/12-IMSCOLL919.

[87] Hans Reichenbach. *The Theory of Relativity and A Priori Knowledge*. University of California Press, Berkeley, 1965.

[88] Herbert Robbins. An empirical Bayes approach to statistics. In Jerzy Neyman, editor, *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Contributions to the Theory of Statistics*, pages 157–163. University of California Press, Berkeley, 1956. Checked against the Project Euclid chapter record.

[89] Imanol Schlag, Kazuki Irie, and Jürgen Schmidhuber. Linear transformers are secretly fast weight programmers. In Marina Meila and Tong Zhang, editors, *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 9355–9366. PMLR, 2021. URL https://proceedings.mlr.press/v139/schlag21a.html. Equation and section numbers cited refer to arXiv:2102.11174v3.

[90] Jürgen Schmidhuber. Learning to control fast-weight memories: An alternative to dynamic recurrent networks. *Neural Computation*, 4(1):131–139, 1992. doi: 10.1162/neco.1992.4.1.131.

[91] Biswa Sengupta, Arturo Tozzi, Gerald K. Cooray, Pamela K. Douglas, and Karl J. Friston. Towards a neuronal gauge theory. *PLoS Biology*, 14(3):e1002400, 2016. doi: 10.1371/journal.pbio.1002400.

[92] C.E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27: 379–423, 623–656, 1948.

[93] Amit Singer and Hau-Tieng Wu. Vector diffusion maps and the connection Laplacian. *Communications on Pure and Applied Mathematics*, 65(8):1067–1144, 2012. doi: 10.1002/cpa.21395.

[94] Morris Skibinsky. Adequate subfields and sufficiency. *The Annals of Mathematical Statistics*, 38(1):155–161, 1967. doi: 10.1214/aoms/1177699065.

[95] A. J. Stam. Some inequalities satisfied by the quantities of information of Fisher and Shannon. *Information and Control*, 2(2):101–112, 1959. doi: 10.1016/S0019-9958(59)90348-1.

[96] Peter Thestrup Waade, Christoffer Lundbak Olesen, Jonathan Ehrenreich Laursen, Samuel William Nehrer, Conor Heins, Karl Friston, and Christoph Mathys. As one and many: Relating individual and emergent group-level generative models in active inference. *Entropy*, 27(2):143, 2025. doi: 10.3390/e27020143.

[97] A. N. Tikhonov. Systems of differential equations containing small parameters in the derivatives. *Matematicheskii Sbornik (N.S.)*, 31(73)(3):575–586, 1952. In Russian.

[98] Jakub Tomczak and Max Welling. VAE with a VampPrior. In Amos Storkey and Fernando Perez-Cruz, editors, *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics*, volume 84 of *Proceedings of Machine Learning Research*, pages 1214–1223. PMLR, 2018. URL https://proceedings.mlr.press/v84/tomczak18a.html. Checked against the PMLR page and PDF; the aggregated posterior as ELBO-optimal prior is Eqs. (7)–(8).

[99] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, Oxford, 1980. ISBN 978-0198244271. doi: 10.1093/0198244274.001.0001.

[100] Petr Vaněk, Jan Mandel, and Marian Brezina. Algebraic multigrid by smoothed aggregation for second and fourth order elliptic problems. *Computing*, 56(3):179–196, 1996. doi: 10.1007/BF02238511.

[101] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, pages 5998–6008, 2017.

[102] Nikolai Nikolaevich Čencov. *Statistical Decision Rules and Optimal Inference*, volume 53 of *Translations of Mathematical Monographs*. American Mathematical Society, Providence, RI, 1982. ISBN 978-0821830512. Translated from Russian by Lev J. Leifman; originally published as *Statisticheskie reshayushchie pravila i optimalnye resheniya*, Moscow, Nauka, 1972.

[103] Pablo Villegas, Tommaso Gili, Guido Caldarelli, and Andrea Gabrielli. Laplacian renormalization group for heterogeneous networks. *Nature Physics*, 19(3):445–450, 2023. doi: 10.1038/s41567-022-01866-8.

[104] Christof Wetterich. Linear lattice gauge theory. *Nuclear Physics B*, 884:44–65, 2014. doi: 10.1016/j.nuclphysb.2014.04.002.

[105] John Archibald Wheeler. Information, physics, quantum: The search for links. In Wojciech H. Zurek, editor, *Complexity, Entropy, and the Physics of Information*, pages 3–28. Addison-Wesley, Redwood City, CA, 1990.

[106] K.G. Wilson. Confinement of quarks. *Physical Review D*, 10(8):2445–2459, 1974.

[107] J. Worrall. Structural realism: The best of both worlds? *Dialectica*, 43(1-2):99–124, 1989.

[108] C. F. Jeff Wu. On the convergence properties of the EM algorithm. *The Annals of Statistics*, 11(1):95–103, 1983.

[109] Zhitao Ying, Jiaxuan You, Christopher Morris, Xiang Ren, William L. Hamilton, and Jure Leskovec. Hierarchical graph representation learning with differentiable pooling. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

[110] Robert Zwanzig. Ensemble method in the theory of irreversibility. *The Journal of Chemical Physics*, 33(5):1338–1341, 1960. doi: $10.1063/1.1731409$.